Synopsys® Security IP

MACsec Single Port Automotive Compliant (AC) Security Module

Databook

DWC 1G-25G MACsec AC Security Module - Product Code: H315-0
DWC 10G-100G MACsec AC Security Module - Product Code: H316-0

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March 2025

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Revision History

The following table provides the history of changes to this *Databook*.

Version	Date	Description
6585-001	March 2025	■ Initial release

Preface

This databook defines the structure and usage of the Synopsys® Security IP MACsec Single Port Automotive Compliant (AC) Security Module (DWC_macsec) core.

Audience

This guide is intended for users who are familiar with the following specification:

- FIPS Publication 197 - Advanced Encryption Standard
- NIST SP 800-38A - Block Cipher Modes of Operation
- NIST SP 800-38D - Galois/Counter Mode (GCM) and GMAC
- ARM IHI 0024C - AMBA APB Protocol
- IEEE 802.1AE-2018 Standard
- IEEE 802.3-2018 Standard
- IEEE 802.1Q-2018 Standard

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Databook Organization

The chapters of this databook are organized as follows:

- Chapter 1, “Product Overview”, provides an introduction to the DWC_macsec.
- Chapter 2, “Architecture”, provides architecture details of the DWC_macsec.
- Chapter 3, “Performance”, provides details about the performance settings.
- Chapter 4, “Parameter Descriptions”, provides descriptions of the configuration parameters.
- Chapter 5, “Signal Descriptions”, provides descriptions of the DWC_macsec signals.
- Chapter 6, “Register Descriptions”, provides the memory map of the DWC_macsec and descriptions of the programmable software registers.
- Appendix A, “Acronyms”, provides the list of acronyms used in this document.

Customer Support

Note for Readers

Note The product name *DesignWare Cores* is updated to *Synopsys IP*.

Related Documentation

See the following documents from the corekit installation directory or get the latest version from SolvNetPlus.

- *coreConsultant User's Guide*
- *coreAssembler User's Guide*

Web Resources

The following web links are various Synopsys online resources you may find useful:

- Synopsys Security IP product information: https://www.synopsys.com/designware-ip.html
- Your custom Synopsys Security IP page: https://www.synopsys.com/dw/mydesignware.php
- Documentation through SolvNetPlus: https://solvnetplus.synopsys.com (Synopsys password required)
- Synopsys Common Licensing (SCL): https://www.synopsys.com/keys

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  c. Complete the mandatory fields marked with an asterisk and click **Save**.

  Make sure to include the following:

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  - **Product L2**: (Select the name that best matches your product)

  d. After creating the case, attach any debug files you created.

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same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IP implements industry-standard specifications that are currently under review to remove exclusionary language.

1 Product Overview

This chapter includes the following sections:

- "General Product Description" on page 12
- "Features" on page 13
- "Roadmap Features Candidates Not Supported" on page 13
- "Clock Requirements" on page 14
- "Reset Requirements" on page 15
- "Memory Requirements" on page 15
- "Supported Data Modes" on page 25
- "System Integration" on page 28
- "Module Startup" on page 33
- "Area" on page 35
- "Compliance With Quality Metrics" on page 42

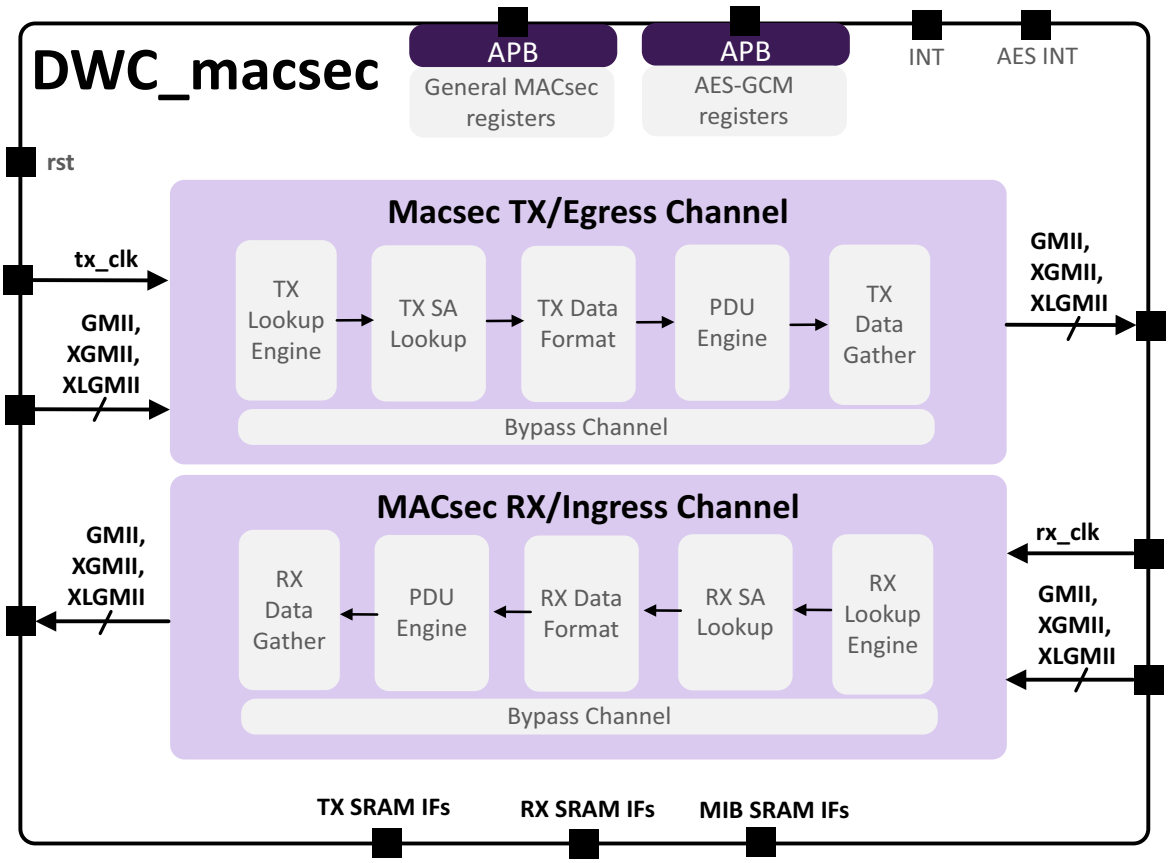
1.1 General Product Description

The DWC_macsec module is used for inline encryption/decryption and authentication of ethernet packets as defined by the *IEEE 802.1AE Standard*. The DWC_macsec module handles full line rate single port support for the *IEEE 802.1AE Standard* up to 100 Gb/s. The DWC_macsec provides encryption and authentication based on the AES-GCM algorithm, as specified in the *NIST Special Publication 800-38D Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC of November 2007*.

The DWC_macsec is developed to be integrated between a MAC (Media Access Controller) and a Physical Coding Sublayer (PCS) or PHY, and it offers GMII, XGMII and XLGMII interfaces as defined in the IEEE 802.3-2018 - IEEE Standard for Ethernet. Reduced interfaces such as RMII and RGMII are also supported and follow the RMII™ specification, RMII consortium, and Reduced Gigabit Media Independent Interface (RGMII) version 2.6 standards.

The DWC_macsec is developed to minimize latency, by offering an optimized AES-GCM cryptographic core for TX/Egress and RX/Ingress paths. Several additional features are also present in the DWC_macsec, such as bypass mode, parametrizable amount of TX and RX secure channels and security associations, support for different VLAN TAG on the clear, separate APB interfaces for general configuration registers and AES critical registers.

Figure 1-1 DWC MACsec AC Product Overview



1.2 Features

- Supports the following:
  - *IEEE 802.1AE-2018 Standard*
  - Frame Preemption (IEEE 802.1Qbu Standard) for lower performance data rates
  - Up to 100 Gb/s for Egress path
  - Up to 100 Gb/s for Ingress path
  - Up to 32 secure channels for TX
  - Up to 256 secure channels for RX
  - Up to 64 secure associations for TX
  - Up to 512 secure associations for RX
  - Up to two VLAN TAGs on the clear (for Cisco Mode)
  - VLAN TAG on the clear, copy and replace modes
  - Jumbo frame
  - Insertion and removal of SecTAG
  - RGMII, RMII, SGMII, MII, GMII, XGMII and XLGMII interfaces
  - APB for register access and key configuration
  - Separate APB for AES-GCM register access and key configuration
  - Asynchronous clock operation between TX and RX
  - Asynchronous clock operation between APB and datapath clocks (TX, RX)
  - Two rounds per clock for AES-GCM
  - Area optimized Galois Field for AES-GCM
  - Ternary Content Addressable Memory (TCAM) for security association search
  - Direct or linear search modes for security association lookup
  - FIPS 140-3 AES-GCM Test Mode
  - Proprietary tagging for correctly received frames
  - Configurable filtering rules for non-MACsec frames
  - SRAM ECC
- Seamless integration with Synopsys Ethernet controllers and PCS modules

1.3 Roadmap Features Candidates Not Supported

The following features are not supported for DWC_macsec in the current release:

- FIFO data streaming interfaces

Note The not supported features could be part of a specific roadmap release. Reach out to Synopsys for a complete roadmap description.

1.4 Clock Requirements

The DWC_macsec has three clock input signals. The clock input signals are connected to the DWC_macsec module sub-parts as shown in Figure 1-2.

- One clock for the APB interfaces
- One clock for the TX datapath
- One clock for the RX datapath

Figure 1-2 Clock Architecture Overview

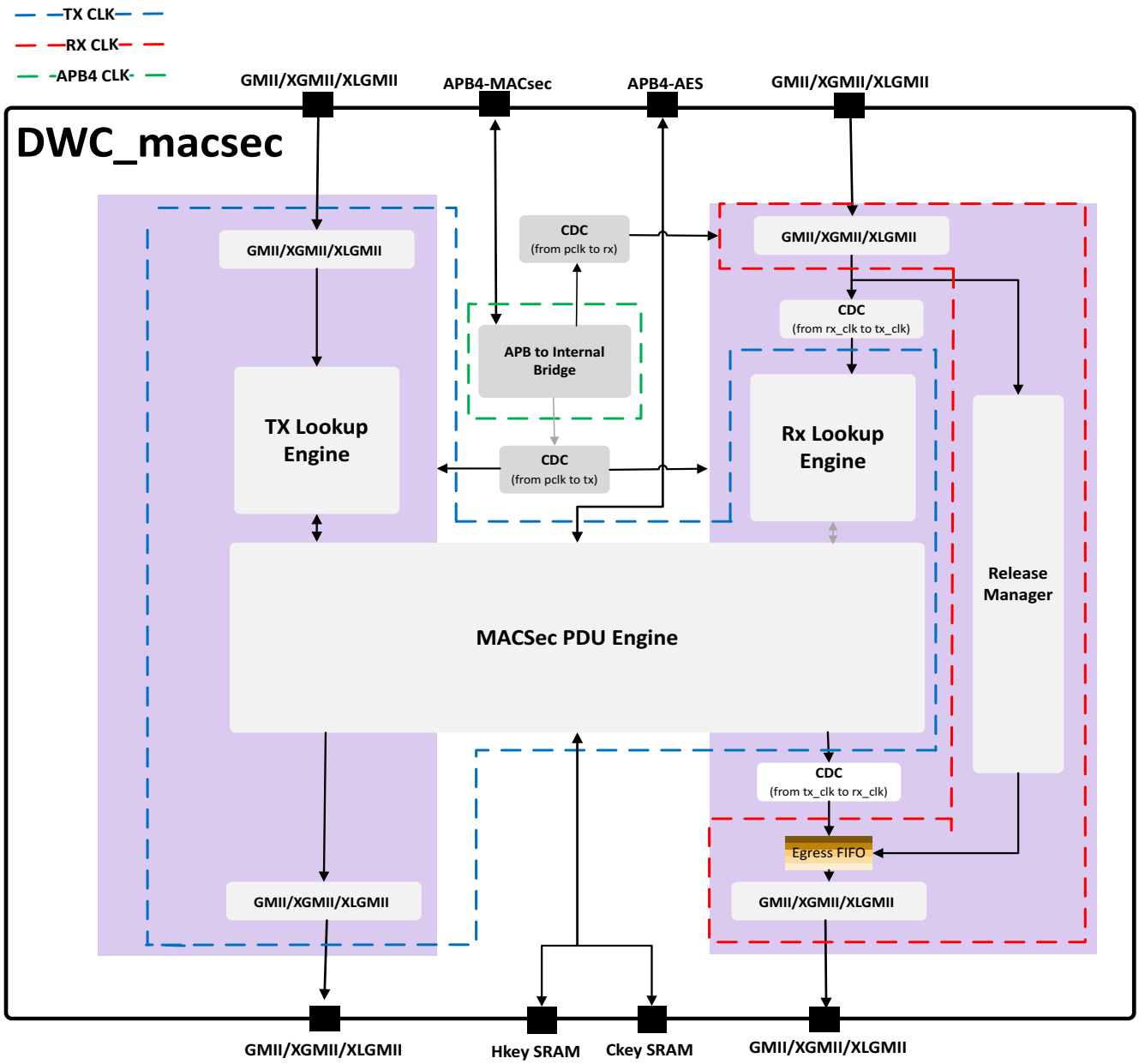


Table 1-1 describes considerations for clock connectivity and operating frequency.

Table 1-1 Clock Summary Description

Clock	Description	Dependency	Maximum Frequency
tx_clk	■ Clock for the TX/Egress path ■ This clock must be synchronous with the MAC TX/Egress clock ■ This clock can be asynchronous to the rx_clk and apb_clk	DWC_MACSEC_DATA_IF	125/312.5 MHz for GMII 781.25 MHz for 128 bits XLGMII
rx_clk	■ Clock for the TX/Egress path ■ This clock must be synchronous with the MAC TX/Egress clock ■ This clock can be asynchronous to the rx_clk and apb_clk	DWC_MACSEC_DATA_IF	125/312.5 MHz for GMII 781.25 MHz for 128 bits XLGMII
apb_clk	■ Clock for the APB Interfaces ■ Can be asynchronous to the tx_clk and rx_clk	-	400 MHz[a]

a. Maximum frequency value **highly** depends on the core configuration and target technology node. For frequencies higher than those provided in the Table 1-1, contact Synopsys Support.

1.5 Reset Requirements

The DWC_macsec has one reset signal, as shown in Figure 1-3.

- Power on Reset (pwr_on_rst): resets the complete IP

Table 1-2 explains considerations for the reset connectivity.

Table 1-2 Reset Summary Description

Reset	Description	Dependencies	Active	Assertion
pwr_on_rst	Resets the complete core. It is synchronized internally to the different clock domains	-	High	■ Asynchronous ■ De-assertion done internally

1.6 Memory Requirements

The DWC_macsec has following memories:

- 1x SRAM or TCAM for the Egress Channel/TX SAI (Secure Association Index) lookup
- 1x SRAM for the Egress Channel/TX SAD (Secure Association) lookup
- 1x SRAM for the Egress Channel/TX SA MIB data
- 1x SRAM for the Egress Channel/TX SC (Secure Channel) MIB data
- 1x SRAM or TCAM for the Ingress Channel/RX SAI lookup

- 1x SRAM for the Ingress Channel/RX SAD lookup
- 1x SRAM for the Ingress Channel/RX SA MIB data
- 2x SRAM for the Ingress Channel/RX SC MIB data
- 1x SRAM for the AES-GCM cipher key data
- 1x SRAM for the AES-GCM hash key data
- 1x SRAM for the AES-GCM hash context data
- 1x SRAM for the AES-GCM cipher context data

All external SRAM have a dedicated interface where the SoC integrator must connect their SRAM instances. All external SRAM interfaces operate in the `tx_clk` domain.

1.6.1 External SRAMs versus Internal Synthesizable Flip-Flops

The memories can be external (SRAMs/TCAMs instantiated by the integrator and connected on the SoC) or internal (synthesizable Flip-Flops). The integrator decides if the memories are implemented in external SRAMs/TCAMs or internal synthesizable Flip-Flops (FFs).

The internal/external implementation is configured by the parameters mentioned in the Table 1-3:

Table 1-3 Types of SRAM [a] Memories

Memory Name	Memory Type	Parameter Dependency
TX SAD	TP SRAM [b] or FFs	DWC_MACSEC_XPN_SEL == 1 DWC_MACSEC_TX_SAD_MEM_TYPE
		DWC_MACSEC_XPN_SEL == 0 DWC_MACSEC_TX_SAD_MEM_TYPE
TX SALT	TP SRAM or FFs	DWC_MACSEC_XPN_SEL == 1 DWC_MACSEC_TX_SALT_MEM_TYPE
TX SAI	TP SRAM, TCAM or FFs	DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==2 DWC_MACSEC_TX_SAI_MEM_TYPE DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD
		DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==1 DWC_MACSEC_TX_SAI_MEM_TYPE DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD
		DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==0 DWC_MACSEC_TX_SAI_MEM_TYPE DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD

Table 1-3 Types of SRAM [a] Memories

Memory Name	Memory Type	Parameter Dependency
RX SAI	TP SRAM, TCAM or FFs	DWC_MACSEC_RX_SAI_MEM_TYPE DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==2 DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD
		DWC_MACSEC_RX_SAI_MEM_TYPE DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==1 DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD
		DWC_MACSEC_RX_SAI_MEM_TYPE DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==0 DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD
RX SAD	DP SRAM [2] or FFs	DWC_MACSEC_XPN_SEL == 1 DWC_MACSEC_RX_SAD_MEM_TYPE
		DWC_MACSEC_XPN_SEL == 0 DWC_MACSEC_RX_SAD_MEM_TYPE
RX SALT	TP SRAM or FFs	DWC_MACSEC_XPN_SEL == 1 DWC_MACSEC_RX_SALT_MEM_TYPE
RX SAI SC CORRELATION TABLE	FFs	-
RX ANRW SRAM	TP SRAM or FFs	DWC_MACSEC_WIN_SIZE_WIDTH DWC_MACSEC_RX_ARW_MEM_TYPE
AES-GCM Cipher Key	TP SRAM or FFs	DWC_MACSEC_MAX_KEY_SIZE == 0 DWC_UAES_CFG_SYNTH_CKEY_MEM DWC_MACSEC_AES_FIPS_EN
		DWC_MACSEC_MAX_KEY_SIZE == 1 DWC_UAES_CFG_SYNTH_CKEY_MEM DWC_MACSEC_AES_FIPS_EN
AES-GCM Hash Key	TP SRAM or FFs	DWC_MACSEC_PERF_RATE <= 2 DWC_UAES_CFG_SYNTH_HKEY_MEM DWC_MACSEC_AES_FIPS_EN
		DWC_MACSEC_PERF_RATE >= 3 DWC_UAES_CFG_SYNTH_HKEY_MEM DWC_MACSEC_AES_FIPS_EN
AES-GCM Cipher Context	TP SRAM or FFs	DWC_UAES_CFG_SYNTH_CCTX_MEM DWC_MACSEC_AES_FIPS_EN
TX MIB GBL	FFs	-
TX MIB SA	FFs or TP SRAM	DWC_MACSEC_TX_MIB_SA_MEM_TYPE

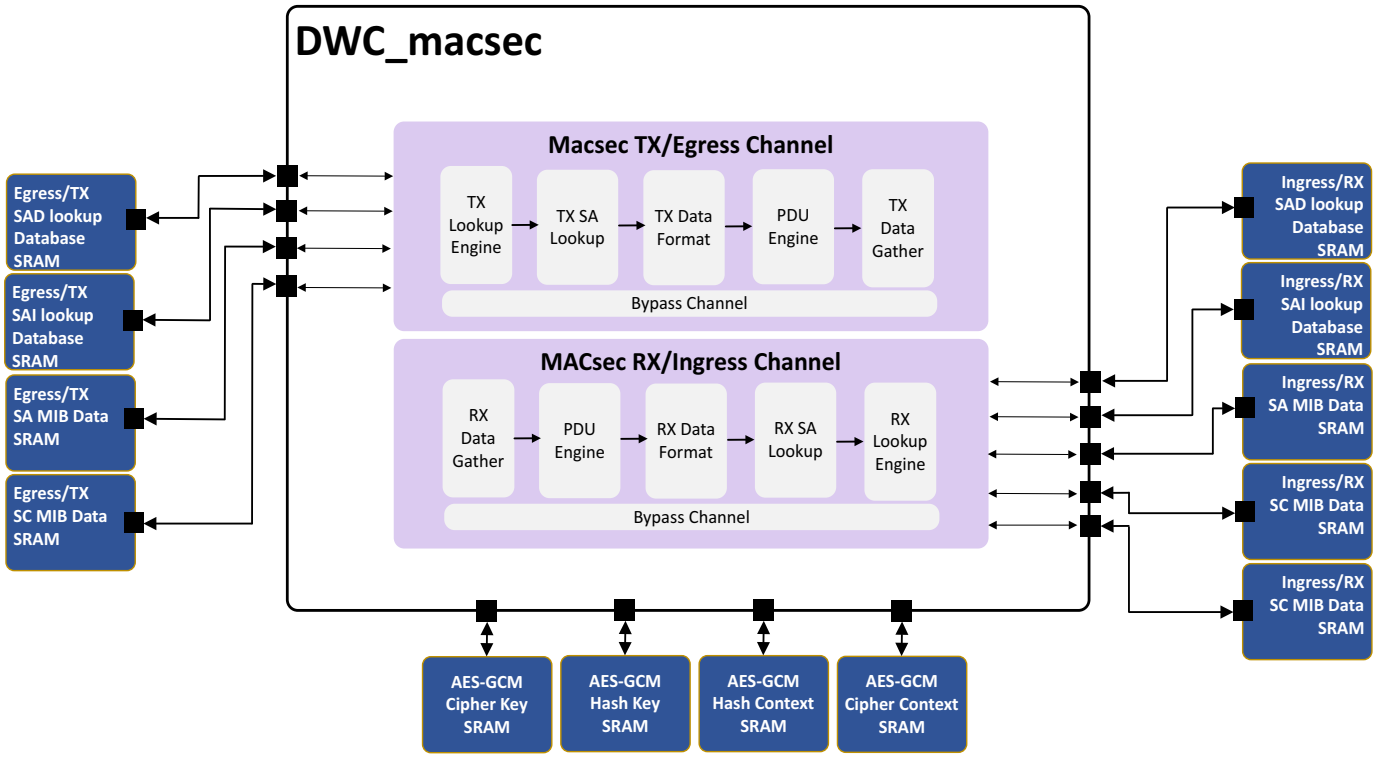
Table 1-3 Types of SRAM [a] Memories

Memory Name	Memory Type	Parameter Dependency
TX MIB SC	FFs	-
RX MIB GBL	FFs	-
RX MIB SA	FFs or TP SRAM	DWC_MACSEC_RX_MIB_SA_MEM_TYPE
RX MIB SC	FFs or TP SRAM	DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_MEM_TYPE
	FFs or TP SRAM	DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_MEM_TYPE

a. The SRAM instances are not part of the DWC_macsec.
b. TP SRAM means Two Port SRAM (one read port and one write port). DP SRAM means Dual Port SRAM (two ports with read/write capability).

A trade-off that needs to be evaluated by integrator to decide upon the selection is area versus latency. For example, if the amount of data to be stored in the memories is small, then implementing it in DWC_macsec internal synthesizable Flip-Flops is the best solution.

Figure 1-3 DWC MACsec AC Memory Overview



1.6.2 Memory Sizes

The memory requirements are dependent by the following parameters:

- **DWC_MACSEC_AES_MAX_KEY_SIZE:** The maximum AES-GCM key size selection is 128 or 256 bits.
- **DWC_MACSEC_TX_NUM_SC_SUPPORTED:** The maximum amount of TX Secure Channels. This influences the SRAMs depth associated with the Egress/TX.
- **DWC_MACSEC_RX_NUM_SA_SUPPORTED:** The maximum amount of RX secure associations. This influences the SRAMs depth associated with the Ingress/RX.

Table 1-4 describes the width and depth for all the DWC_macsec SRAMs. The integrator can via CoreConsultant to enable internal flip-flops. Table 1-4 also describes memory sizes calculation that can be used as Flip-Flop count estimate.

Note In the DWC_macsec CoreConsultant view, the SRAMs sizes are calculated automatically and shown in the GUI.

Table 1-4 Memory Sizes Calculation

Memory	Dependencies	Depth (rows)	Width (bit)
AES-GCM Hash Key Data SRAM	DWC_MACSEC_AES_FIPS_EN=0 && DWC_MACSEC_PREEMPTION_EN=0	Memory Baseline: DWC_MACSEC_TX_NUM_SA_SUPPORTED + DWC_MACSEC_RX_NUM_SA_SUPPORTED + 1	DWC_MACSEC_SRAM_ECC_SUPPORT=1 -> 294 DWC_MACSEC_SRAM_ECC_SUPPORT=0 -> 256
	(DWC_MACSEC_AES_FIPS_EN=1 || DWC_MACSEC_PREEMPTION_EN=1)	Add +1 bit to baseline	
AES-GCM Cipher Key Data SRAM	DWC_MACSEC_AES_FIPS_EN=0 && DWC_MACSEC_PREEMPTION_EN=0	Memory Baseline: DWC_MACSEC_TX_NUM_SA_SUPPORTED + DWC_MACSEC_RX_NUM_SA_SUPPORTED + 1	DWC_MACSEC_SRAM_ECC_SUPPORT=1: DWC_MACSEC_MAX_KEY_SIZE == 128 -> 151 DWC_MACSEC_MAX_KEY_SIZE ==256 -> 297 DWC_MACSEC_SRAM_ECC_SUPPORT=0: DWC_MACSEC_MAX_KEY_SIZE == 128 -> 129 DWC_MACSEC_MAX_KEY_SIZE ==256 -> 259
	(DWC_MACSEC_AES_FIPS_EN=1 || DWC_MACSEC_PREEMPTION_EN=1)	Add +1 bit to baseline	
AES-GCM Cipher Context Data SRAM	DWC_MACSEC_AUTOMOTIVE=0 && DWC_MACSEC_AES_FIPS_EN=0 && DWC_MACSEC_PREEMPTION_EN=0	Memory Baseline: 2*DWC_MACSEC_TX_NUM_SA_SUPPORTED + 2*DWC_MACSEC_RX_NUM_SA_SUPPORTED	DWC_MACSEC_SRAM_ECC_SUPPORT=1 -> 150 DWC_MACSEC_SRAM_ECC_SUPPORT=0 -> 128
	(DWC_MACSEC_AES_FIPS_EN=1 || DWC_MACSEC_PREEMPTION_EN=1)	Add +1 bit to baseline	

1.6.3 Memory Access Timings

Figure 1-4 shows memory timings that must be adhered when instantiating technology-specific memory instances. Different memories can be configured depending on the selected configuration of the DWC_macsec.

- **Single Port SRAM**
  - External SRAM containing just one single port, that is used for reading or writing.
- **Two Port SRAM**
  - External SRAM containing one port for reading and one port for writing.
  - Read and write ports use the tx_clk.
- **Dual Port SRAM**
  - External SRAM containing two ports that can be used for Read+Read, Read+Write or Write+Write parallel operations.
  - Read and write ports use the tx_clk.
- Ternary Content Addressable Memory (TCAM)
  - External SRAM that contains a Key input to be matched port, a Mask input port, a read/write port and a Key search hit output port.
  - Key search and read/write port use tx_clk.

Figure 1-4 Memory Access Timings (non-TCAM SRAMs)

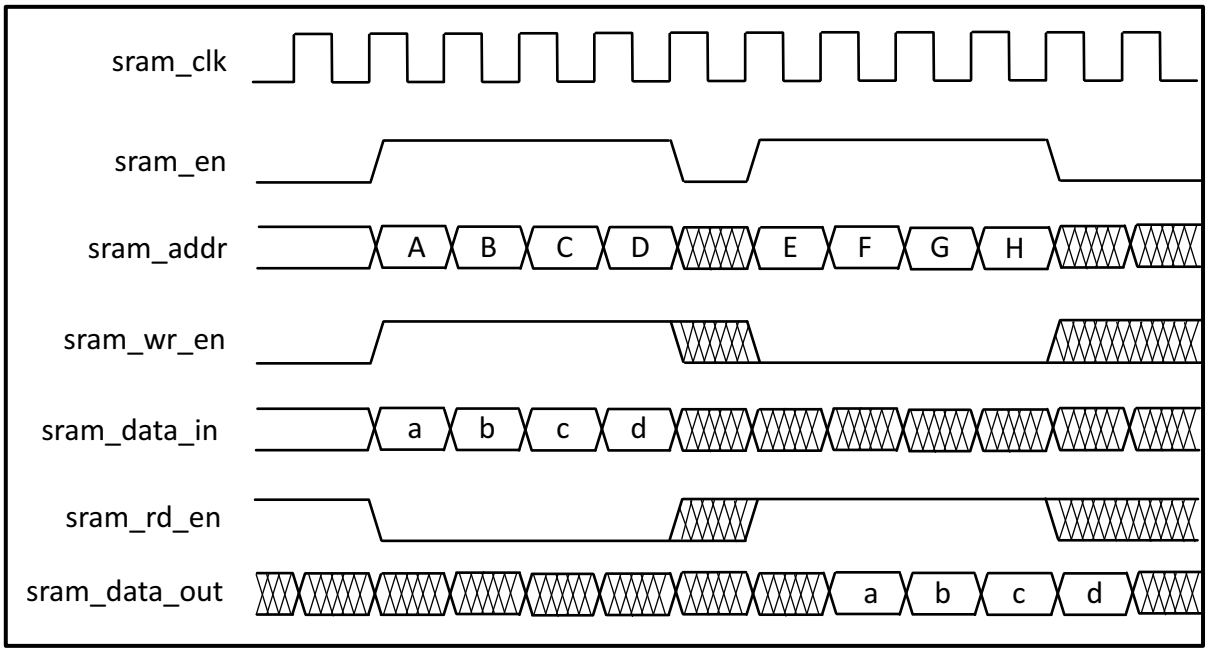


Figure 1-5 Memory Access Timings - TCAM SRAM Reset, Zeroization and Flush

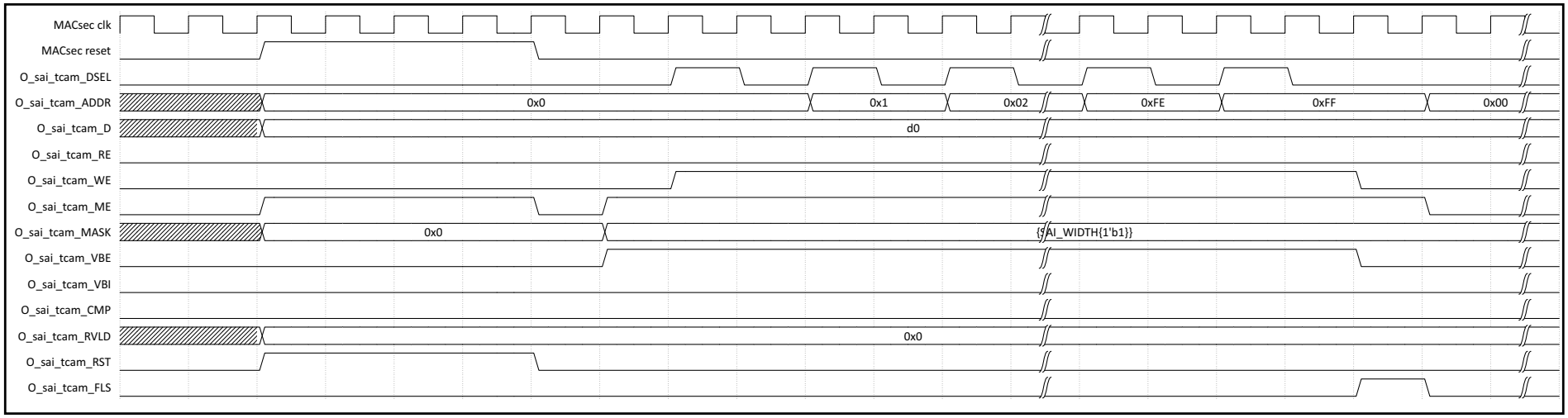


Figure 1-6 Memory Access Timings - TCAM SRAM Write

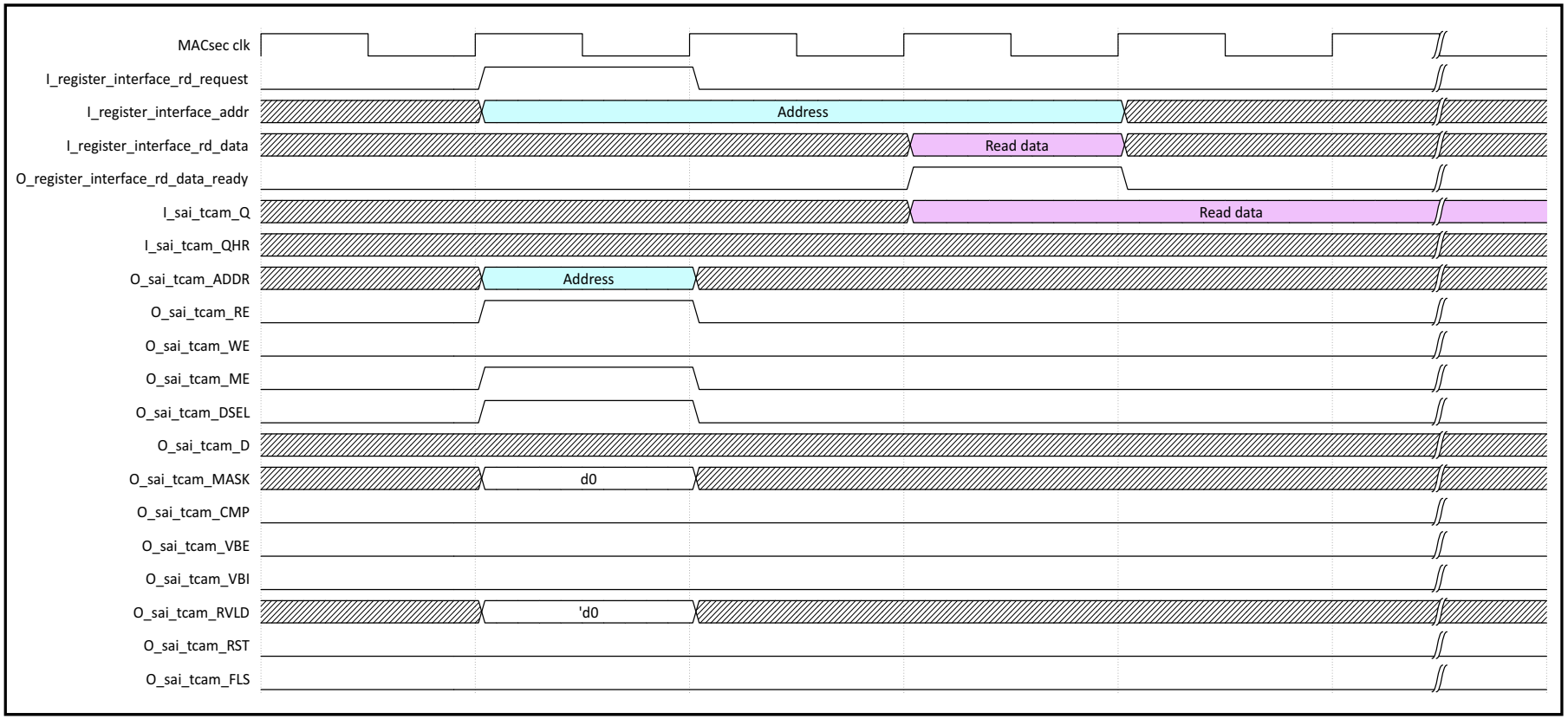


Figure 1-7 Memory Access Timings - TCAM SRAM Read

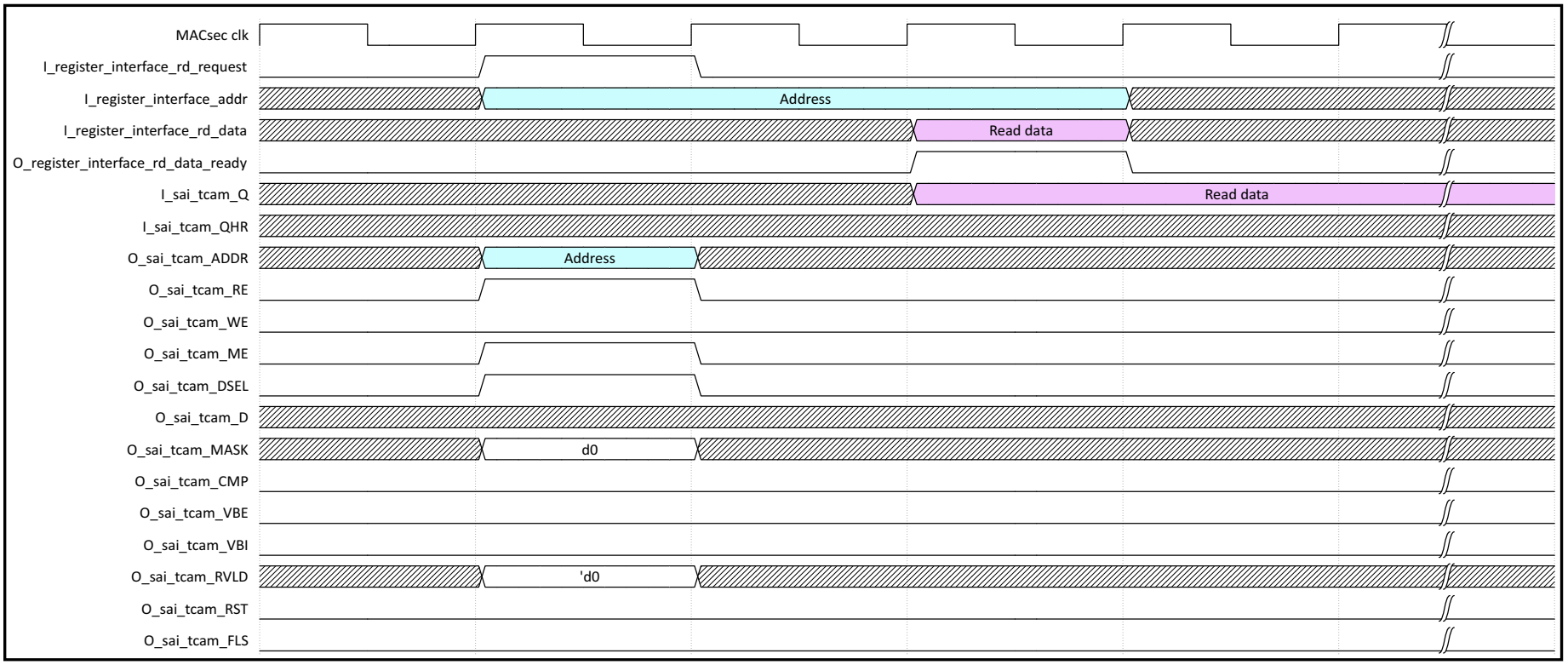
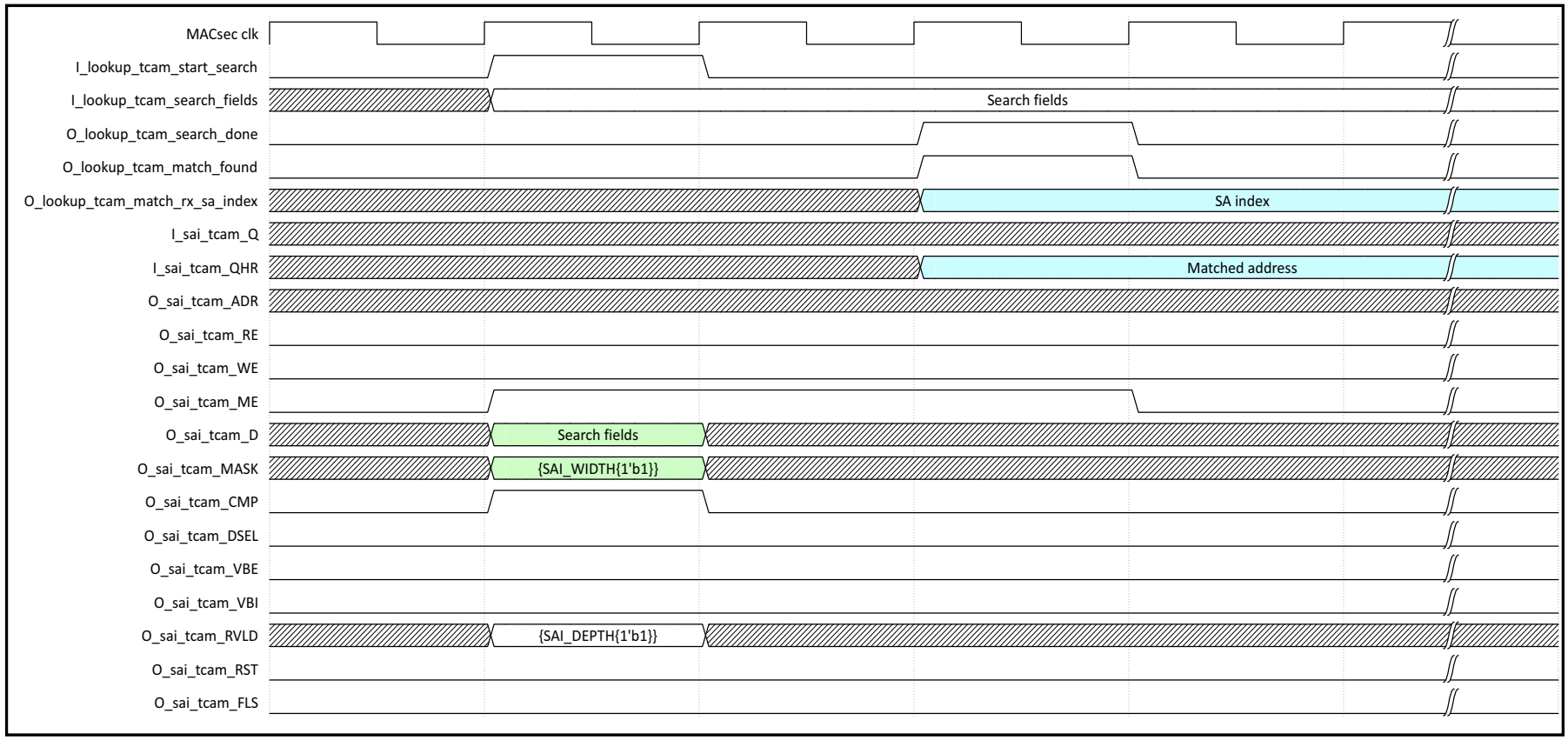


Figure 1-8 Memory Access Timings - TCAM SRAM Compare



1.7 Supported Data Modes

DWC_macsec supports the data modes mentioned in the Table 1-5.

Table 1-5 Types of Data Modes

Data Mode Configuration Value	Data Mode Description	Data Mode available for the following configuration parameters
5'd0	Use the 4-bit MII interface over the 8-bit GMII interface at 10Mbps, 2.5MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd1	Use the 4-bit MII interface over the 8-bit GMII interface at 100Mbps, 25MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd2	Use the full 8-bit GMII interface at 1Gbps, 125MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd3	Use the full 8-bit GMII interface at 2.5Gbps, 312.5MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd4	Use a 32-bit XGMII interface over the 128-bit XLGMII interface at 2.5Gbps, 78.125MHz	DWC_MACSEC_PERF_RATE >= 1));
5'd5	Use a 32-bit XGMII interface over the 128-bit XLGMII interface at 5Gbps, 156.25MHz	DWC_MACSEC_PERF_RATE >= 1));
5'd6	Use a 32-bit XGMII interface over the 128-bit XLGMII interface at 10Gbps, 312.5MHz	DWC_MACSEC_PERF_RATE >= 1));
5'd7	Use a 32-bit XGMII interface over the 128-bit XLGMII interface at 25Gbps, 781.25MHz	DWC_MACSEC_PERF_RATE >= 2));
5'd8	Use a 64-bit XLGMII interface over the 128-bit XLGMII interface at 10Gbps, 156.25MHz	DWC_MACSEC_PERF_RATE >= 2) && !DWC_MACSEC_AUTOMOTIVE)));
5'd9	Use a 64-bit XLGMII interface over the 128-bit XLGMII interface at 25Gbps, 390MHz/395MHz	DWC_MACSEC_PERF_RATE >= 2) && !DWC_MACSEC_AUTOMOTIVE)));
5'd10	Use a 64-bit XLGMII interface over the 128-bit XLGMII interface at 40Gbps, 625MHz	DWC_MACSEC_PERF_RATE >= 3) && !DWC_MACSEC_AUTOMOTIVE)));
5'd11	Use a 64-bit XLGMII interface over the 128-bit XLGMII interface at 50Gbps, 781MHz	DWC_MACSEC_PERF_RATE >= 3) && !DWC_MACSEC_AUTOMOTIVE)));
5'd12	Use the full XLGMII 128-bit interface at 25Gbps, 195.3125MHz	DWC_MACSEC_PERF_RATE >= 2) && !DWC_MACSEC_AUTOMOTIVE)));

Table 1-5 Types of Data Modes (Continued)

Data Mode Configuration Value	Data Mode Description	Data Mode available for the following configuration parameters
5'd13	Use the full XLGMII 128-bit interface at 25Gbps, 390.625MHz alternate cycle mode	DWC_MACSEC_PERF_RATE >= 2) && !DWC_MACSEC_AUTOMOTIVE)));
5'd14	Use the full XLGMII 128-bit interface at 40Gbps, 312.5MHz	DWC_MACSEC_PERF_RATE >= 3) && !DWC_MACSEC_AUTOMOTIVE)));
5'd15	Use the full XLGMII 128-bit interface at 50Gbps, 390.625MHz	DWC_MACSEC_PERF_RATE >= 3) && !DWC_MACSEC_AUTOMOTIVE)));
5'd16	Use the full XLGMII 128-bit interface at 100Gbps, 781.25MHz	DWC_MACSEC_PERF_RATE == 4) && !DWC_MACSEC_AUTOMOTIVE)));
5'd17	Use the 2-bit RMII interface over the 8-bit GMII interface at 10Mbps, 50MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd18	Use the 2-bit RMII interface over the 8-bit GMII interface at 100Mbps, 50MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd19	Use the 4-bit RGMII interface over the 8-bit GMII interface at 10Mbps, 2.5MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd20	Use the 4-bit RGMII interface over the 8-bit GMII interface at 100Mbps, 25MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));
5'd21	Use the 4-bit RGMII interface over the 8-bit GMII interface at 1Gbps, 125MHz	DWC_MACSEC_PERF_RATE = 0) || DWC_MACSEC_PERF_RATE > 0) && DWC_MACSEC_DUAL_IF = 1))));

The availability of data modes is dependent on the IP configuration.

1.8 Frame Preemption Usage Requirements

1.8.1 Initial setup

When the `PHY_MAP` bit in the `MIB_VALIDATE` register is set, all non-MACsec frames are poisoned. Therefore, do not set this bit in the initial setup phase so that response frames can pass through MACsec, which enables frame preemption.

1.8.2 Data Modes

Frame Preemption is supported for all performance modes.

1.8.3 Fragment Sizes

Frame Preemption is only supported for ingress Tx fragments size equal to or higher than 70 bytes (including CRC).

Rx MAC controller must accept fragment sizes equal to or higher than 48 bytes (including CRC). For more information on configuring fragment sizes, see *Ethernet MAC controller Databook*.

1.9 System Integration

This section describes the integration of the DWC_macsec within a SoC.

1.9.1 Parameter Dependency versus the Ethernet Controller

Table 1-6 describes the DWC_macsec parameter dependencies towards the Ethernet Controller.

Table 1-6 Parameter with Ethernet Controller Dependencies

Parameter	Description
DWC_MACSEC_PERF_RATE	This parameter selects the maximum throughput supported by the DWC_macsec and the associated data interface. The interface needs to match the one interface used in the Ethernet Controller.
DWC_MACSEC_DUAL_IF	This parameter when enabled, generates a DWC_macsec IP that contains two interfaces: GMII and XGMII. This allows the integrator to connect both to the MAC and PCS, and then at boot time select which interface (and associated throughput) is to be used.

1.9.2 Interfaces Integration Overview

The signal list described in Chapter 5, “Signal Descriptions” is grouped into functional interfaces, so it is easier to depict the complete module integration. The interface grouping shown in Figure 1-9 follows the same interface/signal grouping described in Chapter 5, “Signal Descriptions”.

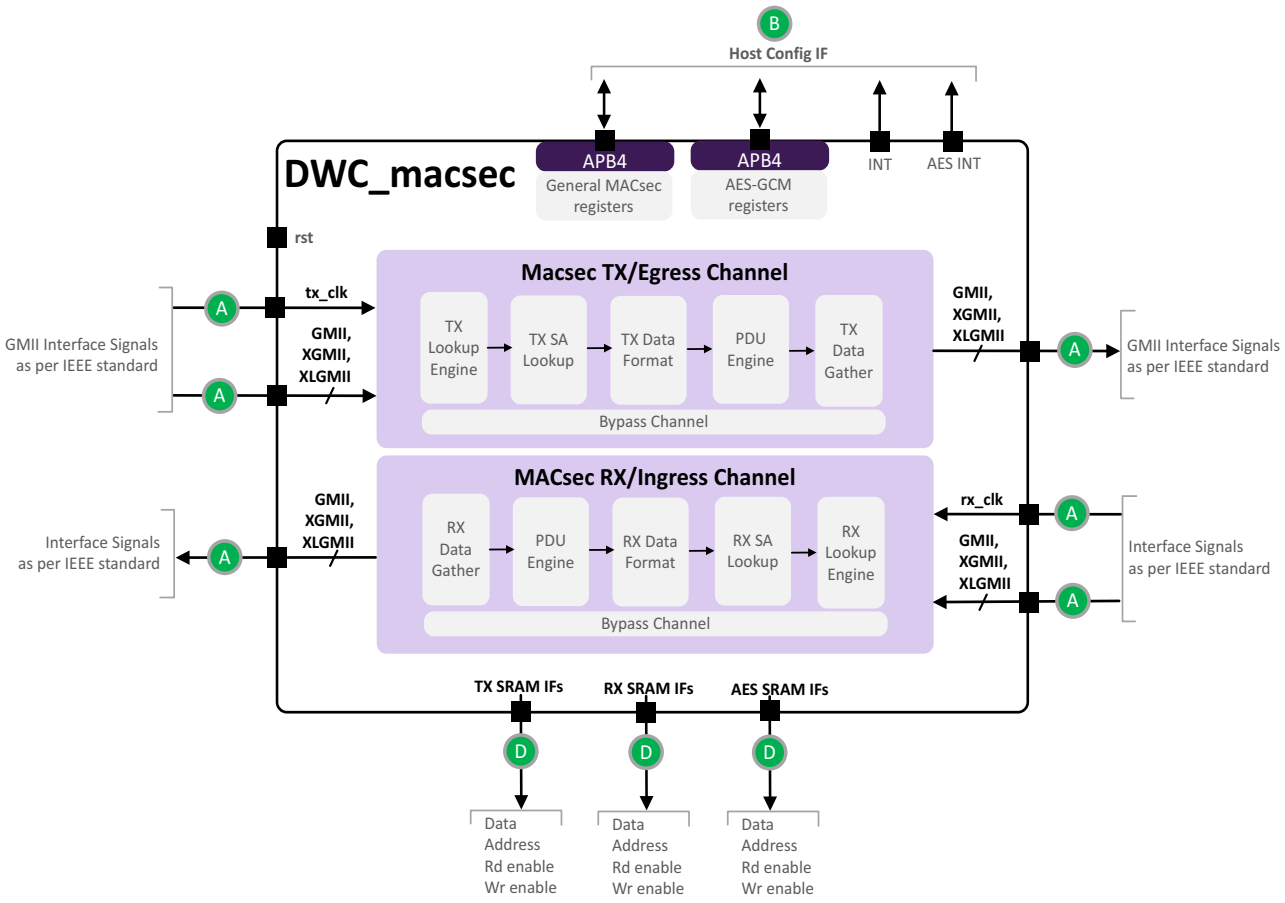
This integration example describes an inline MACsec AC operation solution between the Ethernet Controller (MAC) and an Ethernet PHY or PCS.

Figure 1-9 shows the types of interfaces avaliable for the DWC_macsec integration.

- **A - Interfaces for the Ethernet Controller and PHY (or PCS):** It is shown in Figure 1-9 as the direct connections to the MAC and PCS.
- **B - Functional Interfaces to the System:** All the interfaces that are connected to the SoC infrastructure, containing APB interfaces and interrupts and register access lock signals. The integrator finds the most suitable connection scheme for these interfaces for the targeted SoC.
- **C- Clocks and Resets:** It is shown in Figure 1-9 as the Clock and Reset Signals group containing all clocks and reset signals. The integrator follows the considerations described in “Clock Requirements” on page 14 and “Reset Requirements” on page 15.

- **D - Interfaces for the External Memories:** It is shown in Figure 1-9 as AES-GCM SRAM Interfaces, TX/Egress SRAM Interfaces and RX/Ingress SRAM Interfaces (available when memories are selected as not synthesizable logic). The integrator must connect these specific interfaces to their specific memory/SRAM components, if internal synthesizable flip-flops are not selected by configuration options.

Figure 1-9 DWC_macsec AC Interfaces Integration Overview



1.9.3 System Integration Overview

Figure 1-10 shows the complete integration of the DWC_macsec with a Synopsys Ethernet MAC and Synopsys PCS.

1.9.3.1 Data Interfaces

The Ethernet data interfaces are connected directly to the Ethernet MAC and Ethernet PCS/PHY. The interfaces used are standard IEEE type of interfaces. The clock used for the `rx_clk` is the one provided by the PHY while the clock used for the `tx_clk` is under control by the DWC_macsec (but must be the same clock used by the Ethernet MAC TX).

1.9.3.2 Clock and Reset

The DWC_macsec operates most of the logic with the `tx_clk`. The `tx_clk` is used for the Egress GMII interface. Therefore, it must be the same source as the one used to generate the GMII, XGMII, XLGMII data for the Ethernet MAC Egress path.

The `rx_clk` is provided by the PHY and with this clock the Ingress GMII, XGMII, XLGMII interface is processed and passed to the Ethernet MAC. Therefore, the same clock source must be used for the Ingress path of the Ethernet MAC.

The DWC_macsec has one reset source that resets the complete IP.

The reset source should not be connected to any bypass selector, to avoid any security pitfalls. For example, it is not possible to reset the IP and at the same time select a bypass path that allows the data to be propagated on-the-clear/without encryption/authentication.

1.9.3.3 System Interfaces

The APB interfaces can be directly connected to the system. These are used to configure the DWC_macsec control registers and program the keys used with the AES-GCM. It is suggested that the AES-GCM APB is connected directly to a trusted module such as embedded HSM/SE to mitigate security issues.

The associated PPROT signals of the APB must be used to control the access to the DWC_macsec registers, and all accesses must be secure (meaning PPROT signals driven with 0).

1.9.3.4 SRAM Interfaces

The DWC_macsec products needs several SRAMs to store AES-GCM specific hash and cipher intermediate data, as well SRAMs to store Egress/Ingress lookup databases and MIB data. The DWC_masec provides the SRAM interfaces, but the SRAMs must be instantiated by the integrator.

1.9.4 Automotive Support Overview

The DWC_macsec AC is an ASIL-B for random hardware faults product. It implements the following safety packages:

- Internal Safety Mechanisms (SMs)
- Failure Modes, Effects, and Diagnostic Analysis (FMEDA)
- Design Failure Mode and Effect Analysis (DFMEA)

The MACsec automotive-compliant security module safety documentation includes the following (but is not limited to):

- FMEDA report
- DFMEA report
- Safety manual

1.9.4.1 DWC_macsec Automotive Safety Mechanisms

The following section describes the external and internal safety mechanisms.

1.9.4.1.1 Internal Safety Mechanism

The DWC_macsec ASIL-B has several internal safety mechanisms that monitor transient and permanent faults. The following list provides a summary of the safety mechanisms:

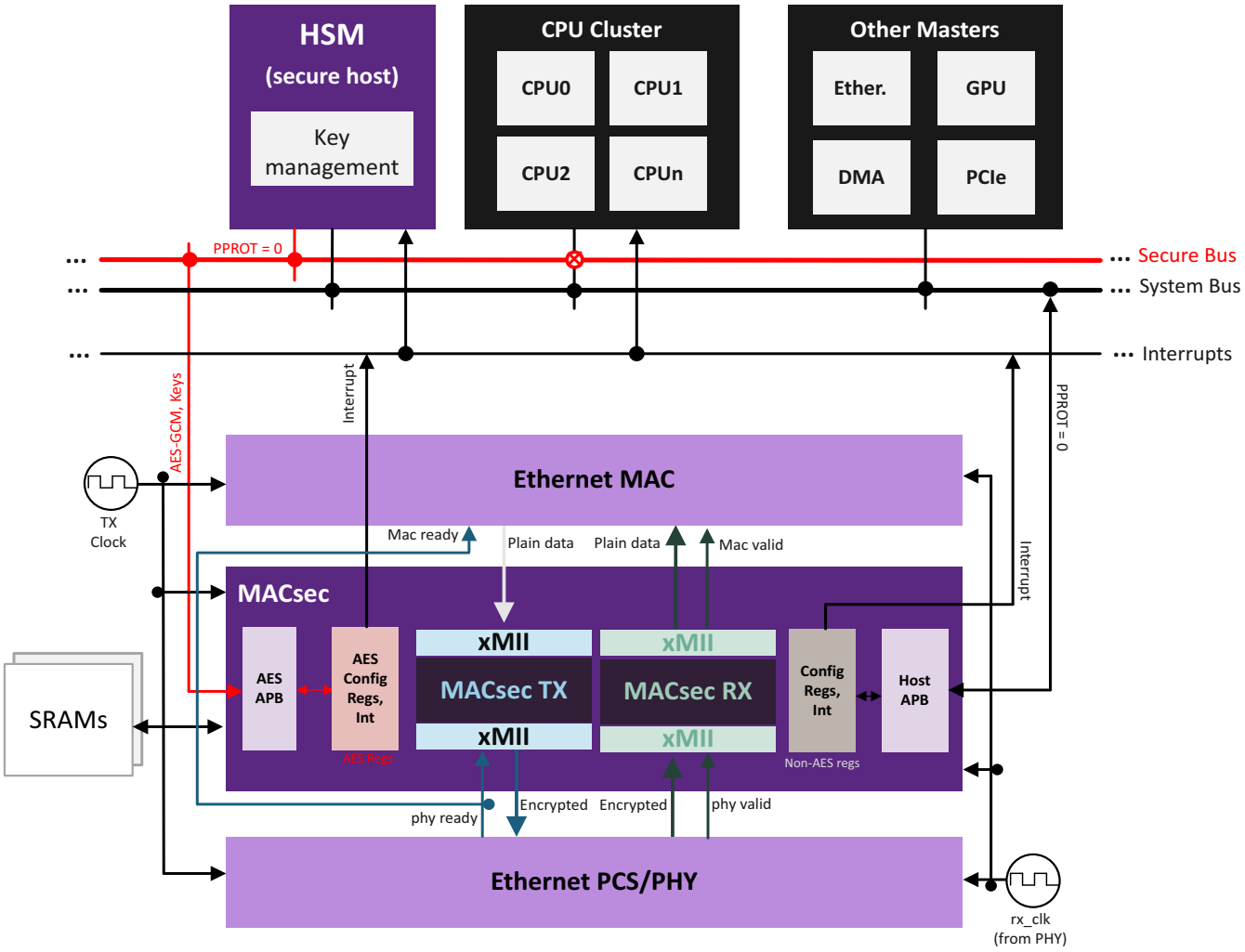
- Parity protection for the main data and control signals.
- Parity protection for the APB transactions
- Configuration register duplications for all available registers.
- SRAM ECC for writing and reading SRAM data and addresses.
- Latency managers monitor datapath stalling errors.
- CRC imposes a standard feature that is used for ingress or egress ethernet frame validation.
- RX Message Authentication Code (MAC) imposes a standard feature that is used as end-to-end protection for AES-GCM.
- AES-GCM Built-in Self-Test (BIST) for detecting latent permanent faults.
- Lock step for detecting single permanent faults in AES-GCM.
- Timing redundancy (TR) for detecting single permanent faults in AES-GCM.
- AES-GCM FIPS firmware-driven test for detecting latent permanent faults.
- Dual modular redundancy for critical blocks.
- Triple modular redundancy for certain features or blocks.
- Dual rail protection for detecting faults in critical input signals.
- Alarms dual rail for detecting faults in critical output signals.
- Several error injection mechanisms for parity, ECC, duplication, and so on.

1.9.4.1.2 External Safety Mechanism

The DWC_macsec ASIL-B depends on several external safety mechanisms to obtain the ASIL-B random fault metrics. The following list provides a summary of the safety mechanisms:

- SRAM ECC check to detect any ECC errors when DWC_macsec is accessing the external SRAMs.
- Parity protection for the APB transactions.
- Logical BIST detects permanent faults through logical testing at startup.
- APB register read back to detect faults in configuration registers.
- APB watchdog timer detects faults in configuration registers.
- Key monitors (TX or RX external key monitors).
- Error injection mechanisms for all safety mechanisms.

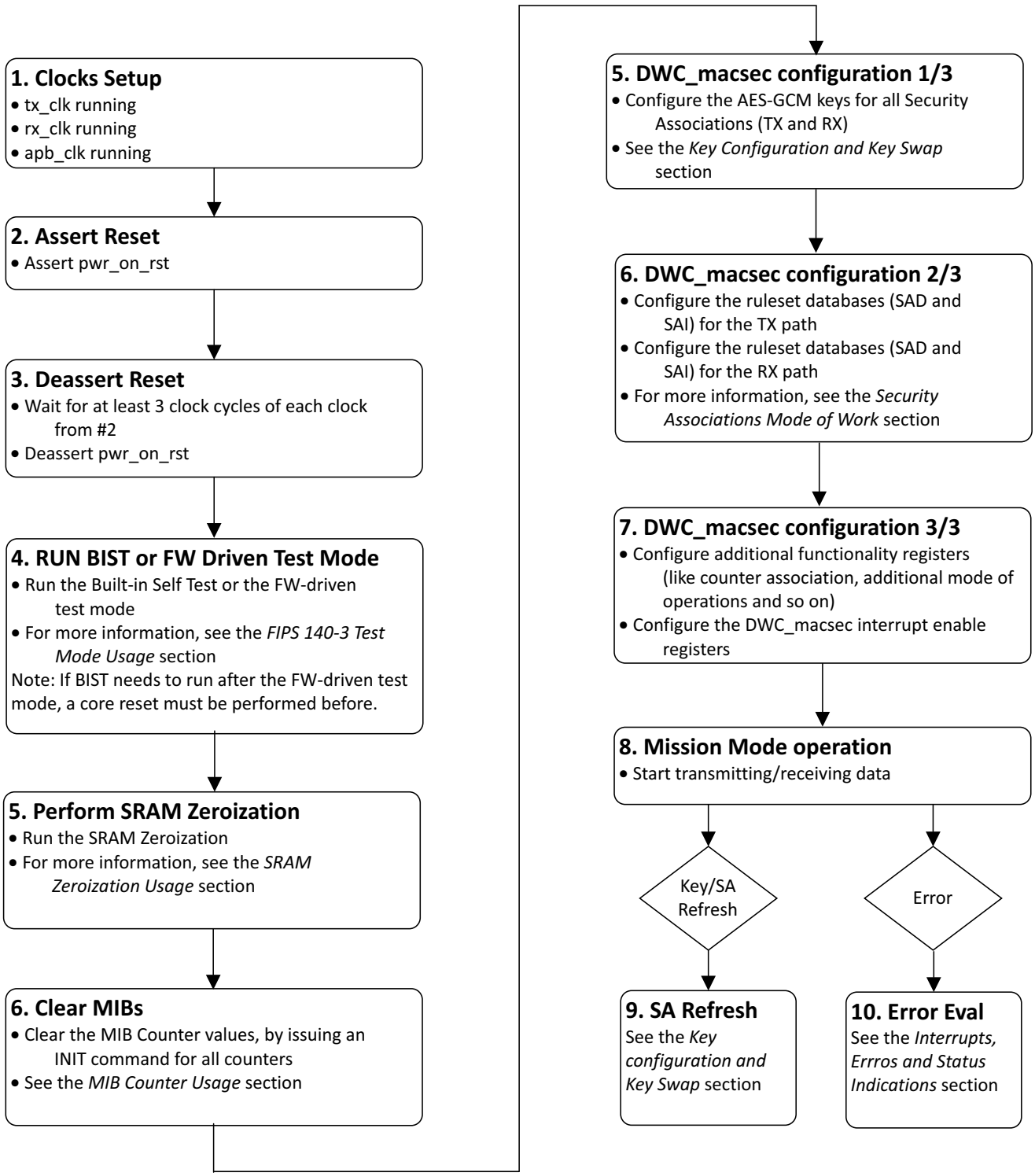
Figure 1-10 DWC_macsec System Integration



1.10 Module Startup

Figure 1-11 shows how the DWC_macsec start up within a system. The Ethernet MAC and PHY/PCS initialization steps are not included in the flow, as this should be described in the specific product documentation.

Figure 1-11 DWC_macsec Startup



1.11 Area

Note The synthesis is done using 5nm technology and following the Synopsys reference flow, and targeting the specific datapath interface frequencies, needed for that specific performance rate. All the parameters that are not defined in the tables are set taking the default configuration provided by the Core Consultant.

Table 1-7 Synthesis Area Numbers for MACsec up to 2.5Gbps

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AUTOMOTIVE 1 ■ set_configuration_parameter DWC_MACSEC_PERF_RATE 0 ■ set_configuration_parameter DWC_MACSEC_PREEMPTION_EN 1 ■ set_configuration_parameter DWC_MACSEC_RXT0_NEXT_SUPPORT 0 ■ set_configuration_parameter DWC_DUAL_IF 0 ■ set_configuration_parameter DWC_MACSEC_TX_NUM_SC_SUPPORTED 8 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SA_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SC_SUPPORTED 8 ■ set_configuration_parameter DWC_MACSEC_MAX_KEY_SIZE 1 ■ set_configuration_parameter DWC_MACSEC_JUMBO_FRAMES_SUPPORTED 1 ■ set_configuration_parameter DWC_MACSEC_WIN_SIZE_MODE 1 ■ set_configuration_parameter DWC_MACSEC_XPN_SEL 1 ■ set_configuration_parameter DWC_MACSEC_SYNC_DEPTH 3 ■ set_configuration_parameter DWC_MACSEC_SRAM_ZERO_EN 1 ■ set_configuration_parameter DWC_MACSEC_APB_VER 2 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_ME THOD 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS 0 ■ set_configuration_parameter DWC_LOOKUP_SEARCH_METHOD 0 ■ set_configuration_parameter DWC_MACSEC_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS 0 ■ set_configuration_parameter DWC_MACSEC_TX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_arw_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_TX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SA_MEM_TYPE 1	2.75 M

Table 1-7 Synthesis Area Numbers for MACsec up to 2.5Gbps (Continued)

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_RX_MIB_SC_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_AES_SBOX_ARCH 1 ■ set_configuration_parameter DWC_MACSEC_AES_2_ROUNDS_PER_CYCLE_EN 1 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_FIPS_EN 1	

Table 1-8 Synthesis Area Numbers for MACsec up to 2.5Gbps with VLAN Tags Support

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AUTOMOTIVE 1 ■ set_configuration_parameter DWC_MACSEC_PERF_RATE 0 ■ set_configuration_parameter DWC_MACSEC_PREEMPTION_EN 1 ■ set_configuration_parameter DWC_MACSEC_RXT0_NEXT_SUPPORT 0 ■ set_configuration_parameter DWC_DUAL_IF 0 ■ set_configuration_parameter DWC_MACSEC_TX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SA_SUPPORTED 32 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_MAX_KEY_SIZE 1 ■ set_configuration_parameter DWC_MACSEC_JUMBO_FRAMES_SUPPORTED 1 ■ set_configuration_parameter DWC_MACSEC_WIN_SIZE_MODE 1 ■ set_configuration_parameter DWC_MACSEC_XPN_SEL 1 ■ set_configuration_parameter DWC_MACSEC_SYNC_DEPTH 2 ■ set_configuration_parameter DWC_MACSEC_SRAM_ZERO_EN 1 ■ set_configuration_parameter DWC_MACSEC_APB_VER 2 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_ME THOD 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS 2 ■ set_configuration_parameter DWC_LOOKUP_SEARCH_METHOD 0 ■ set_configuration_parameter DWC_MACSEC_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES 0	2.9 M

Table 1-8 Synthesis Area Numbers for MACsec up to 2.5Gbps with VLAN Tags Support (Continued)

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS 2 ■ set_configuration_parameter DWC_MACSEC_TX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_arw_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_TX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SC_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_AES_SBOX_ARCH 1 ■ set_configuration_parameter DWC_MACSEC_AES_2_ROUNDS_PER_CYCLE_EN 1 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_FIPS_EN 1	

Table 1-9 Synthesis Area Numbers for MACsec up to 10 Gbps

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AUTOMOTIVE 1 ■ set_configuration_parameter DWC_MACSEC_PERF_RATE 1 ■ set_configuration_parameter DWC_MACSEC_PREEMPTION_EN 1 ■ set_configuration_parameter DWC_MACSEC_RXT0_NEXT_SUPPORT 0 ■ set_configuration_parameter DWC_DUAL_IF 0 ■ set_configuration_parameter DWC_MACSEC_TX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SA_SUPPORTED 32 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_MAX_KEY_SIZE 1 ■ set_configuration_parameter DWC_MACSEC_JUMBO_FRAMES_SUPPORTED 1 ■ set_configuration_parameter DWC_MACSEC_WIN_SIZE_MODE 1 ■ set_configuration_parameter DWC_MACSEC_XPN_SEL 1 ■ set_configuration_parameter DWC_MACSEC_SYNC_DEPTH 3 ■ set_configuration_parameter DWC_MACSEC_SRAM_ZERO_EN 1 ■ set_configuration_parameter DWC_MACSEC_APB_VER 2	2.73 M

Table 1-9 Synthesis Area Numbers for MACsec up to 10 Gbps (Continued)

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_ME THOD 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS 0 ■ set_configuration_parameter DWC_LOOKUP_SEARCH_METHOD 0 ■ set_configuration_parameter DWC_MACSEC_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS 0 ■ set_configuration_parameter DWC_MACSEC_TX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_ARW_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_TX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SC_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_AES_SBOX_ARCH 1 ■ set_configuration_parameter DWC_MACSEC_AES_2_ROUNDS_PER_CYCLE_EN 1 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_FIPS_EN 1	

Table 1-10 Synthesis Area Numbers for MACsec up to 10 Gbps with Dual Interface

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AUTOMOTIVE 1 ■ set_configuration_parameter DWC_MACSEC_PERF_RATE 1 ■ set_configuration_parameter DWC_MACSEC_PREEMPTION_EN 1 ■ set_configuration_parameter DWC_MACSEC_RXT0_NEXT_SUPPORT 0 ■ set_configuration_parameter DWC_DUAL_IF 1 ■ set_configuration_parameter DWC_MACSEC_TX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SA_SUPPORTED 32 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SC_SUPPORTED 16	3.14 M

Table 1-10 Synthesis Area Numbers for MACsec up to 10 Gbps with Dual Interface (Continued)

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_MAX_KEY_SIZE 1 ■ set_configuration_parameter DWC_MACSEC_JUMBO_FRAMES_SUPPORTED 1 ■ set_configuration_parameter DWC_MACSEC_WIN_SIZE_MODE 1 ■ set_configuration_parameter DWC_MACSEC_XPN_SEL 1 ■ set_configuration_parameter DWC_MACSEC_SYNC_DEPTH 3 ■ set_configuration_parameter DWC_MACSEC_SRAM_ZERO_EN 1 ■ set_configuration_parameter DWC_MACSEC_APB_VER 2 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_ME THOD 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS 2 ■ set_configuration_parameter DWC_LOOKUP_SEARCH_METHOD 0 ■ set_configuration_parameter DWC_MACSEC_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS 2 ■ set_configuration_parameter DWC_MACSEC_TX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_ARW_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_TX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SC_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_AES_SBOX_ARCH 1 ■ set_configuration_parameter DWC_MACSEC_AES_2_ROUNDS_PER_CYCLE_EN 1 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_FIPS_EN 1	

Table 1-11 Synthesis Area Numbers for MACsec up to 25 Gbps

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AUTOMOTIVE 1 ■ set_configuration_parameter DWC_MACSEC_PERF_RATE 2 ■ set_configuration_parameter DWC_MACSEC_PREEMPTION_EN 1 ■ set_configuration_parameter DWC_MACSEC_RXT0_NEXT_SUPPORT 0 ■ set_configuration_parameter DWC_DUAL_IF 0 ■ set_configuration_parameter DWC_MACSEC_TX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SA_SUPPORTED 32 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_MAX_KEY_SIZE 1 ■ set_configuration_parameter DWC_MACSEC_JUMBO_FRAMES_SUPPORTED 1 ■ set_configuration_parameter DWC_MACSEC_WIN_SIZE_MODE 1 ■ set_configuration_parameter DWC_MACSEC_XPN_SEL 1 ■ set_configuration_parameter DWC_MACSEC_SYNC_DEPTH 3 ■ set_configuration_parameter DWC_MACSEC_SRAM_ZERO_EN 1 ■ set_configuration_parameter DWC_MACSEC_APB_VER 2 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_ME THOD 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS 0 ■ set_configuration_parameter DWC_LOOKUP_SEARCH_METHOD 0 ■ set_configuration_parameter DWC_MACSEC_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS 0 ■ set_configuration_parameter DWC_MACSEC_TX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_ARW_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_TX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SA_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_MIB_SC_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_AES_SBOX_ARCH 1 ■ set_configuration_parameter DWC_MACSEC_AES_2_ROUNDS_PER_CYCLE_EN 1 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CKEY_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HKEY_MEM 0	2.79 Million

Table 1-11 Synthesis Area Numbers for MACsec up to 25 Gbps (Continued)

Parameter Configuration	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_FIPS_EN 1	

Table 1-12 Synthesis Area Numbers for MACsec up to 25 Gbps with Dual Interface

Parameter Description	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AUTOMOTIVE 1 ■ set_configuration_parameter DWC_MACSEC_PERF_RATE 2 ■ set_configuration_parameter DWC_MACSEC_PREEMPTION_EN 1 ■ set_configuration_parameter DWC_MACSEC_RXT0_NEXT_SUPPORT 0 ■ set_configuration_parameter DWC_DUAL_IF 1 ■ set_configuration_parameter DWC_MACSEC_TX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SA_SUPPORTED 32 ■ set_configuration_parameter DWC_MACSEC_RX_NUM_SC_SUPPORTED 16 ■ set_configuration_parameter DWC_MACSEC_MAX_KEY_SIZE 1 ■ set_configuration_parameter DWC_MACSEC_JUMBO_FRAMES_SUPPORTED 1 ■ set_configuration_parameter DWC_MACSEC_WIN_SIZE_MODE 1 ■ set_configuration_parameter DWC_MACSEC_XPN_SEL 1 ■ set_configuration_parameter DWC_MACSEC_SYNC_DEPTH 3 ■ set_configuration_parameter DWC_MACSEC_SRAM_ZERO_EN 1 ■ set_configuration_parameter DWC_MACSEC_APB_VER 2 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_ME THOD 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS 2 ■ set_configuration_parameter DWC_LOOKUP_SEARCH_METHOD 0 ■ set_configuration_parameter DWC_MACSEC_SAI_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS 2 ■ set_configuration_parameter DWC_MACSEC_TX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_SALT_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_RX_ARW_MEM_TYPE 1 ■ set_configuration_parameter DWC_MACSEC_TX_SAD_MEM_TYPE 0 ■ set_configuration_parameter DWC_MACSEC_RX_SAD_MEM_TYPE 0	3.33M

Table 1-12 Synthesis Area Numbers for MACsec up to 25 Gbps with Dual Interface

Parameter Description	Area (NAND2 Gates)
■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_CCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_SYNTH_HCTX_MEM 0 ■ set_configuration_parameter DWC_MACSEC_AES_FIPS_EN 1	

1.12 Compliance With Quality Metrics

Table 1-13 shows coverage results achieved through simulations with various configurations.

Table 1-13 Total Coverage Summary[a]

Score	Line	Toggle
100%	100%	100%

a. Toggle coverage numbers at top-level; Waivers and exclusion may apply to the metrics;

2 Architecture

This chapter includes the following sections:

- "Architecture Overview" on page 48
- "Data Interfaces" on page 51
- "VLAN on the Clear Mode Support" on page 63
- "Inter Packet Gap (IPG) Considerations and MAC Programming" on page 69
- "KaY and Unknown Cipher Packet Handling" on page 71
- "MIB Counters Usage" on page 72
- "Key Configuration and Key Swap" on page 77
- "Host Configuration Interface - APB Interface" on page 79
- "Parity and ECC Support" on page 79
- "PHY Link Fault Signal and LPI Messages Handling" on page 79
- "LPI Indication for MII/GMII" on page 80

2.1 Architecture Overview

2.1.1 General Description

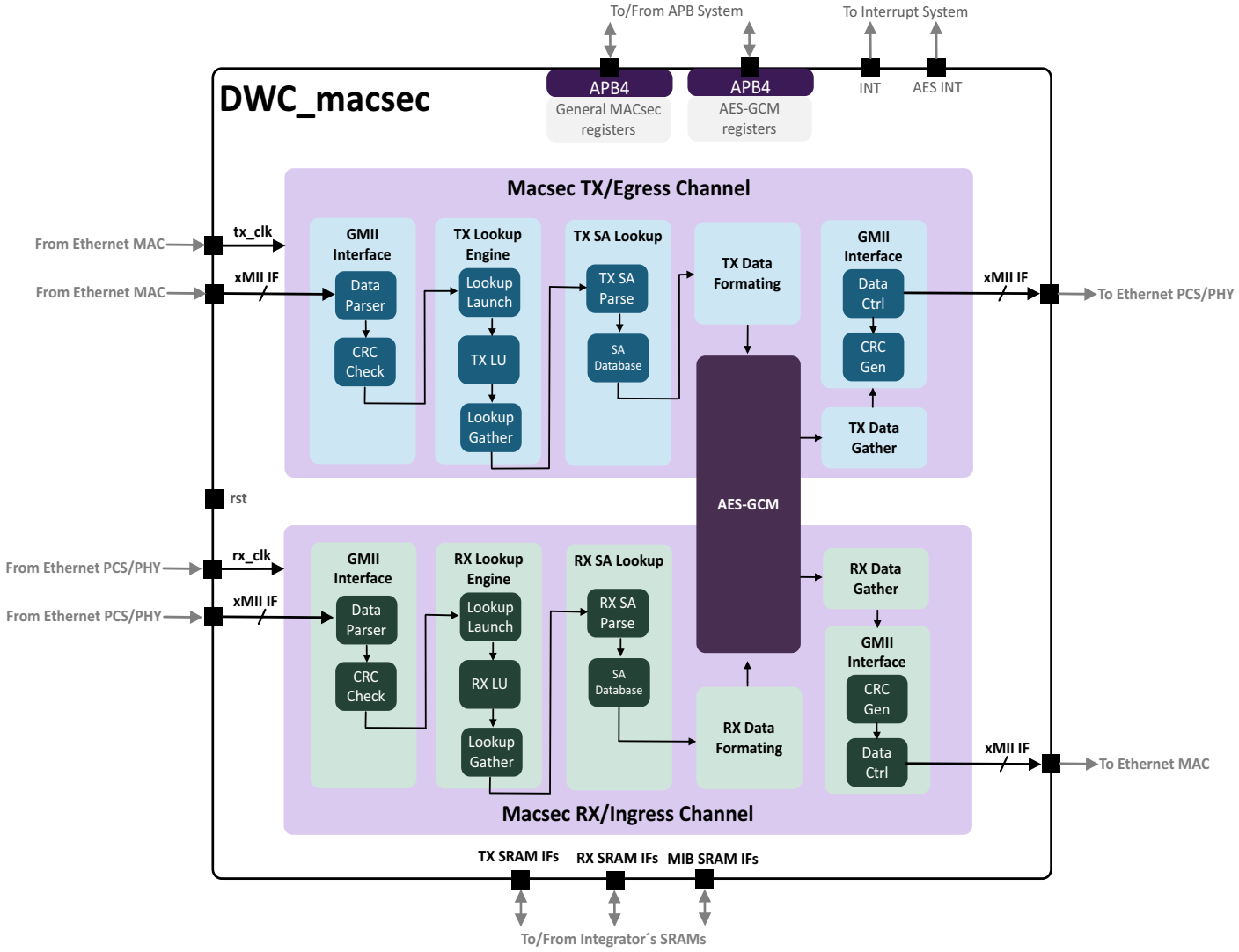
The DWC_macsec module contain the following main sub-parts: TX/Egress channel, RX/Ingress channel, and the AES-GCM cryptographic core, as shown in Figure 2-1. The AES-GCM cryptographic core throughput is dimensioned to serve both TX/Egress and RX/Ingress channels without any performance drop or stalling.

The DWC_macsec data interfaces are GMII, XGMII or XLGMII. These interfaces can be plug-and-play in a seamless way into a system with an Ethernet MAC and Ethernet PHY/PCS. The configuration of the DWC_macsec is done by two APB interfaces: one for the general purpose MACsec AC registers and one for the specific AES-GCM registers and keys.

The system features two independent interrupt lines that can be connected to an interrupt system, which can be used to monitor several error conditions.

All the SRAM components need to be integrated at SoC level as the DWC_macsec does not provide built-in SRAM wrappers or SRAM components. There is nonetheless the possibility of replacing, by configuration options, some SRAMs by DWC_macsec internal Flip-Flop banks.

Figure 2-1 DWC_macsec Architecture Overview



2.1.2 AES-GCM Cryptographic Core

The DWC_macsec contains one 128-bit or 256-bit key size AES-GCM cryptographic engine. The datapath width for the AES-GCM core is linked to the overall throughput required for the selected MACsec AC performance. The integrator selects the wanted MACsec AC throughput, and the AES-GCM is dimensioned accordingly.

The value set for the DWC_MACSEC_PERF_RATE automatically configures the datapath width of the AES-GCM engine, so it can encrypt/decrypt and authenticate the wanted line rate without performance drop or stalling.

The AES-GCM can be configured to support 128-bit key size or 256-bit key size by the DWC_MACSEC_MAX_KEY_SIZE parameter.

2.1.3 TX/Egress Channel

The TX/Egress channel receives the GMII, XGMII or XLGMII interface data from the Ethernet MAC.

The data CRC is checked against any errors; errors are removed before data is fed to the AES-GCM. After the data is encrypted and the authentication Tag is generated by the AES-GCM, the packet CRC is again generated and appended to the data.

The TX/Egress channel output GMII, XGMII or XLGMII interface should be connected to an Ethernet PCS or PHY. The TX/Egress channel performs the packet parsing rules as described in the *IEEE 802.1AE Standard*.

2.1.4 RX/Ingress Channel

The RX/Ingress channel receives the GMII, XGMII or XLGMII interface data from the Ethernet PCS or PHY.

The data CRC is checked against any errors; errors are removed before data is fed to the AES-GCM. After the data is decrypted and authenticated by the AES-GCM, the packet CRC is again generated and appended to the data.

The RX/Ingress channel output GMII, XGMII or XLGMII interface should be connected to an Ethernet MAC. The RX/Ingress channel performs the packet parsing rules as described in the *IEEE 802.1AE Standard*.

2.1.5 Key Handling and Swapping

The security association keys are installed into the AES-GCM core by the specific AES-GCM APB interface. The key installation bit and the integrator enable a key installation interrupt that is triggered when the AES-GCM finishes installing a key. Typically, a key installation on the AES-GCM is fast and is done under 30 TX clock cycles.

Swapping keys for the AES-GCM is based on the principles of security association keys defined in the *IEEE 802.1AE Standard*. The system is responsible for handling the active/inactive values for each security association. Programming a new key into an active security association of a specific security channel leads to unexpected behavior.

2.1.6 Bypass Encryption/Decryption

The DWC_macsec bypass function allows unencrypted data to pass through the core. The DWC_macsec supports following types of bypass:

- Frame bypass when it does not match any Security Association on TX/RX.
  - For example, if the packet does not have a match against the security association lookups, then the packet is not encrypted/decrypted and/or authenticated.
- Static bypass of all frames when the specific bypass registers are set on TX/RX.
  - This mode is used to perform system integration debugging; you can leave the SA databases empty and pass traffic between the MAC and PCS.

2.1.7 Programming Interfaces

The programming of DWC_macsec, including the AES keys, is done by two APB interfaces.

2.1.8 Register Access

The registers are accessed by a host APB interface, and an additional protection control exists to prevent non-secure accesses:

- The APB *pprot[0]* signals must be driven low so the APB interface can access any DWC_macsec registers.

2.2 Data Interfaces

The DWC_macsec is connected to the Ethernet MAC and Ethernet PCS/PHY as shown in Figure 2-10. The data interfaces are divided in:

- **xGMII Interfaces from Ethernet MAC:** These contain the RMII, RGMII, MII, GMII, XGMII or XLGMII plain data interface for the Egress data and the RMII, RGMII, MII, GMII, XGMII or XLGMII plain data interface for the ingress data, that needs to be connected to the MAC.
- x**GMII Interfaces to Ethernet PCS/PHY:** These contain the RMII, RGMII, MII, GMII, XGMII or XLGMII secure data interface for the Egress data and the RMII, RGMII, MII, GMII, XGMII or XLGMII secure data interface for the ingress data, that need to be connected to the PCS/PHY.

2.2.1 Data Interfaces Signaling

All the data interfaces are based on the xMII interface definition of the *IEEE 802.3 Standard*. Figure 2-10 shows the DWC_macsec xGMII interface signals. The COL and CRS signals are not used, as the DWC_macsec always operates in full duplex mode.

2.2.2 Data Interfaces and CRC

The DWC_macsec is integrated between an Ethernet MAC and an Ethernet PCS/PHY. The CRC calculated by the MAC needs to be re-calculated in the Egress and Ingress paths after encryption and/or authentication takes place.

The DWC_macsec recalculates the CRC as defined in the *IEEE 802.3 Standard,* and if a CRC error is identified, the packet is poised with a CRC error towards the Ethernet MAC. An interrupt can be triggered by the global interrupt line to the system (if enabled).

The DWC_macsec does not drop frames when a CRC error is detected.

There are several variants included in the xGMII spectrum. The 802.3 standard defines the following:

- MII using a nominal 4-bit interface
- GMII using a nominal 8-bit interface
- XGMII using a nominal 32-bit interface
- XLGMII/CGMII using a nominal 64-bit interface

But as these interfaces follow the same protocol, implementations of 128-bit XLGMII are also available to reduce clock frequencies.

Note

- RGMII and RMII interfaces are widely used, but they are not covered by the IEEE 802.3 standard.
- CGMII is another notation for 100 Gb/s XLGMII, which implies the use of a 128-bit interface.

As mentioned in the "General Product Description" section, the DWC_MACsec module is integrated between the MAC and PHY/PCS blocks, and communication is made through the xGMII data interfaces. The interfaces are split between the TX and RX paths, each having its associated control signals.

In Figure 2-3, Figure 2-3, Figure 2-4, Figure 2-5, Figure 2-6, Figure 2-7, and Figure 2-8, the signal interface section details all the individual signals and buses that make up these interfaces.

Figure 2-2 DWC_macsec MII 4-bit signal interface

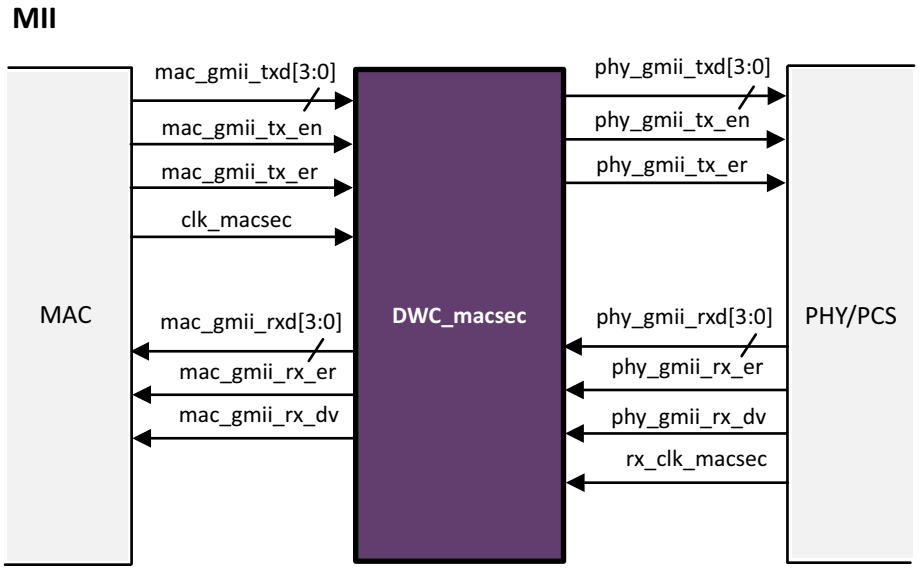
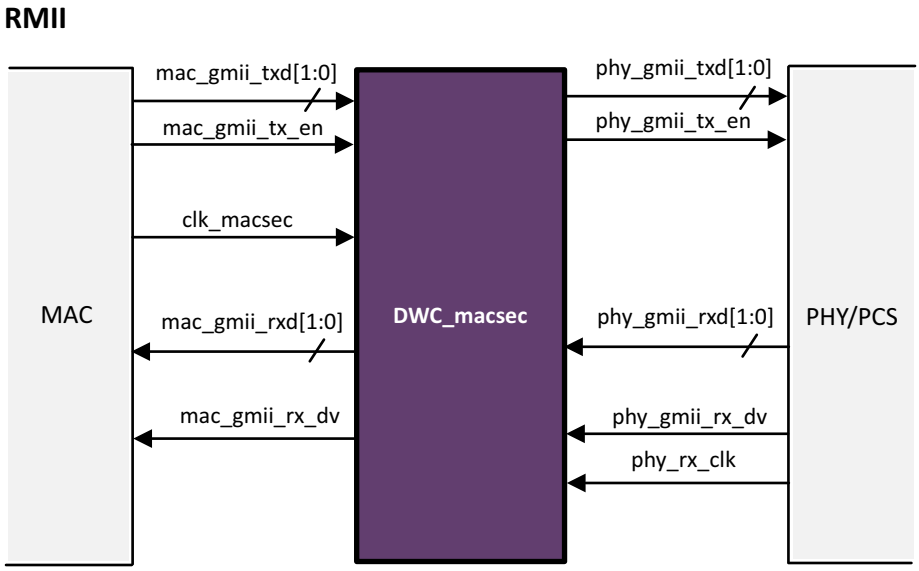


Figure 2-3 DWC_macsec RMII 2-bit Signal Interface



Note When using RMII modes, the `mac_gmii_tx_er` and `phy_gmii_rx_er` signals should be driven to zero.

Figure 2-4 DWC_macsec GMII 8-bit Signal Interface

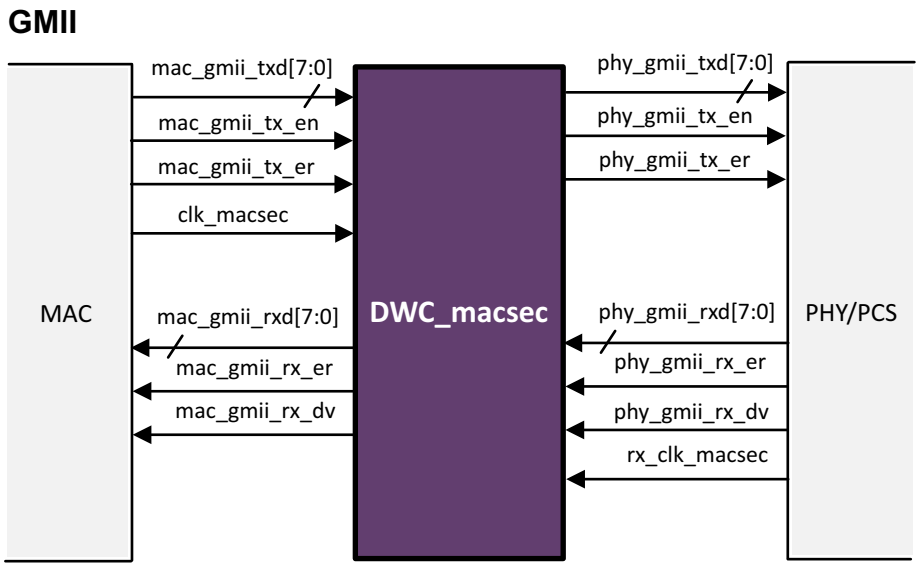
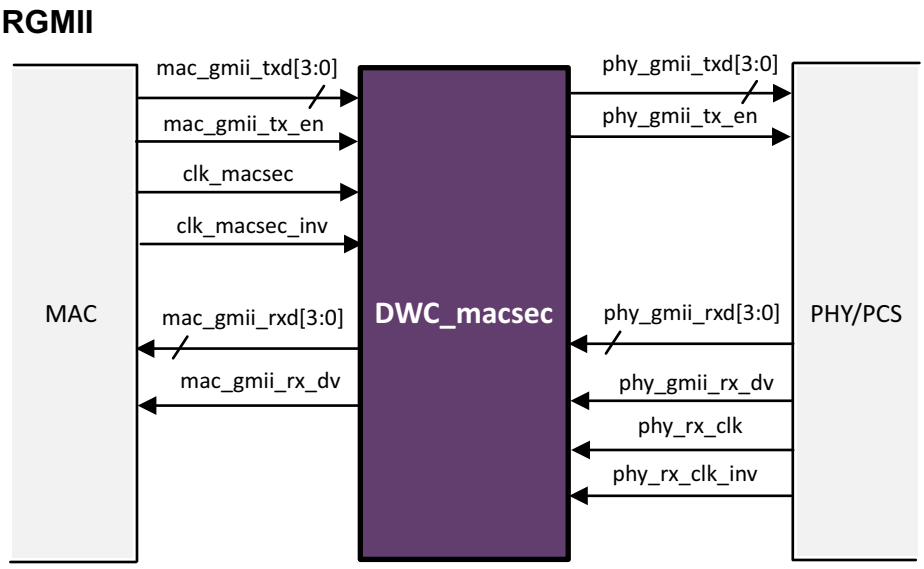


Figure 2-5 DWC_macsec RGMII 4-bit Signal Interface



Note

- When using RMII modes, the `mac_gmii_tx_er` and `phy_gmii_rx_er` signals should be driven to 0.
- As mentioned in the Reduced Gigabit Media Independent Interface (RGMII) specification, a data-to-clock input skew (at the receiver) must exist based on the Timing Specifications of the standard. The DWC_MACSec RGMII interface behavior is indeterminate if the data-to-clock input skew at the receiver is non-compliant.

Figure 2-6 DWC_macsec XGMII 32-bit Signal Interface

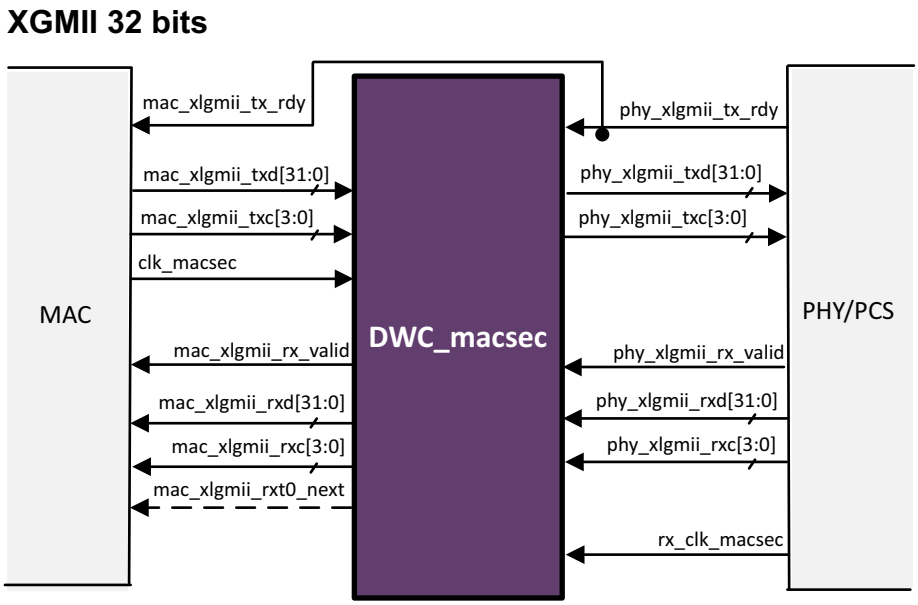


Figure 2-7 DWC_macsec XLGMII 64-bit Signal Interface

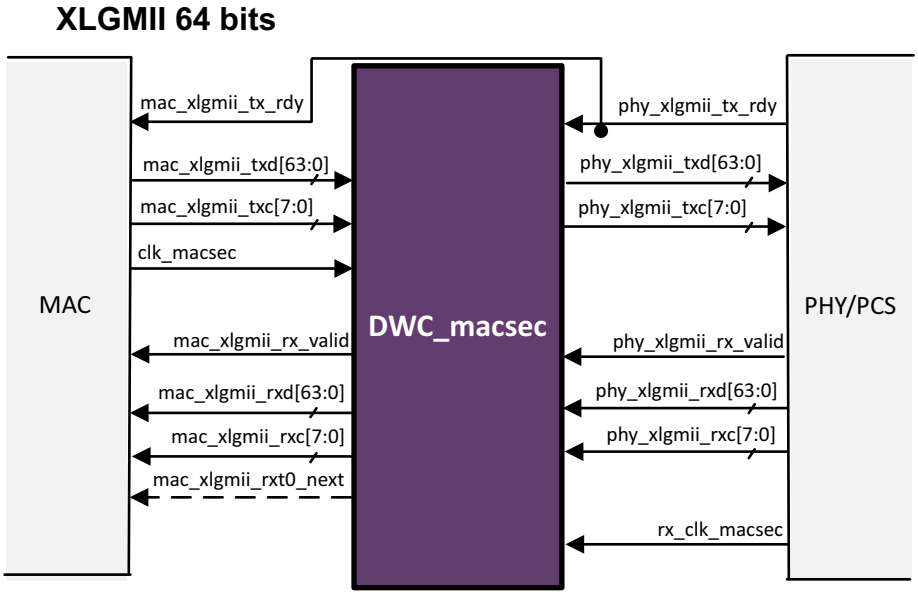
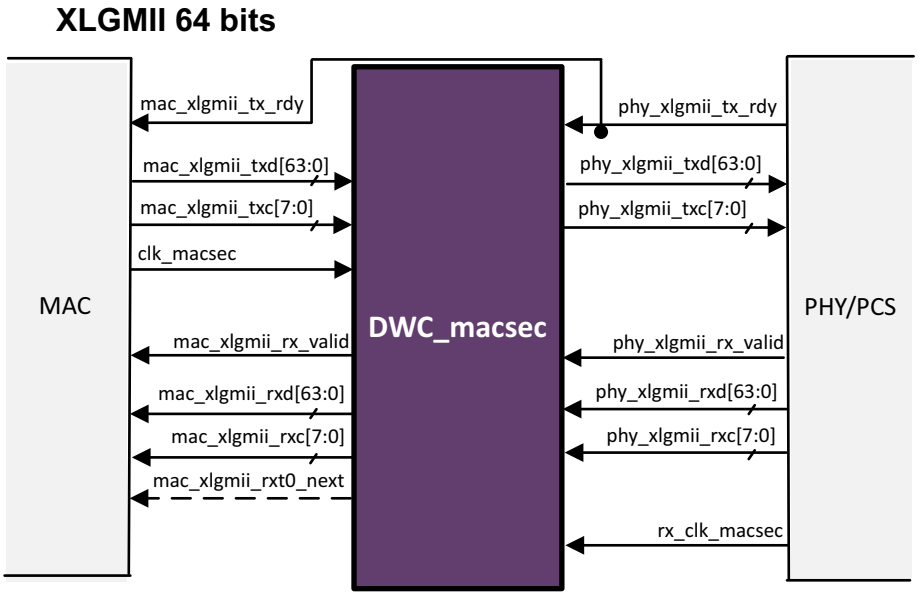


Figure 2-8 DWC_macsec XLGMII/CGMII 128-bit Signal Interface

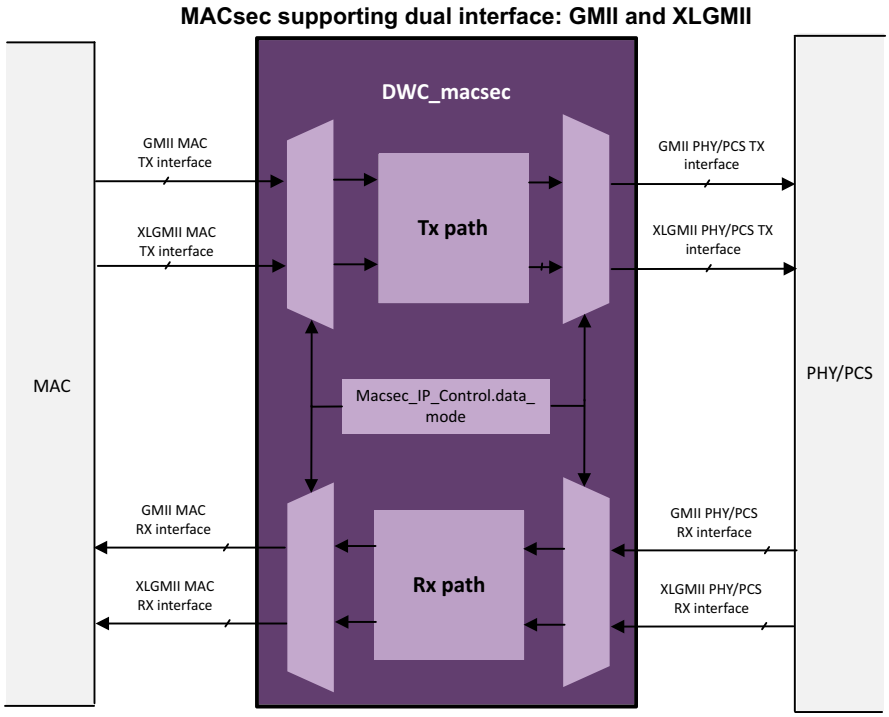


2.3 Dual Interface configuration

DWC_MACsec supports dual interface mode. When set at boot time, it enables the use of either an 8-bit GMII or a 128-bit XLGMII interface through the `MACsec_IP_Control.data_mode` configuration register.

Depending on the selected performance rate (DWC_MACSEC_PERF_RATE), this allows for the support of different speed modes and, consequently, sub-interfaces of GMII (MII, RMII, RGMII) or XLGMII (32-bit XGMII, 64-bit XLGMII) respectively.

Figure 2-9 MACsec AC Supporting Dual Interface: GMII and XLGMII



2.4 Security Associations Mode of Work

This section describes how the security associations and security channels are parsed and evaluated, as per the values programmed into the SAI and SAD databases.

2.4.1 TX/Egress Security Associations Mode of Work

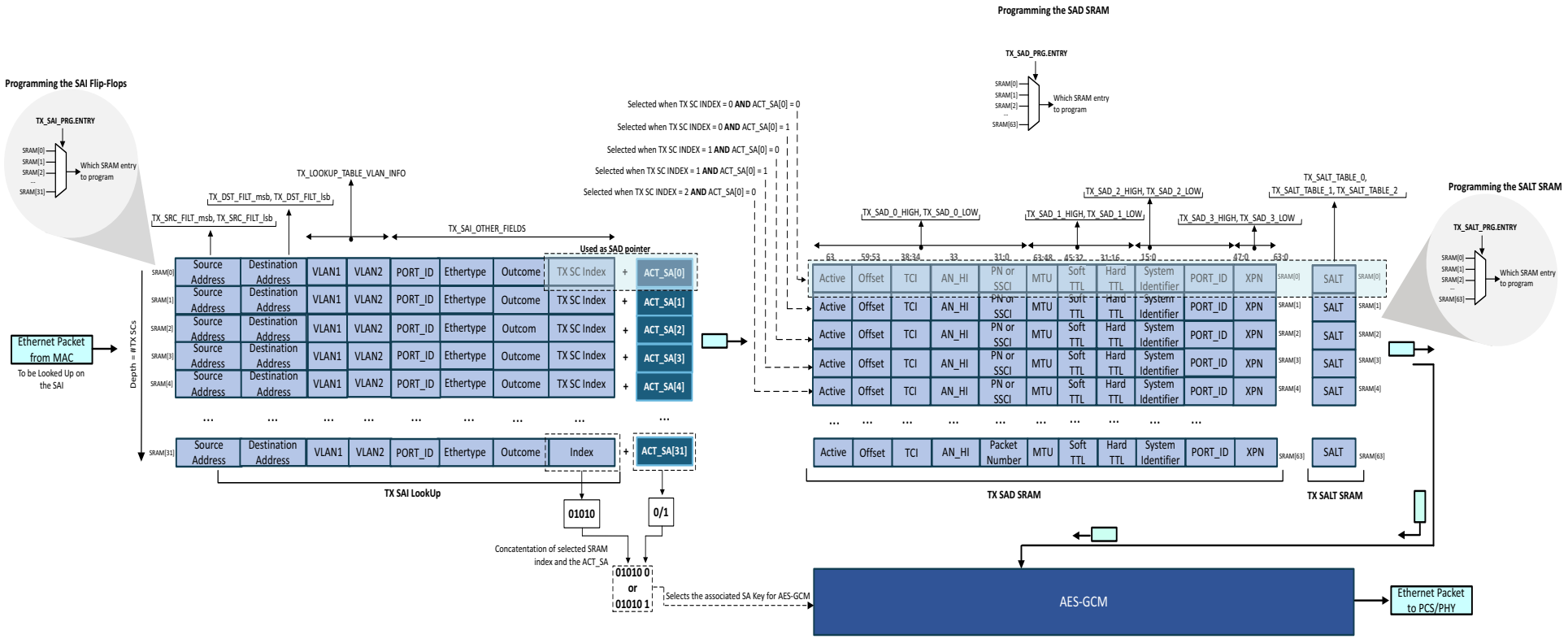
The DWC_macsec TX/Egress SAI and SAD lookup behavior is shown in Figure 2-10. The packet received from the MAC is initially parsed against the SAI lookup table, and an associated TX SC index is extracted.

This TX SC index is concatenated with the ACT_SA[x], and the final pointer to the TX SAD is created. The packet is then parsed against the configuration set on the TX SAD, that is pointed by the concatenated index.

Note The concatenation of the TX SC Index and ACT_SA[x] is nothing else than creating selecting which SA is active for a given Security Channel.

The AES-GCM key pointer is also extracted from the TX SC Index with the ACT_SA[x] concatenation. For more information on AES-GCM key programming and mapping, see "Programming a Key Context" on page 77.

Figure 2-10 MACsec TX/Egress SAI and SAD Lookups Operation



2.4.1.1 Programming the TX SAI, SAD and SALT Tables

This section describes how to program the TX SAD, SAI, and SALT tables.

To program the TX SAD, the following steps must be performed:

1. Program the TX_SAD_0_LOW.
2. Program the TX_SAD_0_HIGH.
3. Program the TX_SAD_1_LOW.

4. Program the TX_SAD_1_HIGH.

Note The SAD register allocation versus bit-fields is described in Figure 2-10.

5. Program the TX_SAD_2_LOW.
6. Program the TX_SAD_2_HIGH.
7. Program the TX_SAD_3_LOW.
8. Program the TX_SAD_3_HIGH.
9. Program the TX_PRG_SAD:
   a. Set the SRAM entry number that must be loaded with the above values, by the TX_PRG_SAD.ENTRY.
   b. Set the indication in the TX_PRG_SAD.TYPE to indicate a write operation.

Note The TX_ SAD_PRG needs to be programmed in a single APB write.

10. If a new entry needs to be programmed.
    a. Poll the TX_SAD_PG.WR_BUSY and if 0, then go to step 1.

To program the TX SALT, the following steps must be performed:

1. Program the TX_SALT_TABLE_0.
2. Program the TX_SALT_TABLE_1.
3. Program the TX_SALT_TABLE_2.
4. Program the TX_SALT_PRG:
   a. Set the SRAM entry number that must be loaded with the above values, by the TX_SALT_PRG.ENTRY.
   b. Set the indication in the TX_SALT_PRG.TYPE to indicate a write operation.
5. If a new entry needs to be programmed
   a. Poll the TX_SALT_PG.WR_BUSY and if 0, then go to step 1.

To program the TX SAI, the following steps must be performed:

1. Program the TX_SRC_FILT_0.
2. Program the TX_SRC_FILT_1.
3. Program the TX_DST_FILT_0.
4. Program the TX_DST_FILT_1.

5. Program the TX_LOOKUP_TABLE_VLAN_INFO.
6. Program the TX_SAI_OTHER_FIELDS register.
7. Program the TX_SAI_PRG:
   a. Set the SRAM entry number that must be loaded with the above values, by the TX_SAI_PRG.ENTRY.
   b. Set the indication in the TX_SAI_PRG.TYPE to indicate a write operation.
8. If a new entry needs to be programmed.
   a. Poll the TX_SAD_PG.WR_BUSY and if 0, then go to step #1.
9. Program the TX_ACTIVE_AN register.
   a. This sets which TX ANs are active for the session, as shown in Figure 2-10.
   b. This can be modified during mission mode with the objective of changing keys or to bring an AN/SA down.

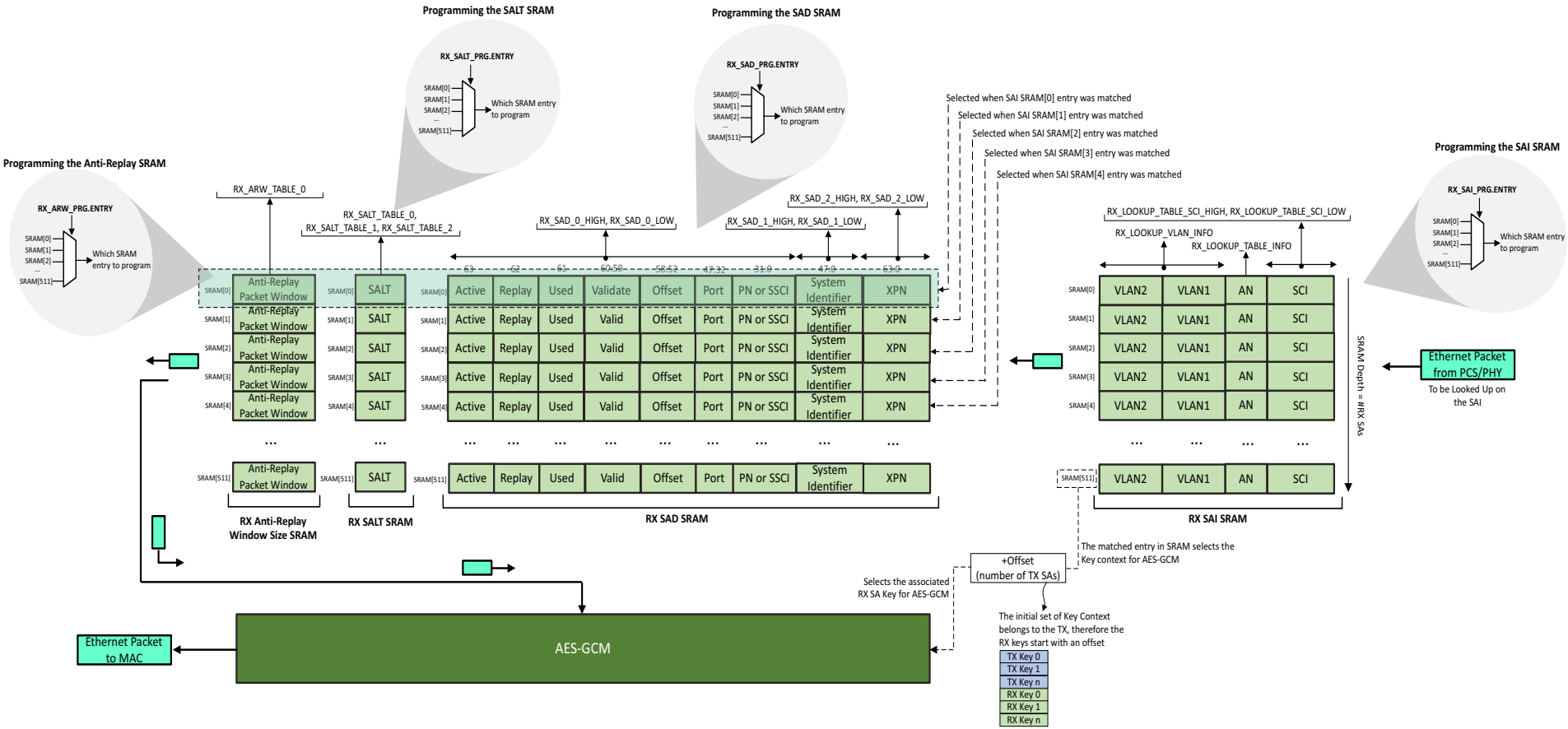
Confidentiality offset is not supported in AES-GCM-XPN mode as per definition of the *IEEE 802.1AE Standards*. Therefore, the OFFSET field is not used when AES-GCM-XPN is enabled.

2.4.2 RX/Ingress Security Associations Mode of Work

The DWC_macsec RX/Ingress SAI and SAD lookup behavior is shown in Figure 2-11. The packet received from the PCS/PHY is initially parsed against the SAI lookup table. If a match exists for a specific entry, then same entry number is taken for the SAD, SALT and Anti-Replay Window for further packet inspection.

The AES-GCM key pointer is constructed taking into consideration the SAI SRAM entry hit plus a fixed offset (that is imposed by the amount of TX Security Associations). For more information on AES-GCM key programming and mapping, see “Programming a Key Context” on page 77.

Figure 2-11 MACsec AC RX/Ingress SAI and SAD Lookups Operation



2.4.2.1 Programming the RX SAI, SAD, SALT and Anti-Replay Window Tables

This section describes how to program the RX SAI, SAD, SALT and Anti-Replay Packet Window Lookup tables.

To program the TX SAI, the following steps must be performed:

1. Program the RX_SAD_0_LOW.
2. Program the RX_SAD_0_HIGH.
3. Program the RX_SAD_1_LOW.
4. Program the RX_SAD_1_HIGH.

Note The tables register allocation versus bit-fields is described in Figure 2-11.

---

5. Program the RX_SAD_2_LOW.
6. Program the RX_SAD_2_HIGH.
7. Program the RX_PRG_SAD:
   a. Set the SRAM entry number that must be loaded with the above values, by the RX_PRG_SAD.ENTRY.
   b. Set the indication in the RX_PRG_SAD.TYPE to indicate a write operation.

Note The RX_SAD_PRG needs to be programmed in a single APB write.

---

8. If a new entry needs to be programmed:
   a. Poll the RX_SAD_PG.WR_BUSY and if 0, then go to step 1.

To program the RX SALT, the following steps must be performed:

1. Program the RX_SALT_TABLE_0.
2. Program the RX_SALT_TABLE_1.
3. Program the RX_SALT_TABLE_2.
4. Program the RX_SALT_PRG:
   a. Set the SRAM entry number that must be loaded with the above values, by the RX_SALT_PRG.ENTRY.
   b. Set the indication in the RX_SALT_PRG.TYPE to indicate a write operation.
5. If a new entry needs to be programmed
   a. Poll the RX_SALT_PG.WR_BUSY and if 0, then go to step 1.

To program the RX Anti-Replay Window Size, the following steps must be performed:

1. Program the RX_ARW_TABLE_0.
2. Program the RX_ARW_PRG:
   a. Set the SRAM entry number that must be loaded with the above values, by the RX_ARW_PRG.ENTRY.
   b. Set the indication in the RX_ARW_PRG.TYPE to indicate a write operation.
3. If a new entry needs to be programmed:
   a. Poll the RX_ARW_PG.WR_BUSY and if 0, then go to step 1.

To program the RX SAI, the following steps must be executed

1. Program the RX_LOOKUP_TABLE_SCI_LOW.
2. Program the RX_LOOKUP_TABLE_SCI_HIGH.
3. Program the RX_LOOKUP_TABLE_INFO.
4. Program the RX_VLAN_LOOKUP_INFO.
5. Program the RX_SAI_PRG:
   a. Set the SRAM entry number that must be loaded with the above values, by the RX_ARW_PRG.ENTRY.
   b. Set the indication in the RX_ARW_PRG.TYPE to indicate a write operation.
6. If a new entry needs to be programmed:
   a. Poll the RX_ARW_PG.WR_BUSY and if 0, then go to step 1.

Note Confidentiality offset is not supported in AES-GCM-XPN mode as per definition of the *IEEE 802.1AE Standards*. Therefore, the OFFSET field is not used when AES-GCM-XPN is enabled.

2.5 VLAN on the Clear Mode Support

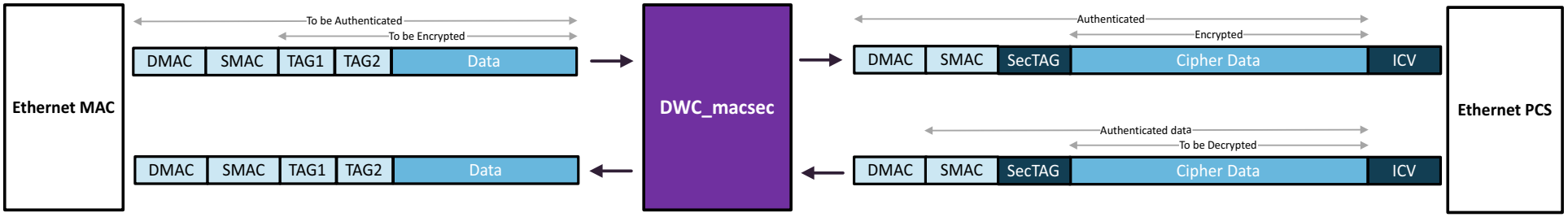
The DWC_macsec supports several VLAN on the clear modes. These modes allow the MACsec functionality to be used in a system that comprises multiple LANs/VLANs.

The VLAN on the clear modes are configured by the MACsec_IP_Control.vlanmode register field.

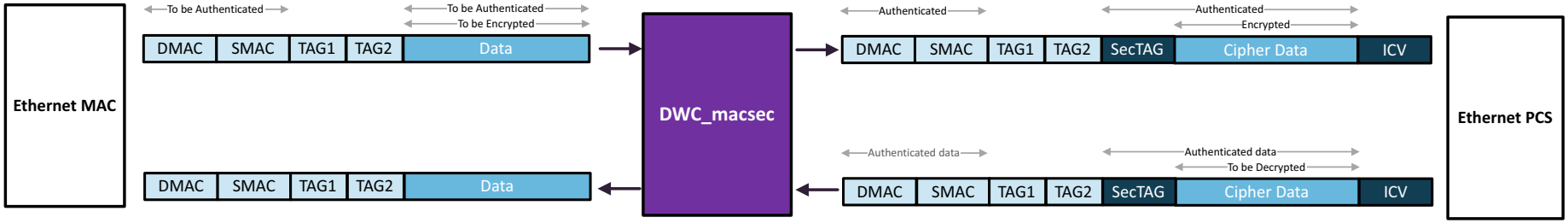
This section describes the supported modes by the DWC_macsec.

Note
- The ethernet packet fields are simplified for reducing figure complexity.
- When a TAG is encrypted, this means that the MACsec AC frames are configured to be authenticated and encrypted.

- **Up to two VLAN TAGs not on the clear mode: vlanmode = 3'b000**
  - The DWC_macsec accepts up to two VLAN TAGs
  - The DWC_macsec encrypts and authenticates the VLAN TAGs

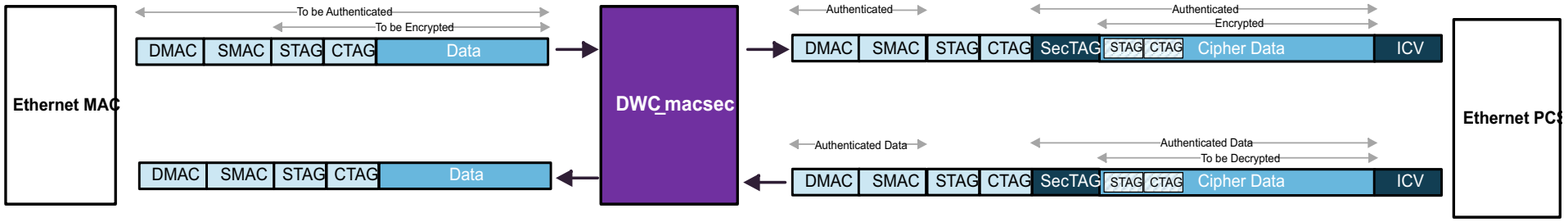


- **Up to two VLAN TAGs on the clear mode: vlanmode = 3'b001**
  - The DWC_macsec accepts up to two VLAN TAGs
  - The TAGs are transmitted on the clear
  - The TAGs are not part of authentication/ICV
  - TPID checks are performed to ensure TPID 0x8100 or 0x88a8 is present following the 802.1Q C-TAG/S-TAG order (if the check fails, the DWC_macsec encrypts everything - TAGs are not transmitted on the clear)

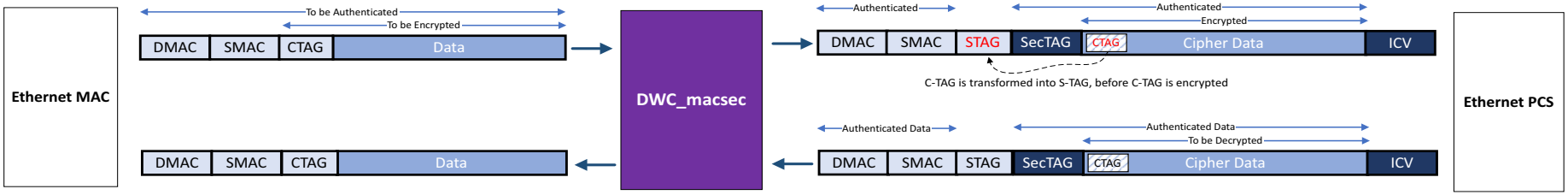


- **S-TAG and C-TAG on the clear dual copy mode: vlanmode = 3'b010**
  - The DWC_macsec accepts up to two VLAN TAGs
  - The two TAGs are transmitted on the clean (not part of the encrypted and authenticated data) but also copied and transmitted as part of the encrypted and authenticated data

- TPID checks are performed to ensure TPID 0x8100 or 0x88a8 is present following the 802.1Q C-TAG/S-TAG order (if the check fails, the DWC_macsec encrypts everything - TAGs are not transmitted on the clear)

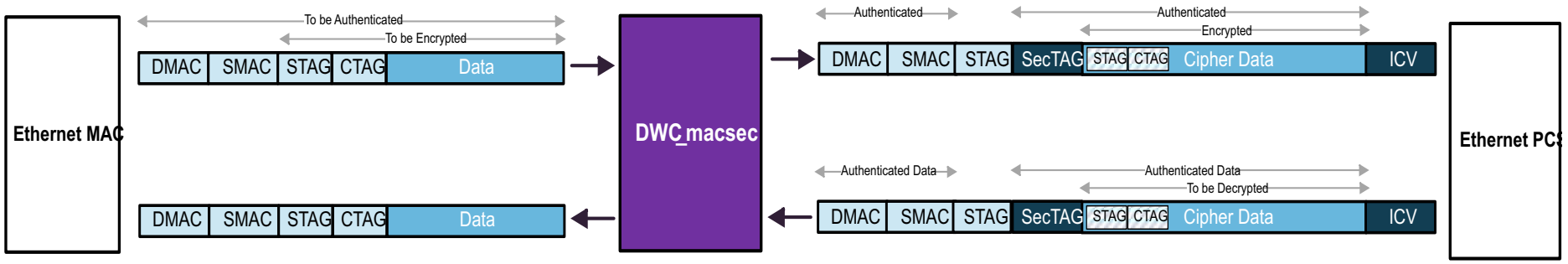


- **C-TAG to S-TAG replace mode: vlanmode = 3'b011**
  - The DWC_macsec accepts one VLAN TAGs (C-TAG)
  - The C-TAG is transformed into an S-TAG and transmitted on the clear (not part of the encrypted and authenticated data)
  - The original C-TAG is transmitted encrypted and authenticated
  - TPID checks are performed to ensure TPID 0x8100 is present (if check fails, the DWC_macsec encrypts everything - TAG is not transmitted on the clear)

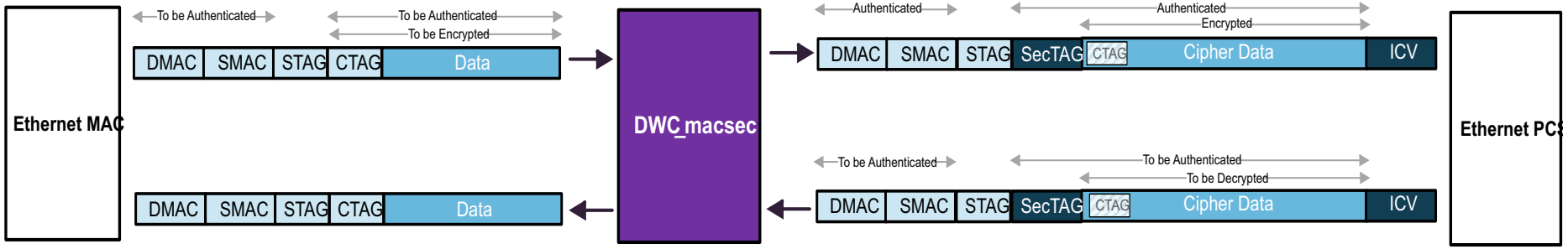


- **S-TAG on the clear single copy mode: vlanmode = 3'b100**
  - The DWC_macsec accepts two VLAN TAGs (S-TAG, C-TAG)
  - The S-TAG is copied and transmitted on the clear (not part of the encrypted and authenticated data)
  - Both, the original S-TAG and C-TAGs are transmitted, encrypted and authenticated

- TPID checks are performed to ensure TPID 0x8100 and 0x88a8 are present (if check fails, the DWC_macsec encrypts everything - TAG is not transmitted on the clear)

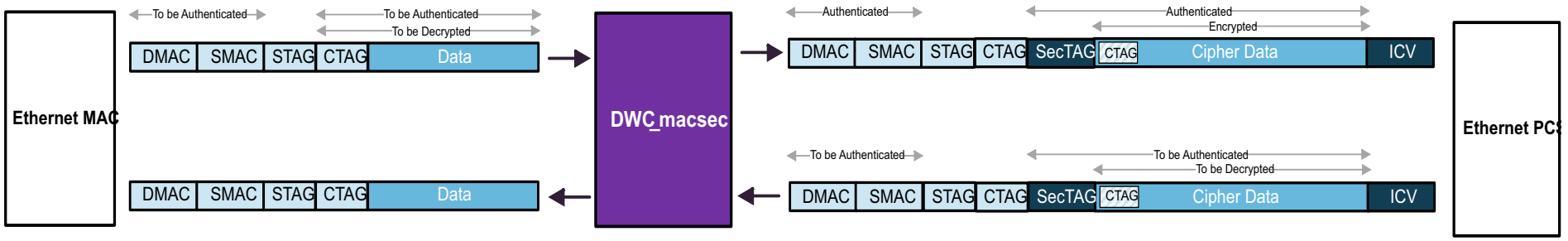


- **Two TAGs with S-TAG on the clear mode: vlanmode = 3'b101**
  - The DWC_macsec accepts two VLAN TAGs (S-TAG, C-TAG)
  - The S-TAG is transmitted on the clear (not being part of the encrypted and authenticated data)
  - The C-TAG is transmitted encrypted and authenticated
  - TPID checks are performed to ensure TPID 0x8100 and 0x88a8 are present (if check fails, the DWC_macsec encrypts everything - TAG are not transmitted on the clear)



- **Two TAGs with C-TAG copy mode mode: vlanmode = 3'b110**
  - The DWC_macsec accepts two VLAN TAGs (S-TAG, C-TAG)
  - The S-TAG and C-TAGs are transmitted on the clear (not part of the encrypted and authenticated data)

- The C-TAG is also transmitted encrypted and authenticated
- TPID checks are performed to ensure TPID 0x8100 and 0x88a8 are present (if check fails, the DWC_macsec encrypts everything - TAGs are not transmitted on the clear).



In the DWC_macsec RX if a packet is a MACsec AC packet, and the VLAN check fails against the programmed mode, the VLAN Tags are on the clear, and the frame is processed normally.

2.5.1 Security Association Lookup search using VLAN Values

The DWC_macsec offers the functionality of using the VLAN as a Security Association LookUp search. This is done by programming the 16 bits associated with the PRI, CFI, and VID on the TX and RX SAI lookups as explained in "Security Associations Mode of Work".

Table 2-1 lists the security association match/no match behavior considering the possibility of using the wildcard bits for VLAN values on the SAI tables:

Table 2-1 Security Association Lookup

	Programmed fields on SAIs				
VLANS on frame	Wildcard bit for VLAN2	VLAN2 Value	Wildcard bit for VLAN1	VLAN1 Value	SA Match Result
Frame has zero VLAN tags	0	Value2	0	Value1	No match
	0	Value2	1	Don´t care	No match
	1	Don´t care	0	Value1	No match
	1	Don´t care	1	Don´t care	Match

Table 2-1 Security Association Lookup (Continued)

	Programmed fields on SAIs				
Frame has one VLAN tag (VLAN1 is present)	0	Value2	0	Value1	No match
	0	Value2	1	Don´t care	No match
	1	Don´t care	0	Value1	Match
	1	Don´t care	0	Value1	No match
	1	Don´t care	1	Don´t care	Match
Frame has one VLAN tag (VLAN2 is present)	0	Value2	0	Value1	No match
	0	Value2	1	Don´t care	Match
	0	Value2	1	Don´t care	No match
	1	Don’t care	0	Value1	No match
	1	Don´t care	1	Don´t care	Match
Frame has two VLAN tags (VLAN1 and VLAN2 present)	0	Value2	0	Value1	No match
	0	Value2	1	Don´t care	Match
	0	Value2	1	Don´t care	No match
	1	Don´t care	0	Value	No match
	1	Don´t care	1	Don´t care	Match
	0	Value2	0	Value1	Match
	0	Value2	0	Value1	No match
	0	Value2	1	Don´t care	Match
	0	Value2	1	Don´t care	No match
	1	Don´t care	0	Value1	Match

Table 2-1 Security Association Lookup (Continued)

	Programmed fields on SAIs				
	1	Don´t care	0	Value1	No match
	1	Don´t care	1	Don´t care	Match

2.6 Inter Packet Gap (IPG) Considerations and MAC Programming

The DWC_macsec is placed between an Ethernet MAC and an Ethernet PCS/PHY, the Inter Packet Gap at the output of the Ethernet MAC, is large enough for the DWC_macsec to add the SecTAG and ICV. Additionally, if the DWC_macsec is configured to operate with a VLAN on the clear mode, it may be necessary (depending on the mode) to increase even further the minimum TX IPG at the inputs of the DWC_macsec.

Table 2-2 shows the input values of the DWC_macsec (column 6 and column 7), depending on the configured operation mode. The column of the *VLAN Copy Mode* indicates the following:

- **None:** no VLAN TAG is added by the DWC_macsec IP (represents the configuration space of the register MACsec_IP_Control.vlanmode = 3'b000, 3'b001, 3'b101)
- **Single:** one additional VLAN TAG is inserted by the DWC_macsec IP (represents the configuration space of the register MACsec_IP_Control.vlanmode = 3'b011, 3'b100, 3'b110)
- **Dual:** two additional VLAN TAG are inserted by the DWC_macsec IP (represents the configuration space of the register MACsec_IP_Control.vlanmode = 3'b010)

Table 2-2 Inter Packet Gap (IPG) Considerations of the DWC_macsec

Modes	Standard MinTX IPG	Standard Min RX IPG	Minimum TX IPG at the inputs of the DWC_macsec (Force Bypass [a])	Minimum TX IPG at the inputs of the DWC_macsec [b] (VLAN mode: None)	Minimum TX IPG at the inputs of the DW-C_macsec [b] (VLAN mode: Single)	Minimum TX IPG at the inputs of the DW-C_macsec [b] (VLAN mode: Dual)	Minimum RX IPG at the inputs of the DWC_macsec
0,1, 17-20	12	5	5 bytes	37 bytes	41 bytes	45 bytes	3 bytes [c]
2,3,21	12	5	12 bytes	44 bytes	48 bytes	52 bytes	5 bytes
4-9, 12,13	9 [d]	5	9 bytes	41 bytes	45 bytes	49 bytes	5 bytes

Table 2-2 Inter Packet Gap (IPG) Considerations of the DWC_macsec (Continued)

Modes	Standard MinTX IPG	Standard Min RX IPG	Minimum TX IPG at the inputs of the DWC_macsec (Force Bypass [a])	Minimum TX IPG at the inputs of the DWC_macsec [b] (VLAN mode: None)	Minimum TX IPG at the inputs of the DW-C_macsec [b] (VLAN mode: Single)	Minimum TX IPG at the inputs of the DW-C_macsec [b] (VLAN mode: Dual)	Minimum RX IPG at the inputs of the DWC_macsec
10,11,14-16	5 [d]	1	5 bytes	37 bytes	41 bytes	45 bytes	1 byte

a. Force Bypass represents the mode where the DWC_macsec is active but is configured to bypass all frames. This is configured by the `MACsec_IP_Control.rx_bypass` and `MACsec_IP_Control.tx_bypass` registers.
b. Consider an ICV of 16 bytes.
c. For these modes, although configuration is always done with byte resolution, frames with IPGs of 1 nibble less than the configured value are also allowed. For example, with a minimum IPG configured to 3 bytes, frames presenting an IPG of 2.5 bytes are accepted by DWC_macsec.
d. Consider the minimum IPG when a Deficit Idle Counter algorithm is used.

Attention

- The MACSec_CFG_MIN_IPG.min_ipg_tx and MACSec_CFG_MIN_IPG.min_ipg_rx registers are configured to match the minimum IPG as defined in Table 2-2. If the Ethernet MAC cannot respect the above values and transmits packets to the DWC_macsec that have an IPG below those numbers (except RX modes 0,1,17-20), the processing of the frames in the DWC_macsec may result in an unexpected behavior.
- Additionally, if the minimum TX and RX IPG values supported by the Ethernet MAC and Ethernet PCS/PHY, respectively are higher than the listed in the Table 2-2, then the MACSec_CFG_MIN_IPG.min_ipg_tx and MACSec_CFG_MIN_IPG.min_ipg_rx registers can be configured to match the exact values.

2.6.1 Synopsys XLGMAC or XGMAC Configuration Example

When pairing the DWC_macsec with a Synopsys DWC_xlgmac or DWC_xgmac, one can use the table below as a configuration baseline for the IPs.

Attention

The XLGMAC and XGMAC need an upgrade patch for allowing the extra configuration of 44 bytes for xgmii and 52 bytes for xlgmii. Please contact Synopsys in case this patch is needed.

Table 2-3 XLGMAC or XGMAC Configuration - Example

Mode	VLAN Copy Mode	Standard Min TX IPG	Standard Min RX IPG	DWC_xlgmac IPG Configuration	DWC_macsec IPG Configuration
Force Bypass	NA	5 bytes	1 byte	DDIC = 0'b0 IFP = 0'b0 IPG = 000'b	min_ipg_tx = 8'd5 min_ipg_rx = 8'd1
50 Gbps	None	5 bytes	1 byte	DDIC = 1'b IFP = 1'b1 IPG = 100'b	min_ipg_tx = 8'd40 min_ipg_rx = 8'd1
	Single	5 bytes	1 byte	DDIC = 1'b IFP = 1'b1 IPG = 100'b	min_ipg_tx = 8'd40 min_ipg_rx = 8'd1;
	Dual	5 bytes	1 byte	DDIC = 1'b IFP = 1'b1 IPG = 101'b	min_ipg_tx = 8'd52 min_ipg_rx = 8'd1;
25 Gbps	None	9 bytes	5 bytes	DDIC = 1'b IFP = 1'b1 IPG = 100'b	min_ipg_tx = 8'd40 min_ipg_rx = 8'd5
	Single	9 bytes	5 bytes	DDIC = 1'b IFP = 1'b1 IPG = 101'b	min_ipg_tx = 8'd52 min_ipg_rx = 8'd5
	Dual	9 bytes	5 bytes	DDIC = 1'b IFP = 1'b1 IPG = 101'b	min_ipg_tx = 8'd52 min_ipg_rx = 8'd5
10 Gbps	None	9 bytes	5 bytes	DDIC = 1'b IFP = 1'b1 IPG = 101'b	min_ipg_tx = 8'd40 min_ipg_rx = 8'd5
	Dual	9 bytes	5 bytes	DDIC = 1'b IFP = 1'b1 IPG = 110'b	min_ipg_tx = 8'd52 min_ipg_rx = 8'd5
	Single	9 bytes	5 bytes	DDIC = 1'b IFP = 1'b1 IPG = 110'b	min_ipg_tx = 8'd52 min_ipg_rx = 8'd5

2.7 KaY and Unknown Cipher Packet Handling

The DWC MACsec standard defines the following modes of handling packets by a controlled port, imposed by TCI (Tag Control Information) field:

- **Case 1:** E = 0, C = 0 - packet to be authenticated with standard defined AES-GCM algorithm
- **Case 2:** E = 0, C = 1 - packet using other cipher suite
- **Case 3:** E = 1, C = 0 - KaY frames

- **Case 4:** E = 1, C = 1 - packet to be authenticated and encrypted with standard defined AES-GCM algorithm

Case 1 and Case 4 are handled by the DWC_macsec as defined by the standard:

- On the TX an association with an SA is complete, a SecTAG is created, ICV is calculated and if E = 1, the data payload is encrypted
- On the RX the packet is matched against an SA, the SecTAG is removed, ICV is checked and if E = 1, the data payload is decrypted

Case 2 and Case 4 are handled as following:

- KaY frames are handled as per defined in the standard and the frames are not processed on the TX or RX. This allows the application to create KaY frames and transmit them unencrypted (with perspective of MACsec AC). When a frame is marked as KaY frame (E= 1, C = 0) on the TX or RX SAD, it is simply bypassed and sent to the PCS or MAC, respectively.
- Unknown Cipher frames are also handled as bypass frames. While there is not a direct applicability currently at the 802.1AE standard for an unknown cipher frame, it could be used to pass data between Uncontrolled Ports. For this, it is the responsibility of the application to create and parse these type of frames.

2.8 MIB Counters Usage

The DWC_macsec contains several MIB (Management Information Base) counters. The following are the main types of MIB Counters:

- Security Association (SA) Monitoring Counters
- Security Channel (SA) Monitoring Counters
- Global Monitoring Counters

All the counters need to be addressed indirectly by a command interface. This command interface is controlled by the following APB addressable registers:

- **MIB_CMD**
  - This is the main command register, where you should configure the required action (for example, read back TxUntaggedPkts_GBL).
- **MIB_CMD_STAT**
  - This register indicates an ongoing operation.
- **MIB_WR_DATA_HIGH**
  - This includes the MSB data registers where you can set a value that is written to the selected counter. You can not only read but also set a counter value.
  - Counter selection is done by the MIB_CMD register.
- **MIB_WR_DATA_LOW**
  - This includes the MSB data registers where you can set a value that is written into the selected counter. You can not only read but also set a counter value.

- Counter selection is done by the MIB_CMD register.

- **MIB_CMD_D_BUF_HIGH/LOW_x (with x from 0 to 11)**
  - These registers read back data from the selected counter
    - Counter selection is done by the MIB_CMD register.
  - Most of the counters are 64-bit wide. The data is divided into *_LOW and *_HIGH registers (if a counter is only 32 bit, then only the *_LOW has valid data).
  - In case of a counter read back: For a 64 bit counter, both _HIGH_ and _LOW_ have valid data, and for a 32 bit counter only the _LOW_ has valid data.

Attention The MIB counters should be all cleared after reset, by issuing an INIT command (see the sections below about the commands). This will set all the MIB stored in SRAMs to 0x0.

2.8.1 List of MIB Counters

Table 2-4 and Table 2-5 lists the complete available set of MIB counters. The CTR value indicates the value that needs to be written into the MIB_CMD.CTR_ID register field, so the specific individual counter can be addressed (write, read).

Table 2-4 TX/Egress List of MIB Counters

TX/Egress MIB Counter	CTR Value (hex)	Counter Description
Global Counters		
TxUntaggedPkts_GBL	0x0	Global counter for TX untagged packets (bypass with no SecTAG)
TxLookupDisPkts_GBL	0x1	Global counter for TX lookup discarded packets (increment only when hard TTL is hit or when an inactive SA is hit)
TxTooLongPkts_GBL	0x2	Global counter for TX packets that are too long (higher than MTU)
TxOctetsProtected_GBL	0x5	Global counter for TX number of protected octets
TxOctetsEncrypted_GBL	0x6	Global counter for TX number of encrypted octets
Security Channel Counters		
TxProtectedPkts _SC	0x20	TX Security Channel number of protected packets
TxEncryptedPkts_SC	0x21	TX Security Channel number of protected packets
TxOctetsProtected_SC	0x22	TX Security Channel number of protected octets
TxOctetsEncrypted_SC	0x23	TX Security Channel number of encrypted octets
Security Association Counters		
TxProtectedPkts_SA	0x40	TX Security Association number of protected packets
TxEncryptedPkts_SA	0x41	TX Security Association number of encrypted packets

Table 2-5 RX/Ingress List of MIB Counters

RX/Ingress MIB Counter	CTR Value (hex)	Counter Description
Global Counters		
RxUntaggedPkts_GBL	0x0	Global counter for all received RX packets without a SecTAG when Protect Frames is enabled
RxBadTagPkts_GBL	0x1	Global counter for all received RX bad SecTAG packets
RxNoTagPkts_GBL	0x4	Global counter for all received RX packets without a SecTAG when protect frames are disabled
RxOctetsValidated_GBL	0x5	Global counter for RX number of protected octets
RxOctetsDecrypted_GBL	0x6	Global counter for RX number of decrypted octets
RxOverrunPkts_GBL	0x7	Global counter for RX packets that are discarded because they fail replay checks and SA replay bit is enabled
RxNoSAErrorPkts_GBL	0x8	Global counter for the number of RX packets received with an inactive SA
RxNoSAPkts_GBL	0x9	Global counter for the number of RX packets received with SecTAG that do not get a lookup match and are by-passed
Security Channel Counters		
RxOKPkts_SC	0x20	Number of packets that are properly processed on the RX for a specific Security Channel
RxInvalidPkts_SC	0x21	Number of packets that are discarded when ICV validation fails in a specific security channel and both the validate frames are checked and C bit is disabled
RxNotValidPkts_SC	0x22	Number of packets that are discarded when ICV validation fails in a specific security channel and either validate frames is Strict or C bit is enabled
RxDelayedPkts_SC	0x25	Number of packets that fail replay check and Replay Protect Enable is set to false
RxLatePkts_SC	0x26	Number of packets discarded when replay check fails and Replay Protect Enable is set to true
RxUnchecked_SC	0x27	Number of packets that are unchecked by SC
RxOctetsValidated_SC	0x28	Number of octets that are validated for a specific Security Channel
RxOctetsDecrypted_SC	0x29	Number of octets that are decrypted for a specific Security Channel
Security Association Counters		
RxOKPkts_SA	0x40	Number of packets that are properly processed on the RX for a specific Security Association

Table 2-5 RX/Ingress List of MIB Counters (Continued)

RX/Ingress MIB Counter	CTR Value (hex)	Counter Description
RxInvalidPkts_SA	0x41	Number of packets discarded when ICV validation fails in a specific security association and both the validate frames are checked and C bit is disabled
RxNotValidPkts_SA	0x42	Number of packets discarded when ICV validation fails in a specific security association and either of the validate frames is Strict or C bit is enabled
RxNotUsingSAPkts_SA	0x43	Number of packets discarded in a specific security association with an inactive SA, and have either the C bit is en-abled or validate frames set to strict
RxUnusedSAPkts_SA	0x44	Number of packets discarded in a specific security association with an inactive SA, and have both the C bit is disa-bled or validate frames set to check

Note The MIB counters do not wrap up to 0 in case of an overflow. If an overflow occurs, the application software must clear the counters to 0.

2.8.2 Accessing the MIB Counters

The MIB Counters are addressed by a command interface, which is accessible by the Host ABP interface.

To execute a read command, the firmware must:

1. Write to the MIB_CMD register with the command details.
2. Poll the BUSY field of MIB_CMD_STAT register. When the register field is low, the operation is finished.
3. Read back the results from the MIB_CMD_D_BUF_LOW/HIGH_x registers. Not all *_LOW/HIGH_x register indexes may contain valid data (depending on the executed command).

To execute a write command, the firmware must:

1. Write the MIB_WR_DATA_LOW/HIGH with the wanted data.
2. Write to the MIB_CMD register with the command details.
3. Poll the BUSY field of MIB_CMD_STAT register. When the register field is low, the operation is finished.

Table 2-6 describes the possible commands that can be requested by the `MIB_CMD` register.

Table 2-6 MIB_CMD Command Configuration

Register Name	Address	Field	Description
MIB_CMD	0x0800	CMD	■ This field contains the Command Number to be executed. ■ The command numbers are described in the Table 2-7.
		DIR	■ Specify the direction: 0 = Rx/Ingress, 1 = Tx/Egress.
		CLR	■ If this bit is set, the corresponding registers are cleared atomically with the read transaction.
		CTR_ID	■ ID of the counter to access. The value of this field is defined in the column *CTR* of the Table 2-4 and Table 2-5.
		OBJ_ID	■ Counter can be either on a global, SA or SC basis. ■ If the counter to be read/written is a SA counter you need to specify the SA index, if the counter is on a SC basis you need to specify the SC index. ■ In the case of the counter being on a global basis this field is not required.

Table 2-7 Type of Available MIB Commands

Command Name	Command Number	Description	Field Usage			
			DIR	CLR	CTR_ID	OBJ_ID
INIT	0	Clears all counters for a particular direction. On system initialization this command must be issued once for each direction.	0,1	NA	NA	NA
RD_SGL	1	Reads an individual counter. Results are returned in the first location of MIB_CMD_D_BUF.	0,1	0,1	Counter ID Value	SC or SA Index
Reserved	2	Reserved	0,1	0,1	NA	NA
Reserved	3	Reserved	0,1	0,1	NA	NA
Reserved	4	Reserved	0,1	0,1	NA	NA
WR_SGL	5	Writes (initializes) a single counter with the value stored in the MIB_WR_DATA register. Any data stored within the counter will be overwritten.	0,1	NA	Counter ID Value	SC or SA Index

2.8.3 MIB Counters Linking to Security Associations (SA) and Security Channels

The MIB SA counters associated with TX and/or RX SAs, follow a direct mapping. The SAs programmed in the TX SAI in the Lookup entry 0, corresponds to the MIB Counters for Security Association 0. The same mapping method is applicable for RX SAI Lookup. Nonetheless, the RX Security Associations need to be mapped into specific MIB SC Counters, as for RX there isn´t any notion of which Security Association belongs to which security channel.

The DWC_macsec allows you to select which RX Security Association maps to which security channel.

To perform the MIB Counter security channel linking with the respective Security Association, you must configure the association table:

1. Program the RX_SAI_SC_CORR_TABLE_SC_VALUE: with a value of a Security Channel.
2. Program the RX_SAI_SC_CORR_TABLE_PRG.
   a. TYPE: set this field to 1'b1 to indicate a write access.
   b. ENTRY: in this field one must write the Security Association number, that should be associated with the Security Channel set on the _VALUE register above. As an example:
      - RX_SAI_SC_CORR_TABLE_SC_VALUE = 5 and ENTRY = 3.
      - This means that RX Security Association #3 belongs to the RX Security Channel #5. Thus, when an event occurs on Security Association #3, the MIB counter(s) of Security Channel #5 are going to be incremented.
3. If another entry needs to be programmed:
   a. Poll RX_SAI_SC_CORR_TABLE_PR.BUSY.
   b. If 1'b0, then jump to Step 1.

2.9 Key Configuration and Key Swap

This section describes how to program the keys into the AES cores and how to perform a key swap. Each key is considered as a context (also known as *key context*).

2.9.1 Programming a Key Context

The keys are programmed into the AES core by an APB interface. The following steps describe how to program a key context to the firmware:

1. Write the Key Value into the KEY_x registers.

Key must be written in big endian format.

2. Select the key context which will be programmed by writing the KEY_IDX.
3. Poll KEY_LOAD_STAT.BUSY register. This is only asserted during key installation that is roundabout 30 core clock cycles, until it is 0. Alternatively, use key done interrupt to check if key installation is finished.
4. To program more keys go to step 2.
5. Go to step 3, if all keys are already programmed.

2.9.1.1 Key Context Mapping

The mapping of the AES-GCM contexts versus the RX and TX security association index, is performed as follows:

- KEY_IDX = 0, ENCRYPT = 1: programs the key used for the TX SC#0, SA#0 (SA#0, corresponds to AN = 0)

- KEY_IDX = 1, ENCRYPT = 1: programs the key used for the TX SC#0, SA#1 (SA#1, corresponds to AN = 1)
- KEY_IDX = 2, ENCRYPT = 1: programs the key used for the TX SC#1, SA#0
- KEY_IDX = 3, ENCRYPT = 1: programs the key used for the TX SC#1, SA#1
- …
- KEY_IDX = N, ENCRYPT = 1: programs the key used for the TX SC#N, SA#1
- KEY_IDX = N+1, ENCRYPT = 0: programs the key used for the RX SA#0
- KEY_IDX = N+2, ENCRYPT = 0: programs the key used for the RX SA#1
- KEY_IDX = N+3, ENCRYPT = 0: programs the key used for the RX SA#2
- KEY_IDX = N+3, ENCRYPT = 0: programs the key used for the RX SA#3
- ...

The application must program the CTRL register of the AES-GCM core as per the described mapping. The application must program not only the KEY_IDX, but also the ENCRYPT bit, which indicates if the context is used for encryption or decryption and the KEY_SZ field that will indicate the size of the key.

Note The complete CTRL register must be programmed as indicated above for every context/SA.

2.9.2 Changing a Key for an SA

The DWC_macsec allows key swapping through multiple ways. You can select any one of the following options.

- **Swapping between ANs for a given SA:**
  - The application has one AN that is active and the other that is inactive.
  - Program a new key the inactive SA AN. For detailed steps, see "Programming a Key Context" on page 77.
  - Set this newly programmed SA AN as active.
  - Set the initially active SA AN as inactive.
- **Not swapping between ANs for a given SA:**
  - The application has all ANs active for a given SA.
  - Application halts the traffic associated to the specific SA AN.
  - Set the specific SA AN to inactive.
  - Program a new key for that specific SA AN. For detailed steps, see "Programming a Key Context" on page 77.
  - Set the specific SA AN to active.

2.10 Host Configuration Interface - APB Interface

The APB4 interface follows the AMBA defined standard. The integrator addresses the *ARM IHI 0024C - AMBA 4 APB Protocol*.

The APB interface for the DWC_macsec, uses the `PPROT[1]` bit to protect access to the complete address space.

2.11 Parity and ECC Support

2.11.1 Parity Support

The AES-GCM present in the DWC_macsec, offers parity protection on the configuration registers. For the integrator to make usage of this parity feature, it need to enable the specific interrupt by the AES-GCM APB interface. The parity failure interrupt is forwarded to the `O_aes_hcfg_irq`.

2.12 PHY Link Fault Signal and LPI Messages Handling

2.12.1 Link Fault Signaling

The DWC_macsec allows the transmission of Link Fault Signaling from/to the Ethernet MAC and Ethernet PHY, when using the XLGMII/XGMII interface.

Table 2-8 shows an example of messages starting at Lane 0. The messages are passed non-encrypted from/to the MAC and the PHY.

Table 2-8 Messages Starting at Lane 0 - Example

Lane 0	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Description
0x9C	0x00	0x00	0x00	0x00	0x00	0x00	0x00	Reserved
0x9C	0x00	0x00	0x01	0x00	0x00	0x00	0x00	Local Fault
0x9C	0x00	0x00	0x02	0x00	0x00	0x00	0x00	Remote Fault
0x9C	0x00	0x00	0x03	0x00	0x00	0x00	0x00	Link Interruption

Table 2-9 shows an example of messages starting at Lane 8 (only applicable for XLGMII with 128 bits). The messages are passed non-encrypted from/to the MAC and the PHY.

Table 2-9 Messages Starting at Lane 8 - Example

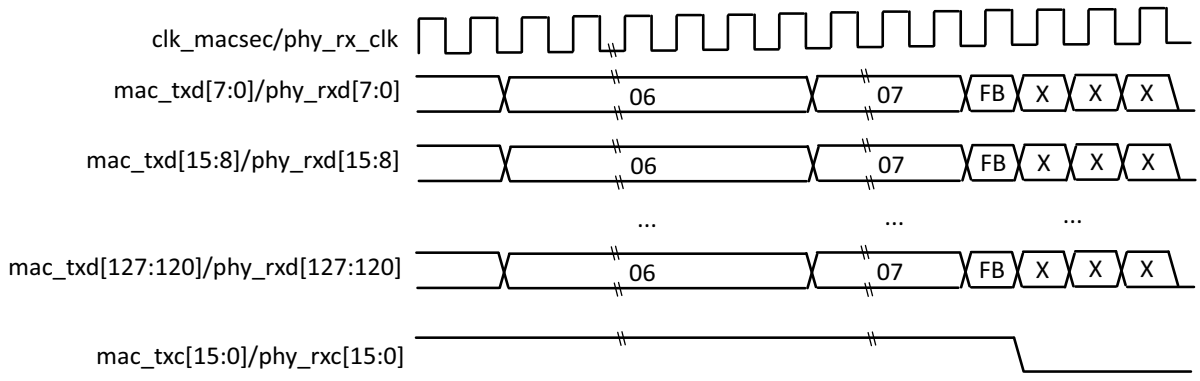
Lane 8	Lane 9	Lane 10	Lane 11	Lane 12	Lane 13	Lane 14	Lane 15	Description
0x9C	0x00	0x00	0x00	0x00	0x00	0x00	0x00	Reserved
0x9C	0x00	0x00	0x01	0x00	0x00	0x00	0x00	Local Fault
0x9C	0x00	0x00	0x02	0x00	0x00	0x00	0x00	Remote Fault
0x9C	0x00	0x00	0x03	0x00	0x00	0x00	0x00	Link Interruption

Caution Using XLGMII/XGMII requires a recovery time of 160 IDLE bytes after a FAULT sequence is transmitted or received, before the transmission or reception of a new frame. Not respecting this time results in unexpected behavior for the transmitted or received data.

2.12.2 LPI Indication for XGMII/XLGMII

The DWC_macsec supports the indication of LPI between the MAC and the PHY, when a XLGMII/XGMII interface is used. Figure 2-12 shows the LPI indication timing diagram for a message starting at Lane 0.

Figure 2-12 Timing Diagram for a Message Starting at Lane 0 – XLGMII/XGMII Interfaces



Note The DWC_macsec discards any LPI frames below 64 bytes, as these are interpreted as noise on the interface.

2.12.3 LPI Indication for MII/GMII

The duration of the LPI signaling for MII/GMII is depicted in the following figures:

Figure 2-13 Timing Diagram for an MII Interface

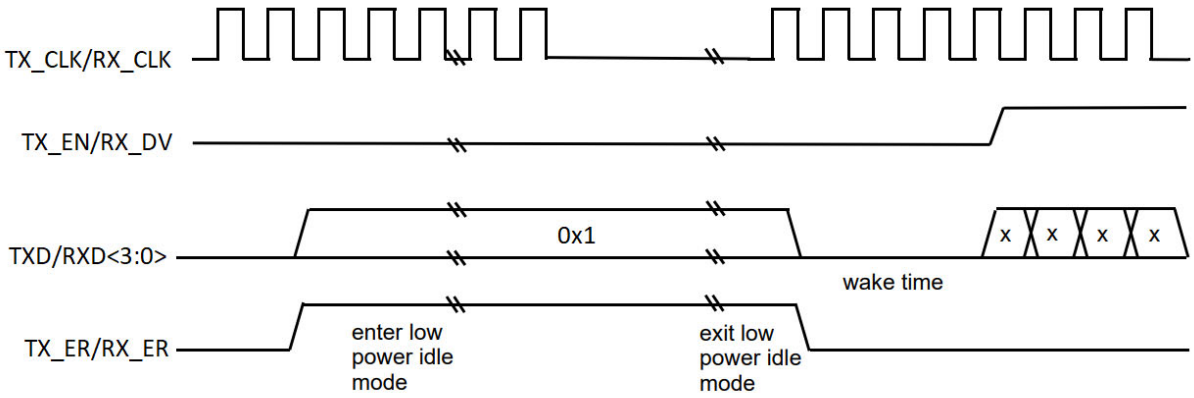
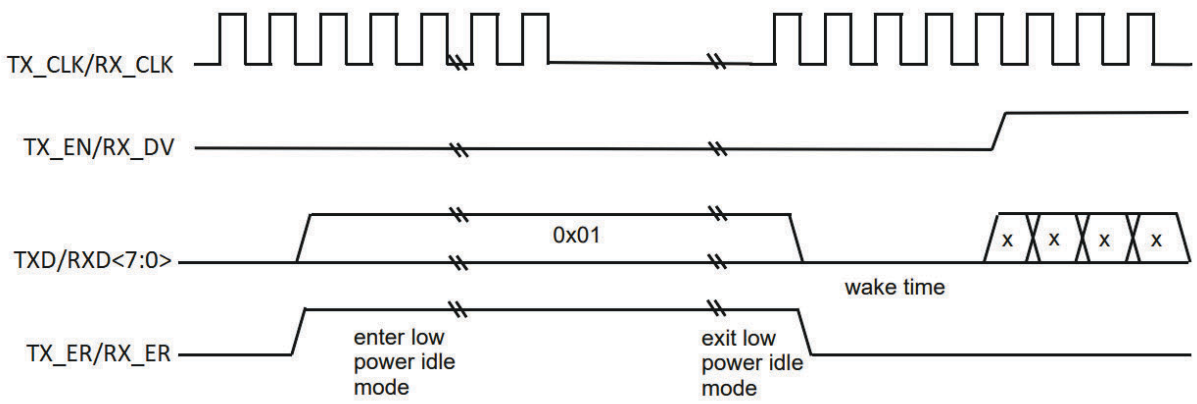


Figure 2-14 Timing Diagram for an GMII Interface



Note Like XGMII/XLGMII interfaces, for MII/GMII and reduced interfaces, the DWC_macsec discards any LPI frames below 64 bytes, as these are interpreted as noise on the interface.

2.13 Interrupts, Errors and Status Indications

This section describes the list of interrupts/error monitoring conditions that the DWC_macsec contains and additionally the behavior dependent on certain error conditions.

2.13.1 Interrupts and Error Sources

The DWC_macsec has several indications and interrupts to monitor the module status.

Table 2-10 describes the set of indications/interrupts available in the DWC_macsec.

Table 2-10 Interrupts and Status Indications

Name	Trigger When	Output Signal	Monitored by/Connected to
AES key done	A key context installation is finished	O_aes_hcfg_irq	Hardware Security Module
AES key programming error	Key context index selected is not valid		
AES register parity error	A parity error in register is spotted		

Table 2-10 Interrupts and Status Indications (Continued)

Name	Trigger When	Output Signal	Monitored by/Connected to
MACsec AC frame too long	Processing of the TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded	macsec_irq	Hardware Security Module, Application
MACsec AC frame too short	If a frame shorter than a certain amount of bytes is received (for more information about frame lengths, see section “Controlled and Uncontrolled Ports Frame Processing” on page 83)		
MACsec AC inactive SA	Ethernet packet targets an SA that is not active		
MACsec AC soft TTL	Ethernet packet has a packet number equivalent to the Soft TTL limit set for the SA		
MACsec AC hard TTL	Ethernet packet has a packet number equivalent to the Hard TTL limit set for the SA		
MACsec AC bad sectag	Ethernet packet has a bad sectag		
MACsec AC unknown SC	Ethernet packet is targeting an unknown Security Channel		
MACsec AC pre replay	Ethernet packet failed the preliminary replay check		
MACsec AC post replay	Ethernet packet failed the post decryption replay check		
MACsec AC RX check fail	Ethernet packet failed the ICV check		
MACsec AC new SA detected	Ethernet packet received with indication of a new SA	macsec_irq	Hardware Security Module, Application
MACsec AC unknown cipher	Ethernet packet received indicates an unknown algorithm (via E and C bits)		
MACsec AC lookup discard	The result of the lookup indicated that the packet should be discarded		
MACsec AC CRC frame error	CRC error found in a packet		
MACsec AC RX illegal IPG	Ethernet packet with illegal IPG value		

2.13.2 Controlled and Uncontrolled Ports Frame Processing

In the 802.1AE standard, the processing behavior of MACsec AC and non-MACsec frames are described depending if a controlled or uncontrolled port is addressed.

Since the DWC_macsec is unaware if the physical port is connected only to a controlled port or to both controlled and uncontrolled ports, several configuration options increase the flexibility for the integrator to deal with both scenarios.

In addition to the regular security association rule checking imposed by the TX/RX SAI and TX/RX SAD lookups, the DWC_macsec contains the following options when non-MACsec frames are mentioned (this is any type of frame that does not have the 0x88e5 ethertype):

- Global Validate Configuration - configured through the `MIB_VALIDATE` register
  - This allows the user to apply a rule check for the following:
    - non-MACsec frames
    - MACsec AC frames that do not match a lookup search and have the C bit = 0
  - If it is set to non-strict, any of the above-mentioned frames are bypassed (without being modified).
  - If it is set to strict, the following happens:
    - the non-MACsec frames with ethertypes that match the `ETHERTYPE_FILTER_x` are bypassed
    - the non-MACsec frames with ethertypes that do not match the `ETHERTYPE_FILTER_x` are discarded
    - the MACsec AC frames that do not match a lookup search and have the C bit = 0 are discarded
- Non-MACsec frame valid Ethertype - configured through the `ETHERTYPE_FILTER_x` and `ethertype_filter_en` register fields
  - This is a lookup that contains up to 8 programmable ethertypes.
  - If the Global Validate Configuration is set to strict, then the non-MACsec frames are parsed against this lookup. If there is a match, the frame is bypassed; if there is no match the frame is dropped.
- Global MACsec AC Tagging Enable - configured through the `replace_ethertype` register field
  - When enabled, it replaces the 0x88e5 ethertype with a proprietary ethertype one of the MACsec AC frames that are processed in the RX, and passes all checks.
  - The frame is received by the application on the RX with the SecTAG. Within the SecTAG, the application checks this proprietary ethertype (ICV is nonetheless removed).

Table 2-11 represents the behavior corresponding to the global configuration along with the MACsec AC frames protocol parsing (the values configured on the RX security association lookup):

Table 2-11 Global configuration along with the MACsec AC frames protocol parsing

		Global Configuration		MACsec AC Frames Protocol Parameters					
	Frame Type	MIB_VALIDATE = Strict	MIB_VALIDATE = Non-Strict	SA Validate = Strict	SA Validate = Disabled	SA Validate = Check	SA Replay Protect =1	SA Replay Protect = 0	replace_ethertype = 1
Non-MACsec Frames	Non-Macsec Valid Ethertype (ETHERTYPE_FILTER_(0 to 3) & ethertype_filter_en = 1)	Bypass	Bypass	N/A	N/A	N/A	N/A	N/A	N/A
	Non-Macsec Invalid Ethertype	Discard	Bypass	N/A	N/A	N/A	N/A	N/A	N/A
MACsec AC Frame with Bad SecTAG	Macsec Bad Tag	Discard	Discard	N/A	N/A	N/A	N/A	N/A	N/A
MACsec AC KaY Frame	Macsec C bit = 0 & E bit = 1 (KaY)	Bypass	Bypass	N/A	N/A	N/A	N/A	N/A	N/A
MACsec AC Frames without SA Match	Macsec no lookup match C bit = 1	Discard	Discard	N/A	N/A	N/A	N/A	N/A	N/A
	Macsec no lookup match C bit = 0	Discard	Bypass (SecTag and ICV removed)	N/A	N/A	N/A	N/A	N/A	N/A

Table 2-11 Global configuration along with the MACsec AC frames protocol parsing (Continued)

MACsec AC Frames with SA Match (processed frames accordingly with the defined protocol)	Macsec inactive sa & C bit = 0	N/A	N/A	Discard	Decrypt	Decrypt	N/A	N/A	N/A
	Macsec inactive sa & C bit = 1	N/A	N/A	Discard	Discard	Discard	Discard	Discard	N/A
	Macsec replay check fail	N/A	N/A	N/A	N/A	N/A	Discard	Decrypt	N/A
	Macsec icv check fail & C bit = 0	N/A	N/A	Discard	Decrypt	Decrypt	N/A	N/A	N/A
	Macsec icv check fail & C bit = 1	N/A	N/A	Discard	Discard	Discard	N/A	N/A	N/A
	Macsec check all OK	N/A[a]	N/A	Decrypt	Decrypt	Decrypt	Decrypt	Decrypt	0x88e5 replaced by Proprietary EtherType

a. The global configuration is only applicable for non-processed MACsec frames and non-MACsec frames. MACsec Frames Protocol Parameters are only applicable for MACsec Frames that Match an SA.

2.13.3 Behavior Under Error Conditions

The DWC_macsec implements the behavior described in this section that is applicable for the error conditions specified in Table 2-10 (that may occur due to protocol errors or wrong programming):

2.13.3.1 On TX/Egress

- Frame is poisoned with CRC error if HARD_TTL limit is reached.
- A RUNT frame is discarded or dropped if it has less than 64 bytes (excluding preamble and terminate control character).
- Frame is truncated, and CRC is poisoned if the MTU limit is hit.
- Frame is encrypted and CRC sent to the PCS is poisoned, if a frame is received from the MAC with a CRC error.
- Frame is discarded or dropped, if a frame matches an Security Association that is inactive.

- Frame is encrypted but CRC poisoned if the size is larger than what has been programmed for the specific Security Association.

2.13.3.2 On RX/Ingress

- Frame CRC is poisoned, if an ICV error is detected.
- Frame CRC is poisoned, if the Security Association is inactive.
- Frame CRC is poisoned, if a SecTAG error is detected.
- Frame CRC is poisoned, if a Replay Error is detected.
- Frame CRC is poisoned, if an AES-GCM Security Association does not have a key installed (due to ICV check failure).
- Frame is discarded/dropped if an IPG violation is detected.
- A RUNT frame is discarded/dropped if it has less than 64 bytes (excluding preamble and terminate control character).
- Frame is passed through to the MAC if the Security Association is not present in the RX SAI LUT.
- Frame is decrypted and CRC sent to the MAC is poisoned, if a frame is received from the PCS with a CRC error.

Ethernet standard guarantees an Ethernet minimum frame total size of 64 bytes, hence if a frame is smaller and needs nonetheless to be transmitted, the MAC or application must add padding. The DWC_macsec IP does not add padding to the frames or process frames with an SL byte different from 0.

2.14 FIPS 140-3 Test Mode Usage

The DWC_macsec contains an AES-centric test mode, that enables the integrator to evaluate or test the algorithm implementation of the cryptographic core. This test mode is used by the integrator to perform the algorithm testing which is required to achieve a FIPS certification. The test mode is available when the `DWC_MACSEC_AES_FIPS_EN` parameter is set to 1.

The DWC_macsec has the following test modes:

- BIST mode
- Firmware driven (FW) test mode

2.14.1 BIST Mode

The BIST mode runs automatically after the core reset is released. To use the BIST mode, the integrator must perform the following:

1. Reset the DWC_macsec.
2. Poll the done flag through the following options:
   - Monitor the `O_aes_fips_self_test_done` output signal until it is asserted to 1'b1.
   - Poll the `FIPS_SELF_TEST_STAT.SELF_TEST_DONE_APB` field until it is asserted to 1'b1.

3. Check the fail status through the following options:
   - Monitor the O_aes_fips_self_test_fail output signal.
   - Read the FIPS_SELF_TEST_STAT.FAIL register field.
4. If a failure occurred, read the FIPS_SELF_TEST_STAT.FAIL_CAUSE register to identify the reason for the failure.
5. In case of failure, it is possible to inhibit the output of the AES-GCM (making the AES-GCM unusable) through the following options:
   - Assert the I_aes_fips_inhib input pin to 1'b1.
   - Set the MISC_CONFIG.INHIB_OUTPUT register field to 1'b1.

It is possible to re-run the BIST after the core is in mission mode.

Note The DWC_macsec does not perform data back pressuring automatically, therefore it is up to the integrator's responsibility to guarantee that there is no data corruption in the process (BIST must run when there is no application data).

To re-trigger the BIST, the integrator must perform the following steps:

1. Inhibit AES-GCM output through one of the two options:
   - assert the I_aes_fips_inhibit input signal to 1'1b
   - write 1'b1 to the MISC_CONFIG.INHIBIT_OUTPUT register field
2. Enable test mode by writing 1'b1 to the FIPS_SELF_TEST_CTL.FIPS_SELF_TEST_ENA register field.
3. Setup BIST_VECT_MODE register:
   - The FUNCT field selects the test operation.
     - 0: Batch (runs all tests)
     - 1: CTR tests
     - 2: GCM tests
     - 3: ECB

Note If the Batch operation is selected, the remaining fields of this register are irrelevant.

   - The KEY_SIZE field sets the key size to use.
     - 0: 128-bit key
     - 1: 256-bit. This field is only used if 256-bit keys are enabled (DWC_MACSEC_MAX_KEY_SIZE==1)
   - The ENCRYPT sets the operation direction. Set it to 0 to decrypt and set it to 1 to encrypt.
4. To test the error injection, set the BIST_VECT_ERR_INJ register.

- By setting the ECB_ERR_INJ, MAC_ERR_INJ, GCM_ERR_INJ, and CTR_ERR_INJ fields, a bit-flip is inserted in the respective test vectors to execute the error testing.

5. Clear the FIPS_SELF_TEST_STAT.SELF_TEST_DONE flag and write 1'b1 to the FIPS_SELF_TEST_STAT.SELF_TEST_FAIL register field.
6. Trigger the test by writing 1'b1 to the BIST_VECT_CTL.BIST_TEST_GO field (this field is self-clearing).
7. Poll the done status through the following options:
   - Monitor the O_aes_fips_self_test_done output signal until it is asserted to 1'b1.
   - Poll the FIPS_SELF_TEST_STAT.SELF_TEST_DONE APB register field until is asserted to 1'b1.
8. Check the fail status through the following options:
   - Monitor the O_aes_fips_self_test_fail output.
   - Read the FIPS_SELF_TEST_STAT.FAIL register field.
9. In case of failure, read the FIPS_SELF_TEST_STAT.FAIL_CAUSE register to identify the reason for the failure.
   - If the GCM decrypt test or batch test is run, read also the FIPS_SELF_TEST_STAT.SELF_TEST_MATCH_RES register. A 1'b1 indicates a proper match for the MAC. If it is 1'b0, this means there is a failure in the MAC comparison.
10. Disable test mode by setting the FIPS_SELF_TEST_CTL.FIPS_SELF_TEST_ENA register field to 1'b0.
11. If the test is passed, deassert the output inhibition in the same way it is asserted:
    - By deasserting the I_aes_fips_inhibit input signal to 1'b0.
    - By writing MISC_CONFIG.INHIBIT_OUTPUT register field to 1'b0.

2.14.2 Firmware (FW) Driven Test Mode

The FW-driven test mode allows the integrator to run its specific test vectors. In this test mode, the AES-GCM inputs are driven by a set of APB-accessible registers, where the integrator can drive the required test vectors.

Perform the following steps to use the FW-driven test mode:

1. Inhibit the AES output through one of the following options:
   - Assert the I_aes_fips_inhibit input signal to 1'b1.
   - Write 1'b1 to the MISC_CONFIG.INHIBIT_OUTPUT register field.
2. Enable test mode by writing 1'b1 to the FIPS_SELF_TEST_CTL.FIPS_SELF_TEST_ENA register field.
3. (Optional) Enable the internal self-checker by setting the FIPS_SELF_TEST_CTL.FIPS_SELF_TEST_CHECK register field.

4. Load the AES key in the same way as in mission mode:

Note

- Load the test mode key into a specific context. Both the FIPS_SELF_TEST_VECT_CMD_CTX register and the DWC_MACSEC_AES_DEFAULT_FIPS_CTX_IDX parameter indicate which context to load the key into.
- For CTR mode tests the key must be loaded with the CTRL.MODE register field set to 1.

5. Load the IV into the FIPS_SELF_TEST_IVx registers (GCM and CTR modes only):
   - The most significant word goes into the IV0 register.
   - In GCM mode, the IV is 96-bit wide, so write the IV3 register to 0.
6. Load the Additional Authentication Data (AAD) and message (MSG) vectors:
   - Write the AAD (fulfilling the size indicated by the DWC_MACSEC_AES_DP_WIDTH bits) into FIPS_SELF_TEST_IDATAx registers (most significant word goes into the IDATA0 register, AAD is only present in GCM mode).
   - Write MSG (fulfilling the size indicated by the DWC_MACSEC_AES_MSG_WIDTH bits) into the remaining FIPS_SELF_TEST_IDATAx registers (MSG and AAD must not share the same 128-bit block, the most significant word goes into register IDATA0, 4, 8, and so on).

Note

- AAD+MSG must not exceed DWC_MACSEC_AES_DP_WIDTH bits.
- The number of AAD or MSG bits should be byte-aligned.

7. Load the FIPS_SELF_TEST_ATT register.
   - Set the SELF_TEST_AAD field to 1'b1 if the data path contains AAD.
   - Set the SELF_TEST_MSG field to 1'b1 if the data path contains MSG.
   - (Optional) Set the SELF_TEST_MAC_MATCH field to 1'b1 to use the MAC checker (if the checker is enabled in step 3). This checker is used during the last round (SELF_TEST_EOP ==1) of GCM decryption operations.
   - Set the SELF_TEST_ECB field to 1'b1 when in ECB mode.
   - Set the SELF_TEST_SOP field to 1'b1 during the first round of the packet (first portion of the vector).
   - Set the SELF_TEST_EOP field to 1'b1 during the last round (the entire vector has been written).
8. (Optional) Load AAD and expected cipher or plain text vectors only if the internal self-checker is activated in step 3.
   - Load expected AAD and cipher or plain text vectors into FIPS_SELF_TEST_ODATAx registers the same way as in step 6.
9. (Optional) Load expected MAC vector only if the internal MAC checker is enabled in step 3.
10. Load AAD and MSG sizes:

- If the current round contains AAD (`FIPS_SELF_TEST_ATT.SELF_TEST_AAD == 1`), load the AAD size (in bytes) into the `FIPS_SELF_TEST_AAD_SIZE` register. A zero in this register means `DWC_MACSEC_AES_DP_WIDTH` of 8 bytes.
- If the current round contains MSG (`FIPS_SELF_TEST_ATT.SELF_TEST_MSG == 1`), load the MSG size (in bytes) into the `FIPS_SELF_TEST_MSG_SIZE` register.

11. Load total AAD and MSG length (GCM mode only):
    - This is only required during the last round of a multi-round packet (`eop==1 && sop==0`).
    - Write the following into the `FIPS_SELF_TEST_IVx: {tot_aad_len[47:0], 12'h0, tot_msg_len[35:0]}` registers.
    - Most significant word goes into the `IV0` register. `IV3` register is set to all zeros.
12. Clear the done and fail status fields by writing 1'b1 into the following fields:
    - `FIPS_SELF_TEST_STAT.SELF_TEST_DONE`
    - `FIPS_SELF_TEST_STAT.SELF_TEST_FAIL`
13. Trigger the test by writing 1'b1 to `FIPS_SELF_TEST_VECT_CTL.SELF_TEST_GO` field (this field is self-clearing).
14. Poll the done status through the following options:
    - Monitor the `O_aes_fips_self_test_done` output signal until is set to 1'b1.
    - Poll the `FIPS_SELF_TEST_STAT.SELF_TEST_DONE` register field until is set to 1'b1.
15. Read fail status through the following options:
    - `O_aes_fips_self_test_fail` output signal
    - `FIPS_SELF_TEST_STAT.SELF_TEST_FAIL` register field
    - In case the MAC checker is enabled, the `FIPS_SELF_TEST_STAT.SELF_TEST_MAC_MATCH_RES` register field indicates its result: 1'b0 means that it did not match, 1'b1 means that it did match
16. Read the `FIPS_SELF_TEST_ODATAx` registers and check if the cipher or plain is the expected one.
17. Read the `FIPS_SELF_TEST_MACx` registers and check if the MAC is the expected (GCM only). This is only done during the last round of the packet (`eop==1`).
18. If an error occurs, the test ends, and the output is left inhibited (disabling the AES usage).
19. Clear the `FIPS_SELF_TEST_STAT` register by writing 1'b1 to the fields (fields are W1C).
20. Repeat from step 6 until the entire vectors are injected.
21. Repeat from step 5 until all vectors are tested.
22. Disable test mode by setting the `FIPS_SELF_TEST_CTL.FIPS_SELF_TEST_ENA` register field to 1'b0.
23. If the test is passed, deassert the output inhibition in the same way it is asserted:
    - By setting the `I_aes_fips_inhibit` input signal to '1b0.

- By writing the MISC_CONFIG.INHIBIT_OUTPUT register field to 1'b0.

Note An On-demand BIST test must be executed before any FW-driven test. If the integrator needs to execute a new BIST test after a FW-driven one, the DWC_macsec must be reset.

2.15 SRAM Zeroization

When external SRAMs are used, the DWC_macsec contains an SRAM zeroization functionality that allows for all external SRAMs to be written with 0s. This feature is meant to be used during the initialization of the DWC_macsec (for example, to be done after BIST finishes). The zeroization feature is available when DWC_MACSEC_SRAM_ZERO_EN is set.

To perform the SRAM zeroization, the following steps should be performed:

1. Reset the DWC_macsec.
2. Wait for the reset phase to complete.
3. Assert the I_sram_zero_init input signal to 1'b1.
4. Wait until the output signal O_sram_zero_done gets deasserted to 1'b0 and then deassert the input signal I_sram_zero_init to 1'b0.
5. Wait for the signal O_sram_zero_done to be asserted to 1'b1 (signaling that the zeroization process is terminated).

3 Performance

This chapter includes the following sections:

- "Datapath Latency" on page 94
- "Performance Considerations" on page 98

The DWC_macsec architecture maintains full-duplex data (two separate channels - one TX and one RX channel) throughput to match the associated line rate.

3.1 Datapath Latency

The DWC_macsec has a fixed latency for each specific operating mode. This fixed latency is applied for any MACsec or non-MACsec frame that is not dropped internally (due to protocol violations).

The Table 3-1 tables describes the latency numbers for all modes and all possible memory types.

Table 3-1 Latency for All Modes

Data Mode	Description	Parameter Configuration	SRAM with Linear Search [a]	SRAM/FF Memory Enabled [b]	SRAM/FF Memory Enabled [c]	TCAMs Enabled [d]
0	■ Use the 4-bit MII interface over the 8-bit GMII interface @10Mbps, 2.5MHz	(DWC_MACSEC_PERF_RATE = 0) ||	TX: 473	TX: 267	TX: 473	TX: 569
		(DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	RX: 720	RX: 378	RX: 720	RX: 720
1	■ Use the 4-bit MII interface over the 8-bit GMII interface @100Mbps, 25MHz	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 473	TX: 267	TX: 473	TX: 569
			RX: 720	RX: 378	RX: 720	RX: 720
2	■ Use the full 8-bit GMII interface @ 1Gbps, 125MHz	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 255	TX: 149	TX: 255	TX: 303
			RX: 375	RX: 192	RX: 375	RX: 375

Table 3-1 Latency for All Modes (Continued)

Data Mode	Description	Parameter Configuration	SRAM with Linear Search [a]	SRAM/FF Memory Enabled [b]	SRAM/FF Memory Enabled [c]	TCAMs Enabled [d]
3	■ Use the full 8-bit GMII interface @ 2.5Gbps, 312.5MHz	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 255	TX: 149	TX: 255	TX: 303
			RX: 375	RX: 192	RX: 375	RX: 375
4	■ Use a 32-bit XGMII interface over the 128-bit XLGMII interface @2.5Gbps, 78.125MHz	(DWC_MACSEC_PERF_RATE >= 1)	-	TX: 60	TX: 126	TX: 138
				RX: 85 [e]	RX: 85 [e]	RX: 85 [e]
5	■ Use a 32-bit XGMII interface over the 128-bit XLGMII interface @5Gbps, 156.25MHz	(DWC_MACSEC_PERF_RATE >= 1)	-	TX: 60	TX: 126	TX: 138
				RX: 85 [e]	RX: 85 [e]	RX: 85 [e]
6	■ Use a 32-bit XGMII interface over the 128-bit XLGMII interface @10Gbps, 312.5MHz	(DWC_MACSEC_PERF_RATE >= 1)	-	TX: 60	TX: 126	TX: 138
				RX: 85 [e]	RX: 85 [e]	RX: 85 [e]
7	■ Use a 32-bit XGMII interface over the 128-bit XLGMII interface @25Gbps, 781.25MHz	(DWC_MACSEC_PERF_RATE >= 2)	-	TX: 60	TX: 126	TX: 138
				RX: 85 [e]	RX: 85 [e]	RX: 85 [e]
8	■ Use a 64-bit XLGMII interface over the 128-bit XLGMII interface @10Gbps, 156.25MHz	(DWC_MACSEC_PERF_RATE >= 2) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 98	TX: 98	TX: 104
				RX: 128 [e]	RX: 128 [e]	RX: 128 [e]

Table 3-1 Latency for All Modes (Continued)

Data Mode	Description	Parameter Configuration	SRAM with Linear Search [a]	SRAM/FF Memory Enabled [b]	SRAM/FF Memory Enabled [c]	TCAMs Enabled [d]
9	■ Use a 64-bit XLGMII interface over the 128-bit XLGMII interface @25Gbps, 390MHz/395MHz	(DWC_MACSEC_PERF_RATE >= 2) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 98	TX: 98	TX: 104
				RX: 128 [e]	RX: 128 [e]	RX: 128 [e]
10	■ Use a 64-bit XLGMII interface over the 128-bit XLGMII interface @40Gbps, 625MHz	(DWC_MACSEC_PERF_RATE >= 3) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 98	TX: 98	TX: 104
				RX: 128 [e]	RX: 128 [e]	RX: 128 [e]
11	■ Use a 64-bit XLGMII interface over the 128-bit XLGMII interface @50Gbps, 781MHz	(DWC_MACSEC_PERF_RATE >= 3) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 98	TX: 98	TX: 104
				RX: 128 [e]	RX: 128 [e]	RX: 128 [e]
12	■ Use the full XLGMII 128-bit interface @25Gbps, 195.3125MHz	(DWC_MACSEC_PERF_RATE >= 2) && (!DWC_MACSEC_AUTOMOTIVE	-	TX: 59	TX: 59	TX: 62
				RX: 72 [e]	RX: 72 [e]	RX: 72 [e]
13	■ Use the full XLGMII 128-bit interface @25Gbps, 390.625MHz (alternate cycle mode)	(DWC_MACSEC_PERF_RATE >= 2) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 88	TX: 88	TX: 94
				RX: 106 [f]	RX: 106 [f]	RX: 106 [f]
14	■ Use the full XLGMII 128-bit interface @40Gbps, 312.5MHz	(DWC_MACSEC_PERF_RATE >= 3) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 59	TX: 59	TX: 62
				RX: 72 [e]	RX: 72 [e]	RX: 72 [e]

Table 3-1 Latency for All Modes (Continued)

Data Mode	Description	Parameter Configuration	SRAM with Linear Search [a]	SRAM/FF Memory Enabled [b]	SRAM/FF Memory Enabled [c]	TCAMs Enabled [d]
15	■ Use the full XLGMII 128-bit interface @50Gbps, 390.625MHz	(DWC_MACSEC_PERF_RATE >= 3) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 59	TX: 59	TX: 62
				RX: 72 [e]	RX: 72 [e]	RX: 72 [e]
16	■ Use the full XLGMII 128-bit interface @100Gbps, 781.25MHz (also known as CGMII)	(DWC_MACSEC_PERF_RATE == 4) && (!DWC_MACSEC_AUTOMOTIVE)	-	TX: 59	TX: 59	TX: 62
				RX: 72 [e]	RX: 72 [e]	RX: 72 [e]
17	■ Use the 2-bit RMII interface over the 8-bit GMII interface @10Mbps, 50MHz. ■ Data is updated every 10 clock cycles.	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 11935	TX: 5106	TX: 11935	TX: 12895
			RX: 15876	RX: 6928	RX: 15876	RX: 15876
18	■ Use the 2-bit RMII interface over the 8-bit GMII interface @100Mbps, 50MHz	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 904	TX: 476	TX: 904	TX: 1096
			RX: 1399	RX: 850	RX: 1399	RX: 1399
19	■ Use the 4-bit RGMII interface over the 8-bit GMII interface @10Mbps, 2.5MHz	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 478	TX: 272	TX: 478	TX: 574
			RX: 725	RX: 383	RX: 725	RX: 725

Table 3-1 Latency for All Modes (Continued)

Data Mode	Description	Parameter Configuration	SRAM with Linear Search [a]	SRAM/FF Memory Enabled [b]	SRAM/FF Memory Enabled [c]	TCAMs Enabled [d]
20	■ Use the 4-bit RGMII interface over the 8-bit GMII interface @100Mbps, 25MHz	(DWC_MACSEC_PERF_RATE = 0) || (DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	TX: 478	TX: 272	TX: 478	TX: 574
			RX: 725	RX: 383	RX: 725	RX: 725
21	■ Use the 4-bit RGMII interface over the 8-bit GMII interface @1Gbps, 125MHz	(DWC_MACSEC_PERF_RATE = 0) ||	TX: 260	TX: 154	TX: 260	TX: 308
		(DWC_MACSEC_PERF_RATE > 0 && DWC_MACSEC_DUAL_IF = 1)	RX: 380	RX: 197	RX: 380	RX: 380

a. `((DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD ==1) && (DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD ==1))` is the parameter configuration to perform the linear search with SRAM.
b. `(DWC_MACSEC_PERF_RATE <= 1) || ((DWC_MACSEC_PERF_RATE == 2) && (DWC_MACSEC_AUTOMOTIVE))` is the parameter configuration to enable the SRAM/FF memory (low performance).
c. `((DWC_MACSEC_PERF_RATE >= 2) && (!DWC_MACSEC_AUTOMOTIVE))` is the parameter configuration to enable the SRAM/FF memory (high performance).
d. `DWC_MACSEC_TX_SAI_MEM_TYPE ==1 || DWC_MACSEC_RX_SAI_MEM_TYPE == 1` is the parameter configuration to enable the TCAMs.
e. Latency value increases by 1 if `DWC_MACSEC_RXT0_NEXT_SUPPORT ==1`.
f. Latency value increases by 2 if `DWC_MACSEC_RXT0_NEXT_SUPPORT ==1`.

3.2 Performance Considerations

The DWC_macsec supports full throughput that matches the link speed. The throughput matching is achieved by selecting the `DWC_MACSEC_PERF_RATE` parameter value.

4 Parameter Descriptions

This chapter details all the configuration parameters. **You can use the coreConsultant GUI configuration reports to determine the actual configured state of the controller.** Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>)** that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

The parameter descriptions in this chapter include the **Enabled:** attribute which indicates the values required to be set on other parameters before you can change the value of this parameter.

These tables define all of the configuration options for this component.

- Config on page 100
- Info on page 110

4.1 Config Parameters

Table 4-1 Config Parameters

Label	Description
Automotive Support Selection	Configure the product to support an automotive or non-automotive version. When using the Synopsys Ethernet Controller, both products need to match on the automotive support requirement. Using automotive or non-automotive versions requires different product licenses. **Values:** ■ Non-Automotive support (0) ■ Automotive support (1) **Default Value:** Default value depends on highest license located. **Enabled:** Always **Parameter Name:** DWC_MACSEC_AUTO_AC_VERS
MACsec Max Performance	MACsec Performance Rate Selection Selection of maximum performance for MACsec This selection is always full duplex, which means that the performance is achieved for both the channels (and not an aggregated performance). **Values:** ■ up to 1Gbps/2.5Gbps with GMII (0) ■ up to 10Gbps (1) ■ up to 25Gbps (2) ■ up to 50Gbps (3) ■ up to 100Gbps (4) **Default Value:** up to 1Gbps/2.5Gbps with GMII **Enabled:** Always **Parameter Name:** DWC_MACSEC_PERF_RATE
Frame Preemption support	MACsec frame preemption support Support for frame preemption as defined in the 802.3br standard Enabled for configurations with 2 or more RX and 2 or more TX Secure channels. **Values:** 0, 1 **Default Value:** 0 **Enabled:** ((DWC_MACSEC_RX_NUM_SC_SUPPORTED >= 2) && (DWC_MACSEC_TX_NUM_SC_SUPPORTED >= 2)) **Parameter Name:** DWC_MACSEC_PREEMPTION_EN

Table 4-1 Config Parameters (Continued)

Label	Description
Data Interface Selection	Selection of the data interface for Ingress and Egress. The selection of the interface is done automatically and is dependent on the selected performance rate (e.g. one cannot have 50Gb with an 8-bit GMII Interface). **Values:** ■ GMII (8-bits) (0) ■ XLGMII (128-bits) (1) **Default Value:** (DWC_MACSEC_PERF_RATE > 0) ? 1 : 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_DATA_IF
Rxt0 support Enabled	When set, it enables the receive advance indication signal mac_xlgmii_rxt0_next. This indicates that in the next valid clk cycle, the terminate control character is located on lane 0 and all other lanes contain the IDLE character. It can be used for all XLGMII data modes, including CGMII (100Gbps 128-bit XLGMII). **Values:** ■ Rxt0 support not enabled (0) ■ Rxt0 support enabled (1) **Default Value:** Rxt0 support not enabled **Enabled:** DWC_MACSEC_DATA_IF == 1 **Parameter Name:** DWC_MACSEC_RXT0_NEXT_SUPPORT
Dual-IF Enabled	When set, it enables the usage of either a GMII or XGMII/XLGMII interface on boot time. This allows to support different modes **Values:** ■ Dual Interface not enabled (0) ■ Dual interface enabled (1) **Default Value:** Dual Interface not enabled **Enabled:** DWC_MACSEC_DATA_IF == 1 **Parameter Name:** DWC_MACSEC_DUAL_IF
Number of Tx Secure Channels	Number of Secure Channels supported by DWC MACSEC on Tx. Supported from 1-32 Secure Channels. **Values:** 1, ..., 32 **Default Value:** 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_NUM_SC_SUPPORTED
Number of Tx Secure Associations	Number of Secure Associations supported by DWC MACSEC on Tx. The number of Secure Associations is automatically calculated as Number of TX Secure Channels * 2 **Values:** 2, ..., 64 **Default Value:** DWC_MACSEC_TX_NUM_SC_SUPPORTED*2 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_NUM_SA_SUPPORTED

Table 4-1 Config Parameters (Continued)

Label	Description
Number of Rx Secure Associations	Number of Secure Associations supported by DWC MACSEC on Rx. Must be equal or greather than DWC_MACSEC_RX_NUM_SC_SUPPORTED **Values:** 1, ..., 512 **Default Value:** 2 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_NUM_SA_SUPPORTED
Number of Rx Secure Channels	Number of Secure Channels supported by DWC MACSEC on Rx. This parameter cannot be greater than DWC_MACSEC_RX_NUM_SA_SUPPORTED since each SC must have at least one SA associated Supported from 1-256 Secure Channels. **Values:** 1, ..., 256 **Default Value:** 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_NUM_SC_SUPPORTED
Maximum Key Size Selection	Maximum Key Size Selection. Selection of 256-bit key or 128-bit key size **Values:** ■ 128-bit (0) ■ 256-bit (1) **Default Value:** 256-bit **Enabled:** Always **Parameter Name:** DWC_MACSEC_MAX_KEY_SIZE
Jumbo Frames Support	Enable Jumbo Frames support up to 9k bytes **Values:** ■ Jumbo Frames NOT supported (0) ■ Jumbo Frames supported (1) **Default Value:** Jumbo Frames supported **Enabled:** Always **Parameter Name:** DWC_MACSEC_JUMBO_FRAMES_SUPPORTED
Anti-Replay Window Type	Defines how anti-replay windows can be defined: in a linear mode or logarithmic mode. The recommendation is to set logarithmic mode, where windows are defined on power of 2 values only, due to the fact that saves significant logic resources. **Values:** ■ Linear (0) ■ Logarithmic (1) **Default Value:** Logarithmic **Enabled:** Always **Parameter Name:** DWC_MACSEC_WIN_SIZE_MODE

Table 4-1 Config Parameters (Continued)

Label	Description
MACsec Extended Packet Numbering Support	MACsec Extended Packet Numbering support This is the AES-GCM-XPN feature that creates additional PN security resilience for high speed ethernet applications (100Gb+). **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_XPN_SEL
Number of stages in CDC Synchronizers	Number of flip-flops in CDC synchronizer **Values:** 2, 3, 4 **Default Value:** 2 **Enabled:** Always **Parameter Name:** DWC_MACSEC_SYNC_DEPTH
Number of stages in RDC Synchronizers	Number of flip-flops in RDC synchronizer **Values:** 2, 3, 4 **Default Value:** 3 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RDC_SYNC_DEPTH
SRAM Zeroization Support	MACsec External SRAM Zeroization Support Enabling this feature, will offer a way for the external SRAMs to be written with 0s The writing of 0s will be done by the MACsec IP automatically dependent on the status of an input control pin **Values:** 0, 1 **Default Value:** 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_SRAM_ZERO_EN

Table 4-1 Config Parameters (Continued)

Label	Description
APB version	Selects the APB version and corresponding features to be supported. The following options can be selected: ■ DWC_MACSEC_APB_VER = 0 | "APB-4" : follows the AMBA APB Specification (Issue C), commonly referred to as APB-4. ■ DWC_MACSEC_APB_VER = 1 | "APB-5 RME" : follows the AMBA APB Specification (Issue E), adding only Realm Management Extension (RME) support. ■ DWC_MACSEC_APB_VER = 2 | "APB-5 Parity" : follows the AMBA APB Specification (Issue D), adding only parity protection and check signals. ■ DWC_MACSEC_APB_VER = 3 | "APB-5 RME Parity": follows the AMBA APB Specification (Issue E), adding only Realm Management Extension (RME) support and parity protection and check signals. **Values:** ■ APB-4 (0) ■ APB-5 RME (1) ■ APB-5 Parity (2) ■ APB-5 RME PARITY (3) **Default Value:** APB-4 **Enabled:** Always **Parameter Name:** DWC_MACSEC_APB_VER
TX SAI Lookup architecture	Selection of TX SAI Lookup architecture for TX Security Association decoding **Values:** ■ Direct decoding (0) ■ Linear search (1) **Default Value:** ((DWC_MACSEC_PERF_RATE > 0) || (DWC_MACSEC_TX_NUM_SA_SUPPORTED >16))? 0 : 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD
TX SAI Memory type	Selection of TX SAI Memory type for TX Lookup process for a MACsec performance selection of more than 1Gbps **Values:** ■ Internal FFs (0) ■ External TCAM memory (1) ■ External SRAM memory (2) **Default Value:** (DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD == 1) ? 2 : 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAI_MEM_TYPE

Table 4-1 Config Parameters (Continued)

Label	Description
Pipeline stages to include in TX SAI Lookup process	Selects (or not) how many pipeline stages will be included in TX SAI Lookup process **Values:** ■ Direct decoding (0 stages) (0) ■ 1 pipeline stage (1) ■ 2 pipeline stages (2) ■ 3 pipeline stages (3) **Default Value:** 1 pipeline stage **Enabled:** DWC_MACSEC_TX_SAI_MEM_TYPE==0 **Parameter Name:** DWC_MACSEC_TX_SAI_LOOKUP_NUM_PIPE_STAGES
VLAN tags include in TX SAI Lookup process	Selects (or not) how many VLAN TAGs are used for TX Security Association decoding **Values:** ■ No VLAN tags (0) ■ One VLAN tag (1) ■ Two VLAN tags (2) **Default Value:** Two VLAN tags **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS
RX SAI Lookup architecture	Selection of RX SAI Lookup architecture for RX Security Association decoding **Values:** ■ Direct decoding (0) ■ Linear search (1) **Default Value:** ((DWC_MACSEC_PERF_RATE > 0) || (DWC_MACSEC_RX_NUM_SA_SUPPORTED >16))? 0 : 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD
RX SAI Memory type	Selection of RX SAI Memory type for RX Lookup process for a MACsec performance selection of more than 1Gbps **Values:** ■ Internal FFs (0) ■ External TCAM memory (1) ■ External SRAM memory (2) **Default Value:** (DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD == 1) ? 2 : 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAI_MEM_TYPE

Table 4-1 Config Parameters (Continued)

Label	Description
Pipeline stages to include in RX SAI Lookup process	Selects (or not) how many pipeline stages will be included in RX SAI Lookup process **Values:** ■ Direct decoding (0 stages) (0) ■ 1 pipeline stage (1) ■ 2 pipeline stages (2) **Default Value:** 2 pipeline stages **Enabled:** DWC_MACSEC_RX_SAI_MEM_TYPE==0 **Parameter Name:** DWC_MACSEC_RX_SAI_LOOKUP_NUM_PIPE_STAGES
VLAN tags include in RX SAI Lookup process	Selects (or not) how many VLAN TAGs are used for RX Security Association decoding **Values:** ■ No VLAN tags (0) ■ One VLAN tag (1) ■ Two VLAN tags (2) **Default Value:** Two VLAN tags **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS
TX SALT memory type	Selection of TX SALT Memory type **Values:** ■ Internal FFs (0) ■ External Two Port SRAM (1) **Default Value:** External Two Port SRAM **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_TX_SALT_MEM_TYPE
RX SALT memory type	Selection of RX SALT Memory type **Values:** ■ Internal FFs (0) ■ External Two Port SRAM (1) **Default Value:** External Two Port SRAM **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_RX_SALT_MEM_TYPE

Table 4-1 Config Parameters (Continued)

Label	Description
RX ARW memory type	Selection of RX ARW Memory type **Values:** ■ Internal FFs (0) ■ External Two Port SRAM (1) **Default Value:** External Two Port SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_ARW_MEM_TYPE
TX SAD memory type	Selection of TX SAD Memory type **Values:** ■ Internal FFs (0) ■ External Dual Port SRAM (1) **Default Value:** External Dual Port SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAD_MEM_TYPE
RX SAD memory type	Selection of RX SAD Memory type **Values:** ■ Internal FFs (0) ■ External Dual Port SRAM (1) **Default Value:** External Dual Port SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAD_MEM_TYPE
TX MIB SA memory type	Selection of TX MIB SA Memory type **Values:** ■ Internal FFs (0) ■ External Two Port SRAM (1) **Default Value:** External Two Port SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_MIB_SA_MEM_TYPE
RX MIB SA memory type	Selection of RX MIB SA Memory type **Values:** ■ Internal FFs (0) ■ External Two Port SRAM (1) **Default Value:** External Two Port SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_MIB_SA_MEM_TYPE

Table 4-1 Config Parameters (Continued)

Label	Description
RX MIB SC memory type	Selection of RX MIB SC Memory type **Values:** ■ Internal FFs (0) ■ External Two Port SRAM (1) **Default Value:** External Two Port SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_MIB_SC_MEM_TYPE
AES SBOX Architecture Selection	Select the architecture of the S-boxes used within the AES cores. Selecting high-speed architecture will offer better results for reaching high operating frequencies, but will increase the size of the IP. Selecting area optimized architecture will offer lower IP area footprint, but lower maximum operating frequency. **Values:** ■ High-Speed Optimized (0) ■ Low-Area Optimized (1) **Default Value:** Low-Area Optimized **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_SBOX_ARCH
AES-GCM with 2 rounds per cycle (less latency)	AES GCM two rounds per cycle enable. When enabled the AES-GCM computes 2 XOR rounds per cycle, minimizing the latency. This impacts the overall timing and for some configs,the target frequency may not be met. **Values:** ■ Disabled (0) ■ Enabled (1) **Default Value:** Enabled **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_2_ROUNDS_PER_CYCLE_EN
AES Cipher Key Memory Selection	AES cores Cipher Key Memory Interface: External SRAM or Internal Synthesized Memory. **Values:** ■ External SRAM (0) ■ Synthesized Memory (1) **Default Value:** External SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_SYNTH_CKEY_MEM

Table 4-1 Config Parameters (Continued)

Label	Description
AES Hash Key Memory Selection	AES cores Hash Key Memory Interface: External SRAM or Internal Synthesized Memory. **Values:** ■ External SRAM (0) ■ Synthesized Memory (1) **Default Value:** External SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_SYNTH_HKEY_MEM
AES Cipher Context Memory Selection	AES cores Cipher Context Memory Interface: External SRAM or Internal Synthesized Memory. **Values:** ■ External SRAM (0) ■ Synthesized Memory (1) **Default Value:** External SRAM **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_SYNTH_CCTX_MEM
AES FIPS140-3 Self Test Support	FIPS140-3 support selection. If FIPS140-3 is enabled, an internal self-test module is availabe and the pass/fail checker logic is utilized. Power-up and on-demand BIST use hard-coded self-test vectors. In FW-driven FIPS test-mode vectors are injected by software registers. **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_FIPS_EN

4.2 Info Parameters

Table 4-2 Info Parameters

Label	Description
Number of AES key contexts	AES number of AES key contexts. **Values:** 0, ..., (64+512+1) **Default Value:** DWC_MACSEC_TX_NUM_SA_SUPPORTED + DWC_MACSEC_RX_NUM_SA_SUPPORTED + (DWC_MACSEC_AES_FIPS_EN ? 1 : (DWC_MACSEC_PREEMPTION_EN ? 1 : 0)) **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_NCTX
Maximum MSG size per round (in bits).	During the FIPS Firmware Driven Test Mode, the number of MSG bits cannot exceed this value in each round. **Values:** -2147483648, ..., 2147483647 **Default Value:** ((DWC_MACSEC_PERF_RATE <= 1) || ((DWC_MACSEC_PERF_RATE == 2) && (DWC_MACSEC_AUTOMOTIVE))) ? 128 : 256 **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_MSG_WIDTH
Maximum AAD+MSG size per round (in bits).	During the FIPS Firmware Driven Test mode, the number of written AAD+MSG bits cannot exceed this value in each round. **Values:** -2147483648, ..., 2147483647 **Default Value:** DWC_MACSEC_AES_MSG_WIDTH + 128 **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_DP_WIDTH
Cipher Key Context Index for FIPS	The context index to load the key into when performing a FIPS test. By default it's the last context. **Values:** 0, ..., (DWC_MACSEC_AES_NCTX - 1) **Default Value:** DWC_MACSEC_AES_FIPS_EN ? (DWC_MACSEC_AES_NCTX - 1) : 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_DEFAULT_FIPS_CTX_IDX
Hash Key Memory Width for AES Core(in bits)	The width of the Hash Key Memory for AES Core. **Values:** 0, ..., 2147483647 **Default Value:** (DWC_MACSEC_AES_SYNTH_HKEY_MEM == 1) ? 0 : (DWC_MACSEC_AES_DP_WIDTH) **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_HKEY_SRAM_WIDTH_INFO

Table 4-2 Info Parameters (Continued)

Label	Description
Hash Key Memory Depth for AES Core(in rows)	The depth of the Hash Key Memory for AES Cores. **Values:** 0, ..., (64+512+1) **Default Value:** (DWC_MACSEC_AES_SYNTH_HKEY_MEM == 1) ? 0 : DWC_MACSEC_AES_NCTX **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_HKEY_SRAM_DEPTH_INFO
Cipher Key Memory Width for AES Core(in bits)	The width of the Cipher Key Memory for AES Core. **Values:** 0, ..., 642 **Default Value:** (DWC_MACSEC_AES_SYNTH_CKEY_MEM == 1) ? 0 : DWC_UAES_CFG_CKEY_MEM_DATA_WIDTH **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_CKEY_SRAM_WIDTH_INFO
Cipher Key Memory Depth for AES Core(in rows)	The depth of the Cipher Key Memory for AES Core. **Values:** 0, ..., (64+512+1) **Default Value:** (DWC_MACSEC_AES_SYNTH_CKEY_MEM == 1) ? 0 : DWC_MACSEC_AES_NCTX **Enabled:** Always **Parameter Name:** DWC_MACSEC_AES_CKEY_SRAM_DEPTH_INFO
Hash Context Memory Width for AES Core(in bits)	The width of the Hash Context Memory for AES Core. **Values:** -2147483648, ..., 2147483647 **Default Value:** (DWC_MACSEC_AES_SYNTH_HCTX_MEM == 1) ? 0 : DWC_UAES_CFG_HCTX_MEM_DATA_WIDTH **Enabled:** ((DWC_MACSEC_AES_SYNTH_HCTX_MEM==0) && (DWC_UAES_CFG_GCM_EN==1) && (DWC_UAES_CFG_ILEAVE_EN==1)) **Parameter Name:** DWC_MACSEC_AES_HCTX_MEM_DATA_SRAM_WIDTH_INFO
Hash Context Memory Depth for AES Core(in rows)	The depth of the Hash Context Memory for AES Core. **Values:** -2147483648, ..., 2147483647 **Default Value:** (DWC_MACSEC_AES_SYNTH_HCTX_MEM == 1) ? 0 : DWC_MACSEC_AES_NCTX **Enabled:** ((DWC_MACSEC_AES_SYNTH_HCTX_MEM==0) && (DWC_UAES_CFG_GCM_EN==1) && (DWC_UAES_CFG_ILEAVE_EN==1)) **Parameter Name:** DWC_MACSEC_AES_HCTX_MEM_DATA_SRAM_DEPTH_INFO

Table 4-2 Info Parameters (Continued)

Label	Description
Cipher Context Memory Width for AES Core(in bits)	The width of the Cipher Context Memory for AES Core. **Values:** 0, ..., 128 **Default Value:** (DWC_MACSEC_AES_SYNTH_CCTX_MEM == 1) ? 0 : DWC_UAES_CFG_CCTX_MEM_DATA_WIDTH **Enabled:** ((DWC_MACSEC_AES_SYNTH_CCTX_MEM==0) && (DWC_UAES_CFG_ILEAVE_EN==1)) **Parameter Name:** DWC_MACSEC_AES_CCTX_MEM_DATA_SRAM_WIDTH_INFO
Cipher Context Memory Depth for AES Core(in rows)	The depth of the Cipher Context Memory for AES Core. **Values:** 0, ..., (64+512+1) **Default Value:** (DWC_MACSEC_AES_SYNTH_CCTX_MEM == 1) ? 0 : DWC_MACSEC_AES_NCTX **Enabled:** ((DWC_MACSEC_AES_SYNTH_CCTX_MEM==0) && (DWC_UAES_CFG_ILEAVE_EN==1)) **Parameter Name:** DWC_MACSEC_AES_CCTX_MEM_DATA_SRAM_DEPTH_INFO
TX SAI Lookup memory width (in bits)	TX SAI Lookup Memory width. It will consider enabled VLAN tags to be used for Security Association decoding. **Values:** -2147483648, ..., 2147483647 **Default Value:** (DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==0) ? (131 + DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED): (DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==1) ? (131 + DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED + 17): (131 + DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED + 34) **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAI_LOOKUP_MEM_WIDTH
RX SAI Lookup memory width (in bits)	RX SAI Lookup Memory width. It will consider enabled VLAN tags to be used for Security Association decoding, plus AN (2 bits) and SCI (64 bits) **Values:** -2147483648, ..., 2147483647 **Default Value:** (DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==0) ? 66 : (DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==1) ? (66+16+1) : (66+32+2) **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAI_LOOKUP_MEM_WIDTH
TX SAD SRAM depth (in rows)	Two Port TX SAD SRAM depth **Values:** 0, ..., DWC_MACSEC_TX_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_TX_NUM_SA_SUPPORTED **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAD_MEM_DEPTH

Table 4-2 Info Parameters (Continued)

Label	Description
TX SAD SRAM width (in bits)	Two Port TX SAD SRAM width **Values:** 124, ..., 156 **Default Value:** DWC_MACSEC_XPN_SEL ? 156 : 124 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAD_MEM_WIDTH
TX SAD SRAM address bus width (in bits)	Two Port TX SAD SRAM address width **Values:** 0, ..., DWC_MACSEC_TX_LOG2_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_TX_LOG2_NUM_SA_SUPPORTED **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SAD_MEM_ADDR_WIDTH
RX SAD SRAM depth (in rows)	Dual Port RX SAD SRAM memory depth **Values:** 0, ..., DWC_MACSEC_RX_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_RX_NUM_SA_SUPPORTED **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAD_MEM_DEPTH
RX SAD SRAM width (in bits)	Dual Port RX SAD SRAM Memory width **Values:** 0, ..., 172 **Default Value:** DWC_MACSEC_XPN_SEL ? 172 : 108 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAD_MEM_WIDTH
RX SAD SRAM address bus width (in bits)	Dual Port RX SAD SRAM address width **Values:** 0, ..., DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SAD_MEM_ADDR_WIDTH
TX/RX SALT SRAMs Memories width (in bits)	Two Port SALT SRAMs Memories width **Values:** 96, ..., 96 **Default Value:** 96 **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_SALT_MEM_WIDTH
RX SALT SRAM depth (in rows)	Two Port RX SALT SRAM Memory depth **Values:** 0, ..., DWC_MACSEC_RX_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_RX_NUM_SA_SUPPORTED **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_RX_SALT_MEM_DEPTH

Table 4-2 Info Parameters (Continued)

Label	Description
RX SALT SRAM address bus width (in bits)	Two Port RX SALT SRAM address width **Values:** 0, ..., DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_RX_SALT_MEM_ADDR_WIDTH
TX SALT SRAM depth (in rows)	Two Port TX SALT SRAM Memory depth **Values:** 0, ..., DWC_MACSEC_TX_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_TX_NUM_SA_SUPPORTED **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_TX_SALT_MEM_DEPTH
TX SALT SRAM address bus width (in bits)	Two Port TX SALT SRAM address width **Values:** 0, ..., DWC_MACSEC_TX_LOG2_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_TX_LOG2_NUM_SA_SUPPORTED **Enabled:** ((DWC_MACSEC_XPN_SEL == 1)) **Parameter Name:** DWC_MACSEC_TX_SALT_MEM_ADDR_WIDTH
RX ARW SRAM width (in bits)	Two Port RX AntiReplay Window SRAM Memory width **Values:** 0, ..., 32 **Default Value:** (DWC_MACSEC_WIN_SIZE_MODE==1) ? 5: 32 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_ARW_MEM_WIDTH
RX ARW SRAM depth (in rows)	Two Port RX AntiReplay Window SRAM Memory depth **Values:** 0, ..., DWC_MACSEC_RX_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_RX_NUM_SA_SUPPORTED **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_ARW_MEM_DEPTH
RX ARW SRAM address bus width (in bits)	Two Port RX ARW SRAM address width **Values:** 0, ..., DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED **Default Value:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_ARW_MEM_ADDR_WIDTH
TX/RX_GLB_MIB SRAM memory width(in bits)	The width of the TX/RX GBL MIB counters. **Values:** 0, ..., 64 **Default Value:** 64 **Enabled:** Always **Parameter Name:** DWC_MACSEC_MIB_GBL_WIDTH

Table 4-2 Info Parameters (Continued)

Label	Description
TX_GLB_MIB SRAM memory depth(in rows)	The depth of the TX GLB MIB memory. These counters are implemented in FFs internally. **Values:** 0, ..., (16 * 6) **Default Value:** (6) + 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_GLB_MIB_SRAM_DEPTH
RX_GLB_MIB SRAM memory depth(in rows)	The depth of the RX GLB MIB memory. These counters are implemented in FFs internally. **Values:** 0, ..., (16 * 6) **Default Value:** (11) + 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_GLB_MIB_SRAM_DEPTH
TX/RX_SA MIB SRAM memory width(in bits)	The width of the TX/RX SA MIB counters. To be noted that 1x SRAM is needed for the TX SA MIB Counters and 1x SRAM is needed for the RX SA MIB Counters **Values:** 0, ..., 32 **Default Value:** 32 **Enabled:** Always **Parameter Name:** DWC_MACSEC_MIB_SA_WIDTH
TX/RX SC MIB SRAM memory width (in bits)	The width of the TX/RX SC MIB counters. To be noted that 1x SRAM is needed for the TX SC MIB Counters and 1x SRAM is needed for the RX SC MIB Counters **Values:** 0, ..., 64 **Default Value:** 64 **Enabled:** Always **Parameter Name:** DWC_MACSEC_MIB_SC_WIDTH
TX SA MIB SRAM memory depth(in rows)	The depth of the TX SA MIB memory (can be SRAM or FFs, depending on DWC_MACSEC_TX_MIB_SA_MEM_TYPE) **Values:** 0, ..., (DWC_MACSEC_TX_NUM_SA_SUPPORTED * 2) **Default Value:** DWC_MACSEC_TX_NUM_SA_SUPPORTED * (DWC_MACSEC_MIB_TX_SA_MAX_ID + 1) **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SA_MIB_SRAM_DEPTH
RX SA MIB SRAM memory depth(in rows)	The depth of the RX SA MIB memory (can be SRAM or FFs, depending on DWC_MACSEC_RX_MIB_SA_MEM_TYPE) **Values:** -2147483648, ..., (DWC_MACSEC_RX_NUM_SA_SUPPORTED * 5) **Default Value:** DWC_MACSEC_RX_NUM_SA_SUPPORTED * (DWC_MACSEC_MIB_RX_SA_MAX_ID + 1) **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SA_MIB_SRAM_DEPTH

Table 4-2 Info Parameters (Continued)

Label	Description
RX SC MIB SRAM bank 1 memory depth(in rows)	The depth of the RX SC MIB SRAM bank 1 memory. Each RX SC bank can be independently placed internally as FFs, or externally in an SRAM **Values:** -2147483648, ..., (DWC_MACSEC_RX_NUM_SC_SUPPORTED*8) **Default Value:** (DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_MEM_TYPE==1) ? (DWC_MACSEC_RX_NUM_SC_SUPPORTED * (DWC_MACSEC_MIB_RX_SC_MAX_ID + 1 - 2)) : 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SC_MIB_SRAM_1_DEPTH
RX SC MIB SRAM bank 2 memory depth(in rows)	The depth of the RX SC MIB SRAM bank 2 memory. Each RX SC bank can be independently placed internally as FFs, or externally in an SRAM **Values:** -2147483648, ..., (DWC_MACSEC_RX_NUM_SC_SUPPORTED*2) **Default Value:** (DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_MEM_TYPE==1) ? (DWC_MACSEC_RX_NUM_SC_SUPPORTED*2) : 0 **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SC_MIB_SRAM_2_DEPTH
RX SC MIB SRAM bank 2 memory depth(in rows)	The depth of the TX SC MIB SRAM memory. Each RX SC bank can be independently placed internally as FFs, or externally in an SRAM **Values:** -2147483648, ..., ((DWC_MACSEC_TX_NUM_SC_SUPPORTED*2) + (DWC_MACSEC_TX_NUM_SC_SUPPORTED*8)) **Default Value:** DWC_MACSEC_TX_NUM_SC_SUPPORTED) * (DWC_MACSEC_MIB_TX_SC_MAX_ID + 1 **Enabled:** Always **Parameter Name:** DWC_MACSEC_TX_SC_MIB_SRAM_DEPTH
RX SC MIB SRAM memory depth(in rows)	The depth of the RX SC MIB SRAM total memory. **Values:** -2147483648, ..., ((DWC_MACSEC_RX_NUM_SC_SUPPORTED*2) + (DWC_MACSEC_RX_NUM_SC_SUPPORTED*8)) **Default Value:** DWC_MACSEC_RX_SC_MIB_SRAM_1_DEPTH) + (DWC_MACSEC_RX_SC_MIB_SRAM_2_DEPTH **Enabled:** Always **Parameter Name:** DWC_MACSEC_RX_SC_MIB_SRAM_DEPTH

5 Signal Descriptions

This chapter details all possible I/O signals in the controller. For configurable IP titles, your actual configuration might not contain all of these signals.

Inputs are on the left of the signal diagrams; outputs are on the right.

Attention: For configurable IP titles, do not use this document to determine the exact I/O footprint of the controller. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the I/O signals for your actual configuration at workspace/report/IO.html or workspace/report/IO.xml after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the I/O signals that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the widths might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>)** that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

Active State: Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).

Registered: Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of No does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the controller. A value of N/A indicates that this information is not provided for this IP title.

Synchronous to: Indicates which clocks in the IP sample this input (drive for an output) when considering all possible configurations. A particular configuration might not have all of the clocks listed. This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook. The presence of the postfix SuperList indicates a list of all possible clocks over all possible configs. Consult coreConsultant report for which clock applies to your specific configuration.

Exists: Names of configuration parameters that populate this signal in your configuration.

Validated by: Assertion or de-assertion of signals that validates the signal being described.

The I/O signals are grouped as follows:

- Clocks Interface on page 119
- Resets Interface on page 121
- DFT Interface on page 122
- Memory Zeroization Interface on page 123
- APB Host Config Interface on page 124
- AES APB Host Config Interface on page 129
- Status Interface on page 134
- Safety Alarms Interface on page 135
- MAC GMII Interface on page 136
- PHY GMII Interface on page 138
- MAC XGMII/XLGMII Interface on page 140
- PHY XGMII/XLGMII Interface on page 142
- TX MIB SA Memory Interface on page 144
- RX MIB SC Bank 1 Memory Interface on page 147
- RX MIB SC Bank 2 Memory Interface on page 151
- RX MIB SA Memory Interface on page 155
- TX SAI Memory Interface on page 158
- RX SAI Memory Interface on page 164
- TX SALT Memory Interface on page 170
- RX SALT Memory Interface on page 174
- RX ARW Memory Interface on page 178
- TX SAD Memory Interface on page 181
- RX SAD DP Memory Interface on page 183
- AES Cipher Key Memory Interface on page 186
- AES Cipher Context Memory Interface on page 188
- AES Hash Key Memory Interface on page 191
- AES Hash Context Memory Interface on page 194
- AES FIPS Interface on page 197

5.1 Clocks Interface Signals

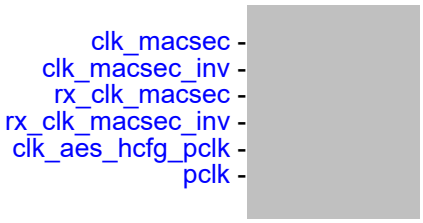


Table 5-1 Clocks Interface Signals

Port Name	I/O	Description
clk_macsec	I	Clock signal from TX GMII/XLGMII used with core MACsec IP blocks. All interface signals are referenced to the rising edge of this signal **Exists:** Always **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
clk_macsec_inv	I	Clock signal from TX RGMII with 180 degrees phase shift when compared to clk_macsec. TX RGMII interface signals are referenced to the rising edge of clk_macsec and this clock **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
rx_clk_macsec	I	Clock signal used in the RX GMII/XLGMII IF. All RX interface signals are referenced to the rising edge of this signal **Exists:** Always **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-1 Clocks Interface Signals (Continued)

Port Name	I/O	Description
rx_clk_macsec_inv	I	Clock signal from RX RGMII with 180 degrees phase shift when compared to rx_clk_macsec. RX RGMII interface signals are referenced to the rising edge of rx_clk_macsec and this clock **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
clk_aes_hcfg_pclk	I	AES Host Config Clock. **Exists:** DWC_UAES_CFG_HOST_CLK_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
pclk	I	APB4 clock **Exists:** Always **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** AHB Clock Signal. This clock times all bus transfers. All signal timings are related to the rising edge of pclk.

5.2 Resets Interface Signals

Table 5-2 Resets Interface Signals

Port Name	I/O	Description
pwr_on_rst	I	Power on Reset. Serves as hardware reset for the core IP functions excluding AXI-4 Stream interface and AXI-4 Lite interface reg This is an asynchronous active-high reset signal. **Exists:** Always **Synchronous To:** None_as **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal

5.3 DFT Interface Signals

scan_mode -

Table 5-3 DFT Interface Signals

Port Name	I/O	Description
scan_mode	I	Scan mode active high enable. **Exists:** Always **Synchronous To:** None_as **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.4 Memory Zeroization Interface Signals

Table 5-4 Memory Zeroization Interface Signals

Port Name	I/O	Description
I_sram_zero_init	I	SRAM Zeroization initialization input. Pulse this signal. Zeroization starts after deassertion. **Exists:** DWC_MACSEC_SRAM_ZERO_EN **Synchronous To:** None_as **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_sram_zero_done	O	SRAM Zeroization done output. This signal stays HIGH while there is no memory zeroization process, or the memory zeroization has finished. Deasserts (LOW) after memory zeroization has been initiated. **Exists:** DWC_MACSEC_SRAM_ZERO_EN **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.5 APB Host Config Interface Signals

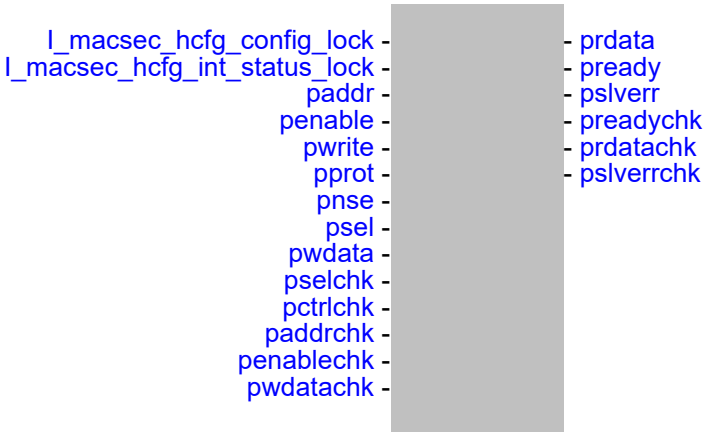


Table 5-5 APB Host Config Interface Signals

Port Name	I/O	Description
I_macsec_hcfg_config_lock	I	APB lock - disables any writes from the APB bus into configuration registers except interrupt status **Exists:** Always **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** AHB Clock Signal. This clock times all bus transfers. All signal timings are related to the rising edge of pclk.
I_macsec_hcfg_int_status_lock	I	Interrupt status register lock - disables any writes from the APB bus into interrupt status registers, stops interrupt clearance. **Exists:** Always **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** AHB Clock Signal. This clock times all bus transfers. All signal timings are related to the rising edge of pclk.

Table 5-5 APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
paddr[(PADDR_WIDTH-1):0]	I	APB4 address bus **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Must e synchronous with pclk.
penable	I	APB4 Enable Strobe **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** high **Validated by:** Asserted to validate APB transfer. It is always_ffdriven low at the end of each APB access.
pwrite	I	APB Read/Write Signal **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** high **Validated by:** When pwrite is high, there is a write transfer and data is broadcast on the write data bus (pwdata). When pwrite is low, a read transfer is performed, and the slave must generate the data on its read data bus (prdata).
pprot[2:0]	I	APB4 Protection type signal **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** HPROT values are mapped to relevant pprot signal if the input signal is included. Else, default values are copied to the pprot signals upon access.
pnse	I	APB5 AMBA APB5 RME PNSE. **Exists:** DWC_MACSEC_APB5_RME **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-5 APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
psel	I	APB Select lines for APB slaves **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** high **Validated by:** Select lines for APB slaves (one per slave)
prdata[(APB_DATA_WIDTH-1):0]	O	Read Data from APB slaves **Exists:** Always **Synchronous To:** pclk **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** The width of each bus is APB_DATA_WIDTH.
pready	O	Indicates whether a APB request cycle was accepted **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** high **Validated by:** Exists only on slave interfaces configured as APB3 or APB4
pslverr	O	Flag for the slave error response from APB **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** high **Validated by:** Exists only on slave interfaces configured as APB3 or APB4
pwdata[(APB_DATA_WIDTH-1):0]	I	APB transfer write data bus shared by all slaves. **Exists:** Always **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** APB transfer write data bus shared by all slaves.

Table 5-5 APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
pselchk	I	APB transfer AMBA APB5 PSEL parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
pctrlchk	I	APB transfer AMBA APB5 PCTRL parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
paddrchk[1:0]	I	APB transfer AMBA APB5 PADDR parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
penablechk	I	APB transfer AMBA APB5 PENABLE parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
pwdatachk[3:0]	I	APB transfer AMBA APB5 PWDATA parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-5 APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
preadychk	O	APB transfer AMBA APB5 PREADY parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
prdatachk[3:0]	O	APB transfer AMBA APB5 PRDATA parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
pslverrchk	O	APB transfer AMBA APB5 PSLVERR parity check. **Exists:** DWC_MACSEC_APB5_PARITY **Synchronous To:** pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.6 AES APB Host Config Interface Signals

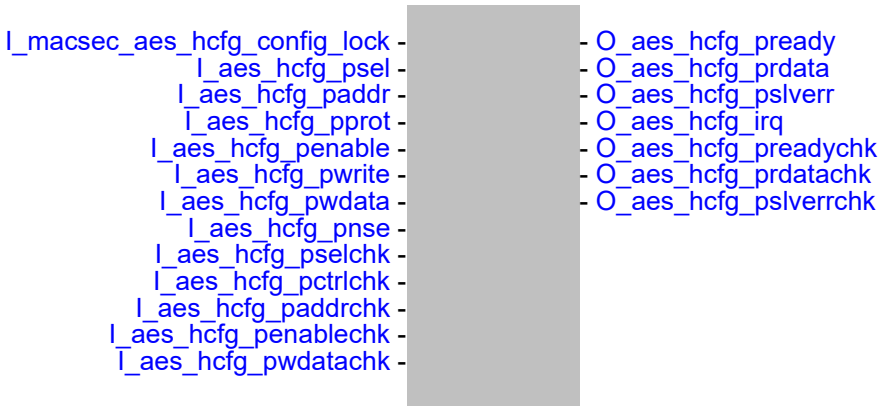


Table 5-6 AES APB Host Config Interface Signals

Port Name	I/O	Description
I_macsec_aes_hcfg_config_lock	I	AES APB lock - disables any writes from the APB bus into AES configuration registers **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** AHB Clock Signal. This clock times all bus transfers. All signal timings are related to the rising edge of pclk.
I_aes_hcfg_psel	I	AES Host Config Interface AMBA APB4 PSEL. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal
I_aes_hcfg_paddr[(DWC_MACSEC_AES_PADDR_WIDTH-1):0]	I	AES Host Config Interface AMBA APB4 PADDR. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-6 AES APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
I_aes_hcfg_pprot[2:0]	I	AES Host Config Interface AMBA APB4 PPROT. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_hcfg_penable	I	AES Host Config Interface AMBA APB4 PENABLE. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal
I_aes_hcfg_pwrite	I	AES Host Config Interface AMBA APB4 PWRITE. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal
I_aes_hcfg_pwdata[31:0]	I	AES Host Config Interface AMBA APB4 PWDATA. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_hcfg_pnse	I	AES Host Config Interface AMBA APB5 RME PNSE. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_RME) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal

Table 5-6 AES APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
O_aes_hcfg_pready	O	AES Host Config Interface AMBA APB4 PREADY. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal
O_aes_hcfg_prdata[31:0]	O	AES Host Config Interface AMBA APB4 PRDATA. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_aes_hcfg_pslverr	O	AES Host Config Interface AMBA APB4 PSLVERR. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High **Validated by:** Not validated by another signal
O_aes_hcfg_irq	O	AES Host Config Interface Interrupt request signal. **Exists:** DWC_UAES_CFG_APB_BUS_EN **Synchronous To:** clk_aes_hcfg_pclk **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_hcfg_pselchk	I	AES Host Config Interface AMBA APB5 PSEL parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-6 AES APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
I_aes_hcfg_pctrlchk	I	AES Host Config Interface AMBA APB5 PCTRL parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_hcfg_paddrchk[1:0]	I	AES Host Config Interface AMBA APB5 PADDR parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_hcfg_penablechk	I	AES Host Config Interface AMBA APB5 PENABLE parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_hcfg_pwdatachk[3:0]	I	AES Host Config Interface AMBA APB5 PWDATA parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_aes_hcfg_preadychk	O	AES Host Config Interface AMBA APB5 PREADY parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-6 AES APB Host Config Interface Signals (Continued)

Port Name	I/O	Description
O_aes_hcfg_prdatachk[3:0]	O	AES Host Config Interface AMBA APB5 PRDATA parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_aes_hcfg_pslverrchk	O	AES Host Config Interface AMBA APB5 PSLVERR parity check. **Exists:** (DWC_UAES_CFG_APB_BUS_EN) && (DWC_MACSEC_APB5_PARITY) **Synchronous To:** clk_aes_hcfg_pclk **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.7 Status Interface Signals

Table 5-7 Status Interface Signals

Port Name	I/O	Description
macsec_irq	O	DWC_macsec Interrupt line **Exists:** Always **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.8 Safety Alarms Interface Signals

Table 5-8 Safety Alarms Interface Signals

Port Name	I/O	Description
O_macsec_safety_alarm	O	MACsec safety alarm. When asserted, indicates the presence of a safety error detected by a safety mechanism. **Exists:** DWC_MACSEC_AUTOMOTIVE **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_macsec_safety_alarm_n	O	MACsec bank safety alarm. When asserted, indicates the presence of a safety error detected by a safety mechanism. Dual rail version of O_macsec_safety_alarm. **Exists:** DWC_MACSEC_AUTOMOTIVE **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.9 MAC GMII Interface Signals

Table 5-9 MAC GMII Interface Signals

Port Name	I/O	Description
mac_gmii_txd[(DWC_MACSEC_GMII_DATA_BUS_WIDTH-1):0]	I	Transmit Data signal coming from the MAC. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_gmii_tx_en	I	Transmit Enable signal coming from the MAC. Send enable signal, only the data transmitted during the valid period of TX_EN is valid For RGMII configurations, this is equivalent to the TX_CTL control signal which multiplexes GMII enable and control data. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_gmii_tx_er	I	Transmit Error signal coming from the MAC. Send data error prompt signal, synchronized to TX_CLK, active high, indicating that the data transmitted during TX_ER validity period is invalid. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-9 MAC GMII Interface Signals (Continued)

Port Name	I/O	Description
mac_gmii_rxd[(DWC_MACSEC_GMII_DATA_BUS_WIDTH-1):0]	O	Receive Data signal going to the MAC. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_gmii_rx_dv	O	Data Valid signal going to the MAC. For RGMII configurations, this is equivalent to the RX_CTL control signal which multiplexes GMII enable and control data. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_gmii_rx_er	O	Receive data error prompt signal going to the MAC. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.10 PHY GMII Interface Signals

Table 5-10 PHY GMII Interface Signals

Port Name	I/O	Description
phy_gmii_txd[(DWC_MACSEC_GMII_DATA_BUS_WIDTH-1):0]	O	Transmit Data signal going to the PHY/PCS. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_gmii_tx_en	O	Transmit Enable signal going to the PHY/PCS. Send enable signal, only the data transmitted during the valid period of TX_EN is valid For RGMII configurations, this is equivalent to the TX_CTL control signal which multiplexes GMII enable and control data. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_gmii_tx_er	O	Transmit Error. Send data error prompt signal going to the PHY/PCS, synchronized to TX_CLK. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-10 PHY GMII Interface Signals (Continued)

Port Name	I/O	Description
phy_gmii_rxd[(DWC_MACSEC_GMII_DATA_BUS_WIDTH-1):0]	I	Receive Data signal coming from the PHY/PCS. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_gmii_rx_dv	I	Receive data valid signal coming from the PHY/PCS. For RGMII configurations, this is equivalent to the RX_CTL control signal which multiplexes GMII enable and control data. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** TX_EN
phy_gmii_rx_er	I	Receive data error prompt signal coming from the PHY/PCS, synchronized to RX_CLK, active high, indicating that the data transmitted during the valid period of RX_ER is invalid. **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.11 MAC XGMII/XLGMII Interface Signals

Table 5-11 MAC XGMII/XLGMII Interface Signals

Port Name	I/O	Description
mac_xlgmii_txd[(DWC_MACSEC_XLGMII_DATA_BUS_WIDTH-1):0]	I	Transmit Data signal (TXD) coming from the MAC. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_xlgmii_txc[(DWC_MACSEC_XLGMII_CTRL_BUS_WIDTH-1):0]	I	TX control signal (TXC) coming from the MAC. One bit for each data lane. Each lane's TXC bit shall be de-asserted when a data byte is being sent on the corresponding lane and asserted when a control character is being sent. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_xlgmii_rxd[(DWC_MACSEC_XLGMII_DATA_BUS_WIDTH-1):0]	O	Receive Data signal (RXD) going to the MAC. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** rx_clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-11 MAC XGMII/XLGMII Interface Signals (Continued)

Port Name	I/O	Description
mac_xlgmii_rxc[(DWC_MACSEC_XLGMII_CTRL_BUS_WIDTH-1):0]	O	RX control signal (RXC) going to the MAC. One bit for each data lane. Each lane's RXC bit shall be de-asserted when a data byte is being received on the corresponding lane and asserted when a control character is being received. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** rx_clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_xlgmii_rx_valid	O	Receive Valid signal going to the MAC. This signal is asserted to indicate that the data on mac_xlgmii_rxd/mac_xlgmii_rxc is valid **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** rx_clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
mac_xlgmii_rxt0_next	O	Receive advance indication signal going to the MAC to indicate that in the next valid clk cycle, the terminate control character is located on lane 0 and all other lanes contain the IDLE control character. This signal can be used in all 128-bit XLGMII configurations of MACsec, including CGMII (100Gbps 128-bit XLGMII). **Exists:** (((DWC_MACSEC_DATA_IF==1))) && ((DWC_MACSEC_RXT0_NEXT_SUPPORT==1)) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.12 PHY XGMII/XLGMII Interface Signals

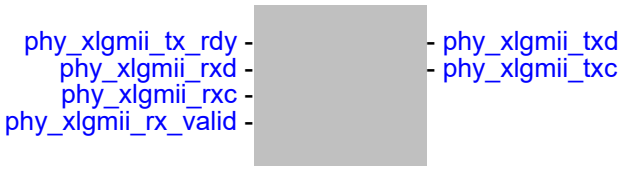


Table 5-12 PHY XGMII/XLGMII Interface Signals

Port Name	I/O	Description
phy_xlgmii_txd[(DWC_MACSEC_XLGMII_DATA_BUS_WIDTH-1):0]	O	Transmit Data signal (TXD) going to the PHY/PCS. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_xlgmii_txc[(DWC_MACSEC_XLGMII_CTRL_BUS_WIDTH-1):0]	O	TX control signal (TXC) going to the PHY/PCS. One bit for each data lane. Each lane's TXC bit shall be de-asserted when a data byte is being sent on the corresponding lane and asserted when a control character is being sent. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_xlgmii_tx_rdy	I	Transmit Ready signal coming from the PHY/PCS. This signal is asserted by the PHY/PCS when its transmitter is ready to receive new data on phy_xlgmii_txd/phy_xlgmii_txc **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-12 PHY XGMII/XLGMII Interface Signals (Continued)

Port Name	I/O	Description
phy_xlgmii_rxd[(DWC_MACSEC_XLGMII_DATA_BUS_WIDTH-1):0]	I	Receive Data signal (RXD) coming from the PHY/PCS. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_xlgmii_rxc[(DWC_MACSEC_XLGMII_CTRL_BUS_WIDTH-1):0]	I	RX control signal (RXC) coming from the PHY/PCS. One bit for each data lane. Each lane's RXC bit shall be de-asserted when a data byte is being received on the corresponding lane and asserted when a control character is being received. **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
phy_xlgmii_rx_valid	I	Receive Valid signal coming from the PHY/PCS. This signal is asserted by the PHY/PCS to indicate that the data on phy_xlgmii_rxd/phy_xlgmii_rxc is valid **Exists:** ((DWC_MACSEC_DATA_IF==1)) **Synchronous To:** rx_clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.13 TX MIB SA Memory Interface Signals

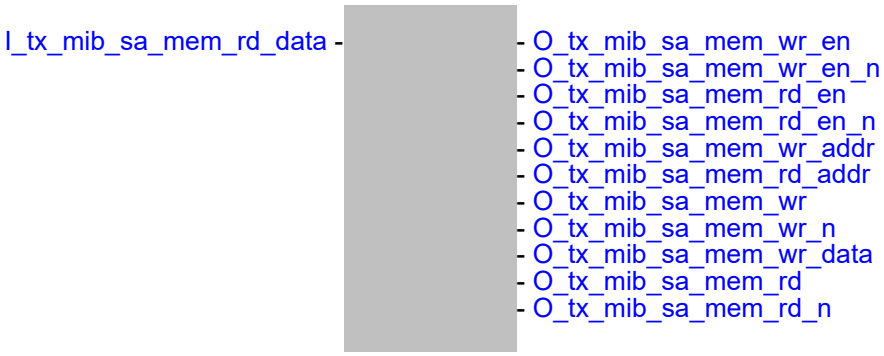


Table 5-13 TX MIB SA Memory Interface Signals

Port Name	I/O	Description
O_tx_mib_sa_mem_wr_en	O	TP TX MIB SA memory write port enable **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_mib_sa_mem_wr_en_n	O	TP TX MIB SA memory write port enable. Dual rail version of O_tx_mib_sa_mem_wr_en **Exists:** (!(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_mib_sa_mem_rd_en	O	TP TX MIB SA memory read port enable **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-13 TX MIB SA Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_mib_sa_mem_rd_en_n	O	TP TX MIB SA memory read port enable. Dual rail version of O_tx_mib_sa_mem_rd_en **Exists:** (!(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_mib_sa_mem_wr_addr[(pTX_MIB_ADDR_WIDTH_SA-1):0]	O	TP TX MIB SA memory write address **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_mib_sa_mem_wr_en
O_tx_mib_sa_mem_rd_addr[(pTX_MIB_ADDR_WIDTH_SA-1):0]	O	TP TX MIB SA memory read address **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_mib_sa_mem_rd_en
O_tx_mib_sa_mem_wr	O	TP TX MIB SA memory write enable **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_mib_sa_mem_wr_n	O	TP TX MIB SA memory write enable. Dual rail version of O_tx_mib_sa_mem_wr. **Exists:** (!(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-13 TX MIB SA Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_mib_sa_mem_wr_data[((pMIB_SA_WIDTH+pDWC_MACSEC_TX_SA_PROT_WIDTH)-1):0]	O	TP TX MIB SA memory write data **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_mib_sa_mem_wr
O_tx_mib_sa_mem_rd	O	TP TX MIB SA memory read enable **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_mib_sa_mem_rd_n	O	TP TX MIB SA memory read enable. Dual rail version of O_tx_mib_sa_mem_rd. **Exists:** (!(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_tx_mib_sa_mem_rd_data[((pMIB_SA_WIDTH+pDWC_MACSEC_TX_SA_PROT_WIDTH)-1):0]	I	TP TX MIB SA memory read data **Exists:** !(DWC_MACSEC_TX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_mib_sa_mem_rd

5.14 RX MIB SC Bank 1 Memory Interface Signals

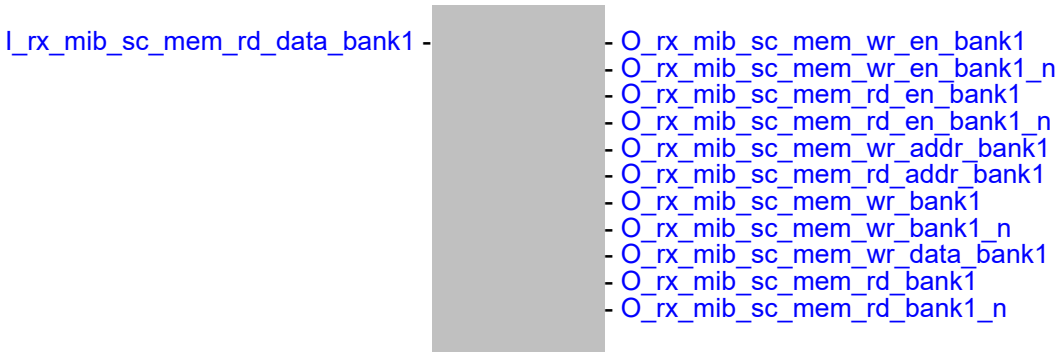


Table 5-14 RX MIB SC Bank 1 Memory Interface Signals

Port Name	I/O	Description
O_rx_mib_sc_mem_wr_en_bank1	O	TP RX SC MIB Bank 1 memory write port enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_en_bank1_n	O	TP RX SC MIB Bank 1 memory write port enable. Dual rail version of O_rx_mib_sc_mem_wr_en_bank1. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-14 RX MIB SC Bank 1 Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sc_mem_rd_en_bank1	O	TP RX SC MIB Bank 1 memory read port enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_rd_en_bank1_n	O	TP RX SC MIB Bank 1 memory read port enable. Dual rail version of O_rx_mib_sc_mem_rd_en_bank1. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_addr_bank1[(pRX_SC_MEM_ADDR_WIDTH_BNK1-1):0]	O	TP RX SC MIB Bank 1 memory write address **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_wr_en_mem_bank1
O_rx_mib_sc_mem_rd_addr_bank1[(pRX_SC_MEM_ADDR_WIDTH_BNK1-1):0]	O	TP RX SC MIB Bank 1 memory read address **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_rd_en_mem_bank1

Table 5-14 RX MIB SC Bank 1 Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sc_mem_wr_bank1	O	TP RX SC MIB Bank 1 memory write enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_bank1_n	O	TP RX SC MIB Bank 1 memory write enable. Dual rail version of O_rx_mib_sc_mem_wr_bank1. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_data_bank1[((pMIB_SC_WIDTH+pDWC_MACSEC_RX_SC_PROT_WIDTH)-1):0]	O	TP RX SC MIB Bank 1 memory write data **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_wr_mem_bank1
O_rx_mib_sc_mem_rd_bank1	O	TP RX SC MIB Bank 1 memory read enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-14 RX MIB SC Bank 1 Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sc_mem_rd_bank1_n	O	TP RX SC MIB Bank 1 memory read enable. Dual rail version of O_rx_mib_sc_mem_rd_bank1. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_rx_mib_sc_mem_rd_data_bank1[((pMIB_SC_WIDTH+pDWC_MACSEC_RX_SC_PROT_WIDTH)-1):0]	I	TP RX SC MIB Bank 1 memory read data **Exists:** DWC_MACSEC_RX_SC_MIB_BANK1_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_rd_mem_bank1

5.15 RX MIB SC Bank 2 Memory Interface Signals

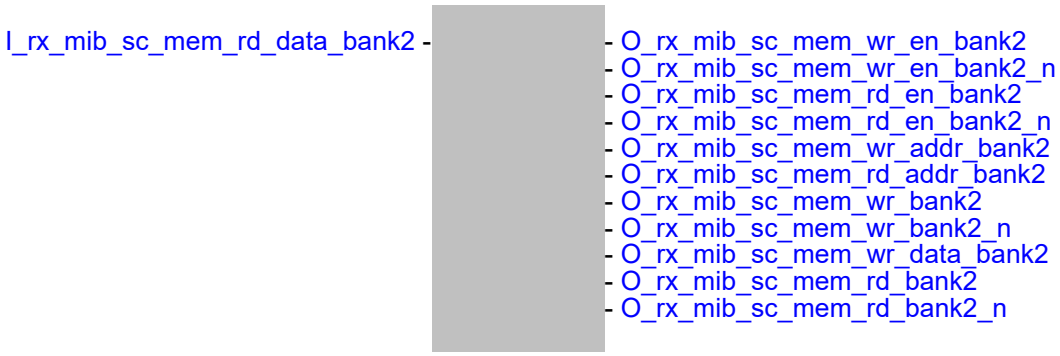


Table 5-15 RX MIB SC Bank 2 Memory Interface Signals

Port Name	I/O	Description
O_rx_mib_sc_mem_wr_en_bank2	O	TP RX SC MIB Bank 2 memory write port enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_en_bank2_n	O	TP RX SC MIB Bank 2 memory write port enable. Dual rail version of O_rx_mib_sc_mem_wr_en_bank2. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-15 RX MIB SC Bank 2 Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sc_mem_rd_en_bank2	O	TP RX SC MIB Bank 2 memory read port enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_rd_en_bank2_n	O	TP RX SC MIB Bank 2 memory read port enable. Dual rail version of O_rx_mib_sc_mem_rd_en_bank2. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_addr_bank2[(pRX_SC_MEM_ADDR_WIDTH_BNK2-1):0]	O	TP RX SC MIB Bank 2 memory write address **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_wr_en_mem_bank1
O_rx_mib_sc_mem_rd_addr_bank2[(pRX_SC_MEM_ADDR_WIDTH_BNK2-1):0]	O	TP RX SC MIB Bank 2 memory read address **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_rd_en_mem_bank1

Table 5-15 RX MIB SC Bank 2 Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sc_mem_wr_bank2	O	TP RX SC MIB Bank 2 memory write enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_bank2_n	O	TP RX SC MIB Bank 2 memory write enable. Dual rail version of O_rx_mib_sc_mem_wr_bank2. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sc_mem_wr_data_bank2[((pMIB_SC_WIDTH+pDWC_MACSEC_RX_SC_PROT_WIDTH)-1):0]	O	TP RX SC MIB Bank 2 memory write data **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_wr_mem_bank1
O_rx_mib_sc_mem_rd_bank2	O	TP RX SC MIB Bank 2 memory read enable **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-15 RX MIB SC Bank 2 Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sc_mem_rd_bank2_n	O	TP RX SC MIB Bank 2 memory read enable. Dual rail version of O_rx_mib_sc_mem_rd_bank2. **Exists:** (DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_rx_mib_sc_mem_rd_data_bank2[((pMIB_SC_WIDTH+pDWC_MACSEC_RX_SC_PROT_WIDTH)-1):0]	I	TP RX SC MIB Bank 2 memory read data **Exists:** DWC_MACSEC_RX_SC_MIB_BANK2_PARTIAL_STORAGE_USE_SRAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sc_mem_rd_mem_bank2

5.16 RX MIB SA Memory Interface Signals

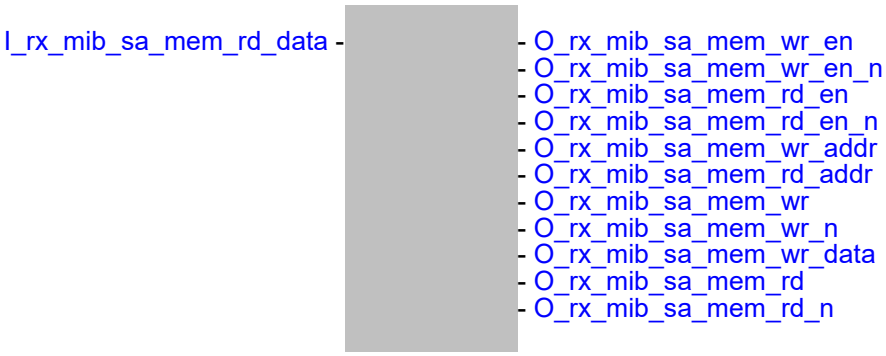


Table 5-16 RX MIB SA Memory Interface Signals

Port Name	I/O	Description
O_rx_mib_sa_mem_wr_en	O	TP RX SA MIB memory write port enable **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sa_mem_wr_en_n	O	TP RX SA MIB memory write port enable. Dual rail version of O_rx_mib_sa_mem_wr_en. **Exists:** (!(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sa_mem_rd_en	O	TP RX SA MIB memory read port enable **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-16 RX MIB SA Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sa_mem_rd_en_n	O	TP RX SA MIB memory read port enable. Dual rail version of O_rx_mib_sa_mem_rd_en. **Exists:** (!(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sa_mem_wr_addr[(pRX_MIB_ADDR_WIDTH_SA-1):0]	O	TP RX SA MIB memory write address **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sa_mem_wr_en
O_rx_mib_sa_mem_rd_addr[(pRX_MIB_ADDR_WIDTH_SA-1):0]	O	TP RX SA MIB memory read address **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sa_mem_rd_en
O_rx_mib_sa_mem_wr	O	TP RX SA MIB memory write enable **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sa_mem_wr_n	O	TP RX SA MIB memory write enable. Dual rail version of O_rx_mib_sa_mem_wr. **Exists:** (!(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-16 RX MIB SA Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_mib_sa_mem_wr_data[((pMIB_SA_WIDTH+pDWC_MACSEC_RX_SA_PROT_WIDTH)-1):0]	O	TP RX SA MIB memory write data **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sa_mem_wr
O_rx_mib_sa_mem_rd	O	TP RX SA MIB memory read enable **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_mib_sa_mem_rd_n	O	TP RX SA MIB memory read enable. Dual rail version of O_rx_mib_sa_mem_rd. **Exists:** (!(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_rx_mib_sa_mem_rd_data[((pMIB_SA_WIDTH+pDWC_MACSEC_RX_SA_PROT_WIDTH)-1):0]	I	TP RX SA MIB memory read data **Exists:** !(DWC_MACSEC_RX_MIB_SA_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_mib_sa_mem_rd

5.17 TX SAI Memory Interface Signals

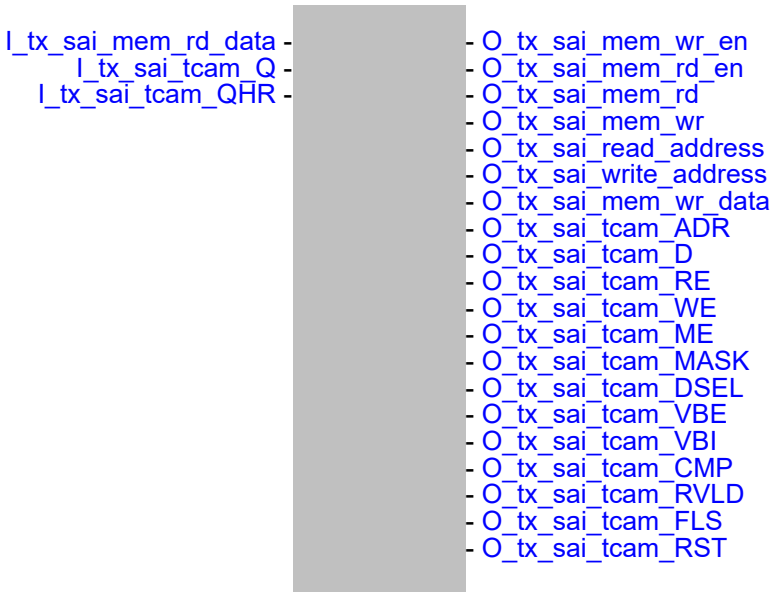


Table 5-17 TX SAI Memory Interface Signals

Port Name	I/O	Description
O_tx_sai_mem_wr_en	O	TP TX SAI memory write port enable **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_mem_rd_en	O	TP TX SAI memory read port enable **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-17 TX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_sai_mem_rd	O	TP TX SAI memory read enable **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_mem_wr	O	TP TX SAI memory write enable **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_read_address[(pTX_SAI_MEM_ADDR_WIDTH-1):0]	O	TP TX SAI memory read address **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sai_mem_rd_en
O_tx_sai_write_address[(pTX_SAI_MEM_ADDR_WIDTH-1):0]	O	TP TX SAI memory write address **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sai_mem_wr_en
O_tx_sai_mem_wr_data[(pTX_SAI_MEM_WIDTH-1):0]	O	TP TX SAI memory write data **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sai_mem_wr

Table 5-17 TX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
I_tx_sai_mem_rd_data[(pTX_SAI_MEM_WIDTH-1):0]	I	TP TX SAI Memory read data **Exists:** DWC_MACSEC_TX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sai_mem_rd
I_tx_sai_tcam_Q[(pTX_SAI_MEM_WIDTH-1):0]	I	TX SAI SP TCAM Memory output bus for READ operation **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_tx_sai_tcam_QHR[(pTX_SAI_MEM_DEPTH-1):0]	I	TX SAI SP TCAM Memory hit row. Without Priority Encoder, this pin will be used to indicate whenever a row matched **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_ADR[(pTX_SAI_MEM_ADDR_WIDTH-1):0]	O	TX SAI SP TCAM Memory Address. This port is used to address the location to be written during the Write cycle and read during the Read cycle **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_D[(pTX_SAI_MEM_WIDTH-1):0]	O	TX SAI SP TCAM Memory Data bus. During write operation, D will be written to DATA storage or MASK storage based on DSEL value. During CMP (search) operation, D is used as search KEY. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-17 TX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_sai_tcam_RE	O	TX SAI SP TCAM Memory Read Enable. When the read enable is Logic High, the memory is in Read cycle. Q, QVLD are updated on next CLK. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_WE	O	TX SAI SP TCAM Memory Write Enable. When the Write Enable is Logic High, the memory is in the Write cycle. Encoded data are written to the address. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_ME	O	TX SAI SP TCAM Memory Enable. When the Memory Enable is Logic High, the memory is enabled and read/write, compare, flush, reset operations can be performed. When the Memory Enable input is Logic Low, the memory is deactivated **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_MASK[(pTX_SAI_MEM_WIDTH-1):0]	O	TX SAI SP TCAM Memory Mask bus. During Write operation, MASK are used as bit write mask, no bit write if MASK is low. During CMP (search) operation, MASK are used as global bit mask, if MASK[i] is low it will exclude D[i] from the key. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-17 TX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_sai_tcam_DSEL	O	TX SAI SP TCAM Memory DATA/MASK storage selector. During memory Read or Write operation, if DSEL is high, DATA storage is selected, otherwise it will select MASK storage. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_VBE	O	TX SAI SP TCAM Memory Valid Bit Enable. Enable to access the Valid bit array during read or write operation. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_VBI	O	TX SAI SP TCAM Memory Valid Bit Array Input. VBI is written to the array when WE and VBE are active. If VBI is high,the contents of the corresponding row is valid for the Key search (compare operation). If VBI is low, the corresponding row is invalid. **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_CMP	O	TX SAI SP TCAM Memory Compare Enable port. When compare enable input is logic High, it will search for the Key (D and MASK), and search outputs QHR are updated with its respective latency **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-17 TX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_sai_tcam_RVLD[(pTX_SAI_MEM_DEPTH-1):0]	O	TX SAI SP TCAM Memory Row Validation Bus. Each RVLD input controls 1 row. Logic Low on RVLD[n] will pull the row[n] hit line outputs to "0" and also makes this TCAM row invalid, disabling the hit line pre-charge. Logic High on RVLD[n] will allow that row to participate in compare operations **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_FLS	O	TX SAI SP TCAM Memory Flush Enable. When flush enable is Logic High, it will reset valid bit array to 'L' (0) during first cycle **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sai_tcam_RST	O	TX SAI SP TCAM Memory Reset. When reset enable input is Logic High, it will reset valid array at the first clock cycle, read port pins (Q, QVLD) at the 2nd clock cycle and Search port pins QHR at 3rd clock cycle **Exists:** DWC_MACSEC_TX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.18 RX SAI Memory Interface Signals

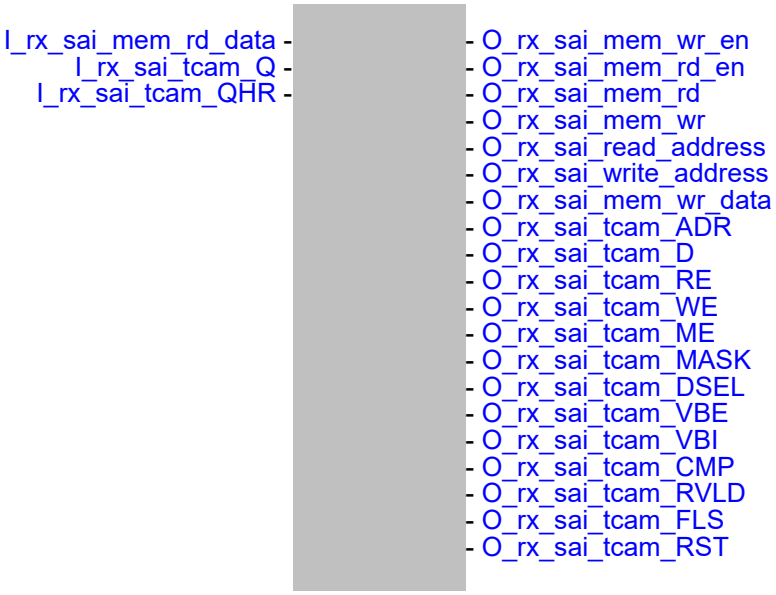


Table 5-18 RX SAI Memory Interface Signals

Port Name	I/O	Description
O_rx_sai_mem_wr_en	O	TP RX SAI memory write port enable **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_mem_rd_en	O	TP RX SAI memory read port enable **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-18 RX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_sai_mem_rd	O	TP RX SAI memory read enable **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_mem_wr	O	TP RX SAI memory write enable **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_read_address[(pRX_SAI_MEM_ADDR_WIDTH-1):0]	O	TP RX SAI memory read address **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sai_mem_rd_en
O_rx_sai_write_address[(pRX_SAI_MEM_ADDR_WIDTH-1):0]	O	TP RX SAI memory write address **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sai_mem_wr_en
O_rx_sai_mem_wr_data[(pRX_SAI_MEM_WIDTH-1):0]	O	TP RX SAI memory write data **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sai_mem_wr

Table 5-18 RX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
I_rx_sai_mem_rd_data[(pRX_SAI_MEM_WIDTH-1):0]	I	TP RX SAI Memory read data **Exists:** DWC_MACSEC_RX_SAI_LOOKUP_LINEAR_SEARCH **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sai_mem_rd
I_rx_sai_tcam_Q[(pRX_SAI_MEM_WIDTH-1):0]	I	RX SAI SP TCAM Memory output bus for READ operation **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_rx_sai_tcam_QHR[(pRX_SAI_MEM_DEPTH-1):0]	I	RX SAI SP TCAM Memory hit row. Without Priority Encoder, this pin will be used to indicate whenever a row matched **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_ADR[(pRX_SAI_MEM_ADDR_WIDTH-1):0]	O	RX SAI SP TCAM Memory Address. This port is used to address the location to be written during the Write cycle and read during the Read cycle **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_D[(pRX_SAI_MEM_WIDTH-1):0]	O	RX SAI SP TCAM Memory Data bus. During write operation, D will be written to DATA storage or MASK storage based on DSEL value. During CMP (search) operation, D is used as search KEY. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-18 RX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_sai_tcam_RE	O	RX SAI SP TCAM Memory Read Enable. When the read enable is Logic High, the memory is in Read cycle. Q, QVLD are updated on next CLK. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_WE	O	RX SAI SP TCAM Memory Write Enable. When the Write Enable is Logic High, the memory is in the Write cycle. Encoded data are written to the address. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_ME	O	RX SAI SP TCAM Memory Enable. When the Memory Enable is Logic High, the memory is enabled and read/write, compare, flush, reset operations can be performed. When the Memory Enable input is Logic Low, the memory is deactivated **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_MASK[(pRX_SAI_MEM_WIDTH-1):0]	O	RX SAI SP TCAM Memory Mask bus. During Write operation, MASK are used as bit write mask, no bit write if MASK is low. During CMP (search) operation, MASK are used as global bit mask, if MASK[i] is low it will exclude D[i] from the key. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-18 RX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_sai_tcam_DSEL	O	RX SAI SP TCAM Memory DATA/MASK storage selector. During memory Read or Write operation, if DSEL is high, DATA storage is selected, otherwise it will select MASK storage. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_VBE	O	RX SAI SP TCAM Memory Valid Bit Enable. Enable to access the Valid bit array during read or write operation. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_VBI	O	RX SAI SP TCAM Memory Valid Bit Array Input. VBI is written to the array when WE and VBE are active. If VBI is high,the contents of the corresponding row is valid for the Key search (compare operation). If VBI is low, the corresponding row is invalid. **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_CMP	O	RX SAI SP TCAM Memory Compare Enable port. When compare enable input is logic High, it will search for the Key (D and MASK), and search outputs QHR are updated together with its respective latency **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-18 RX SAI Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_sai_tcam_RVLD[(pRX_SAI_MEM_DEPTH-1):0]	O	RX SAI SP TCAM Memory Row Validation Bus. Each RVLD input controls 1 row. Logic Low on RVLD[n] will pull the row[n] hit line outputs to "0" and also makes this TCAM row invalid, disabling the hit line pre-charge. Logic High on RVLD[n] will allow that row to participate in compare operations **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_FLS	O	RX SAI SP TCAM Memory Flush Enable. When flush enable is Logic High, it will reset valid bit array to 'L' (0) during first cycle **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sai_tcam_RST	O	RX SAI SP TCAM Memory Reset. When reset enable input is Logic High, it will reset valid array at the first clock cycle, read port pins (Q, QVLD) at the 2nd clock cycle and Search port pins QHR at 3rd clock cycle **Exists:** DWC_MACSEC_RX_SAI_USE_TCAM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.19 TX SALT Memory Interface Signals

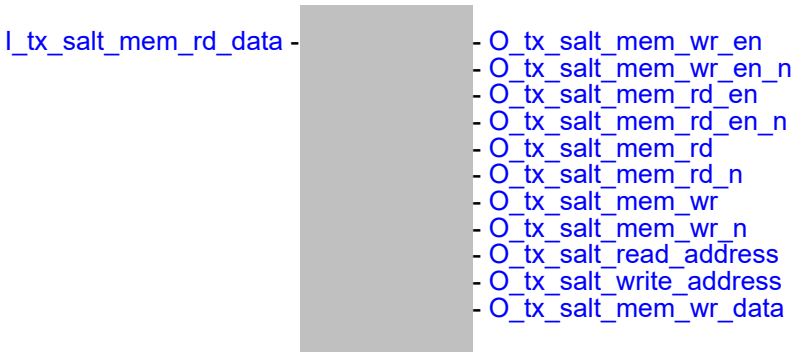


Table 5-19 TX SALT Memory Interface Signals

Port Name	I/O	Description
O_tx_salt_mem_wr_en	O	TP TX SALT memory enable write port **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_salt_mem_wr_en_n	O	TP TX SALT memory enable write port. Dual rail version of O_tx_salt_mem_wr_en **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_salt_mem_rd_en	O	TP TX SALT memory enable read port **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-19 TX SALT Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_salt_mem_rd_en_n	O	TP TX SALT memory enable read port. Dual rail version of O_tx_salt_mem_rd_en **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_salt_mem_rd	O	TP TX SALT memory read enable **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_salt_mem_rd_n	O	TP TX SALT memory read enable. Dual rail version of O_tx_salt_mem_rd **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_salt_mem_wr	O	TP TX SALT memory write port enable **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-19 TX SALT Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_salt_mem_wr_n	O	TP TX SALT memory write port enable. Dual rail version of O_tx_salt_mem_wr **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_salt_read_address[(pTX_SALT_MEM_ADDR_WIDTH-1):0]	O	TP TX SALT memory read address **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_salt_mem_rd_en
O_tx_salt_write_address[(pTX_SALT_MEM_ADDR_WIDTH-1):0]	O	TP TX SALT memory write address **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_salt_mem_wr_en
O_tx_salt_mem_wr_data[((pTX_SALT_MEM_WIDTH+pDWC_MACSEC_TX_SALT_PROT_WIDTH)-1):0]	O	TP TX SALT memory write data **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_salt_mem_wr

Table 5-19 TX SALT Memory Interface Signals (Continued)

Port Name	I/O	Description
I_tx_salt_mem_rd_data[((pTX_SALT_MEM_WIDTH+pDWC_MACSEC_TX_SALT_PROT_WIDTH)-1):0]	I	TP TX SALT Memory read data **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_TX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_salt_mem_rd

5.20 RX SALT Memory Interface Signals

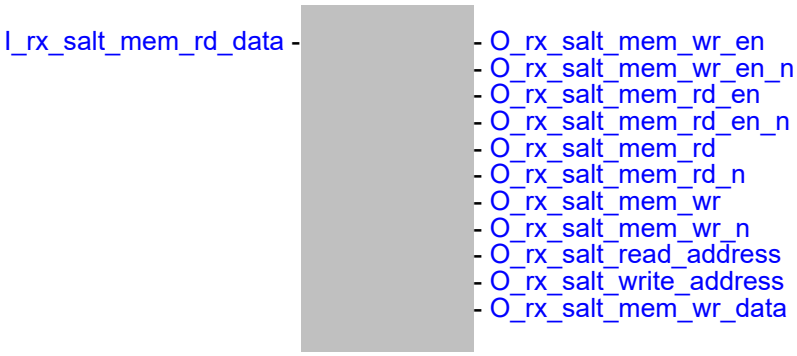


Table 5-20 RX SALT Memory Interface Signals

Port Name	I/O	Description
O_rx_salt_mem_wr_en	O	TP RX SALT memory write port enable **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_salt_mem_wr_en_n	O	TP RX SALT memory write port enable. Dual rail version of O_rx_salt_mem_wr_en. **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_salt_mem_rd_en	O	TP RX SALT memory read port enable **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-20 RX SALT Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_salt_mem_rd_en_n	O	TP RX SALT memory read port enable. Dual rail version of O_rx_salt_mem_rd_en. **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_salt_mem_rd	O	TP RX SALT memory read enable **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_salt_mem_rd_n	O	TP RX SALT memory read enable. Dual rail version of O_rx_salt_mem_rd **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_salt_mem_wr	O	TP RX SALT memory write enable **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-20 RX SALT Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_salt_mem_wr_n	O	TP RX SALT memory write enable. Dual rail version of O_rx_salt_mem_wr. **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_salt_read_address[(pRX_SALT_MEM_ADDR_WIDTH-1):0]	O	TP RX SALT memory read address **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_salt_mem_rd_en
O_rx_salt_write_address[(pRX_SALT_MEM_ADDR_WIDTH-1):0]	O	TP RX SALT memory write address **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_salt_mem_wr_en
O_rx_salt_mem_wr_data[((pRX_SALT_MEM_WIDTH+pDWC_MACSEC_RX_SALT_PROT_WIDTH)-1):0]	O	TP RX SALT memory write data **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_salt_mem_wr

Table 5-20 RX SALT Memory Interface Signals (Continued)

Port Name	I/O	Description
I_rx_salt_mem_rd_data[((pRX_SALT_MEM_WIDTH+pDWC_MACSEC_RX_SALT_PROT_WIDTH)-1):0]	I	TP RX SALT Memory read data **Exists:** ((DWC_MACSEC_XPN_SEL == 1)) && (!(DWC_MACSEC_RX_SALT_MEM_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_salt_mem_rd

5.21 RX ARW Memory Interface Signals

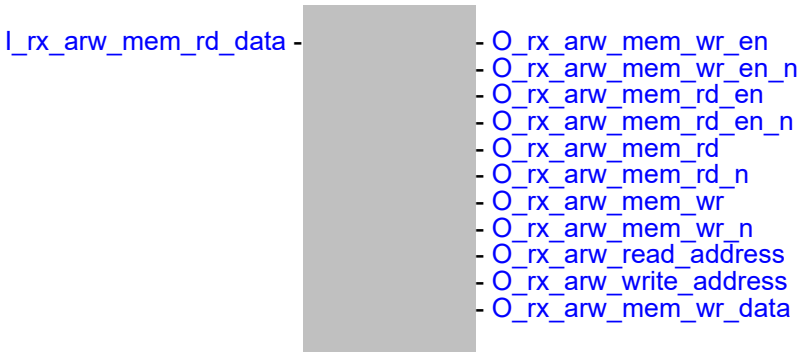


Table 5-21 RX ARW Memory Interface Signals

Port Name	I/O	Description
O_rx_arw_mem_wr_en	O	TP RX ARW memory write port enable **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_arw_mem_wr_en_n	O	TP RX ARW memory write port enable. Dual rail version of O_rx_arw_mem_wr_en. **Exists:** (!(DWC_MACSEC_RX_ARW_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_arw_mem_rd_en	O	TP RX ARW memory read port enable **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-21 RX ARW Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_arw_mem_rd_en_n	O	TP RX ARW memory read port enable. Dual rail version of O_rx_arw_mem_rd_en. **Exists:** (!(DWC_MACSEC_RX_ARW_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_arw_mem_rd	O	TP RX ARW memory read enable **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_arw_mem_rd_n	O	TP RX ARW memory read enable. Dual rail version of O_rx_arw_mem_rd. **Exists:** (!(DWC_MACSEC_RX_ARW_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_arw_mem_wr	O	TP RX ARW memory write enable **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_arw_mem_wr_n	O	TP RX ARW memory write enable. Dual rail version of O_rx_arw_mem_wr. **Exists:** (!(DWC_MACSEC_RX_ARW_MEM_TYPE==0)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-21 RX ARW Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_arw_read_address[(pRX_ARW_MEM_ADDR_WIDTH-1):0]	O	TP RX ARW memory read address **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_arw_mem_rd_en
O_rx_arw_write_address[(pRX_ARW_MEM_ADDR_WIDTH-1):0]	O	TP RX ARW memory write address **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_arw_mem_wr_en
O_rx_arw_mem_wr_data[((pRX_ARW_MEM_WIDTH+pDWC_MACSEC_RX_ARW_PROT_WIDTH)-1):0]	O	TP RX ARW memory write data **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_arw_mem_wr
I_rx_arw_mem_rd_data[((pRX_ARW_MEM_WIDTH+pDWC_MACSEC_RX_ARW_PROT_WIDTH)-1):0]	I	TP RX ARW Memory read data **Exists:** !(DWC_MACSEC_RX_ARW_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_arw_mem_rd

5.22 TX SAD Memory Interface Signals

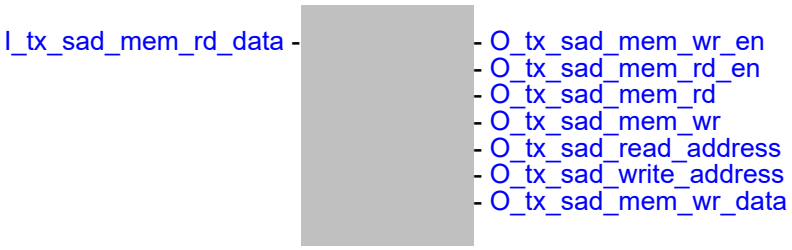


Table 5-22 TX SAD Memory Interface Signals

Port Name	I/O	Description
O_tx_sad_mem_wr_en	O	TP TX SAD memory write port enable **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sad_mem_rd_en	O	TP TX SAD memory read port enable **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sad_mem_rd	O	TP TX SAD memory read enable **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_tx_sad_mem_wr	O	TP TX SAD memory write enable **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-22 TX SAD Memory Interface Signals (Continued)

Port Name	I/O	Description
O_tx_sad_read_address[(pTX_SAD_MEM_ADDR_WIDTH-1):0]	O	TP TX SAD memory read address **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sad_mem_rd_en
O_tx_sad_write_address[(pTX_SAD_MEM_ADDR_WIDTH-1):0]	O	TP TX SAD memory write address **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sad_mem_wr_en
O_tx_sad_mem_wr_data[((pTX_SAD_MEM_WIDTH+pDWC_MACSEC_TX_SAD_PROT_WIDTH)-1):0]	O	TP TX SAD memory write data **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sad_mem_wr
I_tx_sad_mem_rd_data[((pTX_SAD_MEM_WIDTH+pDWC_MACSEC_TX_SAD_PROT_WIDTH)-1):0]	I	TP TX SAD Memory read data **Exists:** !(DWC_MACSEC_TX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_tx_sad_mem_rd

5.23 RX SAD DP Memory Interface Signals

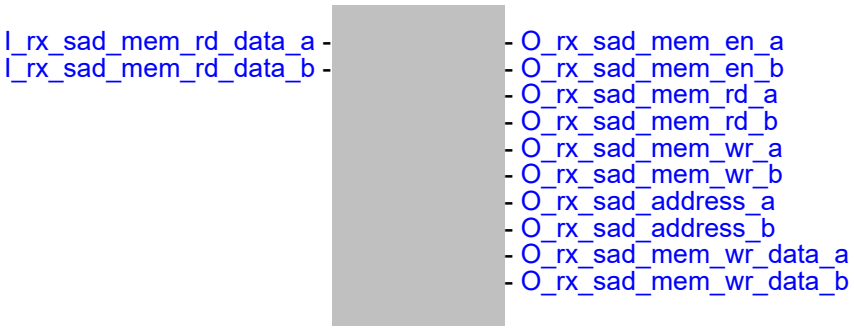


Table 5-23 RX SAD DP Memory Interface Signals

Port Name	I/O	Description
O_rx_sad_mem_en_a	O	DP RX SAD Memory Port A enable **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sad_mem_en_b	O	DP RX SAD Memory Port B enable **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sad_mem_rd_a	O	DP RX SAD Memory Port A read enable **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-23 RX SAD DP Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_sad_mem_rd_b	O	DP RX SAD Memory Port B read enable **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sad_mem_wr_a	O	DP RX SAD Memory Port A write enable **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sad_mem_wr_b	O	DP RX SAD Memory Port B write enable **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_rx_sad_address_a[(pRX_SAD_MEM_ADDR_WIDTH-1):0]	O	DP RX SAD Memory Port A address **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sad_mem_en_a
O_rx_sad_address_b[(pRX_SAD_MEM_ADDR_WIDTH-1):0]	O	DP RX SAD Memory Port B address **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sad_mem_en_b

Table 5-23 RX SAD DP Memory Interface Signals (Continued)

Port Name	I/O	Description
O_rx_sad_mem_wr_data_a[((pRX_SAD_MEM_WIDTH+pDWC_MACSEC_RX_SAD_PROT_WIDTH)-1):0]	O	DP RX SAD Memory Port A write data **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sad_mem_wr_a
O_rx_sad_mem_wr_data_b[((pRX_SAD_MEM_WIDTH+pDWC_MACSEC_RX_SAD_PROT_WIDTH)-1):0]	O	DP RX SAD Memory Port B write data **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sad_mem_wr_b
I_rx_sad_mem_rd_data_a[((pRX_SAD_MEM_WIDTH+pDWC_MACSEC_RX_SAD_PROT_WIDTH)-1):0]	I	DP RX SAD Memory Port A read data **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sad_mem_rd _a
I_rx_sad_mem_rd_data_b[((pRX_SAD_MEM_WIDTH+pDWC_MACSEC_RX_SAD_PROT_WIDTH)-1):0]	I	DP RX SAD Memory Port B read data **Exists:** !(DWC_MACSEC_RX_SAD_MEM_TYPE==0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_rx_sad_mem_rd_b

5.24 AES Cipher Key Memory Interface Signals

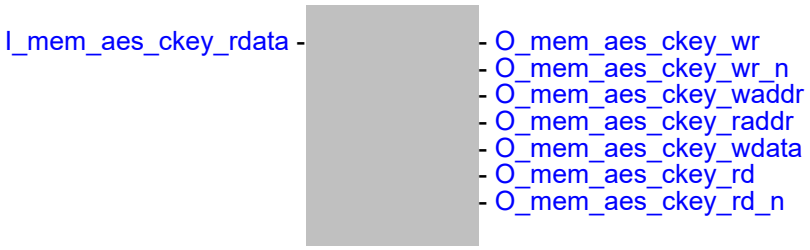


Table 5-24 AES Cipher Key Memory Interface Signals

Port Name	I/O	Description
O_mem_aes_ckey_wr	O	AES Cipher Key SRAM Memory write enable. **Exists:** !DWC_UAES_CFG_SYNTH_CKEY_MEM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_ckey_wr_n	O	AES Cipher Key SRAM Memory write enable. Dual rail version of O_mem_aes_ckey_wr. **Exists:** (!DWC_UAES_CFG_SYNTH_CKEY_MEM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_ckey_waddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Cipher Key SRAM Memory write address. **Exists:** !DWC_UAES_CFG_SYNTH_CKEY_MEM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-24 AES Cipher Key Memory Interface Signals (Continued)

Port Name	I/O	Description
O_mem_aes_ckey_raddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Cipher Key SRAM Memory read address. **Exists:** !DWC_UAES_CFG_SYNTH_CKEY_MEM **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_ckey_wdata[((DWC_UAES_CFG_CKEY_MEM_DATA_WIDTH+pDWC_ULTRA_AES_CKEY_PROT_WIDTH)-1):0]	O	AES Cipher Key SRAM Memory write data. **Exists:** !DWC_UAES_CFG_SYNTH_CKEY_MEM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_ckey_rd	O	AES Cipher Key SRAM Memory read enable. **Exists:** !DWC_UAES_CFG_SYNTH_CKEY_MEM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_ckey_rd_n	O	AES Cipher Key SRAM Memory read enable. Dual rail version of O_mem_aes_ckey_rd. **Exists:** (!DWC_UAES_CFG_SYNTH_CKEY_MEM) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_mem_aes_ckey_rdata[((DWC_UAES_CFG_CKEY_MEM_DATA_WIDTH+pDWC_ULTRA_AES_CKEY_PROT_WIDTH)-1):0]	I	AES Cipher Key SRAM Memory read data. **Exists:** !DWC_UAES_CFG_SYNTH_CKEY_MEM **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.25 AES Cipher Context Memory Interface Signals

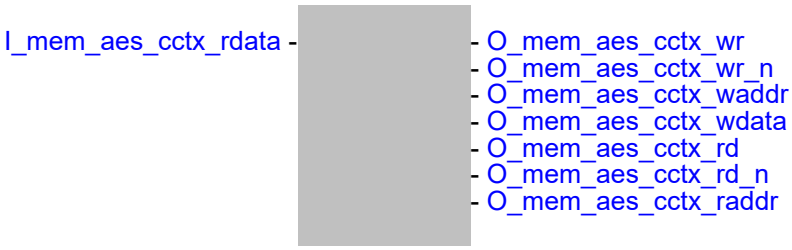


Table 5-25 AES Cipher Context Memory Interface Signals

Port Name	I/O	Description
O_mem_aes_cctx_wr	O	AES Cipher Context SRAM Memory write enable. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_cctx_wr_n	O	AES Cipher Context SRAM Memory write enable. Dual rail version of O_mem_aes_cctx_wr. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_cctx_waddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Cipher Context SRAM Memory write address. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-25 AES Cipher Context Memory Interface Signals (Continued)

Port Name	I/O	Description
O_mem_aes_cctx_wdata[(127+pDWC_ULTRA_AES_CCTX_PROT_WIDTH):0]	O	AES Cipher Context SRAM Memory write data. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_cctx_rd	O	AES Cipher Context SRAM Memory read enable. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_cctx_rd_n	O	AES Cipher Context SRAM Memory read enable. Dual rail version of O_mem_aes_cctx_rd. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_cctx_raddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Cipher Context SRAM Memory read address. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-25 AES Cipher Context Memory Interface Signals (Continued)

Port Name	I/O	Description
I_mem_aes_cctx_rdata[(127+pDWC_ULTRA_AES_CCTX_PROT_WIDTH):0]	I	AES Cipher Context SRAM Memory read data. **Exists:** (!DWC_UAES_CFG_SYNTH_CCTX_MEM) && (DWC_UAES_CFG_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.26 AES Hash Key Memory Interface Signals

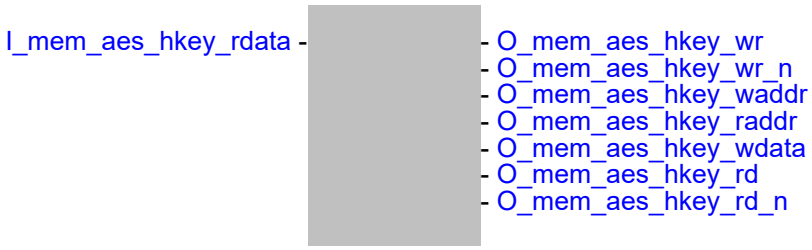


Table 5-26 AES Hash Key Memory Interface Signals

Port Name	I/O	Description
O_mem_aes_hkey_wr	O	AES Hash Key SRAM Memory write enable. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hkey_wr_n	O	AES Hash Key SRAM Memory write enable. Dual rail version of O_mem_aes_hkey_wr. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hkey_waddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Hash Key SRAM Memory write address. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-26 AES Hash Key Memory Interface Signals (Continued)

Port Name	I/O	Description
O_mem_aes_hkey_raddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Hash Key SRAM Memory read address. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hkey_wdata[((DWC_UAES_CFG_HKEY_MEM_DATA_WIDTH+pDWC_ULTRA_AES_HKEY_PROT_WIDTH)-1):0]	O	AES Hash Key SRAM Memory write data. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hkey_rd	O	AES Hash Key SRAM Memory read enable. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hkey_rd_n	O	AES Hash Key SRAM Memory read enable. Dual rail version of O_mem_aes_hkey_rd. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-26 AES Hash Key Memory Interface Signals (Continued)

Port Name	I/O	Description
I_mem_aes_hkey_rdata[((DWC_UAES_CFG_HKEY_MEM_DATA_WIDTH+pDWC_ULTRA_AES_HKEY_PROT_WIDTH)-1):0]	I	AES Hash Key SRAM Memory read data. **Exists:** (!DWC_UAES_CFG_SYNTH_HKEY_MEM) && ((DWC_UAES_CFG_ENCRYPT_MODE==0) || (DWC_UAES_CFG_ENCRYPT_MODE==2)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.27 AES Hash Context Memory Interface Signals

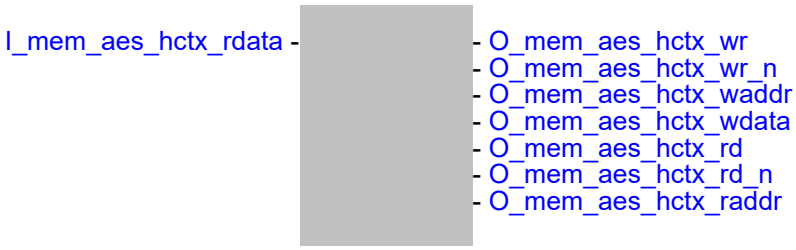


Table 5-27 AES Hash Context Memory Interface Signals

Port Name	I/O	Description
O_mem_aes_hctx_wr	O	AES Hash Context SRAM Memory write enable. **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hctx_wr_n	O	AES Hash Context SRAM Memory write enable. Dual rail version of O_mem_aes_hctx_wr **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hctx_waddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Hash Context SRAM Memory write address. **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-27 AES Hash Context Memory Interface Signals (Continued)

Port Name	I/O	Description
O_mem_aes_hctx_wdata[(127+pDWC_ULTRA_AES_HCTX_PROT_WIDTH):0]	O	AES Hash Context SRAM Memory write data. **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hctx_rd	O	AES Hash Context SRAM Memory read enable. **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hctx_rd_n	O	AES Hash Context SRAM Memory read enable. Dual rail version of O_mem_aes_hctx_rd **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) && (DWC_MACSEC_AUTOMOTIVE) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_mem_aes_hctx_raddr[(DWC_UAES_CFG_CTX_WIDTH-1):0]	O	AES Hash Context SRAM Memory read address. **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-27 AES Hash Context Memory Interface Signals (Continued)

Port Name	I/O	Description
I_mem_aes_hctx_rdata[(127+pDWC_ULTRA_AES_HCTX_PROT_WIDTH):0]	I	AES Hash Context SRAM Memory read data. **Exists:** (!DWC_UAES_CFG_SYNTH_HCTX_MEM) && (DWC_UAES_CFG_HASH_ILEAVE_EN) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

5.28 AES FIPS Interface Signals

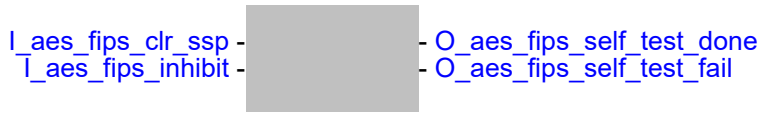


Table 5-28 AES FIPS Interface Signals

Port Name	I/O	Description
I_aes_fips_clr_ssp	I	This is a pulse signal. When asserted , it clears all AES "Key" registers (does not affect memories). **Exists:** ((DWC_UAES_CFG_FIPS_SELF_TEST > 0)) && (!DWC_UAES_CFG_CMD_IF_KEY_EN) && (!(DWC_UAES_CFG_KEY_IF_TYPE==0)) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
I_aes_fips_inhibit	I	This is a level active signal. When asserted, this signal enforces the output cmd/data inhibition (clears all cmd and data outputs to zero). **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST > 0) **Synchronous To:** clk_macsec **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal
O_aes_fips_self_test_done	O	When asserted, it indicates that the Self-Test has completed. **Exists:** ((DWC_UAES_CFG_FIPS_SELF_TEST > 0)) && ((DWC_UAES_CFG_FIPS_SELF_TEST == 2)) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** Not validated by another signal

Table 5-28 AES FIPS Interface Signals (Continued)

Port Name	I/O	Description
O_aes_fips_self_test_fail	O	When asserted, it indicates that the Self-Test has Failed. **Exists:** ((DWC_UAES_CFG_FIPS_SELF_TEST > 0)) && ((DWC_UAES_CFG_FIPS_SELF_TEST == 2)) **Synchronous To:** clk_macsec **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Validated by:** O_aes_fips_self_test_done

6 Register Descriptions

This chapter details all possible registers in the IP. They are arranged hierarchically into maps and blocks (banks).Your actual configuration might not contain all of these registers.

Attention: For configurable IP titles, do not use this document to determine the exact attributes of your register map. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the register attributes for your actual configuration at workspace/report/ComponentRegisters.html or workspace/report/ComponentRegisters.xml after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the registers that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the Offset and Memory Access values might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>)** that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Exists Expressions

These expressions indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address.*

Memory Access Attributes

The Memory Access attribute is defined as <ReadBehavior>/<WriteBehavior> which are defined in the following table.

Table 6-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies this register field in a manner not covered above.
Wo	You can only write once to this register field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	A write modifies this register field in a manner not covered above.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 6-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 6-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the core updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

Component Memory Maps

The following table documents the memory maps visible in this component.

Table 6-4 DWC_macsec Memory Maps

Memory Map	Description
DWC_macsec_apb on page 202	macsec_apb Memory Map **Exists:** 1==1
DWC_macsec_uaes_gcmp_apb on page 202	DWC_macsec Memory Map UAES_GCMP APB-Interface **Exists:** 1==1

Component Banks/Blocks

The following table shows the address blocks for each memory map. Follow the link for an address block to see a table of its registers.

Table 6-5 Address Banks/Blocks for Memory Map: DWC_macsec_apb

Address Block	Description
macsec_apb_cfg on page 203	macsec_apb_cfg **Exists:** Always

Table 6-6 Address Banks/Blocks for Memory Map: DWC_macsec_uaes_gcmp_apb

Address Block	Description
macsec_id_aes_host_cfg on page 3848	AES Host **Exists:** Always

6.1 DWC_macsec_apb/macsec_apb_cfg Registers

macsec_apb_cfg . Follow the link for the register to see a detailed description of the register.

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg

Register	Offset	Description
CORE_VER_NUM on page 270	0x0	IP Version Number Register.
CORE_VER_TYPE on page 271	0x4	IP Version Type Register.
MACsec_IP_Control on page 273	0x8	MACsec IP Control Register
MACSec_CFG_MIN_IPG on page 277	0xc	MACsec Minimum Inter-Packet Configuration
MACsec_IP_VLAN on page 280	0x10	VLAN Configuration register
ETHERTYPE_REPLACE on page 281	0x18	ETHERTYPE_REPLACE Register
RX_LOOKUP_TABLE_SCI_LOW on page 283	0x1c	RX_LOOKUP_TABLE_SCI_LOW
RX_LOOKUP_TABLE_SCI_HIGH on page 284	0x20	RX_LOOKUP_TABLE_SCI_HIGH
RX_LOOKUP_TABLE_VLAN_INFO on page 285	0x24	RX_LOOKUP_TABLE_VLAN_INFO
RX_SAI_PG on page 287	0x28	RX_SAI_PG
RX_SAI_PRG on page 288	0x2c	RX_SAI_PRG
RX_SAI_OTHER_FIELDS on page 289	0x30	RX_SAI_OTHER_FIELDS
RX_SAI_DATA_FETCH on page 291	0x34	RX_SAI_DATA_FETCH
TX_SRC_FILT_msb on page 293	0x38	TX_SRC_FILT_msb
TX_SRC_FILT_lsb on page 294	0x3c	TX_SRC_FILT_lsb
TX_DST_FILT_msb on page 295	0x40	TX_DST_FILT_msb
TX_DST_FILT_lsb on page 296	0x44	TX_DST_FILT_lsb
TX_SAI_PG on page 297	0x48	TX_SAI_PG
TX_SAI_PRG on page 298	0x4c	TX_SAI_PRG
TX_SAI_OTHER_FIELDS on page 299	0x50	TX_SAI_OTHER_FIELDS
TX_SAI_DATA_FETCH on page 301	0x54	TX_SAI_DATA_FETCH
TX_LOOKUP_TABLE_VLAN_INFO on page 303	0x58	TX_LOOKUP_TABLE_VLAN_INFO
TX_SAI_PORT_ID on page 305	0x5c	TX_SAI_PORT_ID

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
TX_ACTIVE_AN on page 306	0x60	TX_ACTIVE_AN
MIB_CMD on page 307	0x64	MIB_CMD
MIB_CMD_STAT on page 309	0x68	MIB_CMD_STAT
MIB_VALIDATE on page 310	0x6c	MIB_VALIDATE
MIB_WR_DATA_HIGH on page 311	0x70	MIB_WR_DATA_HIGH
MIB_WR_DATA_LOW on page 312	0x74	MIB_WR_DATA_LOW
RX_SAI_SC_CORR_TABLE_PG on page 313	0x78	RX_SAI_SC_CORR_TABLE_PG
RX_SAI_SC_CORR_TABLE_PRG on page 314	0x7c	RX_SAI_SC_CORR_TABLE_PRG
RX_SAI_SC_CORR_TABLE_DATA_FETCH on page 315	0x80	RX_SAI_SC_CORR_TABLE_DATA_FETCH
RX_SAI_SC_CORR_TABLE_SC_VALUE on page 316	0x84	RX_SAI_SC_CORR_TABLE_SC_VALUE
RX_SALT_PG on page 317	0x88	RX_SALT_PG
RX_SALT_PRG on page 318	0x8c	RX_SALT_PRG
RX_SALT_DATA_FETCH on page 319	0x90	RX_SALT_DATA_FETCH
RX_SALT_TABLE_0 on page 320	0x94	RX_SALT_TABLE_0
RX_SALT_TABLE_1 on page 321	0x98	RX_SALT_TABLE_1
RX_SALT_TABLE_2 on page 322	0x9c	RX_SALT_TABLE_2
TX_SALT_PG on page 323	0xa0	TX_SALT_PG
TX_SALT_PRG on page 324	0xa4	TX_SALT_PRG
TX_SALT_DATA_FETCH on page 325	0xa8	TX_SALT_DATA_FETCH
TX_SALT_TABLE_0 on page 326	0xac	TX_SALT_TABLE_0
TX_SALT_TABLE_1 on page 327	0xb0	TX_SALT_TABLE_1
TX_SALT_TABLE_2 on page 328	0xb4	TX_SALT_TABLE_2
RX_ARW_PG on page 329	0xb8	RX_ARW_PG
RX_ARW_PRG on page 330	0xbc	RX_ARW_PRG
RX_ARW_DATA_FETCH on page 331	0xc0	RX_ARW_DATA_FETCH
RX_ARW_TABLE_0 on page 332	0xc4	RX_ARW_TABLE_0

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
TX_SAD_PG on page 333	0xc8	TX_SAD_PG
TX_SAD_PRG on page 334	0xcc	TX_SAD_PRG
TX_SAD_DATA_FETCH on page 335	0xd0	TX_SAD_DATA_FETCH
TX_SAD_0_LOW on page 336	0xd4	TX_SAD_0_LOW
TX_SAD_0_HIGH on page 337	0xd8	TX_SAD_0_HIGH
TX_SAD_1_LOW on page 339	0xdc	TX_SAD_1_LOW
TX_SAD_1_HIGH on page 340	0xe0	TX_SAD_1_HIGH
TX_SAD_2_LOW on page 341	0xe4	TX_SAD_2_LOW
TX_SAD_2_HIGH on page 342	0xe8	TX_SAD_2_HIGH
TX_SAD_3_LOW on page 343	0xec	TX_SAD_3_LOW
TX_SAD_3_HIGH on page 344	0xf0	TX_SAD_3_HIGH
RX_SAD_PG on page 345	0xf4	RX_SAD_PG
RX_SAD_PRG on page 346	0xf8	RX_SAD_PRG
RX_SAD_DATA_FETCH on page 347	0xfc	RX_SAD_DATA_FETCH
RX_SAD_0_LOW on page 348	0x100	RX_SAD_0_LOW
RX_SAD_0_HIGH on page 350	0x104	RX_SAD_0_HIGH
RX_SAD_1_LOW on page 352	0x108	RX_SAD_1_LOW
RX_SAD_1_HIGH on page 353	0x10c	RX_SAD_1_HIGH
RX_SAD_2_LOW on page 354	0x110	RX_SAD_2_LOW
RX_SAD_2_HIGH on page 355	0x114	RX_SAD_2_HIGH
MIB_CMD_D_BUF_HIGH_0 on page 356	0x118	MIB_CMD_D_BUF_HIGH_0
MIB_CMD_D_BUF_LOW_0 on page 357	0x11c	MIB_CMD_D_BUF_LOW_0
MIB_CMD_D_BUF_HIGH_1 on page 358	0x120	MIB_CMD_D_BUF_HIGH_1
MIB_CMD_D_BUF_LOW_1 on page 359	0x124	MIB_CMD_D_BUF_LOW_1
MIB_CMD_D_BUF_HIGH_2 on page 360	0x128	MIB_CMD_D_BUF_HIGH_2
MIB_CMD_D_BUF_LOW_2 on page 361	0x12c	MIB_CMD_D_BUF_LOW_2
MIB_CMD_D_BUF_HIGH_3 on page 362	0x130	MIB_CMD_D_BUF_HIGH_3
MIB_CMD_D_BUF_LOW_3 on page 363	0x134	MIB_CMD_D_BUF_LOW_3

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
MIB_CMD_D_BUF_HIGH_4 on page 364	0x138	MIB_CMD_D_BUF_HIGH_4
MIB_CMD_D_BUF_LOW_4 on page 365	0x13c	MIB_CMD_D_BUF_LOW_4
MIB_CMD_D_BUF_HIGH_5 on page 366	0x140	MIB_CMD_D_BUF_HIGH_5
MIB_CMD_D_BUF_LOW_5 on page 367	0x144	MIB_CMD_D_BUF_LOW_5
MIB_CMD_D_BUF_HIGH_6 on page 368	0x148	MIB_CMD_D_BUF_HIGH_6
MIB_CMD_D_BUF_LOW_6 on page 369	0x14c	MIB_CMD_D_BUF_LOW_6
MIB_CMD_D_BUF_HIGH_7 on page 370	0x150	MIB_CMD_D_BUF_HIGH_7
MIB_CMD_D_BUF_LOW_7 on page 371	0x154	MIB_CMD_D_BUF_LOW_7
MIB_CMD_D_BUF_HIGH_8 on page 372	0x158	MIB_CMD_D_BUF_HIGH_8
MIB_CMD_D_BUF_LOW_8 on page 373	0x15c	MIB_CMD_D_BUF_LOW_8
MIB_CMD_D_BUF_HIGH_9 on page 374	0x160	MIB_CMD_D_BUF_HIGH_9
MIB_CMD_D_BUF_LOW_9 on page 375	0x164	MIB_CMD_D_BUF_LOW_9
MIB_CMD_D_BUF_HIGH_10 on page 376	0x168	MIB_CMD_D_BUF_HIGH_10
MIB_CMD_D_BUF_LOW_10 on page 377	0x16c	MIB_CMD_D_BUF_LOW_10
MIB_CMD_D_BUF_HIGH_11 on page 378	0x170	MIB_CMD_D_BUF_HIGH_11
MIB_CMD_D_BUF_LOW_11 on page 379	0x174	MIB_CMD_D_BUF_LOW_11
IRQ_GLBL_EN on page 380	0x178	IRQ_GLBL_EN
IRQ_GLBL_STAT on page 384	0x17c	IRQ_GLBL_STAT
ETHERTYPE_FILTER_0 on page 389	0x180	ETHERTYPE_FILTER_0 Register
ETHERTYPE_FILTER_1 on page 390	0x184	ETHERTYPE_FILTER_1 Register
ETHERTYPE_FILTER_2 on page 391	0x188	ETHERTYPE_FILTER_2 Register
ETHERTYPE_FILTER_3 on page 392	0x18c	ETHERTYPE_FILTER_3 Register
IRQ_TYPE_EN_TX_SA_0 on page 393	0x190	IRQ_TYPE_EN_TX_SA_0
IRQ_TYPE_STAT_TX_SA_0 on page 395	0x194	IRQ_TYPE_STAT_TX_SA_0
IRQ_TYPE_EN_TX_SA_1 on page 397	0x198	IRQ_TYPE_EN_TX_SA_1
IRQ_TYPE_STAT_TX_SA_1 on page 399	0x19c	IRQ_TYPE_STAT_TX_SA_1
IRQ_TYPE_EN_TX_SA_2 on page 401	0x1a0	IRQ_TYPE_EN_TX_SA_2

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_TX_SA_2 on page 403	0x1a4	IRQ_TYPE_STAT_TX_SA_2
IRQ_TYPE_EN_TX_SA_3 on page 405	0x1a8	IRQ_TYPE_EN_TX_SA_3
IRQ_TYPE_STAT_TX_SA_3 on page 407	0x1ac	IRQ_TYPE_STAT_TX_SA_3
IRQ_TYPE_EN_TX_SA_4 on page 409	0x1b0	IRQ_TYPE_EN_TX_SA_4
IRQ_TYPE_STAT_TX_SA_4 on page 411	0x1b4	IRQ_TYPE_STAT_TX_SA_4
IRQ_TYPE_EN_TX_SA_5 on page 413	0x1b8	IRQ_TYPE_EN_TX_SA_5
IRQ_TYPE_STAT_TX_SA_5 on page 415	0x1bc	IRQ_TYPE_STAT_TX_SA_5
IRQ_TYPE_EN_TX_SA_6 on page 417	0x1c0	IRQ_TYPE_EN_TX_SA_6
IRQ_TYPE_STAT_TX_SA_6 on page 419	0x1c4	IRQ_TYPE_STAT_TX_SA_6
IRQ_TYPE_EN_TX_SA_7 on page 421	0x1c8	IRQ_TYPE_EN_TX_SA_7
IRQ_TYPE_STAT_TX_SA_7 on page 423	0x1cc	IRQ_TYPE_STAT_TX_SA_7
IRQ_TYPE_EN_TX_SA_8 on page 425	0x1d0	IRQ_TYPE_EN_TX_SA_8
IRQ_TYPE_STAT_TX_SA_8 on page 427	0x1d4	IRQ_TYPE_STAT_TX_SA_8
IRQ_TYPE_EN_TX_SA_9 on page 429	0x1d8	IRQ_TYPE_EN_TX_SA_9
IRQ_TYPE_STAT_TX_SA_9 on page 431	0x1dc	IRQ_TYPE_STAT_TX_SA_9
IRQ_TYPE_EN_TX_SA_10 on page 433	0x1e0	IRQ_TYPE_EN_TX_SA_10
IRQ_TYPE_STAT_TX_SA_10 on page 435	0x1e4	IRQ_TYPE_STAT_TX_SA_10
IRQ_TYPE_EN_TX_SA_11 on page 437	0x1e8	IRQ_TYPE_EN_TX_SA_11
IRQ_TYPE_STAT_TX_SA_11 on page 439	0x1ec	IRQ_TYPE_STAT_TX_SA_11
IRQ_TYPE_EN_TX_SA_12 on page 441	0x1f0	IRQ_TYPE_EN_TX_SA_12
IRQ_TYPE_STAT_TX_SA_12 on page 443	0x1f4	IRQ_TYPE_STAT_TX_SA_12
IRQ_TYPE_EN_TX_SA_13 on page 445	0x1f8	IRQ_TYPE_EN_TX_SA_13
IRQ_TYPE_STAT_TX_SA_13 on page 447	0x1fc	IRQ_TYPE_STAT_TX_SA_13
IRQ_TYPE_EN_TX_SA_14 on page 449	0x200	IRQ_TYPE_EN_TX_SA_14
IRQ_TYPE_STAT_TX_SA_14 on page 451	0x204	IRQ_TYPE_STAT_TX_SA_14

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_TX_SA_15 on page 453	0x208	IRQ_TYPE_EN_TX_SA_15
IRQ_TYPE_STAT_TX_SA_15 on page 455	0x20c	IRQ_TYPE_STAT_TX_SA_15
IRQ_TYPE_EN_TX_SA_16 on page 457	0x210	IRQ_TYPE_EN_TX_SA_16
IRQ_TYPE_STAT_TX_SA_16 on page 459	0x214	IRQ_TYPE_STAT_TX_SA_16
IRQ_TYPE_EN_TX_SA_17 on page 461	0x218	IRQ_TYPE_EN_TX_SA_17
IRQ_TYPE_STAT_TX_SA_17 on page 463	0x21c	IRQ_TYPE_STAT_TX_SA_17
IRQ_TYPE_EN_TX_SA_18 on page 465	0x220	IRQ_TYPE_EN_TX_SA_18
IRQ_TYPE_STAT_TX_SA_18 on page 467	0x224	IRQ_TYPE_STAT_TX_SA_18
IRQ_TYPE_EN_TX_SA_19 on page 469	0x228	IRQ_TYPE_EN_TX_SA_19
IRQ_TYPE_STAT_TX_SA_19 on page 471	0x22c	IRQ_TYPE_STAT_TX_SA_19
IRQ_TYPE_EN_TX_SA_20 on page 473	0x230	IRQ_TYPE_EN_TX_SA_20
IRQ_TYPE_STAT_TX_SA_20 on page 475	0x234	IRQ_TYPE_STAT_TX_SA_20
IRQ_TYPE_EN_TX_SA_21 on page 477	0x238	IRQ_TYPE_EN_TX_SA_21
IRQ_TYPE_STAT_TX_SA_21 on page 479	0x23c	IRQ_TYPE_STAT_TX_SA_21
IRQ_TYPE_EN_TX_SA_22 on page 481	0x240	IRQ_TYPE_EN_TX_SA_22
IRQ_TYPE_STAT_TX_SA_22 on page 483	0x244	IRQ_TYPE_STAT_TX_SA_22
IRQ_TYPE_EN_TX_SA_23 on page 485	0x248	IRQ_TYPE_EN_TX_SA_23
IRQ_TYPE_STAT_TX_SA_23 on page 487	0x24c	IRQ_TYPE_STAT_TX_SA_23
IRQ_TYPE_EN_TX_SA_24 on page 489	0x250	IRQ_TYPE_EN_TX_SA_24
IRQ_TYPE_STAT_TX_SA_24 on page 491	0x254	IRQ_TYPE_STAT_TX_SA_24
IRQ_TYPE_EN_TX_SA_25 on page 493	0x258	IRQ_TYPE_EN_TX_SA_25
IRQ_TYPE_STAT_TX_SA_25 on page 495	0x25c	IRQ_TYPE_STAT_TX_SA_25

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_TX_SA_26 on page 497	0x260	IRQ_TYPE_EN_TX_SA_26
IRQ_TYPE_STAT_TX_SA_26 on page 499	0x264	IRQ_TYPE_STAT_TX_SA_26
IRQ_TYPE_EN_TX_SA_27 on page 501	0x268	IRQ_TYPE_EN_TX_SA_27
IRQ_TYPE_STAT_TX_SA_27 on page 503	0x26c	IRQ_TYPE_STAT_TX_SA_27
IRQ_TYPE_EN_TX_SA_28 on page 505	0x270	IRQ_TYPE_EN_TX_SA_28
IRQ_TYPE_STAT_TX_SA_28 on page 507	0x274	IRQ_TYPE_STAT_TX_SA_28
IRQ_TYPE_EN_TX_SA_29 on page 509	0x278	IRQ_TYPE_EN_TX_SA_29
IRQ_TYPE_STAT_TX_SA_29 on page 511	0x27c	IRQ_TYPE_STAT_TX_SA_29
IRQ_TYPE_EN_TX_SA_30 on page 513	0x280	IRQ_TYPE_EN_TX_SA_30
IRQ_TYPE_STAT_TX_SA_30 on page 515	0x284	IRQ_TYPE_STAT_TX_SA_30
IRQ_TYPE_EN_TX_SA_31 on page 517	0x288	IRQ_TYPE_EN_TX_SA_31
IRQ_TYPE_STAT_TX_SA_31 on page 519	0x28c	IRQ_TYPE_STAT_TX_SA_31
IRQ_TYPE_EN_TX_SA_32 on page 521	0x290	IRQ_TYPE_EN_TX_SA_32
IRQ_TYPE_STAT_TX_SA_32 on page 523	0x294	IRQ_TYPE_STAT_TX_SA_32
IRQ_TYPE_EN_TX_SA_33 on page 525	0x298	IRQ_TYPE_EN_TX_SA_33
IRQ_TYPE_STAT_TX_SA_33 on page 527	0x29c	IRQ_TYPE_STAT_TX_SA_33
IRQ_TYPE_EN_TX_SA_34 on page 529	0x2a0	IRQ_TYPE_EN_TX_SA_34
IRQ_TYPE_STAT_TX_SA_34 on page 531	0x2a4	IRQ_TYPE_STAT_TX_SA_34
IRQ_TYPE_EN_TX_SA_35 on page 533	0x2a8	IRQ_TYPE_EN_TX_SA_35
IRQ_TYPE_STAT_TX_SA_35 on page 535	0x2ac	IRQ_TYPE_STAT_TX_SA_35
IRQ_TYPE_EN_TX_SA_36 on page 537	0x2b0	IRQ_TYPE_EN_TX_SA_36
IRQ_TYPE_STAT_TX_SA_36 on page 539	0x2b4	IRQ_TYPE_STAT_TX_SA_36

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_TX_SA_37 on page 541	0x2b8	IRQ_TYPE_EN_TX_SA_37
IRQ_TYPE_STAT_TX_SA_37 on page 543	0x2bc	IRQ_TYPE_STAT_TX_SA_37
IRQ_TYPE_EN_TX_SA_38 on page 545	0x2c0	IRQ_TYPE_EN_TX_SA_38
IRQ_TYPE_STAT_TX_SA_38 on page 547	0x2c4	IRQ_TYPE_STAT_TX_SA_38
IRQ_TYPE_EN_TX_SA_39 on page 549	0x2c8	IRQ_TYPE_EN_TX_SA_39
IRQ_TYPE_STAT_TX_SA_39 on page 551	0x2cc	IRQ_TYPE_STAT_TX_SA_39
IRQ_TYPE_EN_TX_SA_40 on page 553	0x2d0	IRQ_TYPE_EN_TX_SA_40
IRQ_TYPE_STAT_TX_SA_40 on page 555	0x2d4	IRQ_TYPE_STAT_TX_SA_40
IRQ_TYPE_EN_TX_SA_41 on page 557	0x2d8	IRQ_TYPE_EN_TX_SA_41
IRQ_TYPE_STAT_TX_SA_41 on page 559	0x2dc	IRQ_TYPE_STAT_TX_SA_41
IRQ_TYPE_EN_TX_SA_42 on page 561	0x2e0	IRQ_TYPE_EN_TX_SA_42
IRQ_TYPE_STAT_TX_SA_42 on page 563	0x2e4	IRQ_TYPE_STAT_TX_SA_42
IRQ_TYPE_EN_TX_SA_43 on page 565	0x2e8	IRQ_TYPE_EN_TX_SA_43
IRQ_TYPE_STAT_TX_SA_43 on page 567	0x2ec	IRQ_TYPE_STAT_TX_SA_43
IRQ_TYPE_EN_TX_SA_44 on page 569	0x2f0	IRQ_TYPE_EN_TX_SA_44
IRQ_TYPE_STAT_TX_SA_44 on page 571	0x2f4	IRQ_TYPE_STAT_TX_SA_44
IRQ_TYPE_EN_TX_SA_45 on page 573	0x2f8	IRQ_TYPE_EN_TX_SA_45
IRQ_TYPE_STAT_TX_SA_45 on page 575	0x2fc	IRQ_TYPE_STAT_TX_SA_45
IRQ_TYPE_EN_TX_SA_46 on page 577	0x300	IRQ_TYPE_EN_TX_SA_46
IRQ_TYPE_STAT_TX_SA_46 on page 579	0x304	IRQ_TYPE_STAT_TX_SA_46
IRQ_TYPE_EN_TX_SA_47 on page 581	0x308	IRQ_TYPE_EN_TX_SA_47
IRQ_TYPE_STAT_TX_SA_47 on page 583	0x30c	IRQ_TYPE_STAT_TX_SA_47

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_TX_SA_48 on page 585	0x310	IRQ_TYPE_EN_TX_SA_48
IRQ_TYPE_STAT_TX_SA_48 on page 587	0x314	IRQ_TYPE_STAT_TX_SA_48
IRQ_TYPE_EN_TX_SA_49 on page 589	0x318	IRQ_TYPE_EN_TX_SA_49
IRQ_TYPE_STAT_TX_SA_49 on page 591	0x31c	IRQ_TYPE_STAT_TX_SA_49
IRQ_TYPE_EN_TX_SA_50 on page 593	0x320	IRQ_TYPE_EN_TX_SA_50
IRQ_TYPE_STAT_TX_SA_50 on page 595	0x324	IRQ_TYPE_STAT_TX_SA_50
IRQ_TYPE_EN_TX_SA_51 on page 597	0x328	IRQ_TYPE_EN_TX_SA_51
IRQ_TYPE_STAT_TX_SA_51 on page 599	0x32c	IRQ_TYPE_STAT_TX_SA_51
IRQ_TYPE_EN_TX_SA_52 on page 601	0x330	IRQ_TYPE_EN_TX_SA_52
IRQ_TYPE_STAT_TX_SA_52 on page 603	0x334	IRQ_TYPE_STAT_TX_SA_52
IRQ_TYPE_EN_TX_SA_53 on page 605	0x338	IRQ_TYPE_EN_TX_SA_53
IRQ_TYPE_STAT_TX_SA_53 on page 607	0x33c	IRQ_TYPE_STAT_TX_SA_53
IRQ_TYPE_EN_TX_SA_54 on page 609	0x340	IRQ_TYPE_EN_TX_SA_54
IRQ_TYPE_STAT_TX_SA_54 on page 611	0x344	IRQ_TYPE_STAT_TX_SA_54
IRQ_TYPE_EN_TX_SA_55 on page 613	0x348	IRQ_TYPE_EN_TX_SA_55
IRQ_TYPE_STAT_TX_SA_55 on page 615	0x34c	IRQ_TYPE_STAT_TX_SA_55
IRQ_TYPE_EN_TX_SA_56 on page 617	0x350	IRQ_TYPE_EN_TX_SA_56
IRQ_TYPE_STAT_TX_SA_56 on page 619	0x354	IRQ_TYPE_STAT_TX_SA_56
IRQ_TYPE_EN_TX_SA_57 on page 621	0x358	IRQ_TYPE_EN_TX_SA_57
IRQ_TYPE_STAT_TX_SA_57 on page 623	0x35c	IRQ_TYPE_STAT_TX_SA_57
IRQ_TYPE_EN_TX_SA_58 on page 625	0x360	IRQ_TYPE_EN_TX_SA_58
IRQ_TYPE_STAT_TX_SA_58 on page 627	0x364	IRQ_TYPE_STAT_TX_SA_58

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_TX_SA_59 on page 629	0x368	IRQ_TYPE_EN_TX_SA_59
IRQ_TYPE_STAT_TX_SA_59 on page 631	0x36c	IRQ_TYPE_STAT_TX_SA_59
IRQ_TYPE_EN_TX_SA_60 on page 633	0x370	IRQ_TYPE_EN_TX_SA_60
IRQ_TYPE_STAT_TX_SA_60 on page 635	0x374	IRQ_TYPE_STAT_TX_SA_60
IRQ_TYPE_EN_TX_SA_61 on page 637	0x378	IRQ_TYPE_EN_TX_SA_61
IRQ_TYPE_STAT_TX_SA_61 on page 639	0x37c	IRQ_TYPE_STAT_TX_SA_61
IRQ_TYPE_EN_TX_SA_62 on page 641	0x380	IRQ_TYPE_EN_TX_SA_62
IRQ_TYPE_STAT_TX_SA_62 on page 643	0x384	IRQ_TYPE_STAT_TX_SA_62
IRQ_TYPE_EN_TX_SA_63 on page 645	0x388	IRQ_TYPE_EN_TX_SA_63
IRQ_TYPE_STAT_TX_SA_63 on page 647	0x38c	IRQ_TYPE_STAT_TX_SA_63
IRQ_TYPE_EN_RX_SA_0 on page 649	0x390	IRQ_TYPE_EN_RX_SA_0
IRQ_TYPE_STAT_RX_SA_0 on page 652	0x394	IRQ_TYPE_STAT_RX_SA_0
IRQ_TYPE_EN_RX_SA_1 on page 655	0x398	IRQ_TYPE_EN_RX_SA_1
IRQ_TYPE_STAT_RX_SA_1 on page 658	0x39c	IRQ_TYPE_STAT_RX_SA_1
IRQ_TYPE_EN_RX_SA_2 on page 661	0x3a0	IRQ_TYPE_EN_RX_SA_2
IRQ_TYPE_STAT_RX_SA_2 on page 664	0x3a4	IRQ_TYPE_STAT_RX_SA_2
IRQ_TYPE_EN_RX_SA_3 on page 667	0x3a8	IRQ_TYPE_EN_RX_SA_3
IRQ_TYPE_STAT_RX_SA_3 on page 670	0x3ac	IRQ_TYPE_STAT_RX_SA_3
IRQ_TYPE_EN_RX_SA_4 on page 673	0x3b0	IRQ_TYPE_EN_RX_SA_4
IRQ_TYPE_STAT_RX_SA_4 on page 676	0x3b4	IRQ_TYPE_STAT_RX_SA_4
IRQ_TYPE_EN_RX_SA_5 on page 679	0x3b8	IRQ_TYPE_EN_RX_SA_5
IRQ_TYPE_STAT_RX_SA_5 on page 682	0x3bc	IRQ_TYPE_STAT_RX_SA_5
IRQ_TYPE_EN_RX_SA_6 on page 685	0x3c0	IRQ_TYPE_EN_RX_SA_6
IRQ_TYPE_STAT_RX_SA_6 on page 688	0x3c4	IRQ_TYPE_STAT_RX_SA_6
IRQ_TYPE_EN_RX_SA_7 on page 691	0x3c8	IRQ_TYPE_EN_RX_SA_7

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_7 on page 694	0x3cc	IRQ_TYPE_STAT_RX_SA_7
IRQ_TYPE_EN_RX_SA_8 on page 697	0x3d0	IRQ_TYPE_EN_RX_SA_8
IRQ_TYPE_STAT_RX_SA_8 on page 700	0x3d4	IRQ_TYPE_STAT_RX_SA_8
IRQ_TYPE_EN_RX_SA_9 on page 703	0x3d8	IRQ_TYPE_EN_RX_SA_9
IRQ_TYPE_STAT_RX_SA_9 on page 706	0x3dc	IRQ_TYPE_STAT_RX_SA_9
IRQ_TYPE_EN_RX_SA_10 on page 709	0x3e0	IRQ_TYPE_EN_RX_SA_10
IRQ_TYPE_STAT_RX_SA_10 on page 712	0x3e4	IRQ_TYPE_STAT_RX_SA_10
IRQ_TYPE_EN_RX_SA_11 on page 715	0x3e8	IRQ_TYPE_EN_RX_SA_11
IRQ_TYPE_STAT_RX_SA_11 on page 718	0x3ec	IRQ_TYPE_STAT_RX_SA_11
IRQ_TYPE_EN_RX_SA_12 on page 721	0x3f0	IRQ_TYPE_EN_RX_SA_12
IRQ_TYPE_STAT_RX_SA_12 on page 724	0x3f4	IRQ_TYPE_STAT_RX_SA_12
IRQ_TYPE_EN_RX_SA_13 on page 727	0x3f8	IRQ_TYPE_EN_RX_SA_13
IRQ_TYPE_STAT_RX_SA_13 on page 730	0x3fc	IRQ_TYPE_STAT_RX_SA_13
IRQ_TYPE_EN_RX_SA_14 on page 733	0x400	IRQ_TYPE_EN_RX_SA_14
IRQ_TYPE_STAT_RX_SA_14 on page 736	0x404	IRQ_TYPE_STAT_RX_SA_14
IRQ_TYPE_EN_RX_SA_15 on page 739	0x408	IRQ_TYPE_EN_RX_SA_15
IRQ_TYPE_STAT_RX_SA_15 on page 742	0x40c	IRQ_TYPE_STAT_RX_SA_15
IRQ_TYPE_EN_RX_SA_16 on page 745	0x410	IRQ_TYPE_EN_RX_SA_16
IRQ_TYPE_STAT_RX_SA_16 on page 748	0x414	IRQ_TYPE_STAT_RX_SA_16
IRQ_TYPE_EN_RX_SA_17 on page 751	0x418	IRQ_TYPE_EN_RX_SA_17
IRQ_TYPE_STAT_RX_SA_17 on page 754	0x41c	IRQ_TYPE_STAT_RX_SA_17
IRQ_TYPE_EN_RX_SA_18 on page 757	0x420	IRQ_TYPE_EN_RX_SA_18
IRQ_TYPE_STAT_RX_SA_18 on page 760	0x424	IRQ_TYPE_STAT_RX_SA_18

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_19 on page 763	0x428	IRQ_TYPE_EN_RX_SA_19
IRQ_TYPE_STAT_RX_SA_19 on page 766	0x42c	IRQ_TYPE_STAT_RX_SA_19
IRQ_TYPE_EN_RX_SA_20 on page 769	0x430	IRQ_TYPE_EN_RX_SA_20
IRQ_TYPE_STAT_RX_SA_20 on page 772	0x434	IRQ_TYPE_STAT_RX_SA_20
IRQ_TYPE_EN_RX_SA_21 on page 775	0x438	IRQ_TYPE_EN_RX_SA_21
IRQ_TYPE_STAT_RX_SA_21 on page 778	0x43c	IRQ_TYPE_STAT_RX_SA_21
IRQ_TYPE_EN_RX_SA_22 on page 781	0x440	IRQ_TYPE_EN_RX_SA_22
IRQ_TYPE_STAT_RX_SA_22 on page 784	0x444	IRQ_TYPE_STAT_RX_SA_22
IRQ_TYPE_EN_RX_SA_23 on page 787	0x448	IRQ_TYPE_EN_RX_SA_23
IRQ_TYPE_STAT_RX_SA_23 on page 790	0x44c	IRQ_TYPE_STAT_RX_SA_23
IRQ_TYPE_EN_RX_SA_24 on page 793	0x450	IRQ_TYPE_EN_RX_SA_24
IRQ_TYPE_STAT_RX_SA_24 on page 796	0x454	IRQ_TYPE_STAT_RX_SA_24
IRQ_TYPE_EN_RX_SA_25 on page 799	0x458	IRQ_TYPE_EN_RX_SA_25
IRQ_TYPE_STAT_RX_SA_25 on page 802	0x45c	IRQ_TYPE_STAT_RX_SA_25
IRQ_TYPE_EN_RX_SA_26 on page 805	0x460	IRQ_TYPE_EN_RX_SA_26
IRQ_TYPE_STAT_RX_SA_26 on page 808	0x464	IRQ_TYPE_STAT_RX_SA_26
IRQ_TYPE_EN_RX_SA_27 on page 811	0x468	IRQ_TYPE_EN_RX_SA_27
IRQ_TYPE_STAT_RX_SA_27 on page 814	0x46c	IRQ_TYPE_STAT_RX_SA_27
IRQ_TYPE_EN_RX_SA_28 on page 817	0x470	IRQ_TYPE_EN_RX_SA_28
IRQ_TYPE_STAT_RX_SA_28 on page 820	0x474	IRQ_TYPE_STAT_RX_SA_28
IRQ_TYPE_EN_RX_SA_29 on page 823	0x478	IRQ_TYPE_EN_RX_SA_29
IRQ_TYPE_STAT_RX_SA_29 on page 826	0x47c	IRQ_TYPE_STAT_RX_SA_29

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_30 on page 829	0x480	IRQ_TYPE_EN_RX_SA_30
IRQ_TYPE_STAT_RX_SA_30 on page 832	0x484	IRQ_TYPE_STAT_RX_SA_30
IRQ_TYPE_EN_RX_SA_31 on page 835	0x488	IRQ_TYPE_EN_RX_SA_31
IRQ_TYPE_STAT_RX_SA_31 on page 838	0x48c	IRQ_TYPE_STAT_RX_SA_31
IRQ_TYPE_EN_RX_SA_32 on page 841	0x490	IRQ_TYPE_EN_RX_SA_32
IRQ_TYPE_STAT_RX_SA_32 on page 844	0x494	IRQ_TYPE_STAT_RX_SA_32
IRQ_TYPE_EN_RX_SA_33 on page 847	0x498	IRQ_TYPE_EN_RX_SA_33
IRQ_TYPE_STAT_RX_SA_33 on page 850	0x49c	IRQ_TYPE_STAT_RX_SA_33
IRQ_TYPE_EN_RX_SA_34 on page 853	0x4a0	IRQ_TYPE_EN_RX_SA_34
IRQ_TYPE_STAT_RX_SA_34 on page 856	0x4a4	IRQ_TYPE_STAT_RX_SA_34
IRQ_TYPE_EN_RX_SA_35 on page 859	0x4a8	IRQ_TYPE_EN_RX_SA_35
IRQ_TYPE_STAT_RX_SA_35 on page 862	0x4ac	IRQ_TYPE_STAT_RX_SA_35
IRQ_TYPE_EN_RX_SA_36 on page 865	0x4b0	IRQ_TYPE_EN_RX_SA_36
IRQ_TYPE_STAT_RX_SA_36 on page 868	0x4b4	IRQ_TYPE_STAT_RX_SA_36
IRQ_TYPE_EN_RX_SA_37 on page 871	0x4b8	IRQ_TYPE_EN_RX_SA_37
IRQ_TYPE_STAT_RX_SA_37 on page 874	0x4bc	IRQ_TYPE_STAT_RX_SA_37
IRQ_TYPE_EN_RX_SA_38 on page 877	0x4c0	IRQ_TYPE_EN_RX_SA_38
IRQ_TYPE_STAT_RX_SA_38 on page 880	0x4c4	IRQ_TYPE_STAT_RX_SA_38
IRQ_TYPE_EN_RX_SA_39 on page 883	0x4c8	IRQ_TYPE_EN_RX_SA_39
IRQ_TYPE_STAT_RX_SA_39 on page 886	0x4cc	IRQ_TYPE_STAT_RX_SA_39
IRQ_TYPE_EN_RX_SA_40 on page 889	0x4d0	IRQ_TYPE_EN_RX_SA_40
IRQ_TYPE_STAT_RX_SA_40 on page 892	0x4d4	IRQ_TYPE_STAT_RX_SA_40

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_41 on page 895	0x4d8	IRQ_TYPE_EN_RX_SA_41
IRQ_TYPE_STAT_RX_SA_41 on page 898	0x4dc	IRQ_TYPE_STAT_RX_SA_41
IRQ_TYPE_EN_RX_SA_42 on page 901	0x4e0	IRQ_TYPE_EN_RX_SA_42
IRQ_TYPE_STAT_RX_SA_42 on page 904	0x4e4	IRQ_TYPE_STAT_RX_SA_42
IRQ_TYPE_EN_RX_SA_43 on page 907	0x4e8	IRQ_TYPE_EN_RX_SA_43
IRQ_TYPE_STAT_RX_SA_43 on page 910	0x4ec	IRQ_TYPE_STAT_RX_SA_43
IRQ_TYPE_EN_RX_SA_44 on page 913	0x4f0	IRQ_TYPE_EN_RX_SA_44
IRQ_TYPE_STAT_RX_SA_44 on page 916	0x4f4	IRQ_TYPE_STAT_RX_SA_44
IRQ_TYPE_EN_RX_SA_45 on page 919	0x4f8	IRQ_TYPE_EN_RX_SA_45
IRQ_TYPE_STAT_RX_SA_45 on page 922	0x4fc	IRQ_TYPE_STAT_RX_SA_45
IRQ_TYPE_EN_RX_SA_46 on page 925	0x500	IRQ_TYPE_EN_RX_SA_46
IRQ_TYPE_STAT_RX_SA_46 on page 928	0x504	IRQ_TYPE_STAT_RX_SA_46
IRQ_TYPE_EN_RX_SA_47 on page 931	0x508	IRQ_TYPE_EN_RX_SA_47
IRQ_TYPE_STAT_RX_SA_47 on page 934	0x50c	IRQ_TYPE_STAT_RX_SA_47
IRQ_TYPE_EN_RX_SA_48 on page 937	0x510	IRQ_TYPE_EN_RX_SA_48
IRQ_TYPE_STAT_RX_SA_48 on page 940	0x514	IRQ_TYPE_STAT_RX_SA_48
IRQ_TYPE_EN_RX_SA_49 on page 943	0x518	IRQ_TYPE_EN_RX_SA_49
IRQ_TYPE_STAT_RX_SA_49 on page 946	0x51c	IRQ_TYPE_STAT_RX_SA_49
IRQ_TYPE_EN_RX_SA_50 on page 949	0x520	IRQ_TYPE_EN_RX_SA_50
IRQ_TYPE_STAT_RX_SA_50 on page 952	0x524	IRQ_TYPE_STAT_RX_SA_50
IRQ_TYPE_EN_RX_SA_51 on page 955	0x528	IRQ_TYPE_EN_RX_SA_51
IRQ_TYPE_STAT_RX_SA_51 on page 958	0x52c	IRQ_TYPE_STAT_RX_SA_51

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_52 on page 961	0x530	IRQ_TYPE_EN_RX_SA_52
IRQ_TYPE_STAT_RX_SA_52 on page 964	0x534	IRQ_TYPE_STAT_RX_SA_52
IRQ_TYPE_EN_RX_SA_53 on page 967	0x538	IRQ_TYPE_EN_RX_SA_53
IRQ_TYPE_STAT_RX_SA_53 on page 970	0x53c	IRQ_TYPE_STAT_RX_SA_53
IRQ_TYPE_EN_RX_SA_54 on page 973	0x540	IRQ_TYPE_EN_RX_SA_54
IRQ_TYPE_STAT_RX_SA_54 on page 976	0x544	IRQ_TYPE_STAT_RX_SA_54
IRQ_TYPE_EN_RX_SA_55 on page 979	0x548	IRQ_TYPE_EN_RX_SA_55
IRQ_TYPE_STAT_RX_SA_55 on page 982	0x54c	IRQ_TYPE_STAT_RX_SA_55
IRQ_TYPE_EN_RX_SA_56 on page 985	0x550	IRQ_TYPE_EN_RX_SA_56
IRQ_TYPE_STAT_RX_SA_56 on page 988	0x554	IRQ_TYPE_STAT_RX_SA_56
IRQ_TYPE_EN_RX_SA_57 on page 991	0x558	IRQ_TYPE_EN_RX_SA_57
IRQ_TYPE_STAT_RX_SA_57 on page 994	0x55c	IRQ_TYPE_STAT_RX_SA_57
IRQ_TYPE_EN_RX_SA_58 on page 997	0x560	IRQ_TYPE_EN_RX_SA_58
IRQ_TYPE_STAT_RX_SA_58 on page 1000	0x564	IRQ_TYPE_STAT_RX_SA_58
IRQ_TYPE_EN_RX_SA_59 on page 1003	0x568	IRQ_TYPE_EN_RX_SA_59
IRQ_TYPE_STAT_RX_SA_59 on page 1006	0x56c	IRQ_TYPE_STAT_RX_SA_59
IRQ_TYPE_EN_RX_SA_60 on page 1009	0x570	IRQ_TYPE_EN_RX_SA_60
IRQ_TYPE_STAT_RX_SA_60 on page 1012	0x574	IRQ_TYPE_STAT_RX_SA_60
IRQ_TYPE_EN_RX_SA_61 on page 1015	0x578	IRQ_TYPE_EN_RX_SA_61
IRQ_TYPE_STAT_RX_SA_61 on page 1018	0x57c	IRQ_TYPE_STAT_RX_SA_61
IRQ_TYPE_EN_RX_SA_62 on page 1021	0x580	IRQ_TYPE_EN_RX_SA_62
IRQ_TYPE_STAT_RX_SA_62 on page 1024	0x584	IRQ_TYPE_STAT_RX_SA_62

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_63 on page 1027	0x588	IRQ_TYPE_EN_RX_SA_63
IRQ_TYPE_STAT_RX_SA_63 on page 1030	0x58c	IRQ_TYPE_STAT_RX_SA_63
IRQ_TYPE_EN_RX_SA_64 on page 1033	0x590	IRQ_TYPE_EN_RX_SA_64
IRQ_TYPE_STAT_RX_SA_64 on page 1036	0x594	IRQ_TYPE_STAT_RX_SA_64
IRQ_TYPE_EN_RX_SA_65 on page 1039	0x598	IRQ_TYPE_EN_RX_SA_65
IRQ_TYPE_STAT_RX_SA_65 on page 1042	0x59c	IRQ_TYPE_STAT_RX_SA_65
IRQ_TYPE_EN_RX_SA_66 on page 1045	0x5a0	IRQ_TYPE_EN_RX_SA_66
IRQ_TYPE_STAT_RX_SA_66 on page 1048	0x5a4	IRQ_TYPE_STAT_RX_SA_66
IRQ_TYPE_EN_RX_SA_67 on page 1051	0x5a8	IRQ_TYPE_EN_RX_SA_67
IRQ_TYPE_STAT_RX_SA_67 on page 1054	0x5ac	IRQ_TYPE_STAT_RX_SA_67
IRQ_TYPE_EN_RX_SA_68 on page 1057	0x5b0	IRQ_TYPE_EN_RX_SA_68
IRQ_TYPE_STAT_RX_SA_68 on page 1060	0x5b4	IRQ_TYPE_STAT_RX_SA_68
IRQ_TYPE_EN_RX_SA_69 on page 1063	0x5b8	IRQ_TYPE_EN_RX_SA_69
IRQ_TYPE_STAT_RX_SA_69 on page 1066	0x5bc	IRQ_TYPE_STAT_RX_SA_69
IRQ_TYPE_EN_RX_SA_70 on page 1069	0x5c0	IRQ_TYPE_EN_RX_SA_70
IRQ_TYPE_STAT_RX_SA_70 on page 1072	0x5c4	IRQ_TYPE_STAT_RX_SA_70
IRQ_TYPE_EN_RX_SA_71 on page 1075	0x5c8	IRQ_TYPE_EN_RX_SA_71
IRQ_TYPE_STAT_RX_SA_71 on page 1078	0x5cc	IRQ_TYPE_STAT_RX_SA_71
IRQ_TYPE_EN_RX_SA_72 on page 1081	0x5d0	IRQ_TYPE_EN_RX_SA_72
IRQ_TYPE_STAT_RX_SA_72 on page 1084	0x5d4	IRQ_TYPE_STAT_RX_SA_72
IRQ_TYPE_EN_RX_SA_73 on page 1087	0x5d8	IRQ_TYPE_EN_RX_SA_73
IRQ_TYPE_STAT_RX_SA_73 on page 1090	0x5dc	IRQ_TYPE_STAT_RX_SA_73

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_74 on page 1093	0x5e0	IRQ_TYPE_EN_RX_SA_74
IRQ_TYPE_STAT_RX_SA_74 on page 1096	0x5e4	IRQ_TYPE_STAT_RX_SA_74
IRQ_TYPE_EN_RX_SA_75 on page 1099	0x5e8	IRQ_TYPE_EN_RX_SA_75
IRQ_TYPE_STAT_RX_SA_75 on page 1102	0x5ec	IRQ_TYPE_STAT_RX_SA_75
IRQ_TYPE_EN_RX_SA_76 on page 1105	0x5f0	IRQ_TYPE_EN_RX_SA_76
IRQ_TYPE_STAT_RX_SA_76 on page 1108	0x5f4	IRQ_TYPE_STAT_RX_SA_76
IRQ_TYPE_EN_RX_SA_77 on page 1111	0x5f8	IRQ_TYPE_EN_RX_SA_77
IRQ_TYPE_STAT_RX_SA_77 on page 1114	0x5fc	IRQ_TYPE_STAT_RX_SA_77
IRQ_TYPE_EN_RX_SA_78 on page 1117	0x600	IRQ_TYPE_EN_RX_SA_78
IRQ_TYPE_STAT_RX_SA_78 on page 1120	0x604	IRQ_TYPE_STAT_RX_SA_78
IRQ_TYPE_EN_RX_SA_79 on page 1123	0x608	IRQ_TYPE_EN_RX_SA_79
IRQ_TYPE_STAT_RX_SA_79 on page 1126	0x60c	IRQ_TYPE_STAT_RX_SA_79
IRQ_TYPE_EN_RX_SA_80 on page 1129	0x610	IRQ_TYPE_EN_RX_SA_80
IRQ_TYPE_STAT_RX_SA_80 on page 1132	0x614	IRQ_TYPE_STAT_RX_SA_80
IRQ_TYPE_EN_RX_SA_81 on page 1135	0x618	IRQ_TYPE_EN_RX_SA_81
IRQ_TYPE_STAT_RX_SA_81 on page 1138	0x61c	IRQ_TYPE_STAT_RX_SA_81
IRQ_TYPE_EN_RX_SA_82 on page 1141	0x620	IRQ_TYPE_EN_RX_SA_82
IRQ_TYPE_STAT_RX_SA_82 on page 1144	0x624	IRQ_TYPE_STAT_RX_SA_82
IRQ_TYPE_EN_RX_SA_83 on page 1147	0x628	IRQ_TYPE_EN_RX_SA_83
IRQ_TYPE_STAT_RX_SA_83 on page 1150	0x62c	IRQ_TYPE_STAT_RX_SA_83
IRQ_TYPE_EN_RX_SA_84 on page 1153	0x630	IRQ_TYPE_EN_RX_SA_84
IRQ_TYPE_STAT_RX_SA_84 on page 1156	0x634	IRQ_TYPE_STAT_RX_SA_84

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_85 on page 1159	0x638	IRQ_TYPE_EN_RX_SA_85
IRQ_TYPE_STAT_RX_SA_85 on page 1162	0x63c	IRQ_TYPE_STAT_RX_SA_85
IRQ_TYPE_EN_RX_SA_86 on page 1165	0x640	IRQ_TYPE_EN_RX_SA_86
IRQ_TYPE_STAT_RX_SA_86 on page 1168	0x644	IRQ_TYPE_STAT_RX_SA_86
IRQ_TYPE_EN_RX_SA_87 on page 1171	0x648	IRQ_TYPE_EN_RX_SA_87
IRQ_TYPE_STAT_RX_SA_87 on page 1174	0x64c	IRQ_TYPE_STAT_RX_SA_87
IRQ_TYPE_EN_RX_SA_88 on page 1177	0x650	IRQ_TYPE_EN_RX_SA_88
IRQ_TYPE_STAT_RX_SA_88 on page 1180	0x654	IRQ_TYPE_STAT_RX_SA_88
IRQ_TYPE_EN_RX_SA_89 on page 1183	0x658	IRQ_TYPE_EN_RX_SA_89
IRQ_TYPE_STAT_RX_SA_89 on page 1186	0x65c	IRQ_TYPE_STAT_RX_SA_89
IRQ_TYPE_EN_RX_SA_90 on page 1189	0x660	IRQ_TYPE_EN_RX_SA_90
IRQ_TYPE_STAT_RX_SA_90 on page 1192	0x664	IRQ_TYPE_STAT_RX_SA_90
IRQ_TYPE_EN_RX_SA_91 on page 1195	0x668	IRQ_TYPE_EN_RX_SA_91
IRQ_TYPE_STAT_RX_SA_91 on page 1198	0x66c	IRQ_TYPE_STAT_RX_SA_91
IRQ_TYPE_EN_RX_SA_92 on page 1201	0x670	IRQ_TYPE_EN_RX_SA_92
IRQ_TYPE_STAT_RX_SA_92 on page 1204	0x674	IRQ_TYPE_STAT_RX_SA_92
IRQ_TYPE_EN_RX_SA_93 on page 1207	0x678	IRQ_TYPE_EN_RX_SA_93
IRQ_TYPE_STAT_RX_SA_93 on page 1210	0x67c	IRQ_TYPE_STAT_RX_SA_93
IRQ_TYPE_EN_RX_SA_94 on page 1213	0x680	IRQ_TYPE_EN_RX_SA_94
IRQ_TYPE_STAT_RX_SA_94 on page 1216	0x684	IRQ_TYPE_STAT_RX_SA_94
IRQ_TYPE_EN_RX_SA_95 on page 1219	0x688	IRQ_TYPE_EN_RX_SA_95
IRQ_TYPE_STAT_RX_SA_95 on page 1222	0x68c	IRQ_TYPE_STAT_RX_SA_95

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_EN_RX_SA_96 on page 1225	0x690	IRQ_TYPE_EN_RX_SA_96
IRQ_TYPE_STAT_RX_SA_96 on page 1228	0x694	IRQ_TYPE_STAT_RX_SA_96
IRQ_TYPE_EN_RX_SA_97 on page 1231	0x698	IRQ_TYPE_EN_RX_SA_97
IRQ_TYPE_STAT_RX_SA_97 on page 1234	0x69c	IRQ_TYPE_STAT_RX_SA_97
IRQ_TYPE_EN_RX_SA_98 on page 1237	0x6a0	IRQ_TYPE_EN_RX_SA_98
IRQ_TYPE_STAT_RX_SA_98 on page 1240	0x6a4	IRQ_TYPE_STAT_RX_SA_98
IRQ_TYPE_EN_RX_SA_99 on page 1243	0x6a8	IRQ_TYPE_EN_RX_SA_99
IRQ_TYPE_STAT_RX_SA_99 on page 1246	0x6ac	IRQ_TYPE_STAT_RX_SA_99
IRQ_TYPE_EN_RX_SA_100 on page 1249	0x6b0	IRQ_TYPE_EN_RX_SA_100
IRQ_TYPE_STAT_RX_SA_100 on page 1252	0x6b4	IRQ_TYPE_STAT_RX_SA_100
IRQ_TYPE_EN_RX_SA_101 on page 1255	0x6b8	IRQ_TYPE_EN_RX_SA_101
IRQ_TYPE_STAT_RX_SA_101 on page 1258	0x6bc	IRQ_TYPE_STAT_RX_SA_101
IRQ_TYPE_EN_RX_SA_102 on page 1261	0x6c0	IRQ_TYPE_EN_RX_SA_102
IRQ_TYPE_STAT_RX_SA_102 on page 1264	0x6c4	IRQ_TYPE_STAT_RX_SA_102
IRQ_TYPE_EN_RX_SA_103 on page 1267	0x6c8	IRQ_TYPE_EN_RX_SA_103
IRQ_TYPE_STAT_RX_SA_103 on page 1270	0x6cc	IRQ_TYPE_STAT_RX_SA_103
IRQ_TYPE_EN_RX_SA_104 on page 1273	0x6d0	IRQ_TYPE_EN_RX_SA_104
IRQ_TYPE_STAT_RX_SA_104 on page 1276	0x6d4	IRQ_TYPE_STAT_RX_SA_104
IRQ_TYPE_EN_RX_SA_105 on page 1279	0x6d8	IRQ_TYPE_EN_RX_SA_105

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_105 on page 1282	0x6dc	IRQ_TYPE_STAT_RX_SA_105
IRQ_TYPE_EN_RX_SA_106 on page 1285	0x6e0	IRQ_TYPE_EN_RX_SA_106
IRQ_TYPE_STAT_RX_SA_106 on page 1288	0x6e4	IRQ_TYPE_STAT_RX_SA_106
IRQ_TYPE_EN_RX_SA_107 on page 1291	0x6e8	IRQ_TYPE_EN_RX_SA_107
IRQ_TYPE_STAT_RX_SA_107 on page 1294	0x6ec	IRQ_TYPE_STAT_RX_SA_107
IRQ_TYPE_EN_RX_SA_108 on page 1297	0x6f0	IRQ_TYPE_EN_RX_SA_108
IRQ_TYPE_STAT_RX_SA_108 on page 1300	0x6f4	IRQ_TYPE_STAT_RX_SA_108
IRQ_TYPE_EN_RX_SA_109 on page 1303	0x6f8	IRQ_TYPE_EN_RX_SA_109
IRQ_TYPE_STAT_RX_SA_109 on page 1306	0x6fc	IRQ_TYPE_STAT_RX_SA_109
IRQ_TYPE_EN_RX_SA_110 on page 1309	0x700	IRQ_TYPE_EN_RX_SA_110
IRQ_TYPE_STAT_RX_SA_110 on page 1312	0x704	IRQ_TYPE_STAT_RX_SA_110
IRQ_TYPE_EN_RX_SA_111 on page 1315	0x708	IRQ_TYPE_EN_RX_SA_111
IRQ_TYPE_STAT_RX_SA_111 on page 1318	0x70c	IRQ_TYPE_STAT_RX_SA_111
IRQ_TYPE_EN_RX_SA_112 on page 1321	0x710	IRQ_TYPE_EN_RX_SA_112
IRQ_TYPE_STAT_RX_SA_112 on page 1324	0x714	IRQ_TYPE_STAT_RX_SA_112
IRQ_TYPE_EN_RX_SA_113 on page 1327	0x718	IRQ_TYPE_EN_RX_SA_113
IRQ_TYPE_STAT_RX_SA_113 on page 1330	0x71c	IRQ_TYPE_STAT_RX_SA_113
IRQ_TYPE_EN_RX_SA_114 on page 1333	0x720	IRQ_TYPE_EN_RX_SA_114

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_114 on page 1336	0x724	IRQ_TYPE_STAT_RX_SA_114
IRQ_TYPE_EN_RX_SA_115 on page 1339	0x728	IRQ_TYPE_EN_RX_SA_115
IRQ_TYPE_STAT_RX_SA_115 on page 1342	0x72c	IRQ_TYPE_STAT_RX_SA_115
IRQ_TYPE_EN_RX_SA_116 on page 1345	0x730	IRQ_TYPE_EN_RX_SA_116
IRQ_TYPE_STAT_RX_SA_116 on page 1348	0x734	IRQ_TYPE_STAT_RX_SA_116
IRQ_TYPE_EN_RX_SA_117 on page 1351	0x738	IRQ_TYPE_EN_RX_SA_117
IRQ_TYPE_STAT_RX_SA_117 on page 1354	0x73c	IRQ_TYPE_STAT_RX_SA_117
IRQ_TYPE_EN_RX_SA_118 on page 1357	0x740	IRQ_TYPE_EN_RX_SA_118
IRQ_TYPE_STAT_RX_SA_118 on page 1360	0x744	IRQ_TYPE_STAT_RX_SA_118
IRQ_TYPE_EN_RX_SA_119 on page 1363	0x748	IRQ_TYPE_EN_RX_SA_119
IRQ_TYPE_STAT_RX_SA_119 on page 1366	0x74c	IRQ_TYPE_STAT_RX_SA_119
IRQ_TYPE_EN_RX_SA_120 on page 1369	0x750	IRQ_TYPE_EN_RX_SA_120
IRQ_TYPE_STAT_RX_SA_120 on page 1372	0x754	IRQ_TYPE_STAT_RX_SA_120
IRQ_TYPE_EN_RX_SA_121 on page 1375	0x758	IRQ_TYPE_EN_RX_SA_121
IRQ_TYPE_STAT_RX_SA_121 on page 1378	0x75c	IRQ_TYPE_STAT_RX_SA_121
IRQ_TYPE_EN_RX_SA_122 on page 1381	0x760	IRQ_TYPE_EN_RX_SA_122
IRQ_TYPE_STAT_RX_SA_122 on page 1384	0x764	IRQ_TYPE_STAT_RX_SA_122
IRQ_TYPE_EN_RX_SA_123 on page 1387	0x768	IRQ_TYPE_EN_RX_SA_123

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_123 on page 1390	0x76c	IRQ_TYPE_STAT_RX_SA_123
IRQ_TYPE_EN_RX_SA_124 on page 1393	0x770	IRQ_TYPE_EN_RX_SA_124
IRQ_TYPE_STAT_RX_SA_124 on page 1396	0x774	IRQ_TYPE_STAT_RX_SA_124
IRQ_TYPE_EN_RX_SA_125 on page 1399	0x778	IRQ_TYPE_EN_RX_SA_125
IRQ_TYPE_STAT_RX_SA_125 on page 1402	0x77c	IRQ_TYPE_STAT_RX_SA_125
IRQ_TYPE_EN_RX_SA_126 on page 1405	0x780	IRQ_TYPE_EN_RX_SA_126
IRQ_TYPE_STAT_RX_SA_126 on page 1408	0x784	IRQ_TYPE_STAT_RX_SA_126
IRQ_TYPE_EN_RX_SA_127 on page 1411	0x788	IRQ_TYPE_EN_RX_SA_127
IRQ_TYPE_STAT_RX_SA_127 on page 1414	0x78c	IRQ_TYPE_STAT_RX_SA_127
IRQ_TYPE_EN_RX_SA_128 on page 1417	0x790	IRQ_TYPE_EN_RX_SA_128
IRQ_TYPE_STAT_RX_SA_128 on page 1420	0x794	IRQ_TYPE_STAT_RX_SA_128
IRQ_TYPE_EN_RX_SA_129 on page 1423	0x798	IRQ_TYPE_EN_RX_SA_129
IRQ_TYPE_STAT_RX_SA_129 on page 1426	0x79c	IRQ_TYPE_STAT_RX_SA_129
IRQ_TYPE_EN_RX_SA_130 on page 1429	0x7a0	IRQ_TYPE_EN_RX_SA_130
IRQ_TYPE_STAT_RX_SA_130 on page 1432	0x7a4	IRQ_TYPE_STAT_RX_SA_130
IRQ_TYPE_EN_RX_SA_131 on page 1435	0x7a8	IRQ_TYPE_EN_RX_SA_131
IRQ_TYPE_STAT_RX_SA_131 on page 1438	0x7ac	IRQ_TYPE_STAT_RX_SA_131
IRQ_TYPE_EN_RX_SA_132 on page 1441	0x7b0	IRQ_TYPE_EN_RX_SA_132

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_132 on page 1444	0x7b4	IRQ_TYPE_STAT_RX_SA_132
IRQ_TYPE_EN_RX_SA_133 on page 1447	0x7b8	IRQ_TYPE_EN_RX_SA_133
IRQ_TYPE_STAT_RX_SA_133 on page 1450	0x7bc	IRQ_TYPE_STAT_RX_SA_133
IRQ_TYPE_EN_RX_SA_134 on page 1453	0x7c0	IRQ_TYPE_EN_RX_SA_134
IRQ_TYPE_STAT_RX_SA_134 on page 1456	0x7c4	IRQ_TYPE_STAT_RX_SA_134
IRQ_TYPE_EN_RX_SA_135 on page 1459	0x7c8	IRQ_TYPE_EN_RX_SA_135
IRQ_TYPE_STAT_RX_SA_135 on page 1462	0x7cc	IRQ_TYPE_STAT_RX_SA_135
IRQ_TYPE_EN_RX_SA_136 on page 1465	0x7d0	IRQ_TYPE_EN_RX_SA_136
IRQ_TYPE_STAT_RX_SA_136 on page 1468	0x7d4	IRQ_TYPE_STAT_RX_SA_136
IRQ_TYPE_EN_RX_SA_137 on page 1471	0x7d8	IRQ_TYPE_EN_RX_SA_137
IRQ_TYPE_STAT_RX_SA_137 on page 1474	0x7dc	IRQ_TYPE_STAT_RX_SA_137
IRQ_TYPE_EN_RX_SA_138 on page 1477	0x7e0	IRQ_TYPE_EN_RX_SA_138
IRQ_TYPE_STAT_RX_SA_138 on page 1480	0x7e4	IRQ_TYPE_STAT_RX_SA_138
IRQ_TYPE_EN_RX_SA_139 on page 1483	0x7e8	IRQ_TYPE_EN_RX_SA_139
IRQ_TYPE_STAT_RX_SA_139 on page 1486	0x7ec	IRQ_TYPE_STAT_RX_SA_139
IRQ_TYPE_EN_RX_SA_140 on page 1489	0x7f0	IRQ_TYPE_EN_RX_SA_140
IRQ_TYPE_STAT_RX_SA_140 on page 1492	0x7f4	IRQ_TYPE_STAT_RX_SA_140
IRQ_TYPE_EN_RX_SA_141 on page 1495	0x7f8	IRQ_TYPE_EN_RX_SA_141

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_141 on page 1498	0x7fc	IRQ_TYPE_STAT_RX_SA_141
IRQ_TYPE_EN_RX_SA_142 on page 1501	0x800	IRQ_TYPE_EN_RX_SA_142
IRQ_TYPE_STAT_RX_SA_142 on page 1504	0x804	IRQ_TYPE_STAT_RX_SA_142
IRQ_TYPE_EN_RX_SA_143 on page 1507	0x808	IRQ_TYPE_EN_RX_SA_143
IRQ_TYPE_STAT_RX_SA_143 on page 1510	0x80c	IRQ_TYPE_STAT_RX_SA_143
IRQ_TYPE_EN_RX_SA_144 on page 1513	0x810	IRQ_TYPE_EN_RX_SA_144
IRQ_TYPE_STAT_RX_SA_144 on page 1516	0x814	IRQ_TYPE_STAT_RX_SA_144
IRQ_TYPE_EN_RX_SA_145 on page 1519	0x818	IRQ_TYPE_EN_RX_SA_145
IRQ_TYPE_STAT_RX_SA_145 on page 1522	0x81c	IRQ_TYPE_STAT_RX_SA_145
IRQ_TYPE_EN_RX_SA_146 on page 1525	0x820	IRQ_TYPE_EN_RX_SA_146
IRQ_TYPE_STAT_RX_SA_146 on page 1528	0x824	IRQ_TYPE_STAT_RX_SA_146
IRQ_TYPE_EN_RX_SA_147 on page 1531	0x828	IRQ_TYPE_EN_RX_SA_147
IRQ_TYPE_STAT_RX_SA_147 on page 1534	0x82c	IRQ_TYPE_STAT_RX_SA_147
IRQ_TYPE_EN_RX_SA_148 on page 1537	0x830	IRQ_TYPE_EN_RX_SA_148
IRQ_TYPE_STAT_RX_SA_148 on page 1540	0x834	IRQ_TYPE_STAT_RX_SA_148
IRQ_TYPE_EN_RX_SA_149 on page 1543	0x838	IRQ_TYPE_EN_RX_SA_149
IRQ_TYPE_STAT_RX_SA_149 on page 1546	0x83c	IRQ_TYPE_STAT_RX_SA_149
IRQ_TYPE_EN_RX_SA_150 on page 1549	0x840	IRQ_TYPE_EN_RX_SA_150

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_150 on page 1552	0x844	IRQ_TYPE_STAT_RX_SA_150
IRQ_TYPE_EN_RX_SA_151 on page 1555	0x848	IRQ_TYPE_EN_RX_SA_151
IRQ_TYPE_STAT_RX_SA_151 on page 1558	0x84c	IRQ_TYPE_STAT_RX_SA_151
IRQ_TYPE_EN_RX_SA_152 on page 1561	0x850	IRQ_TYPE_EN_RX_SA_152
IRQ_TYPE_STAT_RX_SA_152 on page 1564	0x854	IRQ_TYPE_STAT_RX_SA_152
IRQ_TYPE_EN_RX_SA_153 on page 1567	0x858	IRQ_TYPE_EN_RX_SA_153
IRQ_TYPE_STAT_RX_SA_153 on page 1570	0x85c	IRQ_TYPE_STAT_RX_SA_153
IRQ_TYPE_EN_RX_SA_154 on page 1573	0x860	IRQ_TYPE_EN_RX_SA_154
IRQ_TYPE_STAT_RX_SA_154 on page 1576	0x864	IRQ_TYPE_STAT_RX_SA_154
IRQ_TYPE_EN_RX_SA_155 on page 1579	0x868	IRQ_TYPE_EN_RX_SA_155
IRQ_TYPE_STAT_RX_SA_155 on page 1582	0x86c	IRQ_TYPE_STAT_RX_SA_155
IRQ_TYPE_EN_RX_SA_156 on page 1585	0x870	IRQ_TYPE_EN_RX_SA_156
IRQ_TYPE_STAT_RX_SA_156 on page 1588	0x874	IRQ_TYPE_STAT_RX_SA_156
IRQ_TYPE_EN_RX_SA_157 on page 1591	0x878	IRQ_TYPE_EN_RX_SA_157
IRQ_TYPE_STAT_RX_SA_157 on page 1594	0x87c	IRQ_TYPE_STAT_RX_SA_157
IRQ_TYPE_EN_RX_SA_158 on page 1597	0x880	IRQ_TYPE_EN_RX_SA_158
IRQ_TYPE_STAT_RX_SA_158 on page 1600	0x884	IRQ_TYPE_STAT_RX_SA_158
IRQ_TYPE_EN_RX_SA_159 on page 1603	0x888	IRQ_TYPE_EN_RX_SA_159

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_159 on page 1606	0x88c	IRQ_TYPE_STAT_RX_SA_159
IRQ_TYPE_EN_RX_SA_160 on page 1609	0x890	IRQ_TYPE_EN_RX_SA_160
IRQ_TYPE_STAT_RX_SA_160 on page 1612	0x894	IRQ_TYPE_STAT_RX_SA_160
IRQ_TYPE_EN_RX_SA_161 on page 1615	0x898	IRQ_TYPE_EN_RX_SA_161
IRQ_TYPE_STAT_RX_SA_161 on page 1618	0x89c	IRQ_TYPE_STAT_RX_SA_161
IRQ_TYPE_EN_RX_SA_162 on page 1621	0x8a0	IRQ_TYPE_EN_RX_SA_162
IRQ_TYPE_STAT_RX_SA_162 on page 1624	0x8a4	IRQ_TYPE_STAT_RX_SA_162
IRQ_TYPE_EN_RX_SA_163 on page 1627	0x8a8	IRQ_TYPE_EN_RX_SA_163
IRQ_TYPE_STAT_RX_SA_163 on page 1630	0x8ac	IRQ_TYPE_STAT_RX_SA_163
IRQ_TYPE_EN_RX_SA_164 on page 1633	0x8b0	IRQ_TYPE_EN_RX_SA_164
IRQ_TYPE_STAT_RX_SA_164 on page 1636	0x8b4	IRQ_TYPE_STAT_RX_SA_164
IRQ_TYPE_EN_RX_SA_165 on page 1639	0x8b8	IRQ_TYPE_EN_RX_SA_165
IRQ_TYPE_STAT_RX_SA_165 on page 1642	0x8bc	IRQ_TYPE_STAT_RX_SA_165
IRQ_TYPE_EN_RX_SA_166 on page 1645	0x8c0	IRQ_TYPE_EN_RX_SA_166
IRQ_TYPE_STAT_RX_SA_166 on page 1648	0x8c4	IRQ_TYPE_STAT_RX_SA_166
IRQ_TYPE_EN_RX_SA_167 on page 1651	0x8c8	IRQ_TYPE_EN_RX_SA_167
IRQ_TYPE_STAT_RX_SA_167 on page 1654	0x8cc	IRQ_TYPE_STAT_RX_SA_167
IRQ_TYPE_EN_RX_SA_168 on page 1657	0x8d0	IRQ_TYPE_EN_RX_SA_168

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_168 on page 1660	0x8d4	IRQ_TYPE_STAT_RX_SA_168
IRQ_TYPE_EN_RX_SA_169 on page 1663	0x8d8	IRQ_TYPE_EN_RX_SA_169
IRQ_TYPE_STAT_RX_SA_169 on page 1666	0x8dc	IRQ_TYPE_STAT_RX_SA_169
IRQ_TYPE_EN_RX_SA_170 on page 1669	0x8e0	IRQ_TYPE_EN_RX_SA_170
IRQ_TYPE_STAT_RX_SA_170 on page 1672	0x8e4	IRQ_TYPE_STAT_RX_SA_170
IRQ_TYPE_EN_RX_SA_171 on page 1675	0x8e8	IRQ_TYPE_EN_RX_SA_171
IRQ_TYPE_STAT_RX_SA_171 on page 1678	0x8ec	IRQ_TYPE_STAT_RX_SA_171
IRQ_TYPE_EN_RX_SA_172 on page 1681	0x8f0	IRQ_TYPE_EN_RX_SA_172
IRQ_TYPE_STAT_RX_SA_172 on page 1684	0x8f4	IRQ_TYPE_STAT_RX_SA_172
IRQ_TYPE_EN_RX_SA_173 on page 1687	0x8f8	IRQ_TYPE_EN_RX_SA_173
IRQ_TYPE_STAT_RX_SA_173 on page 1690	0x8fc	IRQ_TYPE_STAT_RX_SA_173
IRQ_TYPE_EN_RX_SA_174 on page 1693	0x900	IRQ_TYPE_EN_RX_SA_174
IRQ_TYPE_STAT_RX_SA_174 on page 1696	0x904	IRQ_TYPE_STAT_RX_SA_174
IRQ_TYPE_EN_RX_SA_175 on page 1699	0x908	IRQ_TYPE_EN_RX_SA_175
IRQ_TYPE_STAT_RX_SA_175 on page 1702	0x90c	IRQ_TYPE_STAT_RX_SA_175
IRQ_TYPE_EN_RX_SA_176 on page 1705	0x910	IRQ_TYPE_EN_RX_SA_176
IRQ_TYPE_STAT_RX_SA_176 on page 1708	0x914	IRQ_TYPE_STAT_RX_SA_176
IRQ_TYPE_EN_RX_SA_177 on page 1711	0x918	IRQ_TYPE_EN_RX_SA_177

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_177 on page 1714	0x91c	IRQ_TYPE_STAT_RX_SA_177
IRQ_TYPE_EN_RX_SA_178 on page 1717	0x920	IRQ_TYPE_EN_RX_SA_178
IRQ_TYPE_STAT_RX_SA_178 on page 1720	0x924	IRQ_TYPE_STAT_RX_SA_178
IRQ_TYPE_EN_RX_SA_179 on page 1723	0x928	IRQ_TYPE_EN_RX_SA_179
IRQ_TYPE_STAT_RX_SA_179 on page 1726	0x92c	IRQ_TYPE_STAT_RX_SA_179
IRQ_TYPE_EN_RX_SA_180 on page 1729	0x930	IRQ_TYPE_EN_RX_SA_180
IRQ_TYPE_STAT_RX_SA_180 on page 1732	0x934	IRQ_TYPE_STAT_RX_SA_180
IRQ_TYPE_EN_RX_SA_181 on page 1735	0x938	IRQ_TYPE_EN_RX_SA_181
IRQ_TYPE_STAT_RX_SA_181 on page 1738	0x93c	IRQ_TYPE_STAT_RX_SA_181
IRQ_TYPE_EN_RX_SA_182 on page 1741	0x940	IRQ_TYPE_EN_RX_SA_182
IRQ_TYPE_STAT_RX_SA_182 on page 1744	0x944	IRQ_TYPE_STAT_RX_SA_182
IRQ_TYPE_EN_RX_SA_183 on page 1747	0x948	IRQ_TYPE_EN_RX_SA_183
IRQ_TYPE_STAT_RX_SA_183 on page 1750	0x94c	IRQ_TYPE_STAT_RX_SA_183
IRQ_TYPE_EN_RX_SA_184 on page 1753	0x950	IRQ_TYPE_EN_RX_SA_184
IRQ_TYPE_STAT_RX_SA_184 on page 1756	0x954	IRQ_TYPE_STAT_RX_SA_184
IRQ_TYPE_EN_RX_SA_185 on page 1759	0x958	IRQ_TYPE_EN_RX_SA_185
IRQ_TYPE_STAT_RX_SA_185 on page 1762	0x95c	IRQ_TYPE_STAT_RX_SA_185
IRQ_TYPE_EN_RX_SA_186 on page 1765	0x960	IRQ_TYPE_EN_RX_SA_186

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_186 on page 1768	0x964	IRQ_TYPE_STAT_RX_SA_186
IRQ_TYPE_EN_RX_SA_187 on page 1771	0x968	IRQ_TYPE_EN_RX_SA_187
IRQ_TYPE_STAT_RX_SA_187 on page 1774	0x96c	IRQ_TYPE_STAT_RX_SA_187
IRQ_TYPE_EN_RX_SA_188 on page 1777	0x970	IRQ_TYPE_EN_RX_SA_188
IRQ_TYPE_STAT_RX_SA_188 on page 1780	0x974	IRQ_TYPE_STAT_RX_SA_188
IRQ_TYPE_EN_RX_SA_189 on page 1783	0x978	IRQ_TYPE_EN_RX_SA_189
IRQ_TYPE_STAT_RX_SA_189 on page 1786	0x97c	IRQ_TYPE_STAT_RX_SA_189
IRQ_TYPE_EN_RX_SA_190 on page 1789	0x980	IRQ_TYPE_EN_RX_SA_190
IRQ_TYPE_STAT_RX_SA_190 on page 1792	0x984	IRQ_TYPE_STAT_RX_SA_190
IRQ_TYPE_EN_RX_SA_191 on page 1795	0x988	IRQ_TYPE_EN_RX_SA_191
IRQ_TYPE_STAT_RX_SA_191 on page 1798	0x98c	IRQ_TYPE_STAT_RX_SA_191
IRQ_TYPE_EN_RX_SA_192 on page 1801	0x990	IRQ_TYPE_EN_RX_SA_192
IRQ_TYPE_STAT_RX_SA_192 on page 1804	0x994	IRQ_TYPE_STAT_RX_SA_192
IRQ_TYPE_EN_RX_SA_193 on page 1807	0x998	IRQ_TYPE_EN_RX_SA_193
IRQ_TYPE_STAT_RX_SA_193 on page 1810	0x99c	IRQ_TYPE_STAT_RX_SA_193
IRQ_TYPE_EN_RX_SA_194 on page 1813	0x9a0	IRQ_TYPE_EN_RX_SA_194
IRQ_TYPE_STAT_RX_SA_194 on page 1816	0x9a4	IRQ_TYPE_STAT_RX_SA_194
IRQ_TYPE_EN_RX_SA_195 on page 1819	0x9a8	IRQ_TYPE_EN_RX_SA_195

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_195 on page 1822	0x9ac	IRQ_TYPE_STAT_RX_SA_195
IRQ_TYPE_EN_RX_SA_196 on page 1825	0x9b0	IRQ_TYPE_EN_RX_SA_196
IRQ_TYPE_STAT_RX_SA_196 on page 1828	0x9b4	IRQ_TYPE_STAT_RX_SA_196
IRQ_TYPE_EN_RX_SA_197 on page 1831	0x9b8	IRQ_TYPE_EN_RX_SA_197
IRQ_TYPE_STAT_RX_SA_197 on page 1834	0x9bc	IRQ_TYPE_STAT_RX_SA_197
IRQ_TYPE_EN_RX_SA_198 on page 1837	0x9c0	IRQ_TYPE_EN_RX_SA_198
IRQ_TYPE_STAT_RX_SA_198 on page 1840	0x9c4	IRQ_TYPE_STAT_RX_SA_198
IRQ_TYPE_EN_RX_SA_199 on page 1843	0x9c8	IRQ_TYPE_EN_RX_SA_199
IRQ_TYPE_STAT_RX_SA_199 on page 1846	0x9cc	IRQ_TYPE_STAT_RX_SA_199
IRQ_TYPE_EN_RX_SA_200 on page 1849	0x9d0	IRQ_TYPE_EN_RX_SA_200
IRQ_TYPE_STAT_RX_SA_200 on page 1852	0x9d4	IRQ_TYPE_STAT_RX_SA_200
IRQ_TYPE_EN_RX_SA_201 on page 1855	0x9d8	IRQ_TYPE_EN_RX_SA_201
IRQ_TYPE_STAT_RX_SA_201 on page 1858	0x9dc	IRQ_TYPE_STAT_RX_SA_201
IRQ_TYPE_EN_RX_SA_202 on page 1861	0x9e0	IRQ_TYPE_EN_RX_SA_202
IRQ_TYPE_STAT_RX_SA_202 on page 1864	0x9e4	IRQ_TYPE_STAT_RX_SA_202
IRQ_TYPE_EN_RX_SA_203 on page 1867	0x9e8	IRQ_TYPE_EN_RX_SA_203
IRQ_TYPE_STAT_RX_SA_203 on page 1870	0x9ec	IRQ_TYPE_STAT_RX_SA_203
IRQ_TYPE_EN_RX_SA_204 on page 1873	0x9f0	IRQ_TYPE_EN_RX_SA_204

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_204 on page 1876	0x9f4	IRQ_TYPE_STAT_RX_SA_204
IRQ_TYPE_EN_RX_SA_205 on page 1879	0x9f8	IRQ_TYPE_EN_RX_SA_205
IRQ_TYPE_STAT_RX_SA_205 on page 1882	0x9fc	IRQ_TYPE_STAT_RX_SA_205
IRQ_TYPE_EN_RX_SA_206 on page 1885	0xa00	IRQ_TYPE_EN_RX_SA_206
IRQ_TYPE_STAT_RX_SA_206 on page 1888	0xa04	IRQ_TYPE_STAT_RX_SA_206
IRQ_TYPE_EN_RX_SA_207 on page 1891	0xa08	IRQ_TYPE_EN_RX_SA_207
IRQ_TYPE_STAT_RX_SA_207 on page 1894	0xa0c	IRQ_TYPE_STAT_RX_SA_207
IRQ_TYPE_EN_RX_SA_208 on page 1897	0xa10	IRQ_TYPE_EN_RX_SA_208
IRQ_TYPE_STAT_RX_SA_208 on page 1900	0xa14	IRQ_TYPE_STAT_RX_SA_208
IRQ_TYPE_EN_RX_SA_209 on page 1903	0xa18	IRQ_TYPE_EN_RX_SA_209
IRQ_TYPE_STAT_RX_SA_209 on page 1906	0xa1c	IRQ_TYPE_STAT_RX_SA_209
IRQ_TYPE_EN_RX_SA_210 on page 1909	0xa20	IRQ_TYPE_EN_RX_SA_210
IRQ_TYPE_STAT_RX_SA_210 on page 1912	0xa24	IRQ_TYPE_STAT_RX_SA_210
IRQ_TYPE_EN_RX_SA_211 on page 1915	0xa28	IRQ_TYPE_EN_RX_SA_211
IRQ_TYPE_STAT_RX_SA_211 on page 1918	0xa2c	IRQ_TYPE_STAT_RX_SA_211
IRQ_TYPE_EN_RX_SA_212 on page 1921	0xa30	IRQ_TYPE_EN_RX_SA_212
IRQ_TYPE_STAT_RX_SA_212 on page 1924	0xa34	IRQ_TYPE_STAT_RX_SA_212
IRQ_TYPE_EN_RX_SA_213 on page 1927	0xa38	IRQ_TYPE_EN_RX_SA_213

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_213 on page 1930	0xa3c	IRQ_TYPE_STAT_RX_SA_213
IRQ_TYPE_EN_RX_SA_214 on page 1933	0xa40	IRQ_TYPE_EN_RX_SA_214
IRQ_TYPE_STAT_RX_SA_214 on page 1936	0xa44	IRQ_TYPE_STAT_RX_SA_214
IRQ_TYPE_EN_RX_SA_215 on page 1939	0xa48	IRQ_TYPE_EN_RX_SA_215
IRQ_TYPE_STAT_RX_SA_215 on page 1942	0xa4c	IRQ_TYPE_STAT_RX_SA_215
IRQ_TYPE_EN_RX_SA_216 on page 1945	0xa50	IRQ_TYPE_EN_RX_SA_216
IRQ_TYPE_STAT_RX_SA_216 on page 1948	0xa54	IRQ_TYPE_STAT_RX_SA_216
IRQ_TYPE_EN_RX_SA_217 on page 1951	0xa58	IRQ_TYPE_EN_RX_SA_217
IRQ_TYPE_STAT_RX_SA_217 on page 1954	0xa5c	IRQ_TYPE_STAT_RX_SA_217
IRQ_TYPE_EN_RX_SA_218 on page 1957	0xa60	IRQ_TYPE_EN_RX_SA_218
IRQ_TYPE_STAT_RX_SA_218 on page 1960	0xa64	IRQ_TYPE_STAT_RX_SA_218
IRQ_TYPE_EN_RX_SA_219 on page 1963	0xa68	IRQ_TYPE_EN_RX_SA_219
IRQ_TYPE_STAT_RX_SA_219 on page 1966	0xa6c	IRQ_TYPE_STAT_RX_SA_219
IRQ_TYPE_EN_RX_SA_220 on page 1969	0xa70	IRQ_TYPE_EN_RX_SA_220
IRQ_TYPE_STAT_RX_SA_220 on page 1972	0xa74	IRQ_TYPE_STAT_RX_SA_220
IRQ_TYPE_EN_RX_SA_221 on page 1975	0xa78	IRQ_TYPE_EN_RX_SA_221
IRQ_TYPE_STAT_RX_SA_221 on page 1978	0xa7c	IRQ_TYPE_STAT_RX_SA_221
IRQ_TYPE_EN_RX_SA_222 on page 1981	0xa80	IRQ_TYPE_EN_RX_SA_222

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_222 on page 1984	0xa84	IRQ_TYPE_STAT_RX_SA_222
IRQ_TYPE_EN_RX_SA_223 on page 1987	0xa88	IRQ_TYPE_EN_RX_SA_223
IRQ_TYPE_STAT_RX_SA_223 on page 1990	0xa8c	IRQ_TYPE_STAT_RX_SA_223
IRQ_TYPE_EN_RX_SA_224 on page 1993	0xa90	IRQ_TYPE_EN_RX_SA_224
IRQ_TYPE_STAT_RX_SA_224 on page 1996	0xa94	IRQ_TYPE_STAT_RX_SA_224
IRQ_TYPE_EN_RX_SA_225 on page 1999	0xa98	IRQ_TYPE_EN_RX_SA_225
IRQ_TYPE_STAT_RX_SA_225 on page 2002	0xa9c	IRQ_TYPE_STAT_RX_SA_225
IRQ_TYPE_EN_RX_SA_226 on page 2005	0xaa0	IRQ_TYPE_EN_RX_SA_226
IRQ_TYPE_STAT_RX_SA_226 on page 2008	0xaa4	IRQ_TYPE_STAT_RX_SA_226
IRQ_TYPE_EN_RX_SA_227 on page 2011	0xaa8	IRQ_TYPE_EN_RX_SA_227
IRQ_TYPE_STAT_RX_SA_227 on page 2014	0xaac	IRQ_TYPE_STAT_RX_SA_227
IRQ_TYPE_EN_RX_SA_228 on page 2017	0xab0	IRQ_TYPE_EN_RX_SA_228
IRQ_TYPE_STAT_RX_SA_228 on page 2020	0xab4	IRQ_TYPE_STAT_RX_SA_228
IRQ_TYPE_EN_RX_SA_229 on page 2023	0xab8	IRQ_TYPE_EN_RX_SA_229
IRQ_TYPE_STAT_RX_SA_229 on page 2026	0xabc	IRQ_TYPE_STAT_RX_SA_229
IRQ_TYPE_EN_RX_SA_230 on page 2029	0xac0	IRQ_TYPE_EN_RX_SA_230
IRQ_TYPE_STAT_RX_SA_230 on page 2032	0xac4	IRQ_TYPE_STAT_RX_SA_230
IRQ_TYPE_EN_RX_SA_231 on page 2035	0xac8	IRQ_TYPE_EN_RX_SA_231

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_231 on page 2038	0xacc	IRQ_TYPE_STAT_RX_SA_231
IRQ_TYPE_EN_RX_SA_232 on page 2041	0xad0	IRQ_TYPE_EN_RX_SA_232
IRQ_TYPE_STAT_RX_SA_232 on page 2044	0xad4	IRQ_TYPE_STAT_RX_SA_232
IRQ_TYPE_EN_RX_SA_233 on page 2047	0xad8	IRQ_TYPE_EN_RX_SA_233
IRQ_TYPE_STAT_RX_SA_233 on page 2050	0xadc	IRQ_TYPE_STAT_RX_SA_233
IRQ_TYPE_EN_RX_SA_234 on page 2053	0xae0	IRQ_TYPE_EN_RX_SA_234
IRQ_TYPE_STAT_RX_SA_234 on page 2056	0xae4	IRQ_TYPE_STAT_RX_SA_234
IRQ_TYPE_EN_RX_SA_235 on page 2059	0xae8	IRQ_TYPE_EN_RX_SA_235
IRQ_TYPE_STAT_RX_SA_235 on page 2062	0xaec	IRQ_TYPE_STAT_RX_SA_235
IRQ_TYPE_EN_RX_SA_236 on page 2065	0xaf0	IRQ_TYPE_EN_RX_SA_236
IRQ_TYPE_STAT_RX_SA_236 on page 2068	0xaf4	IRQ_TYPE_STAT_RX_SA_236
IRQ_TYPE_EN_RX_SA_237 on page 2071	0xaf8	IRQ_TYPE_EN_RX_SA_237
IRQ_TYPE_STAT_RX_SA_237 on page 2074	0xafc	IRQ_TYPE_STAT_RX_SA_237
IRQ_TYPE_EN_RX_SA_238 on page 2077	0xb00	IRQ_TYPE_EN_RX_SA_238
IRQ_TYPE_STAT_RX_SA_238 on page 2080	0xb04	IRQ_TYPE_STAT_RX_SA_238
IRQ_TYPE_EN_RX_SA_239 on page 2083	0xb08	IRQ_TYPE_EN_RX_SA_239
IRQ_TYPE_STAT_RX_SA_239 on page 2086	0xb0c	IRQ_TYPE_STAT_RX_SA_239
IRQ_TYPE_EN_RX_SA_240 on page 2089	0xb10	IRQ_TYPE_EN_RX_SA_240

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_240 on page 2092	0xb14	IRQ_TYPE_STAT_RX_SA_240
IRQ_TYPE_EN_RX_SA_241 on page 2095	0xb18	IRQ_TYPE_EN_RX_SA_241
IRQ_TYPE_STAT_RX_SA_241 on page 2098	0xb1c	IRQ_TYPE_STAT_RX_SA_241
IRQ_TYPE_EN_RX_SA_242 on page 2101	0xb20	IRQ_TYPE_EN_RX_SA_242
IRQ_TYPE_STAT_RX_SA_242 on page 2104	0xb24	IRQ_TYPE_STAT_RX_SA_242
IRQ_TYPE_EN_RX_SA_243 on page 2107	0xb28	IRQ_TYPE_EN_RX_SA_243
IRQ_TYPE_STAT_RX_SA_243 on page 2110	0xb2c	IRQ_TYPE_STAT_RX_SA_243
IRQ_TYPE_EN_RX_SA_244 on page 2113	0xb30	IRQ_TYPE_EN_RX_SA_244
IRQ_TYPE_STAT_RX_SA_244 on page 2116	0xb34	IRQ_TYPE_STAT_RX_SA_244
IRQ_TYPE_EN_RX_SA_245 on page 2119	0xb38	IRQ_TYPE_EN_RX_SA_245
IRQ_TYPE_STAT_RX_SA_245 on page 2122	0xb3c	IRQ_TYPE_STAT_RX_SA_245
IRQ_TYPE_EN_RX_SA_246 on page 2125	0xb40	IRQ_TYPE_EN_RX_SA_246
IRQ_TYPE_STAT_RX_SA_246 on page 2128	0xb44	IRQ_TYPE_STAT_RX_SA_246
IRQ_TYPE_EN_RX_SA_247 on page 2131	0xb48	IRQ_TYPE_EN_RX_SA_247
IRQ_TYPE_STAT_RX_SA_247 on page 2134	0xb4c	IRQ_TYPE_STAT_RX_SA_247
IRQ_TYPE_EN_RX_SA_248 on page 2137	0xb50	IRQ_TYPE_EN_RX_SA_248
IRQ_TYPE_STAT_RX_SA_248 on page 2140	0xb54	IRQ_TYPE_STAT_RX_SA_248
IRQ_TYPE_EN_RX_SA_249 on page 2143	0xb58	IRQ_TYPE_EN_RX_SA_249

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_249 on page 2146	0xb5c	IRQ_TYPE_STAT_RX_SA_249
IRQ_TYPE_EN_RX_SA_250 on page 2149	0xb60	IRQ_TYPE_EN_RX_SA_250
IRQ_TYPE_STAT_RX_SA_250 on page 2152	0xb64	IRQ_TYPE_STAT_RX_SA_250
IRQ_TYPE_EN_RX_SA_251 on page 2155	0xb68	IRQ_TYPE_EN_RX_SA_251
IRQ_TYPE_STAT_RX_SA_251 on page 2158	0xb6c	IRQ_TYPE_STAT_RX_SA_251
IRQ_TYPE_EN_RX_SA_252 on page 2161	0xb70	IRQ_TYPE_EN_RX_SA_252
IRQ_TYPE_STAT_RX_SA_252 on page 2164	0xb74	IRQ_TYPE_STAT_RX_SA_252
IRQ_TYPE_EN_RX_SA_253 on page 2167	0xb78	IRQ_TYPE_EN_RX_SA_253
IRQ_TYPE_STAT_RX_SA_253 on page 2170	0xb7c	IRQ_TYPE_STAT_RX_SA_253
IRQ_TYPE_EN_RX_SA_254 on page 2173	0xb80	IRQ_TYPE_EN_RX_SA_254
IRQ_TYPE_STAT_RX_SA_254 on page 2176	0xb84	IRQ_TYPE_STAT_RX_SA_254
IRQ_TYPE_EN_RX_SA_255 on page 2179	0xb88	IRQ_TYPE_EN_RX_SA_255
IRQ_TYPE_STAT_RX_SA_255 on page 2182	0xb8c	IRQ_TYPE_STAT_RX_SA_255
IRQ_TYPE_EN_RX_SA_256 on page 2185	0xb90	IRQ_TYPE_EN_RX_SA_256
IRQ_TYPE_STAT_RX_SA_256 on page 2188	0xb94	IRQ_TYPE_STAT_RX_SA_256
IRQ_TYPE_EN_RX_SA_257 on page 2191	0xb98	IRQ_TYPE_EN_RX_SA_257
IRQ_TYPE_STAT_RX_SA_257 on page 2194	0xb9c	IRQ_TYPE_STAT_RX_SA_257
IRQ_TYPE_EN_RX_SA_258 on page 2197	0xba0	IRQ_TYPE_EN_RX_SA_258

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_258 on page 2200	0xba4	IRQ_TYPE_STAT_RX_SA_258
IRQ_TYPE_EN_RX_SA_259 on page 2203	0xba8	IRQ_TYPE_EN_RX_SA_259
IRQ_TYPE_STAT_RX_SA_259 on page 2206	0xbac	IRQ_TYPE_STAT_RX_SA_259
IRQ_TYPE_EN_RX_SA_260 on page 2209	0xbb0	IRQ_TYPE_EN_RX_SA_260
IRQ_TYPE_STAT_RX_SA_260 on page 2212	0xbb4	IRQ_TYPE_STAT_RX_SA_260
IRQ_TYPE_EN_RX_SA_261 on page 2215	0xbb8	IRQ_TYPE_EN_RX_SA_261
IRQ_TYPE_STAT_RX_SA_261 on page 2218	0xbbc	IRQ_TYPE_STAT_RX_SA_261
IRQ_TYPE_EN_RX_SA_262 on page 2221	0xbc0	IRQ_TYPE_EN_RX_SA_262
IRQ_TYPE_STAT_RX_SA_262 on page 2224	0xbc4	IRQ_TYPE_STAT_RX_SA_262
IRQ_TYPE_EN_RX_SA_263 on page 2227	0xbc8	IRQ_TYPE_EN_RX_SA_263
IRQ_TYPE_STAT_RX_SA_263 on page 2230	0xbcc	IRQ_TYPE_STAT_RX_SA_263
IRQ_TYPE_EN_RX_SA_264 on page 2233	0xbd0	IRQ_TYPE_EN_RX_SA_264
IRQ_TYPE_STAT_RX_SA_264 on page 2236	0xbd4	IRQ_TYPE_STAT_RX_SA_264
IRQ_TYPE_EN_RX_SA_265 on page 2239	0xbd8	IRQ_TYPE_EN_RX_SA_265
IRQ_TYPE_STAT_RX_SA_265 on page 2242	0xbdc	IRQ_TYPE_STAT_RX_SA_265
IRQ_TYPE_EN_RX_SA_266 on page 2245	0xbe0	IRQ_TYPE_EN_RX_SA_266
IRQ_TYPE_STAT_RX_SA_266 on page 2248	0xbe4	IRQ_TYPE_STAT_RX_SA_266
IRQ_TYPE_EN_RX_SA_267 on page 2251	0xbe8	IRQ_TYPE_EN_RX_SA_267

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_267 on page 2254	0xbec	IRQ_TYPE_STAT_RX_SA_267
IRQ_TYPE_EN_RX_SA_268 on page 2257	0xbf0	IRQ_TYPE_EN_RX_SA_268
IRQ_TYPE_STAT_RX_SA_268 on page 2260	0xbf4	IRQ_TYPE_STAT_RX_SA_268
IRQ_TYPE_EN_RX_SA_269 on page 2263	0xbf8	IRQ_TYPE_EN_RX_SA_269
IRQ_TYPE_STAT_RX_SA_269 on page 2266	0xbfc	IRQ_TYPE_STAT_RX_SA_269
IRQ_TYPE_EN_RX_SA_270 on page 2269	0xc00	IRQ_TYPE_EN_RX_SA_270
IRQ_TYPE_STAT_RX_SA_270 on page 2272	0xc04	IRQ_TYPE_STAT_RX_SA_270
IRQ_TYPE_EN_RX_SA_271 on page 2275	0xc08	IRQ_TYPE_EN_RX_SA_271
IRQ_TYPE_STAT_RX_SA_271 on page 2278	0xc0c	IRQ_TYPE_STAT_RX_SA_271
IRQ_TYPE_EN_RX_SA_272 on page 2281	0xc10	IRQ_TYPE_EN_RX_SA_272
IRQ_TYPE_STAT_RX_SA_272 on page 2284	0xc14	IRQ_TYPE_STAT_RX_SA_272
IRQ_TYPE_EN_RX_SA_273 on page 2287	0xc18	IRQ_TYPE_EN_RX_SA_273
IRQ_TYPE_STAT_RX_SA_273 on page 2290	0xc1c	IRQ_TYPE_STAT_RX_SA_273
IRQ_TYPE_EN_RX_SA_274 on page 2293	0xc20	IRQ_TYPE_EN_RX_SA_274
IRQ_TYPE_STAT_RX_SA_274 on page 2296	0xc24	IRQ_TYPE_STAT_RX_SA_274
IRQ_TYPE_EN_RX_SA_275 on page 2299	0xc28	IRQ_TYPE_EN_RX_SA_275
IRQ_TYPE_STAT_RX_SA_275 on page 2302	0xc2c	IRQ_TYPE_STAT_RX_SA_275
IRQ_TYPE_EN_RX_SA_276 on page 2305	0xc30	IRQ_TYPE_EN_RX_SA_276

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_276 on page 2308	0xc34	IRQ_TYPE_STAT_RX_SA_276
IRQ_TYPE_EN_RX_SA_277 on page 2311	0xc38	IRQ_TYPE_EN_RX_SA_277
IRQ_TYPE_STAT_RX_SA_277 on page 2314	0xc3c	IRQ_TYPE_STAT_RX_SA_277
IRQ_TYPE_EN_RX_SA_278 on page 2317	0xc40	IRQ_TYPE_EN_RX_SA_278
IRQ_TYPE_STAT_RX_SA_278 on page 2320	0xc44	IRQ_TYPE_STAT_RX_SA_278
IRQ_TYPE_EN_RX_SA_279 on page 2323	0xc48	IRQ_TYPE_EN_RX_SA_279
IRQ_TYPE_STAT_RX_SA_279 on page 2326	0xc4c	IRQ_TYPE_STAT_RX_SA_279
IRQ_TYPE_EN_RX_SA_280 on page 2329	0xc50	IRQ_TYPE_EN_RX_SA_280
IRQ_TYPE_STAT_RX_SA_280 on page 2332	0xc54	IRQ_TYPE_STAT_RX_SA_280
IRQ_TYPE_EN_RX_SA_281 on page 2335	0xc58	IRQ_TYPE_EN_RX_SA_281
IRQ_TYPE_STAT_RX_SA_281 on page 2338	0xc5c	IRQ_TYPE_STAT_RX_SA_281
IRQ_TYPE_EN_RX_SA_282 on page 2341	0xc60	IRQ_TYPE_EN_RX_SA_282
IRQ_TYPE_STAT_RX_SA_282 on page 2344	0xc64	IRQ_TYPE_STAT_RX_SA_282
IRQ_TYPE_EN_RX_SA_283 on page 2347	0xc68	IRQ_TYPE_EN_RX_SA_283
IRQ_TYPE_STAT_RX_SA_283 on page 2350	0xc6c	IRQ_TYPE_STAT_RX_SA_283
IRQ_TYPE_EN_RX_SA_284 on page 2353	0xc70	IRQ_TYPE_EN_RX_SA_284
IRQ_TYPE_STAT_RX_SA_284 on page 2356	0xc74	IRQ_TYPE_STAT_RX_SA_284
IRQ_TYPE_EN_RX_SA_285 on page 2359	0xc78	IRQ_TYPE_EN_RX_SA_285

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_285 on page 2362	0xc7c	IRQ_TYPE_STAT_RX_SA_285
IRQ_TYPE_EN_RX_SA_286 on page 2365	0xc80	IRQ_TYPE_EN_RX_SA_286
IRQ_TYPE_STAT_RX_SA_286 on page 2368	0xc84	IRQ_TYPE_STAT_RX_SA_286
IRQ_TYPE_EN_RX_SA_287 on page 2371	0xc88	IRQ_TYPE_EN_RX_SA_287
IRQ_TYPE_STAT_RX_SA_287 on page 2374	0xc8c	IRQ_TYPE_STAT_RX_SA_287
IRQ_TYPE_EN_RX_SA_288 on page 2377	0xc90	IRQ_TYPE_EN_RX_SA_288
IRQ_TYPE_STAT_RX_SA_288 on page 2380	0xc94	IRQ_TYPE_STAT_RX_SA_288
IRQ_TYPE_EN_RX_SA_289 on page 2383	0xc98	IRQ_TYPE_EN_RX_SA_289
IRQ_TYPE_STAT_RX_SA_289 on page 2386	0xc9c	IRQ_TYPE_STAT_RX_SA_289
IRQ_TYPE_EN_RX_SA_290 on page 2389	0xca0	IRQ_TYPE_EN_RX_SA_290
IRQ_TYPE_STAT_RX_SA_290 on page 2392	0xca4	IRQ_TYPE_STAT_RX_SA_290
IRQ_TYPE_EN_RX_SA_291 on page 2395	0xca8	IRQ_TYPE_EN_RX_SA_291
IRQ_TYPE_STAT_RX_SA_291 on page 2398	0xcac	IRQ_TYPE_STAT_RX_SA_291
IRQ_TYPE_EN_RX_SA_292 on page 2401	0xcb0	IRQ_TYPE_EN_RX_SA_292
IRQ_TYPE_STAT_RX_SA_292 on page 2404	0xcb4	IRQ_TYPE_STAT_RX_SA_292
IRQ_TYPE_EN_RX_SA_293 on page 2407	0xcb8	IRQ_TYPE_EN_RX_SA_293
IRQ_TYPE_STAT_RX_SA_293 on page 2410	0xcbc	IRQ_TYPE_STAT_RX_SA_293
IRQ_TYPE_EN_RX_SA_294 on page 2413	0xcc0	IRQ_TYPE_EN_RX_SA_294

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_294 on page 2416	0xcc4	IRQ_TYPE_STAT_RX_SA_294
IRQ_TYPE_EN_RX_SA_295 on page 2419	0xcc8	IRQ_TYPE_EN_RX_SA_295
IRQ_TYPE_STAT_RX_SA_295 on page 2422	0xccc	IRQ_TYPE_STAT_RX_SA_295
IRQ_TYPE_EN_RX_SA_296 on page 2425	0xcd0	IRQ_TYPE_EN_RX_SA_296
IRQ_TYPE_STAT_RX_SA_296 on page 2428	0xcd4	IRQ_TYPE_STAT_RX_SA_296
IRQ_TYPE_EN_RX_SA_297 on page 2431	0xcd8	IRQ_TYPE_EN_RX_SA_297
IRQ_TYPE_STAT_RX_SA_297 on page 2434	0xcdc	IRQ_TYPE_STAT_RX_SA_297
IRQ_TYPE_EN_RX_SA_298 on page 2437	0xce0	IRQ_TYPE_EN_RX_SA_298
IRQ_TYPE_STAT_RX_SA_298 on page 2440	0xce4	IRQ_TYPE_STAT_RX_SA_298
IRQ_TYPE_EN_RX_SA_299 on page 2443	0xce8	IRQ_TYPE_EN_RX_SA_299
IRQ_TYPE_STAT_RX_SA_299 on page 2446	0xcec	IRQ_TYPE_STAT_RX_SA_299
IRQ_TYPE_EN_RX_SA_300 on page 2449	0xcf0	IRQ_TYPE_EN_RX_SA_300
IRQ_TYPE_STAT_RX_SA_300 on page 2452	0xcf4	IRQ_TYPE_STAT_RX_SA_300
IRQ_TYPE_EN_RX_SA_301 on page 2455	0xcf8	IRQ_TYPE_EN_RX_SA_301
IRQ_TYPE_STAT_RX_SA_301 on page 2458	0xcfc	IRQ_TYPE_STAT_RX_SA_301
IRQ_TYPE_EN_RX_SA_302 on page 2461	0xd00	IRQ_TYPE_EN_RX_SA_302
IRQ_TYPE_STAT_RX_SA_302 on page 2464	0xd04	IRQ_TYPE_STAT_RX_SA_302
IRQ_TYPE_EN_RX_SA_303 on page 2467	0xd08	IRQ_TYPE_EN_RX_SA_303

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_303 on page 2470	0xd0c	IRQ_TYPE_STAT_RX_SA_303
IRQ_TYPE_EN_RX_SA_304 on page 2473	0xd10	IRQ_TYPE_EN_RX_SA_304
IRQ_TYPE_STAT_RX_SA_304 on page 2476	0xd14	IRQ_TYPE_STAT_RX_SA_304
IRQ_TYPE_EN_RX_SA_305 on page 2479	0xd18	IRQ_TYPE_EN_RX_SA_305
IRQ_TYPE_STAT_RX_SA_305 on page 2482	0xd1c	IRQ_TYPE_STAT_RX_SA_305
IRQ_TYPE_EN_RX_SA_306 on page 2485	0xd20	IRQ_TYPE_EN_RX_SA_306
IRQ_TYPE_STAT_RX_SA_306 on page 2488	0xd24	IRQ_TYPE_STAT_RX_SA_306
IRQ_TYPE_EN_RX_SA_307 on page 2491	0xd28	IRQ_TYPE_EN_RX_SA_307
IRQ_TYPE_STAT_RX_SA_307 on page 2494	0xd2c	IRQ_TYPE_STAT_RX_SA_307
IRQ_TYPE_EN_RX_SA_308 on page 2497	0xd30	IRQ_TYPE_EN_RX_SA_308
IRQ_TYPE_STAT_RX_SA_308 on page 2500	0xd34	IRQ_TYPE_STAT_RX_SA_308
IRQ_TYPE_EN_RX_SA_309 on page 2503	0xd38	IRQ_TYPE_EN_RX_SA_309
IRQ_TYPE_STAT_RX_SA_309 on page 2506	0xd3c	IRQ_TYPE_STAT_RX_SA_309
IRQ_TYPE_EN_RX_SA_310 on page 2509	0xd40	IRQ_TYPE_EN_RX_SA_310
IRQ_TYPE_STAT_RX_SA_310 on page 2512	0xd44	IRQ_TYPE_STAT_RX_SA_310
IRQ_TYPE_EN_RX_SA_311 on page 2515	0xd48	IRQ_TYPE_EN_RX_SA_311
IRQ_TYPE_STAT_RX_SA_311 on page 2518	0xd4c	IRQ_TYPE_STAT_RX_SA_311
IRQ_TYPE_EN_RX_SA_312 on page 2521	0xd50	IRQ_TYPE_EN_RX_SA_312

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_312 on page 2524	0xd54	IRQ_TYPE_STAT_RX_SA_312
IRQ_TYPE_EN_RX_SA_313 on page 2527	0xd58	IRQ_TYPE_EN_RX_SA_313
IRQ_TYPE_STAT_RX_SA_313 on page 2530	0xd5c	IRQ_TYPE_STAT_RX_SA_313
IRQ_TYPE_EN_RX_SA_314 on page 2533	0xd60	IRQ_TYPE_EN_RX_SA_314
IRQ_TYPE_STAT_RX_SA_314 on page 2536	0xd64	IRQ_TYPE_STAT_RX_SA_314
IRQ_TYPE_EN_RX_SA_315 on page 2539	0xd68	IRQ_TYPE_EN_RX_SA_315
IRQ_TYPE_STAT_RX_SA_315 on page 2542	0xd6c	IRQ_TYPE_STAT_RX_SA_315
IRQ_TYPE_EN_RX_SA_316 on page 2545	0xd70	IRQ_TYPE_EN_RX_SA_316
IRQ_TYPE_STAT_RX_SA_316 on page 2548	0xd74	IRQ_TYPE_STAT_RX_SA_316
IRQ_TYPE_EN_RX_SA_317 on page 2551	0xd78	IRQ_TYPE_EN_RX_SA_317
IRQ_TYPE_STAT_RX_SA_317 on page 2554	0xd7c	IRQ_TYPE_STAT_RX_SA_317
IRQ_TYPE_EN_RX_SA_318 on page 2557	0xd80	IRQ_TYPE_EN_RX_SA_318
IRQ_TYPE_STAT_RX_SA_318 on page 2560	0xd84	IRQ_TYPE_STAT_RX_SA_318
IRQ_TYPE_EN_RX_SA_319 on page 2563	0xd88	IRQ_TYPE_EN_RX_SA_319
IRQ_TYPE_STAT_RX_SA_319 on page 2566	0xd8c	IRQ_TYPE_STAT_RX_SA_319
IRQ_TYPE_EN_RX_SA_320 on page 2569	0xd90	IRQ_TYPE_EN_RX_SA_320
IRQ_TYPE_STAT_RX_SA_320 on page 2572	0xd94	IRQ_TYPE_STAT_RX_SA_320
IRQ_TYPE_EN_RX_SA_321 on page 2575	0xd98	IRQ_TYPE_EN_RX_SA_321

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_321 on page 2578	0xd9c	IRQ_TYPE_STAT_RX_SA_321
IRQ_TYPE_EN_RX_SA_322 on page 2581	0xda0	IRQ_TYPE_EN_RX_SA_322
IRQ_TYPE_STAT_RX_SA_322 on page 2584	0xda4	IRQ_TYPE_STAT_RX_SA_322
IRQ_TYPE_EN_RX_SA_323 on page 2587	0xda8	IRQ_TYPE_EN_RX_SA_323
IRQ_TYPE_STAT_RX_SA_323 on page 2590	0xdac	IRQ_TYPE_STAT_RX_SA_323
IRQ_TYPE_EN_RX_SA_324 on page 2593	0xdb0	IRQ_TYPE_EN_RX_SA_324
IRQ_TYPE_STAT_RX_SA_324 on page 2596	0xdb4	IRQ_TYPE_STAT_RX_SA_324
IRQ_TYPE_EN_RX_SA_325 on page 2599	0xdb8	IRQ_TYPE_EN_RX_SA_325
IRQ_TYPE_STAT_RX_SA_325 on page 2602	0xdbc	IRQ_TYPE_STAT_RX_SA_325
IRQ_TYPE_EN_RX_SA_326 on page 2605	0xdc0	IRQ_TYPE_EN_RX_SA_326
IRQ_TYPE_STAT_RX_SA_326 on page 2608	0xdc4	IRQ_TYPE_STAT_RX_SA_326
IRQ_TYPE_EN_RX_SA_327 on page 2611	0xdc8	IRQ_TYPE_EN_RX_SA_327
IRQ_TYPE_STAT_RX_SA_327 on page 2614	0xdcc	IRQ_TYPE_STAT_RX_SA_327
IRQ_TYPE_EN_RX_SA_328 on page 2617	0xdd0	IRQ_TYPE_EN_RX_SA_328
IRQ_TYPE_STAT_RX_SA_328 on page 2620	0xdd4	IRQ_TYPE_STAT_RX_SA_328
IRQ_TYPE_EN_RX_SA_329 on page 2623	0xdd8	IRQ_TYPE_EN_RX_SA_329
IRQ_TYPE_STAT_RX_SA_329 on page 2626	0xddc	IRQ_TYPE_STAT_RX_SA_329
IRQ_TYPE_EN_RX_SA_330 on page 2629	0xde0	IRQ_TYPE_EN_RX_SA_330

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_330 on page 2632	0xde4	IRQ_TYPE_STAT_RX_SA_330
IRQ_TYPE_EN_RX_SA_331 on page 2635	0xde8	IRQ_TYPE_EN_RX_SA_331
IRQ_TYPE_STAT_RX_SA_331 on page 2638	0xdec	IRQ_TYPE_STAT_RX_SA_331
IRQ_TYPE_EN_RX_SA_332 on page 2641	0xdf0	IRQ_TYPE_EN_RX_SA_332
IRQ_TYPE_STAT_RX_SA_332 on page 2644	0xdf4	IRQ_TYPE_STAT_RX_SA_332
IRQ_TYPE_EN_RX_SA_333 on page 2647	0xdf8	IRQ_TYPE_EN_RX_SA_333
IRQ_TYPE_STAT_RX_SA_333 on page 2650	0xdfc	IRQ_TYPE_STAT_RX_SA_333
IRQ_TYPE_EN_RX_SA_334 on page 2653	0xe00	IRQ_TYPE_EN_RX_SA_334
IRQ_TYPE_STAT_RX_SA_334 on page 2656	0xe04	IRQ_TYPE_STAT_RX_SA_334
IRQ_TYPE_EN_RX_SA_335 on page 2659	0xe08	IRQ_TYPE_EN_RX_SA_335
IRQ_TYPE_STAT_RX_SA_335 on page 2662	0xe0c	IRQ_TYPE_STAT_RX_SA_335
IRQ_TYPE_EN_RX_SA_336 on page 2665	0xe10	IRQ_TYPE_EN_RX_SA_336
IRQ_TYPE_STAT_RX_SA_336 on page 2668	0xe14	IRQ_TYPE_STAT_RX_SA_336
IRQ_TYPE_EN_RX_SA_337 on page 2671	0xe18	IRQ_TYPE_EN_RX_SA_337
IRQ_TYPE_STAT_RX_SA_337 on page 2674	0xe1c	IRQ_TYPE_STAT_RX_SA_337
IRQ_TYPE_EN_RX_SA_338 on page 2677	0xe20	IRQ_TYPE_EN_RX_SA_338
IRQ_TYPE_STAT_RX_SA_338 on page 2680	0xe24	IRQ_TYPE_STAT_RX_SA_338
IRQ_TYPE_EN_RX_SA_339 on page 2683	0xe28	IRQ_TYPE_EN_RX_SA_339

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_339 on page 2686	0xe2c	IRQ_TYPE_STAT_RX_SA_339
IRQ_TYPE_EN_RX_SA_340 on page 2689	0xe30	IRQ_TYPE_EN_RX_SA_340
IRQ_TYPE_STAT_RX_SA_340 on page 2692	0xe34	IRQ_TYPE_STAT_RX_SA_340
IRQ_TYPE_EN_RX_SA_341 on page 2695	0xe38	IRQ_TYPE_EN_RX_SA_341
IRQ_TYPE_STAT_RX_SA_341 on page 2698	0xe3c	IRQ_TYPE_STAT_RX_SA_341
IRQ_TYPE_EN_RX_SA_342 on page 2701	0xe40	IRQ_TYPE_EN_RX_SA_342
IRQ_TYPE_STAT_RX_SA_342 on page 2704	0xe44	IRQ_TYPE_STAT_RX_SA_342
IRQ_TYPE_EN_RX_SA_343 on page 2707	0xe48	IRQ_TYPE_EN_RX_SA_343
IRQ_TYPE_STAT_RX_SA_343 on page 2710	0xe4c	IRQ_TYPE_STAT_RX_SA_343
IRQ_TYPE_EN_RX_SA_344 on page 2713	0xe50	IRQ_TYPE_EN_RX_SA_344
IRQ_TYPE_STAT_RX_SA_344 on page 2716	0xe54	IRQ_TYPE_STAT_RX_SA_344
IRQ_TYPE_EN_RX_SA_345 on page 2719	0xe58	IRQ_TYPE_EN_RX_SA_345
IRQ_TYPE_STAT_RX_SA_345 on page 2722	0xe5c	IRQ_TYPE_STAT_RX_SA_345
IRQ_TYPE_EN_RX_SA_346 on page 2725	0xe60	IRQ_TYPE_EN_RX_SA_346
IRQ_TYPE_STAT_RX_SA_346 on page 2728	0xe64	IRQ_TYPE_STAT_RX_SA_346
IRQ_TYPE_EN_RX_SA_347 on page 2731	0xe68	IRQ_TYPE_EN_RX_SA_347
IRQ_TYPE_STAT_RX_SA_347 on page 2734	0xe6c	IRQ_TYPE_STAT_RX_SA_347
IRQ_TYPE_EN_RX_SA_348 on page 2737	0xe70	IRQ_TYPE_EN_RX_SA_348

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_348 on page 2740	0xe74	IRQ_TYPE_STAT_RX_SA_348
IRQ_TYPE_EN_RX_SA_349 on page 2743	0xe78	IRQ_TYPE_EN_RX_SA_349
IRQ_TYPE_STAT_RX_SA_349 on page 2746	0xe7c	IRQ_TYPE_STAT_RX_SA_349
IRQ_TYPE_EN_RX_SA_350 on page 2749	0xe80	IRQ_TYPE_EN_RX_SA_350
IRQ_TYPE_STAT_RX_SA_350 on page 2752	0xe84	IRQ_TYPE_STAT_RX_SA_350
IRQ_TYPE_EN_RX_SA_351 on page 2755	0xe88	IRQ_TYPE_EN_RX_SA_351
IRQ_TYPE_STAT_RX_SA_351 on page 2758	0xe8c	IRQ_TYPE_STAT_RX_SA_351
IRQ_TYPE_EN_RX_SA_352 on page 2761	0xe90	IRQ_TYPE_EN_RX_SA_352
IRQ_TYPE_STAT_RX_SA_352 on page 2764	0xe94	IRQ_TYPE_STAT_RX_SA_352
IRQ_TYPE_EN_RX_SA_353 on page 2767	0xe98	IRQ_TYPE_EN_RX_SA_353
IRQ_TYPE_STAT_RX_SA_353 on page 2770	0xe9c	IRQ_TYPE_STAT_RX_SA_353
IRQ_TYPE_EN_RX_SA_354 on page 2773	0xea0	IRQ_TYPE_EN_RX_SA_354
IRQ_TYPE_STAT_RX_SA_354 on page 2776	0xea4	IRQ_TYPE_STAT_RX_SA_354
IRQ_TYPE_EN_RX_SA_355 on page 2779	0xea8	IRQ_TYPE_EN_RX_SA_355
IRQ_TYPE_STAT_RX_SA_355 on page 2782	0xeac	IRQ_TYPE_STAT_RX_SA_355
IRQ_TYPE_EN_RX_SA_356 on page 2785	0xeb0	IRQ_TYPE_EN_RX_SA_356
IRQ_TYPE_STAT_RX_SA_356 on page 2788	0xeb4	IRQ_TYPE_STAT_RX_SA_356
IRQ_TYPE_EN_RX_SA_357 on page 2791	0xeb8	IRQ_TYPE_EN_RX_SA_357

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_357 on page 2794	0xebc	IRQ_TYPE_STAT_RX_SA_357
IRQ_TYPE_EN_RX_SA_358 on page 2797	0xec0	IRQ_TYPE_EN_RX_SA_358
IRQ_TYPE_STAT_RX_SA_358 on page 2800	0xec4	IRQ_TYPE_STAT_RX_SA_358
IRQ_TYPE_EN_RX_SA_359 on page 2803	0xec8	IRQ_TYPE_EN_RX_SA_359
IRQ_TYPE_STAT_RX_SA_359 on page 2806	0xecc	IRQ_TYPE_STAT_RX_SA_359
IRQ_TYPE_EN_RX_SA_360 on page 2809	0xed0	IRQ_TYPE_EN_RX_SA_360
IRQ_TYPE_STAT_RX_SA_360 on page 2812	0xed4	IRQ_TYPE_STAT_RX_SA_360
IRQ_TYPE_EN_RX_SA_361 on page 2815	0xed8	IRQ_TYPE_EN_RX_SA_361
IRQ_TYPE_STAT_RX_SA_361 on page 2818	0xedc	IRQ_TYPE_STAT_RX_SA_361
IRQ_TYPE_EN_RX_SA_362 on page 2821	0xee0	IRQ_TYPE_EN_RX_SA_362
IRQ_TYPE_STAT_RX_SA_362 on page 2824	0xee4	IRQ_TYPE_STAT_RX_SA_362
IRQ_TYPE_EN_RX_SA_363 on page 2827	0xee8	IRQ_TYPE_EN_RX_SA_363
IRQ_TYPE_STAT_RX_SA_363 on page 2830	0xeec	IRQ_TYPE_STAT_RX_SA_363
IRQ_TYPE_EN_RX_SA_364 on page 2833	0xef0	IRQ_TYPE_EN_RX_SA_364
IRQ_TYPE_STAT_RX_SA_364 on page 2836	0xef4	IRQ_TYPE_STAT_RX_SA_364
IRQ_TYPE_EN_RX_SA_365 on page 2839	0xef8	IRQ_TYPE_EN_RX_SA_365
IRQ_TYPE_STAT_RX_SA_365 on page 2842	0xefc	IRQ_TYPE_STAT_RX_SA_365
IRQ_TYPE_EN_RX_SA_366 on page 2845	0xf00	IRQ_TYPE_EN_RX_SA_366

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_366 on page 2848	0xf04	IRQ_TYPE_STAT_RX_SA_366
IRQ_TYPE_EN_RX_SA_367 on page 2851	0xf08	IRQ_TYPE_EN_RX_SA_367
IRQ_TYPE_STAT_RX_SA_367 on page 2854	0xf0c	IRQ_TYPE_STAT_RX_SA_367
IRQ_TYPE_EN_RX_SA_368 on page 2857	0xf10	IRQ_TYPE_EN_RX_SA_368
IRQ_TYPE_STAT_RX_SA_368 on page 2860	0xf14	IRQ_TYPE_STAT_RX_SA_368
IRQ_TYPE_EN_RX_SA_369 on page 2863	0xf18	IRQ_TYPE_EN_RX_SA_369
IRQ_TYPE_STAT_RX_SA_369 on page 2866	0xf1c	IRQ_TYPE_STAT_RX_SA_369
IRQ_TYPE_EN_RX_SA_370 on page 2869	0xf20	IRQ_TYPE_EN_RX_SA_370
IRQ_TYPE_STAT_RX_SA_370 on page 2872	0xf24	IRQ_TYPE_STAT_RX_SA_370
IRQ_TYPE_EN_RX_SA_371 on page 2875	0xf28	IRQ_TYPE_EN_RX_SA_371
IRQ_TYPE_STAT_RX_SA_371 on page 2878	0xf2c	IRQ_TYPE_STAT_RX_SA_371
IRQ_TYPE_EN_RX_SA_372 on page 2881	0xf30	IRQ_TYPE_EN_RX_SA_372
IRQ_TYPE_STAT_RX_SA_372 on page 2884	0xf34	IRQ_TYPE_STAT_RX_SA_372
IRQ_TYPE_EN_RX_SA_373 on page 2887	0xf38	IRQ_TYPE_EN_RX_SA_373
IRQ_TYPE_STAT_RX_SA_373 on page 2890	0xf3c	IRQ_TYPE_STAT_RX_SA_373
IRQ_TYPE_EN_RX_SA_374 on page 2893	0xf40	IRQ_TYPE_EN_RX_SA_374
IRQ_TYPE_STAT_RX_SA_374 on page 2896	0xf44	IRQ_TYPE_STAT_RX_SA_374
IRQ_TYPE_EN_RX_SA_375 on page 2899	0xf48	IRQ_TYPE_EN_RX_SA_375

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_375 on page 2902	0xf4c	IRQ_TYPE_STAT_RX_SA_375
IRQ_TYPE_EN_RX_SA_376 on page 2905	0xf50	IRQ_TYPE_EN_RX_SA_376
IRQ_TYPE_STAT_RX_SA_376 on page 2908	0xf54	IRQ_TYPE_STAT_RX_SA_376
IRQ_TYPE_EN_RX_SA_377 on page 2911	0xf58	IRQ_TYPE_EN_RX_SA_377
IRQ_TYPE_STAT_RX_SA_377 on page 2914	0xf5c	IRQ_TYPE_STAT_RX_SA_377
IRQ_TYPE_EN_RX_SA_378 on page 2917	0xf60	IRQ_TYPE_EN_RX_SA_378
IRQ_TYPE_STAT_RX_SA_378 on page 2920	0xf64	IRQ_TYPE_STAT_RX_SA_378
IRQ_TYPE_EN_RX_SA_379 on page 2923	0xf68	IRQ_TYPE_EN_RX_SA_379
IRQ_TYPE_STAT_RX_SA_379 on page 2926	0xf6c	IRQ_TYPE_STAT_RX_SA_379
IRQ_TYPE_EN_RX_SA_380 on page 2929	0xf70	IRQ_TYPE_EN_RX_SA_380
IRQ_TYPE_STAT_RX_SA_380 on page 2932	0xf74	IRQ_TYPE_STAT_RX_SA_380
IRQ_TYPE_EN_RX_SA_381 on page 2935	0xf78	IRQ_TYPE_EN_RX_SA_381
IRQ_TYPE_STAT_RX_SA_381 on page 2938	0xf7c	IRQ_TYPE_STAT_RX_SA_381
IRQ_TYPE_EN_RX_SA_382 on page 2941	0xf80	IRQ_TYPE_EN_RX_SA_382
IRQ_TYPE_STAT_RX_SA_382 on page 2944	0xf84	IRQ_TYPE_STAT_RX_SA_382
IRQ_TYPE_EN_RX_SA_383 on page 2947	0xf88	IRQ_TYPE_EN_RX_SA_383
IRQ_TYPE_STAT_RX_SA_383 on page 2950	0xf8c	IRQ_TYPE_STAT_RX_SA_383
IRQ_TYPE_EN_RX_SA_384 on page 2953	0xf90	IRQ_TYPE_EN_RX_SA_384

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_384 on page 2956	0xf94	IRQ_TYPE_STAT_RX_SA_384
IRQ_TYPE_EN_RX_SA_385 on page 2959	0xf98	IRQ_TYPE_EN_RX_SA_385
IRQ_TYPE_STAT_RX_SA_385 on page 2962	0xf9c	IRQ_TYPE_STAT_RX_SA_385
IRQ_TYPE_EN_RX_SA_386 on page 2965	0xfa0	IRQ_TYPE_EN_RX_SA_386
IRQ_TYPE_STAT_RX_SA_386 on page 2968	0xfa4	IRQ_TYPE_STAT_RX_SA_386
IRQ_TYPE_EN_RX_SA_387 on page 2971	0xfa8	IRQ_TYPE_EN_RX_SA_387
IRQ_TYPE_STAT_RX_SA_387 on page 2974	0xfac	IRQ_TYPE_STAT_RX_SA_387
IRQ_TYPE_EN_RX_SA_388 on page 2977	0xfb0	IRQ_TYPE_EN_RX_SA_388
IRQ_TYPE_STAT_RX_SA_388 on page 2980	0xfb4	IRQ_TYPE_STAT_RX_SA_388
IRQ_TYPE_EN_RX_SA_389 on page 2983	0xfb8	IRQ_TYPE_EN_RX_SA_389
IRQ_TYPE_STAT_RX_SA_389 on page 2986	0xfbc	IRQ_TYPE_STAT_RX_SA_389
IRQ_TYPE_EN_RX_SA_390 on page 2989	0xfc0	IRQ_TYPE_EN_RX_SA_390
IRQ_TYPE_STAT_RX_SA_390 on page 2992	0xfc4	IRQ_TYPE_STAT_RX_SA_390
IRQ_TYPE_EN_RX_SA_391 on page 2995	0xfc8	IRQ_TYPE_EN_RX_SA_391
IRQ_TYPE_STAT_RX_SA_391 on page 2998	0xfcc	IRQ_TYPE_STAT_RX_SA_391
IRQ_TYPE_EN_RX_SA_392 on page 3001	0xfd0	IRQ_TYPE_EN_RX_SA_392
IRQ_TYPE_STAT_RX_SA_392 on page 3004	0xfd4	IRQ_TYPE_STAT_RX_SA_392
IRQ_TYPE_EN_RX_SA_393 on page 3007	0xfd8	IRQ_TYPE_EN_RX_SA_393

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_393 on page 3010	0xfdc	IRQ_TYPE_STAT_RX_SA_393
IRQ_TYPE_EN_RX_SA_394 on page 3013	0xfe0	IRQ_TYPE_EN_RX_SA_394
IRQ_TYPE_STAT_RX_SA_394 on page 3016	0xfe4	IRQ_TYPE_STAT_RX_SA_394
IRQ_TYPE_EN_RX_SA_395 on page 3019	0xfe8	IRQ_TYPE_EN_RX_SA_395
IRQ_TYPE_STAT_RX_SA_395 on page 3022	0xfec	IRQ_TYPE_STAT_RX_SA_395
IRQ_TYPE_EN_RX_SA_396 on page 3025	0xff0	IRQ_TYPE_EN_RX_SA_396
IRQ_TYPE_STAT_RX_SA_396 on page 3028	0xff4	IRQ_TYPE_STAT_RX_SA_396
IRQ_TYPE_EN_RX_SA_397 on page 3031	0xff8	IRQ_TYPE_EN_RX_SA_397
IRQ_TYPE_STAT_RX_SA_397 on page 3034	0xffc	IRQ_TYPE_STAT_RX_SA_397
IRQ_TYPE_EN_RX_SA_398 on page 3037	0x1000	IRQ_TYPE_EN_RX_SA_398
IRQ_TYPE_STAT_RX_SA_398 on page 3040	0x1004	IRQ_TYPE_STAT_RX_SA_398
IRQ_TYPE_EN_RX_SA_399 on page 3043	0x1008	IRQ_TYPE_EN_RX_SA_399
IRQ_TYPE_STAT_RX_SA_399 on page 3046	0x100c	IRQ_TYPE_STAT_RX_SA_399
IRQ_TYPE_EN_RX_SA_400 on page 3049	0x1010	IRQ_TYPE_EN_RX_SA_400
IRQ_TYPE_STAT_RX_SA_400 on page 3052	0x1014	IRQ_TYPE_STAT_RX_SA_400
IRQ_TYPE_EN_RX_SA_401 on page 3055	0x1018	IRQ_TYPE_EN_RX_SA_401
IRQ_TYPE_STAT_RX_SA_401 on page 3058	0x101c	IRQ_TYPE_STAT_RX_SA_401
IRQ_TYPE_EN_RX_SA_402 on page 3061	0x1020	IRQ_TYPE_EN_RX_SA_402

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_402 on page 3064	0x1024	IRQ_TYPE_STAT_RX_SA_402
IRQ_TYPE_EN_RX_SA_403 on page 3067	0x1028	IRQ_TYPE_EN_RX_SA_403
IRQ_TYPE_STAT_RX_SA_403 on page 3070	0x102c	IRQ_TYPE_STAT_RX_SA_403
IRQ_TYPE_EN_RX_SA_404 on page 3073	0x1030	IRQ_TYPE_EN_RX_SA_404
IRQ_TYPE_STAT_RX_SA_404 on page 3076	0x1034	IRQ_TYPE_STAT_RX_SA_404
IRQ_TYPE_EN_RX_SA_405 on page 3079	0x1038	IRQ_TYPE_EN_RX_SA_405
IRQ_TYPE_STAT_RX_SA_405 on page 3082	0x103c	IRQ_TYPE_STAT_RX_SA_405
IRQ_TYPE_EN_RX_SA_406 on page 3085	0x1040	IRQ_TYPE_EN_RX_SA_406
IRQ_TYPE_STAT_RX_SA_406 on page 3088	0x1044	IRQ_TYPE_STAT_RX_SA_406
IRQ_TYPE_EN_RX_SA_407 on page 3091	0x1048	IRQ_TYPE_EN_RX_SA_407
IRQ_TYPE_STAT_RX_SA_407 on page 3094	0x104c	IRQ_TYPE_STAT_RX_SA_407
IRQ_TYPE_EN_RX_SA_408 on page 3097	0x1050	IRQ_TYPE_EN_RX_SA_408
IRQ_TYPE_STAT_RX_SA_408 on page 3100	0x1054	IRQ_TYPE_STAT_RX_SA_408
IRQ_TYPE_EN_RX_SA_409 on page 3103	0x1058	IRQ_TYPE_EN_RX_SA_409
IRQ_TYPE_STAT_RX_SA_409 on page 3106	0x105c	IRQ_TYPE_STAT_RX_SA_409
IRQ_TYPE_EN_RX_SA_410 on page 3109	0x1060	IRQ_TYPE_EN_RX_SA_410
IRQ_TYPE_STAT_RX_SA_410 on page 3112	0x1064	IRQ_TYPE_STAT_RX_SA_410
IRQ_TYPE_EN_RX_SA_411 on page 3115	0x1068	IRQ_TYPE_EN_RX_SA_411

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_411 on page 3118	0x106c	IRQ_TYPE_STAT_RX_SA_411
IRQ_TYPE_EN_RX_SA_412 on page 3121	0x1070	IRQ_TYPE_EN_RX_SA_412
IRQ_TYPE_STAT_RX_SA_412 on page 3124	0x1074	IRQ_TYPE_STAT_RX_SA_412
IRQ_TYPE_EN_RX_SA_413 on page 3127	0x1078	IRQ_TYPE_EN_RX_SA_413
IRQ_TYPE_STAT_RX_SA_413 on page 3130	0x107c	IRQ_TYPE_STAT_RX_SA_413
IRQ_TYPE_EN_RX_SA_414 on page 3133	0x1080	IRQ_TYPE_EN_RX_SA_414
IRQ_TYPE_STAT_RX_SA_414 on page 3136	0x1084	IRQ_TYPE_STAT_RX_SA_414
IRQ_TYPE_EN_RX_SA_415 on page 3139	0x1088	IRQ_TYPE_EN_RX_SA_415
IRQ_TYPE_STAT_RX_SA_415 on page 3142	0x108c	IRQ_TYPE_STAT_RX_SA_415
IRQ_TYPE_EN_RX_SA_416 on page 3145	0x1090	IRQ_TYPE_EN_RX_SA_416
IRQ_TYPE_STAT_RX_SA_416 on page 3148	0x1094	IRQ_TYPE_STAT_RX_SA_416
IRQ_TYPE_EN_RX_SA_417 on page 3151	0x1098	IRQ_TYPE_EN_RX_SA_417
IRQ_TYPE_STAT_RX_SA_417 on page 3154	0x109c	IRQ_TYPE_STAT_RX_SA_417
IRQ_TYPE_EN_RX_SA_418 on page 3157	0x10a0	IRQ_TYPE_EN_RX_SA_418
IRQ_TYPE_STAT_RX_SA_418 on page 3160	0x10a4	IRQ_TYPE_STAT_RX_SA_418
IRQ_TYPE_EN_RX_SA_419 on page 3163	0x10a8	IRQ_TYPE_EN_RX_SA_419
IRQ_TYPE_STAT_RX_SA_419 on page 3166	0x10ac	IRQ_TYPE_STAT_RX_SA_419
IRQ_TYPE_EN_RX_SA_420 on page 3169	0x10b0	IRQ_TYPE_EN_RX_SA_420

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_420 on page 3172	0x10b4	IRQ_TYPE_STAT_RX_SA_420
IRQ_TYPE_EN_RX_SA_421 on page 3175	0x10b8	IRQ_TYPE_EN_RX_SA_421
IRQ_TYPE_STAT_RX_SA_421 on page 3178	0x10bc	IRQ_TYPE_STAT_RX_SA_421
IRQ_TYPE_EN_RX_SA_422 on page 3181	0x10c0	IRQ_TYPE_EN_RX_SA_422
IRQ_TYPE_STAT_RX_SA_422 on page 3184	0x10c4	IRQ_TYPE_STAT_RX_SA_422
IRQ_TYPE_EN_RX_SA_423 on page 3187	0x10c8	IRQ_TYPE_EN_RX_SA_423
IRQ_TYPE_STAT_RX_SA_423 on page 3190	0x10cc	IRQ_TYPE_STAT_RX_SA_423
IRQ_TYPE_EN_RX_SA_424 on page 3193	0x10d0	IRQ_TYPE_EN_RX_SA_424
IRQ_TYPE_STAT_RX_SA_424 on page 3196	0x10d4	IRQ_TYPE_STAT_RX_SA_424
IRQ_TYPE_EN_RX_SA_425 on page 3199	0x10d8	IRQ_TYPE_EN_RX_SA_425
IRQ_TYPE_STAT_RX_SA_425 on page 3202	0x10dc	IRQ_TYPE_STAT_RX_SA_425
IRQ_TYPE_EN_RX_SA_426 on page 3205	0x10e0	IRQ_TYPE_EN_RX_SA_426
IRQ_TYPE_STAT_RX_SA_426 on page 3208	0x10e4	IRQ_TYPE_STAT_RX_SA_426
IRQ_TYPE_EN_RX_SA_427 on page 3211	0x10e8	IRQ_TYPE_EN_RX_SA_427
IRQ_TYPE_STAT_RX_SA_427 on page 3214	0x10ec	IRQ_TYPE_STAT_RX_SA_427
IRQ_TYPE_EN_RX_SA_428 on page 3217	0x10f0	IRQ_TYPE_EN_RX_SA_428
IRQ_TYPE_STAT_RX_SA_428 on page 3220	0x10f4	IRQ_TYPE_STAT_RX_SA_428
IRQ_TYPE_EN_RX_SA_429 on page 3223	0x10f8	IRQ_TYPE_EN_RX_SA_429

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_429 on page 3226	0x10fc	IRQ_TYPE_STAT_RX_SA_429
IRQ_TYPE_EN_RX_SA_430 on page 3229	0x1100	IRQ_TYPE_EN_RX_SA_430
IRQ_TYPE_STAT_RX_SA_430 on page 3232	0x1104	IRQ_TYPE_STAT_RX_SA_430
IRQ_TYPE_EN_RX_SA_431 on page 3235	0x1108	IRQ_TYPE_EN_RX_SA_431
IRQ_TYPE_STAT_RX_SA_431 on page 3238	0x110c	IRQ_TYPE_STAT_RX_SA_431
IRQ_TYPE_EN_RX_SA_432 on page 3241	0x1110	IRQ_TYPE_EN_RX_SA_432
IRQ_TYPE_STAT_RX_SA_432 on page 3244	0x1114	IRQ_TYPE_STAT_RX_SA_432
IRQ_TYPE_EN_RX_SA_433 on page 3247	0x1118	IRQ_TYPE_EN_RX_SA_433
IRQ_TYPE_STAT_RX_SA_433 on page 3250	0x111c	IRQ_TYPE_STAT_RX_SA_433
IRQ_TYPE_EN_RX_SA_434 on page 3253	0x1120	IRQ_TYPE_EN_RX_SA_434
IRQ_TYPE_STAT_RX_SA_434 on page 3256	0x1124	IRQ_TYPE_STAT_RX_SA_434
IRQ_TYPE_EN_RX_SA_435 on page 3259	0x1128	IRQ_TYPE_EN_RX_SA_435
IRQ_TYPE_STAT_RX_SA_435 on page 3262	0x112c	IRQ_TYPE_STAT_RX_SA_435
IRQ_TYPE_EN_RX_SA_436 on page 3265	0x1130	IRQ_TYPE_EN_RX_SA_436
IRQ_TYPE_STAT_RX_SA_436 on page 3268	0x1134	IRQ_TYPE_STAT_RX_SA_436
IRQ_TYPE_EN_RX_SA_437 on page 3271	0x1138	IRQ_TYPE_EN_RX_SA_437
IRQ_TYPE_STAT_RX_SA_437 on page 3274	0x113c	IRQ_TYPE_STAT_RX_SA_437
IRQ_TYPE_EN_RX_SA_438 on page 3277	0x1140	IRQ_TYPE_EN_RX_SA_438

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_438 on page 3280	0x1144	IRQ_TYPE_STAT_RX_SA_438
IRQ_TYPE_EN_RX_SA_439 on page 3283	0x1148	IRQ_TYPE_EN_RX_SA_439
IRQ_TYPE_STAT_RX_SA_439 on page 3286	0x114c	IRQ_TYPE_STAT_RX_SA_439
IRQ_TYPE_EN_RX_SA_440 on page 3289	0x1150	IRQ_TYPE_EN_RX_SA_440
IRQ_TYPE_STAT_RX_SA_440 on page 3292	0x1154	IRQ_TYPE_STAT_RX_SA_440
IRQ_TYPE_EN_RX_SA_441 on page 3295	0x1158	IRQ_TYPE_EN_RX_SA_441
IRQ_TYPE_STAT_RX_SA_441 on page 3298	0x115c	IRQ_TYPE_STAT_RX_SA_441
IRQ_TYPE_EN_RX_SA_442 on page 3301	0x1160	IRQ_TYPE_EN_RX_SA_442
IRQ_TYPE_STAT_RX_SA_442 on page 3304	0x1164	IRQ_TYPE_STAT_RX_SA_442
IRQ_TYPE_EN_RX_SA_443 on page 3307	0x1168	IRQ_TYPE_EN_RX_SA_443
IRQ_TYPE_STAT_RX_SA_443 on page 3310	0x116c	IRQ_TYPE_STAT_RX_SA_443
IRQ_TYPE_EN_RX_SA_444 on page 3313	0x1170	IRQ_TYPE_EN_RX_SA_444
IRQ_TYPE_STAT_RX_SA_444 on page 3316	0x1174	IRQ_TYPE_STAT_RX_SA_444
IRQ_TYPE_EN_RX_SA_445 on page 3319	0x1178	IRQ_TYPE_EN_RX_SA_445
IRQ_TYPE_STAT_RX_SA_445 on page 3322	0x117c	IRQ_TYPE_STAT_RX_SA_445
IRQ_TYPE_EN_RX_SA_446 on page 3325	0x1180	IRQ_TYPE_EN_RX_SA_446
IRQ_TYPE_STAT_RX_SA_446 on page 3328	0x1184	IRQ_TYPE_STAT_RX_SA_446
IRQ_TYPE_EN_RX_SA_447 on page 3331	0x1188	IRQ_TYPE_EN_RX_SA_447

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_447 on page 3334	0x118c	IRQ_TYPE_STAT_RX_SA_447
IRQ_TYPE_EN_RX_SA_448 on page 3337	0x1190	IRQ_TYPE_EN_RX_SA_448
IRQ_TYPE_STAT_RX_SA_448 on page 3340	0x1194	IRQ_TYPE_STAT_RX_SA_448
IRQ_TYPE_EN_RX_SA_449 on page 3343	0x1198	IRQ_TYPE_EN_RX_SA_449
IRQ_TYPE_STAT_RX_SA_449 on page 3346	0x119c	IRQ_TYPE_STAT_RX_SA_449
IRQ_TYPE_EN_RX_SA_450 on page 3349	0x11a0	IRQ_TYPE_EN_RX_SA_450
IRQ_TYPE_STAT_RX_SA_450 on page 3352	0x11a4	IRQ_TYPE_STAT_RX_SA_450
IRQ_TYPE_EN_RX_SA_451 on page 3355	0x11a8	IRQ_TYPE_EN_RX_SA_451
IRQ_TYPE_STAT_RX_SA_451 on page 3358	0x11ac	IRQ_TYPE_STAT_RX_SA_451
IRQ_TYPE_EN_RX_SA_452 on page 3361	0x11b0	IRQ_TYPE_EN_RX_SA_452
IRQ_TYPE_STAT_RX_SA_452 on page 3364	0x11b4	IRQ_TYPE_STAT_RX_SA_452
IRQ_TYPE_EN_RX_SA_453 on page 3367	0x11b8	IRQ_TYPE_EN_RX_SA_453
IRQ_TYPE_STAT_RX_SA_453 on page 3370	0x11bc	IRQ_TYPE_STAT_RX_SA_453
IRQ_TYPE_EN_RX_SA_454 on page 3373	0x11c0	IRQ_TYPE_EN_RX_SA_454
IRQ_TYPE_STAT_RX_SA_454 on page 3376	0x11c4	IRQ_TYPE_STAT_RX_SA_454
IRQ_TYPE_EN_RX_SA_455 on page 3379	0x11c8	IRQ_TYPE_EN_RX_SA_455
IRQ_TYPE_STAT_RX_SA_455 on page 3382	0x11cc	IRQ_TYPE_STAT_RX_SA_455
IRQ_TYPE_EN_RX_SA_456 on page 3385	0x11d0	IRQ_TYPE_EN_RX_SA_456

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_456 on page 3388	0x11d4	IRQ_TYPE_STAT_RX_SA_456
IRQ_TYPE_EN_RX_SA_457 on page 3391	0x11d8	IRQ_TYPE_EN_RX_SA_457
IRQ_TYPE_STAT_RX_SA_457 on page 3394	0x11dc	IRQ_TYPE_STAT_RX_SA_457
IRQ_TYPE_EN_RX_SA_458 on page 3397	0x11e0	IRQ_TYPE_EN_RX_SA_458
IRQ_TYPE_STAT_RX_SA_458 on page 3400	0x11e4	IRQ_TYPE_STAT_RX_SA_458
IRQ_TYPE_EN_RX_SA_459 on page 3403	0x11e8	IRQ_TYPE_EN_RX_SA_459
IRQ_TYPE_STAT_RX_SA_459 on page 3406	0x11ec	IRQ_TYPE_STAT_RX_SA_459
IRQ_TYPE_EN_RX_SA_460 on page 3409	0x11f0	IRQ_TYPE_EN_RX_SA_460
IRQ_TYPE_STAT_RX_SA_460 on page 3412	0x11f4	IRQ_TYPE_STAT_RX_SA_460
IRQ_TYPE_EN_RX_SA_461 on page 3415	0x11f8	IRQ_TYPE_EN_RX_SA_461
IRQ_TYPE_STAT_RX_SA_461 on page 3418	0x11fc	IRQ_TYPE_STAT_RX_SA_461
IRQ_TYPE_EN_RX_SA_462 on page 3421	0x1200	IRQ_TYPE_EN_RX_SA_462
IRQ_TYPE_STAT_RX_SA_462 on page 3424	0x1204	IRQ_TYPE_STAT_RX_SA_462
IRQ_TYPE_EN_RX_SA_463 on page 3427	0x1208	IRQ_TYPE_EN_RX_SA_463
IRQ_TYPE_STAT_RX_SA_463 on page 3430	0x120c	IRQ_TYPE_STAT_RX_SA_463
IRQ_TYPE_EN_RX_SA_464 on page 3433	0x1210	IRQ_TYPE_EN_RX_SA_464
IRQ_TYPE_STAT_RX_SA_464 on page 3436	0x1214	IRQ_TYPE_STAT_RX_SA_464
IRQ_TYPE_EN_RX_SA_465 on page 3439	0x1218	IRQ_TYPE_EN_RX_SA_465

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_465 on page 3442	0x121c	IRQ_TYPE_STAT_RX_SA_465
IRQ_TYPE_EN_RX_SA_466 on page 3445	0x1220	IRQ_TYPE_EN_RX_SA_466
IRQ_TYPE_STAT_RX_SA_466 on page 3448	0x1224	IRQ_TYPE_STAT_RX_SA_466
IRQ_TYPE_EN_RX_SA_467 on page 3451	0x1228	IRQ_TYPE_EN_RX_SA_467
IRQ_TYPE_STAT_RX_SA_467 on page 3454	0x122c	IRQ_TYPE_STAT_RX_SA_467
IRQ_TYPE_EN_RX_SA_468 on page 3457	0x1230	IRQ_TYPE_EN_RX_SA_468
IRQ_TYPE_STAT_RX_SA_468 on page 3460	0x1234	IRQ_TYPE_STAT_RX_SA_468
IRQ_TYPE_EN_RX_SA_469 on page 3463	0x1238	IRQ_TYPE_EN_RX_SA_469
IRQ_TYPE_STAT_RX_SA_469 on page 3466	0x123c	IRQ_TYPE_STAT_RX_SA_469
IRQ_TYPE_EN_RX_SA_470 on page 3469	0x1240	IRQ_TYPE_EN_RX_SA_470
IRQ_TYPE_STAT_RX_SA_470 on page 3472	0x1244	IRQ_TYPE_STAT_RX_SA_470
IRQ_TYPE_EN_RX_SA_471 on page 3475	0x1248	IRQ_TYPE_EN_RX_SA_471
IRQ_TYPE_STAT_RX_SA_471 on page 3478	0x124c	IRQ_TYPE_STAT_RX_SA_471
IRQ_TYPE_EN_RX_SA_472 on page 3481	0x1250	IRQ_TYPE_EN_RX_SA_472
IRQ_TYPE_STAT_RX_SA_472 on page 3484	0x1254	IRQ_TYPE_STAT_RX_SA_472
IRQ_TYPE_EN_RX_SA_473 on page 3487	0x1258	IRQ_TYPE_EN_RX_SA_473
IRQ_TYPE_STAT_RX_SA_473 on page 3490	0x125c	IRQ_TYPE_STAT_RX_SA_473
IRQ_TYPE_EN_RX_SA_474 on page 3493	0x1260	IRQ_TYPE_EN_RX_SA_474

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_474 on page 3496	0x1264	IRQ_TYPE_STAT_RX_SA_474
IRQ_TYPE_EN_RX_SA_475 on page 3499	0x1268	IRQ_TYPE_EN_RX_SA_475
IRQ_TYPE_STAT_RX_SA_475 on page 3502	0x126c	IRQ_TYPE_STAT_RX_SA_475
IRQ_TYPE_EN_RX_SA_476 on page 3505	0x1270	IRQ_TYPE_EN_RX_SA_476
IRQ_TYPE_STAT_RX_SA_476 on page 3508	0x1274	IRQ_TYPE_STAT_RX_SA_476
IRQ_TYPE_EN_RX_SA_477 on page 3511	0x1278	IRQ_TYPE_EN_RX_SA_477
IRQ_TYPE_STAT_RX_SA_477 on page 3514	0x127c	IRQ_TYPE_STAT_RX_SA_477
IRQ_TYPE_EN_RX_SA_478 on page 3517	0x1280	IRQ_TYPE_EN_RX_SA_478
IRQ_TYPE_STAT_RX_SA_478 on page 3520	0x1284	IRQ_TYPE_STAT_RX_SA_478
IRQ_TYPE_EN_RX_SA_479 on page 3523	0x1288	IRQ_TYPE_EN_RX_SA_479
IRQ_TYPE_STAT_RX_SA_479 on page 3526	0x128c	IRQ_TYPE_STAT_RX_SA_479
IRQ_TYPE_EN_RX_SA_480 on page 3529	0x1290	IRQ_TYPE_EN_RX_SA_480
IRQ_TYPE_STAT_RX_SA_480 on page 3532	0x1294	IRQ_TYPE_STAT_RX_SA_480
IRQ_TYPE_EN_RX_SA_481 on page 3535	0x1298	IRQ_TYPE_EN_RX_SA_481
IRQ_TYPE_STAT_RX_SA_481 on page 3538	0x129c	IRQ_TYPE_STAT_RX_SA_481
IRQ_TYPE_EN_RX_SA_482 on page 3541	0x12a0	IRQ_TYPE_EN_RX_SA_482
IRQ_TYPE_STAT_RX_SA_482 on page 3544	0x12a4	IRQ_TYPE_STAT_RX_SA_482
IRQ_TYPE_EN_RX_SA_483 on page 3547	0x12a8	IRQ_TYPE_EN_RX_SA_483

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_483 on page 3550	0x12ac	IRQ_TYPE_STAT_RX_SA_483
IRQ_TYPE_EN_RX_SA_484 on page 3553	0x12b0	IRQ_TYPE_EN_RX_SA_484
IRQ_TYPE_STAT_RX_SA_484 on page 3556	0x12b4	IRQ_TYPE_STAT_RX_SA_484
IRQ_TYPE_EN_RX_SA_485 on page 3559	0x12b8	IRQ_TYPE_EN_RX_SA_485
IRQ_TYPE_STAT_RX_SA_485 on page 3562	0x12bc	IRQ_TYPE_STAT_RX_SA_485
IRQ_TYPE_EN_RX_SA_486 on page 3565	0x12c0	IRQ_TYPE_EN_RX_SA_486
IRQ_TYPE_STAT_RX_SA_486 on page 3568	0x12c4	IRQ_TYPE_STAT_RX_SA_486
IRQ_TYPE_EN_RX_SA_487 on page 3571	0x12c8	IRQ_TYPE_EN_RX_SA_487
IRQ_TYPE_STAT_RX_SA_487 on page 3574	0x12cc	IRQ_TYPE_STAT_RX_SA_487
IRQ_TYPE_EN_RX_SA_488 on page 3577	0x12d0	IRQ_TYPE_EN_RX_SA_488
IRQ_TYPE_STAT_RX_SA_488 on page 3580	0x12d4	IRQ_TYPE_STAT_RX_SA_488
IRQ_TYPE_EN_RX_SA_489 on page 3583	0x12d8	IRQ_TYPE_EN_RX_SA_489
IRQ_TYPE_STAT_RX_SA_489 on page 3586	0x12dc	IRQ_TYPE_STAT_RX_SA_489
IRQ_TYPE_EN_RX_SA_490 on page 3589	0x12e0	IRQ_TYPE_EN_RX_SA_490
IRQ_TYPE_STAT_RX_SA_490 on page 3592	0x12e4	IRQ_TYPE_STAT_RX_SA_490
IRQ_TYPE_EN_RX_SA_491 on page 3595	0x12e8	IRQ_TYPE_EN_RX_SA_491
IRQ_TYPE_STAT_RX_SA_491 on page 3598	0x12ec	IRQ_TYPE_STAT_RX_SA_491
IRQ_TYPE_EN_RX_SA_492 on page 3601	0x12f0	IRQ_TYPE_EN_RX_SA_492

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_492 on page 3604	0x12f4	IRQ_TYPE_STAT_RX_SA_492
IRQ_TYPE_EN_RX_SA_493 on page 3607	0x12f8	IRQ_TYPE_EN_RX_SA_493
IRQ_TYPE_STAT_RX_SA_493 on page 3610	0x12fc	IRQ_TYPE_STAT_RX_SA_493
IRQ_TYPE_EN_RX_SA_494 on page 3613	0x1300	IRQ_TYPE_EN_RX_SA_494
IRQ_TYPE_STAT_RX_SA_494 on page 3616	0x1304	IRQ_TYPE_STAT_RX_SA_494
IRQ_TYPE_EN_RX_SA_495 on page 3619	0x1308	IRQ_TYPE_EN_RX_SA_495
IRQ_TYPE_STAT_RX_SA_495 on page 3622	0x130c	IRQ_TYPE_STAT_RX_SA_495
IRQ_TYPE_EN_RX_SA_496 on page 3625	0x1310	IRQ_TYPE_EN_RX_SA_496
IRQ_TYPE_STAT_RX_SA_496 on page 3628	0x1314	IRQ_TYPE_STAT_RX_SA_496
IRQ_TYPE_EN_RX_SA_497 on page 3631	0x1318	IRQ_TYPE_EN_RX_SA_497
IRQ_TYPE_STAT_RX_SA_497 on page 3634	0x131c	IRQ_TYPE_STAT_RX_SA_497
IRQ_TYPE_EN_RX_SA_498 on page 3637	0x1320	IRQ_TYPE_EN_RX_SA_498
IRQ_TYPE_STAT_RX_SA_498 on page 3640	0x1324	IRQ_TYPE_STAT_RX_SA_498
IRQ_TYPE_EN_RX_SA_499 on page 3643	0x1328	IRQ_TYPE_EN_RX_SA_499
IRQ_TYPE_STAT_RX_SA_499 on page 3646	0x132c	IRQ_TYPE_STAT_RX_SA_499
IRQ_TYPE_EN_RX_SA_500 on page 3649	0x1330	IRQ_TYPE_EN_RX_SA_500
IRQ_TYPE_STAT_RX_SA_500 on page 3652	0x1334	IRQ_TYPE_STAT_RX_SA_500
IRQ_TYPE_EN_RX_SA_501 on page 3655	0x1338	IRQ_TYPE_EN_RX_SA_501

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_501 on page 3658	0x133c	IRQ_TYPE_STAT_RX_SA_501
IRQ_TYPE_EN_RX_SA_502 on page 3661	0x1340	IRQ_TYPE_EN_RX_SA_502
IRQ_TYPE_STAT_RX_SA_502 on page 3664	0x1344	IRQ_TYPE_STAT_RX_SA_502
IRQ_TYPE_EN_RX_SA_503 on page 3667	0x1348	IRQ_TYPE_EN_RX_SA_503
IRQ_TYPE_STAT_RX_SA_503 on page 3670	0x134c	IRQ_TYPE_STAT_RX_SA_503
IRQ_TYPE_EN_RX_SA_504 on page 3673	0x1350	IRQ_TYPE_EN_RX_SA_504
IRQ_TYPE_STAT_RX_SA_504 on page 3676	0x1354	IRQ_TYPE_STAT_RX_SA_504
IRQ_TYPE_EN_RX_SA_505 on page 3679	0x1358	IRQ_TYPE_EN_RX_SA_505
IRQ_TYPE_STAT_RX_SA_505 on page 3682	0x135c	IRQ_TYPE_STAT_RX_SA_505
IRQ_TYPE_EN_RX_SA_506 on page 3685	0x1360	IRQ_TYPE_EN_RX_SA_506
IRQ_TYPE_STAT_RX_SA_506 on page 3688	0x1364	IRQ_TYPE_STAT_RX_SA_506
IRQ_TYPE_EN_RX_SA_507 on page 3691	0x1368	IRQ_TYPE_EN_RX_SA_507
IRQ_TYPE_STAT_RX_SA_507 on page 3694	0x136c	IRQ_TYPE_STAT_RX_SA_507
IRQ_TYPE_EN_RX_SA_508 on page 3697	0x1370	IRQ_TYPE_EN_RX_SA_508
IRQ_TYPE_STAT_RX_SA_508 on page 3700	0x1374	IRQ_TYPE_STAT_RX_SA_508
IRQ_TYPE_EN_RX_SA_509 on page 3703	0x1378	IRQ_TYPE_EN_RX_SA_509
IRQ_TYPE_STAT_RX_SA_509 on page 3706	0x137c	IRQ_TYPE_STAT_RX_SA_509
IRQ_TYPE_EN_RX_SA_510 on page 3709	0x1380	IRQ_TYPE_EN_RX_SA_510

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
IRQ_TYPE_STAT_RX_SA_510 on page 3712	0x1384	IRQ_TYPE_STAT_RX_SA_510
IRQ_TYPE_EN_RX_SA_511 on page 3715	0x1388	IRQ_TYPE_EN_RX_SA_511
IRQ_TYPE_STAT_RX_SA_511 on page 3718	0x138c	IRQ_TYPE_STAT_RX_SA_511
MACSEC_SW_INFO on page 3721	0x1390	MACSec SW INFO
IRQ_TYPE_EN_SRAM_ECC on page 3724	0x1394	IRQ_TYPE_EN_SRAM_ECC
IRQ_TYPE_STAT_SRAM_ECC on page 3731	0x1398	IRQ_TYPE_STAT_SRAM_ECC
MACSEC_SAFETY_ALARM_SOURCES_0 on page 3738	0x2000	MACSEC Safety Alarm Sources Register bank 0.
MACSEC_SAFETY_ALARM_SOURCES_1 on page 3745	0x2004	MACSEC Safety Alarm Sources Register bank 1.
MACSEC_SAFETY_ALARM_SOURCES_2 on page 3753	0x2008	MACSEC Safety Alarm Sources Register bank 2.
MACSEC_SAFETY_ALARM_SOURCES_3 on page 3761	0x200c	MACSEC Safety Alarm Sources Register bank 3.
MACSEC_REGBANK_ERROR_INJECTION on page 3766	0x2010	MACsec Register Bank Injection Register.
MACSEC_SRAM_ECC_ERROR_INJECTION on page 3769	0x2014	MACsec SRAM Injection Register.
MACSEC_APB_PARITY_ERROR_INJECTION on page 3772	0x2018	MACsec APB Injection Register.
MACSEC_DMR_ERROR_INJECTION on page 3775	0x201c	MACsec DMR Injection Register.
MACSEC_TOP_BCM_WRAPPER_PARITY_ERROR_INJECTION on page 3787	0x2028	MACsec Top BCM Wrapper FIFOs Error Injection Register.
MACSEC_TX_IN_TO_GMII_PARITY_ERROR_INJECTION on page 3789	0x202c	MACsec TX RMII/MII/RGMII to GMII ingress bridges Error Injection Register.
MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_0 on page 3792	0x2030	MACsec RX RMII/MII to GMII ingress bridges Error Injection Register 0.
MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_1 on page 3796	0x2034	MACsec RX RGMII to GMII ingress bridges Error Injection Register 1.

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
MACSEC_TX_OUT_FROM_GMII_PARITY_ERROR_INJECTION on page 3798	0x2038	MACsec TX GMII to RMII/MII/RGMII egress bridges Error Injection Register.
MACSEC_RX_OUT_FROM_GMII_PARITY_ERROR_INJECTION on page 3802	0x203c	MACsec RX GMII to RMII/MII/RGMII egress bridges Error Injection Register.
MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJECTION on page 3806	0x2040	MACsec TX GMII ingress Error Injection Register.
MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJECTION on page 3810	0x2044	MACsec RX GMII ingress Error Injection Register.
MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJECTION on page 3814	0x2048	MACsec TX GMII egress Error Injection Register.
MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJECTION on page 3817	0x204c	MACsec RX GMII egress Error Injection Register.
MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJECTION on page 3820	0x2050	MACsec TX XGMII ingress Error Injection Register.
MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJECTION on page 3823	0x2054	MACsec RX XGMII ingress Error Injection Register.
MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJECTION on page 3826	0x2058	MACsec TX XGMII egress Error Injection Register.
MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJECTION on page 3828	0x205c	MACsec RX XGMII egress Error Injection Register.
MACSEC_MAC_ERROR_INJ_EN on page 3830	0x2060	MACSEC_MAC_ERROR_INJ_EN
MACSEC_CRC_ERROR_INJECTION on page 3832	0x2064	MACsec CRC Error Injection Register.
MACSEC_TX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION on page 3836	0x2068	MACsec TX ingress VLAN controller Error Injection Register.
MACSEC_RX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION on page 3838	0x206c	MACsec RX ingress VLAN controller Error Injection Register.
MACSEC_TX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION on page 3840	0x2070	MACsec TX egress VLAN controller Error Injection Register.
MACSEC_RX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION on page 3842	0x2074	MACsec RX egress VLAN controller Error Injection Register.

Table 6-7 Registers for Address Block: DWC_macsec_apb/macsec_apb_cfg (Continued)

Register	Offset	Description
MACSEC_RX_PNUP_PARITY_ERROR_INJECTION on page 3844	0x2078	MACSEC_RX_PNUP_PARITY_ERROR_INJECTION.
MACSEC_SAD_INTERNAL_MEM_PARITY_ERR_INJ on page 3846	0x207c	MACSEC_SAD_INTERNAL_MEM_PARITY_ERROR_INJECTION.

6.1.1 CORE_VER_NUM

- **Name:** IP Version Number Register.
- **Description:** IP Version Number.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0x0
- **Exists:** Always

Table 6-8 Fields for Register: CORE_VER_NUM

Bits	Name	Memory Access	Description
31:16	RESERVED_31_16	R	Reserved and read as zero. **Values:** ■ 0x10a (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always
15:0	VERSION_NUM	R	Indicates the coreKit release version number in BCD format. For example: Release number 1.70a is represented as 0x0170. Release number 12.41a is represented as 0x1241. **Values:** ■ 0x105 (VERSION_NUM_REG) **Value After Reset:** DWC_MACSEC_REG_VER_NUM **Exists:** Always

6.1.2 CORE_VER_TYPE

- **Name:** IP Version Type Register.
- **Description:** IP Version Type Register.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0x4
- **Exists:** Always

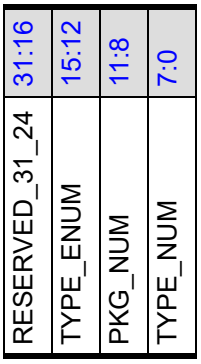


Table 6-9 Fields for Register: CORE_VER_TYPE

Bits	Name	Memory Access	Description
31:16	RESERVED_31_24	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always
15:12	TYPE_ENUM	R	Indicates the coreKit release type. Values, 0x0 (TYPE_ENUM_GA): GA release 0x1 (TYPE_ENUM_LCA): LCA release 0x2 (TYPE_ENUM_EA): EA release 0x3 (TYPE_ENUM_LP): LP release 0x6 (TYPE_ENUM_SOW): SOW release 0x6 (TYPE_ENUM_EC): EC release Others: Reserved **Values:** ■ 0x0 (TYPE_ENUM_REG) **Value After Reset:** DWC_MACSEC_REG_TYPE_ENUM **Exists:** Always

Table 6-9 Fields for Register: CORE_VER_TYPE (Continued)

Bits	Name	Memory Access	Description
11:8	PKG_NUM	R	Indicates the alphabetical packaging number of coreKit ID. Values 4'b1010: 'a' 4'b1011: 'b' 4'b1100: 'c' 4'b1101: 'd' 4'b1110: 'e' 4'b1111: 'f' 4'b0000: 'g' -------- 4'b1000: 'o' 4'b1001: 'p' NOTE: PKG_NUM can have a value from 'a' to 'p' only. **Values:** ■ 0xa (PKG_NUM_REG) **Value After Reset:** DWC_MACSEC_REG_PKG_NUM **Exists:** Always
7:0	TYPE_NUM	R	Indicates the coreKit release version type number in BCD format. For example: Release number `1.70a-lp04` is represented as 0x04. GA releases have a value of 0. **Values:** ■ 0x0 (TYPE_NUM_REG) **Value After Reset:** DWC_MACSEC_REG_TYPE_NUM **Exists:** Always

6.1.3 MACsec_IP_Control

- **Name:** MACsec IP Control Register
- **Description:** MACsec IP Control
- **Size:** 32 bits
- **Offset:** 0x8
- **Exists:** Always

31:29	28	27:12	11	10	9:5	4	3	2	1	0
vlanmode	PNSE_CHECK_EN	Reserved_27_12	kay_frames_drop_en	ethertype_filter_en	data_mode	rx_bypass	tx_bypass	usgmii	Reserved_1	xpnsel

Table 6-10 Fields for Register: MACsec_IP_Control

Bits	Name	Memory Access	Description
31:29	vlanmode	R/W	VLAN Tag modes are encoded as below: 3'b000 : Up to 2 VLAN Tags encrypted/authenticated 3'b001 : Up to 2 VLAN Tags On the Clear 3'b010 : Up to 2 VLAN Tags on the Clear and these VLAN Tags are also encrypted/authenticated 3'b011 : C-TAG is transformed into S-TAG, and C-TAG is encrypted/authenticated 3'b100 : S-TAG copy on the clear and encrypts C-TAG and S-TAG 3'b101 : S-TAG on the clear and C-TAG encrypted/authenticated 3'b110 : S-TAG and C-TAG on the clear and C-TAG encrypted/authenticated 3'b111 : Reserved **Values:** ■ 0x0 (Default): vlanmode **Value After Reset:** 0x0 **Exists:** Always

Table 6-10 Fields for Register: MACsec_IP_Control (Continued)

Bits	Name	Memory Access	Description
28	PNSE_CHECK_EN	R/W	Enables PNSE register access checks on APB5. **Values:** ■ 0x1 (ENABLE_PNSE_CHECK): Enable PNSE register access checks on APB5. ■ 0x0 (DISABLE_PNSE_CHECK): Disable PNSE register access checks on APB5. **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_APB5_RME==1 **Testable:** untestable
27:12	Reserved_27_12	R	Reserved and read-only as zero. **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
11	kay_frames_drop_en	R/W	If this register is enabled, KAY frames coming through the RX path are discarded **Values:** ■ 0x0 (kay_frames_drop_en): kay_frames_drop_en **Value After Reset:** 0x0 **Exists:** Always
10	ethertype_filter_en	R/W	If this register is enabled, frames coming through the RX path with ethertypes equal to one of 8 the programed in the ETHERTYPE_FILTER_* registers are bypassed **Values:** ■ 0x0 (ethertype_filter_en): ethertype_filter_en **Value After Reset:** 0x0 **Exists:** Always
9:5	data_mode	R/W	Value to select the data mode in which the IP is to operate - Please refer to section "Supported Data Modes" of the databook **Values:** ■ 0x0 (data_mode): data_mode **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Table 6-10 Fields for Register: MACsec_IP_Control (Continued)

Bits	Name	Memory Access	Description
4	rx_bypass	R/W	If this register is enabled, frames coming through the RX path are bypassed **Values:** ■ 0x0 (rx_bypass): rx_bypass **Value After Reset:** 0x0 **Exists:** Always
3	tx_bypass	R/W	If this register is enabled, frames coming through the TX path are bypassed **Values:** ■ 0x0 (tx_bypass): tx_bypass **Value After Reset:** 0x0 **Exists:** Always
2	usgmii	R/W	If this register is enabled, usgmii/usxgmii is enabled **Values:** ■ 0x0 (tx_bypass): usgmii **Value After Reset:** 0x0 **Exists:** (((DWC_MACSEC_PERF_RATE_UP_TO_10_EN) && (DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD == 0) && (DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD == 0) && (DWC_MACSEC_TX_SAI_MEM_TYPE==0) && (DWC_MACSEC_RX_SAI_MEM_TYPE==0))==1)
1	Reserved_1	R	Reserved and read-only as zero. **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** (((DWC_MACSEC_PERF_RATE_UP_TO_10_EN) && (DWC_MACSEC_TX_SAI_LOOKUP_SEARCH_METHOD == 0) && (DWC_MACSEC_RX_SAI_LOOKUP_SEARCH_METHOD == 0) && (DWC_MACSEC_TX_SAI_MEM_TYPE==0) && (DWC_MACSEC_RX_SAI_MEM_TYPE==0))==1)

Table 6-10 Fields for Register: MACsec_IP_Control (Continued)

Bits	Name	Memory Access	Description
0	xpnsel	R/W	0 : 32-bit PN (Cipher suite GCM_AES_128 OR GCM_AES_256) 1 : 64-bit PN (Cipher suite GCM_AES_XPN_128 OR GCM_AES_XPN_256) **Values:** ■ 0x0 (xpnsel): xpnsel **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_XPN_SEL==1)

6.1.4 MACSec_CFG_MIN_IPG

- **Name:** MACsec Minimum Inter-Packet Configuration
- **Description:** MACsec Minimum Inter-Packet Gap (IPG) configuration register
- **Size:** 32 bits
- **Offset:** 0xc
- **Exists:** Always

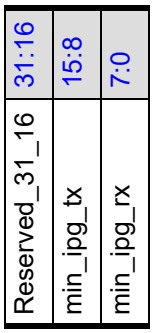


Table 6-11 Fields for Register: MACSec_CFG_MIN_IPG

Bits	Name	Memory Access	Description
31:16	Reserved_31_16	R	Reserved and read-only as zero. **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-11 Fields for Register: MACSec_CFG_MIN_IPG (Continued)

Bits	Name	Memory Access	Description
15:8	min_ipg_tx	R/W	Configuration of Minimum Inter-Packet Gap (IPG) for the TX path. The minimum value to be configured (for force bypass) depends on the selected data mode: ■ Data modes 0, 1, 10-11, 14-20: 5 bytes. ■ Data modes 2, 3, 21: 12 bytes. ■ Data modes 4-9, 12-13: 9 bytes. The minimum value to be configured (for force bypass) depends on the selected data mode: ■ Data modes 0, 1, 10-11, 14-20: 37 bytes (no VLANs), 41 bytes (Single VLAN) or 45 bytes (Dual VLANs). ■ Data modes 2, 3, 21: 44 bytes (no VLANs), 48 bytes (Single VLAN), or 52 bytes (Dual VLANs). ■ Data modes 4-9, 12-13: 41 bytes (no VLANs), 45 bytes (Single VLAN) or 49 bytes (Dual VLANs). **Values:** ■ 0x2c (min_ipg_tx): min_ipg_tx **Value After Reset:** DWC_MACSEC_MIN_IPG_TX **Exists:** Always

Table 6-11 Fields for Register: MACSec_CFG_MIN_IPG (Continued)

Bits	Name	Memory Access	Description
7:0	min_ipg_rx	R/W	Configuration of Minimum Inter-Packet Gap (IPG) for the Rx path. The minimum value to be configured (same values for force bypass or not) depends on the selected data mode: ■ Data modes 0-1, 17-20: 3 bytes (for these modes specifically, though configuration is done with byte resolution, frames with IPGs of 0.5 bytes less than the configured value are also allowed). ■ Data modes 2-9, 12, 13 and 21: 5 bytes. ■ Data modes 10, 11, 14-16: 1 byte. **Values:** ■ 0x5 (min_ipg_rx): min_ipg_rx **Value After Reset:** DWC_MACSEC_MIN_IPG_RX **Exists:** Always

6.1.5 MACsec_IP_VLAN

- **Name:** VLAN Configuration register
- **Description:** VLAN Configuration register 1 and 2
- **Size:** 32 bits
- **Offset:** 0x10
- **Exists:** Always

Table 6-12 Fields for Register: MACsec_IP_VLAN

Bits	Name	Memory Access	Description
31:16	vlantpid2	R	VLAN TPID 2 value IP recognizes the 4-byte field right after MAC SA as VLAN if the TPID value matches any of the values programed in vlantpid1 OR vlantpid2 **Values:** ■ 0x88a8 (vlantpid2): vlantpid2 **Value After Reset:** 0x88a8 **Exists:** Always
15:0	vlantpid1	R	VLAN TPID 1 value IP recognizes the 4-byte field right after MAC SA as VLAN if the TPID value matches any of the values programed in vlantpid1 OR vlantpid2 **Values:** ■ 0x8100 (vlantpid1): vlantpid1 **Value After Reset:** 0x8100 **Exists:** Always

6.1.6 ETHERTYPE_REPLACE

- **Name:** ETHERTYPE_REPLACE Register
- **Description:** ETHERTYPE_REPLACE defines the Ethertype value to replace the original one on a processed Macsec frame. This happens when the field replace_ethertype is set on this register. Frames with this Ethertype will also be poisoned on the RX processing if field replace_ethertype is set.
- **Size:** 32 bits
- **Offset:** 0x18
- **Exists:** Always

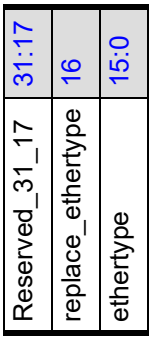


Table 6-13 Fields for Register: ETHERTYPE_REPLACE

Bits	Name	Memory Access	Description
31:17	Reserved_31_17	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
16	replace_ethertype	R/W	If set, defines that processed Macsec frames' Ethertype is to be replaced and if Frames with this Ethertype are poisoned. **Values:** ■ 0x0 (replace_ethertype): replace_ethertype **Value After Reset:** 0x0 **Exists:** Always

Table 6-13 Fields for Register: ETHERTYPE_REPLACE (Continued)

Bits	Name	Memory Access	Description
15:0	ethertype	R/W	Ethertype value to replace the original one on a processed Macsec frame when the field replace_ethertype is set **Values:** ■ 0x0 (ethertype): ethertype **Value After Reset:** 0x0 **Exists:** Always

6.1.7 RX_LOOKUP_TABLE_SCI_LOW

- **Name:** RX_LOOKUP_TABLE_SCI_LOW
- **Description:** RX SAI Lookup SCI Lower bits ([31:0])
- **Size:** 32 bits
- **Offset:** 0x1c
- **Exists:** Always

Table 6-14 Fields for Register: RX_LOOKUP_TABLE_SCI_LOW

Bits	Name	Memory Access	Description
31:0	SCI_LOW	R/W	The least significant bits of the SCI value used in the RX lookup to determine the SAI **Values:** ■ 0x0 (SCI_LOW): SCI_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.8 RX_LOOKUP_TABLE_SCI_HIGH

- **Name:** RX_LOOKUP_TABLE_SCI_HIGH
- **Description:** RX SAI Lookup SCI Higher bits ([63:32])
- **Size:** 32 bits
- **Offset:** 0x20
- **Exists:** Always

Table 6-15 Fields for Register: RX_LOOKUP_TABLE_SCI_HIGH

Bits	Name	Memory Access	Description
31:0	SCI_HIGH	R/W	The most significant bits of the SCI value used in the RX lookup to determine the SAI **Values:** ■ 0x0 (SCI_HIGH): SCI_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.9 RX_LOOKUP_TABLE_VLAN_INFO

- **Name:** RX_LOOKUP_TABLE_VLAN_INFO
- **Description:** RX SAI Lookup VLAN Tags
- **Size:** 32 bits
- **Offset:** 0x24
- **Exists:** Always

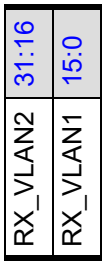


Table 6-16 Fields for Register: RX_LOOKUP_TABLE_VLAN_INFO

Bits	Name	Memory Access	Description
31:16	RX_VLAN2	R/W	RX VLAN TAG 2 information to be programmed in RX Lookup Table for a given SA. This 16 bits will follow 802.1Q tag format related to lower two bytes, which are used for tag control information (TCI) and sub-divided into PCP, DEI, and VID respectively (MSB to LSB). **Values:** ■ 0x0 (RX_VLAN2): RX_VLAN2 **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==2)

Table 6-16 Fields for Register: RX_LOOKUP_TABLE_VLAN_INFO (Continued)

Bits	Name	Memory Access	Description
15:0	RX_VLAN1	R/W	RX VLAN TAG 1 information to be programmed in RX Lookup Table for a given SA. This 16 bits will follow 802.1Q tag format related to lower two bytes, which are used for tag control information (TCI) and sub-divided into PCP, DEI, and VID respectively (MSB to LSB). **Values:** ■ 0x0 (RX_VLAN1): RX_VLAN1 **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==1 || DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==2)

6.1.10 RX_SAI_PG

- **Name:** RX_SAI_PG
- **Description:** Register that indicates if RX SAI Lookup memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0x28
- **Exists:** Always

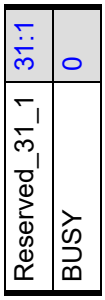


Table 6-17 Fields for Register: RX_SAI_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
0	BUSY	R	When high indicates that RX SAI Lookup Table memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.11 RX_SAI_PRG

- **Name:** RX_SAI_PRG
- **Description:** Writing to this register causes the information contained in the RX_SAI_OTHER_FIELDS, RX_LOOKUP_TABLE_VLAN_INFO, RX_LOOKUP_TABLE_SCI_LOW, RX_LOOKUP_TABLE_SCI_HIGH registers to be loaded in RX SAI Lookup Table memory entry indicated by ENTRY field if TYPE==WRITE.
- **Size:** 32 bits
- **Offset:** 0x2c
- **Exists:** Always

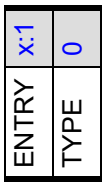


Table 6-18 Fields for Register: RX_SAI_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	RX SAI Lookup Table memory entry to be read or programmed **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	Dictates the type of memory operation, if 0 - read, if 1 - write **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.12 RX_SAI_OTHER_FIELDS

- **Name:** RX_SAI_OTHER_FIELDS
- **Description:** Association Number and VLAN wildcards for RX SAI Lookup Table memory programming.
- **Size:** 32 bits
- **Offset:** 0x30
- **Exists:** Always

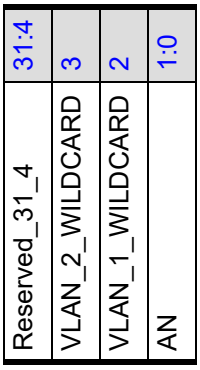


Table 6-19 Fields for Register: RX_SAI_OTHER_FIELDS

Bits	Name	Memory Access	Description
31:4	Reserved_31_4	R	Reserved bit **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
3	VLAN_2_WILDCARD	R/W	VLAN_2 Wildcard. Write this bit to indicate that the RX SAI VLAN Tag 2 field is a wildcard, and all values are matched. **Values:** ■ 0x0 (VLAN_2_WILDCARD): VLAN_2_WILDCARD **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==2)

Table 6-19 Fields for Register: RX_SAI_OTHER_FIELDS (Continued)

Bits	Name	Memory Access	Description
2	VLAN_1_WILDCARD	R/W	VLAN_1 Wildcard. Write this bit to indicate that the RX SAI VLAN Tag 1 field is a wildcard, and all values are matched. **Values:** ■ 0x0 (VLAN_1_WILDCARD): VLAN_1_WILDCARD **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==1 || DWC_MACSEC_RX_SAI_LOOKUP_NUM_VLANTAGS==2)
1:0	AN	R/W	AN value used in the RX lookup to determine the SAI **Values:** ■ 0x0 (AN): AN **Value After Reset:** 0x0 **Exists:** Always

6.1.13 RX_SAI_DATA_FETCH

- **Name:** RX_SAI_DATA_FETCH
- **Description:** When READY is high, it indicates that all RX SAI Lookup related fetch registers were updated with requested memory entry read value.
- **Size:** 32 bits
- **Offset:** 0x34
- **Exists:** Always

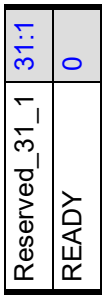


Table 6-20 Fields for Register: RX_SAI_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-20 Fields for Register: RX_SAI_DATA_FETCH (Continued)

Bits	Name	Memory Access	Description
0	READY	R	When high, it indicates that fetched data from requested memory entry is available in RX_SAI_OTHER_FIELDS, RX_LOOKUP_TABLE_VLAN_INFO, RX_LOOKUP_TABLE_SCI_LOW, RX_LOOKUP_TABLE_SCI_HIGH registers. The sequence should be: 1) Program RX_SAI_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. RX_SAI_OTHER_FIELDS, RX_LOOKUP_TABLE_VLAN_INFO, RX_LOOKUP_TABLE_SCI_LOW, RX_LOOKUP_TABLE_SCI_HIGH). **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always

6.1.14 TX_SRC_FILT_msb

- **Name:** TX_SRC_FILT_msb
- **Description:** TX SAI Source MAC Addresss register ([47:32])
- **Size:** 32 bits
- **Offset:** 0x38
- **Exists:** Always

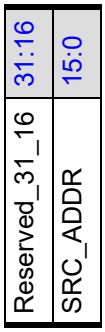


Table 6-21 Fields for Register: TX_SRC_FILT_msb

Bits	Name	Memory Access	Description
31:16	Reserved_31_16	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
15:0	SRC_ADDR	R/W	Filter for the source MAC address used for the TX SAI Lookup. Write 0xFFFFFFFFFFFF (broadcast address) as a wildcard for this field, indicating all values are matched. **Values:** ■ 0x0 (ADDR): ADDR **Value After Reset:** 0x0 **Exists:** Always

6.1.15 TX_SRC_FILT_lsb

- **Name:** TX_SRC_FILT_lsb
- **Description:** TX SAI Source MAC Addresss register ([31:0])
- **Size:** 32 bits
- **Offset:** 0x3c
- **Exists:** Always

Table 6-22 Fields for Register: TX_SRC_FILT_lsb

Bits	Name	Memory Access	Description
31:0	SRC_ADDR	R/W	Filter for the source MAC address used for the TX SAI Lookup. Write 0xFFFFFFFFFFFF (broadcast address) as a wildcard for this field, indicating all values are matched. **Values:** ■ 0x0 (ADR): ADR **Value After Reset:** 0x0 **Exists:** Always

6.1.16 TX_DST_FILT_msb

- **Name:** TX_DST_FILT_msb
- **Description:** TX SAI Destination MAC Addresss register ([47:32])
- **Size:** 32 bits
- **Offset:** 0x40
- **Exists:** Always

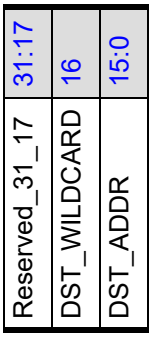


Table 6-23 Fields for Register: TX_DST_FILT_msb

Bits	Name	Memory Access	Description
31:17	Reserved_31_17	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
16	DST_WILDCARD	R/W	Destination Address Wildcard setting **Values:** ■ 0x0 (DST_WILDCARD): DST_WILDCARD **Value After Reset:** 0x0 **Exists:** Always
15:0	DST_ADDR	R/W	Filter for the destination MAC address used for the TX SAI Lookup. **Values:** ■ 0x0 (ADDR): DST_ADDR **Value After Reset:** 0x0 **Exists:** Always

6.1.17 TX_DST_FILT_lsb

- **Name:** TX_DST_FILT_lsb
- **Description:** TX SAI Destination MAC Addresss register ([31:0])
- **Size:** 32 bits
- **Offset:** 0x44
- **Exists:** Always

Table 6-24 Fields for Register: TX_DST_FILT_lsb

Bits	Name	Memory Access	Description
31:0	DST_ADDR	R/W	Filter for the destination MAC address used for the TX SAI Lookup. **Values:** ■ 0x0 (ADR): DST_ADDR **Value After Reset:** 0x0 **Exists:** Always

6.1.18 TX_SAI_PG

- **Name:** TX_SAI_PG
- **Description:** Register that indicates if TX SAI Lookup memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0x48
- **Exists:** Always

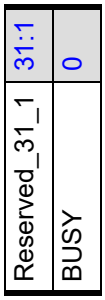


Table 6-25 Fields for Register: TX_SAI_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	BUSY	R	When high indicates that TX SAI Lookup Table memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.19 TX_SAI_PRG

- **Name:** TX_SAI_PRG
- **Description:** Writing to this register causes the information contained in the TX_LOOKUP_TABLE_VLAN_INFO, TX_SAI_PORT_ID (future use for multi port), TX_SAI_OTHER_FIELDS, TX_DST_FILT_lsb, TX_DST_FILT_msb, TX_SRC_FILT_lsb, TX_SRC_FILT_msb registers to be loaded in TX SAI Lookup Table memory entry indicated by ENTRY field if TYPE==WRITE.
- **Size:** 32 bits
- **Offset:** 0x4c
- **Exists:** Always

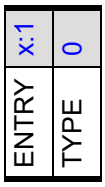


Table 6-26 Fields for Register: TX_SAI_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	TX SAI Lookup Table memory entry to be read or programmed **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED
0	TYPE	R/W	Dictates the type of memory operation, if 0 - read, if 1 - write **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.20 TX_SAI_OTHER_FIELDS

- **Name:** TX_SAI_OTHER_FIELDS
- **Description:** Outcome, Secure Channel Index, Ethernet type and Vlan wildcards fields from the TX SAI Lookup
- **Size:** 32 bits
- **Offset:** 0x50
- **Exists:** Always

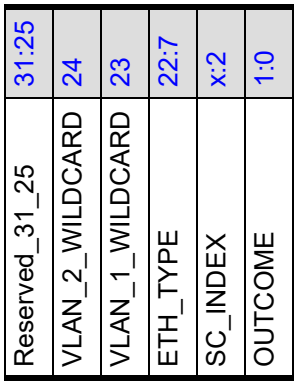


Table 6-27 Fields for Register: TX_SAI_OTHER_FIELDS

Bits	Name	Memory Access	Description
31:25	Reserved_31_25	R	Reserved bit **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
24	VLAN_2_WILDCARD	R/W	VLAN_2 Wildcard. Write this bit to indicate that the TX SAI VLAN Tag 2 field is a wildcard, and all values are matched **Values:** ■ 0x0 (VLAN_2_WILDCARD): VLAN_2_WILDCARD **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==2)

Table 6-27 Fields for Register: TX_SAI_OTHER_FIELDS (Continued)

Bits	Name	Memory Access	Description
23	VLAN_1_WILDCARD	R/W	VLAN_1 Wildcard. Write this bit to indicate that the TX SAI VLAN Tag 1 field is a wildcard, and all values are matched **Values:** ■ 0x0 (VLAN_1_WILDCARD): VLAN_1_WILDCARD **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==1 || DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==2)
22:7	ETH_TYPE	R/W	Filter for the EtherType. **Values:** ■ 0x0 (ETH_TYPE): ETH_TYPE **Value After Reset:** 0x0 **Exists:** Always
x:2	SC_INDEX	R/W	Security Channel (SC) Index **Values:** ■ 0x0 (SC_INDEX): SC_INDEX **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** (DWC_MACSEC_TX_LOG2_NUM_SC_SUPPORTED) + 1
1:0	OUTCOME	R/W	Indicates the outcome when a match occurs for this SC. There are 2 valid values: 0: Bypass 1: Protect frames Other: Reserved **Values:** ■ 0x0 (OUTCOME): OUTCOME **Value After Reset:** 0x0 **Exists:** Always

6.1.21 TX_SAI_DATA_FETCH

- **Name:** TX_SAI_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry is available in TX_LOOKUP_TABLE_VLAN_INFO, TX_SAI_PORT_ID (future use for multi port), TX_SAI_OTHER_FIELDS, TX_DST_FILT_lsb, TX_DST_FILT_msb, TX_SRC_FILT_lsb, TX_SRC_FILT_msb registers.
- **Size:** 32 bits
- **Offset:** 0x54
- **Exists:** Always

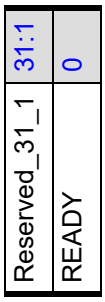


Table 6-28 Fields for Register: TX_SAI_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-28 Fields for Register: TX_SAI_DATA_FETCH (Continued)

Bits	Name	Memory Access	Description
0	READY	R	When high, it indicates that fetched data from requested memory entry is available in TX_LOOKUP_TABLE_VLAN_INFO, TX_SAI_PORT_ID (future use for multi port), TX_SAI_OTHER_FIELDS, TX_DST_FILT_lsb, TX_DST_FILT_msb, TX_SRC_FILT_lsb, TX_SRC_FILT_msb registers. 1) Program TX_SAI_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. TX_LOOKUP_TABLE_VLAN_INFO, TX_SAI_PORT_ID (future use for multi port), TX_SAI_OTHER_FIELDS, TX_DST_FILT_lsb, TX_DST_FILT_msb, TX_SRC_FILT_lsb, TX_SRC_FILT_msb). **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always

6.1.22 TX_LOOKUP_TABLE_VLAN_INFO

- **Name:** TX_LOOKUP_TABLE_VLAN_INFO
- **Description:** TX SAI Lookup VLAN Tags
- **Size:** 32 bits
- **Offset:** 0x58
- **Exists:** Always

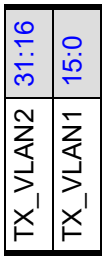


Table 6-29 Fields for Register: TX_LOOKUP_TABLE_VLAN_INFO

Bits	Name	Memory Access	Description
31:16	TX_VLAN2	R/W	TX VLAN TAG 2 information to be programmed in TX SAI Lookup Table for a given SA. This 16 bits will follow 802.1Q tag format related to lower two bytes, which are used for tag control information (TCI) and sub-divided into PCP, DEI, and VID respectively (MSB to LSB). **Values:** ■ 0x0 (TX_VLAN2): TX_VLAN2 **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==2)

Table 6-29 Fields for Register: TX_LOOKUP_TABLE_VLAN_INFO (Continued)

Bits	Name	Memory Access	Description
15:0	TX_VLAN1	R/W	TX VLAN TAG 1 information to be programmed in TX SAI Lookup Table for a given SA. This 16 bits will follow 802.1Q tag format related to lower two bytes, which are used for tag control information (TCI) and sub-divided into PCP, DEI, and VID respectively (MSB to LSB). **Values:** ■ 0x0 (TX_VLAN1): TX_VLAN1 **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==1 || DWC_MACSEC_TX_SAI_LOOKUP_NUM_VLANTAGS==2)

6.1.23 TX_SAI_PORT_ID

- **Name:** TX_SAI_PORT_ID
- **Description:** TX SAI Port ID register
- **Size:** 32 bits
- **Offset:** 0x5c
- **Exists:** Always

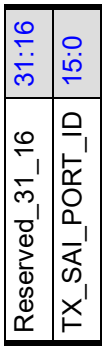


Table 6-30 Fields for Register: TX_SAI_PORT_ID

Bits	Name	Memory Access	Description
31:16	Reserved_31_16	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
15:0	TX_SAI_PORT_ID	R/W	Port Identifier associated with the programmed TX Security Association future use for multi port, in single port not used to lookup. **Values:** ■ 0x0 (TX_SAI_PORT_ID): TX_SAI_PORT_ID **Value After Reset:** 0x0 **Exists:** Always

6.1.24 TX_ACTIVE_AN

- **Name:** TX_ACTIVE_AN
- **Description:** This register has one bit per SC, that indicates which SA of that SC is active. The amount of bits used dependends on the amount of configured SCs.
- **Size:** 32 bits
- **Offset:** 0x60
- **Exists:** Always

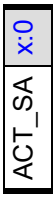
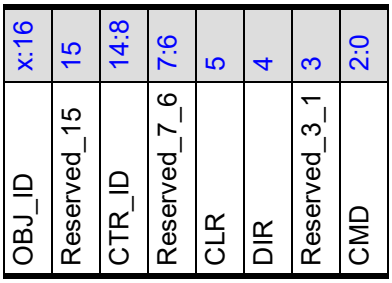


Table 6-31 Fields for Register: TX_ACTIVE_AN

Bits	Name	Memory Access	Description
x:0	ACT_SA	R/W	Bit map of TX active SAs where each bit refers to a given TX Security Channel. If bit 5 of this register is 0, it means that Security Channel 5 active SA is SA 0. **Values:** ■ 0x0 (ACT_SA): ACT_SA **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** (DWC_MACSEC_TX_NUM_SC_SUPPORTED) - 1

6.1.25 MIB_CMD

- **Name:** MIB_CMD
- **Description:** The MIB_CMD register allows a user to indirectly access the MIB counters. This is done through a set of commands
- **Size:** 32 bits
- **Offset:** 0x64
- **Exists:** Always



x:16	15	14:8	7:6	5	4	3	2:0
OBJ_ID	Reserved_15	CTR_ID	Reserved_7_6	CLR	DIR	Reserved_3_1	CMD

Table 6-32 Fields for Register: MIB_CMD

Bits	Name	Memory Access	Description
x:16	OBJ_ID	R/W	Index of the counter's SA or SC to be read/written. That counter can be either on a global, SA or SC basis. If the counter to be read/written is a SA counter you need to specify the SA index, if the counter is on a SC basis you need to specify the SC index. In the case of the counter being on a global basis this field is not required. **Values:** ■ 0x0 (OBJ_ID): OBJ_ID **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** (DWC_MACSEC_LOG2_MAX_SA_SUPPORTED) + 15
15	Reserved_15	R	Reserved bit **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-32 Fields for Register: MIB_CMD (Continued)

Bits	Name	Memory Access	Description
14:8	CTR_ID	R/W	ID of the counter to access . Each counter (for TX and RX) has a specific ID defined in the databook section 2 **Values:** ■ 0x0 (CTR_ID): CTR_ID **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved bit **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
5	CLR	R/W	If this bit is set, the corresponding registers are cleared atomically with the read transaction **Values:** ■ 0x0 (CLR): CLR **Value After Reset:** 0x0 **Exists:** Always
4	DIR	R/W	Specify the direction: 0 = Rx, 1 = TX **Values:** ■ 0x0 (DIR): DIR **Value After Reset:** 0x0 **Exists:** Always
3	Reserved_3_1	R	Reserved bit **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
2:0	CMD	R/W	The MIB_CMD register allows a user to indirectly access the MIB counters. This field contains the Command Number to be executed and its values are decribed in the databook. **Values:** ■ 0x0 (CMD): CMD **Value After Reset:** 0x0 **Exists:** Always

6.1.26 MIB_CMD_STAT

- **Name:** MIB_CMD_STAT
- **Description:** When high indicates that a MIB_CMD operation is underway
- **Size:** 32 bits
- **Offset:** 0x68
- **Exists:** Always

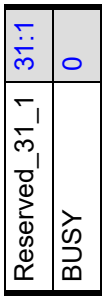


Table 6-33 Fields for Register: MIB_CMD_STAT

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
0	BUSY	R	When high indicates that a MIB_CMD operation is underway **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

6.1.27 MIB_VALIDATE

- **Name:** MIB_VALIDATE
- **Description:** Each bit in this register indicates the default VALIDATE_FRAMES setting for each physical port
- **Size:** 32 bits
- **Offset:** 0x6c
- **Exists:** Always

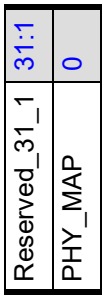


Table 6-34 Fields for Register: MIB_VALIDATE

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	PHY_MAP	R/W	Each bit in this register indicates the default VALIDATE_FRAMES setting for each physical port. If an SA is not resolved during lookup for current frame, the value from this register is used when incrementing the MIB. 0=disabled/check, 1=strict **Values:** ■ 0x0 (PHY_MAP): PHY_MAP **Value After Reset:** 0x0 **Exists:** Always

6.1.28 MIB_WR_DATA_HIGH

- **Name:** MIB_WR_DATA_HIGH
- **Description:** Counter MSB value to be loaded when executing a write command.
- **Size:** 32 bits
- **Offset:** 0x70
- **Exists:** Always

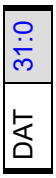


Table 6-35 Fields for Register: MIB_WR_DATA_HIGH

Bits	Name	Memory Access	Description
31:0	DATA	R/W	Counter MSB value to be loaded when executing a write command. **Values:** ■ 0x0 (DATA): DATA **Value After Reset:** 0x0 **Exists:** Always

6.1.29 MIB_WR_DATA_LOW

- **Name:** MIB_WR_DATA_LOW
- **Description:** Counter LSB value to be loaded when executing a write command
- **Size:** 32 bits
- **Offset:** 0x74
- **Exists:** Always

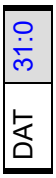


Table 6-36 Fields for Register: MIB_WR_DATA_LOW

Bits	Name	Memory Access	Description
31:0	DATA	R/W	Counter LSB value to be loaded when executing a write command **Values:** ■ 0x0 (DATA): DATA **Value After Reset:** 0x0 **Exists:** Always

6.1.30 RX_SAI_SC_CORR_TABLE_PG

- **Name:** RX_SAI_SC_CORR_TABLE_PG
- **Description:** Register that indicates if RX SAI SC Correlation Table memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0x78
- **Exists:** Always

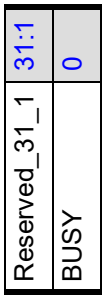


Table 6-37 Fields for Register: RX_SAI_SC_CORR_TABLE_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
0	BUSY	R	When high indicates that RX SAI SC Correlation Table memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.31 RX_SAI_SC_CORR_TABLE_PRG

- **Name:** RX_SAI_SC_CORR_TABLE_PRG
- **Description:** Writing to this register causes the information contained in RX_SAI_SC_CORR_TABLE_SC_VALUE, to be loaded in RX SAI SC Correlation Table memory entry indicated by ENTRY field if TYPE==WRITE
- **Size:** 32 bits
- **Offset:** 0x7c
- **Exists:** Always

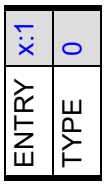


Table 6-38 Fields for Register: RX_SAI_SC_CORR_TABLE_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	Entry field to select which RX SAI SC Correlation Table memory entry will be programmed. This field must be programmed with the Security Association number that should be associated with the Security Channel set RX_SAI_SC_CORR_TABLE_SC_VALUE register. **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	When high indicates that a write operation is to be performed, when low it indicates that a read operation is to be performed **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.32 RX_SAI_SC_CORR_TABLE_DATA_FETCH

- **Name:** RX_SAI_SC_CORR_TABLE_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry is available in RX_SAI_SC_CORR_TABLE_SC_VALUE register.
- **Size:** 32 bits
- **Offset:** 0x80
- **Exists:** Always

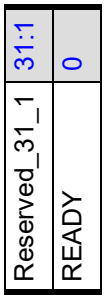


Table 6-39 Fields for Register: RX_SAI_SC_CORR_TABLE_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	READY	R	When high, it indicates it indicates that fetched data from requested memory entry is available in RX_SAI_SC_CORR_TABLE_SC_VALUE register. The sequence should be: 1) Program RX_SAI_SC_CORR_TABLE_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. RX_SAI_SC_CORR_TABLE_SC_VALUE) . **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

6.1.33 RX_SAI_SC_CORR_TABLE_SC_VALUE

- **Name:** RX_SAI_SC_CORR_TABLE_SC_VALUE
- **Description:** Security Channel index value
- **Size:** 32 bits
- **Offset:** 0x84
- **Exists:** Always

Table 6-40 Fields for Register: RX_SAI_SC_CORR_TABLE_SC_VALUE

Bits	Name	Memory Access	Description
x:0	SC_VALUE	R/W	Index of the Security Channel that will be associated with the SA that is being programmed. If RX_SAI_SC_CORR_TABLE_PRG.ENTRY=3 and RX_SAI_SC_CORR_TABLE_SC_VALUE is programmed to be 5, its means that RX Security Association #3 belongs to RX Security Channel #5. **Values:** ■ 0x0 (SC_VALUE): SC_VALUE **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable **Range Variable[x]:** (DWC_MACSEC_RX_LOG2_NUM_SC_SUPPORTED) - 1

6.1.34 RX_SALT_PG

- **Name:** RX_SALT_PG
- **Description:** Register that indicates if RX SALT memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0x88
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

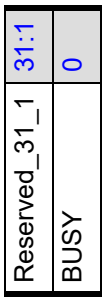


Table 6-41 Fields for Register: RX_SALT_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
0	BUSY	R	When high indicates that RX SALT memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.35 RX_SALT_PRG

- **Name:** RX_SALT_PRG
- **Description:** Writing to this register causes the information contained in RX_SALT_TABLE_0, RX_SALT_TABLE_1 and RX_SALT_TABLE_2 registers, to be loaded in RX SALT memory entry indicated by ENTRY field if TYPE==WRITE
- **Size:** 32 bits
- **Offset:** 0x8c
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

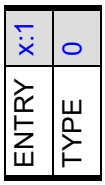


Table 6-42 Fields for Register: RX_SALT_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	Entry field to select which RX SALT memory entry will be programmed. **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	When high indicates that a write operation is to be performed, when low it indicates that a read operation is to be performed **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.36 RX_SALT_DATA_FETCH

- **Name:** RX_SALT_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry is available in RX_SALT_TABLE_0, RX_SALT_TABLE_1 and RX_SALT_TABLE_2 registers.
- **Size:** 32 bits
- **Offset:** 0x90
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

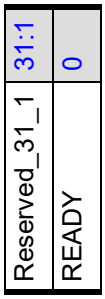


Table 6-43 Fields for Register: RX_SALT_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	READY	R	When high, it indicates that fetched data from requested memory entry is available in RX_SALT_TABLE_0, RX_SALT_TABLE_1 and RX_SALT_TABLE_2 registers. The sequence should be: 1) Program RX_SALT_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. RX_SALT_TABLE_0, RX_SALT_TABLE_1 and RX_SALT_TABLE_2) . **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always

6.1.37 RX_SALT_TABLE_0

- **Name:** RX_SALT_TABLE_0
- **Description:** RX SALT table intermediate register 0 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0x94
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

Table 6-44 Fields for Register: RX_SALT_TABLE_0

Bits	Name	Memory Access	Description
31:0	RX_SALT_VALUE_0	R/W	Lowest 32 bits from RX Salt table with 96-bit Salt value (RX_SALT_VALUE[31:0]) . **Values:** ■ 0x0 (RX_SALT_VALUE_0): RX_SALT_VALUE_0 **Value After Reset:** 0x0 **Exists:** Always

6.1.38 RX_SALT_TABLE_1

- **Name:** RX_SALT_TABLE_1
- **Description:** RX SALT table intermediate register 1 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0x98
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

Table 6-45 Fields for Register: RX_SALT_TABLE_1

Bits	Name	Memory Access	Description
31:0	RX_SALT_VALUE_1	R/W	Middle 32 bits from RX Salt table with 96-bit Salt value (RX_SALT_VALUE[63:32]) . **Values:** ■ 0x0 (RX_SALT_VALUE_1): RX_SALT_VALUE_1 **Value After Reset:** 0x0 **Exists:** Always

6.1.39 RX_SALT_TABLE_2

- **Name:** RX_SALT_TABLE_2
- **Description:** RX SALT table intermediate register 2 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0x9c
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

Table 6-46 Fields for Register: RX_SALT_TABLE_2

Bits	Name	Memory Access	Description
31:0	RX_SALT_VALUE_2	R/W	Higher 32 bits from RX Salt table with 96-bit Salt value (RX_SALT_VALUE[95:64]) . **Values:** ■ 0x0 (RX_SALT_VALUE_2): RX_SALT_VALUE_2 **Value After Reset:** 0x0 **Exists:** Always

6.1.40 TX_SALT_PG

- **Name:** TX_SALT_PG
- **Description:** Register that indicates if TX SALT memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0xa0
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

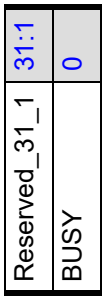


Table 6-47 Fields for Register: TX_SALT_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	BUSY	R	When high indicates that TX SALT memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.41 TX_SALT_PRG

- **Name:** TX_SALT_PRG
- **Description:** Writing to this register causes the information contained in the TX SALT_TABLE_0, TX SALT_TABLE_1 and TX SALT_TABLE_2 registers

  to be loaded in TX SALT memory entry indicated by ENTRY field if TYPE==WRITE
- **Size:** 32 bits
- **Offset:** 0xa4
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

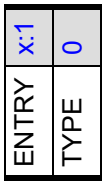


Table 6-48 Fields for Register: TX_SALT_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	Entry field to select which TX SALT memory entry will be programmed. **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_TX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	When high indicates that a write operation is to be performed, when low it indicates that a read operation is to be performed **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.42 TX_SALT_DATA_FETCH

- **Name:** TX_SALT_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry is available in TX_SALT_TABLE_0, TX_SALT_TABLE_1 and TX_SALT_TABLE_2 registers.
- **Size:** 32 bits
- **Offset:** 0xa8
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

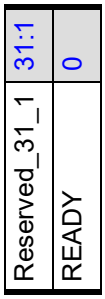


Table 6-49 Fields for Register: TX_SALT_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	READY	R	When high, it indicates that fetched data from requested memory entry is available in TX_SALT_TABLE_0, TX_SALT_TABLE_1 and TX_SALT_TABLE_2 registers. The sequence should be: 1) Program TX_SALT_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. TX_SALT_TABLE_0, TX_SALT_TABLE_1 and TX_SALT_TABLE_2) . **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always

6.1.43 TX_SALT_TABLE_0

- **Name:** TX_SALT_TABLE_0
- **Description:** TX SALT table intermediate register 0 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0xac
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

Table 6-50 Fields for Register: TX_SALT_TABLE_0

Bits	Name	Memory Access	Description
31:0	TX_SALT_VALUE_0	R/W	Lowest 32 bits from TX Salt table with 96-bit Salt value (TX_SALT_VALUE[31:0]) . **Values:** ■ 0x0 (TX_SALT_VALUE_0): TX_SALT_VALUE_0 **Value After Reset:** 0x0 **Exists:** Always

6.1.44 TX_SALT_TABLE_1

- **Name:** TX_SALT_TABLE_1
- **Description:** TX SALT table intermediate register 1 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0xb0
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

Table 6-51 Fields for Register: TX_SALT_TABLE_1

Bits	Name	Memory Access	Description
31:0	TX_SALT_VALUE_1	R/W	Middle 32 bits from TX Salt table with 96-bit Salt value (TX_SALT_VALUE[63:32]) . **Values:** ■ 0x0 (TX_SALT_VALUE_1): TX_SALT_VALUE_1 **Value After Reset:** 0x0 **Exists:** Always

6.1.45 TX_SALT_TABLE_2

- **Name:** TX_SALT_TABLE_2
- **Description:** TX SALT table intermediate register 2 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0xb4
- **Exists:** (DWC_MACSEC_XPN_SEL == 1)

Table 6-52 Fields for Register: TX_SALT_TABLE_2

Bits	Name	Memory Access	Description
31:0	TX_SALT_VALUE_2	R/W	Higher 32 bits from TX Salt table with 96-bit Salt value (TX_SALT_VALUE[95:64]) . **Values:** ■ 0x0 (TX_SALT_VALUE_2): TX_SALT_VALUE_2 **Value After Reset:** 0x0 **Exists:** Always

6.1.46 RX_ARW_PG

- **Name:** RX_ARW_PG
- **Description:** Register that indicates if RX ARW memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0xb8
- **Exists:** Always

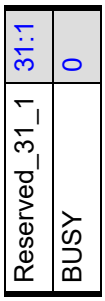


Table 6-53 Fields for Register: RX_ARW_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	BUSY	R	When high indicates that RX Anti Replay Window memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.47 RX_ARW_PRG

- **Name:** RX_ARW_PRG
- **Description:** Writing to this register causes the information contained in RX_ARW_TABLE_0 register, to be loaded in the RX ARW memory entry pointed by ENTRY field if TYPE==WRITE
- **Size:** 32 bits
- **Offset:** 0xbc
- **Exists:** Always

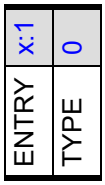


Table 6-54 Fields for Register: RX_ARW_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	Entry field to select which RX ARW memory entry will be programmed. **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	When high indicates that a write operation is to be performed, when low it indicates that a read operation is to be performed. **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.48 RX_ARW_DATA_FETCH

- **Name:** RX_ARW_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry is available in RX_ARW_TABLE_0 register.
- **Size:** 32 bits
- **Offset:** 0xc0
- **Exists:** Always

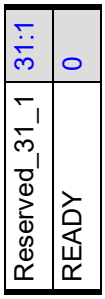


Table 6-55 Fields for Register: RX_ARW_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	READY	R	When high, it indicates that fetched data from requested memory entry is available in RX_ARW_TABLE_0 register. The sequence should be: 1) Program RX_ARW_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. RX_ARW_TABLE_0) . **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always

6.1.49 RX_ARW_TABLE_0

- **Name:** RX_ARW_TABLE_0
- **Description:** RX ARW table intermediate register 0 to hold a given entry read value or to program it
- **Size:** 32 bits
- **Offset:** 0xc4
- **Exists:** Always

Table 6-56 Fields for Register: RX_ARW_TABLE_0

Bits	Name	Memory Access	Description
x:0	RX_ARW_VALUE_0	R/W	Replay value for related RX SA **Values:** ■ 0x0 (RX_ARW_VALUE_0): RX_ARW_VALUE_0 **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** (DWC_MACSEC_RX_ARW_MEM_WIDTH) - 1

6.1.50 TX_SAD_PG

- **Name:** TX_SAD_PG
- **Description:** Register that indicates if TX SAD memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0xc8
- **Exists:** Always

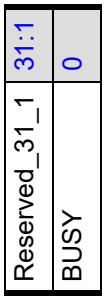


Table 6-57 Fields for Register: TX_SAD_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	BUSY	R	When high indicates that TX SAD memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.51 TX_SAD_PRG

- **Name:** TX_SAD_PRG
- **Description:** Writing to this register causes the information contained in the TX_SAD_*_LOW and TX_SAD_*_HIGH registers, to be loaded in TX SAD memory entry indicated by ENTRY field if TYPE==WRITE
- **Size:** 32 bits
- **Offset:** 0xcc
- **Exists:** Always

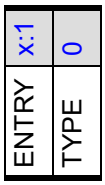


Table 6-58 Fields for Register: TX_SAD_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	Entry field to select which TX SAD memory entry will be programmed. **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_TX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	When high indicates that a write operation is to be performed, when low it indicates that a read operation is to be performed **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.52 TX_SAD_DATA_FETCH

- **Name:** TX_SAD_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry, is available in TX_SAD_*_LOW and TX_SAD_*_HIGH registers.
- **Size:** 32 bits
- **Offset:** 0xd0
- **Exists:** Always

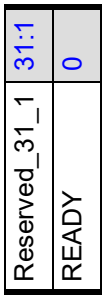


Table 6-59 Fields for Register: TX_SAD_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	READY	R	When high, it indicates that fetched data from requested memory entry is available in TX_SAD_*_LOW and TX_SAD_*_HIGH registers. The sequence should be: 1) Program TX_SAD_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. TX_SAD_*_LOW and TX_SAD_*_HIGH) . **Values:** ■ 0x0 (READY): READY **Value After Reset:** 0x0 **Exists:** Always

6.1.53 TX_SAD_0_LOW

- **Name:** TX_SAD_0_LOW
- **Description:** TX SAD entry programming register 0 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0xd4
- **Exists:** Always

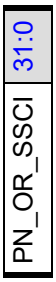


Table 6-60 Fields for Register: TX_SAD_0_LOW

Bits	Name	Memory Access	Description
31:0	PN_OR_SSCI	R/W	If AES-GCM-XPN feature was not selected in CC, this field acts as Next packet number. This field must be set to 32'h1 on creation of the SA. If AES-GCM-XPN feature was selected in CC, then according to XPN/non-XPN mode, this field will be used as follows: ■ In non-XPN mode: this register won't be used, since the packet sequence number will come from XPN[31:0] (TX_SAD_3_LOW) and SSCI won't exist, since it is needed only for XPN mode ■ In XPN mode: This is the SSCI used to identify the the SA. **Values:** ■ 0x0 (PN): PN **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.54 TX_SAD_0_HIGH

- **Name:** TX_SAD_0_HIGH
- **Description:** TX SAD entry programming register 0 (higher 32 bits)
- **Size:** 32 bits
- **Offset:** 0xd8
- **Exists:** Always

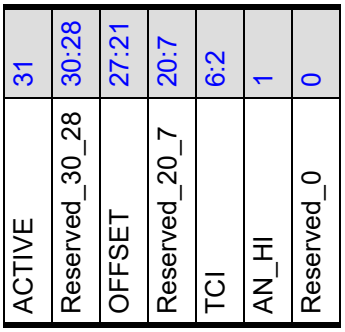


Table 6-61 Fields for Register: TX_SAD_0_HIGH

Bits	Name	Memory Access	Description
31	ACTIVE	R/W	Indicates if the SA is currently active. **Values:** ■ 0x0 (ACTIVE): ACTIVE **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
30:28	Reserved_30_28	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
27:21	OFFSET	R/W	Length in bytes of the confidentiality offset to apply. **Values:** ■ 0x0 (OFFSET): OFFSET **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Table 6-61 Fields for Register: TX_SAD_0_HIGH (Continued)

Bits	Name	Memory Access	Description
20:7	Reserved_20_7	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
6:2	TCI	R/W	The following bits to put in the TCI field in concatenated order {ES, SC, SCB, E, C}. **Values:** ■ 0x0 (TCI): TCI **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
1	AN_HI	R/W	The upper bit of the Association Number for this frame. **Values:** ■ 0x0 (AN_HI): AN_HI **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	Reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

6.1.55 TX_SAD_1_LOW

- **Name:** TX_SAD_1_LOW
- **Description:** TX SAD entry programming register 1 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0xdc
- **Exists:** Always

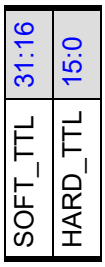


Table 6-62 Fields for Register: TX_SAD_1_LOW

Bits	Name	Memory Access	Description
31:16	SOFT_TTL	R/W	Most significant 16 bits only, the lower 16 bits (or the lower 48 bits when using XPN) are implied as zero. When a packet is transferred with a PN equivalent to the SOFT_TTL, a SOFT_TTL event will occur to indicate the the KaY to begin negotiating a new SAK. **Values:** ■ 0x0 (SOFT_TTL): SOFT_TTL **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
15:0	HARD_TTL	R/W	Most significant 16 bits only. The lower 16 bits (or the lower 48 bits when using XPN) are implied as zero. Generally, this field is set to zero which will cause a HARD_TTL event to occur when the PN number overflows. **Values:** ■ 0x0 (HARD_TTL): HARD_TTL **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.56 TX_SAD_1_HIGH

- **Name:** TX_SAD_1_HIGH
- **Description:** TX SAD entry programming register 1 (higher 32 bits)
- **Size:** 32 bits
- **Offset:** 0xe0
- **Exists:** Always

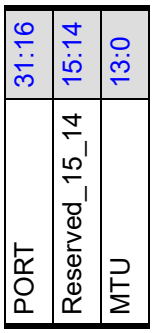


Table 6-63 Fields for Register: TX_SAD_1_HIGH

Bits	Name	Memory Access	Description
31:16	PORT	R/W	Port value to use when creating the SCI. **Values:** ■ 0x0 (PORT): PORT **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
15:14	Reserved_15_14	R	Reserved **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
13:0	MTU	R/W	Maximum Transmission Unit size of the media that this SA controls. This must be in the range of 64-16000 octets. **Values:** ■ 0x0 (MTU): MTU **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.57 TX_SAD_2_LOW

- **Name:** TX_SAD_2_LOW
- **Description:** TX SAD entry programming register 2 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0xe4
- **Exists:** Always

Table 6-64 Fields for Register: TX_SAD_2_LOW

Bits	Name	Memory Access	Description
31:0	SYSTEM_IDENTIFIER	R/W	System Identifier to be used where the ES bit is clear in the TCI. This field is considered RESERVED if the pEND_STATION parameter is set. **Values:** ■ 0x0 (SYSTEM_IDENTIFIER): SYSTEM_IDENTIFIER **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.58 TX_SAD_2_HIGH

- **Name:** TX_SAD_2_HIGH
- **Description:** TX SAD entry programming register 2 (higher 32 bits)
- **Size:** 32 bits
- **Offset:** 0xe8
- **Exists:** Always

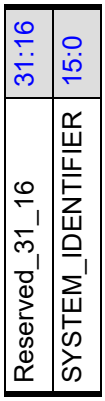


Table 6-65 Fields for Register: TX_SAD_2_HIGH

Bits	Name	Memory Access	Description
31:16	Reserved_31_16	R	Reserved **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
15:0	SYSTEM_IDENTIFIER	R/W	System Identifier to be used where the ES bit is clear in the TCI. This field is considered RESERVED if the pEND_STATION parameter is set. **Values:** ■ 0x0 (SYSTEM_IDENTIFIER): SYSTEM_IDENTIFIER **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.59 TX_SAD_3_LOW

- **Name:** TX_SAD_3_LOW
- **Description:** TX SAD entry programming register 3 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0xec
- **Exists:** ((DWC_MACSEC_XPN_SEL == 1))

Table 6-66 Fields for Register: TX_SAD_3_LOW

Bits	Name	Memory Access	Description
31:0	XPN	R/W	If AES-GCM-XPN feature was not selected in CC, this register is reserved. If AES-GCM-XPN feature was selected in CC, then according to XPN/non-XPN mode, this field will be used as follows: ■ In non-XPN mode: this register will act as Lowest Acceptable PN (i.e PN = XPN[31:0]) ■ In XPN mode: this field corresponds to the 32 lower bits of the extended Lowest Acceptable PN for the TX SA. **Values:** ■ 0x0 (XPN): XPN **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.60 TX_SAD_3_HIGH

- **Name:** TX_SAD_3_HIGH
- **Description:** TX SAD entry 3 Fourth 64-bit word, higher bits.
- **Size:** 32 bits
- **Offset:** 0xf0
- **Exists:** ((DWC_MACSEC_XPN_SEL == 1))

Table 6-67 Fields for Register: TX_SAD_3_HIGH

Bits	Name	Memory Access	Description
31:0	XPN	R/W	If AES-GCM-XPN feature was not selected in CC, this register is reserved. If AES-GCM-XPN feature was selected in CC, then according to XPN/non-XPN mode, this field will be used as follows: ■ In non-XPN mode: this register won't be used, since 32-bit PN mode was selected and thus, Lowest Acceptable PN=XPN[31:0] ■ In XPN mode: this field corresponds to the 32 higher bits of the extended Lowest Acceptable PN for the TX SA. **Values:** ■ 0x0 (XPN): XPN **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.61 RX_SAD_PG

- **Name:** RX_SAD_PG
- **Description:** Register that indicates if RX SAD memory is being accessed by MACsec module.
- **Size:** 32 bits
- **Offset:** 0xf4
- **Exists:** Always

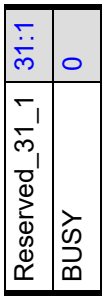


Table 6-68 Fields for Register: RX_SAD_PG

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
0	BUSY	R	When high indicates that RX SAD memory is being accessed by MACsec module. **Values:** ■ 0x0 (BUSY): BUSY **Value After Reset:** 0x0 **Exists:** Always

6.1.62 RX_SAD_PRG

- **Name:** RX_SAD_PRG
- **Description:** Writing to this register causes the information contained in the RX_SAD_*_LOW and RX_SAD_*_HIGH registers, to be loaded in RX SAD memory entry indicated by ENTRY field if TYPE==WRITE
- **Size:** 32 bits
- **Offset:** 0xf8
- **Exists:** Always

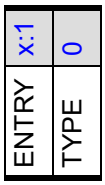


Table 6-69 Fields for Register: RX_SAD_PRG

Bits	Name	Memory Access	Description
x:1	ENTRY	R/W	Entry field to select which RX SAD memory entry will be programmed. **Values:** ■ 0x0 (ENTRY): ENTRY **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_RX_LOG2_NUM_SA_SUPPORTED
0	TYPE	R/W	When high indicates that a write operation is to be performed, when low it indicates that a read operation is to be performed **Values:** ■ 0x0 (TYPE): TYPE **Value After Reset:** 0x0 **Exists:** Always

6.1.63 RX_SAD_DATA_FETCH

- **Name:** RX_SAD_DATA_FETCH
- **Description:** When READY is high, it indicates that fetched data from requested memory entry, is available in RX_SAD_*_LOW and RX_SAD_*_HIGH registers.
- **Size:** 32 bits
- **Offset:** 0xfc
- **Exists:** Always

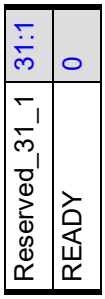


Table 6-70 Fields for Register: RX_SAD_DATA_FETCH

Bits	Name	Memory Access	Description
31:1	Reserved_31_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
0	READY	R	When high, it indicates that fetched data from requested memory entry, is available in RX_SAD_*_LOW and RX_SAD_*_HIGH registers. set_register_field_attribute $field RALBackdoorName "DATA_READY" The sequence should be: 1) Program RX_SAD_PRG to read entry X 2) Poll READY bit 3) Read all registers that hold fetched data after READY is 1 (i.e. RX_SAD_*_LOW and RX_SAD_*_HIGH) . **Value After Reset:** 0x0 **Exists:** Always

6.1.64 RX_SAD_0_LOW

- **Name:** RX_SAD_0_LOW
- **Description:** RX SAD entry programming register 0 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0x100
- **Exists:** Always

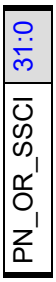


Table 6-71 Fields for Register: RX_SAD_0_LOW

Bits	Name	Memory Access	Description
31:0	PN_OR_SSCI	R/W	If AES-GCM-XPN feature was not selected in CC, this field acts as Lowest Acceptable PN. Frames with a packet number less than this value will be tagged as replays if the REPLAY flag is set. This field must be set to 32'h1 on creation of the SA. This field may be arbitrarily increased by the host at any time to move the replay window forward over time, but the host must use the update interface in the register map instead of writing the SA directly to perform this operation. Frames received with this exact value of PN, or in advance of the receive window will cause an automatic update of the PN field. If AES-GCM-XPN feature was selected in CC, then according to XPN/non-XPN mode, this field will be used as follows: ■ In non-XPN mode: this register won't be used, since the packet sequence number will come from XPN[31:0] (RX_SAD_2_LOW) and SSCI won't exist, since it is needed only for XPN mode ■ In XPN mode: This is the SSCI used to identify the the RX SA. **Values:** ■ 0x0 (PN): PN **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.65 RX_SAD_0_HIGH

- **Name:** RX_SAD_0_HIGH
- **Description:** RX SAD entry programming register 0 (higher 32 bits)
- **Size:** 32 bits
- **Offset:** 0x104
- **Exists:** Always

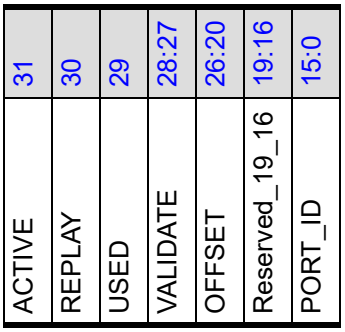


Table 6-72 Fields for Register: RX_SAD_0_HIGH

Bits	Name	Memory Access	Description
31	ACTIVE	R/W	Indicates if the SA is currently active. **Values:** ■ 0x0 (ACTIVE): ACTIVE **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
30	REPLAY	R/W	If set, replay check is enabled. Otherwise replay check will be skipped. **Values:** ■ 0x0 (REPLAY): REPLAY **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Table 6-72 Fields for Register: RX_SAD_0_HIGH (Continued)

Bits	Name	Memory Access	Description
29	USED	R/W	This flag must be clear upon creation of the SA. It is used to allow detection of the first packet processed by a new SA. **Values:** ■ 0x0 (USED): USED **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
28:27	VALIDATE	R/W	Validate field. 0=disabled, 1=check, 2=strict **Values:** ■ 0x0 (VALIDATE): VALIDATE **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
26:20	OFFSET	R/W	Length in bytes of confidentiality offset to apply. Currently only supporting 0 bytes (value 0x00 on this field), 30 bytes (value 0x1E value on this field) and 50 bytes offset (value 0x32 on this field). **Values:** ■ 0x0 (OFFSET): OFFSET **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
19:16	Reserved_19_16	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
15:0	PORT_ID	R/W	Port value to use when processing frames with implicit SCI (SC=0). **Values:** ■ 0x0 (PORT_ID): PORT_ID **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.66 RX_SAD_1_LOW

- **Name:** RX_SAD_1_LOW
- **Description:** RX SAD entry programming register 1 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0x108
- **Exists:** Always

Table 6-73 Fields for Register: RX_SAD_1_LOW

Bits	Name	Memory Access	Description
31:0	SYSTEM_IDENTIFIER	R/W	System Identifier to be used for receipt of frames with ES=0 and SC=0. **Values:** ■ 0x0 (SYSTEM_IDENTIFIER): SYSTEM_IDENTIFIER **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.67 RX_SAD_1_HIGH

- **Name:** RX_SAD_1_HIGH
- **Description:** RX SAD entry programming register 1 (higher 32 bits)
- **Size:** 32 bits
- **Offset:** 0x10c
- **Exists:** Always

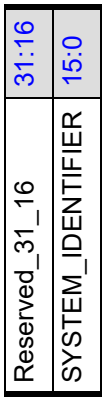


Table 6-74 Fields for Register: RX_SAD_1_HIGH

Bits	Name	Memory Access	Description
31:16	Reserved_31_16	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
15:0	SYSTEM_IDENTIFIER	R/W	System Identifier to be used for receipt of frames with ES=0 and SC=0. **Values:** ■ 0x0 (SYSTEM_IDENTIFIER): SYSTEM_IDENTIFIER **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.68 RX_SAD_2_LOW

- **Name:** RX_SAD_2_LOW
- **Description:** RX SAD entry programming register 2 (lower 32 bits)
- **Size:** 32 bits
- **Offset:** 0x110
- **Exists:** ((DWC_MACSEC_XPN_SEL == 1))

Table 6-75 Fields for Register: RX_SAD_2_LOW

Bits	Name	Memory Access	Description
31:0	XPN	R/W	If AES-GCM-XPN feature was not selected in CC, this register is reserved. If AES-GCM-XPN feature was selected in CC, then according to XPN/non-XPN mode, this field will be used as follows: ■ In non-XPN mode: this register will act as Lowest Acceptable PN (32-bit PN, i.e PN = XPN[31:0]) ■ In XPN mode: this field corresponds to the 32 lower bits of the extended Lowest Acceptable packet number for the Rx SA. **Values:** ■ 0x0 (XPN): XPN **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.69 RX_SAD_2_HIGH

- **Name:** RX_SAD_2_HIGH
- **Description:** RX SAD entry programming register 2 (higher 32 bits)
- **Size:** 32 bits
- **Offset:** 0x114
- **Exists:** ((DWC_MACSEC_XPN_SEL == 1))

Table 6-76 Fields for Register: RX_SAD_2_HIGH

Bits	Name	Memory Access	Description
31:0	XPN	R/W	If AES-GCM-XPN feature was not selected in CC, this register is reserved. If AES-GCM-XPN feature was selected in CC, then according to XPN/non-XPN mode, this field will be used as follows: ■ In non-XPN mode: this register won't be used, since 32-bit PN mode was selected and thus, PN=XPN[31:0] ■ In XPN mode: this field corresponds to the 32 higher bits of the extended packet number for the Rx SA. **Values:** ■ 0x0 (XPN): XPN **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

6.1.70 MIB_CMD_D_BUF_HIGH_0

- **Name:** MIB_CMD_D_BUF_HIGH_0
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x118
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 0

Table 6-77 Fields for Register: MIB_CMD_D_BUF_HIGH_0

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.71 MIB_CMD_D_BUF_LOW_0

- **Name:** MIB_CMD_D_BUF_LOW_0
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x11c
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 0

Table 6-78 Fields for Register: MIB_CMD_D_BUF_LOW_0

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.72 MIB_CMD_D_BUF_HIGH_1

- **Name:** MIB_CMD_D_BUF_HIGH_1
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x120
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 1

Table 6-79 Fields for Register: MIB_CMD_D_BUF_HIGH_1

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.73 MIB_CMD_D_BUF_LOW_1

- **Name:** MIB_CMD_D_BUF_LOW_1
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x124
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 1

Table 6-80 Fields for Register: MIB_CMD_D_BUF_LOW_1

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.74 MIB_CMD_D_BUF_HIGH_2

- **Name:** MIB_CMD_D_BUF_HIGH_2
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x128
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 2

Table 6-81 Fields for Register: MIB_CMD_D_BUF_HIGH_2

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.75 MIB_CMD_D_BUF_LOW_2

- **Name:** MIB_CMD_D_BUF_LOW_2
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x12c
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 2

Table 6-82 Fields for Register: MIB_CMD_D_BUF_LOW_2

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.76 MIB_CMD_D_BUF_HIGH_3

- **Name:** MIB_CMD_D_BUF_HIGH_3
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x130
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 3

Table 6-83 Fields for Register: MIB_CMD_D_BUF_HIGH_3

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.77 MIB_CMD_D_BUF_LOW_3

- **Name:** MIB_CMD_D_BUF_LOW_3
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x134
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 3

Table 6-84 Fields for Register: MIB_CMD_D_BUF_LOW_3

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.78 MIB_CMD_D_BUF_HIGH_4

- **Name:** MIB_CMD_D_BUF_HIGH_4
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x138
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 4

Table 6-85 Fields for Register: MIB_CMD_D_BUF_HIGH_4

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.79 MIB_CMD_D_BUF_LOW_4

- **Name:** MIB_CMD_D_BUF_LOW_4
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x13c
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 4

Table 6-86 Fields for Register: MIB_CMD_D_BUF_LOW_4

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.80 MIB_CMD_D_BUF_HIGH_5

- **Name:** MIB_CMD_D_BUF_HIGH_5
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x140
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 5

Table 6-87 Fields for Register: MIB_CMD_D_BUF_HIGH_5

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.81 MIB_CMD_D_BUF_LOW_5

- **Name:** MIB_CMD_D_BUF_LOW_5
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x144
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 5

Table 6-88 Fields for Register: MIB_CMD_D_BUF_LOW_5

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.82 MIB_CMD_D_BUF_HIGH_6

- **Name:** MIB_CMD_D_BUF_HIGH_6
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x148
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 6

Table 6-89 Fields for Register: MIB_CMD_D_BUF_HIGH_6

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.83 MIB_CMD_D_BUF_LOW_6

- **Name:** MIB_CMD_D_BUF_LOW_6
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x14c
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 6

Table 6-90 Fields for Register: MIB_CMD_D_BUF_LOW_6

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.84 MIB_CMD_D_BUF_HIGH_7

- **Name:** MIB_CMD_D_BUF_HIGH_7
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x150
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 7

Table 6-91 Fields for Register: MIB_CMD_D_BUF_HIGH_7

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.85 MIB_CMD_D_BUF_LOW_7

- **Name:** MIB_CMD_D_BUF_LOW_7
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x154
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 7

Table 6-92 Fields for Register: MIB_CMD_D_BUF_LOW_7

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.86 MIB_CMD_D_BUF_HIGH_8

- **Name:** MIB_CMD_D_BUF_HIGH_8
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x158
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 8

Table 6-93 Fields for Register: MIB_CMD_D_BUF_HIGH_8

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.87 MIB_CMD_D_BUF_LOW_8

- **Name:** MIB_CMD_D_BUF_LOW_8
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x15c
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 8

Table 6-94 Fields for Register: MIB_CMD_D_BUF_LOW_8

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.88 MIB_CMD_D_BUF_HIGH_9

- **Name:** MIB_CMD_D_BUF_HIGH_9
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x160
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 9

Table 6-95 Fields for Register: MIB_CMD_D_BUF_HIGH_9

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.89 MIB_CMD_D_BUF_LOW_9

- **Name:** MIB_CMD_D_BUF_LOW_9
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x164
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 9

Table 6-96 Fields for Register: MIB_CMD_D_BUF_LOW_9

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.90 MIB_CMD_D_BUF_HIGH_10

- **Name:** MIB_CMD_D_BUF_HIGH_10
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x168
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 10

Table 6-97 Fields for Register: MIB_CMD_D_BUF_HIGH_10

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.91 MIB_CMD_D_BUF_LOW_10

- **Name:** MIB_CMD_D_BUF_LOW_10
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x16c
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 10

Table 6-98 Fields for Register: MIB_CMD_D_BUF_LOW_10

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.92 MIB_CMD_D_BUF_HIGH_11

- **Name:** MIB_CMD_D_BUF_HIGH_11
- **Description:** Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values). For single counter read only the MIB_CMD_D_BUF_HIGH_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_HIGH_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x170
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 11

Table 6-99 Fields for Register: MIB_CMD_D_BUF_HIGH_11

Bits	Name	Memory Access	Description
31:0	DATA_HIGH	R	Contains the return MSB value of the 64 bit MIB Counter (Not applicable for SA counters as they are 32 bit values) **Values:** ■ 0x0 (DATA_HIGH): DATA_HIGH **Value After Reset:** 0x0 **Exists:** Always

6.1.93 MIB_CMD_D_BUF_LOW_11

- **Name:** MIB_CMD_D_BUF_LOW_11
- **Description:** Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter). For single counter read only the MIB_CMD_D_BUF_LOW_0 is used. IF all global, SC or SA counters are read at once, the rest of MIB_CMD_D_BUF_LOW_* registers will be used in the order specified on the section 2 of the databook.
- **Size:** 32 bits
- **Offset:** 0x174
- **Exists:** ((DWC_MACSEC_MIB_SC_SA_MAX_ID > DWC_MACSEC_MIB_SA_GBL_MAX_ID) ? DWC_MACSEC_MIB_SC_SA_MAX_ID : DWC_MACSEC_MIB_SA_GBL_MAX_ID) >= 11

Table 6-100 Fields for Register: MIB_CMD_D_BUF_LOW_11

Bits	Name	Memory Access	Description
31:0	DATA_LOW	R	Contains the return LSB value of the 64 bit MIB Counter (32 bit in a SA counter) **Values:** ■ 0x0 (DATA_LOW): DATA_LOW **Value After Reset:** 0x0 **Exists:** Always

6.1.94 IRQ_GLBL_EN

- **Name:** IRQ_GLBL_EN
- **Description:** System global interrupt register enable
- **Size:** 32 bits
- **Offset:** 0x178
- **Exists:** Always

31	30:16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GLBL_EN	reserved_0	AES_APB_PTCOL_ERR_OK_IRQ_EN	APB_PTCOL_ERR_OK_IRQ_EN	AES_APB_PARITY_NOT_OK_IRQ_EN	APB_PARITY_NOT_OK_IRQ_EN	AES_KEY_ERROR_EN	RX_PREAMBLE_ERROR_EN	TX_PREAMBLE_ERROR_EN	RX_IF_ERROR_EN	TX_IF_ERROR_EN	RX_ILLEGAL_IPG_EN	TX_ILLEGAL_IPG_EN	RX_FRAME_TOO_SHORT_EN	TX_FRAME_TOO_SHORT_EN	RX_FRAME_CRC_ERROR_EN	TX_FRAME_CRC_ERROR_EN	RX_UNKNOWN_SC_EN

Table 6-101 Fields for Register: IRQ_GLBL_EN

Bits	Name	Memory Access	Description
31	GLBL_EN	R/W	Global enable control for interrupt sources. If 0, all system interrupts are disabled. **Values:** ■ 0x0 (GLBL_EN): GLBL_EN **Value After Reset:** 0x0 **Exists:** Always
30:16	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-101 Fields for Register: IRQ_GLBL_EN (Continued)

Bits	Name	Memory Access	Description
15	AES_APB_PTCOL_ERR_OK_IRQ_EN	R/W	Enables AES APB Protocol Check IRQ, indicating that the APB protocol was not respected on a MACsec APB5 transfer. **Values:** ■ 0x1 (ENABLE_AES_APB_PTCOL_ERR_OK_EN): Enable AES APB Protocol Check IRQ. ■ 0x0 (DISABLE_AES_APB_PARITY_NOT_OK_EN): Disable AES APB Protocol Check IRQ. **Value After Reset:** 0x0 **Exists:** Always
14	APB_PTCOL_ERR_OK_IRQ_EN	R/W	Enables MACsec APB Protocol Check IRQ, indicating that the APB protocol was not respected on a MACsec APB5 transfer. **Values:** ■ 0x1 (ENABLE_APB_PTCOL_ERR_OK_EN): Enable MACsec APB Protocol Check IRQ. ■ 0x0 (DISABLE_APB_PTCOL_ERR_OK_EN): Disable MACsec APB Protocol Check IRQ. **Value After Reset:** 0x0 **Exists:** Always
13	AES_APB_PARITY_NOT_OK_IRQ_EN	R/W	Enables AES APB Parity Check IRQ, indicating parity is not okay on a AES APB5 transfer. **Values:** ■ 0x1 (ENABLE_AES_APB_PARITY_NOT_OK_EN): Enable AES APB Parity Check IRQ. ■ 0x0 (DISABLE_AES_APB_PARITY_NOT_OK_EN): Disable AES APB Parity Check IRQ. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_APB5_PARITY_SUPPORTED==1)
12	APB_PARITY_NOT_OK_IRQ_EN	R/W	Enables MACsec APB Parity Check IRQ, indicating parity is not okay on a MACsec APB5 transfer. **Values:** ■ 0x1 (ENABLE_APB_PARITY_NOT_OK_EN): Enable MACsec APB Parity Check IRQ. ■ 0x0 (DISABLE_APB_PARITY_NOT_OK_EN): Disable MACsec APB Parity Check IRQ. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_APB5_PARITY_SUPPORTED==1)

Table 6-101 Fields for Register: IRQ_GLBL_EN (Continued)

Bits	Name	Memory Access	Description
11	AES_KEY_ERROR_EN	R/W	Enables the AES key error interrupt. **Values:** ■ 0x0 (AES_KEY_ERROR_EN): AES_KEY_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
10	RX_PREAMBLE_ERROR_EN	R/W	Enables the preamble error interrupt (RX Side). **Values:** ■ 0x0 (RX_PREAMBLE_ERROR_EN): RX_PREAMBLE_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
9	TX_PREAMBLE_ERROR_EN	R/W	Enables the preamble error interrupt (TX Side). **Values:** ■ 0x0 (TX_PREAMBLE_ERROR_EN): TX_PREAMBLE_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
8	RX_IF_ERROR_EN	R/W	Enables the interface error interrupt (RX Side). **Values:** ■ 0x0 (RX_IF_ERROR_EN): RX_IF_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
7	TX_IF_ERROR_EN	R/W	Enables the interface error interrupt (TX Side). **Values:** ■ 0x0 (TX_IF_ERROR_EN): TX_IF_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
6	RX_ILLEGAL_IPG_EN	R/W	Enables the Illegal IPG interrupt (RX Side). **Values:** ■ 0x0 (RX_ILLEGAL_IPG_EN): RX_ILLEGAL_IPG_EN **Value After Reset:** 0x0 **Exists:** Always
5	TX_ILLEGAL_IPG_EN	R/W	Enables the Illegal IPG interrupt (TX Side). **Values:** ■ 0x0 (TX_ILLEGAL_IPG_EN): TX_ILLEGAL_IPG_EN **Value After Reset:** 0x0 **Exists:** Always

Table 6-101 Fields for Register: IRQ_GLBL_EN (Continued)

Bits	Name	Memory Access	Description
4	RX_FRAME_TOO_SHORT_EN	R/W	Enables the Frame too short contribution to IRQ signal. (RX side) **Values:** ■ 0x0 (RX_FRAME_TOO_LONG_EN): RX_FRAME_TOO_SHORT_EN **Value After Reset:** 0x0 **Exists:** Always
3	TX_FRAME_TOO_SHORT_EN	R/W	Enables the Frame too short contribution to IRQ signal. (TX Side) **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_SHORT_EN **Value After Reset:** 0x0 **Exists:** Always
2	RX_FRAME_CRC_ERROR_EN	R/W	Enables the CRC error contribution to IRQ signal (RX Side). **Values:** ■ 0x0 (RX_FRAME_CRC_ERROR_EN): RX_FRAME_CRC_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_CRC_ERROR_EN	R/W	Enables the CRC error contribution to IRQ signal (TX Side). **Values:** ■ 0x0 (TX_FRAME_CRC_ERROR_EN): TX_FRAME_CRC_ERROR_EN **Value After Reset:** 0x0 **Exists:** Always
0	RX_UNKNOWN_SC_EN	R/W	Enables the Global RX Unknown SC contribution to the IRQ signal. If triggered, the secure channel could not be identified. **Values:** ■ 0x0 (RX_UNKNOWN_SC_EN): RX_UNKNOWN_SC_EN **Value After Reset:** 0x0 **Exists:** Always

6.1.95 IRQ_GLBL_STAT

- **Name:** IRQ_GLBL_STAT
- **Description:** System global interrupt register status
- **Size:** 32 bits
- **Offset:** 0x17c
- **Exists:** Always

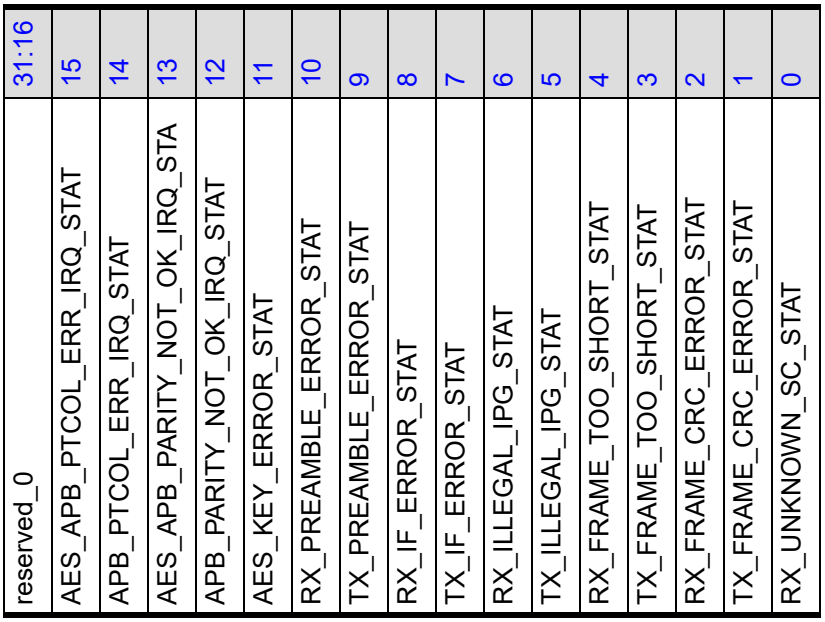


Table 6-102 Fields for Register: IRQ_GLBL_STAT

Bits	Name	Memory Access	Description
31:16	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-102 Fields for Register: IRQ_GLBL_STAT (Continued)

Bits	Name	Memory Access	Description
15	AES_APB_PTCOL_ERR_IRQ_STAT	R/W1C	AES APB Protocol Check IRQ, indicating that the APB5 protocol was not respected on a certain AES APB transfer for APB5. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x1 (AES_APB_PTCOL_ERR): Parity Not OK IRQ. ■ 0x0 (AES_APB_PARITY_OK): Parity OK. **Value After Reset:** 0x0 **Exists:** (DWC_UAES_CFG_APB_BUS_EN==1)
14	APB_PTCOL_ERR_IRQ_STAT	R/W1C	MACsec APB Protocol Check IRQ, indicating that the APB5 protocol was not respected on a certain APB transfer for APB5. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x1 (APB_PTCOL_ERR): Parity Not OK IRQ. ■ 0x0 (APB_PARITY_OK): Parity OK. **Value After Reset:** 0x0 **Exists:** Always
13	AES_APB_PARITY_NOT_OK_IRQ_STAT	R/W1C	AES APB Parity Check IRQ, indicating parity is not okay on a certain AES APB transfer for APB5. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x1 (AES_APB_PARITY_NOT_OK): Parity Not OK IRQ. ■ 0x0 (AES_APB_PARITY_OK): Parity OK. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_APB5_PARITY_SUPPORTED==1)
12	APB_PARITY_NOT_OK_IRQ_STAT	R/W1C	MACsec APB Parity Check IRQ, indicating parity is not okay on a certain MACsec APB transfer for APB5. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x1 (APB_PARITY_NOT_OK): Parity Not OK IRQ. ■ 0x0 (APB_PARITY_OK): Parity OK. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_APB5_PARITY_SUPPORTED==1)

Table 6-102 Fields for Register: IRQ_GLBL_STAT (Continued)

Bits	Name	Memory Access	Description
11	AES_KEY_ERROR_STAT	R/W1C	AES key error interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x1 (AES_KEY_ERROR_STAT): AES_KEY_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always
10	RX_PREAMBLE_ERROR_STAT	R/W1C	Preamble error. (RX Side). Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PREAMBLE_ERROR_STAT): RX_PREAMBLE_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
9	TX_PREAMBLE_ERROR_STAT	R/W1C	Preamble error. (TX Side). Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_PREAMBLE_ERROR_STAT): TX_PREAMBLE_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
8	RX_IF_ERROR_STAT	R/W1C	Current status of the interface error interrupt. If 1, an error has occured on RX. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_IF_ERROR_STAT): RX_IF_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-102 Fields for Register: IRQ_GLBL_STAT (Continued)

Bits	Name	Memory Access	Description
7	TX_IF_ERROR_STAT	R/W1C	Current status of the interface error interrupt. If 1, an error has occured on TX. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_IF_ERROR_STAT): TX_IF_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always
6	RX_ILLEGAL_IPG_STAT	R/W1C	Current status of Illegal IPG interrupt. If 1, IPG was lower than the RX IPG minimum set. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_ILLEGAL_IPG_STAT): RX_ILLEGAL_IPG_STAT **Value After Reset:** 0x0 **Exists:** Always
5	TX_ILLEGAL_IPG_STAT	R/W1C	Current status of Illegal IPG interrupt. If 1, IPG was lower than the TX IPG minimum set. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_ILLEGAL_IPG_STAT): TX_ILLEGAL_IPG_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_FRAME_TOO_SHORT_STAT	R/W1C	Current status of Frame too short interrupt. If 1, a frame (RX side) is <64 bytes. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_FRAME_TOO_SHORT_STAT): RX_FRAME_TOO_SHORT_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-102 Fields for Register: IRQ_GLBL_STAT (Continued)

Bits	Name	Memory Access	Description
3	TX_FRAME_TOO_SHORT_STAT	R/W1C	Current status of Frame too short interrupt. If 1, a frame (TX side) is <64 bytes. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_SHORT_STAT): TX_FRAME_TOO_SHORT_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_FRAME_CRC_ERROR_STAT	R/W1C	Current status of CRC error on frame interrupt (RX side). If 1, CRC resulted in an error for involved frame. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_FRAME_CRC_ERROR_STAT): RX_FRAME_CRC_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_CRC_ERROR_STAT	R/W1C	Current status of CRC error on frame interrupt (TX side). If 1, CRC resulted in an error for involved frame. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_CRC_ERROR_STAT): TX_FRAME_CRC_ERROR_STAT **Value After Reset:** 0x0 **Exists:** Always
0	RX_UNKNOWN_SC_STAT	R/W1C	Current status of Global RX Unknown SC contribution to the IRQ signal. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_SC_STAT): RX_UNKNOWN_SC_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.96 ETHERTYPE_FILTER_0

- **Name:** ETHERTYPE_FILTER_0 Register
- **Description:** ETHERTYPE_FILTER_0 defines the Ethertype values to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted. Frames with other Ethertype values and non-Macsec will be discarded if MIB_VALIDATE is on strict mode.
- **Size:** 32 bits
- **Offset:** 0x180
- **Exists:** Always

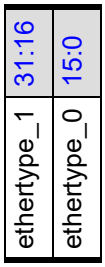


Table 6-103 Fields for Register: ETHERTYPE_FILTER_0

Bits	Name	Memory Access	Description
31:16	ethertype_1	R/W	Ethertype value 1 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_1): ethertype_1 **Value After Reset:** 0x0 **Exists:** Always
15:0	ethertype_0	R/W	Ethertype value 0 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_0): ethertype_0 **Value After Reset:** 0x0 **Exists:** Always

6.1.97 ETHERTYPE_FILTER_1

- **Name:** ETHERTYPE_FILTER_1 Register
- **Description:** ETHERTYPE_FILTER_1 defines the Ethertype values to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted. Frames with other Ethertype values and non-Macsec will be discarded if MIB_VALIDATE is on strict mode.
- **Size:** 32 bits
- **Offset:** 0x184
- **Exists:** Always

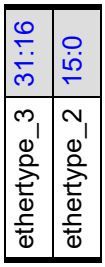


Table 6-104 Fields for Register: ETHERTYPE_FILTER_1

Bits	Name	Memory Access	Description
31:16	ethertype_3	R/W	Ethertype value 3 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_3): ethertype_3 **Value After Reset:** 0x0 **Exists:** Always
15:0	ethertype_2	R/W	Ethertype value 2 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_2): ethertype_2 **Value After Reset:** 0x0 **Exists:** Always

6.1.98 ETHERTYPE_FILTER_2

- **Name:** ETHERTYPE_FILTER_2 Register
- **Description:** ETHERTYPE_FILTER_2 defines the Ethertype values to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted. Frames with other Ethertype values and non-Macsec will be discarded if MIB_VALIDATE is on strict mode.
- **Size:** 32 bits
- **Offset:** 0x188
- **Exists:** Always

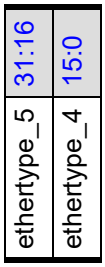


Table 6-105 Fields for Register: ETHERTYPE_FILTER_2

Bits	Name	Memory Access	Description
31:16	ethertype_5	R/W	Ethertype value 5 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_5): ethertype_5 **Value After Reset:** 0x0 **Exists:** Always
15:0	ethertype_4	R/W	Ethertype value 4 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_4): ethertype_4 **Value After Reset:** 0x0 **Exists:** Always

6.1.99 ETHERTYPE_FILTER_3

- **Name:** ETHERTYPE_FILTER_3 Register
- **Description:** ETHERTYPE_FILTER_3 defines the Ethertype values to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted. Frames with other Ethertype values and non-Macsec will be discarded if MIB_VALIDATE is on strict mode.
- **Size:** 32 bits
- **Offset:** 0x18c
- **Exists:** Always

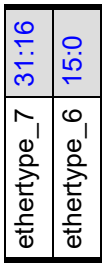


Table 6-106 Fields for Register: ETHERTYPE_FILTER_3

Bits	Name	Memory Access	Description
31:16	ethertype_7	R/W	Ethertype value 7 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_7): ethertype_7 **Value After Reset:** 0x0 **Exists:** Always
15:0	ethertype_6	R/W	Ethertype value 6 to bypass when MIB_VALIDATE register is on strict mode (1) for the phy port and the ethertype_filter_en register is asserted **Values:** ■ 0x0 (ethertype_6): ethertype_6 **Value After Reset:** 0x0 **Exists:** Always

6.1.100 IRQ_TYPE_EN_TX_SA_0

- **Name:** IRQ_TYPE_EN_TX_SA_0
- **Description:** Interrupt Enable for each IRQ type of TX SA 0
- **Size:** 32 bits
- **Offset:** 0x190
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 0

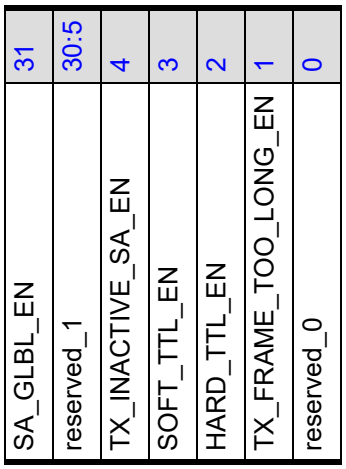


Table 6-107 Fields for Register: IRQ_TYPE_EN_TX_SA_0

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-107 Fields for Register: IRQ_TYPE_EN_TX_SA_0 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.101 IRQ_TYPE_STAT_TX_SA_0

- **Name:** IRQ_TYPE_STAT_TX_SA_0
- **Description:** Interrupt Status for each IRQ type of TX SA 0
- **Size:** 32 bits
- **Offset:** 0x194
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 0

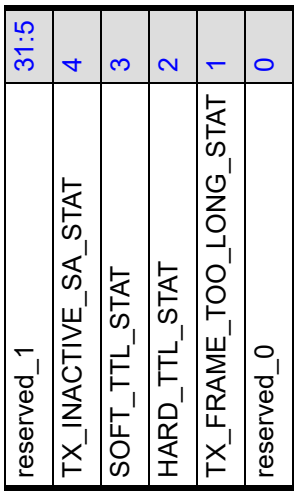


Table 6-108 Fields for Register: IRQ_TYPE_STAT_TX_SA_0

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-108 Fields for Register: IRQ_TYPE_STAT_TX_SA_0 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.102 IRQ_TYPE_EN_TX_SA_1

- **Name:** IRQ_TYPE_EN_TX_SA_1
- **Description:** Interrupt Enable for each IRQ type of TX SA 1
- **Size:** 32 bits
- **Offset:** 0x198
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 1

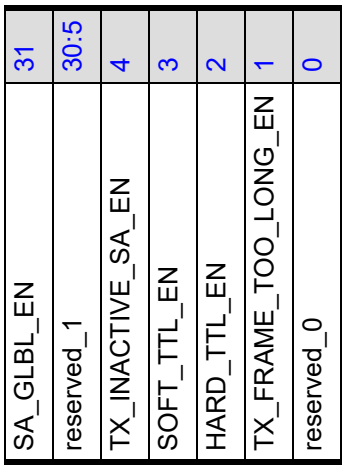


Table 6-109 Fields for Register: IRQ_TYPE_EN_TX_SA_1

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-109 Fields for Register: IRQ_TYPE_EN_TX_SA_1 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.103 IRQ_TYPE_STAT_TX_SA_1

- **Name:** IRQ_TYPE_STAT_TX_SA_1
- **Description:** Interrupt Status for each IRQ type of TX SA 1
- **Size:** 32 bits
- **Offset:** 0x19c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 1

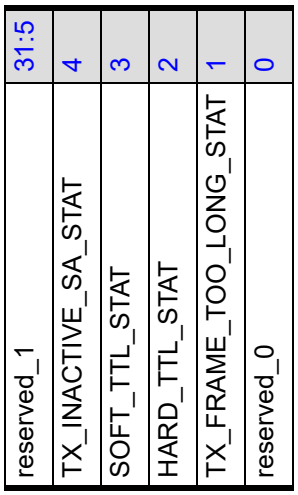


Table 6-110 Fields for Register: IRQ_TYPE_STAT_TX_SA_1

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-110 Fields for Register: IRQ_TYPE_STAT_TX_SA_1 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.104 IRQ_TYPE_EN_TX_SA_2

- **Name:** IRQ_TYPE_EN_TX_SA_2
- **Description:** Interrupt Enable for each IRQ type of TX SA 2
- **Size:** 32 bits
- **Offset:** 0x1a0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 2

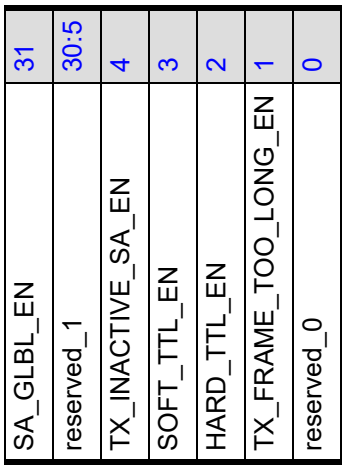


Table 6-111 Fields for Register: IRQ_TYPE_EN_TX_SA_2

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-111 Fields for Register: IRQ_TYPE_EN_TX_SA_2 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.105 IRQ_TYPE_STAT_TX_SA_2

- **Name:** IRQ_TYPE_STAT_TX_SA_2
- **Description:** Interrupt Status for each IRQ type of TX SA 2
- **Size:** 32 bits
- **Offset:** 0x1a4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 2

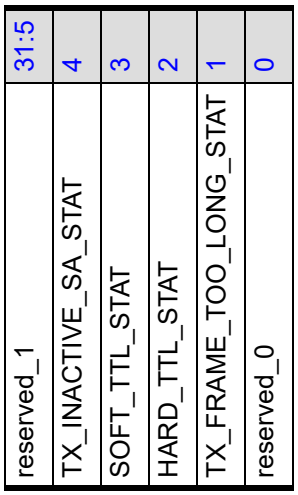


Table 6-112 Fields for Register: IRQ_TYPE_STAT_TX_SA_2

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-112 Fields for Register: IRQ_TYPE_STAT_TX_SA_2 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.106 IRQ_TYPE_EN_TX_SA_3

- **Name:** IRQ_TYPE_EN_TX_SA_3
- **Description:** Interrupt Enable for each IRQ type of TX SA 3
- **Size:** 32 bits
- **Offset:** 0x1a8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 3

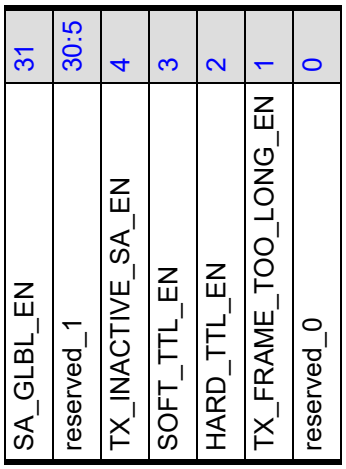


Table 6-113 Fields for Register: IRQ_TYPE_EN_TX_SA_3

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-113 Fields for Register: IRQ_TYPE_EN_TX_SA_3 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.107 IRQ_TYPE_STAT_TX_SA_3

- **Name:** IRQ_TYPE_STAT_TX_SA_3
- **Description:** Interrupt Status for each IRQ type of TX SA 3
- **Size:** 32 bits
- **Offset:** 0x1ac
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 3

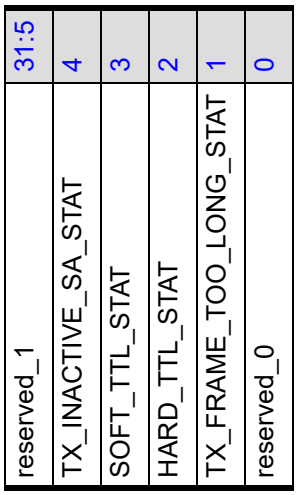


Table 6-114 Fields for Register: IRQ_TYPE_STAT_TX_SA_3

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-114 Fields for Register: IRQ_TYPE_STAT_TX_SA_3 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.108 IRQ_TYPE_EN_TX_SA_4

- **Name:** IRQ_TYPE_EN_TX_SA_4
- **Description:** Interrupt Enable for each IRQ type of TX SA 4
- **Size:** 32 bits
- **Offset:** 0x1b0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 4

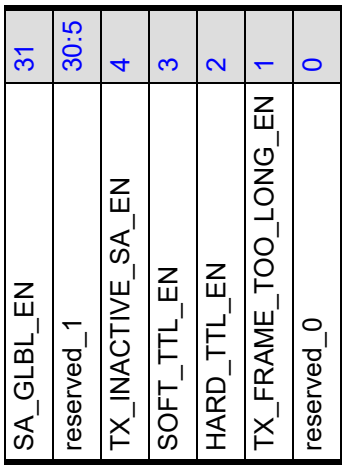


Table 6-115 Fields for Register: IRQ_TYPE_EN_TX_SA_4

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-115 Fields for Register: IRQ_TYPE_EN_TX_SA_4 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.109 IRQ_TYPE_STAT_TX_SA_4

- **Name:** IRQ_TYPE_STAT_TX_SA_4
- **Description:** Interrupt Status for each IRQ type of TX SA 4
- **Size:** 32 bits
- **Offset:** 0x1b4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 4

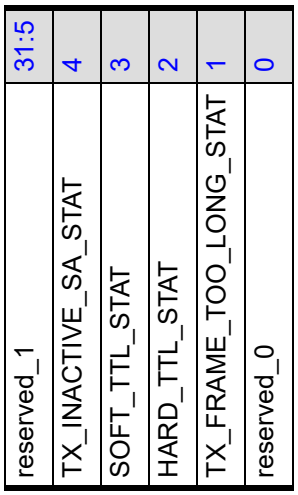


Table 6-116 Fields for Register: IRQ_TYPE_STAT_TX_SA_4

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-116 Fields for Register: IRQ_TYPE_STAT_TX_SA_4 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.110 IRQ_TYPE_EN_TX_SA_5

- **Name:** IRQ_TYPE_EN_TX_SA_5
- **Description:** Interrupt Enable for each IRQ type of TX SA 5
- **Size:** 32 bits
- **Offset:** 0x1b8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 5

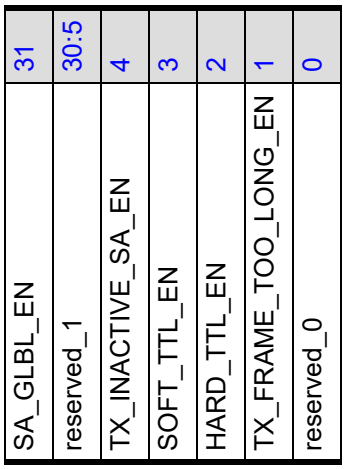


Table 6-117 Fields for Register: IRQ_TYPE_EN_TX_SA_5

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-117 Fields for Register: IRQ_TYPE_EN_TX_SA_5 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.111 IRQ_TYPE_STAT_TX_SA_5

- **Name:** IRQ_TYPE_STAT_TX_SA_5
- **Description:** Interrupt Status for each IRQ type of TX SA 5
- **Size:** 32 bits
- **Offset:** 0x1bc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 5

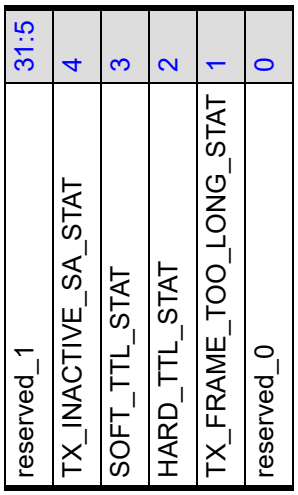


Table 6-118 Fields for Register: IRQ_TYPE_STAT_TX_SA_5

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-118 Fields for Register: IRQ_TYPE_STAT_TX_SA_5 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.112 IRQ_TYPE_EN_TX_SA_6

- **Name:** IRQ_TYPE_EN_TX_SA_6
- **Description:** Interrupt Enable for each IRQ type of TX SA 6
- **Size:** 32 bits
- **Offset:** 0x1c0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 6

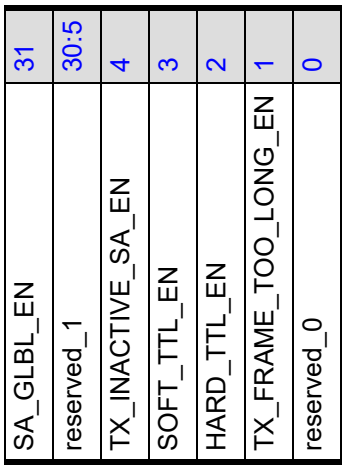


Table 6-119 Fields for Register: IRQ_TYPE_EN_TX_SA_6

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-119 Fields for Register: IRQ_TYPE_EN_TX_SA_6 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.113 IRQ_TYPE_STAT_TX_SA_6

- **Name:** IRQ_TYPE_STAT_TX_SA_6
- **Description:** Interrupt Status for each IRQ type of TX SA 6
- **Size:** 32 bits
- **Offset:** 0x1c4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 6

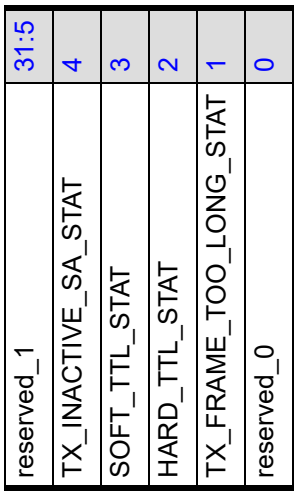


Table 6-120 Fields for Register: IRQ_TYPE_STAT_TX_SA_6

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-120 Fields for Register: IRQ_TYPE_STAT_TX_SA_6 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.114 IRQ_TYPE_EN_TX_SA_7

- **Name:** IRQ_TYPE_EN_TX_SA_7
- **Description:** Interrupt Enable for each IRQ type of TX SA 7
- **Size:** 32 bits
- **Offset:** 0x1c8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 7

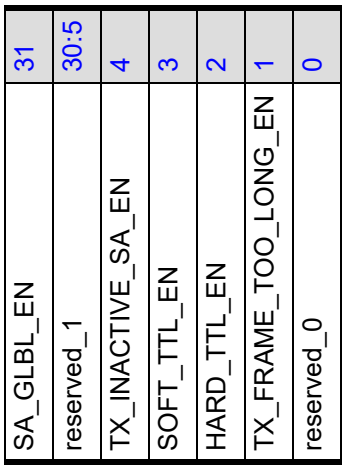


Table 6-121 Fields for Register: IRQ_TYPE_EN_TX_SA_7

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-121 Fields for Register: IRQ_TYPE_EN_TX_SA_7 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.115 IRQ_TYPE_STAT_TX_SA_7

- **Name:** IRQ_TYPE_STAT_TX_SA_7
- **Description:** Interrupt Status for each IRQ type of TX SA 7
- **Size:** 32 bits
- **Offset:** 0x1cc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 7

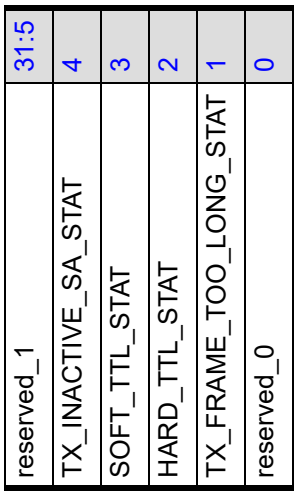


Table 6-122 Fields for Register: IRQ_TYPE_STAT_TX_SA_7

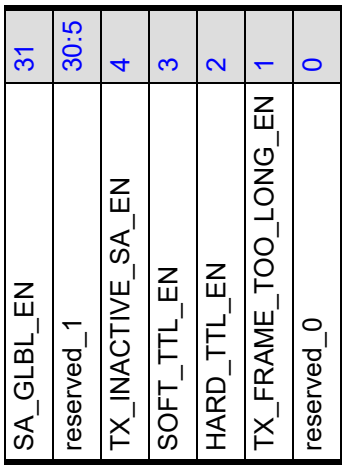
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-122 Fields for Register: IRQ_TYPE_STAT_TX_SA_7 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.116 IRQ_TYPE_EN_TX_SA_8

- **Name:** IRQ_TYPE_EN_TX_SA_8
- **Description:** Interrupt Enable for each IRQ type of TX SA 8
- **Size:** 32 bits
- **Offset:** 0x1d0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 8



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-123 Fields for Register: IRQ_TYPE_EN_TX_SA_8

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-123 Fields for Register: IRQ_TYPE_EN_TX_SA_8 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.117 IRQ_TYPE_STAT_TX_SA_8

- **Name:** IRQ_TYPE_STAT_TX_SA_8
- **Description:** Interrupt Status for each IRQ type of TX SA 8
- **Size:** 32 bits
- **Offset:** 0x1d4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 8

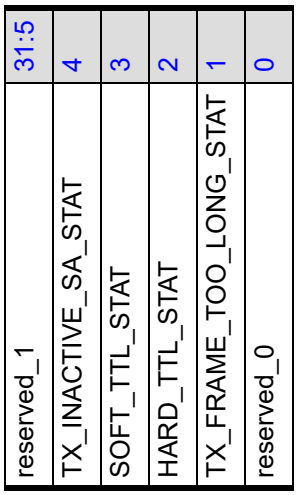


Table 6-124 Fields for Register: IRQ_TYPE_STAT_TX_SA_8

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-124 Fields for Register: IRQ_TYPE_STAT_TX_SA_8 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.118 IRQ_TYPE_EN_TX_SA_9

- **Name:** IRQ_TYPE_EN_TX_SA_9
- **Description:** Interrupt Enable for each IRQ type of TX SA 9
- **Size:** 32 bits
- **Offset:** 0x1d8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 9

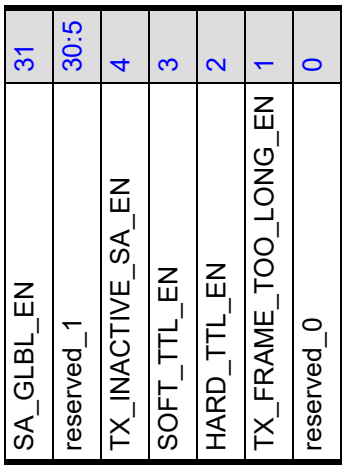


Table 6-125 Fields for Register: IRQ_TYPE_EN_TX_SA_9

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-125 Fields for Register: IRQ_TYPE_EN_TX_SA_9 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.119 IRQ_TYPE_STAT_TX_SA_9

- **Name:** IRQ_TYPE_STAT_TX_SA_9
- **Description:** Interrupt Status for each IRQ type of TX SA 9
- **Size:** 32 bits
- **Offset:** 0x1dc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 9

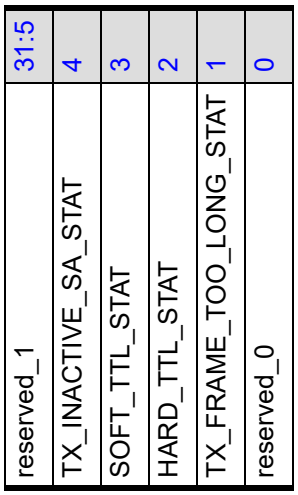


Table 6-126 Fields for Register: IRQ_TYPE_STAT_TX_SA_9

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-126 Fields for Register: IRQ_TYPE_STAT_TX_SA_9 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.120 IRQ_TYPE_EN_TX_SA_10

- **Name:** IRQ_TYPE_EN_TX_SA_10
- **Description:** Interrupt Enable for each IRQ type of TX SA 10
- **Size:** 32 bits
- **Offset:** 0x1e0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 10

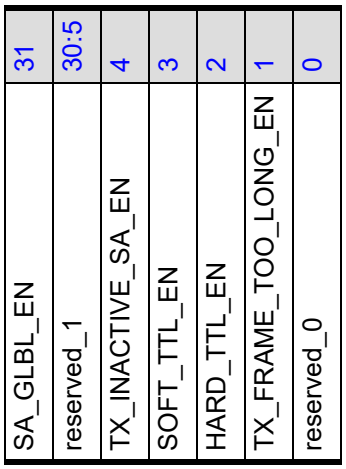


Table 6-127 Fields for Register: IRQ_TYPE_EN_TX_SA_10

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-127 Fields for Register: IRQ_TYPE_EN_TX_SA_10 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.121 IRQ_TYPE_STAT_TX_SA_10

- **Name:** IRQ_TYPE_STAT_TX_SA_10
- **Description:** Interrupt Status for each IRQ type of TX SA 10
- **Size:** 32 bits
- **Offset:** 0x1e4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 10

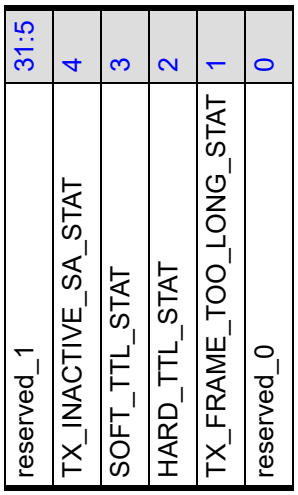


Table 6-128 Fields for Register: IRQ_TYPE_STAT_TX_SA_10

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-128 Fields for Register: IRQ_TYPE_STAT_TX_SA_10 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.122 IRQ_TYPE_EN_TX_SA_11

- **Name:** IRQ_TYPE_EN_TX_SA_11
- **Description:** Interrupt Enable for each IRQ type of TX SA 11
- **Size:** 32 bits
- **Offset:** 0x1e8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 11

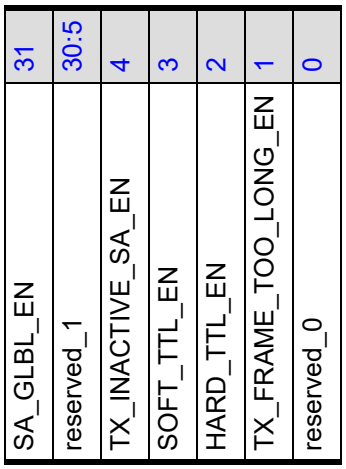


Table 6-129 Fields for Register: IRQ_TYPE_EN_TX_SA_11

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-129 Fields for Register: IRQ_TYPE_EN_TX_SA_11 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.123 IRQ_TYPE_STAT_TX_SA_11

- **Name:** IRQ_TYPE_STAT_TX_SA_11
- **Description:** Interrupt Status for each IRQ type of TX SA 11
- **Size:** 32 bits
- **Offset:** 0x1ec
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 11

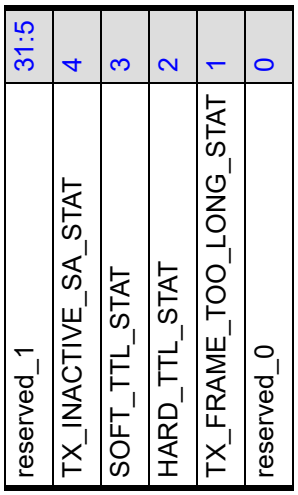


Table 6-130 Fields for Register: IRQ_TYPE_STAT_TX_SA_11

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-130 Fields for Register: IRQ_TYPE_STAT_TX_SA_11 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.124 IRQ_TYPE_EN_TX_SA_12

- **Name:** IRQ_TYPE_EN_TX_SA_12
- **Description:** Interrupt Enable for each IRQ type of TX SA 12
- **Size:** 32 bits
- **Offset:** 0x1f0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 12

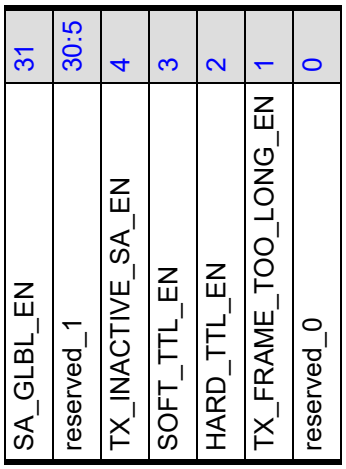


Table 6-131 Fields for Register: IRQ_TYPE_EN_TX_SA_12

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-131 Fields for Register: IRQ_TYPE_EN_TX_SA_12 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.125 IRQ_TYPE_STAT_TX_SA_12

- **Name:** IRQ_TYPE_STAT_TX_SA_12
- **Description:** Interrupt Status for each IRQ type of TX SA 12
- **Size:** 32 bits
- **Offset:** 0x1f4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 12

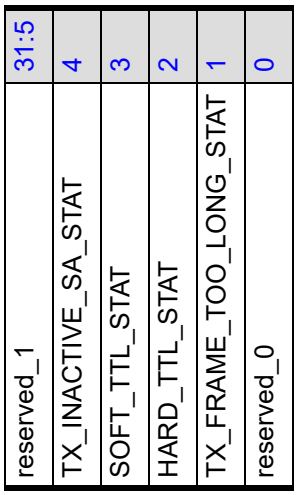


Table 6-132 Fields for Register: IRQ_TYPE_STAT_TX_SA_12

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-132 Fields for Register: IRQ_TYPE_STAT_TX_SA_12 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.126 IRQ_TYPE_EN_TX_SA_13

- **Name:** IRQ_TYPE_EN_TX_SA_13
- **Description:** Interrupt Enable for each IRQ type of TX SA 13
- **Size:** 32 bits
- **Offset:** 0x1f8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 13

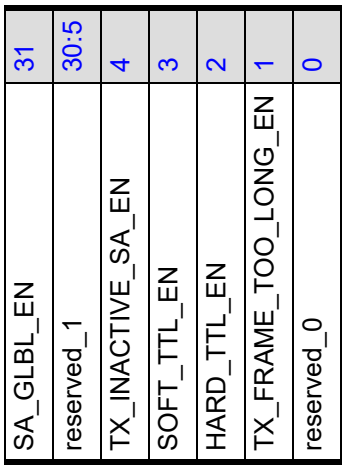


Table 6-133 Fields for Register: IRQ_TYPE_EN_TX_SA_13

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-133 Fields for Register: IRQ_TYPE_EN_TX_SA_13 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.127 IRQ_TYPE_STAT_TX_SA_13

- **Name:** IRQ_TYPE_STAT_TX_SA_13
- **Description:** Interrupt Status for each IRQ type of TX SA 13
- **Size:** 32 bits
- **Offset:** 0x1fc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 13

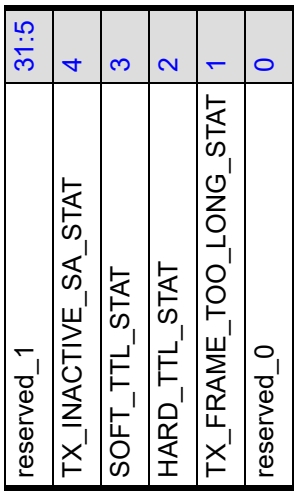


Table 6-134 Fields for Register: IRQ_TYPE_STAT_TX_SA_13

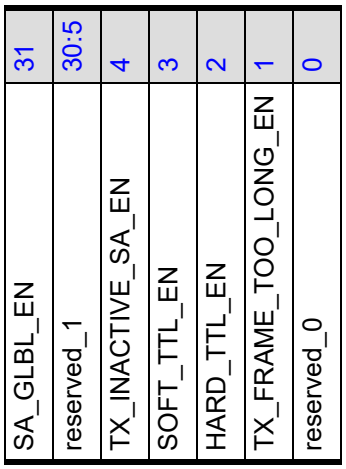
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-134 Fields for Register: IRQ_TYPE_STAT_TX_SA_13 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.128 IRQ_TYPE_EN_TX_SA_14

- **Name:** IRQ_TYPE_EN_TX_SA_14
- **Description:** Interrupt Enable for each IRQ type of TX SA 14
- **Size:** 32 bits
- **Offset:** 0x200
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 14



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-135 Fields for Register: IRQ_TYPE_EN_TX_SA_14

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-135 Fields for Register: IRQ_TYPE_EN_TX_SA_14 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.129 IRQ_TYPE_STAT_TX_SA_14

- **Name:** IRQ_TYPE_STAT_TX_SA_14
- **Description:** Interrupt Status for each IRQ type of TX SA 14
- **Size:** 32 bits
- **Offset:** 0x204
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 14

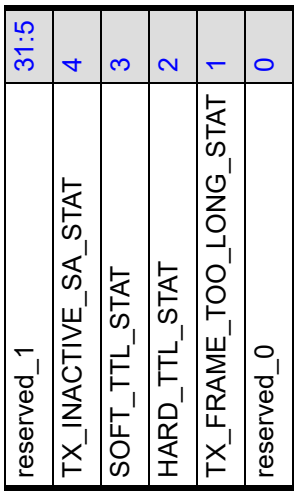


Table 6-136 Fields for Register: IRQ_TYPE_STAT_TX_SA_14

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-136 Fields for Register: IRQ_TYPE_STAT_TX_SA_14 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.130 IRQ_TYPE_EN_TX_SA_15

- **Name:** IRQ_TYPE_EN_TX_SA_15
- **Description:** Interrupt Enable for each IRQ type of TX SA 15
- **Size:** 32 bits
- **Offset:** 0x208
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 15

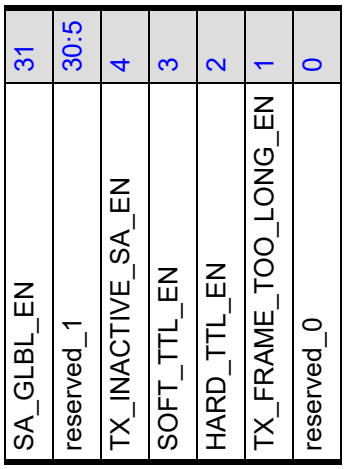


Table 6-137 Fields for Register: IRQ_TYPE_EN_TX_SA_15

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-137 Fields for Register: IRQ_TYPE_EN_TX_SA_15 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.131 IRQ_TYPE_STAT_TX_SA_15

- **Name:** IRQ_TYPE_STAT_TX_SA_15
- **Description:** Interrupt Status for each IRQ type of TX SA 15
- **Size:** 32 bits
- **Offset:** 0x20c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 15

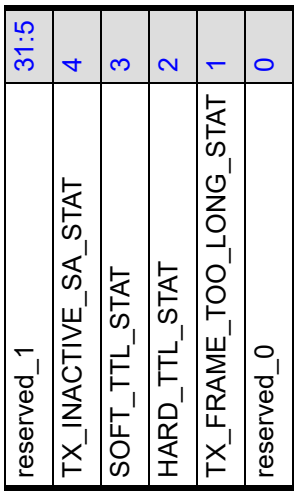


Table 6-138 Fields for Register: IRQ_TYPE_STAT_TX_SA_15

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-138 Fields for Register: IRQ_TYPE_STAT_TX_SA_15 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.132 IRQ_TYPE_EN_TX_SA_16

- **Name:** IRQ_TYPE_EN_TX_SA_16
- **Description:** Interrupt Enable for each IRQ type of TX SA 16
- **Size:** 32 bits
- **Offset:** 0x210
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 16

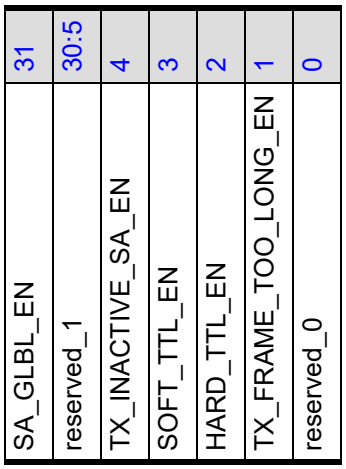


Table 6-139 Fields for Register: IRQ_TYPE_EN_TX_SA_16

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-139 Fields for Register: IRQ_TYPE_EN_TX_SA_16 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.133 IRQ_TYPE_STAT_TX_SA_16

- **Name:** IRQ_TYPE_STAT_TX_SA_16
- **Description:** Interrupt Status for each IRQ type of TX SA 16
- **Size:** 32 bits
- **Offset:** 0x214
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 16

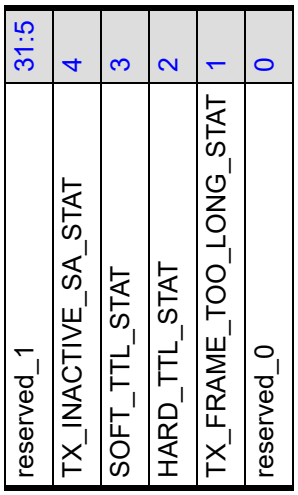


Table 6-140 Fields for Register: IRQ_TYPE_STAT_TX_SA_16

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-140 Fields for Register: IRQ_TYPE_STAT_TX_SA_16 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.134 IRQ_TYPE_EN_TX_SA_17

- **Name:** IRQ_TYPE_EN_TX_SA_17
- **Description:** Interrupt Enable for each IRQ type of TX SA 17
- **Size:** 32 bits
- **Offset:** 0x218
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 17

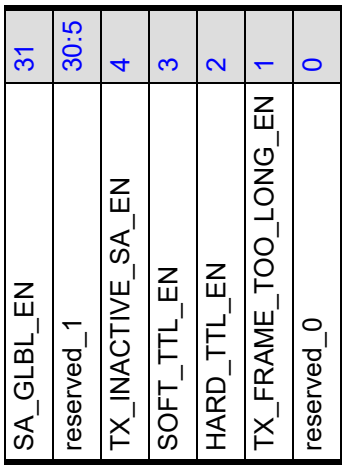


Table 6-141 Fields for Register: IRQ_TYPE_EN_TX_SA_17

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-141 Fields for Register: IRQ_TYPE_EN_TX_SA_17 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.135 IRQ_TYPE_STAT_TX_SA_17

- **Name:** IRQ_TYPE_STAT_TX_SA_17
- **Description:** Interrupt Status for each IRQ type of TX SA 17
- **Size:** 32 bits
- **Offset:** 0x21c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 17

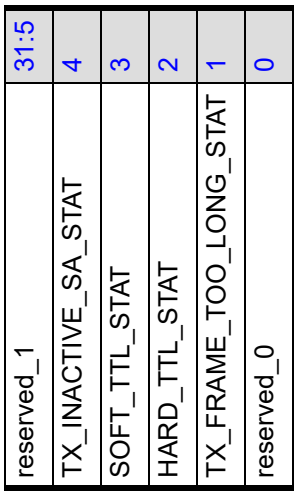


Table 6-142 Fields for Register: IRQ_TYPE_STAT_TX_SA_17

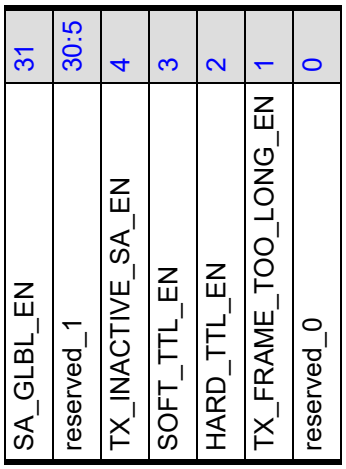
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-142 Fields for Register: IRQ_TYPE_STAT_TX_SA_17 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.136 IRQ_TYPE_EN_TX_SA_18

- **Name:** IRQ_TYPE_EN_TX_SA_18
- **Description:** Interrupt Enable for each IRQ type of TX SA 18
- **Size:** 32 bits
- **Offset:** 0x220
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 18



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-143 Fields for Register: IRQ_TYPE_EN_TX_SA_18

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-143 Fields for Register: IRQ_TYPE_EN_TX_SA_18 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.137 IRQ_TYPE_STAT_TX_SA_18

- **Name:** IRQ_TYPE_STAT_TX_SA_18
- **Description:** Interrupt Status for each IRQ type of TX SA 18
- **Size:** 32 bits
- **Offset:** 0x224
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 18

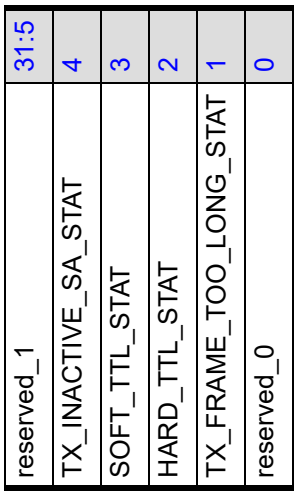


Table 6-144 Fields for Register: IRQ_TYPE_STAT_TX_SA_18

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-144 Fields for Register: IRQ_TYPE_STAT_TX_SA_18 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.138 IRQ_TYPE_EN_TX_SA_19

- **Name:** IRQ_TYPE_EN_TX_SA_19
- **Description:** Interrupt Enable for each IRQ type of TX SA 19
- **Size:** 32 bits
- **Offset:** 0x228
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 19

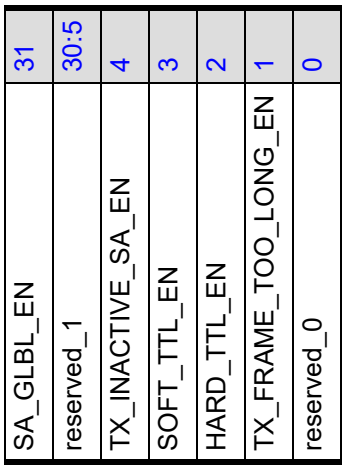


Table 6-145 Fields for Register: IRQ_TYPE_EN_TX_SA_19

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-145 Fields for Register: IRQ_TYPE_EN_TX_SA_19 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.139 IRQ_TYPE_STAT_TX_SA_19

- **Name:** IRQ_TYPE_STAT_TX_SA_19
- **Description:** Interrupt Status for each IRQ type of TX SA 19
- **Size:** 32 bits
- **Offset:** 0x22c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 19

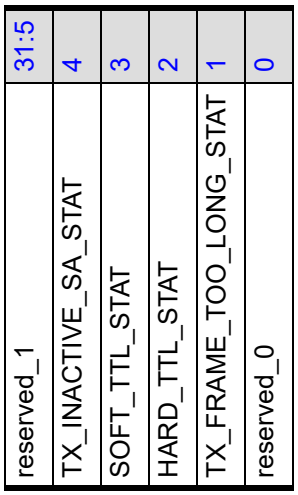


Table 6-146 Fields for Register: IRQ_TYPE_STAT_TX_SA_19

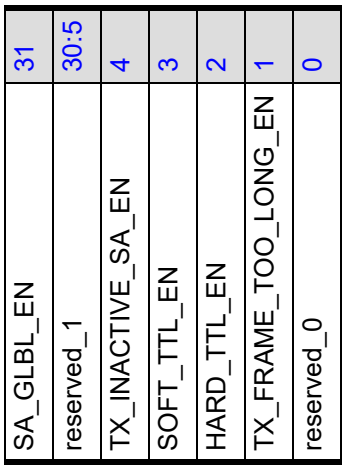
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-146 Fields for Register: IRQ_TYPE_STAT_TX_SA_19 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.140 IRQ_TYPE_EN_TX_SA_20

- **Name:** IRQ_TYPE_EN_TX_SA_20
- **Description:** Interrupt Enable for each IRQ type of TX SA 20
- **Size:** 32 bits
- **Offset:** 0x230
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 20



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-147 Fields for Register: IRQ_TYPE_EN_TX_SA_20

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-147 Fields for Register: IRQ_TYPE_EN_TX_SA_20 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.141 IRQ_TYPE_STAT_TX_SA_20

- **Name:** IRQ_TYPE_STAT_TX_SA_20
- **Description:** Interrupt Status for each IRQ type of TX SA 20
- **Size:** 32 bits
- **Offset:** 0x234
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 20

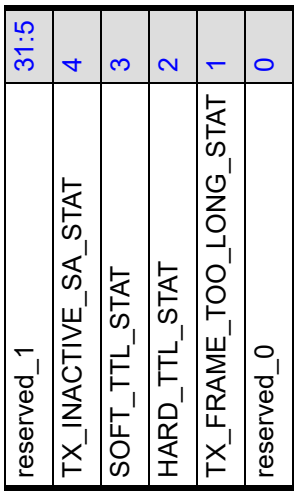


Table 6-148 Fields for Register: IRQ_TYPE_STAT_TX_SA_20

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-148 Fields for Register: IRQ_TYPE_STAT_TX_SA_20 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.142 IRQ_TYPE_EN_TX_SA_21

- **Name:** IRQ_TYPE_EN_TX_SA_21
- **Description:** Interrupt Enable for each IRQ type of TX SA 21
- **Size:** 32 bits
- **Offset:** 0x238
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 21

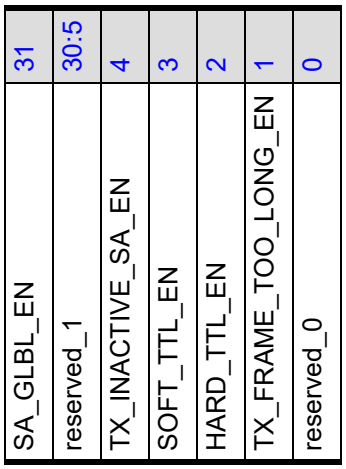


Table 6-149 Fields for Register: IRQ_TYPE_EN_TX_SA_21

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-149 Fields for Register: IRQ_TYPE_EN_TX_SA_21 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.143 IRQ_TYPE_STAT_TX_SA_21

- **Name:** IRQ_TYPE_STAT_TX_SA_21
- **Description:** Interrupt Status for each IRQ type of TX SA 21
- **Size:** 32 bits
- **Offset:** 0x23c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 21

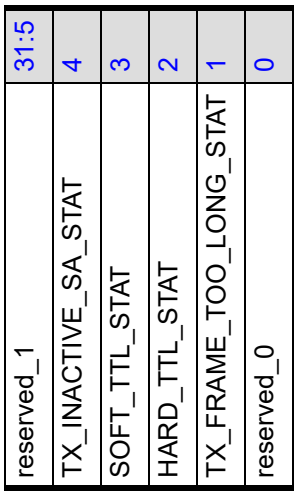


Table 6-150 Fields for Register: IRQ_TYPE_STAT_TX_SA_21

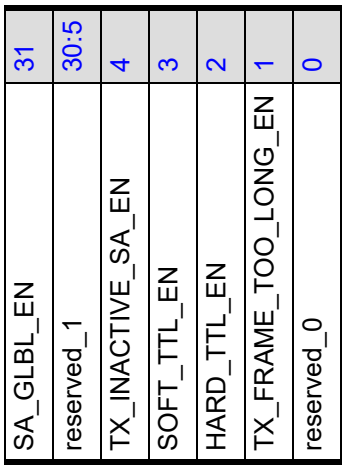
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-150 Fields for Register: IRQ_TYPE_STAT_TX_SA_21 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.144 IRQ_TYPE_EN_TX_SA_22

- **Name:** IRQ_TYPE_EN_TX_SA_22
- **Description:** Interrupt Enable for each IRQ type of TX SA 22
- **Size:** 32 bits
- **Offset:** 0x240
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 22



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-151 Fields for Register: IRQ_TYPE_EN_TX_SA_22

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-151 Fields for Register: IRQ_TYPE_EN_TX_SA_22 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.145 IRQ_TYPE_STAT_TX_SA_22

- **Name:** IRQ_TYPE_STAT_TX_SA_22
- **Description:** Interrupt Status for each IRQ type of TX SA 22
- **Size:** 32 bits
- **Offset:** 0x244
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 22

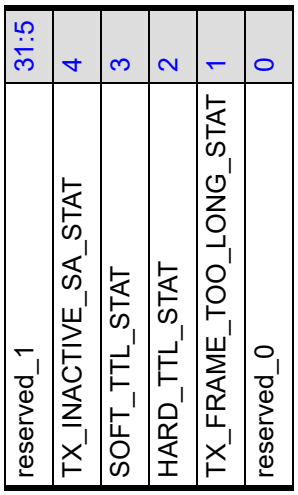


Table 6-152 Fields for Register: IRQ_TYPE_STAT_TX_SA_22

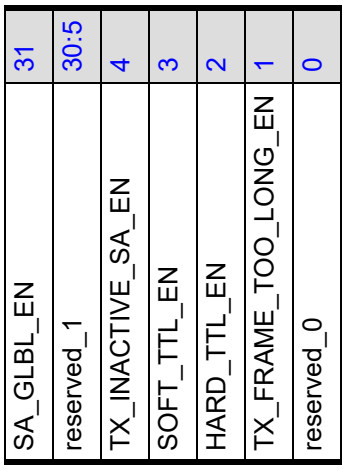
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-152 Fields for Register: IRQ_TYPE_STAT_TX_SA_22 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.146 IRQ_TYPE_EN_TX_SA_23

- **Name:** IRQ_TYPE_EN_TX_SA_23
- **Description:** Interrupt Enable for each IRQ type of TX SA 23
- **Size:** 32 bits
- **Offset:** 0x248
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 23



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-153 Fields for Register: IRQ_TYPE_EN_TX_SA_23

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-153 Fields for Register: IRQ_TYPE_EN_TX_SA_23 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.147 IRQ_TYPE_STAT_TX_SA_23

- **Name:** IRQ_TYPE_STAT_TX_SA_23
- **Description:** Interrupt Status for each IRQ type of TX SA 23
- **Size:** 32 bits
- **Offset:** 0x24c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 23

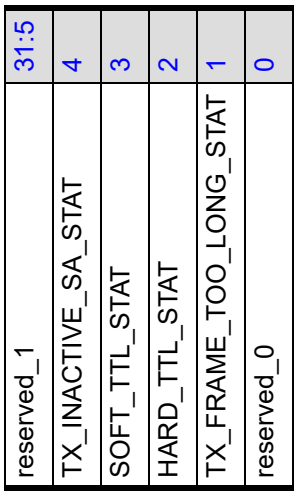


Table 6-154 Fields for Register: IRQ_TYPE_STAT_TX_SA_23

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-154 Fields for Register: IRQ_TYPE_STAT_TX_SA_23 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.148 IRQ_TYPE_EN_TX_SA_24

- **Name:** IRQ_TYPE_EN_TX_SA_24
- **Description:** Interrupt Enable for each IRQ type of TX SA 24
- **Size:** 32 bits
- **Offset:** 0x250
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 24

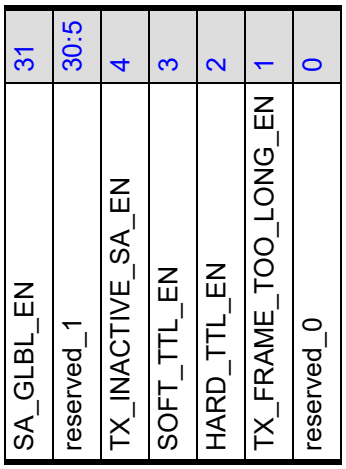


Table 6-155 Fields for Register: IRQ_TYPE_EN_TX_SA_24

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-155 Fields for Register: IRQ_TYPE_EN_TX_SA_24 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.149 IRQ_TYPE_STAT_TX_SA_24

- **Name:** IRQ_TYPE_STAT_TX_SA_24
- **Description:** Interrupt Status for each IRQ type of TX SA 24
- **Size:** 32 bits
- **Offset:** 0x254
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 24

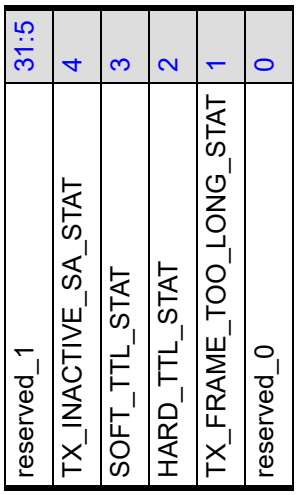


Table 6-156 Fields for Register: IRQ_TYPE_STAT_TX_SA_24

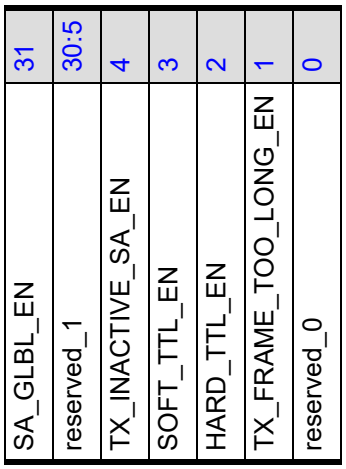
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-156 Fields for Register: IRQ_TYPE_STAT_TX_SA_24 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.150 IRQ_TYPE_EN_TX_SA_25

- **Name:** IRQ_TYPE_EN_TX_SA_25
- **Description:** Interrupt Enable for each IRQ type of TX SA 25
- **Size:** 32 bits
- **Offset:** 0x258
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 25



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-157 Fields for Register: IRQ_TYPE_EN_TX_SA_25

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-157 Fields for Register: IRQ_TYPE_EN_TX_SA_25 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.151 IRQ_TYPE_STAT_TX_SA_25

- **Name:** IRQ_TYPE_STAT_TX_SA_25
- **Description:** Interrupt Status for each IRQ type of TX SA 25
- **Size:** 32 bits
- **Offset:** 0x25c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 25

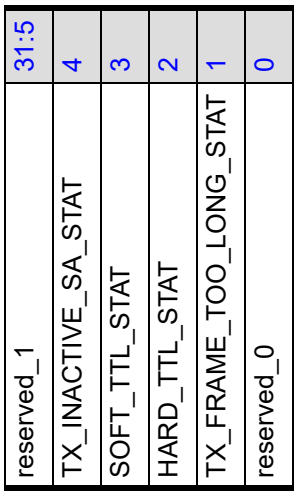


Table 6-158 Fields for Register: IRQ_TYPE_STAT_TX_SA_25

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-158 Fields for Register: IRQ_TYPE_STAT_TX_SA_25 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.152 IRQ_TYPE_EN_TX_SA_26

- **Name:** IRQ_TYPE_EN_TX_SA_26
- **Description:** Interrupt Enable for each IRQ type of TX SA 26
- **Size:** 32 bits
- **Offset:** 0x260
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 26

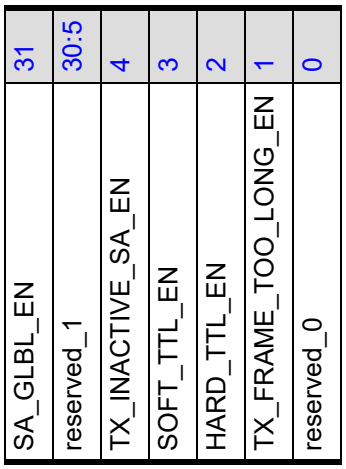


Table 6-159 Fields for Register: IRQ_TYPE_EN_TX_SA_26

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-159 Fields for Register: IRQ_TYPE_EN_TX_SA_26 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.153 IRQ_TYPE_STAT_TX_SA_26

- **Name:** IRQ_TYPE_STAT_TX_SA_26
- **Description:** Interrupt Status for each IRQ type of TX SA 26
- **Size:** 32 bits
- **Offset:** 0x264
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 26

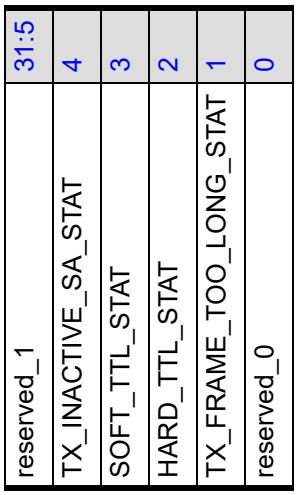


Table 6-160 Fields for Register: IRQ_TYPE_STAT_TX_SA_26

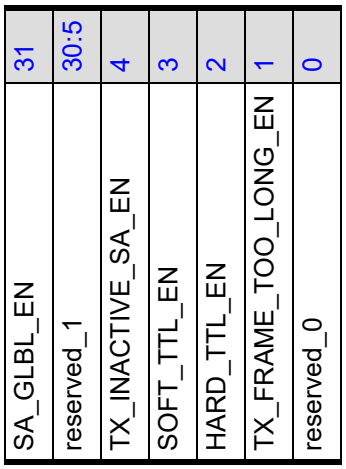
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-160 Fields for Register: IRQ_TYPE_STAT_TX_SA_26 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.154 IRQ_TYPE_EN_TX_SA_27

- **Name:** IRQ_TYPE_EN_TX_SA_27
- **Description:** Interrupt Enable for each IRQ type of TX SA 27
- **Size:** 32 bits
- **Offset:** 0x268
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 27



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-161 Fields for Register: IRQ_TYPE_EN_TX_SA_27

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-161 Fields for Register: IRQ_TYPE_EN_TX_SA_27 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.155 IRQ_TYPE_STAT_TX_SA_27

- **Name:** IRQ_TYPE_STAT_TX_SA_27
- **Description:** Interrupt Status for each IRQ type of TX SA 27
- **Size:** 32 bits
- **Offset:** 0x26c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 27

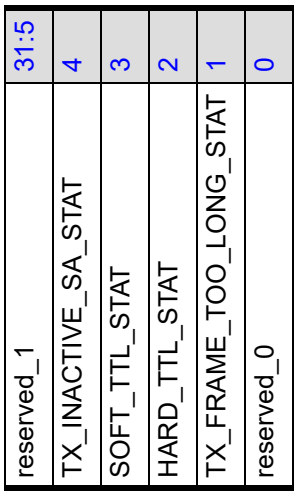


Table 6-162 Fields for Register: IRQ_TYPE_STAT_TX_SA_27

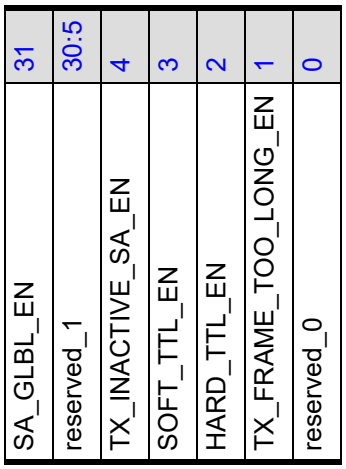
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-162 Fields for Register: IRQ_TYPE_STAT_TX_SA_27 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.156 IRQ_TYPE_EN_TX_SA_28

- **Name:** IRQ_TYPE_EN_TX_SA_28
- **Description:** Interrupt Enable for each IRQ type of TX SA 28
- **Size:** 32 bits
- **Offset:** 0x270
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 28



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-163 Fields for Register: IRQ_TYPE_EN_TX_SA_28

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-163 Fields for Register: IRQ_TYPE_EN_TX_SA_28 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.157 IRQ_TYPE_STAT_TX_SA_28

- **Name:** IRQ_TYPE_STAT_TX_SA_28
- **Description:** Interrupt Status for each IRQ type of TX SA 28
- **Size:** 32 bits
- **Offset:** 0x274
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 28

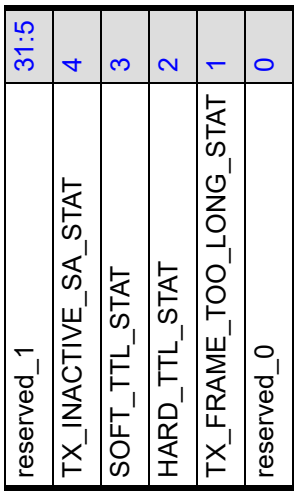


Table 6-164 Fields for Register: IRQ_TYPE_STAT_TX_SA_28

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-164 Fields for Register: IRQ_TYPE_STAT_TX_SA_28 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.158 IRQ_TYPE_EN_TX_SA_29

- **Name:** IRQ_TYPE_EN_TX_SA_29
- **Description:** Interrupt Enable for each IRQ type of TX SA 29
- **Size:** 32 bits
- **Offset:** 0x278
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 29

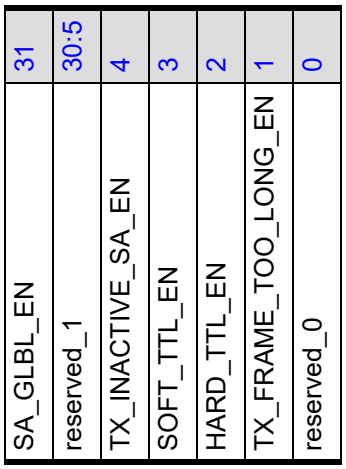


Table 6-165 Fields for Register: IRQ_TYPE_EN_TX_SA_29

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-165 Fields for Register: IRQ_TYPE_EN_TX_SA_29 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.159 IRQ_TYPE_STAT_TX_SA_29

- **Name:** IRQ_TYPE_STAT_TX_SA_29
- **Description:** Interrupt Status for each IRQ type of TX SA 29
- **Size:** 32 bits
- **Offset:** 0x27c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 29

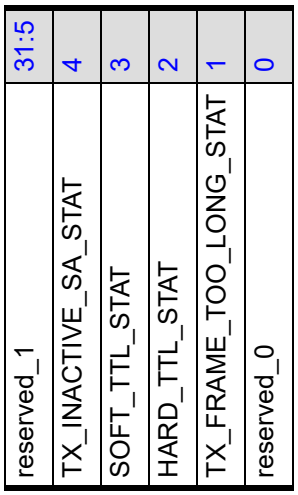


Table 6-166 Fields for Register: IRQ_TYPE_STAT_TX_SA_29

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-166 Fields for Register: IRQ_TYPE_STAT_TX_SA_29 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.160 IRQ_TYPE_EN_TX_SA_30

- **Name:** IRQ_TYPE_EN_TX_SA_30
- **Description:** Interrupt Enable for each IRQ type of TX SA 30
- **Size:** 32 bits
- **Offset:** 0x280
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 30

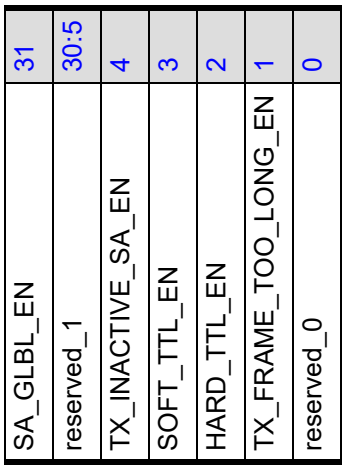


Table 6-167 Fields for Register: IRQ_TYPE_EN_TX_SA_30

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-167 Fields for Register: IRQ_TYPE_EN_TX_SA_30 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.161 IRQ_TYPE_STAT_TX_SA_30

- **Name:** IRQ_TYPE_STAT_TX_SA_30
- **Description:** Interrupt Status for each IRQ type of TX SA 30
- **Size:** 32 bits
- **Offset:** 0x284
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 30

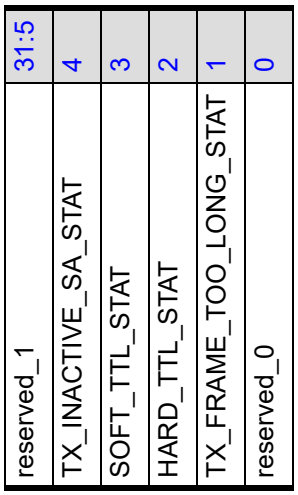


Table 6-168 Fields for Register: IRQ_TYPE_STAT_TX_SA_30

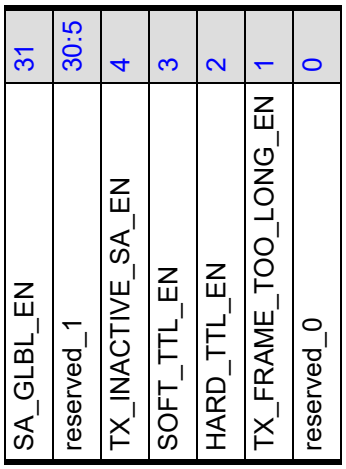
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-168 Fields for Register: IRQ_TYPE_STAT_TX_SA_30 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.162 IRQ_TYPE_EN_TX_SA_31

- **Name:** IRQ_TYPE_EN_TX_SA_31
- **Description:** Interrupt Enable for each IRQ type of TX SA 31
- **Size:** 32 bits
- **Offset:** 0x288
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 31



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-169 Fields for Register: IRQ_TYPE_EN_TX_SA_31

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-169 Fields for Register: IRQ_TYPE_EN_TX_SA_31 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.163 IRQ_TYPE_STAT_TX_SA_31

- **Name:** IRQ_TYPE_STAT_TX_SA_31
- **Description:** Interrupt Status for each IRQ type of TX SA 31
- **Size:** 32 bits
- **Offset:** 0x28c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 31

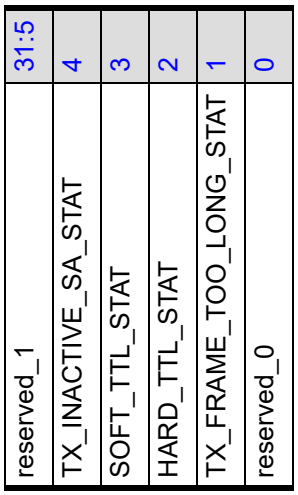


Table 6-170 Fields for Register: IRQ_TYPE_STAT_TX_SA_31

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-170 Fields for Register: IRQ_TYPE_STAT_TX_SA_31 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.164 IRQ_TYPE_EN_TX_SA_32

- **Name:** IRQ_TYPE_EN_TX_SA_32
- **Description:** Interrupt Enable for each IRQ type of TX SA 32
- **Size:** 32 bits
- **Offset:** 0x290
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 32

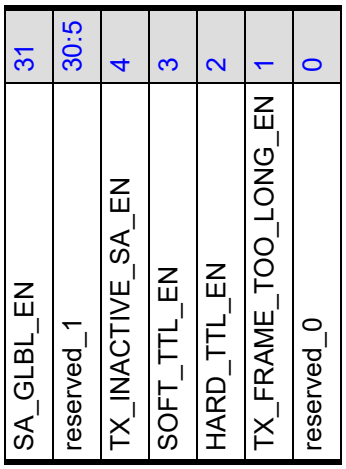


Table 6-171 Fields for Register: IRQ_TYPE_EN_TX_SA_32

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-171 Fields for Register: IRQ_TYPE_EN_TX_SA_32 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.165 IRQ_TYPE_STAT_TX_SA_32

- **Name:** IRQ_TYPE_STAT_TX_SA_32
- **Description:** Interrupt Status for each IRQ type of TX SA 32
- **Size:** 32 bits
- **Offset:** 0x294
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 32

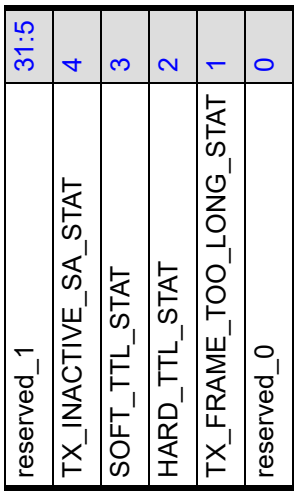


Table 6-172 Fields for Register: IRQ_TYPE_STAT_TX_SA_32

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-172 Fields for Register: IRQ_TYPE_STAT_TX_SA_32 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.166 IRQ_TYPE_EN_TX_SA_33

- **Name:** IRQ_TYPE_EN_TX_SA_33
- **Description:** Interrupt Enable for each IRQ type of TX SA 33
- **Size:** 32 bits
- **Offset:** 0x298
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 33

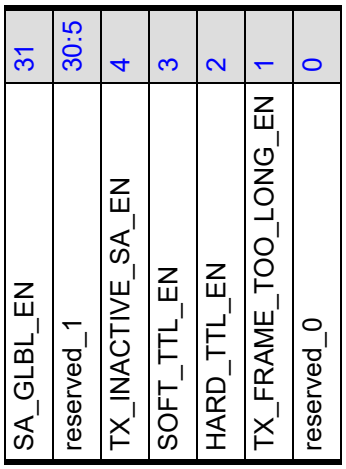


Table 6-173 Fields for Register: IRQ_TYPE_EN_TX_SA_33

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-173 Fields for Register: IRQ_TYPE_EN_TX_SA_33 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.167 IRQ_TYPE_STAT_TX_SA_33

- **Name:** IRQ_TYPE_STAT_TX_SA_33
- **Description:** Interrupt Status for each IRQ type of TX SA 33
- **Size:** 32 bits
- **Offset:** 0x29c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 33

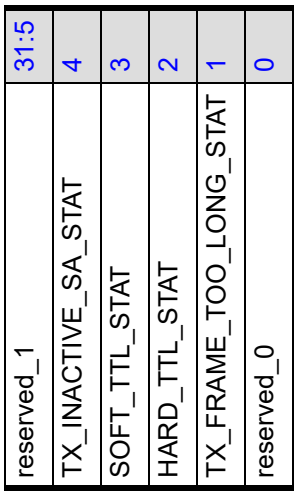


Table 6-174 Fields for Register: IRQ_TYPE_STAT_TX_SA_33

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-174 Fields for Register: IRQ_TYPE_STAT_TX_SA_33 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.168 IRQ_TYPE_EN_TX_SA_34

- **Name:** IRQ_TYPE_EN_TX_SA_34
- **Description:** Interrupt Enable for each IRQ type of TX SA 34
- **Size:** 32 bits
- **Offset:** 0x2a0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 34

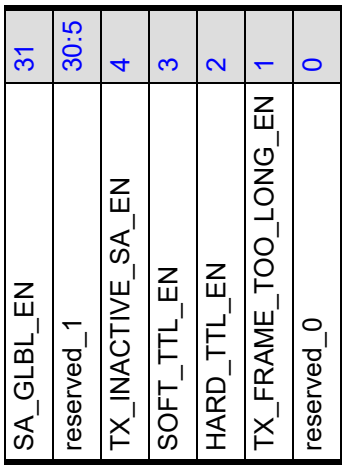


Table 6-175 Fields for Register: IRQ_TYPE_EN_TX_SA_34

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-175 Fields for Register: IRQ_TYPE_EN_TX_SA_34 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.169 IRQ_TYPE_STAT_TX_SA_34

- **Name:** IRQ_TYPE_STAT_TX_SA_34
- **Description:** Interrupt Status for each IRQ type of TX SA 34
- **Size:** 32 bits
- **Offset:** 0x2a4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 34

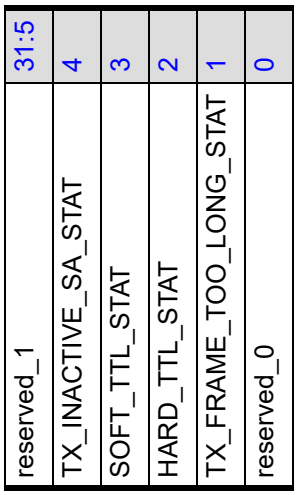


Table 6-176 Fields for Register: IRQ_TYPE_STAT_TX_SA_34

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-176 Fields for Register: IRQ_TYPE_STAT_TX_SA_34 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.170 IRQ_TYPE_EN_TX_SA_35

- **Name:** IRQ_TYPE_EN_TX_SA_35
- **Description:** Interrupt Enable for each IRQ type of TX SA 35
- **Size:** 32 bits
- **Offset:** 0x2a8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 35

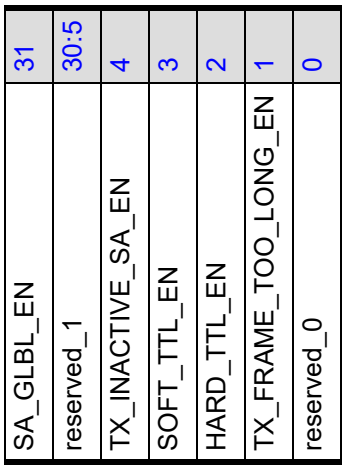


Table 6-177 Fields for Register: IRQ_TYPE_EN_TX_SA_35

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-177 Fields for Register: IRQ_TYPE_EN_TX_SA_35 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.171 IRQ_TYPE_STAT_TX_SA_35

- **Name:** IRQ_TYPE_STAT_TX_SA_35
- **Description:** Interrupt Status for each IRQ type of TX SA 35
- **Size:** 32 bits
- **Offset:** 0x2ac
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 35

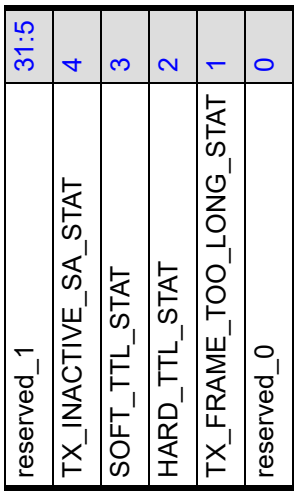


Table 6-178 Fields for Register: IRQ_TYPE_STAT_TX_SA_35

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-178 Fields for Register: IRQ_TYPE_STAT_TX_SA_35 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.172 IRQ_TYPE_EN_TX_SA_36

- **Name:** IRQ_TYPE_EN_TX_SA_36
- **Description:** Interrupt Enable for each IRQ type of TX SA 36
- **Size:** 32 bits
- **Offset:** 0x2b0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 36

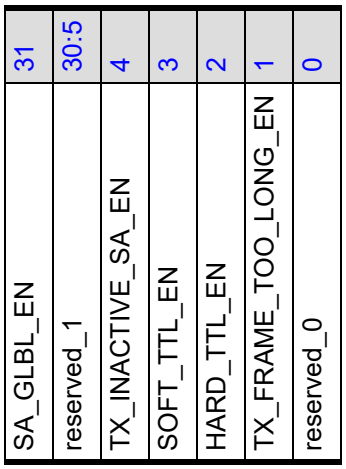


Table 6-179 Fields for Register: IRQ_TYPE_EN_TX_SA_36

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-179 Fields for Register: IRQ_TYPE_EN_TX_SA_36 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.173 IRQ_TYPE_STAT_TX_SA_36

- **Name:** IRQ_TYPE_STAT_TX_SA_36
- **Description:** Interrupt Status for each IRQ type of TX SA 36
- **Size:** 32 bits
- **Offset:** 0x2b4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 36

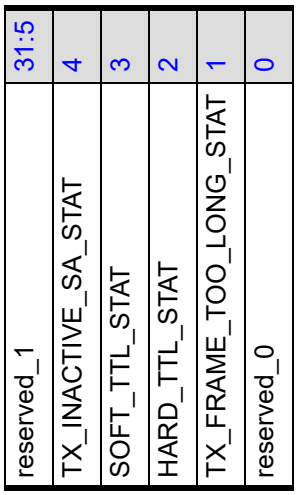


Table 6-180 Fields for Register: IRQ_TYPE_STAT_TX_SA_36

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-180 Fields for Register: IRQ_TYPE_STAT_TX_SA_36 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.174 IRQ_TYPE_EN_TX_SA_37

- **Name:** IRQ_TYPE_EN_TX_SA_37
- **Description:** Interrupt Enable for each IRQ type of TX SA 37
- **Size:** 32 bits
- **Offset:** 0x2b8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 37

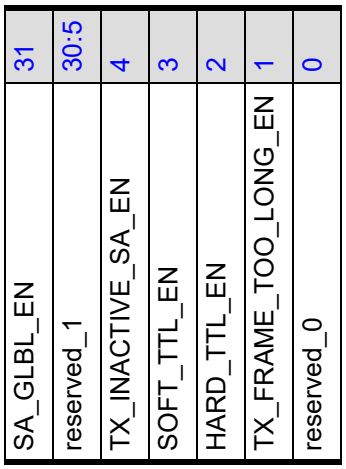


Table 6-181 Fields for Register: IRQ_TYPE_EN_TX_SA_37

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-181 Fields for Register: IRQ_TYPE_EN_TX_SA_37 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.175 IRQ_TYPE_STAT_TX_SA_37

- **Name:** IRQ_TYPE_STAT_TX_SA_37
- **Description:** Interrupt Status for each IRQ type of TX SA 37
- **Size:** 32 bits
- **Offset:** 0x2bc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 37

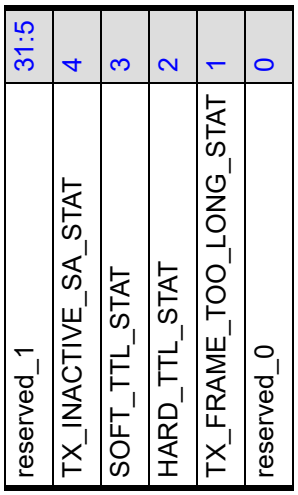


Table 6-182 Fields for Register: IRQ_TYPE_STAT_TX_SA_37

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-182 Fields for Register: IRQ_TYPE_STAT_TX_SA_37 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.176 IRQ_TYPE_EN_TX_SA_38

- **Name:** IRQ_TYPE_EN_TX_SA_38
- **Description:** Interrupt Enable for each IRQ type of TX SA 38
- **Size:** 32 bits
- **Offset:** 0x2c0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 38

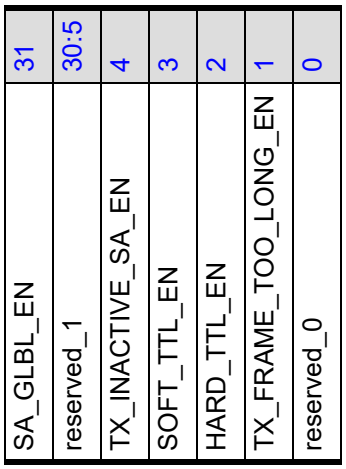


Table 6-183 Fields for Register: IRQ_TYPE_EN_TX_SA_38

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-183 Fields for Register: IRQ_TYPE_EN_TX_SA_38 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.177 IRQ_TYPE_STAT_TX_SA_38

- **Name:** IRQ_TYPE_STAT_TX_SA_38
- **Description:** Interrupt Status for each IRQ type of TX SA 38
- **Size:** 32 bits
- **Offset:** 0x2c4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 38

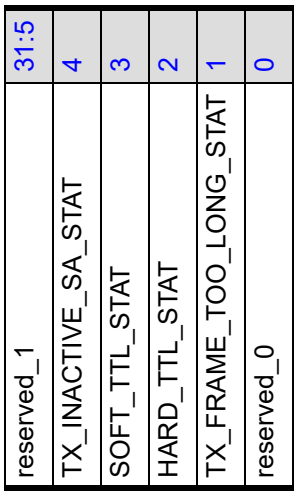


Table 6-184 Fields for Register: IRQ_TYPE_STAT_TX_SA_38

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-184 Fields for Register: IRQ_TYPE_STAT_TX_SA_38 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.178 IRQ_TYPE_EN_TX_SA_39

- **Name:** IRQ_TYPE_EN_TX_SA_39
- **Description:** Interrupt Enable for each IRQ type of TX SA 39
- **Size:** 32 bits
- **Offset:** 0x2c8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 39

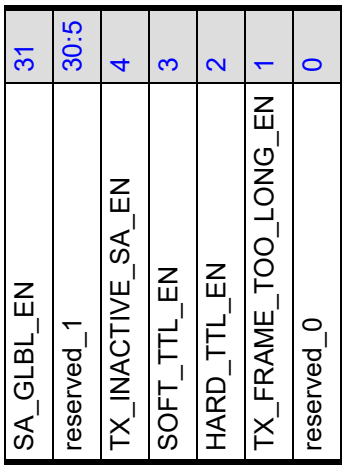


Table 6-185 Fields for Register: IRQ_TYPE_EN_TX_SA_39

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-185 Fields for Register: IRQ_TYPE_EN_TX_SA_39 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.179 IRQ_TYPE_STAT_TX_SA_39

- **Name:** IRQ_TYPE_STAT_TX_SA_39
- **Description:** Interrupt Status for each IRQ type of TX SA 39
- **Size:** 32 bits
- **Offset:** 0x2cc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 39

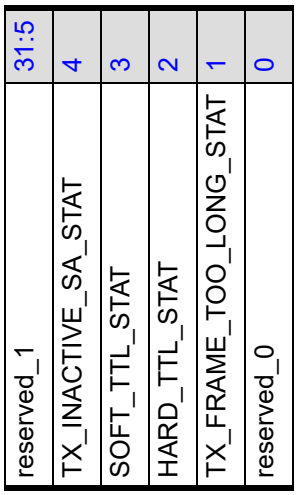


Table 6-186 Fields for Register: IRQ_TYPE_STAT_TX_SA_39

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-186 Fields for Register: IRQ_TYPE_STAT_TX_SA_39 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.180 IRQ_TYPE_EN_TX_SA_40

- **Name:** IRQ_TYPE_EN_TX_SA_40
- **Description:** Interrupt Enable for each IRQ type of TX SA 40
- **Size:** 32 bits
- **Offset:** 0x2d0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 40

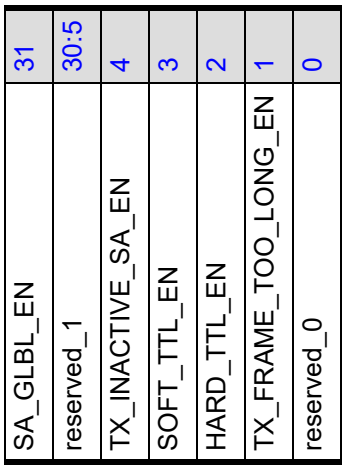


Table 6-187 Fields for Register: IRQ_TYPE_EN_TX_SA_40

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-187 Fields for Register: IRQ_TYPE_EN_TX_SA_40 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.181 IRQ_TYPE_STAT_TX_SA_40

- **Name:** IRQ_TYPE_STAT_TX_SA_40
- **Description:** Interrupt Status for each IRQ type of TX SA 40
- **Size:** 32 bits
- **Offset:** 0x2d4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 40

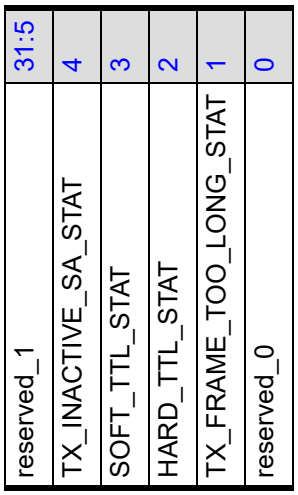


Table 6-188 Fields for Register: IRQ_TYPE_STAT_TX_SA_40

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-188 Fields for Register: IRQ_TYPE_STAT_TX_SA_40 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.182 IRQ_TYPE_EN_TX_SA_41

- **Name:** IRQ_TYPE_EN_TX_SA_41
- **Description:** Interrupt Enable for each IRQ type of TX SA 41
- **Size:** 32 bits
- **Offset:** 0x2d8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 41

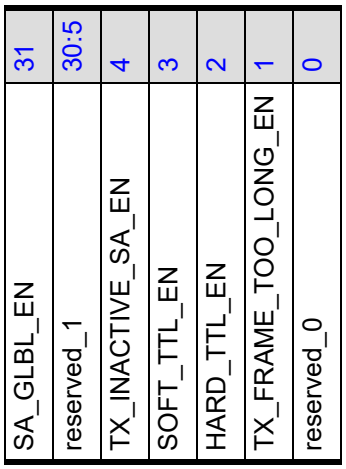


Table 6-189 Fields for Register: IRQ_TYPE_EN_TX_SA_41

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-189 Fields for Register: IRQ_TYPE_EN_TX_SA_41 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.183 IRQ_TYPE_STAT_TX_SA_41

- **Name:** IRQ_TYPE_STAT_TX_SA_41
- **Description:** Interrupt Status for each IRQ type of TX SA 41
- **Size:** 32 bits
- **Offset:** 0x2dc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 41

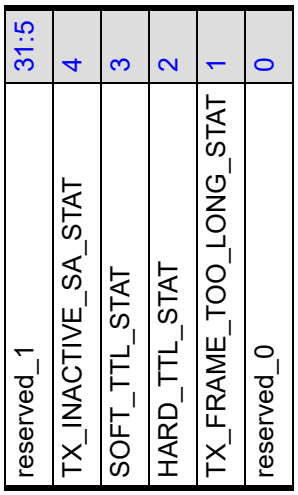


Table 6-190 Fields for Register: IRQ_TYPE_STAT_TX_SA_41

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-190 Fields for Register: IRQ_TYPE_STAT_TX_SA_41 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.184 IRQ_TYPE_EN_TX_SA_42

- **Name:** IRQ_TYPE_EN_TX_SA_42
- **Description:** Interrupt Enable for each IRQ type of TX SA 42
- **Size:** 32 bits
- **Offset:** 0x2e0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 42

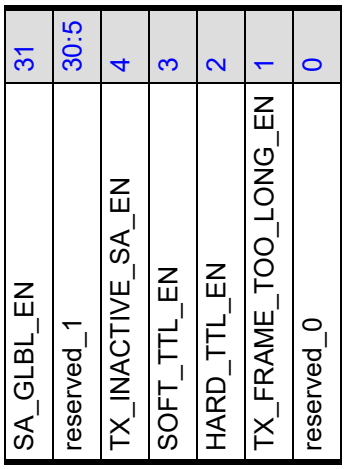


Table 6-191 Fields for Register: IRQ_TYPE_EN_TX_SA_42

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-191 Fields for Register: IRQ_TYPE_EN_TX_SA_42 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.185 IRQ_TYPE_STAT_TX_SA_42

- **Name:** IRQ_TYPE_STAT_TX_SA_42
- **Description:** Interrupt Status for each IRQ type of TX SA 42
- **Size:** 32 bits
- **Offset:** 0x2e4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 42

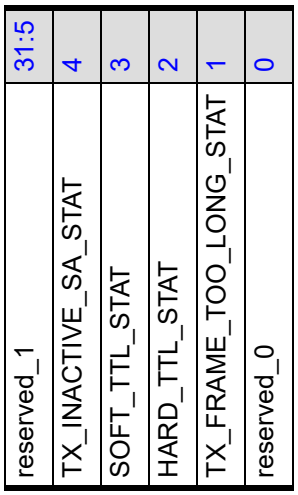


Table 6-192 Fields for Register: IRQ_TYPE_STAT_TX_SA_42

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-192 Fields for Register: IRQ_TYPE_STAT_TX_SA_42 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.186 IRQ_TYPE_EN_TX_SA_43

- **Name:** IRQ_TYPE_EN_TX_SA_43
- **Description:** Interrupt Enable for each IRQ type of TX SA 43
- **Size:** 32 bits
- **Offset:** 0x2e8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 43

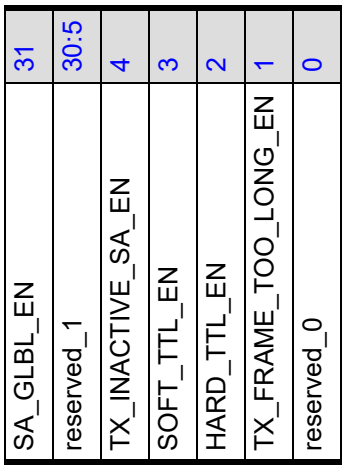


Table 6-193 Fields for Register: IRQ_TYPE_EN_TX_SA_43

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-193 Fields for Register: IRQ_TYPE_EN_TX_SA_43 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.187 IRQ_TYPE_STAT_TX_SA_43

- **Name:** IRQ_TYPE_STAT_TX_SA_43
- **Description:** Interrupt Status for each IRQ type of TX SA 43
- **Size:** 32 bits
- **Offset:** 0x2ec
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 43

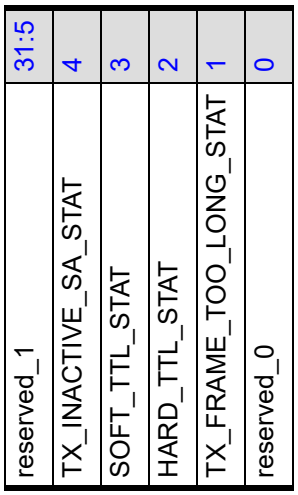


Table 6-194 Fields for Register: IRQ_TYPE_STAT_TX_SA_43

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-194 Fields for Register: IRQ_TYPE_STAT_TX_SA_43 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.188 IRQ_TYPE_EN_TX_SA_44

- **Name:** IRQ_TYPE_EN_TX_SA_44
- **Description:** Interrupt Enable for each IRQ type of TX SA 44
- **Size:** 32 bits
- **Offset:** 0x2f0
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 44

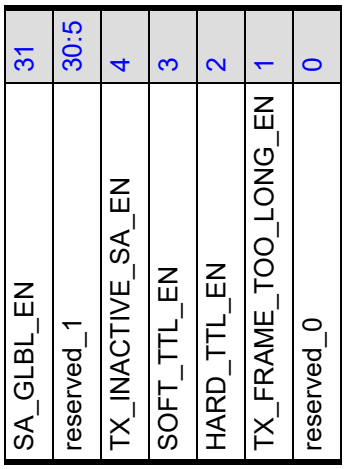


Table 6-195 Fields for Register: IRQ_TYPE_EN_TX_SA_44

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-195 Fields for Register: IRQ_TYPE_EN_TX_SA_44 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.189 IRQ_TYPE_STAT_TX_SA_44

- **Name:** IRQ_TYPE_STAT_TX_SA_44
- **Description:** Interrupt Status for each IRQ type of TX SA 44
- **Size:** 32 bits
- **Offset:** 0x2f4
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 44

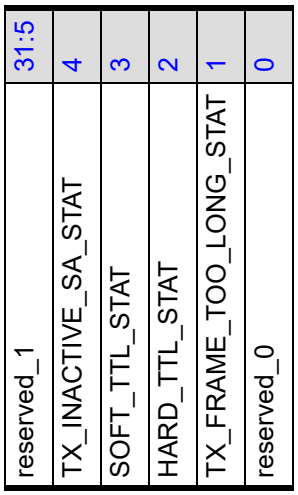


Table 6-196 Fields for Register: IRQ_TYPE_STAT_TX_SA_44

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-196 Fields for Register: IRQ_TYPE_STAT_TX_SA_44 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.190 IRQ_TYPE_EN_TX_SA_45

- **Name:** IRQ_TYPE_EN_TX_SA_45
- **Description:** Interrupt Enable for each IRQ type of TX SA 45
- **Size:** 32 bits
- **Offset:** 0x2f8
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 45

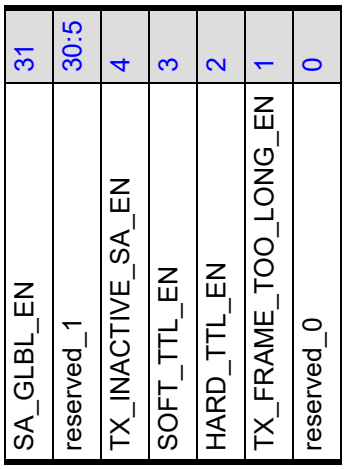


Table 6-197 Fields for Register: IRQ_TYPE_EN_TX_SA_45

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-197 Fields for Register: IRQ_TYPE_EN_TX_SA_45 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.191 IRQ_TYPE_STAT_TX_SA_45

- **Name:** IRQ_TYPE_STAT_TX_SA_45
- **Description:** Interrupt Status for each IRQ type of TX SA 45
- **Size:** 32 bits
- **Offset:** 0x2fc
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 45

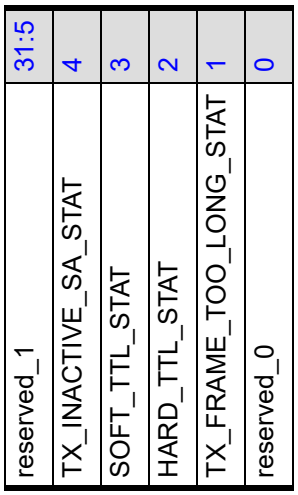


Table 6-198 Fields for Register: IRQ_TYPE_STAT_TX_SA_45

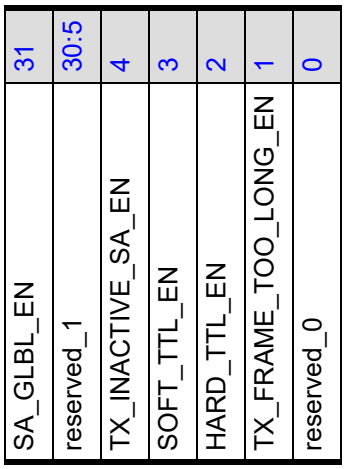
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-198 Fields for Register: IRQ_TYPE_STAT_TX_SA_45 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.192 IRQ_TYPE_EN_TX_SA_46

- **Name:** IRQ_TYPE_EN_TX_SA_46
- **Description:** Interrupt Enable for each IRQ type of TX SA 46
- **Size:** 32 bits
- **Offset:** 0x300
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 46



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-199 Fields for Register: IRQ_TYPE_EN_TX_SA_46

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-199 Fields for Register: IRQ_TYPE_EN_TX_SA_46 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.193 IRQ_TYPE_STAT_TX_SA_46

- **Name:** IRQ_TYPE_STAT_TX_SA_46
- **Description:** Interrupt Status for each IRQ type of TX SA 46
- **Size:** 32 bits
- **Offset:** 0x304
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 46

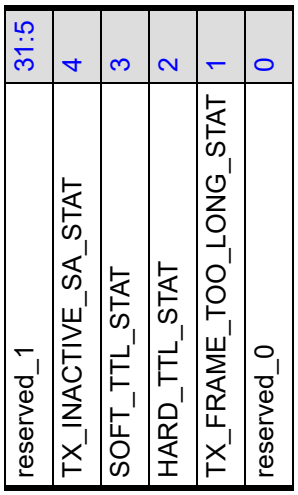


Table 6-200 Fields for Register: IRQ_TYPE_STAT_TX_SA_46

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-200 Fields for Register: IRQ_TYPE_STAT_TX_SA_46 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.194 IRQ_TYPE_EN_TX_SA_47

- **Name:** IRQ_TYPE_EN_TX_SA_47
- **Description:** Interrupt Enable for each IRQ type of TX SA 47
- **Size:** 32 bits
- **Offset:** 0x308
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 47

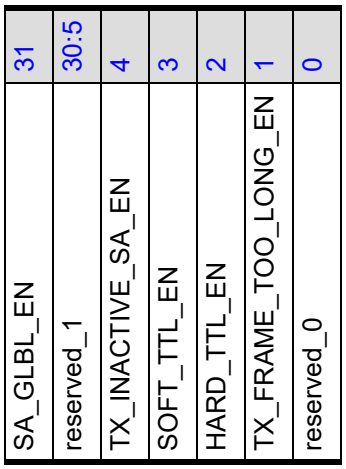


Table 6-201 Fields for Register: IRQ_TYPE_EN_TX_SA_47

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-201 Fields for Register: IRQ_TYPE_EN_TX_SA_47 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.195 IRQ_TYPE_STAT_TX_SA_47

- **Name:** IRQ_TYPE_STAT_TX_SA_47
- **Description:** Interrupt Status for each IRQ type of TX SA 47
- **Size:** 32 bits
- **Offset:** 0x30c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 47

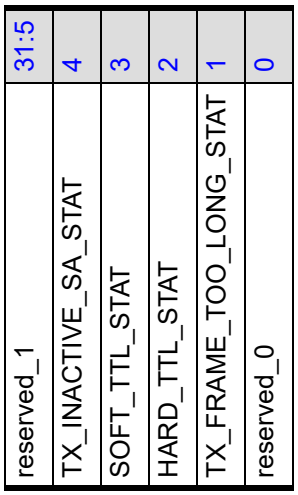


Table 6-202 Fields for Register: IRQ_TYPE_STAT_TX_SA_47

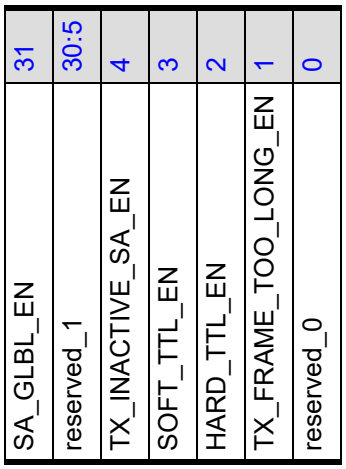
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-202 Fields for Register: IRQ_TYPE_STAT_TX_SA_47 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.196 IRQ_TYPE_EN_TX_SA_48

- **Name:** IRQ_TYPE_EN_TX_SA_48
- **Description:** Interrupt Enable for each IRQ type of TX SA 48
- **Size:** 32 bits
- **Offset:** 0x310
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 48



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-203 Fields for Register: IRQ_TYPE_EN_TX_SA_48

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-203 Fields for Register: IRQ_TYPE_EN_TX_SA_48 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.197 IRQ_TYPE_STAT_TX_SA_48

- **Name:** IRQ_TYPE_STAT_TX_SA_48
- **Description:** Interrupt Status for each IRQ type of TX SA 48
- **Size:** 32 bits
- **Offset:** 0x314
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 48

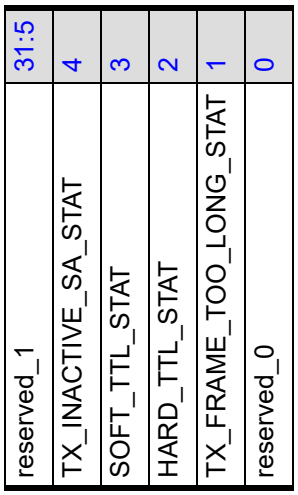


Table 6-204 Fields for Register: IRQ_TYPE_STAT_TX_SA_48

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-204 Fields for Register: IRQ_TYPE_STAT_TX_SA_48 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.198 IRQ_TYPE_EN_TX_SA_49

- **Name:** IRQ_TYPE_EN_TX_SA_49
- **Description:** Interrupt Enable for each IRQ type of TX SA 49
- **Size:** 32 bits
- **Offset:** 0x318
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 49

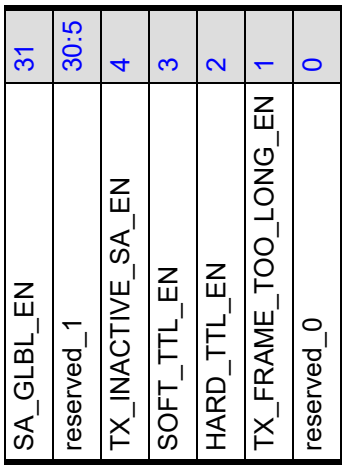


Table 6-205 Fields for Register: IRQ_TYPE_EN_TX_SA_49

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-205 Fields for Register: IRQ_TYPE_EN_TX_SA_49 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.199 IRQ_TYPE_STAT_TX_SA_49

- **Name:** IRQ_TYPE_STAT_TX_SA_49
- **Description:** Interrupt Status for each IRQ type of TX SA 49
- **Size:** 32 bits
- **Offset:** 0x31c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 49

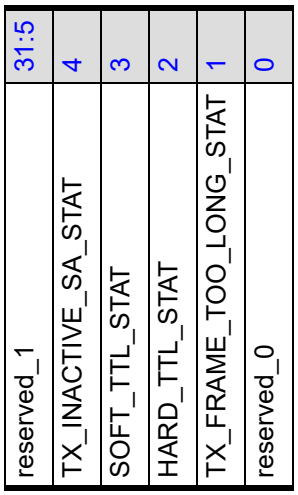


Table 6-206 Fields for Register: IRQ_TYPE_STAT_TX_SA_49

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-206 Fields for Register: IRQ_TYPE_STAT_TX_SA_49 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.200 IRQ_TYPE_EN_TX_SA_50

- **Name:** IRQ_TYPE_EN_TX_SA_50
- **Description:** Interrupt Enable for each IRQ type of TX SA 50
- **Size:** 32 bits
- **Offset:** 0x320
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 50

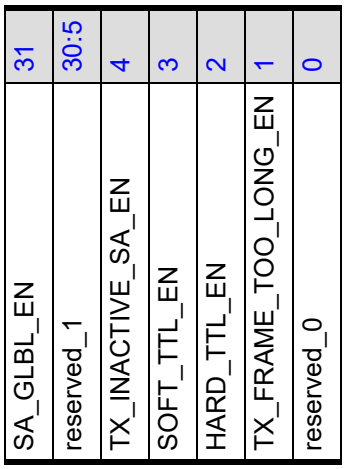


Table 6-207 Fields for Register: IRQ_TYPE_EN_TX_SA_50

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-207 Fields for Register: IRQ_TYPE_EN_TX_SA_50 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.201 IRQ_TYPE_STAT_TX_SA_50

- **Name:** IRQ_TYPE_STAT_TX_SA_50
- **Description:** Interrupt Status for each IRQ type of TX SA 50
- **Size:** 32 bits
- **Offset:** 0x324
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 50

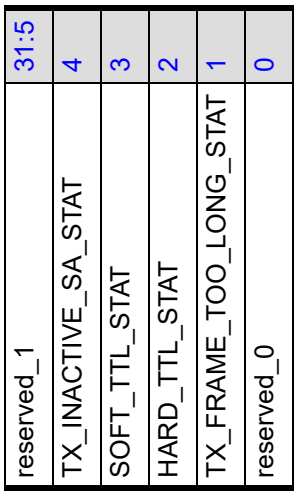


Table 6-208 Fields for Register: IRQ_TYPE_STAT_TX_SA_50

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-208 Fields for Register: IRQ_TYPE_STAT_TX_SA_50 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.202 IRQ_TYPE_EN_TX_SA_51

- **Name:** IRQ_TYPE_EN_TX_SA_51
- **Description:** Interrupt Enable for each IRQ type of TX SA 51
- **Size:** 32 bits
- **Offset:** 0x328
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 51

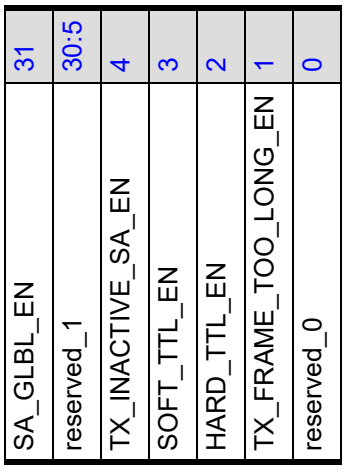


Table 6-209 Fields for Register: IRQ_TYPE_EN_TX_SA_51

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-209 Fields for Register: IRQ_TYPE_EN_TX_SA_51 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.203 IRQ_TYPE_STAT_TX_SA_51

- **Name:** IRQ_TYPE_STAT_TX_SA_51
- **Description:** Interrupt Status for each IRQ type of TX SA 51
- **Size:** 32 bits
- **Offset:** 0x32c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 51

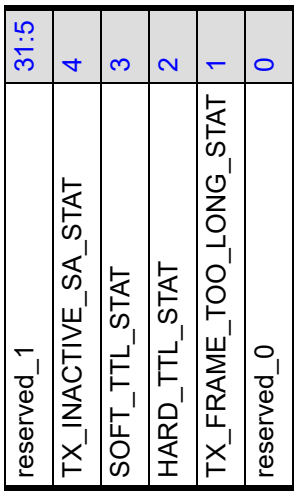


Table 6-210 Fields for Register: IRQ_TYPE_STAT_TX_SA_51

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-210 Fields for Register: IRQ_TYPE_STAT_TX_SA_51 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.204 IRQ_TYPE_EN_TX_SA_52

- **Name:** IRQ_TYPE_EN_TX_SA_52
- **Description:** Interrupt Enable for each IRQ type of TX SA 52
- **Size:** 32 bits
- **Offset:** 0x330
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 52

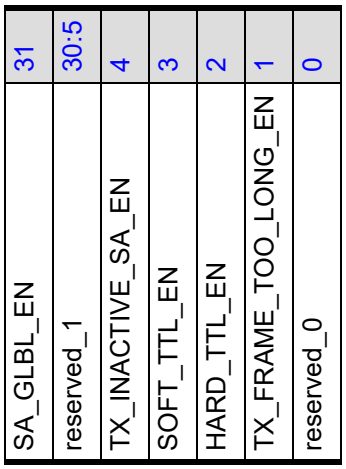


Table 6-211 Fields for Register: IRQ_TYPE_EN_TX_SA_52

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-211 Fields for Register: IRQ_TYPE_EN_TX_SA_52 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.205 IRQ_TYPE_STAT_TX_SA_52

- **Name:** IRQ_TYPE_STAT_TX_SA_52
- **Description:** Interrupt Status for each IRQ type of TX SA 52
- **Size:** 32 bits
- **Offset:** 0x334
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 52

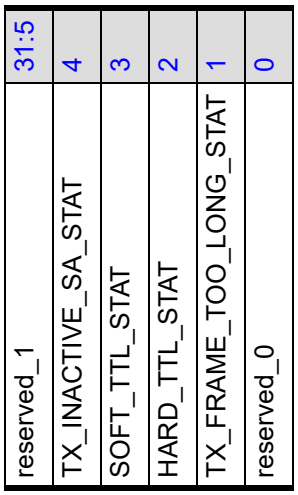


Table 6-212 Fields for Register: IRQ_TYPE_STAT_TX_SA_52

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-212 Fields for Register: IRQ_TYPE_STAT_TX_SA_52 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.206 IRQ_TYPE_EN_TX_SA_53

- **Name:** IRQ_TYPE_EN_TX_SA_53
- **Description:** Interrupt Enable for each IRQ type of TX SA 53
- **Size:** 32 bits
- **Offset:** 0x338
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 53

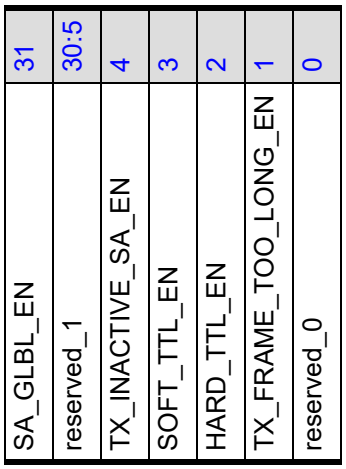


Table 6-213 Fields for Register: IRQ_TYPE_EN_TX_SA_53

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-213 Fields for Register: IRQ_TYPE_EN_TX_SA_53 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.207 IRQ_TYPE_STAT_TX_SA_53

- **Name:** IRQ_TYPE_STAT_TX_SA_53
- **Description:** Interrupt Status for each IRQ type of TX SA 53
- **Size:** 32 bits
- **Offset:** 0x33c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 53

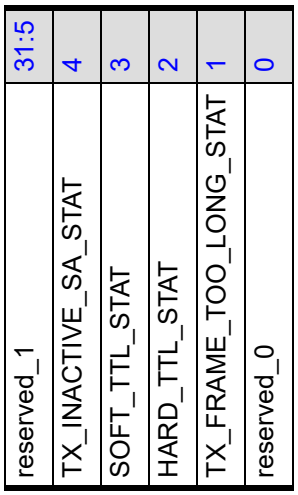


Table 6-214 Fields for Register: IRQ_TYPE_STAT_TX_SA_53

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-214 Fields for Register: IRQ_TYPE_STAT_TX_SA_53 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.208 IRQ_TYPE_EN_TX_SA_54

- **Name:** IRQ_TYPE_EN_TX_SA_54
- **Description:** Interrupt Enable for each IRQ type of TX SA 54
- **Size:** 32 bits
- **Offset:** 0x340
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 54

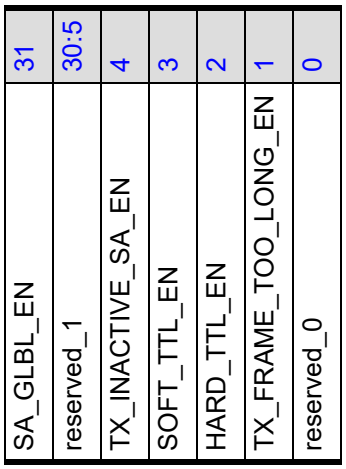


Table 6-215 Fields for Register: IRQ_TYPE_EN_TX_SA_54

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-215 Fields for Register: IRQ_TYPE_EN_TX_SA_54 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.209 IRQ_TYPE_STAT_TX_SA_54

- **Name:** IRQ_TYPE_STAT_TX_SA_54
- **Description:** Interrupt Status for each IRQ type of TX SA 54
- **Size:** 32 bits
- **Offset:** 0x344
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 54

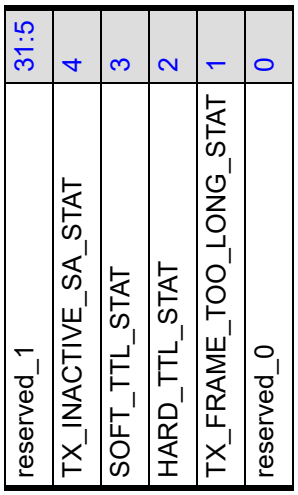


Table 6-216 Fields for Register: IRQ_TYPE_STAT_TX_SA_54

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-216 Fields for Register: IRQ_TYPE_STAT_TX_SA_54 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.210 IRQ_TYPE_EN_TX_SA_55

- **Name:** IRQ_TYPE_EN_TX_SA_55
- **Description:** Interrupt Enable for each IRQ type of TX SA 55
- **Size:** 32 bits
- **Offset:** 0x348
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 55

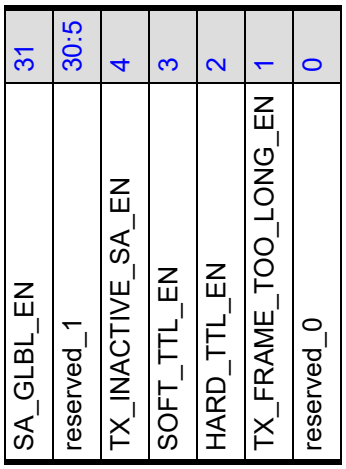


Table 6-217 Fields for Register: IRQ_TYPE_EN_TX_SA_55

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-217 Fields for Register: IRQ_TYPE_EN_TX_SA_55 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.211 IRQ_TYPE_STAT_TX_SA_55

- **Name:** IRQ_TYPE_STAT_TX_SA_55
- **Description:** Interrupt Status for each IRQ type of TX SA 55
- **Size:** 32 bits
- **Offset:** 0x34c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 55

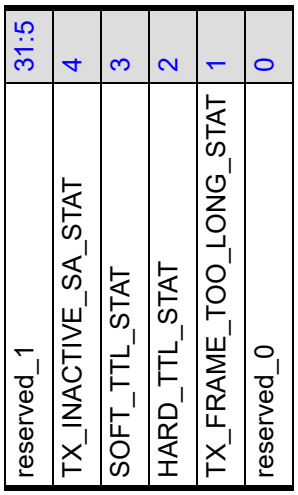


Table 6-218 Fields for Register: IRQ_TYPE_STAT_TX_SA_55

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-218 Fields for Register: IRQ_TYPE_STAT_TX_SA_55 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.212 IRQ_TYPE_EN_TX_SA_56

- **Name:** IRQ_TYPE_EN_TX_SA_56
- **Description:** Interrupt Enable for each IRQ type of TX SA 56
- **Size:** 32 bits
- **Offset:** 0x350
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 56

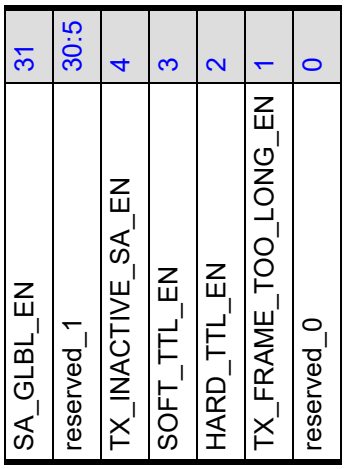


Table 6-219 Fields for Register: IRQ_TYPE_EN_TX_SA_56

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-219 Fields for Register: IRQ_TYPE_EN_TX_SA_56 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.213 IRQ_TYPE_STAT_TX_SA_56

- **Name:** IRQ_TYPE_STAT_TX_SA_56
- **Description:** Interrupt Status for each IRQ type of TX SA 56
- **Size:** 32 bits
- **Offset:** 0x354
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 56

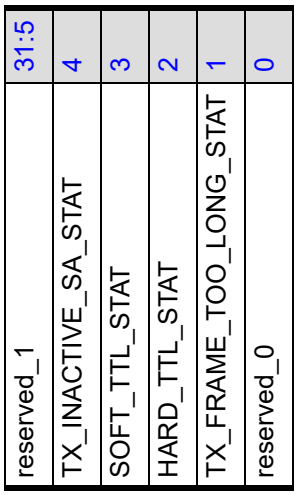


Table 6-220 Fields for Register: IRQ_TYPE_STAT_TX_SA_56

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-220 Fields for Register: IRQ_TYPE_STAT_TX_SA_56 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.214 IRQ_TYPE_EN_TX_SA_57

- **Name:** IRQ_TYPE_EN_TX_SA_57
- **Description:** Interrupt Enable for each IRQ type of TX SA 57
- **Size:** 32 bits
- **Offset:** 0x358
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 57

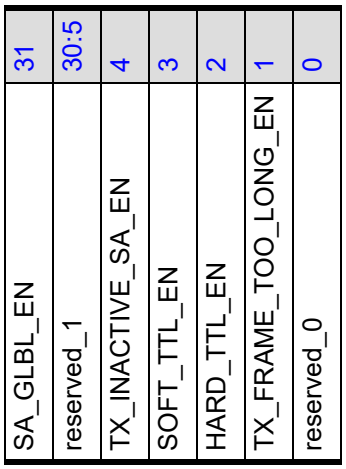


Table 6-221 Fields for Register: IRQ_TYPE_EN_TX_SA_57

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-221 Fields for Register: IRQ_TYPE_EN_TX_SA_57 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.215 IRQ_TYPE_STAT_TX_SA_57

- **Name:** IRQ_TYPE_STAT_TX_SA_57
- **Description:** Interrupt Status for each IRQ type of TX SA 57
- **Size:** 32 bits
- **Offset:** 0x35c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 57

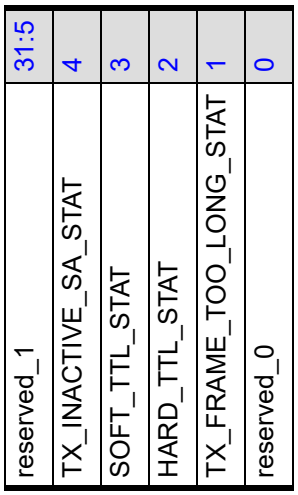


Table 6-222 Fields for Register: IRQ_TYPE_STAT_TX_SA_57

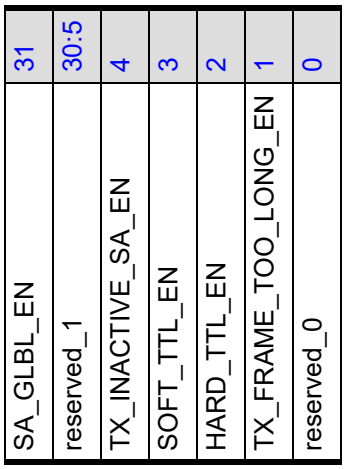
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-222 Fields for Register: IRQ_TYPE_STAT_TX_SA_57 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.216 IRQ_TYPE_EN_TX_SA_58

- **Name:** IRQ_TYPE_EN_TX_SA_58
- **Description:** Interrupt Enable for each IRQ type of TX SA 58
- **Size:** 32 bits
- **Offset:** 0x360
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 58



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-223 Fields for Register: IRQ_TYPE_EN_TX_SA_58

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-223 Fields for Register: IRQ_TYPE_EN_TX_SA_58 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.217 IRQ_TYPE_STAT_TX_SA_58

- **Name:** IRQ_TYPE_STAT_TX_SA_58
- **Description:** Interrupt Status for each IRQ type of TX SA 58
- **Size:** 32 bits
- **Offset:** 0x364
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 58

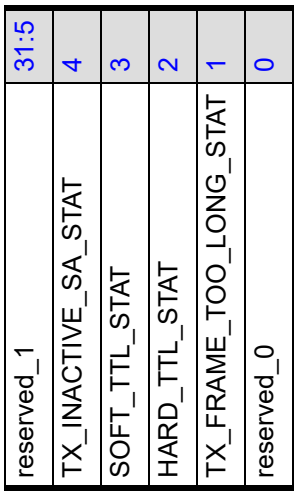


Table 6-224 Fields for Register: IRQ_TYPE_STAT_TX_SA_58

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-224 Fields for Register: IRQ_TYPE_STAT_TX_SA_58 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.218 IRQ_TYPE_EN_TX_SA_59

- **Name:** IRQ_TYPE_EN_TX_SA_59
- **Description:** Interrupt Enable for each IRQ type of TX SA 59
- **Size:** 32 bits
- **Offset:** 0x368
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 59

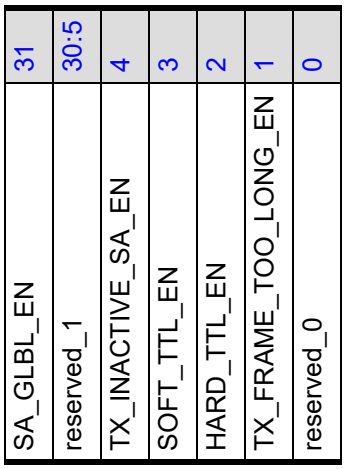


Table 6-225 Fields for Register: IRQ_TYPE_EN_TX_SA_59

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-225 Fields for Register: IRQ_TYPE_EN_TX_SA_59 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.219 IRQ_TYPE_STAT_TX_SA_59

- **Name:** IRQ_TYPE_STAT_TX_SA_59
- **Description:** Interrupt Status for each IRQ type of TX SA 59
- **Size:** 32 bits
- **Offset:** 0x36c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 59

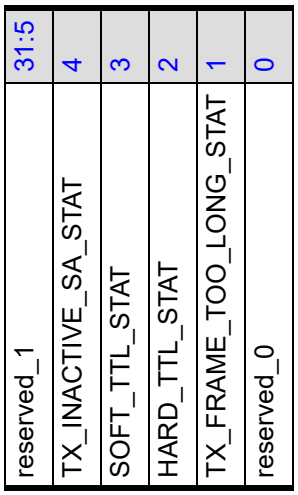


Table 6-226 Fields for Register: IRQ_TYPE_STAT_TX_SA_59

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-226 Fields for Register: IRQ_TYPE_STAT_TX_SA_59 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.220 IRQ_TYPE_EN_TX_SA_60

- **Name:** IRQ_TYPE_EN_TX_SA_60
- **Description:** Interrupt Enable for each IRQ type of TX SA 60
- **Size:** 32 bits
- **Offset:** 0x370
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 60

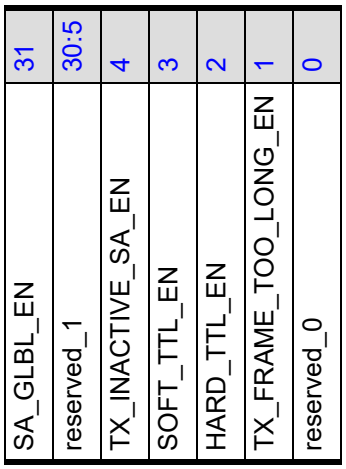


Table 6-227 Fields for Register: IRQ_TYPE_EN_TX_SA_60

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-227 Fields for Register: IRQ_TYPE_EN_TX_SA_60 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.221 IRQ_TYPE_STAT_TX_SA_60

- **Name:** IRQ_TYPE_STAT_TX_SA_60
- **Description:** Interrupt Status for each IRQ type of TX SA 60
- **Size:** 32 bits
- **Offset:** 0x374
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 60

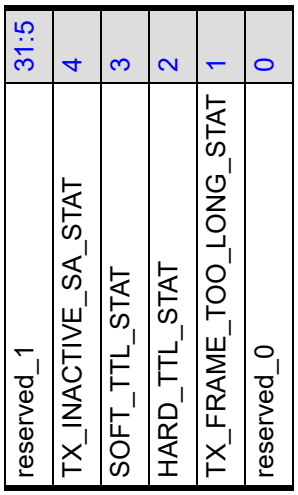


Table 6-228 Fields for Register: IRQ_TYPE_STAT_TX_SA_60

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-228 Fields for Register: IRQ_TYPE_STAT_TX_SA_60 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.222 IRQ_TYPE_EN_TX_SA_61

- **Name:** IRQ_TYPE_EN_TX_SA_61
- **Description:** Interrupt Enable for each IRQ type of TX SA 61
- **Size:** 32 bits
- **Offset:** 0x378
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 61

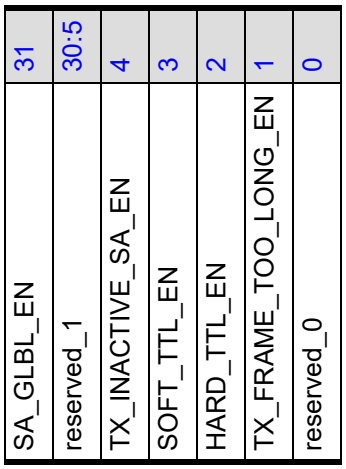


Table 6-229 Fields for Register: IRQ_TYPE_EN_TX_SA_61

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-229 Fields for Register: IRQ_TYPE_EN_TX_SA_61 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.223 IRQ_TYPE_STAT_TX_SA_61

- **Name:** IRQ_TYPE_STAT_TX_SA_61
- **Description:** Interrupt Status for each IRQ type of TX SA 61
- **Size:** 32 bits
- **Offset:** 0x37c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 61

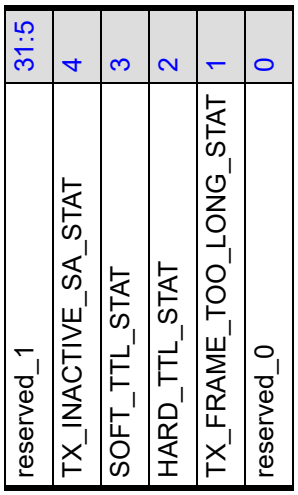


Table 6-230 Fields for Register: IRQ_TYPE_STAT_TX_SA_61

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-230 Fields for Register: IRQ_TYPE_STAT_TX_SA_61 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.224 IRQ_TYPE_EN_TX_SA_62

- **Name:** IRQ_TYPE_EN_TX_SA_62
- **Description:** Interrupt Enable for each IRQ type of TX SA 62
- **Size:** 32 bits
- **Offset:** 0x380
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 62

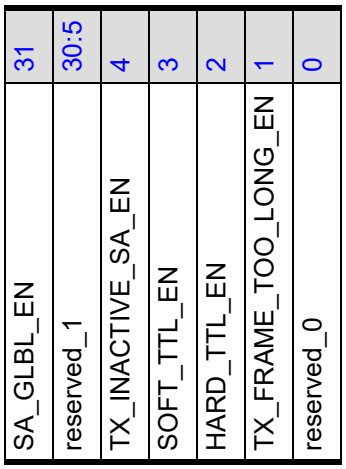


Table 6-231 Fields for Register: IRQ_TYPE_EN_TX_SA_62

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-231 Fields for Register: IRQ_TYPE_EN_TX_SA_62 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.225 IRQ_TYPE_STAT_TX_SA_62

- **Name:** IRQ_TYPE_STAT_TX_SA_62
- **Description:** Interrupt Status for each IRQ type of TX SA 62
- **Size:** 32 bits
- **Offset:** 0x384
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 62

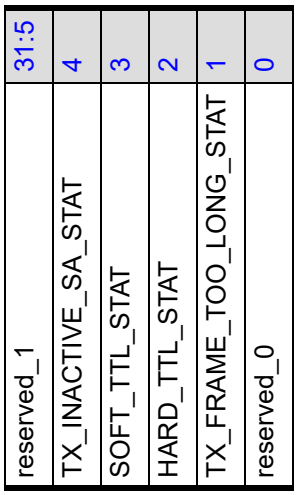


Table 6-232 Fields for Register: IRQ_TYPE_STAT_TX_SA_62

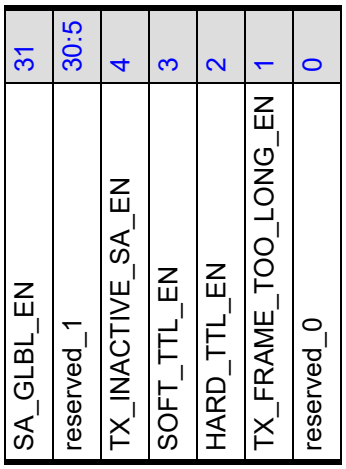
Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-232 Fields for Register: IRQ_TYPE_STAT_TX_SA_62 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.226 IRQ_TYPE_EN_TX_SA_63

- **Name:** IRQ_TYPE_EN_TX_SA_63
- **Description:** Interrupt Enable for each IRQ type of TX SA 63
- **Size:** 32 bits
- **Offset:** 0x388
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 63



31	30:5	4	3	2	1	0
SA_GLBL_EN	reserved_1	TX_INACTIVE_SA_EN	SOFT_TTL_EN	HARD_TTL_EN	TX_FRAME_TOO_LONG_EN	reserved_0

Table 6-233 Fields for Register: IRQ_TYPE_EN_TX_SA_63

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x1 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-233 Fields for Register: IRQ_TYPE_EN_TX_SA_63 (Continued)

Bits	Name	Memory Access	Description
4	TX_INACTIVE_SA_EN	R/W	Enables the TX Inactive SA interrupt. This interrupt means that this SA does not have the active flag set. **Values:** ■ 0x1 (TX_INACTIVE_SA_EN): TX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
3	SOFT_TTL_EN	R/W	Enables the Soft TTL interrupt. A TX frame has a packet number equivalent to the SOFT_TTL limit set in the SA. **Values:** ■ 0x1 (SOFT_TTL_EN): SOFT_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
2	HARD_TTL_EN	R/W	Enables the Hard TTL interrupt. A TX frame has a packet number equivalent to the HARD_TTL limit set in the SA. **Values:** ■ 0x1 (HARD_TTL_EN): HARD_TTL_EN **Value After Reset:** 0x1 **Exists:** Always
1	TX_FRAME_TOO_LONG_EN	R/W	Enables the TX Frame too long interrupt. Security processing on a TX frame caused the frame length to increase such that the MTU of the transmission medium was exceeded. **Values:** ■ 0x1 (TX_FRAME_TOO_LONG_EN): TX_FRAME_TOO_LONG_EN **Value After Reset:** 0x1 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.227 IRQ_TYPE_STAT_TX_SA_63

- **Name:** IRQ_TYPE_STAT_TX_SA_63
- **Description:** Interrupt Status for each IRQ type of TX SA 63
- **Size:** 32 bits
- **Offset:** 0x38c
- **Exists:** DWC_MACSEC_TX_NUM_SA_SUPPORTED > 63

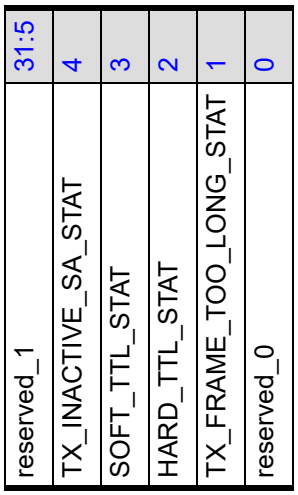


Table 6-234 Fields for Register: IRQ_TYPE_STAT_TX_SA_63

Bits	Name	Memory Access	Description
31:5	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INACTIVE_SA_STAT	R/W1C	Current status of TX Inactive SA interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_INACTIVE_SA_STAT): TX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-234 Fields for Register: IRQ_TYPE_STAT_TX_SA_63 (Continued)

Bits	Name	Memory Access	Description
3	SOFT_TTL_STAT	R/W1C	Current status of Soft TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (SOFT_TTL_STAT): SOFT_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
2	HARD_TTL_STAT	R/W1C	Current status of Hard TTL interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (HARD_TTL_STAT): HARD_TTL_STAT **Value After Reset:** 0x0 **Exists:** Always
1	TX_FRAME_TOO_LONG_STAT	R/W1C	Current status of TX Frame too long interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (TX_FRAME_TOO_LONG_STAT): TX_FRAME_TOO_LONG_STAT **Value After Reset:** 0x0 **Exists:** Always
0	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

6.1.228 IRQ_TYPE_EN_RX_SA_0

- **Name:** IRQ_TYPE_EN_RX_SA_0
- **Description:** Interrupt Enable for each IRQ type of Rx SA 0
- **Size:** 32 bits
- **Offset:** 0x390
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 0

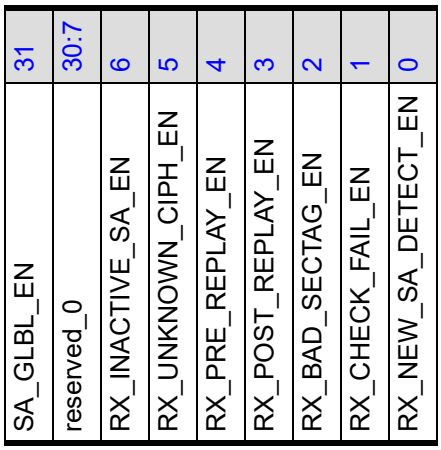


Table 6-235 Fields for Register: IRQ_TYPE_EN_RX_SA_0

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-235 Fields for Register: IRQ_TYPE_EN_RX_SA_0 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-235 Fields for Register: IRQ_TYPE_EN_RX_SA_0 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.229 IRQ_TYPE_STAT_RX_SA_0

- **Name:** IRQ_TYPE_STAT_RX_SA_0
- **Description:** Interrupt Status for each IRQ type of Rx SA 0
- **Size:** 32 bits
- **Offset:** 0x394
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 0

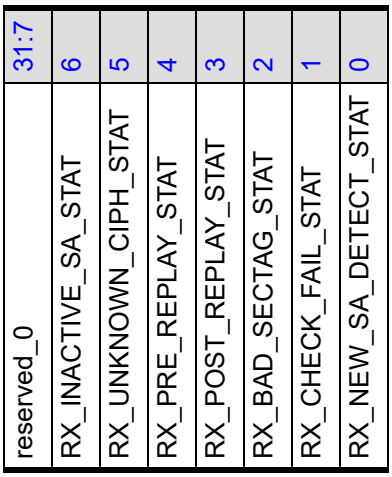


Table 6-236 Fields for Register: IRQ_TYPE_STAT_RX_SA_0

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-236 Fields for Register: IRQ_TYPE_STAT_RX_SA_0 (Continued)

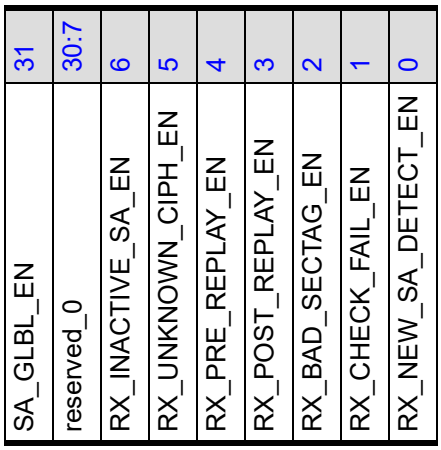
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-236 Fields for Register: IRQ_TYPE_STAT_RX_SA_0 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.230 IRQ_TYPE_EN_RX_SA_1

- **Name:** IRQ_TYPE_EN_RX_SA_1
- **Description:** Interrupt Enable for each IRQ type of Rx SA 1
- **Size:** 32 bits
- **Offset:** 0x398
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 1



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-237 Fields for Register: IRQ_TYPE_EN_RX_SA_1

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-237 Fields for Register: IRQ_TYPE_EN_RX_SA_1 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-237 Fields for Register: IRQ_TYPE_EN_RX_SA_1 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.231 IRQ_TYPE_STAT_RX_SA_1

- **Name:** IRQ_TYPE_STAT_RX_SA_1
- **Description:** Interrupt Status for each IRQ type of Rx SA 1
- **Size:** 32 bits
- **Offset:** 0x39c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 1

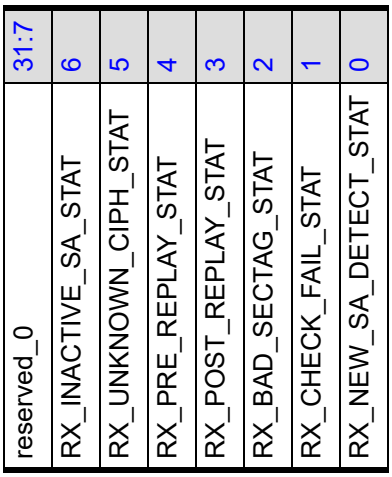


Table 6-238 Fields for Register: IRQ_TYPE_STAT_RX_SA_1

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-238 Fields for Register: IRQ_TYPE_STAT_RX_SA_1 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-238 Fields for Register: IRQ_TYPE_STAT_RX_SA_1 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.232 IRQ_TYPE_EN_RX_SA_2

- **Name:** IRQ_TYPE_EN_RX_SA_2
- **Description:** Interrupt Enable for each IRQ type of Rx SA 2
- **Size:** 32 bits
- **Offset:** 0x3a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 2

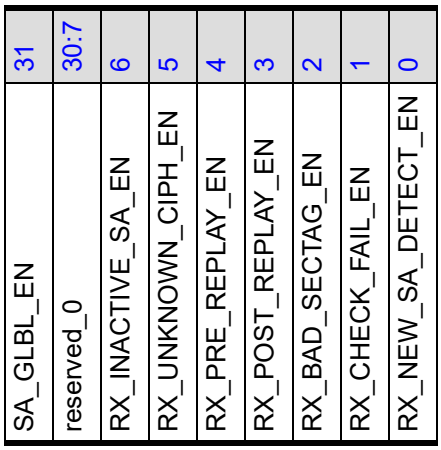


Table 6-239 Fields for Register: IRQ_TYPE_EN_RX_SA_2

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-239 Fields for Register: IRQ_TYPE_EN_RX_SA_2 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-239 Fields for Register: IRQ_TYPE_EN_RX_SA_2 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.233 IRQ_TYPE_STAT_RX_SA_2

- **Name:** IRQ_TYPE_STAT_RX_SA_2
- **Description:** Interrupt Status for each IRQ type of Rx SA 2
- **Size:** 32 bits
- **Offset:** 0x3a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 2

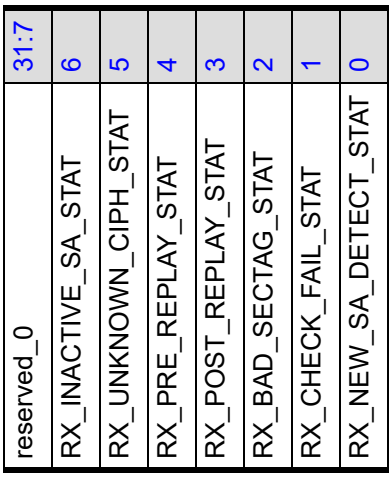


Table 6-240 Fields for Register: IRQ_TYPE_STAT_RX_SA_2

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-240 Fields for Register: IRQ_TYPE_STAT_RX_SA_2 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-240 Fields for Register: IRQ_TYPE_STAT_RX_SA_2 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.234 IRQ_TYPE_EN_RX_SA_3

- **Name:** IRQ_TYPE_EN_RX_SA_3
- **Description:** Interrupt Enable for each IRQ type of Rx SA 3
- **Size:** 32 bits
- **Offset:** 0x3a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 3

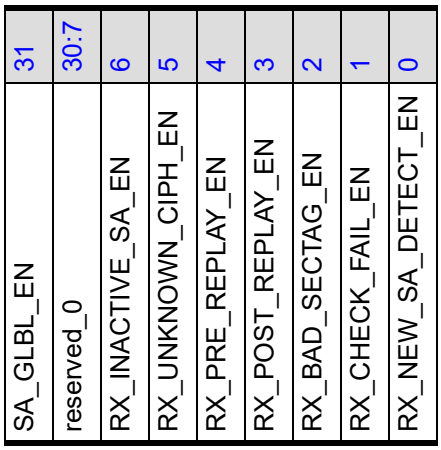


Table 6-241 Fields for Register: IRQ_TYPE_EN_RX_SA_3

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-241 Fields for Register: IRQ_TYPE_EN_RX_SA_3 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-241 Fields for Register: IRQ_TYPE_EN_RX_SA_3 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.235 IRQ_TYPE_STAT_RX_SA_3

- **Name:** IRQ_TYPE_STAT_RX_SA_3
- **Description:** Interrupt Status for each IRQ type of Rx SA 3
- **Size:** 32 bits
- **Offset:** 0x3ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 3

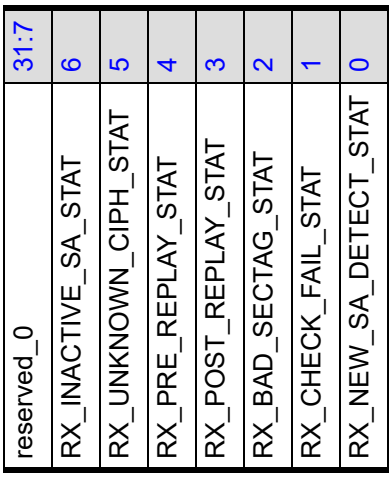


Table 6-242 Fields for Register: IRQ_TYPE_STAT_RX_SA_3

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-242 Fields for Register: IRQ_TYPE_STAT_RX_SA_3 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-242 Fields for Register: IRQ_TYPE_STAT_RX_SA_3 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.236 IRQ_TYPE_EN_RX_SA_4

- **Name:** IRQ_TYPE_EN_RX_SA_4
- **Description:** Interrupt Enable for each IRQ type of Rx SA 4
- **Size:** 32 bits
- **Offset:** 0x3b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 4

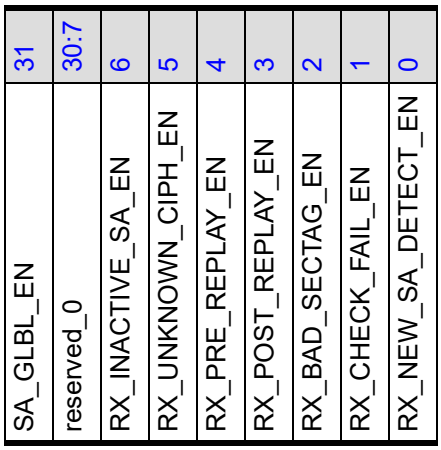


Table 6-243 Fields for Register: IRQ_TYPE_EN_RX_SA_4

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-243 Fields for Register: IRQ_TYPE_EN_RX_SA_4 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-243 Fields for Register: IRQ_TYPE_EN_RX_SA_4 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.237 IRQ_TYPE_STAT_RX_SA_4

- **Name:** IRQ_TYPE_STAT_RX_SA_4
- **Description:** Interrupt Status for each IRQ type of Rx SA 4
- **Size:** 32 bits
- **Offset:** 0x3b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 4

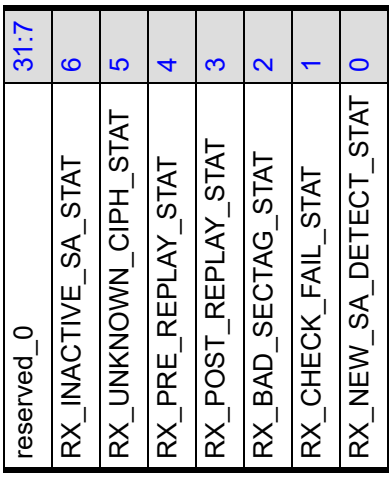


Table 6-244 Fields for Register: IRQ_TYPE_STAT_RX_SA_4

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-244 Fields for Register: IRQ_TYPE_STAT_RX_SA_4 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-244 Fields for Register: IRQ_TYPE_STAT_RX_SA_4 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.238 IRQ_TYPE_EN_RX_SA_5

- **Name:** IRQ_TYPE_EN_RX_SA_5
- **Description:** Interrupt Enable for each IRQ type of Rx SA 5
- **Size:** 32 bits
- **Offset:** 0x3b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 5

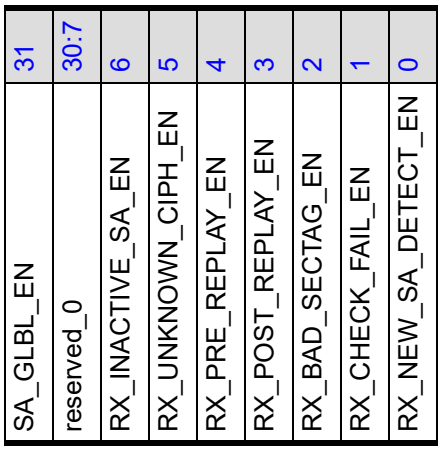


Table 6-245 Fields for Register: IRQ_TYPE_EN_RX_SA_5

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-245 Fields for Register: IRQ_TYPE_EN_RX_SA_5 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-245 Fields for Register: IRQ_TYPE_EN_RX_SA_5 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.239 IRQ_TYPE_STAT_RX_SA_5

- **Name:** IRQ_TYPE_STAT_RX_SA_5
- **Description:** Interrupt Status for each IRQ type of Rx SA 5
- **Size:** 32 bits
- **Offset:** 0x3bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 5

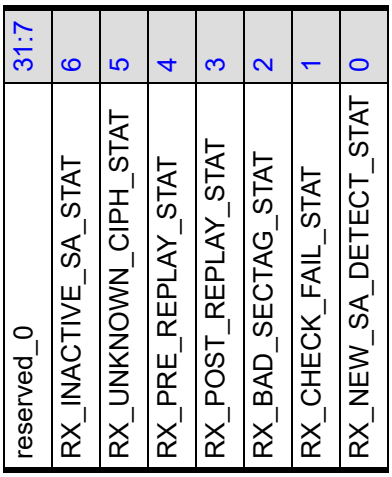


Table 6-246 Fields for Register: IRQ_TYPE_STAT_RX_SA_5

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-246 Fields for Register: IRQ_TYPE_STAT_RX_SA_5 (Continued)

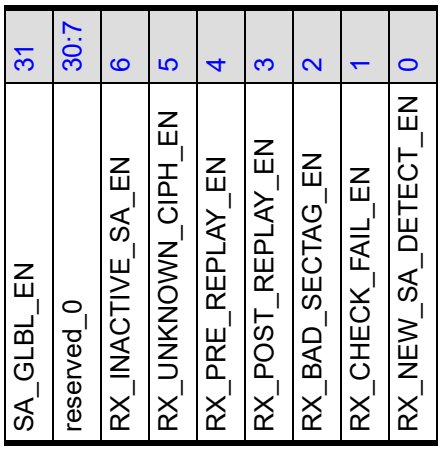
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-246 Fields for Register: IRQ_TYPE_STAT_RX_SA_5 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.240 IRQ_TYPE_EN_RX_SA_6

- **Name:** IRQ_TYPE_EN_RX_SA_6
- **Description:** Interrupt Enable for each IRQ type of Rx SA 6
- **Size:** 32 bits
- **Offset:** 0x3c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 6



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-247 Fields for Register: IRQ_TYPE_EN_RX_SA_6

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-247 Fields for Register: IRQ_TYPE_EN_RX_SA_6 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-247 Fields for Register: IRQ_TYPE_EN_RX_SA_6 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.241 IRQ_TYPE_STAT_RX_SA_6

- **Name:** IRQ_TYPE_STAT_RX_SA_6
- **Description:** Interrupt Status for each IRQ type of Rx SA 6
- **Size:** 32 bits
- **Offset:** 0x3c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 6

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-248 Fields for Register: IRQ_TYPE_STAT_RX_SA_6

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-248 Fields for Register: IRQ_TYPE_STAT_RX_SA_6 (Continued)

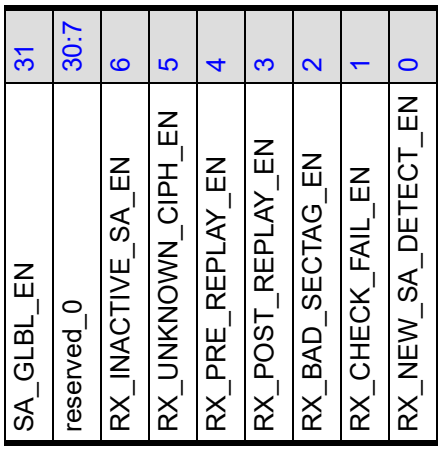
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-248 Fields for Register: IRQ_TYPE_STAT_RX_SA_6 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.242 IRQ_TYPE_EN_RX_SA_7

- **Name:** IRQ_TYPE_EN_RX_SA_7
- **Description:** Interrupt Enable for each IRQ type of Rx SA 7
- **Size:** 32 bits
- **Offset:** 0x3c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 7



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-249 Fields for Register: IRQ_TYPE_EN_RX_SA_7

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-249 Fields for Register: IRQ_TYPE_EN_RX_SA_7 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-249 Fields for Register: IRQ_TYPE_EN_RX_SA_7 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.243 IRQ_TYPE_STAT_RX_SA_7

- **Name:** IRQ_TYPE_STAT_RX_SA_7
- **Description:** Interrupt Status for each IRQ type of Rx SA 7
- **Size:** 32 bits
- **Offset:** 0x3cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 7

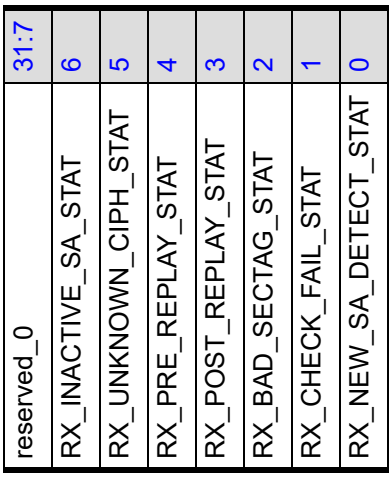


Table 6-250 Fields for Register: IRQ_TYPE_STAT_RX_SA_7

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-250 Fields for Register: IRQ_TYPE_STAT_RX_SA_7 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-250 Fields for Register: IRQ_TYPE_STAT_RX_SA_7 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.244 IRQ_TYPE_EN_RX_SA_8

- **Name:** IRQ_TYPE_EN_RX_SA_8
- **Description:** Interrupt Enable for each IRQ type of Rx SA 8
- **Size:** 32 bits
- **Offset:** 0x3d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 8

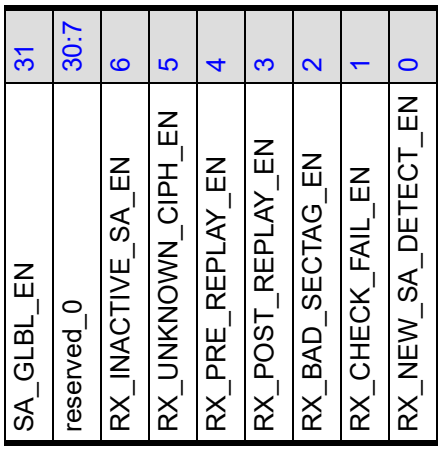


Table 6-251 Fields for Register: IRQ_TYPE_EN_RX_SA_8

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-251 Fields for Register: IRQ_TYPE_EN_RX_SA_8 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-251 Fields for Register: IRQ_TYPE_EN_RX_SA_8 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.245 IRQ_TYPE_STAT_RX_SA_8

- **Name:** IRQ_TYPE_STAT_RX_SA_8
- **Description:** Interrupt Status for each IRQ type of Rx SA 8
- **Size:** 32 bits
- **Offset:** 0x3d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 8

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-252 Fields for Register: IRQ_TYPE_STAT_RX_SA_8

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-252 Fields for Register: IRQ_TYPE_STAT_RX_SA_8 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-252 Fields for Register: IRQ_TYPE_STAT_RX_SA_8 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.246 IRQ_TYPE_EN_RX_SA_9

- **Name:** IRQ_TYPE_EN_RX_SA_9
- **Description:** Interrupt Enable for each IRQ type of Rx SA 9
- **Size:** 32 bits
- **Offset:** 0x3d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 9

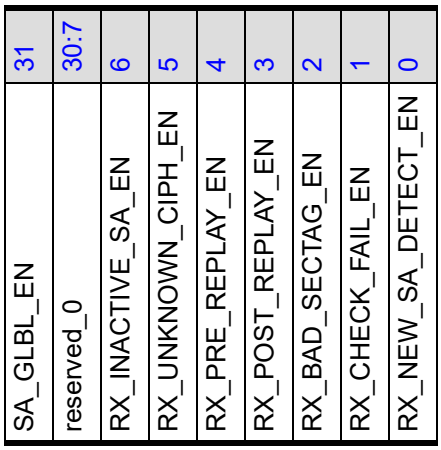


Table 6-253 Fields for Register: IRQ_TYPE_EN_RX_SA_9

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-253 Fields for Register: IRQ_TYPE_EN_RX_SA_9 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-253 Fields for Register: IRQ_TYPE_EN_RX_SA_9 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.247 IRQ_TYPE_STAT_RX_SA_9

- **Name:** IRQ_TYPE_STAT_RX_SA_9
- **Description:** Interrupt Status for each IRQ type of Rx SA 9
- **Size:** 32 bits
- **Offset:** 0x3dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 9

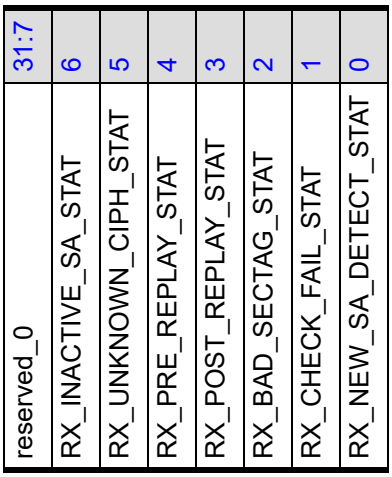


Table 6-254 Fields for Register: IRQ_TYPE_STAT_RX_SA_9

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-254 Fields for Register: IRQ_TYPE_STAT_RX_SA_9 (Continued)

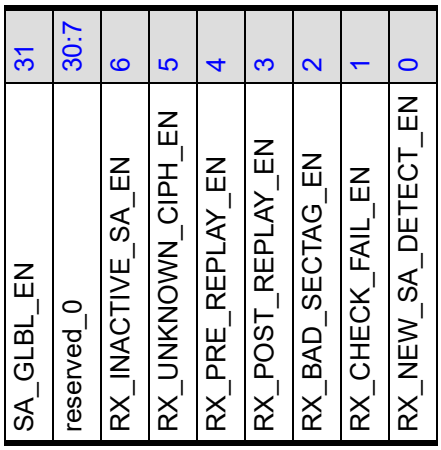
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-254 Fields for Register: IRQ_TYPE_STAT_RX_SA_9 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.248 IRQ_TYPE_EN_RX_SA_10

- **Name:** IRQ_TYPE_EN_RX_SA_10
- **Description:** Interrupt Enable for each IRQ type of Rx SA 10
- **Size:** 32 bits
- **Offset:** 0x3e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 10



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-255 Fields for Register: IRQ_TYPE_EN_RX_SA_10

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-255 Fields for Register: IRQ_TYPE_EN_RX_SA_10 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-255 Fields for Register: IRQ_TYPE_EN_RX_SA_10 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.249 IRQ_TYPE_STAT_RX_SA_10

- **Name:** IRQ_TYPE_STAT_RX_SA_10
- **Description:** Interrupt Status for each IRQ type of Rx SA 10
- **Size:** 32 bits
- **Offset:** 0x3e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 10

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-256 Fields for Register: IRQ_TYPE_STAT_RX_SA_10

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-256 Fields for Register: IRQ_TYPE_STAT_RX_SA_10 (Continued)

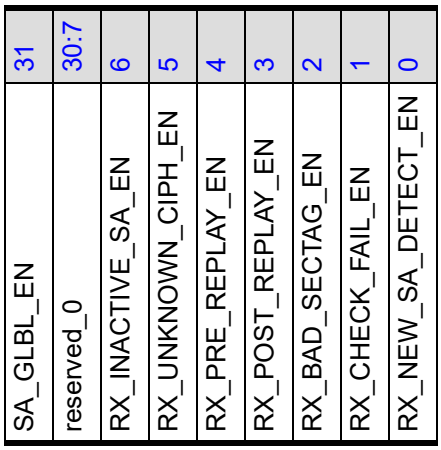
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-256 Fields for Register: IRQ_TYPE_STAT_RX_SA_10 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.250 IRQ_TYPE_EN_RX_SA_11

- **Name:** IRQ_TYPE_EN_RX_SA_11
- **Description:** Interrupt Enable for each IRQ type of Rx SA 11
- **Size:** 32 bits
- **Offset:** 0x3e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 11



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-257 Fields for Register: IRQ_TYPE_EN_RX_SA_11

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-257 Fields for Register: IRQ_TYPE_EN_RX_SA_11 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-257 Fields for Register: IRQ_TYPE_EN_RX_SA_11 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.251 IRQ_TYPE_STAT_RX_SA_11

- **Name:** IRQ_TYPE_STAT_RX_SA_11
- **Description:** Interrupt Status for each IRQ type of Rx SA 11
- **Size:** 32 bits
- **Offset:** 0x3ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 11

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-258 Fields for Register: IRQ_TYPE_STAT_RX_SA_11

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-258 Fields for Register: IRQ_TYPE_STAT_RX_SA_11 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-258 Fields for Register: IRQ_TYPE_STAT_RX_SA_11 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.252 IRQ_TYPE_EN_RX_SA_12

- **Name:** IRQ_TYPE_EN_RX_SA_12
- **Description:** Interrupt Enable for each IRQ type of Rx SA 12
- **Size:** 32 bits
- **Offset:** 0x3f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 12

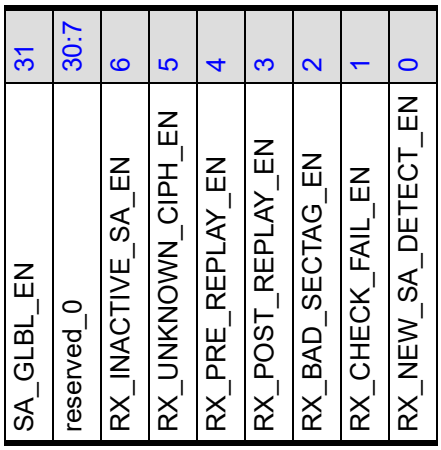


Table 6-259 Fields for Register: IRQ_TYPE_EN_RX_SA_12

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-259 Fields for Register: IRQ_TYPE_EN_RX_SA_12 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-259 Fields for Register: IRQ_TYPE_EN_RX_SA_12 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.253 IRQ_TYPE_STAT_RX_SA_12

- **Name:** IRQ_TYPE_STAT_RX_SA_12
- **Description:** Interrupt Status for each IRQ type of Rx SA 12
- **Size:** 32 bits
- **Offset:** 0x3f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 12

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-260 Fields for Register: IRQ_TYPE_STAT_RX_SA_12

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-260 Fields for Register: IRQ_TYPE_STAT_RX_SA_12 (Continued)

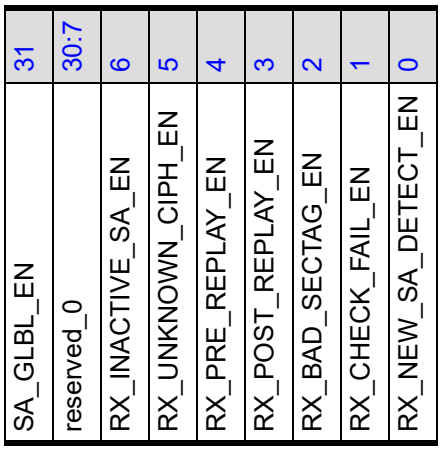
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-260 Fields for Register: IRQ_TYPE_STAT_RX_SA_12 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.254 IRQ_TYPE_EN_RX_SA_13

- **Name:** IRQ_TYPE_EN_RX_SA_13
- **Description:** Interrupt Enable for each IRQ type of Rx SA 13
- **Size:** 32 bits
- **Offset:** 0x3f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 13



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-261 Fields for Register: IRQ_TYPE_EN_RX_SA_13

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-261 Fields for Register: IRQ_TYPE_EN_RX_SA_13 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-261 Fields for Register: IRQ_TYPE_EN_RX_SA_13 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.255 IRQ_TYPE_STAT_RX_SA_13

- **Name:** IRQ_TYPE_STAT_RX_SA_13
- **Description:** Interrupt Status for each IRQ type of Rx SA 13
- **Size:** 32 bits
- **Offset:** 0x3fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 13

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-262 Fields for Register: IRQ_TYPE_STAT_RX_SA_13

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-262 Fields for Register: IRQ_TYPE_STAT_RX_SA_13 (Continued)

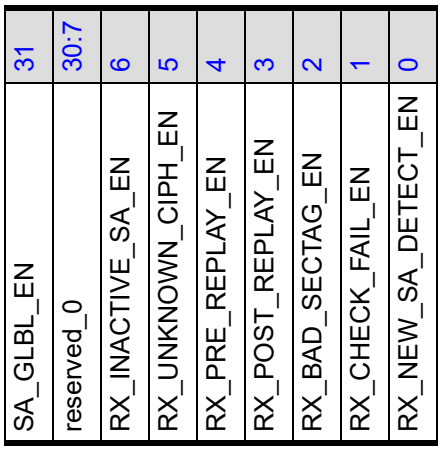
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-262 Fields for Register: IRQ_TYPE_STAT_RX_SA_13 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.256 IRQ_TYPE_EN_RX_SA_14

- **Name:** IRQ_TYPE_EN_RX_SA_14
- **Description:** Interrupt Enable for each IRQ type of Rx SA 14
- **Size:** 32 bits
- **Offset:** 0x400
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 14



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-263 Fields for Register: IRQ_TYPE_EN_RX_SA_14

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-263 Fields for Register: IRQ_TYPE_EN_RX_SA_14 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-263 Fields for Register: IRQ_TYPE_EN_RX_SA_14 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.257 IRQ_TYPE_STAT_RX_SA_14

- **Name:** IRQ_TYPE_STAT_RX_SA_14
- **Description:** Interrupt Status for each IRQ type of Rx SA 14
- **Size:** 32 bits
- **Offset:** 0x404
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 14

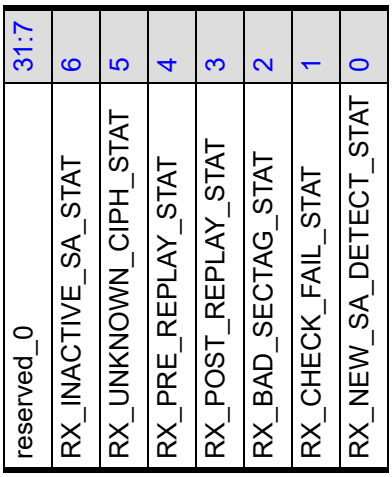


Table 6-264 Fields for Register: IRQ_TYPE_STAT_RX_SA_14

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-264 Fields for Register: IRQ_TYPE_STAT_RX_SA_14 (Continued)

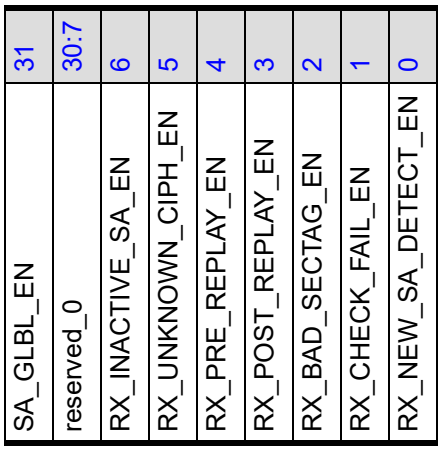
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-264 Fields for Register: IRQ_TYPE_STAT_RX_SA_14 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.258 IRQ_TYPE_EN_RX_SA_15

- **Name:** IRQ_TYPE_EN_RX_SA_15
- **Description:** Interrupt Enable for each IRQ type of Rx SA 15
- **Size:** 32 bits
- **Offset:** 0x408
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 15



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-265 Fields for Register: IRQ_TYPE_EN_RX_SA_15

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-265 Fields for Register: IRQ_TYPE_EN_RX_SA_15 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-265 Fields for Register: IRQ_TYPE_EN_RX_SA_15 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.259 IRQ_TYPE_STAT_RX_SA_15

- **Name:** IRQ_TYPE_STAT_RX_SA_15
- **Description:** Interrupt Status for each IRQ type of Rx SA 15
- **Size:** 32 bits
- **Offset:** 0x40c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 15

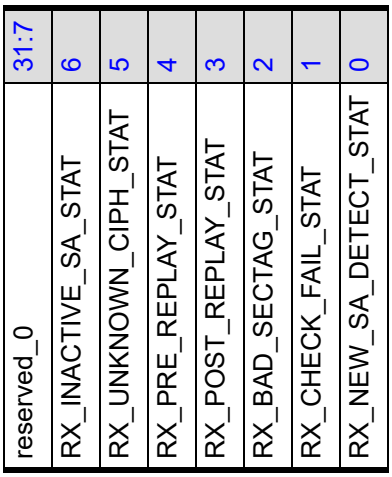


Table 6-266 Fields for Register: IRQ_TYPE_STAT_RX_SA_15

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-266 Fields for Register: IRQ_TYPE_STAT_RX_SA_15 (Continued)

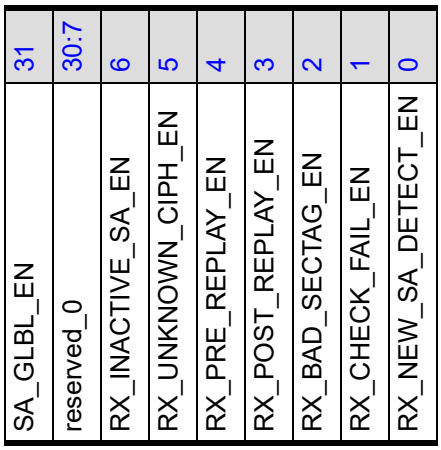
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-266 Fields for Register: IRQ_TYPE_STAT_RX_SA_15 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.260 IRQ_TYPE_EN_RX_SA_16

- **Name:** IRQ_TYPE_EN_RX_SA_16
- **Description:** Interrupt Enable for each IRQ type of Rx SA 16
- **Size:** 32 bits
- **Offset:** 0x410
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 16



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-267 Fields for Register: IRQ_TYPE_EN_RX_SA_16

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-267 Fields for Register: IRQ_TYPE_EN_RX_SA_16 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-267 Fields for Register: IRQ_TYPE_EN_RX_SA_16 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.261 IRQ_TYPE_STAT_RX_SA_16

- **Name:** IRQ_TYPE_STAT_RX_SA_16
- **Description:** Interrupt Status for each IRQ type of Rx SA 16
- **Size:** 32 bits
- **Offset:** 0x414
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 16

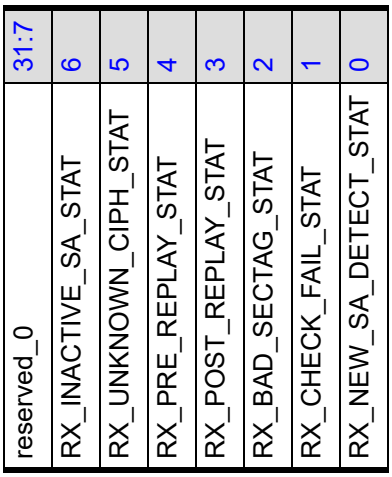


Table 6-268 Fields for Register: IRQ_TYPE_STAT_RX_SA_16

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-268 Fields for Register: IRQ_TYPE_STAT_RX_SA_16 (Continued)

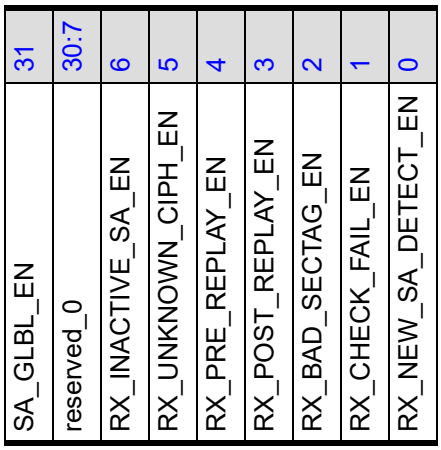
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-268 Fields for Register: IRQ_TYPE_STAT_RX_SA_16 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.262 IRQ_TYPE_EN_RX_SA_17

- **Name:** IRQ_TYPE_EN_RX_SA_17
- **Description:** Interrupt Enable for each IRQ type of Rx SA 17
- **Size:** 32 bits
- **Offset:** 0x418
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 17



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-269 Fields for Register: IRQ_TYPE_EN_RX_SA_17

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-269 Fields for Register: IRQ_TYPE_EN_RX_SA_17 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-269 Fields for Register: IRQ_TYPE_EN_RX_SA_17 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.263 IRQ_TYPE_STAT_RX_SA_17

- **Name:** IRQ_TYPE_STAT_RX_SA_17
- **Description:** Interrupt Status for each IRQ type of Rx SA 17
- **Size:** 32 bits
- **Offset:** 0x41c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 17

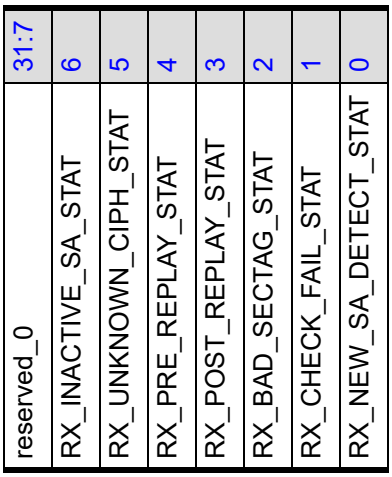


Table 6-270 Fields for Register: IRQ_TYPE_STAT_RX_SA_17

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-270 Fields for Register: IRQ_TYPE_STAT_RX_SA_17 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-270 Fields for Register: IRQ_TYPE_STAT_RX_SA_17 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.264 IRQ_TYPE_EN_RX_SA_18

- **Name:** IRQ_TYPE_EN_RX_SA_18
- **Description:** Interrupt Enable for each IRQ type of Rx SA 18
- **Size:** 32 bits
- **Offset:** 0x420
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 18

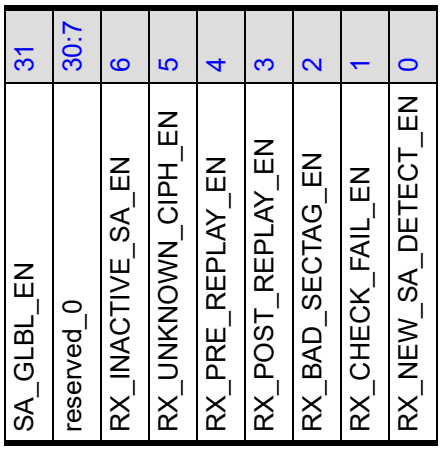


Table 6-271 Fields for Register: IRQ_TYPE_EN_RX_SA_18

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-271 Fields for Register: IRQ_TYPE_EN_RX_SA_18 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-271 Fields for Register: IRQ_TYPE_EN_RX_SA_18 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.265 IRQ_TYPE_STAT_RX_SA_18

- **Name:** IRQ_TYPE_STAT_RX_SA_18
- **Description:** Interrupt Status for each IRQ type of Rx SA 18
- **Size:** 32 bits
- **Offset:** 0x424
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 18

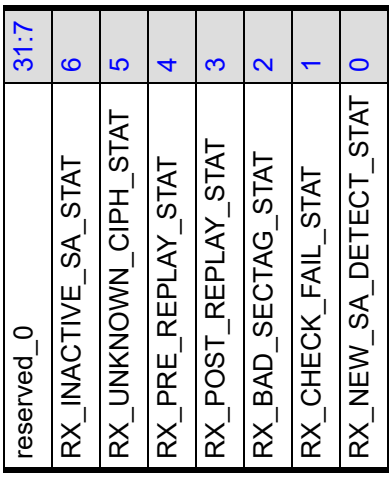


Table 6-272 Fields for Register: IRQ_TYPE_STAT_RX_SA_18

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-272 Fields for Register: IRQ_TYPE_STAT_RX_SA_18 (Continued)

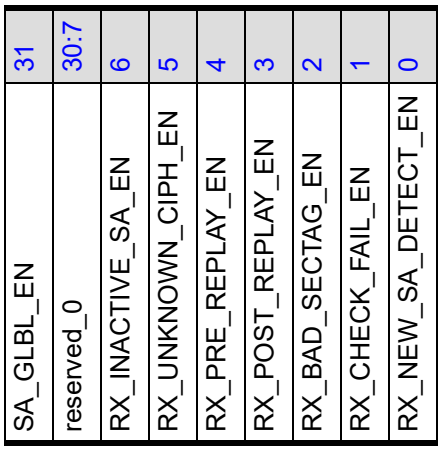
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-272 Fields for Register: IRQ_TYPE_STAT_RX_SA_18 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.266 IRQ_TYPE_EN_RX_SA_19

- **Name:** IRQ_TYPE_EN_RX_SA_19
- **Description:** Interrupt Enable for each IRQ type of Rx SA 19
- **Size:** 32 bits
- **Offset:** 0x428
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 19



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-273 Fields for Register: IRQ_TYPE_EN_RX_SA_19

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-273 Fields for Register: IRQ_TYPE_EN_RX_SA_19 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-273 Fields for Register: IRQ_TYPE_EN_RX_SA_19 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.267 IRQ_TYPE_STAT_RX_SA_19

- **Name:** IRQ_TYPE_STAT_RX_SA_19
- **Description:** Interrupt Status for each IRQ type of Rx SA 19
- **Size:** 32 bits
- **Offset:** 0x42c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 19

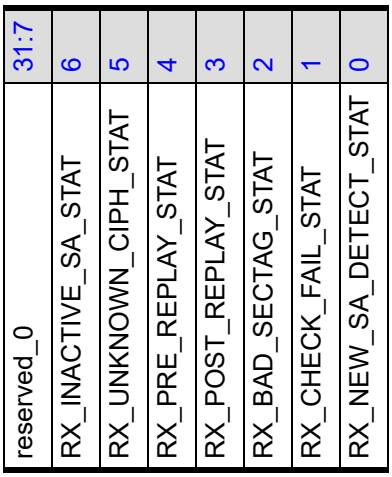


Table 6-274 Fields for Register: IRQ_TYPE_STAT_RX_SA_19

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-274 Fields for Register: IRQ_TYPE_STAT_RX_SA_19 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-274 Fields for Register: IRQ_TYPE_STAT_RX_SA_19 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.268 IRQ_TYPE_EN_RX_SA_20

- **Name:** IRQ_TYPE_EN_RX_SA_20
- **Description:** Interrupt Enable for each IRQ type of Rx SA 20
- **Size:** 32 bits
- **Offset:** 0x430
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 20

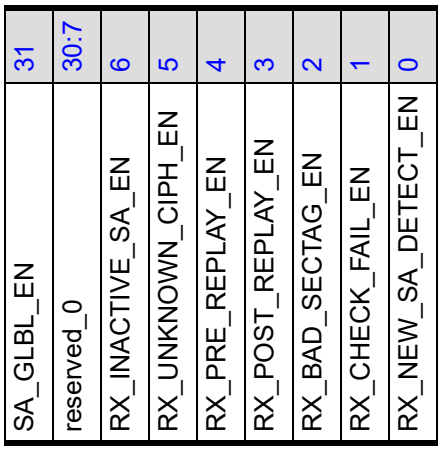


Table 6-275 Fields for Register: IRQ_TYPE_EN_RX_SA_20

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-275 Fields for Register: IRQ_TYPE_EN_RX_SA_20 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-275 Fields for Register: IRQ_TYPE_EN_RX_SA_20 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.269 IRQ_TYPE_STAT_RX_SA_20

- **Name:** IRQ_TYPE_STAT_RX_SA_20
- **Description:** Interrupt Status for each IRQ type of Rx SA 20
- **Size:** 32 bits
- **Offset:** 0x434
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 20

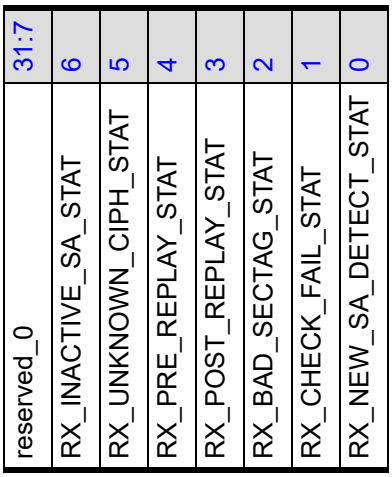


Table 6-276 Fields for Register: IRQ_TYPE_STAT_RX_SA_20

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-276 Fields for Register: IRQ_TYPE_STAT_RX_SA_20 (Continued)

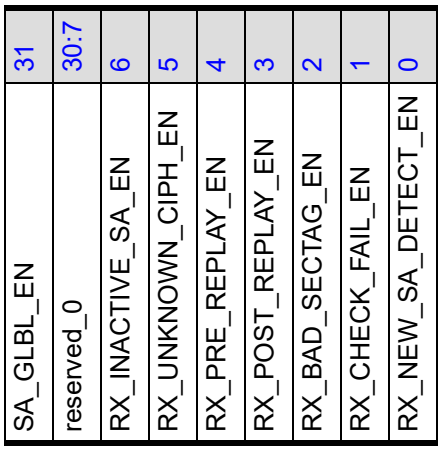
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-276 Fields for Register: IRQ_TYPE_STAT_RX_SA_20 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.270 IRQ_TYPE_EN_RX_SA_21

- **Name:** IRQ_TYPE_EN_RX_SA_21
- **Description:** Interrupt Enable for each IRQ type of Rx SA 21
- **Size:** 32 bits
- **Offset:** 0x438
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 21



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-277 Fields for Register: IRQ_TYPE_EN_RX_SA_21

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-277 Fields for Register: IRQ_TYPE_EN_RX_SA_21 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-277 Fields for Register: IRQ_TYPE_EN_RX_SA_21 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.271 IRQ_TYPE_STAT_RX_SA_21

- **Name:** IRQ_TYPE_STAT_RX_SA_21
- **Description:** Interrupt Status for each IRQ type of Rx SA 21
- **Size:** 32 bits
- **Offset:** 0x43c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 21

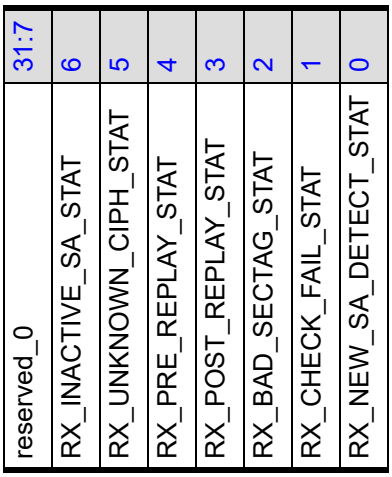


Table 6-278 Fields for Register: IRQ_TYPE_STAT_RX_SA_21

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-278 Fields for Register: IRQ_TYPE_STAT_RX_SA_21 (Continued)

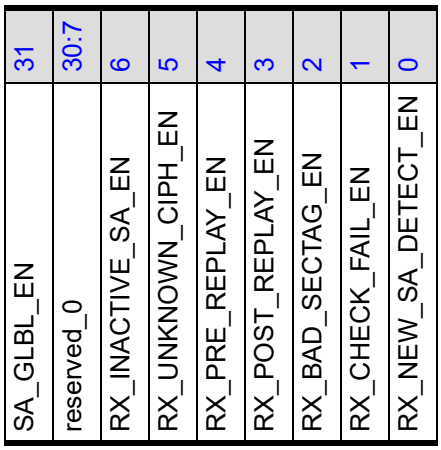
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-278 Fields for Register: IRQ_TYPE_STAT_RX_SA_21 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.272 IRQ_TYPE_EN_RX_SA_22

- **Name:** IRQ_TYPE_EN_RX_SA_22
- **Description:** Interrupt Enable for each IRQ type of Rx SA 22
- **Size:** 32 bits
- **Offset:** 0x440
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 22



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-279 Fields for Register: IRQ_TYPE_EN_RX_SA_22

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-279 Fields for Register: IRQ_TYPE_EN_RX_SA_22 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-279 Fields for Register: IRQ_TYPE_EN_RX_SA_22 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.273 IRQ_TYPE_STAT_RX_SA_22

- **Name:** IRQ_TYPE_STAT_RX_SA_22
- **Description:** Interrupt Status for each IRQ type of Rx SA 22
- **Size:** 32 bits
- **Offset:** 0x444
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 22

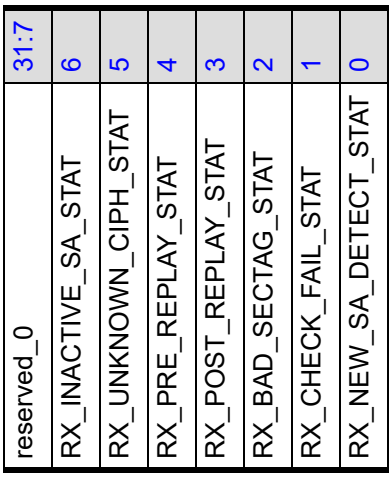


Table 6-280 Fields for Register: IRQ_TYPE_STAT_RX_SA_22

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-280 Fields for Register: IRQ_TYPE_STAT_RX_SA_22 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-280 Fields for Register: IRQ_TYPE_STAT_RX_SA_22 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.274 IRQ_TYPE_EN_RX_SA_23

- **Name:** IRQ_TYPE_EN_RX_SA_23
- **Description:** Interrupt Enable for each IRQ type of Rx SA 23
- **Size:** 32 bits
- **Offset:** 0x448
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 23

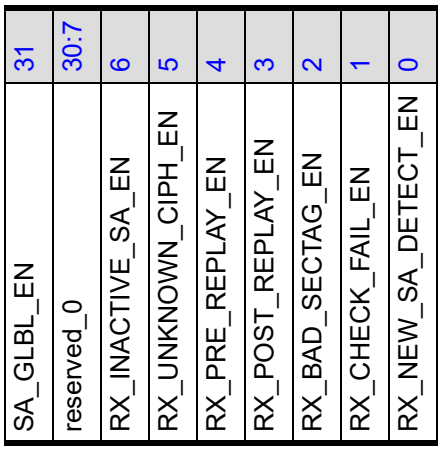


Table 6-281 Fields for Register: IRQ_TYPE_EN_RX_SA_23

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-281 Fields for Register: IRQ_TYPE_EN_RX_SA_23 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-281 Fields for Register: IRQ_TYPE_EN_RX_SA_23 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.275 IRQ_TYPE_STAT_RX_SA_23

- **Name:** IRQ_TYPE_STAT_RX_SA_23
- **Description:** Interrupt Status for each IRQ type of Rx SA 23
- **Size:** 32 bits
- **Offset:** 0x44c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 23

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-282 Fields for Register: IRQ_TYPE_STAT_RX_SA_23

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-282 Fields for Register: IRQ_TYPE_STAT_RX_SA_23 (Continued)

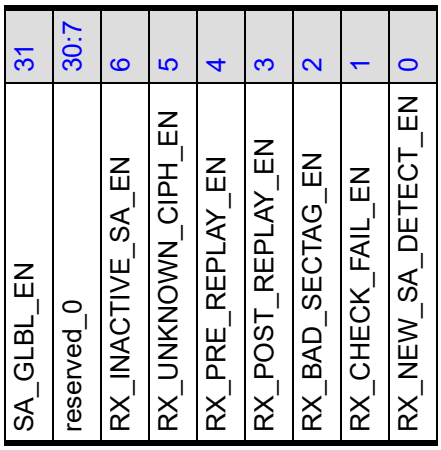
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-282 Fields for Register: IRQ_TYPE_STAT_RX_SA_23 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.276 IRQ_TYPE_EN_RX_SA_24

- **Name:** IRQ_TYPE_EN_RX_SA_24
- **Description:** Interrupt Enable for each IRQ type of Rx SA 24
- **Size:** 32 bits
- **Offset:** 0x450
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 24



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-283 Fields for Register: IRQ_TYPE_EN_RX_SA_24

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-283 Fields for Register: IRQ_TYPE_EN_RX_SA_24 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-283 Fields for Register: IRQ_TYPE_EN_RX_SA_24 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.277 IRQ_TYPE_STAT_RX_SA_24

- **Name:** IRQ_TYPE_STAT_RX_SA_24
- **Description:** Interrupt Status for each IRQ type of Rx SA 24
- **Size:** 32 bits
- **Offset:** 0x454
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 24

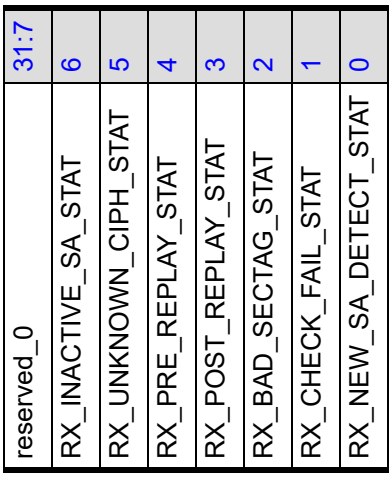


Table 6-284 Fields for Register: IRQ_TYPE_STAT_RX_SA_24

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-284 Fields for Register: IRQ_TYPE_STAT_RX_SA_24 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-284 Fields for Register: IRQ_TYPE_STAT_RX_SA_24 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.278 IRQ_TYPE_EN_RX_SA_25

- **Name:** IRQ_TYPE_EN_RX_SA_25
- **Description:** Interrupt Enable for each IRQ type of Rx SA 25
- **Size:** 32 bits
- **Offset:** 0x458
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 25

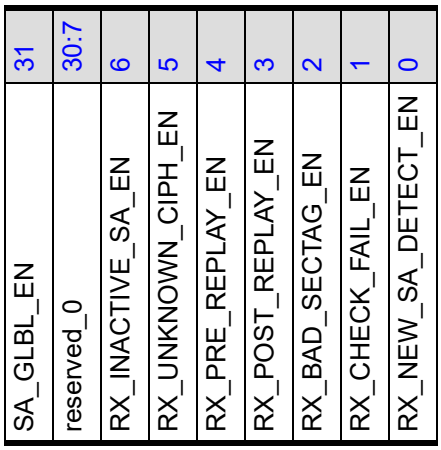


Table 6-285 Fields for Register: IRQ_TYPE_EN_RX_SA_25

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-285 Fields for Register: IRQ_TYPE_EN_RX_SA_25 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-285 Fields for Register: IRQ_TYPE_EN_RX_SA_25 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.279 IRQ_TYPE_STAT_RX_SA_25

- **Name:** IRQ_TYPE_STAT_RX_SA_25
- **Description:** Interrupt Status for each IRQ type of Rx SA 25
- **Size:** 32 bits
- **Offset:** 0x45c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 25

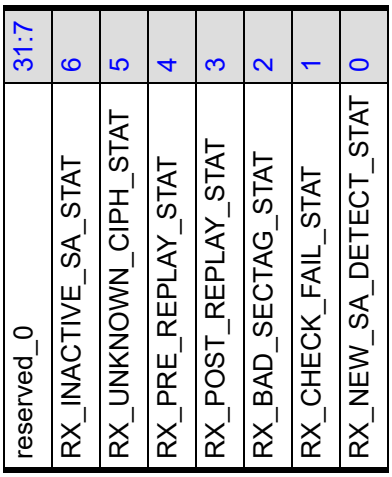


Table 6-286 Fields for Register: IRQ_TYPE_STAT_RX_SA_25

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-286 Fields for Register: IRQ_TYPE_STAT_RX_SA_25 (Continued)

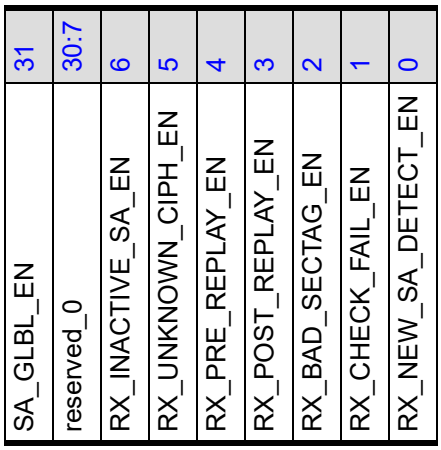
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-286 Fields for Register: IRQ_TYPE_STAT_RX_SA_25 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.280 IRQ_TYPE_EN_RX_SA_26

- **Name:** IRQ_TYPE_EN_RX_SA_26
- **Description:** Interrupt Enable for each IRQ type of Rx SA 26
- **Size:** 32 bits
- **Offset:** 0x460
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 26



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-287 Fields for Register: IRQ_TYPE_EN_RX_SA_26

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-287 Fields for Register: IRQ_TYPE_EN_RX_SA_26 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-287 Fields for Register: IRQ_TYPE_EN_RX_SA_26 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.281 IRQ_TYPE_STAT_RX_SA_26

- **Name:** IRQ_TYPE_STAT_RX_SA_26
- **Description:** Interrupt Status for each IRQ type of Rx SA 26
- **Size:** 32 bits
- **Offset:** 0x464
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 26

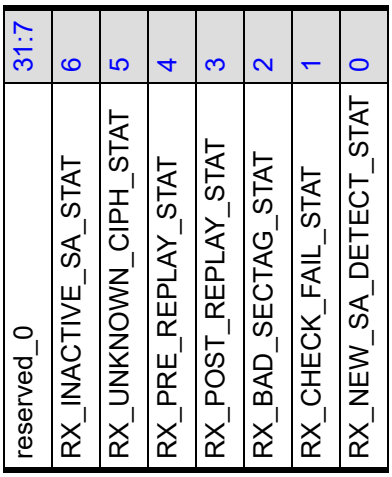


Table 6-288 Fields for Register: IRQ_TYPE_STAT_RX_SA_26

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-288 Fields for Register: IRQ_TYPE_STAT_RX_SA_26 (Continued)

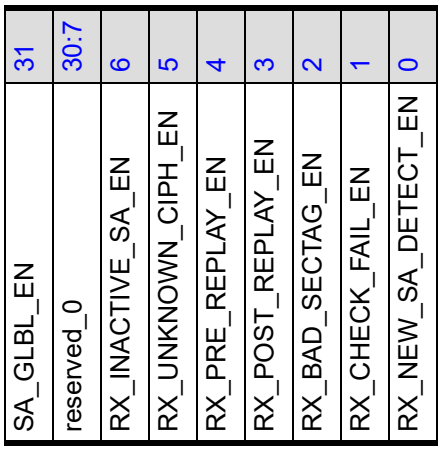
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-288 Fields for Register: IRQ_TYPE_STAT_RX_SA_26 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.282 IRQ_TYPE_EN_RX_SA_27

- **Name:** IRQ_TYPE_EN_RX_SA_27
- **Description:** Interrupt Enable for each IRQ type of Rx SA 27
- **Size:** 32 bits
- **Offset:** 0x468
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 27



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-289 Fields for Register: IRQ_TYPE_EN_RX_SA_27

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-289 Fields for Register: IRQ_TYPE_EN_RX_SA_27 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-289 Fields for Register: IRQ_TYPE_EN_RX_SA_27 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.283 IRQ_TYPE_STAT_RX_SA_27

- **Name:** IRQ_TYPE_STAT_RX_SA_27
- **Description:** Interrupt Status for each IRQ type of Rx SA 27
- **Size:** 32 bits
- **Offset:** 0x46c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 27

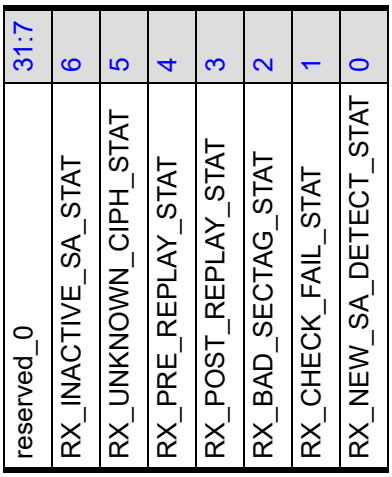


Table 6-290 Fields for Register: IRQ_TYPE_STAT_RX_SA_27

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-290 Fields for Register: IRQ_TYPE_STAT_RX_SA_27 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-290 Fields for Register: IRQ_TYPE_STAT_RX_SA_27 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.284 IRQ_TYPE_EN_RX_SA_28

- **Name:** IRQ_TYPE_EN_RX_SA_28
- **Description:** Interrupt Enable for each IRQ type of Rx SA 28
- **Size:** 32 bits
- **Offset:** 0x470
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 28

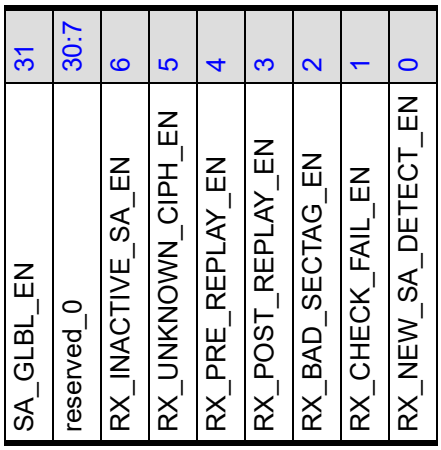


Table 6-291 Fields for Register: IRQ_TYPE_EN_RX_SA_28

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-291 Fields for Register: IRQ_TYPE_EN_RX_SA_28 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-291 Fields for Register: IRQ_TYPE_EN_RX_SA_28 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.285 IRQ_TYPE_STAT_RX_SA_28

- **Name:** IRQ_TYPE_STAT_RX_SA_28
- **Description:** Interrupt Status for each IRQ type of Rx SA 28
- **Size:** 32 bits
- **Offset:** 0x474
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 28

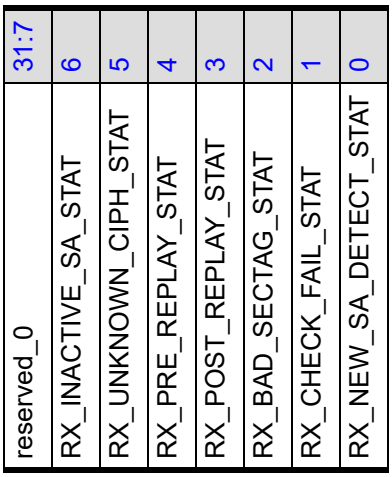


Table 6-292 Fields for Register: IRQ_TYPE_STAT_RX_SA_28

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-292 Fields for Register: IRQ_TYPE_STAT_RX_SA_28 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-292 Fields for Register: IRQ_TYPE_STAT_RX_SA_28 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.286 IRQ_TYPE_EN_RX_SA_29

- **Name:** IRQ_TYPE_EN_RX_SA_29
- **Description:** Interrupt Enable for each IRQ type of Rx SA 29
- **Size:** 32 bits
- **Offset:** 0x478
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 29

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-293 Fields for Register: IRQ_TYPE_EN_RX_SA_29

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-293 Fields for Register: IRQ_TYPE_EN_RX_SA_29 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-293 Fields for Register: IRQ_TYPE_EN_RX_SA_29 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.287 IRQ_TYPE_STAT_RX_SA_29

- **Name:** IRQ_TYPE_STAT_RX_SA_29
- **Description:** Interrupt Status for each IRQ type of Rx SA 29
- **Size:** 32 bits
- **Offset:** 0x47c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 29

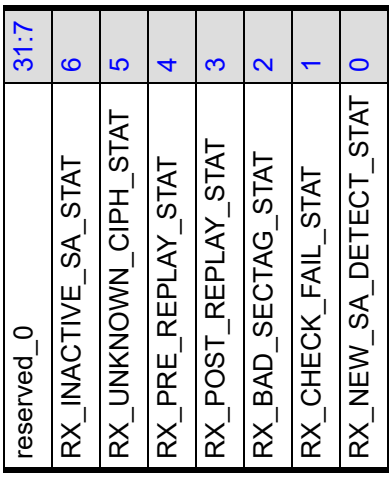


Table 6-294 Fields for Register: IRQ_TYPE_STAT_RX_SA_29

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-294 Fields for Register: IRQ_TYPE_STAT_RX_SA_29 (Continued)

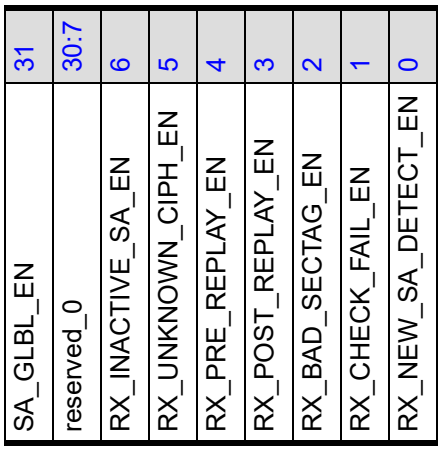
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-294 Fields for Register: IRQ_TYPE_STAT_RX_SA_29 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.288 IRQ_TYPE_EN_RX_SA_30

- **Name:** IRQ_TYPE_EN_RX_SA_30
- **Description:** Interrupt Enable for each IRQ type of Rx SA 30
- **Size:** 32 bits
- **Offset:** 0x480
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 30



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-295 Fields for Register: IRQ_TYPE_EN_RX_SA_30

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-295 Fields for Register: IRQ_TYPE_EN_RX_SA_30 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-295 Fields for Register: IRQ_TYPE_EN_RX_SA_30 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.289 IRQ_TYPE_STAT_RX_SA_30

- **Name:** IRQ_TYPE_STAT_RX_SA_30
- **Description:** Interrupt Status for each IRQ type of Rx SA 30
- **Size:** 32 bits
- **Offset:** 0x484
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 30

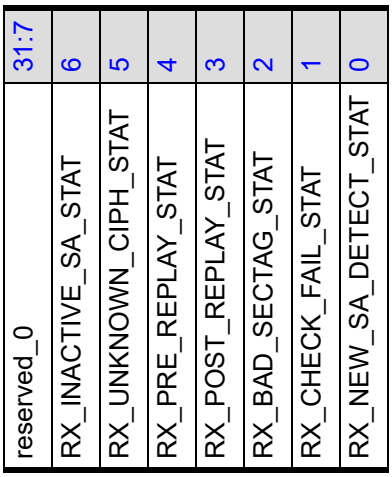


Table 6-296 Fields for Register: IRQ_TYPE_STAT_RX_SA_30

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-296 Fields for Register: IRQ_TYPE_STAT_RX_SA_30 (Continued)

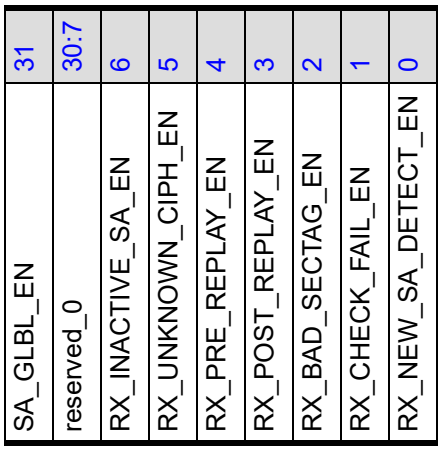
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-296 Fields for Register: IRQ_TYPE_STAT_RX_SA_30 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.290 IRQ_TYPE_EN_RX_SA_31

- **Name:** IRQ_TYPE_EN_RX_SA_31
- **Description:** Interrupt Enable for each IRQ type of Rx SA 31
- **Size:** 32 bits
- **Offset:** 0x488
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 31



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-297 Fields for Register: IRQ_TYPE_EN_RX_SA_31

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-297 Fields for Register: IRQ_TYPE_EN_RX_SA_31 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-297 Fields for Register: IRQ_TYPE_EN_RX_SA_31 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.291 IRQ_TYPE_STAT_RX_SA_31

- **Name:** IRQ_TYPE_STAT_RX_SA_31
- **Description:** Interrupt Status for each IRQ type of Rx SA 31
- **Size:** 32 bits
- **Offset:** 0x48c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 31

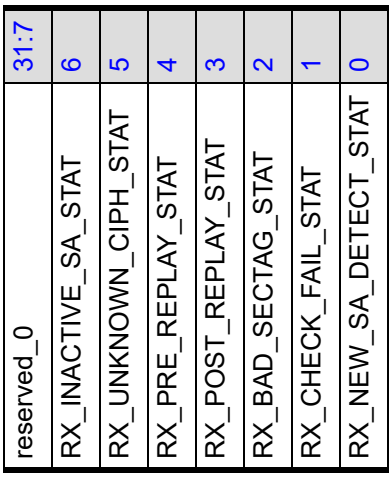


Table 6-298 Fields for Register: IRQ_TYPE_STAT_RX_SA_31

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-298 Fields for Register: IRQ_TYPE_STAT_RX_SA_31 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-298 Fields for Register: IRQ_TYPE_STAT_RX_SA_31 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.292 IRQ_TYPE_EN_RX_SA_32

- **Name:** IRQ_TYPE_EN_RX_SA_32
- **Description:** Interrupt Enable for each IRQ type of Rx SA 32
- **Size:** 32 bits
- **Offset:** 0x490
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 32

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-299 Fields for Register: IRQ_TYPE_EN_RX_SA_32

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-299 Fields for Register: IRQ_TYPE_EN_RX_SA_32 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-299 Fields for Register: IRQ_TYPE_EN_RX_SA_32 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.293 IRQ_TYPE_STAT_RX_SA_32

- **Name:** IRQ_TYPE_STAT_RX_SA_32
- **Description:** Interrupt Status for each IRQ type of Rx SA 32
- **Size:** 32 bits
- **Offset:** 0x494
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 32

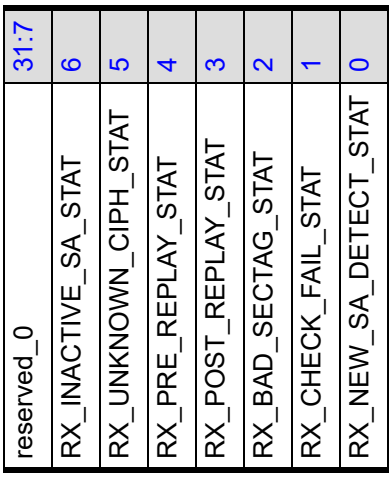


Table 6-300 Fields for Register: IRQ_TYPE_STAT_RX_SA_32

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-300 Fields for Register: IRQ_TYPE_STAT_RX_SA_32 (Continued)

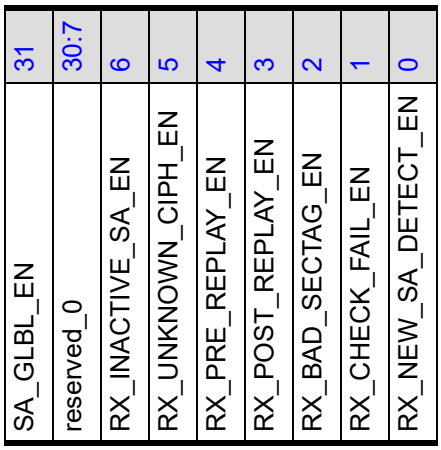
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-300 Fields for Register: IRQ_TYPE_STAT_RX_SA_32 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.294 IRQ_TYPE_EN_RX_SA_33

- **Name:** IRQ_TYPE_EN_RX_SA_33
- **Description:** Interrupt Enable for each IRQ type of Rx SA 33
- **Size:** 32 bits
- **Offset:** 0x498
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 33



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-301 Fields for Register: IRQ_TYPE_EN_RX_SA_33

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-301 Fields for Register: IRQ_TYPE_EN_RX_SA_33 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-301 Fields for Register: IRQ_TYPE_EN_RX_SA_33 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.295 IRQ_TYPE_STAT_RX_SA_33

- **Name:** IRQ_TYPE_STAT_RX_SA_33
- **Description:** Interrupt Status for each IRQ type of Rx SA 33
- **Size:** 32 bits
- **Offset:** 0x49c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 33

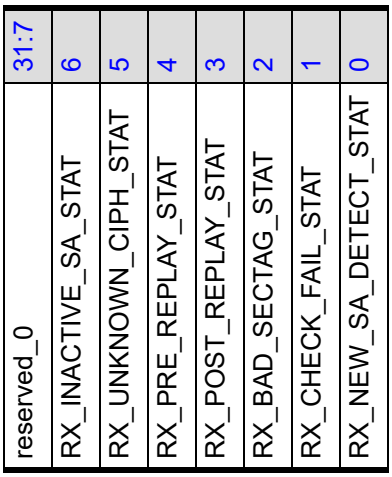


Table 6-302 Fields for Register: IRQ_TYPE_STAT_RX_SA_33

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-302 Fields for Register: IRQ_TYPE_STAT_RX_SA_33 (Continued)

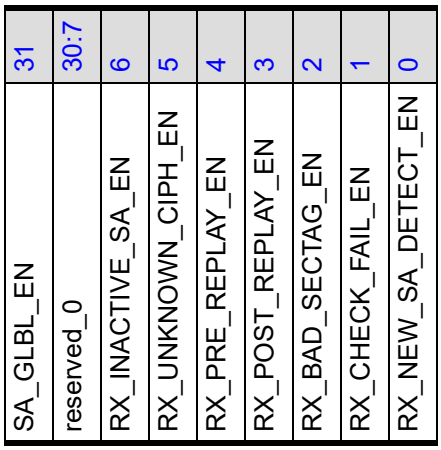
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-302 Fields for Register: IRQ_TYPE_STAT_RX_SA_33 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.296 IRQ_TYPE_EN_RX_SA_34

- **Name:** IRQ_TYPE_EN_RX_SA_34
- **Description:** Interrupt Enable for each IRQ type of Rx SA 34
- **Size:** 32 bits
- **Offset:** 0x4a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 34



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-303 Fields for Register: IRQ_TYPE_EN_RX_SA_34

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-303 Fields for Register: IRQ_TYPE_EN_RX_SA_34 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-303 Fields for Register: IRQ_TYPE_EN_RX_SA_34 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.297 IRQ_TYPE_STAT_RX_SA_34

- **Name:** IRQ_TYPE_STAT_RX_SA_34
- **Description:** Interrupt Status for each IRQ type of Rx SA 34
- **Size:** 32 bits
- **Offset:** 0x4a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 34

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-304 Fields for Register: IRQ_TYPE_STAT_RX_SA_34

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-304 Fields for Register: IRQ_TYPE_STAT_RX_SA_34 (Continued)

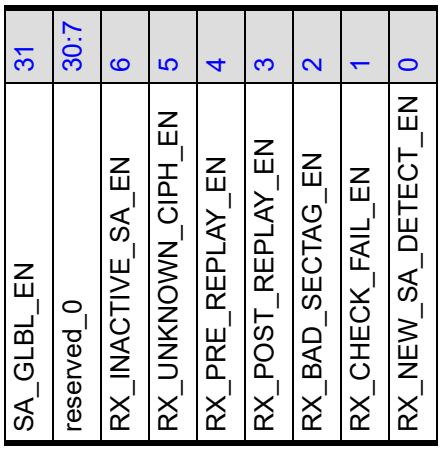
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-304 Fields for Register: IRQ_TYPE_STAT_RX_SA_34 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.298 IRQ_TYPE_EN_RX_SA_35

- **Name:** IRQ_TYPE_EN_RX_SA_35
- **Description:** Interrupt Enable for each IRQ type of Rx SA 35
- **Size:** 32 bits
- **Offset:** 0x4a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 35



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-305 Fields for Register: IRQ_TYPE_EN_RX_SA_35

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-305 Fields for Register: IRQ_TYPE_EN_RX_SA_35 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-305 Fields for Register: IRQ_TYPE_EN_RX_SA_35 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.299 IRQ_TYPE_STAT_RX_SA_35

- **Name:** IRQ_TYPE_STAT_RX_SA_35
- **Description:** Interrupt Status for each IRQ type of Rx SA 35
- **Size:** 32 bits
- **Offset:** 0x4ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 35

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-306 Fields for Register: IRQ_TYPE_STAT_RX_SA_35

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-306 Fields for Register: IRQ_TYPE_STAT_RX_SA_35 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-306 Fields for Register: IRQ_TYPE_STAT_RX_SA_35 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.300 IRQ_TYPE_EN_RX_SA_36

- **Name:** IRQ_TYPE_EN_RX_SA_36
- **Description:** Interrupt Enable for each IRQ type of Rx SA 36
- **Size:** 32 bits
- **Offset:** 0x4b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 36

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-307 Fields for Register: IRQ_TYPE_EN_RX_SA_36

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-307 Fields for Register: IRQ_TYPE_EN_RX_SA_36 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-307 Fields for Register: IRQ_TYPE_EN_RX_SA_36 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.301 IRQ_TYPE_STAT_RX_SA_36

- **Name:** IRQ_TYPE_STAT_RX_SA_36
- **Description:** Interrupt Status for each IRQ type of Rx SA 36
- **Size:** 32 bits
- **Offset:** 0x4b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 36

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-308 Fields for Register: IRQ_TYPE_STAT_RX_SA_36

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-308 Fields for Register: IRQ_TYPE_STAT_RX_SA_36 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-308 Fields for Register: IRQ_TYPE_STAT_RX_SA_36 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.302 IRQ_TYPE_EN_RX_SA_37

- **Name:** IRQ_TYPE_EN_RX_SA_37
- **Description:** Interrupt Enable for each IRQ type of Rx SA 37
- **Size:** 32 bits
- **Offset:** 0x4b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 37

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-309 Fields for Register: IRQ_TYPE_EN_RX_SA_37

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-309 Fields for Register: IRQ_TYPE_EN_RX_SA_37 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-309 Fields for Register: IRQ_TYPE_EN_RX_SA_37 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.303 IRQ_TYPE_STAT_RX_SA_37

- **Name:** IRQ_TYPE_STAT_RX_SA_37
- **Description:** Interrupt Status for each IRQ type of Rx SA 37
- **Size:** 32 bits
- **Offset:** 0x4bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 37

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-310 Fields for Register: IRQ_TYPE_STAT_RX_SA_37

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-310 Fields for Register: IRQ_TYPE_STAT_RX_SA_37 (Continued)

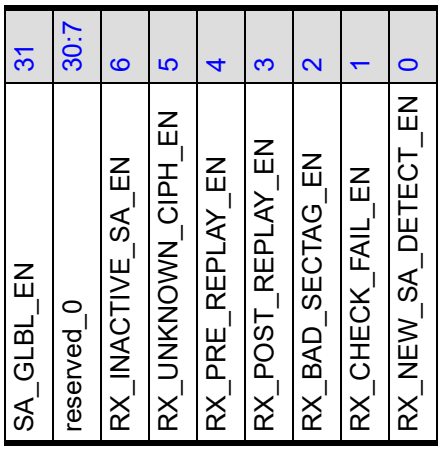
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-310 Fields for Register: IRQ_TYPE_STAT_RX_SA_37 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.304 IRQ_TYPE_EN_RX_SA_38

- **Name:** IRQ_TYPE_EN_RX_SA_38
- **Description:** Interrupt Enable for each IRQ type of Rx SA 38
- **Size:** 32 bits
- **Offset:** 0x4c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 38



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-311 Fields for Register: IRQ_TYPE_EN_RX_SA_38

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-311 Fields for Register: IRQ_TYPE_EN_RX_SA_38 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-311 Fields for Register: IRQ_TYPE_EN_RX_SA_38 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.305 IRQ_TYPE_STAT_RX_SA_38

- **Name:** IRQ_TYPE_STAT_RX_SA_38
- **Description:** Interrupt Status for each IRQ type of Rx SA 38
- **Size:** 32 bits
- **Offset:** 0x4c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 38

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-312 Fields for Register: IRQ_TYPE_STAT_RX_SA_38

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-312 Fields for Register: IRQ_TYPE_STAT_RX_SA_38 (Continued)

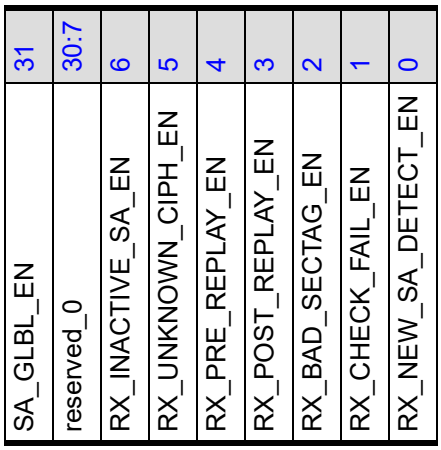
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-312 Fields for Register: IRQ_TYPE_STAT_RX_SA_38 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.306 IRQ_TYPE_EN_RX_SA_39

- **Name:** IRQ_TYPE_EN_RX_SA_39
- **Description:** Interrupt Enable for each IRQ type of Rx SA 39
- **Size:** 32 bits
- **Offset:** 0x4c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 39



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-313 Fields for Register: IRQ_TYPE_EN_RX_SA_39

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-313 Fields for Register: IRQ_TYPE_EN_RX_SA_39 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-313 Fields for Register: IRQ_TYPE_EN_RX_SA_39 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.307 IRQ_TYPE_STAT_RX_SA_39

- **Name:** IRQ_TYPE_STAT_RX_SA_39
- **Description:** Interrupt Status for each IRQ type of Rx SA 39
- **Size:** 32 bits
- **Offset:** 0x4cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 39

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-314 Fields for Register: IRQ_TYPE_STAT_RX_SA_39

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-314 Fields for Register: IRQ_TYPE_STAT_RX_SA_39 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-314 Fields for Register: IRQ_TYPE_STAT_RX_SA_39 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.308 IRQ_TYPE_EN_RX_SA_40

- **Name:** IRQ_TYPE_EN_RX_SA_40
- **Description:** Interrupt Enable for each IRQ type of Rx SA 40
- **Size:** 32 bits
- **Offset:** 0x4d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 40

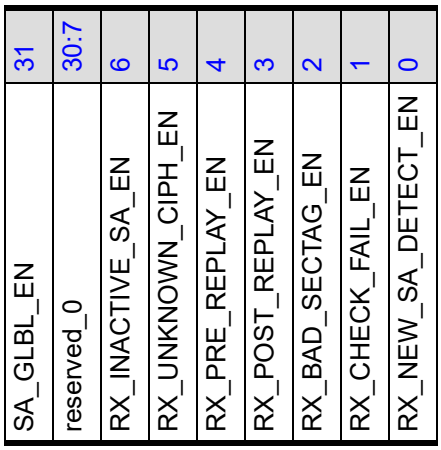


Table 6-315 Fields for Register: IRQ_TYPE_EN_RX_SA_40

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-315 Fields for Register: IRQ_TYPE_EN_RX_SA_40 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-315 Fields for Register: IRQ_TYPE_EN_RX_SA_40 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.309 IRQ_TYPE_STAT_RX_SA_40

- **Name:** IRQ_TYPE_STAT_RX_SA_40
- **Description:** Interrupt Status for each IRQ type of Rx SA 40
- **Size:** 32 bits
- **Offset:** 0x4d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 40

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-316 Fields for Register: IRQ_TYPE_STAT_RX_SA_40

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-316 Fields for Register: IRQ_TYPE_STAT_RX_SA_40 (Continued)

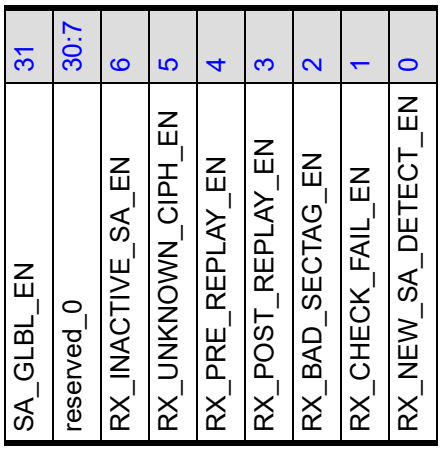
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-316 Fields for Register: IRQ_TYPE_STAT_RX_SA_40 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.310 IRQ_TYPE_EN_RX_SA_41

- **Name:** IRQ_TYPE_EN_RX_SA_41
- **Description:** Interrupt Enable for each IRQ type of Rx SA 41
- **Size:** 32 bits
- **Offset:** 0x4d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 41



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-317 Fields for Register: IRQ_TYPE_EN_RX_SA_41

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-317 Fields for Register: IRQ_TYPE_EN_RX_SA_41 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-317 Fields for Register: IRQ_TYPE_EN_RX_SA_41 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.311 IRQ_TYPE_STAT_RX_SA_41

- **Name:** IRQ_TYPE_STAT_RX_SA_41
- **Description:** Interrupt Status for each IRQ type of Rx SA 41
- **Size:** 32 bits
- **Offset:** 0x4dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 41

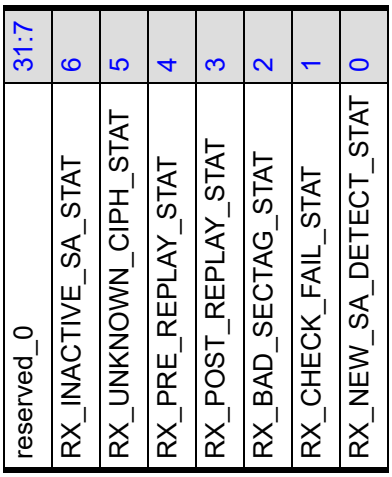


Table 6-318 Fields for Register: IRQ_TYPE_STAT_RX_SA_41

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-318 Fields for Register: IRQ_TYPE_STAT_RX_SA_41 (Continued)

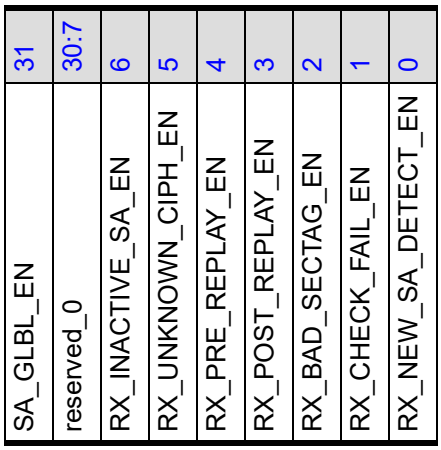
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-318 Fields for Register: IRQ_TYPE_STAT_RX_SA_41 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.312 IRQ_TYPE_EN_RX_SA_42

- **Name:** IRQ_TYPE_EN_RX_SA_42
- **Description:** Interrupt Enable for each IRQ type of Rx SA 42
- **Size:** 32 bits
- **Offset:** 0x4e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 42



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-319 Fields for Register: IRQ_TYPE_EN_RX_SA_42

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-319 Fields for Register: IRQ_TYPE_EN_RX_SA_42 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-319 Fields for Register: IRQ_TYPE_EN_RX_SA_42 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.313 IRQ_TYPE_STAT_RX_SA_42

- **Name:** IRQ_TYPE_STAT_RX_SA_42
- **Description:** Interrupt Status for each IRQ type of Rx SA 42
- **Size:** 32 bits
- **Offset:** 0x4e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 42

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-320 Fields for Register: IRQ_TYPE_STAT_RX_SA_42

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-320 Fields for Register: IRQ_TYPE_STAT_RX_SA_42 (Continued)

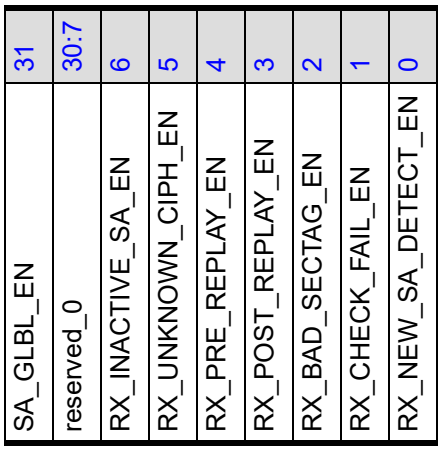
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-320 Fields for Register: IRQ_TYPE_STAT_RX_SA_42 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.314 IRQ_TYPE_EN_RX_SA_43

- **Name:** IRQ_TYPE_EN_RX_SA_43
- **Description:** Interrupt Enable for each IRQ type of Rx SA 43
- **Size:** 32 bits
- **Offset:** 0x4e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 43



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-321 Fields for Register: IRQ_TYPE_EN_RX_SA_43

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-321 Fields for Register: IRQ_TYPE_EN_RX_SA_43 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-321 Fields for Register: IRQ_TYPE_EN_RX_SA_43 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.315 IRQ_TYPE_STAT_RX_SA_43

- **Name:** IRQ_TYPE_STAT_RX_SA_43
- **Description:** Interrupt Status for each IRQ type of Rx SA 43
- **Size:** 32 bits
- **Offset:** 0x4ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 43

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-322 Fields for Register: IRQ_TYPE_STAT_RX_SA_43

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-322 Fields for Register: IRQ_TYPE_STAT_RX_SA_43 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-322 Fields for Register: IRQ_TYPE_STAT_RX_SA_43 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.316 IRQ_TYPE_EN_RX_SA_44

- **Name:** IRQ_TYPE_EN_RX_SA_44
- **Description:** Interrupt Enable for each IRQ type of Rx SA 44
- **Size:** 32 bits
- **Offset:** 0x4f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 44

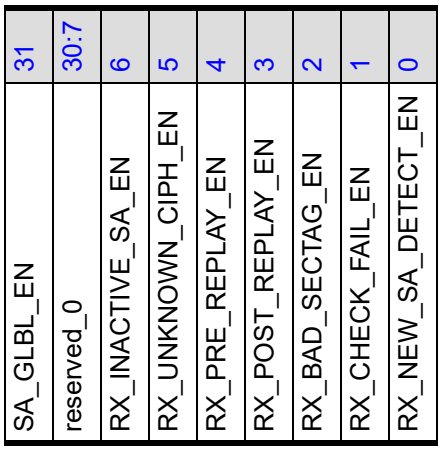


Table 6-323 Fields for Register: IRQ_TYPE_EN_RX_SA_44

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-323 Fields for Register: IRQ_TYPE_EN_RX_SA_44 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-323 Fields for Register: IRQ_TYPE_EN_RX_SA_44 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.317 IRQ_TYPE_STAT_RX_SA_44

- **Name:** IRQ_TYPE_STAT_RX_SA_44
- **Description:** Interrupt Status for each IRQ type of Rx SA 44
- **Size:** 32 bits
- **Offset:** 0x4f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 44

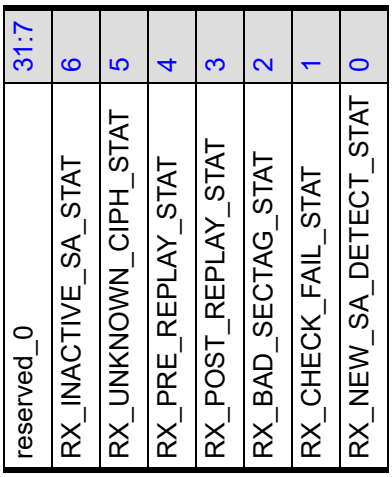


Table 6-324 Fields for Register: IRQ_TYPE_STAT_RX_SA_44

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-324 Fields for Register: IRQ_TYPE_STAT_RX_SA_44 (Continued)

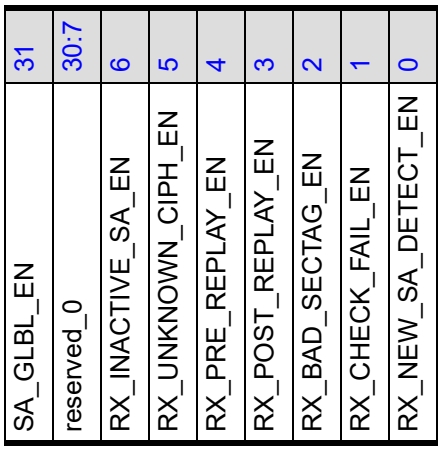
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-324 Fields for Register: IRQ_TYPE_STAT_RX_SA_44 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.318 IRQ_TYPE_EN_RX_SA_45

- **Name:** IRQ_TYPE_EN_RX_SA_45
- **Description:** Interrupt Enable for each IRQ type of Rx SA 45
- **Size:** 32 bits
- **Offset:** 0x4f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 45



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-325 Fields for Register: IRQ_TYPE_EN_RX_SA_45

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-325 Fields for Register: IRQ_TYPE_EN_RX_SA_45 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-325 Fields for Register: IRQ_TYPE_EN_RX_SA_45 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.319 IRQ_TYPE_STAT_RX_SA_45

- **Name:** IRQ_TYPE_STAT_RX_SA_45
- **Description:** Interrupt Status for each IRQ type of Rx SA 45
- **Size:** 32 bits
- **Offset:** 0x4fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 45

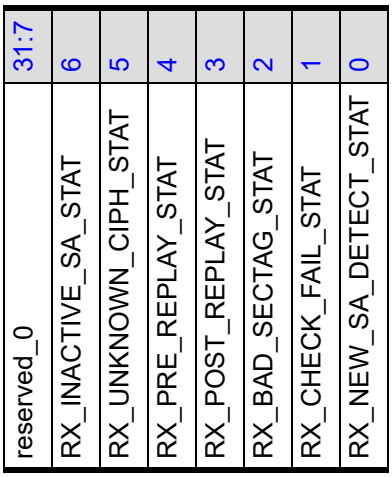


Table 6-326 Fields for Register: IRQ_TYPE_STAT_RX_SA_45

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-326 Fields for Register: IRQ_TYPE_STAT_RX_SA_45 (Continued)

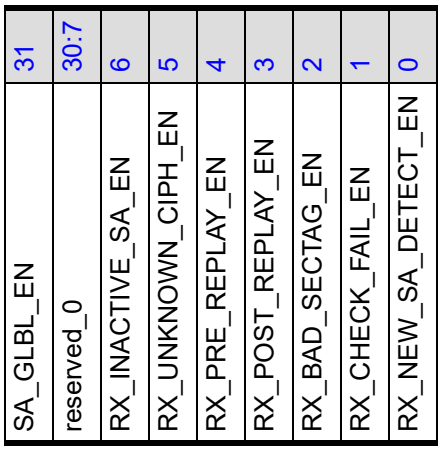
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-326 Fields for Register: IRQ_TYPE_STAT_RX_SA_45 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.320 IRQ_TYPE_EN_RX_SA_46

- **Name:** IRQ_TYPE_EN_RX_SA_46
- **Description:** Interrupt Enable for each IRQ type of Rx SA 46
- **Size:** 32 bits
- **Offset:** 0x500
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 46



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-327 Fields for Register: IRQ_TYPE_EN_RX_SA_46

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-327 Fields for Register: IRQ_TYPE_EN_RX_SA_46 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-327 Fields for Register: IRQ_TYPE_EN_RX_SA_46 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.321 IRQ_TYPE_STAT_RX_SA_46

- **Name:** IRQ_TYPE_STAT_RX_SA_46
- **Description:** Interrupt Status for each IRQ type of Rx SA 46
- **Size:** 32 bits
- **Offset:** 0x504
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 46

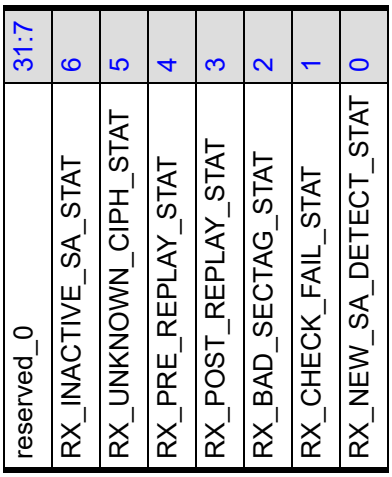


Table 6-328 Fields for Register: IRQ_TYPE_STAT_RX_SA_46

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-328 Fields for Register: IRQ_TYPE_STAT_RX_SA_46 (Continued)

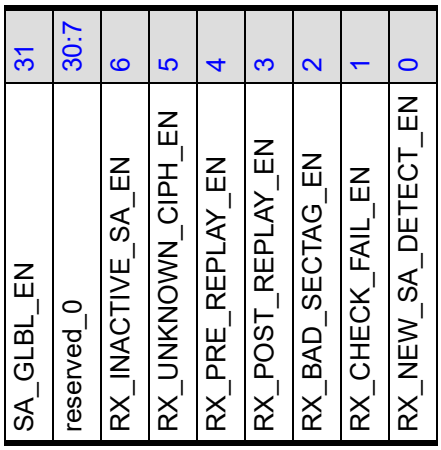
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-328 Fields for Register: IRQ_TYPE_STAT_RX_SA_46 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.322 IRQ_TYPE_EN_RX_SA_47

- **Name:** IRQ_TYPE_EN_RX_SA_47
- **Description:** Interrupt Enable for each IRQ type of Rx SA 47
- **Size:** 32 bits
- **Offset:** 0x508
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 47



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-329 Fields for Register: IRQ_TYPE_EN_RX_SA_47

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-329 Fields for Register: IRQ_TYPE_EN_RX_SA_47 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-329 Fields for Register: IRQ_TYPE_EN_RX_SA_47 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.323 IRQ_TYPE_STAT_RX_SA_47

- **Name:** IRQ_TYPE_STAT_RX_SA_47
- **Description:** Interrupt Status for each IRQ type of Rx SA 47
- **Size:** 32 bits
- **Offset:** 0x50c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 47

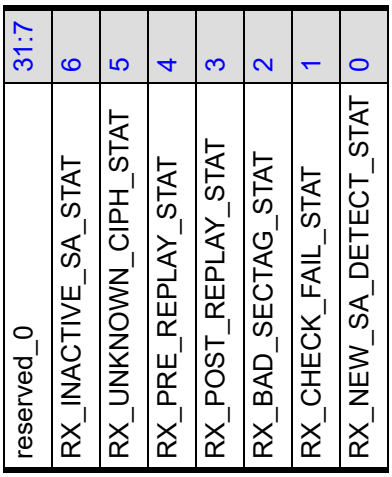


Table 6-330 Fields for Register: IRQ_TYPE_STAT_RX_SA_47

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-330 Fields for Register: IRQ_TYPE_STAT_RX_SA_47 (Continued)

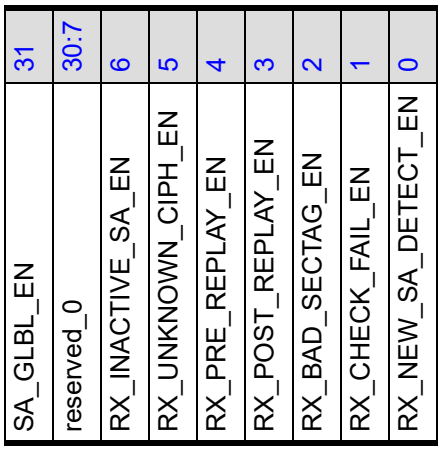
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-330 Fields for Register: IRQ_TYPE_STAT_RX_SA_47 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.324 IRQ_TYPE_EN_RX_SA_48

- **Name:** IRQ_TYPE_EN_RX_SA_48
- **Description:** Interrupt Enable for each IRQ type of Rx SA 48
- **Size:** 32 bits
- **Offset:** 0x510
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 48



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-331 Fields for Register: IRQ_TYPE_EN_RX_SA_48

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-331 Fields for Register: IRQ_TYPE_EN_RX_SA_48 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-331 Fields for Register: IRQ_TYPE_EN_RX_SA_48 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.325 IRQ_TYPE_STAT_RX_SA_48

- **Name:** IRQ_TYPE_STAT_RX_SA_48
- **Description:** Interrupt Status for each IRQ type of Rx SA 48
- **Size:** 32 bits
- **Offset:** 0x514
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 48

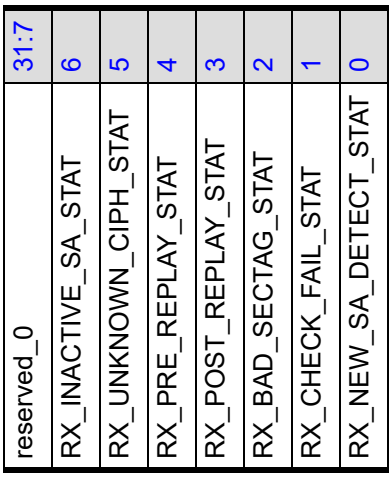


Table 6-332 Fields for Register: IRQ_TYPE_STAT_RX_SA_48

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-332 Fields for Register: IRQ_TYPE_STAT_RX_SA_48 (Continued)

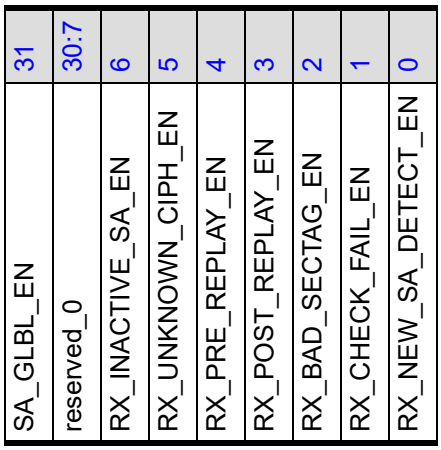
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-332 Fields for Register: IRQ_TYPE_STAT_RX_SA_48 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.326 IRQ_TYPE_EN_RX_SA_49

- **Name:** IRQ_TYPE_EN_RX_SA_49
- **Description:** Interrupt Enable for each IRQ type of Rx SA 49
- **Size:** 32 bits
- **Offset:** 0x518
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 49



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-333 Fields for Register: IRQ_TYPE_EN_RX_SA_49

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-333 Fields for Register: IRQ_TYPE_EN_RX_SA_49 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-333 Fields for Register: IRQ_TYPE_EN_RX_SA_49 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.327 IRQ_TYPE_STAT_RX_SA_49

- **Name:** IRQ_TYPE_STAT_RX_SA_49
- **Description:** Interrupt Status for each IRQ type of Rx SA 49
- **Size:** 32 bits
- **Offset:** 0x51c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 49

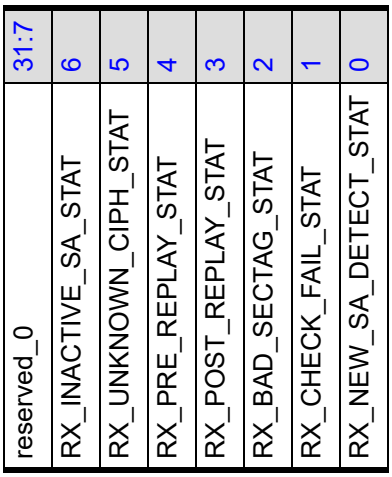


Table 6-334 Fields for Register: IRQ_TYPE_STAT_RX_SA_49

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-334 Fields for Register: IRQ_TYPE_STAT_RX_SA_49 (Continued)

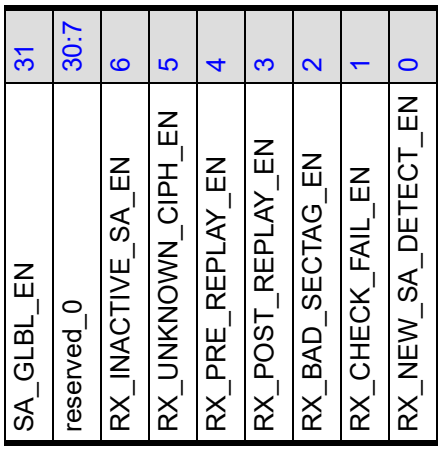
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-334 Fields for Register: IRQ_TYPE_STAT_RX_SA_49 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.328 IRQ_TYPE_EN_RX_SA_50

- **Name:** IRQ_TYPE_EN_RX_SA_50
- **Description:** Interrupt Enable for each IRQ type of Rx SA 50
- **Size:** 32 bits
- **Offset:** 0x520
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 50



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-335 Fields for Register: IRQ_TYPE_EN_RX_SA_50

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-335 Fields for Register: IRQ_TYPE_EN_RX_SA_50 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-335 Fields for Register: IRQ_TYPE_EN_RX_SA_50 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.329 IRQ_TYPE_STAT_RX_SA_50

- **Name:** IRQ_TYPE_STAT_RX_SA_50
- **Description:** Interrupt Status for each IRQ type of Rx SA 50
- **Size:** 32 bits
- **Offset:** 0x524
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 50

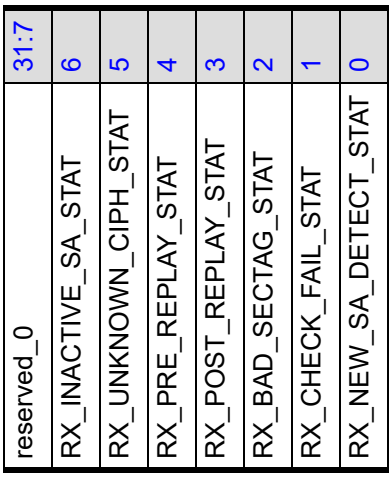


Table 6-336 Fields for Register: IRQ_TYPE_STAT_RX_SA_50

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-336 Fields for Register: IRQ_TYPE_STAT_RX_SA_50 (Continued)

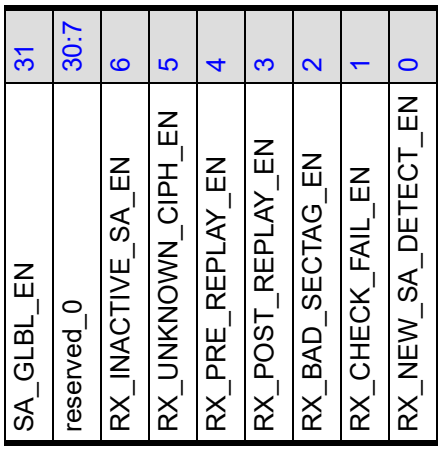
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-336 Fields for Register: IRQ_TYPE_STAT_RX_SA_50 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.330 IRQ_TYPE_EN_RX_SA_51

- **Name:** IRQ_TYPE_EN_RX_SA_51
- **Description:** Interrupt Enable for each IRQ type of Rx SA 51
- **Size:** 32 bits
- **Offset:** 0x528
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 51



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-337 Fields for Register: IRQ_TYPE_EN_RX_SA_51

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-337 Fields for Register: IRQ_TYPE_EN_RX_SA_51 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-337 Fields for Register: IRQ_TYPE_EN_RX_SA_51 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.331 IRQ_TYPE_STAT_RX_SA_51

- **Name:** IRQ_TYPE_STAT_RX_SA_51
- **Description:** Interrupt Status for each IRQ type of Rx SA 51
- **Size:** 32 bits
- **Offset:** 0x52c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 51

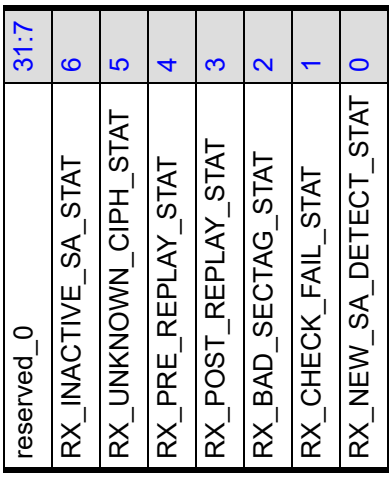


Table 6-338 Fields for Register: IRQ_TYPE_STAT_RX_SA_51

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-338 Fields for Register: IRQ_TYPE_STAT_RX_SA_51 (Continued)

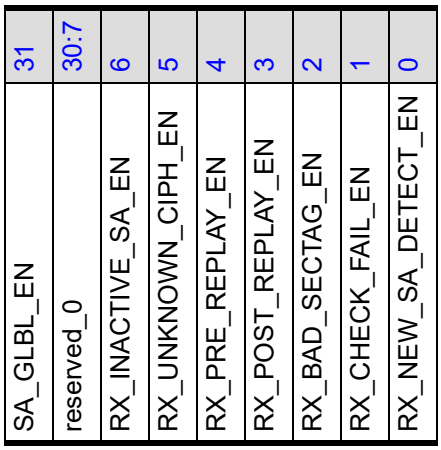
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-338 Fields for Register: IRQ_TYPE_STAT_RX_SA_51 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.332 IRQ_TYPE_EN_RX_SA_52

- **Name:** IRQ_TYPE_EN_RX_SA_52
- **Description:** Interrupt Enable for each IRQ type of Rx SA 52
- **Size:** 32 bits
- **Offset:** 0x530
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 52



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-339 Fields for Register: IRQ_TYPE_EN_RX_SA_52

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-339 Fields for Register: IRQ_TYPE_EN_RX_SA_52 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-339 Fields for Register: IRQ_TYPE_EN_RX_SA_52 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.333 IRQ_TYPE_STAT_RX_SA_52

- **Name:** IRQ_TYPE_STAT_RX_SA_52
- **Description:** Interrupt Status for each IRQ type of Rx SA 52
- **Size:** 32 bits
- **Offset:** 0x534
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 52

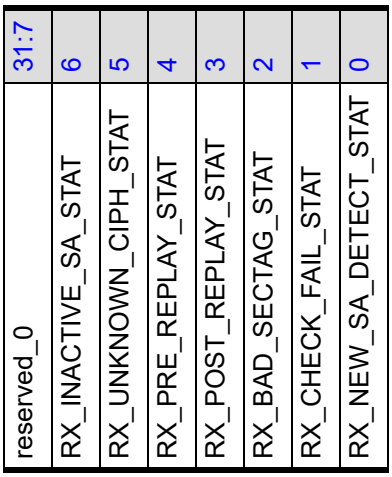


Table 6-340 Fields for Register: IRQ_TYPE_STAT_RX_SA_52

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-340 Fields for Register: IRQ_TYPE_STAT_RX_SA_52 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-340 Fields for Register: IRQ_TYPE_STAT_RX_SA_52 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.334 IRQ_TYPE_EN_RX_SA_53

- **Name:** IRQ_TYPE_EN_RX_SA_53
- **Description:** Interrupt Enable for each IRQ type of Rx SA 53
- **Size:** 32 bits
- **Offset:** 0x538
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 53

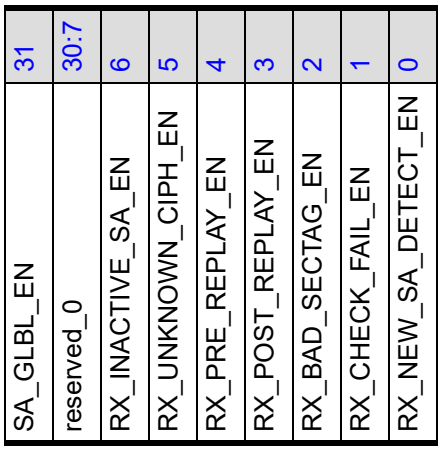


Table 6-341 Fields for Register: IRQ_TYPE_EN_RX_SA_53

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-341 Fields for Register: IRQ_TYPE_EN_RX_SA_53 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-341 Fields for Register: IRQ_TYPE_EN_RX_SA_53 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.335 IRQ_TYPE_STAT_RX_SA_53

- **Name:** IRQ_TYPE_STAT_RX_SA_53
- **Description:** Interrupt Status for each IRQ type of Rx SA 53
- **Size:** 32 bits
- **Offset:** 0x53c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 53

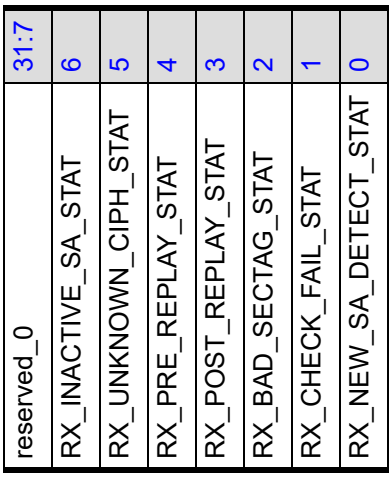


Table 6-342 Fields for Register: IRQ_TYPE_STAT_RX_SA_53

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-342 Fields for Register: IRQ_TYPE_STAT_RX_SA_53 (Continued)

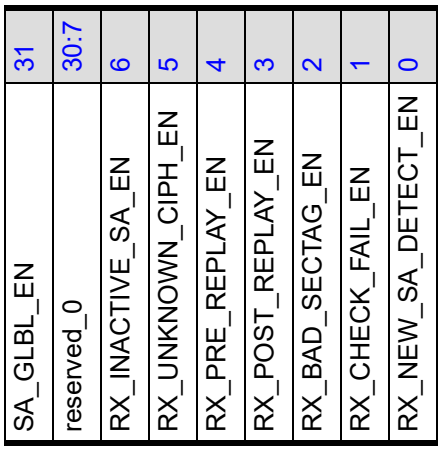
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-342 Fields for Register: IRQ_TYPE_STAT_RX_SA_53 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.336 IRQ_TYPE_EN_RX_SA_54

- **Name:** IRQ_TYPE_EN_RX_SA_54
- **Description:** Interrupt Enable for each IRQ type of Rx SA 54
- **Size:** 32 bits
- **Offset:** 0x540
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 54



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-343 Fields for Register: IRQ_TYPE_EN_RX_SA_54

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-343 Fields for Register: IRQ_TYPE_EN_RX_SA_54 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-343 Fields for Register: IRQ_TYPE_EN_RX_SA_54 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.337 IRQ_TYPE_STAT_RX_SA_54

- **Name:** IRQ_TYPE_STAT_RX_SA_54
- **Description:** Interrupt Status for each IRQ type of Rx SA 54
- **Size:** 32 bits
- **Offset:** 0x544
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 54

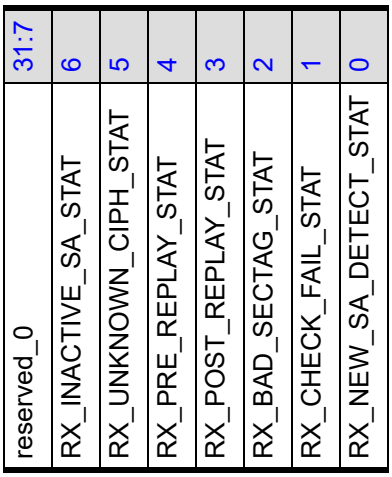


Table 6-344 Fields for Register: IRQ_TYPE_STAT_RX_SA_54

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-344 Fields for Register: IRQ_TYPE_STAT_RX_SA_54 (Continued)

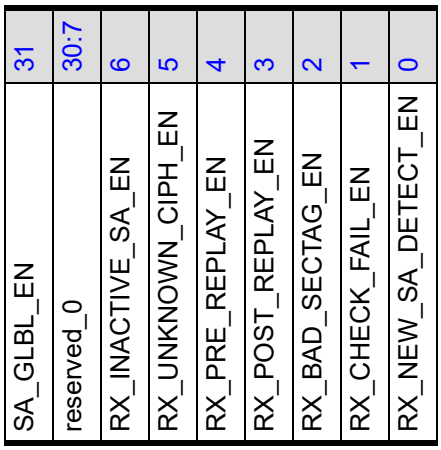
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-344 Fields for Register: IRQ_TYPE_STAT_RX_SA_54 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.338 IRQ_TYPE_EN_RX_SA_55

- **Name:** IRQ_TYPE_EN_RX_SA_55
- **Description:** Interrupt Enable for each IRQ type of Rx SA 55
- **Size:** 32 bits
- **Offset:** 0x548
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 55



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-345 Fields for Register: IRQ_TYPE_EN_RX_SA_55

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-345 Fields for Register: IRQ_TYPE_EN_RX_SA_55 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-345 Fields for Register: IRQ_TYPE_EN_RX_SA_55 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.339 IRQ_TYPE_STAT_RX_SA_55

- **Name:** IRQ_TYPE_STAT_RX_SA_55
- **Description:** Interrupt Status for each IRQ type of Rx SA 55
- **Size:** 32 bits
- **Offset:** 0x54c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 55

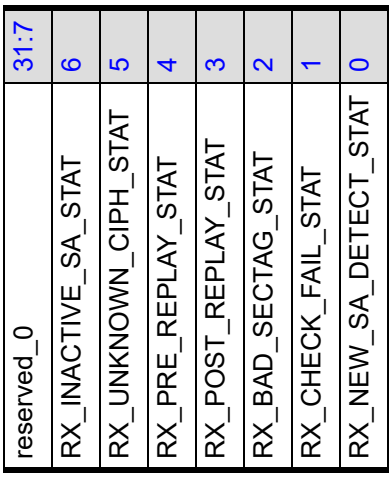


Table 6-346 Fields for Register: IRQ_TYPE_STAT_RX_SA_55

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-346 Fields for Register: IRQ_TYPE_STAT_RX_SA_55 (Continued)

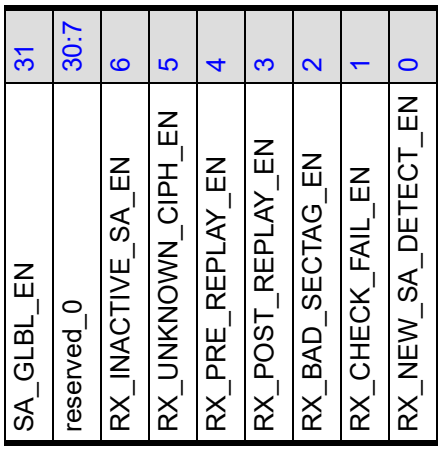
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-346 Fields for Register: IRQ_TYPE_STAT_RX_SA_55 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.340 IRQ_TYPE_EN_RX_SA_56

- **Name:** IRQ_TYPE_EN_RX_SA_56
- **Description:** Interrupt Enable for each IRQ type of Rx SA 56
- **Size:** 32 bits
- **Offset:** 0x550
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 56



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-347 Fields for Register: IRQ_TYPE_EN_RX_SA_56

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-347 Fields for Register: IRQ_TYPE_EN_RX_SA_56 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-347 Fields for Register: IRQ_TYPE_EN_RX_SA_56 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.341 IRQ_TYPE_STAT_RX_SA_56

- **Name:** IRQ_TYPE_STAT_RX_SA_56
- **Description:** Interrupt Status for each IRQ type of Rx SA 56
- **Size:** 32 bits
- **Offset:** 0x554
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 56

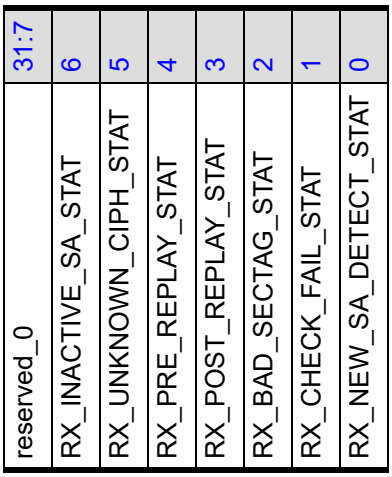


Table 6-348 Fields for Register: IRQ_TYPE_STAT_RX_SA_56

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-348 Fields for Register: IRQ_TYPE_STAT_RX_SA_56 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-348 Fields for Register: IRQ_TYPE_STAT_RX_SA_56 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.342 IRQ_TYPE_EN_RX_SA_57

- **Name:** IRQ_TYPE_EN_RX_SA_57
- **Description:** Interrupt Enable for each IRQ type of Rx SA 57
- **Size:** 32 bits
- **Offset:** 0x558
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 57

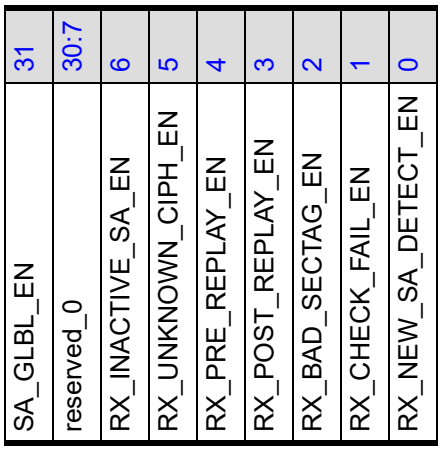


Table 6-349 Fields for Register: IRQ_TYPE_EN_RX_SA_57

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-349 Fields for Register: IRQ_TYPE_EN_RX_SA_57 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-349 Fields for Register: IRQ_TYPE_EN_RX_SA_57 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.343 IRQ_TYPE_STAT_RX_SA_57

- **Name:** IRQ_TYPE_STAT_RX_SA_57
- **Description:** Interrupt Status for each IRQ type of Rx SA 57
- **Size:** 32 bits
- **Offset:** 0x55c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 57

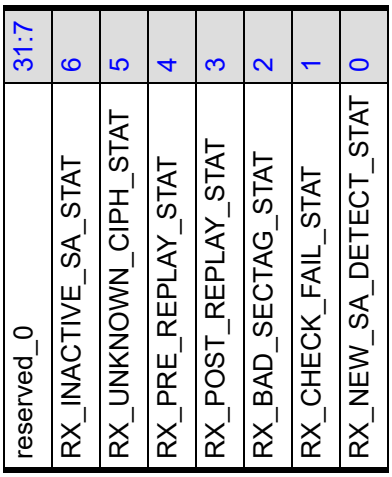


Table 6-350 Fields for Register: IRQ_TYPE_STAT_RX_SA_57

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-350 Fields for Register: IRQ_TYPE_STAT_RX_SA_57 (Continued)

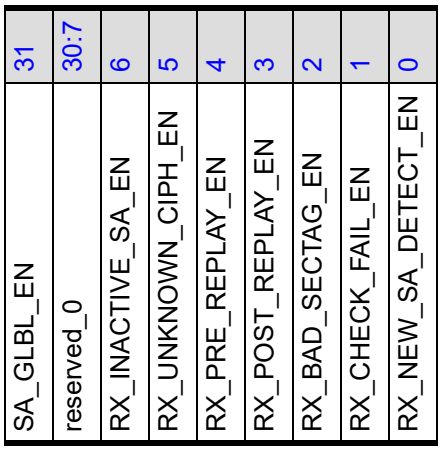
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-350 Fields for Register: IRQ_TYPE_STAT_RX_SA_57 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.344 IRQ_TYPE_EN_RX_SA_58

- **Name:** IRQ_TYPE_EN_RX_SA_58
- **Description:** Interrupt Enable for each IRQ type of Rx SA 58
- **Size:** 32 bits
- **Offset:** 0x560
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 58



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-351 Fields for Register: IRQ_TYPE_EN_RX_SA_58

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-351 Fields for Register: IRQ_TYPE_EN_RX_SA_58 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-351 Fields for Register: IRQ_TYPE_EN_RX_SA_58 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.345 IRQ_TYPE_STAT_RX_SA_58

- **Name:** IRQ_TYPE_STAT_RX_SA_58
- **Description:** Interrupt Status for each IRQ type of Rx SA 58
- **Size:** 32 bits
- **Offset:** 0x564
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 58

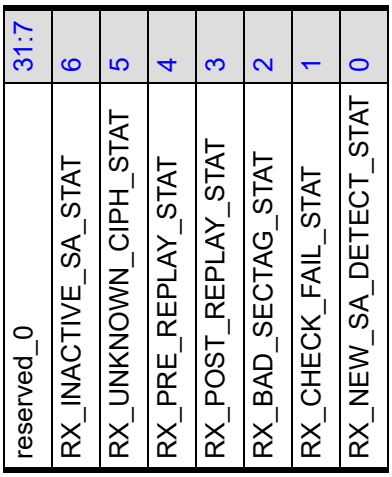


Table 6-352 Fields for Register: IRQ_TYPE_STAT_RX_SA_58

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-352 Fields for Register: IRQ_TYPE_STAT_RX_SA_58 (Continued)

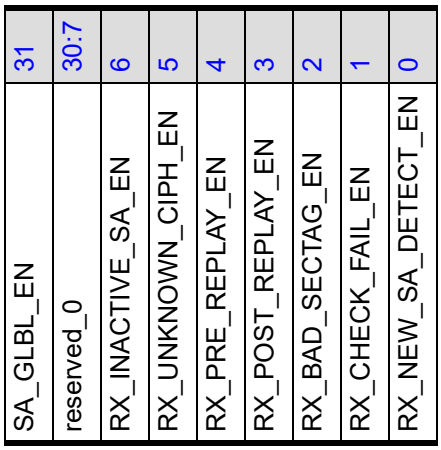
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-352 Fields for Register: IRQ_TYPE_STAT_RX_SA_58 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.346 IRQ_TYPE_EN_RX_SA_59

- **Name:** IRQ_TYPE_EN_RX_SA_59
- **Description:** Interrupt Enable for each IRQ type of Rx SA 59
- **Size:** 32 bits
- **Offset:** 0x568
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 59



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-353 Fields for Register: IRQ_TYPE_EN_RX_SA_59

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-353 Fields for Register: IRQ_TYPE_EN_RX_SA_59 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-353 Fields for Register: IRQ_TYPE_EN_RX_SA_59 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.347 IRQ_TYPE_STAT_RX_SA_59

- **Name:** IRQ_TYPE_STAT_RX_SA_59
- **Description:** Interrupt Status for each IRQ type of Rx SA 59
- **Size:** 32 bits
- **Offset:** 0x56c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 59

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-354 Fields for Register: IRQ_TYPE_STAT_RX_SA_59

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-354 Fields for Register: IRQ_TYPE_STAT_RX_SA_59 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-354 Fields for Register: IRQ_TYPE_STAT_RX_SA_59 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.348 IRQ_TYPE_EN_RX_SA_60

- **Name:** IRQ_TYPE_EN_RX_SA_60
- **Description:** Interrupt Enable for each IRQ type of Rx SA 60
- **Size:** 32 bits
- **Offset:** 0x570
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 60

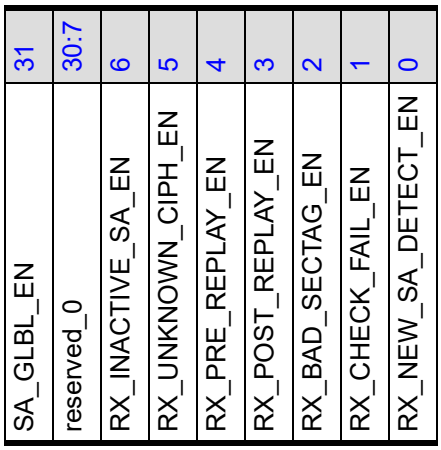


Table 6-355 Fields for Register: IRQ_TYPE_EN_RX_SA_60

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-355 Fields for Register: IRQ_TYPE_EN_RX_SA_60 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-355 Fields for Register: IRQ_TYPE_EN_RX_SA_60 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.349 IRQ_TYPE_STAT_RX_SA_60

- **Name:** IRQ_TYPE_STAT_RX_SA_60
- **Description:** Interrupt Status for each IRQ type of Rx SA 60
- **Size:** 32 bits
- **Offset:** 0x574
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 60

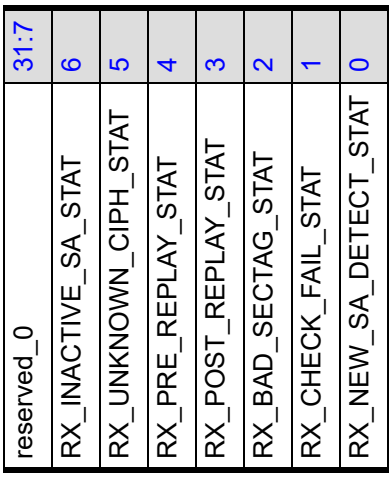


Table 6-356 Fields for Register: IRQ_TYPE_STAT_RX_SA_60

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-356 Fields for Register: IRQ_TYPE_STAT_RX_SA_60 (Continued)

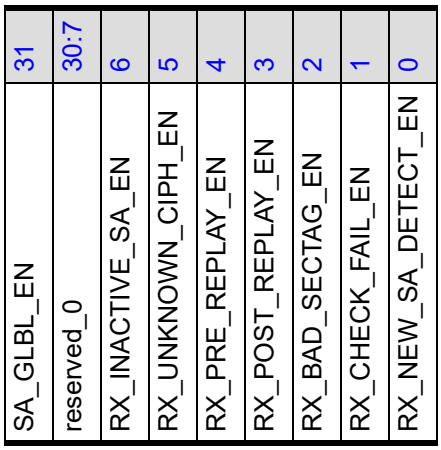
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-356 Fields for Register: IRQ_TYPE_STAT_RX_SA_60 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.350 IRQ_TYPE_EN_RX_SA_61

- **Name:** IRQ_TYPE_EN_RX_SA_61
- **Description:** Interrupt Enable for each IRQ type of Rx SA 61
- **Size:** 32 bits
- **Offset:** 0x578
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 61



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-357 Fields for Register: IRQ_TYPE_EN_RX_SA_61

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-357 Fields for Register: IRQ_TYPE_EN_RX_SA_61 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-357 Fields for Register: IRQ_TYPE_EN_RX_SA_61 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.351 IRQ_TYPE_STAT_RX_SA_61

- **Name:** IRQ_TYPE_STAT_RX_SA_61
- **Description:** Interrupt Status for each IRQ type of Rx SA 61
- **Size:** 32 bits
- **Offset:** 0x57c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 61

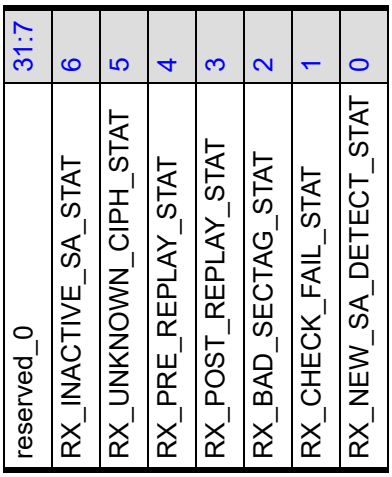


Table 6-358 Fields for Register: IRQ_TYPE_STAT_RX_SA_61

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-358 Fields for Register: IRQ_TYPE_STAT_RX_SA_61 (Continued)

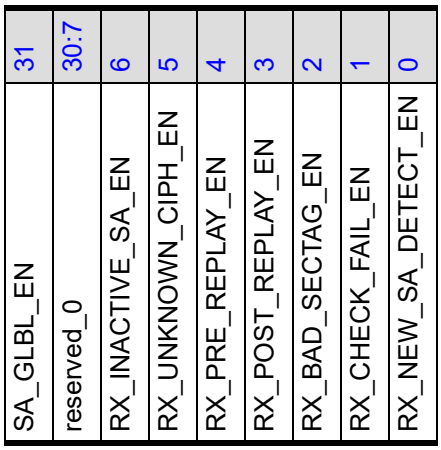
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-358 Fields for Register: IRQ_TYPE_STAT_RX_SA_61 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.352 IRQ_TYPE_EN_RX_SA_62

- **Name:** IRQ_TYPE_EN_RX_SA_62
- **Description:** Interrupt Enable for each IRQ type of Rx SA 62
- **Size:** 32 bits
- **Offset:** 0x580
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 62



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-359 Fields for Register: IRQ_TYPE_EN_RX_SA_62

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-359 Fields for Register: IRQ_TYPE_EN_RX_SA_62 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-359 Fields for Register: IRQ_TYPE_EN_RX_SA_62 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.353 IRQ_TYPE_STAT_RX_SA_62

- **Name:** IRQ_TYPE_STAT_RX_SA_62
- **Description:** Interrupt Status for each IRQ type of Rx SA 62
- **Size:** 32 bits
- **Offset:** 0x584
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 62

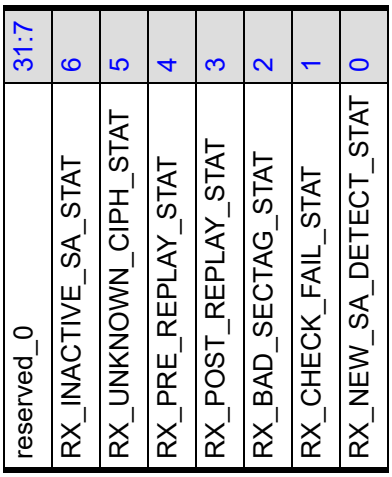


Table 6-360 Fields for Register: IRQ_TYPE_STAT_RX_SA_62

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-360 Fields for Register: IRQ_TYPE_STAT_RX_SA_62 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-360 Fields for Register: IRQ_TYPE_STAT_RX_SA_62 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.354 IRQ_TYPE_EN_RX_SA_63

- **Name:** IRQ_TYPE_EN_RX_SA_63
- **Description:** Interrupt Enable for each IRQ type of Rx SA 63
- **Size:** 32 bits
- **Offset:** 0x588
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 63

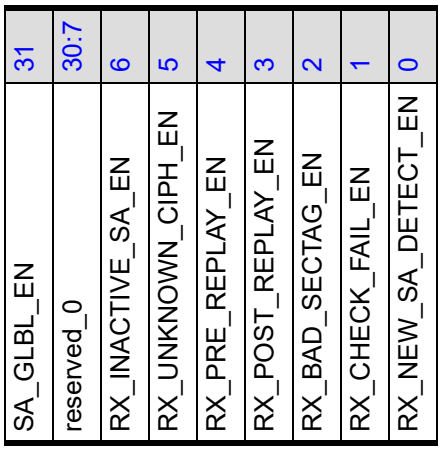


Table 6-361 Fields for Register: IRQ_TYPE_EN_RX_SA_63

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-361 Fields for Register: IRQ_TYPE_EN_RX_SA_63 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-361 Fields for Register: IRQ_TYPE_EN_RX_SA_63 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.355 IRQ_TYPE_STAT_RX_SA_63

- **Name:** IRQ_TYPE_STAT_RX_SA_63
- **Description:** Interrupt Status for each IRQ type of Rx SA 63
- **Size:** 32 bits
- **Offset:** 0x58c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 63

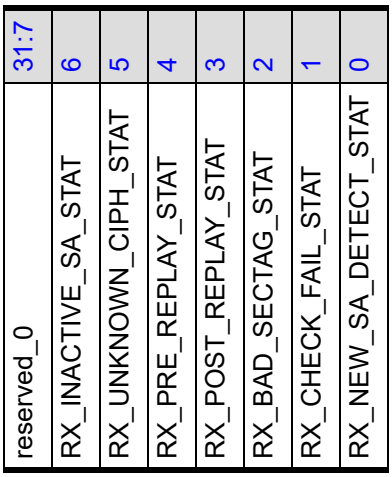


Table 6-362 Fields for Register: IRQ_TYPE_STAT_RX_SA_63

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-362 Fields for Register: IRQ_TYPE_STAT_RX_SA_63 (Continued)

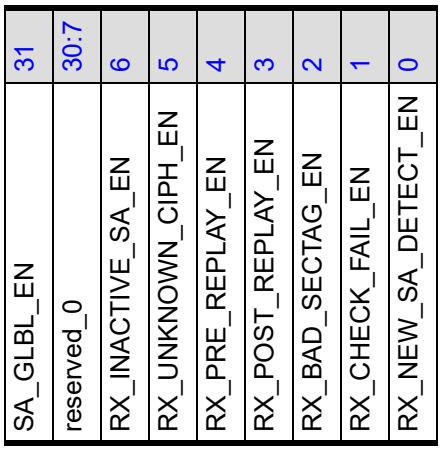
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-362 Fields for Register: IRQ_TYPE_STAT_RX_SA_63 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.356 IRQ_TYPE_EN_RX_SA_64

- **Name:** IRQ_TYPE_EN_RX_SA_64
- **Description:** Interrupt Enable for each IRQ type of Rx SA 64
- **Size:** 32 bits
- **Offset:** 0x590
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 64



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-363 Fields for Register: IRQ_TYPE_EN_RX_SA_64

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-363 Fields for Register: IRQ_TYPE_EN_RX_SA_64 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-363 Fields for Register: IRQ_TYPE_EN_RX_SA_64 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.357 IRQ_TYPE_STAT_RX_SA_64

- **Name:** IRQ_TYPE_STAT_RX_SA_64
- **Description:** Interrupt Status for each IRQ type of Rx SA 64
- **Size:** 32 bits
- **Offset:** 0x594
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 64

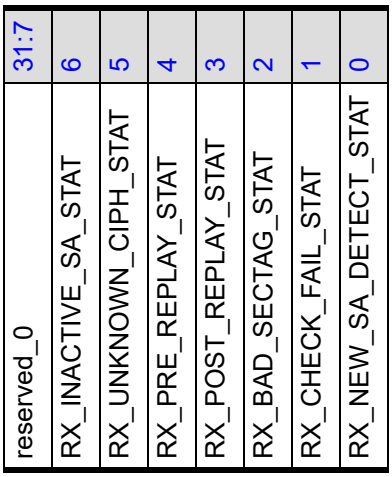


Table 6-364 Fields for Register: IRQ_TYPE_STAT_RX_SA_64

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-364 Fields for Register: IRQ_TYPE_STAT_RX_SA_64 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-364 Fields for Register: IRQ_TYPE_STAT_RX_SA_64 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.358 IRQ_TYPE_EN_RX_SA_65

- **Name:** IRQ_TYPE_EN_RX_SA_65
- **Description:** Interrupt Enable for each IRQ type of Rx SA 65
- **Size:** 32 bits
- **Offset:** 0x598
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 65

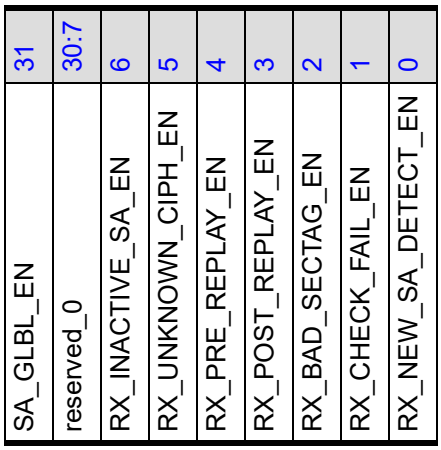


Table 6-365 Fields for Register: IRQ_TYPE_EN_RX_SA_65

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-365 Fields for Register: IRQ_TYPE_EN_RX_SA_65 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-365 Fields for Register: IRQ_TYPE_EN_RX_SA_65 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.359 IRQ_TYPE_STAT_RX_SA_65

- **Name:** IRQ_TYPE_STAT_RX_SA_65
- **Description:** Interrupt Status for each IRQ type of Rx SA 65
- **Size:** 32 bits
- **Offset:** 0x59c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 65

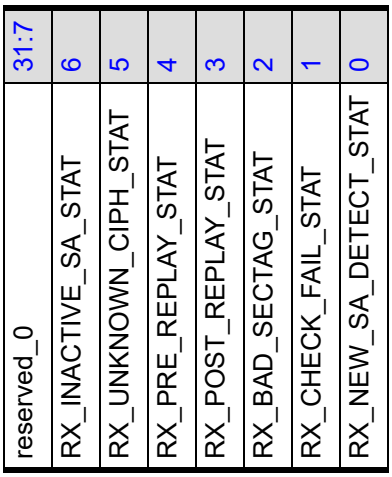


Table 6-366 Fields for Register: IRQ_TYPE_STAT_RX_SA_65

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-366 Fields for Register: IRQ_TYPE_STAT_RX_SA_65 (Continued)

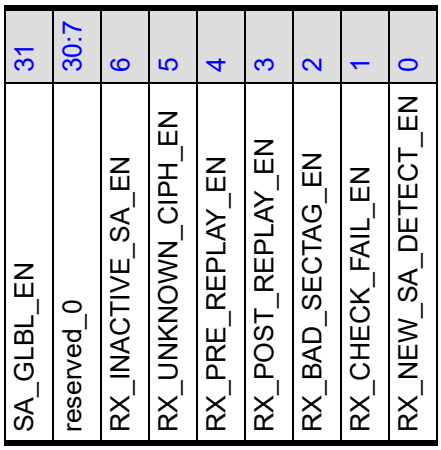
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-366 Fields for Register: IRQ_TYPE_STAT_RX_SA_65 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.360 IRQ_TYPE_EN_RX_SA_66

- **Name:** IRQ_TYPE_EN_RX_SA_66
- **Description:** Interrupt Enable for each IRQ type of Rx SA 66
- **Size:** 32 bits
- **Offset:** 0x5a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 66



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-367 Fields for Register: IRQ_TYPE_EN_RX_SA_66

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-367 Fields for Register: IRQ_TYPE_EN_RX_SA_66 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-367 Fields for Register: IRQ_TYPE_EN_RX_SA_66 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.361 IRQ_TYPE_STAT_RX_SA_66

- **Name:** IRQ_TYPE_STAT_RX_SA_66
- **Description:** Interrupt Status for each IRQ type of Rx SA 66
- **Size:** 32 bits
- **Offset:** 0x5a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 66

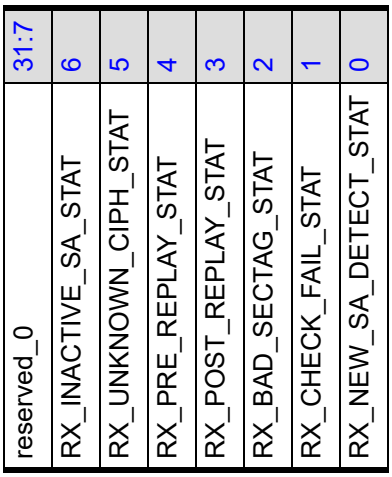


Table 6-368 Fields for Register: IRQ_TYPE_STAT_RX_SA_66

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-368 Fields for Register: IRQ_TYPE_STAT_RX_SA_66 (Continued)

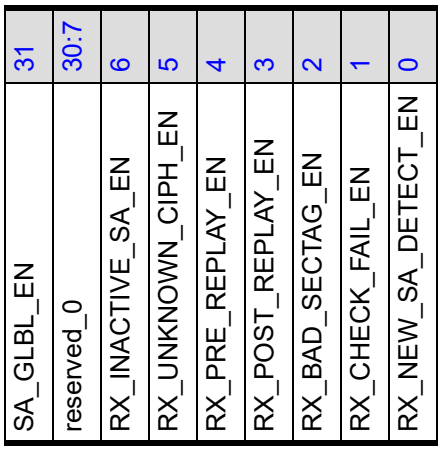
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-368 Fields for Register: IRQ_TYPE_STAT_RX_SA_66 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.362 IRQ_TYPE_EN_RX_SA_67

- **Name:** IRQ_TYPE_EN_RX_SA_67
- **Description:** Interrupt Enable for each IRQ type of Rx SA 67
- **Size:** 32 bits
- **Offset:** 0x5a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 67



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-369 Fields for Register: IRQ_TYPE_EN_RX_SA_67

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-369 Fields for Register: IRQ_TYPE_EN_RX_SA_67 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-369 Fields for Register: IRQ_TYPE_EN_RX_SA_67 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.363 IRQ_TYPE_STAT_RX_SA_67

- **Name:** IRQ_TYPE_STAT_RX_SA_67
- **Description:** Interrupt Status for each IRQ type of Rx SA 67
- **Size:** 32 bits
- **Offset:** 0x5ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 67

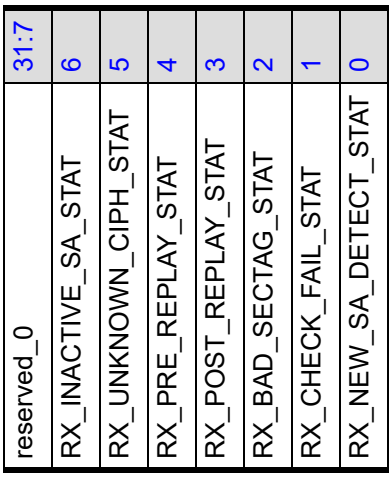


Table 6-370 Fields for Register: IRQ_TYPE_STAT_RX_SA_67

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-370 Fields for Register: IRQ_TYPE_STAT_RX_SA_67 (Continued)

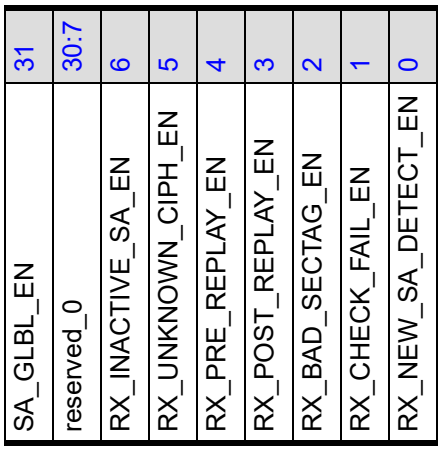
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-370 Fields for Register: IRQ_TYPE_STAT_RX_SA_67 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.364 IRQ_TYPE_EN_RX_SA_68

- **Name:** IRQ_TYPE_EN_RX_SA_68
- **Description:** Interrupt Enable for each IRQ type of Rx SA 68
- **Size:** 32 bits
- **Offset:** 0x5b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 68



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-371 Fields for Register: IRQ_TYPE_EN_RX_SA_68

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-371 Fields for Register: IRQ_TYPE_EN_RX_SA_68 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-371 Fields for Register: IRQ_TYPE_EN_RX_SA_68 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.365 IRQ_TYPE_STAT_RX_SA_68

- **Name:** IRQ_TYPE_STAT_RX_SA_68
- **Description:** Interrupt Status for each IRQ type of Rx SA 68
- **Size:** 32 bits
- **Offset:** 0x5b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 68

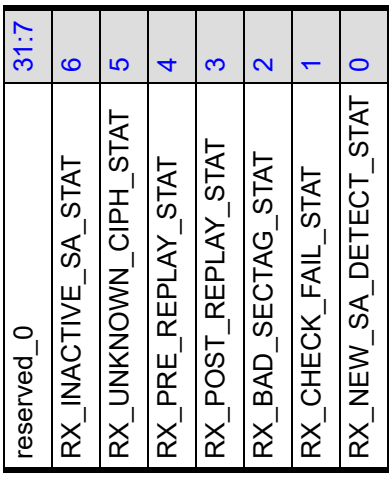


Table 6-372 Fields for Register: IRQ_TYPE_STAT_RX_SA_68

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-372 Fields for Register: IRQ_TYPE_STAT_RX_SA_68 (Continued)

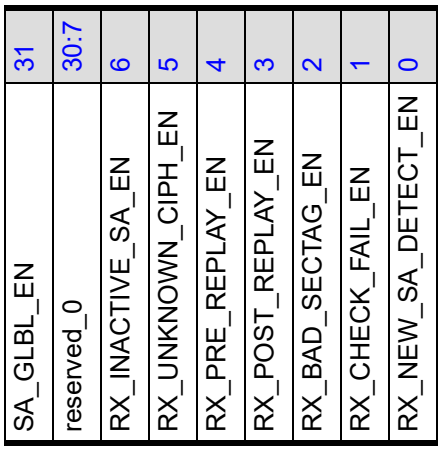
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-372 Fields for Register: IRQ_TYPE_STAT_RX_SA_68 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.366 IRQ_TYPE_EN_RX_SA_69

- **Name:** IRQ_TYPE_EN_RX_SA_69
- **Description:** Interrupt Enable for each IRQ type of Rx SA 69
- **Size:** 32 bits
- **Offset:** 0x5b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 69



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-373 Fields for Register: IRQ_TYPE_EN_RX_SA_69

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-373 Fields for Register: IRQ_TYPE_EN_RX_SA_69 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-373 Fields for Register: IRQ_TYPE_EN_RX_SA_69 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.367 IRQ_TYPE_STAT_RX_SA_69

- **Name:** IRQ_TYPE_STAT_RX_SA_69
- **Description:** Interrupt Status for each IRQ type of Rx SA 69
- **Size:** 32 bits
- **Offset:** 0x5bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 69

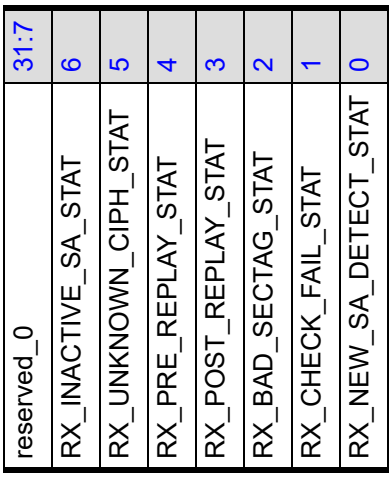


Table 6-374 Fields for Register: IRQ_TYPE_STAT_RX_SA_69

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-374 Fields for Register: IRQ_TYPE_STAT_RX_SA_69 (Continued)

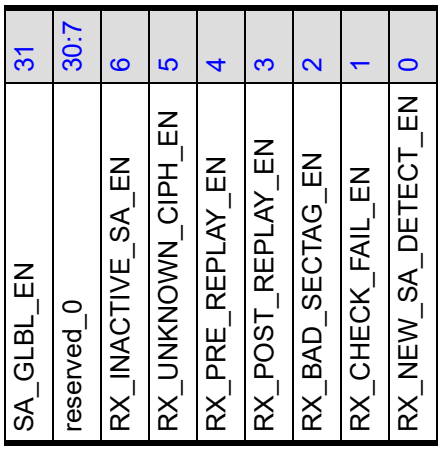
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-374 Fields for Register: IRQ_TYPE_STAT_RX_SA_69 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.368 IRQ_TYPE_EN_RX_SA_70

- **Name:** IRQ_TYPE_EN_RX_SA_70
- **Description:** Interrupt Enable for each IRQ type of Rx SA 70
- **Size:** 32 bits
- **Offset:** 0x5c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 70



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-375 Fields for Register: IRQ_TYPE_EN_RX_SA_70

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-375 Fields for Register: IRQ_TYPE_EN_RX_SA_70 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-375 Fields for Register: IRQ_TYPE_EN_RX_SA_70 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.369 IRQ_TYPE_STAT_RX_SA_70

- **Name:** IRQ_TYPE_STAT_RX_SA_70
- **Description:** Interrupt Status for each IRQ type of Rx SA 70
- **Size:** 32 bits
- **Offset:** 0x5c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 70

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-376 Fields for Register: IRQ_TYPE_STAT_RX_SA_70

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-376 Fields for Register: IRQ_TYPE_STAT_RX_SA_70 (Continued)

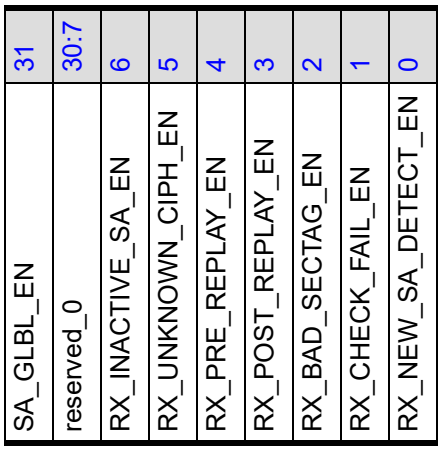
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-376 Fields for Register: IRQ_TYPE_STAT_RX_SA_70 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.370 IRQ_TYPE_EN_RX_SA_71

- **Name:** IRQ_TYPE_EN_RX_SA_71
- **Description:** Interrupt Enable for each IRQ type of Rx SA 71
- **Size:** 32 bits
- **Offset:** 0x5c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 71



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-377 Fields for Register: IRQ_TYPE_EN_RX_SA_71

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-377 Fields for Register: IRQ_TYPE_EN_RX_SA_71 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-377 Fields for Register: IRQ_TYPE_EN_RX_SA_71 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.371 IRQ_TYPE_STAT_RX_SA_71

- **Name:** IRQ_TYPE_STAT_RX_SA_71
- **Description:** Interrupt Status for each IRQ type of Rx SA 71
- **Size:** 32 bits
- **Offset:** 0x5cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 71

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-378 Fields for Register: IRQ_TYPE_STAT_RX_SA_71

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-378 Fields for Register: IRQ_TYPE_STAT_RX_SA_71 (Continued)

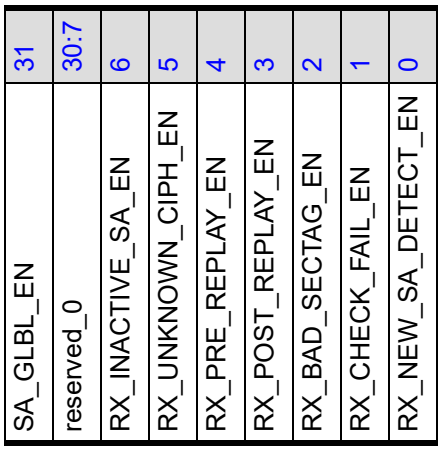
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-378 Fields for Register: IRQ_TYPE_STAT_RX_SA_71 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.372 IRQ_TYPE_EN_RX_SA_72

- **Name:** IRQ_TYPE_EN_RX_SA_72
- **Description:** Interrupt Enable for each IRQ type of Rx SA 72
- **Size:** 32 bits
- **Offset:** 0x5d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 72



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-379 Fields for Register: IRQ_TYPE_EN_RX_SA_72

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-379 Fields for Register: IRQ_TYPE_EN_RX_SA_72 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-379 Fields for Register: IRQ_TYPE_EN_RX_SA_72 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.373 IRQ_TYPE_STAT_RX_SA_72

- **Name:** IRQ_TYPE_STAT_RX_SA_72
- **Description:** Interrupt Status for each IRQ type of Rx SA 72
- **Size:** 32 bits
- **Offset:** 0x5d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 72

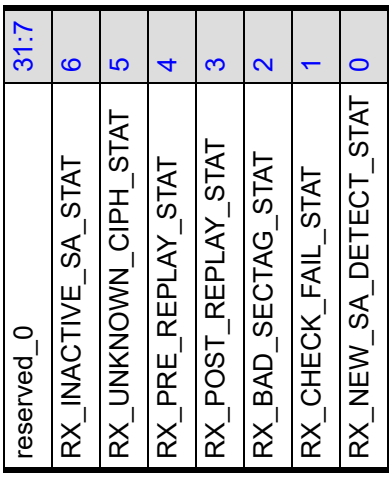


Table 6-380 Fields for Register: IRQ_TYPE_STAT_RX_SA_72

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-380 Fields for Register: IRQ_TYPE_STAT_RX_SA_72 (Continued)

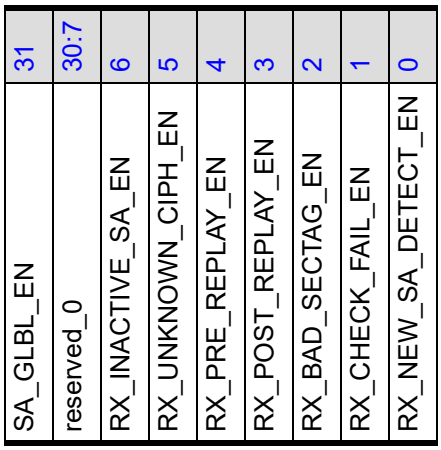
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-380 Fields for Register: IRQ_TYPE_STAT_RX_SA_72 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.374 IRQ_TYPE_EN_RX_SA_73

- **Name:** IRQ_TYPE_EN_RX_SA_73
- **Description:** Interrupt Enable for each IRQ type of Rx SA 73
- **Size:** 32 bits
- **Offset:** 0x5d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 73



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-381 Fields for Register: IRQ_TYPE_EN_RX_SA_73

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-381 Fields for Register: IRQ_TYPE_EN_RX_SA_73 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-381 Fields for Register: IRQ_TYPE_EN_RX_SA_73 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.375 IRQ_TYPE_STAT_RX_SA_73

- **Name:** IRQ_TYPE_STAT_RX_SA_73
- **Description:** Interrupt Status for each IRQ type of Rx SA 73
- **Size:** 32 bits
- **Offset:** 0x5dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 73

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-382 Fields for Register: IRQ_TYPE_STAT_RX_SA_73

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-382 Fields for Register: IRQ_TYPE_STAT_RX_SA_73 (Continued)

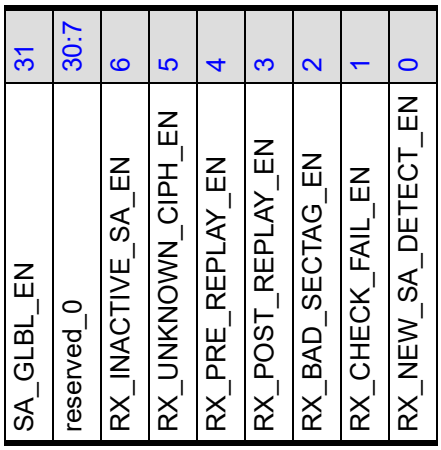
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-382 Fields for Register: IRQ_TYPE_STAT_RX_SA_73 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.376 IRQ_TYPE_EN_RX_SA_74

- **Name:** IRQ_TYPE_EN_RX_SA_74
- **Description:** Interrupt Enable for each IRQ type of Rx SA 74
- **Size:** 32 bits
- **Offset:** 0x5e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 74



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-383 Fields for Register: IRQ_TYPE_EN_RX_SA_74

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-383 Fields for Register: IRQ_TYPE_EN_RX_SA_74 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-383 Fields for Register: IRQ_TYPE_EN_RX_SA_74 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.377 IRQ_TYPE_STAT_RX_SA_74

- **Name:** IRQ_TYPE_STAT_RX_SA_74
- **Description:** Interrupt Status for each IRQ type of Rx SA 74
- **Size:** 32 bits
- **Offset:** 0x5e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 74

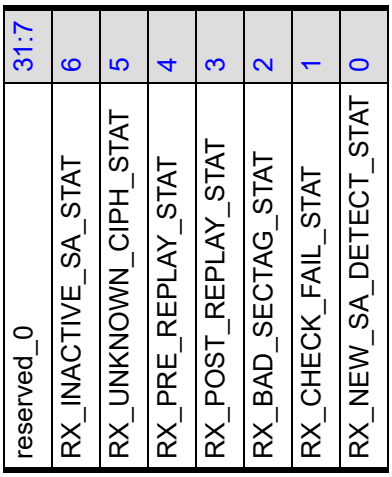


Table 6-384 Fields for Register: IRQ_TYPE_STAT_RX_SA_74

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-384 Fields for Register: IRQ_TYPE_STAT_RX_SA_74 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-384 Fields for Register: IRQ_TYPE_STAT_RX_SA_74 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.378 IRQ_TYPE_EN_RX_SA_75

- **Name:** IRQ_TYPE_EN_RX_SA_75
- **Description:** Interrupt Enable for each IRQ type of Rx SA 75
- **Size:** 32 bits
- **Offset:** 0x5e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 75

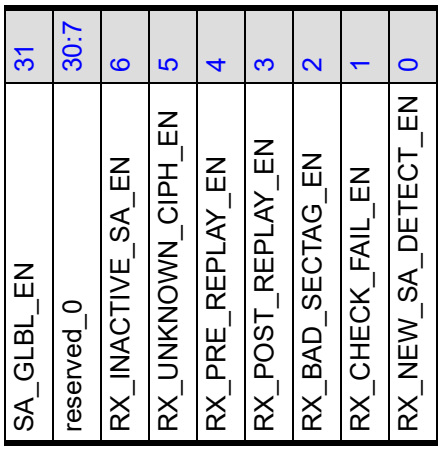


Table 6-385 Fields for Register: IRQ_TYPE_EN_RX_SA_75

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-385 Fields for Register: IRQ_TYPE_EN_RX_SA_75 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-385 Fields for Register: IRQ_TYPE_EN_RX_SA_75 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.379 IRQ_TYPE_STAT_RX_SA_75

- **Name:** IRQ_TYPE_STAT_RX_SA_75
- **Description:** Interrupt Status for each IRQ type of Rx SA 75
- **Size:** 32 bits
- **Offset:** 0x5ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 75

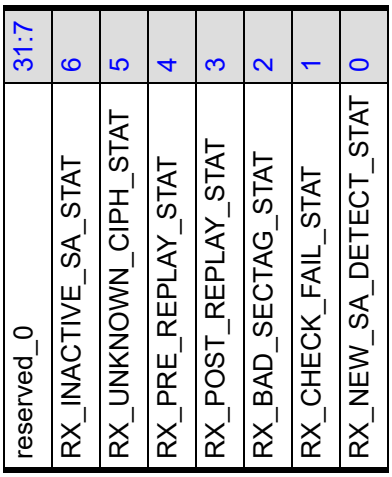


Table 6-386 Fields for Register: IRQ_TYPE_STAT_RX_SA_75

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-386 Fields for Register: IRQ_TYPE_STAT_RX_SA_75 (Continued)

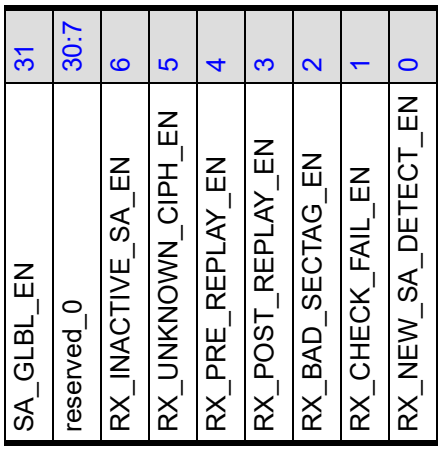
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-386 Fields for Register: IRQ_TYPE_STAT_RX_SA_75 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.380 IRQ_TYPE_EN_RX_SA_76

- **Name:** IRQ_TYPE_EN_RX_SA_76
- **Description:** Interrupt Enable for each IRQ type of Rx SA 76
- **Size:** 32 bits
- **Offset:** 0x5f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 76



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-387 Fields for Register: IRQ_TYPE_EN_RX_SA_76

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-387 Fields for Register: IRQ_TYPE_EN_RX_SA_76 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-387 Fields for Register: IRQ_TYPE_EN_RX_SA_76 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.381 IRQ_TYPE_STAT_RX_SA_76

- **Name:** IRQ_TYPE_STAT_RX_SA_76
- **Description:** Interrupt Status for each IRQ type of Rx SA 76
- **Size:** 32 bits
- **Offset:** 0x5f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 76

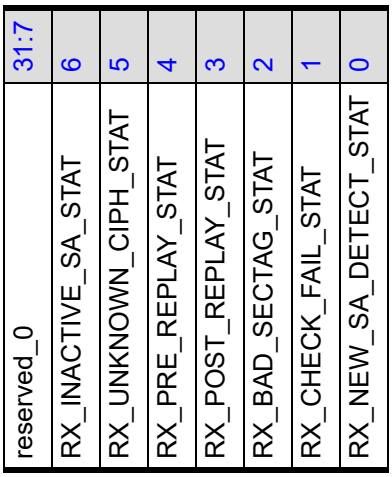


Table 6-388 Fields for Register: IRQ_TYPE_STAT_RX_SA_76

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-388 Fields for Register: IRQ_TYPE_STAT_RX_SA_76 (Continued)

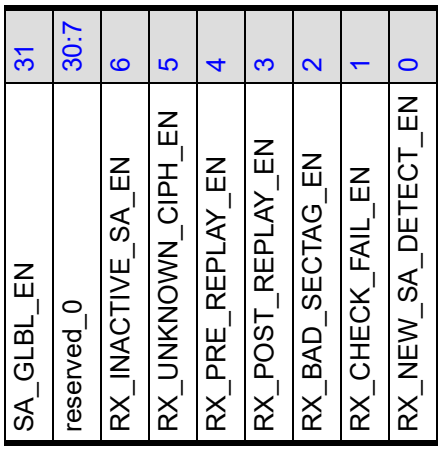
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-388 Fields for Register: IRQ_TYPE_STAT_RX_SA_76 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.382 IRQ_TYPE_EN_RX_SA_77

- **Name:** IRQ_TYPE_EN_RX_SA_77
- **Description:** Interrupt Enable for each IRQ type of Rx SA 77
- **Size:** 32 bits
- **Offset:** 0x5f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 77



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-389 Fields for Register: IRQ_TYPE_EN_RX_SA_77

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-389 Fields for Register: IRQ_TYPE_EN_RX_SA_77 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-389 Fields for Register: IRQ_TYPE_EN_RX_SA_77 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.383 IRQ_TYPE_STAT_RX_SA_77

- **Name:** IRQ_TYPE_STAT_RX_SA_77
- **Description:** Interrupt Status for each IRQ type of Rx SA 77
- **Size:** 32 bits
- **Offset:** 0x5fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 77

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-390 Fields for Register: IRQ_TYPE_STAT_RX_SA_77

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-390 Fields for Register: IRQ_TYPE_STAT_RX_SA_77 (Continued)

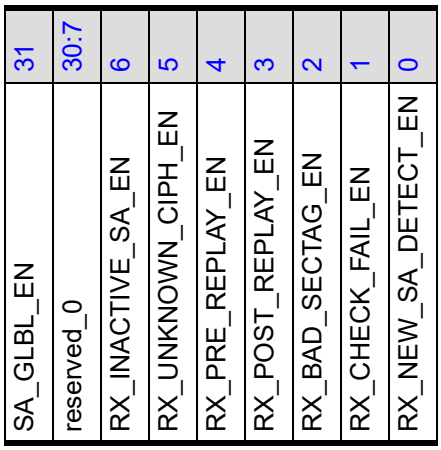
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-390 Fields for Register: IRQ_TYPE_STAT_RX_SA_77 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.384 IRQ_TYPE_EN_RX_SA_78

- **Name:** IRQ_TYPE_EN_RX_SA_78
- **Description:** Interrupt Enable for each IRQ type of Rx SA 78
- **Size:** 32 bits
- **Offset:** 0x600
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 78



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-391 Fields for Register: IRQ_TYPE_EN_RX_SA_78

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-391 Fields for Register: IRQ_TYPE_EN_RX_SA_78 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-391 Fields for Register: IRQ_TYPE_EN_RX_SA_78 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.385 IRQ_TYPE_STAT_RX_SA_78

- **Name:** IRQ_TYPE_STAT_RX_SA_78
- **Description:** Interrupt Status for each IRQ type of Rx SA 78
- **Size:** 32 bits
- **Offset:** 0x604
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 78

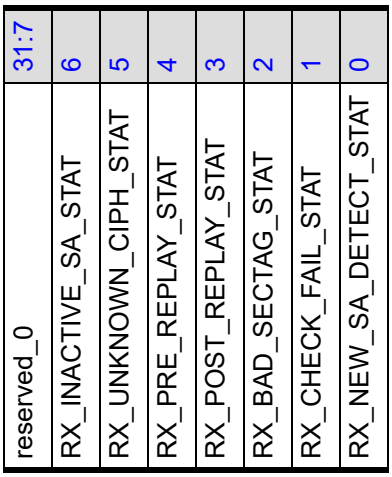


Table 6-392 Fields for Register: IRQ_TYPE_STAT_RX_SA_78

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-392 Fields for Register: IRQ_TYPE_STAT_RX_SA_78 (Continued)

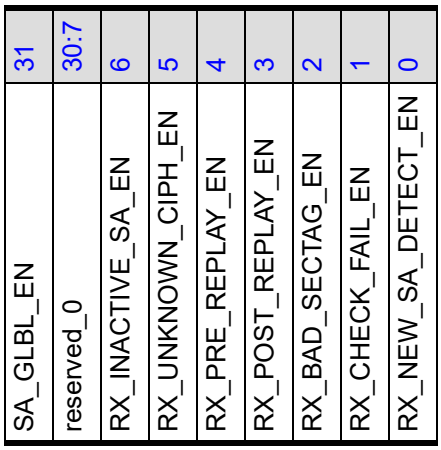
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-392 Fields for Register: IRQ_TYPE_STAT_RX_SA_78 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.386 IRQ_TYPE_EN_RX_SA_79

- **Name:** IRQ_TYPE_EN_RX_SA_79
- **Description:** Interrupt Enable for each IRQ type of Rx SA 79
- **Size:** 32 bits
- **Offset:** 0x608
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 79



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-393 Fields for Register: IRQ_TYPE_EN_RX_SA_79

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-393 Fields for Register: IRQ_TYPE_EN_RX_SA_79 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-393 Fields for Register: IRQ_TYPE_EN_RX_SA_79 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.387 IRQ_TYPE_STAT_RX_SA_79

- **Name:** IRQ_TYPE_STAT_RX_SA_79
- **Description:** Interrupt Status for each IRQ type of Rx SA 79
- **Size:** 32 bits
- **Offset:** 0x60c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 79

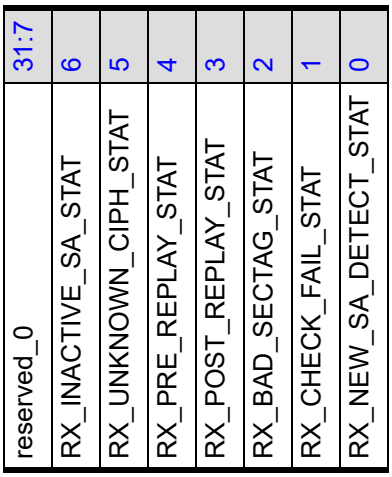


Table 6-394 Fields for Register: IRQ_TYPE_STAT_RX_SA_79

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-394 Fields for Register: IRQ_TYPE_STAT_RX_SA_79 (Continued)

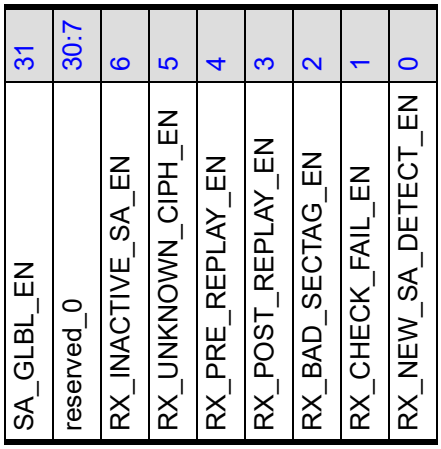
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-394 Fields for Register: IRQ_TYPE_STAT_RX_SA_79 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.388 IRQ_TYPE_EN_RX_SA_80

- **Name:** IRQ_TYPE_EN_RX_SA_80
- **Description:** Interrupt Enable for each IRQ type of Rx SA 80
- **Size:** 32 bits
- **Offset:** 0x610
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 80



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-395 Fields for Register: IRQ_TYPE_EN_RX_SA_80

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-395 Fields for Register: IRQ_TYPE_EN_RX_SA_80 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-395 Fields for Register: IRQ_TYPE_EN_RX_SA_80 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.389 IRQ_TYPE_STAT_RX_SA_80

- **Name:** IRQ_TYPE_STAT_RX_SA_80
- **Description:** Interrupt Status for each IRQ type of Rx SA 80
- **Size:** 32 bits
- **Offset:** 0x614
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 80

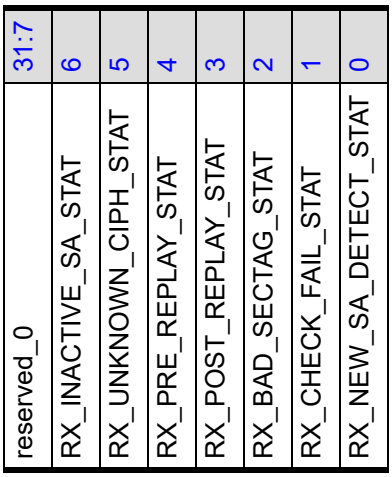


Table 6-396 Fields for Register: IRQ_TYPE_STAT_RX_SA_80

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-396 Fields for Register: IRQ_TYPE_STAT_RX_SA_80 (Continued)

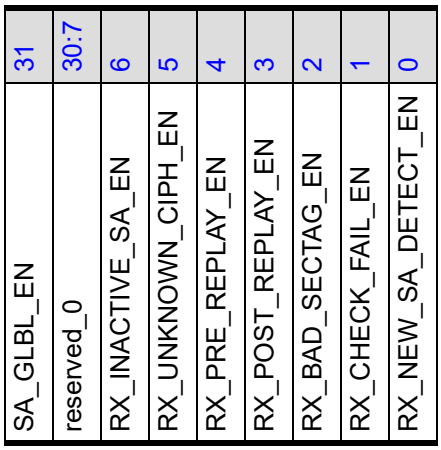
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-396 Fields for Register: IRQ_TYPE_STAT_RX_SA_80 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.390 IRQ_TYPE_EN_RX_SA_81

- **Name:** IRQ_TYPE_EN_RX_SA_81
- **Description:** Interrupt Enable for each IRQ type of Rx SA 81
- **Size:** 32 bits
- **Offset:** 0x618
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 81



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-397 Fields for Register: IRQ_TYPE_EN_RX_SA_81

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-397 Fields for Register: IRQ_TYPE_EN_RX_SA_81 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-397 Fields for Register: IRQ_TYPE_EN_RX_SA_81 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.391 IRQ_TYPE_STAT_RX_SA_81

- **Name:** IRQ_TYPE_STAT_RX_SA_81
- **Description:** Interrupt Status for each IRQ type of Rx SA 81
- **Size:** 32 bits
- **Offset:** 0x61c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 81

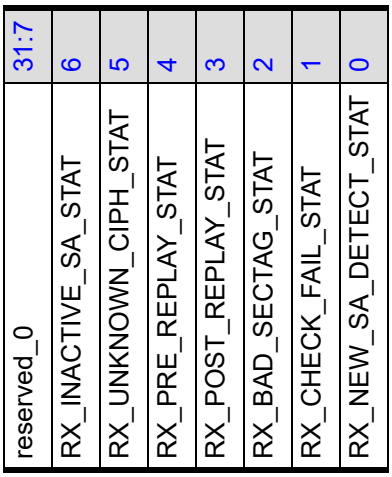


Table 6-398 Fields for Register: IRQ_TYPE_STAT_RX_SA_81

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-398 Fields for Register: IRQ_TYPE_STAT_RX_SA_81 (Continued)

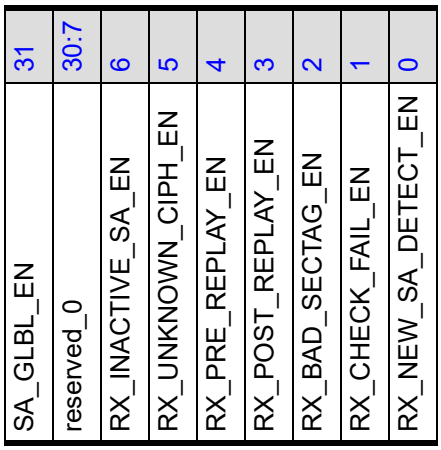
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-398 Fields for Register: IRQ_TYPE_STAT_RX_SA_81 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.392 IRQ_TYPE_EN_RX_SA_82

- **Name:** IRQ_TYPE_EN_RX_SA_82
- **Description:** Interrupt Enable for each IRQ type of Rx SA 82
- **Size:** 32 bits
- **Offset:** 0x620
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 82



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-399 Fields for Register: IRQ_TYPE_EN_RX_SA_82

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-399 Fields for Register: IRQ_TYPE_EN_RX_SA_82 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-399 Fields for Register: IRQ_TYPE_EN_RX_SA_82 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.393 IRQ_TYPE_STAT_RX_SA_82

- **Name:** IRQ_TYPE_STAT_RX_SA_82
- **Description:** Interrupt Status for each IRQ type of Rx SA 82
- **Size:** 32 bits
- **Offset:** 0x624
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 82

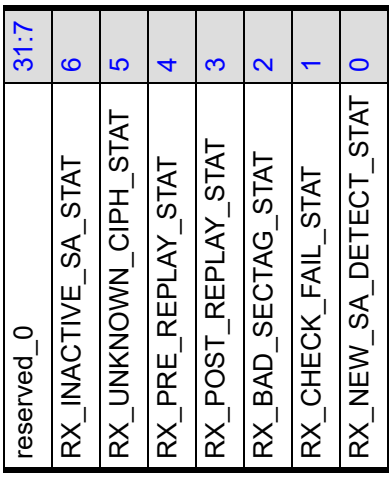


Table 6-400 Fields for Register: IRQ_TYPE_STAT_RX_SA_82

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-400 Fields for Register: IRQ_TYPE_STAT_RX_SA_82 (Continued)

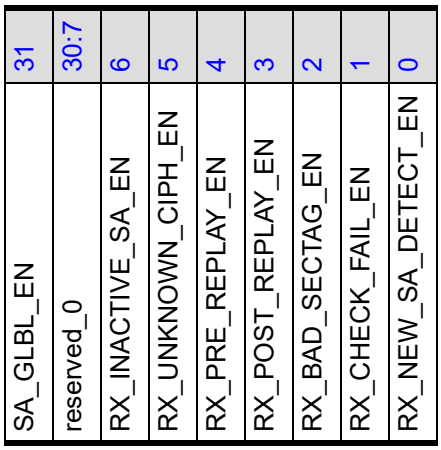
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-400 Fields for Register: IRQ_TYPE_STAT_RX_SA_82 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.394 IRQ_TYPE_EN_RX_SA_83

- **Name:** IRQ_TYPE_EN_RX_SA_83
- **Description:** Interrupt Enable for each IRQ type of Rx SA 83
- **Size:** 32 bits
- **Offset:** 0x628
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 83



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-401 Fields for Register: IRQ_TYPE_EN_RX_SA_83

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-401 Fields for Register: IRQ_TYPE_EN_RX_SA_83 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-401 Fields for Register: IRQ_TYPE_EN_RX_SA_83 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.395 IRQ_TYPE_STAT_RX_SA_83

- **Name:** IRQ_TYPE_STAT_RX_SA_83
- **Description:** Interrupt Status for each IRQ type of Rx SA 83
- **Size:** 32 bits
- **Offset:** 0x62c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 83

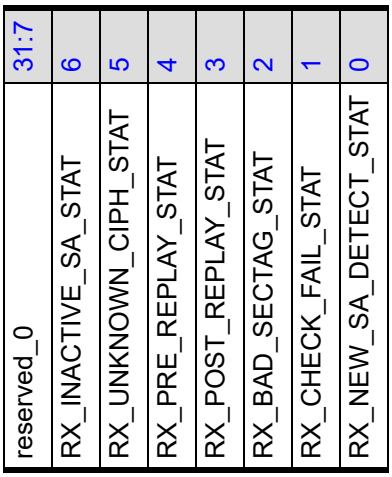


Table 6-402 Fields for Register: IRQ_TYPE_STAT_RX_SA_83

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-402 Fields for Register: IRQ_TYPE_STAT_RX_SA_83 (Continued)

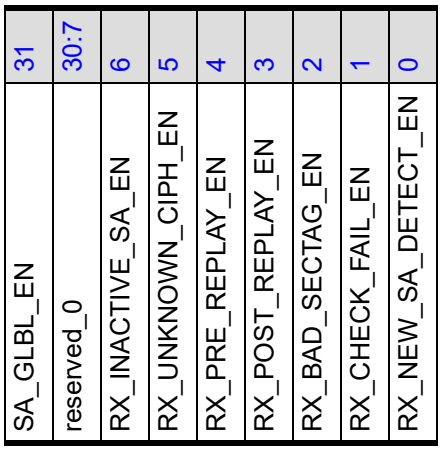
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-402 Fields for Register: IRQ_TYPE_STAT_RX_SA_83 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.396 IRQ_TYPE_EN_RX_SA_84

- **Name:** IRQ_TYPE_EN_RX_SA_84
- **Description:** Interrupt Enable for each IRQ type of Rx SA 84
- **Size:** 32 bits
- **Offset:** 0x630
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 84



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-403 Fields for Register: IRQ_TYPE_EN_RX_SA_84

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-403 Fields for Register: IRQ_TYPE_EN_RX_SA_84 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-403 Fields for Register: IRQ_TYPE_EN_RX_SA_84 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.397 IRQ_TYPE_STAT_RX_SA_84

- **Name:** IRQ_TYPE_STAT_RX_SA_84
- **Description:** Interrupt Status for each IRQ type of Rx SA 84
- **Size:** 32 bits
- **Offset:** 0x634
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 84

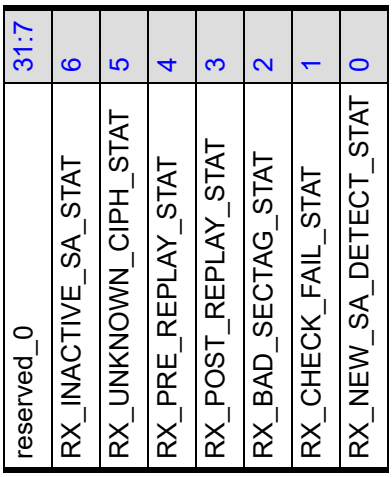


Table 6-404 Fields for Register: IRQ_TYPE_STAT_RX_SA_84

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-404 Fields for Register: IRQ_TYPE_STAT_RX_SA_84 (Continued)

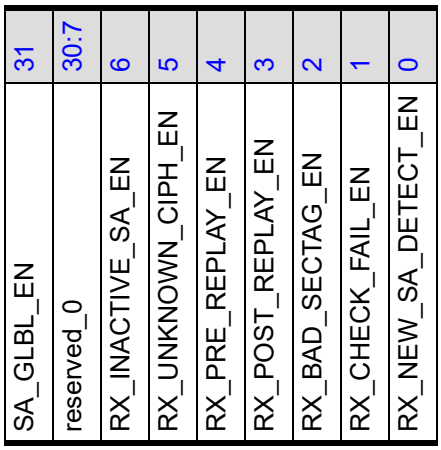
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-404 Fields for Register: IRQ_TYPE_STAT_RX_SA_84 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.398 IRQ_TYPE_EN_RX_SA_85

- **Name:** IRQ_TYPE_EN_RX_SA_85
- **Description:** Interrupt Enable for each IRQ type of Rx SA 85
- **Size:** 32 bits
- **Offset:** 0x638
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 85



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-405 Fields for Register: IRQ_TYPE_EN_RX_SA_85

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-405 Fields for Register: IRQ_TYPE_EN_RX_SA_85 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-405 Fields for Register: IRQ_TYPE_EN_RX_SA_85 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.399 IRQ_TYPE_STAT_RX_SA_85

- **Name:** IRQ_TYPE_STAT_RX_SA_85
- **Description:** Interrupt Status for each IRQ type of Rx SA 85
- **Size:** 32 bits
- **Offset:** 0x63c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 85

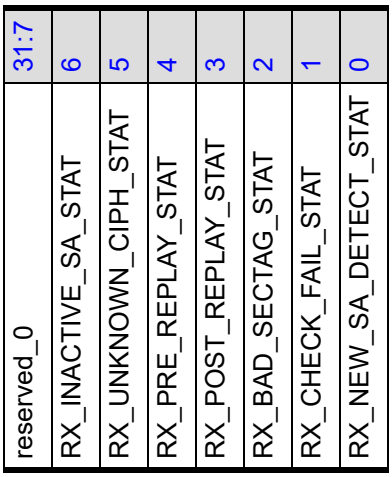


Table 6-406 Fields for Register: IRQ_TYPE_STAT_RX_SA_85

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-406 Fields for Register: IRQ_TYPE_STAT_RX_SA_85 (Continued)

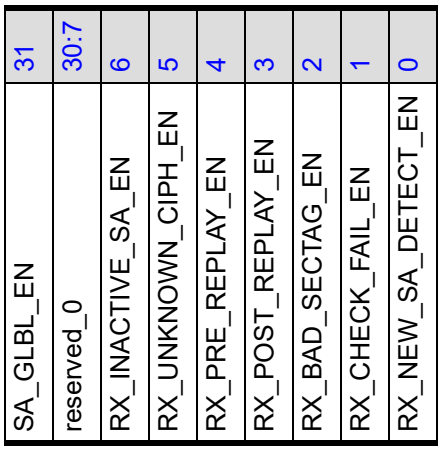
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-406 Fields for Register: IRQ_TYPE_STAT_RX_SA_85 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.400 IRQ_TYPE_EN_RX_SA_86

- **Name:** IRQ_TYPE_EN_RX_SA_86
- **Description:** Interrupt Enable for each IRQ type of Rx SA 86
- **Size:** 32 bits
- **Offset:** 0x640
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 86



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-407 Fields for Register: IRQ_TYPE_EN_RX_SA_86

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-407 Fields for Register: IRQ_TYPE_EN_RX_SA_86 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-407 Fields for Register: IRQ_TYPE_EN_RX_SA_86 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.401 IRQ_TYPE_STAT_RX_SA_86

- **Name:** IRQ_TYPE_STAT_RX_SA_86
- **Description:** Interrupt Status for each IRQ type of Rx SA 86
- **Size:** 32 bits
- **Offset:** 0x644
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 86

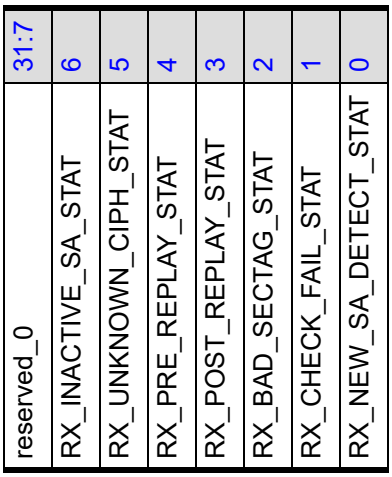


Table 6-408 Fields for Register: IRQ_TYPE_STAT_RX_SA_86

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-408 Fields for Register: IRQ_TYPE_STAT_RX_SA_86 (Continued)

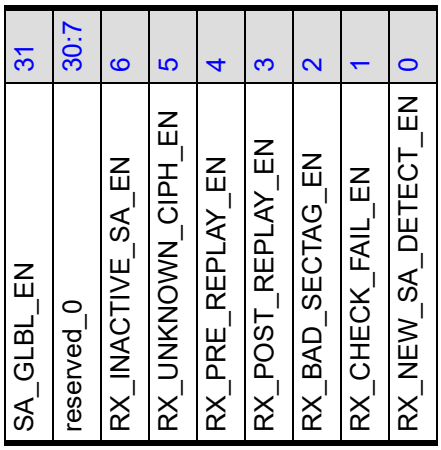
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-408 Fields for Register: IRQ_TYPE_STAT_RX_SA_86 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.402 IRQ_TYPE_EN_RX_SA_87

- **Name:** IRQ_TYPE_EN_RX_SA_87
- **Description:** Interrupt Enable for each IRQ type of Rx SA 87
- **Size:** 32 bits
- **Offset:** 0x648
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 87



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-409 Fields for Register: IRQ_TYPE_EN_RX_SA_87

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-409 Fields for Register: IRQ_TYPE_EN_RX_SA_87 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-409 Fields for Register: IRQ_TYPE_EN_RX_SA_87 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.403 IRQ_TYPE_STAT_RX_SA_87

- **Name:** IRQ_TYPE_STAT_RX_SA_87
- **Description:** Interrupt Status for each IRQ type of Rx SA 87
- **Size:** 32 bits
- **Offset:** 0x64c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 87

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-410 Fields for Register: IRQ_TYPE_STAT_RX_SA_87

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-410 Fields for Register: IRQ_TYPE_STAT_RX_SA_87 (Continued)

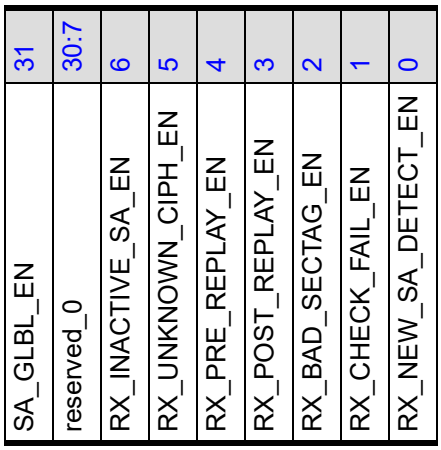
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-410 Fields for Register: IRQ_TYPE_STAT_RX_SA_87 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.404 IRQ_TYPE_EN_RX_SA_88

- **Name:** IRQ_TYPE_EN_RX_SA_88
- **Description:** Interrupt Enable for each IRQ type of Rx SA 88
- **Size:** 32 bits
- **Offset:** 0x650
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 88



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-411 Fields for Register: IRQ_TYPE_EN_RX_SA_88

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-411 Fields for Register: IRQ_TYPE_EN_RX_SA_88 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-411 Fields for Register: IRQ_TYPE_EN_RX_SA_88 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.405 IRQ_TYPE_STAT_RX_SA_88

- **Name:** IRQ_TYPE_STAT_RX_SA_88
- **Description:** Interrupt Status for each IRQ type of Rx SA 88
- **Size:** 32 bits
- **Offset:** 0x654
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 88

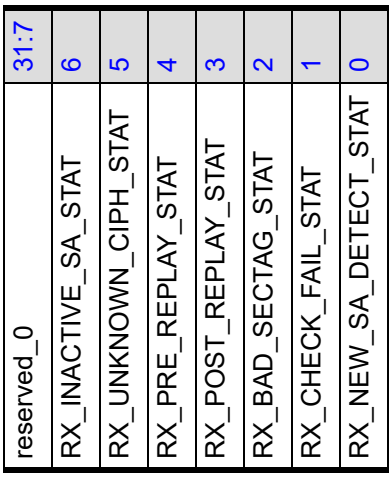


Table 6-412 Fields for Register: IRQ_TYPE_STAT_RX_SA_88

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-412 Fields for Register: IRQ_TYPE_STAT_RX_SA_88 (Continued)

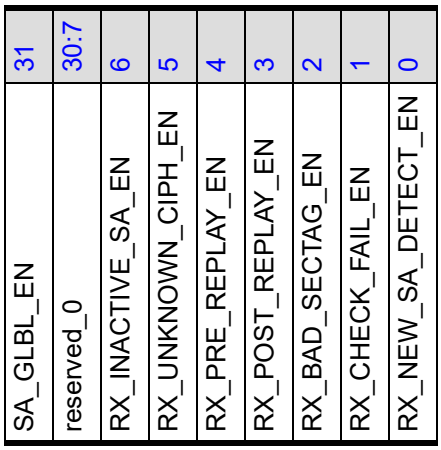
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-412 Fields for Register: IRQ_TYPE_STAT_RX_SA_88 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.406 IRQ_TYPE_EN_RX_SA_89

- **Name:** IRQ_TYPE_EN_RX_SA_89
- **Description:** Interrupt Enable for each IRQ type of Rx SA 89
- **Size:** 32 bits
- **Offset:** 0x658
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 89



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-413 Fields for Register: IRQ_TYPE_EN_RX_SA_89

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-413 Fields for Register: IRQ_TYPE_EN_RX_SA_89 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-413 Fields for Register: IRQ_TYPE_EN_RX_SA_89 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.407 IRQ_TYPE_STAT_RX_SA_89

- **Name:** IRQ_TYPE_STAT_RX_SA_89
- **Description:** Interrupt Status for each IRQ type of Rx SA 89
- **Size:** 32 bits
- **Offset:** 0x65c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 89

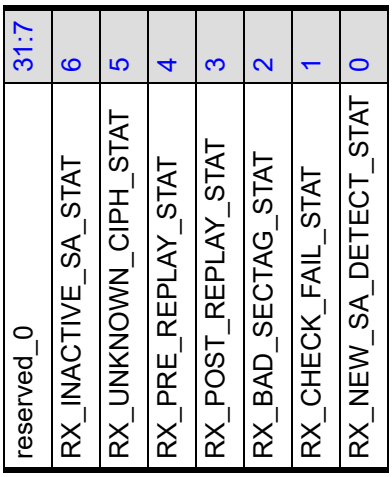


Table 6-414 Fields for Register: IRQ_TYPE_STAT_RX_SA_89

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-414 Fields for Register: IRQ_TYPE_STAT_RX_SA_89 (Continued)

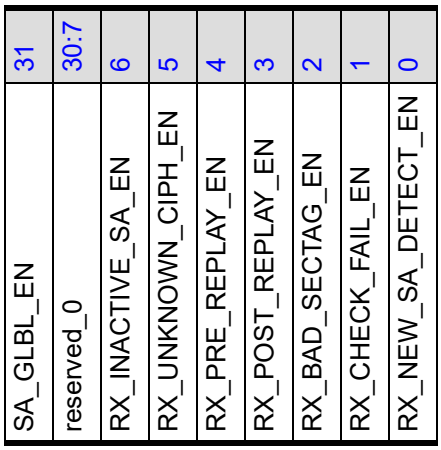
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-414 Fields for Register: IRQ_TYPE_STAT_RX_SA_89 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.408 IRQ_TYPE_EN_RX_SA_90

- **Name:** IRQ_TYPE_EN_RX_SA_90
- **Description:** Interrupt Enable for each IRQ type of Rx SA 90
- **Size:** 32 bits
- **Offset:** 0x660
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 90



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-415 Fields for Register: IRQ_TYPE_EN_RX_SA_90

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-415 Fields for Register: IRQ_TYPE_EN_RX_SA_90 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-415 Fields for Register: IRQ_TYPE_EN_RX_SA_90 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.409 IRQ_TYPE_STAT_RX_SA_90

- **Name:** IRQ_TYPE_STAT_RX_SA_90
- **Description:** Interrupt Status for each IRQ type of Rx SA 90
- **Size:** 32 bits
- **Offset:** 0x664
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 90

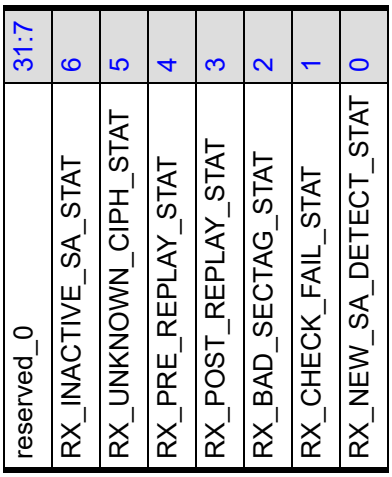


Table 6-416 Fields for Register: IRQ_TYPE_STAT_RX_SA_90

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-416 Fields for Register: IRQ_TYPE_STAT_RX_SA_90 (Continued)

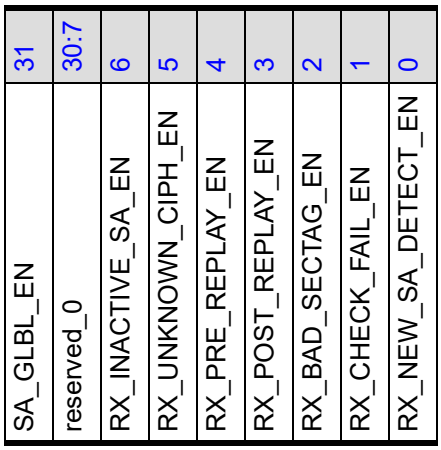
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-416 Fields for Register: IRQ_TYPE_STAT_RX_SA_90 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.410 IRQ_TYPE_EN_RX_SA_91

- **Name:** IRQ_TYPE_EN_RX_SA_91
- **Description:** Interrupt Enable for each IRQ type of Rx SA 91
- **Size:** 32 bits
- **Offset:** 0x668
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 91



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-417 Fields for Register: IRQ_TYPE_EN_RX_SA_91

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-417 Fields for Register: IRQ_TYPE_EN_RX_SA_91 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-417 Fields for Register: IRQ_TYPE_EN_RX_SA_91 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.411 IRQ_TYPE_STAT_RX_SA_91

- **Name:** IRQ_TYPE_STAT_RX_SA_91
- **Description:** Interrupt Status for each IRQ type of Rx SA 91
- **Size:** 32 bits
- **Offset:** 0x66c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 91

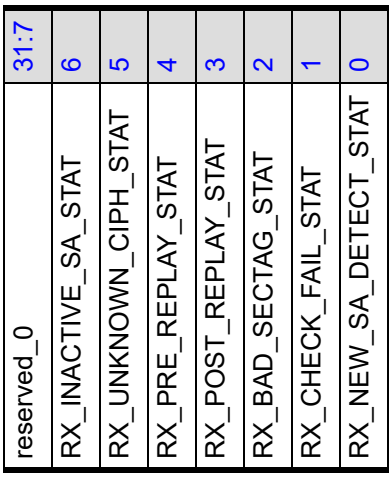


Table 6-418 Fields for Register: IRQ_TYPE_STAT_RX_SA_91

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-418 Fields for Register: IRQ_TYPE_STAT_RX_SA_91 (Continued)

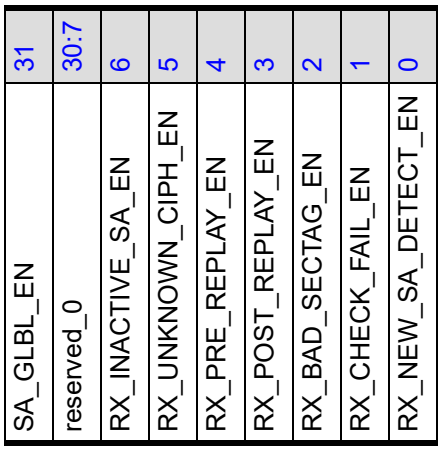
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-418 Fields for Register: IRQ_TYPE_STAT_RX_SA_91 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.412 IRQ_TYPE_EN_RX_SA_92

- **Name:** IRQ_TYPE_EN_RX_SA_92
- **Description:** Interrupt Enable for each IRQ type of Rx SA 92
- **Size:** 32 bits
- **Offset:** 0x670
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 92



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-419 Fields for Register: IRQ_TYPE_EN_RX_SA_92

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-419 Fields for Register: IRQ_TYPE_EN_RX_SA_92 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-419 Fields for Register: IRQ_TYPE_EN_RX_SA_92 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.413 IRQ_TYPE_STAT_RX_SA_92

- **Name:** IRQ_TYPE_STAT_RX_SA_92
- **Description:** Interrupt Status for each IRQ type of Rx SA 92
- **Size:** 32 bits
- **Offset:** 0x674
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 92

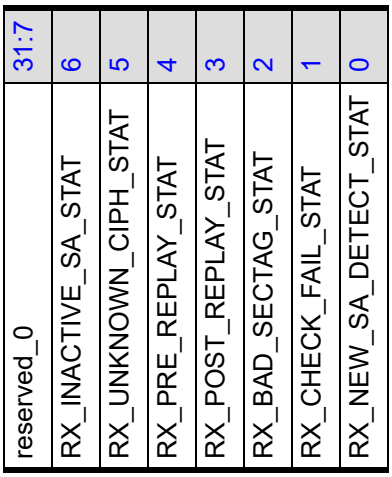


Table 6-420 Fields for Register: IRQ_TYPE_STAT_RX_SA_92

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-420 Fields for Register: IRQ_TYPE_STAT_RX_SA_92 (Continued)

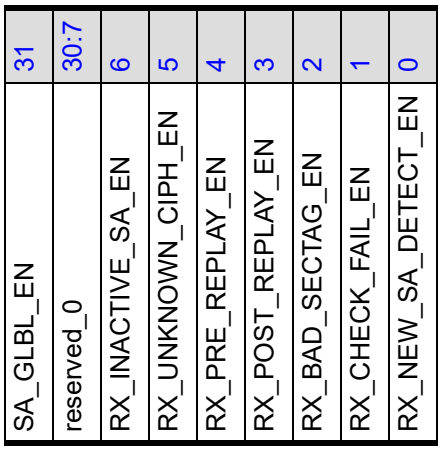
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-420 Fields for Register: IRQ_TYPE_STAT_RX_SA_92 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.414 IRQ_TYPE_EN_RX_SA_93

- **Name:** IRQ_TYPE_EN_RX_SA_93
- **Description:** Interrupt Enable for each IRQ type of Rx SA 93
- **Size:** 32 bits
- **Offset:** 0x678
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 93



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-421 Fields for Register: IRQ_TYPE_EN_RX_SA_93

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-421 Fields for Register: IRQ_TYPE_EN_RX_SA_93 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-421 Fields for Register: IRQ_TYPE_EN_RX_SA_93 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.415 IRQ_TYPE_STAT_RX_SA_93

- **Name:** IRQ_TYPE_STAT_RX_SA_93
- **Description:** Interrupt Status for each IRQ type of Rx SA 93
- **Size:** 32 bits
- **Offset:** 0x67c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 93

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-422 Fields for Register: IRQ_TYPE_STAT_RX_SA_93

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-422 Fields for Register: IRQ_TYPE_STAT_RX_SA_93 (Continued)

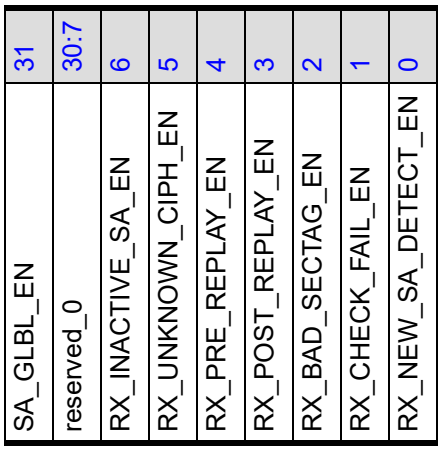
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-422 Fields for Register: IRQ_TYPE_STAT_RX_SA_93 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.416 IRQ_TYPE_EN_RX_SA_94

- **Name:** IRQ_TYPE_EN_RX_SA_94
- **Description:** Interrupt Enable for each IRQ type of Rx SA 94
- **Size:** 32 bits
- **Offset:** 0x680
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 94



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-423 Fields for Register: IRQ_TYPE_EN_RX_SA_94

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-423 Fields for Register: IRQ_TYPE_EN_RX_SA_94 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-423 Fields for Register: IRQ_TYPE_EN_RX_SA_94 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.417 IRQ_TYPE_STAT_RX_SA_94

- **Name:** IRQ_TYPE_STAT_RX_SA_94
- **Description:** Interrupt Status for each IRQ type of Rx SA 94
- **Size:** 32 bits
- **Offset:** 0x684
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 94

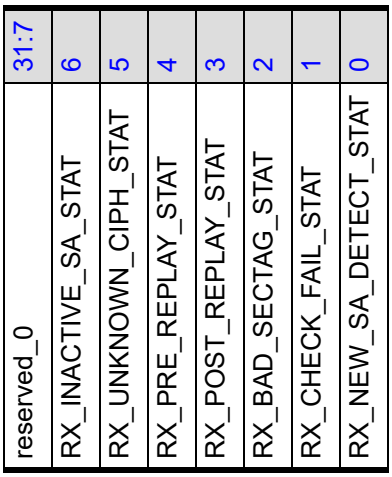


Table 6-424 Fields for Register: IRQ_TYPE_STAT_RX_SA_94

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-424 Fields for Register: IRQ_TYPE_STAT_RX_SA_94 (Continued)

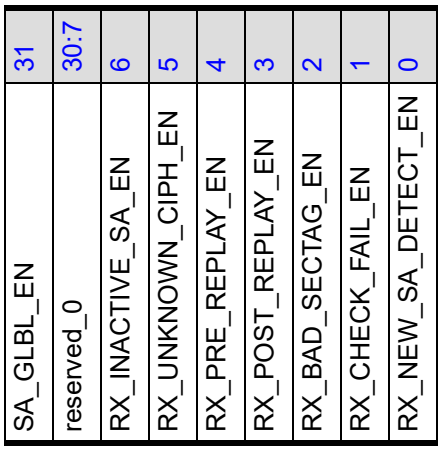
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-424 Fields for Register: IRQ_TYPE_STAT_RX_SA_94 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.418 IRQ_TYPE_EN_RX_SA_95

- **Name:** IRQ_TYPE_EN_RX_SA_95
- **Description:** Interrupt Enable for each IRQ type of Rx SA 95
- **Size:** 32 bits
- **Offset:** 0x688
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 95



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-425 Fields for Register: IRQ_TYPE_EN_RX_SA_95

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-425 Fields for Register: IRQ_TYPE_EN_RX_SA_95 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-425 Fields for Register: IRQ_TYPE_EN_RX_SA_95 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.419 IRQ_TYPE_STAT_RX_SA_95

- **Name:** IRQ_TYPE_STAT_RX_SA_95
- **Description:** Interrupt Status for each IRQ type of Rx SA 95
- **Size:** 32 bits
- **Offset:** 0x68c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 95

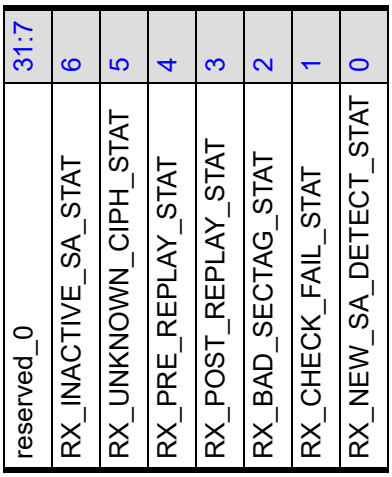


Table 6-426 Fields for Register: IRQ_TYPE_STAT_RX_SA_95

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-426 Fields for Register: IRQ_TYPE_STAT_RX_SA_95 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-426 Fields for Register: IRQ_TYPE_STAT_RX_SA_95 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.420 IRQ_TYPE_EN_RX_SA_96

- **Name:** IRQ_TYPE_EN_RX_SA_96
- **Description:** Interrupt Enable for each IRQ type of Rx SA 96
- **Size:** 32 bits
- **Offset:** 0x690
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 96

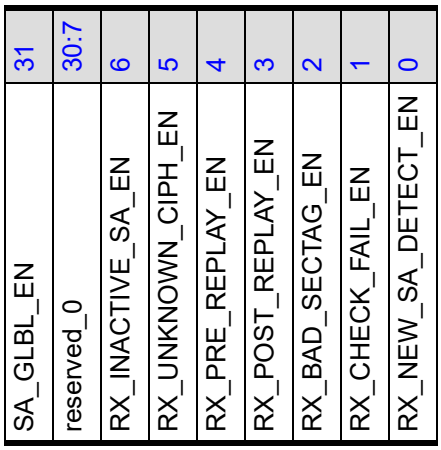


Table 6-427 Fields for Register: IRQ_TYPE_EN_RX_SA_96

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-427 Fields for Register: IRQ_TYPE_EN_RX_SA_96 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-427 Fields for Register: IRQ_TYPE_EN_RX_SA_96 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.421 IRQ_TYPE_STAT_RX_SA_96

- **Name:** IRQ_TYPE_STAT_RX_SA_96
- **Description:** Interrupt Status for each IRQ type of Rx SA 96
- **Size:** 32 bits
- **Offset:** 0x694
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 96

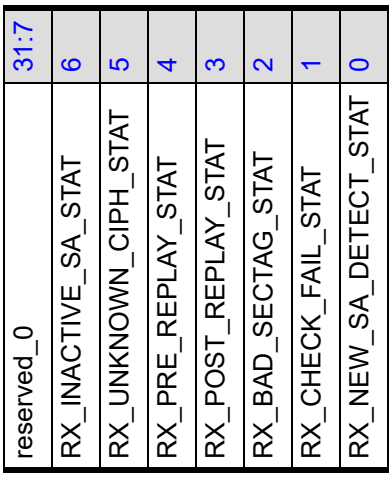


Table 6-428 Fields for Register: IRQ_TYPE_STAT_RX_SA_96

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-428 Fields for Register: IRQ_TYPE_STAT_RX_SA_96 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-428 Fields for Register: IRQ_TYPE_STAT_RX_SA_96 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.422 IRQ_TYPE_EN_RX_SA_97

- **Name:** IRQ_TYPE_EN_RX_SA_97
- **Description:** Interrupt Enable for each IRQ type of Rx SA 97
- **Size:** 32 bits
- **Offset:** 0x698
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 97

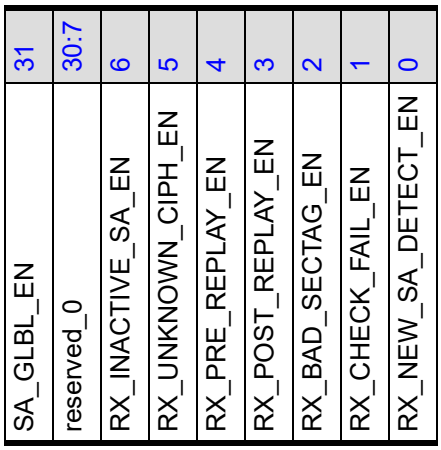


Table 6-429 Fields for Register: IRQ_TYPE_EN_RX_SA_97

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-429 Fields for Register: IRQ_TYPE_EN_RX_SA_97 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-429 Fields for Register: IRQ_TYPE_EN_RX_SA_97 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.423 IRQ_TYPE_STAT_RX_SA_97

- **Name:** IRQ_TYPE_STAT_RX_SA_97
- **Description:** Interrupt Status for each IRQ type of Rx SA 97
- **Size:** 32 bits
- **Offset:** 0x69c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 97

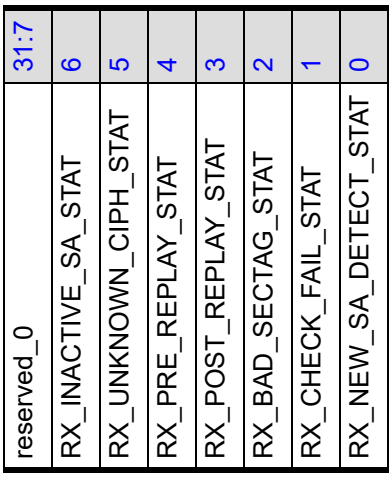


Table 6-430 Fields for Register: IRQ_TYPE_STAT_RX_SA_97

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-430 Fields for Register: IRQ_TYPE_STAT_RX_SA_97 (Continued)

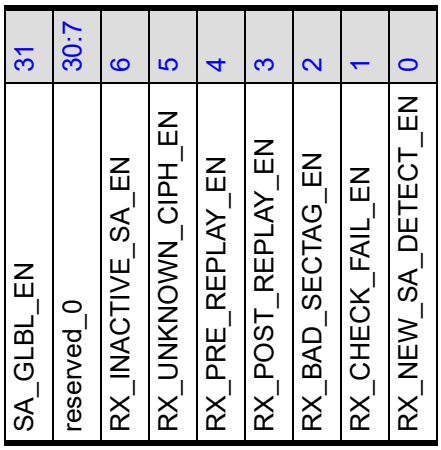
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-430 Fields for Register: IRQ_TYPE_STAT_RX_SA_97 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.424 IRQ_TYPE_EN_RX_SA_98

- **Name:** IRQ_TYPE_EN_RX_SA_98
- **Description:** Interrupt Enable for each IRQ type of Rx SA 98
- **Size:** 32 bits
- **Offset:** 0x6a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 98



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-431 Fields for Register: IRQ_TYPE_EN_RX_SA_98

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-431 Fields for Register: IRQ_TYPE_EN_RX_SA_98 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-431 Fields for Register: IRQ_TYPE_EN_RX_SA_98 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.425 IRQ_TYPE_STAT_RX_SA_98

- **Name:** IRQ_TYPE_STAT_RX_SA_98
- **Description:** Interrupt Status for each IRQ type of Rx SA 98
- **Size:** 32 bits
- **Offset:** 0x6a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 98

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-432 Fields for Register: IRQ_TYPE_STAT_RX_SA_98

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-432 Fields for Register: IRQ_TYPE_STAT_RX_SA_98 (Continued)

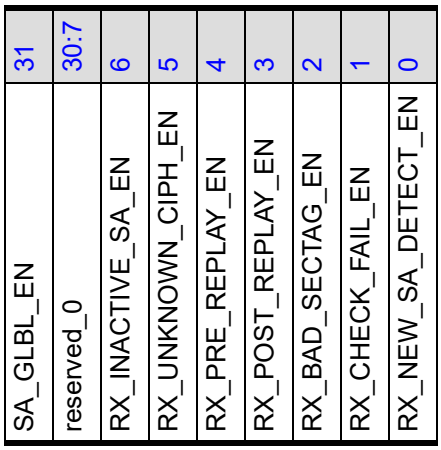
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-432 Fields for Register: IRQ_TYPE_STAT_RX_SA_98 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.426 IRQ_TYPE_EN_RX_SA_99

- **Name:** IRQ_TYPE_EN_RX_SA_99
- **Description:** Interrupt Enable for each IRQ type of Rx SA 99
- **Size:** 32 bits
- **Offset:** 0x6a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 99



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-433 Fields for Register: IRQ_TYPE_EN_RX_SA_99

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-433 Fields for Register: IRQ_TYPE_EN_RX_SA_99 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-433 Fields for Register: IRQ_TYPE_EN_RX_SA_99 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.427 IRQ_TYPE_STAT_RX_SA_99

- **Name:** IRQ_TYPE_STAT_RX_SA_99
- **Description:** Interrupt Status for each IRQ type of Rx SA 99
- **Size:** 32 bits
- **Offset:** 0x6ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 99

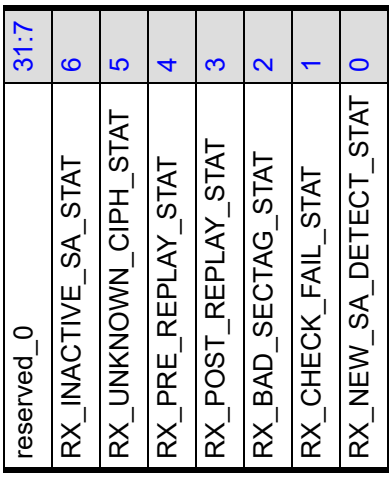


Table 6-434 Fields for Register: IRQ_TYPE_STAT_RX_SA_99

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-434 Fields for Register: IRQ_TYPE_STAT_RX_SA_99 (Continued)

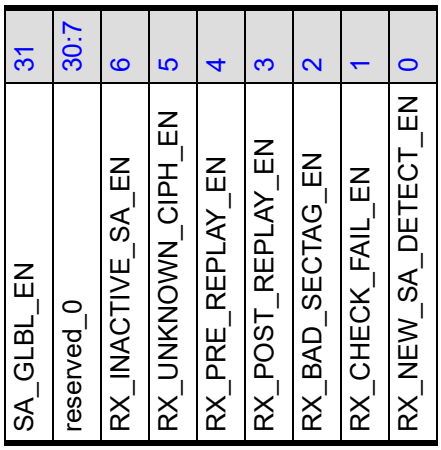
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-434 Fields for Register: IRQ_TYPE_STAT_RX_SA_99 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.428 IRQ_TYPE_EN_RX_SA_100

- **Name:** IRQ_TYPE_EN_RX_SA_100
- **Description:** Interrupt Enable for each IRQ type of Rx SA 100
- **Size:** 32 bits
- **Offset:** 0x6b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 100



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-435 Fields for Register: IRQ_TYPE_EN_RX_SA_100

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-435 Fields for Register: IRQ_TYPE_EN_RX_SA_100 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-435 Fields for Register: IRQ_TYPE_EN_RX_SA_100 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.429 IRQ_TYPE_STAT_RX_SA_100

- **Name:** IRQ_TYPE_STAT_RX_SA_100
- **Description:** Interrupt Status for each IRQ type of Rx SA 100
- **Size:** 32 bits
- **Offset:** 0x6b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 100

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-436 Fields for Register: IRQ_TYPE_STAT_RX_SA_100

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-436 Fields for Register: IRQ_TYPE_STAT_RX_SA_100 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-436 Fields for Register: IRQ_TYPE_STAT_RX_SA_100 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.430 IRQ_TYPE_EN_RX_SA_101

- **Name:** IRQ_TYPE_EN_RX_SA_101
- **Description:** Interrupt Enable for each IRQ type of Rx SA 101
- **Size:** 32 bits
- **Offset:** 0x6b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 101

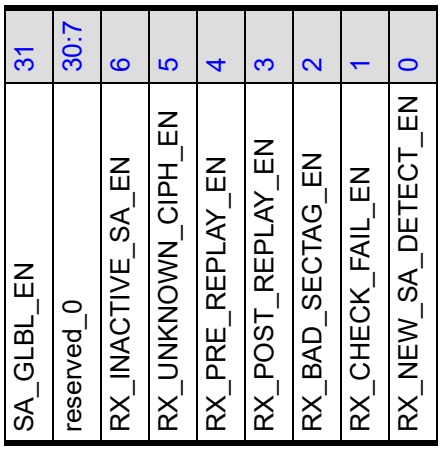


Table 6-437 Fields for Register: IRQ_TYPE_EN_RX_SA_101

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-437 Fields for Register: IRQ_TYPE_EN_RX_SA_101 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-437 Fields for Register: IRQ_TYPE_EN_RX_SA_101 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.431 IRQ_TYPE_STAT_RX_SA_101

- **Name:** IRQ_TYPE_STAT_RX_SA_101
- **Description:** Interrupt Status for each IRQ type of Rx SA 101
- **Size:** 32 bits
- **Offset:** 0x6bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 101

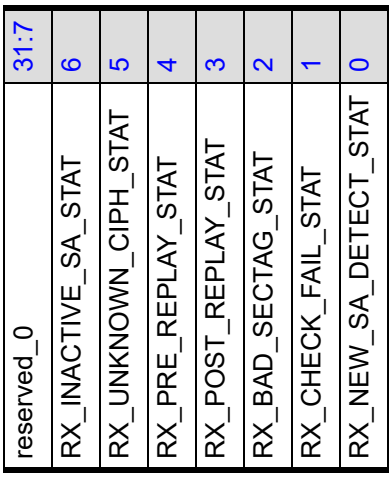


Table 6-438 Fields for Register: IRQ_TYPE_STAT_RX_SA_101

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-438 Fields for Register: IRQ_TYPE_STAT_RX_SA_101 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-438 Fields for Register: IRQ_TYPE_STAT_RX_SA_101 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.432 IRQ_TYPE_EN_RX_SA_102

- **Name:** IRQ_TYPE_EN_RX_SA_102
- **Description:** Interrupt Enable for each IRQ type of Rx SA 102
- **Size:** 32 bits
- **Offset:** 0x6c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 102

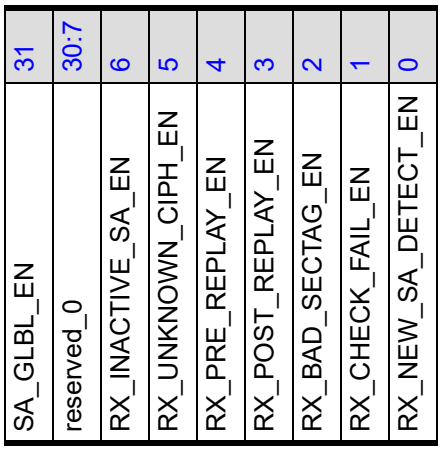


Table 6-439 Fields for Register: IRQ_TYPE_EN_RX_SA_102

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-439 Fields for Register: IRQ_TYPE_EN_RX_SA_102 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-439 Fields for Register: IRQ_TYPE_EN_RX_SA_102 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.433 IRQ_TYPE_STAT_RX_SA_102

- **Name:** IRQ_TYPE_STAT_RX_SA_102
- **Description:** Interrupt Status for each IRQ type of Rx SA 102
- **Size:** 32 bits
- **Offset:** 0x6c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 102

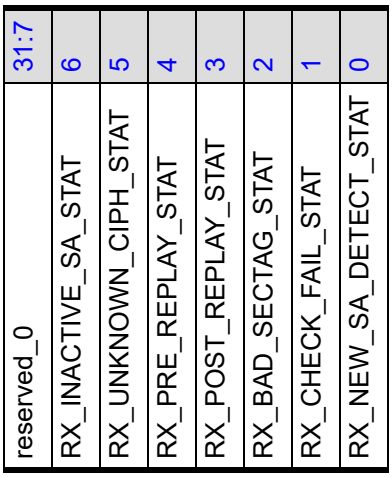


Table 6-440 Fields for Register: IRQ_TYPE_STAT_RX_SA_102

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-440 Fields for Register: IRQ_TYPE_STAT_RX_SA_102 (Continued)

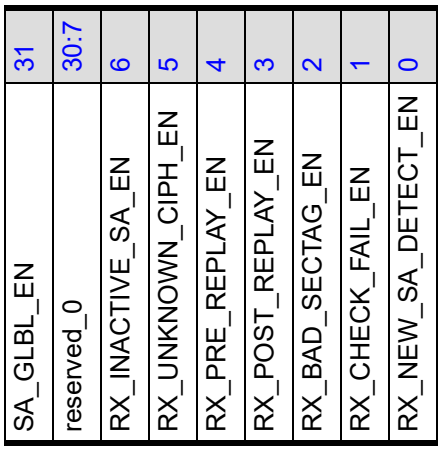
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-440 Fields for Register: IRQ_TYPE_STAT_RX_SA_102 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.434 IRQ_TYPE_EN_RX_SA_103

- **Name:** IRQ_TYPE_EN_RX_SA_103
- **Description:** Interrupt Enable for each IRQ type of Rx SA 103
- **Size:** 32 bits
- **Offset:** 0x6c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 103



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-441 Fields for Register: IRQ_TYPE_EN_RX_SA_103

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-441 Fields for Register: IRQ_TYPE_EN_RX_SA_103 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-441 Fields for Register: IRQ_TYPE_EN_RX_SA_103 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.435 IRQ_TYPE_STAT_RX_SA_103

- **Name:** IRQ_TYPE_STAT_RX_SA_103
- **Description:** Interrupt Status for each IRQ type of Rx SA 103
- **Size:** 32 bits
- **Offset:** 0x6cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 103

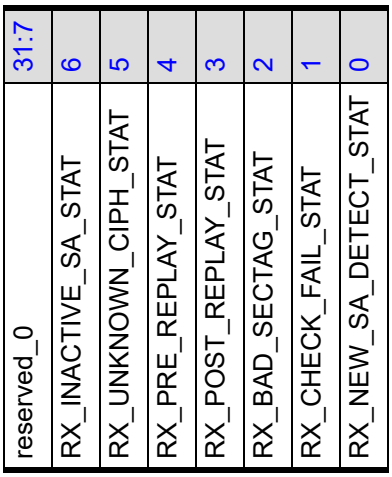


Table 6-442 Fields for Register: IRQ_TYPE_STAT_RX_SA_103

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-442 Fields for Register: IRQ_TYPE_STAT_RX_SA_103 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-442 Fields for Register: IRQ_TYPE_STAT_RX_SA_103 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.436 IRQ_TYPE_EN_RX_SA_104

- **Name:** IRQ_TYPE_EN_RX_SA_104
- **Description:** Interrupt Enable for each IRQ type of Rx SA 104
- **Size:** 32 bits
- **Offset:** 0x6d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 104

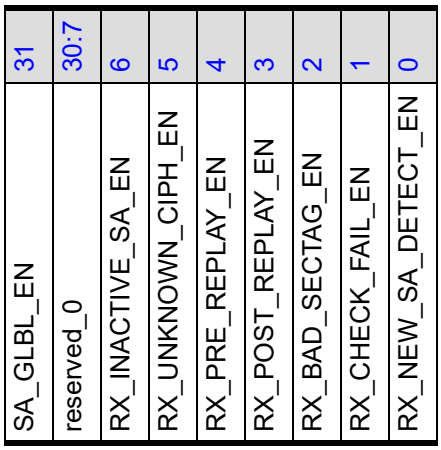


Table 6-443 Fields for Register: IRQ_TYPE_EN_RX_SA_104

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-443 Fields for Register: IRQ_TYPE_EN_RX_SA_104 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-443 Fields for Register: IRQ_TYPE_EN_RX_SA_104 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.437 IRQ_TYPE_STAT_RX_SA_104

- **Name:** IRQ_TYPE_STAT_RX_SA_104
- **Description:** Interrupt Status for each IRQ type of Rx SA 104
- **Size:** 32 bits
- **Offset:** 0x6d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 104

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-444 Fields for Register: IRQ_TYPE_STAT_RX_SA_104

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-444 Fields for Register: IRQ_TYPE_STAT_RX_SA_104 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-444 Fields for Register: IRQ_TYPE_STAT_RX_SA_104 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.438 IRQ_TYPE_EN_RX_SA_105

- **Name:** IRQ_TYPE_EN_RX_SA_105
- **Description:** Interrupt Enable for each IRQ type of Rx SA 105
- **Size:** 32 bits
- **Offset:** 0x6d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 105

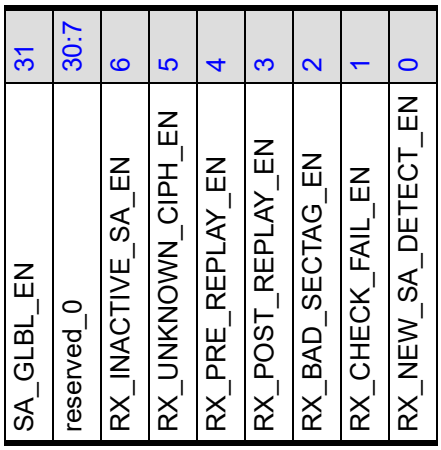


Table 6-445 Fields for Register: IRQ_TYPE_EN_RX_SA_105

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-445 Fields for Register: IRQ_TYPE_EN_RX_SA_105 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-445 Fields for Register: IRQ_TYPE_EN_RX_SA_105 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.439 IRQ_TYPE_STAT_RX_SA_105

- **Name:** IRQ_TYPE_STAT_RX_SA_105
- **Description:** Interrupt Status for each IRQ type of Rx SA 105
- **Size:** 32 bits
- **Offset:** 0x6dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 105

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-446 Fields for Register: IRQ_TYPE_STAT_RX_SA_105

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-446 Fields for Register: IRQ_TYPE_STAT_RX_SA_105 (Continued)

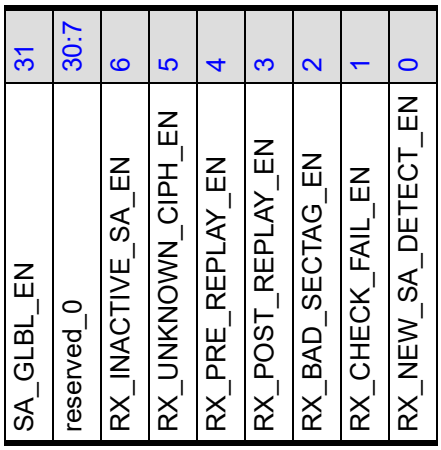
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-446 Fields for Register: IRQ_TYPE_STAT_RX_SA_105 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.440 IRQ_TYPE_EN_RX_SA_106

- **Name:** IRQ_TYPE_EN_RX_SA_106
- **Description:** Interrupt Enable for each IRQ type of Rx SA 106
- **Size:** 32 bits
- **Offset:** 0x6e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 106



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-447 Fields for Register: IRQ_TYPE_EN_RX_SA_106

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-447 Fields for Register: IRQ_TYPE_EN_RX_SA_106 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-447 Fields for Register: IRQ_TYPE_EN_RX_SA_106 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.441 IRQ_TYPE_STAT_RX_SA_106

- **Name:** IRQ_TYPE_STAT_RX_SA_106
- **Description:** Interrupt Status for each IRQ type of Rx SA 106
- **Size:** 32 bits
- **Offset:** 0x6e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 106

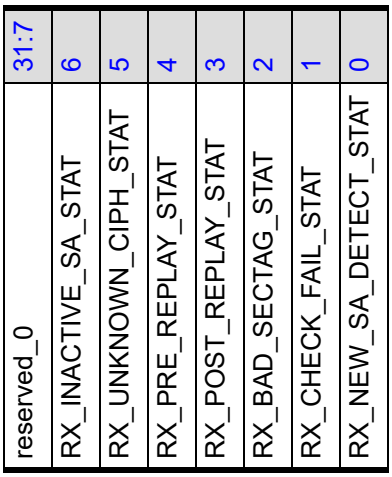


Table 6-448 Fields for Register: IRQ_TYPE_STAT_RX_SA_106

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-448 Fields for Register: IRQ_TYPE_STAT_RX_SA_106 (Continued)

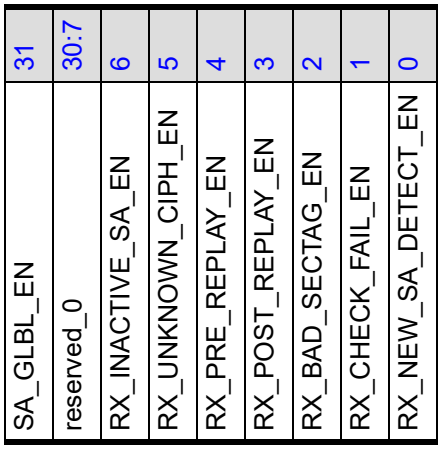
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-448 Fields for Register: IRQ_TYPE_STAT_RX_SA_106 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.442 IRQ_TYPE_EN_RX_SA_107

- **Name:** IRQ_TYPE_EN_RX_SA_107
- **Description:** Interrupt Enable for each IRQ type of Rx SA 107
- **Size:** 32 bits
- **Offset:** 0x6e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 107



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-449 Fields for Register: IRQ_TYPE_EN_RX_SA_107

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-449 Fields for Register: IRQ_TYPE_EN_RX_SA_107 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-449 Fields for Register: IRQ_TYPE_EN_RX_SA_107 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.443 IRQ_TYPE_STAT_RX_SA_107

- **Name:** IRQ_TYPE_STAT_RX_SA_107
- **Description:** Interrupt Status for each IRQ type of Rx SA 107
- **Size:** 32 bits
- **Offset:** 0x6ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 107

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-450 Fields for Register: IRQ_TYPE_STAT_RX_SA_107

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-450 Fields for Register: IRQ_TYPE_STAT_RX_SA_107 (Continued)

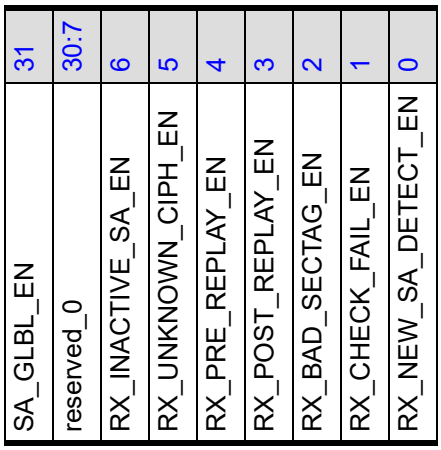
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-450 Fields for Register: IRQ_TYPE_STAT_RX_SA_107 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.444 IRQ_TYPE_EN_RX_SA_108

- **Name:** IRQ_TYPE_EN_RX_SA_108
- **Description:** Interrupt Enable for each IRQ type of Rx SA 108
- **Size:** 32 bits
- **Offset:** 0x6f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 108



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-451 Fields for Register: IRQ_TYPE_EN_RX_SA_108

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-451 Fields for Register: IRQ_TYPE_EN_RX_SA_108 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-451 Fields for Register: IRQ_TYPE_EN_RX_SA_108 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.445 IRQ_TYPE_STAT_RX_SA_108

- **Name:** IRQ_TYPE_STAT_RX_SA_108
- **Description:** Interrupt Status for each IRQ type of Rx SA 108
- **Size:** 32 bits
- **Offset:** 0x6f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 108

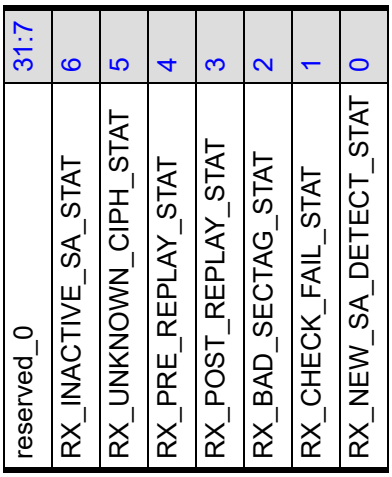


Table 6-452 Fields for Register: IRQ_TYPE_STAT_RX_SA_108

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-452 Fields for Register: IRQ_TYPE_STAT_RX_SA_108 (Continued)

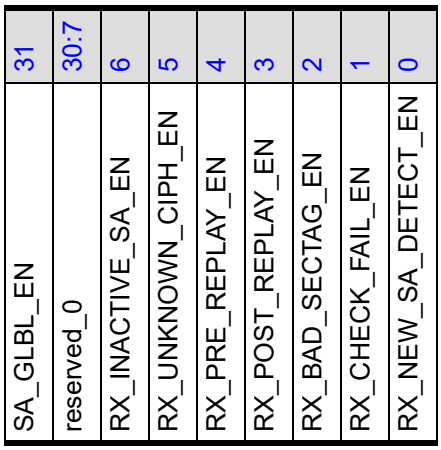
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-452 Fields for Register: IRQ_TYPE_STAT_RX_SA_108 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.446 IRQ_TYPE_EN_RX_SA_109

- **Name:** IRQ_TYPE_EN_RX_SA_109
- **Description:** Interrupt Enable for each IRQ type of Rx SA 109
- **Size:** 32 bits
- **Offset:** 0x6f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 109



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-453 Fields for Register: IRQ_TYPE_EN_RX_SA_109

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-453 Fields for Register: IRQ_TYPE_EN_RX_SA_109 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-453 Fields for Register: IRQ_TYPE_EN_RX_SA_109 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.447 IRQ_TYPE_STAT_RX_SA_109

- **Name:** IRQ_TYPE_STAT_RX_SA_109
- **Description:** Interrupt Status for each IRQ type of Rx SA 109
- **Size:** 32 bits
- **Offset:** 0x6fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 109

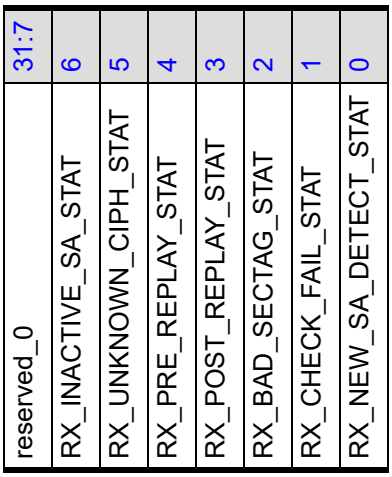


Table 6-454 Fields for Register: IRQ_TYPE_STAT_RX_SA_109

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-454 Fields for Register: IRQ_TYPE_STAT_RX_SA_109 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-454 Fields for Register: IRQ_TYPE_STAT_RX_SA_109 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.448 IRQ_TYPE_EN_RX_SA_110

- **Name:** IRQ_TYPE_EN_RX_SA_110
- **Description:** Interrupt Enable for each IRQ type of Rx SA 110
- **Size:** 32 bits
- **Offset:** 0x700
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 110

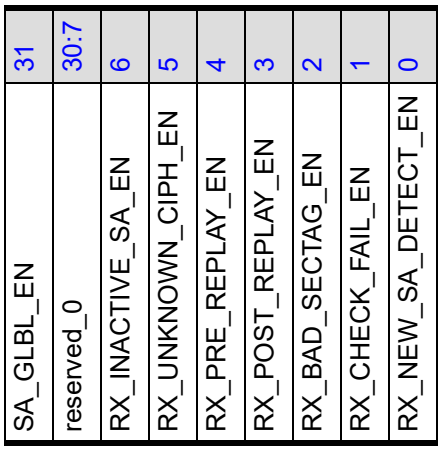


Table 6-455 Fields for Register: IRQ_TYPE_EN_RX_SA_110

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-455 Fields for Register: IRQ_TYPE_EN_RX_SA_110 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-455 Fields for Register: IRQ_TYPE_EN_RX_SA_110 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.449 IRQ_TYPE_STAT_RX_SA_110

- **Name:** IRQ_TYPE_STAT_RX_SA_110
- **Description:** Interrupt Status for each IRQ type of Rx SA 110
- **Size:** 32 bits
- **Offset:** 0x704
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 110

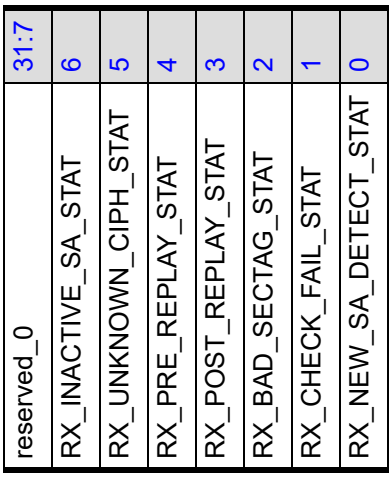


Table 6-456 Fields for Register: IRQ_TYPE_STAT_RX_SA_110

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-456 Fields for Register: IRQ_TYPE_STAT_RX_SA_110 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-456 Fields for Register: IRQ_TYPE_STAT_RX_SA_110 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.450 IRQ_TYPE_EN_RX_SA_111

- **Name:** IRQ_TYPE_EN_RX_SA_111
- **Description:** Interrupt Enable for each IRQ type of Rx SA 111
- **Size:** 32 bits
- **Offset:** 0x708
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 111

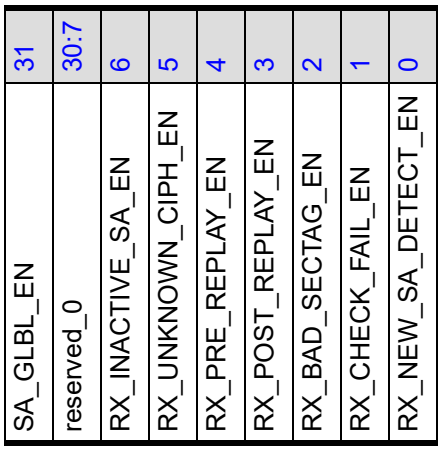


Table 6-457 Fields for Register: IRQ_TYPE_EN_RX_SA_111

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-457 Fields for Register: IRQ_TYPE_EN_RX_SA_111 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-457 Fields for Register: IRQ_TYPE_EN_RX_SA_111 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.451 IRQ_TYPE_STAT_RX_SA_111

- **Name:** IRQ_TYPE_STAT_RX_SA_111
- **Description:** Interrupt Status for each IRQ type of Rx SA 111
- **Size:** 32 bits
- **Offset:** 0x70c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 111

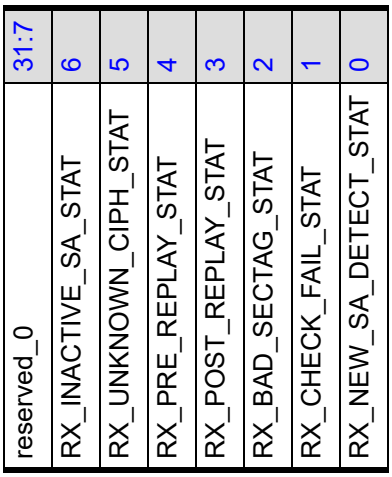


Table 6-458 Fields for Register: IRQ_TYPE_STAT_RX_SA_111

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-458 Fields for Register: IRQ_TYPE_STAT_RX_SA_111 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-458 Fields for Register: IRQ_TYPE_STAT_RX_SA_111 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.452 IRQ_TYPE_EN_RX_SA_112

- **Name:** IRQ_TYPE_EN_RX_SA_112
- **Description:** Interrupt Enable for each IRQ type of Rx SA 112
- **Size:** 32 bits
- **Offset:** 0x710
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 112

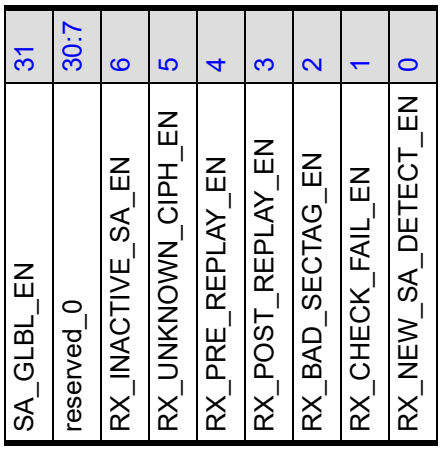


Table 6-459 Fields for Register: IRQ_TYPE_EN_RX_SA_112

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-459 Fields for Register: IRQ_TYPE_EN_RX_SA_112 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-459 Fields for Register: IRQ_TYPE_EN_RX_SA_112 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.453 IRQ_TYPE_STAT_RX_SA_112

- **Name:** IRQ_TYPE_STAT_RX_SA_112
- **Description:** Interrupt Status for each IRQ type of Rx SA 112
- **Size:** 32 bits
- **Offset:** 0x714
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 112

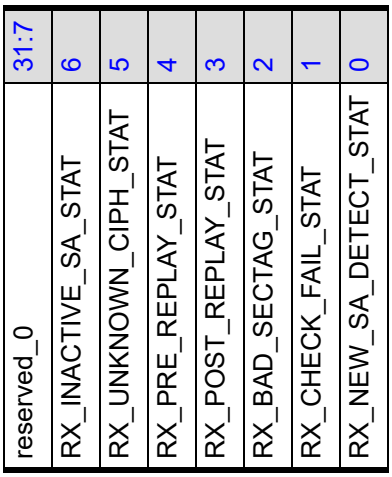


Table 6-460 Fields for Register: IRQ_TYPE_STAT_RX_SA_112

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-460 Fields for Register: IRQ_TYPE_STAT_RX_SA_112 (Continued)

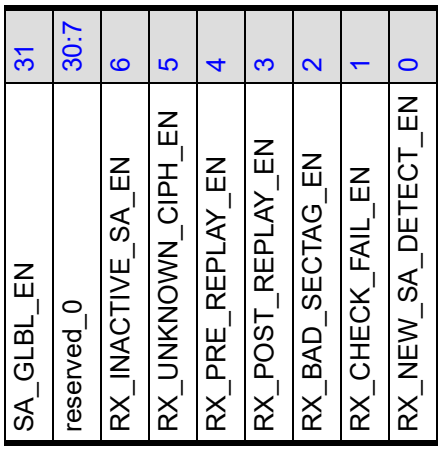
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-460 Fields for Register: IRQ_TYPE_STAT_RX_SA_112 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.454 IRQ_TYPE_EN_RX_SA_113

- **Name:** IRQ_TYPE_EN_RX_SA_113
- **Description:** Interrupt Enable for each IRQ type of Rx SA 113
- **Size:** 32 bits
- **Offset:** 0x718
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 113



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-461 Fields for Register: IRQ_TYPE_EN_RX_SA_113

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-461 Fields for Register: IRQ_TYPE_EN_RX_SA_113 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-461 Fields for Register: IRQ_TYPE_EN_RX_SA_113 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.455 IRQ_TYPE_STAT_RX_SA_113

- **Name:** IRQ_TYPE_STAT_RX_SA_113
- **Description:** Interrupt Status for each IRQ type of Rx SA 113
- **Size:** 32 bits
- **Offset:** 0x71c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 113

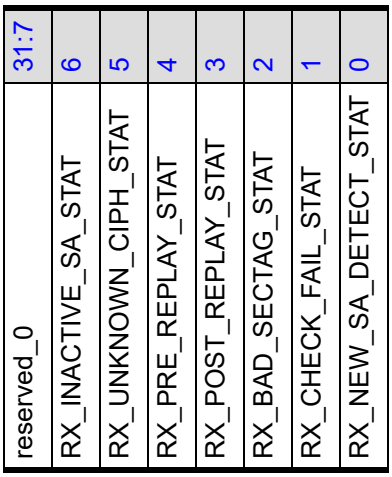


Table 6-462 Fields for Register: IRQ_TYPE_STAT_RX_SA_113

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-462 Fields for Register: IRQ_TYPE_STAT_RX_SA_113 (Continued)

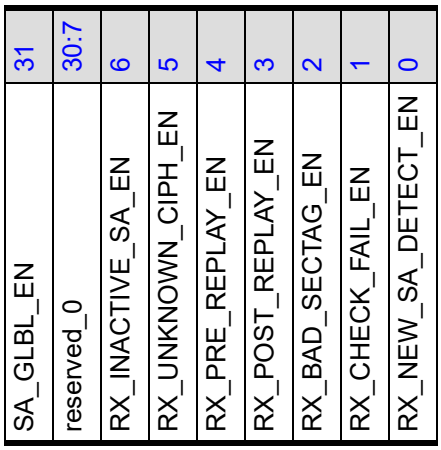
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-462 Fields for Register: IRQ_TYPE_STAT_RX_SA_113 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.456 IRQ_TYPE_EN_RX_SA_114

- **Name:** IRQ_TYPE_EN_RX_SA_114
- **Description:** Interrupt Enable for each IRQ type of Rx SA 114
- **Size:** 32 bits
- **Offset:** 0x720
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 114



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-463 Fields for Register: IRQ_TYPE_EN_RX_SA_114

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-463 Fields for Register: IRQ_TYPE_EN_RX_SA_114 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-463 Fields for Register: IRQ_TYPE_EN_RX_SA_114 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.457 IRQ_TYPE_STAT_RX_SA_114

- **Name:** IRQ_TYPE_STAT_RX_SA_114
- **Description:** Interrupt Status for each IRQ type of Rx SA 114
- **Size:** 32 bits
- **Offset:** 0x724
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 114

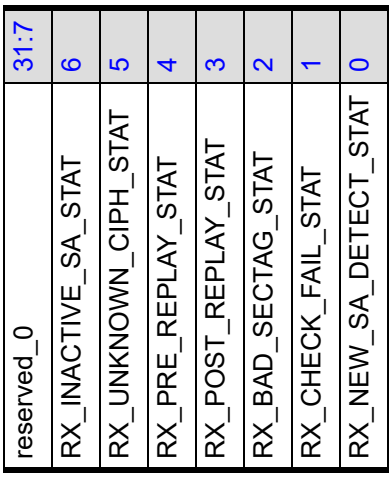


Table 6-464 Fields for Register: IRQ_TYPE_STAT_RX_SA_114

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-464 Fields for Register: IRQ_TYPE_STAT_RX_SA_114 (Continued)

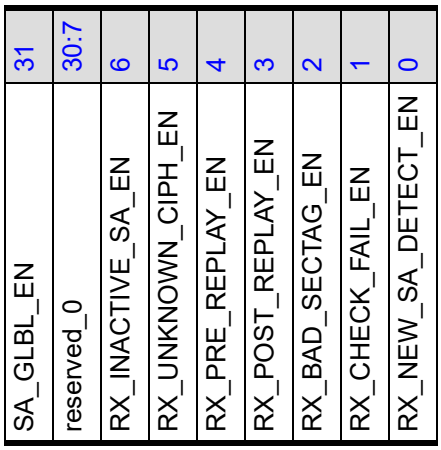
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-464 Fields for Register: IRQ_TYPE_STAT_RX_SA_114 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.458 IRQ_TYPE_EN_RX_SA_115

- **Name:** IRQ_TYPE_EN_RX_SA_115
- **Description:** Interrupt Enable for each IRQ type of Rx SA 115
- **Size:** 32 bits
- **Offset:** 0x728
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 115



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-465 Fields for Register: IRQ_TYPE_EN_RX_SA_115

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-465 Fields for Register: IRQ_TYPE_EN_RX_SA_115 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-465 Fields for Register: IRQ_TYPE_EN_RX_SA_115 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.459 IRQ_TYPE_STAT_RX_SA_115

- **Name:** IRQ_TYPE_STAT_RX_SA_115
- **Description:** Interrupt Status for each IRQ type of Rx SA 115
- **Size:** 32 bits
- **Offset:** 0x72c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 115

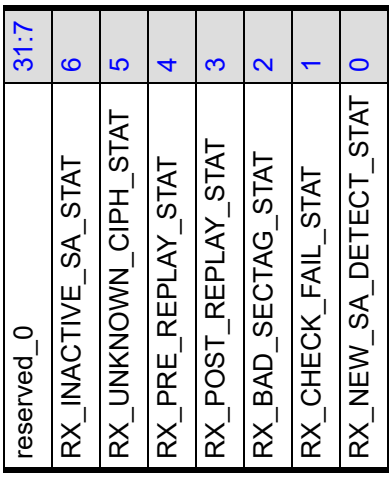


Table 6-466 Fields for Register: IRQ_TYPE_STAT_RX_SA_115

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-466 Fields for Register: IRQ_TYPE_STAT_RX_SA_115 (Continued)

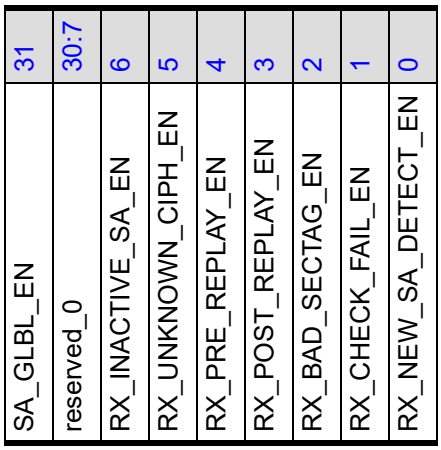
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-466 Fields for Register: IRQ_TYPE_STAT_RX_SA_115 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.460 IRQ_TYPE_EN_RX_SA_116

- **Name:** IRQ_TYPE_EN_RX_SA_116
- **Description:** Interrupt Enable for each IRQ type of Rx SA 116
- **Size:** 32 bits
- **Offset:** 0x730
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 116



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-467 Fields for Register: IRQ_TYPE_EN_RX_SA_116

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-467 Fields for Register: IRQ_TYPE_EN_RX_SA_116 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-467 Fields for Register: IRQ_TYPE_EN_RX_SA_116 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.461 IRQ_TYPE_STAT_RX_SA_116

- **Name:** IRQ_TYPE_STAT_RX_SA_116
- **Description:** Interrupt Status for each IRQ type of Rx SA 116
- **Size:** 32 bits
- **Offset:** 0x734
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 116

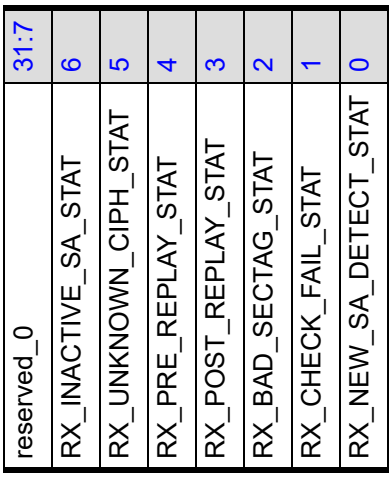


Table 6-468 Fields for Register: IRQ_TYPE_STAT_RX_SA_116

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-468 Fields for Register: IRQ_TYPE_STAT_RX_SA_116 (Continued)

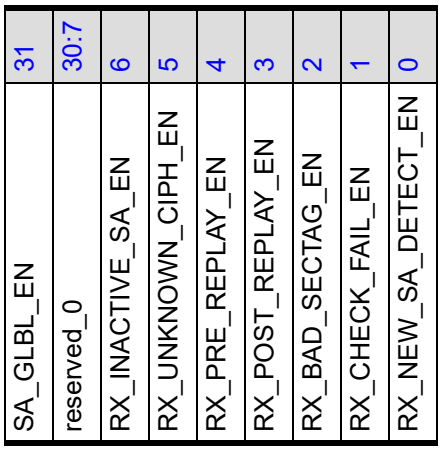
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-468 Fields for Register: IRQ_TYPE_STAT_RX_SA_116 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.462 IRQ_TYPE_EN_RX_SA_117

- **Name:** IRQ_TYPE_EN_RX_SA_117
- **Description:** Interrupt Enable for each IRQ type of Rx SA 117
- **Size:** 32 bits
- **Offset:** 0x738
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 117



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-469 Fields for Register: IRQ_TYPE_EN_RX_SA_117

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-469 Fields for Register: IRQ_TYPE_EN_RX_SA_117 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-469 Fields for Register: IRQ_TYPE_EN_RX_SA_117 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.463 IRQ_TYPE_STAT_RX_SA_117

- **Name:** IRQ_TYPE_STAT_RX_SA_117
- **Description:** Interrupt Status for each IRQ type of Rx SA 117
- **Size:** 32 bits
- **Offset:** 0x73c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 117

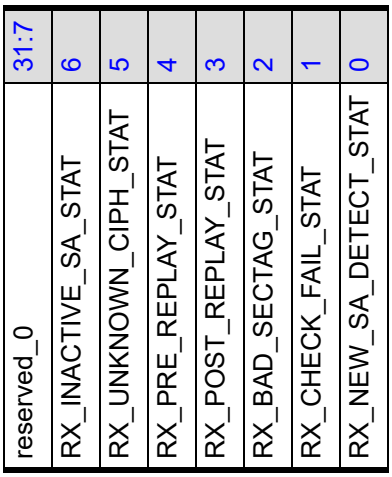


Table 6-470 Fields for Register: IRQ_TYPE_STAT_RX_SA_117

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-470 Fields for Register: IRQ_TYPE_STAT_RX_SA_117 (Continued)

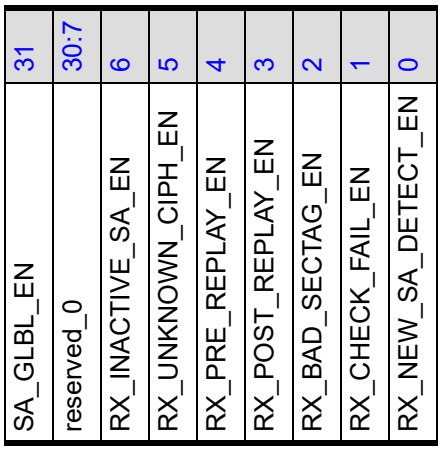
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-470 Fields for Register: IRQ_TYPE_STAT_RX_SA_117 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.464 IRQ_TYPE_EN_RX_SA_118

- **Name:** IRQ_TYPE_EN_RX_SA_118
- **Description:** Interrupt Enable for each IRQ type of Rx SA 118
- **Size:** 32 bits
- **Offset:** 0x740
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 118



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-471 Fields for Register: IRQ_TYPE_EN_RX_SA_118

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-471 Fields for Register: IRQ_TYPE_EN_RX_SA_118 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-471 Fields for Register: IRQ_TYPE_EN_RX_SA_118 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.465 IRQ_TYPE_STAT_RX_SA_118

- **Name:** IRQ_TYPE_STAT_RX_SA_118
- **Description:** Interrupt Status for each IRQ type of Rx SA 118
- **Size:** 32 bits
- **Offset:** 0x744
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 118

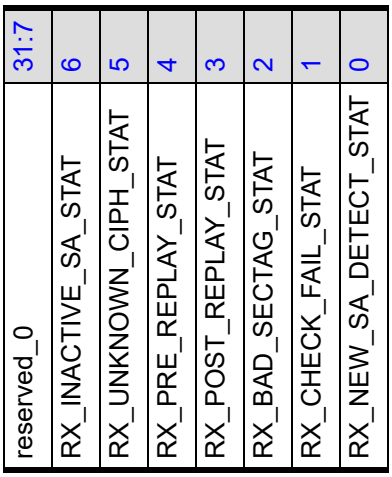


Table 6-472 Fields for Register: IRQ_TYPE_STAT_RX_SA_118

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-472 Fields for Register: IRQ_TYPE_STAT_RX_SA_118 (Continued)

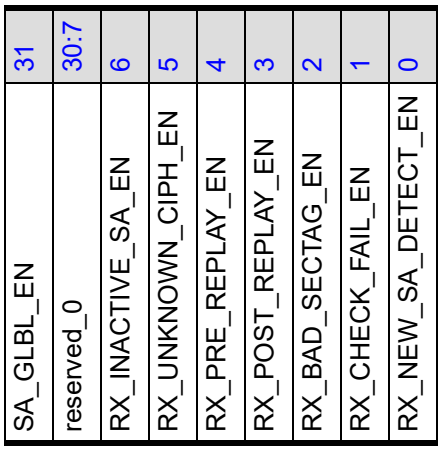
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-472 Fields for Register: IRQ_TYPE_STAT_RX_SA_118 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.466 IRQ_TYPE_EN_RX_SA_119

- **Name:** IRQ_TYPE_EN_RX_SA_119
- **Description:** Interrupt Enable for each IRQ type of Rx SA 119
- **Size:** 32 bits
- **Offset:** 0x748
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 119



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-473 Fields for Register: IRQ_TYPE_EN_RX_SA_119

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-473 Fields for Register: IRQ_TYPE_EN_RX_SA_119 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-473 Fields for Register: IRQ_TYPE_EN_RX_SA_119 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.467 IRQ_TYPE_STAT_RX_SA_119

- **Name:** IRQ_TYPE_STAT_RX_SA_119
- **Description:** Interrupt Status for each IRQ type of Rx SA 119
- **Size:** 32 bits
- **Offset:** 0x74c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 119

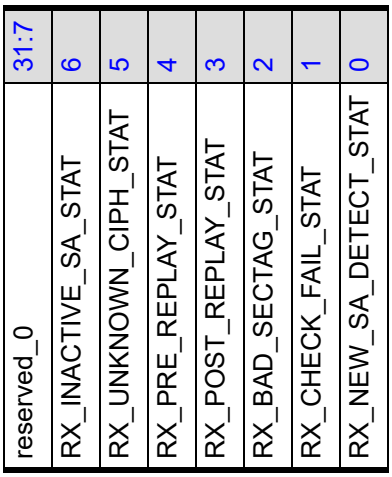


Table 6-474 Fields for Register: IRQ_TYPE_STAT_RX_SA_119

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-474 Fields for Register: IRQ_TYPE_STAT_RX_SA_119 (Continued)

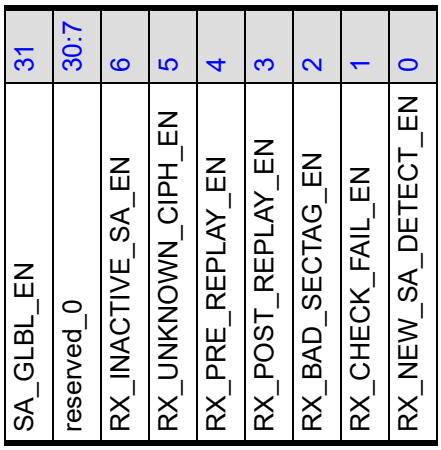
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-474 Fields for Register: IRQ_TYPE_STAT_RX_SA_119 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.468 IRQ_TYPE_EN_RX_SA_120

- **Name:** IRQ_TYPE_EN_RX_SA_120
- **Description:** Interrupt Enable for each IRQ type of Rx SA 120
- **Size:** 32 bits
- **Offset:** 0x750
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 120



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-475 Fields for Register: IRQ_TYPE_EN_RX_SA_120

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-475 Fields for Register: IRQ_TYPE_EN_RX_SA_120 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-475 Fields for Register: IRQ_TYPE_EN_RX_SA_120 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.469 IRQ_TYPE_STAT_RX_SA_120

- **Name:** IRQ_TYPE_STAT_RX_SA_120
- **Description:** Interrupt Status for each IRQ type of Rx SA 120
- **Size:** 32 bits
- **Offset:** 0x754
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 120

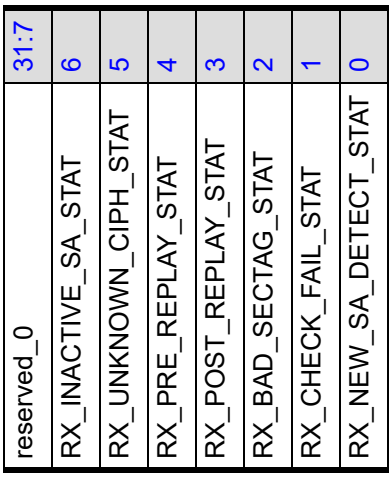


Table 6-476 Fields for Register: IRQ_TYPE_STAT_RX_SA_120

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-476 Fields for Register: IRQ_TYPE_STAT_RX_SA_120 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-476 Fields for Register: IRQ_TYPE_STAT_RX_SA_120 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.470 IRQ_TYPE_EN_RX_SA_121

- **Name:** IRQ_TYPE_EN_RX_SA_121
- **Description:** Interrupt Enable for each IRQ type of Rx SA 121
- **Size:** 32 bits
- **Offset:** 0x758
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 121

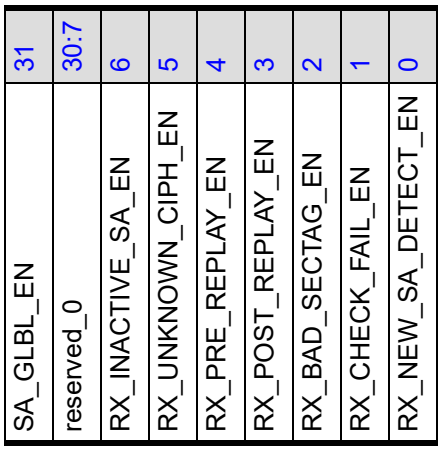


Table 6-477 Fields for Register: IRQ_TYPE_EN_RX_SA_121

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-477 Fields for Register: IRQ_TYPE_EN_RX_SA_121 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-477 Fields for Register: IRQ_TYPE_EN_RX_SA_121 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.471 IRQ_TYPE_STAT_RX_SA_121

- **Name:** IRQ_TYPE_STAT_RX_SA_121
- **Description:** Interrupt Status for each IRQ type of Rx SA 121
- **Size:** 32 bits
- **Offset:** 0x75c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 121

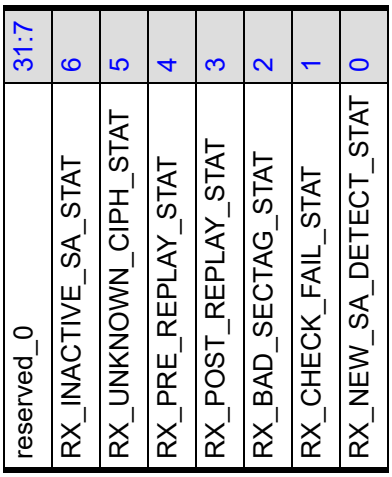


Table 6-478 Fields for Register: IRQ_TYPE_STAT_RX_SA_121

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-478 Fields for Register: IRQ_TYPE_STAT_RX_SA_121 (Continued)

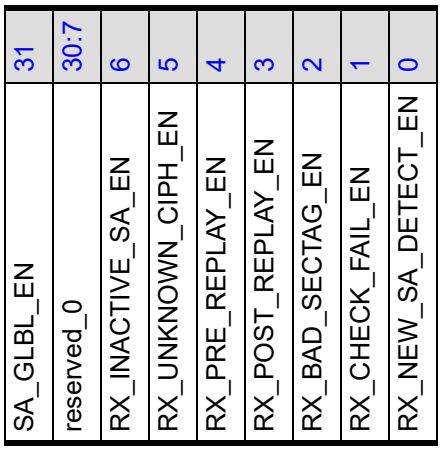
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-478 Fields for Register: IRQ_TYPE_STAT_RX_SA_121 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.472 IRQ_TYPE_EN_RX_SA_122

- **Name:** IRQ_TYPE_EN_RX_SA_122
- **Description:** Interrupt Enable for each IRQ type of Rx SA 122
- **Size:** 32 bits
- **Offset:** 0x760
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 122



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-479 Fields for Register: IRQ_TYPE_EN_RX_SA_122

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-479 Fields for Register: IRQ_TYPE_EN_RX_SA_122 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-479 Fields for Register: IRQ_TYPE_EN_RX_SA_122 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.473 IRQ_TYPE_STAT_RX_SA_122

- **Name:** IRQ_TYPE_STAT_RX_SA_122
- **Description:** Interrupt Status for each IRQ type of Rx SA 122
- **Size:** 32 bits
- **Offset:** 0x764
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 122

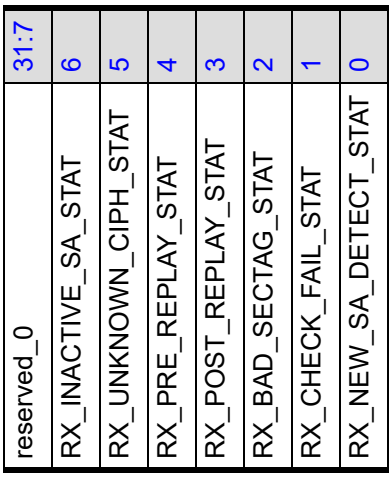


Table 6-480 Fields for Register: IRQ_TYPE_STAT_RX_SA_122

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-480 Fields for Register: IRQ_TYPE_STAT_RX_SA_122 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-480 Fields for Register: IRQ_TYPE_STAT_RX_SA_122 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.474 IRQ_TYPE_EN_RX_SA_123

- **Name:** IRQ_TYPE_EN_RX_SA_123
- **Description:** Interrupt Enable for each IRQ type of Rx SA 123
- **Size:** 32 bits
- **Offset:** 0x768
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 123

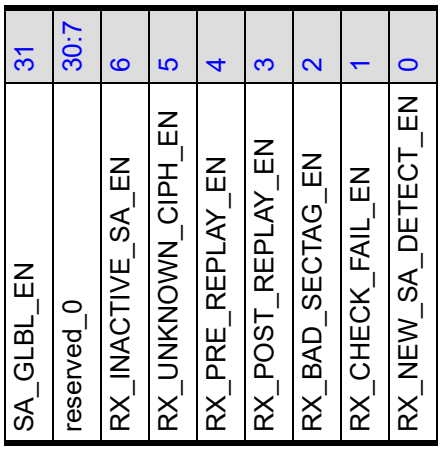


Table 6-481 Fields for Register: IRQ_TYPE_EN_RX_SA_123

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-481 Fields for Register: IRQ_TYPE_EN_RX_SA_123 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-481 Fields for Register: IRQ_TYPE_EN_RX_SA_123 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.475 IRQ_TYPE_STAT_RX_SA_123

- **Name:** IRQ_TYPE_STAT_RX_SA_123
- **Description:** Interrupt Status for each IRQ type of Rx SA 123
- **Size:** 32 bits
- **Offset:** 0x76c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 123

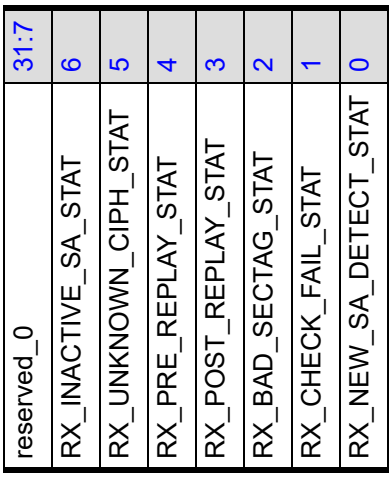


Table 6-482 Fields for Register: IRQ_TYPE_STAT_RX_SA_123

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-482 Fields for Register: IRQ_TYPE_STAT_RX_SA_123 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-482 Fields for Register: IRQ_TYPE_STAT_RX_SA_123 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.476 IRQ_TYPE_EN_RX_SA_124

- **Name:** IRQ_TYPE_EN_RX_SA_124
- **Description:** Interrupt Enable for each IRQ type of Rx SA 124
- **Size:** 32 bits
- **Offset:** 0x770
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 124

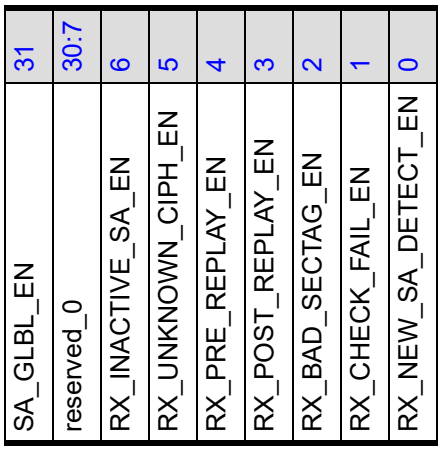


Table 6-483 Fields for Register: IRQ_TYPE_EN_RX_SA_124

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-483 Fields for Register: IRQ_TYPE_EN_RX_SA_124 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-483 Fields for Register: IRQ_TYPE_EN_RX_SA_124 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.477 IRQ_TYPE_STAT_RX_SA_124

- **Name:** IRQ_TYPE_STAT_RX_SA_124
- **Description:** Interrupt Status for each IRQ type of Rx SA 124
- **Size:** 32 bits
- **Offset:** 0x774
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 124

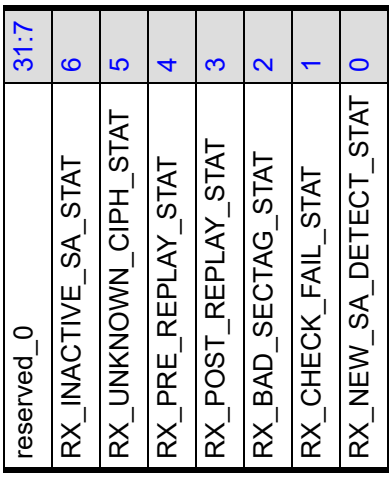


Table 6-484 Fields for Register: IRQ_TYPE_STAT_RX_SA_124

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-484 Fields for Register: IRQ_TYPE_STAT_RX_SA_124 (Continued)

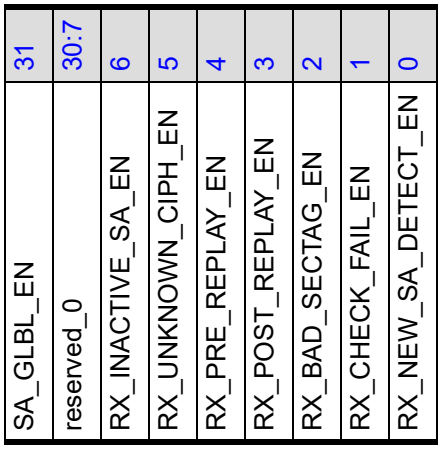
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-484 Fields for Register: IRQ_TYPE_STAT_RX_SA_124 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.478 IRQ_TYPE_EN_RX_SA_125

- **Name:** IRQ_TYPE_EN_RX_SA_125
- **Description:** Interrupt Enable for each IRQ type of Rx SA 125
- **Size:** 32 bits
- **Offset:** 0x778
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 125



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-485 Fields for Register: IRQ_TYPE_EN_RX_SA_125

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-485 Fields for Register: IRQ_TYPE_EN_RX_SA_125 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-485 Fields for Register: IRQ_TYPE_EN_RX_SA_125 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.479 IRQ_TYPE_STAT_RX_SA_125

- **Name:** IRQ_TYPE_STAT_RX_SA_125
- **Description:** Interrupt Status for each IRQ type of Rx SA 125
- **Size:** 32 bits
- **Offset:** 0x77c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 125

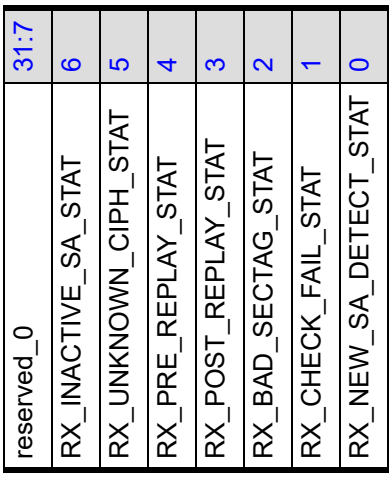


Table 6-486 Fields for Register: IRQ_TYPE_STAT_RX_SA_125

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-486 Fields for Register: IRQ_TYPE_STAT_RX_SA_125 (Continued)

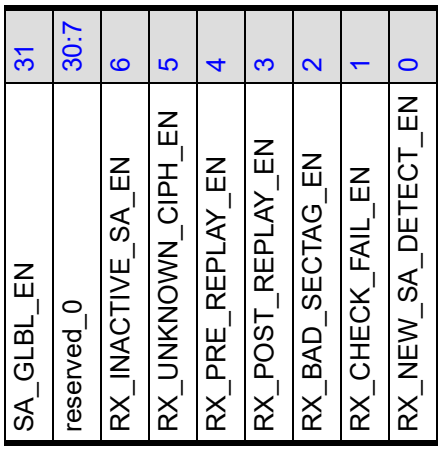
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-486 Fields for Register: IRQ_TYPE_STAT_RX_SA_125 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.480 IRQ_TYPE_EN_RX_SA_126

- **Name:** IRQ_TYPE_EN_RX_SA_126
- **Description:** Interrupt Enable for each IRQ type of Rx SA 126
- **Size:** 32 bits
- **Offset:** 0x780
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 126



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-487 Fields for Register: IRQ_TYPE_EN_RX_SA_126

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-487 Fields for Register: IRQ_TYPE_EN_RX_SA_126 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-487 Fields for Register: IRQ_TYPE_EN_RX_SA_126 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.481 IRQ_TYPE_STAT_RX_SA_126

- **Name:** IRQ_TYPE_STAT_RX_SA_126
- **Description:** Interrupt Status for each IRQ type of Rx SA 126
- **Size:** 32 bits
- **Offset:** 0x784
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 126

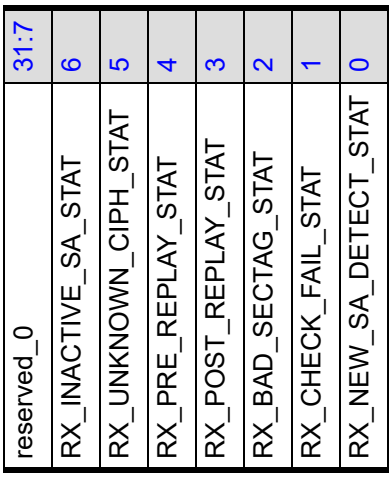


Table 6-488 Fields for Register: IRQ_TYPE_STAT_RX_SA_126

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-488 Fields for Register: IRQ_TYPE_STAT_RX_SA_126 (Continued)

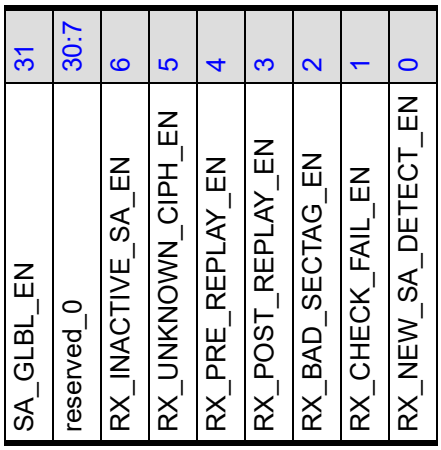
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-488 Fields for Register: IRQ_TYPE_STAT_RX_SA_126 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.482 IRQ_TYPE_EN_RX_SA_127

- **Name:** IRQ_TYPE_EN_RX_SA_127
- **Description:** Interrupt Enable for each IRQ type of Rx SA 127
- **Size:** 32 bits
- **Offset:** 0x788
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 127



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-489 Fields for Register: IRQ_TYPE_EN_RX_SA_127

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-489 Fields for Register: IRQ_TYPE_EN_RX_SA_127 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-489 Fields for Register: IRQ_TYPE_EN_RX_SA_127 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.483 IRQ_TYPE_STAT_RX_SA_127

- **Name:** IRQ_TYPE_STAT_RX_SA_127
- **Description:** Interrupt Status for each IRQ type of Rx SA 127
- **Size:** 32 bits
- **Offset:** 0x78c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 127

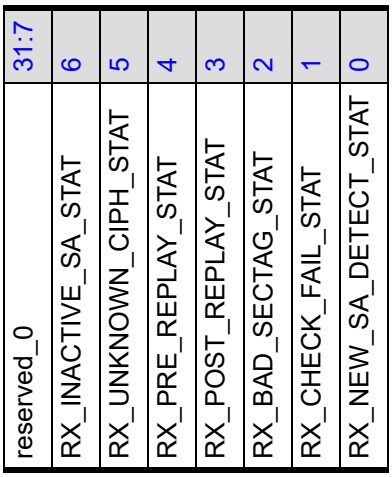


Table 6-490 Fields for Register: IRQ_TYPE_STAT_RX_SA_127

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-490 Fields for Register: IRQ_TYPE_STAT_RX_SA_127 (Continued)

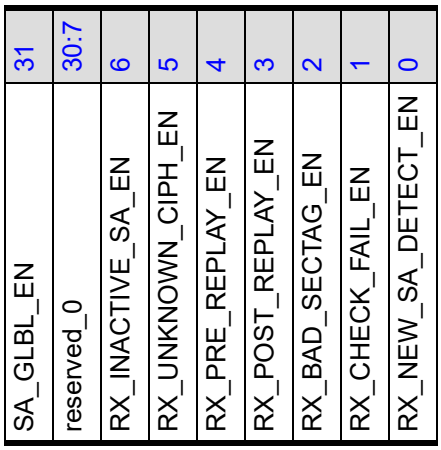
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-490 Fields for Register: IRQ_TYPE_STAT_RX_SA_127 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.484 IRQ_TYPE_EN_RX_SA_128

- **Name:** IRQ_TYPE_EN_RX_SA_128
- **Description:** Interrupt Enable for each IRQ type of Rx SA 128
- **Size:** 32 bits
- **Offset:** 0x790
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 128



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-491 Fields for Register: IRQ_TYPE_EN_RX_SA_128

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-491 Fields for Register: IRQ_TYPE_EN_RX_SA_128 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-491 Fields for Register: IRQ_TYPE_EN_RX_SA_128 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.485 IRQ_TYPE_STAT_RX_SA_128

- **Name:** IRQ_TYPE_STAT_RX_SA_128
- **Description:** Interrupt Status for each IRQ type of Rx SA 128
- **Size:** 32 bits
- **Offset:** 0x794
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 128

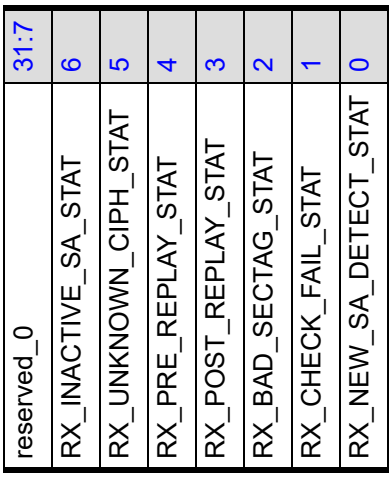


Table 6-492 Fields for Register: IRQ_TYPE_STAT_RX_SA_128

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-492 Fields for Register: IRQ_TYPE_STAT_RX_SA_128 (Continued)

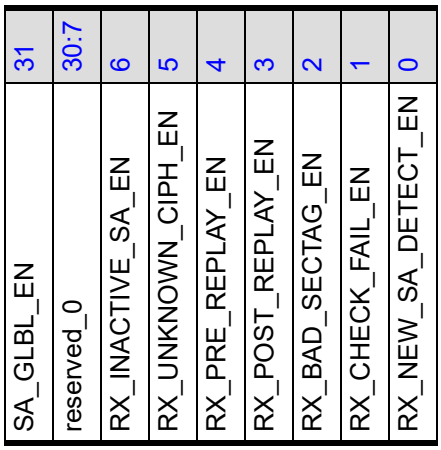
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-492 Fields for Register: IRQ_TYPE_STAT_RX_SA_128 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.486 IRQ_TYPE_EN_RX_SA_129

- **Name:** IRQ_TYPE_EN_RX_SA_129
- **Description:** Interrupt Enable for each IRQ type of Rx SA 129
- **Size:** 32 bits
- **Offset:** 0x798
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 129



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-493 Fields for Register: IRQ_TYPE_EN_RX_SA_129

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-493 Fields for Register: IRQ_TYPE_EN_RX_SA_129 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-493 Fields for Register: IRQ_TYPE_EN_RX_SA_129 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.487 IRQ_TYPE_STAT_RX_SA_129

- **Name:** IRQ_TYPE_STAT_RX_SA_129
- **Description:** Interrupt Status for each IRQ type of Rx SA 129
- **Size:** 32 bits
- **Offset:** 0x79c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 129

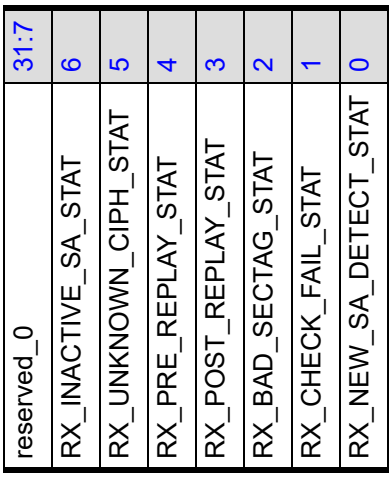


Table 6-494 Fields for Register: IRQ_TYPE_STAT_RX_SA_129

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-494 Fields for Register: IRQ_TYPE_STAT_RX_SA_129 (Continued)

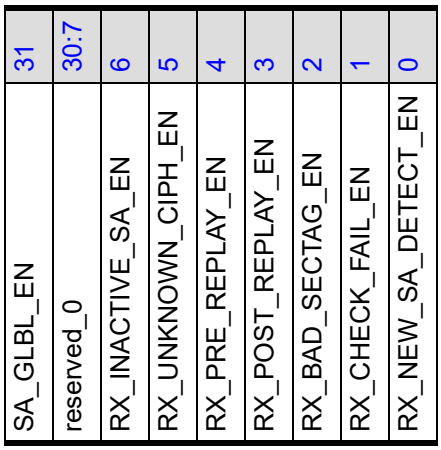
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-494 Fields for Register: IRQ_TYPE_STAT_RX_SA_129 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.488 IRQ_TYPE_EN_RX_SA_130

- **Name:** IRQ_TYPE_EN_RX_SA_130
- **Description:** Interrupt Enable for each IRQ type of Rx SA 130
- **Size:** 32 bits
- **Offset:** 0x7a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 130



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-495 Fields for Register: IRQ_TYPE_EN_RX_SA_130

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-495 Fields for Register: IRQ_TYPE_EN_RX_SA_130 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-495 Fields for Register: IRQ_TYPE_EN_RX_SA_130 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.489 IRQ_TYPE_STAT_RX_SA_130

- **Name:** IRQ_TYPE_STAT_RX_SA_130
- **Description:** Interrupt Status for each IRQ type of Rx SA 130
- **Size:** 32 bits
- **Offset:** 0x7a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 130

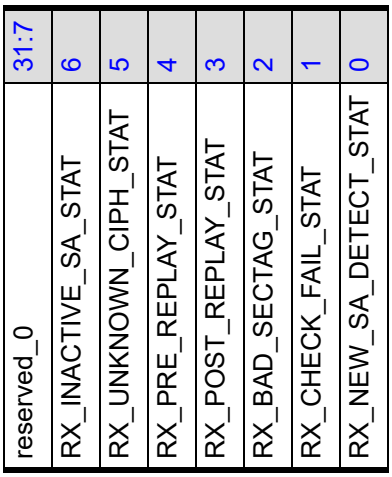


Table 6-496 Fields for Register: IRQ_TYPE_STAT_RX_SA_130

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-496 Fields for Register: IRQ_TYPE_STAT_RX_SA_130 (Continued)

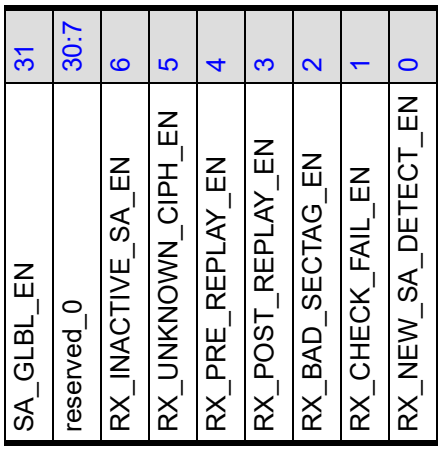
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-496 Fields for Register: IRQ_TYPE_STAT_RX_SA_130 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.490 IRQ_TYPE_EN_RX_SA_131

- **Name:** IRQ_TYPE_EN_RX_SA_131
- **Description:** Interrupt Enable for each IRQ type of Rx SA 131
- **Size:** 32 bits
- **Offset:** 0x7a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 131



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-497 Fields for Register: IRQ_TYPE_EN_RX_SA_131

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-497 Fields for Register: IRQ_TYPE_EN_RX_SA_131 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-497 Fields for Register: IRQ_TYPE_EN_RX_SA_131 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.491 IRQ_TYPE_STAT_RX_SA_131

- **Name:** IRQ_TYPE_STAT_RX_SA_131
- **Description:** Interrupt Status for each IRQ type of Rx SA 131
- **Size:** 32 bits
- **Offset:** 0x7ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 131

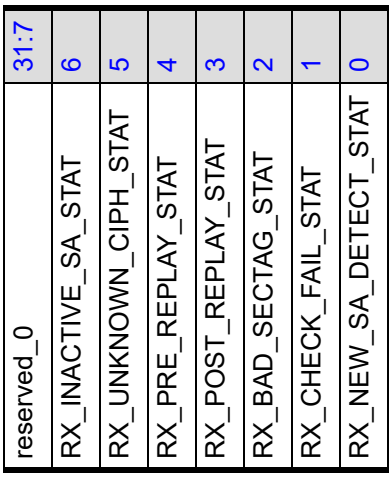


Table 6-498 Fields for Register: IRQ_TYPE_STAT_RX_SA_131

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-498 Fields for Register: IRQ_TYPE_STAT_RX_SA_131 (Continued)

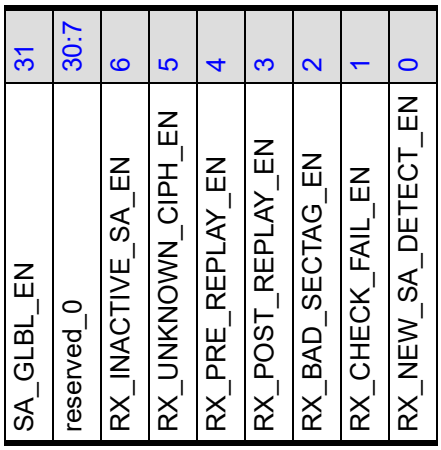
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-498 Fields for Register: IRQ_TYPE_STAT_RX_SA_131 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.492 IRQ_TYPE_EN_RX_SA_132

- **Name:** IRQ_TYPE_EN_RX_SA_132
- **Description:** Interrupt Enable for each IRQ type of Rx SA 132
- **Size:** 32 bits
- **Offset:** 0x7b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 132



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-499 Fields for Register: IRQ_TYPE_EN_RX_SA_132

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-499 Fields for Register: IRQ_TYPE_EN_RX_SA_132 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-499 Fields for Register: IRQ_TYPE_EN_RX_SA_132 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.493 IRQ_TYPE_STAT_RX_SA_132

- **Name:** IRQ_TYPE_STAT_RX_SA_132
- **Description:** Interrupt Status for each IRQ type of Rx SA 132
- **Size:** 32 bits
- **Offset:** 0x7b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 132

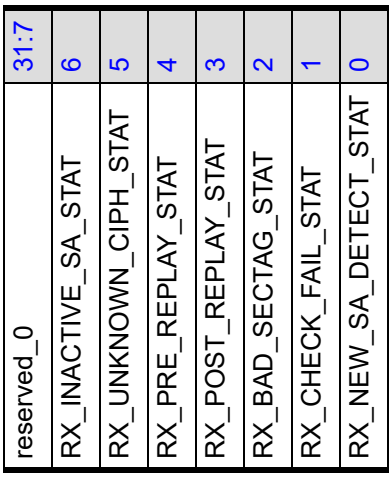


Table 6-500 Fields for Register: IRQ_TYPE_STAT_RX_SA_132

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-500 Fields for Register: IRQ_TYPE_STAT_RX_SA_132 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-500 Fields for Register: IRQ_TYPE_STAT_RX_SA_132 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.494 IRQ_TYPE_EN_RX_SA_133

- **Name:** IRQ_TYPE_EN_RX_SA_133
- **Description:** Interrupt Enable for each IRQ type of Rx SA 133
- **Size:** 32 bits
- **Offset:** 0x7b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 133

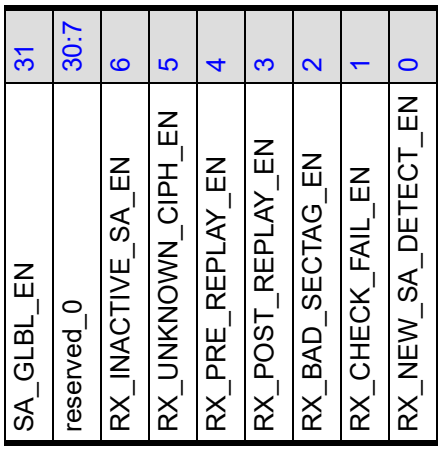


Table 6-501 Fields for Register: IRQ_TYPE_EN_RX_SA_133

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-501 Fields for Register: IRQ_TYPE_EN_RX_SA_133 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-501 Fields for Register: IRQ_TYPE_EN_RX_SA_133 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.495 IRQ_TYPE_STAT_RX_SA_133

- **Name:** IRQ_TYPE_STAT_RX_SA_133
- **Description:** Interrupt Status for each IRQ type of Rx SA 133
- **Size:** 32 bits
- **Offset:** 0x7bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 133

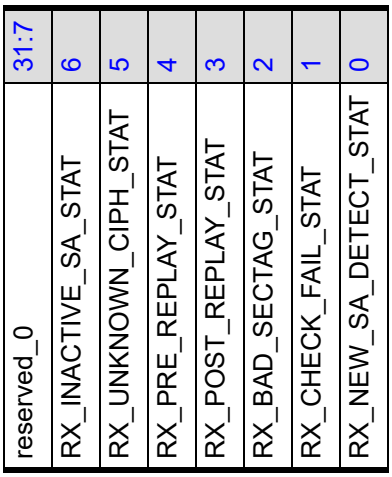


Table 6-502 Fields for Register: IRQ_TYPE_STAT_RX_SA_133

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-502 Fields for Register: IRQ_TYPE_STAT_RX_SA_133 (Continued)

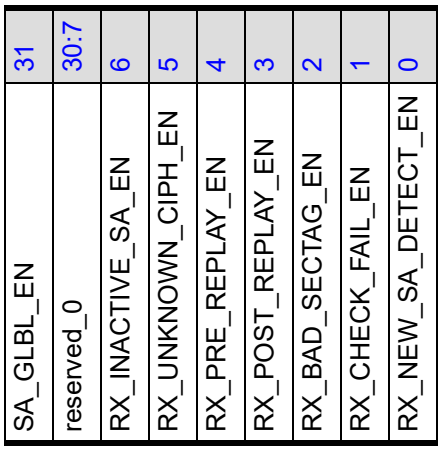
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-502 Fields for Register: IRQ_TYPE_STAT_RX_SA_133 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.496 IRQ_TYPE_EN_RX_SA_134

- **Name:** IRQ_TYPE_EN_RX_SA_134
- **Description:** Interrupt Enable for each IRQ type of Rx SA 134
- **Size:** 32 bits
- **Offset:** 0x7c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 134



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-503 Fields for Register: IRQ_TYPE_EN_RX_SA_134

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-503 Fields for Register: IRQ_TYPE_EN_RX_SA_134 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-503 Fields for Register: IRQ_TYPE_EN_RX_SA_134 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.497 IRQ_TYPE_STAT_RX_SA_134

- **Name:** IRQ_TYPE_STAT_RX_SA_134
- **Description:** Interrupt Status for each IRQ type of Rx SA 134
- **Size:** 32 bits
- **Offset:** 0x7c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 134

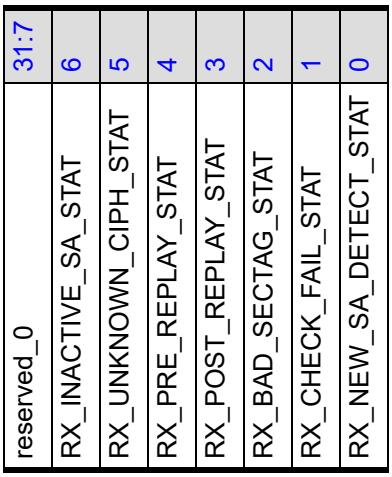


Table 6-504 Fields for Register: IRQ_TYPE_STAT_RX_SA_134

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-504 Fields for Register: IRQ_TYPE_STAT_RX_SA_134 (Continued)

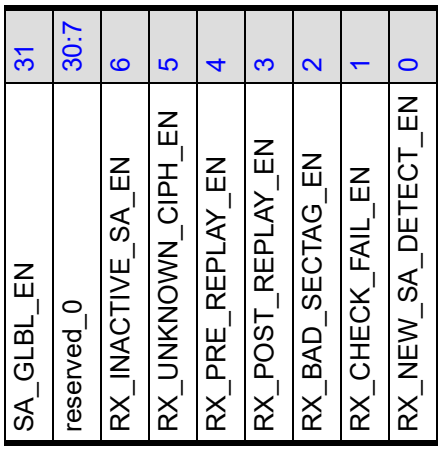
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-504 Fields for Register: IRQ_TYPE_STAT_RX_SA_134 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.498 IRQ_TYPE_EN_RX_SA_135

- **Name:** IRQ_TYPE_EN_RX_SA_135
- **Description:** Interrupt Enable for each IRQ type of Rx SA 135
- **Size:** 32 bits
- **Offset:** 0x7c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 135



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-505 Fields for Register: IRQ_TYPE_EN_RX_SA_135

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-505 Fields for Register: IRQ_TYPE_EN_RX_SA_135 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-505 Fields for Register: IRQ_TYPE_EN_RX_SA_135 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.499 IRQ_TYPE_STAT_RX_SA_135

- **Name:** IRQ_TYPE_STAT_RX_SA_135
- **Description:** Interrupt Status for each IRQ type of Rx SA 135
- **Size:** 32 bits
- **Offset:** 0x7cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 135

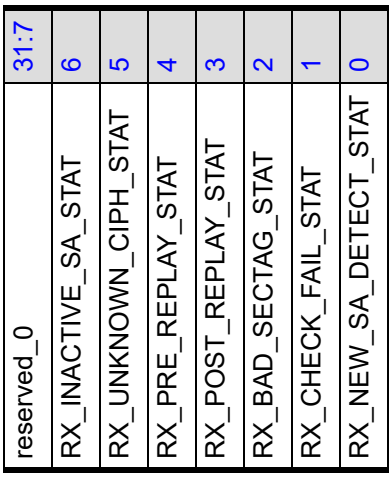


Table 6-506 Fields for Register: IRQ_TYPE_STAT_RX_SA_135

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-506 Fields for Register: IRQ_TYPE_STAT_RX_SA_135 (Continued)

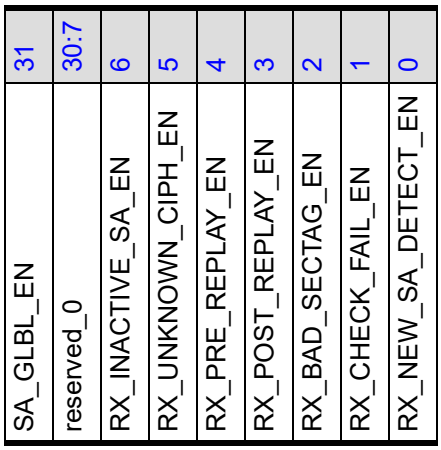
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-506 Fields for Register: IRQ_TYPE_STAT_RX_SA_135 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.500 IRQ_TYPE_EN_RX_SA_136

- **Name:** IRQ_TYPE_EN_RX_SA_136
- **Description:** Interrupt Enable for each IRQ type of Rx SA 136
- **Size:** 32 bits
- **Offset:** 0x7d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 136



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-507 Fields for Register: IRQ_TYPE_EN_RX_SA_136

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-507 Fields for Register: IRQ_TYPE_EN_RX_SA_136 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-507 Fields for Register: IRQ_TYPE_EN_RX_SA_136 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.501 IRQ_TYPE_STAT_RX_SA_136

- **Name:** IRQ_TYPE_STAT_RX_SA_136
- **Description:** Interrupt Status for each IRQ type of Rx SA 136
- **Size:** 32 bits
- **Offset:** 0x7d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 136

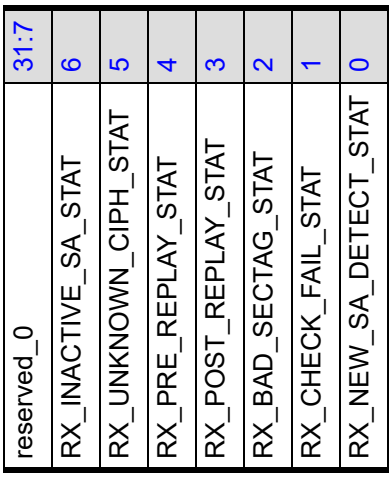


Table 6-508 Fields for Register: IRQ_TYPE_STAT_RX_SA_136

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-508 Fields for Register: IRQ_TYPE_STAT_RX_SA_136 (Continued)

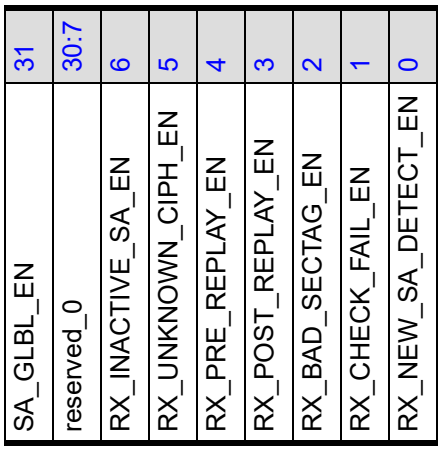
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-508 Fields for Register: IRQ_TYPE_STAT_RX_SA_136 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.502 IRQ_TYPE_EN_RX_SA_137

- **Name:** IRQ_TYPE_EN_RX_SA_137
- **Description:** Interrupt Enable for each IRQ type of Rx SA 137
- **Size:** 32 bits
- **Offset:** 0x7d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 137



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-509 Fields for Register: IRQ_TYPE_EN_RX_SA_137

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-509 Fields for Register: IRQ_TYPE_EN_RX_SA_137 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-509 Fields for Register: IRQ_TYPE_EN_RX_SA_137 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.503 IRQ_TYPE_STAT_RX_SA_137

- **Name:** IRQ_TYPE_STAT_RX_SA_137
- **Description:** Interrupt Status for each IRQ type of Rx SA 137
- **Size:** 32 bits
- **Offset:** 0x7dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 137

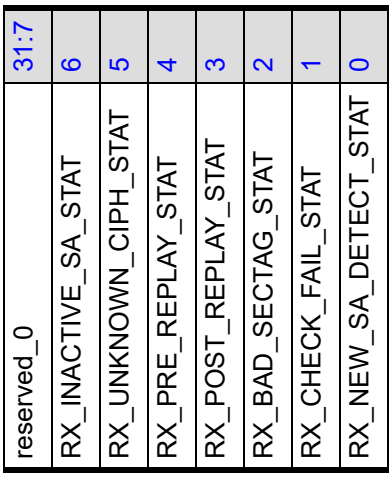


Table 6-510 Fields for Register: IRQ_TYPE_STAT_RX_SA_137

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-510 Fields for Register: IRQ_TYPE_STAT_RX_SA_137 (Continued)

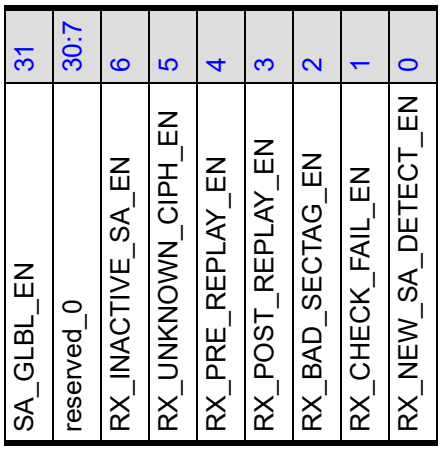
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-510 Fields for Register: IRQ_TYPE_STAT_RX_SA_137 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.504 IRQ_TYPE_EN_RX_SA_138

- **Name:** IRQ_TYPE_EN_RX_SA_138
- **Description:** Interrupt Enable for each IRQ type of Rx SA 138
- **Size:** 32 bits
- **Offset:** 0x7e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 138



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-511 Fields for Register: IRQ_TYPE_EN_RX_SA_138

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-511 Fields for Register: IRQ_TYPE_EN_RX_SA_138 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-511 Fields for Register: IRQ_TYPE_EN_RX_SA_138 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.505 IRQ_TYPE_STAT_RX_SA_138

- **Name:** IRQ_TYPE_STAT_RX_SA_138
- **Description:** Interrupt Status for each IRQ type of Rx SA 138
- **Size:** 32 bits
- **Offset:** 0x7e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 138

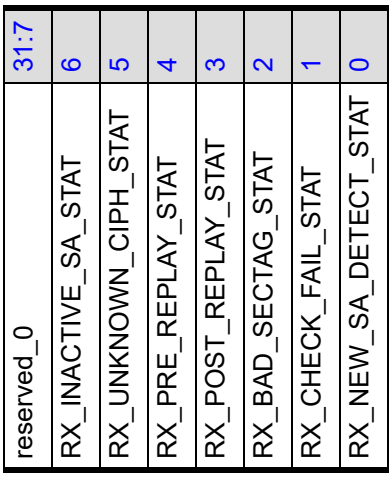


Table 6-512 Fields for Register: IRQ_TYPE_STAT_RX_SA_138

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-512 Fields for Register: IRQ_TYPE_STAT_RX_SA_138 (Continued)

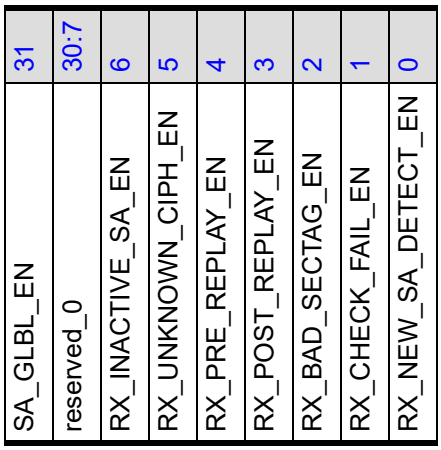
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-512 Fields for Register: IRQ_TYPE_STAT_RX_SA_138 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.506 IRQ_TYPE_EN_RX_SA_139

- **Name:** IRQ_TYPE_EN_RX_SA_139
- **Description:** Interrupt Enable for each IRQ type of Rx SA 139
- **Size:** 32 bits
- **Offset:** 0x7e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 139



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-513 Fields for Register: IRQ_TYPE_EN_RX_SA_139

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-513 Fields for Register: IRQ_TYPE_EN_RX_SA_139 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-513 Fields for Register: IRQ_TYPE_EN_RX_SA_139 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.507 IRQ_TYPE_STAT_RX_SA_139

- **Name:** IRQ_TYPE_STAT_RX_SA_139
- **Description:** Interrupt Status for each IRQ type of Rx SA 139
- **Size:** 32 bits
- **Offset:** 0x7ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 139

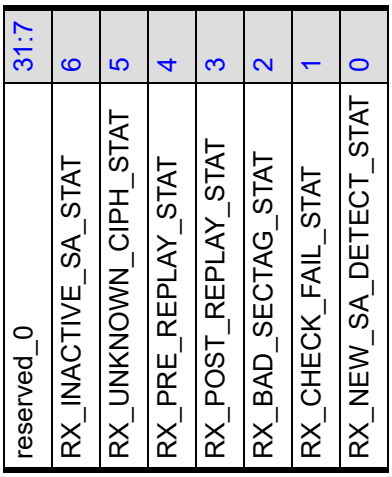


Table 6-514 Fields for Register: IRQ_TYPE_STAT_RX_SA_139

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-514 Fields for Register: IRQ_TYPE_STAT_RX_SA_139 (Continued)

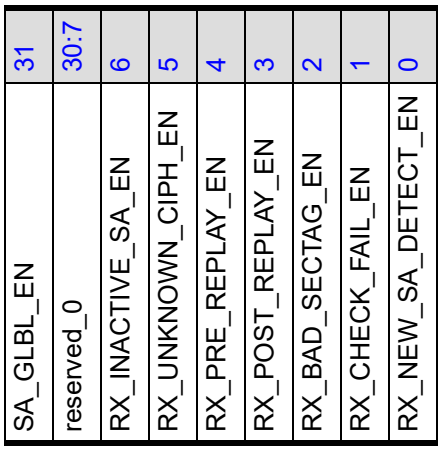
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-514 Fields for Register: IRQ_TYPE_STAT_RX_SA_139 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.508 IRQ_TYPE_EN_RX_SA_140

- **Name:** IRQ_TYPE_EN_RX_SA_140
- **Description:** Interrupt Enable for each IRQ type of Rx SA 140
- **Size:** 32 bits
- **Offset:** 0x7f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 140



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-515 Fields for Register: IRQ_TYPE_EN_RX_SA_140

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-515 Fields for Register: IRQ_TYPE_EN_RX_SA_140 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-515 Fields for Register: IRQ_TYPE_EN_RX_SA_140 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.509 IRQ_TYPE_STAT_RX_SA_140

- **Name:** IRQ_TYPE_STAT_RX_SA_140
- **Description:** Interrupt Status for each IRQ type of Rx SA 140
- **Size:** 32 bits
- **Offset:** 0x7f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 140

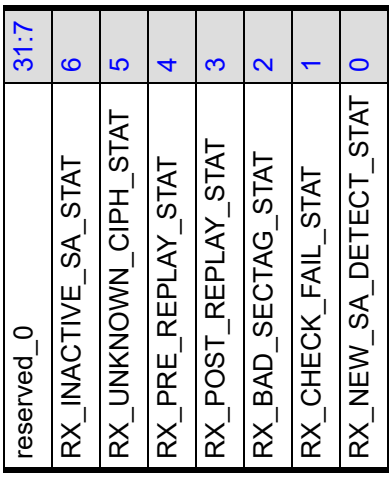


Table 6-516 Fields for Register: IRQ_TYPE_STAT_RX_SA_140

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-516 Fields for Register: IRQ_TYPE_STAT_RX_SA_140 (Continued)

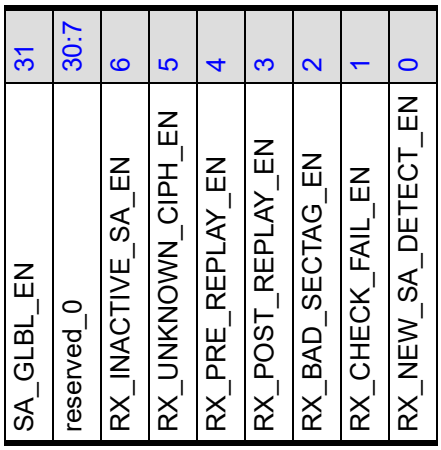
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-516 Fields for Register: IRQ_TYPE_STAT_RX_SA_140 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.510 IRQ_TYPE_EN_RX_SA_141

- **Name:** IRQ_TYPE_EN_RX_SA_141
- **Description:** Interrupt Enable for each IRQ type of Rx SA 141
- **Size:** 32 bits
- **Offset:** 0x7f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 141



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-517 Fields for Register: IRQ_TYPE_EN_RX_SA_141

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-517 Fields for Register: IRQ_TYPE_EN_RX_SA_141 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-517 Fields for Register: IRQ_TYPE_EN_RX_SA_141 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.511 IRQ_TYPE_STAT_RX_SA_141

- **Name:** IRQ_TYPE_STAT_RX_SA_141
- **Description:** Interrupt Status for each IRQ type of Rx SA 141
- **Size:** 32 bits
- **Offset:** 0x7fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 141

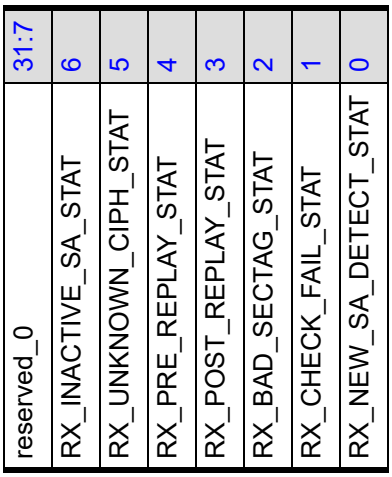


Table 6-518 Fields for Register: IRQ_TYPE_STAT_RX_SA_141

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-518 Fields for Register: IRQ_TYPE_STAT_RX_SA_141 (Continued)

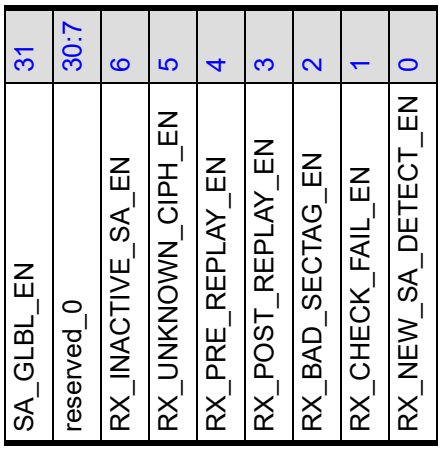
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-518 Fields for Register: IRQ_TYPE_STAT_RX_SA_141 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.512 IRQ_TYPE_EN_RX_SA_142

- **Name:** IRQ_TYPE_EN_RX_SA_142
- **Description:** Interrupt Enable for each IRQ type of Rx SA 142
- **Size:** 32 bits
- **Offset:** 0x800
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 142



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-519 Fields for Register: IRQ_TYPE_EN_RX_SA_142

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-519 Fields for Register: IRQ_TYPE_EN_RX_SA_142 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-519 Fields for Register: IRQ_TYPE_EN_RX_SA_142 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.513 IRQ_TYPE_STAT_RX_SA_142

- **Name:** IRQ_TYPE_STAT_RX_SA_142
- **Description:** Interrupt Status for each IRQ type of Rx SA 142
- **Size:** 32 bits
- **Offset:** 0x804
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 142

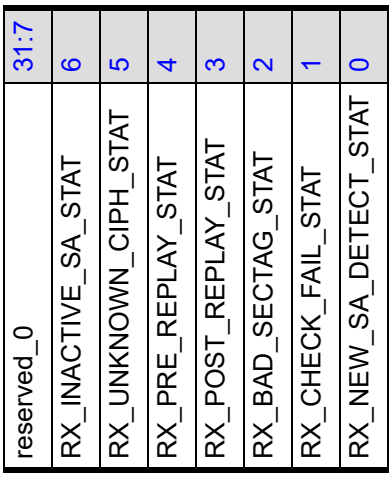


Table 6-520 Fields for Register: IRQ_TYPE_STAT_RX_SA_142

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-520 Fields for Register: IRQ_TYPE_STAT_RX_SA_142 (Continued)

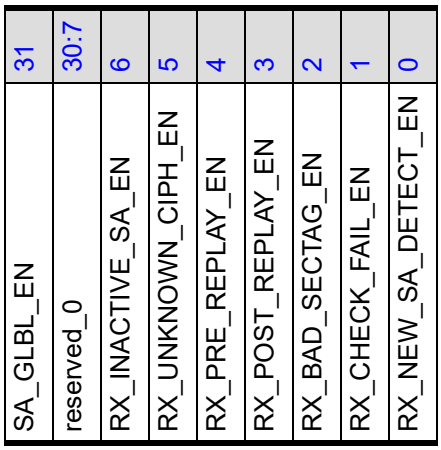
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-520 Fields for Register: IRQ_TYPE_STAT_RX_SA_142 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.514 IRQ_TYPE_EN_RX_SA_143

- **Name:** IRQ_TYPE_EN_RX_SA_143
- **Description:** Interrupt Enable for each IRQ type of Rx SA 143
- **Size:** 32 bits
- **Offset:** 0x808
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 143



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-521 Fields for Register: IRQ_TYPE_EN_RX_SA_143

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-521 Fields for Register: IRQ_TYPE_EN_RX_SA_143 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-521 Fields for Register: IRQ_TYPE_EN_RX_SA_143 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.515 IRQ_TYPE_STAT_RX_SA_143

- **Name:** IRQ_TYPE_STAT_RX_SA_143
- **Description:** Interrupt Status for each IRQ type of Rx SA 143
- **Size:** 32 bits
- **Offset:** 0x80c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 143

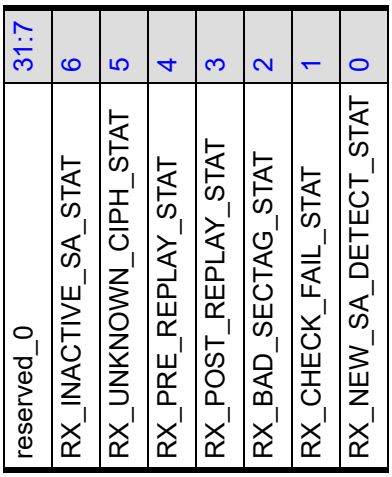


Table 6-522 Fields for Register: IRQ_TYPE_STAT_RX_SA_143

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-522 Fields for Register: IRQ_TYPE_STAT_RX_SA_143 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-522 Fields for Register: IRQ_TYPE_STAT_RX_SA_143 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.516 IRQ_TYPE_EN_RX_SA_144

- **Name:** IRQ_TYPE_EN_RX_SA_144
- **Description:** Interrupt Enable for each IRQ type of Rx SA 144
- **Size:** 32 bits
- **Offset:** 0x810
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 144

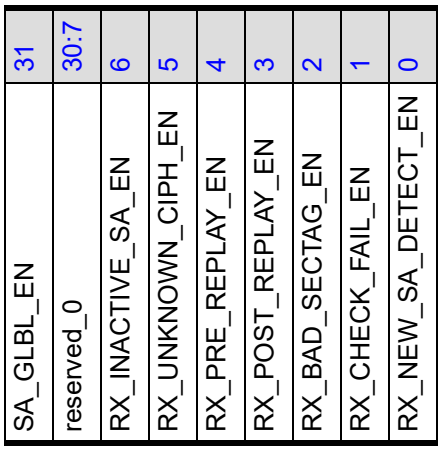


Table 6-523 Fields for Register: IRQ_TYPE_EN_RX_SA_144

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-523 Fields for Register: IRQ_TYPE_EN_RX_SA_144 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-523 Fields for Register: IRQ_TYPE_EN_RX_SA_144 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.517 IRQ_TYPE_STAT_RX_SA_144

- **Name:** IRQ_TYPE_STAT_RX_SA_144
- **Description:** Interrupt Status for each IRQ type of Rx SA 144
- **Size:** 32 bits
- **Offset:** 0x814
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 144

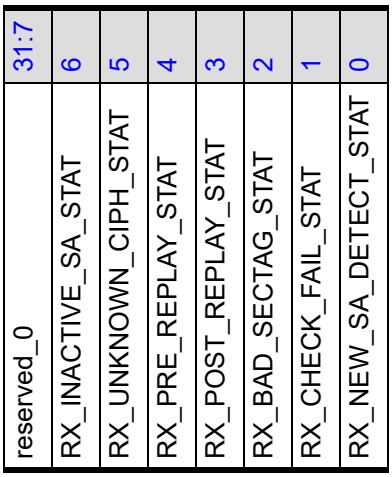


Table 6-524 Fields for Register: IRQ_TYPE_STAT_RX_SA_144

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-524 Fields for Register: IRQ_TYPE_STAT_RX_SA_144 (Continued)

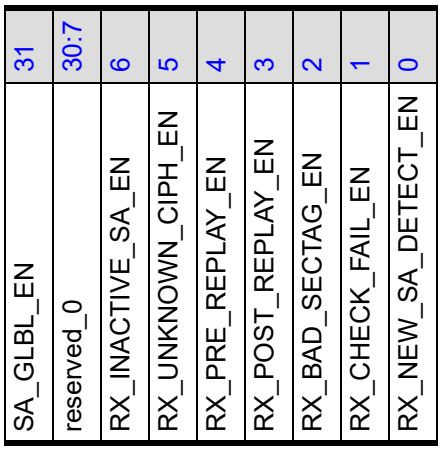
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-524 Fields for Register: IRQ_TYPE_STAT_RX_SA_144 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.518 IRQ_TYPE_EN_RX_SA_145

- **Name:** IRQ_TYPE_EN_RX_SA_145
- **Description:** Interrupt Enable for each IRQ type of Rx SA 145
- **Size:** 32 bits
- **Offset:** 0x818
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 145



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-525 Fields for Register: IRQ_TYPE_EN_RX_SA_145

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-525 Fields for Register: IRQ_TYPE_EN_RX_SA_145 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-525 Fields for Register: IRQ_TYPE_EN_RX_SA_145 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.519 IRQ_TYPE_STAT_RX_SA_145

- **Name:** IRQ_TYPE_STAT_RX_SA_145
- **Description:** Interrupt Status for each IRQ type of Rx SA 145
- **Size:** 32 bits
- **Offset:** 0x81c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 145

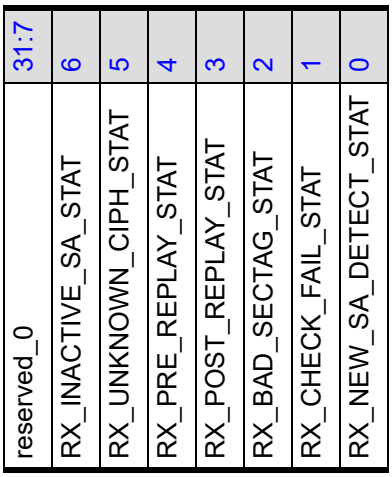


Table 6-526 Fields for Register: IRQ_TYPE_STAT_RX_SA_145

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-526 Fields for Register: IRQ_TYPE_STAT_RX_SA_145 (Continued)

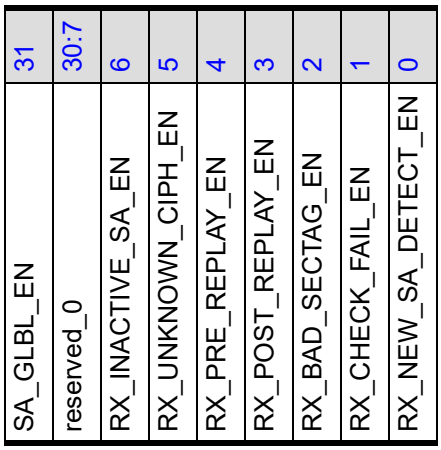
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-526 Fields for Register: IRQ_TYPE_STAT_RX_SA_145 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.520 IRQ_TYPE_EN_RX_SA_146

- **Name:** IRQ_TYPE_EN_RX_SA_146
- **Description:** Interrupt Enable for each IRQ type of Rx SA 146
- **Size:** 32 bits
- **Offset:** 0x820
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 146



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-527 Fields for Register: IRQ_TYPE_EN_RX_SA_146

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-527 Fields for Register: IRQ_TYPE_EN_RX_SA_146 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-527 Fields for Register: IRQ_TYPE_EN_RX_SA_146 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.521 IRQ_TYPE_STAT_RX_SA_146

- **Name:** IRQ_TYPE_STAT_RX_SA_146
- **Description:** Interrupt Status for each IRQ type of Rx SA 146
- **Size:** 32 bits
- **Offset:** 0x824
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 146

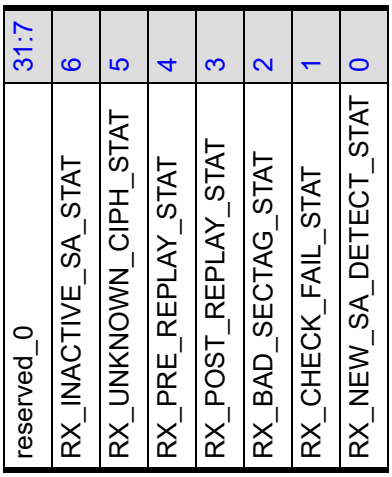


Table 6-528 Fields for Register: IRQ_TYPE_STAT_RX_SA_146

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-528 Fields for Register: IRQ_TYPE_STAT_RX_SA_146 (Continued)

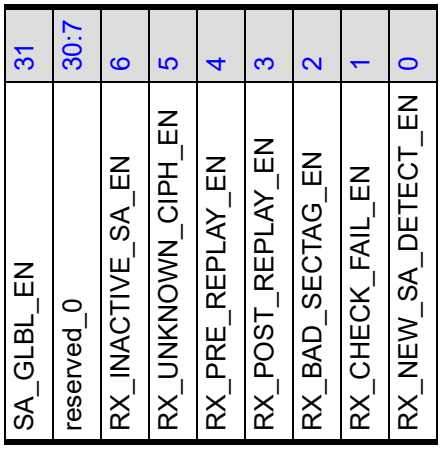
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-528 Fields for Register: IRQ_TYPE_STAT_RX_SA_146 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.522 IRQ_TYPE_EN_RX_SA_147

- **Name:** IRQ_TYPE_EN_RX_SA_147
- **Description:** Interrupt Enable for each IRQ type of Rx SA 147
- **Size:** 32 bits
- **Offset:** 0x828
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 147



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-529 Fields for Register: IRQ_TYPE_EN_RX_SA_147

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-529 Fields for Register: IRQ_TYPE_EN_RX_SA_147 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-529 Fields for Register: IRQ_TYPE_EN_RX_SA_147 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.523 IRQ_TYPE_STAT_RX_SA_147

- **Name:** IRQ_TYPE_STAT_RX_SA_147
- **Description:** Interrupt Status for each IRQ type of Rx SA 147
- **Size:** 32 bits
- **Offset:** 0x82c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 147

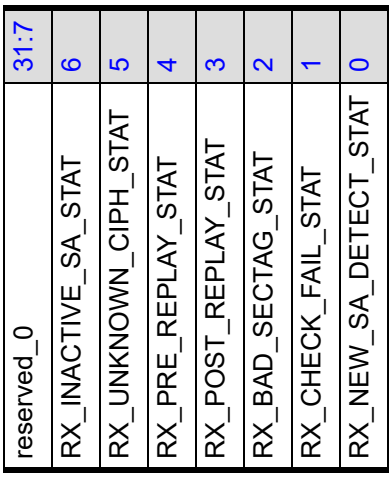


Table 6-530 Fields for Register: IRQ_TYPE_STAT_RX_SA_147

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-530 Fields for Register: IRQ_TYPE_STAT_RX_SA_147 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-530 Fields for Register: IRQ_TYPE_STAT_RX_SA_147 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.524 IRQ_TYPE_EN_RX_SA_148

- **Name:** IRQ_TYPE_EN_RX_SA_148
- **Description:** Interrupt Enable for each IRQ type of Rx SA 148
- **Size:** 32 bits
- **Offset:** 0x830
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 148

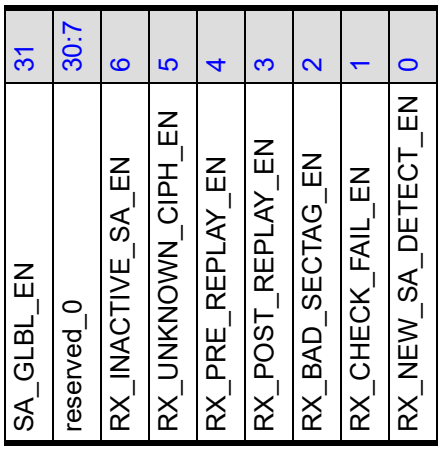


Table 6-531 Fields for Register: IRQ_TYPE_EN_RX_SA_148

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-531 Fields for Register: IRQ_TYPE_EN_RX_SA_148 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-531 Fields for Register: IRQ_TYPE_EN_RX_SA_148 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.525 IRQ_TYPE_STAT_RX_SA_148

- **Name:** IRQ_TYPE_STAT_RX_SA_148
- **Description:** Interrupt Status for each IRQ type of Rx SA 148
- **Size:** 32 bits
- **Offset:** 0x834
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 148

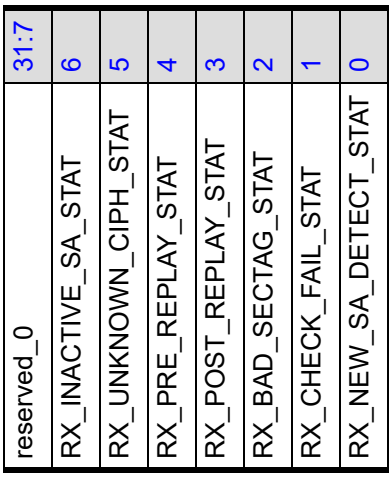


Table 6-532 Fields for Register: IRQ_TYPE_STAT_RX_SA_148

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-532 Fields for Register: IRQ_TYPE_STAT_RX_SA_148 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-532 Fields for Register: IRQ_TYPE_STAT_RX_SA_148 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.526 IRQ_TYPE_EN_RX_SA_149

- **Name:** IRQ_TYPE_EN_RX_SA_149
- **Description:** Interrupt Enable for each IRQ type of Rx SA 149
- **Size:** 32 bits
- **Offset:** 0x838
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 149

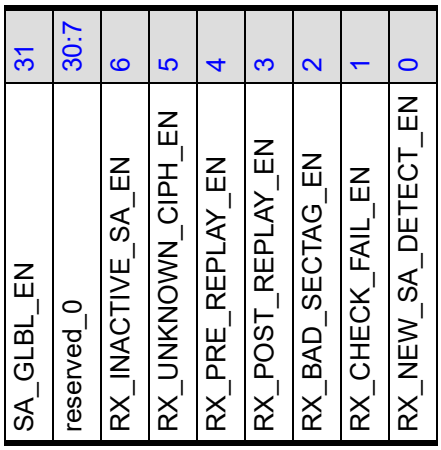


Table 6-533 Fields for Register: IRQ_TYPE_EN_RX_SA_149

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-533 Fields for Register: IRQ_TYPE_EN_RX_SA_149 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-533 Fields for Register: IRQ_TYPE_EN_RX_SA_149 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.527 IRQ_TYPE_STAT_RX_SA_149

- **Name:** IRQ_TYPE_STAT_RX_SA_149
- **Description:** Interrupt Status for each IRQ type of Rx SA 149
- **Size:** 32 bits
- **Offset:** 0x83c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 149

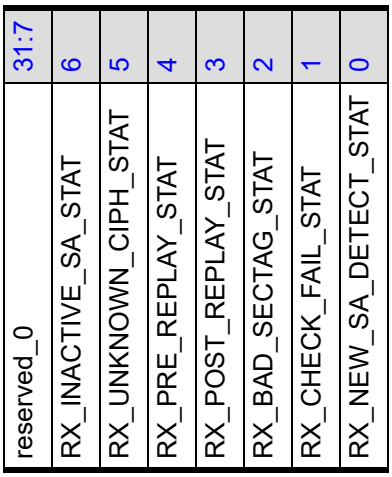


Table 6-534 Fields for Register: IRQ_TYPE_STAT_RX_SA_149

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-534 Fields for Register: IRQ_TYPE_STAT_RX_SA_149 (Continued)

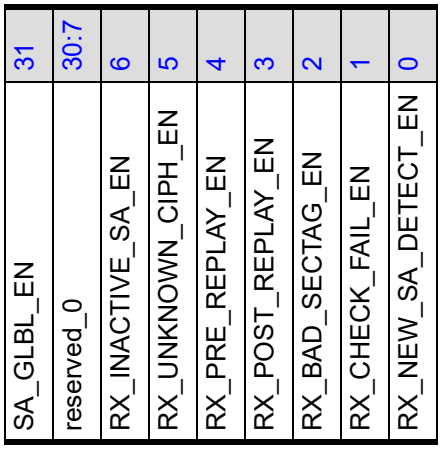
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-534 Fields for Register: IRQ_TYPE_STAT_RX_SA_149 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.528 IRQ_TYPE_EN_RX_SA_150

- **Name:** IRQ_TYPE_EN_RX_SA_150
- **Description:** Interrupt Enable for each IRQ type of Rx SA 150
- **Size:** 32 bits
- **Offset:** 0x840
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 150



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-535 Fields for Register: IRQ_TYPE_EN_RX_SA_150

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-535 Fields for Register: IRQ_TYPE_EN_RX_SA_150 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-535 Fields for Register: IRQ_TYPE_EN_RX_SA_150 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.529 IRQ_TYPE_STAT_RX_SA_150

- **Name:** IRQ_TYPE_STAT_RX_SA_150
- **Description:** Interrupt Status for each IRQ type of Rx SA 150
- **Size:** 32 bits
- **Offset:** 0x844
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 150

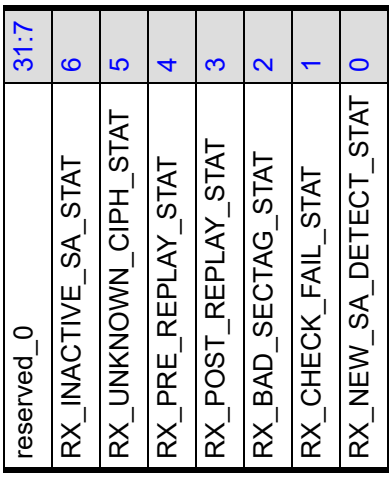


Table 6-536 Fields for Register: IRQ_TYPE_STAT_RX_SA_150

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-536 Fields for Register: IRQ_TYPE_STAT_RX_SA_150 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-536 Fields for Register: IRQ_TYPE_STAT_RX_SA_150 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.530 IRQ_TYPE_EN_RX_SA_151

- **Name:** IRQ_TYPE_EN_RX_SA_151
- **Description:** Interrupt Enable for each IRQ type of Rx SA 151
- **Size:** 32 bits
- **Offset:** 0x848
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 151

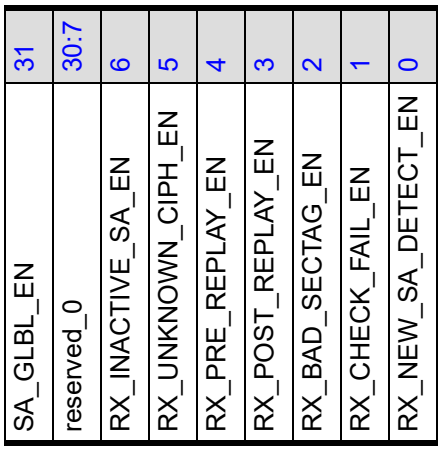


Table 6-537 Fields for Register: IRQ_TYPE_EN_RX_SA_151

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-537 Fields for Register: IRQ_TYPE_EN_RX_SA_151 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-537 Fields for Register: IRQ_TYPE_EN_RX_SA_151 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.531 IRQ_TYPE_STAT_RX_SA_151

- **Name:** IRQ_TYPE_STAT_RX_SA_151
- **Description:** Interrupt Status for each IRQ type of Rx SA 151
- **Size:** 32 bits
- **Offset:** 0x84c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 151

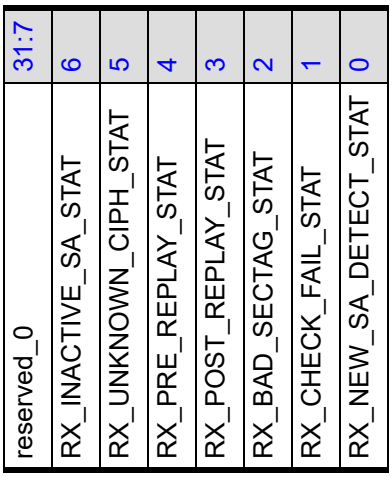


Table 6-538 Fields for Register: IRQ_TYPE_STAT_RX_SA_151

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-538 Fields for Register: IRQ_TYPE_STAT_RX_SA_151 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-538 Fields for Register: IRQ_TYPE_STAT_RX_SA_151 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.532 IRQ_TYPE_EN_RX_SA_152

- **Name:** IRQ_TYPE_EN_RX_SA_152
- **Description:** Interrupt Enable for each IRQ type of Rx SA 152
- **Size:** 32 bits
- **Offset:** 0x850
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 152

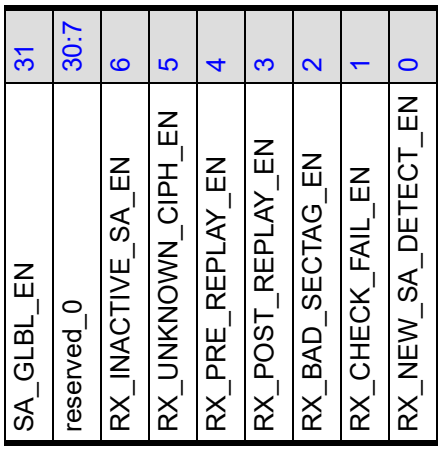


Table 6-539 Fields for Register: IRQ_TYPE_EN_RX_SA_152

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-539 Fields for Register: IRQ_TYPE_EN_RX_SA_152 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-539 Fields for Register: IRQ_TYPE_EN_RX_SA_152 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.533 IRQ_TYPE_STAT_RX_SA_152

- **Name:** IRQ_TYPE_STAT_RX_SA_152
- **Description:** Interrupt Status for each IRQ type of Rx SA 152
- **Size:** 32 bits
- **Offset:** 0x854
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 152

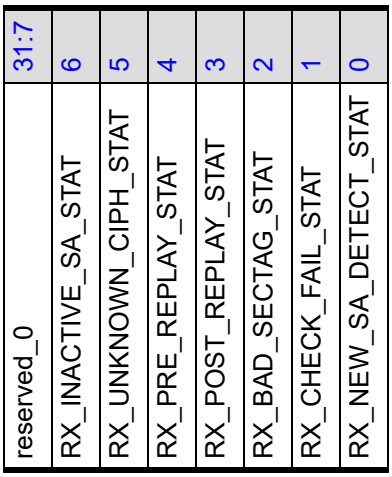


Table 6-540 Fields for Register: IRQ_TYPE_STAT_RX_SA_152

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-540 Fields for Register: IRQ_TYPE_STAT_RX_SA_152 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-540 Fields for Register: IRQ_TYPE_STAT_RX_SA_152 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.534 IRQ_TYPE_EN_RX_SA_153

- **Name:** IRQ_TYPE_EN_RX_SA_153
- **Description:** Interrupt Enable for each IRQ type of Rx SA 153
- **Size:** 32 bits
- **Offset:** 0x858
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 153

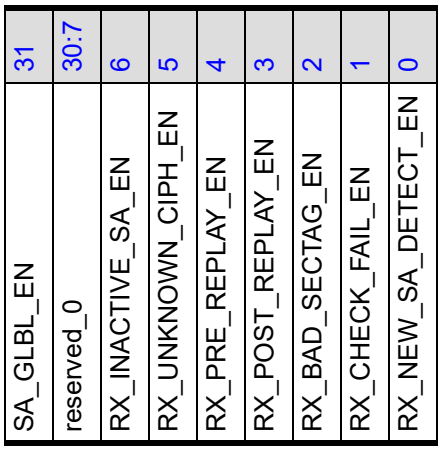


Table 6-541 Fields for Register: IRQ_TYPE_EN_RX_SA_153

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-541 Fields for Register: IRQ_TYPE_EN_RX_SA_153 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-541 Fields for Register: IRQ_TYPE_EN_RX_SA_153 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.535 IRQ_TYPE_STAT_RX_SA_153

- **Name:** IRQ_TYPE_STAT_RX_SA_153
- **Description:** Interrupt Status for each IRQ type of Rx SA 153
- **Size:** 32 bits
- **Offset:** 0x85c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 153

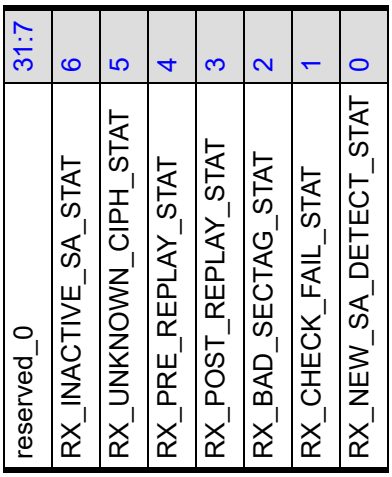


Table 6-542 Fields for Register: IRQ_TYPE_STAT_RX_SA_153

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-542 Fields for Register: IRQ_TYPE_STAT_RX_SA_153 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-542 Fields for Register: IRQ_TYPE_STAT_RX_SA_153 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.536 IRQ_TYPE_EN_RX_SA_154

- **Name:** IRQ_TYPE_EN_RX_SA_154
- **Description:** Interrupt Enable for each IRQ type of Rx SA 154
- **Size:** 32 bits
- **Offset:** 0x860
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 154

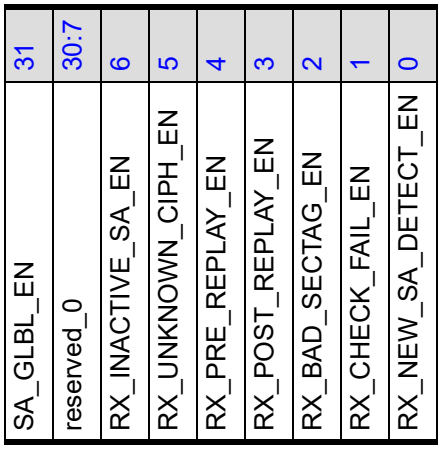


Table 6-543 Fields for Register: IRQ_TYPE_EN_RX_SA_154

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-543 Fields for Register: IRQ_TYPE_EN_RX_SA_154 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-543 Fields for Register: IRQ_TYPE_EN_RX_SA_154 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.537 IRQ_TYPE_STAT_RX_SA_154

- **Name:** IRQ_TYPE_STAT_RX_SA_154
- **Description:** Interrupt Status for each IRQ type of Rx SA 154
- **Size:** 32 bits
- **Offset:** 0x864
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 154

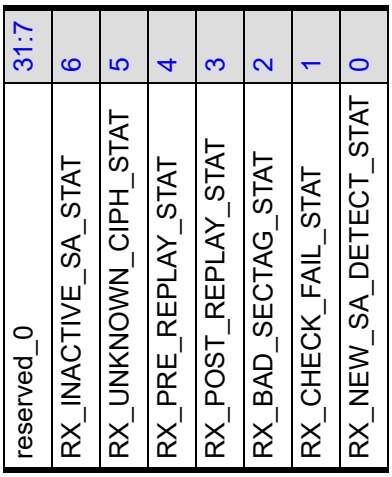


Table 6-544 Fields for Register: IRQ_TYPE_STAT_RX_SA_154

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-544 Fields for Register: IRQ_TYPE_STAT_RX_SA_154 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-544 Fields for Register: IRQ_TYPE_STAT_RX_SA_154 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.538 IRQ_TYPE_EN_RX_SA_155

- **Name:** IRQ_TYPE_EN_RX_SA_155
- **Description:** Interrupt Enable for each IRQ type of Rx SA 155
- **Size:** 32 bits
- **Offset:** 0x868
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 155

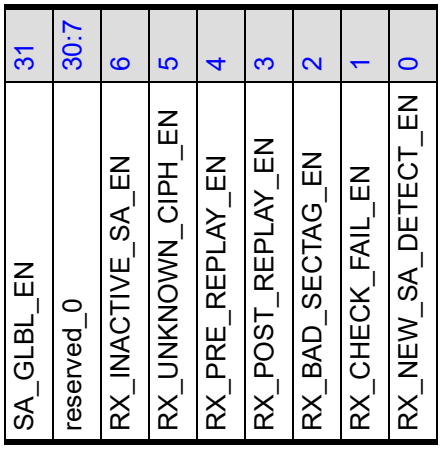


Table 6-545 Fields for Register: IRQ_TYPE_EN_RX_SA_155

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-545 Fields for Register: IRQ_TYPE_EN_RX_SA_155 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-545 Fields for Register: IRQ_TYPE_EN_RX_SA_155 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.539 IRQ_TYPE_STAT_RX_SA_155

- **Name:** IRQ_TYPE_STAT_RX_SA_155
- **Description:** Interrupt Status for each IRQ type of Rx SA 155
- **Size:** 32 bits
- **Offset:** 0x86c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 155

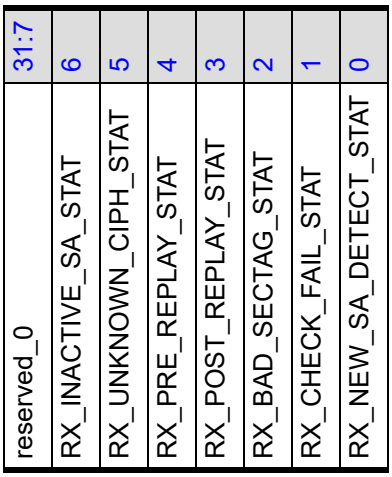


Table 6-546 Fields for Register: IRQ_TYPE_STAT_RX_SA_155

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-546 Fields for Register: IRQ_TYPE_STAT_RX_SA_155 (Continued)

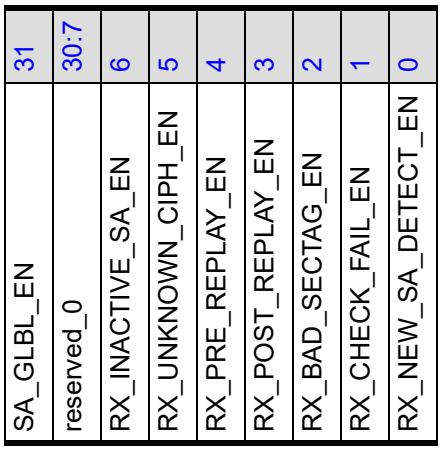
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-546 Fields for Register: IRQ_TYPE_STAT_RX_SA_155 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.540 IRQ_TYPE_EN_RX_SA_156

- **Name:** IRQ_TYPE_EN_RX_SA_156
- **Description:** Interrupt Enable for each IRQ type of Rx SA 156
- **Size:** 32 bits
- **Offset:** 0x870
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 156



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-547 Fields for Register: IRQ_TYPE_EN_RX_SA_156

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-547 Fields for Register: IRQ_TYPE_EN_RX_SA_156 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-547 Fields for Register: IRQ_TYPE_EN_RX_SA_156 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.541 IRQ_TYPE_STAT_RX_SA_156

- **Name:** IRQ_TYPE_STAT_RX_SA_156
- **Description:** Interrupt Status for each IRQ type of Rx SA 156
- **Size:** 32 bits
- **Offset:** 0x874
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 156

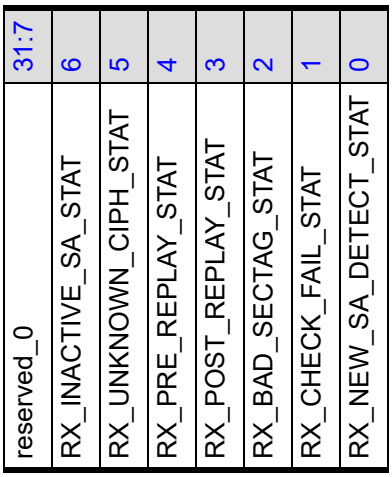


Table 6-548 Fields for Register: IRQ_TYPE_STAT_RX_SA_156

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-548 Fields for Register: IRQ_TYPE_STAT_RX_SA_156 (Continued)

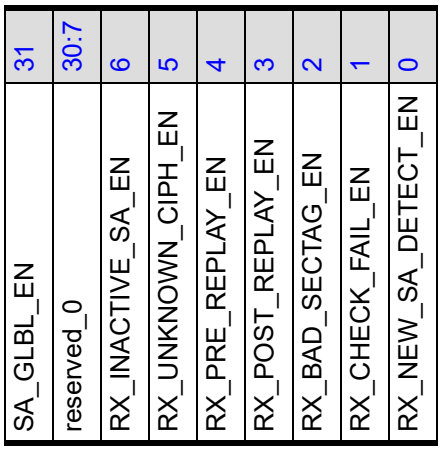
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-548 Fields for Register: IRQ_TYPE_STAT_RX_SA_156 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.542 IRQ_TYPE_EN_RX_SA_157

- **Name:** IRQ_TYPE_EN_RX_SA_157
- **Description:** Interrupt Enable for each IRQ type of Rx SA 157
- **Size:** 32 bits
- **Offset:** 0x878
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 157



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-549 Fields for Register: IRQ_TYPE_EN_RX_SA_157

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-549 Fields for Register: IRQ_TYPE_EN_RX_SA_157 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-549 Fields for Register: IRQ_TYPE_EN_RX_SA_157 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.543 IRQ_TYPE_STAT_RX_SA_157

- **Name:** IRQ_TYPE_STAT_RX_SA_157
- **Description:** Interrupt Status for each IRQ type of Rx SA 157
- **Size:** 32 bits
- **Offset:** 0x87c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 157

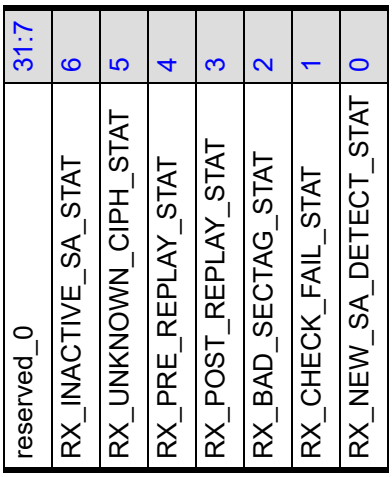


Table 6-550 Fields for Register: IRQ_TYPE_STAT_RX_SA_157

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-550 Fields for Register: IRQ_TYPE_STAT_RX_SA_157 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-550 Fields for Register: IRQ_TYPE_STAT_RX_SA_157 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.544 IRQ_TYPE_EN_RX_SA_158

- **Name:** IRQ_TYPE_EN_RX_SA_158
- **Description:** Interrupt Enable for each IRQ type of Rx SA 158
- **Size:** 32 bits
- **Offset:** 0x880
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 158

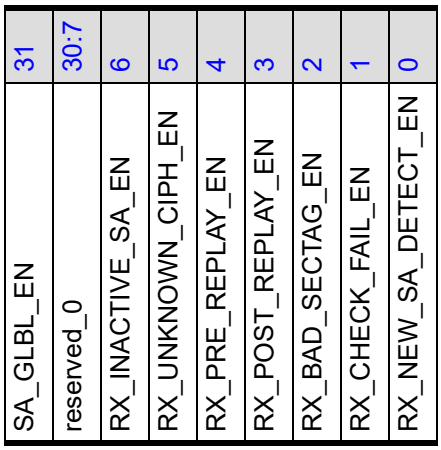


Table 6-551 Fields for Register: IRQ_TYPE_EN_RX_SA_158

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-551 Fields for Register: IRQ_TYPE_EN_RX_SA_158 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-551 Fields for Register: IRQ_TYPE_EN_RX_SA_158 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.545 IRQ_TYPE_STAT_RX_SA_158

- **Name:** IRQ_TYPE_STAT_RX_SA_158
- **Description:** Interrupt Status for each IRQ type of Rx SA 158
- **Size:** 32 bits
- **Offset:** 0x884
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 158

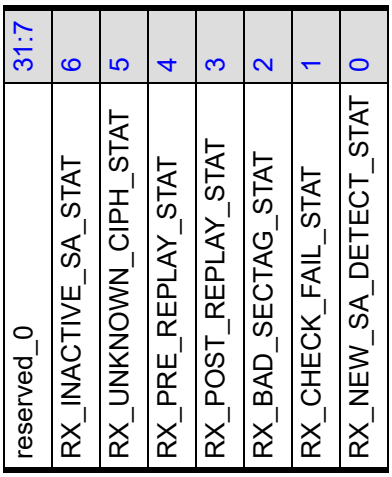


Table 6-552 Fields for Register: IRQ_TYPE_STAT_RX_SA_158

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-552 Fields for Register: IRQ_TYPE_STAT_RX_SA_158 (Continued)

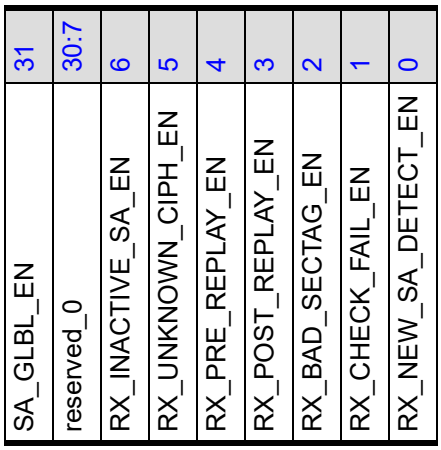
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-552 Fields for Register: IRQ_TYPE_STAT_RX_SA_158 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.546 IRQ_TYPE_EN_RX_SA_159

- **Name:** IRQ_TYPE_EN_RX_SA_159
- **Description:** Interrupt Enable for each IRQ type of Rx SA 159
- **Size:** 32 bits
- **Offset:** 0x888
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 159



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-553 Fields for Register: IRQ_TYPE_EN_RX_SA_159

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-553 Fields for Register: IRQ_TYPE_EN_RX_SA_159 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-553 Fields for Register: IRQ_TYPE_EN_RX_SA_159 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.547 IRQ_TYPE_STAT_RX_SA_159

- **Name:** IRQ_TYPE_STAT_RX_SA_159
- **Description:** Interrupt Status for each IRQ type of Rx SA 159
- **Size:** 32 bits
- **Offset:** 0x88c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 159

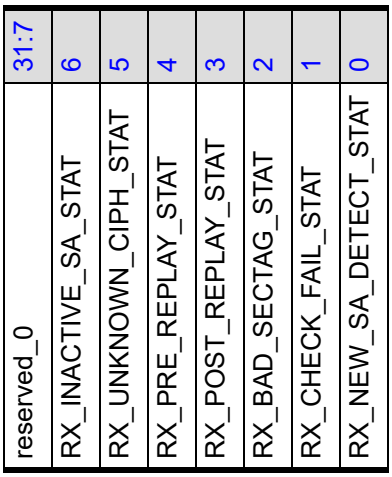


Table 6-554 Fields for Register: IRQ_TYPE_STAT_RX_SA_159

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-554 Fields for Register: IRQ_TYPE_STAT_RX_SA_159 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-554 Fields for Register: IRQ_TYPE_STAT_RX_SA_159 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.548 IRQ_TYPE_EN_RX_SA_160

- **Name:** IRQ_TYPE_EN_RX_SA_160
- **Description:** Interrupt Enable for each IRQ type of Rx SA 160
- **Size:** 32 bits
- **Offset:** 0x890
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 160

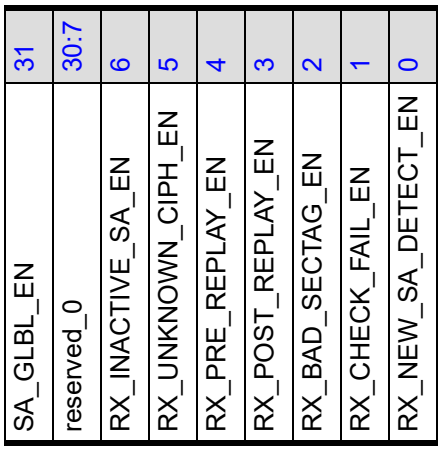


Table 6-555 Fields for Register: IRQ_TYPE_EN_RX_SA_160

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-555 Fields for Register: IRQ_TYPE_EN_RX_SA_160 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-555 Fields for Register: IRQ_TYPE_EN_RX_SA_160 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.549 IRQ_TYPE_STAT_RX_SA_160

- **Name:** IRQ_TYPE_STAT_RX_SA_160
- **Description:** Interrupt Status for each IRQ type of Rx SA 160
- **Size:** 32 bits
- **Offset:** 0x894
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 160

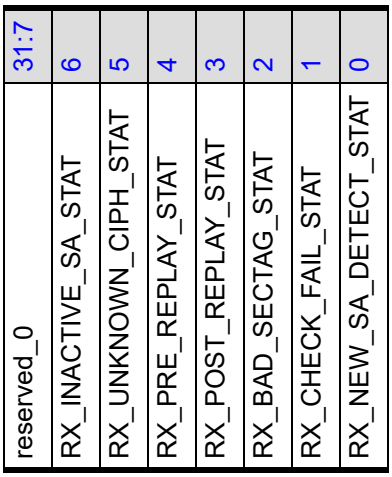


Table 6-556 Fields for Register: IRQ_TYPE_STAT_RX_SA_160

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-556 Fields for Register: IRQ_TYPE_STAT_RX_SA_160 (Continued)

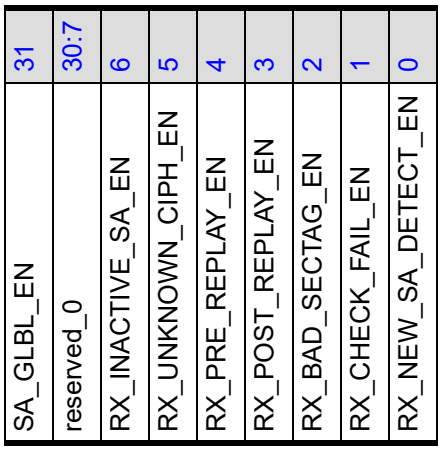
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-556 Fields for Register: IRQ_TYPE_STAT_RX_SA_160 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.550 IRQ_TYPE_EN_RX_SA_161

- **Name:** IRQ_TYPE_EN_RX_SA_161
- **Description:** Interrupt Enable for each IRQ type of Rx SA 161
- **Size:** 32 bits
- **Offset:** 0x898
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 161



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-557 Fields for Register: IRQ_TYPE_EN_RX_SA_161

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-557 Fields for Register: IRQ_TYPE_EN_RX_SA_161 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-557 Fields for Register: IRQ_TYPE_EN_RX_SA_161 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.551 IRQ_TYPE_STAT_RX_SA_161

- **Name:** IRQ_TYPE_STAT_RX_SA_161
- **Description:** Interrupt Status for each IRQ type of Rx SA 161
- **Size:** 32 bits
- **Offset:** 0x89c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 161

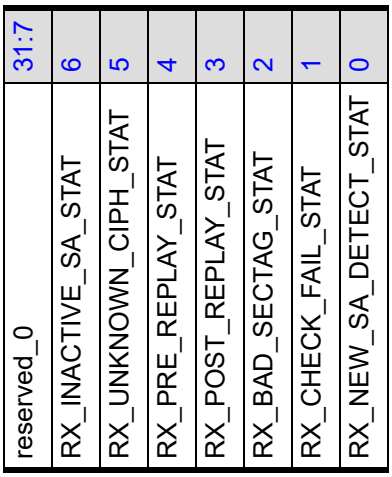


Table 6-558 Fields for Register: IRQ_TYPE_STAT_RX_SA_161

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-558 Fields for Register: IRQ_TYPE_STAT_RX_SA_161 (Continued)

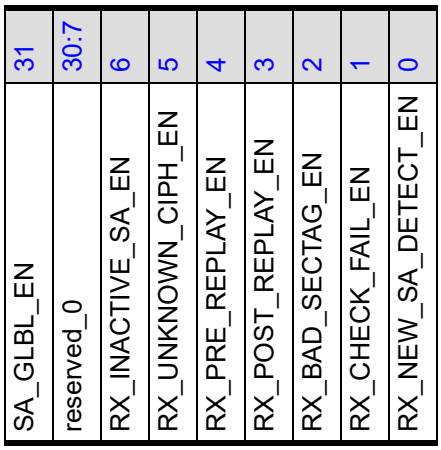
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-558 Fields for Register: IRQ_TYPE_STAT_RX_SA_161 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.552 IRQ_TYPE_EN_RX_SA_162

- **Name:** IRQ_TYPE_EN_RX_SA_162
- **Description:** Interrupt Enable for each IRQ type of Rx SA 162
- **Size:** 32 bits
- **Offset:** 0x8a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 162



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-559 Fields for Register: IRQ_TYPE_EN_RX_SA_162

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-559 Fields for Register: IRQ_TYPE_EN_RX_SA_162 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-559 Fields for Register: IRQ_TYPE_EN_RX_SA_162 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.553 IRQ_TYPE_STAT_RX_SA_162

- **Name:** IRQ_TYPE_STAT_RX_SA_162
- **Description:** Interrupt Status for each IRQ type of Rx SA 162
- **Size:** 32 bits
- **Offset:** 0x8a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 162

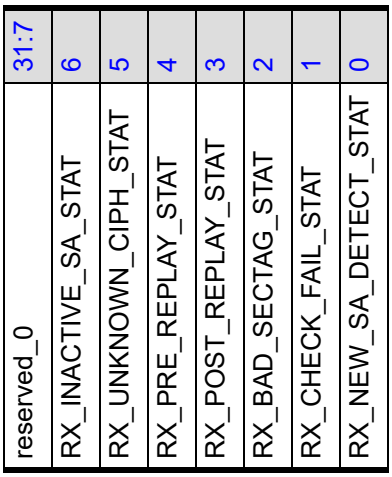


Table 6-560 Fields for Register: IRQ_TYPE_STAT_RX_SA_162

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-560 Fields for Register: IRQ_TYPE_STAT_RX_SA_162 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-560 Fields for Register: IRQ_TYPE_STAT_RX_SA_162 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.554 IRQ_TYPE_EN_RX_SA_163

- **Name:** IRQ_TYPE_EN_RX_SA_163
- **Description:** Interrupt Enable for each IRQ type of Rx SA 163
- **Size:** 32 bits
- **Offset:** 0x8a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 163

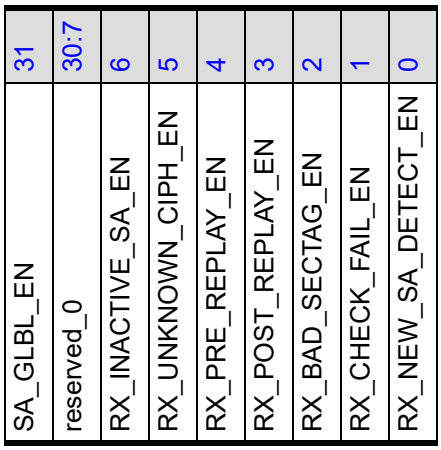


Table 6-561 Fields for Register: IRQ_TYPE_EN_RX_SA_163

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-561 Fields for Register: IRQ_TYPE_EN_RX_SA_163 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-561 Fields for Register: IRQ_TYPE_EN_RX_SA_163 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.555 IRQ_TYPE_STAT_RX_SA_163

- **Name:** IRQ_TYPE_STAT_RX_SA_163
- **Description:** Interrupt Status for each IRQ type of Rx SA 163
- **Size:** 32 bits
- **Offset:** 0x8ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 163

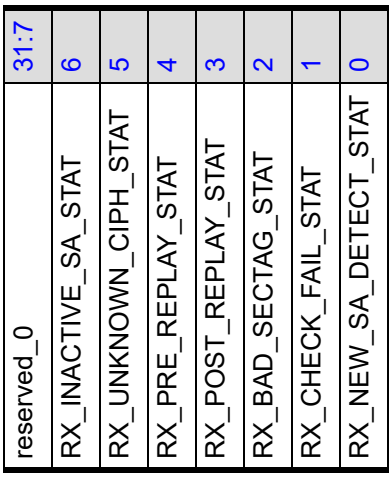


Table 6-562 Fields for Register: IRQ_TYPE_STAT_RX_SA_163

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-562 Fields for Register: IRQ_TYPE_STAT_RX_SA_163 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-562 Fields for Register: IRQ_TYPE_STAT_RX_SA_163 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.556 IRQ_TYPE_EN_RX_SA_164

- **Name:** IRQ_TYPE_EN_RX_SA_164
- **Description:** Interrupt Enable for each IRQ type of Rx SA 164
- **Size:** 32 bits
- **Offset:** 0x8b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 164

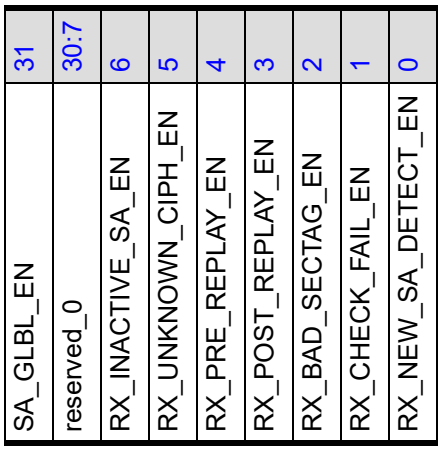


Table 6-563 Fields for Register: IRQ_TYPE_EN_RX_SA_164

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-563 Fields for Register: IRQ_TYPE_EN_RX_SA_164 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-563 Fields for Register: IRQ_TYPE_EN_RX_SA_164 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.557 IRQ_TYPE_STAT_RX_SA_164

- **Name:** IRQ_TYPE_STAT_RX_SA_164
- **Description:** Interrupt Status for each IRQ type of Rx SA 164
- **Size:** 32 bits
- **Offset:** 0x8b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 164

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-564 Fields for Register: IRQ_TYPE_STAT_RX_SA_164

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-564 Fields for Register: IRQ_TYPE_STAT_RX_SA_164 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-564 Fields for Register: IRQ_TYPE_STAT_RX_SA_164 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.558 IRQ_TYPE_EN_RX_SA_165

- **Name:** IRQ_TYPE_EN_RX_SA_165
- **Description:** Interrupt Enable for each IRQ type of Rx SA 165
- **Size:** 32 bits
- **Offset:** 0x8b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 165

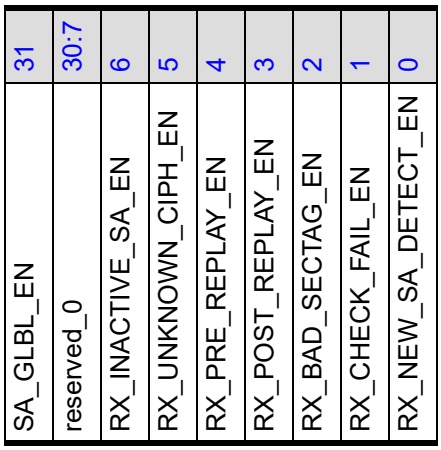


Table 6-565 Fields for Register: IRQ_TYPE_EN_RX_SA_165

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-565 Fields for Register: IRQ_TYPE_EN_RX_SA_165 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-565 Fields for Register: IRQ_TYPE_EN_RX_SA_165 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.559 IRQ_TYPE_STAT_RX_SA_165

- **Name:** IRQ_TYPE_STAT_RX_SA_165
- **Description:** Interrupt Status for each IRQ type of Rx SA 165
- **Size:** 32 bits
- **Offset:** 0x8bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 165

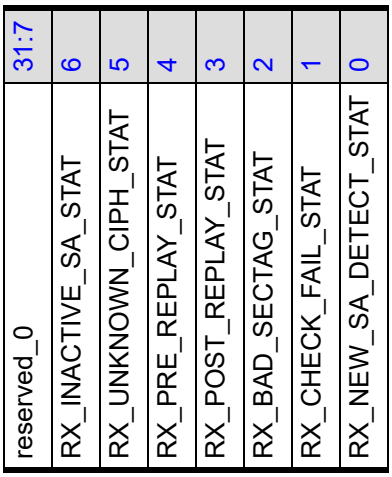


Table 6-566 Fields for Register: IRQ_TYPE_STAT_RX_SA_165

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-566 Fields for Register: IRQ_TYPE_STAT_RX_SA_165 (Continued)

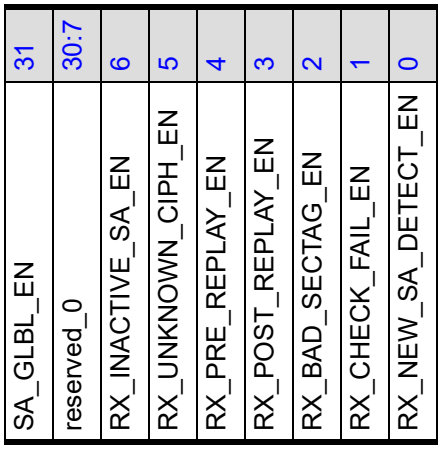
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-566 Fields for Register: IRQ_TYPE_STAT_RX_SA_165 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.560 IRQ_TYPE_EN_RX_SA_166

- **Name:** IRQ_TYPE_EN_RX_SA_166
- **Description:** Interrupt Enable for each IRQ type of Rx SA 166
- **Size:** 32 bits
- **Offset:** 0x8c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 166



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-567 Fields for Register: IRQ_TYPE_EN_RX_SA_166

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-567 Fields for Register: IRQ_TYPE_EN_RX_SA_166 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-567 Fields for Register: IRQ_TYPE_EN_RX_SA_166 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.561 IRQ_TYPE_STAT_RX_SA_166

- **Name:** IRQ_TYPE_STAT_RX_SA_166
- **Description:** Interrupt Status for each IRQ type of Rx SA 166
- **Size:** 32 bits
- **Offset:** 0x8c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 166

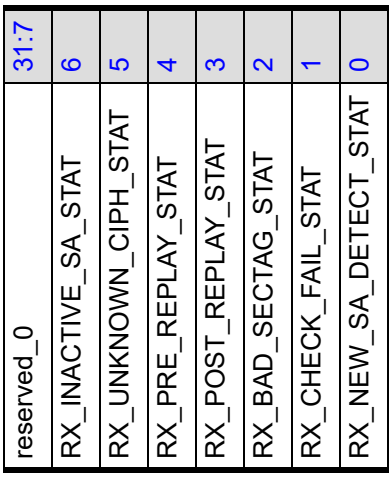


Table 6-568 Fields for Register: IRQ_TYPE_STAT_RX_SA_166

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-568 Fields for Register: IRQ_TYPE_STAT_RX_SA_166 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-568 Fields for Register: IRQ_TYPE_STAT_RX_SA_166 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.562 IRQ_TYPE_EN_RX_SA_167

- **Name:** IRQ_TYPE_EN_RX_SA_167
- **Description:** Interrupt Enable for each IRQ type of Rx SA 167
- **Size:** 32 bits
- **Offset:** 0x8c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 167

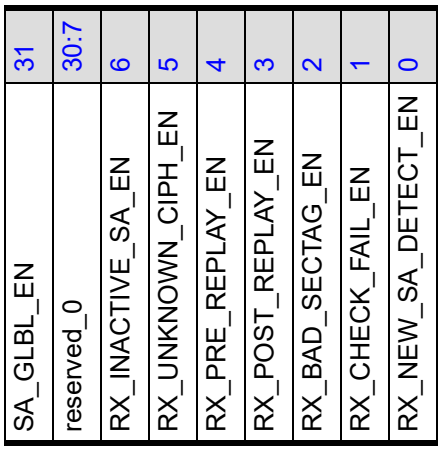


Table 6-569 Fields for Register: IRQ_TYPE_EN_RX_SA_167

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-569 Fields for Register: IRQ_TYPE_EN_RX_SA_167 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-569 Fields for Register: IRQ_TYPE_EN_RX_SA_167 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.563 IRQ_TYPE_STAT_RX_SA_167

- **Name:** IRQ_TYPE_STAT_RX_SA_167
- **Description:** Interrupt Status for each IRQ type of Rx SA 167
- **Size:** 32 bits
- **Offset:** 0x8cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 167

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-570 Fields for Register: IRQ_TYPE_STAT_RX_SA_167

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-570 Fields for Register: IRQ_TYPE_STAT_RX_SA_167 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-570 Fields for Register: IRQ_TYPE_STAT_RX_SA_167 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.564 IRQ_TYPE_EN_RX_SA_168

- **Name:** IRQ_TYPE_EN_RX_SA_168
- **Description:** Interrupt Enable for each IRQ type of Rx SA 168
- **Size:** 32 bits
- **Offset:** 0x8d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 168

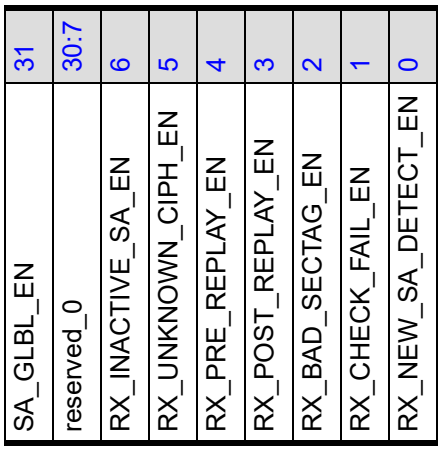


Table 6-571 Fields for Register: IRQ_TYPE_EN_RX_SA_168

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-571 Fields for Register: IRQ_TYPE_EN_RX_SA_168 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-571 Fields for Register: IRQ_TYPE_EN_RX_SA_168 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.565 IRQ_TYPE_STAT_RX_SA_168

- **Name:** IRQ_TYPE_STAT_RX_SA_168
- **Description:** Interrupt Status for each IRQ type of Rx SA 168
- **Size:** 32 bits
- **Offset:** 0x8d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 168

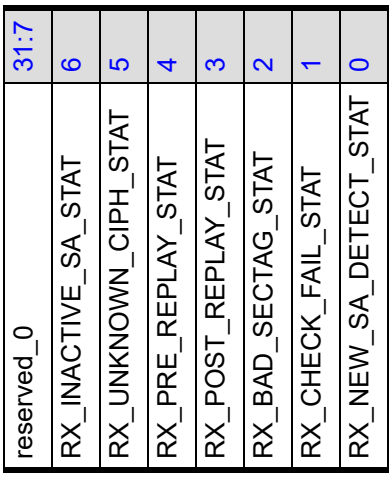


Table 6-572 Fields for Register: IRQ_TYPE_STAT_RX_SA_168

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-572 Fields for Register: IRQ_TYPE_STAT_RX_SA_168 (Continued)

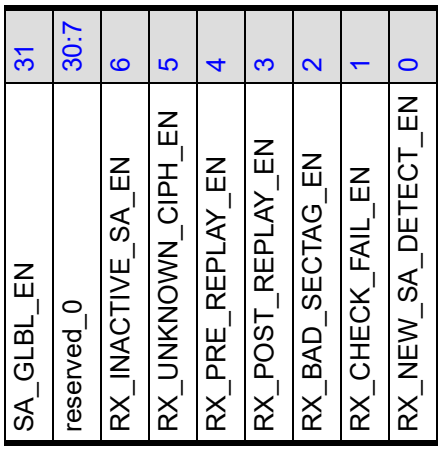
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-572 Fields for Register: IRQ_TYPE_STAT_RX_SA_168 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.566 IRQ_TYPE_EN_RX_SA_169

- **Name:** IRQ_TYPE_EN_RX_SA_169
- **Description:** Interrupt Enable for each IRQ type of Rx SA 169
- **Size:** 32 bits
- **Offset:** 0x8d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 169



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-573 Fields for Register: IRQ_TYPE_EN_RX_SA_169

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-573 Fields for Register: IRQ_TYPE_EN_RX_SA_169 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-573 Fields for Register: IRQ_TYPE_EN_RX_SA_169 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.567 IRQ_TYPE_STAT_RX_SA_169

- **Name:** IRQ_TYPE_STAT_RX_SA_169
- **Description:** Interrupt Status for each IRQ type of Rx SA 169
- **Size:** 32 bits
- **Offset:** 0x8dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 169

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-574 Fields for Register: IRQ_TYPE_STAT_RX_SA_169

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-574 Fields for Register: IRQ_TYPE_STAT_RX_SA_169 (Continued)

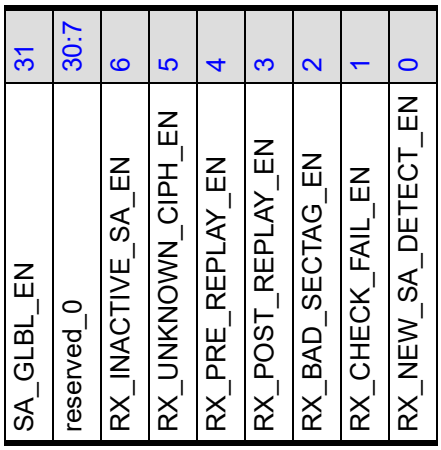
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-574 Fields for Register: IRQ_TYPE_STAT_RX_SA_169 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.568 IRQ_TYPE_EN_RX_SA_170

- **Name:** IRQ_TYPE_EN_RX_SA_170
- **Description:** Interrupt Enable for each IRQ type of Rx SA 170
- **Size:** 32 bits
- **Offset:** 0x8e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 170



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-575 Fields for Register: IRQ_TYPE_EN_RX_SA_170

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-575 Fields for Register: IRQ_TYPE_EN_RX_SA_170 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-575 Fields for Register: IRQ_TYPE_EN_RX_SA_170 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.569 IRQ_TYPE_STAT_RX_SA_170

- **Name:** IRQ_TYPE_STAT_RX_SA_170
- **Description:** Interrupt Status for each IRQ type of Rx SA 170
- **Size:** 32 bits
- **Offset:** 0x8e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 170

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-576 Fields for Register: IRQ_TYPE_STAT_RX_SA_170

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-576 Fields for Register: IRQ_TYPE_STAT_RX_SA_170 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-576 Fields for Register: IRQ_TYPE_STAT_RX_SA_170 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.570 IRQ_TYPE_EN_RX_SA_171

- **Name:** IRQ_TYPE_EN_RX_SA_171
- **Description:** Interrupt Enable for each IRQ type of Rx SA 171
- **Size:** 32 bits
- **Offset:** 0x8e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 171

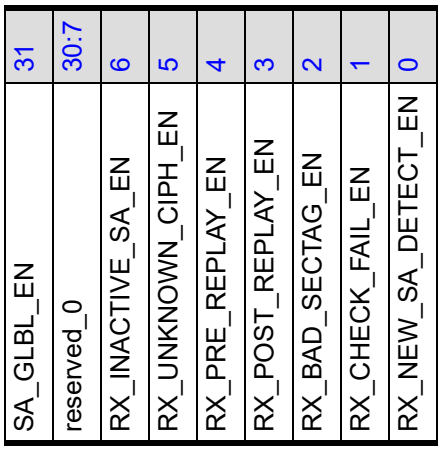


Table 6-577 Fields for Register: IRQ_TYPE_EN_RX_SA_171

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-577 Fields for Register: IRQ_TYPE_EN_RX_SA_171 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-577 Fields for Register: IRQ_TYPE_EN_RX_SA_171 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.571 IRQ_TYPE_STAT_RX_SA_171

- **Name:** IRQ_TYPE_STAT_RX_SA_171
- **Description:** Interrupt Status for each IRQ type of Rx SA 171
- **Size:** 32 bits
- **Offset:** 0x8ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 171

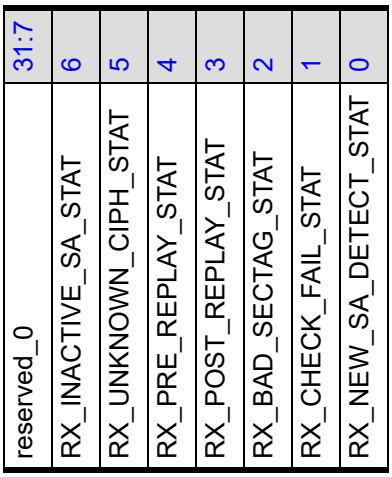


Table 6-578 Fields for Register: IRQ_TYPE_STAT_RX_SA_171

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-578 Fields for Register: IRQ_TYPE_STAT_RX_SA_171 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-578 Fields for Register: IRQ_TYPE_STAT_RX_SA_171 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.572 IRQ_TYPE_EN_RX_SA_172

- **Name:** IRQ_TYPE_EN_RX_SA_172
- **Description:** Interrupt Enable for each IRQ type of Rx SA 172
- **Size:** 32 bits
- **Offset:** 0x8f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 172

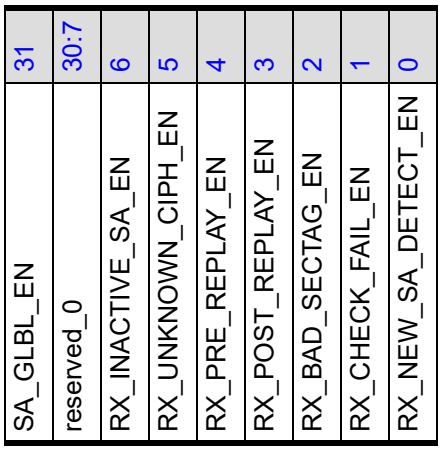


Table 6-579 Fields for Register: IRQ_TYPE_EN_RX_SA_172

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-579 Fields for Register: IRQ_TYPE_EN_RX_SA_172 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-579 Fields for Register: IRQ_TYPE_EN_RX_SA_172 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.573 IRQ_TYPE_STAT_RX_SA_172

- **Name:** IRQ_TYPE_STAT_RX_SA_172
- **Description:** Interrupt Status for each IRQ type of Rx SA 172
- **Size:** 32 bits
- **Offset:** 0x8f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 172

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-580 Fields for Register: IRQ_TYPE_STAT_RX_SA_172

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-580 Fields for Register: IRQ_TYPE_STAT_RX_SA_172 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-580 Fields for Register: IRQ_TYPE_STAT_RX_SA_172 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.574 IRQ_TYPE_EN_RX_SA_173

- **Name:** IRQ_TYPE_EN_RX_SA_173
- **Description:** Interrupt Enable for each IRQ type of Rx SA 173
- **Size:** 32 bits
- **Offset:** 0x8f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 173

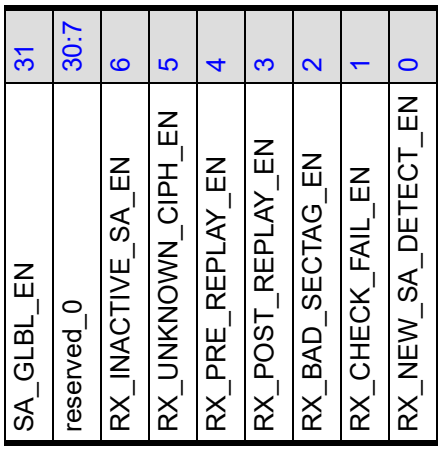


Table 6-581 Fields for Register: IRQ_TYPE_EN_RX_SA_173

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-581 Fields for Register: IRQ_TYPE_EN_RX_SA_173 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-581 Fields for Register: IRQ_TYPE_EN_RX_SA_173 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.575 IRQ_TYPE_STAT_RX_SA_173

- **Name:** IRQ_TYPE_STAT_RX_SA_173
- **Description:** Interrupt Status for each IRQ type of Rx SA 173
- **Size:** 32 bits
- **Offset:** 0x8fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 173

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-582 Fields for Register: IRQ_TYPE_STAT_RX_SA_173

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-582 Fields for Register: IRQ_TYPE_STAT_RX_SA_173 (Continued)

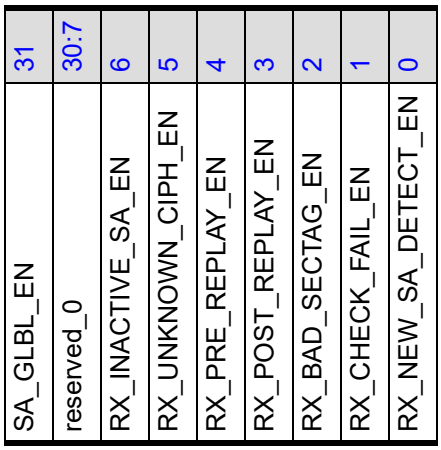
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-582 Fields for Register: IRQ_TYPE_STAT_RX_SA_173 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.576 IRQ_TYPE_EN_RX_SA_174

- **Name:** IRQ_TYPE_EN_RX_SA_174
- **Description:** Interrupt Enable for each IRQ type of Rx SA 174
- **Size:** 32 bits
- **Offset:** 0x900
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 174



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-583 Fields for Register: IRQ_TYPE_EN_RX_SA_174

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-583 Fields for Register: IRQ_TYPE_EN_RX_SA_174 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-583 Fields for Register: IRQ_TYPE_EN_RX_SA_174 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.577 IRQ_TYPE_STAT_RX_SA_174

- **Name:** IRQ_TYPE_STAT_RX_SA_174
- **Description:** Interrupt Status for each IRQ type of Rx SA 174
- **Size:** 32 bits
- **Offset:** 0x904
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 174

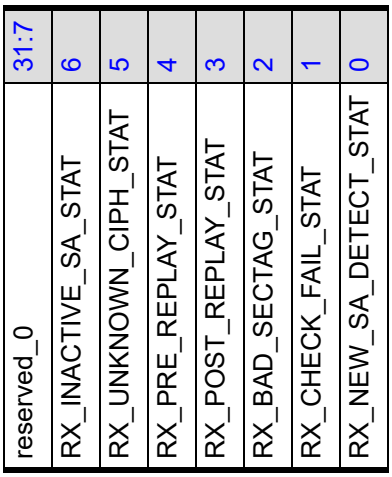


Table 6-584 Fields for Register: IRQ_TYPE_STAT_RX_SA_174

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-584 Fields for Register: IRQ_TYPE_STAT_RX_SA_174 (Continued)

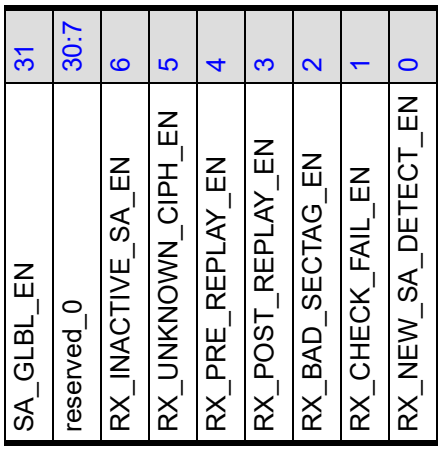
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-584 Fields for Register: IRQ_TYPE_STAT_RX_SA_174 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.578 IRQ_TYPE_EN_RX_SA_175

- **Name:** IRQ_TYPE_EN_RX_SA_175
- **Description:** Interrupt Enable for each IRQ type of Rx SA 175
- **Size:** 32 bits
- **Offset:** 0x908
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 175



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-585 Fields for Register: IRQ_TYPE_EN_RX_SA_175

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-585 Fields for Register: IRQ_TYPE_EN_RX_SA_175 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-585 Fields for Register: IRQ_TYPE_EN_RX_SA_175 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.579 IRQ_TYPE_STAT_RX_SA_175

- **Name:** IRQ_TYPE_STAT_RX_SA_175
- **Description:** Interrupt Status for each IRQ type of Rx SA 175
- **Size:** 32 bits
- **Offset:** 0x90c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 175

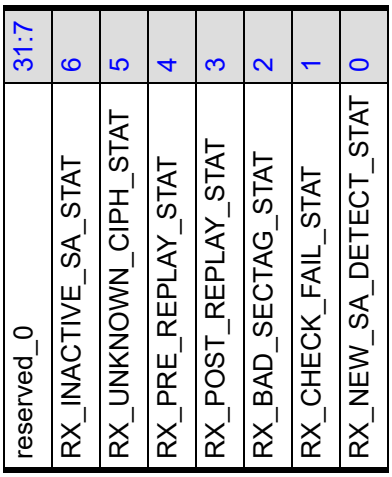


Table 6-586 Fields for Register: IRQ_TYPE_STAT_RX_SA_175

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-586 Fields for Register: IRQ_TYPE_STAT_RX_SA_175 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-586 Fields for Register: IRQ_TYPE_STAT_RX_SA_175 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.580 IRQ_TYPE_EN_RX_SA_176

- **Name:** IRQ_TYPE_EN_RX_SA_176
- **Description:** Interrupt Enable for each IRQ type of Rx SA 176
- **Size:** 32 bits
- **Offset:** 0x910
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 176

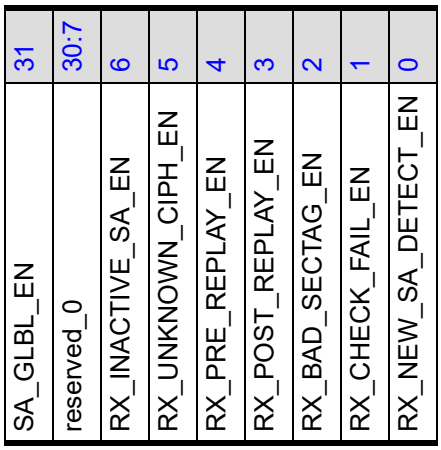


Table 6-587 Fields for Register: IRQ_TYPE_EN_RX_SA_176

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-587 Fields for Register: IRQ_TYPE_EN_RX_SA_176 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-587 Fields for Register: IRQ_TYPE_EN_RX_SA_176 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.581 IRQ_TYPE_STAT_RX_SA_176

- **Name:** IRQ_TYPE_STAT_RX_SA_176
- **Description:** Interrupt Status for each IRQ type of Rx SA 176
- **Size:** 32 bits
- **Offset:** 0x914
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 176

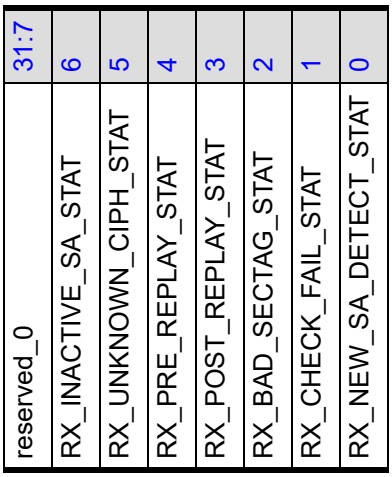


Table 6-588 Fields for Register: IRQ_TYPE_STAT_RX_SA_176

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-588 Fields for Register: IRQ_TYPE_STAT_RX_SA_176 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-588 Fields for Register: IRQ_TYPE_STAT_RX_SA_176 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.582 IRQ_TYPE_EN_RX_SA_177

- **Name:** IRQ_TYPE_EN_RX_SA_177
- **Description:** Interrupt Enable for each IRQ type of Rx SA 177
- **Size:** 32 bits
- **Offset:** 0x918
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 177

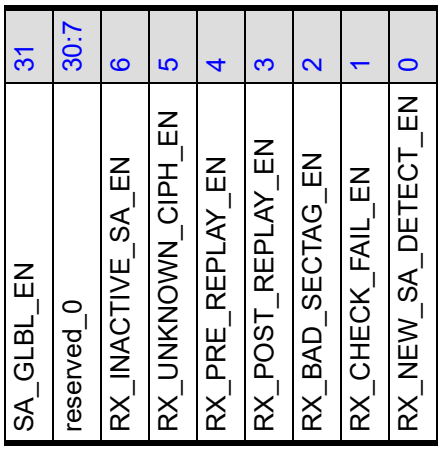


Table 6-589 Fields for Register: IRQ_TYPE_EN_RX_SA_177

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-589 Fields for Register: IRQ_TYPE_EN_RX_SA_177 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-589 Fields for Register: IRQ_TYPE_EN_RX_SA_177 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.583 IRQ_TYPE_STAT_RX_SA_177

- **Name:** IRQ_TYPE_STAT_RX_SA_177
- **Description:** Interrupt Status for each IRQ type of Rx SA 177
- **Size:** 32 bits
- **Offset:** 0x91c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 177

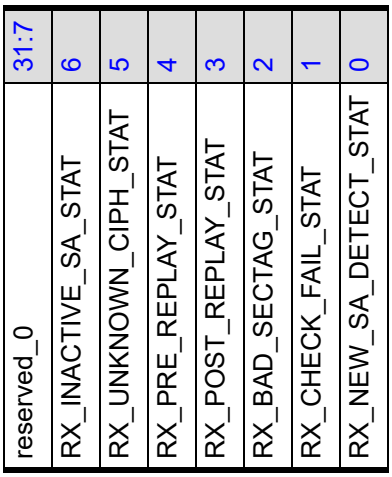


Table 6-590 Fields for Register: IRQ_TYPE_STAT_RX_SA_177

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-590 Fields for Register: IRQ_TYPE_STAT_RX_SA_177 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-590 Fields for Register: IRQ_TYPE_STAT_RX_SA_177 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.584 IRQ_TYPE_EN_RX_SA_178

- **Name:** IRQ_TYPE_EN_RX_SA_178
- **Description:** Interrupt Enable for each IRQ type of Rx SA 178
- **Size:** 32 bits
- **Offset:** 0x920
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 178

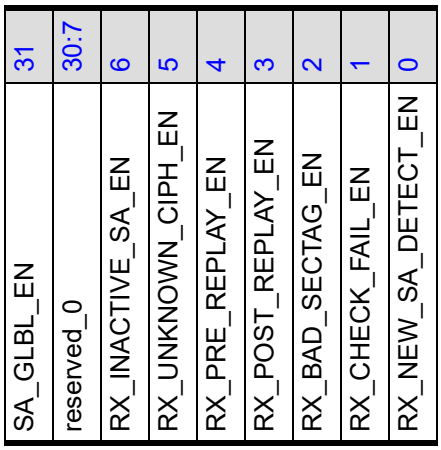


Table 6-591 Fields for Register: IRQ_TYPE_EN_RX_SA_178

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-591 Fields for Register: IRQ_TYPE_EN_RX_SA_178 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-591 Fields for Register: IRQ_TYPE_EN_RX_SA_178 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.585 IRQ_TYPE_STAT_RX_SA_178

- **Name:** IRQ_TYPE_STAT_RX_SA_178
- **Description:** Interrupt Status for each IRQ type of Rx SA 178
- **Size:** 32 bits
- **Offset:** 0x924
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 178

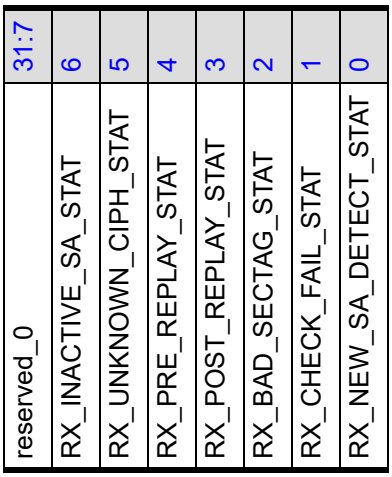


Table 6-592 Fields for Register: IRQ_TYPE_STAT_RX_SA_178

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-592 Fields for Register: IRQ_TYPE_STAT_RX_SA_178 (Continued)

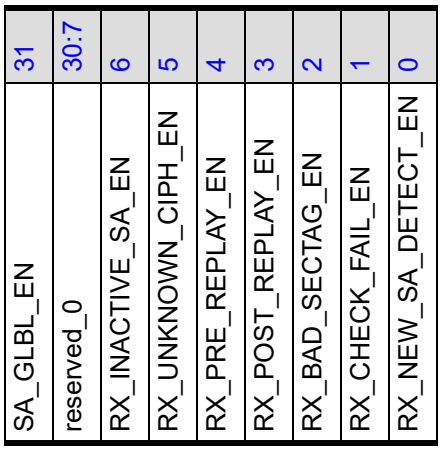
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-592 Fields for Register: IRQ_TYPE_STAT_RX_SA_178 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.586 IRQ_TYPE_EN_RX_SA_179

- **Name:** IRQ_TYPE_EN_RX_SA_179
- **Description:** Interrupt Enable for each IRQ type of Rx SA 179
- **Size:** 32 bits
- **Offset:** 0x928
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 179



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-593 Fields for Register: IRQ_TYPE_EN_RX_SA_179

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-593 Fields for Register: IRQ_TYPE_EN_RX_SA_179 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-593 Fields for Register: IRQ_TYPE_EN_RX_SA_179 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.587 IRQ_TYPE_STAT_RX_SA_179

- **Name:** IRQ_TYPE_STAT_RX_SA_179
- **Description:** Interrupt Status for each IRQ type of Rx SA 179
- **Size:** 32 bits
- **Offset:** 0x92c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 179

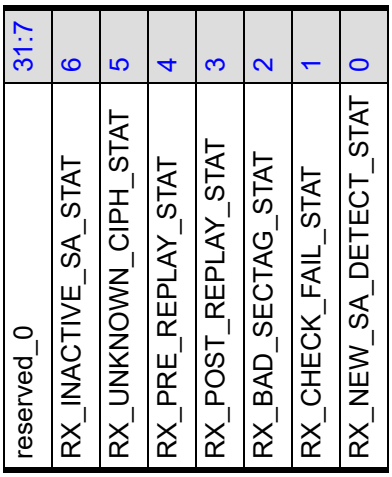


Table 6-594 Fields for Register: IRQ_TYPE_STAT_RX_SA_179

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-594 Fields for Register: IRQ_TYPE_STAT_RX_SA_179 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-594 Fields for Register: IRQ_TYPE_STAT_RX_SA_179 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.588 IRQ_TYPE_EN_RX_SA_180

- **Name:** IRQ_TYPE_EN_RX_SA_180
- **Description:** Interrupt Enable for each IRQ type of Rx SA 180
- **Size:** 32 bits
- **Offset:** 0x930
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 180

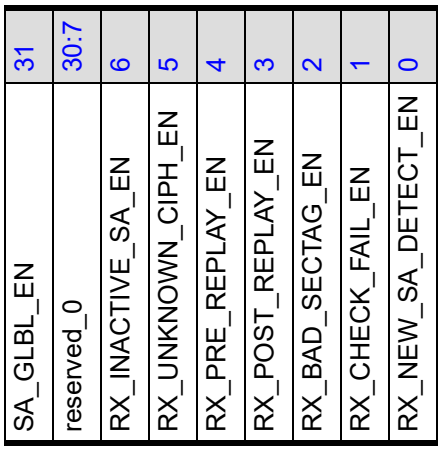


Table 6-595 Fields for Register: IRQ_TYPE_EN_RX_SA_180

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-595 Fields for Register: IRQ_TYPE_EN_RX_SA_180 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-595 Fields for Register: IRQ_TYPE_EN_RX_SA_180 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.589 IRQ_TYPE_STAT_RX_SA_180

- **Name:** IRQ_TYPE_STAT_RX_SA_180
- **Description:** Interrupt Status for each IRQ type of Rx SA 180
- **Size:** 32 bits
- **Offset:** 0x934
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 180

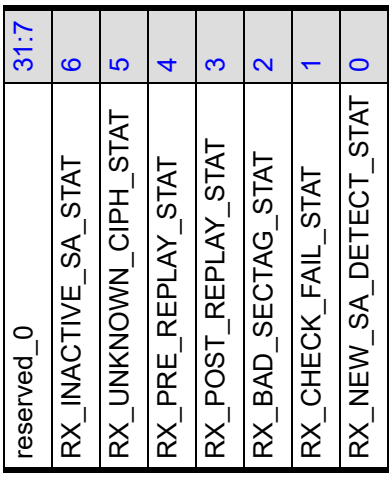


Table 6-596 Fields for Register: IRQ_TYPE_STAT_RX_SA_180

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-596 Fields for Register: IRQ_TYPE_STAT_RX_SA_180 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-596 Fields for Register: IRQ_TYPE_STAT_RX_SA_180 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.590 IRQ_TYPE_EN_RX_SA_181

- **Name:** IRQ_TYPE_EN_RX_SA_181
- **Description:** Interrupt Enable for each IRQ type of Rx SA 181
- **Size:** 32 bits
- **Offset:** 0x938
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 181

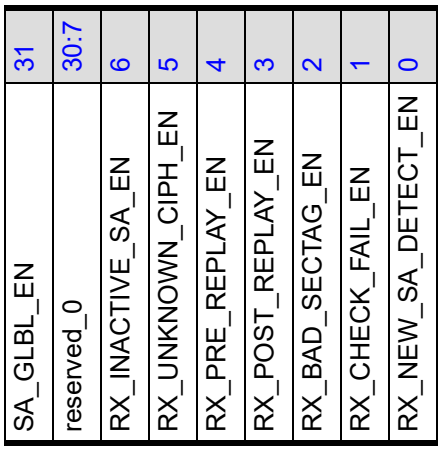


Table 6-597 Fields for Register: IRQ_TYPE_EN_RX_SA_181

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-597 Fields for Register: IRQ_TYPE_EN_RX_SA_181 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-597 Fields for Register: IRQ_TYPE_EN_RX_SA_181 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.591 IRQ_TYPE_STAT_RX_SA_181

- **Name:** IRQ_TYPE_STAT_RX_SA_181
- **Description:** Interrupt Status for each IRQ type of Rx SA 181
- **Size:** 32 bits
- **Offset:** 0x93c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 181

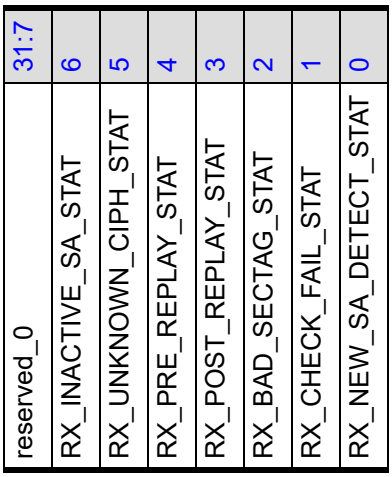


Table 6-598 Fields for Register: IRQ_TYPE_STAT_RX_SA_181

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-598 Fields for Register: IRQ_TYPE_STAT_RX_SA_181 (Continued)

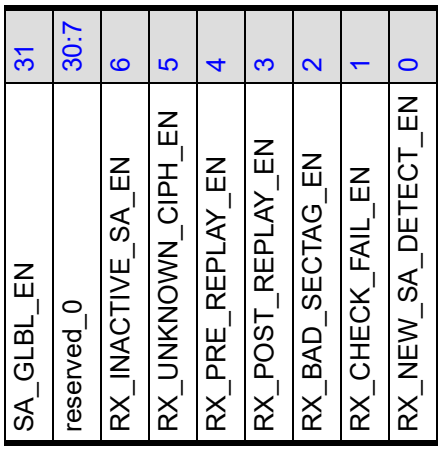
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-598 Fields for Register: IRQ_TYPE_STAT_RX_SA_181 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.592 IRQ_TYPE_EN_RX_SA_182

- **Name:** IRQ_TYPE_EN_RX_SA_182
- **Description:** Interrupt Enable for each IRQ type of Rx SA 182
- **Size:** 32 bits
- **Offset:** 0x940
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 182



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-599 Fields for Register: IRQ_TYPE_EN_RX_SA_182

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-599 Fields for Register: IRQ_TYPE_EN_RX_SA_182 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-599 Fields for Register: IRQ_TYPE_EN_RX_SA_182 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.593 IRQ_TYPE_STAT_RX_SA_182

- **Name:** IRQ_TYPE_STAT_RX_SA_182
- **Description:** Interrupt Status for each IRQ type of Rx SA 182
- **Size:** 32 bits
- **Offset:** 0x944
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 182

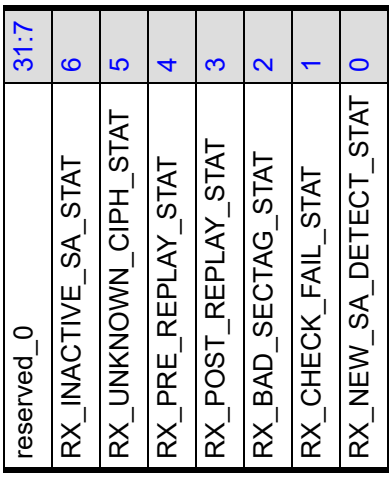


Table 6-600 Fields for Register: IRQ_TYPE_STAT_RX_SA_182

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-600 Fields for Register: IRQ_TYPE_STAT_RX_SA_182 (Continued)

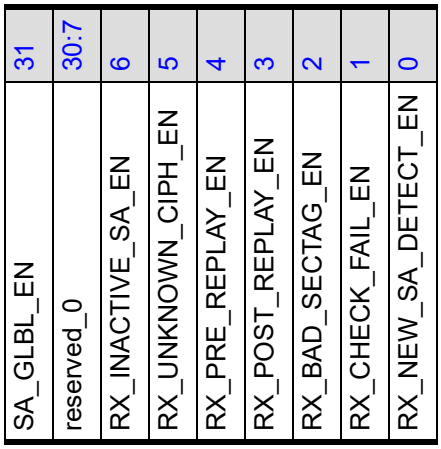
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-600 Fields for Register: IRQ_TYPE_STAT_RX_SA_182 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.594 IRQ_TYPE_EN_RX_SA_183

- **Name:** IRQ_TYPE_EN_RX_SA_183
- **Description:** Interrupt Enable for each IRQ type of Rx SA 183
- **Size:** 32 bits
- **Offset:** 0x948
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 183



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-601 Fields for Register: IRQ_TYPE_EN_RX_SA_183

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-601 Fields for Register: IRQ_TYPE_EN_RX_SA_183 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-601 Fields for Register: IRQ_TYPE_EN_RX_SA_183 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.595 IRQ_TYPE_STAT_RX_SA_183

- **Name:** IRQ_TYPE_STAT_RX_SA_183
- **Description:** Interrupt Status for each IRQ type of Rx SA 183
- **Size:** 32 bits
- **Offset:** 0x94c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 183

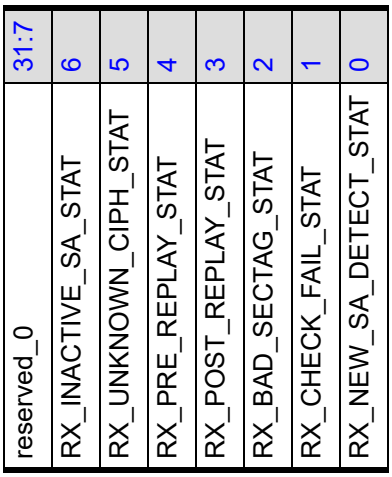


Table 6-602 Fields for Register: IRQ_TYPE_STAT_RX_SA_183

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-602 Fields for Register: IRQ_TYPE_STAT_RX_SA_183 (Continued)

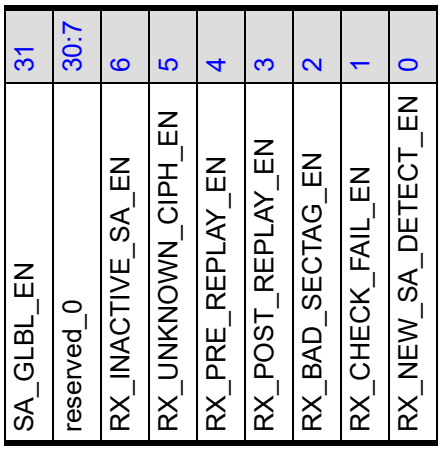
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-602 Fields for Register: IRQ_TYPE_STAT_RX_SA_183 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.596 IRQ_TYPE_EN_RX_SA_184

- **Name:** IRQ_TYPE_EN_RX_SA_184
- **Description:** Interrupt Enable for each IRQ type of Rx SA 184
- **Size:** 32 bits
- **Offset:** 0x950
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 184



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-603 Fields for Register: IRQ_TYPE_EN_RX_SA_184

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-603 Fields for Register: IRQ_TYPE_EN_RX_SA_184 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-603 Fields for Register: IRQ_TYPE_EN_RX_SA_184 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.597 IRQ_TYPE_STAT_RX_SA_184

- **Name:** IRQ_TYPE_STAT_RX_SA_184
- **Description:** Interrupt Status for each IRQ type of Rx SA 184
- **Size:** 32 bits
- **Offset:** 0x954
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 184

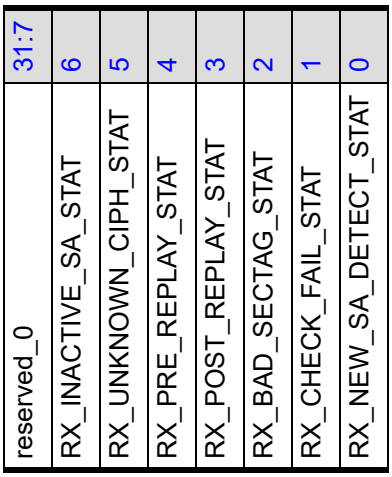


Table 6-604 Fields for Register: IRQ_TYPE_STAT_RX_SA_184

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-604 Fields for Register: IRQ_TYPE_STAT_RX_SA_184 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-604 Fields for Register: IRQ_TYPE_STAT_RX_SA_184 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.598 IRQ_TYPE_EN_RX_SA_185

- **Name:** IRQ_TYPE_EN_RX_SA_185
- **Description:** Interrupt Enable for each IRQ type of Rx SA 185
- **Size:** 32 bits
- **Offset:** 0x958
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 185

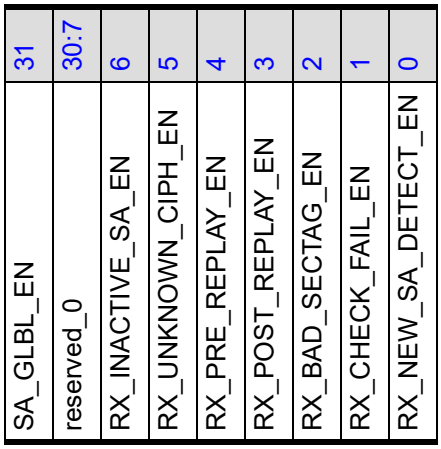


Table 6-605 Fields for Register: IRQ_TYPE_EN_RX_SA_185

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-605 Fields for Register: IRQ_TYPE_EN_RX_SA_185 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-605 Fields for Register: IRQ_TYPE_EN_RX_SA_185 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.599 IRQ_TYPE_STAT_RX_SA_185

- **Name:** IRQ_TYPE_STAT_RX_SA_185
- **Description:** Interrupt Status for each IRQ type of Rx SA 185
- **Size:** 32 bits
- **Offset:** 0x95c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 185

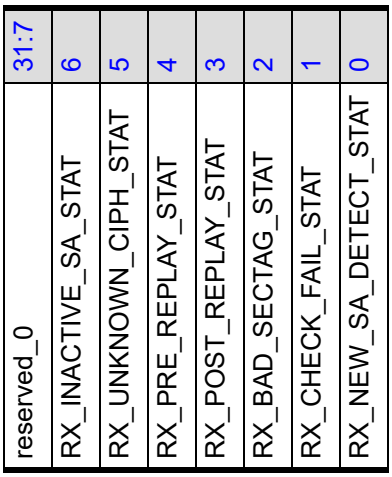


Table 6-606 Fields for Register: IRQ_TYPE_STAT_RX_SA_185

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-606 Fields for Register: IRQ_TYPE_STAT_RX_SA_185 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-606 Fields for Register: IRQ_TYPE_STAT_RX_SA_185 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.600 IRQ_TYPE_EN_RX_SA_186

- **Name:** IRQ_TYPE_EN_RX_SA_186
- **Description:** Interrupt Enable for each IRQ type of Rx SA 186
- **Size:** 32 bits
- **Offset:** 0x960
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 186

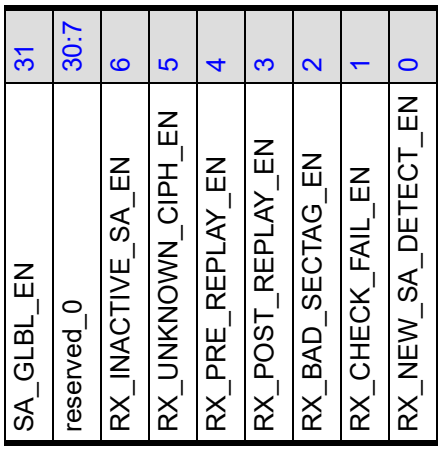


Table 6-607 Fields for Register: IRQ_TYPE_EN_RX_SA_186

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-607 Fields for Register: IRQ_TYPE_EN_RX_SA_186 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-607 Fields for Register: IRQ_TYPE_EN_RX_SA_186 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.601 IRQ_TYPE_STAT_RX_SA_186

- **Name:** IRQ_TYPE_STAT_RX_SA_186
- **Description:** Interrupt Status for each IRQ type of Rx SA 186
- **Size:** 32 bits
- **Offset:** 0x964
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 186

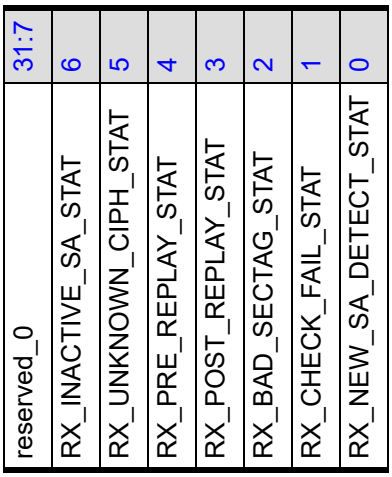


Table 6-608 Fields for Register: IRQ_TYPE_STAT_RX_SA_186

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-608 Fields for Register: IRQ_TYPE_STAT_RX_SA_186 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-608 Fields for Register: IRQ_TYPE_STAT_RX_SA_186 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.602 IRQ_TYPE_EN_RX_SA_187

- **Name:** IRQ_TYPE_EN_RX_SA_187
- **Description:** Interrupt Enable for each IRQ type of Rx SA 187
- **Size:** 32 bits
- **Offset:** 0x968
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 187

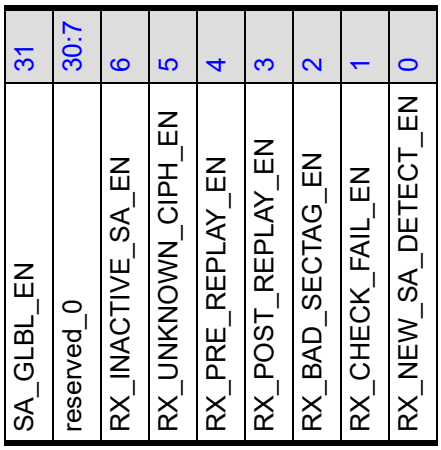


Table 6-609 Fields for Register: IRQ_TYPE_EN_RX_SA_187

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-609 Fields for Register: IRQ_TYPE_EN_RX_SA_187 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-609 Fields for Register: IRQ_TYPE_EN_RX_SA_187 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.603 IRQ_TYPE_STAT_RX_SA_187

- **Name:** IRQ_TYPE_STAT_RX_SA_187
- **Description:** Interrupt Status for each IRQ type of Rx SA 187
- **Size:** 32 bits
- **Offset:** 0x96c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 187

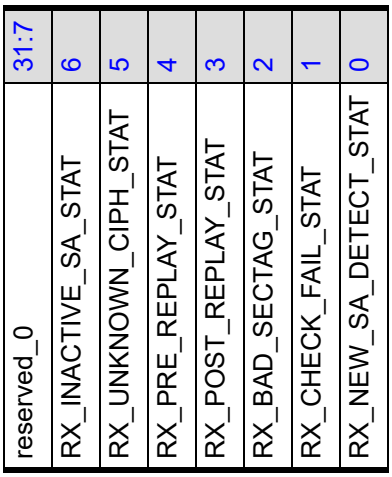


Table 6-610 Fields for Register: IRQ_TYPE_STAT_RX_SA_187

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-610 Fields for Register: IRQ_TYPE_STAT_RX_SA_187 (Continued)

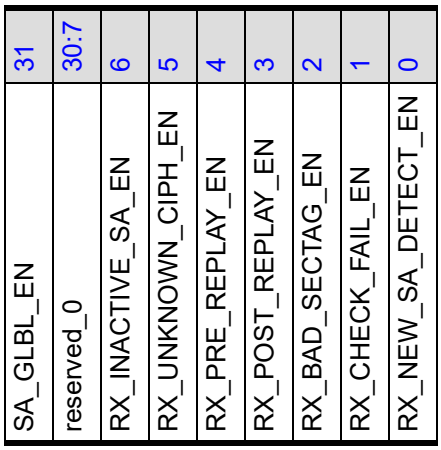
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-610 Fields for Register: IRQ_TYPE_STAT_RX_SA_187 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.604 IRQ_TYPE_EN_RX_SA_188

- **Name:** IRQ_TYPE_EN_RX_SA_188
- **Description:** Interrupt Enable for each IRQ type of Rx SA 188
- **Size:** 32 bits
- **Offset:** 0x970
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 188



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-611 Fields for Register: IRQ_TYPE_EN_RX_SA_188

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-611 Fields for Register: IRQ_TYPE_EN_RX_SA_188 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-611 Fields for Register: IRQ_TYPE_EN_RX_SA_188 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.605 IRQ_TYPE_STAT_RX_SA_188

- **Name:** IRQ_TYPE_STAT_RX_SA_188
- **Description:** Interrupt Status for each IRQ type of Rx SA 188
- **Size:** 32 bits
- **Offset:** 0x974
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 188

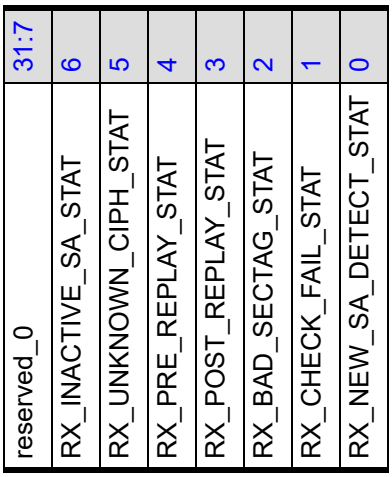


Table 6-612 Fields for Register: IRQ_TYPE_STAT_RX_SA_188

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-612 Fields for Register: IRQ_TYPE_STAT_RX_SA_188 (Continued)

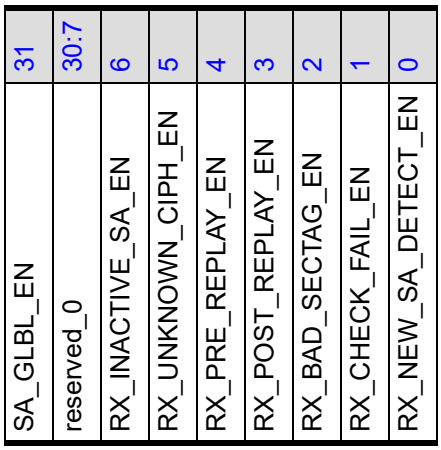
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-612 Fields for Register: IRQ_TYPE_STAT_RX_SA_188 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.606 IRQ_TYPE_EN_RX_SA_189

- **Name:** IRQ_TYPE_EN_RX_SA_189
- **Description:** Interrupt Enable for each IRQ type of Rx SA 189
- **Size:** 32 bits
- **Offset:** 0x978
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 189



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-613 Fields for Register: IRQ_TYPE_EN_RX_SA_189

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-613 Fields for Register: IRQ_TYPE_EN_RX_SA_189 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-613 Fields for Register: IRQ_TYPE_EN_RX_SA_189 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.607 IRQ_TYPE_STAT_RX_SA_189

- **Name:** IRQ_TYPE_STAT_RX_SA_189
- **Description:** Interrupt Status for each IRQ type of Rx SA 189
- **Size:** 32 bits
- **Offset:** 0x97c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 189

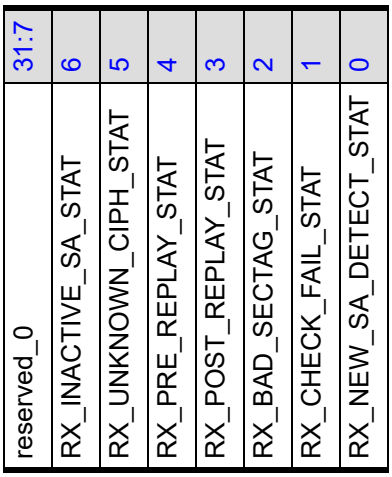


Table 6-614 Fields for Register: IRQ_TYPE_STAT_RX_SA_189

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-614 Fields for Register: IRQ_TYPE_STAT_RX_SA_189 (Continued)

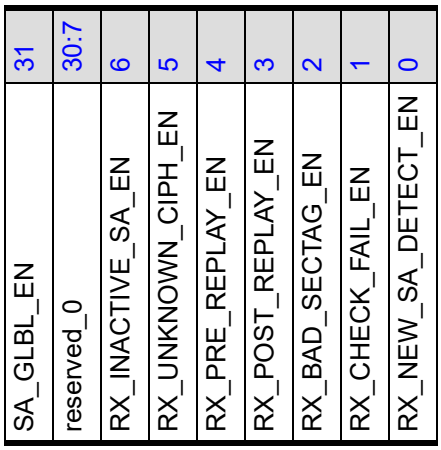
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-614 Fields for Register: IRQ_TYPE_STAT_RX_SA_189 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.608 IRQ_TYPE_EN_RX_SA_190

- **Name:** IRQ_TYPE_EN_RX_SA_190
- **Description:** Interrupt Enable for each IRQ type of Rx SA 190
- **Size:** 32 bits
- **Offset:** 0x980
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 190



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-615 Fields for Register: IRQ_TYPE_EN_RX_SA_190

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-615 Fields for Register: IRQ_TYPE_EN_RX_SA_190 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-615 Fields for Register: IRQ_TYPE_EN_RX_SA_190 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.609 IRQ_TYPE_STAT_RX_SA_190

- **Name:** IRQ_TYPE_STAT_RX_SA_190
- **Description:** Interrupt Status for each IRQ type of Rx SA 190
- **Size:** 32 bits
- **Offset:** 0x984
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 190

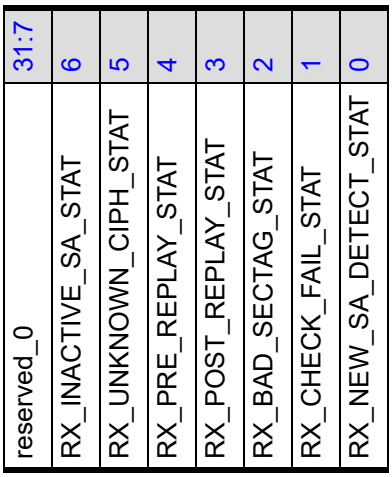


Table 6-616 Fields for Register: IRQ_TYPE_STAT_RX_SA_190

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-616 Fields for Register: IRQ_TYPE_STAT_RX_SA_190 (Continued)

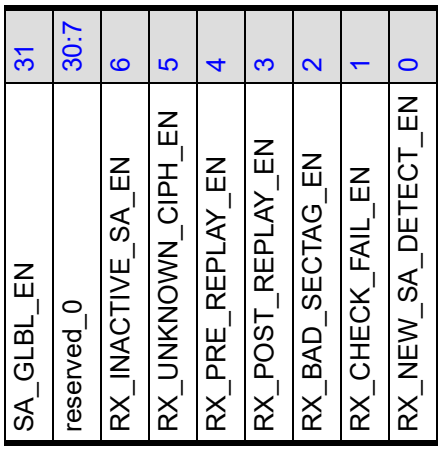
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-616 Fields for Register: IRQ_TYPE_STAT_RX_SA_190 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.610 IRQ_TYPE_EN_RX_SA_191

- **Name:** IRQ_TYPE_EN_RX_SA_191
- **Description:** Interrupt Enable for each IRQ type of Rx SA 191
- **Size:** 32 bits
- **Offset:** 0x988
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 191



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-617 Fields for Register: IRQ_TYPE_EN_RX_SA_191

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-617 Fields for Register: IRQ_TYPE_EN_RX_SA_191 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-617 Fields for Register: IRQ_TYPE_EN_RX_SA_191 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.611 IRQ_TYPE_STAT_RX_SA_191

- **Name:** IRQ_TYPE_STAT_RX_SA_191
- **Description:** Interrupt Status for each IRQ type of Rx SA 191
- **Size:** 32 bits
- **Offset:** 0x98c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 191

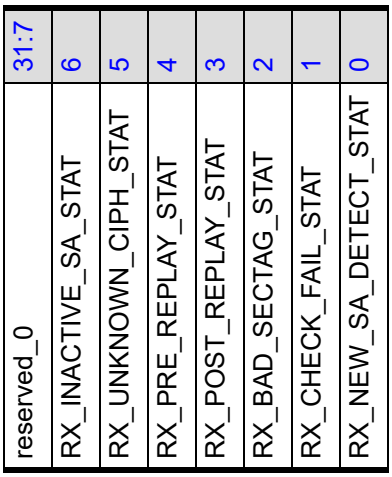


Table 6-618 Fields for Register: IRQ_TYPE_STAT_RX_SA_191

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-618 Fields for Register: IRQ_TYPE_STAT_RX_SA_191 (Continued)

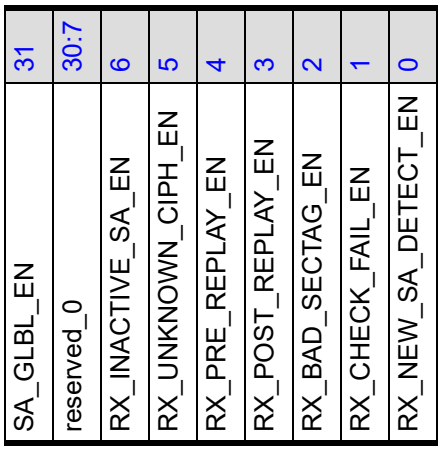
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-618 Fields for Register: IRQ_TYPE_STAT_RX_SA_191 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.612 IRQ_TYPE_EN_RX_SA_192

- **Name:** IRQ_TYPE_EN_RX_SA_192
- **Description:** Interrupt Enable for each IRQ type of Rx SA 192
- **Size:** 32 bits
- **Offset:** 0x990
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 192



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-619 Fields for Register: IRQ_TYPE_EN_RX_SA_192

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-619 Fields for Register: IRQ_TYPE_EN_RX_SA_192 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-619 Fields for Register: IRQ_TYPE_EN_RX_SA_192 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.613 IRQ_TYPE_STAT_RX_SA_192

- **Name:** IRQ_TYPE_STAT_RX_SA_192
- **Description:** Interrupt Status for each IRQ type of Rx SA 192
- **Size:** 32 bits
- **Offset:** 0x994
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 192

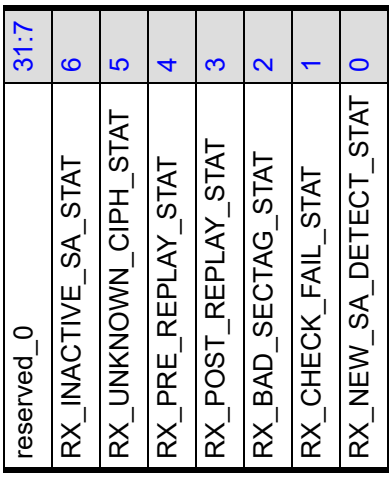


Table 6-620 Fields for Register: IRQ_TYPE_STAT_RX_SA_192

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-620 Fields for Register: IRQ_TYPE_STAT_RX_SA_192 (Continued)

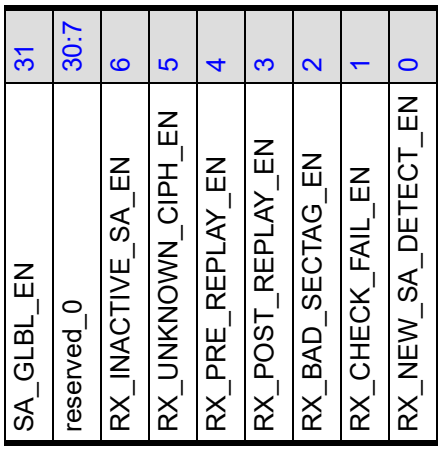
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-620 Fields for Register: IRQ_TYPE_STAT_RX_SA_192 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.614 IRQ_TYPE_EN_RX_SA_193

- **Name:** IRQ_TYPE_EN_RX_SA_193
- **Description:** Interrupt Enable for each IRQ type of Rx SA 193
- **Size:** 32 bits
- **Offset:** 0x998
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 193



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-621 Fields for Register: IRQ_TYPE_EN_RX_SA_193

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-621 Fields for Register: IRQ_TYPE_EN_RX_SA_193 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-621 Fields for Register: IRQ_TYPE_EN_RX_SA_193 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.615 IRQ_TYPE_STAT_RX_SA_193

- **Name:** IRQ_TYPE_STAT_RX_SA_193
- **Description:** Interrupt Status for each IRQ type of Rx SA 193
- **Size:** 32 bits
- **Offset:** 0x99c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 193

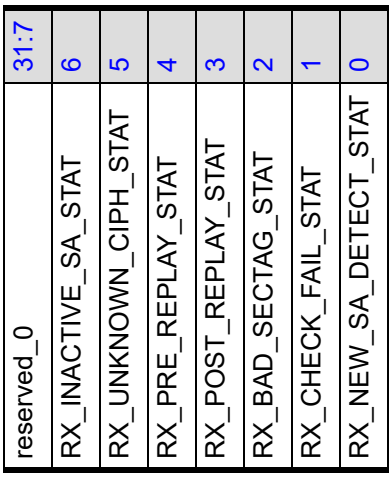


Table 6-622 Fields for Register: IRQ_TYPE_STAT_RX_SA_193

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-622 Fields for Register: IRQ_TYPE_STAT_RX_SA_193 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-622 Fields for Register: IRQ_TYPE_STAT_RX_SA_193 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.616 IRQ_TYPE_EN_RX_SA_194

- **Name:** IRQ_TYPE_EN_RX_SA_194
- **Description:** Interrupt Enable for each IRQ type of Rx SA 194
- **Size:** 32 bits
- **Offset:** 0x9a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 194

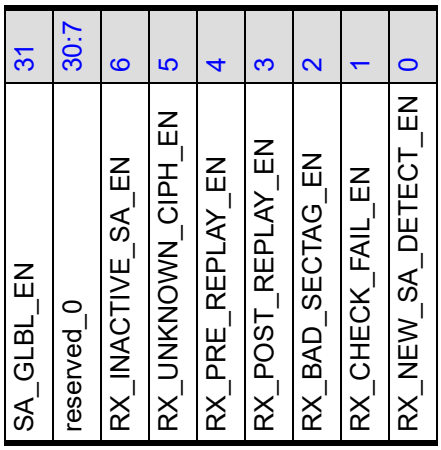


Table 6-623 Fields for Register: IRQ_TYPE_EN_RX_SA_194

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-623 Fields for Register: IRQ_TYPE_EN_RX_SA_194 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-623 Fields for Register: IRQ_TYPE_EN_RX_SA_194 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.617 IRQ_TYPE_STAT_RX_SA_194

- **Name:** IRQ_TYPE_STAT_RX_SA_194
- **Description:** Interrupt Status for each IRQ type of Rx SA 194
- **Size:** 32 bits
- **Offset:** 0x9a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 194

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-624 Fields for Register: IRQ_TYPE_STAT_RX_SA_194

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-624 Fields for Register: IRQ_TYPE_STAT_RX_SA_194 (Continued)

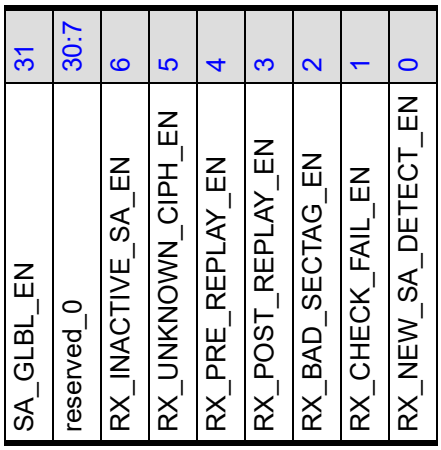
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-624 Fields for Register: IRQ_TYPE_STAT_RX_SA_194 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.618 IRQ_TYPE_EN_RX_SA_195

- **Name:** IRQ_TYPE_EN_RX_SA_195
- **Description:** Interrupt Enable for each IRQ type of Rx SA 195
- **Size:** 32 bits
- **Offset:** 0x9a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 195



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-625 Fields for Register: IRQ_TYPE_EN_RX_SA_195

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-625 Fields for Register: IRQ_TYPE_EN_RX_SA_195 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-625 Fields for Register: IRQ_TYPE_EN_RX_SA_195 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.619 IRQ_TYPE_STAT_RX_SA_195

- **Name:** IRQ_TYPE_STAT_RX_SA_195
- **Description:** Interrupt Status for each IRQ type of Rx SA 195
- **Size:** 32 bits
- **Offset:** 0x9ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 195

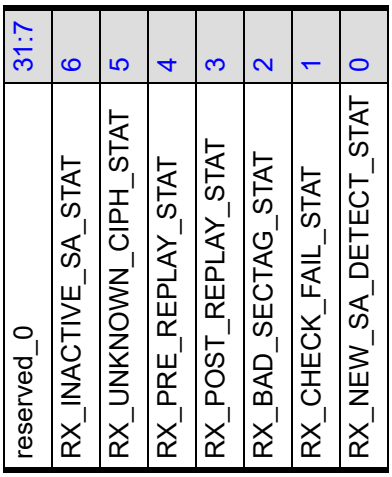


Table 6-626 Fields for Register: IRQ_TYPE_STAT_RX_SA_195

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-626 Fields for Register: IRQ_TYPE_STAT_RX_SA_195 (Continued)

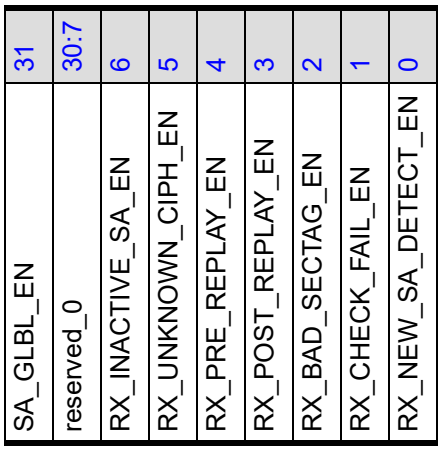
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-626 Fields for Register: IRQ_TYPE_STAT_RX_SA_195 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.620 IRQ_TYPE_EN_RX_SA_196

- **Name:** IRQ_TYPE_EN_RX_SA_196
- **Description:** Interrupt Enable for each IRQ type of Rx SA 196
- **Size:** 32 bits
- **Offset:** 0x9b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 196



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-627 Fields for Register: IRQ_TYPE_EN_RX_SA_196

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-627 Fields for Register: IRQ_TYPE_EN_RX_SA_196 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-627 Fields for Register: IRQ_TYPE_EN_RX_SA_196 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.621 IRQ_TYPE_STAT_RX_SA_196

- **Name:** IRQ_TYPE_STAT_RX_SA_196
- **Description:** Interrupt Status for each IRQ type of Rx SA 196
- **Size:** 32 bits
- **Offset:** 0x9b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 196

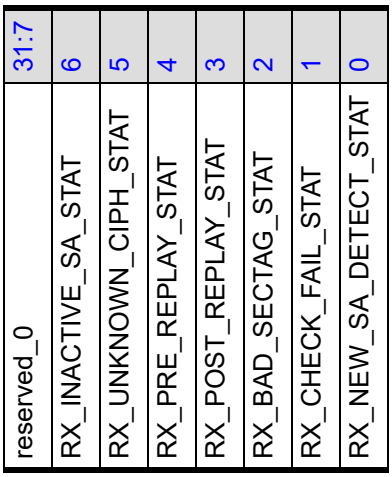


Table 6-628 Fields for Register: IRQ_TYPE_STAT_RX_SA_196

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-628 Fields for Register: IRQ_TYPE_STAT_RX_SA_196 (Continued)

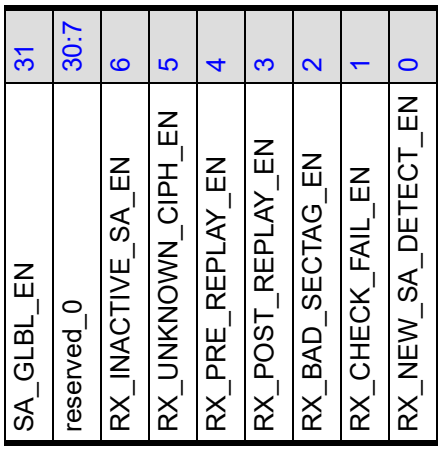
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-628 Fields for Register: IRQ_TYPE_STAT_RX_SA_196 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.622 IRQ_TYPE_EN_RX_SA_197

- **Name:** IRQ_TYPE_EN_RX_SA_197
- **Description:** Interrupt Enable for each IRQ type of Rx SA 197
- **Size:** 32 bits
- **Offset:** 0x9b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 197



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-629 Fields for Register: IRQ_TYPE_EN_RX_SA_197

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-629 Fields for Register: IRQ_TYPE_EN_RX_SA_197 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-629 Fields for Register: IRQ_TYPE_EN_RX_SA_197 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.623 IRQ_TYPE_STAT_RX_SA_197

- **Name:** IRQ_TYPE_STAT_RX_SA_197
- **Description:** Interrupt Status for each IRQ type of Rx SA 197
- **Size:** 32 bits
- **Offset:** 0x9bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 197

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-630 Fields for Register: IRQ_TYPE_STAT_RX_SA_197

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-630 Fields for Register: IRQ_TYPE_STAT_RX_SA_197 (Continued)

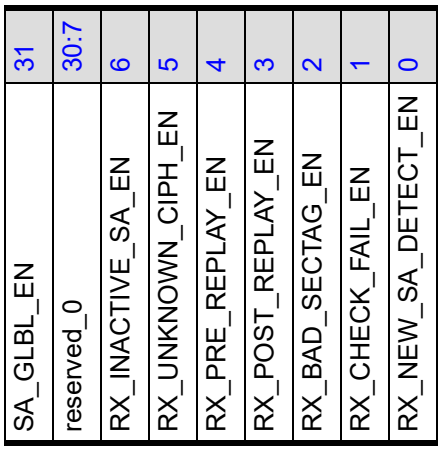
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-630 Fields for Register: IRQ_TYPE_STAT_RX_SA_197 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.624 IRQ_TYPE_EN_RX_SA_198

- **Name:** IRQ_TYPE_EN_RX_SA_198
- **Description:** Interrupt Enable for each IRQ type of Rx SA 198
- **Size:** 32 bits
- **Offset:** 0x9c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 198



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-631 Fields for Register: IRQ_TYPE_EN_RX_SA_198

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-631 Fields for Register: IRQ_TYPE_EN_RX_SA_198 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-631 Fields for Register: IRQ_TYPE_EN_RX_SA_198 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.625 IRQ_TYPE_STAT_RX_SA_198

- **Name:** IRQ_TYPE_STAT_RX_SA_198
- **Description:** Interrupt Status for each IRQ type of Rx SA 198
- **Size:** 32 bits
- **Offset:** 0x9c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 198

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-632 Fields for Register: IRQ_TYPE_STAT_RX_SA_198

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-632 Fields for Register: IRQ_TYPE_STAT_RX_SA_198 (Continued)

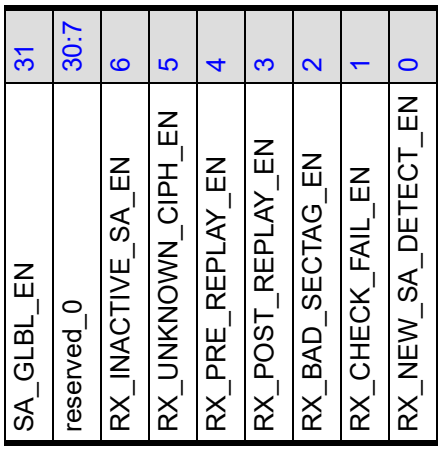
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-632 Fields for Register: IRQ_TYPE_STAT_RX_SA_198 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.626 IRQ_TYPE_EN_RX_SA_199

- **Name:** IRQ_TYPE_EN_RX_SA_199
- **Description:** Interrupt Enable for each IRQ type of Rx SA 199
- **Size:** 32 bits
- **Offset:** 0x9c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 199



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-633 Fields for Register: IRQ_TYPE_EN_RX_SA_199

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-633 Fields for Register: IRQ_TYPE_EN_RX_SA_199 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-633 Fields for Register: IRQ_TYPE_EN_RX_SA_199 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.627 IRQ_TYPE_STAT_RX_SA_199

- **Name:** IRQ_TYPE_STAT_RX_SA_199
- **Description:** Interrupt Status for each IRQ type of Rx SA 199
- **Size:** 32 bits
- **Offset:** 0x9cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 199

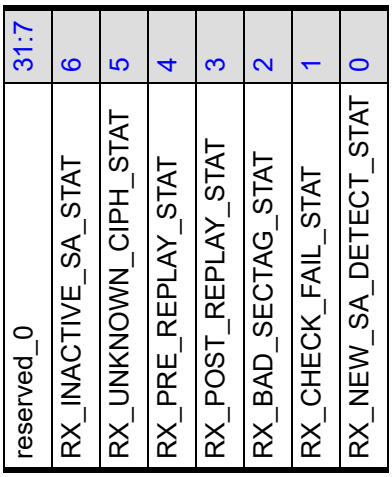


Table 6-634 Fields for Register: IRQ_TYPE_STAT_RX_SA_199

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-634 Fields for Register: IRQ_TYPE_STAT_RX_SA_199 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-634 Fields for Register: IRQ_TYPE_STAT_RX_SA_199 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.628 IRQ_TYPE_EN_RX_SA_200

- **Name:** IRQ_TYPE_EN_RX_SA_200
- **Description:** Interrupt Enable for each IRQ type of Rx SA 200
- **Size:** 32 bits
- **Offset:** 0x9d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 200

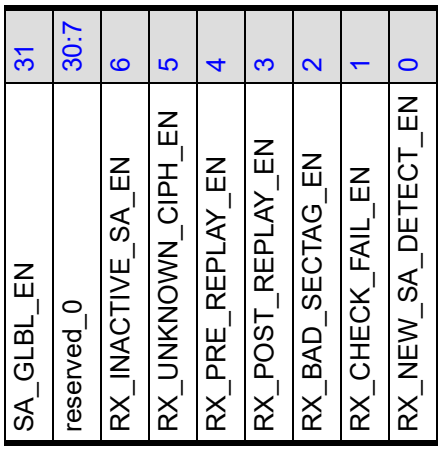


Table 6-635 Fields for Register: IRQ_TYPE_EN_RX_SA_200

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-635 Fields for Register: IRQ_TYPE_EN_RX_SA_200 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-635 Fields for Register: IRQ_TYPE_EN_RX_SA_200 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.629 IRQ_TYPE_STAT_RX_SA_200

- **Name:** IRQ_TYPE_STAT_RX_SA_200
- **Description:** Interrupt Status for each IRQ type of Rx SA 200
- **Size:** 32 bits
- **Offset:** 0x9d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 200

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-636 Fields for Register: IRQ_TYPE_STAT_RX_SA_200

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-636 Fields for Register: IRQ_TYPE_STAT_RX_SA_200 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-636 Fields for Register: IRQ_TYPE_STAT_RX_SA_200 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.630 IRQ_TYPE_EN_RX_SA_201

- **Name:** IRQ_TYPE_EN_RX_SA_201
- **Description:** Interrupt Enable for each IRQ type of Rx SA 201
- **Size:** 32 bits
- **Offset:** 0x9d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 201

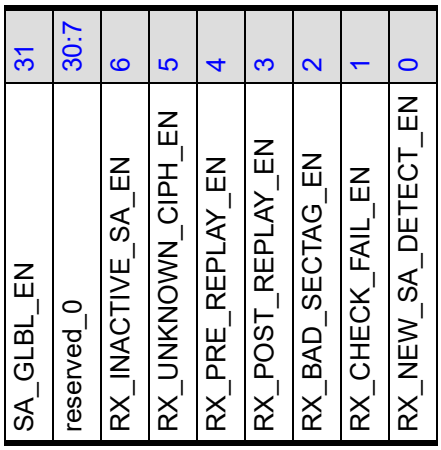


Table 6-637 Fields for Register: IRQ_TYPE_EN_RX_SA_201

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-637 Fields for Register: IRQ_TYPE_EN_RX_SA_201 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-637 Fields for Register: IRQ_TYPE_EN_RX_SA_201 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.631 IRQ_TYPE_STAT_RX_SA_201

- **Name:** IRQ_TYPE_STAT_RX_SA_201
- **Description:** Interrupt Status for each IRQ type of Rx SA 201
- **Size:** 32 bits
- **Offset:** 0x9dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 201

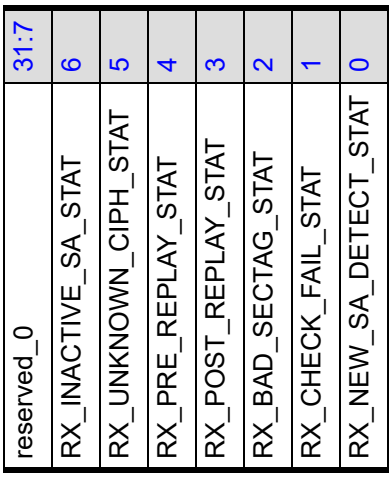


Table 6-638 Fields for Register: IRQ_TYPE_STAT_RX_SA_201

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-638 Fields for Register: IRQ_TYPE_STAT_RX_SA_201 (Continued)

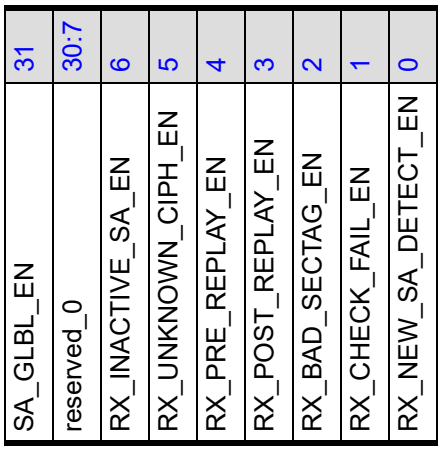
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-638 Fields for Register: IRQ_TYPE_STAT_RX_SA_201 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.632 IRQ_TYPE_EN_RX_SA_202

- **Name:** IRQ_TYPE_EN_RX_SA_202
- **Description:** Interrupt Enable for each IRQ type of Rx SA 202
- **Size:** 32 bits
- **Offset:** 0x9e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 202



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-639 Fields for Register: IRQ_TYPE_EN_RX_SA_202

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-639 Fields for Register: IRQ_TYPE_EN_RX_SA_202 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-639 Fields for Register: IRQ_TYPE_EN_RX_SA_202 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.633 IRQ_TYPE_STAT_RX_SA_202

- **Name:** IRQ_TYPE_STAT_RX_SA_202
- **Description:** Interrupt Status for each IRQ type of Rx SA 202
- **Size:** 32 bits
- **Offset:** 0x9e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 202

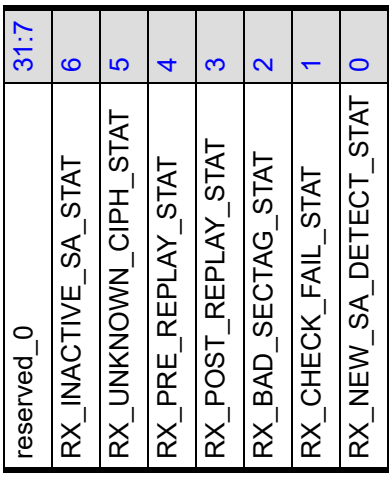


Table 6-640 Fields for Register: IRQ_TYPE_STAT_RX_SA_202

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-640 Fields for Register: IRQ_TYPE_STAT_RX_SA_202 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-640 Fields for Register: IRQ_TYPE_STAT_RX_SA_202 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.634 IRQ_TYPE_EN_RX_SA_203

- **Name:** IRQ_TYPE_EN_RX_SA_203
- **Description:** Interrupt Enable for each IRQ type of Rx SA 203
- **Size:** 32 bits
- **Offset:** 0x9e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 203

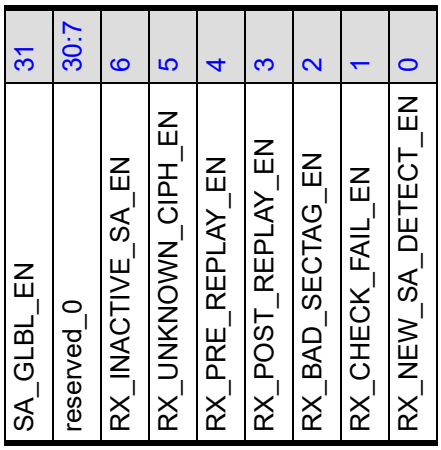


Table 6-641 Fields for Register: IRQ_TYPE_EN_RX_SA_203

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-641 Fields for Register: IRQ_TYPE_EN_RX_SA_203 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-641 Fields for Register: IRQ_TYPE_EN_RX_SA_203 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.635 IRQ_TYPE_STAT_RX_SA_203

- **Name:** IRQ_TYPE_STAT_RX_SA_203
- **Description:** Interrupt Status for each IRQ type of Rx SA 203
- **Size:** 32 bits
- **Offset:** 0x9ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 203

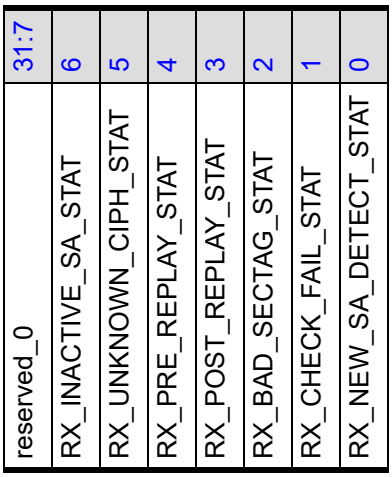


Table 6-642 Fields for Register: IRQ_TYPE_STAT_RX_SA_203

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-642 Fields for Register: IRQ_TYPE_STAT_RX_SA_203 (Continued)

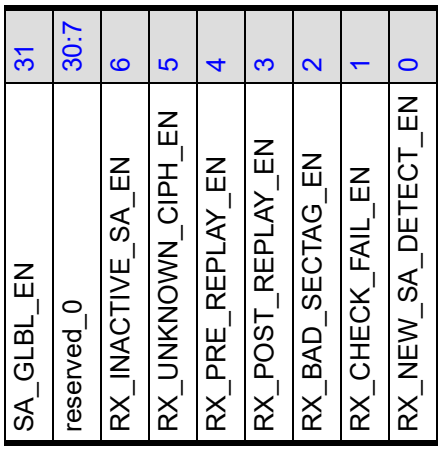
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-642 Fields for Register: IRQ_TYPE_STAT_RX_SA_203 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.636 IRQ_TYPE_EN_RX_SA_204

- **Name:** IRQ_TYPE_EN_RX_SA_204
- **Description:** Interrupt Enable for each IRQ type of Rx SA 204
- **Size:** 32 bits
- **Offset:** 0x9f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 204



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-643 Fields for Register: IRQ_TYPE_EN_RX_SA_204

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-643 Fields for Register: IRQ_TYPE_EN_RX_SA_204 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-643 Fields for Register: IRQ_TYPE_EN_RX_SA_204 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.637 IRQ_TYPE_STAT_RX_SA_204

- **Name:** IRQ_TYPE_STAT_RX_SA_204
- **Description:** Interrupt Status for each IRQ type of Rx SA 204
- **Size:** 32 bits
- **Offset:** 0x9f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 204

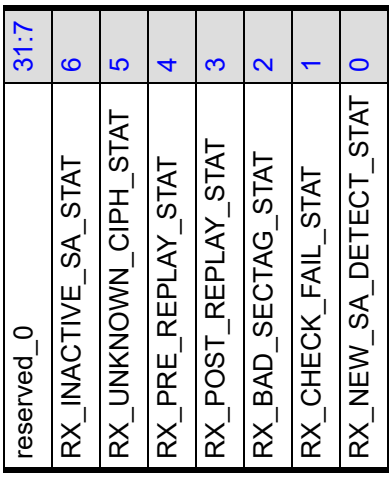


Table 6-644 Fields for Register: IRQ_TYPE_STAT_RX_SA_204

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-644 Fields for Register: IRQ_TYPE_STAT_RX_SA_204 (Continued)

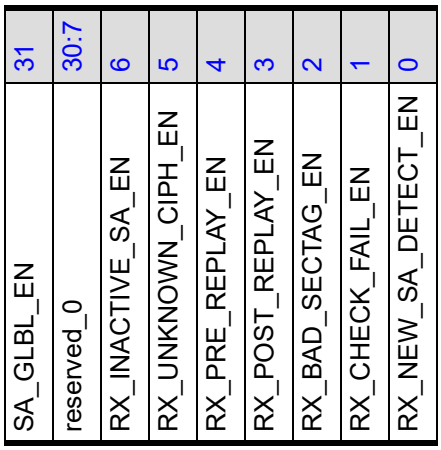
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-644 Fields for Register: IRQ_TYPE_STAT_RX_SA_204 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.638 IRQ_TYPE_EN_RX_SA_205

- **Name:** IRQ_TYPE_EN_RX_SA_205
- **Description:** Interrupt Enable for each IRQ type of Rx SA 205
- **Size:** 32 bits
- **Offset:** 0x9f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 205



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-645 Fields for Register: IRQ_TYPE_EN_RX_SA_205

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-645 Fields for Register: IRQ_TYPE_EN_RX_SA_205 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-645 Fields for Register: IRQ_TYPE_EN_RX_SA_205 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.639 IRQ_TYPE_STAT_RX_SA_205

- **Name:** IRQ_TYPE_STAT_RX_SA_205
- **Description:** Interrupt Status for each IRQ type of Rx SA 205
- **Size:** 32 bits
- **Offset:** 0x9fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 205

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-646 Fields for Register: IRQ_TYPE_STAT_RX_SA_205

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-646 Fields for Register: IRQ_TYPE_STAT_RX_SA_205 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-646 Fields for Register: IRQ_TYPE_STAT_RX_SA_205 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.640 IRQ_TYPE_EN_RX_SA_206

- **Name:** IRQ_TYPE_EN_RX_SA_206
- **Description:** Interrupt Enable for each IRQ type of Rx SA 206
- **Size:** 32 bits
- **Offset:** 0xa00
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 206

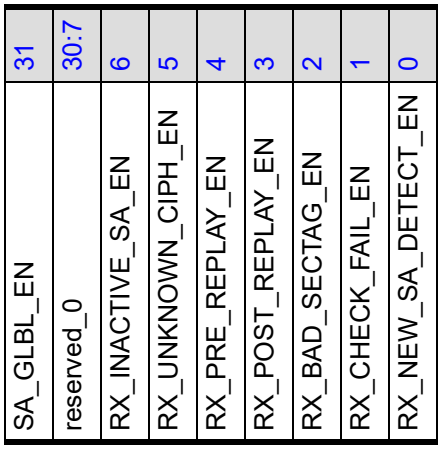


Table 6-647 Fields for Register: IRQ_TYPE_EN_RX_SA_206

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-647 Fields for Register: IRQ_TYPE_EN_RX_SA_206 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-647 Fields for Register: IRQ_TYPE_EN_RX_SA_206 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.641 IRQ_TYPE_STAT_RX_SA_206

- **Name:** IRQ_TYPE_STAT_RX_SA_206
- **Description:** Interrupt Status for each IRQ type of Rx SA 206
- **Size:** 32 bits
- **Offset:** 0xa04
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 206

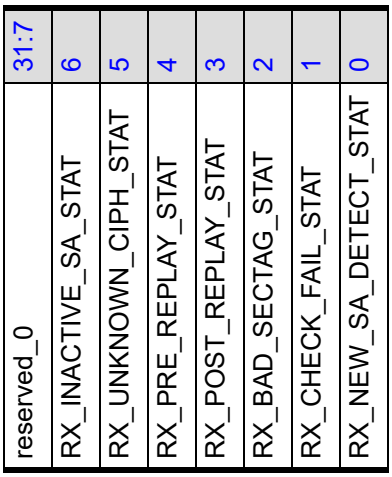


Table 6-648 Fields for Register: IRQ_TYPE_STAT_RX_SA_206

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-648 Fields for Register: IRQ_TYPE_STAT_RX_SA_206 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-648 Fields for Register: IRQ_TYPE_STAT_RX_SA_206 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.642 IRQ_TYPE_EN_RX_SA_207

- **Name:** IRQ_TYPE_EN_RX_SA_207
- **Description:** Interrupt Enable for each IRQ type of Rx SA 207
- **Size:** 32 bits
- **Offset:** 0xa08
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 207

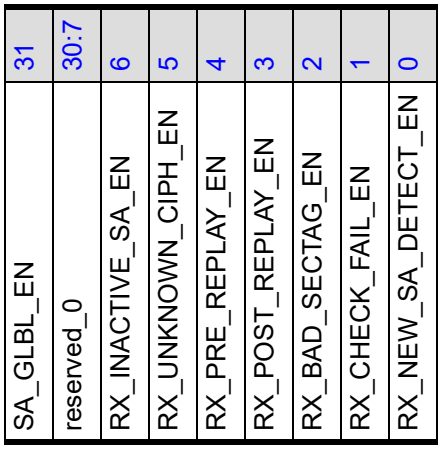


Table 6-649 Fields for Register: IRQ_TYPE_EN_RX_SA_207

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-649 Fields for Register: IRQ_TYPE_EN_RX_SA_207 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-649 Fields for Register: IRQ_TYPE_EN_RX_SA_207 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.643 IRQ_TYPE_STAT_RX_SA_207

- **Name:** IRQ_TYPE_STAT_RX_SA_207
- **Description:** Interrupt Status for each IRQ type of Rx SA 207
- **Size:** 32 bits
- **Offset:** 0xa0c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 207

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-650 Fields for Register: IRQ_TYPE_STAT_RX_SA_207

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-650 Fields for Register: IRQ_TYPE_STAT_RX_SA_207 (Continued)

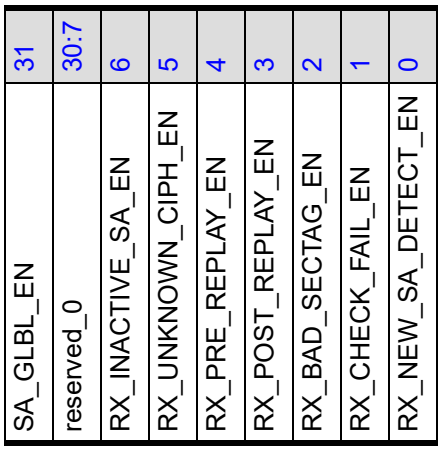
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-650 Fields for Register: IRQ_TYPE_STAT_RX_SA_207 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.644 IRQ_TYPE_EN_RX_SA_208

- **Name:** IRQ_TYPE_EN_RX_SA_208
- **Description:** Interrupt Enable for each IRQ type of Rx SA 208
- **Size:** 32 bits
- **Offset:** 0xa10
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 208



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-651 Fields for Register: IRQ_TYPE_EN_RX_SA_208

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-651 Fields for Register: IRQ_TYPE_EN_RX_SA_208 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-651 Fields for Register: IRQ_TYPE_EN_RX_SA_208 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.645 IRQ_TYPE_STAT_RX_SA_208

- **Name:** IRQ_TYPE_STAT_RX_SA_208
- **Description:** Interrupt Status for each IRQ type of Rx SA 208
- **Size:** 32 bits
- **Offset:** 0xa14
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 208

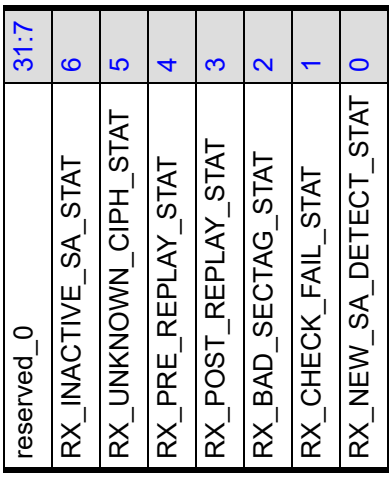


Table 6-652 Fields for Register: IRQ_TYPE_STAT_RX_SA_208

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-652 Fields for Register: IRQ_TYPE_STAT_RX_SA_208 (Continued)

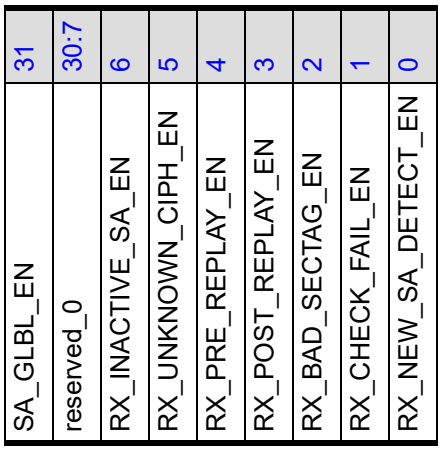
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-652 Fields for Register: IRQ_TYPE_STAT_RX_SA_208 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.646 IRQ_TYPE_EN_RX_SA_209

- **Name:** IRQ_TYPE_EN_RX_SA_209
- **Description:** Interrupt Enable for each IRQ type of Rx SA 209
- **Size:** 32 bits
- **Offset:** 0xa18
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 209



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-653 Fields for Register: IRQ_TYPE_EN_RX_SA_209

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-653 Fields for Register: IRQ_TYPE_EN_RX_SA_209 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-653 Fields for Register: IRQ_TYPE_EN_RX_SA_209 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.647 IRQ_TYPE_STAT_RX_SA_209

- **Name:** IRQ_TYPE_STAT_RX_SA_209
- **Description:** Interrupt Status for each IRQ type of Rx SA 209
- **Size:** 32 bits
- **Offset:** 0xa1c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 209

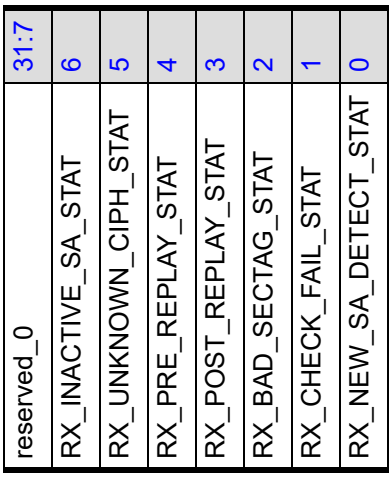


Table 6-654 Fields for Register: IRQ_TYPE_STAT_RX_SA_209

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-654 Fields for Register: IRQ_TYPE_STAT_RX_SA_209 (Continued)

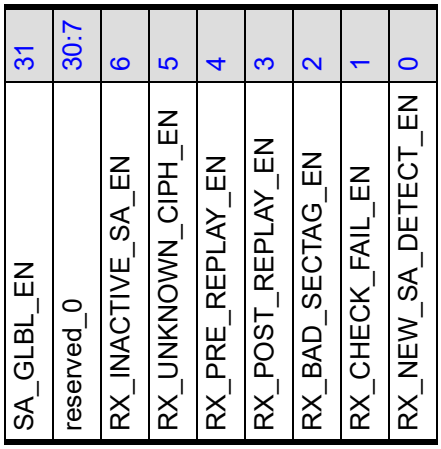
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-654 Fields for Register: IRQ_TYPE_STAT_RX_SA_209 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.648 IRQ_TYPE_EN_RX_SA_210

- **Name:** IRQ_TYPE_EN_RX_SA_210
- **Description:** Interrupt Enable for each IRQ type of Rx SA 210
- **Size:** 32 bits
- **Offset:** 0xa20
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 210



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-655 Fields for Register: IRQ_TYPE_EN_RX_SA_210

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-655 Fields for Register: IRQ_TYPE_EN_RX_SA_210 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-655 Fields for Register: IRQ_TYPE_EN_RX_SA_210 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.649 IRQ_TYPE_STAT_RX_SA_210

- **Name:** IRQ_TYPE_STAT_RX_SA_210
- **Description:** Interrupt Status for each IRQ type of Rx SA 210
- **Size:** 32 bits
- **Offset:** 0xa24
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 210

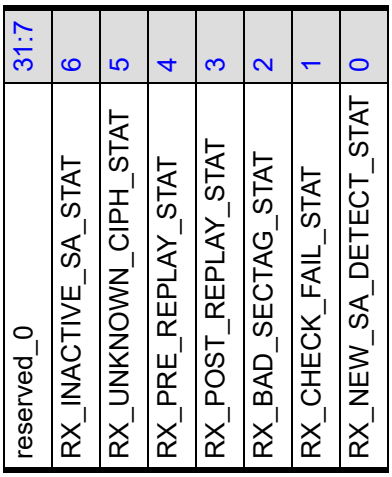


Table 6-656 Fields for Register: IRQ_TYPE_STAT_RX_SA_210

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-656 Fields for Register: IRQ_TYPE_STAT_RX_SA_210 (Continued)

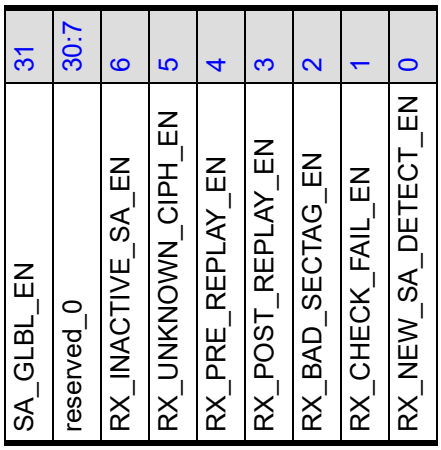
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-656 Fields for Register: IRQ_TYPE_STAT_RX_SA_210 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.650 IRQ_TYPE_EN_RX_SA_211

- **Name:** IRQ_TYPE_EN_RX_SA_211
- **Description:** Interrupt Enable for each IRQ type of Rx SA 211
- **Size:** 32 bits
- **Offset:** 0xa28
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 211



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-657 Fields for Register: IRQ_TYPE_EN_RX_SA_211

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-657 Fields for Register: IRQ_TYPE_EN_RX_SA_211 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-657 Fields for Register: IRQ_TYPE_EN_RX_SA_211 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.651 IRQ_TYPE_STAT_RX_SA_211

- **Name:** IRQ_TYPE_STAT_RX_SA_211
- **Description:** Interrupt Status for each IRQ type of Rx SA 211
- **Size:** 32 bits
- **Offset:** 0xa2c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 211

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-658 Fields for Register: IRQ_TYPE_STAT_RX_SA_211

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-658 Fields for Register: IRQ_TYPE_STAT_RX_SA_211 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-658 Fields for Register: IRQ_TYPE_STAT_RX_SA_211 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.652 IRQ_TYPE_EN_RX_SA_212

- **Name:** IRQ_TYPE_EN_RX_SA_212
- **Description:** Interrupt Enable for each IRQ type of Rx SA 212
- **Size:** 32 bits
- **Offset:** 0xa30
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 212

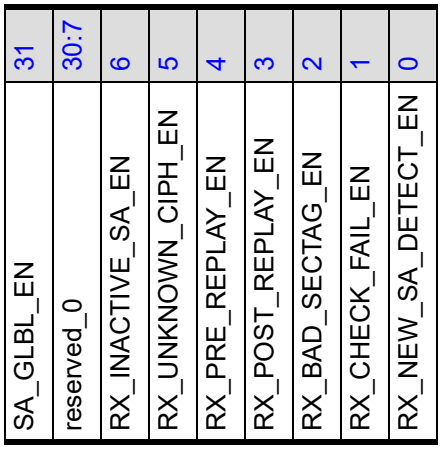


Table 6-659 Fields for Register: IRQ_TYPE_EN_RX_SA_212

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-659 Fields for Register: IRQ_TYPE_EN_RX_SA_212 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-659 Fields for Register: IRQ_TYPE_EN_RX_SA_212 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.653 IRQ_TYPE_STAT_RX_SA_212

- **Name:** IRQ_TYPE_STAT_RX_SA_212
- **Description:** Interrupt Status for each IRQ type of Rx SA 212
- **Size:** 32 bits
- **Offset:** 0xa34
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 212

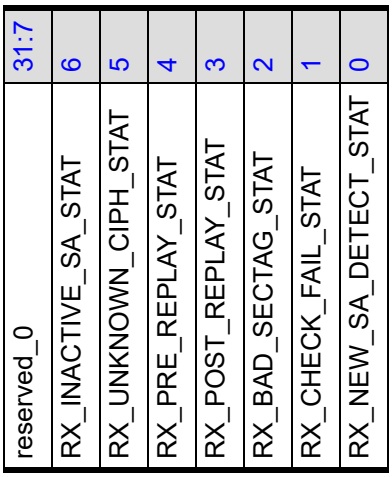


Table 6-660 Fields for Register: IRQ_TYPE_STAT_RX_SA_212

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-660 Fields for Register: IRQ_TYPE_STAT_RX_SA_212 (Continued)

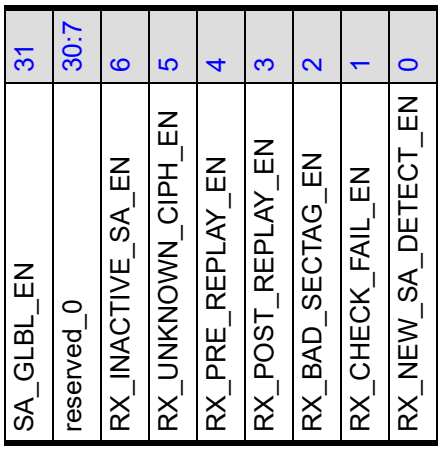
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-660 Fields for Register: IRQ_TYPE_STAT_RX_SA_212 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.654 IRQ_TYPE_EN_RX_SA_213

- **Name:** IRQ_TYPE_EN_RX_SA_213
- **Description:** Interrupt Enable for each IRQ type of Rx SA 213
- **Size:** 32 bits
- **Offset:** 0xa38
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 213



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-661 Fields for Register: IRQ_TYPE_EN_RX_SA_213

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-661 Fields for Register: IRQ_TYPE_EN_RX_SA_213 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-661 Fields for Register: IRQ_TYPE_EN_RX_SA_213 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.655 IRQ_TYPE_STAT_RX_SA_213

- **Name:** IRQ_TYPE_STAT_RX_SA_213
- **Description:** Interrupt Status for each IRQ type of Rx SA 213
- **Size:** 32 bits
- **Offset:** 0xa3c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 213

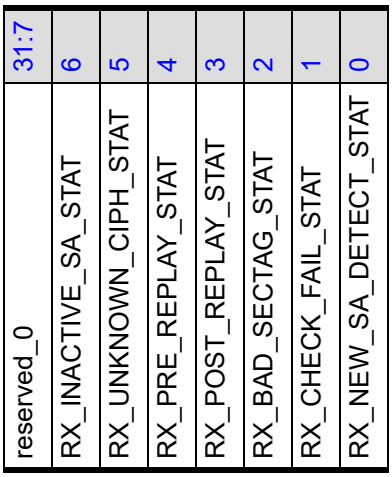


Table 6-662 Fields for Register: IRQ_TYPE_STAT_RX_SA_213

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-662 Fields for Register: IRQ_TYPE_STAT_RX_SA_213 (Continued)

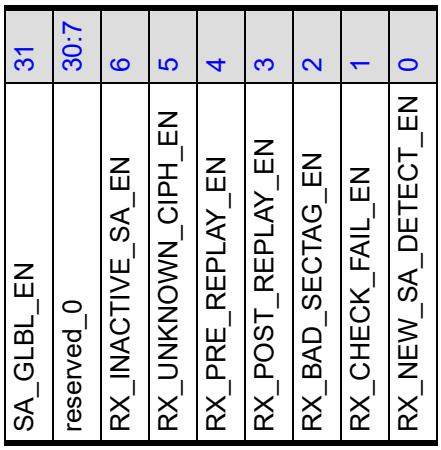
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-662 Fields for Register: IRQ_TYPE_STAT_RX_SA_213 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.656 IRQ_TYPE_EN_RX_SA_214

- **Name:** IRQ_TYPE_EN_RX_SA_214
- **Description:** Interrupt Enable for each IRQ type of Rx SA 214
- **Size:** 32 bits
- **Offset:** 0xa40
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 214



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-663 Fields for Register: IRQ_TYPE_EN_RX_SA_214

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-663 Fields for Register: IRQ_TYPE_EN_RX_SA_214 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-663 Fields for Register: IRQ_TYPE_EN_RX_SA_214 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.657 IRQ_TYPE_STAT_RX_SA_214

- **Name:** IRQ_TYPE_STAT_RX_SA_214
- **Description:** Interrupt Status for each IRQ type of Rx SA 214
- **Size:** 32 bits
- **Offset:** 0xa44
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 214

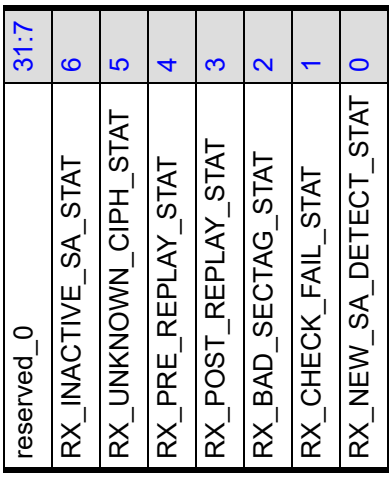


Table 6-664 Fields for Register: IRQ_TYPE_STAT_RX_SA_214

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-664 Fields for Register: IRQ_TYPE_STAT_RX_SA_214 (Continued)

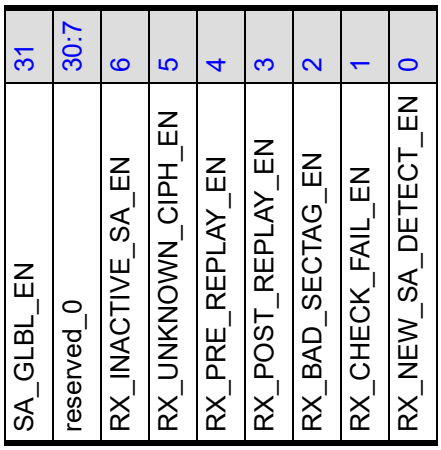
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-664 Fields for Register: IRQ_TYPE_STAT_RX_SA_214 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.658 IRQ_TYPE_EN_RX_SA_215

- **Name:** IRQ_TYPE_EN_RX_SA_215
- **Description:** Interrupt Enable for each IRQ type of Rx SA 215
- **Size:** 32 bits
- **Offset:** 0xa48
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 215



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-665 Fields for Register: IRQ_TYPE_EN_RX_SA_215

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-665 Fields for Register: IRQ_TYPE_EN_RX_SA_215 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-665 Fields for Register: IRQ_TYPE_EN_RX_SA_215 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.659 IRQ_TYPE_STAT_RX_SA_215

- **Name:** IRQ_TYPE_STAT_RX_SA_215
- **Description:** Interrupt Status for each IRQ type of Rx SA 215
- **Size:** 32 bits
- **Offset:** 0xa4c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 215

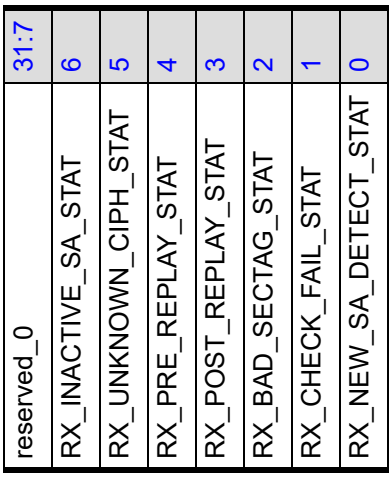


Table 6-666 Fields for Register: IRQ_TYPE_STAT_RX_SA_215

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-666 Fields for Register: IRQ_TYPE_STAT_RX_SA_215 (Continued)

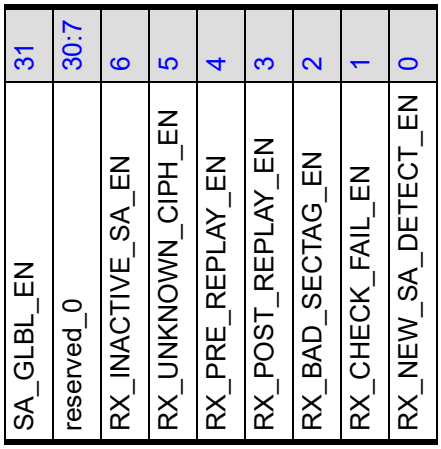
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-666 Fields for Register: IRQ_TYPE_STAT_RX_SA_215 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.660 IRQ_TYPE_EN_RX_SA_216

- **Name:** IRQ_TYPE_EN_RX_SA_216
- **Description:** Interrupt Enable for each IRQ type of Rx SA 216
- **Size:** 32 bits
- **Offset:** 0xa50
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 216



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-667 Fields for Register: IRQ_TYPE_EN_RX_SA_216

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-667 Fields for Register: IRQ_TYPE_EN_RX_SA_216 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-667 Fields for Register: IRQ_TYPE_EN_RX_SA_216 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.661 IRQ_TYPE_STAT_RX_SA_216

- **Name:** IRQ_TYPE_STAT_RX_SA_216
- **Description:** Interrupt Status for each IRQ type of Rx SA 216
- **Size:** 32 bits
- **Offset:** 0xa54
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 216

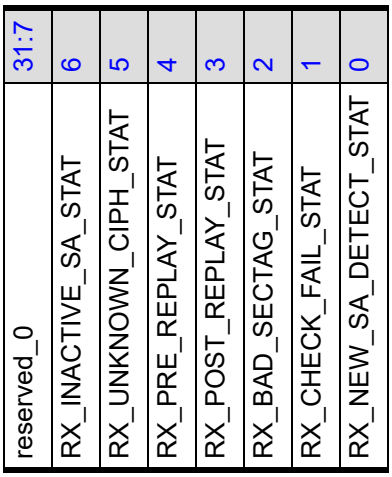


Table 6-668 Fields for Register: IRQ_TYPE_STAT_RX_SA_216

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-668 Fields for Register: IRQ_TYPE_STAT_RX_SA_216 (Continued)

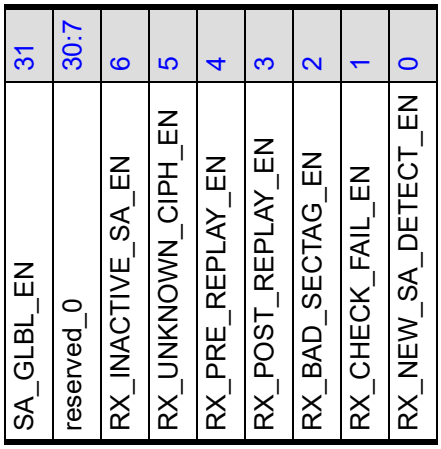
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-668 Fields for Register: IRQ_TYPE_STAT_RX_SA_216 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.662 IRQ_TYPE_EN_RX_SA_217

- **Name:** IRQ_TYPE_EN_RX_SA_217
- **Description:** Interrupt Enable for each IRQ type of Rx SA 217
- **Size:** 32 bits
- **Offset:** 0xa58
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 217



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-669 Fields for Register: IRQ_TYPE_EN_RX_SA_217

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-669 Fields for Register: IRQ_TYPE_EN_RX_SA_217 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-669 Fields for Register: IRQ_TYPE_EN_RX_SA_217 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.663 IRQ_TYPE_STAT_RX_SA_217

- **Name:** IRQ_TYPE_STAT_RX_SA_217
- **Description:** Interrupt Status for each IRQ type of Rx SA 217
- **Size:** 32 bits
- **Offset:** 0xa5c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 217

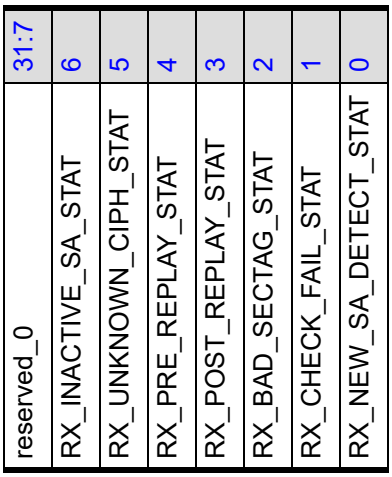


Table 6-670 Fields for Register: IRQ_TYPE_STAT_RX_SA_217

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-670 Fields for Register: IRQ_TYPE_STAT_RX_SA_217 (Continued)

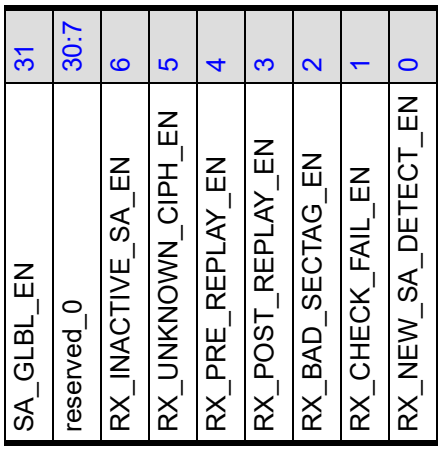
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-670 Fields for Register: IRQ_TYPE_STAT_RX_SA_217 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.664 IRQ_TYPE_EN_RX_SA_218

- **Name:** IRQ_TYPE_EN_RX_SA_218
- **Description:** Interrupt Enable for each IRQ type of Rx SA 218
- **Size:** 32 bits
- **Offset:** 0xa60
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 218



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-671 Fields for Register: IRQ_TYPE_EN_RX_SA_218

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-671 Fields for Register: IRQ_TYPE_EN_RX_SA_218 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-671 Fields for Register: IRQ_TYPE_EN_RX_SA_218 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.665 IRQ_TYPE_STAT_RX_SA_218

- **Name:** IRQ_TYPE_STAT_RX_SA_218
- **Description:** Interrupt Status for each IRQ type of Rx SA 218
- **Size:** 32 bits
- **Offset:** 0xa64
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 218

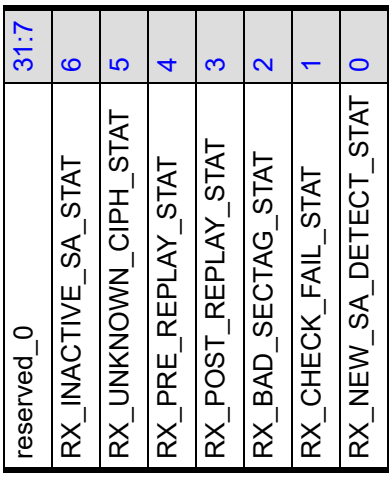


Table 6-672 Fields for Register: IRQ_TYPE_STAT_RX_SA_218

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-672 Fields for Register: IRQ_TYPE_STAT_RX_SA_218 (Continued)

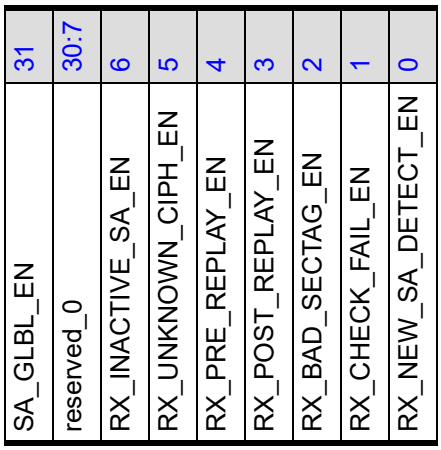
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-672 Fields for Register: IRQ_TYPE_STAT_RX_SA_218 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.666 IRQ_TYPE_EN_RX_SA_219

- **Name:** IRQ_TYPE_EN_RX_SA_219
- **Description:** Interrupt Enable for each IRQ type of Rx SA 219
- **Size:** 32 bits
- **Offset:** 0xa68
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 219



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-673 Fields for Register: IRQ_TYPE_EN_RX_SA_219

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-673 Fields for Register: IRQ_TYPE_EN_RX_SA_219 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-673 Fields for Register: IRQ_TYPE_EN_RX_SA_219 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.667 IRQ_TYPE_STAT_RX_SA_219

- **Name:** IRQ_TYPE_STAT_RX_SA_219
- **Description:** Interrupt Status for each IRQ type of Rx SA 219
- **Size:** 32 bits
- **Offset:** 0xa6c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 219

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-674 Fields for Register: IRQ_TYPE_STAT_RX_SA_219

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-674 Fields for Register: IRQ_TYPE_STAT_RX_SA_219 (Continued)

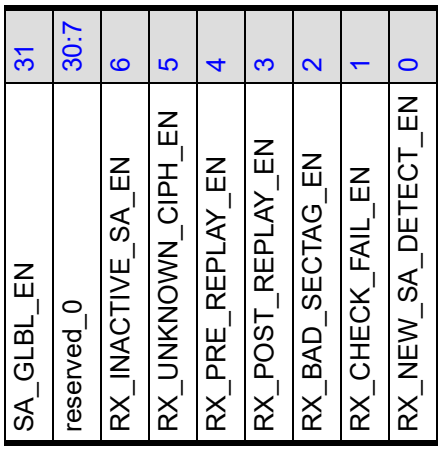
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-674 Fields for Register: IRQ_TYPE_STAT_RX_SA_219 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.668 IRQ_TYPE_EN_RX_SA_220

- **Name:** IRQ_TYPE_EN_RX_SA_220
- **Description:** Interrupt Enable for each IRQ type of Rx SA 220
- **Size:** 32 bits
- **Offset:** 0xa70
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 220



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-675 Fields for Register: IRQ_TYPE_EN_RX_SA_220

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-675 Fields for Register: IRQ_TYPE_EN_RX_SA_220 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-675 Fields for Register: IRQ_TYPE_EN_RX_SA_220 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.669 IRQ_TYPE_STAT_RX_SA_220

- **Name:** IRQ_TYPE_STAT_RX_SA_220
- **Description:** Interrupt Status for each IRQ type of Rx SA 220
- **Size:** 32 bits
- **Offset:** 0xa74
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 220

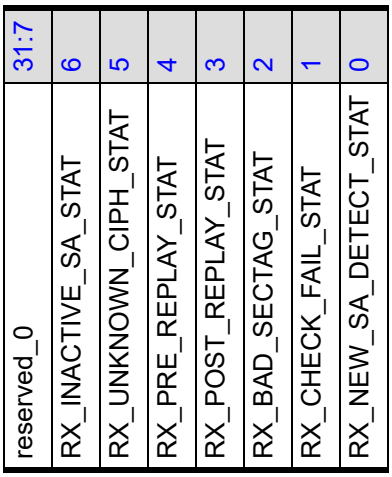


Table 6-676 Fields for Register: IRQ_TYPE_STAT_RX_SA_220

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-676 Fields for Register: IRQ_TYPE_STAT_RX_SA_220 (Continued)

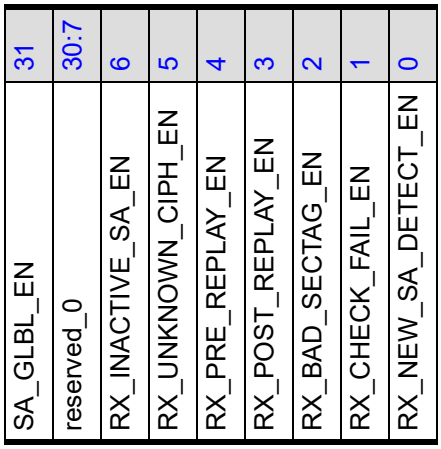
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-676 Fields for Register: IRQ_TYPE_STAT_RX_SA_220 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.670 IRQ_TYPE_EN_RX_SA_221

- **Name:** IRQ_TYPE_EN_RX_SA_221
- **Description:** Interrupt Enable for each IRQ type of Rx SA 221
- **Size:** 32 bits
- **Offset:** 0xa78
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 221



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-677 Fields for Register: IRQ_TYPE_EN_RX_SA_221

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-677 Fields for Register: IRQ_TYPE_EN_RX_SA_221 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-677 Fields for Register: IRQ_TYPE_EN_RX_SA_221 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.671 IRQ_TYPE_STAT_RX_SA_221

- **Name:** IRQ_TYPE_STAT_RX_SA_221
- **Description:** Interrupt Status for each IRQ type of Rx SA 221
- **Size:** 32 bits
- **Offset:** 0xa7c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 221

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-678 Fields for Register: IRQ_TYPE_STAT_RX_SA_221

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-678 Fields for Register: IRQ_TYPE_STAT_RX_SA_221 (Continued)

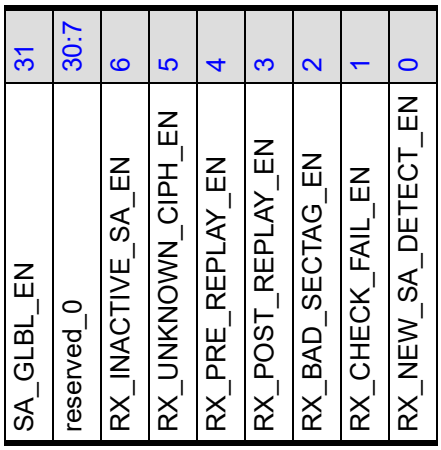
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-678 Fields for Register: IRQ_TYPE_STAT_RX_SA_221 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.672 IRQ_TYPE_EN_RX_SA_222

- **Name:** IRQ_TYPE_EN_RX_SA_222
- **Description:** Interrupt Enable for each IRQ type of Rx SA 222
- **Size:** 32 bits
- **Offset:** 0xa80
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 222



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-679 Fields for Register: IRQ_TYPE_EN_RX_SA_222

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-679 Fields for Register: IRQ_TYPE_EN_RX_SA_222 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-679 Fields for Register: IRQ_TYPE_EN_RX_SA_222 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.673 IRQ_TYPE_STAT_RX_SA_222

- **Name:** IRQ_TYPE_STAT_RX_SA_222
- **Description:** Interrupt Status for each IRQ type of Rx SA 222
- **Size:** 32 bits
- **Offset:** 0xa84
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 222

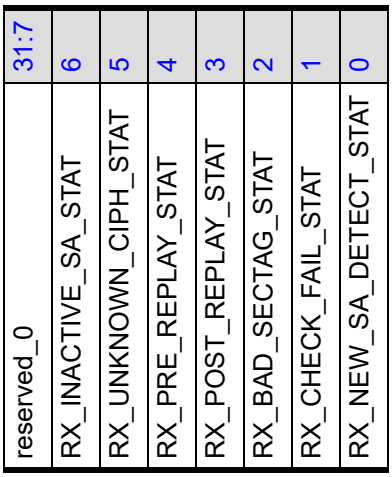


Table 6-680 Fields for Register: IRQ_TYPE_STAT_RX_SA_222

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-680 Fields for Register: IRQ_TYPE_STAT_RX_SA_222 (Continued)

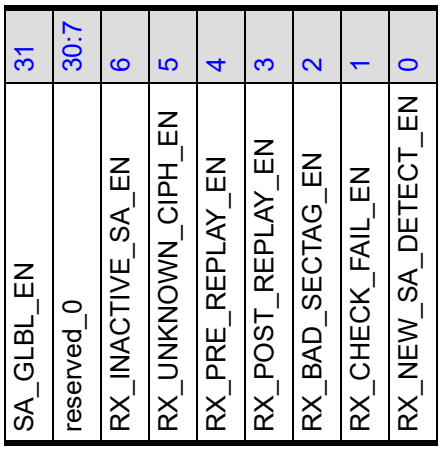
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-680 Fields for Register: IRQ_TYPE_STAT_RX_SA_222 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.674 IRQ_TYPE_EN_RX_SA_223

- **Name:** IRQ_TYPE_EN_RX_SA_223
- **Description:** Interrupt Enable for each IRQ type of Rx SA 223
- **Size:** 32 bits
- **Offset:** 0xa88
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 223



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-681 Fields for Register: IRQ_TYPE_EN_RX_SA_223

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-681 Fields for Register: IRQ_TYPE_EN_RX_SA_223 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-681 Fields for Register: IRQ_TYPE_EN_RX_SA_223 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.675 IRQ_TYPE_STAT_RX_SA_223

- **Name:** IRQ_TYPE_STAT_RX_SA_223
- **Description:** Interrupt Status for each IRQ type of Rx SA 223
- **Size:** 32 bits
- **Offset:** 0xa8c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 223

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-682 Fields for Register: IRQ_TYPE_STAT_RX_SA_223

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-682 Fields for Register: IRQ_TYPE_STAT_RX_SA_223 (Continued)

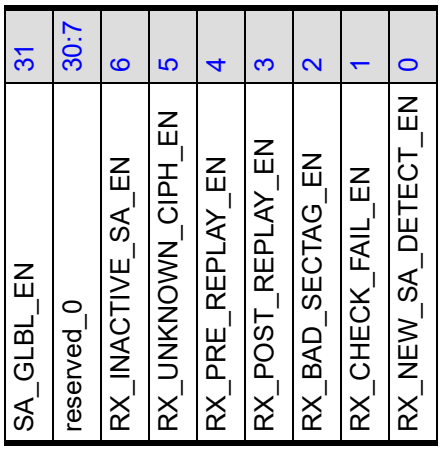
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-682 Fields for Register: IRQ_TYPE_STAT_RX_SA_223 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.676 IRQ_TYPE_EN_RX_SA_224

- **Name:** IRQ_TYPE_EN_RX_SA_224
- **Description:** Interrupt Enable for each IRQ type of Rx SA 224
- **Size:** 32 bits
- **Offset:** 0xa90
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 224



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-683 Fields for Register: IRQ_TYPE_EN_RX_SA_224

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-683 Fields for Register: IRQ_TYPE_EN_RX_SA_224 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-683 Fields for Register: IRQ_TYPE_EN_RX_SA_224 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.677 IRQ_TYPE_STAT_RX_SA_224

- **Name:** IRQ_TYPE_STAT_RX_SA_224
- **Description:** Interrupt Status for each IRQ type of Rx SA 224
- **Size:** 32 bits
- **Offset:** 0xa94
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 224

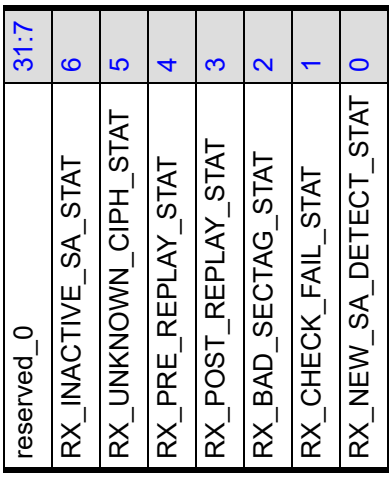


Table 6-684 Fields for Register: IRQ_TYPE_STAT_RX_SA_224

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-684 Fields for Register: IRQ_TYPE_STAT_RX_SA_224 (Continued)

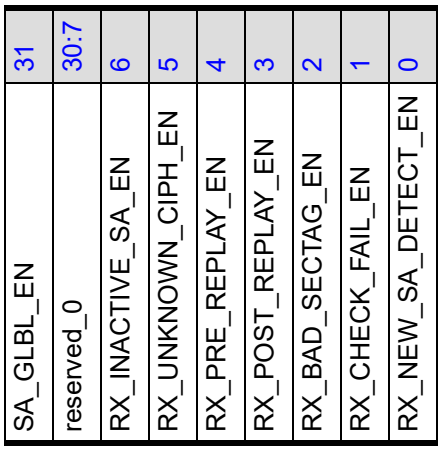
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-684 Fields for Register: IRQ_TYPE_STAT_RX_SA_224 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.678 IRQ_TYPE_EN_RX_SA_225

- **Name:** IRQ_TYPE_EN_RX_SA_225
- **Description:** Interrupt Enable for each IRQ type of Rx SA 225
- **Size:** 32 bits
- **Offset:** 0xa98
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 225



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-685 Fields for Register: IRQ_TYPE_EN_RX_SA_225

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-685 Fields for Register: IRQ_TYPE_EN_RX_SA_225 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-685 Fields for Register: IRQ_TYPE_EN_RX_SA_225 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.679 IRQ_TYPE_STAT_RX_SA_225

- **Name:** IRQ_TYPE_STAT_RX_SA_225
- **Description:** Interrupt Status for each IRQ type of Rx SA 225
- **Size:** 32 bits
- **Offset:** 0xa9c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 225

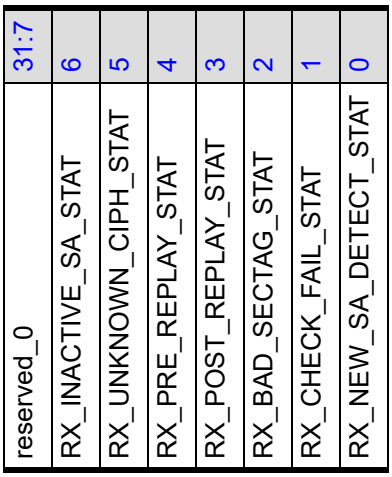


Table 6-686 Fields for Register: IRQ_TYPE_STAT_RX_SA_225

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-686 Fields for Register: IRQ_TYPE_STAT_RX_SA_225 (Continued)

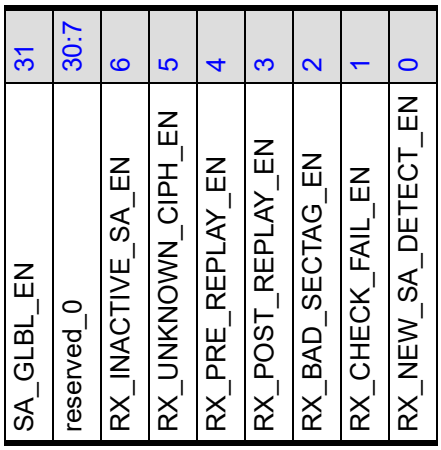
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-686 Fields for Register: IRQ_TYPE_STAT_RX_SA_225 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.680 IRQ_TYPE_EN_RX_SA_226

- **Name:** IRQ_TYPE_EN_RX_SA_226
- **Description:** Interrupt Enable for each IRQ type of Rx SA 226
- **Size:** 32 bits
- **Offset:** 0xaa0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 226



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-687 Fields for Register: IRQ_TYPE_EN_RX_SA_226

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-687 Fields for Register: IRQ_TYPE_EN_RX_SA_226 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-687 Fields for Register: IRQ_TYPE_EN_RX_SA_226 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.681 IRQ_TYPE_STAT_RX_SA_226

- **Name:** IRQ_TYPE_STAT_RX_SA_226
- **Description:** Interrupt Status for each IRQ type of Rx SA 226
- **Size:** 32 bits
- **Offset:** 0xaa4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 226

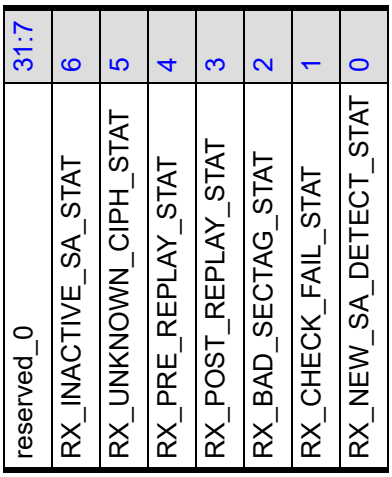


Table 6-688 Fields for Register: IRQ_TYPE_STAT_RX_SA_226

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-688 Fields for Register: IRQ_TYPE_STAT_RX_SA_226 (Continued)

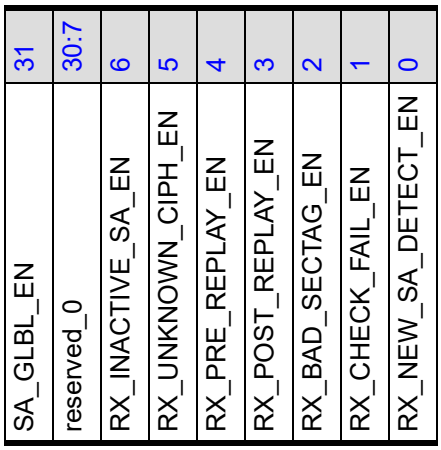
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-688 Fields for Register: IRQ_TYPE_STAT_RX_SA_226 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.682 IRQ_TYPE_EN_RX_SA_227

- **Name:** IRQ_TYPE_EN_RX_SA_227
- **Description:** Interrupt Enable for each IRQ type of Rx SA 227
- **Size:** 32 bits
- **Offset:** 0xaa8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 227



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-689 Fields for Register: IRQ_TYPE_EN_RX_SA_227

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-689 Fields for Register: IRQ_TYPE_EN_RX_SA_227 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-689 Fields for Register: IRQ_TYPE_EN_RX_SA_227 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.683 IRQ_TYPE_STAT_RX_SA_227

- **Name:** IRQ_TYPE_STAT_RX_SA_227
- **Description:** Interrupt Status for each IRQ type of Rx SA 227
- **Size:** 32 bits
- **Offset:** 0xaac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 227

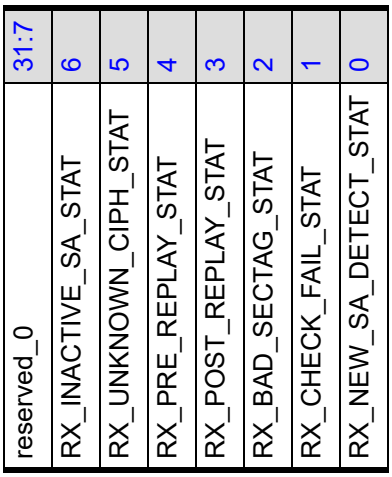


Table 6-690 Fields for Register: IRQ_TYPE_STAT_RX_SA_227

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-690 Fields for Register: IRQ_TYPE_STAT_RX_SA_227 (Continued)

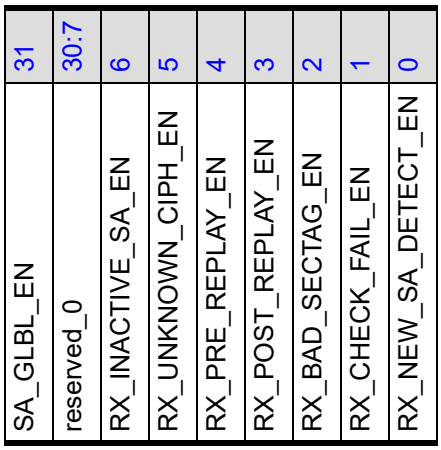
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-690 Fields for Register: IRQ_TYPE_STAT_RX_SA_227 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.684 IRQ_TYPE_EN_RX_SA_228

- **Name:** IRQ_TYPE_EN_RX_SA_228
- **Description:** Interrupt Enable for each IRQ type of Rx SA 228
- **Size:** 32 bits
- **Offset:** 0xab0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 228



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-691 Fields for Register: IRQ_TYPE_EN_RX_SA_228

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-691 Fields for Register: IRQ_TYPE_EN_RX_SA_228 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-691 Fields for Register: IRQ_TYPE_EN_RX_SA_228 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.685 IRQ_TYPE_STAT_RX_SA_228

- **Name:** IRQ_TYPE_STAT_RX_SA_228
- **Description:** Interrupt Status for each IRQ type of Rx SA 228
- **Size:** 32 bits
- **Offset:** 0xab4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 228

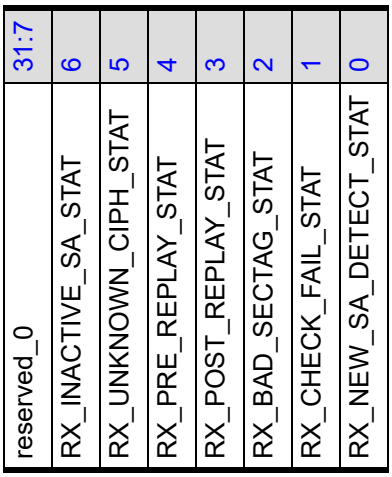


Table 6-692 Fields for Register: IRQ_TYPE_STAT_RX_SA_228

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-692 Fields for Register: IRQ_TYPE_STAT_RX_SA_228 (Continued)

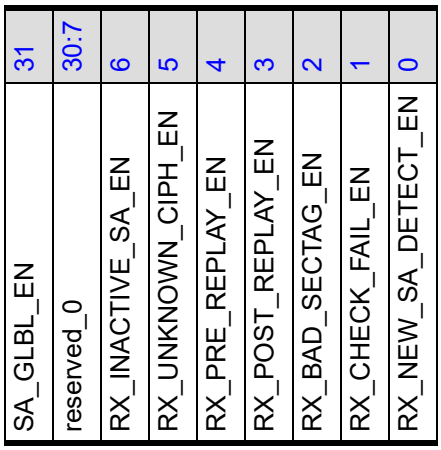
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-692 Fields for Register: IRQ_TYPE_STAT_RX_SA_228 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.686 IRQ_TYPE_EN_RX_SA_229

- **Name:** IRQ_TYPE_EN_RX_SA_229
- **Description:** Interrupt Enable for each IRQ type of Rx SA 229
- **Size:** 32 bits
- **Offset:** 0xab8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 229



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-693 Fields for Register: IRQ_TYPE_EN_RX_SA_229

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-693 Fields for Register: IRQ_TYPE_EN_RX_SA_229 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-693 Fields for Register: IRQ_TYPE_EN_RX_SA_229 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.687 IRQ_TYPE_STAT_RX_SA_229

- **Name:** IRQ_TYPE_STAT_RX_SA_229
- **Description:** Interrupt Status for each IRQ type of Rx SA 229
- **Size:** 32 bits
- **Offset:** 0xabc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 229

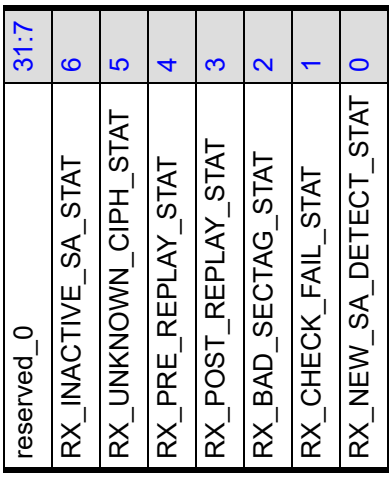


Table 6-694 Fields for Register: IRQ_TYPE_STAT_RX_SA_229

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-694 Fields for Register: IRQ_TYPE_STAT_RX_SA_229 (Continued)

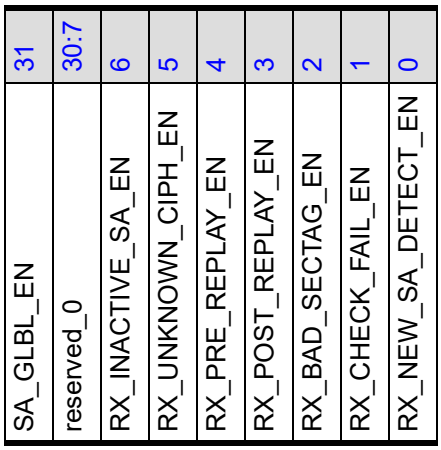
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-694 Fields for Register: IRQ_TYPE_STAT_RX_SA_229 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.688 IRQ_TYPE_EN_RX_SA_230

- **Name:** IRQ_TYPE_EN_RX_SA_230
- **Description:** Interrupt Enable for each IRQ type of Rx SA 230
- **Size:** 32 bits
- **Offset:** 0xac0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 230



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-695 Fields for Register: IRQ_TYPE_EN_RX_SA_230

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-695 Fields for Register: IRQ_TYPE_EN_RX_SA_230 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-695 Fields for Register: IRQ_TYPE_EN_RX_SA_230 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.689 IRQ_TYPE_STAT_RX_SA_230

- **Name:** IRQ_TYPE_STAT_RX_SA_230
- **Description:** Interrupt Status for each IRQ type of Rx SA 230
- **Size:** 32 bits
- **Offset:** 0xac4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 230

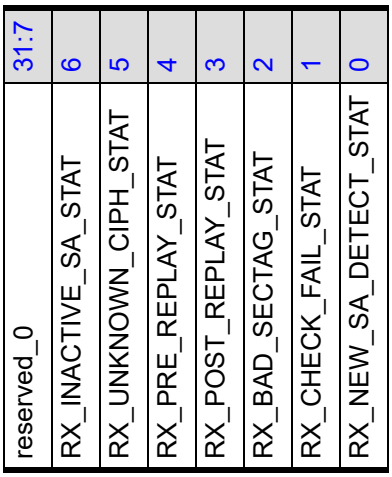


Table 6-696 Fields for Register: IRQ_TYPE_STAT_RX_SA_230

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-696 Fields for Register: IRQ_TYPE_STAT_RX_SA_230 (Continued)

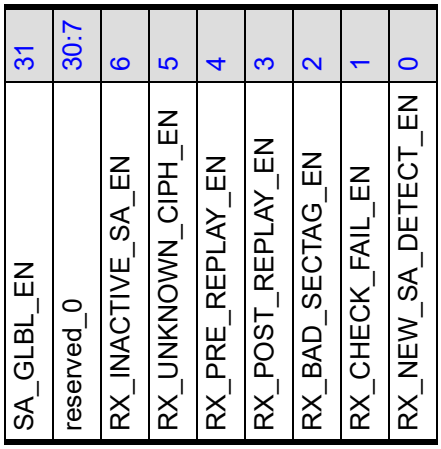
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-696 Fields for Register: IRQ_TYPE_STAT_RX_SA_230 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.690 IRQ_TYPE_EN_RX_SA_231

- **Name:** IRQ_TYPE_EN_RX_SA_231
- **Description:** Interrupt Enable for each IRQ type of Rx SA 231
- **Size:** 32 bits
- **Offset:** 0xac8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 231



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-697 Fields for Register: IRQ_TYPE_EN_RX_SA_231

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-697 Fields for Register: IRQ_TYPE_EN_RX_SA_231 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-697 Fields for Register: IRQ_TYPE_EN_RX_SA_231 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.691 IRQ_TYPE_STAT_RX_SA_231

- **Name:** IRQ_TYPE_STAT_RX_SA_231
- **Description:** Interrupt Status for each IRQ type of Rx SA 231
- **Size:** 32 bits
- **Offset:** 0xacc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 231

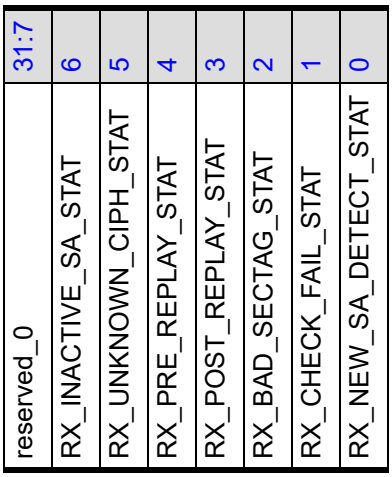


Table 6-698 Fields for Register: IRQ_TYPE_STAT_RX_SA_231

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-698 Fields for Register: IRQ_TYPE_STAT_RX_SA_231 (Continued)

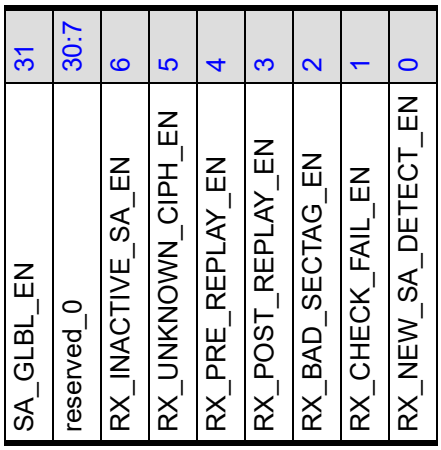
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-698 Fields for Register: IRQ_TYPE_STAT_RX_SA_231 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.692 IRQ_TYPE_EN_RX_SA_232

- **Name:** IRQ_TYPE_EN_RX_SA_232
- **Description:** Interrupt Enable for each IRQ type of Rx SA 232
- **Size:** 32 bits
- **Offset:** 0xad0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 232



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-699 Fields for Register: IRQ_TYPE_EN_RX_SA_232

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-699 Fields for Register: IRQ_TYPE_EN_RX_SA_232 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-699 Fields for Register: IRQ_TYPE_EN_RX_SA_232 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.693 IRQ_TYPE_STAT_RX_SA_232

- **Name:** IRQ_TYPE_STAT_RX_SA_232
- **Description:** Interrupt Status for each IRQ type of Rx SA 232
- **Size:** 32 bits
- **Offset:** 0xad4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 232

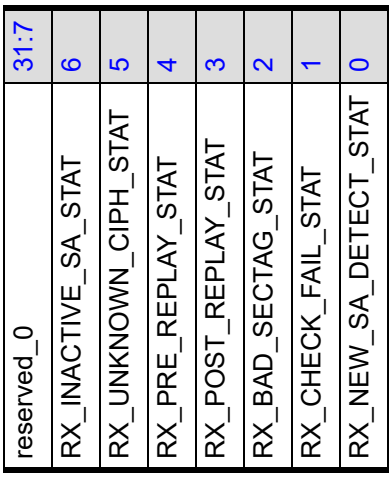


Table 6-700 Fields for Register: IRQ_TYPE_STAT_RX_SA_232

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-700 Fields for Register: IRQ_TYPE_STAT_RX_SA_232 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-700 Fields for Register: IRQ_TYPE_STAT_RX_SA_232 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.694 IRQ_TYPE_EN_RX_SA_233

- **Name:** IRQ_TYPE_EN_RX_SA_233
- **Description:** Interrupt Enable for each IRQ type of Rx SA 233
- **Size:** 32 bits
- **Offset:** 0xad8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 233

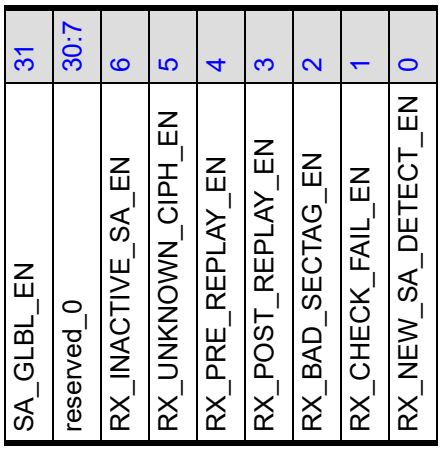


Table 6-701 Fields for Register: IRQ_TYPE_EN_RX_SA_233

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-701 Fields for Register: IRQ_TYPE_EN_RX_SA_233 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-701 Fields for Register: IRQ_TYPE_EN_RX_SA_233 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.695 IRQ_TYPE_STAT_RX_SA_233

- **Name:** IRQ_TYPE_STAT_RX_SA_233
- **Description:** Interrupt Status for each IRQ type of Rx SA 233
- **Size:** 32 bits
- **Offset:** 0xadc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 233

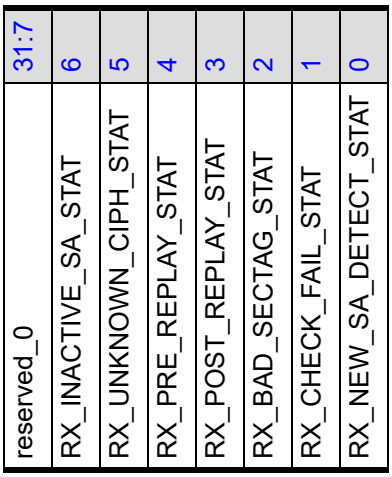


Table 6-702 Fields for Register: IRQ_TYPE_STAT_RX_SA_233

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-702 Fields for Register: IRQ_TYPE_STAT_RX_SA_233 (Continued)

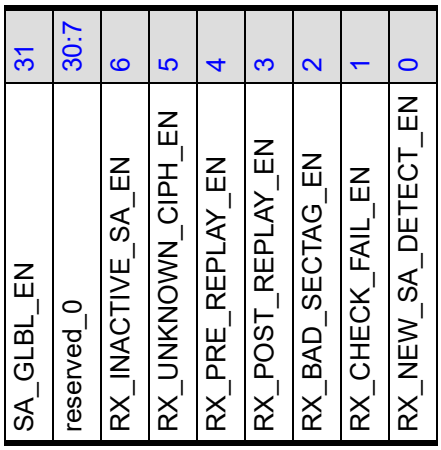
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-702 Fields for Register: IRQ_TYPE_STAT_RX_SA_233 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.696 IRQ_TYPE_EN_RX_SA_234

- **Name:** IRQ_TYPE_EN_RX_SA_234
- **Description:** Interrupt Enable for each IRQ type of Rx SA 234
- **Size:** 32 bits
- **Offset:** 0xae0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 234



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-703 Fields for Register: IRQ_TYPE_EN_RX_SA_234

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-703 Fields for Register: IRQ_TYPE_EN_RX_SA_234 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-703 Fields for Register: IRQ_TYPE_EN_RX_SA_234 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.697 IRQ_TYPE_STAT_RX_SA_234

- **Name:** IRQ_TYPE_STAT_RX_SA_234
- **Description:** Interrupt Status for each IRQ type of Rx SA 234
- **Size:** 32 bits
- **Offset:** 0xae4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 234

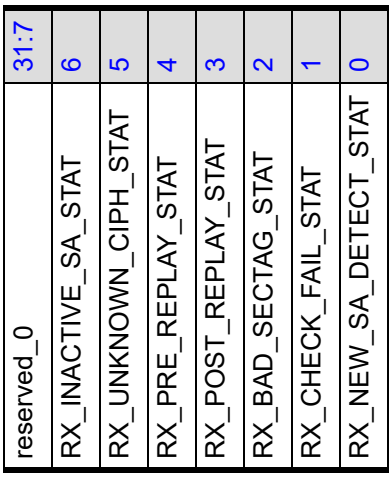


Table 6-704 Fields for Register: IRQ_TYPE_STAT_RX_SA_234

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-704 Fields for Register: IRQ_TYPE_STAT_RX_SA_234 (Continued)

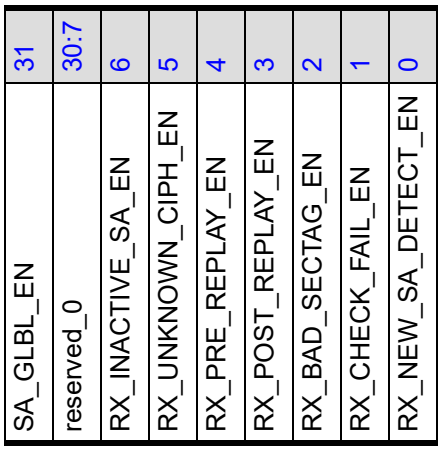
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-704 Fields for Register: IRQ_TYPE_STAT_RX_SA_234 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.698 IRQ_TYPE_EN_RX_SA_235

- **Name:** IRQ_TYPE_EN_RX_SA_235
- **Description:** Interrupt Enable for each IRQ type of Rx SA 235
- **Size:** 32 bits
- **Offset:** 0xae8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 235



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-705 Fields for Register: IRQ_TYPE_EN_RX_SA_235

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-705 Fields for Register: IRQ_TYPE_EN_RX_SA_235 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-705 Fields for Register: IRQ_TYPE_EN_RX_SA_235 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.699 IRQ_TYPE_STAT_RX_SA_235

- **Name:** IRQ_TYPE_STAT_RX_SA_235
- **Description:** Interrupt Status for each IRQ type of Rx SA 235
- **Size:** 32 bits
- **Offset:** 0xaec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 235

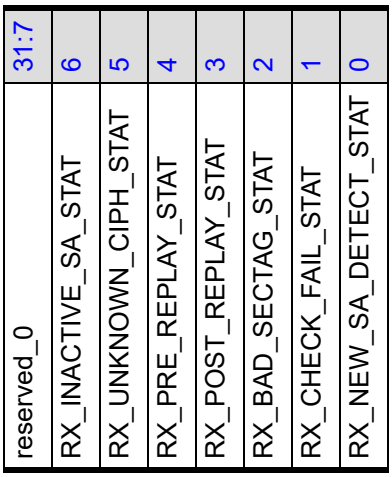


Table 6-706 Fields for Register: IRQ_TYPE_STAT_RX_SA_235

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-706 Fields for Register: IRQ_TYPE_STAT_RX_SA_235 (Continued)

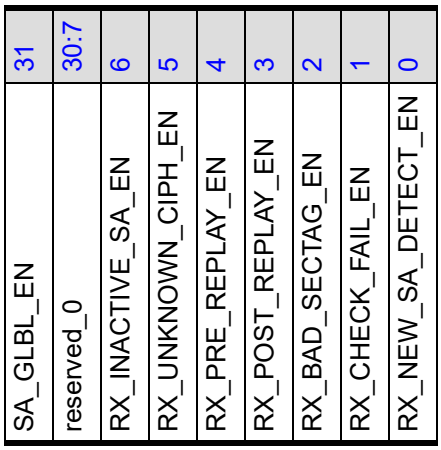
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-706 Fields for Register: IRQ_TYPE_STAT_RX_SA_235 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.700 IRQ_TYPE_EN_RX_SA_236

- **Name:** IRQ_TYPE_EN_RX_SA_236
- **Description:** Interrupt Enable for each IRQ type of Rx SA 236
- **Size:** 32 bits
- **Offset:** 0xaf0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 236



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-707 Fields for Register: IRQ_TYPE_EN_RX_SA_236

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-707 Fields for Register: IRQ_TYPE_EN_RX_SA_236 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-707 Fields for Register: IRQ_TYPE_EN_RX_SA_236 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.701 IRQ_TYPE_STAT_RX_SA_236

- **Name:** IRQ_TYPE_STAT_RX_SA_236
- **Description:** Interrupt Status for each IRQ type of Rx SA 236
- **Size:** 32 bits
- **Offset:** 0xaf4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 236

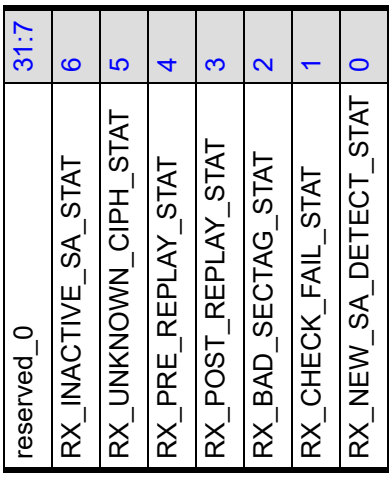


Table 6-708 Fields for Register: IRQ_TYPE_STAT_RX_SA_236

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-708 Fields for Register: IRQ_TYPE_STAT_RX_SA_236 (Continued)

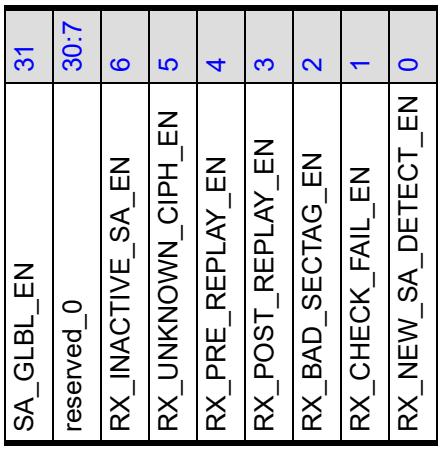
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-708 Fields for Register: IRQ_TYPE_STAT_RX_SA_236 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.702 IRQ_TYPE_EN_RX_SA_237

- **Name:** IRQ_TYPE_EN_RX_SA_237
- **Description:** Interrupt Enable for each IRQ type of Rx SA 237
- **Size:** 32 bits
- **Offset:** 0xaf8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 237



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-709 Fields for Register: IRQ_TYPE_EN_RX_SA_237

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-709 Fields for Register: IRQ_TYPE_EN_RX_SA_237 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-709 Fields for Register: IRQ_TYPE_EN_RX_SA_237 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.703 IRQ_TYPE_STAT_RX_SA_237

- **Name:** IRQ_TYPE_STAT_RX_SA_237
- **Description:** Interrupt Status for each IRQ type of Rx SA 237
- **Size:** 32 bits
- **Offset:** 0xafc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 237

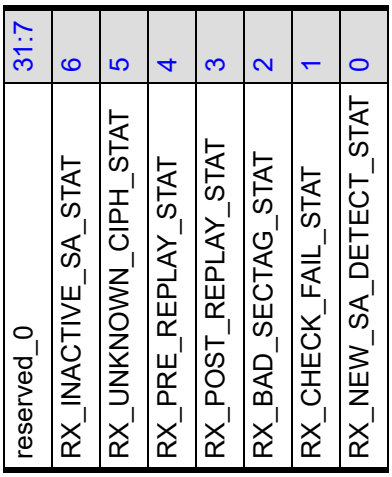


Table 6-710 Fields for Register: IRQ_TYPE_STAT_RX_SA_237

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-710 Fields for Register: IRQ_TYPE_STAT_RX_SA_237 (Continued)

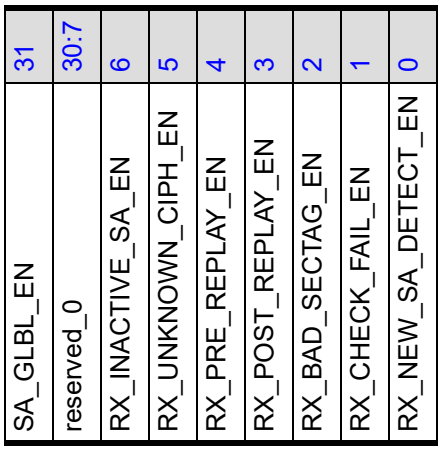
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-710 Fields for Register: IRQ_TYPE_STAT_RX_SA_237 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.704 IRQ_TYPE_EN_RX_SA_238

- **Name:** IRQ_TYPE_EN_RX_SA_238
- **Description:** Interrupt Enable for each IRQ type of Rx SA 238
- **Size:** 32 bits
- **Offset:** 0xb00
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 238



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-711 Fields for Register: IRQ_TYPE_EN_RX_SA_238

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-711 Fields for Register: IRQ_TYPE_EN_RX_SA_238 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-711 Fields for Register: IRQ_TYPE_EN_RX_SA_238 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.705 IRQ_TYPE_STAT_RX_SA_238

- **Name:** IRQ_TYPE_STAT_RX_SA_238
- **Description:** Interrupt Status for each IRQ type of Rx SA 238
- **Size:** 32 bits
- **Offset:** 0xb04
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 238

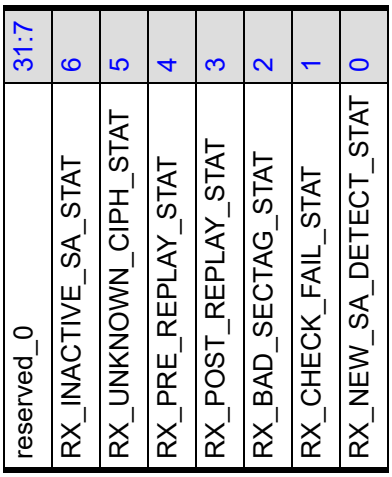


Table 6-712 Fields for Register: IRQ_TYPE_STAT_RX_SA_238

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-712 Fields for Register: IRQ_TYPE_STAT_RX_SA_238 (Continued)

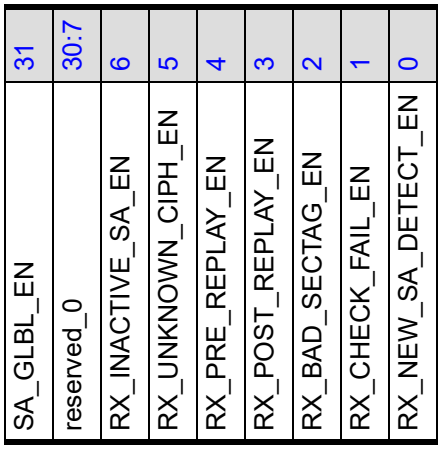
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-712 Fields for Register: IRQ_TYPE_STAT_RX_SA_238 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.706 IRQ_TYPE_EN_RX_SA_239

- **Name:** IRQ_TYPE_EN_RX_SA_239
- **Description:** Interrupt Enable for each IRQ type of Rx SA 239
- **Size:** 32 bits
- **Offset:** 0xb08
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 239



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-713 Fields for Register: IRQ_TYPE_EN_RX_SA_239

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-713 Fields for Register: IRQ_TYPE_EN_RX_SA_239 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-713 Fields for Register: IRQ_TYPE_EN_RX_SA_239 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.707 IRQ_TYPE_STAT_RX_SA_239

- **Name:** IRQ_TYPE_STAT_RX_SA_239
- **Description:** Interrupt Status for each IRQ type of Rx SA 239
- **Size:** 32 bits
- **Offset:** 0xb0c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 239

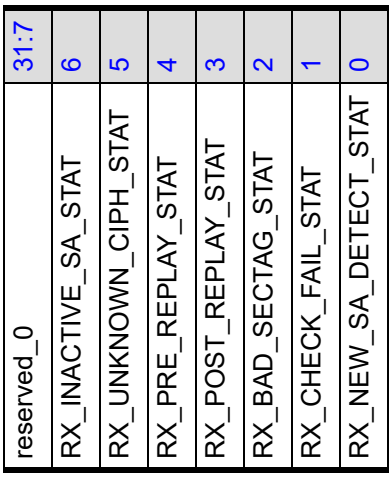


Table 6-714 Fields for Register: IRQ_TYPE_STAT_RX_SA_239

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-714 Fields for Register: IRQ_TYPE_STAT_RX_SA_239 (Continued)

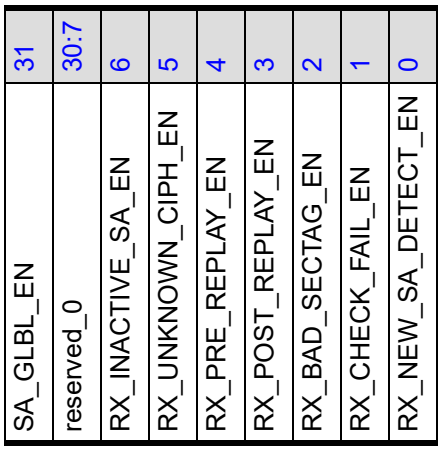
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-714 Fields for Register: IRQ_TYPE_STAT_RX_SA_239 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.708 IRQ_TYPE_EN_RX_SA_240

- **Name:** IRQ_TYPE_EN_RX_SA_240
- **Description:** Interrupt Enable for each IRQ type of Rx SA 240
- **Size:** 32 bits
- **Offset:** 0xb10
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 240



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-715 Fields for Register: IRQ_TYPE_EN_RX_SA_240

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-715 Fields for Register: IRQ_TYPE_EN_RX_SA_240 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-715 Fields for Register: IRQ_TYPE_EN_RX_SA_240 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.709 IRQ_TYPE_STAT_RX_SA_240

- **Name:** IRQ_TYPE_STAT_RX_SA_240
- **Description:** Interrupt Status for each IRQ type of Rx SA 240
- **Size:** 32 bits
- **Offset:** 0xb14
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 240

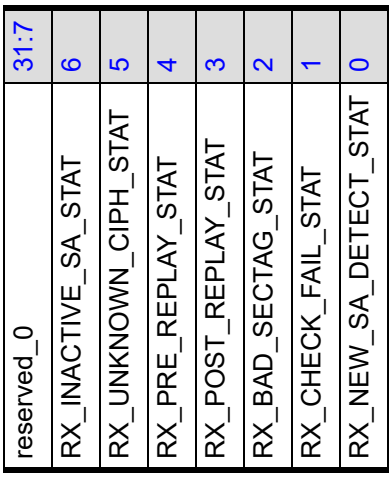


Table 6-716 Fields for Register: IRQ_TYPE_STAT_RX_SA_240

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-716 Fields for Register: IRQ_TYPE_STAT_RX_SA_240 (Continued)

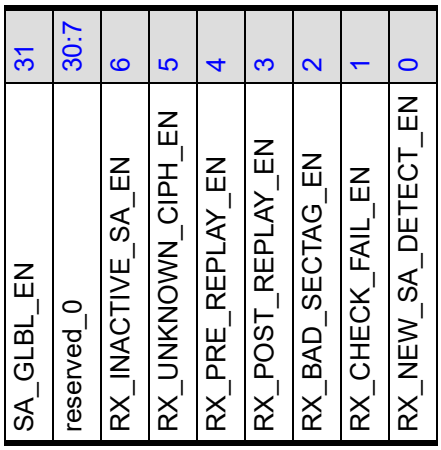
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-716 Fields for Register: IRQ_TYPE_STAT_RX_SA_240 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.710 IRQ_TYPE_EN_RX_SA_241

- **Name:** IRQ_TYPE_EN_RX_SA_241
- **Description:** Interrupt Enable for each IRQ type of Rx SA 241
- **Size:** 32 bits
- **Offset:** 0xb18
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 241



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-717 Fields for Register: IRQ_TYPE_EN_RX_SA_241

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-717 Fields for Register: IRQ_TYPE_EN_RX_SA_241 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-717 Fields for Register: IRQ_TYPE_EN_RX_SA_241 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.711 IRQ_TYPE_STAT_RX_SA_241

- **Name:** IRQ_TYPE_STAT_RX_SA_241
- **Description:** Interrupt Status for each IRQ type of Rx SA 241
- **Size:** 32 bits
- **Offset:** 0xb1c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 241

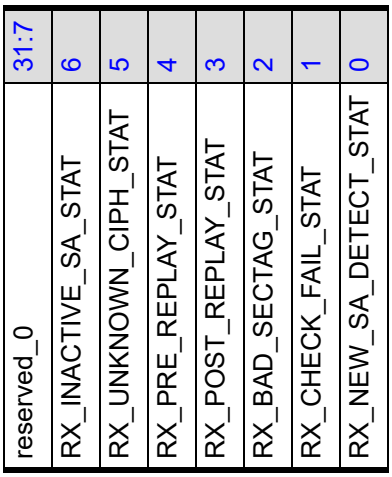


Table 6-718 Fields for Register: IRQ_TYPE_STAT_RX_SA_241

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-718 Fields for Register: IRQ_TYPE_STAT_RX_SA_241 (Continued)

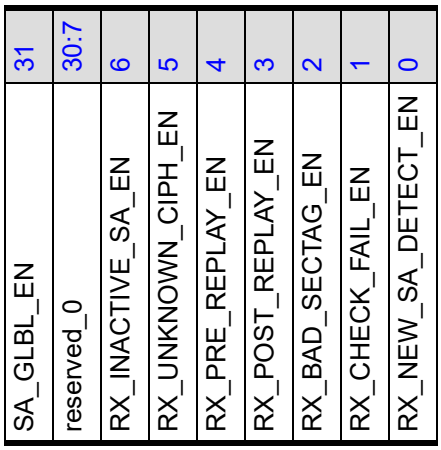
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-718 Fields for Register: IRQ_TYPE_STAT_RX_SA_241 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.712 IRQ_TYPE_EN_RX_SA_242

- **Name:** IRQ_TYPE_EN_RX_SA_242
- **Description:** Interrupt Enable for each IRQ type of Rx SA 242
- **Size:** 32 bits
- **Offset:** 0xb20
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 242



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-719 Fields for Register: IRQ_TYPE_EN_RX_SA_242

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-719 Fields for Register: IRQ_TYPE_EN_RX_SA_242 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-719 Fields for Register: IRQ_TYPE_EN_RX_SA_242 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.713 IRQ_TYPE_STAT_RX_SA_242

- **Name:** IRQ_TYPE_STAT_RX_SA_242
- **Description:** Interrupt Status for each IRQ type of Rx SA 242
- **Size:** 32 bits
- **Offset:** 0xb24
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 242

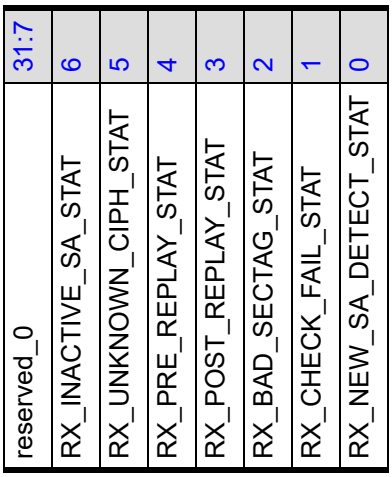


Table 6-720 Fields for Register: IRQ_TYPE_STAT_RX_SA_242

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-720 Fields for Register: IRQ_TYPE_STAT_RX_SA_242 (Continued)

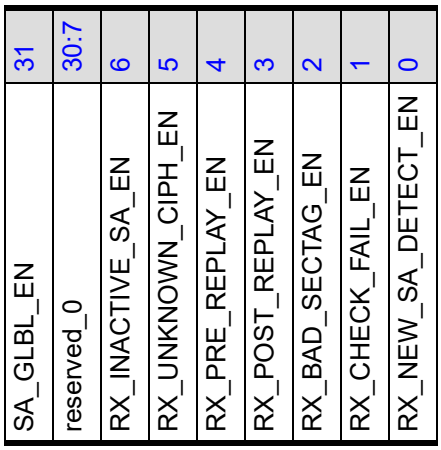
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-720 Fields for Register: IRQ_TYPE_STAT_RX_SA_242 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.714 IRQ_TYPE_EN_RX_SA_243

- **Name:** IRQ_TYPE_EN_RX_SA_243
- **Description:** Interrupt Enable for each IRQ type of Rx SA 243
- **Size:** 32 bits
- **Offset:** 0xb28
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 243



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-721 Fields for Register: IRQ_TYPE_EN_RX_SA_243

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-721 Fields for Register: IRQ_TYPE_EN_RX_SA_243 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-721 Fields for Register: IRQ_TYPE_EN_RX_SA_243 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.715 IRQ_TYPE_STAT_RX_SA_243

- **Name:** IRQ_TYPE_STAT_RX_SA_243
- **Description:** Interrupt Status for each IRQ type of Rx SA 243
- **Size:** 32 bits
- **Offset:** 0xb2c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 243

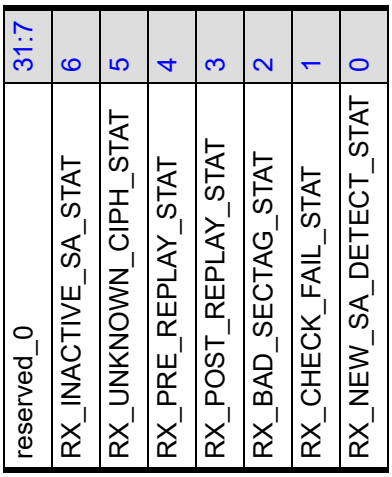


Table 6-722 Fields for Register: IRQ_TYPE_STAT_RX_SA_243

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-722 Fields for Register: IRQ_TYPE_STAT_RX_SA_243 (Continued)

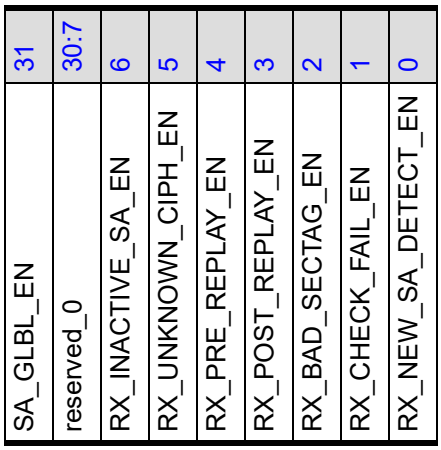
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-722 Fields for Register: IRQ_TYPE_STAT_RX_SA_243 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.716 IRQ_TYPE_EN_RX_SA_244

- **Name:** IRQ_TYPE_EN_RX_SA_244
- **Description:** Interrupt Enable for each IRQ type of Rx SA 244
- **Size:** 32 bits
- **Offset:** 0xb30
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 244



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-723 Fields for Register: IRQ_TYPE_EN_RX_SA_244

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-723 Fields for Register: IRQ_TYPE_EN_RX_SA_244 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-723 Fields for Register: IRQ_TYPE_EN_RX_SA_244 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.717 IRQ_TYPE_STAT_RX_SA_244

- **Name:** IRQ_TYPE_STAT_RX_SA_244
- **Description:** Interrupt Status for each IRQ type of Rx SA 244
- **Size:** 32 bits
- **Offset:** 0xb34
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 244

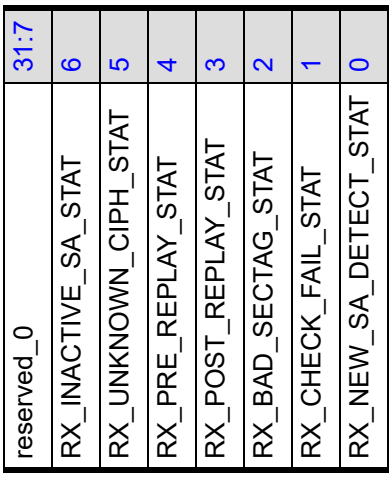


Table 6-724 Fields for Register: IRQ_TYPE_STAT_RX_SA_244

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-724 Fields for Register: IRQ_TYPE_STAT_RX_SA_244 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-724 Fields for Register: IRQ_TYPE_STAT_RX_SA_244 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.718 IRQ_TYPE_EN_RX_SA_245

- **Name:** IRQ_TYPE_EN_RX_SA_245
- **Description:** Interrupt Enable for each IRQ type of Rx SA 245
- **Size:** 32 bits
- **Offset:** 0xb38
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 245

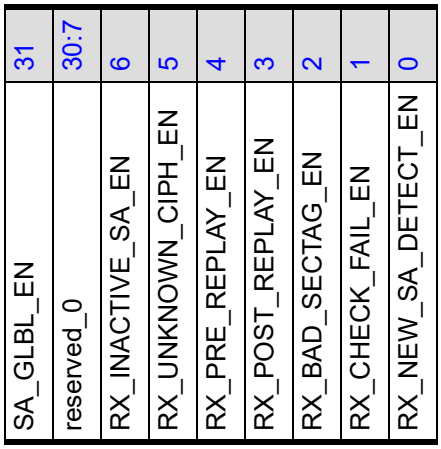


Table 6-725 Fields for Register: IRQ_TYPE_EN_RX_SA_245

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-725 Fields for Register: IRQ_TYPE_EN_RX_SA_245 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-725 Fields for Register: IRQ_TYPE_EN_RX_SA_245 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.719 IRQ_TYPE_STAT_RX_SA_245

- **Name:** IRQ_TYPE_STAT_RX_SA_245
- **Description:** Interrupt Status for each IRQ type of Rx SA 245
- **Size:** 32 bits
- **Offset:** 0xb3c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 245

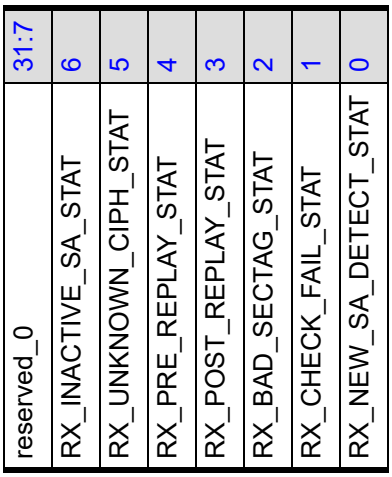


Table 6-726 Fields for Register: IRQ_TYPE_STAT_RX_SA_245

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-726 Fields for Register: IRQ_TYPE_STAT_RX_SA_245 (Continued)

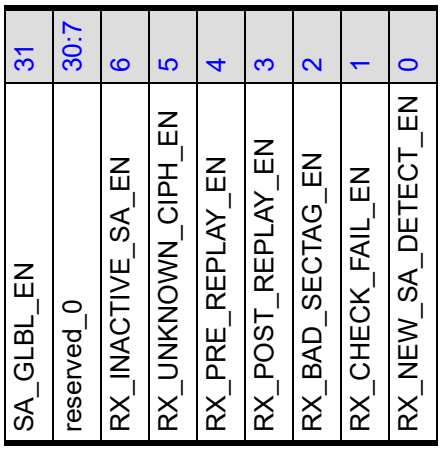
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-726 Fields for Register: IRQ_TYPE_STAT_RX_SA_245 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.720 IRQ_TYPE_EN_RX_SA_246

- **Name:** IRQ_TYPE_EN_RX_SA_246
- **Description:** Interrupt Enable for each IRQ type of Rx SA 246
- **Size:** 32 bits
- **Offset:** 0xb40
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 246



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-727 Fields for Register: IRQ_TYPE_EN_RX_SA_246

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-727 Fields for Register: IRQ_TYPE_EN_RX_SA_246 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-727 Fields for Register: IRQ_TYPE_EN_RX_SA_246 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.721 IRQ_TYPE_STAT_RX_SA_246

- **Name:** IRQ_TYPE_STAT_RX_SA_246
- **Description:** Interrupt Status for each IRQ type of Rx SA 246
- **Size:** 32 bits
- **Offset:** 0xb44
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 246

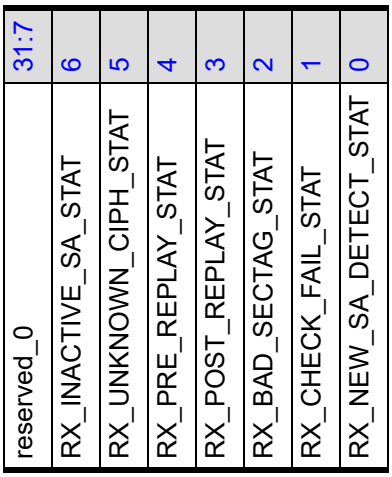


Table 6-728 Fields for Register: IRQ_TYPE_STAT_RX_SA_246

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-728 Fields for Register: IRQ_TYPE_STAT_RX_SA_246 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-728 Fields for Register: IRQ_TYPE_STAT_RX_SA_246 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.722 IRQ_TYPE_EN_RX_SA_247

- **Name:** IRQ_TYPE_EN_RX_SA_247
- **Description:** Interrupt Enable for each IRQ type of Rx SA 247
- **Size:** 32 bits
- **Offset:** 0xb48
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 247

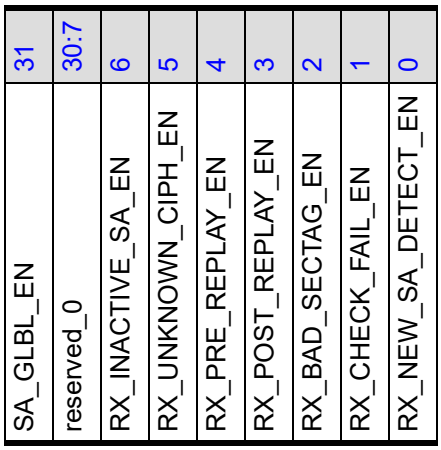


Table 6-729 Fields for Register: IRQ_TYPE_EN_RX_SA_247

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-729 Fields for Register: IRQ_TYPE_EN_RX_SA_247 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-729 Fields for Register: IRQ_TYPE_EN_RX_SA_247 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.723 IRQ_TYPE_STAT_RX_SA_247

- **Name:** IRQ_TYPE_STAT_RX_SA_247
- **Description:** Interrupt Status for each IRQ type of Rx SA 247
- **Size:** 32 bits
- **Offset:** 0xb4c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 247

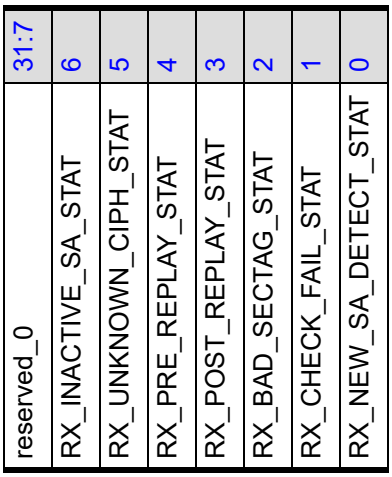


Table 6-730 Fields for Register: IRQ_TYPE_STAT_RX_SA_247

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-730 Fields for Register: IRQ_TYPE_STAT_RX_SA_247 (Continued)

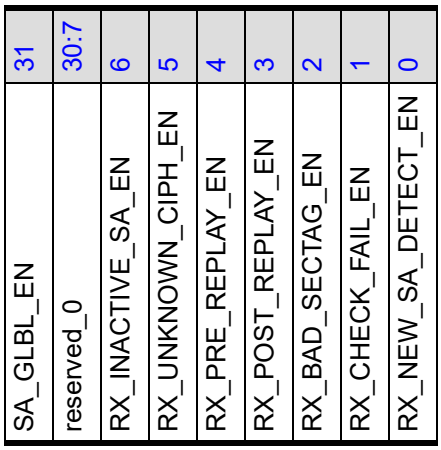
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-730 Fields for Register: IRQ_TYPE_STAT_RX_SA_247 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.724 IRQ_TYPE_EN_RX_SA_248

- **Name:** IRQ_TYPE_EN_RX_SA_248
- **Description:** Interrupt Enable for each IRQ type of Rx SA 248
- **Size:** 32 bits
- **Offset:** 0xb50
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 248



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-731 Fields for Register: IRQ_TYPE_EN_RX_SA_248

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-731 Fields for Register: IRQ_TYPE_EN_RX_SA_248 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-731 Fields for Register: IRQ_TYPE_EN_RX_SA_248 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.725 IRQ_TYPE_STAT_RX_SA_248

- **Name:** IRQ_TYPE_STAT_RX_SA_248
- **Description:** Interrupt Status for each IRQ type of Rx SA 248
- **Size:** 32 bits
- **Offset:** 0xb54
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 248

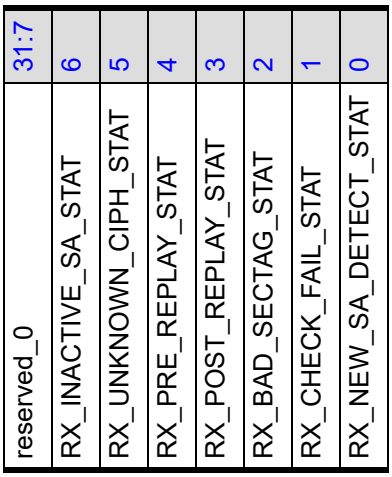


Table 6-732 Fields for Register: IRQ_TYPE_STAT_RX_SA_248

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-732 Fields for Register: IRQ_TYPE_STAT_RX_SA_248 (Continued)

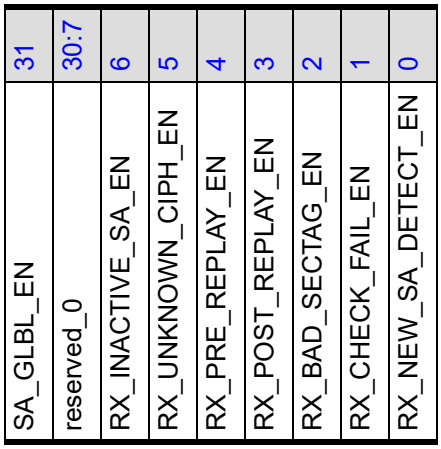
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-732 Fields for Register: IRQ_TYPE_STAT_RX_SA_248 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.726 IRQ_TYPE_EN_RX_SA_249

- **Name:** IRQ_TYPE_EN_RX_SA_249
- **Description:** Interrupt Enable for each IRQ type of Rx SA 249
- **Size:** 32 bits
- **Offset:** 0xb58
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 249



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-733 Fields for Register: IRQ_TYPE_EN_RX_SA_249

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-733 Fields for Register: IRQ_TYPE_EN_RX_SA_249 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-733 Fields for Register: IRQ_TYPE_EN_RX_SA_249 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.727 IRQ_TYPE_STAT_RX_SA_249

- **Name:** IRQ_TYPE_STAT_RX_SA_249
- **Description:** Interrupt Status for each IRQ type of Rx SA 249
- **Size:** 32 bits
- **Offset:** 0xb5c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 249

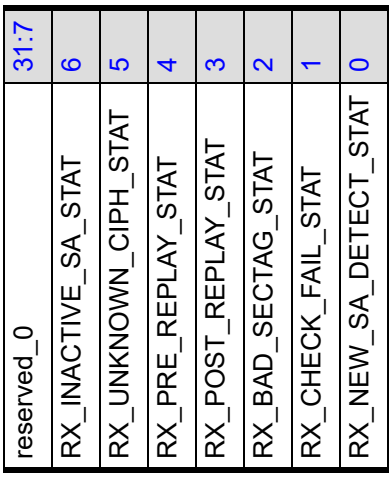


Table 6-734 Fields for Register: IRQ_TYPE_STAT_RX_SA_249

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-734 Fields for Register: IRQ_TYPE_STAT_RX_SA_249 (Continued)

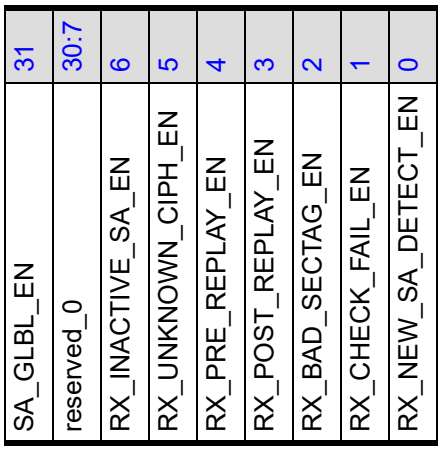
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-734 Fields for Register: IRQ_TYPE_STAT_RX_SA_249 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.728 IRQ_TYPE_EN_RX_SA_250

- **Name:** IRQ_TYPE_EN_RX_SA_250
- **Description:** Interrupt Enable for each IRQ type of Rx SA 250
- **Size:** 32 bits
- **Offset:** 0xb60
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 250



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-735 Fields for Register: IRQ_TYPE_EN_RX_SA_250

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-735 Fields for Register: IRQ_TYPE_EN_RX_SA_250 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-735 Fields for Register: IRQ_TYPE_EN_RX_SA_250 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.729 IRQ_TYPE_STAT_RX_SA_250

- **Name:** IRQ_TYPE_STAT_RX_SA_250
- **Description:** Interrupt Status for each IRQ type of Rx SA 250
- **Size:** 32 bits
- **Offset:** 0xb64
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 250

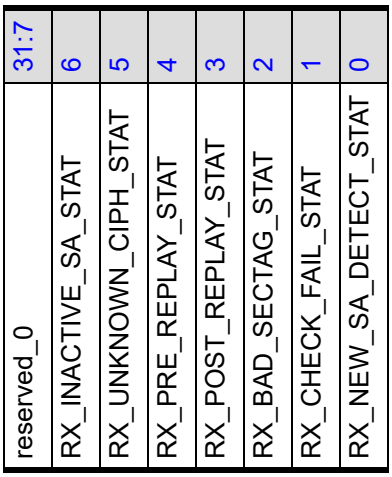


Table 6-736 Fields for Register: IRQ_TYPE_STAT_RX_SA_250

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-736 Fields for Register: IRQ_TYPE_STAT_RX_SA_250 (Continued)

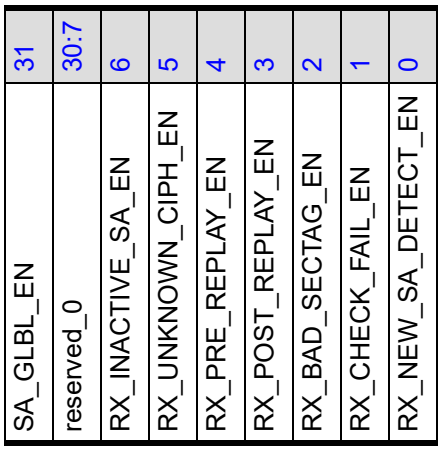
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-736 Fields for Register: IRQ_TYPE_STAT_RX_SA_250 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.730 IRQ_TYPE_EN_RX_SA_251

- **Name:** IRQ_TYPE_EN_RX_SA_251
- **Description:** Interrupt Enable for each IRQ type of Rx SA 251
- **Size:** 32 bits
- **Offset:** 0xb68
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 251



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-737 Fields for Register: IRQ_TYPE_EN_RX_SA_251

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-737 Fields for Register: IRQ_TYPE_EN_RX_SA_251 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-737 Fields for Register: IRQ_TYPE_EN_RX_SA_251 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.731 IRQ_TYPE_STAT_RX_SA_251

- **Name:** IRQ_TYPE_STAT_RX_SA_251
- **Description:** Interrupt Status for each IRQ type of Rx SA 251
- **Size:** 32 bits
- **Offset:** 0xb6c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 251

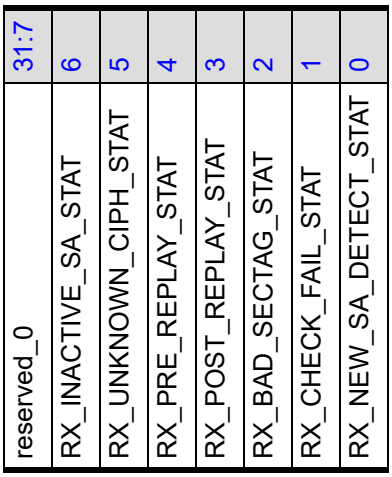


Table 6-738 Fields for Register: IRQ_TYPE_STAT_RX_SA_251

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-738 Fields for Register: IRQ_TYPE_STAT_RX_SA_251 (Continued)

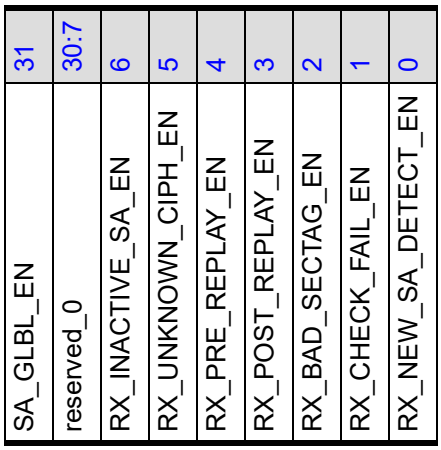
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-738 Fields for Register: IRQ_TYPE_STAT_RX_SA_251 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.732 IRQ_TYPE_EN_RX_SA_252

- **Name:** IRQ_TYPE_EN_RX_SA_252
- **Description:** Interrupt Enable for each IRQ type of Rx SA 252
- **Size:** 32 bits
- **Offset:** 0xb70
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 252



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-739 Fields for Register: IRQ_TYPE_EN_RX_SA_252

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-739 Fields for Register: IRQ_TYPE_EN_RX_SA_252 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-739 Fields for Register: IRQ_TYPE_EN_RX_SA_252 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.733 IRQ_TYPE_STAT_RX_SA_252

- **Name:** IRQ_TYPE_STAT_RX_SA_252
- **Description:** Interrupt Status for each IRQ type of Rx SA 252
- **Size:** 32 bits
- **Offset:** 0xb74
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 252

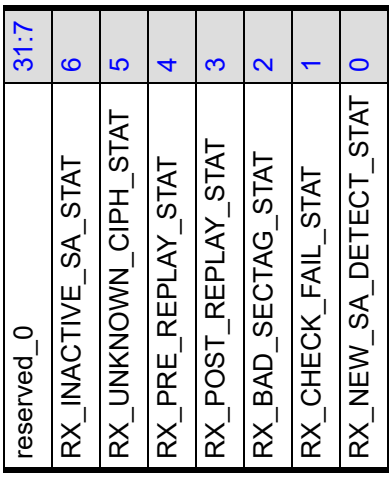


Table 6-740 Fields for Register: IRQ_TYPE_STAT_RX_SA_252

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-740 Fields for Register: IRQ_TYPE_STAT_RX_SA_252 (Continued)

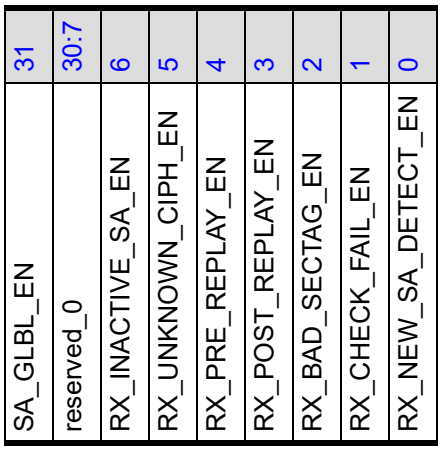
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-740 Fields for Register: IRQ_TYPE_STAT_RX_SA_252 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.734 IRQ_TYPE_EN_RX_SA_253

- **Name:** IRQ_TYPE_EN_RX_SA_253
- **Description:** Interrupt Enable for each IRQ type of Rx SA 253
- **Size:** 32 bits
- **Offset:** 0xb78
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 253



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-741 Fields for Register: IRQ_TYPE_EN_RX_SA_253

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-741 Fields for Register: IRQ_TYPE_EN_RX_SA_253 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-741 Fields for Register: IRQ_TYPE_EN_RX_SA_253 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.735 IRQ_TYPE_STAT_RX_SA_253

- **Name:** IRQ_TYPE_STAT_RX_SA_253
- **Description:** Interrupt Status for each IRQ type of Rx SA 253
- **Size:** 32 bits
- **Offset:** 0xb7c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 253

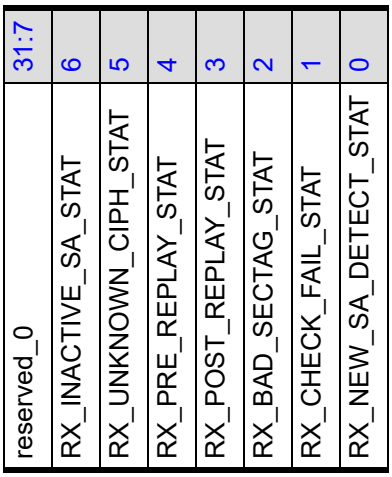


Table 6-742 Fields for Register: IRQ_TYPE_STAT_RX_SA_253

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-742 Fields for Register: IRQ_TYPE_STAT_RX_SA_253 (Continued)

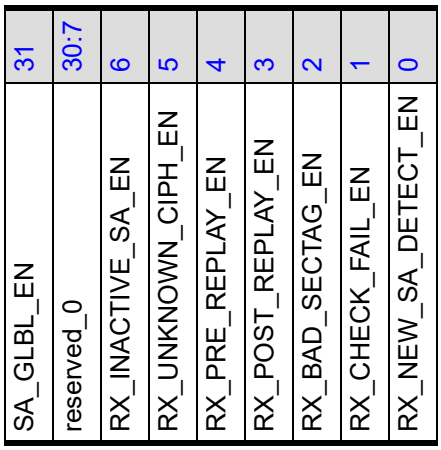
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-742 Fields for Register: IRQ_TYPE_STAT_RX_SA_253 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.736 IRQ_TYPE_EN_RX_SA_254

- **Name:** IRQ_TYPE_EN_RX_SA_254
- **Description:** Interrupt Enable for each IRQ type of Rx SA 254
- **Size:** 32 bits
- **Offset:** 0xb80
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 254



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-743 Fields for Register: IRQ_TYPE_EN_RX_SA_254

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-743 Fields for Register: IRQ_TYPE_EN_RX_SA_254 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-743 Fields for Register: IRQ_TYPE_EN_RX_SA_254 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.737 IRQ_TYPE_STAT_RX_SA_254

- **Name:** IRQ_TYPE_STAT_RX_SA_254
- **Description:** Interrupt Status for each IRQ type of Rx SA 254
- **Size:** 32 bits
- **Offset:** 0xb84
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 254

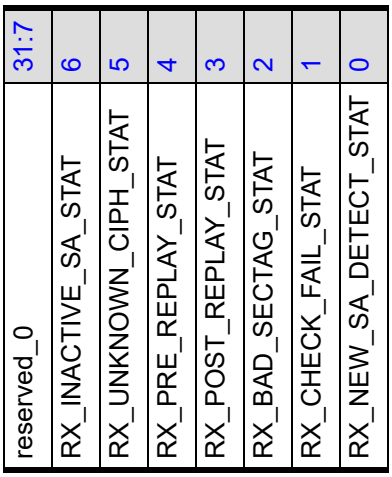


Table 6-744 Fields for Register: IRQ_TYPE_STAT_RX_SA_254

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-744 Fields for Register: IRQ_TYPE_STAT_RX_SA_254 (Continued)

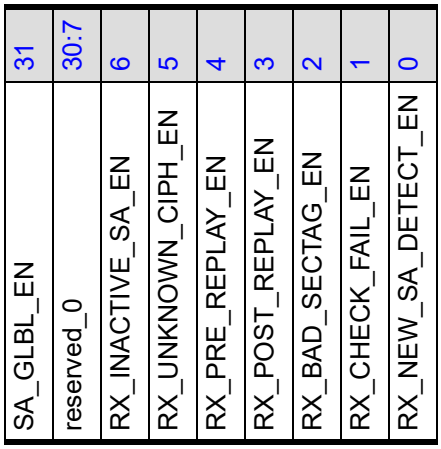
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-744 Fields for Register: IRQ_TYPE_STAT_RX_SA_254 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.738 IRQ_TYPE_EN_RX_SA_255

- **Name:** IRQ_TYPE_EN_RX_SA_255
- **Description:** Interrupt Enable for each IRQ type of Rx SA 255
- **Size:** 32 bits
- **Offset:** 0xb88
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 255



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-745 Fields for Register: IRQ_TYPE_EN_RX_SA_255

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-745 Fields for Register: IRQ_TYPE_EN_RX_SA_255 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-745 Fields for Register: IRQ_TYPE_EN_RX_SA_255 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.739 IRQ_TYPE_STAT_RX_SA_255

- **Name:** IRQ_TYPE_STAT_RX_SA_255
- **Description:** Interrupt Status for each IRQ type of Rx SA 255
- **Size:** 32 bits
- **Offset:** 0xb8c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 255

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-746 Fields for Register: IRQ_TYPE_STAT_RX_SA_255

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-746 Fields for Register: IRQ_TYPE_STAT_RX_SA_255 (Continued)

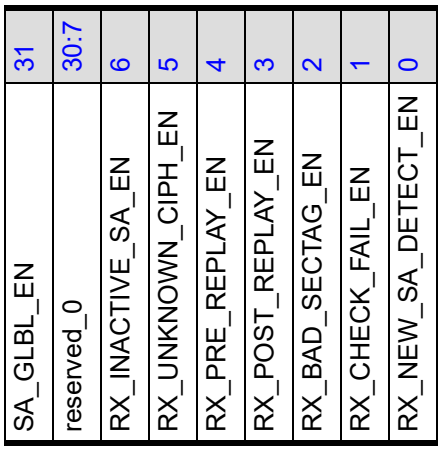
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-746 Fields for Register: IRQ_TYPE_STAT_RX_SA_255 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.740 IRQ_TYPE_EN_RX_SA_256

- **Name:** IRQ_TYPE_EN_RX_SA_256
- **Description:** Interrupt Enable for each IRQ type of Rx SA 256
- **Size:** 32 bits
- **Offset:** 0xb90
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 256



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-747 Fields for Register: IRQ_TYPE_EN_RX_SA_256

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-747 Fields for Register: IRQ_TYPE_EN_RX_SA_256 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-747 Fields for Register: IRQ_TYPE_EN_RX_SA_256 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.741 IRQ_TYPE_STAT_RX_SA_256

- **Name:** IRQ_TYPE_STAT_RX_SA_256
- **Description:** Interrupt Status for each IRQ type of Rx SA 256
- **Size:** 32 bits
- **Offset:** 0xb94
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 256

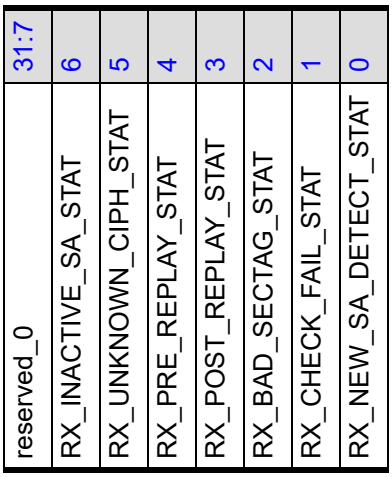


Table 6-748 Fields for Register: IRQ_TYPE_STAT_RX_SA_256

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-748 Fields for Register: IRQ_TYPE_STAT_RX_SA_256 (Continued)

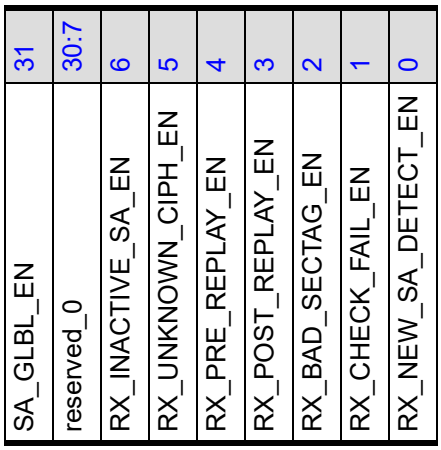
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-748 Fields for Register: IRQ_TYPE_STAT_RX_SA_256 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.742 IRQ_TYPE_EN_RX_SA_257

- **Name:** IRQ_TYPE_EN_RX_SA_257
- **Description:** Interrupt Enable for each IRQ type of Rx SA 257
- **Size:** 32 bits
- **Offset:** 0xb98
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 257



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-749 Fields for Register: IRQ_TYPE_EN_RX_SA_257

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-749 Fields for Register: IRQ_TYPE_EN_RX_SA_257 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-749 Fields for Register: IRQ_TYPE_EN_RX_SA_257 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.743 IRQ_TYPE_STAT_RX_SA_257

- **Name:** IRQ_TYPE_STAT_RX_SA_257
- **Description:** Interrupt Status for each IRQ type of Rx SA 257
- **Size:** 32 bits
- **Offset:** 0xb9c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 257

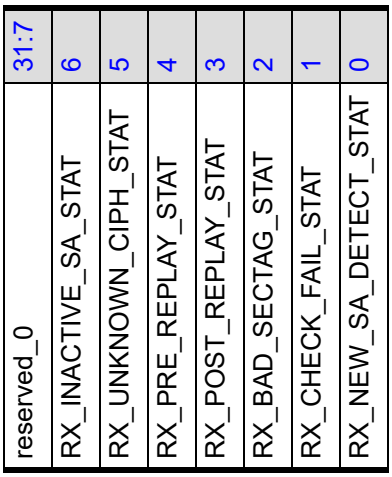


Table 6-750 Fields for Register: IRQ_TYPE_STAT_RX_SA_257

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-750 Fields for Register: IRQ_TYPE_STAT_RX_SA_257 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-750 Fields for Register: IRQ_TYPE_STAT_RX_SA_257 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.744 IRQ_TYPE_EN_RX_SA_258

- **Name:** IRQ_TYPE_EN_RX_SA_258
- **Description:** Interrupt Enable for each IRQ type of Rx SA 258
- **Size:** 32 bits
- **Offset:** 0xba0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 258

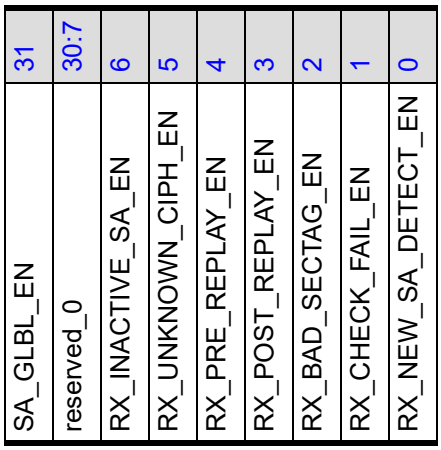


Table 6-751 Fields for Register: IRQ_TYPE_EN_RX_SA_258

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-751 Fields for Register: IRQ_TYPE_EN_RX_SA_258 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-751 Fields for Register: IRQ_TYPE_EN_RX_SA_258 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.745 IRQ_TYPE_STAT_RX_SA_258

- **Name:** IRQ_TYPE_STAT_RX_SA_258
- **Description:** Interrupt Status for each IRQ type of Rx SA 258
- **Size:** 32 bits
- **Offset:** 0xba4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 258

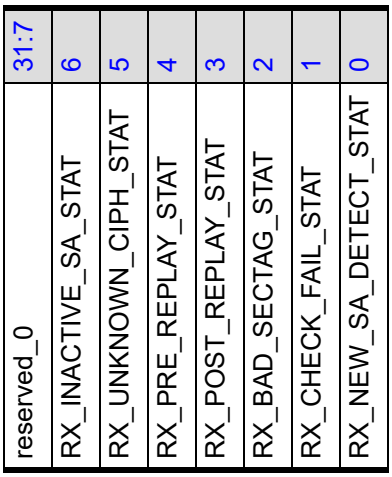


Table 6-752 Fields for Register: IRQ_TYPE_STAT_RX_SA_258

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-752 Fields for Register: IRQ_TYPE_STAT_RX_SA_258 (Continued)

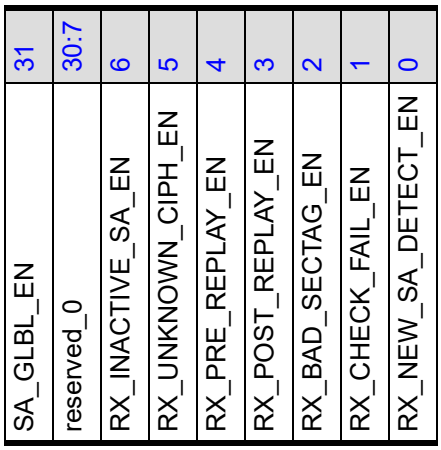
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-752 Fields for Register: IRQ_TYPE_STAT_RX_SA_258 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.746 IRQ_TYPE_EN_RX_SA_259

- **Name:** IRQ_TYPE_EN_RX_SA_259
- **Description:** Interrupt Enable for each IRQ type of Rx SA 259
- **Size:** 32 bits
- **Offset:** 0xba8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 259



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-753 Fields for Register: IRQ_TYPE_EN_RX_SA_259

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-753 Fields for Register: IRQ_TYPE_EN_RX_SA_259 (Continued)

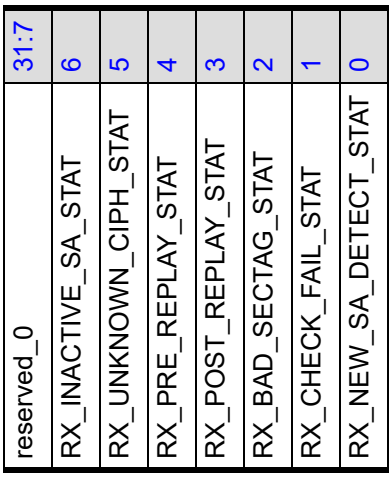
Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-753 Fields for Register: IRQ_TYPE_EN_RX_SA_259 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.747 IRQ_TYPE_STAT_RX_SA_259

- **Name:** IRQ_TYPE_STAT_RX_SA_259
- **Description:** Interrupt Status for each IRQ type of Rx SA 259
- **Size:** 32 bits
- **Offset:** 0xbac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 259



31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-754 Fields for Register: IRQ_TYPE_STAT_RX_SA_259

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-754 Fields for Register: IRQ_TYPE_STAT_RX_SA_259 (Continued)

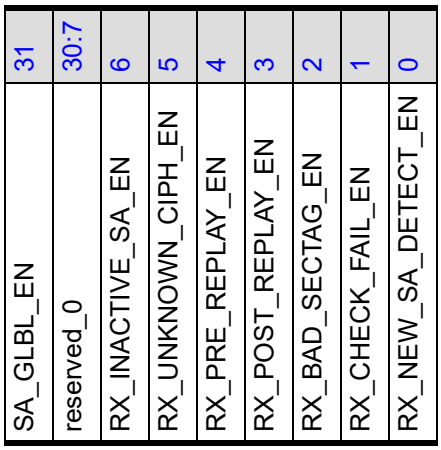
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-754 Fields for Register: IRQ_TYPE_STAT_RX_SA_259 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.748 IRQ_TYPE_EN_RX_SA_260

- **Name:** IRQ_TYPE_EN_RX_SA_260
- **Description:** Interrupt Enable for each IRQ type of Rx SA 260
- **Size:** 32 bits
- **Offset:** 0xbb0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 260



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-755 Fields for Register: IRQ_TYPE_EN_RX_SA_260

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-755 Fields for Register: IRQ_TYPE_EN_RX_SA_260 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-755 Fields for Register: IRQ_TYPE_EN_RX_SA_260 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.749 IRQ_TYPE_STAT_RX_SA_260

- **Name:** IRQ_TYPE_STAT_RX_SA_260
- **Description:** Interrupt Status for each IRQ type of Rx SA 260
- **Size:** 32 bits
- **Offset:** 0xbb4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 260

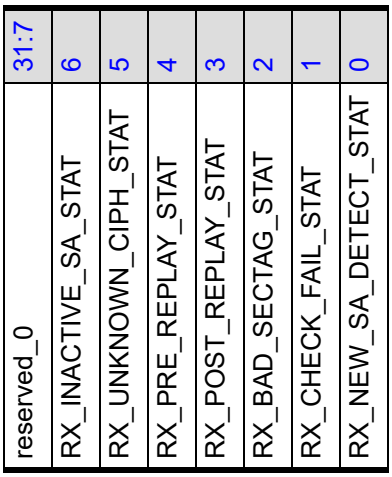


Table 6-756 Fields for Register: IRQ_TYPE_STAT_RX_SA_260

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-756 Fields for Register: IRQ_TYPE_STAT_RX_SA_260 (Continued)

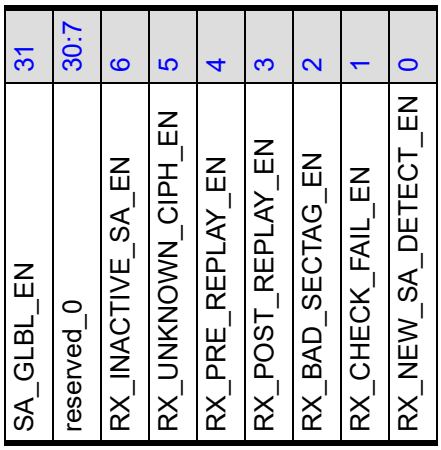
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-756 Fields for Register: IRQ_TYPE_STAT_RX_SA_260 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.750 IRQ_TYPE_EN_RX_SA_261

- **Name:** IRQ_TYPE_EN_RX_SA_261
- **Description:** Interrupt Enable for each IRQ type of Rx SA 261
- **Size:** 32 bits
- **Offset:** 0xbb8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 261



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-757 Fields for Register: IRQ_TYPE_EN_RX_SA_261

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-757 Fields for Register: IRQ_TYPE_EN_RX_SA_261 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-757 Fields for Register: IRQ_TYPE_EN_RX_SA_261 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.751 IRQ_TYPE_STAT_RX_SA_261

- **Name:** IRQ_TYPE_STAT_RX_SA_261
- **Description:** Interrupt Status for each IRQ type of Rx SA 261
- **Size:** 32 bits
- **Offset:** 0xbbc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 261

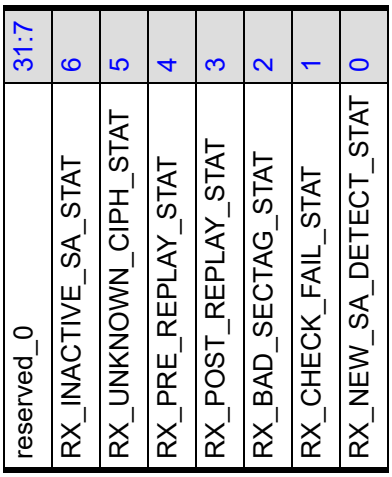


Table 6-758 Fields for Register: IRQ_TYPE_STAT_RX_SA_261

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-758 Fields for Register: IRQ_TYPE_STAT_RX_SA_261 (Continued)

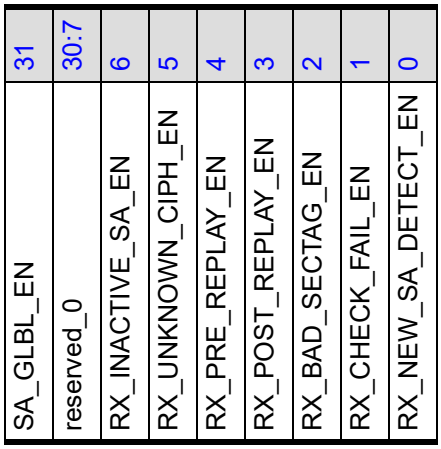
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-758 Fields for Register: IRQ_TYPE_STAT_RX_SA_261 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.752 IRQ_TYPE_EN_RX_SA_262

- **Name:** IRQ_TYPE_EN_RX_SA_262
- **Description:** Interrupt Enable for each IRQ type of Rx SA 262
- **Size:** 32 bits
- **Offset:** 0xbc0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 262



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-759 Fields for Register: IRQ_TYPE_EN_RX_SA_262

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-759 Fields for Register: IRQ_TYPE_EN_RX_SA_262 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-759 Fields for Register: IRQ_TYPE_EN_RX_SA_262 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.753 IRQ_TYPE_STAT_RX_SA_262

- **Name:** IRQ_TYPE_STAT_RX_SA_262
- **Description:** Interrupt Status for each IRQ type of Rx SA 262
- **Size:** 32 bits
- **Offset:** 0xbc4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 262

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-760 Fields for Register: IRQ_TYPE_STAT_RX_SA_262

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-760 Fields for Register: IRQ_TYPE_STAT_RX_SA_262 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-760 Fields for Register: IRQ_TYPE_STAT_RX_SA_262 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.754 IRQ_TYPE_EN_RX_SA_263

- **Name:** IRQ_TYPE_EN_RX_SA_263
- **Description:** Interrupt Enable for each IRQ type of Rx SA 263
- **Size:** 32 bits
- **Offset:** 0xbc8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 263

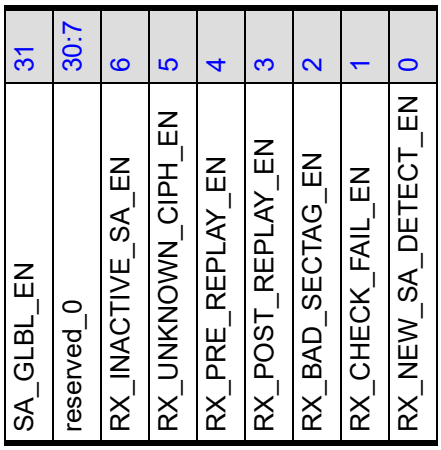


Table 6-761 Fields for Register: IRQ_TYPE_EN_RX_SA_263

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-761 Fields for Register: IRQ_TYPE_EN_RX_SA_263 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-761 Fields for Register: IRQ_TYPE_EN_RX_SA_263 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.755 IRQ_TYPE_STAT_RX_SA_263

- **Name:** IRQ_TYPE_STAT_RX_SA_263
- **Description:** Interrupt Status for each IRQ type of Rx SA 263
- **Size:** 32 bits
- **Offset:** 0xbcc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 263

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-762 Fields for Register: IRQ_TYPE_STAT_RX_SA_263

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-762 Fields for Register: IRQ_TYPE_STAT_RX_SA_263 (Continued)

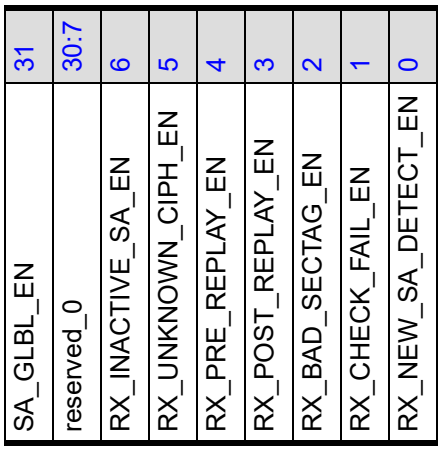
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-762 Fields for Register: IRQ_TYPE_STAT_RX_SA_263 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.756 IRQ_TYPE_EN_RX_SA_264

- **Name:** IRQ_TYPE_EN_RX_SA_264
- **Description:** Interrupt Enable for each IRQ type of Rx SA 264
- **Size:** 32 bits
- **Offset:** 0xbd0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 264



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-763 Fields for Register: IRQ_TYPE_EN_RX_SA_264

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-763 Fields for Register: IRQ_TYPE_EN_RX_SA_264 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-763 Fields for Register: IRQ_TYPE_EN_RX_SA_264 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.757 IRQ_TYPE_STAT_RX_SA_264

- **Name:** IRQ_TYPE_STAT_RX_SA_264
- **Description:** Interrupt Status for each IRQ type of Rx SA 264
- **Size:** 32 bits
- **Offset:** 0xbd4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 264

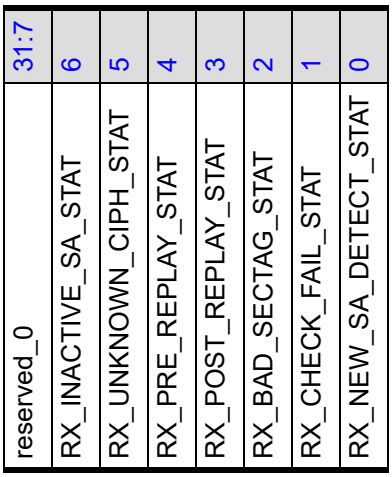


Table 6-764 Fields for Register: IRQ_TYPE_STAT_RX_SA_264

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-764 Fields for Register: IRQ_TYPE_STAT_RX_SA_264 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-764 Fields for Register: IRQ_TYPE_STAT_RX_SA_264 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.758 IRQ_TYPE_EN_RX_SA_265

- **Name:** IRQ_TYPE_EN_RX_SA_265
- **Description:** Interrupt Enable for each IRQ type of Rx SA 265
- **Size:** 32 bits
- **Offset:** 0xbd8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 265

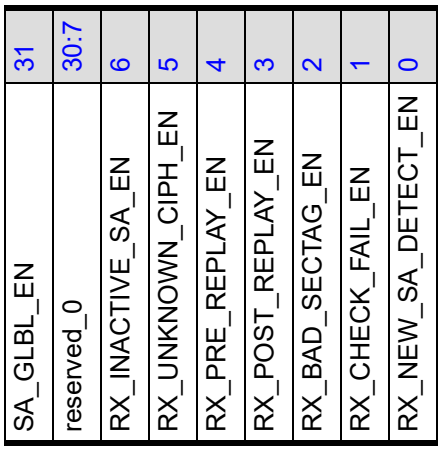


Table 6-765 Fields for Register: IRQ_TYPE_EN_RX_SA_265

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-765 Fields for Register: IRQ_TYPE_EN_RX_SA_265 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-765 Fields for Register: IRQ_TYPE_EN_RX_SA_265 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.759 IRQ_TYPE_STAT_RX_SA_265

- **Name:** IRQ_TYPE_STAT_RX_SA_265
- **Description:** Interrupt Status for each IRQ type of Rx SA 265
- **Size:** 32 bits
- **Offset:** 0xbdc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 265

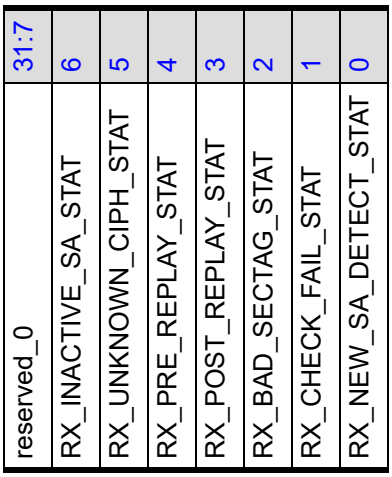


Table 6-766 Fields for Register: IRQ_TYPE_STAT_RX_SA_265

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-766 Fields for Register: IRQ_TYPE_STAT_RX_SA_265 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-766 Fields for Register: IRQ_TYPE_STAT_RX_SA_265 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.760 IRQ_TYPE_EN_RX_SA_266

- **Name:** IRQ_TYPE_EN_RX_SA_266
- **Description:** Interrupt Enable for each IRQ type of Rx SA 266
- **Size:** 32 bits
- **Offset:** 0xbe0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 266

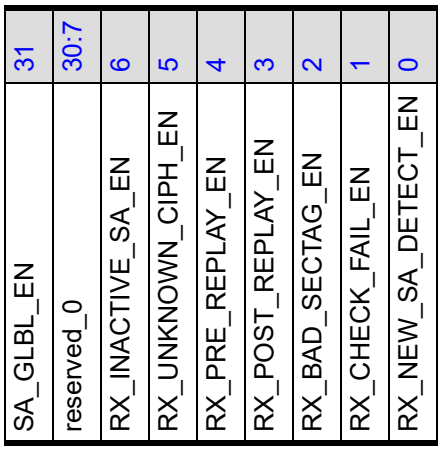


Table 6-767 Fields for Register: IRQ_TYPE_EN_RX_SA_266

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-767 Fields for Register: IRQ_TYPE_EN_RX_SA_266 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-767 Fields for Register: IRQ_TYPE_EN_RX_SA_266 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.761 IRQ_TYPE_STAT_RX_SA_266

- **Name:** IRQ_TYPE_STAT_RX_SA_266
- **Description:** Interrupt Status for each IRQ type of Rx SA 266
- **Size:** 32 bits
- **Offset:** 0xbe4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 266

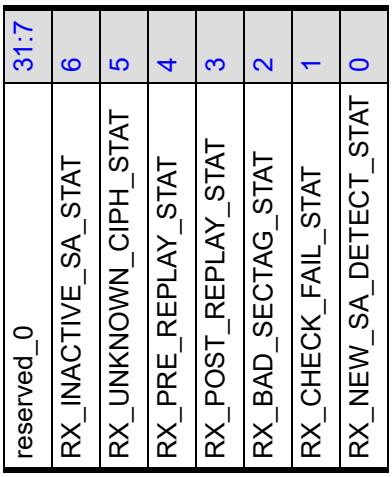


Table 6-768 Fields for Register: IRQ_TYPE_STAT_RX_SA_266

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-768 Fields for Register: IRQ_TYPE_STAT_RX_SA_266 (Continued)

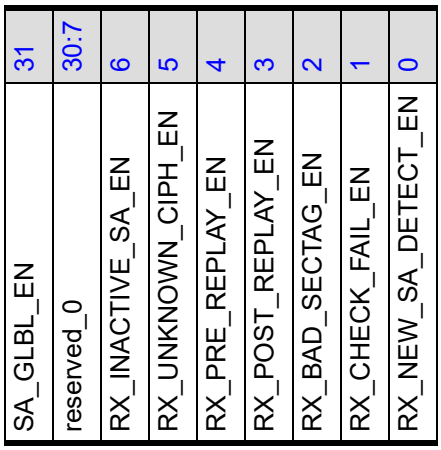
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-768 Fields for Register: IRQ_TYPE_STAT_RX_SA_266 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.762 IRQ_TYPE_EN_RX_SA_267

- **Name:** IRQ_TYPE_EN_RX_SA_267
- **Description:** Interrupt Enable for each IRQ type of Rx SA 267
- **Size:** 32 bits
- **Offset:** 0xbe8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 267



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-769 Fields for Register: IRQ_TYPE_EN_RX_SA_267

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-769 Fields for Register: IRQ_TYPE_EN_RX_SA_267 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-769 Fields for Register: IRQ_TYPE_EN_RX_SA_267 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.763 IRQ_TYPE_STAT_RX_SA_267

- **Name:** IRQ_TYPE_STAT_RX_SA_267
- **Description:** Interrupt Status for each IRQ type of Rx SA 267
- **Size:** 32 bits
- **Offset:** 0xbec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 267

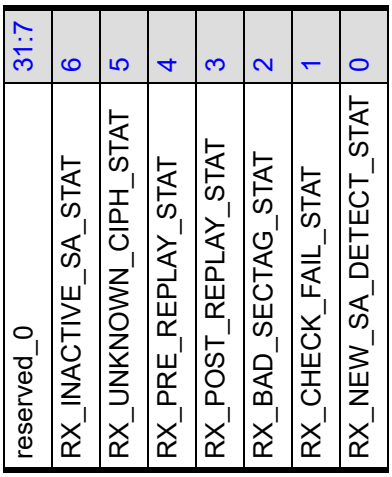


Table 6-770 Fields for Register: IRQ_TYPE_STAT_RX_SA_267

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-770 Fields for Register: IRQ_TYPE_STAT_RX_SA_267 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-770 Fields for Register: IRQ_TYPE_STAT_RX_SA_267 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.764 IRQ_TYPE_EN_RX_SA_268

- **Name:** IRQ_TYPE_EN_RX_SA_268
- **Description:** Interrupt Enable for each IRQ type of Rx SA 268
- **Size:** 32 bits
- **Offset:** 0xbf0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 268

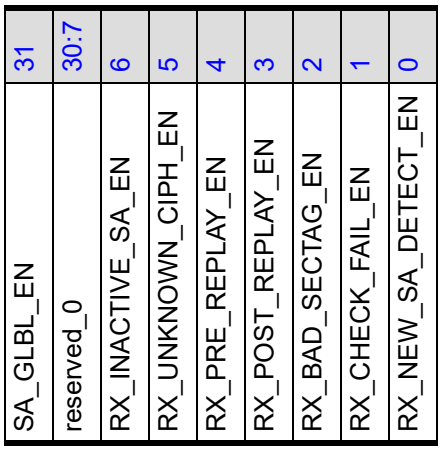


Table 6-771 Fields for Register: IRQ_TYPE_EN_RX_SA_268

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-771 Fields for Register: IRQ_TYPE_EN_RX_SA_268 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-771 Fields for Register: IRQ_TYPE_EN_RX_SA_268 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.765 IRQ_TYPE_STAT_RX_SA_268

- **Name:** IRQ_TYPE_STAT_RX_SA_268
- **Description:** Interrupt Status for each IRQ type of Rx SA 268
- **Size:** 32 bits
- **Offset:** 0xbf4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 268

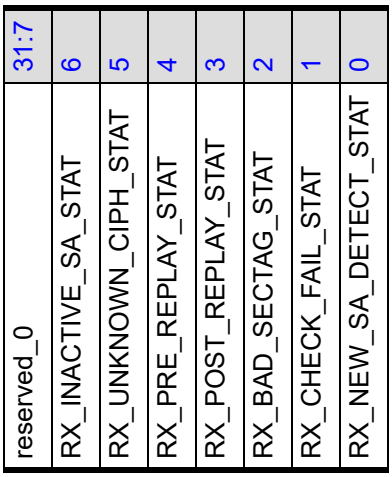


Table 6-772 Fields for Register: IRQ_TYPE_STAT_RX_SA_268

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-772 Fields for Register: IRQ_TYPE_STAT_RX_SA_268 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-772 Fields for Register: IRQ_TYPE_STAT_RX_SA_268 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.766 IRQ_TYPE_EN_RX_SA_269

- **Name:** IRQ_TYPE_EN_RX_SA_269
- **Description:** Interrupt Enable for each IRQ type of Rx SA 269
- **Size:** 32 bits
- **Offset:** 0xbf8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 269

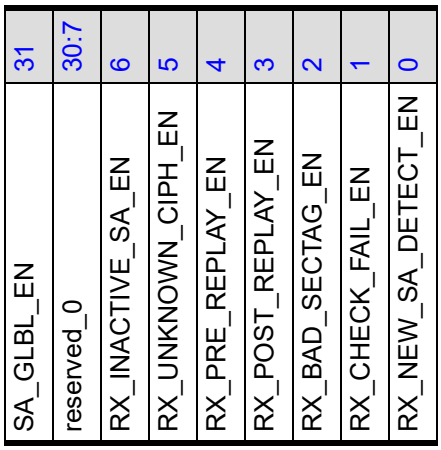


Table 6-773 Fields for Register: IRQ_TYPE_EN_RX_SA_269

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-773 Fields for Register: IRQ_TYPE_EN_RX_SA_269 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-773 Fields for Register: IRQ_TYPE_EN_RX_SA_269 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.767 IRQ_TYPE_STAT_RX_SA_269

- **Name:** IRQ_TYPE_STAT_RX_SA_269
- **Description:** Interrupt Status for each IRQ type of Rx SA 269
- **Size:** 32 bits
- **Offset:** 0xbfc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 269

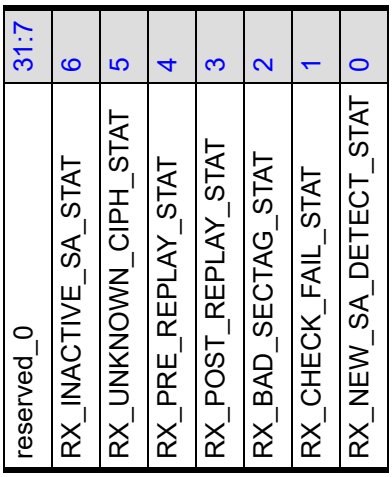


Table 6-774 Fields for Register: IRQ_TYPE_STAT_RX_SA_269

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-774 Fields for Register: IRQ_TYPE_STAT_RX_SA_269 (Continued)

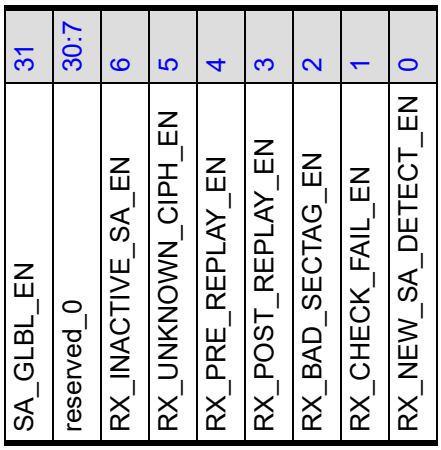
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-774 Fields for Register: IRQ_TYPE_STAT_RX_SA_269 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.768 IRQ_TYPE_EN_RX_SA_270

- **Name:** IRQ_TYPE_EN_RX_SA_270
- **Description:** Interrupt Enable for each IRQ type of Rx SA 270
- **Size:** 32 bits
- **Offset:** 0xc00
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 270



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-775 Fields for Register: IRQ_TYPE_EN_RX_SA_270

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-775 Fields for Register: IRQ_TYPE_EN_RX_SA_270 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-775 Fields for Register: IRQ_TYPE_EN_RX_SA_270 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.769 IRQ_TYPE_STAT_RX_SA_270

- **Name:** IRQ_TYPE_STAT_RX_SA_270
- **Description:** Interrupt Status for each IRQ type of Rx SA 270
- **Size:** 32 bits
- **Offset:** 0xc04
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 270

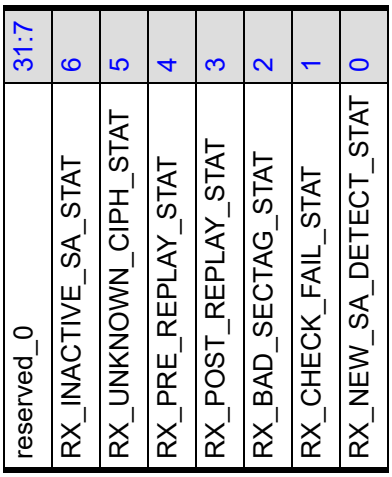


Table 6-776 Fields for Register: IRQ_TYPE_STAT_RX_SA_270

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-776 Fields for Register: IRQ_TYPE_STAT_RX_SA_270 (Continued)

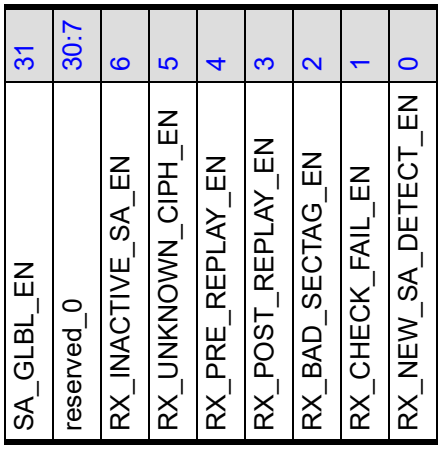
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-776 Fields for Register: IRQ_TYPE_STAT_RX_SA_270 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.770 IRQ_TYPE_EN_RX_SA_271

- **Name:** IRQ_TYPE_EN_RX_SA_271
- **Description:** Interrupt Enable for each IRQ type of Rx SA 271
- **Size:** 32 bits
- **Offset:** 0xc08
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 271



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-777 Fields for Register: IRQ_TYPE_EN_RX_SA_271

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-777 Fields for Register: IRQ_TYPE_EN_RX_SA_271 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-777 Fields for Register: IRQ_TYPE_EN_RX_SA_271 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.771 IRQ_TYPE_STAT_RX_SA_271

- **Name:** IRQ_TYPE_STAT_RX_SA_271
- **Description:** Interrupt Status for each IRQ type of Rx SA 271
- **Size:** 32 bits
- **Offset:** 0xc0c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 271

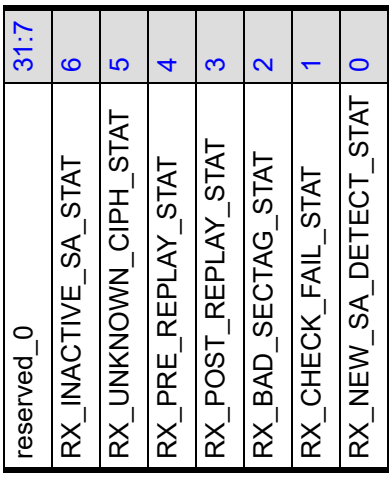


Table 6-778 Fields for Register: IRQ_TYPE_STAT_RX_SA_271

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-778 Fields for Register: IRQ_TYPE_STAT_RX_SA_271 (Continued)

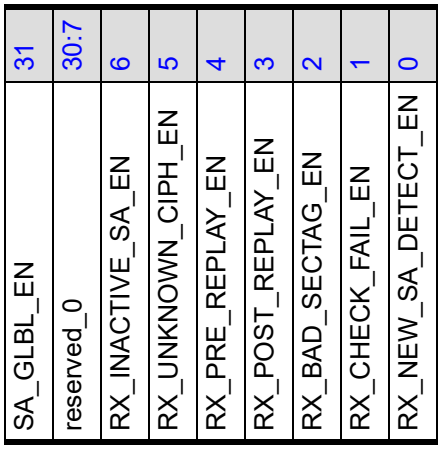
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-778 Fields for Register: IRQ_TYPE_STAT_RX_SA_271 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.772 IRQ_TYPE_EN_RX_SA_272

- **Name:** IRQ_TYPE_EN_RX_SA_272
- **Description:** Interrupt Enable for each IRQ type of Rx SA 272
- **Size:** 32 bits
- **Offset:** 0xc10
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 272



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-779 Fields for Register: IRQ_TYPE_EN_RX_SA_272

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-779 Fields for Register: IRQ_TYPE_EN_RX_SA_272 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-779 Fields for Register: IRQ_TYPE_EN_RX_SA_272 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.773 IRQ_TYPE_STAT_RX_SA_272

- **Name:** IRQ_TYPE_STAT_RX_SA_272
- **Description:** Interrupt Status for each IRQ type of Rx SA 272
- **Size:** 32 bits
- **Offset:** 0xc14
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 272

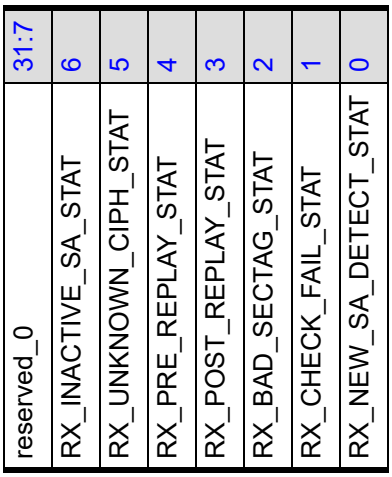


Table 6-780 Fields for Register: IRQ_TYPE_STAT_RX_SA_272

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-780 Fields for Register: IRQ_TYPE_STAT_RX_SA_272 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-780 Fields for Register: IRQ_TYPE_STAT_RX_SA_272 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.774 IRQ_TYPE_EN_RX_SA_273

- **Name:** IRQ_TYPE_EN_RX_SA_273
- **Description:** Interrupt Enable for each IRQ type of Rx SA 273
- **Size:** 32 bits
- **Offset:** 0xc18
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 273

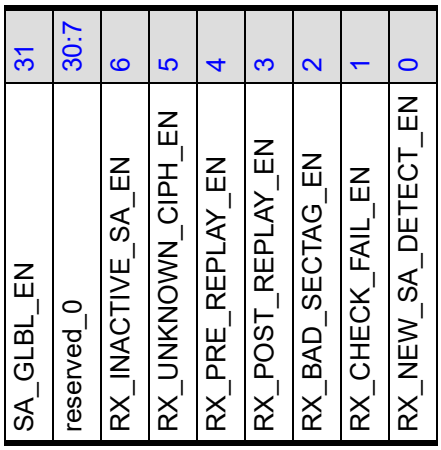


Table 6-781 Fields for Register: IRQ_TYPE_EN_RX_SA_273

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-781 Fields for Register: IRQ_TYPE_EN_RX_SA_273 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-781 Fields for Register: IRQ_TYPE_EN_RX_SA_273 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.775 IRQ_TYPE_STAT_RX_SA_273

- **Name:** IRQ_TYPE_STAT_RX_SA_273
- **Description:** Interrupt Status for each IRQ type of Rx SA 273
- **Size:** 32 bits
- **Offset:** 0xc1c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 273

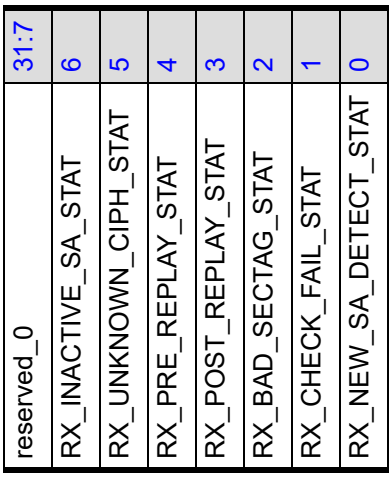


Table 6-782 Fields for Register: IRQ_TYPE_STAT_RX_SA_273

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-782 Fields for Register: IRQ_TYPE_STAT_RX_SA_273 (Continued)

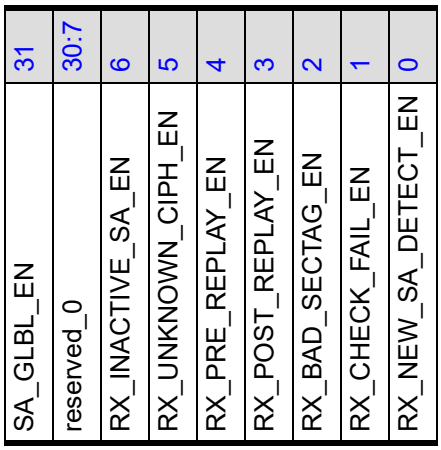
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-782 Fields for Register: IRQ_TYPE_STAT_RX_SA_273 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.776 IRQ_TYPE_EN_RX_SA_274

- **Name:** IRQ_TYPE_EN_RX_SA_274
- **Description:** Interrupt Enable for each IRQ type of Rx SA 274
- **Size:** 32 bits
- **Offset:** 0xc20
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 274



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-783 Fields for Register: IRQ_TYPE_EN_RX_SA_274

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-783 Fields for Register: IRQ_TYPE_EN_RX_SA_274 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-783 Fields for Register: IRQ_TYPE_EN_RX_SA_274 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.777 IRQ_TYPE_STAT_RX_SA_274

- **Name:** IRQ_TYPE_STAT_RX_SA_274
- **Description:** Interrupt Status for each IRQ type of Rx SA 274
- **Size:** 32 bits
- **Offset:** 0xc24
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 274

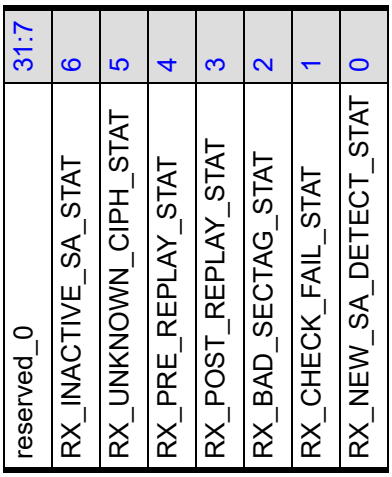


Table 6-784 Fields for Register: IRQ_TYPE_STAT_RX_SA_274

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-784 Fields for Register: IRQ_TYPE_STAT_RX_SA_274 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-784 Fields for Register: IRQ_TYPE_STAT_RX_SA_274 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.778 IRQ_TYPE_EN_RX_SA_275

- **Name:** IRQ_TYPE_EN_RX_SA_275
- **Description:** Interrupt Enable for each IRQ type of Rx SA 275
- **Size:** 32 bits
- **Offset:** 0xc28
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 275

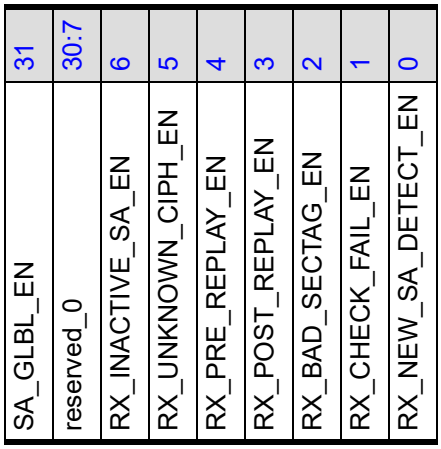


Table 6-785 Fields for Register: IRQ_TYPE_EN_RX_SA_275

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-785 Fields for Register: IRQ_TYPE_EN_RX_SA_275 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-785 Fields for Register: IRQ_TYPE_EN_RX_SA_275 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.779 IRQ_TYPE_STAT_RX_SA_275

- **Name:** IRQ_TYPE_STAT_RX_SA_275
- **Description:** Interrupt Status for each IRQ type of Rx SA 275
- **Size:** 32 bits
- **Offset:** 0xc2c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 275

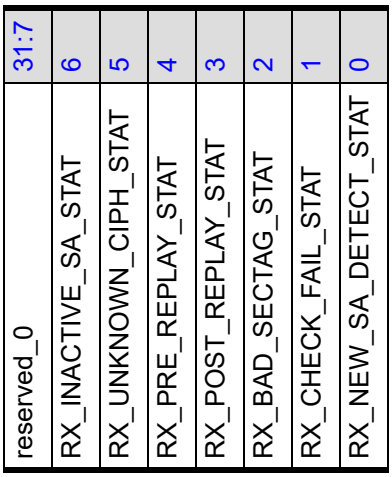


Table 6-786 Fields for Register: IRQ_TYPE_STAT_RX_SA_275

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-786 Fields for Register: IRQ_TYPE_STAT_RX_SA_275 (Continued)

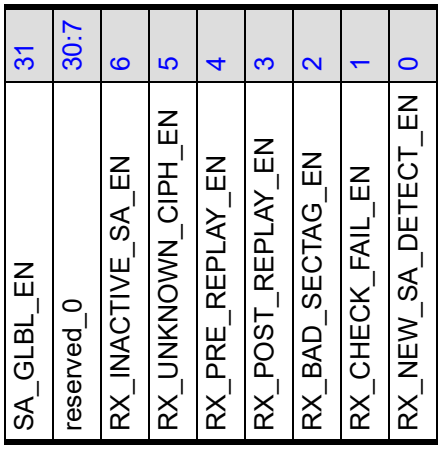
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-786 Fields for Register: IRQ_TYPE_STAT_RX_SA_275 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.780 IRQ_TYPE_EN_RX_SA_276

- **Name:** IRQ_TYPE_EN_RX_SA_276
- **Description:** Interrupt Enable for each IRQ type of Rx SA 276
- **Size:** 32 bits
- **Offset:** 0xc30
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 276



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-787 Fields for Register: IRQ_TYPE_EN_RX_SA_276

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-787 Fields for Register: IRQ_TYPE_EN_RX_SA_276 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-787 Fields for Register: IRQ_TYPE_EN_RX_SA_276 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.781 IRQ_TYPE_STAT_RX_SA_276

- **Name:** IRQ_TYPE_STAT_RX_SA_276
- **Description:** Interrupt Status for each IRQ type of Rx SA 276
- **Size:** 32 bits
- **Offset:** 0xc34
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 276

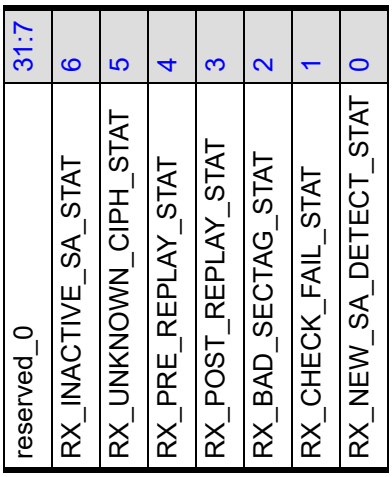


Table 6-788 Fields for Register: IRQ_TYPE_STAT_RX_SA_276

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-788 Fields for Register: IRQ_TYPE_STAT_RX_SA_276 (Continued)

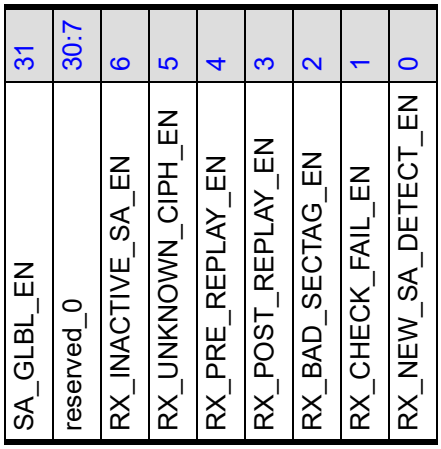
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-788 Fields for Register: IRQ_TYPE_STAT_RX_SA_276 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.782 IRQ_TYPE_EN_RX_SA_277

- **Name:** IRQ_TYPE_EN_RX_SA_277
- **Description:** Interrupt Enable for each IRQ type of Rx SA 277
- **Size:** 32 bits
- **Offset:** 0xc38
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 277



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-789 Fields for Register: IRQ_TYPE_EN_RX_SA_277

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-789 Fields for Register: IRQ_TYPE_EN_RX_SA_277 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-789 Fields for Register: IRQ_TYPE_EN_RX_SA_277 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.783 IRQ_TYPE_STAT_RX_SA_277

- **Name:** IRQ_TYPE_STAT_RX_SA_277
- **Description:** Interrupt Status for each IRQ type of Rx SA 277
- **Size:** 32 bits
- **Offset:** 0xc3c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 277

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-790 Fields for Register: IRQ_TYPE_STAT_RX_SA_277

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-790 Fields for Register: IRQ_TYPE_STAT_RX_SA_277 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-790 Fields for Register: IRQ_TYPE_STAT_RX_SA_277 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.784 IRQ_TYPE_EN_RX_SA_278

- **Name:** IRQ_TYPE_EN_RX_SA_278
- **Description:** Interrupt Enable for each IRQ type of Rx SA 278
- **Size:** 32 bits
- **Offset:** 0xc40
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 278

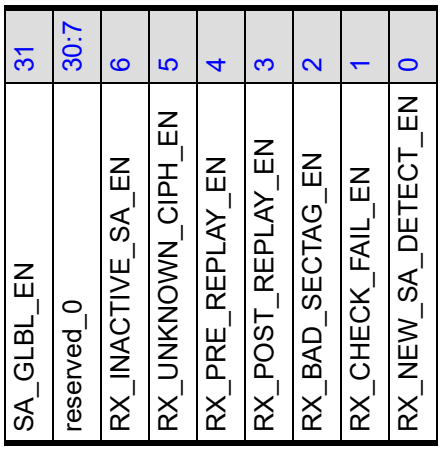


Table 6-791 Fields for Register: IRQ_TYPE_EN_RX_SA_278

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-791 Fields for Register: IRQ_TYPE_EN_RX_SA_278 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-791 Fields for Register: IRQ_TYPE_EN_RX_SA_278 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.785 IRQ_TYPE_STAT_RX_SA_278

- **Name:** IRQ_TYPE_STAT_RX_SA_278
- **Description:** Interrupt Status for each IRQ type of Rx SA 278
- **Size:** 32 bits
- **Offset:** 0xc44
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 278

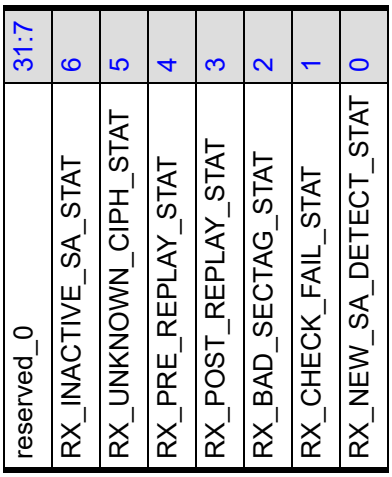


Table 6-792 Fields for Register: IRQ_TYPE_STAT_RX_SA_278

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-792 Fields for Register: IRQ_TYPE_STAT_RX_SA_278 (Continued)

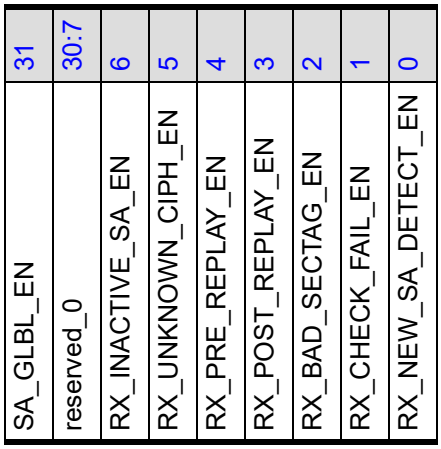
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-792 Fields for Register: IRQ_TYPE_STAT_RX_SA_278 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.786 IRQ_TYPE_EN_RX_SA_279

- **Name:** IRQ_TYPE_EN_RX_SA_279
- **Description:** Interrupt Enable for each IRQ type of Rx SA 279
- **Size:** 32 bits
- **Offset:** 0xc48
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 279



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-793 Fields for Register: IRQ_TYPE_EN_RX_SA_279

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-793 Fields for Register: IRQ_TYPE_EN_RX_SA_279 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-793 Fields for Register: IRQ_TYPE_EN_RX_SA_279 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.787 IRQ_TYPE_STAT_RX_SA_279

- **Name:** IRQ_TYPE_STAT_RX_SA_279
- **Description:** Interrupt Status for each IRQ type of Rx SA 279
- **Size:** 32 bits
- **Offset:** 0xc4c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 279

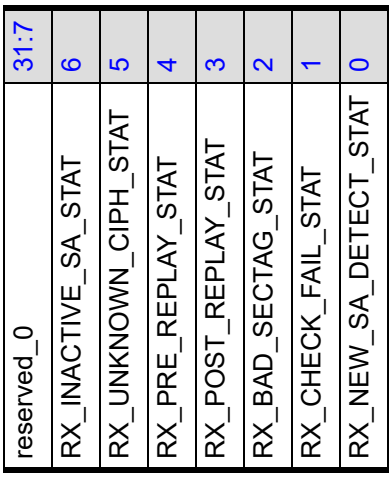


Table 6-794 Fields for Register: IRQ_TYPE_STAT_RX_SA_279

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-794 Fields for Register: IRQ_TYPE_STAT_RX_SA_279 (Continued)

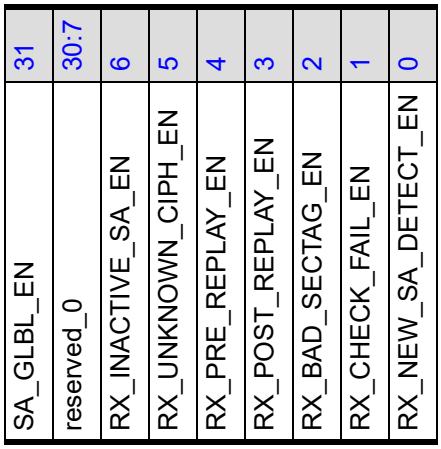
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-794 Fields for Register: IRQ_TYPE_STAT_RX_SA_279 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.788 IRQ_TYPE_EN_RX_SA_280

- **Name:** IRQ_TYPE_EN_RX_SA_280
- **Description:** Interrupt Enable for each IRQ type of Rx SA 280
- **Size:** 32 bits
- **Offset:** 0xc50
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 280



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-795 Fields for Register: IRQ_TYPE_EN_RX_SA_280

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-795 Fields for Register: IRQ_TYPE_EN_RX_SA_280 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-795 Fields for Register: IRQ_TYPE_EN_RX_SA_280 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.789 IRQ_TYPE_STAT_RX_SA_280

- **Name:** IRQ_TYPE_STAT_RX_SA_280
- **Description:** Interrupt Status for each IRQ type of Rx SA 280
- **Size:** 32 bits
- **Offset:** 0xc54
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 280

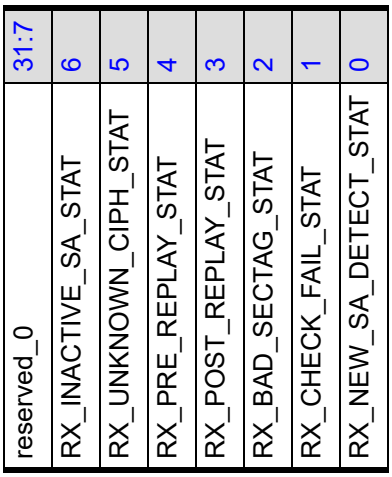


Table 6-796 Fields for Register: IRQ_TYPE_STAT_RX_SA_280

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-796 Fields for Register: IRQ_TYPE_STAT_RX_SA_280 (Continued)

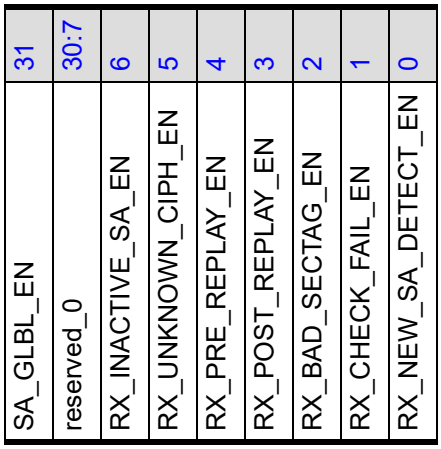
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-796 Fields for Register: IRQ_TYPE_STAT_RX_SA_280 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.790 IRQ_TYPE_EN_RX_SA_281

- **Name:** IRQ_TYPE_EN_RX_SA_281
- **Description:** Interrupt Enable for each IRQ type of Rx SA 281
- **Size:** 32 bits
- **Offset:** 0xc58
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 281



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-797 Fields for Register: IRQ_TYPE_EN_RX_SA_281

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-797 Fields for Register: IRQ_TYPE_EN_RX_SA_281 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-797 Fields for Register: IRQ_TYPE_EN_RX_SA_281 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.791 IRQ_TYPE_STAT_RX_SA_281

- **Name:** IRQ_TYPE_STAT_RX_SA_281
- **Description:** Interrupt Status for each IRQ type of Rx SA 281
- **Size:** 32 bits
- **Offset:** 0xc5c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 281

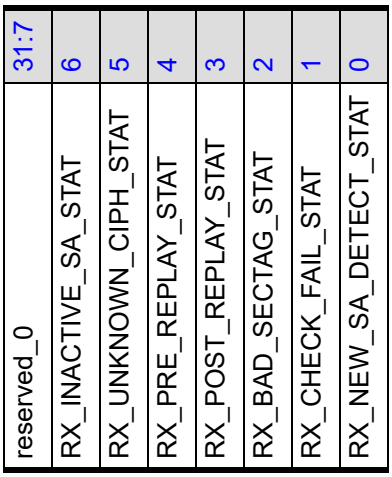


Table 6-798 Fields for Register: IRQ_TYPE_STAT_RX_SA_281

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-798 Fields for Register: IRQ_TYPE_STAT_RX_SA_281 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-798 Fields for Register: IRQ_TYPE_STAT_RX_SA_281 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.792 IRQ_TYPE_EN_RX_SA_282

- **Name:** IRQ_TYPE_EN_RX_SA_282
- **Description:** Interrupt Enable for each IRQ type of Rx SA 282
- **Size:** 32 bits
- **Offset:** 0xc60
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 282

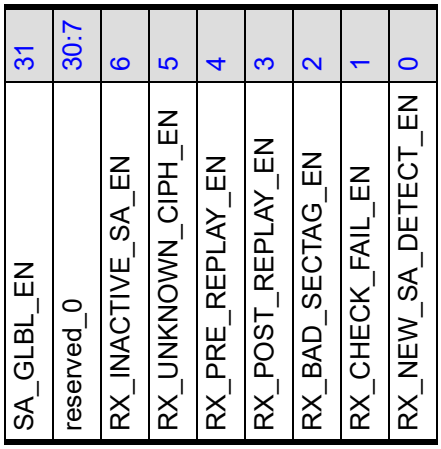


Table 6-799 Fields for Register: IRQ_TYPE_EN_RX_SA_282

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-799 Fields for Register: IRQ_TYPE_EN_RX_SA_282 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-799 Fields for Register: IRQ_TYPE_EN_RX_SA_282 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.793 IRQ_TYPE_STAT_RX_SA_282

- **Name:** IRQ_TYPE_STAT_RX_SA_282
- **Description:** Interrupt Status for each IRQ type of Rx SA 282
- **Size:** 32 bits
- **Offset:** 0xc64
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 282

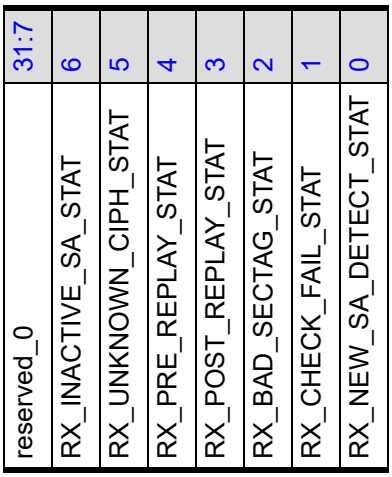


Table 6-800 Fields for Register: IRQ_TYPE_STAT_RX_SA_282

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-800 Fields for Register: IRQ_TYPE_STAT_RX_SA_282 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-800 Fields for Register: IRQ_TYPE_STAT_RX_SA_282 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.794 IRQ_TYPE_EN_RX_SA_283

- **Name:** IRQ_TYPE_EN_RX_SA_283
- **Description:** Interrupt Enable for each IRQ type of Rx SA 283
- **Size:** 32 bits
- **Offset:** 0xc68
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 283

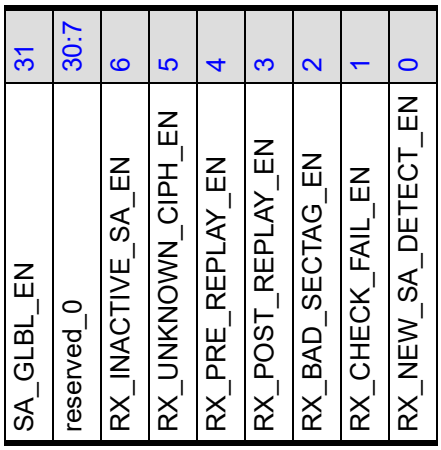


Table 6-801 Fields for Register: IRQ_TYPE_EN_RX_SA_283

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-801 Fields for Register: IRQ_TYPE_EN_RX_SA_283 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-801 Fields for Register: IRQ_TYPE_EN_RX_SA_283 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.795 IRQ_TYPE_STAT_RX_SA_283

- **Name:** IRQ_TYPE_STAT_RX_SA_283
- **Description:** Interrupt Status for each IRQ type of Rx SA 283
- **Size:** 32 bits
- **Offset:** 0xc6c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 283

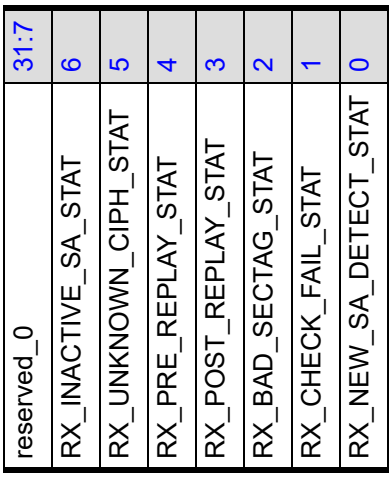


Table 6-802 Fields for Register: IRQ_TYPE_STAT_RX_SA_283

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-802 Fields for Register: IRQ_TYPE_STAT_RX_SA_283 (Continued)

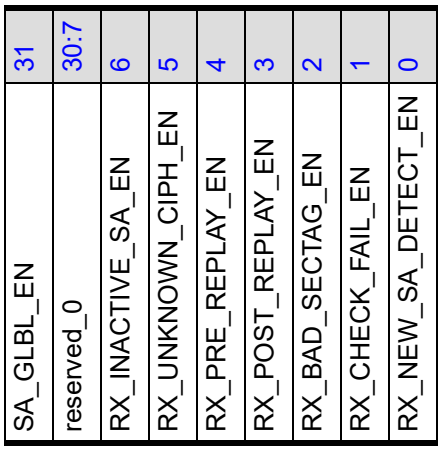
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-802 Fields for Register: IRQ_TYPE_STAT_RX_SA_283 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.796 IRQ_TYPE_EN_RX_SA_284

- **Name:** IRQ_TYPE_EN_RX_SA_284
- **Description:** Interrupt Enable for each IRQ type of Rx SA 284
- **Size:** 32 bits
- **Offset:** 0xc70
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 284



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-803 Fields for Register: IRQ_TYPE_EN_RX_SA_284

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-803 Fields for Register: IRQ_TYPE_EN_RX_SA_284 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-803 Fields for Register: IRQ_TYPE_EN_RX_SA_284 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.797 IRQ_TYPE_STAT_RX_SA_284

- **Name:** IRQ_TYPE_STAT_RX_SA_284
- **Description:** Interrupt Status for each IRQ type of Rx SA 284
- **Size:** 32 bits
- **Offset:** 0xc74
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 284

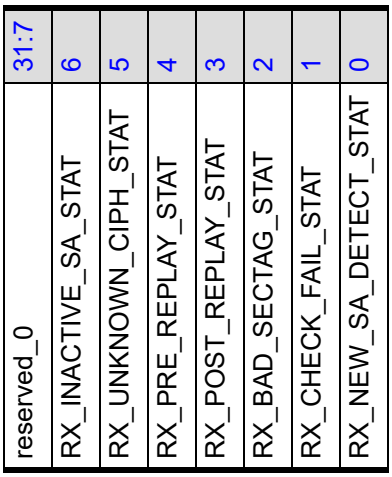


Table 6-804 Fields for Register: IRQ_TYPE_STAT_RX_SA_284

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-804 Fields for Register: IRQ_TYPE_STAT_RX_SA_284 (Continued)

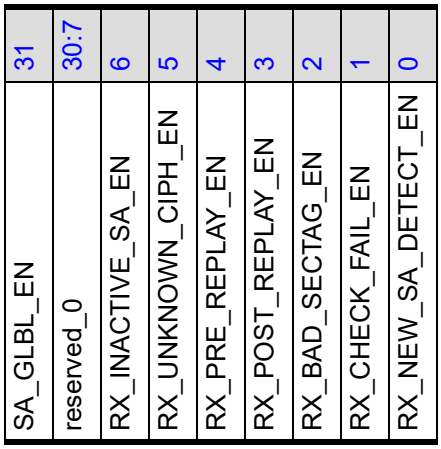
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-804 Fields for Register: IRQ_TYPE_STAT_RX_SA_284 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.798 IRQ_TYPE_EN_RX_SA_285

- **Name:** IRQ_TYPE_EN_RX_SA_285
- **Description:** Interrupt Enable for each IRQ type of Rx SA 285
- **Size:** 32 bits
- **Offset:** 0xc78
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 285



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-805 Fields for Register: IRQ_TYPE_EN_RX_SA_285

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-805 Fields for Register: IRQ_TYPE_EN_RX_SA_285 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-805 Fields for Register: IRQ_TYPE_EN_RX_SA_285 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.799 IRQ_TYPE_STAT_RX_SA_285

- **Name:** IRQ_TYPE_STAT_RX_SA_285
- **Description:** Interrupt Status for each IRQ type of Rx SA 285
- **Size:** 32 bits
- **Offset:** 0xc7c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 285

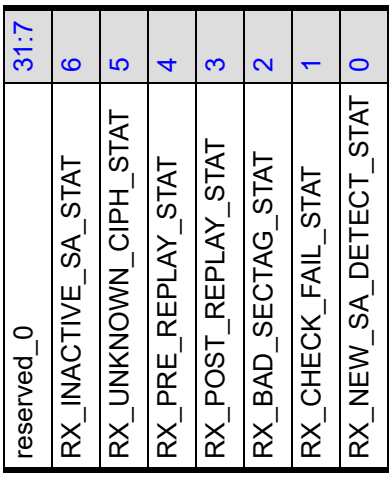


Table 6-806 Fields for Register: IRQ_TYPE_STAT_RX_SA_285

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-806 Fields for Register: IRQ_TYPE_STAT_RX_SA_285 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-806 Fields for Register: IRQ_TYPE_STAT_RX_SA_285 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.800 IRQ_TYPE_EN_RX_SA_286

- **Name:** IRQ_TYPE_EN_RX_SA_286
- **Description:** Interrupt Enable for each IRQ type of Rx SA 286
- **Size:** 32 bits
- **Offset:** 0xc80
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 286

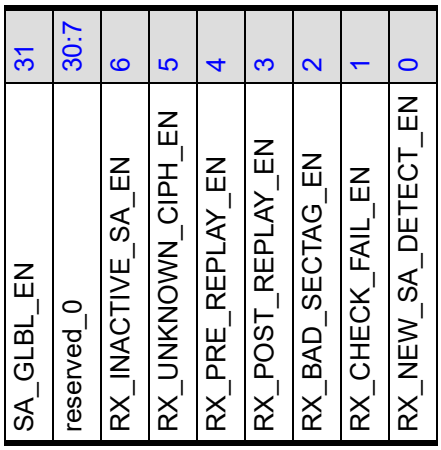


Table 6-807 Fields for Register: IRQ_TYPE_EN_RX_SA_286

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-807 Fields for Register: IRQ_TYPE_EN_RX_SA_286 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-807 Fields for Register: IRQ_TYPE_EN_RX_SA_286 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.801 IRQ_TYPE_STAT_RX_SA_286

- **Name:** IRQ_TYPE_STAT_RX_SA_286
- **Description:** Interrupt Status for each IRQ type of Rx SA 286
- **Size:** 32 bits
- **Offset:** 0xc84
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 286

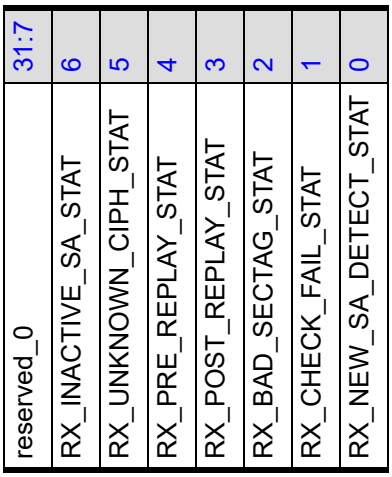


Table 6-808 Fields for Register: IRQ_TYPE_STAT_RX_SA_286

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-808 Fields for Register: IRQ_TYPE_STAT_RX_SA_286 (Continued)

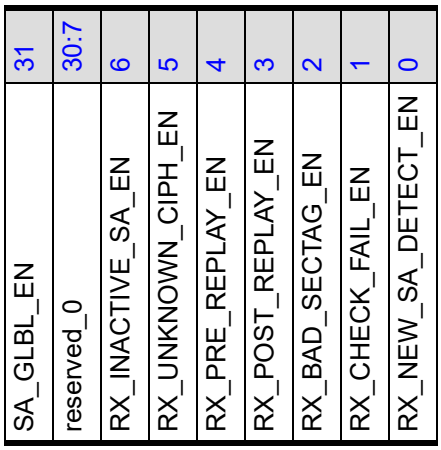
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-808 Fields for Register: IRQ_TYPE_STAT_RX_SA_286 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.802 IRQ_TYPE_EN_RX_SA_287

- **Name:** IRQ_TYPE_EN_RX_SA_287
- **Description:** Interrupt Enable for each IRQ type of Rx SA 287
- **Size:** 32 bits
- **Offset:** 0xc88
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 287



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-809 Fields for Register: IRQ_TYPE_EN_RX_SA_287

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-809 Fields for Register: IRQ_TYPE_EN_RX_SA_287 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-809 Fields for Register: IRQ_TYPE_EN_RX_SA_287 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.803 IRQ_TYPE_STAT_RX_SA_287

- **Name:** IRQ_TYPE_STAT_RX_SA_287
- **Description:** Interrupt Status for each IRQ type of Rx SA 287
- **Size:** 32 bits
- **Offset:** 0xc8c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 287

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-810 Fields for Register: IRQ_TYPE_STAT_RX_SA_287

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-810 Fields for Register: IRQ_TYPE_STAT_RX_SA_287 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-810 Fields for Register: IRQ_TYPE_STAT_RX_SA_287 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.804 IRQ_TYPE_EN_RX_SA_288

- **Name:** IRQ_TYPE_EN_RX_SA_288
- **Description:** Interrupt Enable for each IRQ type of Rx SA 288
- **Size:** 32 bits
- **Offset:** 0xc90
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 288

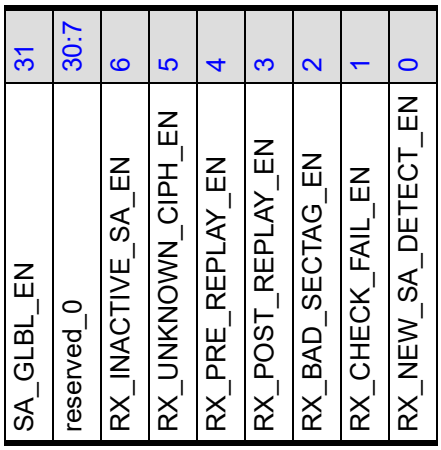


Table 6-811 Fields for Register: IRQ_TYPE_EN_RX_SA_288

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-811 Fields for Register: IRQ_TYPE_EN_RX_SA_288 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-811 Fields for Register: IRQ_TYPE_EN_RX_SA_288 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.805 IRQ_TYPE_STAT_RX_SA_288

- **Name:** IRQ_TYPE_STAT_RX_SA_288
- **Description:** Interrupt Status for each IRQ type of Rx SA 288
- **Size:** 32 bits
- **Offset:** 0xc94
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 288

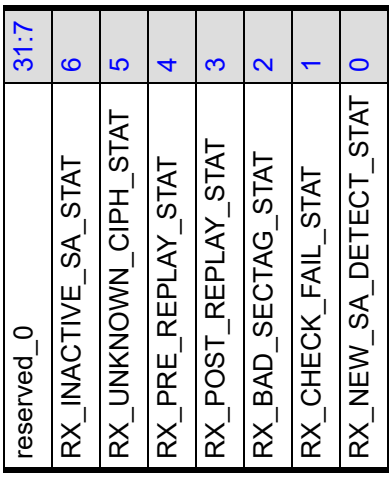


Table 6-812 Fields for Register: IRQ_TYPE_STAT_RX_SA_288

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-812 Fields for Register: IRQ_TYPE_STAT_RX_SA_288 (Continued)

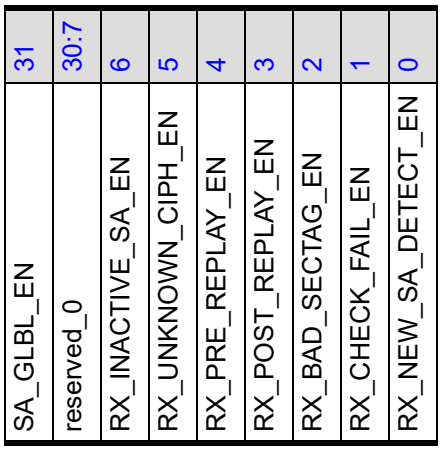
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-812 Fields for Register: IRQ_TYPE_STAT_RX_SA_288 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.806 IRQ_TYPE_EN_RX_SA_289

- **Name:** IRQ_TYPE_EN_RX_SA_289
- **Description:** Interrupt Enable for each IRQ type of Rx SA 289
- **Size:** 32 bits
- **Offset:** 0xc98
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 289



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-813 Fields for Register: IRQ_TYPE_EN_RX_SA_289

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-813 Fields for Register: IRQ_TYPE_EN_RX_SA_289 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-813 Fields for Register: IRQ_TYPE_EN_RX_SA_289 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.807 IRQ_TYPE_STAT_RX_SA_289

- **Name:** IRQ_TYPE_STAT_RX_SA_289
- **Description:** Interrupt Status for each IRQ type of Rx SA 289
- **Size:** 32 bits
- **Offset:** 0xc9c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 289

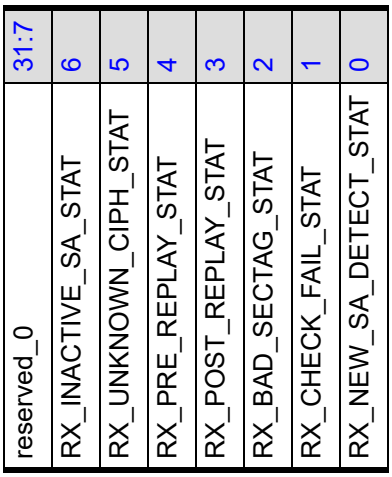


Table 6-814 Fields for Register: IRQ_TYPE_STAT_RX_SA_289

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-814 Fields for Register: IRQ_TYPE_STAT_RX_SA_289 (Continued)

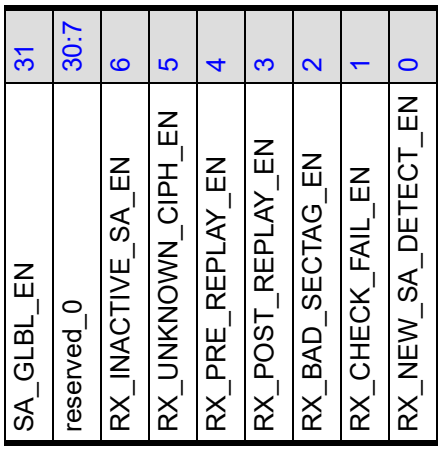
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-814 Fields for Register: IRQ_TYPE_STAT_RX_SA_289 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.808 IRQ_TYPE_EN_RX_SA_290

- **Name:** IRQ_TYPE_EN_RX_SA_290
- **Description:** Interrupt Enable for each IRQ type of Rx SA 290
- **Size:** 32 bits
- **Offset:** 0xca0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 290



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-815 Fields for Register: IRQ_TYPE_EN_RX_SA_290

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-815 Fields for Register: IRQ_TYPE_EN_RX_SA_290 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-815 Fields for Register: IRQ_TYPE_EN_RX_SA_290 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.809 IRQ_TYPE_STAT_RX_SA_290

- **Name:** IRQ_TYPE_STAT_RX_SA_290
- **Description:** Interrupt Status for each IRQ type of Rx SA 290
- **Size:** 32 bits
- **Offset:** 0xca4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 290

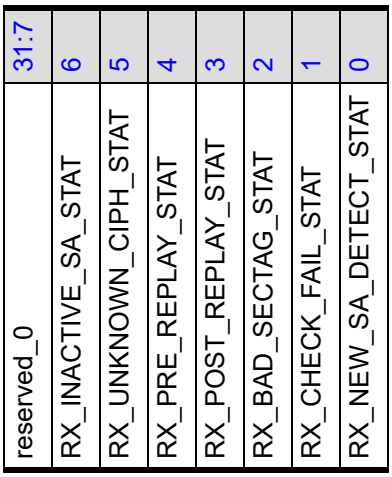


Table 6-816 Fields for Register: IRQ_TYPE_STAT_RX_SA_290

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-816 Fields for Register: IRQ_TYPE_STAT_RX_SA_290 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-816 Fields for Register: IRQ_TYPE_STAT_RX_SA_290 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.810 IRQ_TYPE_EN_RX_SA_291

- **Name:** IRQ_TYPE_EN_RX_SA_291
- **Description:** Interrupt Enable for each IRQ type of Rx SA 291
- **Size:** 32 bits
- **Offset:** 0xca8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 291

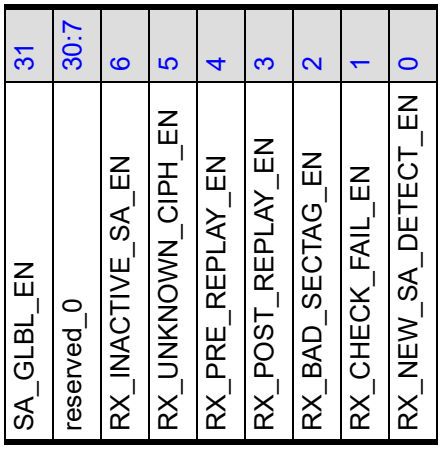


Table 6-817 Fields for Register: IRQ_TYPE_EN_RX_SA_291

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-817 Fields for Register: IRQ_TYPE_EN_RX_SA_291 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-817 Fields for Register: IRQ_TYPE_EN_RX_SA_291 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.811 IRQ_TYPE_STAT_RX_SA_291

- **Name:** IRQ_TYPE_STAT_RX_SA_291
- **Description:** Interrupt Status for each IRQ type of Rx SA 291
- **Size:** 32 bits
- **Offset:** 0xcac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 291

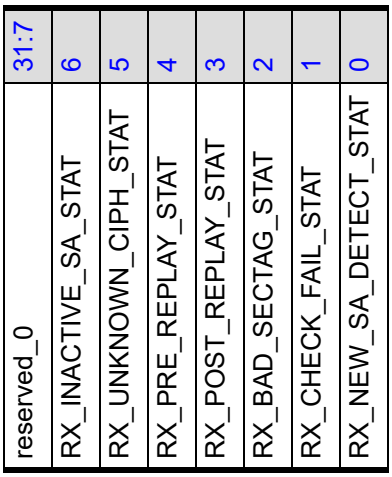


Table 6-818 Fields for Register: IRQ_TYPE_STAT_RX_SA_291

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-818 Fields for Register: IRQ_TYPE_STAT_RX_SA_291 (Continued)

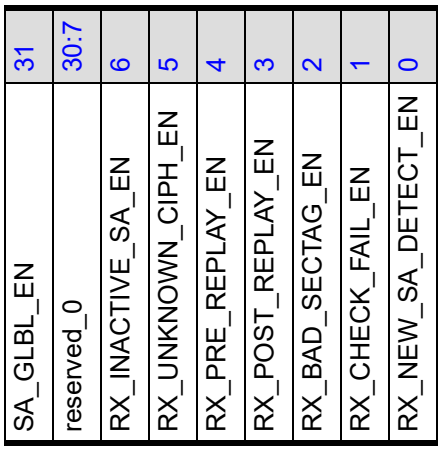
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-818 Fields for Register: IRQ_TYPE_STAT_RX_SA_291 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.812 IRQ_TYPE_EN_RX_SA_292

- **Name:** IRQ_TYPE_EN_RX_SA_292
- **Description:** Interrupt Enable for each IRQ type of Rx SA 292
- **Size:** 32 bits
- **Offset:** 0xcb0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 292



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-819 Fields for Register: IRQ_TYPE_EN_RX_SA_292

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-819 Fields for Register: IRQ_TYPE_EN_RX_SA_292 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-819 Fields for Register: IRQ_TYPE_EN_RX_SA_292 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.813 IRQ_TYPE_STAT_RX_SA_292

- **Name:** IRQ_TYPE_STAT_RX_SA_292
- **Description:** Interrupt Status for each IRQ type of Rx SA 292
- **Size:** 32 bits
- **Offset:** 0xcb4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 292

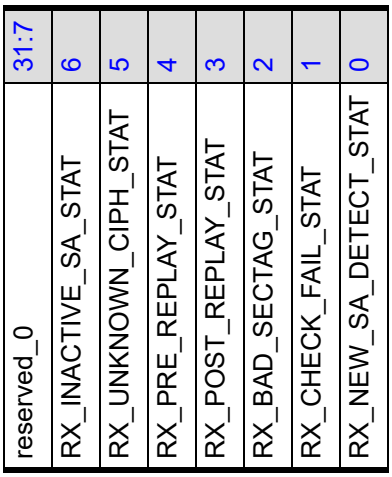


Table 6-820 Fields for Register: IRQ_TYPE_STAT_RX_SA_292

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-820 Fields for Register: IRQ_TYPE_STAT_RX_SA_292 (Continued)

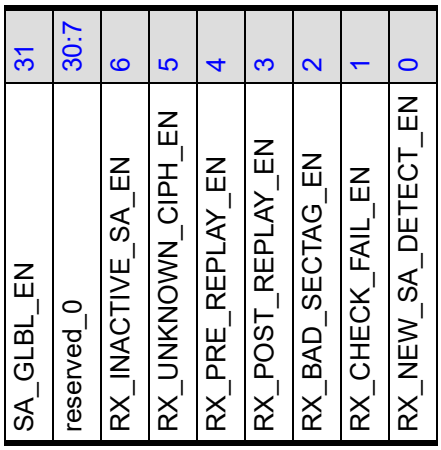
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-820 Fields for Register: IRQ_TYPE_STAT_RX_SA_292 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.814 IRQ_TYPE_EN_RX_SA_293

- **Name:** IRQ_TYPE_EN_RX_SA_293
- **Description:** Interrupt Enable for each IRQ type of Rx SA 293
- **Size:** 32 bits
- **Offset:** 0xcb8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 293



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-821 Fields for Register: IRQ_TYPE_EN_RX_SA_293

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-821 Fields for Register: IRQ_TYPE_EN_RX_SA_293 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-821 Fields for Register: IRQ_TYPE_EN_RX_SA_293 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.815 IRQ_TYPE_STAT_RX_SA_293

- **Name:** IRQ_TYPE_STAT_RX_SA_293
- **Description:** Interrupt Status for each IRQ type of Rx SA 293
- **Size:** 32 bits
- **Offset:** 0xcbc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 293

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-822 Fields for Register: IRQ_TYPE_STAT_RX_SA_293

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-822 Fields for Register: IRQ_TYPE_STAT_RX_SA_293 (Continued)

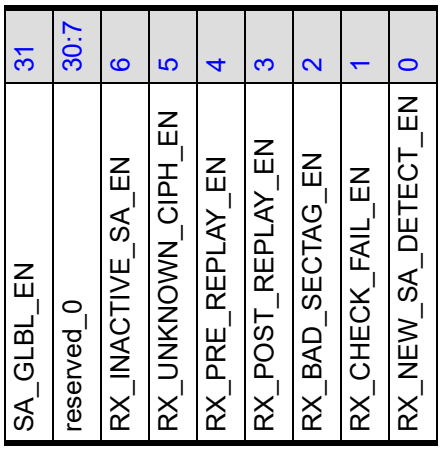
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-822 Fields for Register: IRQ_TYPE_STAT_RX_SA_293 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.816 IRQ_TYPE_EN_RX_SA_294

- **Name:** IRQ_TYPE_EN_RX_SA_294
- **Description:** Interrupt Enable for each IRQ type of Rx SA 294
- **Size:** 32 bits
- **Offset:** 0xcc0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 294



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-823 Fields for Register: IRQ_TYPE_EN_RX_SA_294

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-823 Fields for Register: IRQ_TYPE_EN_RX_SA_294 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-823 Fields for Register: IRQ_TYPE_EN_RX_SA_294 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.817 IRQ_TYPE_STAT_RX_SA_294

- **Name:** IRQ_TYPE_STAT_RX_SA_294
- **Description:** Interrupt Status for each IRQ type of Rx SA 294
- **Size:** 32 bits
- **Offset:** 0xcc4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 294

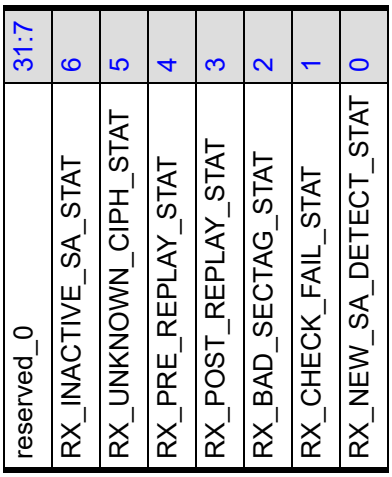


Table 6-824 Fields for Register: IRQ_TYPE_STAT_RX_SA_294

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-824 Fields for Register: IRQ_TYPE_STAT_RX_SA_294 (Continued)

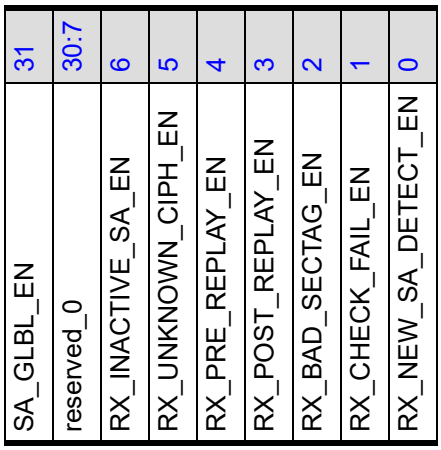
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-824 Fields for Register: IRQ_TYPE_STAT_RX_SA_294 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.818 IRQ_TYPE_EN_RX_SA_295

- **Name:** IRQ_TYPE_EN_RX_SA_295
- **Description:** Interrupt Enable for each IRQ type of Rx SA 295
- **Size:** 32 bits
- **Offset:** 0xcc8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 295



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-825 Fields for Register: IRQ_TYPE_EN_RX_SA_295

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-825 Fields for Register: IRQ_TYPE_EN_RX_SA_295 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-825 Fields for Register: IRQ_TYPE_EN_RX_SA_295 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.819 IRQ_TYPE_STAT_RX_SA_295

- **Name:** IRQ_TYPE_STAT_RX_SA_295
- **Description:** Interrupt Status for each IRQ type of Rx SA 295
- **Size:** 32 bits
- **Offset:** 0xccc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 295

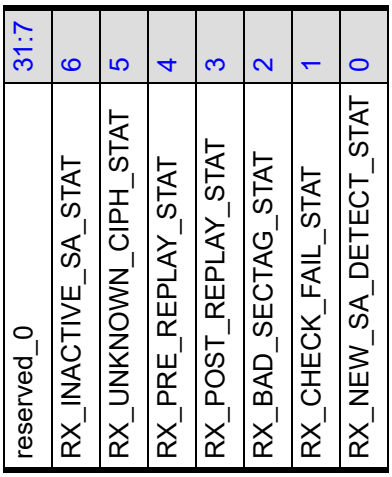


Table 6-826 Fields for Register: IRQ_TYPE_STAT_RX_SA_295

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-826 Fields for Register: IRQ_TYPE_STAT_RX_SA_295 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-826 Fields for Register: IRQ_TYPE_STAT_RX_SA_295 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.820 IRQ_TYPE_EN_RX_SA_296

- **Name:** IRQ_TYPE_EN_RX_SA_296
- **Description:** Interrupt Enable for each IRQ type of Rx SA 296
- **Size:** 32 bits
- **Offset:** 0xcd0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 296

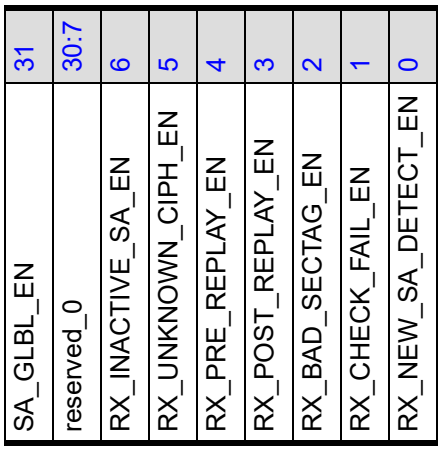


Table 6-827 Fields for Register: IRQ_TYPE_EN_RX_SA_296

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-827 Fields for Register: IRQ_TYPE_EN_RX_SA_296 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-827 Fields for Register: IRQ_TYPE_EN_RX_SA_296 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.821 IRQ_TYPE_STAT_RX_SA_296

- **Name:** IRQ_TYPE_STAT_RX_SA_296
- **Description:** Interrupt Status for each IRQ type of Rx SA 296
- **Size:** 32 bits
- **Offset:** 0xcd4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 296

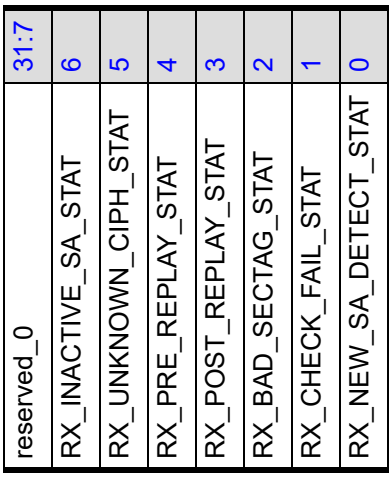


Table 6-828 Fields for Register: IRQ_TYPE_STAT_RX_SA_296

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-828 Fields for Register: IRQ_TYPE_STAT_RX_SA_296 (Continued)

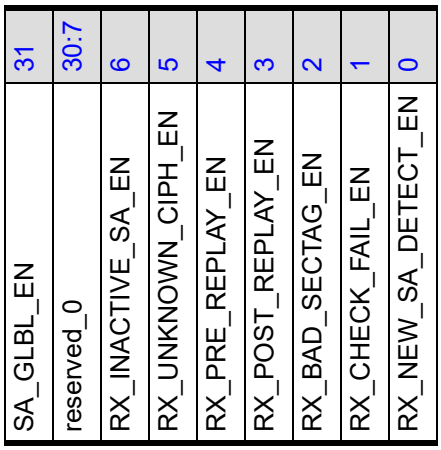
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-828 Fields for Register: IRQ_TYPE_STAT_RX_SA_296 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.822 IRQ_TYPE_EN_RX_SA_297

- **Name:** IRQ_TYPE_EN_RX_SA_297
- **Description:** Interrupt Enable for each IRQ type of Rx SA 297
- **Size:** 32 bits
- **Offset:** 0xcd8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 297



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-829 Fields for Register: IRQ_TYPE_EN_RX_SA_297

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-829 Fields for Register: IRQ_TYPE_EN_RX_SA_297 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-829 Fields for Register: IRQ_TYPE_EN_RX_SA_297 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.823 IRQ_TYPE_STAT_RX_SA_297

- **Name:** IRQ_TYPE_STAT_RX_SA_297
- **Description:** Interrupt Status for each IRQ type of Rx SA 297
- **Size:** 32 bits
- **Offset:** 0xcdc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 297

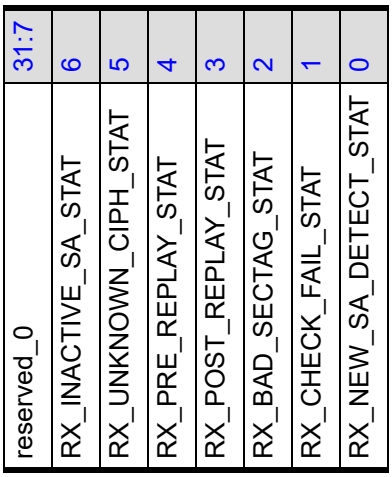


Table 6-830 Fields for Register: IRQ_TYPE_STAT_RX_SA_297

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-830 Fields for Register: IRQ_TYPE_STAT_RX_SA_297 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-830 Fields for Register: IRQ_TYPE_STAT_RX_SA_297 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.824 IRQ_TYPE_EN_RX_SA_298

- **Name:** IRQ_TYPE_EN_RX_SA_298
- **Description:** Interrupt Enable for each IRQ type of Rx SA 298
- **Size:** 32 bits
- **Offset:** 0xce0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 298

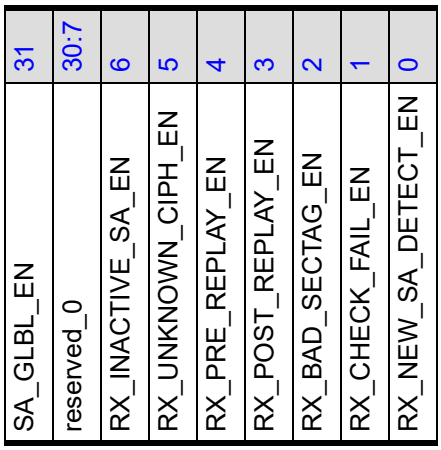


Table 6-831 Fields for Register: IRQ_TYPE_EN_RX_SA_298

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-831 Fields for Register: IRQ_TYPE_EN_RX_SA_298 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-831 Fields for Register: IRQ_TYPE_EN_RX_SA_298 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.825 IRQ_TYPE_STAT_RX_SA_298

- **Name:** IRQ_TYPE_STAT_RX_SA_298
- **Description:** Interrupt Status for each IRQ type of Rx SA 298
- **Size:** 32 bits
- **Offset:** 0xce4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 298

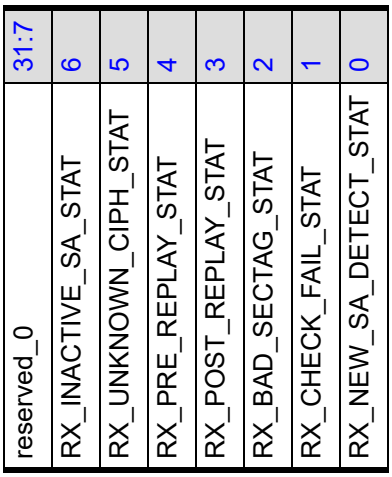


Table 6-832 Fields for Register: IRQ_TYPE_STAT_RX_SA_298

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-832 Fields for Register: IRQ_TYPE_STAT_RX_SA_298 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-832 Fields for Register: IRQ_TYPE_STAT_RX_SA_298 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.826 IRQ_TYPE_EN_RX_SA_299

- **Name:** IRQ_TYPE_EN_RX_SA_299
- **Description:** Interrupt Enable for each IRQ type of Rx SA 299
- **Size:** 32 bits
- **Offset:** 0xce8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 299

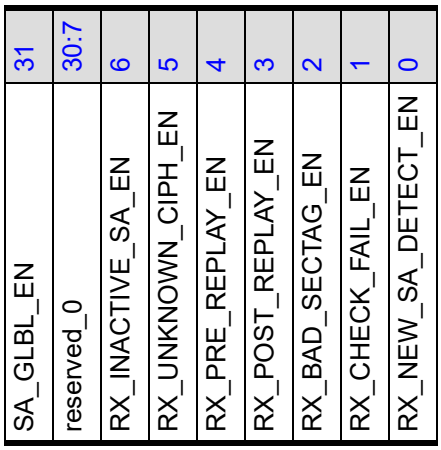


Table 6-833 Fields for Register: IRQ_TYPE_EN_RX_SA_299

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-833 Fields for Register: IRQ_TYPE_EN_RX_SA_299 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-833 Fields for Register: IRQ_TYPE_EN_RX_SA_299 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.827 IRQ_TYPE_STAT_RX_SA_299

- **Name:** IRQ_TYPE_STAT_RX_SA_299
- **Description:** Interrupt Status for each IRQ type of Rx SA 299
- **Size:** 32 bits
- **Offset:** 0xcec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 299

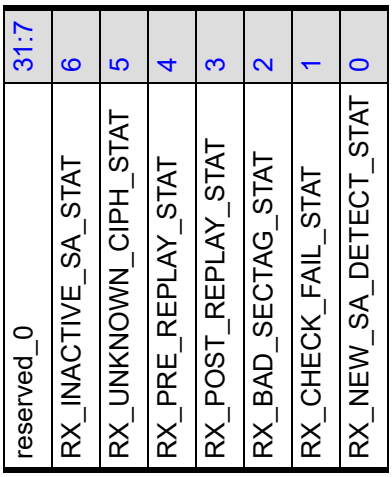


Table 6-834 Fields for Register: IRQ_TYPE_STAT_RX_SA_299

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-834 Fields for Register: IRQ_TYPE_STAT_RX_SA_299 (Continued)

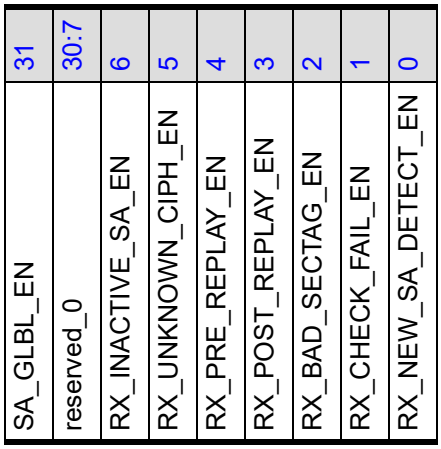
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-834 Fields for Register: IRQ_TYPE_STAT_RX_SA_299 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.828 IRQ_TYPE_EN_RX_SA_300

- **Name:** IRQ_TYPE_EN_RX_SA_300
- **Description:** Interrupt Enable for each IRQ type of Rx SA 300
- **Size:** 32 bits
- **Offset:** 0xcf0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 300



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-835 Fields for Register: IRQ_TYPE_EN_RX_SA_300

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-835 Fields for Register: IRQ_TYPE_EN_RX_SA_300 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-835 Fields for Register: IRQ_TYPE_EN_RX_SA_300 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.829 IRQ_TYPE_STAT_RX_SA_300

- **Name:** IRQ_TYPE_STAT_RX_SA_300
- **Description:** Interrupt Status for each IRQ type of Rx SA 300
- **Size:** 32 bits
- **Offset:** 0xcf4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 300

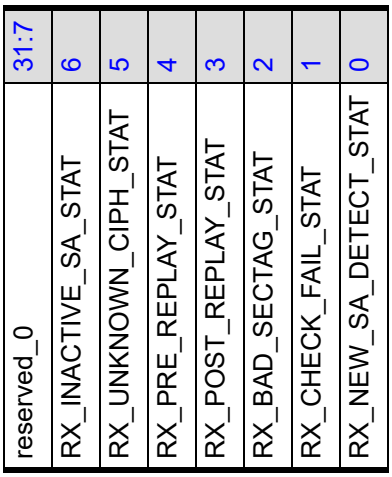


Table 6-836 Fields for Register: IRQ_TYPE_STAT_RX_SA_300

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-836 Fields for Register: IRQ_TYPE_STAT_RX_SA_300 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-836 Fields for Register: IRQ_TYPE_STAT_RX_SA_300 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.830 IRQ_TYPE_EN_RX_SA_301

- **Name:** IRQ_TYPE_EN_RX_SA_301
- **Description:** Interrupt Enable for each IRQ type of Rx SA 301
- **Size:** 32 bits
- **Offset:** 0xcf8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 301

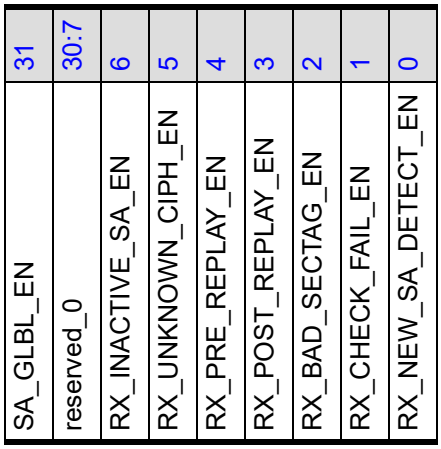


Table 6-837 Fields for Register: IRQ_TYPE_EN_RX_SA_301

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-837 Fields for Register: IRQ_TYPE_EN_RX_SA_301 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-837 Fields for Register: IRQ_TYPE_EN_RX_SA_301 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.831 IRQ_TYPE_STAT_RX_SA_301

- **Name:** IRQ_TYPE_STAT_RX_SA_301
- **Description:** Interrupt Status for each IRQ type of Rx SA 301
- **Size:** 32 bits
- **Offset:** 0xcfc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 301

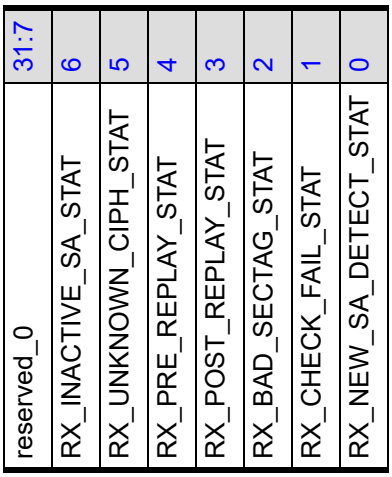


Table 6-838 Fields for Register: IRQ_TYPE_STAT_RX_SA_301

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-838 Fields for Register: IRQ_TYPE_STAT_RX_SA_301 (Continued)

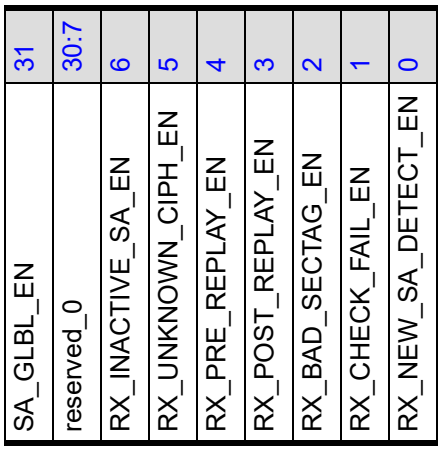
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-838 Fields for Register: IRQ_TYPE_STAT_RX_SA_301 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.832 IRQ_TYPE_EN_RX_SA_302

- **Name:** IRQ_TYPE_EN_RX_SA_302
- **Description:** Interrupt Enable for each IRQ type of Rx SA 302
- **Size:** 32 bits
- **Offset:** 0xd00
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 302



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-839 Fields for Register: IRQ_TYPE_EN_RX_SA_302

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-839 Fields for Register: IRQ_TYPE_EN_RX_SA_302 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-839 Fields for Register: IRQ_TYPE_EN_RX_SA_302 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.833 IRQ_TYPE_STAT_RX_SA_302

- **Name:** IRQ_TYPE_STAT_RX_SA_302
- **Description:** Interrupt Status for each IRQ type of Rx SA 302
- **Size:** 32 bits
- **Offset:** 0xd04
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 302

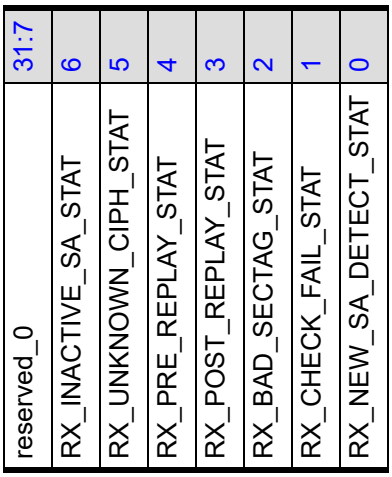


Table 6-840 Fields for Register: IRQ_TYPE_STAT_RX_SA_302

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-840 Fields for Register: IRQ_TYPE_STAT_RX_SA_302 (Continued)

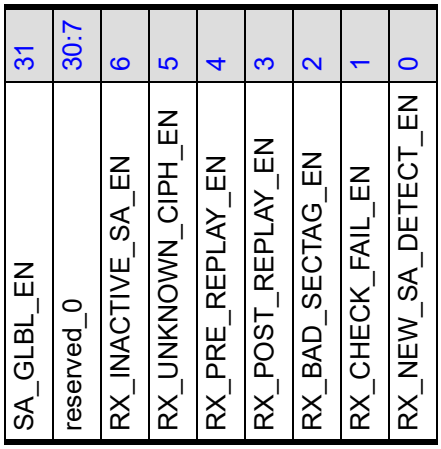
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-840 Fields for Register: IRQ_TYPE_STAT_RX_SA_302 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.834 IRQ_TYPE_EN_RX_SA_303

- **Name:** IRQ_TYPE_EN_RX_SA_303
- **Description:** Interrupt Enable for each IRQ type of Rx SA 303
- **Size:** 32 bits
- **Offset:** 0xd08
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 303



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-841 Fields for Register: IRQ_TYPE_EN_RX_SA_303

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-841 Fields for Register: IRQ_TYPE_EN_RX_SA_303 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-841 Fields for Register: IRQ_TYPE_EN_RX_SA_303 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.835 IRQ_TYPE_STAT_RX_SA_303

- **Name:** IRQ_TYPE_STAT_RX_SA_303
- **Description:** Interrupt Status for each IRQ type of Rx SA 303
- **Size:** 32 bits
- **Offset:** 0xd0c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 303

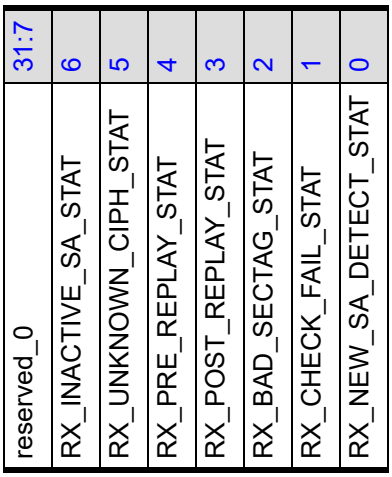


Table 6-842 Fields for Register: IRQ_TYPE_STAT_RX_SA_303

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-842 Fields for Register: IRQ_TYPE_STAT_RX_SA_303 (Continued)

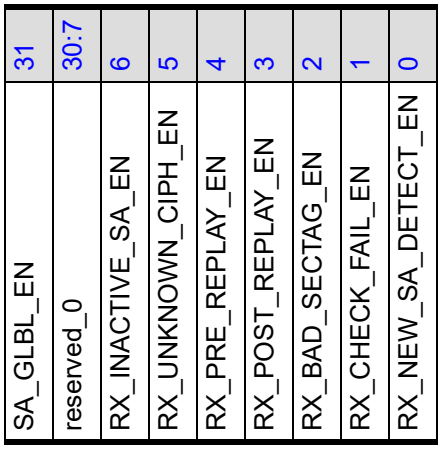
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-842 Fields for Register: IRQ_TYPE_STAT_RX_SA_303 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.836 IRQ_TYPE_EN_RX_SA_304

- **Name:** IRQ_TYPE_EN_RX_SA_304
- **Description:** Interrupt Enable for each IRQ type of Rx SA 304
- **Size:** 32 bits
- **Offset:** 0xd10
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 304



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-843 Fields for Register: IRQ_TYPE_EN_RX_SA_304

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-843 Fields for Register: IRQ_TYPE_EN_RX_SA_304 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-843 Fields for Register: IRQ_TYPE_EN_RX_SA_304 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.837 IRQ_TYPE_STAT_RX_SA_304

- **Name:** IRQ_TYPE_STAT_RX_SA_304
- **Description:** Interrupt Status for each IRQ type of Rx SA 304
- **Size:** 32 bits
- **Offset:** 0xd14
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 304

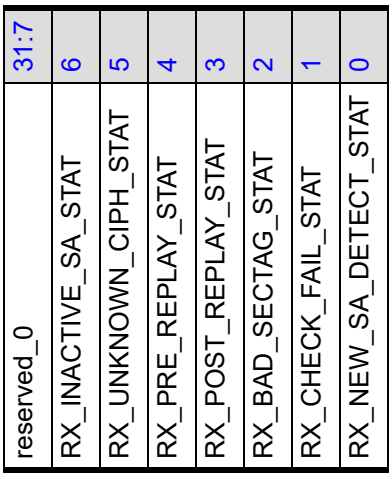


Table 6-844 Fields for Register: IRQ_TYPE_STAT_RX_SA_304

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-844 Fields for Register: IRQ_TYPE_STAT_RX_SA_304 (Continued)

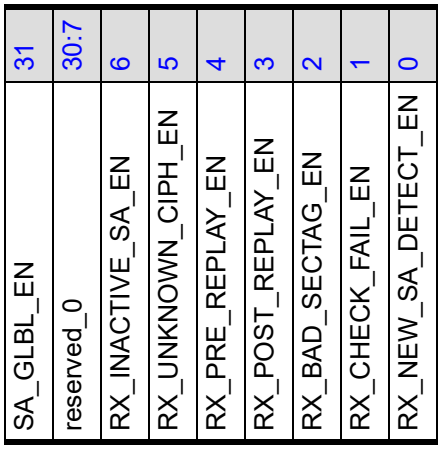
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-844 Fields for Register: IRQ_TYPE_STAT_RX_SA_304 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.838 IRQ_TYPE_EN_RX_SA_305

- **Name:** IRQ_TYPE_EN_RX_SA_305
- **Description:** Interrupt Enable for each IRQ type of Rx SA 305
- **Size:** 32 bits
- **Offset:** 0xd18
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 305



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-845 Fields for Register: IRQ_TYPE_EN_RX_SA_305

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-845 Fields for Register: IRQ_TYPE_EN_RX_SA_305 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-845 Fields for Register: IRQ_TYPE_EN_RX_SA_305 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.839 IRQ_TYPE_STAT_RX_SA_305

- **Name:** IRQ_TYPE_STAT_RX_SA_305
- **Description:** Interrupt Status for each IRQ type of Rx SA 305
- **Size:** 32 bits
- **Offset:** 0xd1c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 305

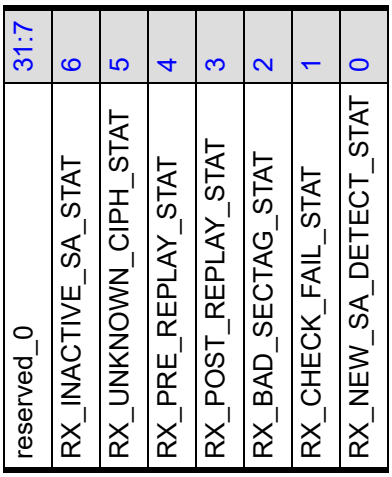


Table 6-846 Fields for Register: IRQ_TYPE_STAT_RX_SA_305

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-846 Fields for Register: IRQ_TYPE_STAT_RX_SA_305 (Continued)

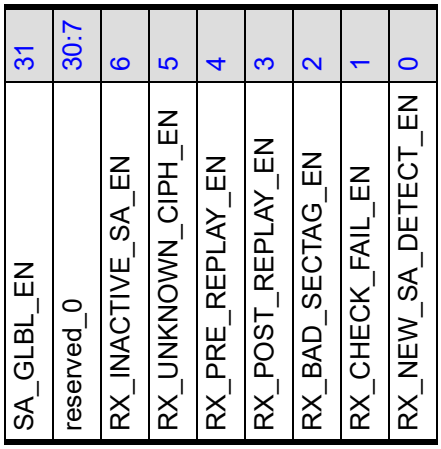
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-846 Fields for Register: IRQ_TYPE_STAT_RX_SA_305 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.840 IRQ_TYPE_EN_RX_SA_306

- **Name:** IRQ_TYPE_EN_RX_SA_306
- **Description:** Interrupt Enable for each IRQ type of Rx SA 306
- **Size:** 32 bits
- **Offset:** 0xd20
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 306



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-847 Fields for Register: IRQ_TYPE_EN_RX_SA_306

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-847 Fields for Register: IRQ_TYPE_EN_RX_SA_306 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-847 Fields for Register: IRQ_TYPE_EN_RX_SA_306 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.841 IRQ_TYPE_STAT_RX_SA_306

- **Name:** IRQ_TYPE_STAT_RX_SA_306
- **Description:** Interrupt Status for each IRQ type of Rx SA 306
- **Size:** 32 bits
- **Offset:** 0xd24
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 306

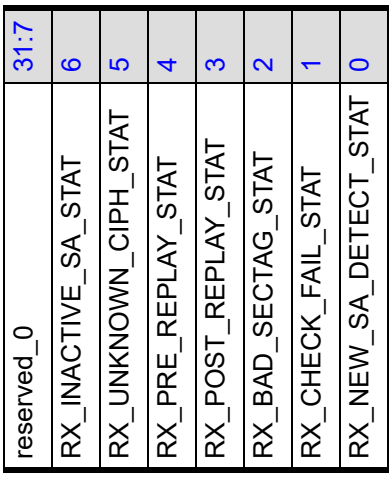


Table 6-848 Fields for Register: IRQ_TYPE_STAT_RX_SA_306

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-848 Fields for Register: IRQ_TYPE_STAT_RX_SA_306 (Continued)

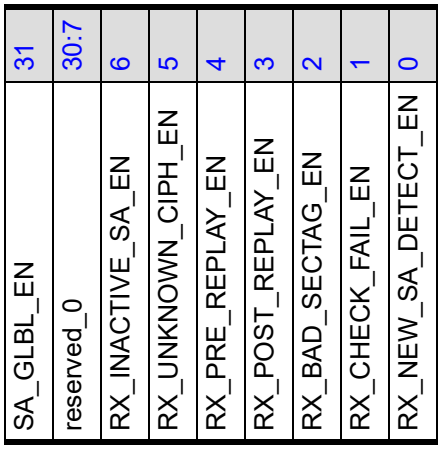
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-848 Fields for Register: IRQ_TYPE_STAT_RX_SA_306 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.842 IRQ_TYPE_EN_RX_SA_307

- **Name:** IRQ_TYPE_EN_RX_SA_307
- **Description:** Interrupt Enable for each IRQ type of Rx SA 307
- **Size:** 32 bits
- **Offset:** 0xd28
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 307



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-849 Fields for Register: IRQ_TYPE_EN_RX_SA_307

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-849 Fields for Register: IRQ_TYPE_EN_RX_SA_307 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-849 Fields for Register: IRQ_TYPE_EN_RX_SA_307 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.843 IRQ_TYPE_STAT_RX_SA_307

- **Name:** IRQ_TYPE_STAT_RX_SA_307
- **Description:** Interrupt Status for each IRQ type of Rx SA 307
- **Size:** 32 bits
- **Offset:** 0xd2c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 307

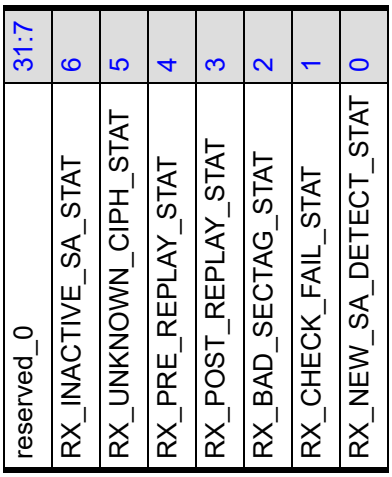


Table 6-850 Fields for Register: IRQ_TYPE_STAT_RX_SA_307

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-850 Fields for Register: IRQ_TYPE_STAT_RX_SA_307 (Continued)

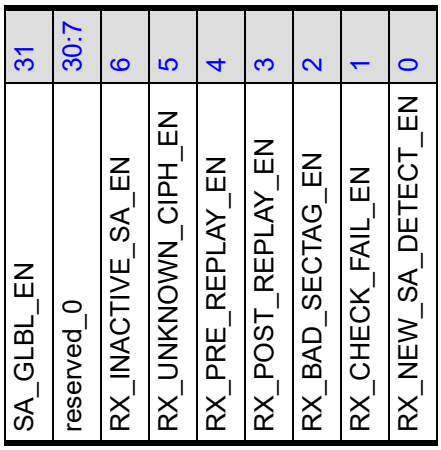
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-850 Fields for Register: IRQ_TYPE_STAT_RX_SA_307 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.844 IRQ_TYPE_EN_RX_SA_308

- **Name:** IRQ_TYPE_EN_RX_SA_308
- **Description:** Interrupt Enable for each IRQ type of Rx SA 308
- **Size:** 32 bits
- **Offset:** 0xd30
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 308



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-851 Fields for Register: IRQ_TYPE_EN_RX_SA_308

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-851 Fields for Register: IRQ_TYPE_EN_RX_SA_308 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-851 Fields for Register: IRQ_TYPE_EN_RX_SA_308 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.845 IRQ_TYPE_STAT_RX_SA_308

- **Name:** IRQ_TYPE_STAT_RX_SA_308
- **Description:** Interrupt Status for each IRQ type of Rx SA 308
- **Size:** 32 bits
- **Offset:** 0xd34
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 308

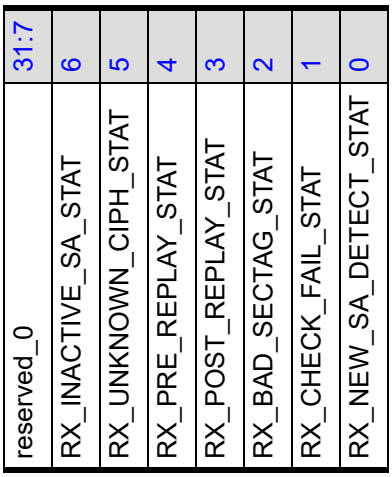


Table 6-852 Fields for Register: IRQ_TYPE_STAT_RX_SA_308

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-852 Fields for Register: IRQ_TYPE_STAT_RX_SA_308 (Continued)

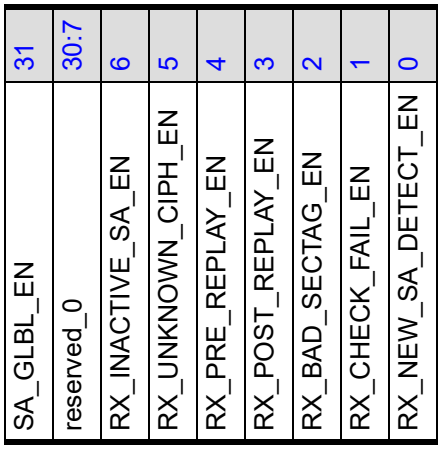
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-852 Fields for Register: IRQ_TYPE_STAT_RX_SA_308 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.846 IRQ_TYPE_EN_RX_SA_309

- **Name:** IRQ_TYPE_EN_RX_SA_309
- **Description:** Interrupt Enable for each IRQ type of Rx SA 309
- **Size:** 32 bits
- **Offset:** 0xd38
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 309



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-853 Fields for Register: IRQ_TYPE_EN_RX_SA_309

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-853 Fields for Register: IRQ_TYPE_EN_RX_SA_309 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-853 Fields for Register: IRQ_TYPE_EN_RX_SA_309 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.847 IRQ_TYPE_STAT_RX_SA_309

- **Name:** IRQ_TYPE_STAT_RX_SA_309
- **Description:** Interrupt Status for each IRQ type of Rx SA 309
- **Size:** 32 bits
- **Offset:** 0xd3c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 309

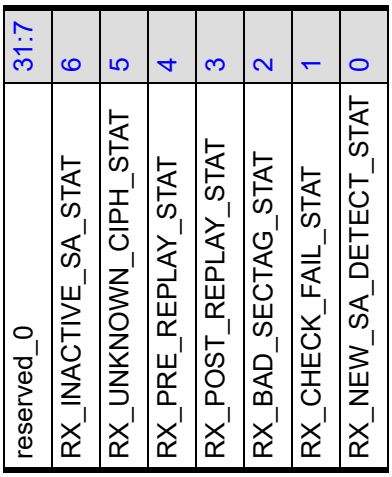


Table 6-854 Fields for Register: IRQ_TYPE_STAT_RX_SA_309

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-854 Fields for Register: IRQ_TYPE_STAT_RX_SA_309 (Continued)

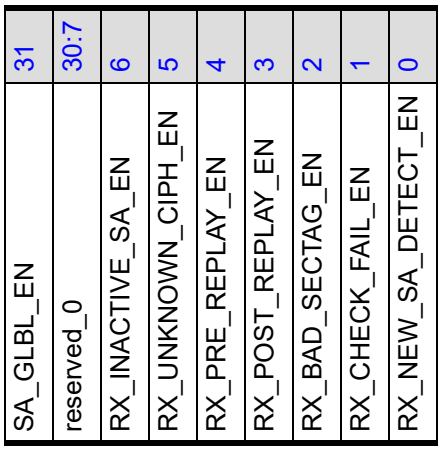
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-854 Fields for Register: IRQ_TYPE_STAT_RX_SA_309 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.848 IRQ_TYPE_EN_RX_SA_310

- **Name:** IRQ_TYPE_EN_RX_SA_310
- **Description:** Interrupt Enable for each IRQ type of Rx SA 310
- **Size:** 32 bits
- **Offset:** 0xd40
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 310



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-855 Fields for Register: IRQ_TYPE_EN_RX_SA_310

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-855 Fields for Register: IRQ_TYPE_EN_RX_SA_310 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-855 Fields for Register: IRQ_TYPE_EN_RX_SA_310 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.849 IRQ_TYPE_STAT_RX_SA_310

- **Name:** IRQ_TYPE_STAT_RX_SA_310
- **Description:** Interrupt Status for each IRQ type of Rx SA 310
- **Size:** 32 bits
- **Offset:** 0xd44
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 310

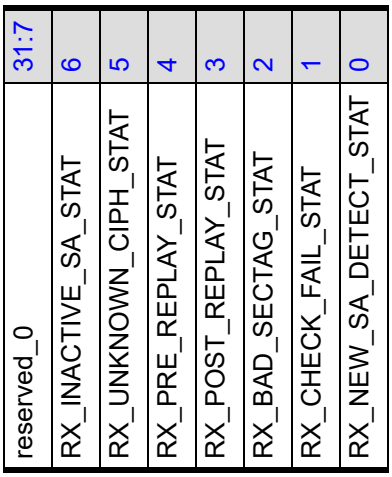


Table 6-856 Fields for Register: IRQ_TYPE_STAT_RX_SA_310

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-856 Fields for Register: IRQ_TYPE_STAT_RX_SA_310 (Continued)

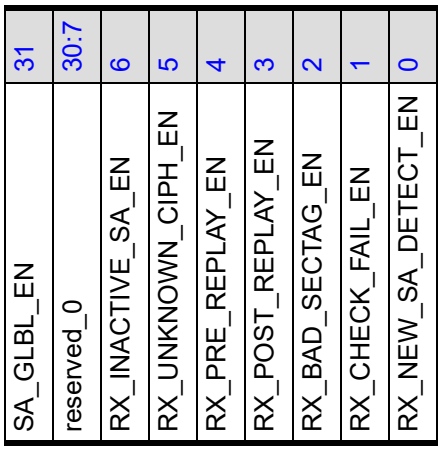
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-856 Fields for Register: IRQ_TYPE_STAT_RX_SA_310 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.850 IRQ_TYPE_EN_RX_SA_311

- **Name:** IRQ_TYPE_EN_RX_SA_311
- **Description:** Interrupt Enable for each IRQ type of Rx SA 311
- **Size:** 32 bits
- **Offset:** 0xd48
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 311



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-857 Fields for Register: IRQ_TYPE_EN_RX_SA_311

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-857 Fields for Register: IRQ_TYPE_EN_RX_SA_311 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-857 Fields for Register: IRQ_TYPE_EN_RX_SA_311 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.851 IRQ_TYPE_STAT_RX_SA_311

- **Name:** IRQ_TYPE_STAT_RX_SA_311
- **Description:** Interrupt Status for each IRQ type of Rx SA 311
- **Size:** 32 bits
- **Offset:** 0xd4c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 311

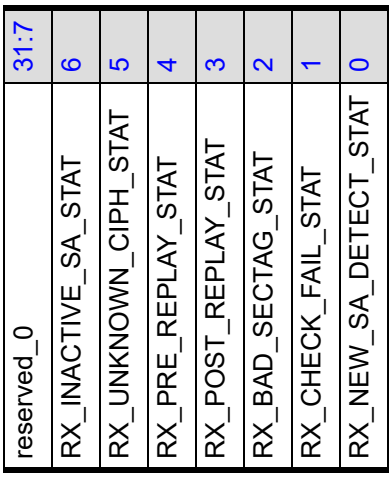


Table 6-858 Fields for Register: IRQ_TYPE_STAT_RX_SA_311

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-858 Fields for Register: IRQ_TYPE_STAT_RX_SA_311 (Continued)

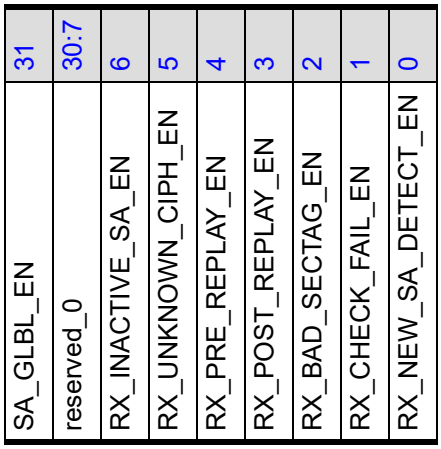
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-858 Fields for Register: IRQ_TYPE_STAT_RX_SA_311 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.852 IRQ_TYPE_EN_RX_SA_312

- **Name:** IRQ_TYPE_EN_RX_SA_312
- **Description:** Interrupt Enable for each IRQ type of Rx SA 312
- **Size:** 32 bits
- **Offset:** 0xd50
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 312



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-859 Fields for Register: IRQ_TYPE_EN_RX_SA_312

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-859 Fields for Register: IRQ_TYPE_EN_RX_SA_312 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-859 Fields for Register: IRQ_TYPE_EN_RX_SA_312 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.853 IRQ_TYPE_STAT_RX_SA_312

- **Name:** IRQ_TYPE_STAT_RX_SA_312
- **Description:** Interrupt Status for each IRQ type of Rx SA 312
- **Size:** 32 bits
- **Offset:** 0xd54
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 312

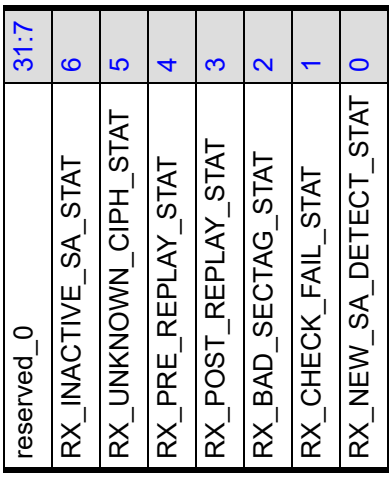


Table 6-860 Fields for Register: IRQ_TYPE_STAT_RX_SA_312

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-860 Fields for Register: IRQ_TYPE_STAT_RX_SA_312 (Continued)

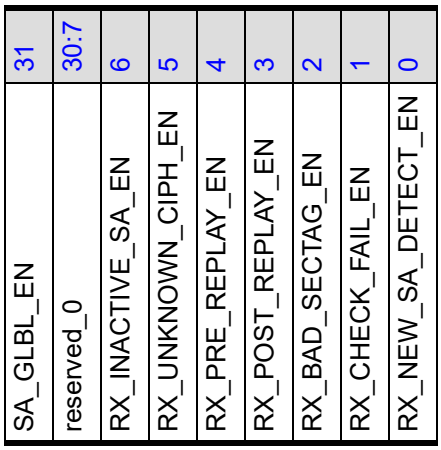
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-860 Fields for Register: IRQ_TYPE_STAT_RX_SA_312 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.854 IRQ_TYPE_EN_RX_SA_313

- **Name:** IRQ_TYPE_EN_RX_SA_313
- **Description:** Interrupt Enable for each IRQ type of Rx SA 313
- **Size:** 32 bits
- **Offset:** 0xd58
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 313



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-861 Fields for Register: IRQ_TYPE_EN_RX_SA_313

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-861 Fields for Register: IRQ_TYPE_EN_RX_SA_313 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-861 Fields for Register: IRQ_TYPE_EN_RX_SA_313 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.855 IRQ_TYPE_STAT_RX_SA_313

- **Name:** IRQ_TYPE_STAT_RX_SA_313
- **Description:** Interrupt Status for each IRQ type of Rx SA 313
- **Size:** 32 bits
- **Offset:** 0xd5c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 313

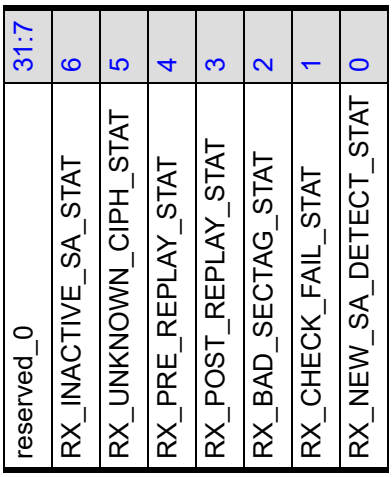


Table 6-862 Fields for Register: IRQ_TYPE_STAT_RX_SA_313

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-862 Fields for Register: IRQ_TYPE_STAT_RX_SA_313 (Continued)

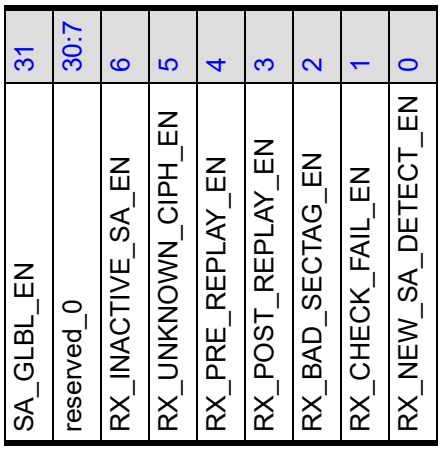
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-862 Fields for Register: IRQ_TYPE_STAT_RX_SA_313 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.856 IRQ_TYPE_EN_RX_SA_314

- **Name:** IRQ_TYPE_EN_RX_SA_314
- **Description:** Interrupt Enable for each IRQ type of Rx SA 314
- **Size:** 32 bits
- **Offset:** 0xd60
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 314



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-863 Fields for Register: IRQ_TYPE_EN_RX_SA_314

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-863 Fields for Register: IRQ_TYPE_EN_RX_SA_314 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-863 Fields for Register: IRQ_TYPE_EN_RX_SA_314 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.857 IRQ_TYPE_STAT_RX_SA_314

- **Name:** IRQ_TYPE_STAT_RX_SA_314
- **Description:** Interrupt Status for each IRQ type of Rx SA 314
- **Size:** 32 bits
- **Offset:** 0xd64
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 314

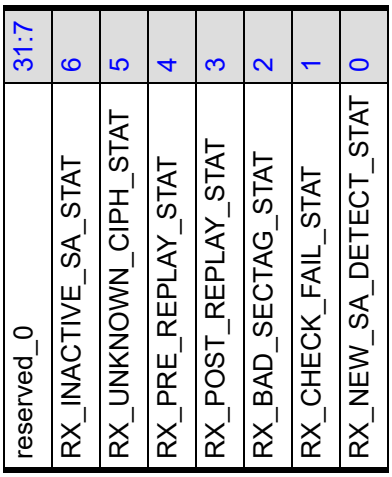


Table 6-864 Fields for Register: IRQ_TYPE_STAT_RX_SA_314

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-864 Fields for Register: IRQ_TYPE_STAT_RX_SA_314 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-864 Fields for Register: IRQ_TYPE_STAT_RX_SA_314 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.858 IRQ_TYPE_EN_RX_SA_315

- **Name:** IRQ_TYPE_EN_RX_SA_315
- **Description:** Interrupt Enable for each IRQ type of Rx SA 315
- **Size:** 32 bits
- **Offset:** 0xd68
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 315

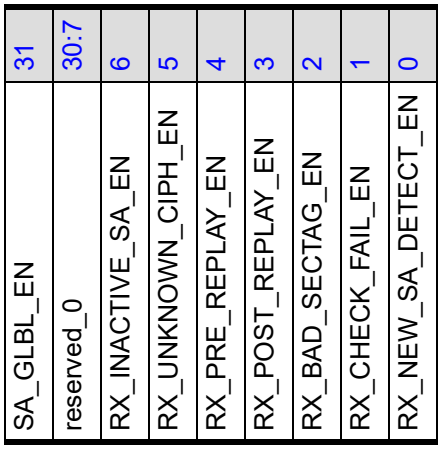


Table 6-865 Fields for Register: IRQ_TYPE_EN_RX_SA_315

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-865 Fields for Register: IRQ_TYPE_EN_RX_SA_315 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-865 Fields for Register: IRQ_TYPE_EN_RX_SA_315 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.859 IRQ_TYPE_STAT_RX_SA_315

- **Name:** IRQ_TYPE_STAT_RX_SA_315
- **Description:** Interrupt Status for each IRQ type of Rx SA 315
- **Size:** 32 bits
- **Offset:** 0xd6c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 315

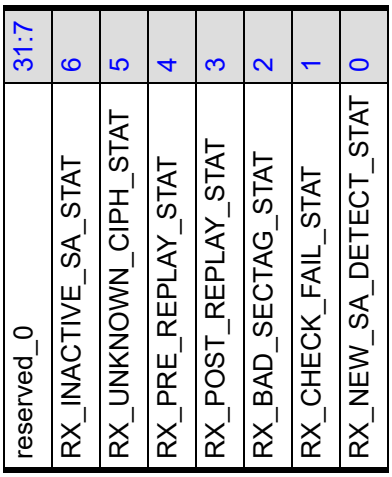


Table 6-866 Fields for Register: IRQ_TYPE_STAT_RX_SA_315

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-866 Fields for Register: IRQ_TYPE_STAT_RX_SA_315 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-866 Fields for Register: IRQ_TYPE_STAT_RX_SA_315 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.860 IRQ_TYPE_EN_RX_SA_316

- **Name:** IRQ_TYPE_EN_RX_SA_316
- **Description:** Interrupt Enable for each IRQ type of Rx SA 316
- **Size:** 32 bits
- **Offset:** 0xd70
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 316

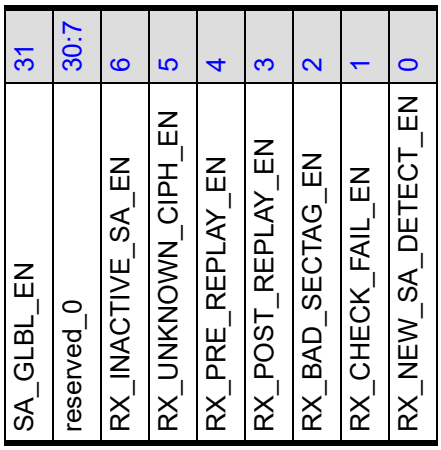


Table 6-867 Fields for Register: IRQ_TYPE_EN_RX_SA_316

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-867 Fields for Register: IRQ_TYPE_EN_RX_SA_316 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-867 Fields for Register: IRQ_TYPE_EN_RX_SA_316 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.861 IRQ_TYPE_STAT_RX_SA_316

- **Name:** IRQ_TYPE_STAT_RX_SA_316
- **Description:** Interrupt Status for each IRQ type of Rx SA 316
- **Size:** 32 bits
- **Offset:** 0xd74
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 316

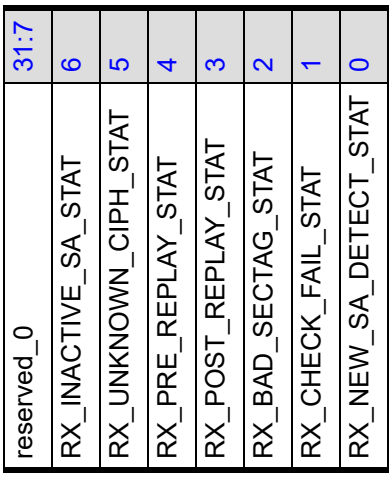


Table 6-868 Fields for Register: IRQ_TYPE_STAT_RX_SA_316

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-868 Fields for Register: IRQ_TYPE_STAT_RX_SA_316 (Continued)

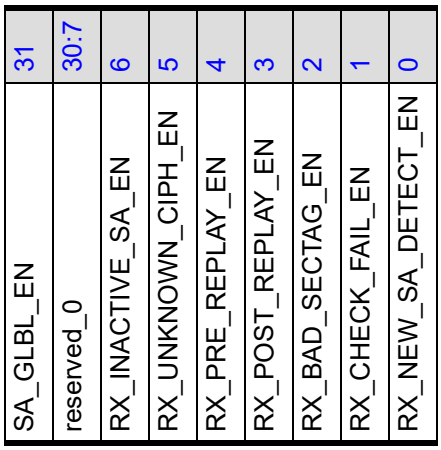
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-868 Fields for Register: IRQ_TYPE_STAT_RX_SA_316 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.862 IRQ_TYPE_EN_RX_SA_317

- **Name:** IRQ_TYPE_EN_RX_SA_317
- **Description:** Interrupt Enable for each IRQ type of Rx SA 317
- **Size:** 32 bits
- **Offset:** 0xd78
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 317



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-869 Fields for Register: IRQ_TYPE_EN_RX_SA_317

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-869 Fields for Register: IRQ_TYPE_EN_RX_SA_317 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-869 Fields for Register: IRQ_TYPE_EN_RX_SA_317 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.863 IRQ_TYPE_STAT_RX_SA_317

- **Name:** IRQ_TYPE_STAT_RX_SA_317
- **Description:** Interrupt Status for each IRQ type of Rx SA 317
- **Size:** 32 bits
- **Offset:** 0xd7c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 317

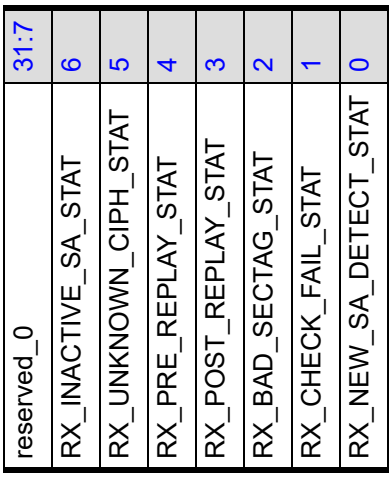


Table 6-870 Fields for Register: IRQ_TYPE_STAT_RX_SA_317

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-870 Fields for Register: IRQ_TYPE_STAT_RX_SA_317 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-870 Fields for Register: IRQ_TYPE_STAT_RX_SA_317 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.864 IRQ_TYPE_EN_RX_SA_318

- **Name:** IRQ_TYPE_EN_RX_SA_318
- **Description:** Interrupt Enable for each IRQ type of Rx SA 318
- **Size:** 32 bits
- **Offset:** 0xd80
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 318

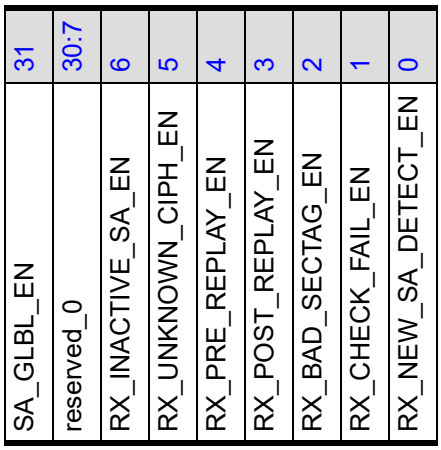


Table 6-871 Fields for Register: IRQ_TYPE_EN_RX_SA_318

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-871 Fields for Register: IRQ_TYPE_EN_RX_SA_318 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-871 Fields for Register: IRQ_TYPE_EN_RX_SA_318 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.865 IRQ_TYPE_STAT_RX_SA_318

- **Name:** IRQ_TYPE_STAT_RX_SA_318
- **Description:** Interrupt Status for each IRQ type of Rx SA 318
- **Size:** 32 bits
- **Offset:** 0xd84
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 318

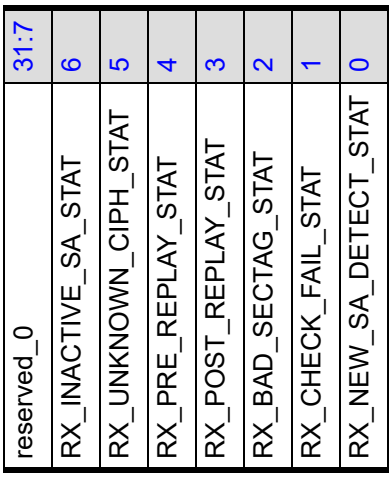


Table 6-872 Fields for Register: IRQ_TYPE_STAT_RX_SA_318

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-872 Fields for Register: IRQ_TYPE_STAT_RX_SA_318 (Continued)

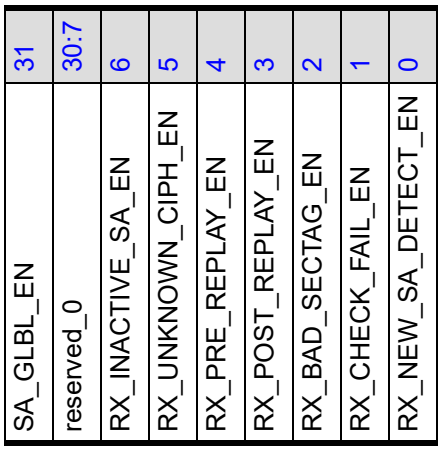
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-872 Fields for Register: IRQ_TYPE_STAT_RX_SA_318 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.866 IRQ_TYPE_EN_RX_SA_319

- **Name:** IRQ_TYPE_EN_RX_SA_319
- **Description:** Interrupt Enable for each IRQ type of Rx SA 319
- **Size:** 32 bits
- **Offset:** 0xd88
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 319



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-873 Fields for Register: IRQ_TYPE_EN_RX_SA_319

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-873 Fields for Register: IRQ_TYPE_EN_RX_SA_319 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-873 Fields for Register: IRQ_TYPE_EN_RX_SA_319 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.867 IRQ_TYPE_STAT_RX_SA_319

- **Name:** IRQ_TYPE_STAT_RX_SA_319
- **Description:** Interrupt Status for each IRQ type of Rx SA 319
- **Size:** 32 bits
- **Offset:** 0xd8c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 319

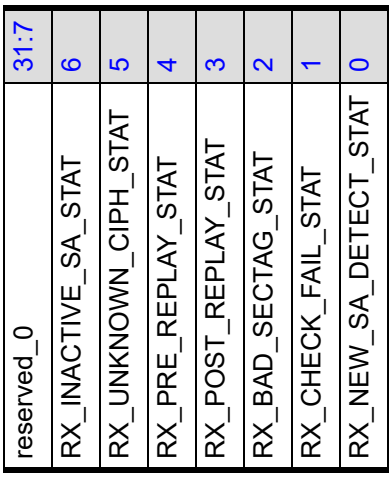


Table 6-874 Fields for Register: IRQ_TYPE_STAT_RX_SA_319

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-874 Fields for Register: IRQ_TYPE_STAT_RX_SA_319 (Continued)

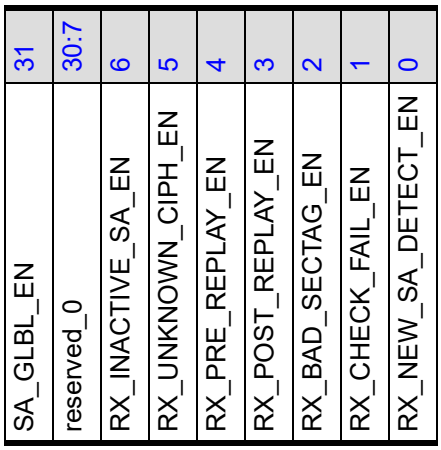
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-874 Fields for Register: IRQ_TYPE_STAT_RX_SA_319 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.868 IRQ_TYPE_EN_RX_SA_320

- **Name:** IRQ_TYPE_EN_RX_SA_320
- **Description:** Interrupt Enable for each IRQ type of Rx SA 320
- **Size:** 32 bits
- **Offset:** 0xd90
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 320



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-875 Fields for Register: IRQ_TYPE_EN_RX_SA_320

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-875 Fields for Register: IRQ_TYPE_EN_RX_SA_320 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-875 Fields for Register: IRQ_TYPE_EN_RX_SA_320 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.869 IRQ_TYPE_STAT_RX_SA_320

- **Name:** IRQ_TYPE_STAT_RX_SA_320
- **Description:** Interrupt Status for each IRQ type of Rx SA 320
- **Size:** 32 bits
- **Offset:** 0xd94
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 320

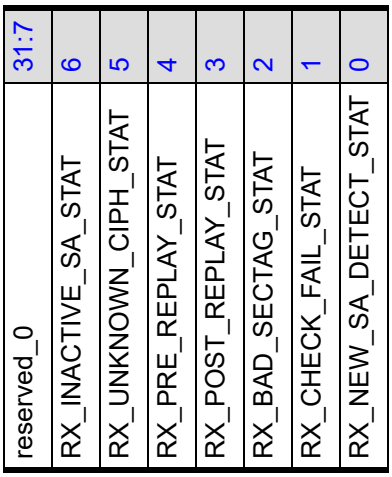


Table 6-876 Fields for Register: IRQ_TYPE_STAT_RX_SA_320

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-876 Fields for Register: IRQ_TYPE_STAT_RX_SA_320 (Continued)

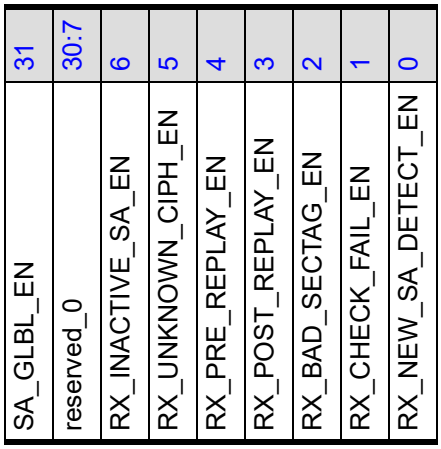
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-876 Fields for Register: IRQ_TYPE_STAT_RX_SA_320 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.870 IRQ_TYPE_EN_RX_SA_321

- **Name:** IRQ_TYPE_EN_RX_SA_321
- **Description:** Interrupt Enable for each IRQ type of Rx SA 321
- **Size:** 32 bits
- **Offset:** 0xd98
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 321



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-877 Fields for Register: IRQ_TYPE_EN_RX_SA_321

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-877 Fields for Register: IRQ_TYPE_EN_RX_SA_321 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-877 Fields for Register: IRQ_TYPE_EN_RX_SA_321 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.871 IRQ_TYPE_STAT_RX_SA_321

- **Name:** IRQ_TYPE_STAT_RX_SA_321
- **Description:** Interrupt Status for each IRQ type of Rx SA 321
- **Size:** 32 bits
- **Offset:** 0xd9c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 321

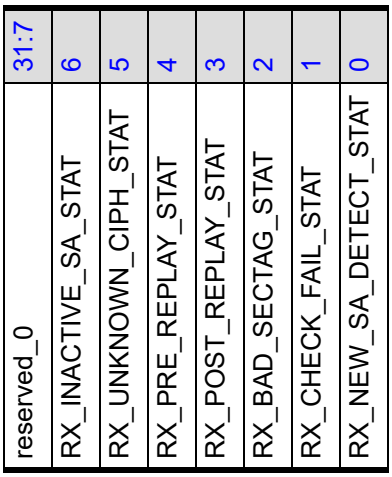


Table 6-878 Fields for Register: IRQ_TYPE_STAT_RX_SA_321

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-878 Fields for Register: IRQ_TYPE_STAT_RX_SA_321 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-878 Fields for Register: IRQ_TYPE_STAT_RX_SA_321 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.872 IRQ_TYPE_EN_RX_SA_322

- **Name:** IRQ_TYPE_EN_RX_SA_322
- **Description:** Interrupt Enable for each IRQ type of Rx SA 322
- **Size:** 32 bits
- **Offset:** 0xda0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 322

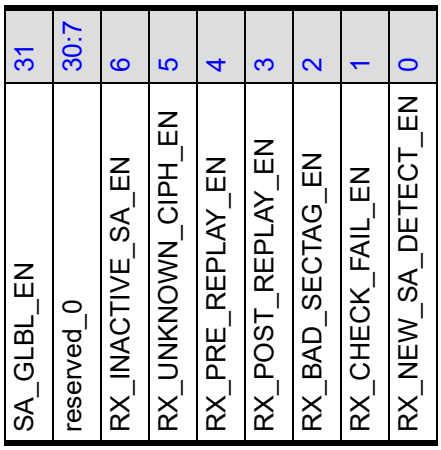


Table 6-879 Fields for Register: IRQ_TYPE_EN_RX_SA_322

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-879 Fields for Register: IRQ_TYPE_EN_RX_SA_322 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-879 Fields for Register: IRQ_TYPE_EN_RX_SA_322 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.873 IRQ_TYPE_STAT_RX_SA_322

- **Name:** IRQ_TYPE_STAT_RX_SA_322
- **Description:** Interrupt Status for each IRQ type of Rx SA 322
- **Size:** 32 bits
- **Offset:** 0xda4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 322

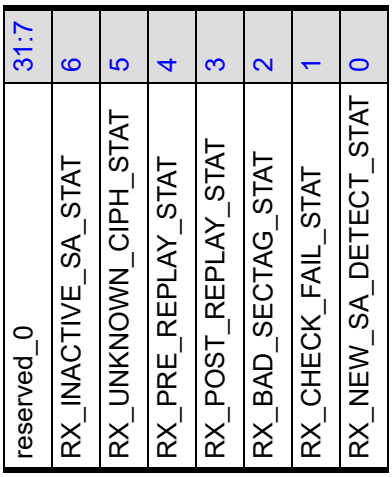


Table 6-880 Fields for Register: IRQ_TYPE_STAT_RX_SA_322

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-880 Fields for Register: IRQ_TYPE_STAT_RX_SA_322 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-880 Fields for Register: IRQ_TYPE_STAT_RX_SA_322 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.874 IRQ_TYPE_EN_RX_SA_323

- **Name:** IRQ_TYPE_EN_RX_SA_323
- **Description:** Interrupt Enable for each IRQ type of Rx SA 323
- **Size:** 32 bits
- **Offset:** 0xda8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 323

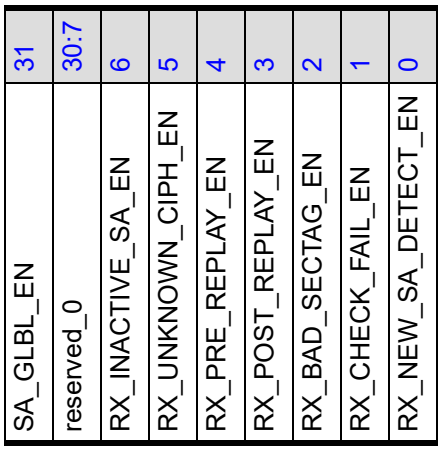


Table 6-881 Fields for Register: IRQ_TYPE_EN_RX_SA_323

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-881 Fields for Register: IRQ_TYPE_EN_RX_SA_323 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-881 Fields for Register: IRQ_TYPE_EN_RX_SA_323 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.875 IRQ_TYPE_STAT_RX_SA_323

- **Name:** IRQ_TYPE_STAT_RX_SA_323
- **Description:** Interrupt Status for each IRQ type of Rx SA 323
- **Size:** 32 bits
- **Offset:** 0xdac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 323

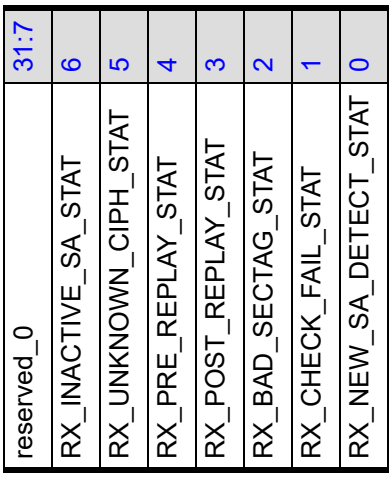


Table 6-882 Fields for Register: IRQ_TYPE_STAT_RX_SA_323

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-882 Fields for Register: IRQ_TYPE_STAT_RX_SA_323 (Continued)

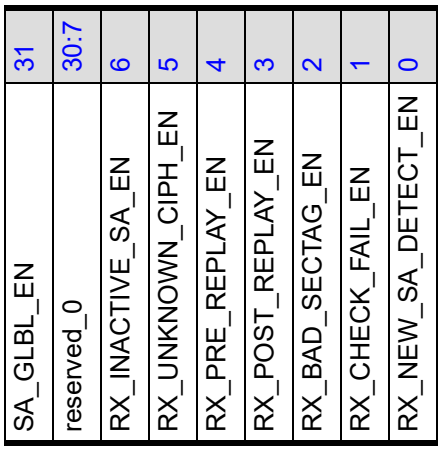
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-882 Fields for Register: IRQ_TYPE_STAT_RX_SA_323 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.876 IRQ_TYPE_EN_RX_SA_324

- **Name:** IRQ_TYPE_EN_RX_SA_324
- **Description:** Interrupt Enable for each IRQ type of Rx SA 324
- **Size:** 32 bits
- **Offset:** 0xdb0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 324



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-883 Fields for Register: IRQ_TYPE_EN_RX_SA_324

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-883 Fields for Register: IRQ_TYPE_EN_RX_SA_324 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-883 Fields for Register: IRQ_TYPE_EN_RX_SA_324 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.877 IRQ_TYPE_STAT_RX_SA_324

- **Name:** IRQ_TYPE_STAT_RX_SA_324
- **Description:** Interrupt Status for each IRQ type of Rx SA 324
- **Size:** 32 bits
- **Offset:** 0xdb4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 324

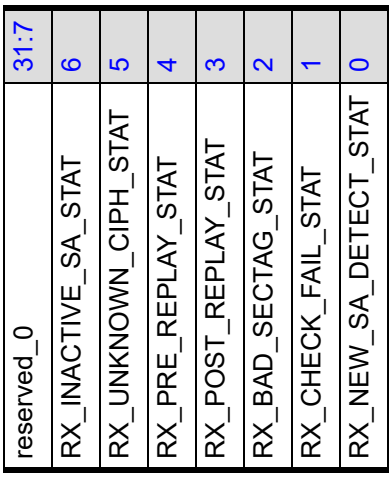


Table 6-884 Fields for Register: IRQ_TYPE_STAT_RX_SA_324

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-884 Fields for Register: IRQ_TYPE_STAT_RX_SA_324 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-884 Fields for Register: IRQ_TYPE_STAT_RX_SA_324 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.878 IRQ_TYPE_EN_RX_SA_325

- **Name:** IRQ_TYPE_EN_RX_SA_325
- **Description:** Interrupt Enable for each IRQ type of Rx SA 325
- **Size:** 32 bits
- **Offset:** 0xdb8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 325

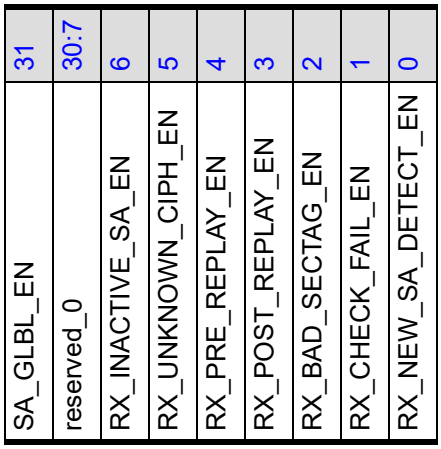


Table 6-885 Fields for Register: IRQ_TYPE_EN_RX_SA_325

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-885 Fields for Register: IRQ_TYPE_EN_RX_SA_325 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-885 Fields for Register: IRQ_TYPE_EN_RX_SA_325 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.879 IRQ_TYPE_STAT_RX_SA_325

- **Name:** IRQ_TYPE_STAT_RX_SA_325
- **Description:** Interrupt Status for each IRQ type of Rx SA 325
- **Size:** 32 bits
- **Offset:** 0xdbc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 325

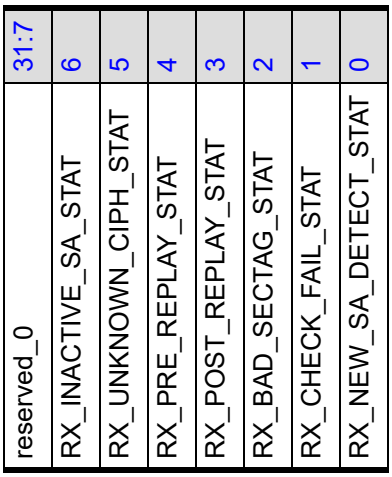


Table 6-886 Fields for Register: IRQ_TYPE_STAT_RX_SA_325

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-886 Fields for Register: IRQ_TYPE_STAT_RX_SA_325 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-886 Fields for Register: IRQ_TYPE_STAT_RX_SA_325 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.880 IRQ_TYPE_EN_RX_SA_326

- **Name:** IRQ_TYPE_EN_RX_SA_326
- **Description:** Interrupt Enable for each IRQ type of Rx SA 326
- **Size:** 32 bits
- **Offset:** 0xdc0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 326

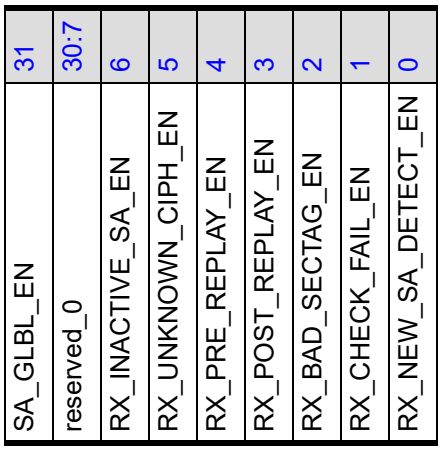


Table 6-887 Fields for Register: IRQ_TYPE_EN_RX_SA_326

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-887 Fields for Register: IRQ_TYPE_EN_RX_SA_326 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-887 Fields for Register: IRQ_TYPE_EN_RX_SA_326 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.881 IRQ_TYPE_STAT_RX_SA_326

- **Name:** IRQ_TYPE_STAT_RX_SA_326
- **Description:** Interrupt Status for each IRQ type of Rx SA 326
- **Size:** 32 bits
- **Offset:** 0xdc4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 326

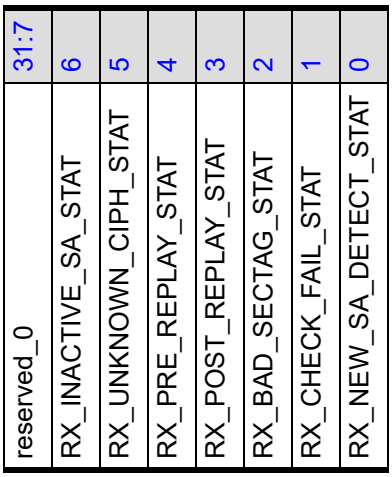


Table 6-888 Fields for Register: IRQ_TYPE_STAT_RX_SA_326

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-888 Fields for Register: IRQ_TYPE_STAT_RX_SA_326 (Continued)

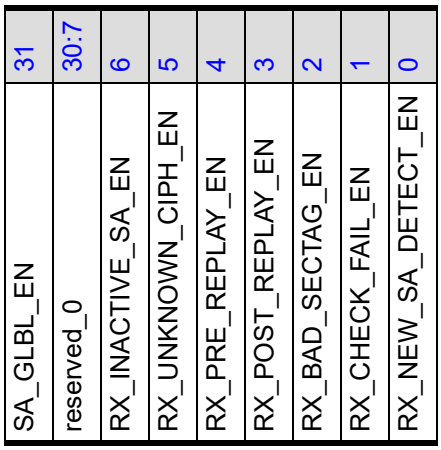
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-888 Fields for Register: IRQ_TYPE_STAT_RX_SA_326 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.882 IRQ_TYPE_EN_RX_SA_327

- **Name:** IRQ_TYPE_EN_RX_SA_327
- **Description:** Interrupt Enable for each IRQ type of Rx SA 327
- **Size:** 32 bits
- **Offset:** 0xdc8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 327



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-889 Fields for Register: IRQ_TYPE_EN_RX_SA_327

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-889 Fields for Register: IRQ_TYPE_EN_RX_SA_327 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-889 Fields for Register: IRQ_TYPE_EN_RX_SA_327 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.883 IRQ_TYPE_STAT_RX_SA_327

- **Name:** IRQ_TYPE_STAT_RX_SA_327
- **Description:** Interrupt Status for each IRQ type of Rx SA 327
- **Size:** 32 bits
- **Offset:** 0xdcc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 327

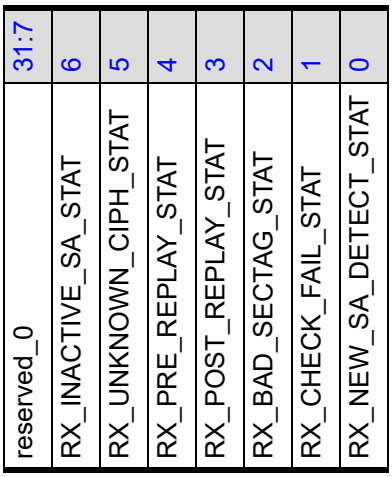


Table 6-890 Fields for Register: IRQ_TYPE_STAT_RX_SA_327

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-890 Fields for Register: IRQ_TYPE_STAT_RX_SA_327 (Continued)

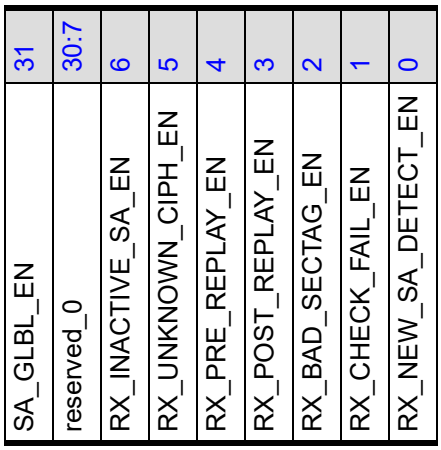
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-890 Fields for Register: IRQ_TYPE_STAT_RX_SA_327 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.884 IRQ_TYPE_EN_RX_SA_328

- **Name:** IRQ_TYPE_EN_RX_SA_328
- **Description:** Interrupt Enable for each IRQ type of Rx SA 328
- **Size:** 32 bits
- **Offset:** 0xdd0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 328



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-891 Fields for Register: IRQ_TYPE_EN_RX_SA_328

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-891 Fields for Register: IRQ_TYPE_EN_RX_SA_328 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-891 Fields for Register: IRQ_TYPE_EN_RX_SA_328 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.885 IRQ_TYPE_STAT_RX_SA_328

- **Name:** IRQ_TYPE_STAT_RX_SA_328
- **Description:** Interrupt Status for each IRQ type of Rx SA 328
- **Size:** 32 bits
- **Offset:** 0xdd4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 328

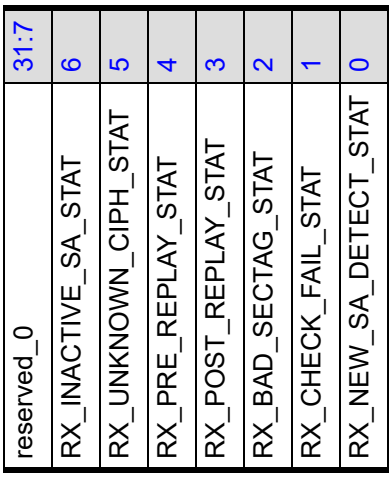


Table 6-892 Fields for Register: IRQ_TYPE_STAT_RX_SA_328

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-892 Fields for Register: IRQ_TYPE_STAT_RX_SA_328 (Continued)

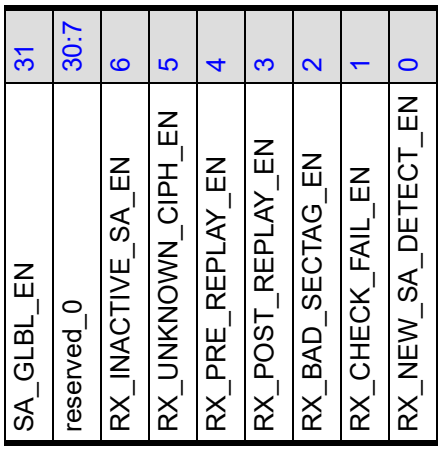
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-892 Fields for Register: IRQ_TYPE_STAT_RX_SA_328 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.886 IRQ_TYPE_EN_RX_SA_329

- **Name:** IRQ_TYPE_EN_RX_SA_329
- **Description:** Interrupt Enable for each IRQ type of Rx SA 329
- **Size:** 32 bits
- **Offset:** 0xdd8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 329



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-893 Fields for Register: IRQ_TYPE_EN_RX_SA_329

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-893 Fields for Register: IRQ_TYPE_EN_RX_SA_329 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-893 Fields for Register: IRQ_TYPE_EN_RX_SA_329 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.887 IRQ_TYPE_STAT_RX_SA_329

- **Name:** IRQ_TYPE_STAT_RX_SA_329
- **Description:** Interrupt Status for each IRQ type of Rx SA 329
- **Size:** 32 bits
- **Offset:** 0xddc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 329

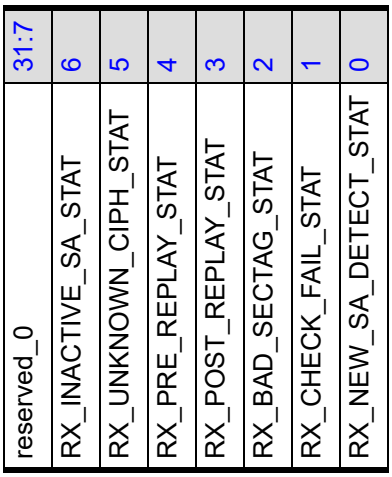


Table 6-894 Fields for Register: IRQ_TYPE_STAT_RX_SA_329

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-894 Fields for Register: IRQ_TYPE_STAT_RX_SA_329 (Continued)

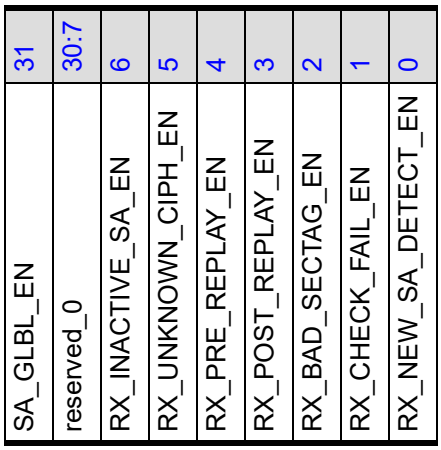
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-894 Fields for Register: IRQ_TYPE_STAT_RX_SA_329 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.888 IRQ_TYPE_EN_RX_SA_330

- **Name:** IRQ_TYPE_EN_RX_SA_330
- **Description:** Interrupt Enable for each IRQ type of Rx SA 330
- **Size:** 32 bits
- **Offset:** 0xde0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 330



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-895 Fields for Register: IRQ_TYPE_EN_RX_SA_330

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-895 Fields for Register: IRQ_TYPE_EN_RX_SA_330 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-895 Fields for Register: IRQ_TYPE_EN_RX_SA_330 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.889 IRQ_TYPE_STAT_RX_SA_330

- **Name:** IRQ_TYPE_STAT_RX_SA_330
- **Description:** Interrupt Status for each IRQ type of Rx SA 330
- **Size:** 32 bits
- **Offset:** 0xde4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 330

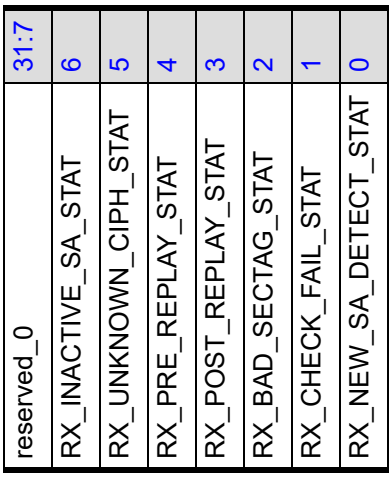


Table 6-896 Fields for Register: IRQ_TYPE_STAT_RX_SA_330

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-896 Fields for Register: IRQ_TYPE_STAT_RX_SA_330 (Continued)

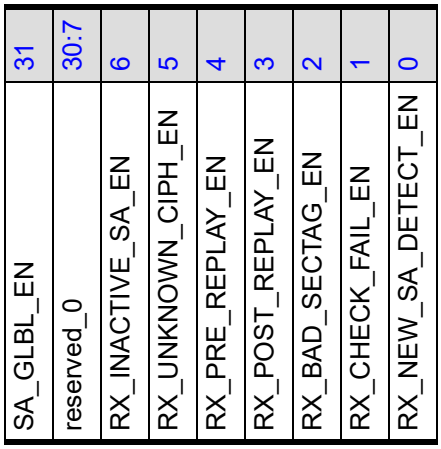
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-896 Fields for Register: IRQ_TYPE_STAT_RX_SA_330 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.890 IRQ_TYPE_EN_RX_SA_331

- **Name:** IRQ_TYPE_EN_RX_SA_331
- **Description:** Interrupt Enable for each IRQ type of Rx SA 331
- **Size:** 32 bits
- **Offset:** 0xde8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 331



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-897 Fields for Register: IRQ_TYPE_EN_RX_SA_331

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-897 Fields for Register: IRQ_TYPE_EN_RX_SA_331 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-897 Fields for Register: IRQ_TYPE_EN_RX_SA_331 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.891 IRQ_TYPE_STAT_RX_SA_331

- **Name:** IRQ_TYPE_STAT_RX_SA_331
- **Description:** Interrupt Status for each IRQ type of Rx SA 331
- **Size:** 32 bits
- **Offset:** 0xdec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 331

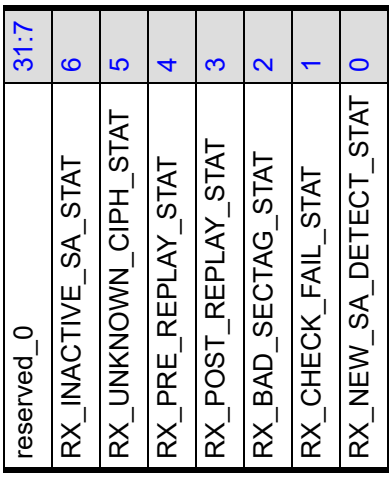


Table 6-898 Fields for Register: IRQ_TYPE_STAT_RX_SA_331

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-898 Fields for Register: IRQ_TYPE_STAT_RX_SA_331 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-898 Fields for Register: IRQ_TYPE_STAT_RX_SA_331 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.892 IRQ_TYPE_EN_RX_SA_332

- **Name:** IRQ_TYPE_EN_RX_SA_332
- **Description:** Interrupt Enable for each IRQ type of Rx SA 332
- **Size:** 32 bits
- **Offset:** 0xdf0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 332

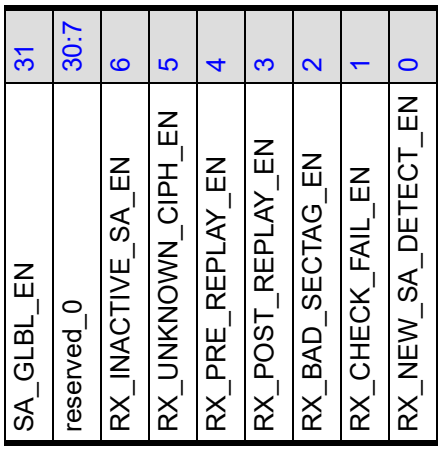


Table 6-899 Fields for Register: IRQ_TYPE_EN_RX_SA_332

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-899 Fields for Register: IRQ_TYPE_EN_RX_SA_332 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-899 Fields for Register: IRQ_TYPE_EN_RX_SA_332 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.893 IRQ_TYPE_STAT_RX_SA_332

- **Name:** IRQ_TYPE_STAT_RX_SA_332
- **Description:** Interrupt Status for each IRQ type of Rx SA 332
- **Size:** 32 bits
- **Offset:** 0xdf4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 332

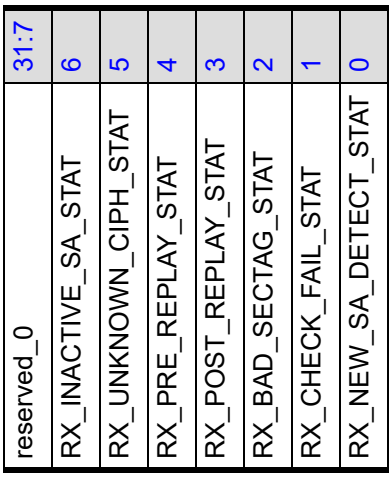


Table 6-900 Fields for Register: IRQ_TYPE_STAT_RX_SA_332

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-900 Fields for Register: IRQ_TYPE_STAT_RX_SA_332 (Continued)

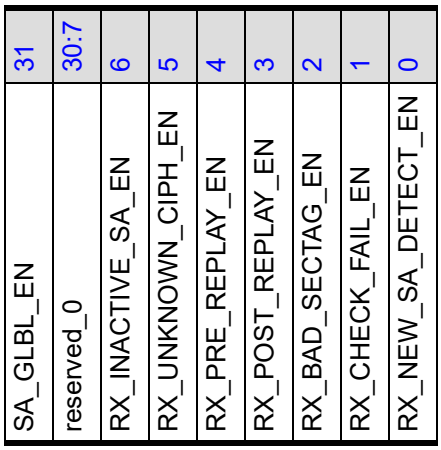
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-900 Fields for Register: IRQ_TYPE_STAT_RX_SA_332 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.894 IRQ_TYPE_EN_RX_SA_333

- **Name:** IRQ_TYPE_EN_RX_SA_333
- **Description:** Interrupt Enable for each IRQ type of Rx SA 333
- **Size:** 32 bits
- **Offset:** 0xdf8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 333



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-901 Fields for Register: IRQ_TYPE_EN_RX_SA_333

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-901 Fields for Register: IRQ_TYPE_EN_RX_SA_333 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-901 Fields for Register: IRQ_TYPE_EN_RX_SA_333 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.895 IRQ_TYPE_STAT_RX_SA_333

- **Name:** IRQ_TYPE_STAT_RX_SA_333
- **Description:** Interrupt Status for each IRQ type of Rx SA 333
- **Size:** 32 bits
- **Offset:** 0xdfc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 333

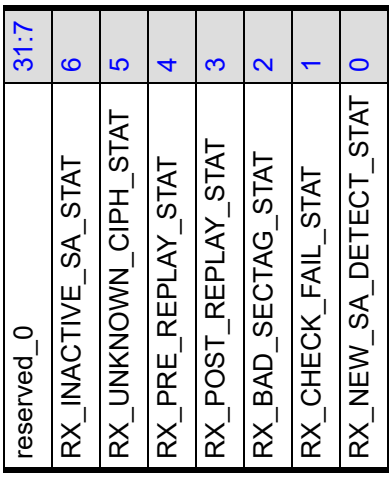


Table 6-902 Fields for Register: IRQ_TYPE_STAT_RX_SA_333

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-902 Fields for Register: IRQ_TYPE_STAT_RX_SA_333 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-902 Fields for Register: IRQ_TYPE_STAT_RX_SA_333 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.896 IRQ_TYPE_EN_RX_SA_334

- **Name:** IRQ_TYPE_EN_RX_SA_334
- **Description:** Interrupt Enable for each IRQ type of Rx SA 334
- **Size:** 32 bits
- **Offset:** 0xe00
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 334

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-903 Fields for Register: IRQ_TYPE_EN_RX_SA_334

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-903 Fields for Register: IRQ_TYPE_EN_RX_SA_334 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-903 Fields for Register: IRQ_TYPE_EN_RX_SA_334 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.897 IRQ_TYPE_STAT_RX_SA_334

- **Name:** IRQ_TYPE_STAT_RX_SA_334
- **Description:** Interrupt Status for each IRQ type of Rx SA 334
- **Size:** 32 bits
- **Offset:** 0xe04
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 334

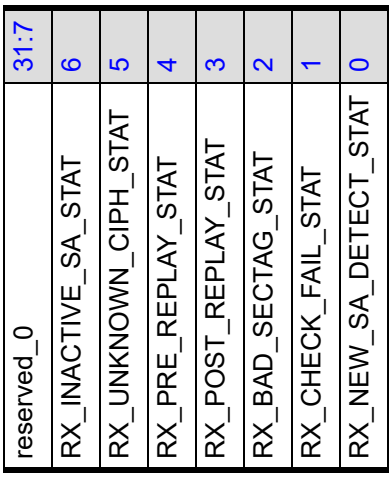


Table 6-904 Fields for Register: IRQ_TYPE_STAT_RX_SA_334

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-904 Fields for Register: IRQ_TYPE_STAT_RX_SA_334 (Continued)

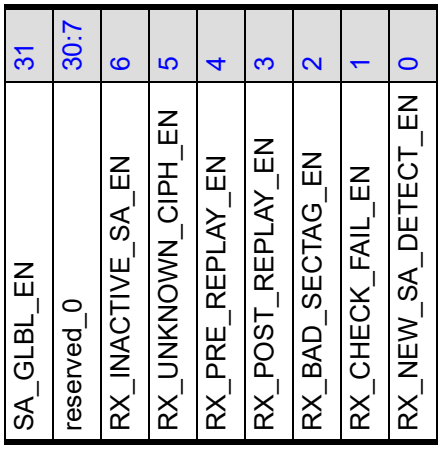
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-904 Fields for Register: IRQ_TYPE_STAT_RX_SA_334 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.898 IRQ_TYPE_EN_RX_SA_335

- **Name:** IRQ_TYPE_EN_RX_SA_335
- **Description:** Interrupt Enable for each IRQ type of Rx SA 335
- **Size:** 32 bits
- **Offset:** 0xe08
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 335



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-905 Fields for Register: IRQ_TYPE_EN_RX_SA_335

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-905 Fields for Register: IRQ_TYPE_EN_RX_SA_335 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-905 Fields for Register: IRQ_TYPE_EN_RX_SA_335 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.899 IRQ_TYPE_STAT_RX_SA_335

- **Name:** IRQ_TYPE_STAT_RX_SA_335
- **Description:** Interrupt Status for each IRQ type of Rx SA 335
- **Size:** 32 bits
- **Offset:** 0xe0c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 335

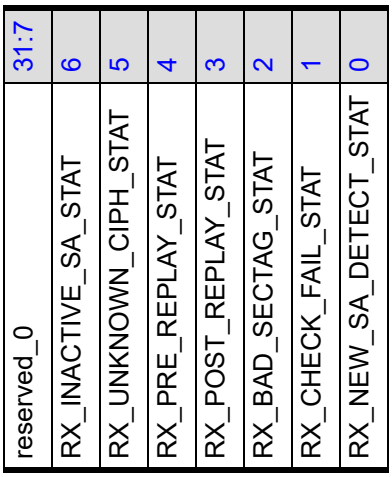


Table 6-906 Fields for Register: IRQ_TYPE_STAT_RX_SA_335

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-906 Fields for Register: IRQ_TYPE_STAT_RX_SA_335 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-906 Fields for Register: IRQ_TYPE_STAT_RX_SA_335 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.900 IRQ_TYPE_EN_RX_SA_336

- **Name:** IRQ_TYPE_EN_RX_SA_336
- **Description:** Interrupt Enable for each IRQ type of Rx SA 336
- **Size:** 32 bits
- **Offset:** 0xe10
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 336

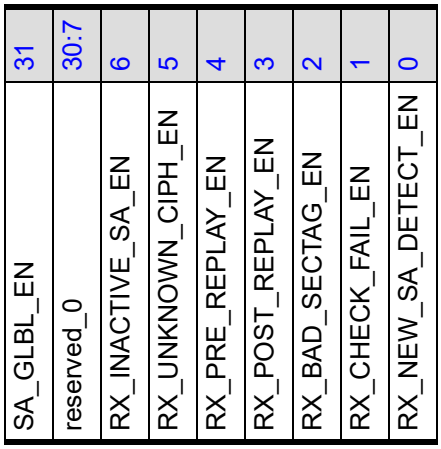


Table 6-907 Fields for Register: IRQ_TYPE_EN_RX_SA_336

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-907 Fields for Register: IRQ_TYPE_EN_RX_SA_336 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-907 Fields for Register: IRQ_TYPE_EN_RX_SA_336 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.901 IRQ_TYPE_STAT_RX_SA_336

- **Name:** IRQ_TYPE_STAT_RX_SA_336
- **Description:** Interrupt Status for each IRQ type of Rx SA 336
- **Size:** 32 bits
- **Offset:** 0xe14
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 336

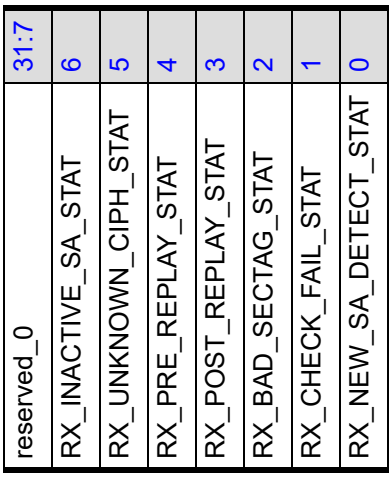


Table 6-908 Fields for Register: IRQ_TYPE_STAT_RX_SA_336

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-908 Fields for Register: IRQ_TYPE_STAT_RX_SA_336 (Continued)

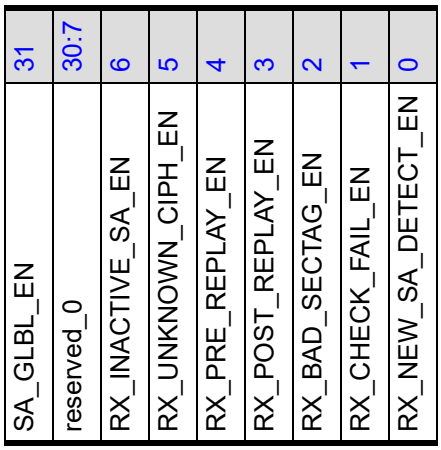
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-908 Fields for Register: IRQ_TYPE_STAT_RX_SA_336 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.902 IRQ_TYPE_EN_RX_SA_337

- **Name:** IRQ_TYPE_EN_RX_SA_337
- **Description:** Interrupt Enable for each IRQ type of Rx SA 337
- **Size:** 32 bits
- **Offset:** 0xe18
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 337



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-909 Fields for Register: IRQ_TYPE_EN_RX_SA_337

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-909 Fields for Register: IRQ_TYPE_EN_RX_SA_337 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-909 Fields for Register: IRQ_TYPE_EN_RX_SA_337 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.903 IRQ_TYPE_STAT_RX_SA_337

- **Name:** IRQ_TYPE_STAT_RX_SA_337
- **Description:** Interrupt Status for each IRQ type of Rx SA 337
- **Size:** 32 bits
- **Offset:** 0xe1c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 337

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-910 Fields for Register: IRQ_TYPE_STAT_RX_SA_337

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-910 Fields for Register: IRQ_TYPE_STAT_RX_SA_337 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-910 Fields for Register: IRQ_TYPE_STAT_RX_SA_337 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.904 IRQ_TYPE_EN_RX_SA_338

- **Name:** IRQ_TYPE_EN_RX_SA_338
- **Description:** Interrupt Enable for each IRQ type of Rx SA 338
- **Size:** 32 bits
- **Offset:** 0xe20
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 338

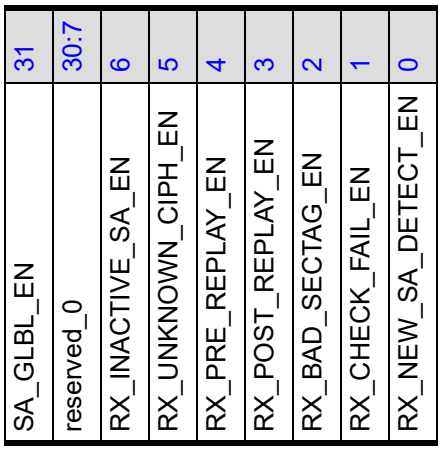


Table 6-911 Fields for Register: IRQ_TYPE_EN_RX_SA_338

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-911 Fields for Register: IRQ_TYPE_EN_RX_SA_338 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-911 Fields for Register: IRQ_TYPE_EN_RX_SA_338 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.905 IRQ_TYPE_STAT_RX_SA_338

- **Name:** IRQ_TYPE_STAT_RX_SA_338
- **Description:** Interrupt Status for each IRQ type of Rx SA 338
- **Size:** 32 bits
- **Offset:** 0xe24
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 338

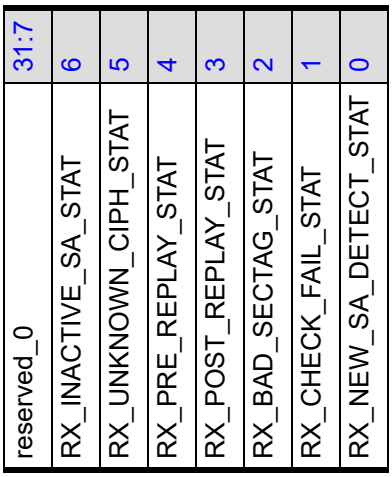


Table 6-912 Fields for Register: IRQ_TYPE_STAT_RX_SA_338

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-912 Fields for Register: IRQ_TYPE_STAT_RX_SA_338 (Continued)

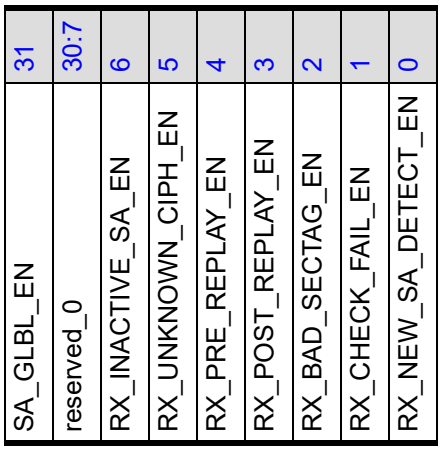
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-912 Fields for Register: IRQ_TYPE_STAT_RX_SA_338 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.906 IRQ_TYPE_EN_RX_SA_339

- **Name:** IRQ_TYPE_EN_RX_SA_339
- **Description:** Interrupt Enable for each IRQ type of Rx SA 339
- **Size:** 32 bits
- **Offset:** 0xe28
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 339



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-913 Fields for Register: IRQ_TYPE_EN_RX_SA_339

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-913 Fields for Register: IRQ_TYPE_EN_RX_SA_339 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-913 Fields for Register: IRQ_TYPE_EN_RX_SA_339 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.907 IRQ_TYPE_STAT_RX_SA_339

- **Name:** IRQ_TYPE_STAT_RX_SA_339
- **Description:** Interrupt Status for each IRQ type of Rx SA 339
- **Size:** 32 bits
- **Offset:** 0xe2c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 339

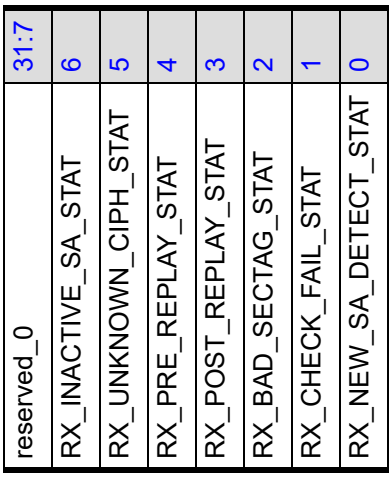


Table 6-914 Fields for Register: IRQ_TYPE_STAT_RX_SA_339

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-914 Fields for Register: IRQ_TYPE_STAT_RX_SA_339 (Continued)

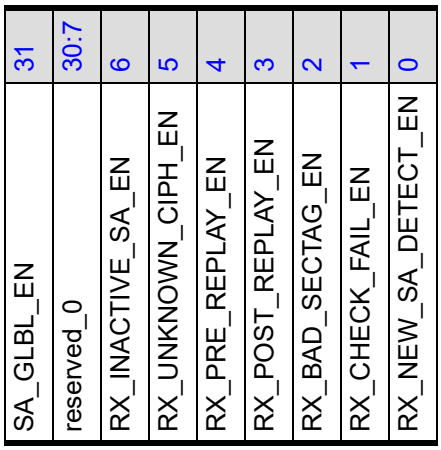
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-914 Fields for Register: IRQ_TYPE_STAT_RX_SA_339 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.908 IRQ_TYPE_EN_RX_SA_340

- **Name:** IRQ_TYPE_EN_RX_SA_340
- **Description:** Interrupt Enable for each IRQ type of Rx SA 340
- **Size:** 32 bits
- **Offset:** 0xe30
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 340



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-915 Fields for Register: IRQ_TYPE_EN_RX_SA_340

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-915 Fields for Register: IRQ_TYPE_EN_RX_SA_340 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-915 Fields for Register: IRQ_TYPE_EN_RX_SA_340 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.909 IRQ_TYPE_STAT_RX_SA_340

- **Name:** IRQ_TYPE_STAT_RX_SA_340
- **Description:** Interrupt Status for each IRQ type of Rx SA 340
- **Size:** 32 bits
- **Offset:** 0xe34
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 340

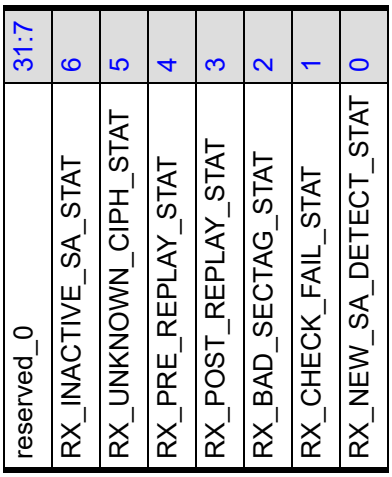


Table 6-916 Fields for Register: IRQ_TYPE_STAT_RX_SA_340

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-916 Fields for Register: IRQ_TYPE_STAT_RX_SA_340 (Continued)

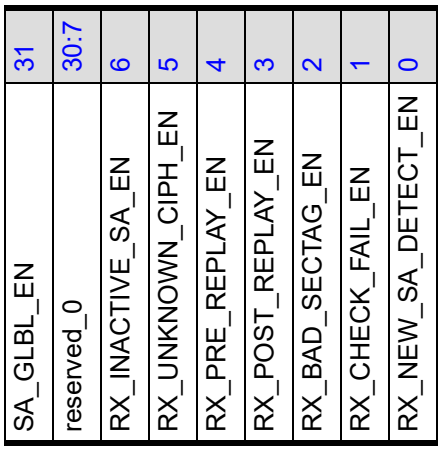
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-916 Fields for Register: IRQ_TYPE_STAT_RX_SA_340 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.910 IRQ_TYPE_EN_RX_SA_341

- **Name:** IRQ_TYPE_EN_RX_SA_341
- **Description:** Interrupt Enable for each IRQ type of Rx SA 341
- **Size:** 32 bits
- **Offset:** 0xe38
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 341



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-917 Fields for Register: IRQ_TYPE_EN_RX_SA_341

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-917 Fields for Register: IRQ_TYPE_EN_RX_SA_341 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-917 Fields for Register: IRQ_TYPE_EN_RX_SA_341 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.911 IRQ_TYPE_STAT_RX_SA_341

- **Name:** IRQ_TYPE_STAT_RX_SA_341
- **Description:** Interrupt Status for each IRQ type of Rx SA 341
- **Size:** 32 bits
- **Offset:** 0xe3c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 341

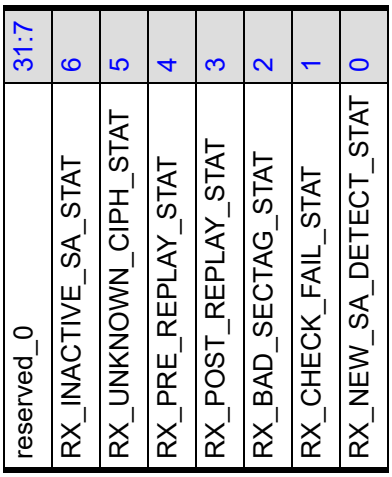


Table 6-918 Fields for Register: IRQ_TYPE_STAT_RX_SA_341

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-918 Fields for Register: IRQ_TYPE_STAT_RX_SA_341 (Continued)

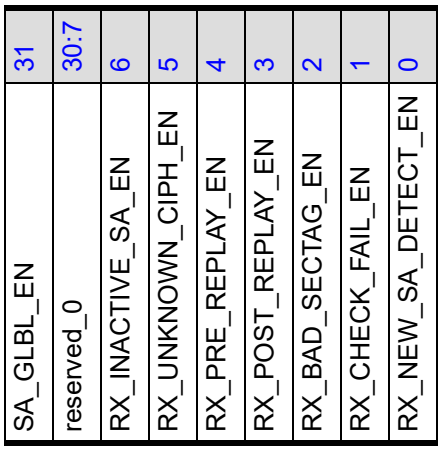
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-918 Fields for Register: IRQ_TYPE_STAT_RX_SA_341 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.912 IRQ_TYPE_EN_RX_SA_342

- **Name:** IRQ_TYPE_EN_RX_SA_342
- **Description:** Interrupt Enable for each IRQ type of Rx SA 342
- **Size:** 32 bits
- **Offset:** 0xe40
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 342



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-919 Fields for Register: IRQ_TYPE_EN_RX_SA_342

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-919 Fields for Register: IRQ_TYPE_EN_RX_SA_342 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-919 Fields for Register: IRQ_TYPE_EN_RX_SA_342 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.913 IRQ_TYPE_STAT_RX_SA_342

- **Name:** IRQ_TYPE_STAT_RX_SA_342
- **Description:** Interrupt Status for each IRQ type of Rx SA 342
- **Size:** 32 bits
- **Offset:** 0xe44
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 342

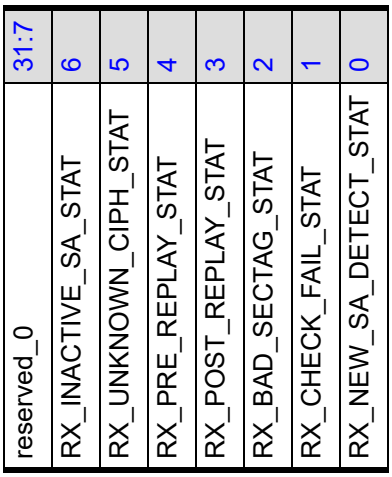


Table 6-920 Fields for Register: IRQ_TYPE_STAT_RX_SA_342

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-920 Fields for Register: IRQ_TYPE_STAT_RX_SA_342 (Continued)

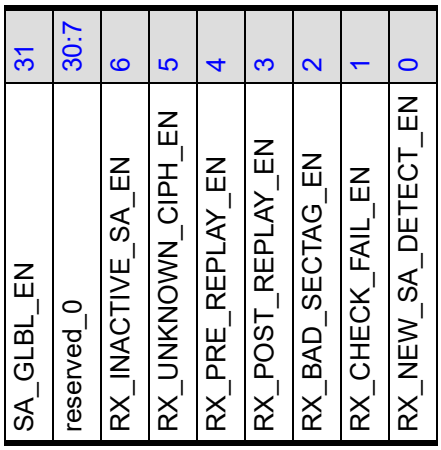
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-920 Fields for Register: IRQ_TYPE_STAT_RX_SA_342 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.914 IRQ_TYPE_EN_RX_SA_343

- **Name:** IRQ_TYPE_EN_RX_SA_343
- **Description:** Interrupt Enable for each IRQ type of Rx SA 343
- **Size:** 32 bits
- **Offset:** 0xe48
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 343



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-921 Fields for Register: IRQ_TYPE_EN_RX_SA_343

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-921 Fields for Register: IRQ_TYPE_EN_RX_SA_343 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-921 Fields for Register: IRQ_TYPE_EN_RX_SA_343 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.915 IRQ_TYPE_STAT_RX_SA_343

- **Name:** IRQ_TYPE_STAT_RX_SA_343
- **Description:** Interrupt Status for each IRQ type of Rx SA 343
- **Size:** 32 bits
- **Offset:** 0xe4c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 343

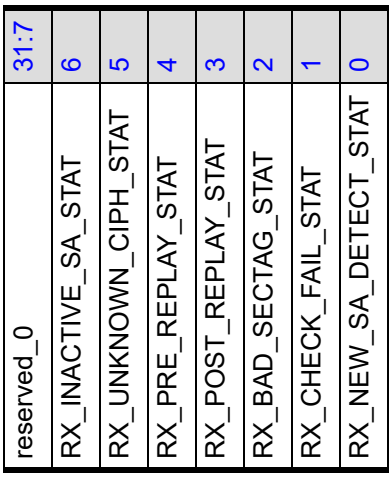


Table 6-922 Fields for Register: IRQ_TYPE_STAT_RX_SA_343

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-922 Fields for Register: IRQ_TYPE_STAT_RX_SA_343 (Continued)

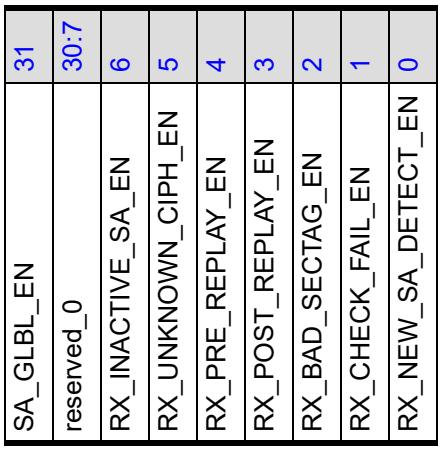
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-922 Fields for Register: IRQ_TYPE_STAT_RX_SA_343 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.916 IRQ_TYPE_EN_RX_SA_344

- **Name:** IRQ_TYPE_EN_RX_SA_344
- **Description:** Interrupt Enable for each IRQ type of Rx SA 344
- **Size:** 32 bits
- **Offset:** 0xe50
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 344



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-923 Fields for Register: IRQ_TYPE_EN_RX_SA_344

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-923 Fields for Register: IRQ_TYPE_EN_RX_SA_344 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-923 Fields for Register: IRQ_TYPE_EN_RX_SA_344 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.917 IRQ_TYPE_STAT_RX_SA_344

- **Name:** IRQ_TYPE_STAT_RX_SA_344
- **Description:** Interrupt Status for each IRQ type of Rx SA 344
- **Size:** 32 bits
- **Offset:** 0xe54
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 344

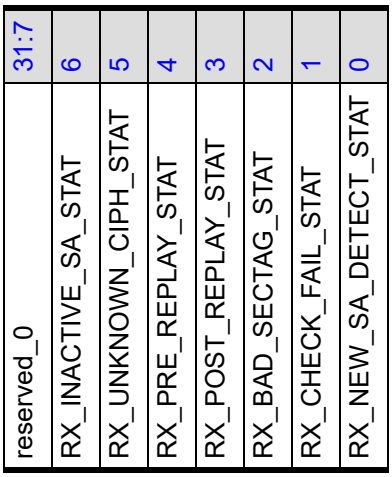


Table 6-924 Fields for Register: IRQ_TYPE_STAT_RX_SA_344

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-924 Fields for Register: IRQ_TYPE_STAT_RX_SA_344 (Continued)

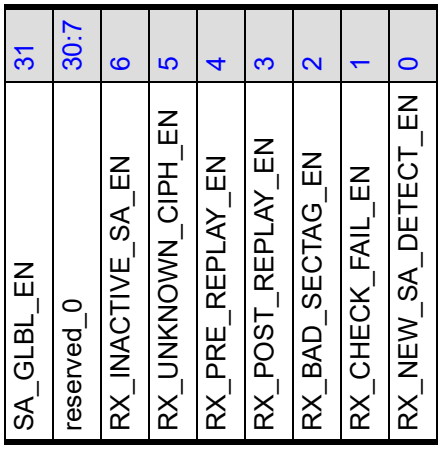
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-924 Fields for Register: IRQ_TYPE_STAT_RX_SA_344 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.918 IRQ_TYPE_EN_RX_SA_345

- **Name:** IRQ_TYPE_EN_RX_SA_345
- **Description:** Interrupt Enable for each IRQ type of Rx SA 345
- **Size:** 32 bits
- **Offset:** 0xe58
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 345



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-925 Fields for Register: IRQ_TYPE_EN_RX_SA_345

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-925 Fields for Register: IRQ_TYPE_EN_RX_SA_345 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-925 Fields for Register: IRQ_TYPE_EN_RX_SA_345 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.919 IRQ_TYPE_STAT_RX_SA_345

- **Name:** IRQ_TYPE_STAT_RX_SA_345
- **Description:** Interrupt Status for each IRQ type of Rx SA 345
- **Size:** 32 bits
- **Offset:** 0xe5c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 345

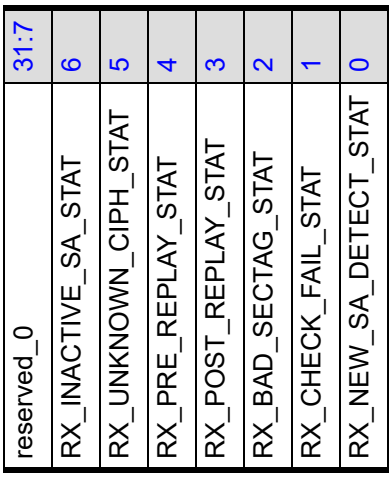


Table 6-926 Fields for Register: IRQ_TYPE_STAT_RX_SA_345

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-926 Fields for Register: IRQ_TYPE_STAT_RX_SA_345 (Continued)

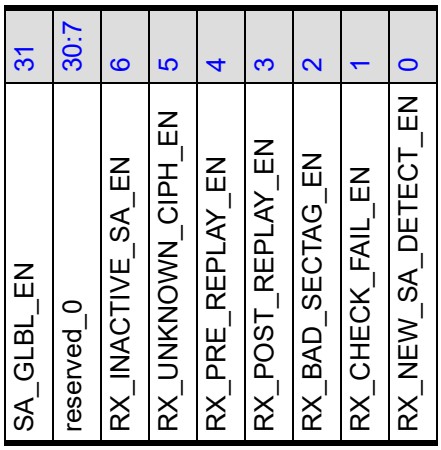
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-926 Fields for Register: IRQ_TYPE_STAT_RX_SA_345 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.920 IRQ_TYPE_EN_RX_SA_346

- **Name:** IRQ_TYPE_EN_RX_SA_346
- **Description:** Interrupt Enable for each IRQ type of Rx SA 346
- **Size:** 32 bits
- **Offset:** 0xe60
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 346



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-927 Fields for Register: IRQ_TYPE_EN_RX_SA_346

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-927 Fields for Register: IRQ_TYPE_EN_RX_SA_346 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-927 Fields for Register: IRQ_TYPE_EN_RX_SA_346 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.921 IRQ_TYPE_STAT_RX_SA_346

- **Name:** IRQ_TYPE_STAT_RX_SA_346
- **Description:** Interrupt Status for each IRQ type of Rx SA 346
- **Size:** 32 bits
- **Offset:** 0xe64
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 346

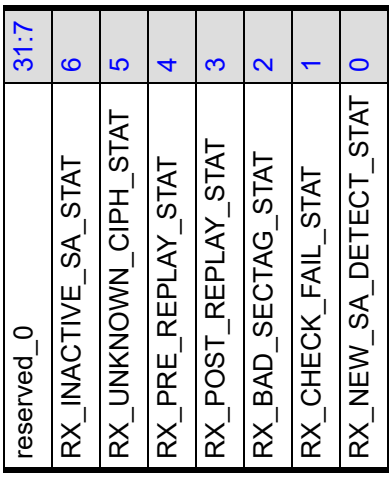


Table 6-928 Fields for Register: IRQ_TYPE_STAT_RX_SA_346

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-928 Fields for Register: IRQ_TYPE_STAT_RX_SA_346 (Continued)

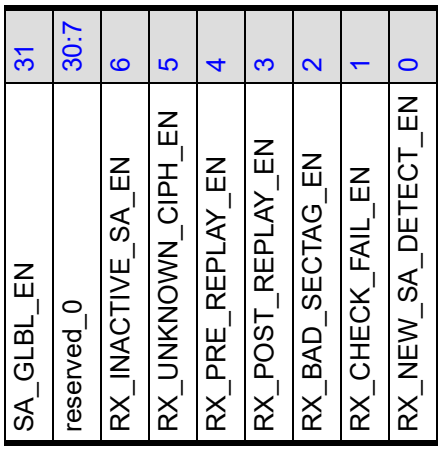
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-928 Fields for Register: IRQ_TYPE_STAT_RX_SA_346 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.922 IRQ_TYPE_EN_RX_SA_347

- **Name:** IRQ_TYPE_EN_RX_SA_347
- **Description:** Interrupt Enable for each IRQ type of Rx SA 347
- **Size:** 32 bits
- **Offset:** 0xe68
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 347



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-929 Fields for Register: IRQ_TYPE_EN_RX_SA_347

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-929 Fields for Register: IRQ_TYPE_EN_RX_SA_347 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-929 Fields for Register: IRQ_TYPE_EN_RX_SA_347 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.923 IRQ_TYPE_STAT_RX_SA_347

- **Name:** IRQ_TYPE_STAT_RX_SA_347
- **Description:** Interrupt Status for each IRQ type of Rx SA 347
- **Size:** 32 bits
- **Offset:** 0xe6c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 347

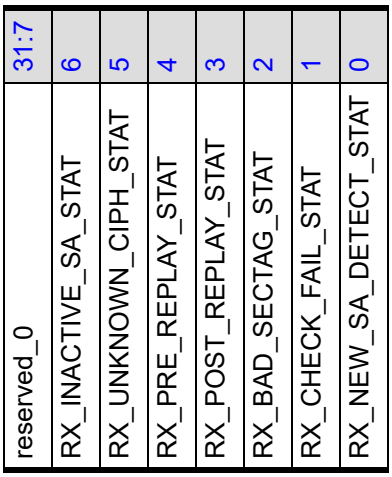


Table 6-930 Fields for Register: IRQ_TYPE_STAT_RX_SA_347

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-930 Fields for Register: IRQ_TYPE_STAT_RX_SA_347 (Continued)

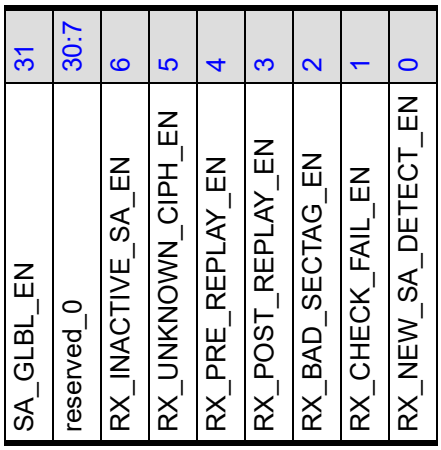
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-930 Fields for Register: IRQ_TYPE_STAT_RX_SA_347 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.924 IRQ_TYPE_EN_RX_SA_348

- **Name:** IRQ_TYPE_EN_RX_SA_348
- **Description:** Interrupt Enable for each IRQ type of Rx SA 348
- **Size:** 32 bits
- **Offset:** 0xe70
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 348



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-931 Fields for Register: IRQ_TYPE_EN_RX_SA_348

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-931 Fields for Register: IRQ_TYPE_EN_RX_SA_348 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-931 Fields for Register: IRQ_TYPE_EN_RX_SA_348 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.925 IRQ_TYPE_STAT_RX_SA_348

- **Name:** IRQ_TYPE_STAT_RX_SA_348
- **Description:** Interrupt Status for each IRQ type of Rx SA 348
- **Size:** 32 bits
- **Offset:** 0xe74
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 348

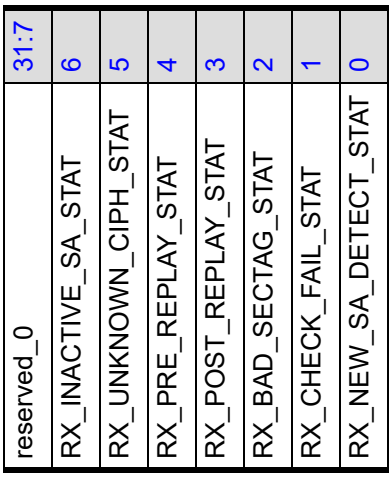


Table 6-932 Fields for Register: IRQ_TYPE_STAT_RX_SA_348

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-932 Fields for Register: IRQ_TYPE_STAT_RX_SA_348 (Continued)

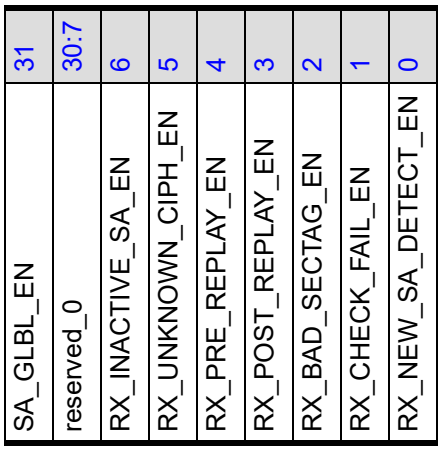
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-932 Fields for Register: IRQ_TYPE_STAT_RX_SA_348 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.926 IRQ_TYPE_EN_RX_SA_349

- **Name:** IRQ_TYPE_EN_RX_SA_349
- **Description:** Interrupt Enable for each IRQ type of Rx SA 349
- **Size:** 32 bits
- **Offset:** 0xe78
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 349



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-933 Fields for Register: IRQ_TYPE_EN_RX_SA_349

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-933 Fields for Register: IRQ_TYPE_EN_RX_SA_349 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-933 Fields for Register: IRQ_TYPE_EN_RX_SA_349 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.927 IRQ_TYPE_STAT_RX_SA_349

- **Name:** IRQ_TYPE_STAT_RX_SA_349
- **Description:** Interrupt Status for each IRQ type of Rx SA 349
- **Size:** 32 bits
- **Offset:** 0xe7c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 349

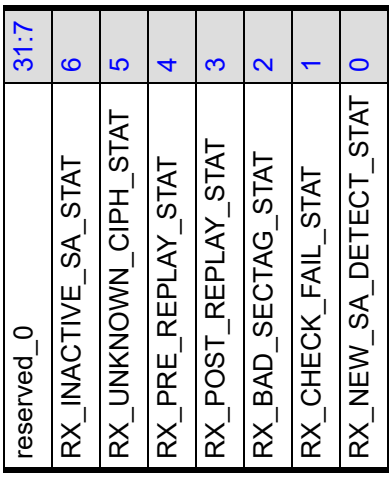


Table 6-934 Fields for Register: IRQ_TYPE_STAT_RX_SA_349

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-934 Fields for Register: IRQ_TYPE_STAT_RX_SA_349 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-934 Fields for Register: IRQ_TYPE_STAT_RX_SA_349 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.928 IRQ_TYPE_EN_RX_SA_350

- **Name:** IRQ_TYPE_EN_RX_SA_350
- **Description:** Interrupt Enable for each IRQ type of Rx SA 350
- **Size:** 32 bits
- **Offset:** 0xe80
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 350

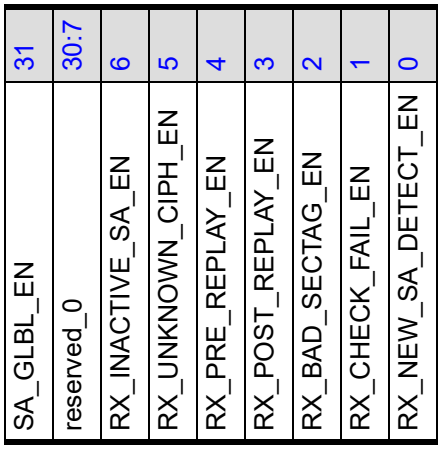


Table 6-935 Fields for Register: IRQ_TYPE_EN_RX_SA_350

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-935 Fields for Register: IRQ_TYPE_EN_RX_SA_350 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-935 Fields for Register: IRQ_TYPE_EN_RX_SA_350 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.929 IRQ_TYPE_STAT_RX_SA_350

- **Name:** IRQ_TYPE_STAT_RX_SA_350
- **Description:** Interrupt Status for each IRQ type of Rx SA 350
- **Size:** 32 bits
- **Offset:** 0xe84
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 350

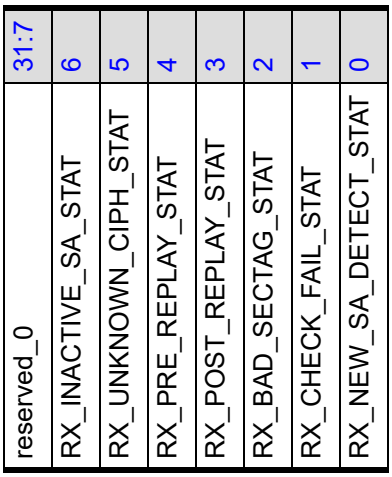


Table 6-936 Fields for Register: IRQ_TYPE_STAT_RX_SA_350

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-936 Fields for Register: IRQ_TYPE_STAT_RX_SA_350 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-936 Fields for Register: IRQ_TYPE_STAT_RX_SA_350 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.930 IRQ_TYPE_EN_RX_SA_351

- **Name:** IRQ_TYPE_EN_RX_SA_351
- **Description:** Interrupt Enable for each IRQ type of Rx SA 351
- **Size:** 32 bits
- **Offset:** 0xe88
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 351

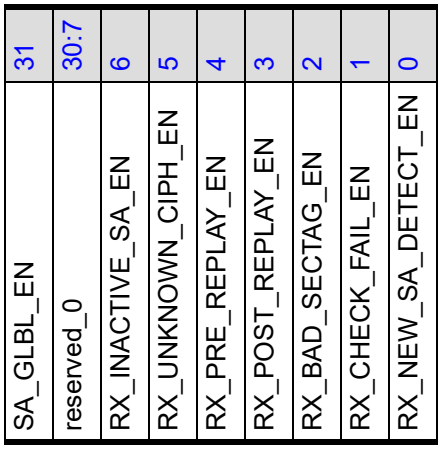


Table 6-937 Fields for Register: IRQ_TYPE_EN_RX_SA_351

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-937 Fields for Register: IRQ_TYPE_EN_RX_SA_351 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-937 Fields for Register: IRQ_TYPE_EN_RX_SA_351 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.931 IRQ_TYPE_STAT_RX_SA_351

- **Name:** IRQ_TYPE_STAT_RX_SA_351
- **Description:** Interrupt Status for each IRQ type of Rx SA 351
- **Size:** 32 bits
- **Offset:** 0xe8c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 351

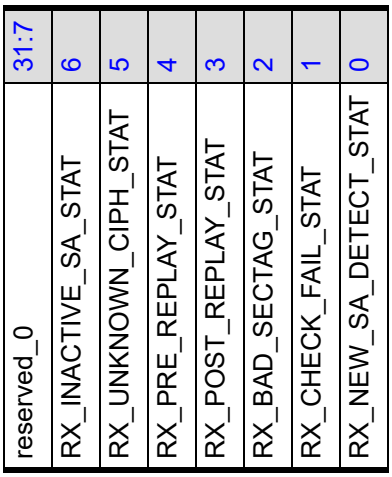


Table 6-938 Fields for Register: IRQ_TYPE_STAT_RX_SA_351

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-938 Fields for Register: IRQ_TYPE_STAT_RX_SA_351 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-938 Fields for Register: IRQ_TYPE_STAT_RX_SA_351 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.932 IRQ_TYPE_EN_RX_SA_352

- **Name:** IRQ_TYPE_EN_RX_SA_352
- **Description:** Interrupt Enable for each IRQ type of Rx SA 352
- **Size:** 32 bits
- **Offset:** 0xe90
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 352

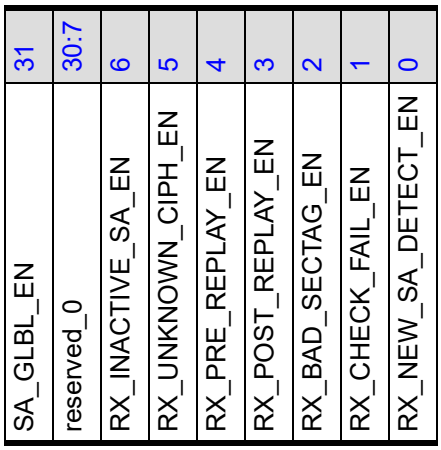


Table 6-939 Fields for Register: IRQ_TYPE_EN_RX_SA_352

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-939 Fields for Register: IRQ_TYPE_EN_RX_SA_352 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-939 Fields for Register: IRQ_TYPE_EN_RX_SA_352 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.933 IRQ_TYPE_STAT_RX_SA_352

- **Name:** IRQ_TYPE_STAT_RX_SA_352
- **Description:** Interrupt Status for each IRQ type of Rx SA 352
- **Size:** 32 bits
- **Offset:** 0xe94
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 352

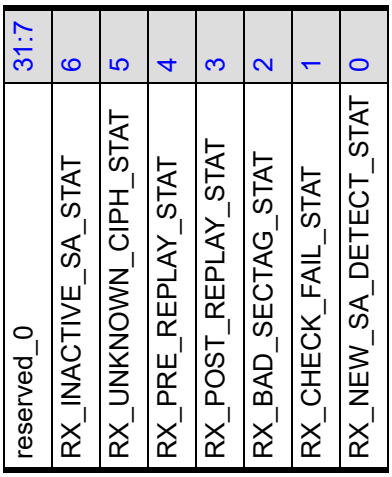


Table 6-940 Fields for Register: IRQ_TYPE_STAT_RX_SA_352

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-940 Fields for Register: IRQ_TYPE_STAT_RX_SA_352 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-940 Fields for Register: IRQ_TYPE_STAT_RX_SA_352 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.934 IRQ_TYPE_EN_RX_SA_353

- **Name:** IRQ_TYPE_EN_RX_SA_353
- **Description:** Interrupt Enable for each IRQ type of Rx SA 353
- **Size:** 32 bits
- **Offset:** 0xe98
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 353

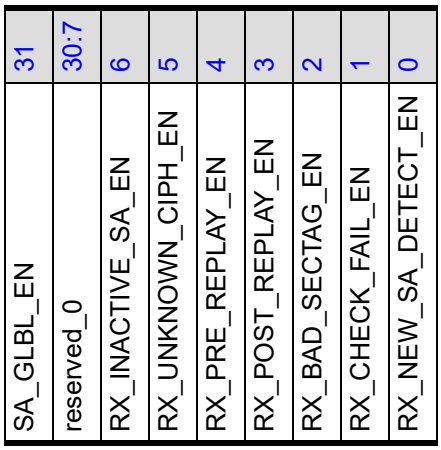


Table 6-941 Fields for Register: IRQ_TYPE_EN_RX_SA_353

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-941 Fields for Register: IRQ_TYPE_EN_RX_SA_353 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-941 Fields for Register: IRQ_TYPE_EN_RX_SA_353 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.935 IRQ_TYPE_STAT_RX_SA_353

- **Name:** IRQ_TYPE_STAT_RX_SA_353
- **Description:** Interrupt Status for each IRQ type of Rx SA 353
- **Size:** 32 bits
- **Offset:** 0xe9c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 353

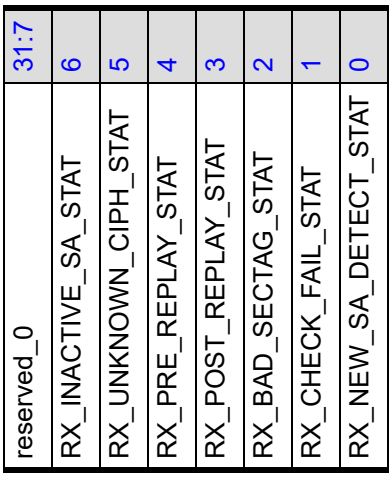


Table 6-942 Fields for Register: IRQ_TYPE_STAT_RX_SA_353

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-942 Fields for Register: IRQ_TYPE_STAT_RX_SA_353 (Continued)

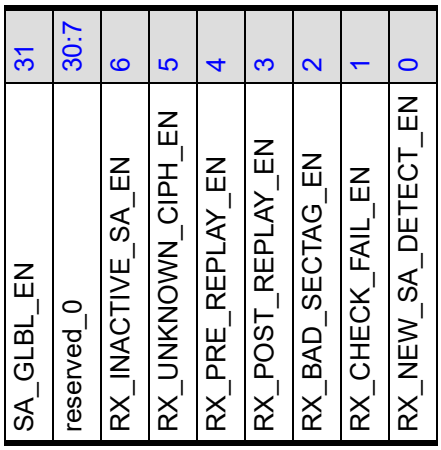
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-942 Fields for Register: IRQ_TYPE_STAT_RX_SA_353 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.936 IRQ_TYPE_EN_RX_SA_354

- **Name:** IRQ_TYPE_EN_RX_SA_354
- **Description:** Interrupt Enable for each IRQ type of Rx SA 354
- **Size:** 32 bits
- **Offset:** 0xea0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 354



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-943 Fields for Register: IRQ_TYPE_EN_RX_SA_354

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-943 Fields for Register: IRQ_TYPE_EN_RX_SA_354 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-943 Fields for Register: IRQ_TYPE_EN_RX_SA_354 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.937 IRQ_TYPE_STAT_RX_SA_354

- **Name:** IRQ_TYPE_STAT_RX_SA_354
- **Description:** Interrupt Status for each IRQ type of Rx SA 354
- **Size:** 32 bits
- **Offset:** 0xea4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 354

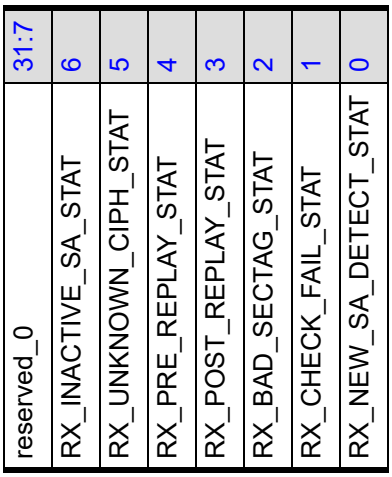


Table 6-944 Fields for Register: IRQ_TYPE_STAT_RX_SA_354

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-944 Fields for Register: IRQ_TYPE_STAT_RX_SA_354 (Continued)

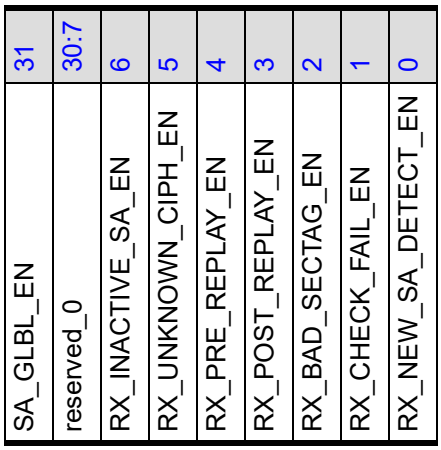
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-944 Fields for Register: IRQ_TYPE_STAT_RX_SA_354 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.938 IRQ_TYPE_EN_RX_SA_355

- **Name:** IRQ_TYPE_EN_RX_SA_355
- **Description:** Interrupt Enable for each IRQ type of Rx SA 355
- **Size:** 32 bits
- **Offset:** 0xea8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 355



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-945 Fields for Register: IRQ_TYPE_EN_RX_SA_355

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-945 Fields for Register: IRQ_TYPE_EN_RX_SA_355 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-945 Fields for Register: IRQ_TYPE_EN_RX_SA_355 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.939 IRQ_TYPE_STAT_RX_SA_355

- **Name:** IRQ_TYPE_STAT_RX_SA_355
- **Description:** Interrupt Status for each IRQ type of Rx SA 355
- **Size:** 32 bits
- **Offset:** 0xeac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 355

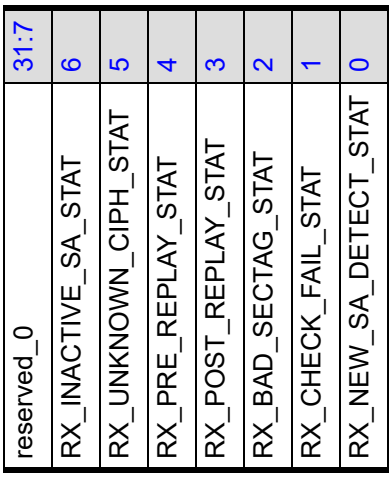


Table 6-946 Fields for Register: IRQ_TYPE_STAT_RX_SA_355

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-946 Fields for Register: IRQ_TYPE_STAT_RX_SA_355 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-946 Fields for Register: IRQ_TYPE_STAT_RX_SA_355 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.940 IRQ_TYPE_EN_RX_SA_356

- **Name:** IRQ_TYPE_EN_RX_SA_356
- **Description:** Interrupt Enable for each IRQ type of Rx SA 356
- **Size:** 32 bits
- **Offset:** 0xeb0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 356

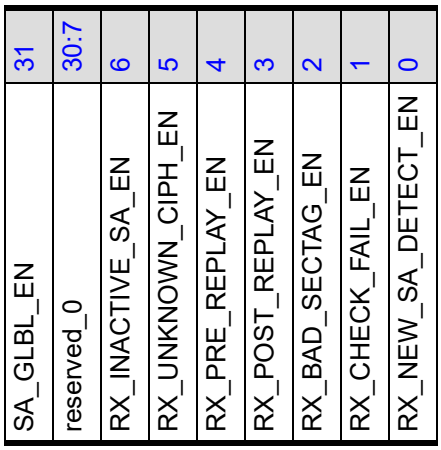


Table 6-947 Fields for Register: IRQ_TYPE_EN_RX_SA_356

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-947 Fields for Register: IRQ_TYPE_EN_RX_SA_356 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-947 Fields for Register: IRQ_TYPE_EN_RX_SA_356 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.941 IRQ_TYPE_STAT_RX_SA_356

- **Name:** IRQ_TYPE_STAT_RX_SA_356
- **Description:** Interrupt Status for each IRQ type of Rx SA 356
- **Size:** 32 bits
- **Offset:** 0xeb4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 356

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-948 Fields for Register: IRQ_TYPE_STAT_RX_SA_356

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-948 Fields for Register: IRQ_TYPE_STAT_RX_SA_356 (Continued)

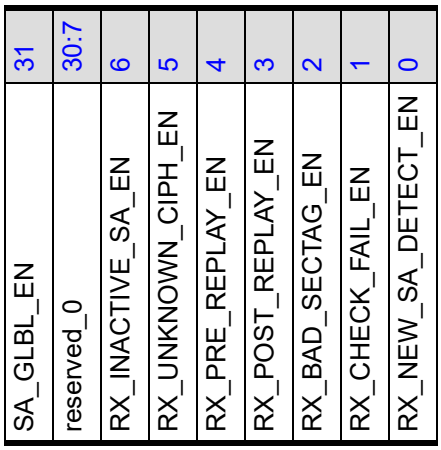
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-948 Fields for Register: IRQ_TYPE_STAT_RX_SA_356 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.942 IRQ_TYPE_EN_RX_SA_357

- **Name:** IRQ_TYPE_EN_RX_SA_357
- **Description:** Interrupt Enable for each IRQ type of Rx SA 357
- **Size:** 32 bits
- **Offset:** 0xeb8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 357



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-949 Fields for Register: IRQ_TYPE_EN_RX_SA_357

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-949 Fields for Register: IRQ_TYPE_EN_RX_SA_357 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-949 Fields for Register: IRQ_TYPE_EN_RX_SA_357 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.943 IRQ_TYPE_STAT_RX_SA_357

- **Name:** IRQ_TYPE_STAT_RX_SA_357
- **Description:** Interrupt Status for each IRQ type of Rx SA 357
- **Size:** 32 bits
- **Offset:** 0xebc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 357

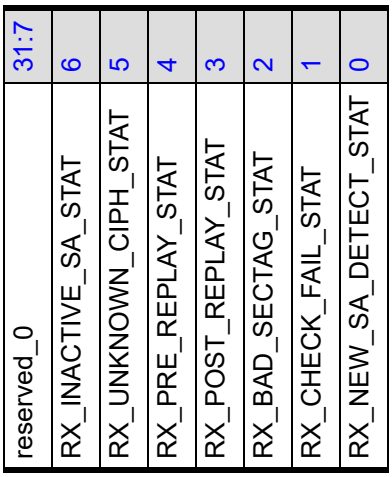


Table 6-950 Fields for Register: IRQ_TYPE_STAT_RX_SA_357

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-950 Fields for Register: IRQ_TYPE_STAT_RX_SA_357 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-950 Fields for Register: IRQ_TYPE_STAT_RX_SA_357 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.944 IRQ_TYPE_EN_RX_SA_358

- **Name:** IRQ_TYPE_EN_RX_SA_358
- **Description:** Interrupt Enable for each IRQ type of Rx SA 358
- **Size:** 32 bits
- **Offset:** 0xec0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 358

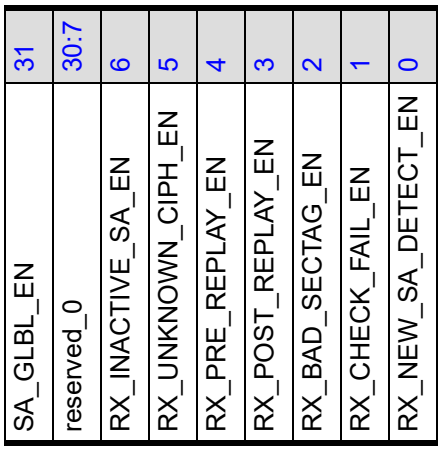


Table 6-951 Fields for Register: IRQ_TYPE_EN_RX_SA_358

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-951 Fields for Register: IRQ_TYPE_EN_RX_SA_358 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-951 Fields for Register: IRQ_TYPE_EN_RX_SA_358 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.945 IRQ_TYPE_STAT_RX_SA_358

- **Name:** IRQ_TYPE_STAT_RX_SA_358
- **Description:** Interrupt Status for each IRQ type of Rx SA 358
- **Size:** 32 bits
- **Offset:** 0xec4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 358

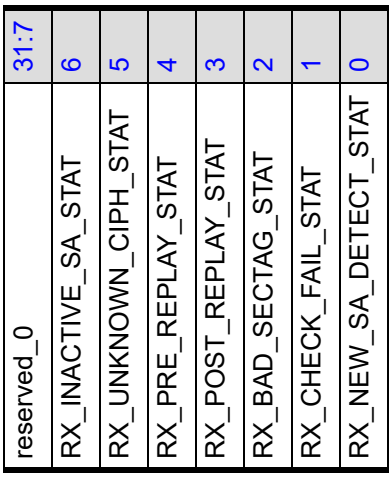


Table 6-952 Fields for Register: IRQ_TYPE_STAT_RX_SA_358

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-952 Fields for Register: IRQ_TYPE_STAT_RX_SA_358 (Continued)

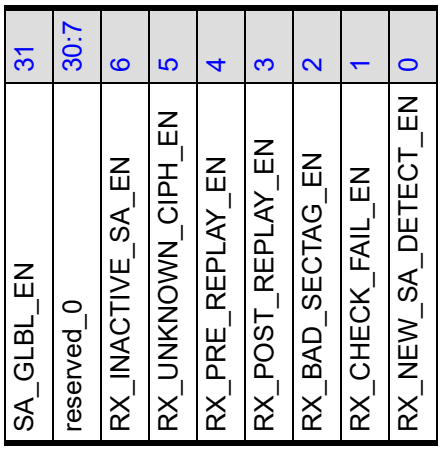
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-952 Fields for Register: IRQ_TYPE_STAT_RX_SA_358 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.946 IRQ_TYPE_EN_RX_SA_359

- **Name:** IRQ_TYPE_EN_RX_SA_359
- **Description:** Interrupt Enable for each IRQ type of Rx SA 359
- **Size:** 32 bits
- **Offset:** 0xec8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 359



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-953 Fields for Register: IRQ_TYPE_EN_RX_SA_359

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-953 Fields for Register: IRQ_TYPE_EN_RX_SA_359 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-953 Fields for Register: IRQ_TYPE_EN_RX_SA_359 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.947 IRQ_TYPE_STAT_RX_SA_359

- **Name:** IRQ_TYPE_STAT_RX_SA_359
- **Description:** Interrupt Status for each IRQ type of Rx SA 359
- **Size:** 32 bits
- **Offset:** 0xecc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 359

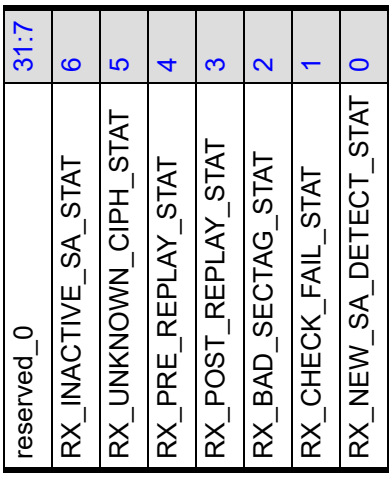


Table 6-954 Fields for Register: IRQ_TYPE_STAT_RX_SA_359

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-954 Fields for Register: IRQ_TYPE_STAT_RX_SA_359 (Continued)

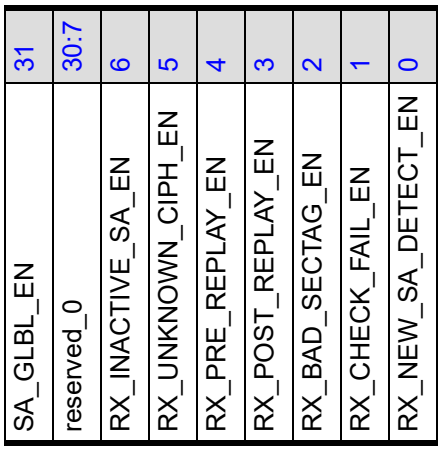
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-954 Fields for Register: IRQ_TYPE_STAT_RX_SA_359 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.948 IRQ_TYPE_EN_RX_SA_360

- **Name:** IRQ_TYPE_EN_RX_SA_360
- **Description:** Interrupt Enable for each IRQ type of Rx SA 360
- **Size:** 32 bits
- **Offset:** 0xed0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 360



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-955 Fields for Register: IRQ_TYPE_EN_RX_SA_360

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-955 Fields for Register: IRQ_TYPE_EN_RX_SA_360 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-955 Fields for Register: IRQ_TYPE_EN_RX_SA_360 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.949 IRQ_TYPE_STAT_RX_SA_360

- **Name:** IRQ_TYPE_STAT_RX_SA_360
- **Description:** Interrupt Status for each IRQ type of Rx SA 360
- **Size:** 32 bits
- **Offset:** 0xed4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 360

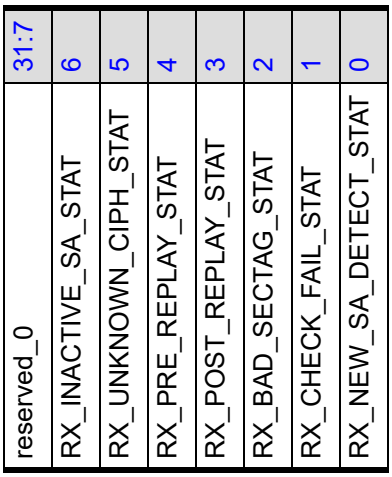


Table 6-956 Fields for Register: IRQ_TYPE_STAT_RX_SA_360

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-956 Fields for Register: IRQ_TYPE_STAT_RX_SA_360 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-956 Fields for Register: IRQ_TYPE_STAT_RX_SA_360 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.950 IRQ_TYPE_EN_RX_SA_361

- **Name:** IRQ_TYPE_EN_RX_SA_361
- **Description:** Interrupt Enable for each IRQ type of Rx SA 361
- **Size:** 32 bits
- **Offset:** 0xed8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 361

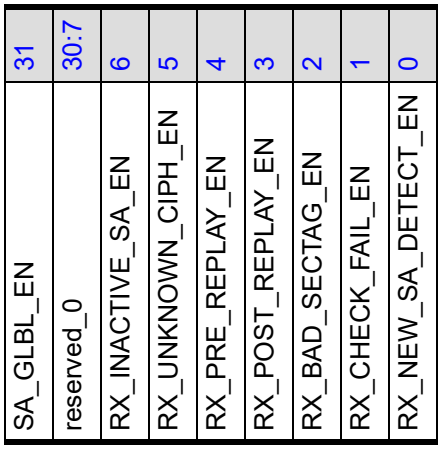


Table 6-957 Fields for Register: IRQ_TYPE_EN_RX_SA_361

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-957 Fields for Register: IRQ_TYPE_EN_RX_SA_361 (Continued)

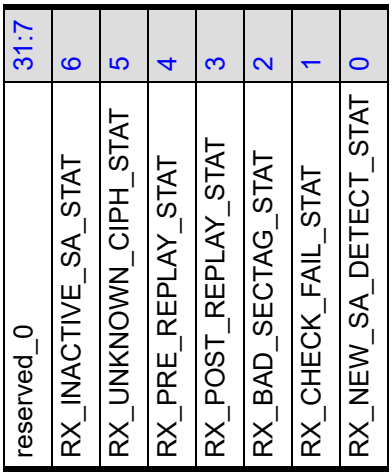
Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-957 Fields for Register: IRQ_TYPE_EN_RX_SA_361 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.951 IRQ_TYPE_STAT_RX_SA_361

- **Name:** IRQ_TYPE_STAT_RX_SA_361
- **Description:** Interrupt Status for each IRQ type of Rx SA 361
- **Size:** 32 bits
- **Offset:** 0xedc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 361



31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-958 Fields for Register: IRQ_TYPE_STAT_RX_SA_361

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-958 Fields for Register: IRQ_TYPE_STAT_RX_SA_361 (Continued)

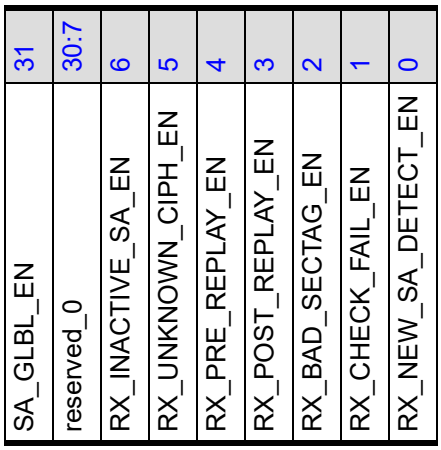
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-958 Fields for Register: IRQ_TYPE_STAT_RX_SA_361 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.952 IRQ_TYPE_EN_RX_SA_362

- **Name:** IRQ_TYPE_EN_RX_SA_362
- **Description:** Interrupt Enable for each IRQ type of Rx SA 362
- **Size:** 32 bits
- **Offset:** 0xee0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 362



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-959 Fields for Register: IRQ_TYPE_EN_RX_SA_362

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-959 Fields for Register: IRQ_TYPE_EN_RX_SA_362 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-959 Fields for Register: IRQ_TYPE_EN_RX_SA_362 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.953 IRQ_TYPE_STAT_RX_SA_362

- **Name:** IRQ_TYPE_STAT_RX_SA_362
- **Description:** Interrupt Status for each IRQ type of Rx SA 362
- **Size:** 32 bits
- **Offset:** 0xee4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 362

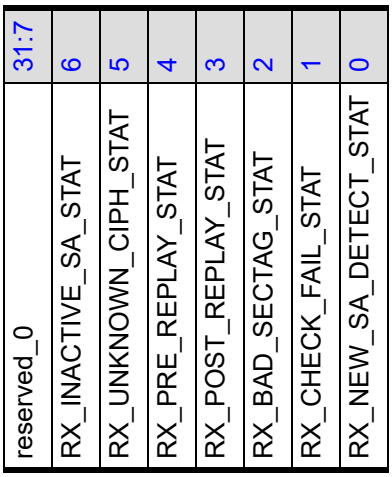


Table 6-960 Fields for Register: IRQ_TYPE_STAT_RX_SA_362

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-960 Fields for Register: IRQ_TYPE_STAT_RX_SA_362 (Continued)

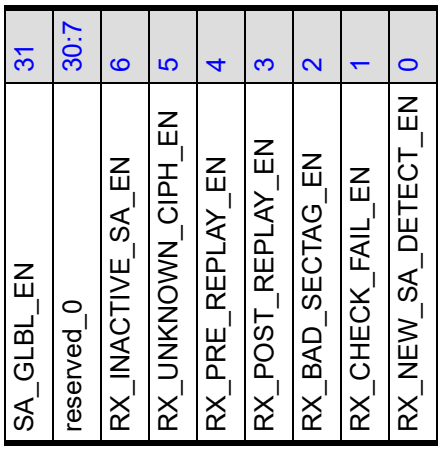
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-960 Fields for Register: IRQ_TYPE_STAT_RX_SA_362 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.954 IRQ_TYPE_EN_RX_SA_363

- **Name:** IRQ_TYPE_EN_RX_SA_363
- **Description:** Interrupt Enable for each IRQ type of Rx SA 363
- **Size:** 32 bits
- **Offset:** 0xee8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 363



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-961 Fields for Register: IRQ_TYPE_EN_RX_SA_363

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-961 Fields for Register: IRQ_TYPE_EN_RX_SA_363 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-961 Fields for Register: IRQ_TYPE_EN_RX_SA_363 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.955 IRQ_TYPE_STAT_RX_SA_363

- **Name:** IRQ_TYPE_STAT_RX_SA_363
- **Description:** Interrupt Status for each IRQ type of Rx SA 363
- **Size:** 32 bits
- **Offset:** 0xeec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 363

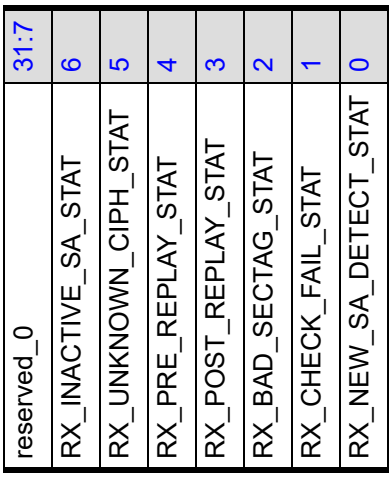


Table 6-962 Fields for Register: IRQ_TYPE_STAT_RX_SA_363

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-962 Fields for Register: IRQ_TYPE_STAT_RX_SA_363 (Continued)

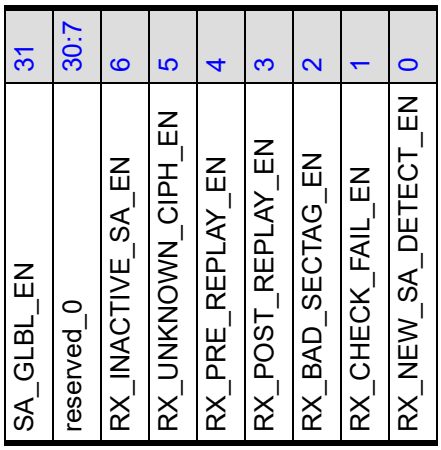
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-962 Fields for Register: IRQ_TYPE_STAT_RX_SA_363 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.956 IRQ_TYPE_EN_RX_SA_364

- **Name:** IRQ_TYPE_EN_RX_SA_364
- **Description:** Interrupt Enable for each IRQ type of Rx SA 364
- **Size:** 32 bits
- **Offset:** 0xef0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 364



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-963 Fields for Register: IRQ_TYPE_EN_RX_SA_364

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-963 Fields for Register: IRQ_TYPE_EN_RX_SA_364 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-963 Fields for Register: IRQ_TYPE_EN_RX_SA_364 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.957 IRQ_TYPE_STAT_RX_SA_364

- **Name:** IRQ_TYPE_STAT_RX_SA_364
- **Description:** Interrupt Status for each IRQ type of Rx SA 364
- **Size:** 32 bits
- **Offset:** 0xef4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 364

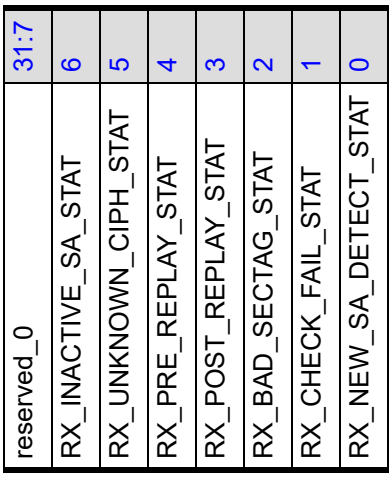


Table 6-964 Fields for Register: IRQ_TYPE_STAT_RX_SA_364

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-964 Fields for Register: IRQ_TYPE_STAT_RX_SA_364 (Continued)

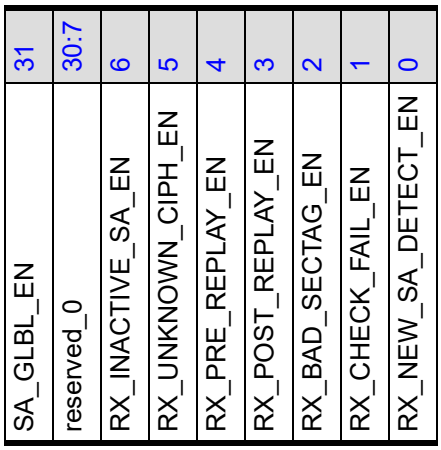
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-964 Fields for Register: IRQ_TYPE_STAT_RX_SA_364 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.958 IRQ_TYPE_EN_RX_SA_365

- **Name:** IRQ_TYPE_EN_RX_SA_365
- **Description:** Interrupt Enable for each IRQ type of Rx SA 365
- **Size:** 32 bits
- **Offset:** 0xef8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 365



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-965 Fields for Register: IRQ_TYPE_EN_RX_SA_365

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-965 Fields for Register: IRQ_TYPE_EN_RX_SA_365 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-965 Fields for Register: IRQ_TYPE_EN_RX_SA_365 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.959 IRQ_TYPE_STAT_RX_SA_365

- **Name:** IRQ_TYPE_STAT_RX_SA_365
- **Description:** Interrupt Status for each IRQ type of Rx SA 365
- **Size:** 32 bits
- **Offset:** 0xefc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 365

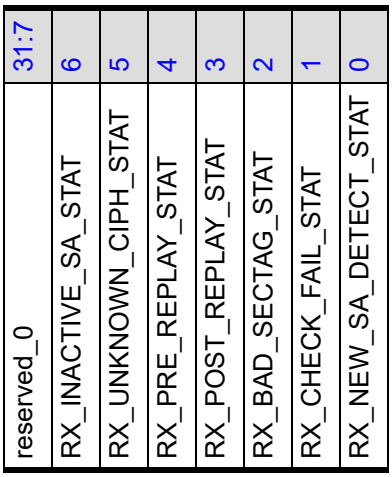


Table 6-966 Fields for Register: IRQ_TYPE_STAT_RX_SA_365

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-966 Fields for Register: IRQ_TYPE_STAT_RX_SA_365 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-966 Fields for Register: IRQ_TYPE_STAT_RX_SA_365 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.960 IRQ_TYPE_EN_RX_SA_366

- **Name:** IRQ_TYPE_EN_RX_SA_366
- **Description:** Interrupt Enable for each IRQ type of Rx SA 366
- **Size:** 32 bits
- **Offset:** 0xf00
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 366

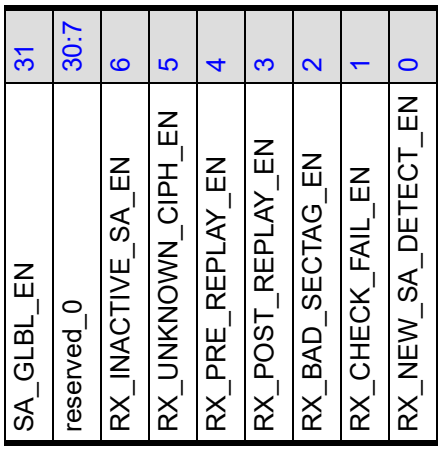


Table 6-967 Fields for Register: IRQ_TYPE_EN_RX_SA_366

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-967 Fields for Register: IRQ_TYPE_EN_RX_SA_366 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-967 Fields for Register: IRQ_TYPE_EN_RX_SA_366 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.961 IRQ_TYPE_STAT_RX_SA_366

- **Name:** IRQ_TYPE_STAT_RX_SA_366
- **Description:** Interrupt Status for each IRQ type of Rx SA 366
- **Size:** 32 bits
- **Offset:** 0xf04
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 366

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-968 Fields for Register: IRQ_TYPE_STAT_RX_SA_366

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-968 Fields for Register: IRQ_TYPE_STAT_RX_SA_366 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-968 Fields for Register: IRQ_TYPE_STAT_RX_SA_366 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.962 IRQ_TYPE_EN_RX_SA_367

- **Name:** IRQ_TYPE_EN_RX_SA_367
- **Description:** Interrupt Enable for each IRQ type of Rx SA 367
- **Size:** 32 bits
- **Offset:** 0xf08
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 367

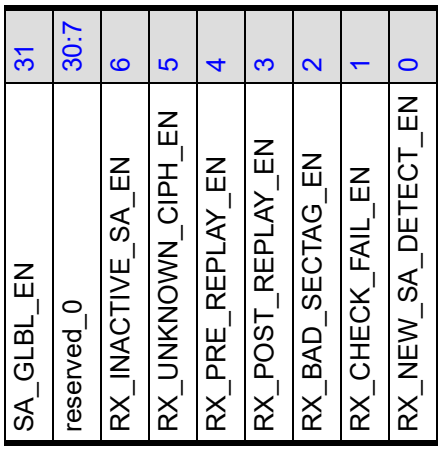


Table 6-969 Fields for Register: IRQ_TYPE_EN_RX_SA_367

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-969 Fields for Register: IRQ_TYPE_EN_RX_SA_367 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-969 Fields for Register: IRQ_TYPE_EN_RX_SA_367 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.963 IRQ_TYPE_STAT_RX_SA_367

- **Name:** IRQ_TYPE_STAT_RX_SA_367
- **Description:** Interrupt Status for each IRQ type of Rx SA 367
- **Size:** 32 bits
- **Offset:** 0xf0c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 367

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-970 Fields for Register: IRQ_TYPE_STAT_RX_SA_367

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-970 Fields for Register: IRQ_TYPE_STAT_RX_SA_367 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-970 Fields for Register: IRQ_TYPE_STAT_RX_SA_367 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.964 IRQ_TYPE_EN_RX_SA_368

- **Name:** IRQ_TYPE_EN_RX_SA_368
- **Description:** Interrupt Enable for each IRQ type of Rx SA 368
- **Size:** 32 bits
- **Offset:** 0xf10
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 368

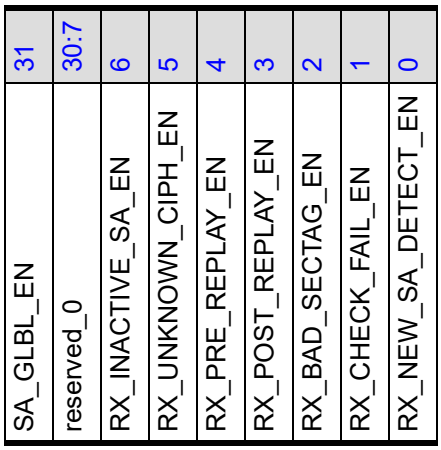


Table 6-971 Fields for Register: IRQ_TYPE_EN_RX_SA_368

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-971 Fields for Register: IRQ_TYPE_EN_RX_SA_368 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-971 Fields for Register: IRQ_TYPE_EN_RX_SA_368 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.965 IRQ_TYPE_STAT_RX_SA_368

- **Name:** IRQ_TYPE_STAT_RX_SA_368
- **Description:** Interrupt Status for each IRQ type of Rx SA 368
- **Size:** 32 bits
- **Offset:** 0xf14
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 368

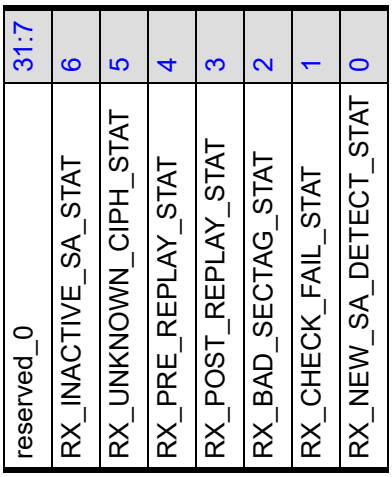


Table 6-972 Fields for Register: IRQ_TYPE_STAT_RX_SA_368

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-972 Fields for Register: IRQ_TYPE_STAT_RX_SA_368 (Continued)

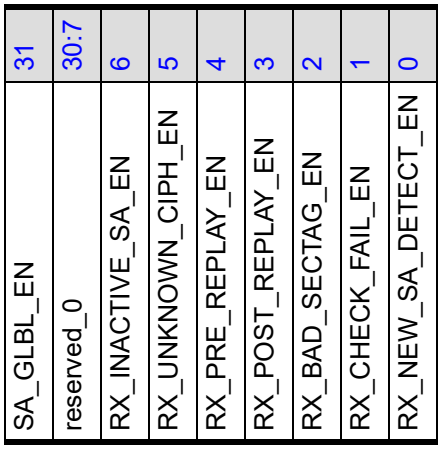
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-972 Fields for Register: IRQ_TYPE_STAT_RX_SA_368 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.966 IRQ_TYPE_EN_RX_SA_369

- **Name:** IRQ_TYPE_EN_RX_SA_369
- **Description:** Interrupt Enable for each IRQ type of Rx SA 369
- **Size:** 32 bits
- **Offset:** 0xf18
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 369



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-973 Fields for Register: IRQ_TYPE_EN_RX_SA_369

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-973 Fields for Register: IRQ_TYPE_EN_RX_SA_369 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-973 Fields for Register: IRQ_TYPE_EN_RX_SA_369 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.967 IRQ_TYPE_STAT_RX_SA_369

- **Name:** IRQ_TYPE_STAT_RX_SA_369
- **Description:** Interrupt Status for each IRQ type of Rx SA 369
- **Size:** 32 bits
- **Offset:** 0xf1c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 369

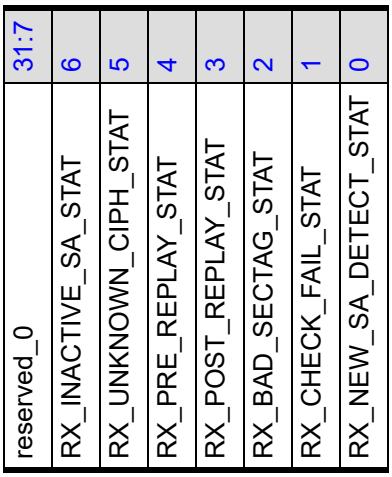


Table 6-974 Fields for Register: IRQ_TYPE_STAT_RX_SA_369

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-974 Fields for Register: IRQ_TYPE_STAT_RX_SA_369 (Continued)

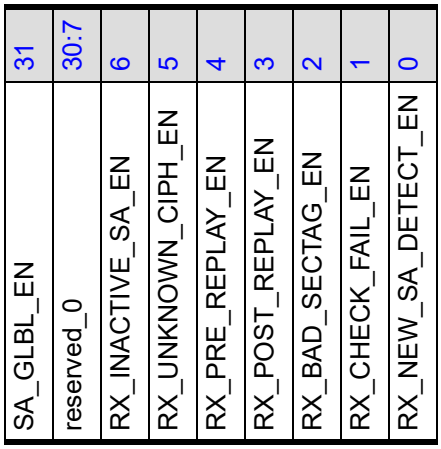
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-974 Fields for Register: IRQ_TYPE_STAT_RX_SA_369 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.968 IRQ_TYPE_EN_RX_SA_370

- **Name:** IRQ_TYPE_EN_RX_SA_370
- **Description:** Interrupt Enable for each IRQ type of Rx SA 370
- **Size:** 32 bits
- **Offset:** 0xf20
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 370



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-975 Fields for Register: IRQ_TYPE_EN_RX_SA_370

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-975 Fields for Register: IRQ_TYPE_EN_RX_SA_370 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-975 Fields for Register: IRQ_TYPE_EN_RX_SA_370 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.969 IRQ_TYPE_STAT_RX_SA_370

- **Name:** IRQ_TYPE_STAT_RX_SA_370
- **Description:** Interrupt Status for each IRQ type of Rx SA 370
- **Size:** 32 bits
- **Offset:** 0xf24
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 370

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-976 Fields for Register: IRQ_TYPE_STAT_RX_SA_370

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-976 Fields for Register: IRQ_TYPE_STAT_RX_SA_370 (Continued)

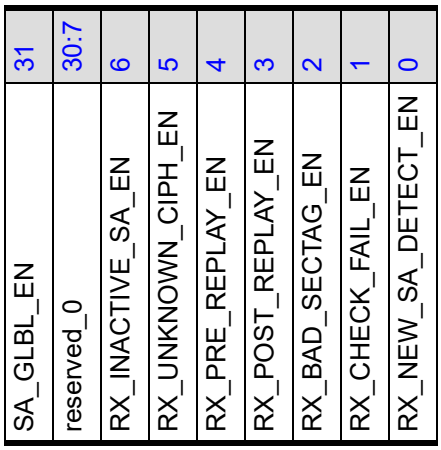
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-976 Fields for Register: IRQ_TYPE_STAT_RX_SA_370 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.970 IRQ_TYPE_EN_RX_SA_371

- **Name:** IRQ_TYPE_EN_RX_SA_371
- **Description:** Interrupt Enable for each IRQ type of Rx SA 371
- **Size:** 32 bits
- **Offset:** 0xf28
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 371



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-977 Fields for Register: IRQ_TYPE_EN_RX_SA_371

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-977 Fields for Register: IRQ_TYPE_EN_RX_SA_371 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-977 Fields for Register: IRQ_TYPE_EN_RX_SA_371 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.971 IRQ_TYPE_STAT_RX_SA_371

- **Name:** IRQ_TYPE_STAT_RX_SA_371
- **Description:** Interrupt Status for each IRQ type of Rx SA 371
- **Size:** 32 bits
- **Offset:** 0xf2c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 371

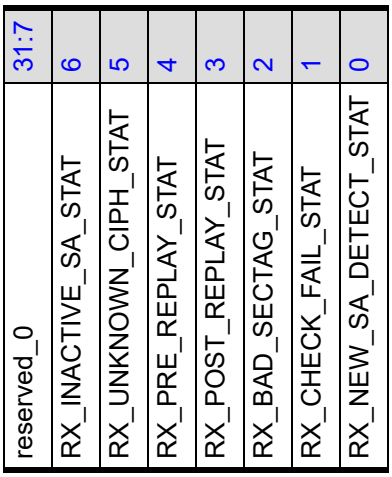


Table 6-978 Fields for Register: IRQ_TYPE_STAT_RX_SA_371

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-978 Fields for Register: IRQ_TYPE_STAT_RX_SA_371 (Continued)

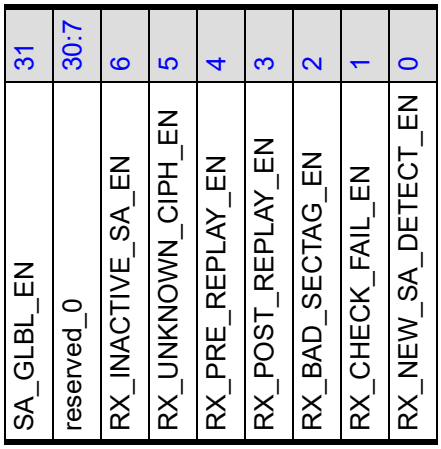
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-978 Fields for Register: IRQ_TYPE_STAT_RX_SA_371 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.972 IRQ_TYPE_EN_RX_SA_372

- **Name:** IRQ_TYPE_EN_RX_SA_372
- **Description:** Interrupt Enable for each IRQ type of Rx SA 372
- **Size:** 32 bits
- **Offset:** 0xf30
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 372



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-979 Fields for Register: IRQ_TYPE_EN_RX_SA_372

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-979 Fields for Register: IRQ_TYPE_EN_RX_SA_372 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-979 Fields for Register: IRQ_TYPE_EN_RX_SA_372 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.973 IRQ_TYPE_STAT_RX_SA_372

- **Name:** IRQ_TYPE_STAT_RX_SA_372
- **Description:** Interrupt Status for each IRQ type of Rx SA 372
- **Size:** 32 bits
- **Offset:** 0xf34
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 372

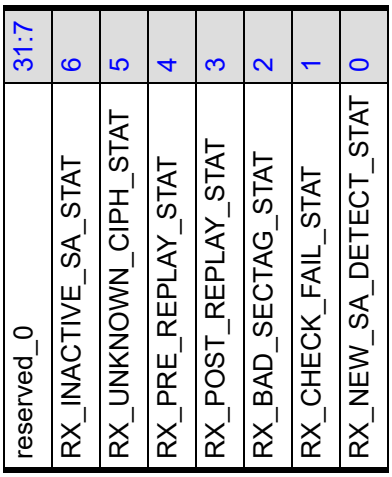


Table 6-980 Fields for Register: IRQ_TYPE_STAT_RX_SA_372

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-980 Fields for Register: IRQ_TYPE_STAT_RX_SA_372 (Continued)

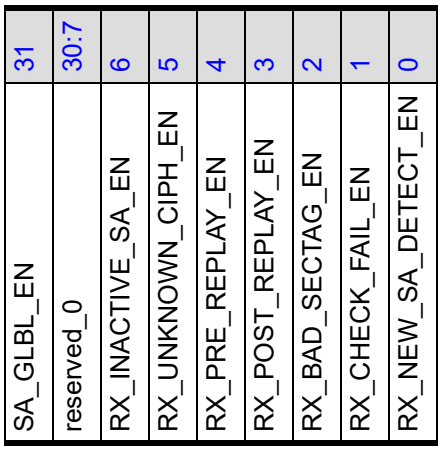
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-980 Fields for Register: IRQ_TYPE_STAT_RX_SA_372 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.974 IRQ_TYPE_EN_RX_SA_373

- **Name:** IRQ_TYPE_EN_RX_SA_373
- **Description:** Interrupt Enable for each IRQ type of Rx SA 373
- **Size:** 32 bits
- **Offset:** 0xf38
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 373



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-981 Fields for Register: IRQ_TYPE_EN_RX_SA_373

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-981 Fields for Register: IRQ_TYPE_EN_RX_SA_373 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-981 Fields for Register: IRQ_TYPE_EN_RX_SA_373 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.975 IRQ_TYPE_STAT_RX_SA_373

- **Name:** IRQ_TYPE_STAT_RX_SA_373
- **Description:** Interrupt Status for each IRQ type of Rx SA 373
- **Size:** 32 bits
- **Offset:** 0xf3c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 373

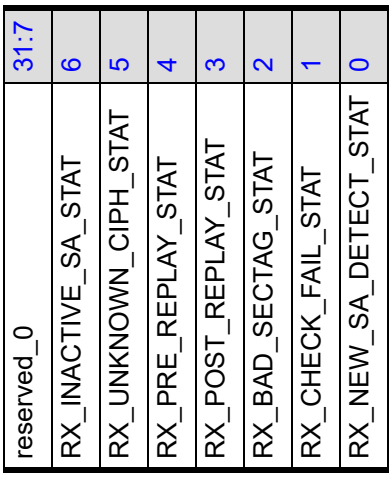


Table 6-982 Fields for Register: IRQ_TYPE_STAT_RX_SA_373

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-982 Fields for Register: IRQ_TYPE_STAT_RX_SA_373 (Continued)

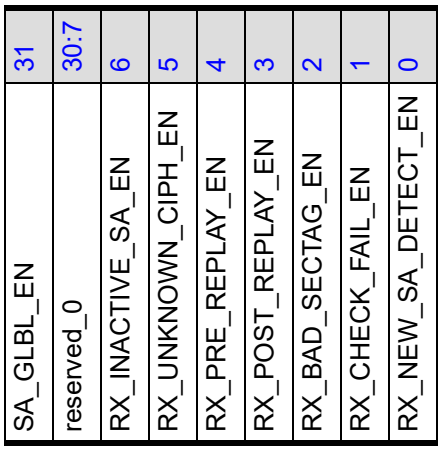
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-982 Fields for Register: IRQ_TYPE_STAT_RX_SA_373 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.976 IRQ_TYPE_EN_RX_SA_374

- **Name:** IRQ_TYPE_EN_RX_SA_374
- **Description:** Interrupt Enable for each IRQ type of Rx SA 374
- **Size:** 32 bits
- **Offset:** 0xf40
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 374



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-983 Fields for Register: IRQ_TYPE_EN_RX_SA_374

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-983 Fields for Register: IRQ_TYPE_EN_RX_SA_374 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-983 Fields for Register: IRQ_TYPE_EN_RX_SA_374 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.977 IRQ_TYPE_STAT_RX_SA_374

- **Name:** IRQ_TYPE_STAT_RX_SA_374
- **Description:** Interrupt Status for each IRQ type of Rx SA 374
- **Size:** 32 bits
- **Offset:** 0xf44
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 374

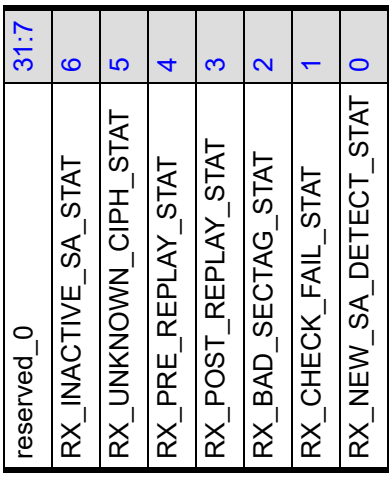


Table 6-984 Fields for Register: IRQ_TYPE_STAT_RX_SA_374

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-984 Fields for Register: IRQ_TYPE_STAT_RX_SA_374 (Continued)

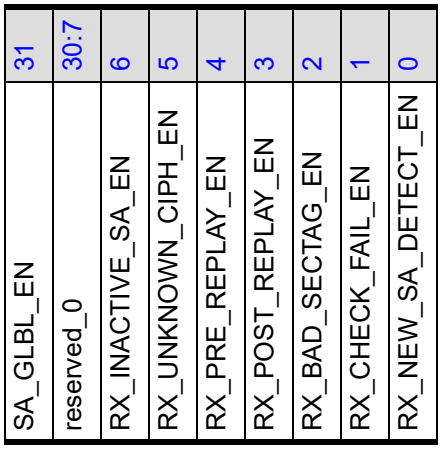
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-984 Fields for Register: IRQ_TYPE_STAT_RX_SA_374 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.978 IRQ_TYPE_EN_RX_SA_375

- **Name:** IRQ_TYPE_EN_RX_SA_375
- **Description:** Interrupt Enable for each IRQ type of Rx SA 375
- **Size:** 32 bits
- **Offset:** 0xf48
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 375



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-985 Fields for Register: IRQ_TYPE_EN_RX_SA_375

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-985 Fields for Register: IRQ_TYPE_EN_RX_SA_375 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-985 Fields for Register: IRQ_TYPE_EN_RX_SA_375 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.979 IRQ_TYPE_STAT_RX_SA_375

- **Name:** IRQ_TYPE_STAT_RX_SA_375
- **Description:** Interrupt Status for each IRQ type of Rx SA 375
- **Size:** 32 bits
- **Offset:** 0xf4c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 375

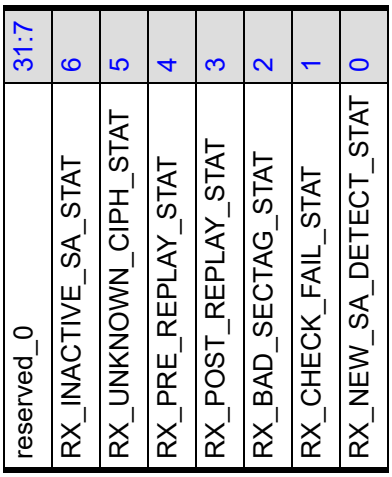


Table 6-986 Fields for Register: IRQ_TYPE_STAT_RX_SA_375

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-986 Fields for Register: IRQ_TYPE_STAT_RX_SA_375 (Continued)

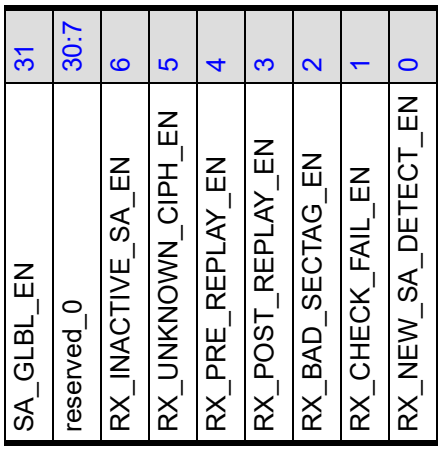
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-986 Fields for Register: IRQ_TYPE_STAT_RX_SA_375 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.980 IRQ_TYPE_EN_RX_SA_376

- **Name:** IRQ_TYPE_EN_RX_SA_376
- **Description:** Interrupt Enable for each IRQ type of Rx SA 376
- **Size:** 32 bits
- **Offset:** 0xf50
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 376



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-987 Fields for Register: IRQ_TYPE_EN_RX_SA_376

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-987 Fields for Register: IRQ_TYPE_EN_RX_SA_376 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-987 Fields for Register: IRQ_TYPE_EN_RX_SA_376 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.981 IRQ_TYPE_STAT_RX_SA_376

- **Name:** IRQ_TYPE_STAT_RX_SA_376
- **Description:** Interrupt Status for each IRQ type of Rx SA 376
- **Size:** 32 bits
- **Offset:** 0xf54
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 376

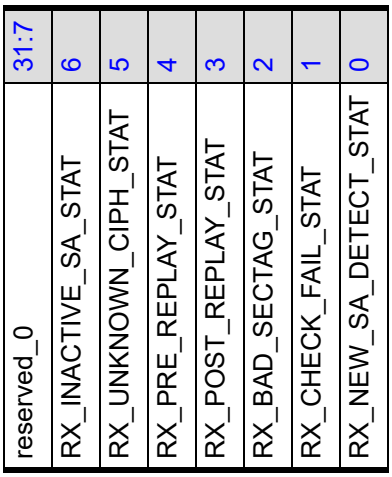


Table 6-988 Fields for Register: IRQ_TYPE_STAT_RX_SA_376

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-988 Fields for Register: IRQ_TYPE_STAT_RX_SA_376 (Continued)

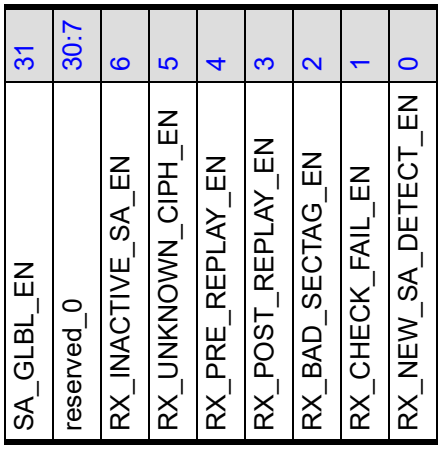
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-988 Fields for Register: IRQ_TYPE_STAT_RX_SA_376 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.982 IRQ_TYPE_EN_RX_SA_377

- **Name:** IRQ_TYPE_EN_RX_SA_377
- **Description:** Interrupt Enable for each IRQ type of Rx SA 377
- **Size:** 32 bits
- **Offset:** 0xf58
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 377



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-989 Fields for Register: IRQ_TYPE_EN_RX_SA_377

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-989 Fields for Register: IRQ_TYPE_EN_RX_SA_377 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-989 Fields for Register: IRQ_TYPE_EN_RX_SA_377 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.983 IRQ_TYPE_STAT_RX_SA_377

- **Name:** IRQ_TYPE_STAT_RX_SA_377
- **Description:** Interrupt Status for each IRQ type of Rx SA 377
- **Size:** 32 bits
- **Offset:** 0xf5c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 377

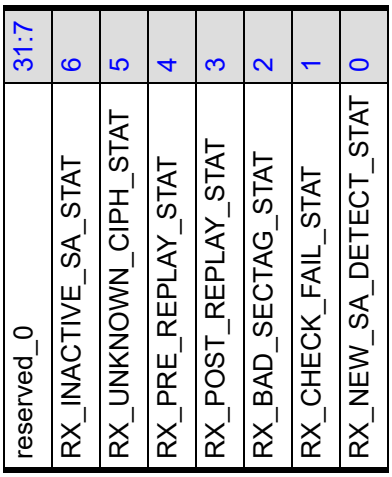


Table 6-990 Fields for Register: IRQ_TYPE_STAT_RX_SA_377

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-990 Fields for Register: IRQ_TYPE_STAT_RX_SA_377 (Continued)

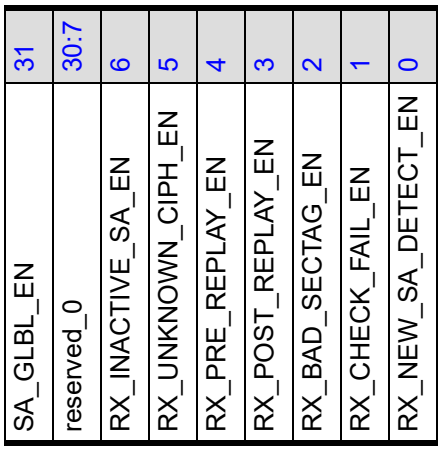
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-990 Fields for Register: IRQ_TYPE_STAT_RX_SA_377 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.984 IRQ_TYPE_EN_RX_SA_378

- **Name:** IRQ_TYPE_EN_RX_SA_378
- **Description:** Interrupt Enable for each IRQ type of Rx SA 378
- **Size:** 32 bits
- **Offset:** 0xf60
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 378



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-991 Fields for Register: IRQ_TYPE_EN_RX_SA_378

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-991 Fields for Register: IRQ_TYPE_EN_RX_SA_378 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-991 Fields for Register: IRQ_TYPE_EN_RX_SA_378 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.985 IRQ_TYPE_STAT_RX_SA_378

- **Name:** IRQ_TYPE_STAT_RX_SA_378
- **Description:** Interrupt Status for each IRQ type of Rx SA 378
- **Size:** 32 bits
- **Offset:** 0xf64
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 378

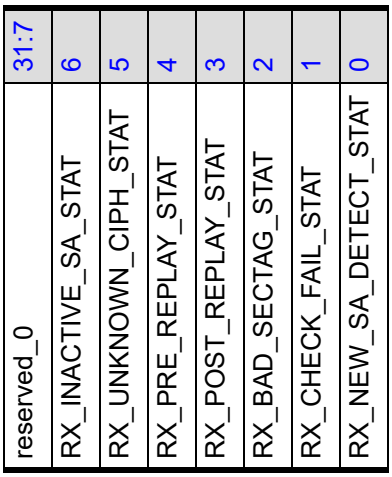


Table 6-992 Fields for Register: IRQ_TYPE_STAT_RX_SA_378

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-992 Fields for Register: IRQ_TYPE_STAT_RX_SA_378 (Continued)

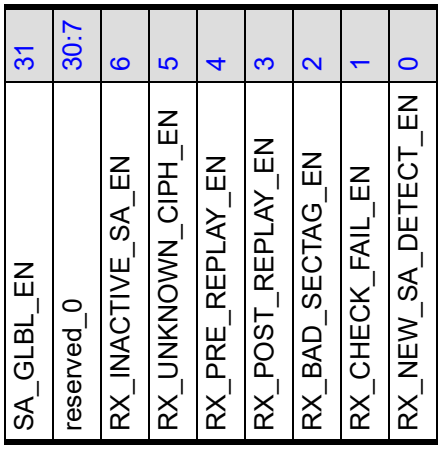
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-992 Fields for Register: IRQ_TYPE_STAT_RX_SA_378 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.986 IRQ_TYPE_EN_RX_SA_379

- **Name:** IRQ_TYPE_EN_RX_SA_379
- **Description:** Interrupt Enable for each IRQ type of Rx SA 379
- **Size:** 32 bits
- **Offset:** 0xf68
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 379



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-993 Fields for Register: IRQ_TYPE_EN_RX_SA_379

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-993 Fields for Register: IRQ_TYPE_EN_RX_SA_379 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-993 Fields for Register: IRQ_TYPE_EN_RX_SA_379 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.987 IRQ_TYPE_STAT_RX_SA_379

- **Name:** IRQ_TYPE_STAT_RX_SA_379
- **Description:** Interrupt Status for each IRQ type of Rx SA 379
- **Size:** 32 bits
- **Offset:** 0xf6c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 379

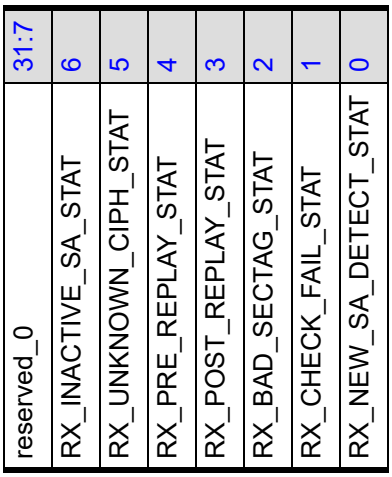


Table 6-994 Fields for Register: IRQ_TYPE_STAT_RX_SA_379

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-994 Fields for Register: IRQ_TYPE_STAT_RX_SA_379 (Continued)

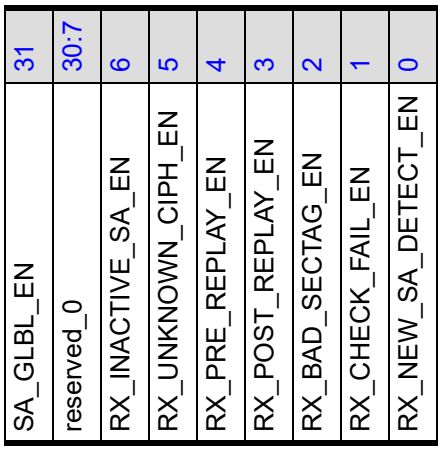
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-994 Fields for Register: IRQ_TYPE_STAT_RX_SA_379 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.988 IRQ_TYPE_EN_RX_SA_380

- **Name:** IRQ_TYPE_EN_RX_SA_380
- **Description:** Interrupt Enable for each IRQ type of Rx SA 380
- **Size:** 32 bits
- **Offset:** 0xf70
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 380



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-995 Fields for Register: IRQ_TYPE_EN_RX_SA_380

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-995 Fields for Register: IRQ_TYPE_EN_RX_SA_380 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-995 Fields for Register: IRQ_TYPE_EN_RX_SA_380 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.989 IRQ_TYPE_STAT_RX_SA_380

- **Name:** IRQ_TYPE_STAT_RX_SA_380
- **Description:** Interrupt Status for each IRQ type of Rx SA 380
- **Size:** 32 bits
- **Offset:** 0xf74
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 380

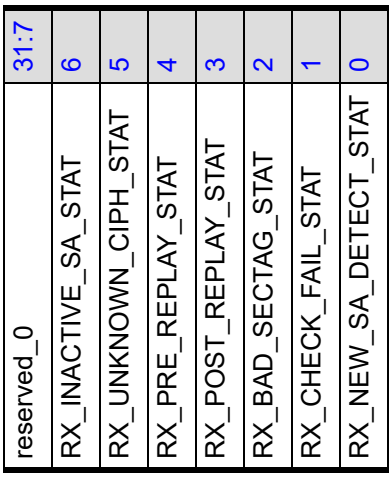


Table 6-996 Fields for Register: IRQ_TYPE_STAT_RX_SA_380

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-996 Fields for Register: IRQ_TYPE_STAT_RX_SA_380 (Continued)

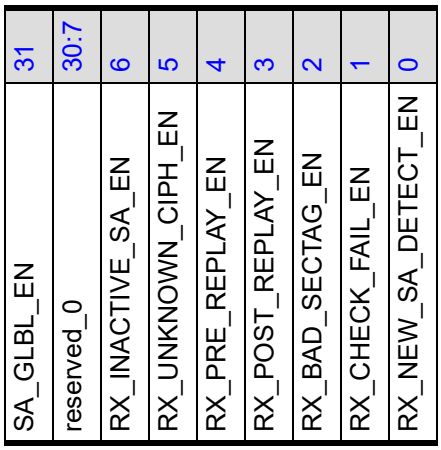
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-996 Fields for Register: IRQ_TYPE_STAT_RX_SA_380 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.990 IRQ_TYPE_EN_RX_SA_381

- **Name:** IRQ_TYPE_EN_RX_SA_381
- **Description:** Interrupt Enable for each IRQ type of Rx SA 381
- **Size:** 32 bits
- **Offset:** 0xf78
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 381



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-997 Fields for Register: IRQ_TYPE_EN_RX_SA_381

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-997 Fields for Register: IRQ_TYPE_EN_RX_SA_381 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-997 Fields for Register: IRQ_TYPE_EN_RX_SA_381 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.991 IRQ_TYPE_STAT_RX_SA_381

- **Name:** IRQ_TYPE_STAT_RX_SA_381
- **Description:** Interrupt Status for each IRQ type of Rx SA 381
- **Size:** 32 bits
- **Offset:** 0xf7c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 381

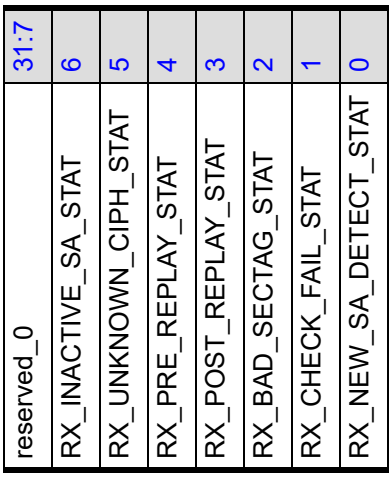


Table 6-998 Fields for Register: IRQ_TYPE_STAT_RX_SA_381

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-998 Fields for Register: IRQ_TYPE_STAT_RX_SA_381 (Continued)

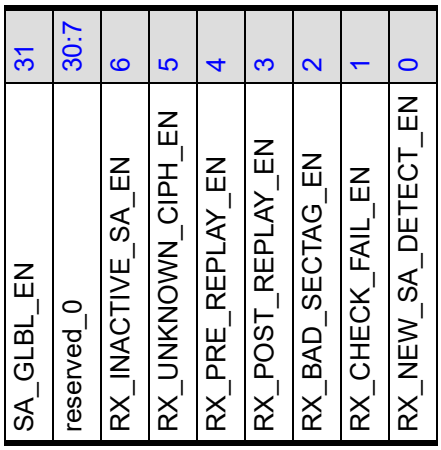
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-998 Fields for Register: IRQ_TYPE_STAT_RX_SA_381 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.992 IRQ_TYPE_EN_RX_SA_382

- **Name:** IRQ_TYPE_EN_RX_SA_382
- **Description:** Interrupt Enable for each IRQ type of Rx SA 382
- **Size:** 32 bits
- **Offset:** 0xf80
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 382



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-999 Fields for Register: IRQ_TYPE_EN_RX_SA_382

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-999 Fields for Register: IRQ_TYPE_EN_RX_SA_382 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-999 Fields for Register: IRQ_TYPE_EN_RX_SA_382 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.993 IRQ_TYPE_STAT_RX_SA_382

- **Name:** IRQ_TYPE_STAT_RX_SA_382
- **Description:** Interrupt Status for each IRQ type of Rx SA 382
- **Size:** 32 bits
- **Offset:** 0xf84
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 382

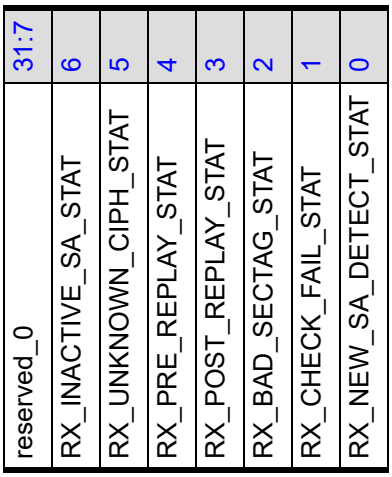


Table 6-1000 Fields for Register: IRQ_TYPE_STAT_RX_SA_382

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1000 Fields for Register: IRQ_TYPE_STAT_RX_SA_382 (Continued)

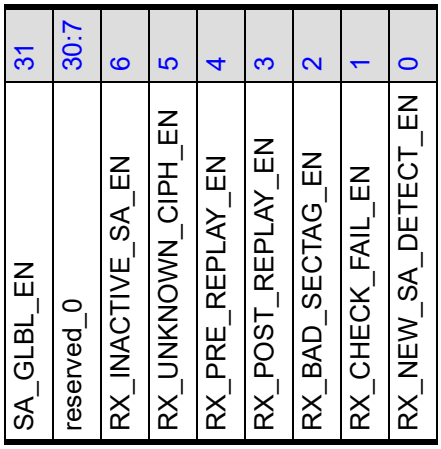
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1000 Fields for Register: IRQ_TYPE_STAT_RX_SA_382 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.994 IRQ_TYPE_EN_RX_SA_383

- **Name:** IRQ_TYPE_EN_RX_SA_383
- **Description:** Interrupt Enable for each IRQ type of Rx SA 383
- **Size:** 32 bits
- **Offset:** 0xf88
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 383



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1001 Fields for Register: IRQ_TYPE_EN_RX_SA_383

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1001 Fields for Register: IRQ_TYPE_EN_RX_SA_383 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1001 Fields for Register: IRQ_TYPE_EN_RX_SA_383 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.995 IRQ_TYPE_STAT_RX_SA_383

- **Name:** IRQ_TYPE_STAT_RX_SA_383
- **Description:** Interrupt Status for each IRQ type of Rx SA 383
- **Size:** 32 bits
- **Offset:** 0xf8c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 383

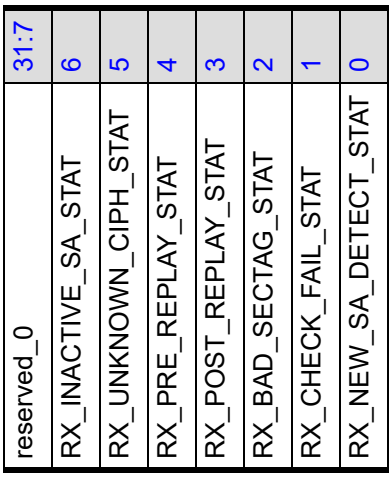


Table 6-1002 Fields for Register: IRQ_TYPE_STAT_RX_SA_383

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1002 Fields for Register: IRQ_TYPE_STAT_RX_SA_383 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1002 Fields for Register: IRQ_TYPE_STAT_RX_SA_383 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.996 IRQ_TYPE_EN_RX_SA_384

- **Name:** IRQ_TYPE_EN_RX_SA_384
- **Description:** Interrupt Enable for each IRQ type of Rx SA 384
- **Size:** 32 bits
- **Offset:** 0xf90
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 384

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1003 Fields for Register: IRQ_TYPE_EN_RX_SA_384

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1003 Fields for Register: IRQ_TYPE_EN_RX_SA_384 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1003 Fields for Register: IRQ_TYPE_EN_RX_SA_384 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.997 IRQ_TYPE_STAT_RX_SA_384

- **Name:** IRQ_TYPE_STAT_RX_SA_384
- **Description:** Interrupt Status for each IRQ type of Rx SA 384
- **Size:** 32 bits
- **Offset:** 0xf94
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 384

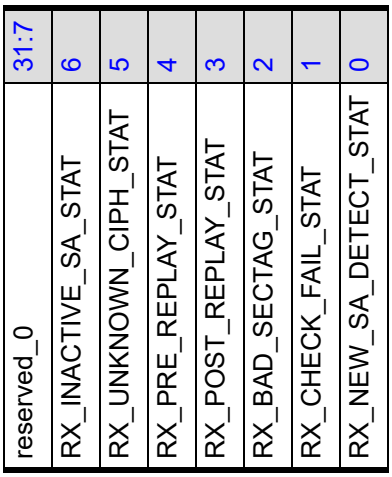


Table 6-1004 Fields for Register: IRQ_TYPE_STAT_RX_SA_384

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1004 Fields for Register: IRQ_TYPE_STAT_RX_SA_384 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1004 Fields for Register: IRQ_TYPE_STAT_RX_SA_384 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.998 IRQ_TYPE_EN_RX_SA_385

- **Name:** IRQ_TYPE_EN_RX_SA_385
- **Description:** Interrupt Enable for each IRQ type of Rx SA 385
- **Size:** 32 bits
- **Offset:** 0xf98
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 385

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1005 Fields for Register: IRQ_TYPE_EN_RX_SA_385

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1005 Fields for Register: IRQ_TYPE_EN_RX_SA_385 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1005 Fields for Register: IRQ_TYPE_EN_RX_SA_385 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.999 IRQ_TYPE_STAT_RX_SA_385

- **Name:** IRQ_TYPE_STAT_RX_SA_385
- **Description:** Interrupt Status for each IRQ type of Rx SA 385
- **Size:** 32 bits
- **Offset:** 0xf9c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 385

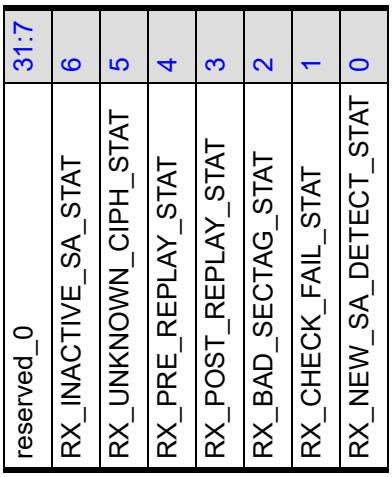


Table 6-1006 Fields for Register: IRQ_TYPE_STAT_RX_SA_385

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1006 Fields for Register: IRQ_TYPE_STAT_RX_SA_385 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1006 Fields for Register: IRQ_TYPE_STAT_RX_SA_385 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1000 IRQ_TYPE_EN_RX_SA_386

- **Name:** IRQ_TYPE_EN_RX_SA_386
- **Description:** Interrupt Enable for each IRQ type of Rx SA 386
- **Size:** 32 bits
- **Offset:** 0xfa0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 386

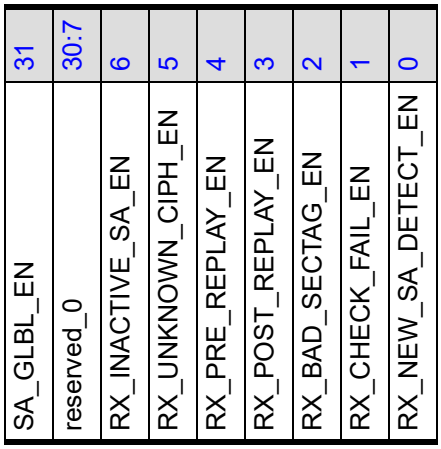


Table 6-1007 Fields for Register: IRQ_TYPE_EN_RX_SA_386

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1007 Fields for Register: IRQ_TYPE_EN_RX_SA_386 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1007 Fields for Register: IRQ_TYPE_EN_RX_SA_386 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1001 IRQ_TYPE_STAT_RX_SA_386

- **Name:** IRQ_TYPE_STAT_RX_SA_386
- **Description:** Interrupt Status for each IRQ type of Rx SA 386
- **Size:** 32 bits
- **Offset:** 0xfa4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 386

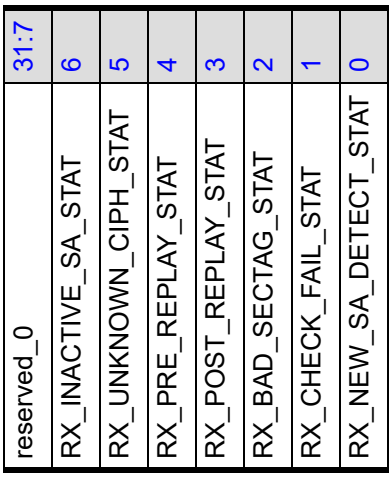


Table 6-1008 Fields for Register: IRQ_TYPE_STAT_RX_SA_386

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1008 Fields for Register: IRQ_TYPE_STAT_RX_SA_386 (Continued)

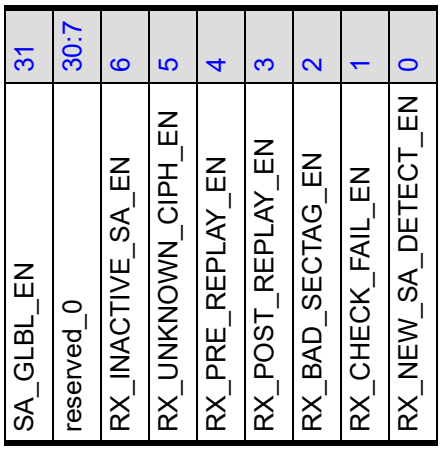
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1008 Fields for Register: IRQ_TYPE_STAT_RX_SA_386 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1002 IRQ_TYPE_EN_RX_SA_387

- **Name:** IRQ_TYPE_EN_RX_SA_387
- **Description:** Interrupt Enable for each IRQ type of Rx SA 387
- **Size:** 32 bits
- **Offset:** 0xfa8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 387



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1009 Fields for Register: IRQ_TYPE_EN_RX_SA_387

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1009 Fields for Register: IRQ_TYPE_EN_RX_SA_387 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1009 Fields for Register: IRQ_TYPE_EN_RX_SA_387 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1003 IRQ_TYPE_STAT_RX_SA_387

- **Name:** IRQ_TYPE_STAT_RX_SA_387
- **Description:** Interrupt Status for each IRQ type of Rx SA 387
- **Size:** 32 bits
- **Offset:** 0xfac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 387

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1010 Fields for Register: IRQ_TYPE_STAT_RX_SA_387

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1010 Fields for Register: IRQ_TYPE_STAT_RX_SA_387 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1010 Fields for Register: IRQ_TYPE_STAT_RX_SA_387 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1004 IRQ_TYPE_EN_RX_SA_388

- **Name:** IRQ_TYPE_EN_RX_SA_388
- **Description:** Interrupt Enable for each IRQ type of Rx SA 388
- **Size:** 32 bits
- **Offset:** 0xfb0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 388

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1011 Fields for Register: IRQ_TYPE_EN_RX_SA_388

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1011 Fields for Register: IRQ_TYPE_EN_RX_SA_388 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1011 Fields for Register: IRQ_TYPE_EN_RX_SA_388 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1005 IRQ_TYPE_STAT_RX_SA_388

- **Name:** IRQ_TYPE_STAT_RX_SA_388
- **Description:** Interrupt Status for each IRQ type of Rx SA 388
- **Size:** 32 bits
- **Offset:** 0xfb4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 388

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1012 Fields for Register: IRQ_TYPE_STAT_RX_SA_388

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1012 Fields for Register: IRQ_TYPE_STAT_RX_SA_388 (Continued)

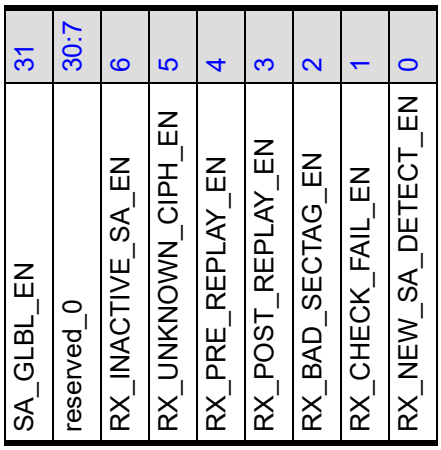
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1012 Fields for Register: IRQ_TYPE_STAT_RX_SA_388 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1006 IRQ_TYPE_EN_RX_SA_389

- **Name:** IRQ_TYPE_EN_RX_SA_389
- **Description:** Interrupt Enable for each IRQ type of Rx SA 389
- **Size:** 32 bits
- **Offset:** 0xfb8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 389



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1013 Fields for Register: IRQ_TYPE_EN_RX_SA_389

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1013 Fields for Register: IRQ_TYPE_EN_RX_SA_389 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1013 Fields for Register: IRQ_TYPE_EN_RX_SA_389 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1007 IRQ_TYPE_STAT_RX_SA_389

- **Name:** IRQ_TYPE_STAT_RX_SA_389
- **Description:** Interrupt Status for each IRQ type of Rx SA 389
- **Size:** 32 bits
- **Offset:** 0xfbc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 389

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1014 Fields for Register: IRQ_TYPE_STAT_RX_SA_389

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1014 Fields for Register: IRQ_TYPE_STAT_RX_SA_389 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1014 Fields for Register: IRQ_TYPE_STAT_RX_SA_389 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1008 IRQ_TYPE_EN_RX_SA_390

- **Name:** IRQ_TYPE_EN_RX_SA_390
- **Description:** Interrupt Enable for each IRQ type of Rx SA 390
- **Size:** 32 bits
- **Offset:** 0xfc0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 390

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1015 Fields for Register: IRQ_TYPE_EN_RX_SA_390

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1015 Fields for Register: IRQ_TYPE_EN_RX_SA_390 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1015 Fields for Register: IRQ_TYPE_EN_RX_SA_390 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1009 IRQ_TYPE_STAT_RX_SA_390

- **Name:** IRQ_TYPE_STAT_RX_SA_390
- **Description:** Interrupt Status for each IRQ type of Rx SA 390
- **Size:** 32 bits
- **Offset:** 0xfc4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 390

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1016 Fields for Register: IRQ_TYPE_STAT_RX_SA_390

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1016 Fields for Register: IRQ_TYPE_STAT_RX_SA_390 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1016 Fields for Register: IRQ_TYPE_STAT_RX_SA_390 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1010 IRQ_TYPE_EN_RX_SA_391

- **Name:** IRQ_TYPE_EN_RX_SA_391
- **Description:** Interrupt Enable for each IRQ type of Rx SA 391
- **Size:** 32 bits
- **Offset:** 0xfc8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 391

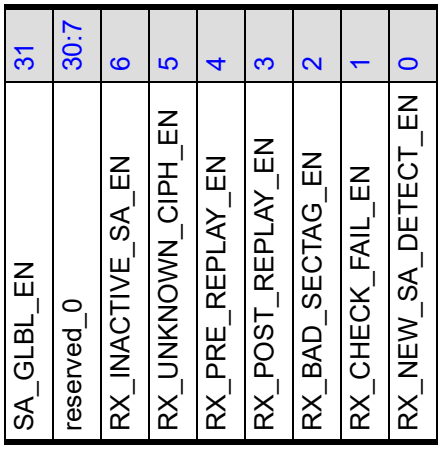


Table 6-1017 Fields for Register: IRQ_TYPE_EN_RX_SA_391

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1017 Fields for Register: IRQ_TYPE_EN_RX_SA_391 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1017 Fields for Register: IRQ_TYPE_EN_RX_SA_391 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1011 IRQ_TYPE_STAT_RX_SA_391

- **Name:** IRQ_TYPE_STAT_RX_SA_391
- **Description:** Interrupt Status for each IRQ type of Rx SA 391
- **Size:** 32 bits
- **Offset:** 0xfcc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 391

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1018 Fields for Register: IRQ_TYPE_STAT_RX_SA_391

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1018 Fields for Register: IRQ_TYPE_STAT_RX_SA_391 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1018 Fields for Register: IRQ_TYPE_STAT_RX_SA_391 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1012 IRQ_TYPE_EN_RX_SA_392

- **Name:** IRQ_TYPE_EN_RX_SA_392
- **Description:** Interrupt Enable for each IRQ type of Rx SA 392
- **Size:** 32 bits
- **Offset:** 0xfd0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 392

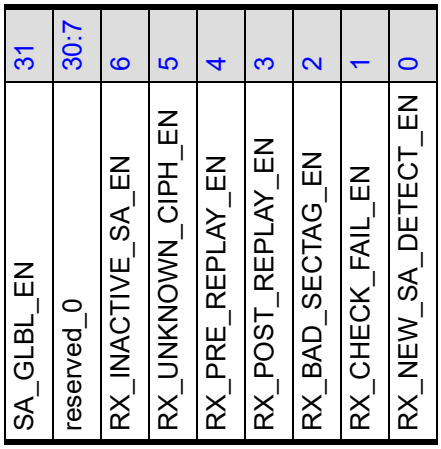


Table 6-1019 Fields for Register: IRQ_TYPE_EN_RX_SA_392

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1019 Fields for Register: IRQ_TYPE_EN_RX_SA_392 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1019 Fields for Register: IRQ_TYPE_EN_RX_SA_392 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1013 IRQ_TYPE_STAT_RX_SA_392

- **Name:** IRQ_TYPE_STAT_RX_SA_392
- **Description:** Interrupt Status for each IRQ type of Rx SA 392
- **Size:** 32 bits
- **Offset:** 0xfd4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 392

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1020 Fields for Register: IRQ_TYPE_STAT_RX_SA_392

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1020 Fields for Register: IRQ_TYPE_STAT_RX_SA_392 (Continued)

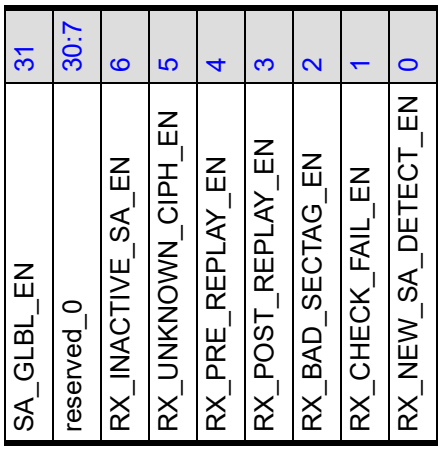
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1020 Fields for Register: IRQ_TYPE_STAT_RX_SA_392 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1014 IRQ_TYPE_EN_RX_SA_393

- **Name:** IRQ_TYPE_EN_RX_SA_393
- **Description:** Interrupt Enable for each IRQ type of Rx SA 393
- **Size:** 32 bits
- **Offset:** 0xfd8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 393



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1021 Fields for Register: IRQ_TYPE_EN_RX_SA_393

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1021 Fields for Register: IRQ_TYPE_EN_RX_SA_393 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1021 Fields for Register: IRQ_TYPE_EN_RX_SA_393 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1015 IRQ_TYPE_STAT_RX_SA_393

- **Name:** IRQ_TYPE_STAT_RX_SA_393
- **Description:** Interrupt Status for each IRQ type of Rx SA 393
- **Size:** 32 bits
- **Offset:** 0xfdc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 393

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1022 Fields for Register: IRQ_TYPE_STAT_RX_SA_393

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1022 Fields for Register: IRQ_TYPE_STAT_RX_SA_393 (Continued)

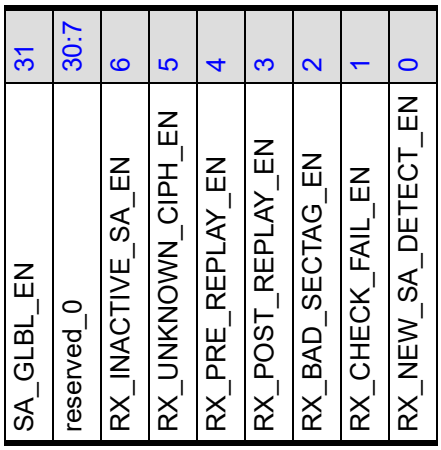
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1022 Fields for Register: IRQ_TYPE_STAT_RX_SA_393 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1016 IRQ_TYPE_EN_RX_SA_394

- **Name:** IRQ_TYPE_EN_RX_SA_394
- **Description:** Interrupt Enable for each IRQ type of Rx SA 394
- **Size:** 32 bits
- **Offset:** 0xfe0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 394



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1023 Fields for Register: IRQ_TYPE_EN_RX_SA_394

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1023 Fields for Register: IRQ_TYPE_EN_RX_SA_394 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1023 Fields for Register: IRQ_TYPE_EN_RX_SA_394 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1017 IRQ_TYPE_STAT_RX_SA_394

- **Name:** IRQ_TYPE_STAT_RX_SA_394
- **Description:** Interrupt Status for each IRQ type of Rx SA 394
- **Size:** 32 bits
- **Offset:** 0xfe4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 394

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1024 Fields for Register: IRQ_TYPE_STAT_RX_SA_394

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1024 Fields for Register: IRQ_TYPE_STAT_RX_SA_394 (Continued)

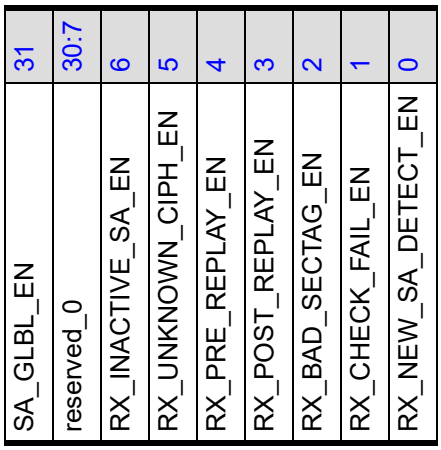
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1024 Fields for Register: IRQ_TYPE_STAT_RX_SA_394 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1018 IRQ_TYPE_EN_RX_SA_395

- **Name:** IRQ_TYPE_EN_RX_SA_395
- **Description:** Interrupt Enable for each IRQ type of Rx SA 395
- **Size:** 32 bits
- **Offset:** 0xfe8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 395



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1025 Fields for Register: IRQ_TYPE_EN_RX_SA_395

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1025 Fields for Register: IRQ_TYPE_EN_RX_SA_395 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1025 Fields for Register: IRQ_TYPE_EN_RX_SA_395 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1019 IRQ_TYPE_STAT_RX_SA_395

- **Name:** IRQ_TYPE_STAT_RX_SA_395
- **Description:** Interrupt Status for each IRQ type of Rx SA 395
- **Size:** 32 bits
- **Offset:** 0xfec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 395

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1026 Fields for Register: IRQ_TYPE_STAT_RX_SA_395

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1026 Fields for Register: IRQ_TYPE_STAT_RX_SA_395 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1026 Fields for Register: IRQ_TYPE_STAT_RX_SA_395 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1020 IRQ_TYPE_EN_RX_SA_396

- **Name:** IRQ_TYPE_EN_RX_SA_396
- **Description:** Interrupt Enable for each IRQ type of Rx SA 396
- **Size:** 32 bits
- **Offset:** 0xff0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 396

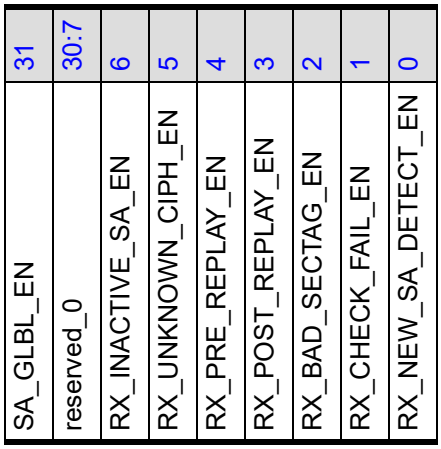


Table 6-1027 Fields for Register: IRQ_TYPE_EN_RX_SA_396

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1027 Fields for Register: IRQ_TYPE_EN_RX_SA_396 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1027 Fields for Register: IRQ_TYPE_EN_RX_SA_396 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1021 IRQ_TYPE_STAT_RX_SA_396

- **Name:** IRQ_TYPE_STAT_RX_SA_396
- **Description:** Interrupt Status for each IRQ type of Rx SA 396
- **Size:** 32 bits
- **Offset:** 0xff4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 396

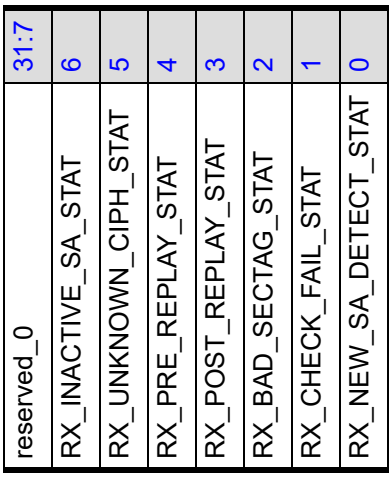


Table 6-1028 Fields for Register: IRQ_TYPE_STAT_RX_SA_396

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1028 Fields for Register: IRQ_TYPE_STAT_RX_SA_396 (Continued)

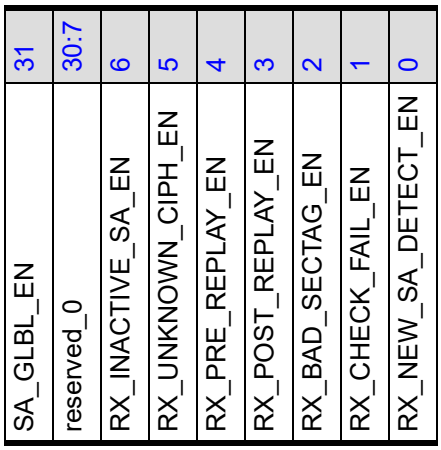
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1028 Fields for Register: IRQ_TYPE_STAT_RX_SA_396 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1022 IRQ_TYPE_EN_RX_SA_397

- **Name:** IRQ_TYPE_EN_RX_SA_397
- **Description:** Interrupt Enable for each IRQ type of Rx SA 397
- **Size:** 32 bits
- **Offset:** 0xff8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 397



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1029 Fields for Register: IRQ_TYPE_EN_RX_SA_397

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1029 Fields for Register: IRQ_TYPE_EN_RX_SA_397 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1029 Fields for Register: IRQ_TYPE_EN_RX_SA_397 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1023 IRQ_TYPE_STAT_RX_SA_397

- **Name:** IRQ_TYPE_STAT_RX_SA_397
- **Description:** Interrupt Status for each IRQ type of Rx SA 397
- **Size:** 32 bits
- **Offset:** 0xffc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 397

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1030 Fields for Register: IRQ_TYPE_STAT_RX_SA_397

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1030 Fields for Register: IRQ_TYPE_STAT_RX_SA_397 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1030 Fields for Register: IRQ_TYPE_STAT_RX_SA_397 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1024 IRQ_TYPE_EN_RX_SA_398

- **Name:** IRQ_TYPE_EN_RX_SA_398
- **Description:** Interrupt Enable for each IRQ type of Rx SA 398
- **Size:** 32 bits
- **Offset:** 0x1000
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 398

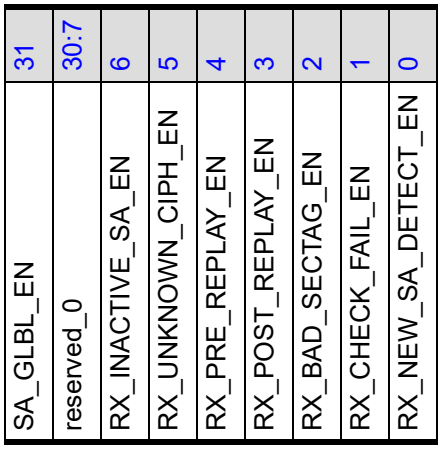


Table 6-1031 Fields for Register: IRQ_TYPE_EN_RX_SA_398

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1031 Fields for Register: IRQ_TYPE_EN_RX_SA_398 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1031 Fields for Register: IRQ_TYPE_EN_RX_SA_398 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1025 IRQ_TYPE_STAT_RX_SA_398

- **Name:** IRQ_TYPE_STAT_RX_SA_398
- **Description:** Interrupt Status for each IRQ type of Rx SA 398
- **Size:** 32 bits
- **Offset:** 0x1004
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 398

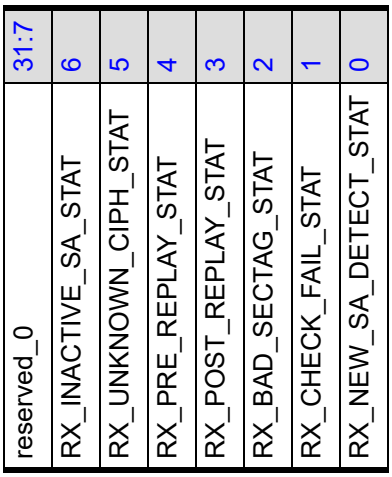


Table 6-1032 Fields for Register: IRQ_TYPE_STAT_RX_SA_398

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1032 Fields for Register: IRQ_TYPE_STAT_RX_SA_398 (Continued)

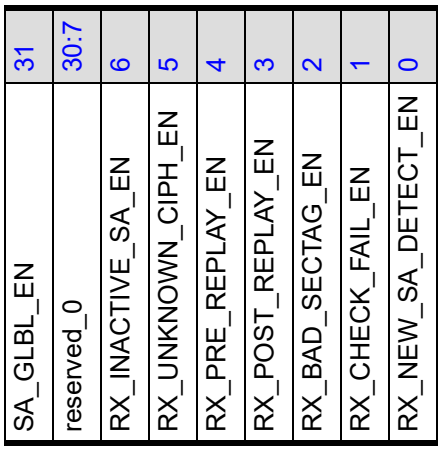
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1032 Fields for Register: IRQ_TYPE_STAT_RX_SA_398 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1026 IRQ_TYPE_EN_RX_SA_399

- **Name:** IRQ_TYPE_EN_RX_SA_399
- **Description:** Interrupt Enable for each IRQ type of Rx SA 399
- **Size:** 32 bits
- **Offset:** 0x1008
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 399



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1033 Fields for Register: IRQ_TYPE_EN_RX_SA_399

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1033 Fields for Register: IRQ_TYPE_EN_RX_SA_399 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1033 Fields for Register: IRQ_TYPE_EN_RX_SA_399 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1027 IRQ_TYPE_STAT_RX_SA_399

- **Name:** IRQ_TYPE_STAT_RX_SA_399
- **Description:** Interrupt Status for each IRQ type of Rx SA 399
- **Size:** 32 bits
- **Offset:** 0x100c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 399

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1034 Fields for Register: IRQ_TYPE_STAT_RX_SA_399

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1034 Fields for Register: IRQ_TYPE_STAT_RX_SA_399 (Continued)

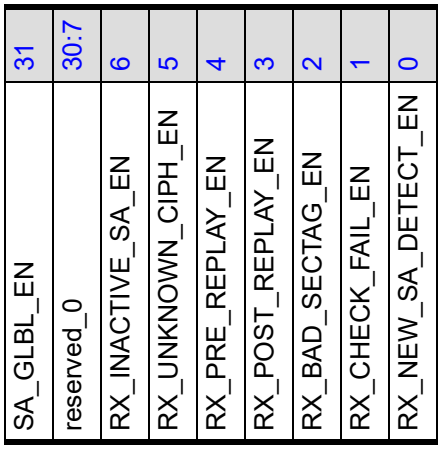
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1034 Fields for Register: IRQ_TYPE_STAT_RX_SA_399 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1028 IRQ_TYPE_EN_RX_SA_400

- **Name:** IRQ_TYPE_EN_RX_SA_400
- **Description:** Interrupt Enable for each IRQ type of Rx SA 400
- **Size:** 32 bits
- **Offset:** 0x1010
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 400



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1035 Fields for Register: IRQ_TYPE_EN_RX_SA_400

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1035 Fields for Register: IRQ_TYPE_EN_RX_SA_400 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1035 Fields for Register: IRQ_TYPE_EN_RX_SA_400 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1029 IRQ_TYPE_STAT_RX_SA_400

- **Name:** IRQ_TYPE_STAT_RX_SA_400
- **Description:** Interrupt Status for each IRQ type of Rx SA 400
- **Size:** 32 bits
- **Offset:** 0x1014
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 400

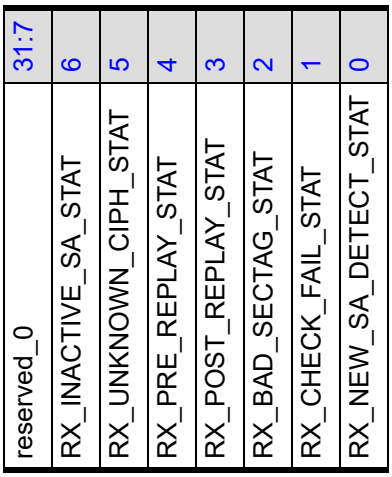


Table 6-1036 Fields for Register: IRQ_TYPE_STAT_RX_SA_400

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1036 Fields for Register: IRQ_TYPE_STAT_RX_SA_400 (Continued)

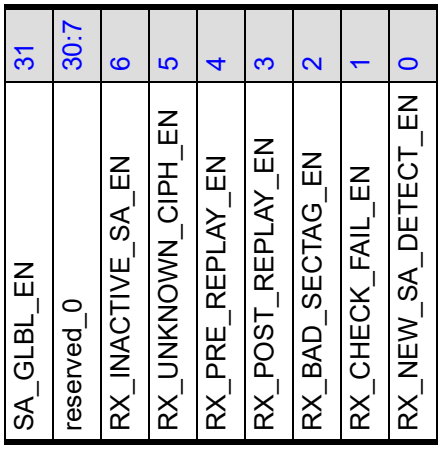
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1036 Fields for Register: IRQ_TYPE_STAT_RX_SA_400 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1030 IRQ_TYPE_EN_RX_SA_401

- **Name:** IRQ_TYPE_EN_RX_SA_401
- **Description:** Interrupt Enable for each IRQ type of Rx SA 401
- **Size:** 32 bits
- **Offset:** 0x1018
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 401



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1037 Fields for Register: IRQ_TYPE_EN_RX_SA_401

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1037 Fields for Register: IRQ_TYPE_EN_RX_SA_401 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1037 Fields for Register: IRQ_TYPE_EN_RX_SA_401 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1031 IRQ_TYPE_STAT_RX_SA_401

- **Name:** IRQ_TYPE_STAT_RX_SA_401
- **Description:** Interrupt Status for each IRQ type of Rx SA 401
- **Size:** 32 bits
- **Offset:** 0x101c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 401

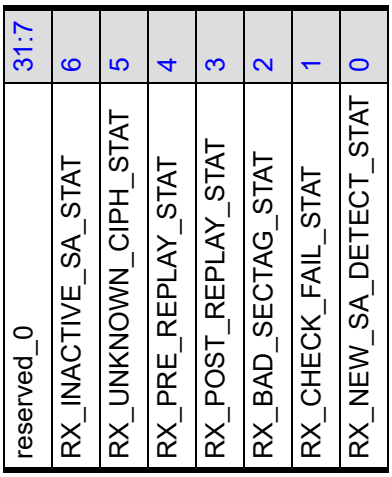


Table 6-1038 Fields for Register: IRQ_TYPE_STAT_RX_SA_401

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1038 Fields for Register: IRQ_TYPE_STAT_RX_SA_401 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1038 Fields for Register: IRQ_TYPE_STAT_RX_SA_401 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1032 IRQ_TYPE_EN_RX_SA_402

- **Name:** IRQ_TYPE_EN_RX_SA_402
- **Description:** Interrupt Enable for each IRQ type of Rx SA 402
- **Size:** 32 bits
- **Offset:** 0x1020
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 402

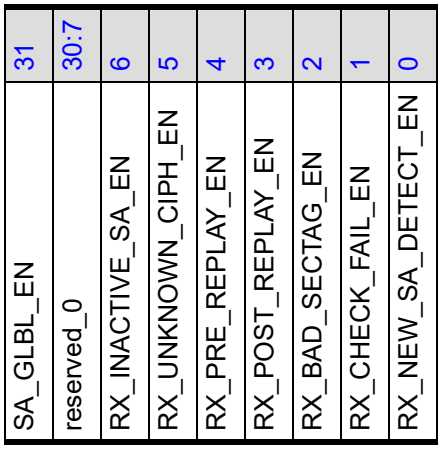


Table 6-1039 Fields for Register: IRQ_TYPE_EN_RX_SA_402

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1039 Fields for Register: IRQ_TYPE_EN_RX_SA_402 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1039 Fields for Register: IRQ_TYPE_EN_RX_SA_402 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1033 IRQ_TYPE_STAT_RX_SA_402

- **Name:** IRQ_TYPE_STAT_RX_SA_402
- **Description:** Interrupt Status for each IRQ type of Rx SA 402
- **Size:** 32 bits
- **Offset:** 0x1024
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 402

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1040 Fields for Register: IRQ_TYPE_STAT_RX_SA_402

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1040 Fields for Register: IRQ_TYPE_STAT_RX_SA_402 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1040 Fields for Register: IRQ_TYPE_STAT_RX_SA_402 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1034 IRQ_TYPE_EN_RX_SA_403

- **Name:** IRQ_TYPE_EN_RX_SA_403
- **Description:** Interrupt Enable for each IRQ type of Rx SA 403
- **Size:** 32 bits
- **Offset:** 0x1028
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 403

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1041 Fields for Register: IRQ_TYPE_EN_RX_SA_403

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1041 Fields for Register: IRQ_TYPE_EN_RX_SA_403 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1041 Fields for Register: IRQ_TYPE_EN_RX_SA_403 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1035 IRQ_TYPE_STAT_RX_SA_403

- **Name:** IRQ_TYPE_STAT_RX_SA_403
- **Description:** Interrupt Status for each IRQ type of Rx SA 403
- **Size:** 32 bits
- **Offset:** 0x102c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 403

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1042 Fields for Register: IRQ_TYPE_STAT_RX_SA_403

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1042 Fields for Register: IRQ_TYPE_STAT_RX_SA_403 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1042 Fields for Register: IRQ_TYPE_STAT_RX_SA_403 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1036 IRQ_TYPE_EN_RX_SA_404

- **Name:** IRQ_TYPE_EN_RX_SA_404
- **Description:** Interrupt Enable for each IRQ type of Rx SA 404
- **Size:** 32 bits
- **Offset:** 0x1030
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 404

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1043 Fields for Register: IRQ_TYPE_EN_RX_SA_404

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1043 Fields for Register: IRQ_TYPE_EN_RX_SA_404 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1043 Fields for Register: IRQ_TYPE_EN_RX_SA_404 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1037 IRQ_TYPE_STAT_RX_SA_404

- **Name:** IRQ_TYPE_STAT_RX_SA_404
- **Description:** Interrupt Status for each IRQ type of Rx SA 404
- **Size:** 32 bits
- **Offset:** 0x1034
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 404

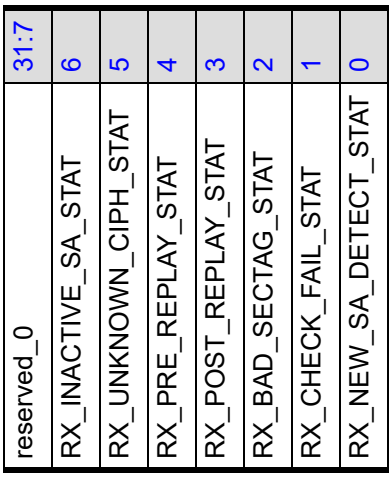


Table 6-1044 Fields for Register: IRQ_TYPE_STAT_RX_SA_404

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1044 Fields for Register: IRQ_TYPE_STAT_RX_SA_404 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1044 Fields for Register: IRQ_TYPE_STAT_RX_SA_404 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1038 IRQ_TYPE_EN_RX_SA_405

- **Name:** IRQ_TYPE_EN_RX_SA_405
- **Description:** Interrupt Enable for each IRQ type of Rx SA 405
- **Size:** 32 bits
- **Offset:** 0x1038
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 405

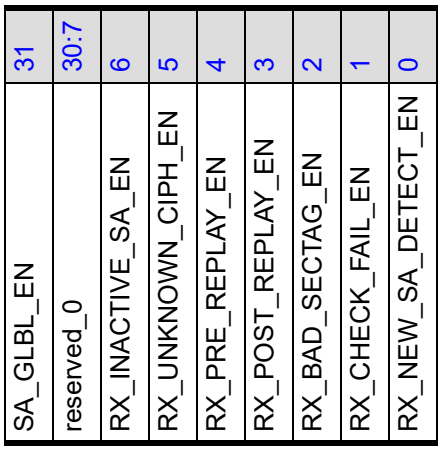


Table 6-1045 Fields for Register: IRQ_TYPE_EN_RX_SA_405

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1045 Fields for Register: IRQ_TYPE_EN_RX_SA_405 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1045 Fields for Register: IRQ_TYPE_EN_RX_SA_405 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1039 IRQ_TYPE_STAT_RX_SA_405

- **Name:** IRQ_TYPE_STAT_RX_SA_405
- **Description:** Interrupt Status for each IRQ type of Rx SA 405
- **Size:** 32 bits
- **Offset:** 0x103c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 405

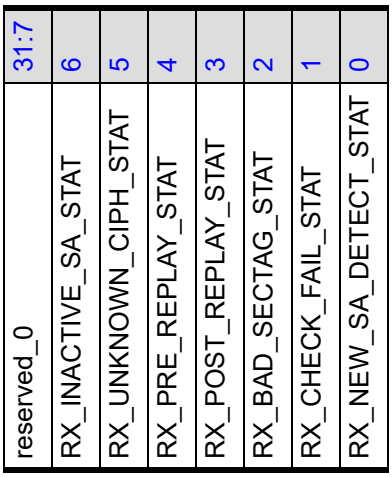


Table 6-1046 Fields for Register: IRQ_TYPE_STAT_RX_SA_405

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1046 Fields for Register: IRQ_TYPE_STAT_RX_SA_405 (Continued)

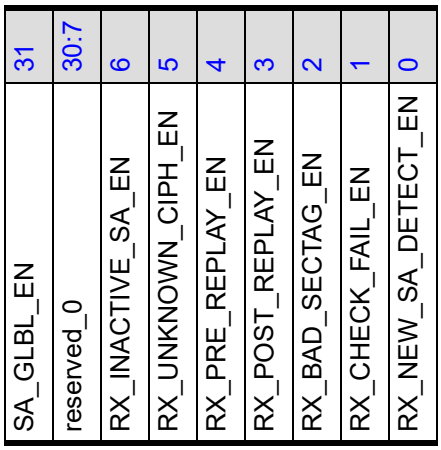
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1046 Fields for Register: IRQ_TYPE_STAT_RX_SA_405 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1040 IRQ_TYPE_EN_RX_SA_406

- **Name:** IRQ_TYPE_EN_RX_SA_406
- **Description:** Interrupt Enable for each IRQ type of Rx SA 406
- **Size:** 32 bits
- **Offset:** 0x1040
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 406



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1047 Fields for Register: IRQ_TYPE_EN_RX_SA_406

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1047 Fields for Register: IRQ_TYPE_EN_RX_SA_406 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1047 Fields for Register: IRQ_TYPE_EN_RX_SA_406 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1041 IRQ_TYPE_STAT_RX_SA_406

- **Name:** IRQ_TYPE_STAT_RX_SA_406
- **Description:** Interrupt Status for each IRQ type of Rx SA 406
- **Size:** 32 bits
- **Offset:** 0x1044
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 406

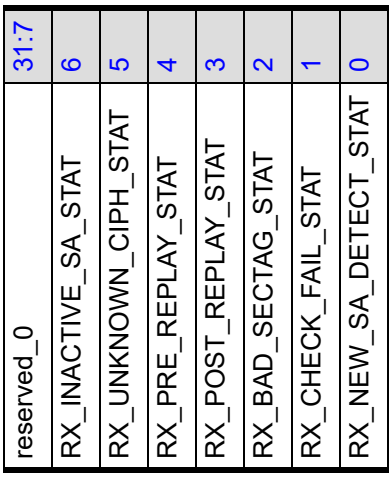


Table 6-1048 Fields for Register: IRQ_TYPE_STAT_RX_SA_406

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1048 Fields for Register: IRQ_TYPE_STAT_RX_SA_406 (Continued)

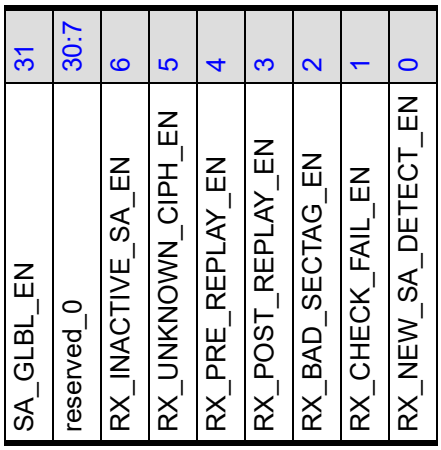
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1048 Fields for Register: IRQ_TYPE_STAT_RX_SA_406 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1042 IRQ_TYPE_EN_RX_SA_407

- **Name:** IRQ_TYPE_EN_RX_SA_407
- **Description:** Interrupt Enable for each IRQ type of Rx SA 407
- **Size:** 32 bits
- **Offset:** 0x1048
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 407



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1049 Fields for Register: IRQ_TYPE_EN_RX_SA_407

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1049 Fields for Register: IRQ_TYPE_EN_RX_SA_407 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1049 Fields for Register: IRQ_TYPE_EN_RX_SA_407 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1043 IRQ_TYPE_STAT_RX_SA_407

- **Name:** IRQ_TYPE_STAT_RX_SA_407
- **Description:** Interrupt Status for each IRQ type of Rx SA 407
- **Size:** 32 bits
- **Offset:** 0x104c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 407

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1050 Fields for Register: IRQ_TYPE_STAT_RX_SA_407

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1050 Fields for Register: IRQ_TYPE_STAT_RX_SA_407 (Continued)

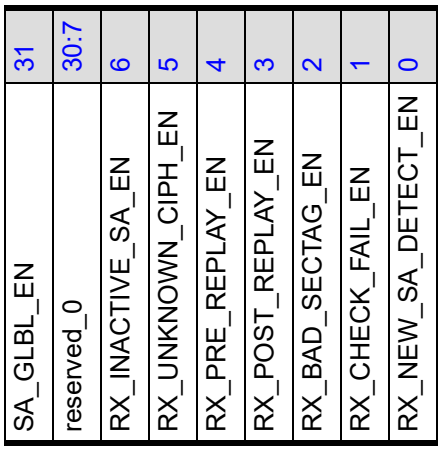
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1050 Fields for Register: IRQ_TYPE_STAT_RX_SA_407 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1044 IRQ_TYPE_EN_RX_SA_408

- **Name:** IRQ_TYPE_EN_RX_SA_408
- **Description:** Interrupt Enable for each IRQ type of Rx SA 408
- **Size:** 32 bits
- **Offset:** 0x1050
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 408



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1051 Fields for Register: IRQ_TYPE_EN_RX_SA_408

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1051 Fields for Register: IRQ_TYPE_EN_RX_SA_408 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1051 Fields for Register: IRQ_TYPE_EN_RX_SA_408 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1045 IRQ_TYPE_STAT_RX_SA_408

- **Name:** IRQ_TYPE_STAT_RX_SA_408
- **Description:** Interrupt Status for each IRQ type of Rx SA 408
- **Size:** 32 bits
- **Offset:** 0x1054
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 408

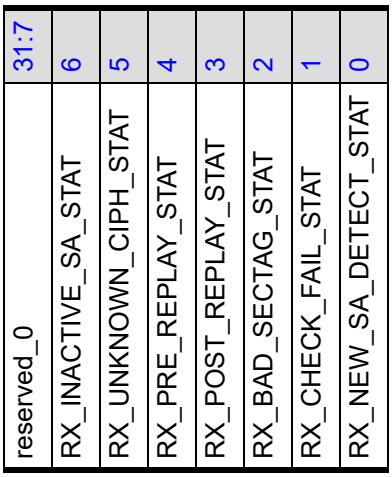


Table 6-1052 Fields for Register: IRQ_TYPE_STAT_RX_SA_408

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1052 Fields for Register: IRQ_TYPE_STAT_RX_SA_408 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1052 Fields for Register: IRQ_TYPE_STAT_RX_SA_408 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1046 IRQ_TYPE_EN_RX_SA_409

- **Name:** IRQ_TYPE_EN_RX_SA_409
- **Description:** Interrupt Enable for each IRQ type of Rx SA 409
- **Size:** 32 bits
- **Offset:** 0x1058
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 409

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1053 Fields for Register: IRQ_TYPE_EN_RX_SA_409

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1053 Fields for Register: IRQ_TYPE_EN_RX_SA_409 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1053 Fields for Register: IRQ_TYPE_EN_RX_SA_409 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1047 IRQ_TYPE_STAT_RX_SA_409

- **Name:** IRQ_TYPE_STAT_RX_SA_409
- **Description:** Interrupt Status for each IRQ type of Rx SA 409
- **Size:** 32 bits
- **Offset:** 0x105c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 409

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1054 Fields for Register: IRQ_TYPE_STAT_RX_SA_409

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1054 Fields for Register: IRQ_TYPE_STAT_RX_SA_409 (Continued)

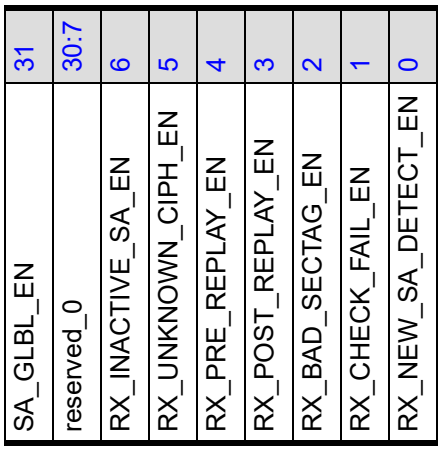
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1054 Fields for Register: IRQ_TYPE_STAT_RX_SA_409 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1048 IRQ_TYPE_EN_RX_SA_410

- **Name:** IRQ_TYPE_EN_RX_SA_410
- **Description:** Interrupt Enable for each IRQ type of Rx SA 410
- **Size:** 32 bits
- **Offset:** 0x1060
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 410



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1055 Fields for Register: IRQ_TYPE_EN_RX_SA_410

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1055 Fields for Register: IRQ_TYPE_EN_RX_SA_410 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1055 Fields for Register: IRQ_TYPE_EN_RX_SA_410 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1049 IRQ_TYPE_STAT_RX_SA_410

- **Name:** IRQ_TYPE_STAT_RX_SA_410
- **Description:** Interrupt Status for each IRQ type of Rx SA 410
- **Size:** 32 bits
- **Offset:** 0x1064
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 410

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1056 Fields for Register: IRQ_TYPE_STAT_RX_SA_410

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1056 Fields for Register: IRQ_TYPE_STAT_RX_SA_410 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1056 Fields for Register: IRQ_TYPE_STAT_RX_SA_410 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1050 IRQ_TYPE_EN_RX_SA_411

- **Name:** IRQ_TYPE_EN_RX_SA_411
- **Description:** Interrupt Enable for each IRQ type of Rx SA 411
- **Size:** 32 bits
- **Offset:** 0x1068
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 411

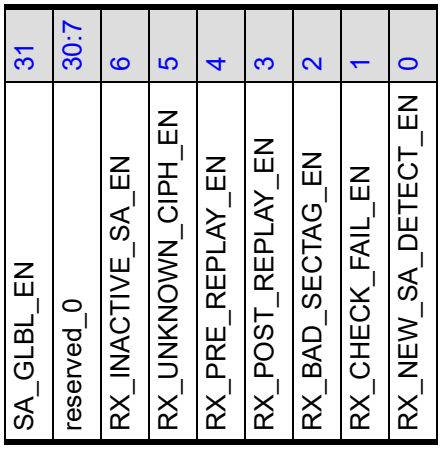


Table 6-1057 Fields for Register: IRQ_TYPE_EN_RX_SA_411

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1057 Fields for Register: IRQ_TYPE_EN_RX_SA_411 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1057 Fields for Register: IRQ_TYPE_EN_RX_SA_411 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1051 IRQ_TYPE_STAT_RX_SA_411

- **Name:** IRQ_TYPE_STAT_RX_SA_411
- **Description:** Interrupt Status for each IRQ type of Rx SA 411
- **Size:** 32 bits
- **Offset:** 0x106c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 411

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1058 Fields for Register: IRQ_TYPE_STAT_RX_SA_411

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1058 Fields for Register: IRQ_TYPE_STAT_RX_SA_411 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1058 Fields for Register: IRQ_TYPE_STAT_RX_SA_411 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1052 IRQ_TYPE_EN_RX_SA_412

- **Name:** IRQ_TYPE_EN_RX_SA_412
- **Description:** Interrupt Enable for each IRQ type of Rx SA 412
- **Size:** 32 bits
- **Offset:** 0x1070
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 412

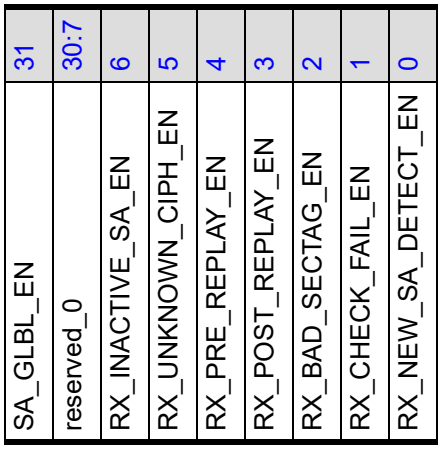


Table 6-1059 Fields for Register: IRQ_TYPE_EN_RX_SA_412

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1059 Fields for Register: IRQ_TYPE_EN_RX_SA_412 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1059 Fields for Register: IRQ_TYPE_EN_RX_SA_412 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1053 IRQ_TYPE_STAT_RX_SA_412

- **Name:** IRQ_TYPE_STAT_RX_SA_412
- **Description:** Interrupt Status for each IRQ type of Rx SA 412
- **Size:** 32 bits
- **Offset:** 0x1074
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 412

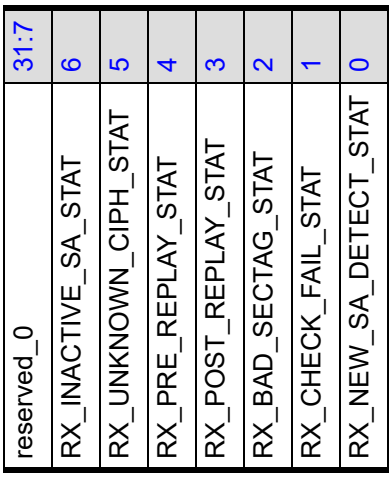


Table 6-1060 Fields for Register: IRQ_TYPE_STAT_RX_SA_412

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1060 Fields for Register: IRQ_TYPE_STAT_RX_SA_412 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1060 Fields for Register: IRQ_TYPE_STAT_RX_SA_412 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1054 IRQ_TYPE_EN_RX_SA_413

- **Name:** IRQ_TYPE_EN_RX_SA_413
- **Description:** Interrupt Enable for each IRQ type of Rx SA 413
- **Size:** 32 bits
- **Offset:** 0x1078
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 413

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1061 Fields for Register: IRQ_TYPE_EN_RX_SA_413

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1061 Fields for Register: IRQ_TYPE_EN_RX_SA_413 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1061 Fields for Register: IRQ_TYPE_EN_RX_SA_413 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1055 IRQ_TYPE_STAT_RX_SA_413

- **Name:** IRQ_TYPE_STAT_RX_SA_413
- **Description:** Interrupt Status for each IRQ type of Rx SA 413
- **Size:** 32 bits
- **Offset:** 0x107c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 413

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1062 Fields for Register: IRQ_TYPE_STAT_RX_SA_413

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1062 Fields for Register: IRQ_TYPE_STAT_RX_SA_413 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1062 Fields for Register: IRQ_TYPE_STAT_RX_SA_413 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1056 IRQ_TYPE_EN_RX_SA_414

- **Name:** IRQ_TYPE_EN_RX_SA_414
- **Description:** Interrupt Enable for each IRQ type of Rx SA 414
- **Size:** 32 bits
- **Offset:** 0x1080
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 414

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1063 Fields for Register: IRQ_TYPE_EN_RX_SA_414

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1063 Fields for Register: IRQ_TYPE_EN_RX_SA_414 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1063 Fields for Register: IRQ_TYPE_EN_RX_SA_414 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1057 IRQ_TYPE_STAT_RX_SA_414

- **Name:** IRQ_TYPE_STAT_RX_SA_414
- **Description:** Interrupt Status for each IRQ type of Rx SA 414
- **Size:** 32 bits
- **Offset:** 0x1084
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 414

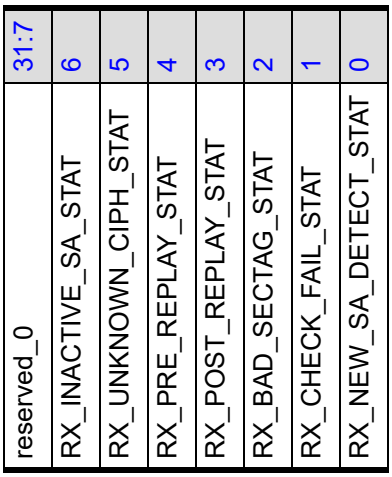


Table 6-1064 Fields for Register: IRQ_TYPE_STAT_RX_SA_414

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1064 Fields for Register: IRQ_TYPE_STAT_RX_SA_414 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1064 Fields for Register: IRQ_TYPE_STAT_RX_SA_414 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1058 IRQ_TYPE_EN_RX_SA_415

- **Name:** IRQ_TYPE_EN_RX_SA_415
- **Description:** Interrupt Enable for each IRQ type of Rx SA 415
- **Size:** 32 bits
- **Offset:** 0x1088
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 415

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1065 Fields for Register: IRQ_TYPE_EN_RX_SA_415

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1065 Fields for Register: IRQ_TYPE_EN_RX_SA_415 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1065 Fields for Register: IRQ_TYPE_EN_RX_SA_415 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1059 IRQ_TYPE_STAT_RX_SA_415

- **Name:** IRQ_TYPE_STAT_RX_SA_415
- **Description:** Interrupt Status for each IRQ type of Rx SA 415
- **Size:** 32 bits
- **Offset:** 0x108c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 415

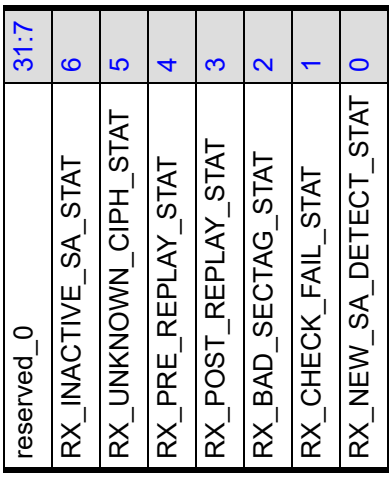


Table 6-1066 Fields for Register: IRQ_TYPE_STAT_RX_SA_415

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1066 Fields for Register: IRQ_TYPE_STAT_RX_SA_415 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1066 Fields for Register: IRQ_TYPE_STAT_RX_SA_415 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1060 IRQ_TYPE_EN_RX_SA_416

- **Name:** IRQ_TYPE_EN_RX_SA_416
- **Description:** Interrupt Enable for each IRQ type of Rx SA 416
- **Size:** 32 bits
- **Offset:** 0x1090
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 416

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1067 Fields for Register: IRQ_TYPE_EN_RX_SA_416

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1067 Fields for Register: IRQ_TYPE_EN_RX_SA_416 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1067 Fields for Register: IRQ_TYPE_EN_RX_SA_416 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1061 IRQ_TYPE_STAT_RX_SA_416

- **Name:** IRQ_TYPE_STAT_RX_SA_416
- **Description:** Interrupt Status for each IRQ type of Rx SA 416
- **Size:** 32 bits
- **Offset:** 0x1094
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 416

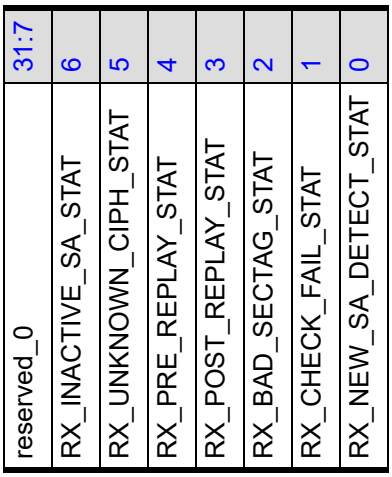


Table 6-1068 Fields for Register: IRQ_TYPE_STAT_RX_SA_416

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1068 Fields for Register: IRQ_TYPE_STAT_RX_SA_416 (Continued)

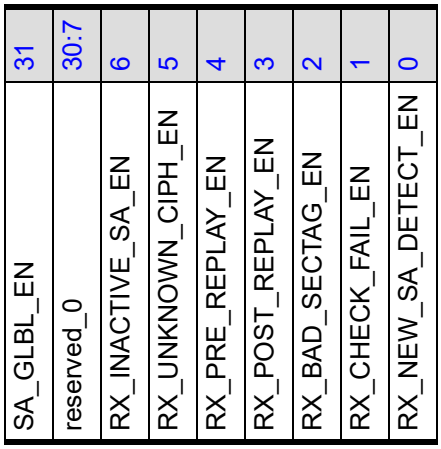
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1068 Fields for Register: IRQ_TYPE_STAT_RX_SA_416 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1062 IRQ_TYPE_EN_RX_SA_417

- **Name:** IRQ_TYPE_EN_RX_SA_417
- **Description:** Interrupt Enable for each IRQ type of Rx SA 417
- **Size:** 32 bits
- **Offset:** 0x1098
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 417



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1069 Fields for Register: IRQ_TYPE_EN_RX_SA_417

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1069 Fields for Register: IRQ_TYPE_EN_RX_SA_417 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1069 Fields for Register: IRQ_TYPE_EN_RX_SA_417 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1063 IRQ_TYPE_STAT_RX_SA_417

- **Name:** IRQ_TYPE_STAT_RX_SA_417
- **Description:** Interrupt Status for each IRQ type of Rx SA 417
- **Size:** 32 bits
- **Offset:** 0x109c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 417

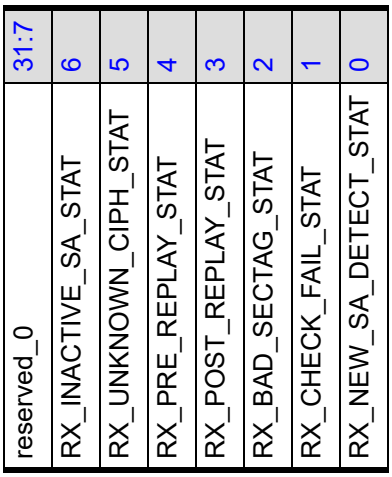


Table 6-1070 Fields for Register: IRQ_TYPE_STAT_RX_SA_417

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1070 Fields for Register: IRQ_TYPE_STAT_RX_SA_417 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1070 Fields for Register: IRQ_TYPE_STAT_RX_SA_417 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1064 IRQ_TYPE_EN_RX_SA_418

- **Name:** IRQ_TYPE_EN_RX_SA_418
- **Description:** Interrupt Enable for each IRQ type of Rx SA 418
- **Size:** 32 bits
- **Offset:** 0x10a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 418

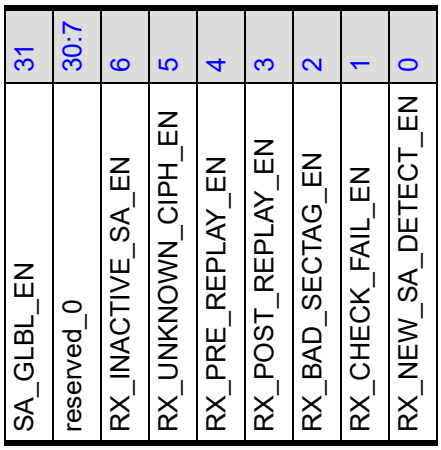


Table 6-1071 Fields for Register: IRQ_TYPE_EN_RX_SA_418

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1071 Fields for Register: IRQ_TYPE_EN_RX_SA_418 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1071 Fields for Register: IRQ_TYPE_EN_RX_SA_418 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1065 IRQ_TYPE_STAT_RX_SA_418

- **Name:** IRQ_TYPE_STAT_RX_SA_418
- **Description:** Interrupt Status for each IRQ type of Rx SA 418
- **Size:** 32 bits
- **Offset:** 0x10a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 418

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1072 Fields for Register: IRQ_TYPE_STAT_RX_SA_418

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1072 Fields for Register: IRQ_TYPE_STAT_RX_SA_418 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1072 Fields for Register: IRQ_TYPE_STAT_RX_SA_418 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1066 IRQ_TYPE_EN_RX_SA_419

- **Name:** IRQ_TYPE_EN_RX_SA_419
- **Description:** Interrupt Enable for each IRQ type of Rx SA 419
- **Size:** 32 bits
- **Offset:** 0x10a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 419

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1073 Fields for Register: IRQ_TYPE_EN_RX_SA_419

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1073 Fields for Register: IRQ_TYPE_EN_RX_SA_419 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1073 Fields for Register: IRQ_TYPE_EN_RX_SA_419 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1067 IRQ_TYPE_STAT_RX_SA_419

- **Name:** IRQ_TYPE_STAT_RX_SA_419
- **Description:** Interrupt Status for each IRQ type of Rx SA 419
- **Size:** 32 bits
- **Offset:** 0x10ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 419

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1074 Fields for Register: IRQ_TYPE_STAT_RX_SA_419

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1074 Fields for Register: IRQ_TYPE_STAT_RX_SA_419 (Continued)

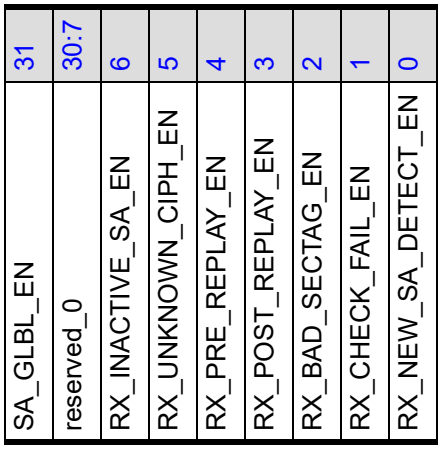
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1074 Fields for Register: IRQ_TYPE_STAT_RX_SA_419 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1068 IRQ_TYPE_EN_RX_SA_420

- **Name:** IRQ_TYPE_EN_RX_SA_420
- **Description:** Interrupt Enable for each IRQ type of Rx SA 420
- **Size:** 32 bits
- **Offset:** 0x10b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 420



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1075 Fields for Register: IRQ_TYPE_EN_RX_SA_420

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1075 Fields for Register: IRQ_TYPE_EN_RX_SA_420 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1075 Fields for Register: IRQ_TYPE_EN_RX_SA_420 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1069 IRQ_TYPE_STAT_RX_SA_420

- **Name:** IRQ_TYPE_STAT_RX_SA_420
- **Description:** Interrupt Status for each IRQ type of Rx SA 420
- **Size:** 32 bits
- **Offset:** 0x10b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 420

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1076 Fields for Register: IRQ_TYPE_STAT_RX_SA_420

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1076 Fields for Register: IRQ_TYPE_STAT_RX_SA_420 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1076 Fields for Register: IRQ_TYPE_STAT_RX_SA_420 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1070 IRQ_TYPE_EN_RX_SA_421

- **Name:** IRQ_TYPE_EN_RX_SA_421
- **Description:** Interrupt Enable for each IRQ type of Rx SA 421
- **Size:** 32 bits
- **Offset:** 0x10b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 421

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1077 Fields for Register: IRQ_TYPE_EN_RX_SA_421

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1077 Fields for Register: IRQ_TYPE_EN_RX_SA_421 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1077 Fields for Register: IRQ_TYPE_EN_RX_SA_421 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1071 IRQ_TYPE_STAT_RX_SA_421

- **Name:** IRQ_TYPE_STAT_RX_SA_421
- **Description:** Interrupt Status for each IRQ type of Rx SA 421
- **Size:** 32 bits
- **Offset:** 0x10bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 421

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1078 Fields for Register: IRQ_TYPE_STAT_RX_SA_421

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1078 Fields for Register: IRQ_TYPE_STAT_RX_SA_421 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1078 Fields for Register: IRQ_TYPE_STAT_RX_SA_421 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1072 IRQ_TYPE_EN_RX_SA_422

- **Name:** IRQ_TYPE_EN_RX_SA_422
- **Description:** Interrupt Enable for each IRQ type of Rx SA 422
- **Size:** 32 bits
- **Offset:** 0x10c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 422

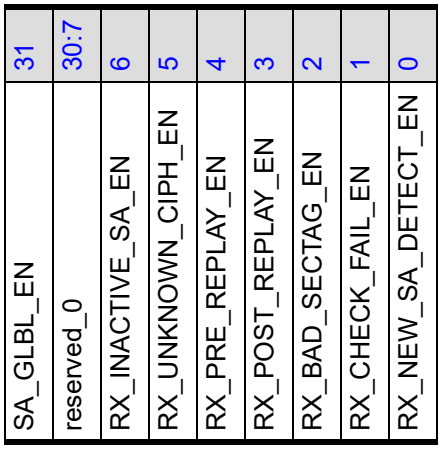


Table 6-1079 Fields for Register: IRQ_TYPE_EN_RX_SA_422

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1079 Fields for Register: IRQ_TYPE_EN_RX_SA_422 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1079 Fields for Register: IRQ_TYPE_EN_RX_SA_422 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1073 IRQ_TYPE_STAT_RX_SA_422

- **Name:** IRQ_TYPE_STAT_RX_SA_422
- **Description:** Interrupt Status for each IRQ type of Rx SA 422
- **Size:** 32 bits
- **Offset:** 0x10c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 422

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1080 Fields for Register: IRQ_TYPE_STAT_RX_SA_422

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1080 Fields for Register: IRQ_TYPE_STAT_RX_SA_422 (Continued)

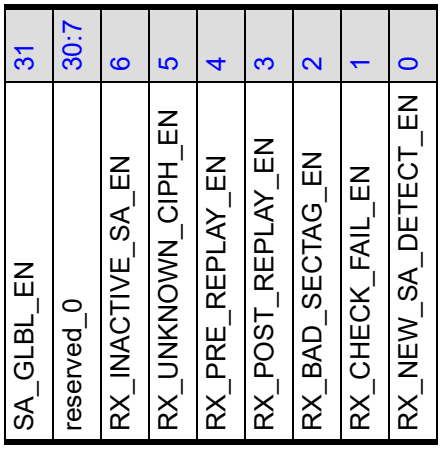
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1080 Fields for Register: IRQ_TYPE_STAT_RX_SA_422 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1074 IRQ_TYPE_EN_RX_SA_423

- **Name:** IRQ_TYPE_EN_RX_SA_423
- **Description:** Interrupt Enable for each IRQ type of Rx SA 423
- **Size:** 32 bits
- **Offset:** 0x10c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 423



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1081 Fields for Register: IRQ_TYPE_EN_RX_SA_423

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1081 Fields for Register: IRQ_TYPE_EN_RX_SA_423 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1081 Fields for Register: IRQ_TYPE_EN_RX_SA_423 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1075 IRQ_TYPE_STAT_RX_SA_423

- **Name:** IRQ_TYPE_STAT_RX_SA_423
- **Description:** Interrupt Status for each IRQ type of Rx SA 423
- **Size:** 32 bits
- **Offset:** 0x10cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 423

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1082 Fields for Register: IRQ_TYPE_STAT_RX_SA_423

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1082 Fields for Register: IRQ_TYPE_STAT_RX_SA_423 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1082 Fields for Register: IRQ_TYPE_STAT_RX_SA_423 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1076 IRQ_TYPE_EN_RX_SA_424

- **Name:** IRQ_TYPE_EN_RX_SA_424
- **Description:** Interrupt Enable for each IRQ type of Rx SA 424
- **Size:** 32 bits
- **Offset:** 0x10d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 424

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1083 Fields for Register: IRQ_TYPE_EN_RX_SA_424

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1083 Fields for Register: IRQ_TYPE_EN_RX_SA_424 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1083 Fields for Register: IRQ_TYPE_EN_RX_SA_424 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1077 IRQ_TYPE_STAT_RX_SA_424

- **Name:** IRQ_TYPE_STAT_RX_SA_424
- **Description:** Interrupt Status for each IRQ type of Rx SA 424
- **Size:** 32 bits
- **Offset:** 0x10d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 424

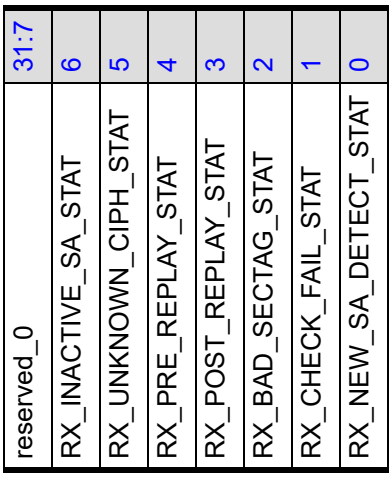


Table 6-1084 Fields for Register: IRQ_TYPE_STAT_RX_SA_424

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1084 Fields for Register: IRQ_TYPE_STAT_RX_SA_424 (Continued)

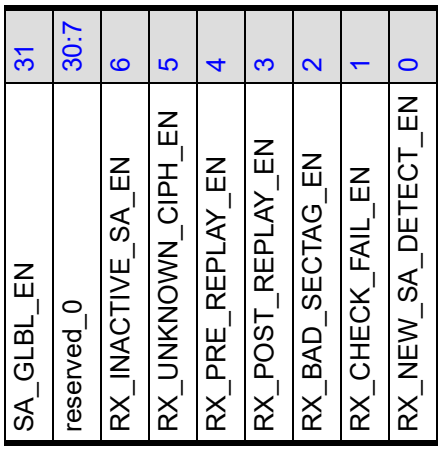
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1084 Fields for Register: IRQ_TYPE_STAT_RX_SA_424 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1078 IRQ_TYPE_EN_RX_SA_425

- **Name:** IRQ_TYPE_EN_RX_SA_425
- **Description:** Interrupt Enable for each IRQ type of Rx SA 425
- **Size:** 32 bits
- **Offset:** 0x10d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 425



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1085 Fields for Register: IRQ_TYPE_EN_RX_SA_425

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1085 Fields for Register: IRQ_TYPE_EN_RX_SA_425 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1085 Fields for Register: IRQ_TYPE_EN_RX_SA_425 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1079 IRQ_TYPE_STAT_RX_SA_425

- **Name:** IRQ_TYPE_STAT_RX_SA_425
- **Description:** Interrupt Status for each IRQ type of Rx SA 425
- **Size:** 32 bits
- **Offset:** 0x10dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 425

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1086 Fields for Register: IRQ_TYPE_STAT_RX_SA_425

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1086 Fields for Register: IRQ_TYPE_STAT_RX_SA_425 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1086 Fields for Register: IRQ_TYPE_STAT_RX_SA_425 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1080 IRQ_TYPE_EN_RX_SA_426

- **Name:** IRQ_TYPE_EN_RX_SA_426
- **Description:** Interrupt Enable for each IRQ type of Rx SA 426
- **Size:** 32 bits
- **Offset:** 0x10e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 426

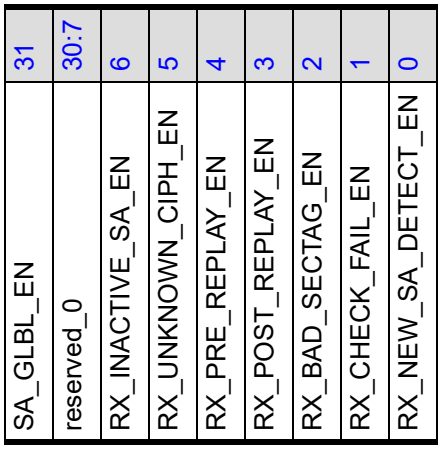


Table 6-1087 Fields for Register: IRQ_TYPE_EN_RX_SA_426

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1087 Fields for Register: IRQ_TYPE_EN_RX_SA_426 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1087 Fields for Register: IRQ_TYPE_EN_RX_SA_426 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1081 IRQ_TYPE_STAT_RX_SA_426

- **Name:** IRQ_TYPE_STAT_RX_SA_426
- **Description:** Interrupt Status for each IRQ type of Rx SA 426
- **Size:** 32 bits
- **Offset:** 0x10e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 426

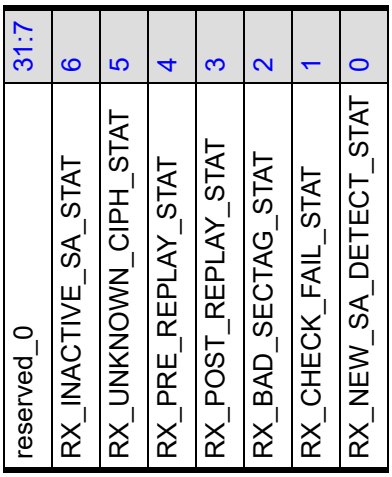


Table 6-1088 Fields for Register: IRQ_TYPE_STAT_RX_SA_426

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1088 Fields for Register: IRQ_TYPE_STAT_RX_SA_426 (Continued)

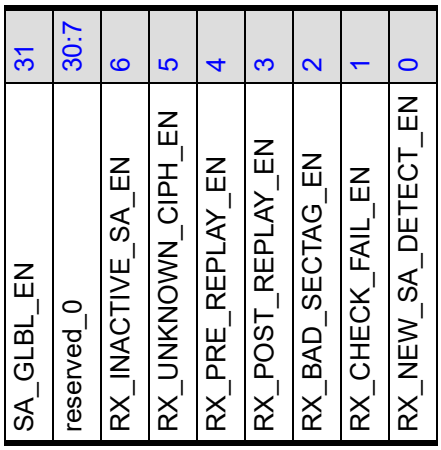
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1088 Fields for Register: IRQ_TYPE_STAT_RX_SA_426 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1082 IRQ_TYPE_EN_RX_SA_427

- **Name:** IRQ_TYPE_EN_RX_SA_427
- **Description:** Interrupt Enable for each IRQ type of Rx SA 427
- **Size:** 32 bits
- **Offset:** 0x10e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 427



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1089 Fields for Register: IRQ_TYPE_EN_RX_SA_427

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1089 Fields for Register: IRQ_TYPE_EN_RX_SA_427 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1089 Fields for Register: IRQ_TYPE_EN_RX_SA_427 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1083 IRQ_TYPE_STAT_RX_SA_427

- **Name:** IRQ_TYPE_STAT_RX_SA_427
- **Description:** Interrupt Status for each IRQ type of Rx SA 427
- **Size:** 32 bits
- **Offset:** 0x10ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 427

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1090 Fields for Register: IRQ_TYPE_STAT_RX_SA_427

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1090 Fields for Register: IRQ_TYPE_STAT_RX_SA_427 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1090 Fields for Register: IRQ_TYPE_STAT_RX_SA_427 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1084 IRQ_TYPE_EN_RX_SA_428

- **Name:** IRQ_TYPE_EN_RX_SA_428
- **Description:** Interrupt Enable for each IRQ type of Rx SA 428
- **Size:** 32 bits
- **Offset:** 0x10f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 428

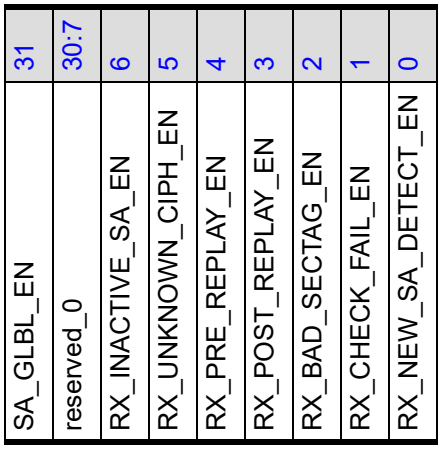


Table 6-1091 Fields for Register: IRQ_TYPE_EN_RX_SA_428

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1091 Fields for Register: IRQ_TYPE_EN_RX_SA_428 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1091 Fields for Register: IRQ_TYPE_EN_RX_SA_428 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1085 IRQ_TYPE_STAT_RX_SA_428

- **Name:** IRQ_TYPE_STAT_RX_SA_428
- **Description:** Interrupt Status for each IRQ type of Rx SA 428
- **Size:** 32 bits
- **Offset:** 0x10f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 428

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1092 Fields for Register: IRQ_TYPE_STAT_RX_SA_428

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1092 Fields for Register: IRQ_TYPE_STAT_RX_SA_428 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1092 Fields for Register: IRQ_TYPE_STAT_RX_SA_428 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1086 IRQ_TYPE_EN_RX_SA_429

- **Name:** IRQ_TYPE_EN_RX_SA_429
- **Description:** Interrupt Enable for each IRQ type of Rx SA 429
- **Size:** 32 bits
- **Offset:** 0x10f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 429

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1093 Fields for Register: IRQ_TYPE_EN_RX_SA_429

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1093 Fields for Register: IRQ_TYPE_EN_RX_SA_429 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1093 Fields for Register: IRQ_TYPE_EN_RX_SA_429 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1087 IRQ_TYPE_STAT_RX_SA_429

- **Name:** IRQ_TYPE_STAT_RX_SA_429
- **Description:** Interrupt Status for each IRQ type of Rx SA 429
- **Size:** 32 bits
- **Offset:** 0x10fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 429

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1094 Fields for Register: IRQ_TYPE_STAT_RX_SA_429

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1094 Fields for Register: IRQ_TYPE_STAT_RX_SA_429 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1094 Fields for Register: IRQ_TYPE_STAT_RX_SA_429 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1088 IRQ_TYPE_EN_RX_SA_430

- **Name:** IRQ_TYPE_EN_RX_SA_430
- **Description:** Interrupt Enable for each IRQ type of Rx SA 430
- **Size:** 32 bits
- **Offset:** 0x1100
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 430

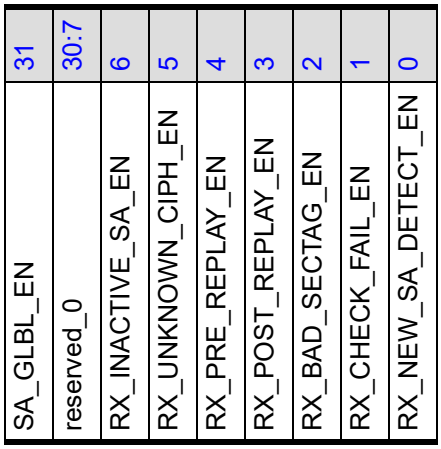


Table 6-1095 Fields for Register: IRQ_TYPE_EN_RX_SA_430

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1095 Fields for Register: IRQ_TYPE_EN_RX_SA_430 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1095 Fields for Register: IRQ_TYPE_EN_RX_SA_430 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1089 IRQ_TYPE_STAT_RX_SA_430

- **Name:** IRQ_TYPE_STAT_RX_SA_430
- **Description:** Interrupt Status for each IRQ type of Rx SA 430
- **Size:** 32 bits
- **Offset:** 0x1104
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 430

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1096 Fields for Register: IRQ_TYPE_STAT_RX_SA_430

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1096 Fields for Register: IRQ_TYPE_STAT_RX_SA_430 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1096 Fields for Register: IRQ_TYPE_STAT_RX_SA_430 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1090 IRQ_TYPE_EN_RX_SA_431

- **Name:** IRQ_TYPE_EN_RX_SA_431
- **Description:** Interrupt Enable for each IRQ type of Rx SA 431
- **Size:** 32 bits
- **Offset:** 0x1108
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 431

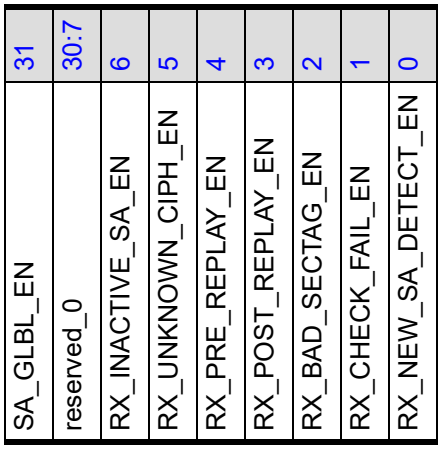


Table 6-1097 Fields for Register: IRQ_TYPE_EN_RX_SA_431

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1097 Fields for Register: IRQ_TYPE_EN_RX_SA_431 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1097 Fields for Register: IRQ_TYPE_EN_RX_SA_431 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1091 IRQ_TYPE_STAT_RX_SA_431

- **Name:** IRQ_TYPE_STAT_RX_SA_431
- **Description:** Interrupt Status for each IRQ type of Rx SA 431
- **Size:** 32 bits
- **Offset:** 0x110c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 431

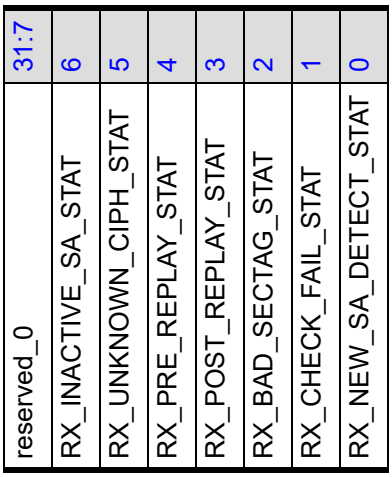


Table 6-1098 Fields for Register: IRQ_TYPE_STAT_RX_SA_431

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1098 Fields for Register: IRQ_TYPE_STAT_RX_SA_431 (Continued)

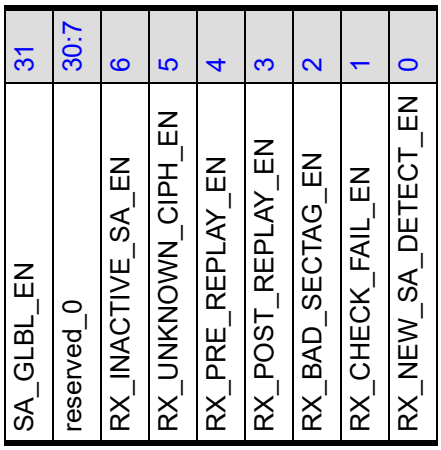
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1098 Fields for Register: IRQ_TYPE_STAT_RX_SA_431 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1092 IRQ_TYPE_EN_RX_SA_432

- **Name:** IRQ_TYPE_EN_RX_SA_432
- **Description:** Interrupt Enable for each IRQ type of Rx SA 432
- **Size:** 32 bits
- **Offset:** 0x1110
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 432



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1099 Fields for Register: IRQ_TYPE_EN_RX_SA_432

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1099 Fields for Register: IRQ_TYPE_EN_RX_SA_432 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1099 Fields for Register: IRQ_TYPE_EN_RX_SA_432 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1093 IRQ_TYPE_STAT_RX_SA_432

- **Name:** IRQ_TYPE_STAT_RX_SA_432
- **Description:** Interrupt Status for each IRQ type of Rx SA 432
- **Size:** 32 bits
- **Offset:** 0x1114
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 432

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1100 Fields for Register: IRQ_TYPE_STAT_RX_SA_432

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1100 Fields for Register: IRQ_TYPE_STAT_RX_SA_432 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1100 Fields for Register: IRQ_TYPE_STAT_RX_SA_432 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1094 IRQ_TYPE_EN_RX_SA_433

- **Name:** IRQ_TYPE_EN_RX_SA_433
- **Description:** Interrupt Enable for each IRQ type of Rx SA 433
- **Size:** 32 bits
- **Offset:** 0x1118
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 433

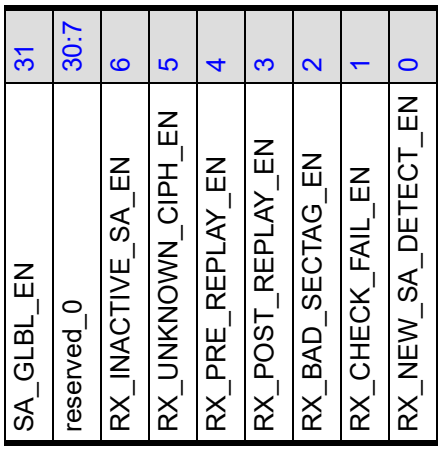


Table 6-1101 Fields for Register: IRQ_TYPE_EN_RX_SA_433

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1101 Fields for Register: IRQ_TYPE_EN_RX_SA_433 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1101 Fields for Register: IRQ_TYPE_EN_RX_SA_433 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1095 IRQ_TYPE_STAT_RX_SA_433

- **Name:** IRQ_TYPE_STAT_RX_SA_433
- **Description:** Interrupt Status for each IRQ type of Rx SA 433
- **Size:** 32 bits
- **Offset:** 0x111c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 433

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1102 Fields for Register: IRQ_TYPE_STAT_RX_SA_433

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1102 Fields for Register: IRQ_TYPE_STAT_RX_SA_433 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1102 Fields for Register: IRQ_TYPE_STAT_RX_SA_433 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1096 IRQ_TYPE_EN_RX_SA_434

- **Name:** IRQ_TYPE_EN_RX_SA_434
- **Description:** Interrupt Enable for each IRQ type of Rx SA 434
- **Size:** 32 bits
- **Offset:** 0x1120
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 434

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1103 Fields for Register: IRQ_TYPE_EN_RX_SA_434

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1103 Fields for Register: IRQ_TYPE_EN_RX_SA_434 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1103 Fields for Register: IRQ_TYPE_EN_RX_SA_434 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1097 IRQ_TYPE_STAT_RX_SA_434

- **Name:** IRQ_TYPE_STAT_RX_SA_434
- **Description:** Interrupt Status for each IRQ type of Rx SA 434
- **Size:** 32 bits
- **Offset:** 0x1124
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 434

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1104 Fields for Register: IRQ_TYPE_STAT_RX_SA_434

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1104 Fields for Register: IRQ_TYPE_STAT_RX_SA_434 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1104 Fields for Register: IRQ_TYPE_STAT_RX_SA_434 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1098 IRQ_TYPE_EN_RX_SA_435

- **Name:** IRQ_TYPE_EN_RX_SA_435
- **Description:** Interrupt Enable for each IRQ type of Rx SA 435
- **Size:** 32 bits
- **Offset:** 0x1128
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 435

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1105 Fields for Register: IRQ_TYPE_EN_RX_SA_435

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1105 Fields for Register: IRQ_TYPE_EN_RX_SA_435 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1105 Fields for Register: IRQ_TYPE_EN_RX_SA_435 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1099 IRQ_TYPE_STAT_RX_SA_435

- **Name:** IRQ_TYPE_STAT_RX_SA_435
- **Description:** Interrupt Status for each IRQ type of Rx SA 435
- **Size:** 32 bits
- **Offset:** 0x112c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 435

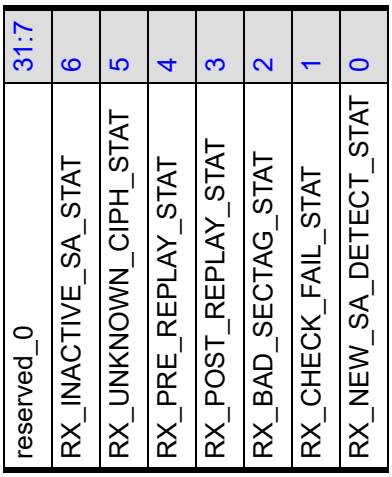


Table 6-1106 Fields for Register: IRQ_TYPE_STAT_RX_SA_435

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1106 Fields for Register: IRQ_TYPE_STAT_RX_SA_435 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1106 Fields for Register: IRQ_TYPE_STAT_RX_SA_435 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1100 IRQ_TYPE_EN_RX_SA_436

- **Name:** IRQ_TYPE_EN_RX_SA_436
- **Description:** Interrupt Enable for each IRQ type of Rx SA 436
- **Size:** 32 bits
- **Offset:** 0x1130
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 436

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1107 Fields for Register: IRQ_TYPE_EN_RX_SA_436

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1107 Fields for Register: IRQ_TYPE_EN_RX_SA_436 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1107 Fields for Register: IRQ_TYPE_EN_RX_SA_436 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1101 IRQ_TYPE_STAT_RX_SA_436

- **Name:** IRQ_TYPE_STAT_RX_SA_436
- **Description:** Interrupt Status for each IRQ type of Rx SA 436
- **Size:** 32 bits
- **Offset:** 0x1134
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 436

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1108 Fields for Register: IRQ_TYPE_STAT_RX_SA_436

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1108 Fields for Register: IRQ_TYPE_STAT_RX_SA_436 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1108 Fields for Register: IRQ_TYPE_STAT_RX_SA_436 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1102 IRQ_TYPE_EN_RX_SA_437

- **Name:** IRQ_TYPE_EN_RX_SA_437
- **Description:** Interrupt Enable for each IRQ type of Rx SA 437
- **Size:** 32 bits
- **Offset:** 0x1138
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 437

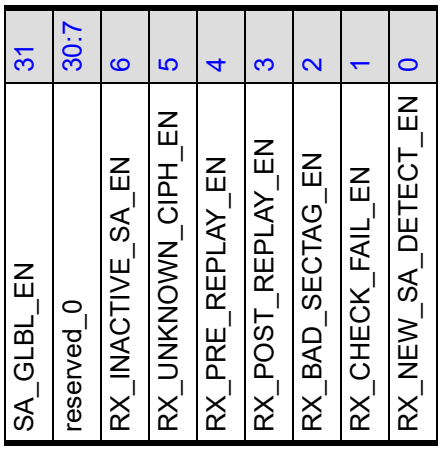


Table 6-1109 Fields for Register: IRQ_TYPE_EN_RX_SA_437

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1109 Fields for Register: IRQ_TYPE_EN_RX_SA_437 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1109 Fields for Register: IRQ_TYPE_EN_RX_SA_437 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1103 IRQ_TYPE_STAT_RX_SA_437

- **Name:** IRQ_TYPE_STAT_RX_SA_437
- **Description:** Interrupt Status for each IRQ type of Rx SA 437
- **Size:** 32 bits
- **Offset:** 0x113c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 437

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1110 Fields for Register: IRQ_TYPE_STAT_RX_SA_437

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1110 Fields for Register: IRQ_TYPE_STAT_RX_SA_437 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1110 Fields for Register: IRQ_TYPE_STAT_RX_SA_437 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1104 IRQ_TYPE_EN_RX_SA_438

- **Name:** IRQ_TYPE_EN_RX_SA_438
- **Description:** Interrupt Enable for each IRQ type of Rx SA 438
- **Size:** 32 bits
- **Offset:** 0x1140
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 438

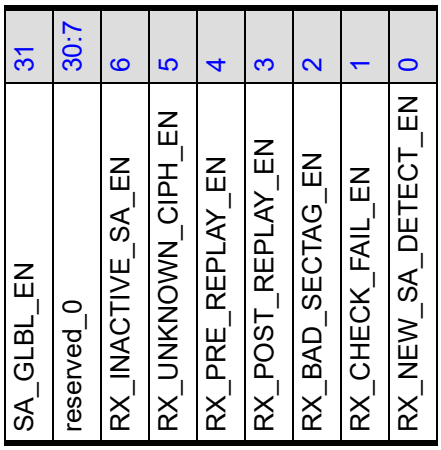


Table 6-1111 Fields for Register: IRQ_TYPE_EN_RX_SA_438

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1111 Fields for Register: IRQ_TYPE_EN_RX_SA_438 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1111 Fields for Register: IRQ_TYPE_EN_RX_SA_438 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1105 IRQ_TYPE_STAT_RX_SA_438

- **Name:** IRQ_TYPE_STAT_RX_SA_438
- **Description:** Interrupt Status for each IRQ type of Rx SA 438
- **Size:** 32 bits
- **Offset:** 0x1144
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 438

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1112 Fields for Register: IRQ_TYPE_STAT_RX_SA_438

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1112 Fields for Register: IRQ_TYPE_STAT_RX_SA_438 (Continued)

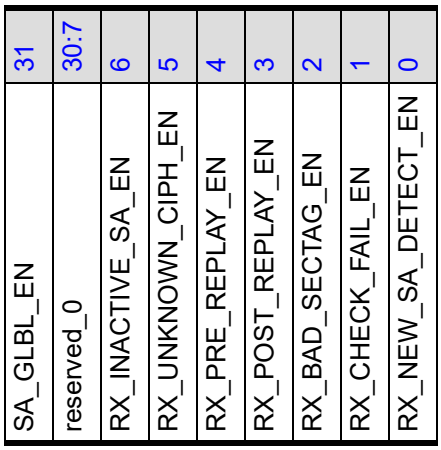
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1112 Fields for Register: IRQ_TYPE_STAT_RX_SA_438 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1106 IRQ_TYPE_EN_RX_SA_439

- **Name:** IRQ_TYPE_EN_RX_SA_439
- **Description:** Interrupt Enable for each IRQ type of Rx SA 439
- **Size:** 32 bits
- **Offset:** 0x1148
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 439



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1113 Fields for Register: IRQ_TYPE_EN_RX_SA_439

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1113 Fields for Register: IRQ_TYPE_EN_RX_SA_439 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1113 Fields for Register: IRQ_TYPE_EN_RX_SA_439 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1107 IRQ_TYPE_STAT_RX_SA_439

- **Name:** IRQ_TYPE_STAT_RX_SA_439
- **Description:** Interrupt Status for each IRQ type of Rx SA 439
- **Size:** 32 bits
- **Offset:** 0x114c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 439

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1114 Fields for Register: IRQ_TYPE_STAT_RX_SA_439

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1114 Fields for Register: IRQ_TYPE_STAT_RX_SA_439 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1114 Fields for Register: IRQ_TYPE_STAT_RX_SA_439 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1108 IRQ_TYPE_EN_RX_SA_440

- **Name:** IRQ_TYPE_EN_RX_SA_440
- **Description:** Interrupt Enable for each IRQ type of Rx SA 440
- **Size:** 32 bits
- **Offset:** 0x1150
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 440

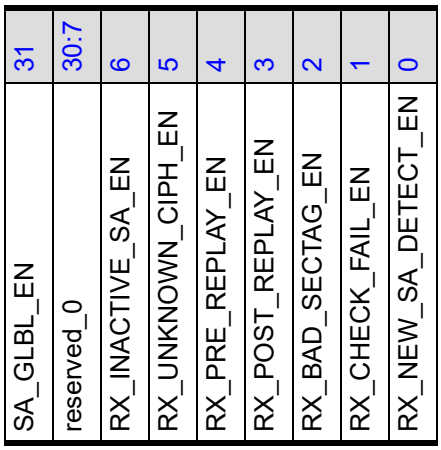


Table 6-1115 Fields for Register: IRQ_TYPE_EN_RX_SA_440

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1115 Fields for Register: IRQ_TYPE_EN_RX_SA_440 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1115 Fields for Register: IRQ_TYPE_EN_RX_SA_440 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1109 IRQ_TYPE_STAT_RX_SA_440

- **Name:** IRQ_TYPE_STAT_RX_SA_440
- **Description:** Interrupt Status for each IRQ type of Rx SA 440
- **Size:** 32 bits
- **Offset:** 0x1154
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 440

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1116 Fields for Register: IRQ_TYPE_STAT_RX_SA_440

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1116 Fields for Register: IRQ_TYPE_STAT_RX_SA_440 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1116 Fields for Register: IRQ_TYPE_STAT_RX_SA_440 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1110 IRQ_TYPE_EN_RX_SA_441

- **Name:** IRQ_TYPE_EN_RX_SA_441
- **Description:** Interrupt Enable for each IRQ type of Rx SA 441
- **Size:** 32 bits
- **Offset:** 0x1158
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 441

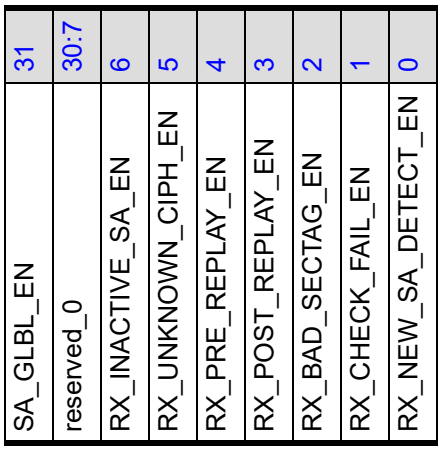


Table 6-1117 Fields for Register: IRQ_TYPE_EN_RX_SA_441

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1117 Fields for Register: IRQ_TYPE_EN_RX_SA_441 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1117 Fields for Register: IRQ_TYPE_EN_RX_SA_441 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1111 IRQ_TYPE_STAT_RX_SA_441

- **Name:** IRQ_TYPE_STAT_RX_SA_441
- **Description:** Interrupt Status for each IRQ type of Rx SA 441
- **Size:** 32 bits
- **Offset:** 0x115c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 441

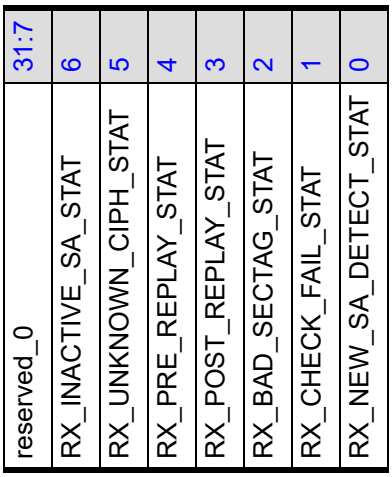


Table 6-1118 Fields for Register: IRQ_TYPE_STAT_RX_SA_441

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1118 Fields for Register: IRQ_TYPE_STAT_RX_SA_441 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1118 Fields for Register: IRQ_TYPE_STAT_RX_SA_441 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1112 IRQ_TYPE_EN_RX_SA_442

- **Name:** IRQ_TYPE_EN_RX_SA_442
- **Description:** Interrupt Enable for each IRQ type of Rx SA 442
- **Size:** 32 bits
- **Offset:** 0x1160
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 442

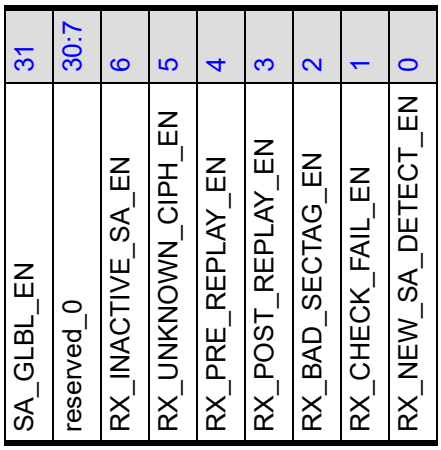


Table 6-1119 Fields for Register: IRQ_TYPE_EN_RX_SA_442

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1119 Fields for Register: IRQ_TYPE_EN_RX_SA_442 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1119 Fields for Register: IRQ_TYPE_EN_RX_SA_442 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1113 IRQ_TYPE_STAT_RX_SA_442

- **Name:** IRQ_TYPE_STAT_RX_SA_442
- **Description:** Interrupt Status for each IRQ type of Rx SA 442
- **Size:** 32 bits
- **Offset:** 0x1164
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 442

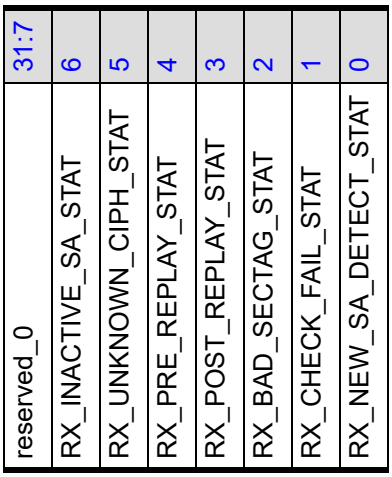


Table 6-1120 Fields for Register: IRQ_TYPE_STAT_RX_SA_442

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1120 Fields for Register: IRQ_TYPE_STAT_RX_SA_442 (Continued)

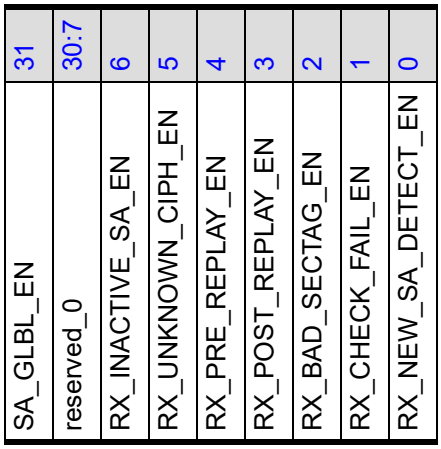
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1120 Fields for Register: IRQ_TYPE_STAT_RX_SA_442 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1114 IRQ_TYPE_EN_RX_SA_443

- **Name:** IRQ_TYPE_EN_RX_SA_443
- **Description:** Interrupt Enable for each IRQ type of Rx SA 443
- **Size:** 32 bits
- **Offset:** 0x1168
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 443



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1121 Fields for Register: IRQ_TYPE_EN_RX_SA_443

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1121 Fields for Register: IRQ_TYPE_EN_RX_SA_443 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1121 Fields for Register: IRQ_TYPE_EN_RX_SA_443 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1115 IRQ_TYPE_STAT_RX_SA_443

- **Name:** IRQ_TYPE_STAT_RX_SA_443
- **Description:** Interrupt Status for each IRQ type of Rx SA 443
- **Size:** 32 bits
- **Offset:** 0x116c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 443

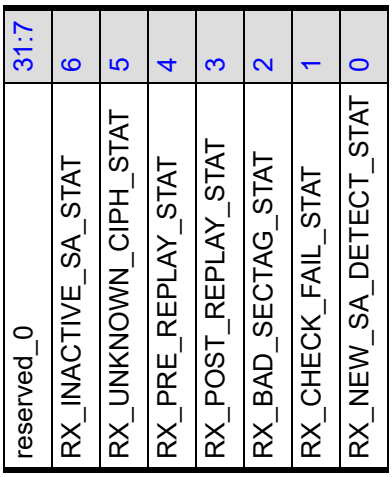


Table 6-1122 Fields for Register: IRQ_TYPE_STAT_RX_SA_443

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1122 Fields for Register: IRQ_TYPE_STAT_RX_SA_443 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1122 Fields for Register: IRQ_TYPE_STAT_RX_SA_443 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1116 IRQ_TYPE_EN_RX_SA_444

- **Name:** IRQ_TYPE_EN_RX_SA_444
- **Description:** Interrupt Enable for each IRQ type of Rx SA 444
- **Size:** 32 bits
- **Offset:** 0x1170
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 444

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1123 Fields for Register: IRQ_TYPE_EN_RX_SA_444

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1123 Fields for Register: IRQ_TYPE_EN_RX_SA_444 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1123 Fields for Register: IRQ_TYPE_EN_RX_SA_444 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1117 IRQ_TYPE_STAT_RX_SA_444

- **Name:** IRQ_TYPE_STAT_RX_SA_444
- **Description:** Interrupt Status for each IRQ type of Rx SA 444
- **Size:** 32 bits
- **Offset:** 0x1174
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 444

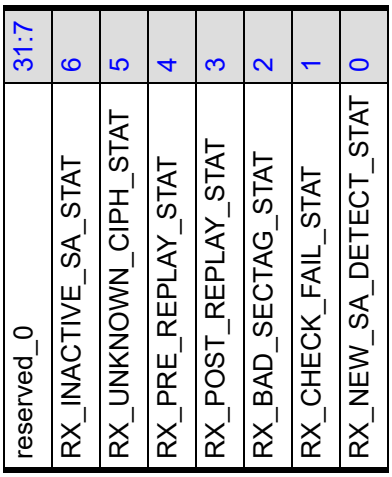


Table 6-1124 Fields for Register: IRQ_TYPE_STAT_RX_SA_444

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1124 Fields for Register: IRQ_TYPE_STAT_RX_SA_444 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1124 Fields for Register: IRQ_TYPE_STAT_RX_SA_444 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1118 IRQ_TYPE_EN_RX_SA_445

- **Name:** IRQ_TYPE_EN_RX_SA_445
- **Description:** Interrupt Enable for each IRQ type of Rx SA 445
- **Size:** 32 bits
- **Offset:** 0x1178
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 445

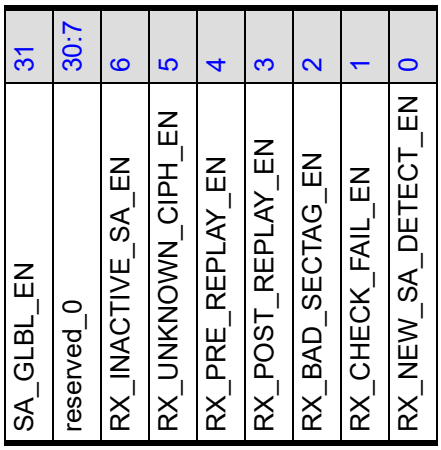


Table 6-1125 Fields for Register: IRQ_TYPE_EN_RX_SA_445

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1125 Fields for Register: IRQ_TYPE_EN_RX_SA_445 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1125 Fields for Register: IRQ_TYPE_EN_RX_SA_445 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1119 IRQ_TYPE_STAT_RX_SA_445

- **Name:** IRQ_TYPE_STAT_RX_SA_445
- **Description:** Interrupt Status for each IRQ type of Rx SA 445
- **Size:** 32 bits
- **Offset:** 0x117c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 445

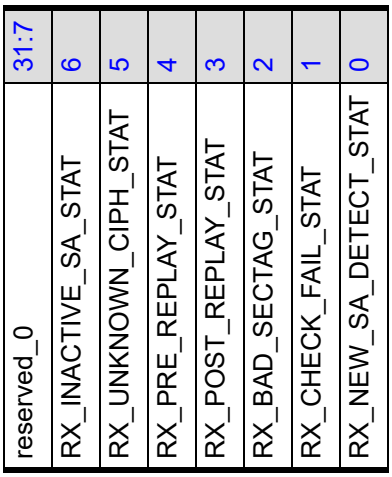


Table 6-1126 Fields for Register: IRQ_TYPE_STAT_RX_SA_445

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1126 Fields for Register: IRQ_TYPE_STAT_RX_SA_445 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1126 Fields for Register: IRQ_TYPE_STAT_RX_SA_445 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1120 IRQ_TYPE_EN_RX_SA_446

- **Name:** IRQ_TYPE_EN_RX_SA_446
- **Description:** Interrupt Enable for each IRQ type of Rx SA 446
- **Size:** 32 bits
- **Offset:** 0x1180
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 446

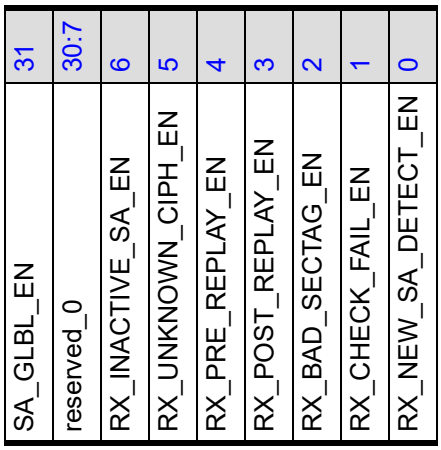


Table 6-1127 Fields for Register: IRQ_TYPE_EN_RX_SA_446

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1127 Fields for Register: IRQ_TYPE_EN_RX_SA_446 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1127 Fields for Register: IRQ_TYPE_EN_RX_SA_446 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1121 IRQ_TYPE_STAT_RX_SA_446

- **Name:** IRQ_TYPE_STAT_RX_SA_446
- **Description:** Interrupt Status for each IRQ type of Rx SA 446
- **Size:** 32 bits
- **Offset:** 0x1184
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 446

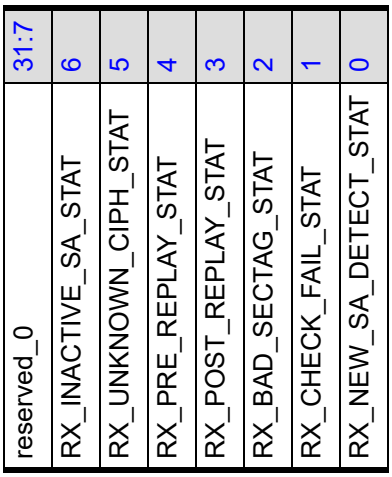


Table 6-1128 Fields for Register: IRQ_TYPE_STAT_RX_SA_446

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1128 Fields for Register: IRQ_TYPE_STAT_RX_SA_446 (Continued)

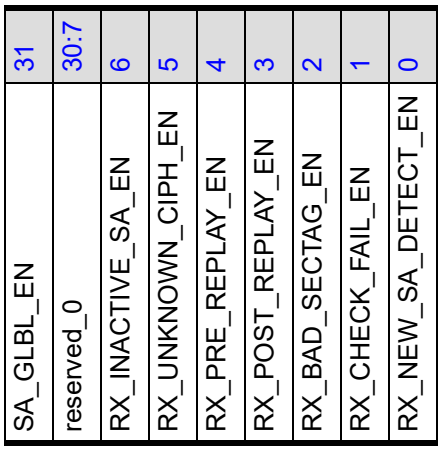
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1128 Fields for Register: IRQ_TYPE_STAT_RX_SA_446 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1122 IRQ_TYPE_EN_RX_SA_447

- **Name:** IRQ_TYPE_EN_RX_SA_447
- **Description:** Interrupt Enable for each IRQ type of Rx SA 447
- **Size:** 32 bits
- **Offset:** 0x1188
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 447



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1129 Fields for Register: IRQ_TYPE_EN_RX_SA_447

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1129 Fields for Register: IRQ_TYPE_EN_RX_SA_447 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1129 Fields for Register: IRQ_TYPE_EN_RX_SA_447 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1123 IRQ_TYPE_STAT_RX_SA_447

- **Name:** IRQ_TYPE_STAT_RX_SA_447
- **Description:** Interrupt Status for each IRQ type of Rx SA 447
- **Size:** 32 bits
- **Offset:** 0x118c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 447

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1130 Fields for Register: IRQ_TYPE_STAT_RX_SA_447

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1130 Fields for Register: IRQ_TYPE_STAT_RX_SA_447 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1130 Fields for Register: IRQ_TYPE_STAT_RX_SA_447 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1124 IRQ_TYPE_EN_RX_SA_448

- **Name:** IRQ_TYPE_EN_RX_SA_448
- **Description:** Interrupt Enable for each IRQ type of Rx SA 448
- **Size:** 32 bits
- **Offset:** 0x1190
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 448

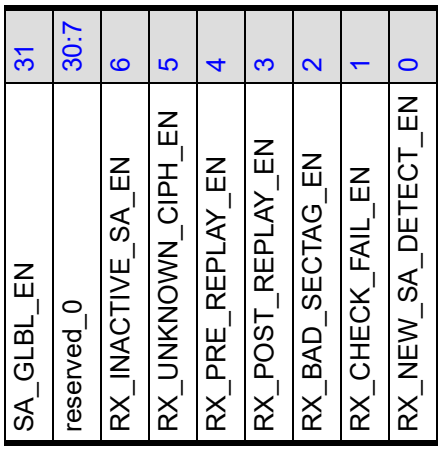


Table 6-1131 Fields for Register: IRQ_TYPE_EN_RX_SA_448

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1131 Fields for Register: IRQ_TYPE_EN_RX_SA_448 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1131 Fields for Register: IRQ_TYPE_EN_RX_SA_448 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1125 IRQ_TYPE_STAT_RX_SA_448

- **Name:** IRQ_TYPE_STAT_RX_SA_448
- **Description:** Interrupt Status for each IRQ type of Rx SA 448
- **Size:** 32 bits
- **Offset:** 0x1194
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 448

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1132 Fields for Register: IRQ_TYPE_STAT_RX_SA_448

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1132 Fields for Register: IRQ_TYPE_STAT_RX_SA_448 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1132 Fields for Register: IRQ_TYPE_STAT_RX_SA_448 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1126 IRQ_TYPE_EN_RX_SA_449

- **Name:** IRQ_TYPE_EN_RX_SA_449
- **Description:** Interrupt Enable for each IRQ type of Rx SA 449
- **Size:** 32 bits
- **Offset:** 0x1198
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 449

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1133 Fields for Register: IRQ_TYPE_EN_RX_SA_449

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1133 Fields for Register: IRQ_TYPE_EN_RX_SA_449 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1133 Fields for Register: IRQ_TYPE_EN_RX_SA_449 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1127 IRQ_TYPE_STAT_RX_SA_449

- **Name:** IRQ_TYPE_STAT_RX_SA_449
- **Description:** Interrupt Status for each IRQ type of Rx SA 449
- **Size:** 32 bits
- **Offset:** 0x119c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 449

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1134 Fields for Register: IRQ_TYPE_STAT_RX_SA_449

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1134 Fields for Register: IRQ_TYPE_STAT_RX_SA_449 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1134 Fields for Register: IRQ_TYPE_STAT_RX_SA_449 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1128 IRQ_TYPE_EN_RX_SA_450

- **Name:** IRQ_TYPE_EN_RX_SA_450
- **Description:** Interrupt Enable for each IRQ type of Rx SA 450
- **Size:** 32 bits
- **Offset:** 0x11a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 450

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1135 Fields for Register: IRQ_TYPE_EN_RX_SA_450

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1135 Fields for Register: IRQ_TYPE_EN_RX_SA_450 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1135 Fields for Register: IRQ_TYPE_EN_RX_SA_450 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1129 IRQ_TYPE_STAT_RX_SA_450

- **Name:** IRQ_TYPE_STAT_RX_SA_450
- **Description:** Interrupt Status for each IRQ type of Rx SA 450
- **Size:** 32 bits
- **Offset:** 0x11a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 450

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1136 Fields for Register: IRQ_TYPE_STAT_RX_SA_450

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1136 Fields for Register: IRQ_TYPE_STAT_RX_SA_450 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1136 Fields for Register: IRQ_TYPE_STAT_RX_SA_450 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1130 IRQ_TYPE_EN_RX_SA_451

- **Name:** IRQ_TYPE_EN_RX_SA_451
- **Description:** Interrupt Enable for each IRQ type of Rx SA 451
- **Size:** 32 bits
- **Offset:** 0x11a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 451

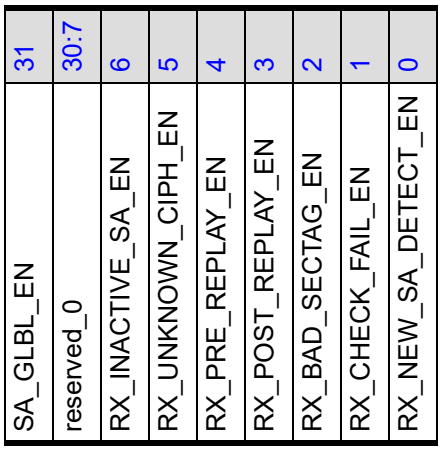


Table 6-1137 Fields for Register: IRQ_TYPE_EN_RX_SA_451

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1137 Fields for Register: IRQ_TYPE_EN_RX_SA_451 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1137 Fields for Register: IRQ_TYPE_EN_RX_SA_451 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1131 IRQ_TYPE_STAT_RX_SA_451

- **Name:** IRQ_TYPE_STAT_RX_SA_451
- **Description:** Interrupt Status for each IRQ type of Rx SA 451
- **Size:** 32 bits
- **Offset:** 0x11ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 451

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1138 Fields for Register: IRQ_TYPE_STAT_RX_SA_451

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1138 Fields for Register: IRQ_TYPE_STAT_RX_SA_451 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1138 Fields for Register: IRQ_TYPE_STAT_RX_SA_451 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1132 IRQ_TYPE_EN_RX_SA_452

- **Name:** IRQ_TYPE_EN_RX_SA_452
- **Description:** Interrupt Enable for each IRQ type of Rx SA 452
- **Size:** 32 bits
- **Offset:** 0x11b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 452

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1139 Fields for Register: IRQ_TYPE_EN_RX_SA_452

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1139 Fields for Register: IRQ_TYPE_EN_RX_SA_452 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1139 Fields for Register: IRQ_TYPE_EN_RX_SA_452 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1133 IRQ_TYPE_STAT_RX_SA_452

- **Name:** IRQ_TYPE_STAT_RX_SA_452
- **Description:** Interrupt Status for each IRQ type of Rx SA 452
- **Size:** 32 bits
- **Offset:** 0x11b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 452

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1140 Fields for Register: IRQ_TYPE_STAT_RX_SA_452

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1140 Fields for Register: IRQ_TYPE_STAT_RX_SA_452 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1140 Fields for Register: IRQ_TYPE_STAT_RX_SA_452 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1134 IRQ_TYPE_EN_RX_SA_453

- **Name:** IRQ_TYPE_EN_RX_SA_453
- **Description:** Interrupt Enable for each IRQ type of Rx SA 453
- **Size:** 32 bits
- **Offset:** 0x11b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 453

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1141 Fields for Register: IRQ_TYPE_EN_RX_SA_453

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1141 Fields for Register: IRQ_TYPE_EN_RX_SA_453 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1141 Fields for Register: IRQ_TYPE_EN_RX_SA_453 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1135 IRQ_TYPE_STAT_RX_SA_453

- **Name:** IRQ_TYPE_STAT_RX_SA_453
- **Description:** Interrupt Status for each IRQ type of Rx SA 453
- **Size:** 32 bits
- **Offset:** 0x11bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 453

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1142 Fields for Register: IRQ_TYPE_STAT_RX_SA_453

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1142 Fields for Register: IRQ_TYPE_STAT_RX_SA_453 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1142 Fields for Register: IRQ_TYPE_STAT_RX_SA_453 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1136 IRQ_TYPE_EN_RX_SA_454

- **Name:** IRQ_TYPE_EN_RX_SA_454
- **Description:** Interrupt Enable for each IRQ type of Rx SA 454
- **Size:** 32 bits
- **Offset:** 0x11c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 454

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1143 Fields for Register: IRQ_TYPE_EN_RX_SA_454

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1143 Fields for Register: IRQ_TYPE_EN_RX_SA_454 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1143 Fields for Register: IRQ_TYPE_EN_RX_SA_454 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1137 IRQ_TYPE_STAT_RX_SA_454

- **Name:** IRQ_TYPE_STAT_RX_SA_454
- **Description:** Interrupt Status for each IRQ type of Rx SA 454
- **Size:** 32 bits
- **Offset:** 0x11c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 454

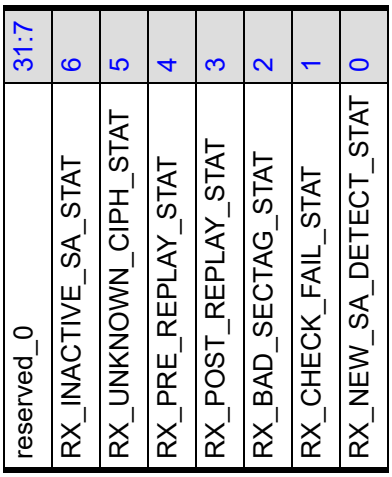


Table 6-1144 Fields for Register: IRQ_TYPE_STAT_RX_SA_454

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1144 Fields for Register: IRQ_TYPE_STAT_RX_SA_454 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1144 Fields for Register: IRQ_TYPE_STAT_RX_SA_454 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1138 IRQ_TYPE_EN_RX_SA_455

- **Name:** IRQ_TYPE_EN_RX_SA_455
- **Description:** Interrupt Enable for each IRQ type of Rx SA 455
- **Size:** 32 bits
- **Offset:** 0x11c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 455

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1145 Fields for Register: IRQ_TYPE_EN_RX_SA_455

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1145 Fields for Register: IRQ_TYPE_EN_RX_SA_455 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1145 Fields for Register: IRQ_TYPE_EN_RX_SA_455 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1139 IRQ_TYPE_STAT_RX_SA_455

- **Name:** IRQ_TYPE_STAT_RX_SA_455
- **Description:** Interrupt Status for each IRQ type of Rx SA 455
- **Size:** 32 bits
- **Offset:** 0x11cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 455

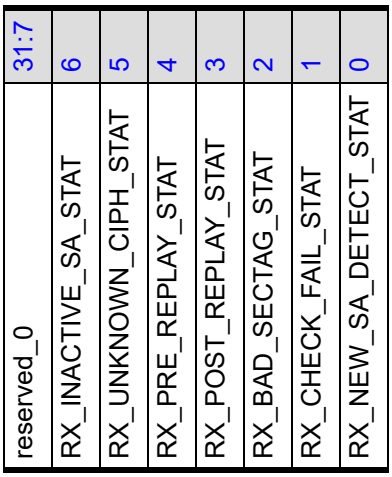


Table 6-1146 Fields for Register: IRQ_TYPE_STAT_RX_SA_455

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1146 Fields for Register: IRQ_TYPE_STAT_RX_SA_455 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1146 Fields for Register: IRQ_TYPE_STAT_RX_SA_455 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1140 IRQ_TYPE_EN_RX_SA_456

- **Name:** IRQ_TYPE_EN_RX_SA_456
- **Description:** Interrupt Enable for each IRQ type of Rx SA 456
- **Size:** 32 bits
- **Offset:** 0x11d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 456

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1147 Fields for Register: IRQ_TYPE_EN_RX_SA_456

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1147 Fields for Register: IRQ_TYPE_EN_RX_SA_456 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1147 Fields for Register: IRQ_TYPE_EN_RX_SA_456 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1141 IRQ_TYPE_STAT_RX_SA_456

- **Name:** IRQ_TYPE_STAT_RX_SA_456
- **Description:** Interrupt Status for each IRQ type of Rx SA 456
- **Size:** 32 bits
- **Offset:** 0x11d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 456

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1148 Fields for Register: IRQ_TYPE_STAT_RX_SA_456

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1148 Fields for Register: IRQ_TYPE_STAT_RX_SA_456 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1148 Fields for Register: IRQ_TYPE_STAT_RX_SA_456 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1142 IRQ_TYPE_EN_RX_SA_457

- **Name:** IRQ_TYPE_EN_RX_SA_457
- **Description:** Interrupt Enable for each IRQ type of Rx SA 457
- **Size:** 32 bits
- **Offset:** 0x11d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 457

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1149 Fields for Register: IRQ_TYPE_EN_RX_SA_457

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1149 Fields for Register: IRQ_TYPE_EN_RX_SA_457 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1149 Fields for Register: IRQ_TYPE_EN_RX_SA_457 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1143 IRQ_TYPE_STAT_RX_SA_457

- **Name:** IRQ_TYPE_STAT_RX_SA_457
- **Description:** Interrupt Status for each IRQ type of Rx SA 457
- **Size:** 32 bits
- **Offset:** 0x11dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 457

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1150 Fields for Register: IRQ_TYPE_STAT_RX_SA_457

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1150 Fields for Register: IRQ_TYPE_STAT_RX_SA_457 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1150 Fields for Register: IRQ_TYPE_STAT_RX_SA_457 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1144 IRQ_TYPE_EN_RX_SA_458

- **Name:** IRQ_TYPE_EN_RX_SA_458
- **Description:** Interrupt Enable for each IRQ type of Rx SA 458
- **Size:** 32 bits
- **Offset:** 0x11e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 458

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1151 Fields for Register: IRQ_TYPE_EN_RX_SA_458

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1151 Fields for Register: IRQ_TYPE_EN_RX_SA_458 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1151 Fields for Register: IRQ_TYPE_EN_RX_SA_458 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1145 IRQ_TYPE_STAT_RX_SA_458

- **Name:** IRQ_TYPE_STAT_RX_SA_458
- **Description:** Interrupt Status for each IRQ type of Rx SA 458
- **Size:** 32 bits
- **Offset:** 0x11e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 458

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1152 Fields for Register: IRQ_TYPE_STAT_RX_SA_458

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1152 Fields for Register: IRQ_TYPE_STAT_RX_SA_458 (Continued)

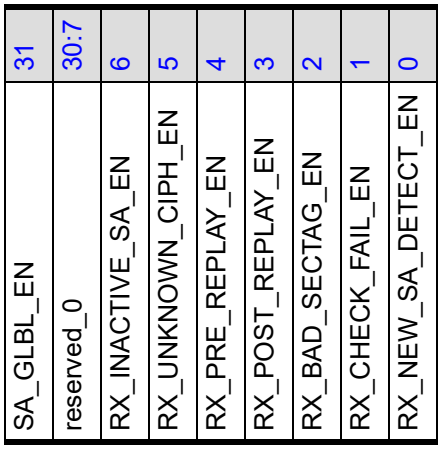
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1152 Fields for Register: IRQ_TYPE_STAT_RX_SA_458 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1146 IRQ_TYPE_EN_RX_SA_459

- **Name:** IRQ_TYPE_EN_RX_SA_459
- **Description:** Interrupt Enable for each IRQ type of Rx SA 459
- **Size:** 32 bits
- **Offset:** 0x11e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 459



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1153 Fields for Register: IRQ_TYPE_EN_RX_SA_459

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1153 Fields for Register: IRQ_TYPE_EN_RX_SA_459 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1153 Fields for Register: IRQ_TYPE_EN_RX_SA_459 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1147 IRQ_TYPE_STAT_RX_SA_459

- **Name:** IRQ_TYPE_STAT_RX_SA_459
- **Description:** Interrupt Status for each IRQ type of Rx SA 459
- **Size:** 32 bits
- **Offset:** 0x11ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 459

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1154 Fields for Register: IRQ_TYPE_STAT_RX_SA_459

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1154 Fields for Register: IRQ_TYPE_STAT_RX_SA_459 (Continued)

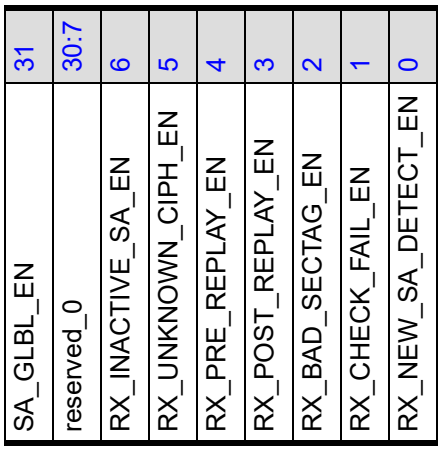
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1154 Fields for Register: IRQ_TYPE_STAT_RX_SA_459 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1148 IRQ_TYPE_EN_RX_SA_460

- **Name:** IRQ_TYPE_EN_RX_SA_460
- **Description:** Interrupt Enable for each IRQ type of Rx SA 460
- **Size:** 32 bits
- **Offset:** 0x11f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 460



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1155 Fields for Register: IRQ_TYPE_EN_RX_SA_460

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1155 Fields for Register: IRQ_TYPE_EN_RX_SA_460 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1155 Fields for Register: IRQ_TYPE_EN_RX_SA_460 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1149 IRQ_TYPE_STAT_RX_SA_460

- **Name:** IRQ_TYPE_STAT_RX_SA_460
- **Description:** Interrupt Status for each IRQ type of Rx SA 460
- **Size:** 32 bits
- **Offset:** 0x11f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 460

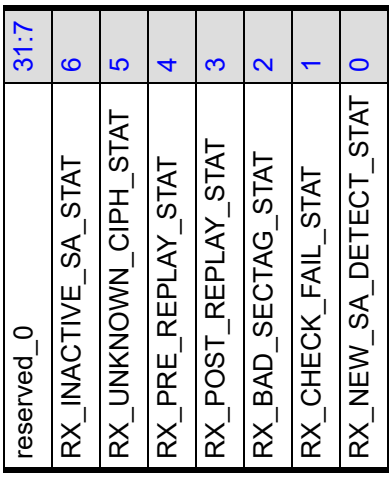


Table 6-1156 Fields for Register: IRQ_TYPE_STAT_RX_SA_460

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1156 Fields for Register: IRQ_TYPE_STAT_RX_SA_460 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1156 Fields for Register: IRQ_TYPE_STAT_RX_SA_460 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1150 IRQ_TYPE_EN_RX_SA_461

- **Name:** IRQ_TYPE_EN_RX_SA_461
- **Description:** Interrupt Enable for each IRQ type of Rx SA 461
- **Size:** 32 bits
- **Offset:** 0x11f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 461

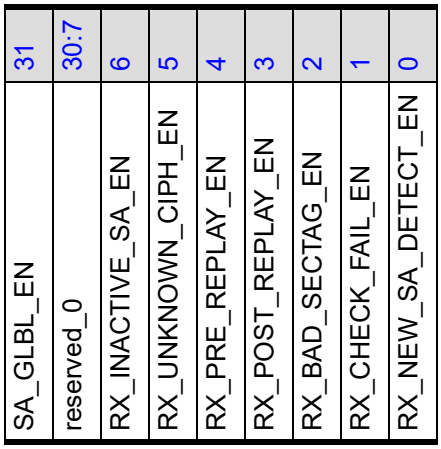


Table 6-1157 Fields for Register: IRQ_TYPE_EN_RX_SA_461

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1157 Fields for Register: IRQ_TYPE_EN_RX_SA_461 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1157 Fields for Register: IRQ_TYPE_EN_RX_SA_461 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1151 IRQ_TYPE_STAT_RX_SA_461

- **Name:** IRQ_TYPE_STAT_RX_SA_461
- **Description:** Interrupt Status for each IRQ type of Rx SA 461
- **Size:** 32 bits
- **Offset:** 0x11fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 461

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1158 Fields for Register: IRQ_TYPE_STAT_RX_SA_461

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1158 Fields for Register: IRQ_TYPE_STAT_RX_SA_461 (Continued)

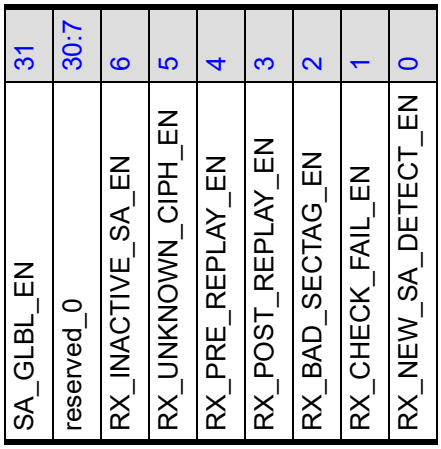
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1158 Fields for Register: IRQ_TYPE_STAT_RX_SA_461 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1152 IRQ_TYPE_EN_RX_SA_462

- **Name:** IRQ_TYPE_EN_RX_SA_462
- **Description:** Interrupt Enable for each IRQ type of Rx SA 462
- **Size:** 32 bits
- **Offset:** 0x1200
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 462



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1159 Fields for Register: IRQ_TYPE_EN_RX_SA_462

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1159 Fields for Register: IRQ_TYPE_EN_RX_SA_462 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1159 Fields for Register: IRQ_TYPE_EN_RX_SA_462 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1153 IRQ_TYPE_STAT_RX_SA_462

- **Name:** IRQ_TYPE_STAT_RX_SA_462
- **Description:** Interrupt Status for each IRQ type of Rx SA 462
- **Size:** 32 bits
- **Offset:** 0x1204
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 462

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1160 Fields for Register: IRQ_TYPE_STAT_RX_SA_462

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1160 Fields for Register: IRQ_TYPE_STAT_RX_SA_462 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1160 Fields for Register: IRQ_TYPE_STAT_RX_SA_462 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1154 IRQ_TYPE_EN_RX_SA_463

- **Name:** IRQ_TYPE_EN_RX_SA_463
- **Description:** Interrupt Enable for each IRQ type of Rx SA 463
- **Size:** 32 bits
- **Offset:** 0x1208
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 463

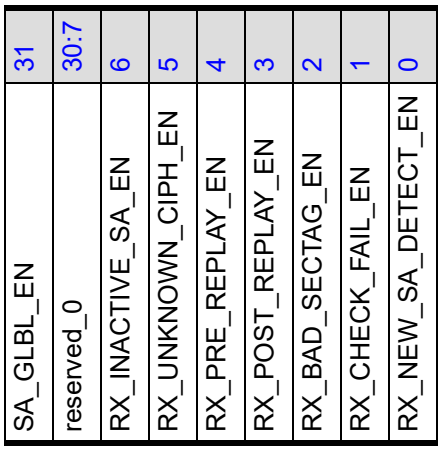


Table 6-1161 Fields for Register: IRQ_TYPE_EN_RX_SA_463

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1161 Fields for Register: IRQ_TYPE_EN_RX_SA_463 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1161 Fields for Register: IRQ_TYPE_EN_RX_SA_463 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1155 IRQ_TYPE_STAT_RX_SA_463

- **Name:** IRQ_TYPE_STAT_RX_SA_463
- **Description:** Interrupt Status for each IRQ type of Rx SA 463
- **Size:** 32 bits
- **Offset:** 0x120c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 463

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1162 Fields for Register: IRQ_TYPE_STAT_RX_SA_463

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1162 Fields for Register: IRQ_TYPE_STAT_RX_SA_463 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1162 Fields for Register: IRQ_TYPE_STAT_RX_SA_463 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1156 IRQ_TYPE_EN_RX_SA_464

- **Name:** IRQ_TYPE_EN_RX_SA_464
- **Description:** Interrupt Enable for each IRQ type of Rx SA 464
- **Size:** 32 bits
- **Offset:** 0x1210
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 464

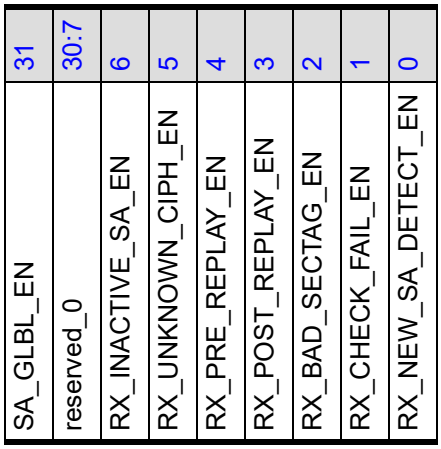


Table 6-1163 Fields for Register: IRQ_TYPE_EN_RX_SA_464

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1163 Fields for Register: IRQ_TYPE_EN_RX_SA_464 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1163 Fields for Register: IRQ_TYPE_EN_RX_SA_464 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1157 IRQ_TYPE_STAT_RX_SA_464

- **Name:** IRQ_TYPE_STAT_RX_SA_464
- **Description:** Interrupt Status for each IRQ type of Rx SA 464
- **Size:** 32 bits
- **Offset:** 0x1214
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 464

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1164 Fields for Register: IRQ_TYPE_STAT_RX_SA_464

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1164 Fields for Register: IRQ_TYPE_STAT_RX_SA_464 (Continued)

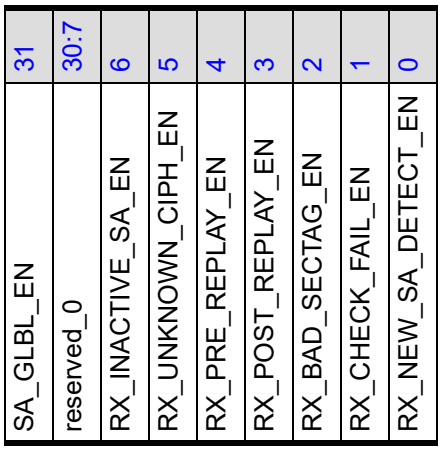
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1164 Fields for Register: IRQ_TYPE_STAT_RX_SA_464 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1158 IRQ_TYPE_EN_RX_SA_465

- **Name:** IRQ_TYPE_EN_RX_SA_465
- **Description:** Interrupt Enable for each IRQ type of Rx SA 465
- **Size:** 32 bits
- **Offset:** 0x1218
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 465



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1165 Fields for Register: IRQ_TYPE_EN_RX_SA_465

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1165 Fields for Register: IRQ_TYPE_EN_RX_SA_465 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1165 Fields for Register: IRQ_TYPE_EN_RX_SA_465 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1159 IRQ_TYPE_STAT_RX_SA_465

- **Name:** IRQ_TYPE_STAT_RX_SA_465
- **Description:** Interrupt Status for each IRQ type of Rx SA 465
- **Size:** 32 bits
- **Offset:** 0x121c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 465

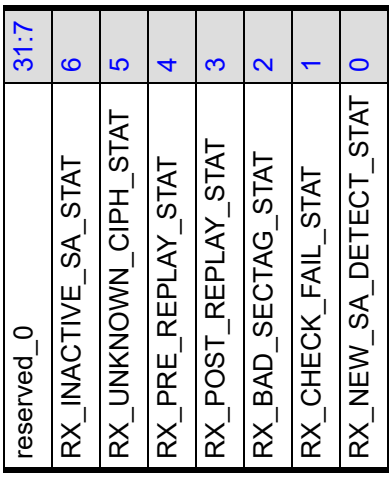


Table 6-1166 Fields for Register: IRQ_TYPE_STAT_RX_SA_465

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1166 Fields for Register: IRQ_TYPE_STAT_RX_SA_465 (Continued)

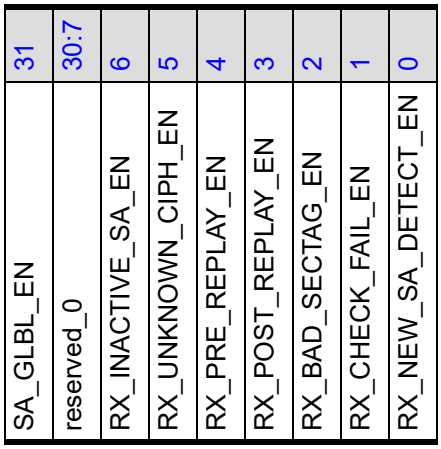
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1166 Fields for Register: IRQ_TYPE_STAT_RX_SA_465 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1160 IRQ_TYPE_EN_RX_SA_466

- **Name:** IRQ_TYPE_EN_RX_SA_466
- **Description:** Interrupt Enable for each IRQ type of Rx SA 466
- **Size:** 32 bits
- **Offset:** 0x1220
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 466



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1167 Fields for Register: IRQ_TYPE_EN_RX_SA_466

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1167 Fields for Register: IRQ_TYPE_EN_RX_SA_466 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1167 Fields for Register: IRQ_TYPE_EN_RX_SA_466 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1161 IRQ_TYPE_STAT_RX_SA_466

- **Name:** IRQ_TYPE_STAT_RX_SA_466
- **Description:** Interrupt Status for each IRQ type of Rx SA 466
- **Size:** 32 bits
- **Offset:** 0x1224
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 466

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1168 Fields for Register: IRQ_TYPE_STAT_RX_SA_466

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1168 Fields for Register: IRQ_TYPE_STAT_RX_SA_466 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1168 Fields for Register: IRQ_TYPE_STAT_RX_SA_466 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1162 IRQ_TYPE_EN_RX_SA_467

- **Name:** IRQ_TYPE_EN_RX_SA_467
- **Description:** Interrupt Enable for each IRQ type of Rx SA 467
- **Size:** 32 bits
- **Offset:** 0x1228
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 467

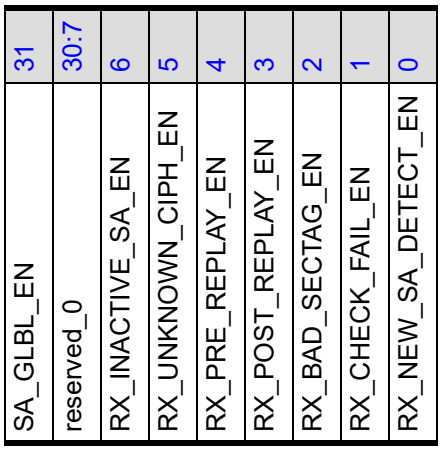


Table 6-1169 Fields for Register: IRQ_TYPE_EN_RX_SA_467

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1169 Fields for Register: IRQ_TYPE_EN_RX_SA_467 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1169 Fields for Register: IRQ_TYPE_EN_RX_SA_467 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1163 IRQ_TYPE_STAT_RX_SA_467

- **Name:** IRQ_TYPE_STAT_RX_SA_467
- **Description:** Interrupt Status for each IRQ type of Rx SA 467
- **Size:** 32 bits
- **Offset:** 0x122c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 467

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1170 Fields for Register: IRQ_TYPE_STAT_RX_SA_467

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1170 Fields for Register: IRQ_TYPE_STAT_RX_SA_467 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1170 Fields for Register: IRQ_TYPE_STAT_RX_SA_467 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1164 IRQ_TYPE_EN_RX_SA_468

- **Name:** IRQ_TYPE_EN_RX_SA_468
- **Description:** Interrupt Enable for each IRQ type of Rx SA 468
- **Size:** 32 bits
- **Offset:** 0x1230
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 468

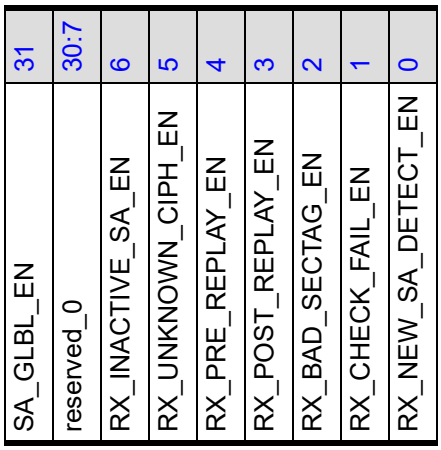


Table 6-1171 Fields for Register: IRQ_TYPE_EN_RX_SA_468

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1171 Fields for Register: IRQ_TYPE_EN_RX_SA_468 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1171 Fields for Register: IRQ_TYPE_EN_RX_SA_468 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1165 IRQ_TYPE_STAT_RX_SA_468

- **Name:** IRQ_TYPE_STAT_RX_SA_468
- **Description:** Interrupt Status for each IRQ type of Rx SA 468
- **Size:** 32 bits
- **Offset:** 0x1234
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 468

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1172 Fields for Register: IRQ_TYPE_STAT_RX_SA_468

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1172 Fields for Register: IRQ_TYPE_STAT_RX_SA_468 (Continued)

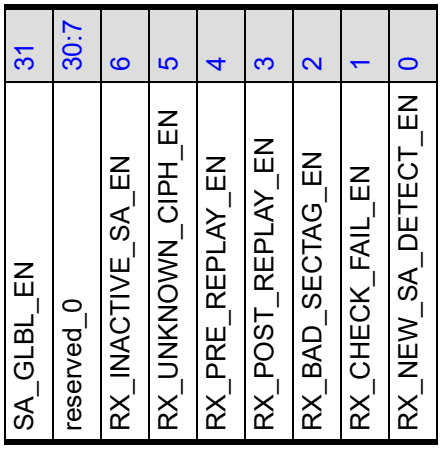
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1172 Fields for Register: IRQ_TYPE_STAT_RX_SA_468 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1166 IRQ_TYPE_EN_RX_SA_469

- **Name:** IRQ_TYPE_EN_RX_SA_469
- **Description:** Interrupt Enable for each IRQ type of Rx SA 469
- **Size:** 32 bits
- **Offset:** 0x1238
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 469



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1173 Fields for Register: IRQ_TYPE_EN_RX_SA_469

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1173 Fields for Register: IRQ_TYPE_EN_RX_SA_469 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1173 Fields for Register: IRQ_TYPE_EN_RX_SA_469 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1167 IRQ_TYPE_STAT_RX_SA_469

- **Name:** IRQ_TYPE_STAT_RX_SA_469
- **Description:** Interrupt Status for each IRQ type of Rx SA 469
- **Size:** 32 bits
- **Offset:** 0x123c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 469

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1174 Fields for Register: IRQ_TYPE_STAT_RX_SA_469

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1174 Fields for Register: IRQ_TYPE_STAT_RX_SA_469 (Continued)

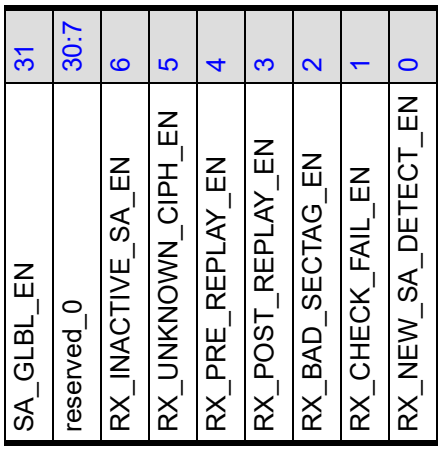
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1174 Fields for Register: IRQ_TYPE_STAT_RX_SA_469 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1168 IRQ_TYPE_EN_RX_SA_470

- **Name:** IRQ_TYPE_EN_RX_SA_470
- **Description:** Interrupt Enable for each IRQ type of Rx SA 470
- **Size:** 32 bits
- **Offset:** 0x1240
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 470



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1175 Fields for Register: IRQ_TYPE_EN_RX_SA_470

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1175 Fields for Register: IRQ_TYPE_EN_RX_SA_470 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1175 Fields for Register: IRQ_TYPE_EN_RX_SA_470 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1169 IRQ_TYPE_STAT_RX_SA_470

- **Name:** IRQ_TYPE_STAT_RX_SA_470
- **Description:** Interrupt Status for each IRQ type of Rx SA 470
- **Size:** 32 bits
- **Offset:** 0x1244
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 470

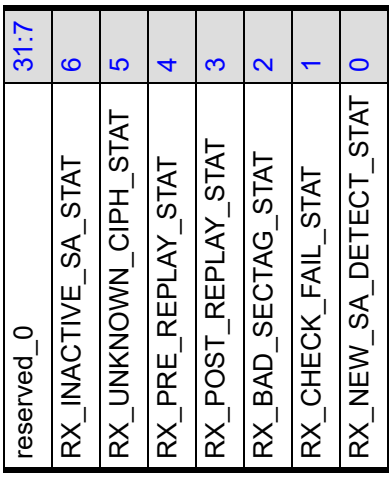


Table 6-1176 Fields for Register: IRQ_TYPE_STAT_RX_SA_470

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1176 Fields for Register: IRQ_TYPE_STAT_RX_SA_470 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1176 Fields for Register: IRQ_TYPE_STAT_RX_SA_470 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1170 IRQ_TYPE_EN_RX_SA_471

- **Name:** IRQ_TYPE_EN_RX_SA_471
- **Description:** Interrupt Enable for each IRQ type of Rx SA 471
- **Size:** 32 bits
- **Offset:** 0x1248
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 471

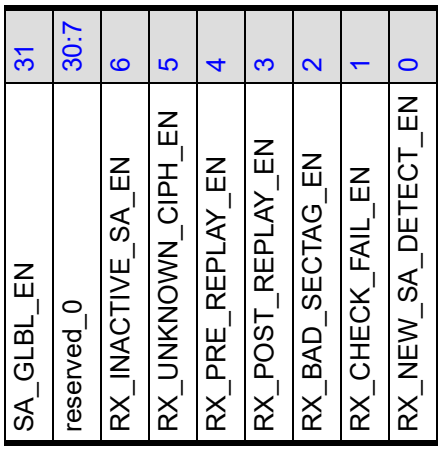


Table 6-1177 Fields for Register: IRQ_TYPE_EN_RX_SA_471

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1177 Fields for Register: IRQ_TYPE_EN_RX_SA_471 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1177 Fields for Register: IRQ_TYPE_EN_RX_SA_471 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1171 IRQ_TYPE_STAT_RX_SA_471

- **Name:** IRQ_TYPE_STAT_RX_SA_471
- **Description:** Interrupt Status for each IRQ type of Rx SA 471
- **Size:** 32 bits
- **Offset:** 0x124c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 471

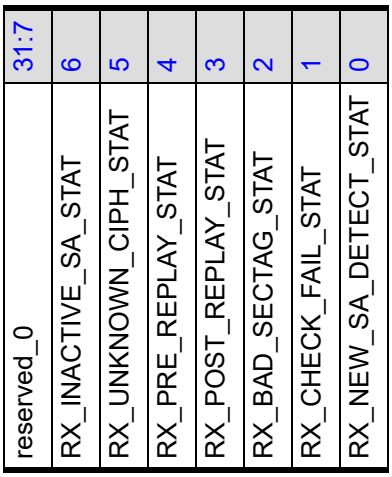


Table 6-1178 Fields for Register: IRQ_TYPE_STAT_RX_SA_471

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1178 Fields for Register: IRQ_TYPE_STAT_RX_SA_471 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1178 Fields for Register: IRQ_TYPE_STAT_RX_SA_471 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1172 IRQ_TYPE_EN_RX_SA_472

- **Name:** IRQ_TYPE_EN_RX_SA_472
- **Description:** Interrupt Enable for each IRQ type of Rx SA 472
- **Size:** 32 bits
- **Offset:** 0x1250
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 472

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1179 Fields for Register: IRQ_TYPE_EN_RX_SA_472

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1179 Fields for Register: IRQ_TYPE_EN_RX_SA_472 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1179 Fields for Register: IRQ_TYPE_EN_RX_SA_472 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1173 IRQ_TYPE_STAT_RX_SA_472

- **Name:** IRQ_TYPE_STAT_RX_SA_472
- **Description:** Interrupt Status for each IRQ type of Rx SA 472
- **Size:** 32 bits
- **Offset:** 0x1254
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 472

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1180 Fields for Register: IRQ_TYPE_STAT_RX_SA_472

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1180 Fields for Register: IRQ_TYPE_STAT_RX_SA_472 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1180 Fields for Register: IRQ_TYPE_STAT_RX_SA_472 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1174 IRQ_TYPE_EN_RX_SA_473

- **Name:** IRQ_TYPE_EN_RX_SA_473
- **Description:** Interrupt Enable for each IRQ type of Rx SA 473
- **Size:** 32 bits
- **Offset:** 0x1258
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 473

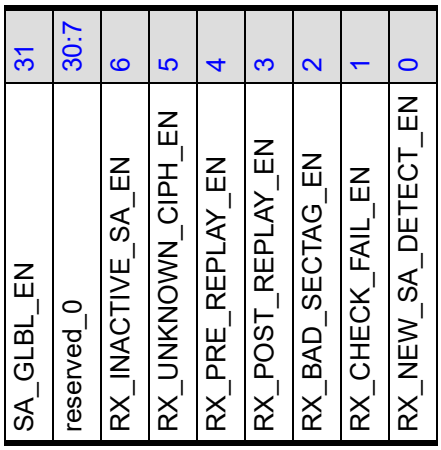


Table 6-1181 Fields for Register: IRQ_TYPE_EN_RX_SA_473

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1181 Fields for Register: IRQ_TYPE_EN_RX_SA_473 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1181 Fields for Register: IRQ_TYPE_EN_RX_SA_473 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1175 IRQ_TYPE_STAT_RX_SA_473

- **Name:** IRQ_TYPE_STAT_RX_SA_473
- **Description:** Interrupt Status for each IRQ type of Rx SA 473
- **Size:** 32 bits
- **Offset:** 0x125c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 473

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1182 Fields for Register: IRQ_TYPE_STAT_RX_SA_473

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1182 Fields for Register: IRQ_TYPE_STAT_RX_SA_473 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1182 Fields for Register: IRQ_TYPE_STAT_RX_SA_473 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1176 IRQ_TYPE_EN_RX_SA_474

- **Name:** IRQ_TYPE_EN_RX_SA_474
- **Description:** Interrupt Enable for each IRQ type of Rx SA 474
- **Size:** 32 bits
- **Offset:** 0x1260
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 474

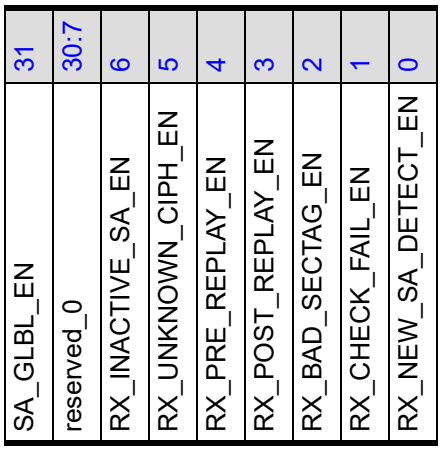


Table 6-1183 Fields for Register: IRQ_TYPE_EN_RX_SA_474

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1183 Fields for Register: IRQ_TYPE_EN_RX_SA_474 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1183 Fields for Register: IRQ_TYPE_EN_RX_SA_474 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1177 IRQ_TYPE_STAT_RX_SA_474

- **Name:** IRQ_TYPE_STAT_RX_SA_474
- **Description:** Interrupt Status for each IRQ type of Rx SA 474
- **Size:** 32 bits
- **Offset:** 0x1264
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 474

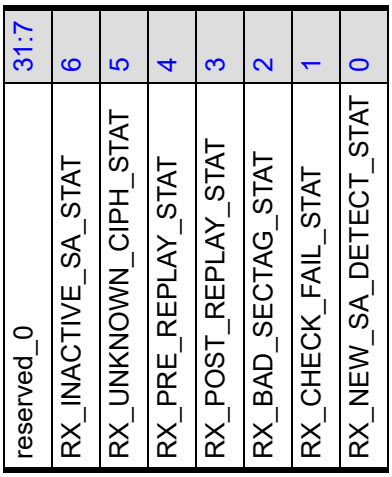


Table 6-1184 Fields for Register: IRQ_TYPE_STAT_RX_SA_474

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1184 Fields for Register: IRQ_TYPE_STAT_RX_SA_474 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1184 Fields for Register: IRQ_TYPE_STAT_RX_SA_474 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1178 IRQ_TYPE_EN_RX_SA_475

- **Name:** IRQ_TYPE_EN_RX_SA_475
- **Description:** Interrupt Enable for each IRQ type of Rx SA 475
- **Size:** 32 bits
- **Offset:** 0x1268
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 475

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1185 Fields for Register: IRQ_TYPE_EN_RX_SA_475

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1185 Fields for Register: IRQ_TYPE_EN_RX_SA_475 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1185 Fields for Register: IRQ_TYPE_EN_RX_SA_475 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1179 IRQ_TYPE_STAT_RX_SA_475

- **Name:** IRQ_TYPE_STAT_RX_SA_475
- **Description:** Interrupt Status for each IRQ type of Rx SA 475
- **Size:** 32 bits
- **Offset:** 0x126c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 475

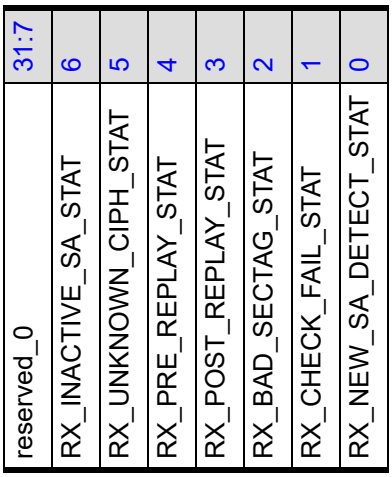


Table 6-1186 Fields for Register: IRQ_TYPE_STAT_RX_SA_475

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1186 Fields for Register: IRQ_TYPE_STAT_RX_SA_475 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1186 Fields for Register: IRQ_TYPE_STAT_RX_SA_475 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1180 IRQ_TYPE_EN_RX_SA_476

- **Name:** IRQ_TYPE_EN_RX_SA_476
- **Description:** Interrupt Enable for each IRQ type of Rx SA 476
- **Size:** 32 bits
- **Offset:** 0x1270
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 476

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1187 Fields for Register: IRQ_TYPE_EN_RX_SA_476

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1187 Fields for Register: IRQ_TYPE_EN_RX_SA_476 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1187 Fields for Register: IRQ_TYPE_EN_RX_SA_476 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1181 IRQ_TYPE_STAT_RX_SA_476

- **Name:** IRQ_TYPE_STAT_RX_SA_476
- **Description:** Interrupt Status for each IRQ type of Rx SA 476
- **Size:** 32 bits
- **Offset:** 0x1274
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 476

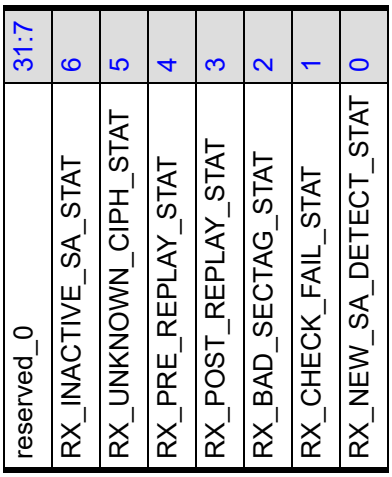


Table 6-1188 Fields for Register: IRQ_TYPE_STAT_RX_SA_476

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1188 Fields for Register: IRQ_TYPE_STAT_RX_SA_476 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1188 Fields for Register: IRQ_TYPE_STAT_RX_SA_476 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1182 IRQ_TYPE_EN_RX_SA_477

- **Name:** IRQ_TYPE_EN_RX_SA_477
- **Description:** Interrupt Enable for each IRQ type of Rx SA 477
- **Size:** 32 bits
- **Offset:** 0x1278
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 477

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1189 Fields for Register: IRQ_TYPE_EN_RX_SA_477

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1189 Fields for Register: IRQ_TYPE_EN_RX_SA_477 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1189 Fields for Register: IRQ_TYPE_EN_RX_SA_477 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1183 IRQ_TYPE_STAT_RX_SA_477

- **Name:** IRQ_TYPE_STAT_RX_SA_477
- **Description:** Interrupt Status for each IRQ type of Rx SA 477
- **Size:** 32 bits
- **Offset:** 0x127c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 477

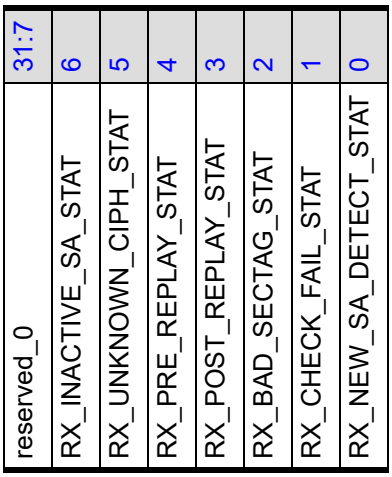


Table 6-1190 Fields for Register: IRQ_TYPE_STAT_RX_SA_477

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1190 Fields for Register: IRQ_TYPE_STAT_RX_SA_477 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1190 Fields for Register: IRQ_TYPE_STAT_RX_SA_477 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1184 IRQ_TYPE_EN_RX_SA_478

- **Name:** IRQ_TYPE_EN_RX_SA_478
- **Description:** Interrupt Enable for each IRQ type of Rx SA 478
- **Size:** 32 bits
- **Offset:** 0x1280
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 478

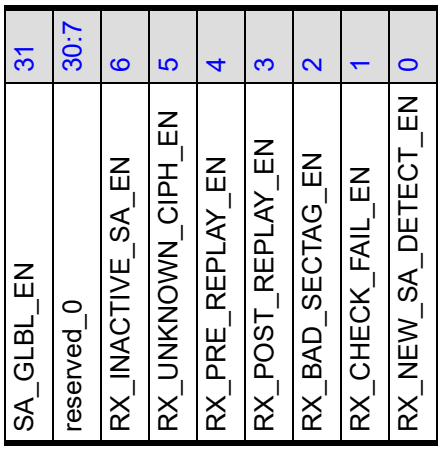


Table 6-1191 Fields for Register: IRQ_TYPE_EN_RX_SA_478

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1191 Fields for Register: IRQ_TYPE_EN_RX_SA_478 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1191 Fields for Register: IRQ_TYPE_EN_RX_SA_478 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1185 IRQ_TYPE_STAT_RX_SA_478

- **Name:** IRQ_TYPE_STAT_RX_SA_478
- **Description:** Interrupt Status for each IRQ type of Rx SA 478
- **Size:** 32 bits
- **Offset:** 0x1284
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 478

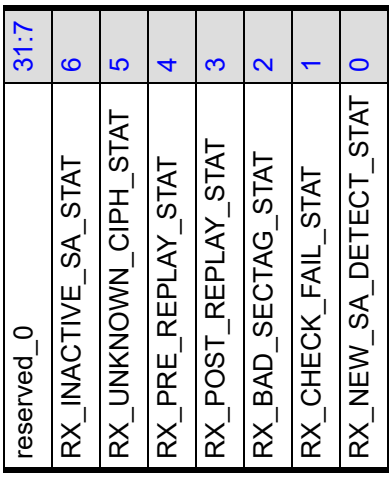


Table 6-1192 Fields for Register: IRQ_TYPE_STAT_RX_SA_478

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1192 Fields for Register: IRQ_TYPE_STAT_RX_SA_478 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1192 Fields for Register: IRQ_TYPE_STAT_RX_SA_478 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1186 IRQ_TYPE_EN_RX_SA_479

- **Name:** IRQ_TYPE_EN_RX_SA_479
- **Description:** Interrupt Enable for each IRQ type of Rx SA 479
- **Size:** 32 bits
- **Offset:** 0x1288
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 479

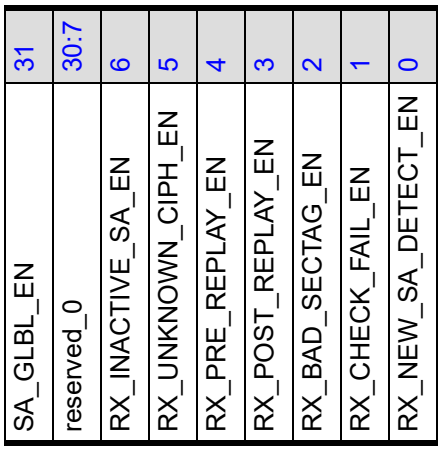


Table 6-1193 Fields for Register: IRQ_TYPE_EN_RX_SA_479

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1193 Fields for Register: IRQ_TYPE_EN_RX_SA_479 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1193 Fields for Register: IRQ_TYPE_EN_RX_SA_479 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1187 IRQ_TYPE_STAT_RX_SA_479

- **Name:** IRQ_TYPE_STAT_RX_SA_479
- **Description:** Interrupt Status for each IRQ type of Rx SA 479
- **Size:** 32 bits
- **Offset:** 0x128c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 479

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1194 Fields for Register: IRQ_TYPE_STAT_RX_SA_479

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1194 Fields for Register: IRQ_TYPE_STAT_RX_SA_479 (Continued)

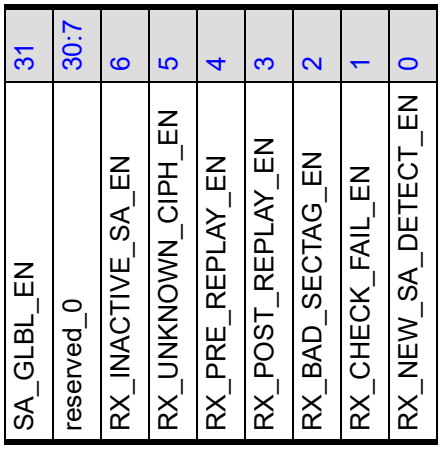
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1194 Fields for Register: IRQ_TYPE_STAT_RX_SA_479 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1188 IRQ_TYPE_EN_RX_SA_480

- **Name:** IRQ_TYPE_EN_RX_SA_480
- **Description:** Interrupt Enable for each IRQ type of Rx SA 480
- **Size:** 32 bits
- **Offset:** 0x1290
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 480



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1195 Fields for Register: IRQ_TYPE_EN_RX_SA_480

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1195 Fields for Register: IRQ_TYPE_EN_RX_SA_480 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1195 Fields for Register: IRQ_TYPE_EN_RX_SA_480 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1189 IRQ_TYPE_STAT_RX_SA_480

- **Name:** IRQ_TYPE_STAT_RX_SA_480
- **Description:** Interrupt Status for each IRQ type of Rx SA 480
- **Size:** 32 bits
- **Offset:** 0x1294
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 480

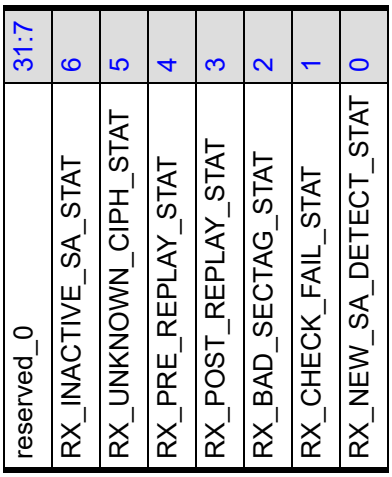


Table 6-1196 Fields for Register: IRQ_TYPE_STAT_RX_SA_480

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1196 Fields for Register: IRQ_TYPE_STAT_RX_SA_480 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1196 Fields for Register: IRQ_TYPE_STAT_RX_SA_480 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1190 IRQ_TYPE_EN_RX_SA_481

- **Name:** IRQ_TYPE_EN_RX_SA_481
- **Description:** Interrupt Enable for each IRQ type of Rx SA 481
- **Size:** 32 bits
- **Offset:** 0x1298
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 481

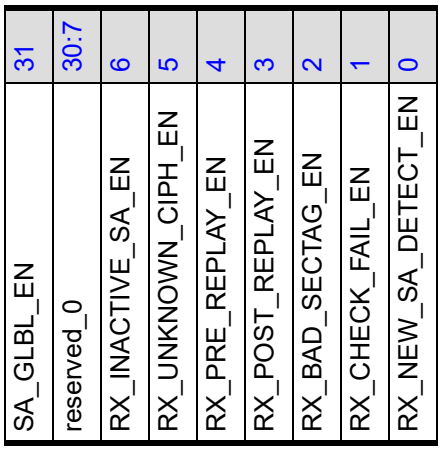


Table 6-1197 Fields for Register: IRQ_TYPE_EN_RX_SA_481

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1197 Fields for Register: IRQ_TYPE_EN_RX_SA_481 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1197 Fields for Register: IRQ_TYPE_EN_RX_SA_481 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1191 IRQ_TYPE_STAT_RX_SA_481

- **Name:** IRQ_TYPE_STAT_RX_SA_481
- **Description:** Interrupt Status for each IRQ type of Rx SA 481
- **Size:** 32 bits
- **Offset:** 0x129c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 481

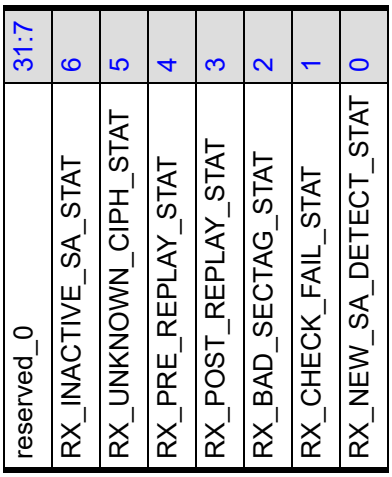


Table 6-1198 Fields for Register: IRQ_TYPE_STAT_RX_SA_481

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1198 Fields for Register: IRQ_TYPE_STAT_RX_SA_481 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1198 Fields for Register: IRQ_TYPE_STAT_RX_SA_481 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1192 IRQ_TYPE_EN_RX_SA_482

- **Name:** IRQ_TYPE_EN_RX_SA_482
- **Description:** Interrupt Enable for each IRQ type of Rx SA 482
- **Size:** 32 bits
- **Offset:** 0x12a0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 482

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1199 Fields for Register: IRQ_TYPE_EN_RX_SA_482

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1199 Fields for Register: IRQ_TYPE_EN_RX_SA_482 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1199 Fields for Register: IRQ_TYPE_EN_RX_SA_482 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1193 IRQ_TYPE_STAT_RX_SA_482

- **Name:** IRQ_TYPE_STAT_RX_SA_482
- **Description:** Interrupt Status for each IRQ type of Rx SA 482
- **Size:** 32 bits
- **Offset:** 0x12a4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 482

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1200 Fields for Register: IRQ_TYPE_STAT_RX_SA_482

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1200 Fields for Register: IRQ_TYPE_STAT_RX_SA_482 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1200 Fields for Register: IRQ_TYPE_STAT_RX_SA_482 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1194 IRQ_TYPE_EN_RX_SA_483

- **Name:** IRQ_TYPE_EN_RX_SA_483
- **Description:** Interrupt Enable for each IRQ type of Rx SA 483
- **Size:** 32 bits
- **Offset:** 0x12a8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 483

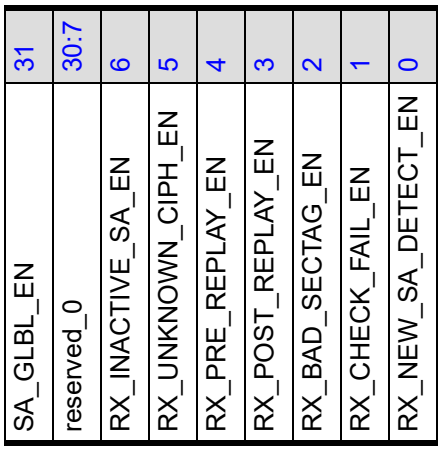


Table 6-1201 Fields for Register: IRQ_TYPE_EN_RX_SA_483

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1201 Fields for Register: IRQ_TYPE_EN_RX_SA_483 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1201 Fields for Register: IRQ_TYPE_EN_RX_SA_483 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1195 IRQ_TYPE_STAT_RX_SA_483

- **Name:** IRQ_TYPE_STAT_RX_SA_483
- **Description:** Interrupt Status for each IRQ type of Rx SA 483
- **Size:** 32 bits
- **Offset:** 0x12ac
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 483

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1202 Fields for Register: IRQ_TYPE_STAT_RX_SA_483

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1202 Fields for Register: IRQ_TYPE_STAT_RX_SA_483 (Continued)

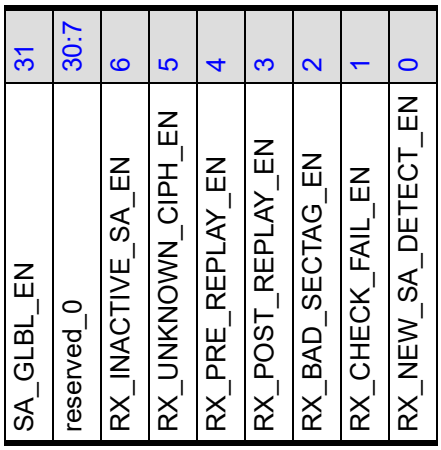
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1202 Fields for Register: IRQ_TYPE_STAT_RX_SA_483 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1196 IRQ_TYPE_EN_RX_SA_484

- **Name:** IRQ_TYPE_EN_RX_SA_484
- **Description:** Interrupt Enable for each IRQ type of Rx SA 484
- **Size:** 32 bits
- **Offset:** 0x12b0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 484



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1203 Fields for Register: IRQ_TYPE_EN_RX_SA_484

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1203 Fields for Register: IRQ_TYPE_EN_RX_SA_484 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1203 Fields for Register: IRQ_TYPE_EN_RX_SA_484 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1197 IRQ_TYPE_STAT_RX_SA_484

- **Name:** IRQ_TYPE_STAT_RX_SA_484
- **Description:** Interrupt Status for each IRQ type of Rx SA 484
- **Size:** 32 bits
- **Offset:** 0x12b4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 484

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1204 Fields for Register: IRQ_TYPE_STAT_RX_SA_484

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1204 Fields for Register: IRQ_TYPE_STAT_RX_SA_484 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1204 Fields for Register: IRQ_TYPE_STAT_RX_SA_484 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1198 IRQ_TYPE_EN_RX_SA_485

- **Name:** IRQ_TYPE_EN_RX_SA_485
- **Description:** Interrupt Enable for each IRQ type of Rx SA 485
- **Size:** 32 bits
- **Offset:** 0x12b8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 485

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1205 Fields for Register: IRQ_TYPE_EN_RX_SA_485

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1205 Fields for Register: IRQ_TYPE_EN_RX_SA_485 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1205 Fields for Register: IRQ_TYPE_EN_RX_SA_485 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1199 IRQ_TYPE_STAT_RX_SA_485

- **Name:** IRQ_TYPE_STAT_RX_SA_485
- **Description:** Interrupt Status for each IRQ type of Rx SA 485
- **Size:** 32 bits
- **Offset:** 0x12bc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 485

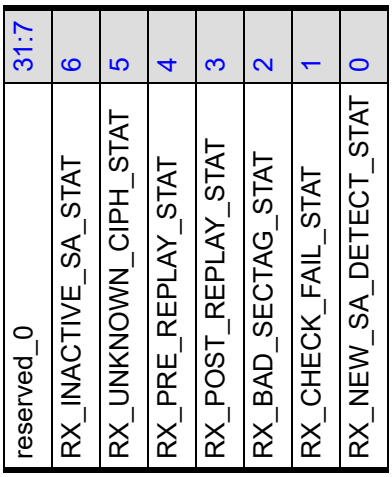


Table 6-1206 Fields for Register: IRQ_TYPE_STAT_RX_SA_485

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1206 Fields for Register: IRQ_TYPE_STAT_RX_SA_485 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1206 Fields for Register: IRQ_TYPE_STAT_RX_SA_485 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1200 IRQ_TYPE_EN_RX_SA_486

- **Name:** IRQ_TYPE_EN_RX_SA_486
- **Description:** Interrupt Enable for each IRQ type of Rx SA 486
- **Size:** 32 bits
- **Offset:** 0x12c0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 486

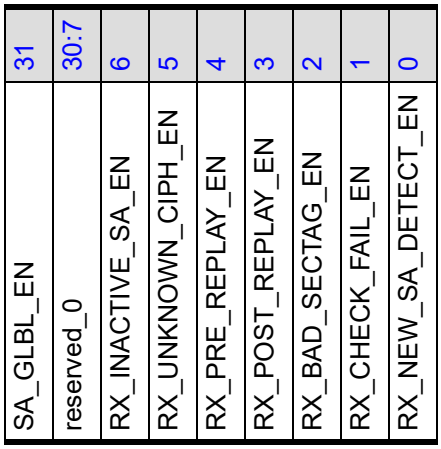


Table 6-1207 Fields for Register: IRQ_TYPE_EN_RX_SA_486

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1207 Fields for Register: IRQ_TYPE_EN_RX_SA_486 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1207 Fields for Register: IRQ_TYPE_EN_RX_SA_486 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1201 IRQ_TYPE_STAT_RX_SA_486

- **Name:** IRQ_TYPE_STAT_RX_SA_486
- **Description:** Interrupt Status for each IRQ type of Rx SA 486
- **Size:** 32 bits
- **Offset:** 0x12c4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 486

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1208 Fields for Register: IRQ_TYPE_STAT_RX_SA_486

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1208 Fields for Register: IRQ_TYPE_STAT_RX_SA_486 (Continued)

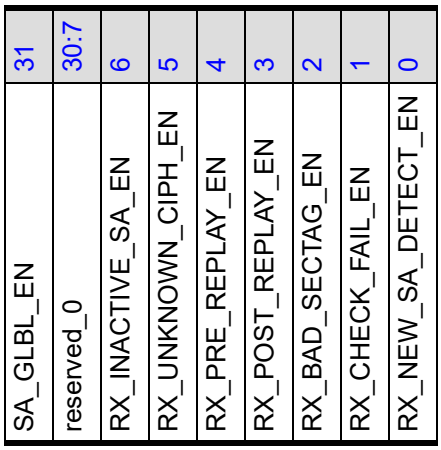
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1208 Fields for Register: IRQ_TYPE_STAT_RX_SA_486 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1202 IRQ_TYPE_EN_RX_SA_487

- **Name:** IRQ_TYPE_EN_RX_SA_487
- **Description:** Interrupt Enable for each IRQ type of Rx SA 487
- **Size:** 32 bits
- **Offset:** 0x12c8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 487



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1209 Fields for Register: IRQ_TYPE_EN_RX_SA_487

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1209 Fields for Register: IRQ_TYPE_EN_RX_SA_487 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1209 Fields for Register: IRQ_TYPE_EN_RX_SA_487 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1203 IRQ_TYPE_STAT_RX_SA_487

- **Name:** IRQ_TYPE_STAT_RX_SA_487
- **Description:** Interrupt Status for each IRQ type of Rx SA 487
- **Size:** 32 bits
- **Offset:** 0x12cc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 487

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1210 Fields for Register: IRQ_TYPE_STAT_RX_SA_487

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1210 Fields for Register: IRQ_TYPE_STAT_RX_SA_487 (Continued)

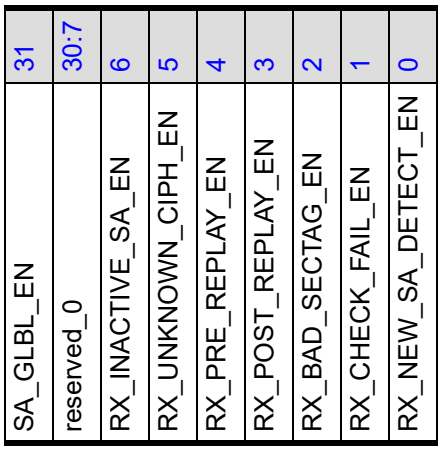
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1210 Fields for Register: IRQ_TYPE_STAT_RX_SA_487 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1204 IRQ_TYPE_EN_RX_SA_488

- **Name:** IRQ_TYPE_EN_RX_SA_488
- **Description:** Interrupt Enable for each IRQ type of Rx SA 488
- **Size:** 32 bits
- **Offset:** 0x12d0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 488



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1211 Fields for Register: IRQ_TYPE_EN_RX_SA_488

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1211 Fields for Register: IRQ_TYPE_EN_RX_SA_488 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1211 Fields for Register: IRQ_TYPE_EN_RX_SA_488 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1205 IRQ_TYPE_STAT_RX_SA_488

- **Name:** IRQ_TYPE_STAT_RX_SA_488
- **Description:** Interrupt Status for each IRQ type of Rx SA 488
- **Size:** 32 bits
- **Offset:** 0x12d4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 488

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1212 Fields for Register: IRQ_TYPE_STAT_RX_SA_488

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1212 Fields for Register: IRQ_TYPE_STAT_RX_SA_488 (Continued)

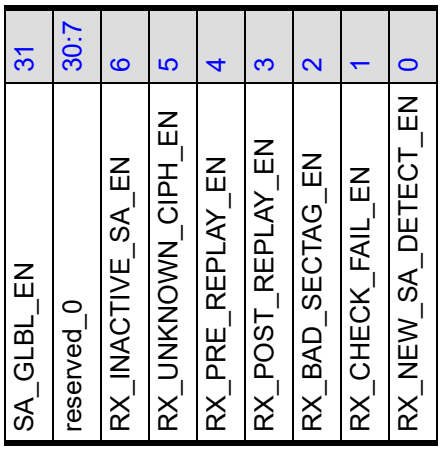
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1212 Fields for Register: IRQ_TYPE_STAT_RX_SA_488 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1206 IRQ_TYPE_EN_RX_SA_489

- **Name:** IRQ_TYPE_EN_RX_SA_489
- **Description:** Interrupt Enable for each IRQ type of Rx SA 489
- **Size:** 32 bits
- **Offset:** 0x12d8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 489



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1213 Fields for Register: IRQ_TYPE_EN_RX_SA_489

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1213 Fields for Register: IRQ_TYPE_EN_RX_SA_489 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1213 Fields for Register: IRQ_TYPE_EN_RX_SA_489 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1207 IRQ_TYPE_STAT_RX_SA_489

- **Name:** IRQ_TYPE_STAT_RX_SA_489
- **Description:** Interrupt Status for each IRQ type of Rx SA 489
- **Size:** 32 bits
- **Offset:** 0x12dc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 489

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1214 Fields for Register: IRQ_TYPE_STAT_RX_SA_489

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1214 Fields for Register: IRQ_TYPE_STAT_RX_SA_489 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1214 Fields for Register: IRQ_TYPE_STAT_RX_SA_489 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1208 IRQ_TYPE_EN_RX_SA_490

- **Name:** IRQ_TYPE_EN_RX_SA_490
- **Description:** Interrupt Enable for each IRQ type of Rx SA 490
- **Size:** 32 bits
- **Offset:** 0x12e0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 490

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1215 Fields for Register: IRQ_TYPE_EN_RX_SA_490

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1215 Fields for Register: IRQ_TYPE_EN_RX_SA_490 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1215 Fields for Register: IRQ_TYPE_EN_RX_SA_490 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1209 IRQ_TYPE_STAT_RX_SA_490

- **Name:** IRQ_TYPE_STAT_RX_SA_490
- **Description:** Interrupt Status for each IRQ type of Rx SA 490
- **Size:** 32 bits
- **Offset:** 0x12e4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 490

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1216 Fields for Register: IRQ_TYPE_STAT_RX_SA_490

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1216 Fields for Register: IRQ_TYPE_STAT_RX_SA_490 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1216 Fields for Register: IRQ_TYPE_STAT_RX_SA_490 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1210 IRQ_TYPE_EN_RX_SA_491

- **Name:** IRQ_TYPE_EN_RX_SA_491
- **Description:** Interrupt Enable for each IRQ type of Rx SA 491
- **Size:** 32 bits
- **Offset:** 0x12e8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 491

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1217 Fields for Register: IRQ_TYPE_EN_RX_SA_491

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1217 Fields for Register: IRQ_TYPE_EN_RX_SA_491 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1217 Fields for Register: IRQ_TYPE_EN_RX_SA_491 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1211 IRQ_TYPE_STAT_RX_SA_491

- **Name:** IRQ_TYPE_STAT_RX_SA_491
- **Description:** Interrupt Status for each IRQ type of Rx SA 491
- **Size:** 32 bits
- **Offset:** 0x12ec
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 491

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1218 Fields for Register: IRQ_TYPE_STAT_RX_SA_491

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1218 Fields for Register: IRQ_TYPE_STAT_RX_SA_491 (Continued)

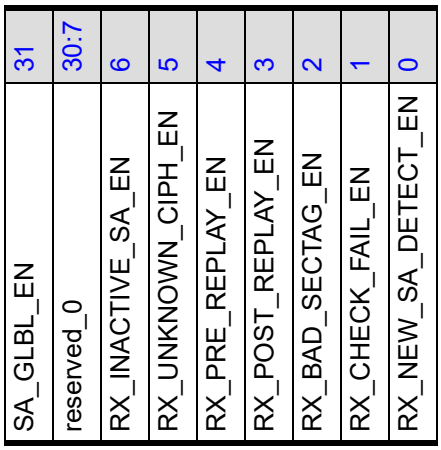
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1218 Fields for Register: IRQ_TYPE_STAT_RX_SA_491 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1212 IRQ_TYPE_EN_RX_SA_492

- **Name:** IRQ_TYPE_EN_RX_SA_492
- **Description:** Interrupt Enable for each IRQ type of Rx SA 492
- **Size:** 32 bits
- **Offset:** 0x12f0
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 492



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1219 Fields for Register: IRQ_TYPE_EN_RX_SA_492

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1219 Fields for Register: IRQ_TYPE_EN_RX_SA_492 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1219 Fields for Register: IRQ_TYPE_EN_RX_SA_492 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1213 IRQ_TYPE_STAT_RX_SA_492

- **Name:** IRQ_TYPE_STAT_RX_SA_492
- **Description:** Interrupt Status for each IRQ type of Rx SA 492
- **Size:** 32 bits
- **Offset:** 0x12f4
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 492

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1220 Fields for Register: IRQ_TYPE_STAT_RX_SA_492

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1220 Fields for Register: IRQ_TYPE_STAT_RX_SA_492 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1220 Fields for Register: IRQ_TYPE_STAT_RX_SA_492 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1214 IRQ_TYPE_EN_RX_SA_493

- **Name:** IRQ_TYPE_EN_RX_SA_493
- **Description:** Interrupt Enable for each IRQ type of Rx SA 493
- **Size:** 32 bits
- **Offset:** 0x12f8
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 493

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1221 Fields for Register: IRQ_TYPE_EN_RX_SA_493

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1221 Fields for Register: IRQ_TYPE_EN_RX_SA_493 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1221 Fields for Register: IRQ_TYPE_EN_RX_SA_493 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1215 IRQ_TYPE_STAT_RX_SA_493

- **Name:** IRQ_TYPE_STAT_RX_SA_493
- **Description:** Interrupt Status for each IRQ type of Rx SA 493
- **Size:** 32 bits
- **Offset:** 0x12fc
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 493

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1222 Fields for Register: IRQ_TYPE_STAT_RX_SA_493

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1222 Fields for Register: IRQ_TYPE_STAT_RX_SA_493 (Continued)

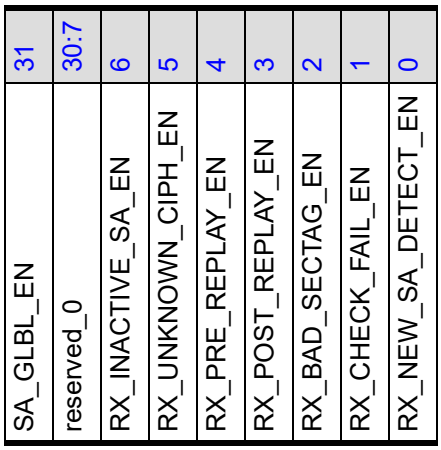
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1222 Fields for Register: IRQ_TYPE_STAT_RX_SA_493 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1216 IRQ_TYPE_EN_RX_SA_494

- **Name:** IRQ_TYPE_EN_RX_SA_494
- **Description:** Interrupt Enable for each IRQ type of Rx SA 494
- **Size:** 32 bits
- **Offset:** 0x1300
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 494



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1223 Fields for Register: IRQ_TYPE_EN_RX_SA_494

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1223 Fields for Register: IRQ_TYPE_EN_RX_SA_494 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1223 Fields for Register: IRQ_TYPE_EN_RX_SA_494 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1217 IRQ_TYPE_STAT_RX_SA_494

- **Name:** IRQ_TYPE_STAT_RX_SA_494
- **Description:** Interrupt Status for each IRQ type of Rx SA 494
- **Size:** 32 bits
- **Offset:** 0x1304
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 494

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1224 Fields for Register: IRQ_TYPE_STAT_RX_SA_494

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1224 Fields for Register: IRQ_TYPE_STAT_RX_SA_494 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1224 Fields for Register: IRQ_TYPE_STAT_RX_SA_494 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1218 IRQ_TYPE_EN_RX_SA_495

- **Name:** IRQ_TYPE_EN_RX_SA_495
- **Description:** Interrupt Enable for each IRQ type of Rx SA 495
- **Size:** 32 bits
- **Offset:** 0x1308
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 495

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1225 Fields for Register: IRQ_TYPE_EN_RX_SA_495

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1225 Fields for Register: IRQ_TYPE_EN_RX_SA_495 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1225 Fields for Register: IRQ_TYPE_EN_RX_SA_495 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1219 IRQ_TYPE_STAT_RX_SA_495

- **Name:** IRQ_TYPE_STAT_RX_SA_495
- **Description:** Interrupt Status for each IRQ type of Rx SA 495
- **Size:** 32 bits
- **Offset:** 0x130c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 495

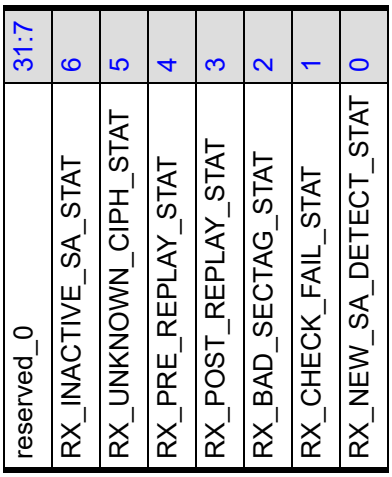


Table 6-1226 Fields for Register: IRQ_TYPE_STAT_RX_SA_495

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1226 Fields for Register: IRQ_TYPE_STAT_RX_SA_495 (Continued)

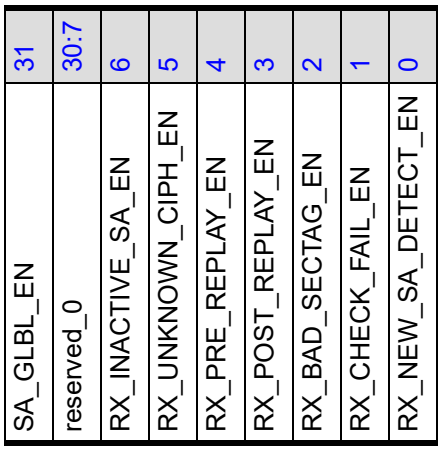
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1226 Fields for Register: IRQ_TYPE_STAT_RX_SA_495 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1220 IRQ_TYPE_EN_RX_SA_496

- **Name:** IRQ_TYPE_EN_RX_SA_496
- **Description:** Interrupt Enable for each IRQ type of Rx SA 496
- **Size:** 32 bits
- **Offset:** 0x1310
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 496



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1227 Fields for Register: IRQ_TYPE_EN_RX_SA_496

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1227 Fields for Register: IRQ_TYPE_EN_RX_SA_496 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1227 Fields for Register: IRQ_TYPE_EN_RX_SA_496 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1221 IRQ_TYPE_STAT_RX_SA_496

- **Name:** IRQ_TYPE_STAT_RX_SA_496
- **Description:** Interrupt Status for each IRQ type of Rx SA 496
- **Size:** 32 bits
- **Offset:** 0x1314
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 496

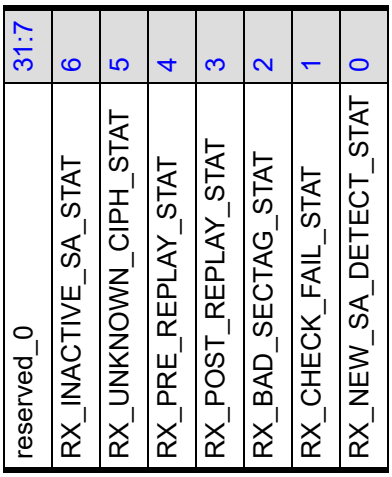


Table 6-1228 Fields for Register: IRQ_TYPE_STAT_RX_SA_496

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1228 Fields for Register: IRQ_TYPE_STAT_RX_SA_496 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1228 Fields for Register: IRQ_TYPE_STAT_RX_SA_496 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1222 IRQ_TYPE_EN_RX_SA_497

- **Name:** IRQ_TYPE_EN_RX_SA_497
- **Description:** Interrupt Enable for each IRQ type of Rx SA 497
- **Size:** 32 bits
- **Offset:** 0x1318
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 497

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1229 Fields for Register: IRQ_TYPE_EN_RX_SA_497

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1229 Fields for Register: IRQ_TYPE_EN_RX_SA_497 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1229 Fields for Register: IRQ_TYPE_EN_RX_SA_497 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1223 IRQ_TYPE_STAT_RX_SA_497

- **Name:** IRQ_TYPE_STAT_RX_SA_497
- **Description:** Interrupt Status for each IRQ type of Rx SA 497
- **Size:** 32 bits
- **Offset:** 0x131c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 497

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1230 Fields for Register: IRQ_TYPE_STAT_RX_SA_497

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1230 Fields for Register: IRQ_TYPE_STAT_RX_SA_497 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1230 Fields for Register: IRQ_TYPE_STAT_RX_SA_497 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1224 IRQ_TYPE_EN_RX_SA_498

- **Name:** IRQ_TYPE_EN_RX_SA_498
- **Description:** Interrupt Enable for each IRQ type of Rx SA 498
- **Size:** 32 bits
- **Offset:** 0x1320
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 498

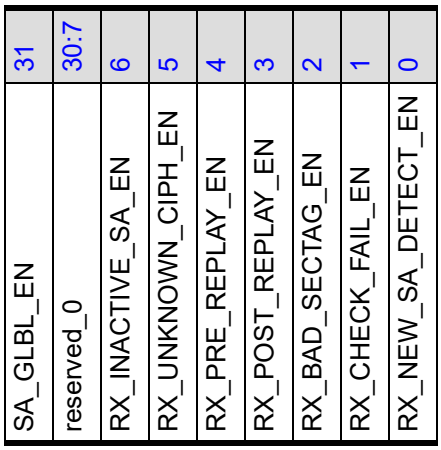


Table 6-1231 Fields for Register: IRQ_TYPE_EN_RX_SA_498

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1231 Fields for Register: IRQ_TYPE_EN_RX_SA_498 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1231 Fields for Register: IRQ_TYPE_EN_RX_SA_498 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1225 IRQ_TYPE_STAT_RX_SA_498

- **Name:** IRQ_TYPE_STAT_RX_SA_498
- **Description:** Interrupt Status for each IRQ type of Rx SA 498
- **Size:** 32 bits
- **Offset:** 0x1324
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 498

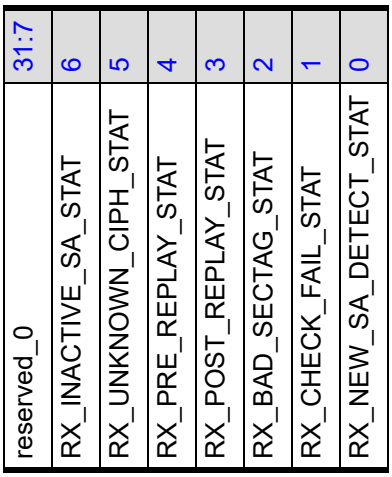


Table 6-1232 Fields for Register: IRQ_TYPE_STAT_RX_SA_498

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1232 Fields for Register: IRQ_TYPE_STAT_RX_SA_498 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1232 Fields for Register: IRQ_TYPE_STAT_RX_SA_498 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1226 IRQ_TYPE_EN_RX_SA_499

- **Name:** IRQ_TYPE_EN_RX_SA_499
- **Description:** Interrupt Enable for each IRQ type of Rx SA 499
- **Size:** 32 bits
- **Offset:** 0x1328
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 499

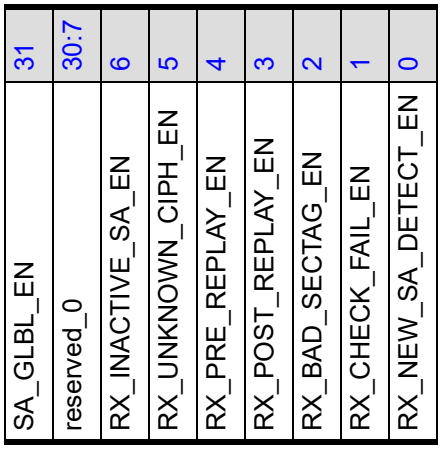


Table 6-1233 Fields for Register: IRQ_TYPE_EN_RX_SA_499

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1233 Fields for Register: IRQ_TYPE_EN_RX_SA_499 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1233 Fields for Register: IRQ_TYPE_EN_RX_SA_499 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1227 IRQ_TYPE_STAT_RX_SA_499

- **Name:** IRQ_TYPE_STAT_RX_SA_499
- **Description:** Interrupt Status for each IRQ type of Rx SA 499
- **Size:** 32 bits
- **Offset:** 0x132c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 499

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1234 Fields for Register: IRQ_TYPE_STAT_RX_SA_499

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1234 Fields for Register: IRQ_TYPE_STAT_RX_SA_499 (Continued)

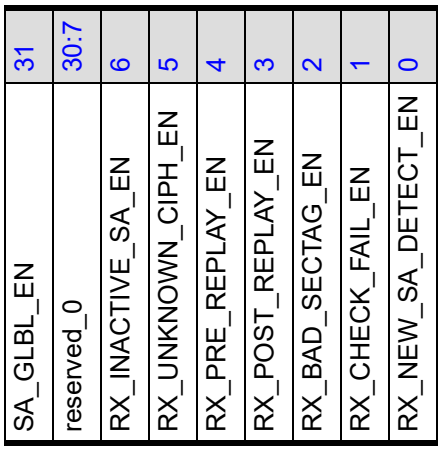
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1234 Fields for Register: IRQ_TYPE_STAT_RX_SA_499 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1228 IRQ_TYPE_EN_RX_SA_500

- **Name:** IRQ_TYPE_EN_RX_SA_500
- **Description:** Interrupt Enable for each IRQ type of Rx SA 500
- **Size:** 32 bits
- **Offset:** 0x1330
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 500



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1235 Fields for Register: IRQ_TYPE_EN_RX_SA_500

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1235 Fields for Register: IRQ_TYPE_EN_RX_SA_500 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1235 Fields for Register: IRQ_TYPE_EN_RX_SA_500 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1229 IRQ_TYPE_STAT_RX_SA_500

- **Name:** IRQ_TYPE_STAT_RX_SA_500
- **Description:** Interrupt Status for each IRQ type of Rx SA 500
- **Size:** 32 bits
- **Offset:** 0x1334
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 500

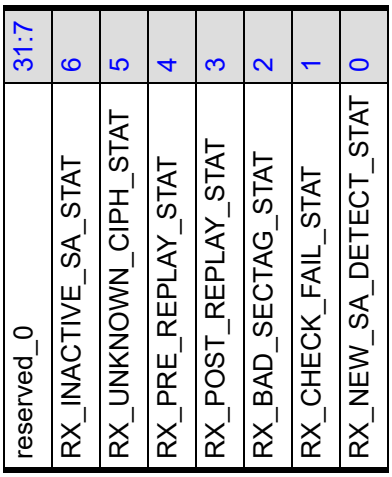


Table 6-1236 Fields for Register: IRQ_TYPE_STAT_RX_SA_500

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1236 Fields for Register: IRQ_TYPE_STAT_RX_SA_500 (Continued)

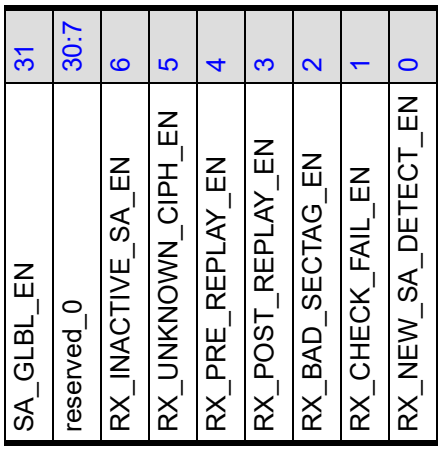
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1236 Fields for Register: IRQ_TYPE_STAT_RX_SA_500 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1230 IRQ_TYPE_EN_RX_SA_501

- **Name:** IRQ_TYPE_EN_RX_SA_501
- **Description:** Interrupt Enable for each IRQ type of Rx SA 501
- **Size:** 32 bits
- **Offset:** 0x1338
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 501



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1237 Fields for Register: IRQ_TYPE_EN_RX_SA_501

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1237 Fields for Register: IRQ_TYPE_EN_RX_SA_501 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1237 Fields for Register: IRQ_TYPE_EN_RX_SA_501 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1231 IRQ_TYPE_STAT_RX_SA_501

- **Name:** IRQ_TYPE_STAT_RX_SA_501
- **Description:** Interrupt Status for each IRQ type of Rx SA 501
- **Size:** 32 bits
- **Offset:** 0x133c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 501

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1238 Fields for Register: IRQ_TYPE_STAT_RX_SA_501

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1238 Fields for Register: IRQ_TYPE_STAT_RX_SA_501 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1238 Fields for Register: IRQ_TYPE_STAT_RX_SA_501 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1232 IRQ_TYPE_EN_RX_SA_502

- **Name:** IRQ_TYPE_EN_RX_SA_502
- **Description:** Interrupt Enable for each IRQ type of Rx SA 502
- **Size:** 32 bits
- **Offset:** 0x1340
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 502

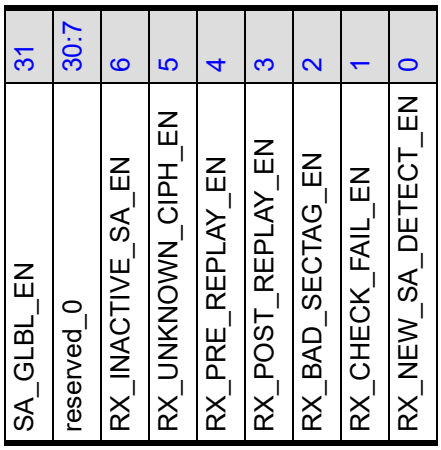


Table 6-1239 Fields for Register: IRQ_TYPE_EN_RX_SA_502

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1239 Fields for Register: IRQ_TYPE_EN_RX_SA_502 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1239 Fields for Register: IRQ_TYPE_EN_RX_SA_502 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1233 IRQ_TYPE_STAT_RX_SA_502

- **Name:** IRQ_TYPE_STAT_RX_SA_502
- **Description:** Interrupt Status for each IRQ type of Rx SA 502
- **Size:** 32 bits
- **Offset:** 0x1344
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 502

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1240 Fields for Register: IRQ_TYPE_STAT_RX_SA_502

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1240 Fields for Register: IRQ_TYPE_STAT_RX_SA_502 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1240 Fields for Register: IRQ_TYPE_STAT_RX_SA_502 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1234 IRQ_TYPE_EN_RX_SA_503

- **Name:** IRQ_TYPE_EN_RX_SA_503
- **Description:** Interrupt Enable for each IRQ type of Rx SA 503
- **Size:** 32 bits
- **Offset:** 0x1348
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 503

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1241 Fields for Register: IRQ_TYPE_EN_RX_SA_503

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1241 Fields for Register: IRQ_TYPE_EN_RX_SA_503 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1241 Fields for Register: IRQ_TYPE_EN_RX_SA_503 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1235 IRQ_TYPE_STAT_RX_SA_503

- **Name:** IRQ_TYPE_STAT_RX_SA_503
- **Description:** Interrupt Status for each IRQ type of Rx SA 503
- **Size:** 32 bits
- **Offset:** 0x134c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 503

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1242 Fields for Register: IRQ_TYPE_STAT_RX_SA_503

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1242 Fields for Register: IRQ_TYPE_STAT_RX_SA_503 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1242 Fields for Register: IRQ_TYPE_STAT_RX_SA_503 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1236 IRQ_TYPE_EN_RX_SA_504

- **Name:** IRQ_TYPE_EN_RX_SA_504
- **Description:** Interrupt Enable for each IRQ type of Rx SA 504
- **Size:** 32 bits
- **Offset:** 0x1350
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 504

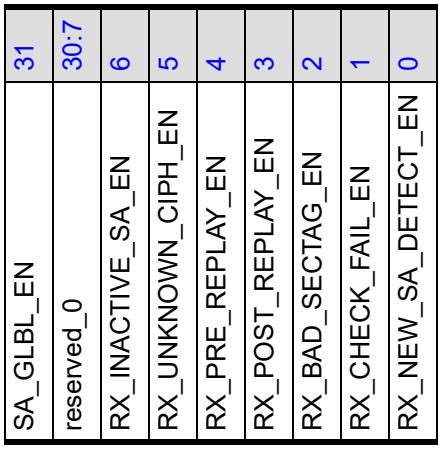


Table 6-1243 Fields for Register: IRQ_TYPE_EN_RX_SA_504

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1243 Fields for Register: IRQ_TYPE_EN_RX_SA_504 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1243 Fields for Register: IRQ_TYPE_EN_RX_SA_504 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1237 IRQ_TYPE_STAT_RX_SA_504

- **Name:** IRQ_TYPE_STAT_RX_SA_504
- **Description:** Interrupt Status for each IRQ type of Rx SA 504
- **Size:** 32 bits
- **Offset:** 0x1354
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 504

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1244 Fields for Register: IRQ_TYPE_STAT_RX_SA_504

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1244 Fields for Register: IRQ_TYPE_STAT_RX_SA_504 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1244 Fields for Register: IRQ_TYPE_STAT_RX_SA_504 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1238 IRQ_TYPE_EN_RX_SA_505

- **Name:** IRQ_TYPE_EN_RX_SA_505
- **Description:** Interrupt Enable for each IRQ type of Rx SA 505
- **Size:** 32 bits
- **Offset:** 0x1358
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 505

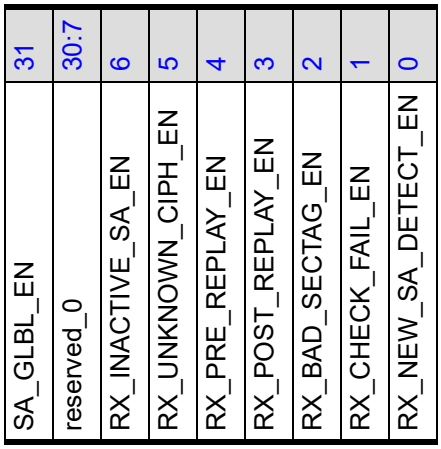


Table 6-1245 Fields for Register: IRQ_TYPE_EN_RX_SA_505

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1245 Fields for Register: IRQ_TYPE_EN_RX_SA_505 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1245 Fields for Register: IRQ_TYPE_EN_RX_SA_505 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1239 IRQ_TYPE_STAT_RX_SA_505

- **Name:** IRQ_TYPE_STAT_RX_SA_505
- **Description:** Interrupt Status for each IRQ type of Rx SA 505
- **Size:** 32 bits
- **Offset:** 0x135c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 505

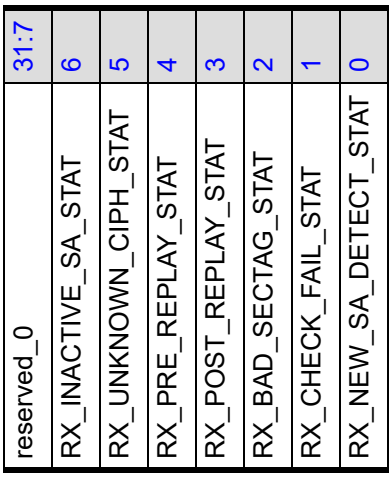


Table 6-1246 Fields for Register: IRQ_TYPE_STAT_RX_SA_505

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1246 Fields for Register: IRQ_TYPE_STAT_RX_SA_505 (Continued)

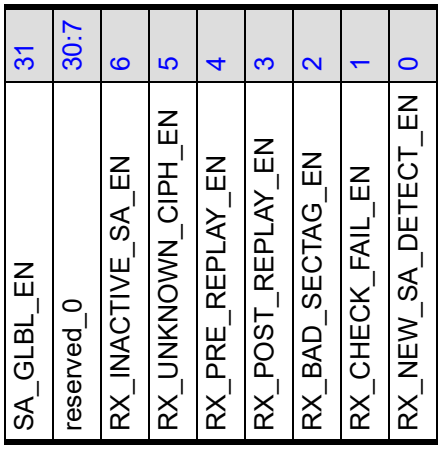
Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1246 Fields for Register: IRQ_TYPE_STAT_RX_SA_505 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1240 IRQ_TYPE_EN_RX_SA_506

- **Name:** IRQ_TYPE_EN_RX_SA_506
- **Description:** Interrupt Enable for each IRQ type of Rx SA 506
- **Size:** 32 bits
- **Offset:** 0x1360
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 506



31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1247 Fields for Register: IRQ_TYPE_EN_RX_SA_506

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1247 Fields for Register: IRQ_TYPE_EN_RX_SA_506 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1247 Fields for Register: IRQ_TYPE_EN_RX_SA_506 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1241 IRQ_TYPE_STAT_RX_SA_506

- **Name:** IRQ_TYPE_STAT_RX_SA_506
- **Description:** Interrupt Status for each IRQ type of Rx SA 506
- **Size:** 32 bits
- **Offset:** 0x1364
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 506

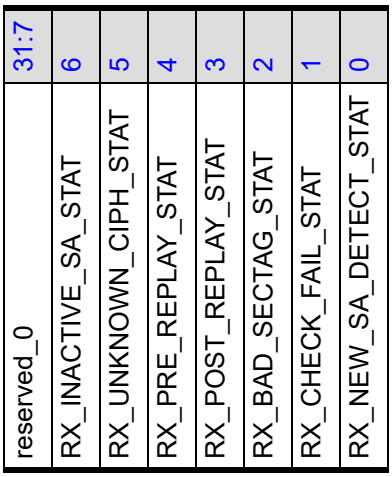


Table 6-1248 Fields for Register: IRQ_TYPE_STAT_RX_SA_506

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1248 Fields for Register: IRQ_TYPE_STAT_RX_SA_506 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1248 Fields for Register: IRQ_TYPE_STAT_RX_SA_506 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1242 IRQ_TYPE_EN_RX_SA_507

- **Name:** IRQ_TYPE_EN_RX_SA_507
- **Description:** Interrupt Enable for each IRQ type of Rx SA 507
- **Size:** 32 bits
- **Offset:** 0x1368
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 507

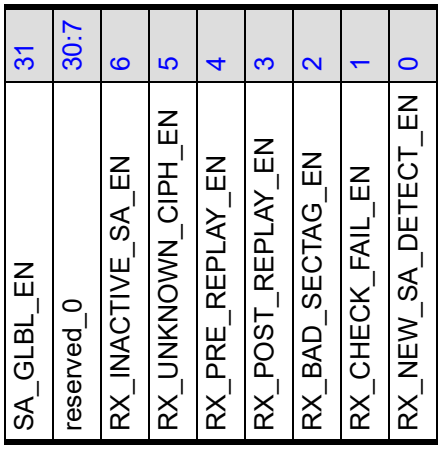


Table 6-1249 Fields for Register: IRQ_TYPE_EN_RX_SA_507

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1249 Fields for Register: IRQ_TYPE_EN_RX_SA_507 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1249 Fields for Register: IRQ_TYPE_EN_RX_SA_507 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1243 IRQ_TYPE_STAT_RX_SA_507

- **Name:** IRQ_TYPE_STAT_RX_SA_507
- **Description:** Interrupt Status for each IRQ type of Rx SA 507
- **Size:** 32 bits
- **Offset:** 0x136c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 507

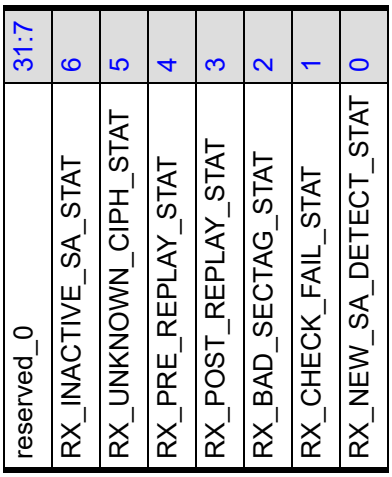


Table 6-1250 Fields for Register: IRQ_TYPE_STAT_RX_SA_507

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1250 Fields for Register: IRQ_TYPE_STAT_RX_SA_507 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1250 Fields for Register: IRQ_TYPE_STAT_RX_SA_507 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1244 IRQ_TYPE_EN_RX_SA_508

- **Name:** IRQ_TYPE_EN_RX_SA_508
- **Description:** Interrupt Enable for each IRQ type of Rx SA 508
- **Size:** 32 bits
- **Offset:** 0x1370
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 508

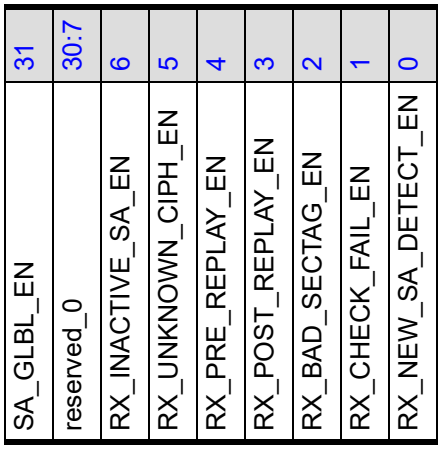


Table 6-1251 Fields for Register: IRQ_TYPE_EN_RX_SA_508

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1251 Fields for Register: IRQ_TYPE_EN_RX_SA_508 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1251 Fields for Register: IRQ_TYPE_EN_RX_SA_508 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1245 IRQ_TYPE_STAT_RX_SA_508

- **Name:** IRQ_TYPE_STAT_RX_SA_508
- **Description:** Interrupt Status for each IRQ type of Rx SA 508
- **Size:** 32 bits
- **Offset:** 0x1374
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 508

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1252 Fields for Register: IRQ_TYPE_STAT_RX_SA_508

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1252 Fields for Register: IRQ_TYPE_STAT_RX_SA_508 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1252 Fields for Register: IRQ_TYPE_STAT_RX_SA_508 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1246 IRQ_TYPE_EN_RX_SA_509

- **Name:** IRQ_TYPE_EN_RX_SA_509
- **Description:** Interrupt Enable for each IRQ type of Rx SA 509
- **Size:** 32 bits
- **Offset:** 0x1378
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 509

31	30:7	6	5	4	3	2	1	0
SA_GLBL_EN	reserved_0	RX_INACTIVE_SA_EN	RX_UNKNOWN_CIPH_EN	RX_PRE_REPLAY_EN	RX_POST_REPLAY_EN	RX_BAD_SECTAG_EN	RX_CHECK_FAIL_EN	RX_NEW_SA_DETECT_EN

Table 6-1253 Fields for Register: IRQ_TYPE_EN_RX_SA_509

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1253 Fields for Register: IRQ_TYPE_EN_RX_SA_509 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1253 Fields for Register: IRQ_TYPE_EN_RX_SA_509 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1247 IRQ_TYPE_STAT_RX_SA_509

- **Name:** IRQ_TYPE_STAT_RX_SA_509
- **Description:** Interrupt Status for each IRQ type of Rx SA 509
- **Size:** 32 bits
- **Offset:** 0x137c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 509

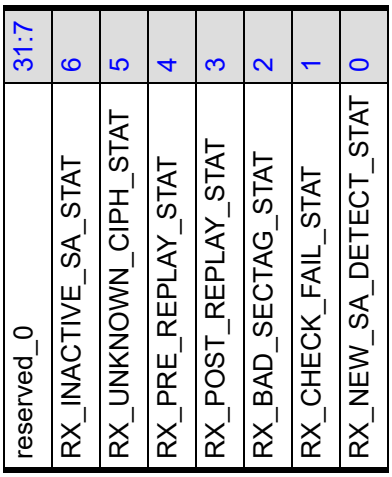


Table 6-1254 Fields for Register: IRQ_TYPE_STAT_RX_SA_509

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1254 Fields for Register: IRQ_TYPE_STAT_RX_SA_509 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1254 Fields for Register: IRQ_TYPE_STAT_RX_SA_509 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1248 IRQ_TYPE_EN_RX_SA_510

- **Name:** IRQ_TYPE_EN_RX_SA_510
- **Description:** Interrupt Enable for each IRQ type of Rx SA 510
- **Size:** 32 bits
- **Offset:** 0x1380
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 510

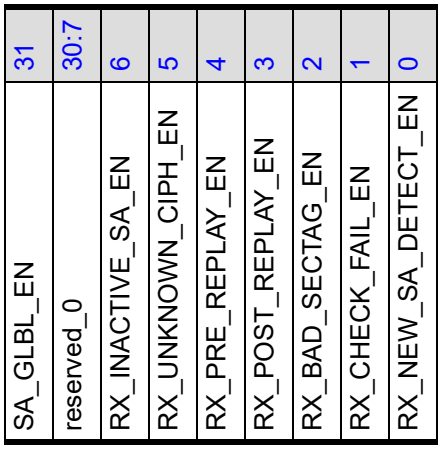


Table 6-1255 Fields for Register: IRQ_TYPE_EN_RX_SA_510

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1255 Fields for Register: IRQ_TYPE_EN_RX_SA_510 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1255 Fields for Register: IRQ_TYPE_EN_RX_SA_510 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1249 IRQ_TYPE_STAT_RX_SA_510

- **Name:** IRQ_TYPE_STAT_RX_SA_510
- **Description:** Interrupt Status for each IRQ type of Rx SA 510
- **Size:** 32 bits
- **Offset:** 0x1384
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 510

31:7	6	5	4	3	2	1	0
reserved_0	RX_INACTIVE_SA_STAT	RX_UNKNOWN_CIPH_STAT	RX_PRE_REPLAY_STAT	RX_POST_REPLAY_STAT	RX_BAD_SECTAG_STAT	RX_CHECK_FAIL_STAT	RX_NEW_SA_DETECT_STAT

Table 6-1256 Fields for Register: IRQ_TYPE_STAT_RX_SA_510

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1256 Fields for Register: IRQ_TYPE_STAT_RX_SA_510 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1256 Fields for Register: IRQ_TYPE_STAT_RX_SA_510 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1250 IRQ_TYPE_EN_RX_SA_511

- **Name:** IRQ_TYPE_EN_RX_SA_511
- **Description:** Interrupt Enable for each IRQ type of Rx SA 511
- **Size:** 32 bits
- **Offset:** 0x1388
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 511

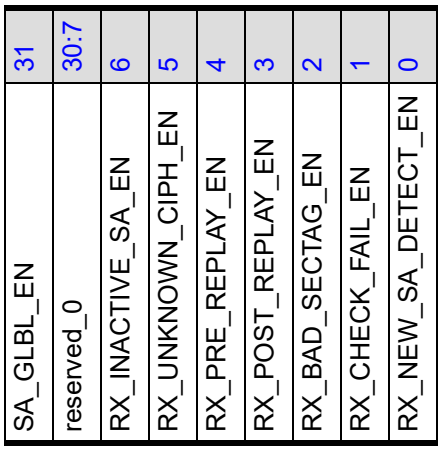


Table 6-1257 Fields for Register: IRQ_TYPE_EN_RX_SA_511

Bits	Name	Memory Access	Description
31	SA_GLBL_EN	R/W	Global SA enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for this SA, otherwise, disabled. **Values:** ■ 0x0 (SA_GLBL_EN): SA_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always
30:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1257 Fields for Register: IRQ_TYPE_EN_RX_SA_511 (Continued)

Bits	Name	Memory Access	Description
6	RX_INACTIVE_SA_EN	R/W	Enables the RX Inactive SA. The SA selected does not have the active flag set. **Values:** ■ 0x1 (RX_INACTIVE_SA_EN): RX_INACTIVE_SA_EN **Value After Reset:** 0x1 **Exists:** Always
5	RX_UNKNOWN_CIPH_EN	R/W	Enables the RX Unknown Cipher interrupt. Unknown cipher-suite. An Rx frame was received with the E bit clear and C bit set in the TCI. **Values:** ■ 0x1 (RX_UNKNOWN_CIPH_EN): RX_UNKNOWN_CIPH_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_PRE_REPLAY_EN	R/W	Enables the RX Pre Replay interrupt. An Rx frame failed the preliminary replay check. **Values:** ■ 0x1 (RX_PRE_REPLAY_EN): RX_PRE_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_POST_REPLAY_EN	R/W	Enables the RX Post Replay interrupt. An Rx frame failed the post decryption replay check. **Values:** ■ 0x1 (RX_POST_REPLAY_EN): RX_POST_REPLAY_EN **Value After Reset:** 0x1 **Exists:** Always
2	RX_BAD_SECTAG_EN	R/W	Enables the RX Bad Sectag interrupt. An Rx frame violated the standard rules, or the packet number was zero, or the frame could not be parsed for some other reason. **Values:** ■ 0x1 (RX_BAD_SECTAG_EN): RX_BAD_SECTAG_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1257 Fields for Register: IRQ_TYPE_EN_RX_SA_511 (Continued)

Bits	Name	Memory Access	Description
1	RX_CHECK_FAIL_EN	R/W	Enables the RX Check Fail interrupt meaning that an Rx frame failed the integrity check. **Values:** ■ 0x1 (RX_CHECK_FAIL_EN): RX_CHECK_FAIL_EN **Value After Reset:** 0x1 **Exists:** Always
0	RX_NEW_SA_DETECT_EN	R/W	Enables the New SA detected interrupt, i.e, An Rx frame was received on an SA for the first time. **Values:** ■ 0x1 (RX_NEW_SA_DETECT_EN): RX_NEW_SA_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1251 IRQ_TYPE_STAT_RX_SA_511

- **Name:** IRQ_TYPE_STAT_RX_SA_511
- **Description:** Interrupt Status for each IRQ type of Rx SA 511
- **Size:** 32 bits
- **Offset:** 0x138c
- **Exists:** DWC_MACSEC_RX_NUM_SA_SUPPORTED > 511

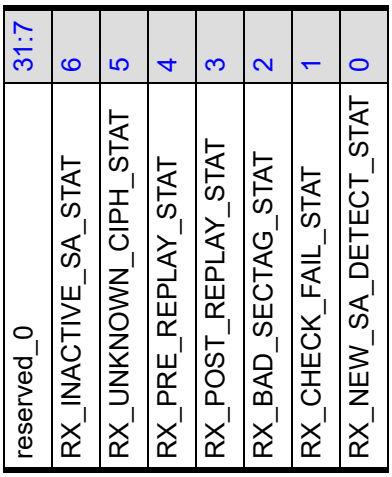


Table 6-1258 Fields for Register: IRQ_TYPE_STAT_RX_SA_511

Bits	Name	Memory Access	Description
31:7	reserved_0	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RX_INACTIVE_SA_STAT	R/W1C	Current status of RX Inactive SA. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_INACTIVE_SA_STAT): RX_INACTIVE_SA_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1258 Fields for Register: IRQ_TYPE_STAT_RX_SA_511 (Continued)

Bits	Name	Memory Access	Description
5	RX_UNKNOWN_CIPH_STAT	R/W1C	Current status of RX Unknown Cipher interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_UNKNOWN_CIPH_STAT): RX_UNKNOWN_CIPH_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_PRE_REPLAY_STAT	R/W1C	Current status of RX Pre Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_PRE_REPLAY_STAT): RX_PRE_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_POST_REPLAY_STAT	R/W1C	Current status of RX Post Replay interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_POST_REPLAY_STAT): RX_POST_REPLAY_STAT **Value After Reset:** 0x0 **Exists:** Always
2	RX_BAD_SECTAG_STAT	R/W1C	Current status of RX Bad Sectag interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_BAD_SECTAG_STAT): RX_BAD_SECTAG_STAT **Value After Reset:** 0x0 **Exists:** Always
1	RX_CHECK_FAIL_STAT	R/W1C	Current status of RX Check Fail interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_CHECK_FAIL_STAT): RX_CHECK_FAIL_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1258 Fields for Register: IRQ_TYPE_STAT_RX_SA_511 (Continued)

Bits	Name	Memory Access	Description
0	RX_NEW_SA_DETECT_STAT	R/W1C	Current status of New SA detected interrupt. Note: The interrupt status fields are Write-One-To-Clear, meaning that their status can be cleared by writing 1 to the respective field. **Values:** ■ 0x0 (RX_NEW_SA_DETECT_STAT): RX_NEW_SA_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1252 MACSEC_SW_INFO

- **Name:** MACSec SW INFO
- **Description:** MACSec information for SW
- **Size:** 32 bits
- **Offset:** 0x1390
- **Exists:** Always

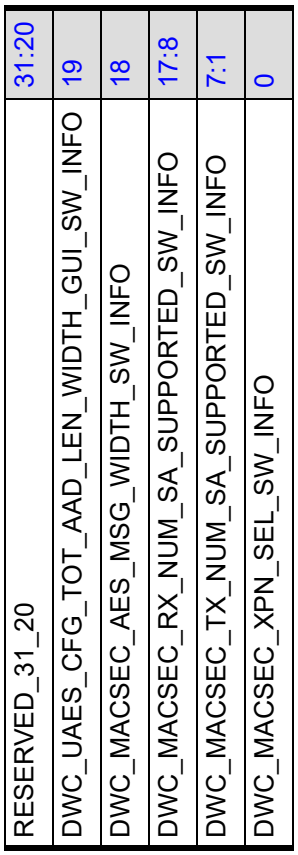


Table 6-1259 Fields for Register: MACSEC_SW_INFO

Bits	Name	Memory Access	Description
31:20	RESERVED_31_20	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1259 Fields for Register: MACSEC_SW_INFO (Continued)

Bits	Name	Memory Access	Description
19	DWC_UAES_CFG_TOT_AAD_LEN_WIDTH_GUI_SW_INFO	R	MACSec AES Width of total AAD length in bytes supported: 0-32 bytes, 1-Unsupported **Values:** ■ 0x0 (DWC_UAES_CFG_TOT_AAD_LEN_WIDTH_GUI_SW_INFO): DWC_UAES_CFG_TOT_AAD_LEN_WIDTH_GUI_SW_INFO **Value After Reset:** DWC_UAES_CFG_TOT_AAD_LEN_WIDTH_GUI_ENUM **Exists:** Always
18	DWC_MACSEC_AES_MSG_WIDTH_SW_INFO	R	MACSec AES maximum message size per round: 0-128 bit, 1-256 bit **Values:** ■ 0x0 (DWC_MACSEC_AES_MSG_WIDTH_SW_INFO): DWC_MACSEC_AES_MSG_WIDTH_SW_INFO **Value After Reset:** DWC_MACSEC_AES_MSG_WIDTH_ENUM **Exists:** Always
17:8	DWC_MACSEC_RX_NUM_SA_SUPPORTED_SW_INFO	R	Number of Tx Secure Associations supported **Values:** ■ 0x0 (DWC_MACSEC_RX_NUM_SA_SUPPORTED_SW_INFO): DWC_MACSEC_RX_NUM_SA_SUPPORTED_SW_INFO **Value After Reset:** DWC_MACSEC_RX_NUM_SA_SUPPORTED **Exists:** Always
7:1	DWC_MACSEC_TX_NUM_SA_SUPPORTED_SW_INFO	R	Number of Tx Secure Associations supported **Values:** ■ 0x0 (RX_CHECK_FADWC_MACSEC_TX_NUM_SA_SUPPORTED_SW_INFO): DWC_MACSEC_TX_NUM_SA_SUPPORTED_SW_INFO **Value After Reset:** DWC_MACSEC_TX_NUM_SA_SUPPORTED **Exists:** Always

Table 6-1259 Fields for Register: MACSEC_SW_INFO (Continued)

Bits	Name	Memory Access	Description
0	DWC_MACSEC_XPN_SEL_SW_INFO	R	Extended Packet Numbering Enabled. 0-Disabled, 1-Enabled **Values:** ■ 0x0 (DWC_MACSEC_XPN_SEL_SW_INFO): DWC_MACSEC_XPN_SEL_SW_INFO **Value After Reset:** DWC_MACSEC_XPN_SEL **Exists:** Always

6.1.1253 IRQ_TYPE_EN_SRAM_ECC

- **Name:** IRQ_TYPE_EN_SRAM_ECC
- **Description:** Interrupt Enable for each IRQ type of SRAM ECC Uncorrectable Errors
- **Size:** 32 bits
- **Offset:** 0x1394
- **Exists:** DWC_MACSEC_SRAM_ECC_SUPPORT==1

31	30:22	21	20	21	20	19	18	17	16	15	14	13	12	11	30:11	10	11	10	9	10	9	8	7	6	5	4	3	2	1	0
SRAM_ECC_GLBL_EN	reserved_2	reserved_1	Rsvd	HCTX_ERR_UNCORRECT_EN	RX_MIB_SA_ERR_UNCORRECT_EN	TX_MIB_SA_ERR_UNCORRECT_EN	RX_ARW_ERR_UNCORRECT_EN	RX_SALT_ERR_UNCORRECT_EN	TX_SALT_ERR_UNCORRECT_EN	RX_MIB_SC_ERR_UNCORRECT_BNK2_EN	RX_MIB_SC_ERR_UNCORRECT_BNK1_EN	CCTX_ERR_UNCORRECT_EN	HKEY_ERR_UNCORRECT_EN	Rsvd	reserved_3	Rsvd	CKEY_ERR_UNCORRECT_EN	reserved_0	Rsvd	HCTX_ERR_DETECT_EN	RX_MIB_SA_ERR_DETECT_EN	TX_MIB_SA_ERR_DETECT_EN	RX_ARW_ERR_DETECT_EN	RX_SALT_ERR_DETECT_EN	TX_SALT_ERR_DETECT_EN	RX_MIB_SC_ERR_DETECT_BNK2_EN	RX_MIB_SC_ERR_DETECT_BNK1_EN	CCTX_ERR_DETECT_EN	HKEY_ERR_DETECT_EN	CKEY_ERR_DETECT_EN

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC

Bits	Name	Memory Access	Description
31	SRAM_ECC_GLBL_EN	R/W	Global SRAM_ECC enable control for interrupt sources. If this bit is set, all supported IRQ are enabled for SRAM ECC related interrupts, otherwise, disabled. **Values:** ■ 0x0 (SRAM_ECC_GLBL_EN): SRAM_ECC_GLBL_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
30:22	reserved_2	R	Reserved Bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
21	reserved_1	R	Reserved Bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0 && DWC_UAES_CFG_HASH_ILEAVE_EN==0
20			**Reserved Field:** Yes
21	HCTX_ERR_UNCORRECT_EN	R/W	Enables the hctx uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the ckey hctx memory. **Values:** ■ 0x1 (HCTX_ERR_UNCORRECT_EN): HCTX_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0 && DWC_UAES_CFG_HASH_ILEAVE_EN==1
20	RX_MIB_SA_ERR_UNCORRECT_EN	R/W	Enables the rx_mib_sa uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the rx_mib_sa SRAM memory. **Values:** ■ 0x1 (RX_MIB_SA_ERR_UNCORRECT_EN): RX_MIB_SA_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
19	TX_MIB_SA_ERR_UNCORRECT_EN	R/W	Enables the tx_mib_sa uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the tx_mib_sa SRAM memory. **Values:** ■ 0x1 (TX_MIB_SA_ERR_UNCORRECT_EN): TX_MIB_SA_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
18	RX_ARW_ERR_UNCORRECT_EN	R/W	Enables the rx_arw uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the rx_arw SRAM memory. **Values:** ■ 0x1 (RX_ARW_ERR_UNCORRECT_EN): RX_ARW_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
17	RX_SALT_ERR_UNCORRECT_EN	R/W	Enables the rx_salt uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the rx_salt SRAM memory. **Values:** ■ 0x1 (RX_SALT_ERR_UNCORRECT_EN): RX_SALT_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
16	TX_SALT_ERR_UNCORRECT_EN	R/W	Enables the tx_salt uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the tx_salt SRAM memory. **Values:** ■ 0x1 (TX_SALT_ERR_UNCORRECT_EN): TX_SALT_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
15	RX_MIB_SC_ERR_UNCORRECT_BNK2_EN	R/W	Enables the rx_mib_sc_bank2 uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the rx_mib_sc_bank2 SRAM memory. **Values:** ■ 0x1 (RX_MIB_SC_ERR_UNCORRECT_BNK2_EN): RX_MIB_SC_ERR_UNCORRECT_BNK2_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
14	RX_MIB_SC_ERR_UNCORRECT_BNK1_EN	R/W	Enables the rx_mib_sc_bank1 uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the rx_mib_sc_bank1 SRAM memory. **Values:** ■ 0x1 (RX_MIB_SC_ERR_UNCORRECT_BNK1_EN): RX_MIB_SC_ERR_UNCORRECT_BNK1_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
13	CCTX_ERR_UNCORRECT_EN	R/W	Enables the cctx uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the cctx SRAM memory. **Values:** ■ 0x1 (CCTX_ERR_UNCORRECT_EN): CCTX_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
12	HKEY_ERR_UNCORRECT_EN	R/W	Enables the hkey uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the hkey SRAM memory. **Values:** ■ 0x1 (HKEY_ERR_UNCORRECT_EN): HKEY_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
11			**Reserved Field:** Yes
30:11	reserved_3	R	Reserved Bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==1
10			**Reserved Field:** Yes
11	CKEY_ERR_UNCORRECT_EN	R/W	Enables the ckey uncorrectable errors detected interrupt, whenever there is at least 2 bit flips in the ckey SRAM memory. **Values:** ■ 0x1 (CKEY_ERR_UNCORRECT_EN): CKEY_ERR_UNCORRECT_EN **Value After Reset:** 0x1 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
10	reserved_0	R	Reserved Bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_UAES_CFG_HASH_ILEAVE_EN==0
9			**Reserved Field:** Yes

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
10	HCTX_ERR_DETECT_EN	R/W	Enables the hctx correctable errors detected interrupt, whenever there is 1 bit flip in the ckey hctx memory. **Values:** ■ 0x1 (HCTX_ERR_DETECT_EN): HCTX_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** DWC_UAES_CFG_HASH_ILEAVE_EN==1
9	RX_MIB_SA_ERR_DETECT_EN	R/W	Enables the rx_mib_sa correctable errors detected interrupt, whenever there is 1 bit flip in the rx_mib_sa SRAM memory. **Values:** ■ 0x1 (RX_MIB_SA_ERR_DETECT_EN): RX_MIB_SA_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always
8	TX_MIB_SA_ERR_DETECT_EN	R/W	Enables the tx_mib_sa correctable errors detected interrupt, whenever there is 1 bit flip in the tx_mib_sa SRAM memory. **Values:** ■ 0x1 (TX_MIB_SA_ERR_DETECT_EN): TX_MIB_SA_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always
7	RX_ARW_ERR_DETECT_EN	R/W	Enables the rx_arw correctable errors detected interrupt, whenever there is 1 bit flip in the rx_arw SRAM memory. **Values:** ■ 0x1 (RX_ARW_ERR_DETECT_EN): RX_ARW_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always
6	RX_SALT_ERR_DETECT_EN	R/W	Enables the rx_salt correctable errors detected interrupt, whenever there is 1 bit flip in the rx_salt SRAM memory. **Values:** ■ 0x1 (RX_SALT_ERR_DETECT_EN): RX_SALT_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
5	TX_SALT_ERR_DETECT_EN	R/W	Enables the tx_salt correctable errors detected interrupt, whenever there is 1 bit flip in the tx_salt SRAM memory. **Values:** ■ 0x1 (TX_SALT_ERR_DETECT_EN): TX_SALT_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always
4	RX_MIB_SC_ERR_DETECT_BNK2_EN	R/W	Enables the rx_mib_sc_bank2 correctable errors detected interrupt, whenever there is 1 bit flip in the rx_mib_sc_bank2 SRAM memory. **Values:** ■ 0x1 (RX_MIB_SC_ERR_DETECT_BNK2_EN): RX_MIB_SC_ERR_DETECT_BNK2_EN **Value After Reset:** 0x1 **Exists:** Always
3	RX_MIB_SC_ERR_DETECT_BNK1_EN	R/W	Enables the rx_mib_sc_bank1 correctable errors detected interrupt, whenever there is 1 bit flip in the rx_mib_sc_bank1 SRAM memory. **Values:** ■ 0x1 (RX_MIB_SC_ERR_DETECT_BNK1_EN): RX_MIB_SC_ERR_DETECT_BNK1_EN **Value After Reset:** 0x1 **Exists:** Always
2	CCTX_ERR_DETECT_EN	R/W	Enables the cctx correctable errors detected interrupt, whenever there is 1 bit flip in the cctx SRAM memory. **Values:** ■ 0x1 (CCTX_ERR_DETECT_EN): CCTX_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always
1	HKEY_ERR_DETECT_EN	R/W	Enables the hkey correctable errors detected interrupt, whenever there is 1 bit flip in the hkey SRAM memory. **Values:** ■ 0x1 (HKEY_ERR_DETECT_EN): HKEY_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

Table 6-1260 Fields for Register: IRQ_TYPE_EN_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
0	CKEY_ERR_DETECT_EN	R/W	Enables the ckey correctable errors detected interrupt, whenever there is 1 bit flip in the ckey SRAM memory. **Values:** ■ 0x1 (CKEY_ERR_DETECT_EN): CKEY_ERR_DETECT_EN **Value After Reset:** 0x1 **Exists:** Always

6.1.1254 IRQ_TYPE_STAT_SRAM_ECC

- **Name:** IRQ_TYPE_STAT_SRAM_ECC
- **Description:** Interrupt Status for each IRQ type of SRAM_ECC
- **Size:** 32 bits
- **Offset:** 0x1398
- **Exists:** DWC_MACSEC_SRAM_ECC_SUPPORT==1

31:22	21	20	21	20	19	18	17	16	15	14	13	12	11	31:11	10	11	10	9	10	9	8	7	6	5	4	3	2	1	0
reserved_2	reserved_1	Rsvd	HCTX_ERR_UNCORRECT_STAT	RX_MIB_SA_ERR_UNCORRECT_STAT	TX_MIB_SA_ERR_UNCORRECT_STAT	RX_ARW_ERR_UNCORRECT_STAT	RX_SALT_ERR_UNCORRECT_STAT	TX_SALT_ERR_UNCORRECT_STAT	RX_MIB_SC_ERR_UNCORRECT_BNK2_STAT	RX_MIB_SC_ERR_UNCORRECT_BNK1_STAT	CCTX_ERR_UNCORRECT_STAT	HKEY_ERR_UNCORRECT_STAT	Rsvd	reserved_3	Rsvd	CKEY_ERR_UNCORRECT_STAT	reserved_0	Rsvd	HCTX_ERR_DETECT_STAT	RX_MIB_SA_ERR_DETECT_STAT	TX_MIB_SA_ERR_DETECT_STAT	RX_ARW_ERR_DETECT_STAT	RX_SALT_ERR_DETECT_STAT	TX_SALT_ERR_DETECT_STAT	RX_MIB_SC_ERR_DETECT_BNK2_STAT	RX_MIB_SC_ERR_DETECT_BNK1_STAT	CCTX_ERR_DETECT_STAT	HKEY_ERR_DETECT_STAT	CKEY_ERR_DETECT_STAT

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC

Bits	Name	Memory Access	Description
31:22	reserved_2	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
21	reserved_1	R	Reserved Bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0 && DWC_UAES_CFG_HASH_ILEAVE_EN==0
20			**Reserved Field:** Yes
21	HCTX_ERR_UNCORRECT_STAT	R/W1C	Current status of a hctx uncorrectable error detected interrupt, i.e, more than one bit flip occured in the hctx SRAM. **Values:** ■ 0x1 (HCTX_ERR_UNCORRECT_STAT): HCTX_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_UAES_CFG_HASH_ILEAVE_EN==1 && DWC_MACSEC_AUTOMOTIVE==0
20	RX_MIB_SA_ERR_UNCORRECT_STAT	R/W1C	Current status of a rx mib sa uncorrectable error detected interrupt, i.e, more than one bit flip occured in the rx mib sa SRAM. **Values:** ■ 0x1 (RX_MIB_SA_ERR_UNCORRECT_STAT): RX_MIB_SA_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
19	TX_MIB_SA_ERR_UNCORRECT_STAT	R/W1C	Current status of a tx mib sa uncorrectable error detected interrupt, i.e, more than one bit flip occured in the tx mib sa SRAM. **Values:** ■ 0x1 (TX_MIB_SA_ERR_UNCORRECT_STAT): TX_MIB_SA_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
18	RX_ARW_ERR_UNCORRECT_STAT	R/W1C	Current status of a rx arw uncorrectable error detected interrupt, i.e, more than one bit flip occured in the rx arw SRAM. **Values:** ■ 0x1 (RX_ARW_ERR_UNCORRECT_STAT): RX_ARW_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
17	RX_SALT_ERR_UNCORRECT_STAT	R/W1C	Current status of a rx salt uncorrectable error detected interrupt, i.e, more than one bit flip occured in the rx salt SRAM. **Values:** ■ 0x1 (RX_SALT_ERR_UNCORRECT_STAT): RX_SALT_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
16	TX_SALT_ERR_UNCORRECT_STAT	R/W1C	Current status of a tx_salt uncorrectable error detected interrupt, i.e, more than one bit flip occured in the tx_salt SRAM. **Values:** ■ 0x1 (TX_SALT_ERR_UNCORRECT_STAT): TX_SALT_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
15	RX_MIB_SC_ERR_UNCORRECT_BNK2_STAT	R/W1C	Current status of a rx mib sc bank 2 uncorrectable error detected interrupt, i.e, more than one bit flip occured in the rx mib sc bank 2 SRAM. **Values:** ■ 0x1 (RX_MIB_SC_ERR_UNCORRECT_BNK2_STAT): RX_MIB_SC_ERR_UNCORRECT_BNK2_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
14	RX_MIB_SC_ERR_UNCORRECT_BNK1_STAT	R/W1C	Current status of a rx mib sc bank 1 uncorrectable error detected interrupt, i.e, more than one bit flip occured in the rx mib sc bank 1 SRAM. **Values:** ■ 0x1 (RX_MIB_SC_ERR_UNCORRECT_BNK1_STAT): RX_MIB_SC_ERR_UNCORRECT_BNK1_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
13	CCTX_ERR_UNCORRECT_STAT	R/W1C	Current status of a cctx uncorrectable error detected interrupt, i.e, more than one bit flip occured in the cctx SRAM. **Values:** ■ 0x1 (CCTX_ERR_UNCORRECT_STAT): CCTX_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
12	HKEY_ERR_UNCORRECT_STAT	R/W1C	Current status of a hkey uncorrectable error detected interrupt, i.e, more than one bit flip occured in the hkey SRAM. **Values:** ■ 0x1 (HKEY_ERR_UNCORRECT_STAT): HKEY_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
11			**Reserved Field:** Yes
31:11	reserved_3	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==1
10			**Reserved Field:** Yes
11	CKEY_ERR_UNCORRECT_STAT	R/W1C	Current status of a ckey uncorrectable error detected interrupt, i.e, more than one bit flip occured in the ckey SRAM. **Values:** ■ 0x1 (CKEY_ERR_UNCORRECT_STAT): CKEY_ERR_UNCORRECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AUTOMOTIVE==0
10	reserved_0	R	Reserved Bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_UAES_CFG_HASH_ILEAVE_EN==0
9			**Reserved Field:** Yes

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
10	HCTX_ERR_DETECT_STAT	R/W1C	Current status of a hctx correctable error detected interrupt, i.e, a single bit flip occured in the hctx SRAM. **Values:** ■ 0x1 (HCTX_ERR_DETECT_STAT): HCTX_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** DWC_UAES_CFG_HASH_ILEAVE_EN==1
9	RX_MIB_SA_ERR_DETECT_STAT	R/W1C	Current status of a rx mib sa correctable error detected interrupt, i.e, a single bit flip occured in the rx mib sa SRAM. **Values:** ■ 0x1 (RX_MIB_SA_ERR_DETECT_STAT): RX_MIB_SA_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always
8	TX_MIB_SA_ERR_DETECT_STAT	R/W1C	Current status of a tx mib sa correctable error detected interrupt, i.e, a single bit flip occured in the tx mib sa SRAM. **Values:** ■ 0x1 (TX_MIB_SA_ERR_DETECT_STAT): TX_MIB_SA_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always
7	RX_ARW_ERR_DETECT_STAT	R/W1C	Current status of a rx arw correctable error detected interrupt, i.e, a single bit flip occured in the rx arw SRAM. **Values:** ■ 0x1 (RX_ARW_ERR_DETECT_STAT): RX_ARW_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always
6	RX_SALT_ERR_DETECT_STAT	R/W1C	Current status of a rx salt correctable error detected interrupt, i.e, a single bit flip occured in the rx salt SRAM. **Values:** ■ 0x1 (RX_SALT_ERR_DETECT_STAT): RX_SALT_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
5	TX_SALT_ERR_DETECT_STAT	R/W1C	Current status of a tx_salt correctable error detected interrupt, i.e, a single bit flip occured in the tx_salt SRAM. **Values:** ■ 0x1 (TX_SALT_ERR_DETECT_STAT): TX_SALT_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always
4	RX_MIB_SC_ERR_DETECT_BNK2_STAT	R/W1C	Current status of a rx mib sc bank 2 correctable error detected interrupt, i.e, a single bit flip occured in the rx mib sc bank 2 SRAM. **Values:** ■ 0x1 (RX_MIB_SC_ERR_DETECT_BNK2_STAT): RX_MIB_SC_ERR_DETECT_BNK2_STAT **Value After Reset:** 0x0 **Exists:** Always
3	RX_MIB_SC_ERR_DETECT_BNK1_STAT	R/W1C	Current status of a rx mib sc bank 1 correctable error detected interrupt, i.e, a single bit flip occured in the rx mib sc bank 1 SRAM. **Values:** ■ 0x1 (RX_MIB_SC_ERR_DETECT_BNK1_STAT): RX_MIB_SC_ERR_DETECT_BNK1_STAT **Value After Reset:** 0x0 **Exists:** Always
2	CCTX_ERR_DETECT_STAT	R/W1C	Current status of a cctx correctable error detected interrupt, i.e, a single bit flip occured in the cctx SRAM. **Values:** ■ 0x1 (CCTX_ERR_DETECT_STAT): CCTX_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always
1	HKEY_ERR_DETECT_STAT	R/W1C	Current status of a hkey correctable error detected interrupt, i.e, a single bit flip occured in the hkey SRAM. **Values:** ■ 0x1 (HKEY_ERR_DETECT_STAT): HKEY_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

Table 6-1261 Fields for Register: IRQ_TYPE_STAT_SRAM_ECC (Continued)

Bits	Name	Memory Access	Description
0	CKEY_ERR_DETECT_STAT	R/W1C	Current status of a ckey correctable error detected interrupt, i.e, a single bit flip occured in the ckey SRAM. **Values:** ■ 0x1 (CKEY_ERR_DETECT_STAT): CKEY_ERR_DETECT_STAT **Value After Reset:** 0x0 **Exists:** Always

6.1.1255 MACSEC_SAFETY_ALARM_SOURCES_0

- **Name:** MACSEC Safety Alarm Sources Register bank 0.
- **Description:** MACSEC Safety Alarm Sources Register bank 0.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2000
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

31	30	29	28	27	26	25	24	23	22	21:19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACSEC_MIB_ERROR	RESERVED_31_30	RX_ARW_MEM_REG_IF_ERROR	RX_ARW_MEM_ACCESS_ERROR	RX_SAD_MEM_REG_IF_ERROR	RX_SAD_MEM_ACCESS_ERROR	RX_SALT_MEM_REG_IF_ERROR	RX_SALT_MEM_ACCESS_ERROR	RX_SAI_MEM_REG_IF_ERROR	RX_SAI_MEM_ACCESS_ERROR	RESERVED_21_19	TX_SAD_MEM_REG_IF_ERROR	TX_SAD_MEM_ACCESS_ERROR	TX_SALT_MEM_REG_IF_ERROR	TX_SALT_MEM_ACCESS_ERROR	TX_SAI_MEM_REG_IF_ERROR	TX_SAI_MEM_ACCESS_ERROR	AES_KEY_DETECT_ERROR	AES_DPLEX_ERROR	AES_MPLEX_ERROR	RX_DUAL_IF_MUX_ERROR	TX_DUAL_IF_MUX_ERROR	RX_GMII_MUX_ERROR	TX_GMII_MUX_ERROR	RX_SIDE_BAND_FIFO_ERROR	TX_SIDE_BAND_FIFO_ERROR	AES_HOST_MEM_REQ_GEN_ERROR	MACSEC_HOST_MEM_REQ_GEN_ERROR	AES_REGBANK_ERROR	MACSEC_REGBANK_ERROR

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0

Bits	Name	Memory Access	Description
31	MACSEC_MIB_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow macsec_mib signals. **Values:** ■ 0x1 (MACSEC_MIB_ERROR): Indicates an output mismatch between the nominal and the shadow macsec_mib signals. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0 (Continued)

Bits	Name	Memory Access	Description
30	RESERVED_31_30	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED_1): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always
29	RX_ARW_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX ARW memory register interfaces signals. **Values:** ■ 0x1 (RX_ARW_MEM_REG_IF_ERROR): Indicates an output mismatch in the RX ARW memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
28	RX_ARW_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX ARW memory access modules signals. **Values:** ■ 0x1 (RX_ARW_MEM_ACCESS_ERROR): Indicates an output mismatch in the RX ARW memory access module signals. **Value After Reset:** 0x0 **Exists:** Always
27	RX_SAD_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX SAD memory register interfaces signals. **Values:** ■ 0x1 (RX_SAD_MEM_REG_IF_ERROR): Indicates an output mismatch in the RX SAD memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
26	RX_SAD_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX SAD memory access modules signals. **Values:** ■ 0x1 (RX_SAD_MEM_ACCESS_ERROR): Indicates an output mismatch in the RX SAD memory access module signals. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0 (Continued)

Bits	Name	Memory Access	Description
25	RX_SALT_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX SALT memory register interfaces signals. **Values:** ■ 0x1 (RX_SALT_MEM_REG_IF_ERROR): Indicates an output mismatch in the RX SALT memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
24	RX_SALT_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX SALT memory access modules signals. **Values:** ■ 0x1 (RX_SALT_MEM_ACCESS_ERROR): Indicates an output mismatch in the RX SALT memory access module signals. **Value After Reset:** 0x0 **Exists:** Always
23	RX_SAI_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX SAI memory register interfaces signals. **Values:** ■ 0x1 (RX_SAI_MEM_REG_IF_ERROR): Indicates an output mismatch in the RX SAI memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
22	RX_SAI_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX SAI memory access modules signals. **Values:** ■ 0x1 (RX_SAI_MEM_ACCESS_ERROR): Indicates an output mismatch in the RX SAI memory access module signals. **Value After Reset:** 0x0 **Exists:** Always
21:19	RESERVED_21_19	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED_1): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0 (Continued)

Bits	Name	Memory Access	Description
18	TX_SAD_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX SAD memory register interfaces signals. **Values:** ■ 0x1 (TX_SAD_MEM_REG_IF_ERROR): Indicates an output mismatch in the TX SAD memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
17	TX_SAD_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX SAD memory access modules signals. **Values:** ■ 0x1 (TX_SAD_MEM_ACCESS_ERROR): Indicates an output mismatch in the TX SAD memory access module signals. **Value After Reset:** 0x0 **Exists:** Always
16	TX_SALT_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX SALT memory register interfaces signals. **Values:** ■ 0x1 (TX_SALT_MEM_REG_IF_ERROR): Indicates an output mismatch in the TX SALT memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
15	TX_SALT_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX SALT memory access modules signals. **Values:** ■ 0x1 (TX_SALT_MEM_ACCESS_ERROR): Indicates an output mismatch in the TX SALT memory access module signals. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0 (Continued)

Bits	Name	Memory Access	Description
14	TX_SAI_MEM_REG_IF_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX SAI memory register interfaces signals. **Values:** ■ 0x1 (TX_SAI_MEM_REG_IF_ERROR): Indicates an output mismatch in the TX SAI memory register interface signals. **Value After Reset:** 0x0 **Exists:** Always
13	TX_SAI_MEM_ACCESS_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX SAI memory access modules signals. **Values:** ■ 0x1 (TX_SAI_MEM_ACCESS_ERROR): Indicates an output mismatch in the TX SAI memory access module signals. **Value After Reset:** 0x0 **Exists:** Always
12	AES_KEY_DETECT_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow AES Key detect modules signals. **Values:** ■ 0x1 (AES_KEY_DETECT_ERROR): Indicates an output mismatch in the AES Key detect modules signals. **Value After Reset:** 0x0 **Exists:** Always
11	AES_DPLEX_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow AES GCM duplexers signals. **Values:** ■ 0x1 (AES_DPLEX_ERROR): Indicates an output mismatch in the AES GCM duplexers signals. **Value After Reset:** 0x0 **Exists:** Always
10	AES_MPLEX_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow AES GCM multipexers signals. **Values:** ■ 0x1 (AES_MPLEX_ERROR): Indicates an output mismatch in the AES GCM multipexers signals. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0 (Continued)

Bits	Name	Memory Access	Description
9	RX_DUAL_IF_MUX_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX dual interface multiplexer signals. **Values:** ■ 0x1 (RX_DUAL_IF_MUX_ERROR): Indicates an output mismatch in the RX DUAL IF multipexers signals. **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_DUAL_IF==1
8	TX_DUAL_IF_MUX_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX dual interface multiplexer signals. **Values:** ■ 0x1 (TX_DUAL_IF_MUX_ERROR): Indicates an output mismatch in the TX DUAL IF multipexers signals. **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_DUAL_IF==1
7	RX_GMII_MUX_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX GMII internal multiplexer signals. **Values:** ■ 0x1 (RX_GMII_MUX_ERROR): Indicates an output mismatch in the RX GMII internal multipexers signals. **Value After Reset:** 0x0 **Exists:** Always
6	TX_GMII_MUX_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX GMII internal multiplexer signals. **Values:** ■ 0x1 (TX_GMII_MUX_ERROR): Indicates an output mismatch in the TX GMII internal multipexers signals. **Value After Reset:** 0x0 **Exists:** Always
5	RX_SIDE_BAND_FIFO_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX side-band FIFO signals. **Values:** ■ 0x1 (RX_SIDE_BAND_FIFO_ERROR): Indicates an output mismatch in the RX Side Band FIFOs. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1262 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_0 (Continued)

Bits	Name	Memory Access	Description
4	TX_SIDE_BAND_FIFO_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX side-band FIFO signals. **Values:** ■ 0x1 (TX_SIDE_BAND_FIFO_ERROR): Indicates an output mismatch in the TX Side Band FIFOs. **Value After Reset:** 0x0 **Exists:** Always
3	AES_HOST_MEM_REQ_GEN_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow AES host memory request generation signals. **Values:** ■ 0x1 (AES_HOST_MEM_REQ_GEN_ERROR): Indicates an output mismatch in the AES host memory request generation modules. **Value After Reset:** 0x0 **Exists:** Always
2	MACSEC_HOST_MEM_REQ_GEN_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow macsec host memory request generation signals. **Values:** ■ 0x1 (MACSEC_HOST_MEM_REQ_GEN_ERROR): Indicates an output mismatch in the macsec host memory request generation modules. **Value After Reset:** 0x0 **Exists:** Always
1	AES_REGBANK_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow AES register banks. **Values:** ■ 0x1 (AES_REGBANK_ERROR): Indicates a register mismatch in the AES Register Banks. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_REGBANK_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow core register banks. **Values:** ■ 0x1 (MACSEC_REGBANK_ERROR): Indicates a register mismatch in the Core Register Banks. **Value After Reset:** 0x0 **Exists:** Always

6.1.1256 MACSEC_SAFETY_ALARM_SOURCES_1

- **Name:** MACSEC Safety Alarm Sources Register bank 1.
- **Description:** MACSEC Safety Alarm Sources Register bank 1.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2004
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RX_LOOKUP_PARITY_ERROR	TX_LOOKUP_PARITY_ERROR	RX_XGMII_EGRESS_IF_PARITY_ERROR	TX_XGMII_EGRESS_IF_PARITY_ERROR	RX_XGMII_INGRESS_IF_PARITY_ERROR	TX_XGMII_INGRESS_IF_PARITY_ERROR	RX_GMII_EGRESS_IF_PARITY_ERROR	TX_GMII_EGRESS_IF_PARITY_ERROR	RX_GMII_INGRESS_IF_PARITY_ERROR	TX_GMII_INGRESS_IF_PARITY_ERROR	RX_GMII_RGMII_PARITY_ERROR	TX_GMII_RGMII_PARITY_ERROR	RX_GMII_MII_PARITY_ERROR	TX_GMII_MII_PARITY_ERROR	RX_MII_RMII_PARITY_ERROR	TX_MII_RMII_PARITY_ERROR	RX_RGMII_GMII_PARITY_ERROR	TX_RGMII_GMII_PARITY_ERROR	RX_MII_GMII_PARITY_ERROR	TX_MII_GMII_PARITY_ERROR	RX_RMII_MII_PARITY_ERROR	TX_RMII_MII_PARITY_ERROR	RX_EGRESS_FIFO_PARITY_ERROR	TX_EGRESS_FIFO_PARITY_ERROR	RX_INGRESS_IRQ_FIFO_PARITY_ERROR	RX_INGRESS_FIFO_PARITY_ERROR	TX_INGRESS_FIFO_PARITY_ERROR	AES_APB_PARITY_ERROR	MACSEC_APB_PARITY_ERROR	reserved_2	TX_PN_REGBANK_ERROR	RESERVED_0

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1

Bits	Name	Memory Access	Description
31	RX_LOOKUP_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX Lookup module outputs. **Values:** ■ 0x1 (RX_LOOKUP_PARITY_ERROR): Indicates a parity mismatch in the RX Lookup. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
30	TX_LOOKUP_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX Lookup module outputs. **Values:** ■ 0x1 (TX_LOOKUP_PARITY_ERROR): Indicates a parity mismatch in the TX Lookup. **Value After Reset:** 0x0 **Exists:** Always
29	RX_XGMII_EGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX egress XGMII IF module outputs. **Values:** ■ 0x1 (RX_EGRESS_XGMII_IF_PARITY_ERROR): Indicates a parity mismatch in the RX egress XGMII IF. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
28	TX_XGMII_EGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX egress XGMII IF module outputs. **Values:** ■ 0x1 (TX_EGRESS_XGMII_IF_PARITY_ERROR): Indicates a parity mismatch in the TX egress XGMII IF. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
27	RX_XGMII_INGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX ingress XGMII IF module outputs. **Values:** ■ 0x1 (RX_INGRESS_XGMII_IF_PARITY_ERROR): Indicates a parity mismatch in the RX ingress XGMII IF. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
26	TX_XGMII_INGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX ingress XGMII IF module outputs. **Values:** ■ 0x1 (TX_INGRESS_XGMII_IF_PARITY_ERROR): Indicates a parity mismatch in the TX ingress GMII IF. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
25	RX_GMII_EGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX egress GMII IF module outputs. **Values:** ■ 0x1 (RX_EGRESS_GMII_IF_PARITY_ERROR): Indicates a parity mismatch in the RX egress GMII IF. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
24	TX_GMII_EGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX egress GMII IF module outputs. **Values:** ■ 0x1 (TX_EGRESS_GMII_IF_PARITY_ERROR): Indicates a parity mismatch in the TX egress GMII IF. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
23	RX_GMII_INGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX ingress GMII IF module outputs. **Values:** ■ 0x1 (RX_INGRESS_GMII_IF_PARITY_ERROR): Indicates a parity mismatch in the RX ingress GMII IF. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
22	TX_GMII_INGRESS_IF_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX ingress GMII IF module outputs. **Values:** ■ 0x1 (TX_INGRESS_GMII_IF_PARITY_ERROR): Indicates a parity mismatch in the TX ingress GMII IF. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
21	RX_GMII_RGMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX GMII to RGMII egress bridge outputs. **Values:** ■ 0x1 (RX_GMII_RGMII_PARITY_ERROR): Indicates a parity mismatch in the RX GMII to RGMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
20	TX_GMII_RGMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX GMII to RGMII egress bridge outputs. **Values:** ■ 0x1 (TX_GMII_RGMII_PARITY_ERROR): Indicates a parity mismatch in the TX GMII to RGMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
19	RX_GMII_MII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX GMII to MII egress bridge outputs. **Values:** ■ 0x1 (RX_GMII_MII_PARITY_ERROR): Indicates a parity mismatch in the RX GMII to MII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
18	TX_GMII_MII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX GMII to MII egress bridge outputs. **Values:** ■ 0x1 (TX_GMII_MII_PARITY_ERROR): Indicates a parity mismatch in the TX GMII to MII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
17	RX_MII_RMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX MII to RMII egress bridge outputs. **Values:** ■ 0x1 (RX_MII_RMII_PARITY_ERROR): Indicates a parity mismatch in the RX MII to RMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
16	TX_MII_RMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX MII to RMII egress bridge outputs. **Values:** ■ 0x1 (TX_MII_RMII_PARITY_ERROR): Indicates a parity mismatch in the TX MII to RMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
15	RX_RGMII_GMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX RGMII to GMII ingress bridge outputs. **Values:** ■ 0x1 (RX_RGMII_GMII_PARITY_ERROR): Indicates a parity mismatch in the RX RGMII to GMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
14	TX_RGMII_GMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX RGMII to GMII ingress bridge outputs. **Values:** ■ 0x1 (TX_RGMII_GMII_PARITY_ERROR): Indicates a parity mismatch in the TX RGMII to GMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
13	RX_MII_GMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX MII to GMII ingress bridge outputs. **Values:** ■ 0x1 (RX_MII_GMII_PARITY_ERROR): Indicates a parity mismatch in the RX MII to GMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
12	TX_MII_GMII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX MII to GMII ingress bridge outputs. **Values:** ■ 0x1 (TX_MII_GMII_PARITY_ERROR): Indicates a parity mismatch in the TX MII to GMII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
11	RX_RMII_MII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX RMII to MII ingress bridge outputs. **Values:** ■ 0x1 (RX_RMII_MII_PARITY_ERROR): Indicates a parity mismatch in the RX RMII to MII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
10	TX_RMII_MII_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX RMII to MII ingress bridge outputs. **Values:** ■ 0x1 (TX_RMII_MII_PARITY_ERROR): Indicates a parity mismatch in the TX RMII to MII bridge. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
9	RX_EGRESS_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX egress FIFO module outputs. **Values:** ■ 0x1 (RX_EGRESS_FIFO_PARITY_ERROR): Indicates a parity mismatch in the RX egress FIFO. **Value After Reset:** 0x0 **Exists:** Always
8	TX_EGRESS_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX egress FIFO module outputs. **Values:** ■ 0x1 (TX_EGRESS_FIFO_PARITY_ERROR): Indicates a parity mismatch in the TX egress FIFO. **Value After Reset:** 0x0 **Exists:** Always
7	RX_INGRESS_IRQ_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX ingress interrupts FIFO module outputs. **Values:** ■ 0x1 (RX_INGRESS_IRQ_FIFO_PARITY_ERROR): Indicates a parity mismatch in the RX IRQ ingress FIFO. **Value After Reset:** 0x0 **Exists:** Always
6	RX_INGRESS_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX ingress FIFO module outputs. **Values:** ■ 0x1 (RX_INGRESS_FIFO_PARITY_ERROR): Indicates a parity mismatch in the RX ingress FIFO. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
5	TX_INGRESS_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX ingress FIFO module outputs. **Values:** ■ 0x1 (TX_INGRESS_FIFO_PARITY_ERROR): Indicates a parity mismatch in the TX ingress FIFO. **Value After Reset:** 0x0 **Exists:** Always
4	AES_APB_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the AES APB module outputs. **Values:** ■ 0x1 (AES_APB_PARITY_ERROR): Indicates a parity mismatch in the AES APB. **Value After Reset:** 0x0 **Exists:** Always
3	MACSEC_APB_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the MACsec APB module outputs. **Values:** ■ 0x1 (MACSEC_APB_PARITY_ERROR): Indicates a parity mismatch in the MACsec APB. **Value After Reset:** 0x0 **Exists:** Always
2	reserved_2	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
1	TX_PN_REGBANK_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX PN register banks. **Values:** ■ 0x1 (TX_PN_REGBANK_ERROR): Indicates reset mismatches in the TX PN register bank. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1263 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_1 (Continued)

Bits	Name	Memory Access	Description
0	RESERVED_0	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED_1): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always

6.1.1257 MACSEC_SAFETY_ALARM_SOURCES_2

- **Name:** MACSEC Safety Alarm Sources Register bank 2.
- **Description:** MACSEC Safety Alarm Sources Register bank 2.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2008
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RX_SC_MIBS_SRAM_BNK1_ECC_ERROR	RX_SA_MIBS_SRAM_ECC_ERROR	TX_SA_MIBS_SRAM_ECC_ERROR	RX_ARW_SRAM_ECC_ERROR	RX_SALT_SRAM_ECC_ERROR	TX_SALT_SRAM_ECC_ERROR	RX_LAT_MNGR_ERROR	TX_LAT_MNGR_ERROR	RX_MISSION_MODE_BIST_ERROR	TX_MISSION_MODE_BIST_ERROR	AES_RX_TIMING_REDUNDANCY_ERROR	AES_TX_TIMING_REDUNDANCY_ERROR	RX_VLAN_EGRESS_PARITY_ERROR	TX_VLAN_EGRESS_PARITY_ERROR	RX_SECTAG_REMOVAL_PARITY_ERROR	RX_PNUP_PARITY_ERROR	RX_DATA_GATHER_PARITY_ERROR	TX_DATA_GATHER_PARITY_ERROR	RX_AES_DATA_CONVERT_PARITY_ERROR	TX_AES_DATA_CONVERT_PARITY_ERROR	RX_DATA_FORMAT_PARITY_ERROR	TX_DATA_FORMAT_PARITY_ERROR	RX_SCI_RECOVERED_XPN_FIFO_PARITY_ERROR	TX_SCI_RECOVERED_XPN_FIFO_PARITY_ERROR	RX_SCPN_SSCI_FIFO_PARITY_ERROR	TX_SCPN_SSCI_FIFO_PARITY_ERROR	RX_SALT_FIFO_PARITY_ERROR	TX_SALT_FIFO_PARITY_ERROR	RX_SA_PARSE_PARITY_ERROR	TX_SA_PARSE_PARITY_ERROR	RX_VLAN_INGRESS_PARITY_ERROR	TX_VLAN_INGRESS_PARITY_ERROR

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2

Bits	Name	Memory Access	Description
31	RX_SC_MIBS_SRAM_BNK1_ECC_ERROR	R/W1C	Indicates if there has been a RX SC MIBS SRAM memory bank 1 ECC error. **Values:** ■ 0x1 (RX_SC_MIBS_SRAM_BNK1_ECC_ERROR): Indicates RX SC MIBS SRAM memory bank 1 ECC errors. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
30	RX_SA_MIBS_SRAM_ECC_ERROR	R/W1C	Indicates if there has been a RX SA MIBS SRAM ECC error. **Values:** ■ 0x1 (RX_SA_MIBS_SRAM_ECC_ERROR): Indicates RX SA MIBS SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
29	TX_SA_MIBS_SRAM_ECC_ERROR	R/W1C	Indicates if there has been a TX SA MIBS SRAM ECC error. **Values:** ■ 0x1 (TX_SA_MIBS_SRAM_ECC_ERROR): Indicates TX SA MIBS SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
28	RX_ARW_SRAM_ECC_ERROR	R/W1C	Indicates if there has been a RX ARW SRAM ECC error. **Values:** ■ 0x1 (RX_ARW_SRAM_ECC_ERROR): Indicates RX ARW SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
27	RX_SALT_SRAM_ECC_ERROR	R/W1C	Indicates if there has been a RX SALT SRAM ECC error. **Values:** ■ 0x1 (RX_SALT_SRAM_ECC_ERROR): Indicates RX SALT SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
26	TX_SALT_SRAM_ECC_ERROR	R/W1C	Indicates if there has been a TX SALT SRAM ECC error. **Values:** ■ 0x1 (TX_SALT_SRAM_ECC_ERROR): Indicates TX SALT SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
25	RX_LAT_MNGR_ERROR	R/W1C	Indicates if there is a RX latency error. **Values:** ■ 0x1 (RX_LAT_MNGR_ERROR): Indicates if the RX latency has been violated. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
24	TX_LAT_MNGR_ERROR	R/W1C	Indicates if there is a TX latency error. **Values:** ■ 0x1 (TX_LAT_MNGR_ERROR): Indicates if the TX latency has been violated. **Value After Reset:** 0x0 **Exists:** Always
23	RX_MISSION_MODE_BIST_ERROR	R/W1C	Indicates if there is an decryption mismatch of the NIST vectors during mission mode BIST (RX KAT). **Values:** ■ 0x1 (RX_MISSION_MODE_BIST_ERROR): Indicates a RX mission mode BIST error. **Value After Reset:** 0x0 **Exists:** Always
22	TX_MISSION_MODE_BIST_ERROR	R/W1C	Indicates if there is an encryption mismatch of the NIST vectors during mission mode BIST (TX KAT). **Values:** ■ 0x1 (TX_MISSION_MODE_BIST_ERROR): Indicates a TX mission mode BIST error. **Value After Reset:** 0x0 **Exists:** Always
21	AES_RX_TIMING_REDUNDANCY_ERROR	R/W1C	Indicates if there is an encryption mismatch between the 'normal' and shadow RX data units coming out of the AES (RX AES timing redundancy). **Values:** ■ 0x1 (AES_RX_TIMING_REDUNDANCY_ERROR): Indicates an encryption AES RX timing redundancy error. **Value After Reset:** 0x0 **Exists:** Always
20	AES_TX_TIMING_REDUNDANCY_ERROR	R/W1C	Indicates if there is an encryption mismatch between the 'normal' and shadow TX data units coming out of the AES (TX AES timing redundancy). **Values:** ■ 0x1 (AES_TX_TIMING_REDUNDANCY_ERROR): Indicates an encryption AES TX timing redundancy error. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
19	RX_VLAN_EGRESS_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX vlan egress controller outputs. **Values:** ■ 0x1 (RX_VLAN_EGRESS_PARITY_ERROR): Indicates a parity mismatch in the RX vlan egress controller. **Value After Reset:** 0x0 **Exists:** Always
18	TX_VLAN_EGRESS_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX vlan egress controller outputs. **Values:** ■ 0x1 (TX_VLAN_EGRESS_PARITY_ERROR): Indicates a parity mismatch in the TX vlan egress controller. **Value After Reset:** 0x0 **Exists:** Always
17	RX_SECTAG_REMOVAL_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX SECtag removal module outputs. **Values:** ■ 0x1 (RX_SECTAG_REMOVAL_PARITY_ERROR): Indicates a parity mismatch in the RX SECtag Removal signals. **Value After Reset:** 0x0 **Exists:** Always
16	RX_PNUP_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX PNUP module outputs. **Values:** ■ 0x1 (RX_PNUP_PARITY_ERROR): Indicates a parity mismatch in the RX PNUP signals. **Value After Reset:** 0x0 **Exists:** Always
15	RX_DATA_GATHER_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX data gather module outputs. **Values:** ■ 0x1 (RX_DATA_GATHER_PARITY_ERROR): Indicates a parity mismatch in the RX data gather signals. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
14	TX_DATA_GATHER_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX data gather module outputs. **Values:** ■ 0x1 (TX_DATA_GATHER_PARITY_ERROR): Indicates a parity mismatch in the TX data gather signals. **Value After Reset:** 0x0 **Exists:** Always
13	RX_AES_DATA_CONVERT_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX AES data convert module outputs. **Values:** ■ 0x1 (RX_AES_DATA_CONVERT_PARITY_ERROR): Indicates a parity mismatch in the RX AES data convert signals. **Value After Reset:** 0x0 **Exists:** Always
12	TX_AES_DATA_CONVERT_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX AES data convert module outputs. **Values:** ■ 0x1 (TX_AES_DATA_CONVERT_PARITY_ERROR): Indicates a parity mismatch in the TX AES data convert signals. **Value After Reset:** 0x0 **Exists:** Always
11	RX_DATA_FORMAT_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX data format module outputs. **Values:** ■ 0x1 (RX_DATA_FORMAT_PARITY_ERROR): Indicates a parity mismatch in the RX data format signals. **Value After Reset:** 0x0 **Exists:** Always
10	TX_DATA_FORMAT_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX data format module outputs. **Values:** ■ 0x1 (TX_DATA_FORMAT_PARITY_ERROR): Indicates a parity mismatch in the TX data format signals. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
9	RX_SCI_RECOVERED_XPN_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX SCI recovered XPN FIFO outputs. **Values:** ■ 0x1 (RX_SCI_RECOVERED_XPN_FIFO_PARITY_ERROR): Indicates a parity mismatch in the RX SCI recovered XPN FIFO outputs. **Value After Reset:** 0x0 **Exists:** Always
8	TX_SCI_RECOVERED_XPN_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX SCI recovered XPN FIFO outputs. **Values:** ■ 0x1 (TX_SCI_RECOVERED_XPN_FIFO_PARITY_ERROR): Indicates a parity mismatch in the TX SCI recovered XPN FIFO outputs. **Value After Reset:** 0x0 **Exists:** Always
7	RX_SCPN_SSCI_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX SCPN SSCI FIFO outputs. **Values:** ■ 0x1 (RX_SCPN_SSCI_FIFO_PARITY_ERROR): Indicates a parity mismatch in the RX SCPN SSCI FIFO outputs. **Value After Reset:** 0x0 **Exists:** Always
6	TX_SCPN_SSCI_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX SCPN SSCI FIFO outputs. **Values:** ■ 0x1 (TX_SCPN_SSCI_FIFO_PARITY_ERROR): Indicates a parity mismatch in the TX SCPN SSCI FIFO outputs. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
5	RX_SALT_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX SALT FIFO outputs. **Values:** ■ 0x1 (RX_SALT_FIFO_PARITY_ERROR): Indicates a parity mismatch in the RX SALT FIFO outputs. **Value After Reset:** 0x0 **Exists:** Always
4	TX_SALT_FIFO_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX SALT FIFO outputs. **Values:** ■ 0x1 (TX_SALT_FIFO_PARITY_ERROR): Indicates a parity mismatch in the TX SALT FIFO outputs. **Value After Reset:** 0x0 **Exists:** Always
3	RX_SA_PARSE_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX SA Parse module outputs. **Values:** ■ 0x1 (RX_SA_PARSE_ERROR): Indicates a parity mismatch in the RX SA Parse signals. **Value After Reset:** 0x0 **Exists:** Always
2	TX_SA_PARSE_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX SA Parse module outputs. **Values:** ■ 0x1 (TX_SA_PARSE_PARITY_ERROR): Indicates a parity mismatch in the TX SA Parse signals. **Value After Reset:** 0x0 **Exists:** Always
1	RX_VLAN_INGRESS_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the RX vlan ingress controller outputs. **Values:** ■ 0x1 (RX_VLAN_INGRESS_PARITY_ERROR): Indicates a parity mismatch in the RX vlan ingress controller. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1264 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_2 (Continued)

Bits	Name	Memory Access	Description
0	TX_VLAN_INGRESS_PARITY_ERROR	R/W1C	Indicates if there is a parity mismatch in the TX vlan ingress controller outputs. **Values:** ■ 0x1 (TX_VLAN_INGRESS_PARITY_ERROR): Indicates a parity mismatch in the TX vlan ingress controller. **Value After Reset:** 0x0 **Exists:** Always

6.1.1258 MACSEC_SAFETY_ALARM_SOURCES_3

- **Name:** MACSEC Safety Alarm Sources Register bank 3.
- **Description:** MACSEC Safety Alarm Sources Register bank 3.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x200c
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

31:22	21	20	19	18	17	16	15	14	13	12	11	10	9	8:7	6	5	4	3	2	1	0
reserved_31_20	RX_SAD_FLOP_BASED_MEM_PARITY_ERROR	TX_SAD_FLOP_BASED_MEM_PARITY_ERROR	MIB_REGISTER_INTERFACE_ERROR	RX_PCH_FIFO_ERROR	TX_PCH_FIFO_ERROR	RX_XGMII_EGRESS_CRC_ERROR	TX_XGMII_EGRESS_CRC_ERROR	RX_XGMII_INGRESS_CRC_ERROR	TX_XGMII_INGRESS_CRC_ERROR	RX_GMII_EGRESS_CRC_ERROR	TX_GMII_EGRESS_CRC_ERROR	RX_GMII_INGRESS_CRC_ERROR	TX_GMII_INGRESS_CRC_ERROR	Rsvd	RX_SAD_MEM_PARITY_ERROR	TX_SAD_MEM_PARITY_ERROR	AES_HCTX_SRAM_ECC_ERROR	AES_CCTX_SRAM_ECC_ERROR	AES_HKEY_SRAM_ECC_ERROR	AES_CKEY_SRAM_ECC_ERROR	RX_SC_MIBS_SRAM_BNK2_ECC_ERROR

Table 6-1265 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_3

Bits	Name	Memory Access	Description
31:22	reserved_31_20	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1265 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_3 (Continued)

Bits	Name	Memory Access	Description
21	RX_SAD_FLOP_BASED_MEM_PARITY_ERROR	R/W1C	Indicates if there is a parity error in the RX SAD flop based memory. **Values:** ■ 0x1 (RX_SAD_FLOP_BASED_MEM_PARITY_ERROR): Indicates if there is a parity error in the RX SAD flop based memory. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_RX_SAD_MEM_TYPE==0)==1
20	TX_SAD_FLOP_BASED_MEM_PARITY_ERROR	R/W1C	Indicates if there is a parity error in the TX SAD flop based memory. **Values:** ■ 0x1 (TX_SAD_FLOP_BASED_MEM_PARITY_ERROR): Indicates if there is a parity error in the TX SAD flop based memory. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_TX_SAD_MEM_TYPE==0)==1
19	MIB_REGISTER_INTERFACE_ERROR	R/W1C	Indicates if an error is found in the DMR of mib_register_interface. **Values:** ■ 0x1 (MIB_REGISTER_INTERFACE_ERROR): Indicates if an error is found in the DMR of mib_register_interface. **Value After Reset:** 0x0 **Exists:** Always
18	RX_PCH_FIFO_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow RX USGMII/USXGMII (PCH) FIFO signals. **Values:** ■ 0x1 (RX_PCH_FIFO_ERROR): Indicates an output mismatch in the RX USGMII/USXGMII (PCH) FIFOs. **Value After Reset:** 0x0 **Exists:** Always
17	TX_PCH_FIFO_ERROR	R/W1C	Indicates if there is an output mismatch between the nominal and the shadow TX USGMII/USXGMII (PCH) FIFO signals. **Values:** ■ 0x1 (TX_PCH_FIFO_ERROR): Indicates an output mismatch in the TX USGMII/USXGMII (PCH) FIFOs. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1265 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_3 (Continued)

Bits	Name	Memory Access	Description
16	RX_XGMII_EGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the RX XGMII egress engine was detected. **Values:** ■ 0x1 (RX_XGMII_EGRESS_CRC_ERROR): Indicates a CRC mismatch in the RX XGMII egress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
15	TX_XGMII_EGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the TX XGMII egress engine was detected. **Values:** ■ 0x1 (TX_XGMII_EGRESS_CRC_ERROR): Indicates a CRC mismatch in the TX XGMII egress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
14	RX_XGMII_INGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the RX XGMII ingress engine was detected. **Values:** ■ 0x1 (RX_XGMII_INGRESS_CRC_ERROR): Indicates a CRC mismatch in the RX XGMII ingress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
13	TX_XGMII_INGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the TX XGMII ingress engine was detected. **Values:** ■ 0x1 (TX_XGMII_INGRESS_CRC_ERROR): Indicates a CRC mismatch in the TX XGMII ingress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
12	RX_GMII_EGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the RX GMII egress engine was detected. **Values:** ■ 0x1 (RX_GMII_EGRESS_CRC_ERROR): Indicates a CRC mismatch in the RX GMII egress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1

Table 6-1265 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_3 (Continued)

Bits	Name	Memory Access	Description
11	TX_GMII_EGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the TX GMII egress engine was detected. **Values:** ■ 0x1 (TX_GMII_EGRESS_CRC_ERROR): Indicates a CRC mismatch in the TX GMII egress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
10	RX_GMII_INGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the RX GMII ingress engine was detected. **Values:** ■ 0x1 (RX_GMII_INGRESS_CRC_ERROR): Indicates a CRC mismatch in the RX GMII ingress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
9	TX_GMII_INGRESS_CRC_ERROR	R/W1C	Indicates if a CRC mismatch in the TX GMII ingress engine was detected. **Values:** ■ 0x1 (TX_GMII_INGRESS_CRC_ERROR): Indicates a CRC mismatch in the TX GMII ingress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
8:7			**Reserved Field:** Yes
6	RX_SAD_MEM_PARITY_ERROR	R/W1C	Indicates if there has been a RX SAD mem parity error. **Values:** ■ 0x1 (RX_SAD_MEM_PARITY_ERROR): Indicates RX SAD mem parity errors. **Value After Reset:** 0x0 **Exists:** Always
5	TX_SAD_MEM_PARITY_ERROR	R/W1C	Indicates if there has been a TX SAD internal memory PARITY error. **Values:** ■ 0x1 (TX_SAD_MEM_PARITY_ERROR): Indicates TX SAD mem parity errors. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1265 Fields for Register: MACSEC_SAFETY_ALARM_SOURCES_3 (Continued)

Bits	Name	Memory Access	Description
4	AES_HCTX_SRAM_ECC_ERROR	R/W1C	Indicates if there has been an AES HCTX SRAM ECC error. **Values:** ■ 0x1 (AES_HCTX_SRAM_ECC_ERROR): Indicates AES HCTX SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** DWC_UAES_CFG_HASH_ILEAVE_EN==1
3	AES_CCTX_SRAM_ECC_ERROR	R/W1C	Indicates if there has been an AES CCTX SRAM ECC error. **Values:** ■ 0x1 (AES_CCTX_SRAM_ECC_ERROR): Indicates AES CCTX SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
2	AES_HKEY_SRAM_ECC_ERROR	R/W1C	Indicates if there has been an AES HKEY SRAM ECC error. **Values:** ■ 0x1 (AES_HKEY_SRAM_ECC_ERROR): Indicates AES HKEY SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
1	AES_CKEY_SRAM_ECC_ERROR	R/W1C	Indicates if there has been an AES CKEY SRAM ECC error. **Values:** ■ 0x1 (AES_CKEY_SRAM_ECC_ERROR): Indicates AES CKEY SRAM ECC errors. **Value After Reset:** 0x0 **Exists:** Always
0	RX_SC_MIBS_SRAM_BNK2_ECC_ERROR	R/W1C	Indicates if there has been a RX SC MIBS SRAM memory bank 2 ECC error. **Values:** ■ 0x1 (RX_SC_MIBS_SRAM_BNK2_ECC_ERROR): Indicates RX SC MIBS SRAM memory bank 2 ECC errors. **Value After Reset:** 0x0 **Exists:** Always

6.1.1259 MACSEC_REGBANK_ERROR_INJECTION

- **Name:** MACsec Register Bank Injection Register.
- **Description:** MACsec Register Bank Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2010
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

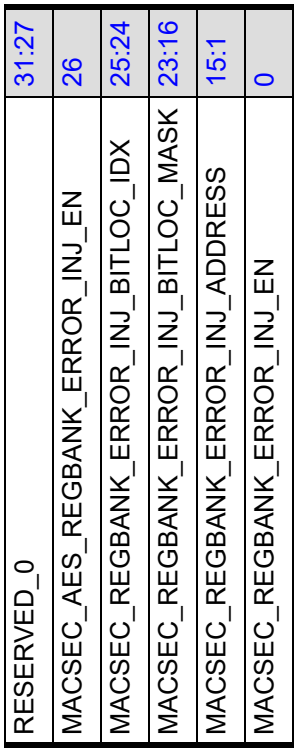


Table 6-1266 Fields for Register: MACSEC_REGBANK_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:27	RESERVED_0	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1266 Fields for Register: MACSEC_REGBANK_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
26	MACSEC_AES_REGBANK_ERROR_INJ_EN	R/W	Enables error injection on the AES configuration register duplication. **Values:** ■ 0x1 (MACSEC_AES_REGBANK_ERROR_INJ_ENABLE): Enables error injection on AES configuration registers. ■ 0x0 (MACSEC_AES_REGBANK_ERROR_INJ_DISABLE): Disables error injection on AES configuration registers. **Value After Reset:** 0x0 **Exists:** Always
25:24	MACSEC_REGBANK_ERROR_INJ_BITLOC_IDX	R/W	Selects the byte be injected by the error Injection bit pattern mask: 0 - register[31:0] = {register[31:8] , register[7:0] ^ bitloc[7:0]} 1 - register[31:0] = {register[31:16] , register[15:8] ^ bitloc[7:0], register[7:0]} 2 - register[31:0] = {register[31:24], register[23:16] ^ bitloc[7:0], register[15:0]} 3 - register[31:0] = {register[31:24] ^ bitloc[7:0], register[7:0]} **Values:** ■ 0x1 (MACSEC_REGBANK_ERROR_INJ_ENABLE): Error Injection byte index. **Value After Reset:** 0x0 **Exists:** Always
23:16	MACSEC_REGBANK_ERROR_INJ_BITLOC_MASK	R/W	Sets up to 8 bits error pattern to be injected. **Values:** ■ 0x1 (MACSEC_REGBANK_ERROR_INJ_ENABLE): Error Injection bit pattern mask. **Value After Reset:** 0x0 **Exists:** Always
15:1	MACSEC_REGBANK_ERROR_INJ_ADDRESS	R/W	Indicates the register address where the error is to be injected. **Values:** ■ 0x1 (MACSEC_REGBANK_ERROR_INJ_ENABLE): Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1266 Fields for Register: MACSEC_REGBANK_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_REGBANK_ERROR_INJ_EN	R/W	Enables error injection on configuration register duplication. **Values:** ■ 0x1 (MACSEC_REGBANK_ERROR_INJ_ENABLE): Enables error injection on configuration registers. ■ 0x0 (MACSEC_REGBANK_ERROR_INJ_DISABLE): Disables error injection on configuration registers. **Value After Reset:** 0x0 **Exists:** Always

6.1.1260 MACSEC_SRAM_ECC_ERROR_INJECTION

- **Name:** MACsec SRAM Injection Register.
- **Description:** MACsec SRAM Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2014
- **Exists:** DWC_MACSEC_SRAM_ECC_SUPPORT==1

31	30:12	11:1	0
MACSEC_SRAM_ECC_ERROR_INJ_UNCORRECTABLE	RESERVED_0	MACSEC_SRAM_ECC_ERROR_INJ_ADDRESS	MACSEC_SRAM_ECC_ERROR_INJ_EN

Table 6-1267 Fields for Register: MACSEC_SRAM_ECC_ERROR_INJECTION

Bits	Name	Memory Access	Description
31	MACSEC_SRAM_ECC_ERROR_INJ_UNCORRECTABLE	R/W	Indicates the type of error to be injected: 0- Injects a correctable error (Only one bit flip on the data) 1- Injects an uncorrectable error (Two or more bit flips on the data) **Values:** ■ 0x1 (MACSEC_SRAM_ECC_ERROR_INJ_ENABLE): SRAM Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
30:12	RESERVED_0	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
11:1	MACSEC_SRAM_ECC_ERROR_INJ_ADDRESS	R/W	Indicates the register address where the error is to be injected: Bit 0 - Injects an error on tx_salt_sram Bit 1 - Injects an error on rx_salt_sram Bit 2 - Injects an error on rx_arw_sram Bit 3 - Injects an error on tx_sa_mibs_sram Bit 4 - Injects an error on rx_sa_mibs_sram Bit 5 - Injects an error on rx_sc_mibs_sram_bnk1 Bit 6 - Injects an error on rx_sc_mibs_sram_bnk2 Bit 7 - Injects an error on aes_ckey_sram Bit 8 - Injects an error on aes_hkey_sram_ecc Bit 9 - Injects an error on aes_cctx_sram Bit 10 - Injects an error on aes_hctx_sram **Values:** ■ 0x1 (MACSEC_SRAM_ECC_ERROR_INJ_ENABLE): Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1267 Fields for Register: MACSEC_SRAM_ECC_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_SRAM_ECC_ERROR_INJ_EN	R/W	Enables error injection on the data/address of an SRAM. **Values:** ■ 0x1 (MACSEC_SRAM_ECC_ERROR_INJ_ENABLE): Enables error injection on the address-selected SRAM. ■ 0x0 (MACSEC_SRAM_ECC_ERROR_INJ_DISABLE): Disables error injection on the address-selected SRAM. **Value After Reset:** 0x0 **Exists:** Always

6.1.1261 MACSEC_APB_PARITY_ERROR_INJECTION

- **Name:** MACsec APB Injection Register.
- **Description:** MACsec APB Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2018
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

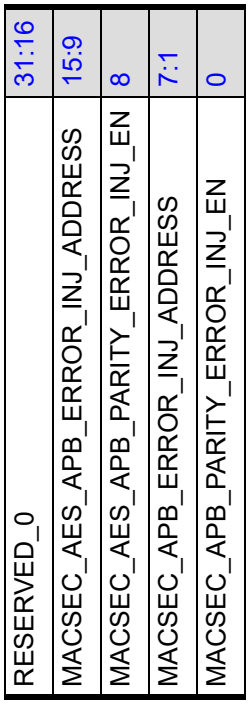


Table 6-1268 Fields for Register: MACSEC_APB_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:16	RESERVED_0	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1268 Fields for Register: MACSEC_APB_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
15:9	MACSEC_AES_APB_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding AES APB signal checker to inject the error: Bit 0 - Injects a parity error on apb_bridge_mem_en Bit 1 - Injects a parity error on apb_bridge_mem_wr Bit 2 - Injects a parity error on apb_bridge_mem_addr Bit 3 - Injects a parity error on apb_bridge_mem_wdata Bit 4 - Injects a parity error on apb_bridge_mem_scure Bit 5 - Injects a parity error on apb_bridge_mem_ptcol_err Bit 6 - Injects a parity error on apb_bridge_mem_ptcol_err_setup **Values:** ■ 0x1 (MACSEC_AES_APB_ERROR_INJ_ADDRESS): AES APB Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
8	MACSEC_AES_APB_PARITY_ERROR_INJ_EN	R/W	Enables error injection on of the AES APB signals. **Values:** ■ 0x1 (MACSEC_AES_APB_PARITY_ERROR_INJ_ENABLE): Enables error injection on the AES APB data. ■ 0x0 (MACSEC_AES_APB_PARITY_ERROR_INJ_DISABLE) : Disables error injection on the AES APB data. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1268 Fields for Register: MACSEC_APB_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
7:1	MACSEC_APB_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding apb signal checker to inject the error: Bit 0 - Injects a parity error on apb_bridge_mem_en Bit 1 - Injects a parity error on apb_bridge_mem_wr Bit 2 - Injects a parity error on apb_bridge_mem_addr Bit 3 - Injects a parity error on apb_bridge_mem_wdata Bit 4 - Injects a parity error on apb_bridge_mem_scure Bit 5 - Injects a parity error on apb_bridge_mem_ptcol_err Bit 6 - Injects a parity error on apb_bridge_mem_ptcol_err_setup **Values:** ■ 0x1 (MACSEC_APB_ERROR_INJ_ADDRESS): APB Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_APB_PARITY_ERROR_INJ_EN	R/W	Enables error injection on of the APB signals. **Values:** ■ 0x1 (MACSEC_APB_PARITY_ERROR_INJ_ENABLE): Enables error injection on the APB data. ■ 0x0 (MACSEC_APB_PARITY_ERROR_INJ_DISABLE): Disables error injection on the APB data. **Value After Reset:** 0x0 **Exists:** Always

6.1.1262 MACSEC_DMR_ERROR_INJECTION

- **Name:** MACsec DMR Injection Register.
- **Description:** MACsec DMR Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x201c
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

MACSEC_DMR_RX_SB_FIFO_ERR_INJ_EN	31
MACSEC_DMR_TX_SB_FIFO_ERR_INJ_EN	30
MACSEC_DMR_RX_DUAL_IF_MUX_ERROR_INJ_EN	29
MACSEC_DMR_TX_DUAL_IF_MUX_ERROR_INJ_EN	28
Rsvd	27
reserved_29_28	29:28
MACSEC_DMR_RX_GMII_MUX_ERROR_INJ_EN	27
MACSEC_DMR_TX_GMII_MUX_ERROR_INJ_EN	26
Rsvd	25
reserved_27_26	27:26
MACSEC_DMR_AES_KEY_DETECT_ERROR_INJ_EN	25
MACSEC_DMR_TX_SAD_PN_REGBANK_ERR_INJ_EN	24
MACSEC_DMR_AES_HOST_MEMORY_REQ_GEN_ERROR_INJ_EN	23
MACSEC_DMR_HOST_MEMORY_REQ_GEN_ERROR_INJ_EN	22
MACSEC_DMR_RX_ARW_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	21
MACSEC_DMR_RX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	20
MACSEC_DMR_RX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	19
MACSEC_DMR_RX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	18
MACSEC_DMR_TX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	17
MACSEC_DMR_TX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	16
MACSEC_DMR_TX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	15
RESERVED_14_14	14
MACSEC_DMR_RX_ARW_ERROR_INJ_EN	13
MACSEC_DMR_RX_SALT_ERROR_INJ_EN	12
MACSEC_DMR_RX_SAI_ERROR_INJ_EN	11
MACSEC_DMR_RX_SAD_ERROR_INJ_EN	10
RESERVED_9_8	9:8
MACSEC_DMR_MIB_REG_IF_ERROR_INJ_EN	7
MACSEC_DMR_TX_SALT_ERROR_INJ_EN	6
MACSEC_DMR_TX_SAD_ERROR_INJ_EN	5
RESERVED_4_4	4
MACSEC_DMR_TX_SAI_ERROR_INJ_EN	3
MACSEC_DMR_RX_PCH_FIFO_ERR_INJ_EN	2
MACSEC_DMR_TX_PCH_FIFO_ERR_INJ_EN	1
MACSEC_DMR_MIB_ERROR_INJ_EN	0

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION

Bits	Name	Memory Access	Description
31	MACSEC_DMR_RX_SB_FIFO_ERR_INJ_EN	R/W	Enables DMR error injection on the GMII/XGMII IF RX Sideband fifo. **Values:** ■ 0x1 (MACSEC_DMR_RX_SB_FIFO_ERR_INJ_ENABLE): Enables dmr error injection on the GMII/XGMII IF RX Sideband fifo. ■ 0x0 (MACSEC_DMR_RX_SB_FIFO_ERR_INJ_DISABLE): Disables dmr error injection on the GMII/XGMII IF RX Sideband fifo. **Value After Reset:** 0x0 **Exists:** Always
30	MACSEC_DMR_TX_SB_FIFO_ERR_INJ_EN	R/W	Enables DMR error injection on the GMII/XGMII IF TX Sideband fifo. **Values:** ■ 0x1 (MACSEC_DMR_TX_SB_FIFO_ERR_INJ_ENABLE): Enables dmr error injection on the GMII/XGMII IF TX Sideband fifo. ■ 0x0 (MACSEC_DMR_TX_SB_FIFO_ERR_INJ_DISABLE): Disables dmr error injection on the GMII/XGMII IF TX Sideband fifo. **Value After Reset:** 0x0 **Exists:** Always
29	MACSEC_DMR_RX_DUAL_IF_MUX_ERROR_INJ_EN	R/W	Enables DMR error injection on the rx side of the u_macsec_dual_if_mux module. **Values:** ■ 0x1 (MACSEC_DMR_RX_DUAL_IF_MUX_ERROR_INJ_ENABLE): Enables dmr error injection on the rx side of the u_macsec_dual_if_mux. ■ 0x0 (MACSEC_DMR_RX_DUAL_IF_MUX_ERROR_INJ_DISABLE): Disables dmr error injection on the rx side of the u_macsec_dual_if_mux. **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_DUAL_IF == 1

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
28	MACSEC_DMR_TX_DUAL_IF_MUX_ERROR_INJ_EN	R/W	Enables DMR error injection on the tx side of the u_macsec_gmii_bridge_mux module. **Values:** ■ 0x1 (MACSEC_DMR_TX_DUAL_IF_MUX_ERROR_INJ_ENABLE): Enables dmr error injection on the tx side of the u_macsec_gmii_bridge_mux. ■ 0x0 (MACSEC_DMR_TX_DUAL_IF_MUX_ERROR_INJ_DISABLE): Disables dmr error injection on the tx side of the u_macsec_gmii_bridge_mux. **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_DUAL_IF == 1
27			**Reserved Field:** Yes
29:28	reserved_29_28	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_DUAL_IF != 1
27	MACSEC_DMR_RX_GMII_MUX_ERROR_INJ_EN	R/W	Enables DMR error injection on the rx side of the u_macsec_gmii_bridge_mux module. **Values:** ■ 0x1 (MACSEC_DMR_RX_GMII_MUX_ERROR_INJ_ENABLE): Enables dmr error injection on the rx side of the u_macsec_gmii_bridge_mux. ■ 0x0 (MACSEC_DMR_RX_GMII_MUX_ERROR_INJ_DISABLE): Disables dmr error injection on the rx side of the u_macsec_gmii_bridge_mux. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) == 1

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
26	MACSEC_DMR_TX_GMII_MUX_ERROR_INJ_EN	R/W	Enables DMR error injection on the tx side of the u_macsec_gmii_bridge_mux module. **Values:** ■ 0x1 (MACSEC_DMR_TX_GMII_MUX_ERROR_INJ_ENABLE): Enables dmr error injection on the tx side of the u_macsec_gmii_bridge_mux. ■ 0x0 (MACSEC_DMR_TX_GMII_MUX_ERROR_INJ_DISABLE): Disables dmr error injection on the tx side of the u_macsec_gmii_bridge_mux. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) == 1
25			**Reserved Field:** Yes
27:26	reserved_27_26	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1) != 1
25	MACSEC_DMR_AES_KEY_DETECT_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_aes_key_detect_duplicate. **Values:** ■ 0x1 (MACSEC_DMR_AES_KEY_DETECT_ERROR_INJ_ENABLE): Enables dmr error injection on the U_aes_key_detect_duplicate. ■ 0x0 (MACSEC_DMR_AES_KEY_DETECT_ERROR_INJ_DISABLE): Disables dmr error injection on the U_aes_key_detect_duplicate. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
24	MACSEC_DMR_TX_SAD_PN_REGBANK_ERR_INJ_EN	R/W	Enables DMR error injection on of the U_tx_sad_pn_reg_bank. **Values:** ■ 0x1 (MACSEC_DMR_TX_SAD_PN_REGBANK_ERR_INJ_ENABLE): Enables dmr error injection on the U_tx_sad_pn_reg_bank. ■ 0x0 (MACSEC_DMR_TX_SAD_PN_REGBANK_ERR_INJ_DISABLE): Disables dmr error injection on the U_tx_sad_pn_reg_bank. **Value After Reset:** 0x0 **Exists:** Always
23	MACSEC_DMR_AES_HOST_MEMORY_REQ_GEN_ERROR_INJ_EN	R/W	Enables DMR error injection on of the u_aes_host_memory_request_generation_duplicate. **Values:** ■ 0x1 (MACSEC_DMR_AES_HOST_MEMORY_REQ_GEN_ERROR_INJ_ENABLE): Enables dmr error injection on the u_aes_host_memory_request_generation_duplicate. ■ 0x0 (MACSEC_DMR_AES_HOST_MEMORY_REQ_GEN_ERROR_INJ_DISABLE): Disables dmr error injection on the u_aes_host_memory_request_generation_duplicate. **Value After Reset:** 0x0 **Exists:** Always
22	MACSEC_DMR_HOST_MEMORY_REQ_GEN_ERROR_INJ_EN	R/W	Enables DMR error injection on of the u_host_memory_request_generation. **Values:** ■ 0x1 (MACSEC_DMR_HOST_MEMORY_REQ_GEN_ERROR_INJ_ENABLE): Enables dmr error injection on the u_host_memory_request_generation. ■ 0x0 (MACSEC_DMR_HOST_MEMORY_REQ_GEN_ERROR_INJ_DISABLE): Disables dmr error injection on the u_host_memory_request_generation. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
21	MACSEC_DMR_RX_ARW_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_arw_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_TX_ARW_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_arw_memory_register_interface. ■ 0x0 (MACSEC_DMR_TX_ARW_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_arw_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always
20	MACSEC_DMR_RX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_salt_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_TX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_salt_memory_register_interface. ■ 0x0 (MACSEC_DMR_TX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_salt_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always
19	MACSEC_DMR_RX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_sad_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_RX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_sad_memory_register_interface. ■ 0x0 (MACSEC_DMR_RX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_sad_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
18	MACSEC_DMR_RX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_sai_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_RX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_sai_memory_register_interface. ■ 0x0 (MACSEC_DMR_RX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_sai_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always
17	MACSEC_DMR_TX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_tx_salt_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_TX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_tx_salt_memory_register_interface. ■ 0x0 (MACSEC_DMR_TX_SALT_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_tx_salt_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always
16	MACSEC_DMR_TX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_tx_sad_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_TX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_tx_sad_memory_register_interface. ■ 0x0 (MACSEC_DMR_TX_SAD_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_tx_sad_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
15	MACSEC_DMR_TX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_tx_sai_memory_register_interface. **Values:** ■ 0x1 (MACSEC_DMR_TX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_ENABLE): Enables dmr error injection on the U_tx_sai_memory_register_interface. ■ 0x0 (MACSEC_DMR_TX_SAI_MEMORY_REGISTER_INTERFACE_ERROR_INJ_DISABLE): Disables dmr error injection on the U_tx_sai_memory_register_interface. **Value After Reset:** 0x0 **Exists:** Always
14	RESERVED_14_14	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED_1): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always
13	MACSEC_DMR_RX_ARW_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_arw_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_RX_ARW_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_arw_memory_access_module. ■ 0x0 (MACSEC_DMR_RX_ARW_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_arw_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
12	MACSEC_DMR_RX_SALT_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_salt_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_RX_SALT_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_salt_memory_access_module. ■ 0x0 (MACSEC_DMR_RX_SALT_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_salt_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always
11	MACSEC_DMR_RX_SAI_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_sai_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_RX_SAI_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_sai_memory_access_module. ■ 0x0 (MACSEC_DMR_RX_SAI_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_sai_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always
10	MACSEC_DMR_RX_SAD_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_rx_sad_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_RX_SAD_ERROR_INJ_ENABLE): Enables dmr error injection on the U_rx_sad_memory_access_module. ■ 0x0 (MACSEC_DMR_RX_SAD_ERROR_INJ_DISABLE): Disables dmr error injection on the U_rx_sad_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always
9:8	RESERVED_9_8	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED_1): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
7	MACSEC_DMR_MIB_REG_IF_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_mib_register_interface module. **Values:** ■ 0x1 (MACSEC_DMR_MIB_REG_IF_ERROR_INJ_ENABLE): Enables dmr error injection on the U_mib_register_interface module. ■ 0x0 (MACSEC_DMR_MIB_REG_IF_ERROR_INJ_DISABLE) : Disables dmr error injection on the U_mib_register_interface module. **Value After Reset:** 0x0 **Exists:** Always
6	MACSEC_DMR_TX_SALT_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_tx_salt_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_TX_SALT_ERROR_INJ_ENABLE): Enables dmr error injection on the U_tx_salt_memory_access_module. ■ 0x0 (MACSEC_DMR_TX_SALT_ERROR_INJ_DISABLE): Disables dmr error injection on the U_tx_salt_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always
5	MACSEC_DMR_TX_SAD_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_tx_sad_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_TX_SAD_ERROR_INJ_ENABLE): Enables dmr error injection on the U_tx_sad_memory_access_module. ■ 0x0 (MACSEC_DMR_TX_SAD_ERROR_INJ_DISABLE): Disables dmr error injection on the U_tx_sad_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
4	RESERVED_4_4	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED_1): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always
3	MACSEC_DMR_TX_SAI_ERRO R_INJ_EN	R/W	Enables DMR error injection on of the U_tx_sai_memory_access_module. **Values:** ■ 0x1 (MACSEC_DMR_TX_SAI_ERROR_INJ_ENABLE): Enables dmr error injection on the U_tx_sai_memory_access_module. ■ 0x0 (MACSEC_DMR_TX_SAI_ERROR_INJ_DISABLE): Disables dmr error injection on the U_tx_sai_memory_access_module. **Value After Reset:** 0x0 **Exists:** Always
2	MACSEC_DMR_RX_PCH_FIFO_ ERR_INJ_EN	R/W	Enables DMR error injection on the USGMII/USXGMII (PCH) RX Sideband fifo. **Values:** ■ 0x1 (MACSEC_DMR_RX_PCH_FIFO_ERR_INJ_ENABLE): Enables dmr error injection on the USGMII/USXGMII (PCH) RX Sideband fifo. ■ 0x0 (MACSEC_DMR_RX_PCH_FIFO_ERR_INJ_DISABLE): Disables dmr error injection on the USGMII/USXGMII (PCH) RX Sideband fifo. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1269 Fields for Register: MACSEC_DMR_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
1	MACSEC_DMR_TX_PCH_FIFO_ERR_INJ_EN	R/W	Enables DMR error injection on the USGMII/USXGMII (PCH) Sideband fifo. **Values:** ■ 0x1 (MACSEC_DMR_TX_PCH_FIFO_ERR_INJ_ENABLE): Enables dmr error injection on the USGMII/USXGMII (PCH) TX Sideband fifo. ■ 0x0 (MACSEC_DMR_TX_PCH_FIFO_ERR_INJ_DISABLE): Disables dmr error injection on the USGMII/USXGMII (PCH) TX Sideband fifo. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_DMR_MIB_ERROR_INJ_EN	R/W	Enables DMR error injection on of the U_mib module. **Values:** ■ 0x1 (MACSEC_DMR_MIB_ERROR_INJ_ENABLE): Enables dmr error injection on the U_mib module. ■ 0x0 (MACSEC_DMR_mib_ERROR_INJ_DISABLE): Disables dmr error injection on the U_mib module. **Value After Reset:** 0x0 **Exists:** Always

6.1.1263 MACSEC_TOP_BCM_WRAPPER_PARITY_ERROR_INJECTION

- **Name:** MACsec Top BCM Wrapper FIFOs Error Injection Register.
- **Description:** MACsec Top BCM Wrapper FIFOs Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2028
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

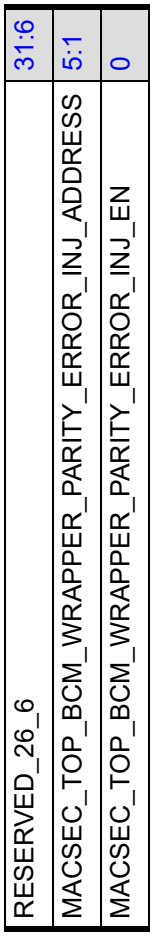


Table 6-1270 Fields for Register: MACSEC_TOP_BCM_WRAPPER_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:6	RESERVED_26_6	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
5:1	MACSEC_TOP_BCM_WRAPPER_PARITY_ERROR_INJ_ADDRESS	R/W	Indicates which FIFO parity checker module to inject the error: Bit 0 - Injects a parity error on if_rx_irq_assrt_dpp_chk Bit 1 - Injects a parity error on if_ingress_rx_to_clk_dpp_chk Bit 2 - Injects a parity error on if_egress_clk_to_rx_dpp_chk Bit 3 - Injects a parity error on tx_ingress_fifo_dpp_chk Bit 4 - Injects a parity error on tx_egress_fifo_dpp_chk **Values:** ■ 0x1 (MACSEC_TOP_BCM_WRAPPER_PARITY_ERROR_INJ_ADDRESS): Top BCM Wrapper Parity Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_TOP_BCM_WRAPPER_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the parity checker modules for the FIFOs located in the Top BCM Wrapper module. **Values:** ■ 0x1 (MACSEC_APB_PARITY_ERROR_INJ_ENABLE): Enables error injection on the parity checker modules for the FIFOs located in the Top BCM Wrapper module. ■ 0x0 (MACSEC_APB_PARITY_ERROR_INJ_DISABLE): Disables error injection on the parity checker modules for the FIFOs located in the Top BCM Wrapper module. **Value After Reset:** 0x0 **Exists:** Always

6.1.1264 MACSEC_TX_IN_TO_GMII_PARITY_ERROR_INJECTION

- **Name:** MACsec TX RMII/MII/RGMII to GMII ingress bridges Error Injection Register.
- **Description:** MACsec TX RMII/MII/RGMII to GMII ingress bridges Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x202c
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

Table 6-1271 Fields for Register: MACSEC_TX_IN_TO_GMII_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31			**Reserved Field:** Yes

Table 6-1271 Fields for Register: MACSEC_TX_IN_TO_GMII_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
30:15	RESERVED_31_15	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
14:1	MACSEC_TX_IN_TO_GMII_BRIDGES_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RMII to MII bridge rmii_en_n_edge_cntr signal Bit 1 - Injects an error on the RMII to MII bridge ipg_signals_to_check signal Bit 2 - Injects an error on the RMII to MII bridge loss_of_carrier_rmii_10M signal Bit 3 - Injects an error on the RMII to MII bridge rmii_10M_sample_data_counter signal Bit 4 - Injects an error on the RMII to MII bridge rmii_10M_post_ipg_violation_counter signal Bit 5 - Injects an error on the RMII to MII bridge rmii_d1 signal Bit 6 - Injects an error on the RMII to MII bridge o_mii_valid signal Bit 7 - Injects an error on the MII to GMII bridge mii_in_delayed signal Bit 8 - Injects an error on the MII to GMII bridge o_gmii_valid_signals_to_check signal Bit 9 - Injects an error on the RGMII to GMII bridge rgmii_ne_fifo_in signal Bit 10 - Injects an error on the RGMII to GMII bridge rgmii_re_fifo_in signal Bit 11 - Injects an error on the RGMII to GMII bridge preamble_error signal Bit 12 - Injects an error on the RGMII to GMII bridge rgmii_ne_post_fifo signal Bit 13 - Injects an error on the RGMII to GMII bridge rgmii_re_post_fifo signal **Values:** ■ 0x1 (MACSEC_TX_IN_TO_GMII_BRIDGES_ERROR_INJ_ADDRESS) **Value After Reset:** 0x0 **Exists:** Always

Table 6-1271 Fields for Register: MACSEC_TX_IN_TO_GMII_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_TX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX ingress bridges to GMII signals. **Values:** ■ 0x1 (MACSEC_TX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_ENABLE): Enables error injection on the TX ingress bridges to GMII. ■ 0x0 (MACSEC_TX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_DISABLE): TX Ingress bridges to GMII Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

6.1.1265 MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_0

- **Name:** MACsec RX RMII/MII to GMII ingress bridges Error Injection Register 0.
- **Description:** MACsec RX RMII/MII to GMII ingress bridges Error Injection Register 0.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2030
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

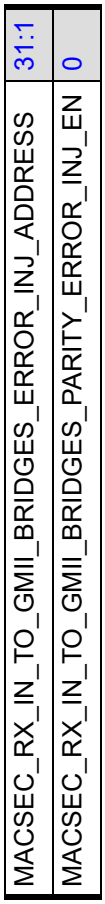


Table 6-1272 Fields for Register: MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_0

Bits	Name	Memory Access	Description
31:1	MACSEC_RX_IN_TO_GMII_BRIDGES_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RMII to MII bridge rmii_en_n_edge_cntr signal Bit 1 - Injects an error on the RMII to MII bridge ipg_signals_to_check signal Bit 2 - Injects an error on the RMII to MII bridge loss_of_carrier_rmii_10M signal Bit 3 - Injects an error on the RMII to MII bridge rmii_10M_sample_data_counter signal Bit 4 - Injects an error on the RMII to MII bridge rmii_10M_post_ipg_violation_counter signal Bit 5 - Injects an error on the RMII to MII bridge rmii_d1 signal Bit 6 - Injects an error on the RMII to MII bridge o_mii_valid signal Bit 7 - Injects an error on the MII to GMII bridge mii_in_delayed signal Bit 8 - Injects an error on the MII to GMII bridge o_gmii_valid_signals_to_check signal Bit 9 - Injects an error on the MII to GMII bridge byte aligner frame_in_progress_d1 signal Bit 10 - Injects an error on the MII to GMII bridge byte aligner mii_fifo_empty_d1 signal Bit 11 - Injects an error on the MII to GMII bridge byte aligner preamble_nibble_cntr signal Bit 12 - Injects an error on the MII to GMII bridge byte aligner I_mii_valid_d1 signal Bit 13 - Injects an error on the MII to GMII bridge byte aligner inter_packet_out_d1 signal Bit 14 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[0] signal Bit 15 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[1] signal Bit 16 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[2] signal

Table 6-1272 Fields for Register: MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_0 (Continued)

Bits	Name	Memory Access	Description
			Bit 17 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[3] signal Bit 18 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[4] signal Bit 19 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[5] signal Bit 20 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[6] signal Bit 21 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[7] signal Bit 22 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[8] signal Bit 23 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[9] signal Bit 24 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[10] signal Bit 25 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[11] signal Bit 26 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[12] signal Bit 27 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[13] signal Bit 28 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[14] signal Bit 29 - Injects an error on the MII to GMII bridge byte aligner pipeline stage mii_aligned_parity_pipe[15] signal Bit 30 - Injects an error on the MII to GMII bridge wrong_frame_start signal **Values:** ■ 0x1 (MACSEC_RX_IN_TO_GMII_BRIDGES_ERROR_INJ_ADDRESS): RX Ingress bridges to GMII Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1272 Fields for Register: MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_0 (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_RX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the RX ingress bridges to GMII signals. **Values:** ■ 0x1 (MACSEC_RX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX ingress bridges to GMII. ■ 0x0 (MACSEC_RX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX ingress bridges to GMII. **Value After Reset:** 0x0 **Exists:** Always

6.1.1266 MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_1

- **Name:** MACsec RX RGMII to GMII ingress bridges Error Injection Register 1.
- **Description:** MACsec RX RGMII to GMII ingress bridges Error Injection Register 1.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2034
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

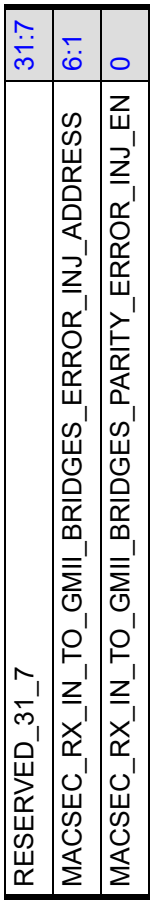


Table 6-1273 Fields for Register: MACSEC_RX_IN_TO_GMII_PARITY_ERROR_INJECTION_1

Bits	Name	Memory Access	Description
31:7	RESERVED_31_7	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
6:1	MACSEC_RX_IN_TO_GMII_BRIDGES_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RGMII to GMII bridge rgmii_ne_fifo_in signal Bit 1 - Injects an error on the RGMII to GMII bridge rgmii_re_fifo_in signal Bit 2 - Injects an error on the RGMII to GMII bridge rgmii_ls_fifo_out_d1 signal Bit 3 - Injects an error on the RGMII to GMII bridge preamble_error signal Bit 4 - Injects an error on the RGMII to GMII bridge rgmii_ne_post_fifo signal Bit 5 - Injects an error on the RGMII to GMII bridge rgmii_re_post_fifo signal **Values:** ■ 0x1 (MACSEC_RX_IN_TO_GMII_BRIDGES_ERROR_INJ_ADDRESS): RX Ingress bridges to GMII Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_RX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the RX ingress bridges to GMII signals. **Values:** ■ 0x1 (MACSEC_RX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX ingress bridges to GMII. ■ 0x0 (MACSEC_RX_IN_TO_GMII_BRIDGES_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX ingress bridges to GMII. **Value After Reset:** 0x0 **Exists:** Always

6.1.1267 MACSEC_TX_OUT_FROM_GMII_PARITY_ERROR_INJECTION

- **Name:** MACsec TX GMII to RMII/MII/RGMII egress bridges Error Injection Register.
- **Description:** MACsec TX GMII to RMII/MII/RGMII egress bridges Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2038
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

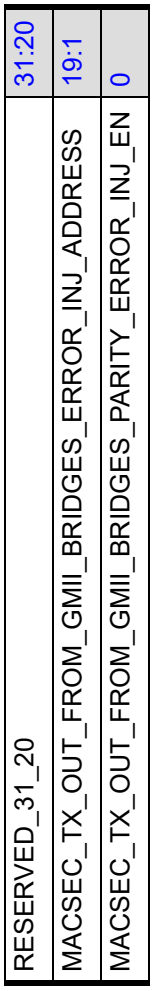


Table 6-1274 Fields for Register: MACSEC_TX_OUT_FROM_GMII_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:20	RESERVED_31_20	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1274 Fields for Register: MACSEC_TX_OUT_FROM_GMII_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
19:1	MACSEC_TX_OUT_FROM_GMII_BRIDGES_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the MII to RMII bridge output_counter signal Bit 1 - Injects an error on the MII to RMII bridge hold_last_di_bit signal Bit 2 - Injects an error on the MII to RMII bridge mii_fifo_empty_d1 signal Bit 3 - Injects an error on the MII to RMII bridge rmii_egress_out_reg signal Bit 4 - Injects an error on the MII to RMII bridge egress_mii_to_rmii_fifo signal Bit 5 - Injects an error on the GMII to MII bridge gmii_fifo_pop_empty_d1 signal Bit 6 - Injects an error on the GMII to MII bridge nibble_counter signal Bit 7 - Injects an error on the GMII to MII bridge mii_out_reg signal Bit 8 - Injects an error on the GMII to MII bridge egress_gmii_to_mii_fifo signal Bit 9 - Injects an error on the RGMII to GMII bridge rgmii_fifo_out_reg signal Bit 10 - Injects an error on the RGMII to GMII bridge egress_gmii_fifo signal Bit 11 - Injects an error on the RGMII to GMII bridge gmii_clk_re_d1 signal Bit 12 - Injects an error on the RGMII to GMII bridge gmii_clk_ne_d1 signal Bit 13 - Injects an error on the RGMII to GMII bridge egress_lsb_rgmii_fifo signal Bit 14 - Injects an error on the RGMII to GMII bridge not_empty signal Bit 15 - Injects an error on the RGMII to GMII bridge lpi_pipe[LPI_PIPELINE_DEPTH-1] signal Bit 16 - Injects an error on the RGMII to GMII bridge rgmii_lsb_fifo_out_reg signal Bit 17 - Injects an error on the RGMII to GMII bridge rgmii_msb_fifo_out_reg signal Bit 18 - Injects an error on the RGMII to GMII bridge egress_msb_rgmii_fifo signal **Values:** ■ 0x1 (MACSEC_TX_OUT_FROM_GMII_BRIDGES_ERROR_INJ_ADDRESS): TX GMII egress bridges Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1274 Fields for Register: MACSEC_TX_OUT_FROM_GMII_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_TX_OUT_FROM_GMII_BRIDGES_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX GMII egress bridges signals. **Values:** ■ 0x1 (MACSEC_TX_OUT_FROM_GMII_BRIDGES_PARITY_ERROR_INJ_ENABLE): Enables error injection on the TX GMII egress. ■ 0x0 (MACSEC_TX_OUT_FROM_GMII_BRIDGES_PARITY_ERROR_INJ_DISABLE): Disables error injection on the TX GMII egress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1268 MACSEC_RX_OUT_FROM_GMII_PARITY_ERROR_INJECTION

- **Name:** MACsec RX GMII to RMII/MII/RGMII egress bridges Error Injection Register.
- **Description:** MACsec RX GMII to RMII/MII/RGMII egress bridges Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x203c
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

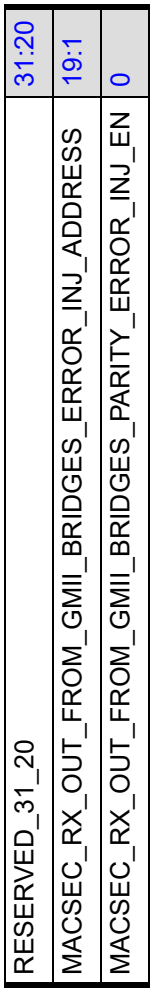


Table 6-1275 Fields for Register: MACSEC_RX_OUT_FROM_GMII_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:20	RESERVED_31_20	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1275 Fields for Register: MACSEC_RX_OUT_FROM_GMII_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
19:1	MACSEC_RX_OUT_FROM_GMII_BRIDGES_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the MII to RMII bridge output_counter signal Bit 1 - Injects an error on the MII to RMII bridge hold_last_di_bit signal Bit 2 - Injects an error on the MII to RMII bridge mii_fifo_empty_d1 signal Bit 3 - Injects an error on the MII to RMII bridge rmii_egress_out_reg signal Bit 4 - Injects an error on the MII to RMII bridge egress_mii_to_rmii_fifo signal Bit 5 - Injects an error on the GMII to MII bridge gmii_fifo_pop_empty_d1 signal Bit 6 - Injects an error on the GMII to MII bridge nibble_counter signal Bit 7 - Injects an error on the GMII to MII bridge mii_out_reg signal Bit 8 - Injects an error on the GMII to MII bridge egress_gmii_to_mii_fifo signal Bit 9 - Injects an error on the RGMII to GMII bridge rgmii_fifo_out_reg signal Bit 10 - Injects an error on the RGMII to GMII bridge egress_gmii_fifo signal Bit 11 - Injects an error on the RGMII to GMII bridge gmii_clk_re_d1 signal Bit 12 - Injects an error on the RGMII to GMII bridge gmii_clk_ne_d1 signal Bit 13 - Injects an error on the RGMII to GMII bridge egress_lsb_rgmii_fifo signal Bit 14 - Injects an error on the RGMII to GMII bridge not_empty signal Bit 15 - Injects an error on the RGMII to GMII bridge lpi_pipe[LPI_PIPELINE_DEPTH-1] signal Bit 16 - Injects an error on the RGMII to GMII bridge rgmii_lsb_fifo_out_reg signal Bit 17 - Injects an error on the RGMII to GMII bridge rgmii_msb_fifo_out_reg signal Bit 18 - Injects an error on the RGMII to GMII bridge egress_msb_rgmii_fifo signal **Values:** ■ 0x1 (MACSEC_RX_OUT_FROM_GMII_BRIDGES_ERROR_INJ_ADDRESS): RX GMII egress bridges Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1275 Fields for Register: MACSEC_RX_OUT_FROM_GMII_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_RX_OUT_FROM_GMII_BRIDGES_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the RX GMII egress bridges signals. **Values:** ■ 0x1 (MACSEC_RX_OUT_FROM_GMII_BRIDGES_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX GMII egress bridges. ■ 0x0 (MACSEC_RX_OUT_FROM_GMII_BRIDGES_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX GMII egress bridges. **Value After Reset:** 0x0 **Exists:** Always

6.1.1269 MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec TX GMII ingress Error Injection Register.
- **Description:** MACsec TX GMII ingress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2040
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

31:18	17:1	0
RESERVED_31_18	MACSEC_TX_GMII_INGRESS_ERROR_INJ_ADDRESS	MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJ_EN

Table 6-1276 Fields for Register: MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:18	RESERVED_31_18	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1276 Fields for Register: MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
17:1	MACSEC_TX_GMII_INGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the TX GMII ingress input_data_parity[0] signal Bit 1 - Injects an error on the TX GMII ingress input_data_parity[1] signal Bit 2 - Injects an error on the TX GMII ingress input_data_parity[2] signal Bit 3 - Injects an error on the TX GMII ingress input_data_parity[3] signal Bit 4 - Injects an error on the TX GMII ingress input_data_parity[4] signal Bit 5 - Injects an error on the TX GMII ingress input_data_parity[5] signal Bit 6 - Injects an error on the TX GMII ingress is_smd_c_reg signal Bit 7 - Injects an error on the TX GMII ingress cfg_min_ipg_reg signal Bit 8 - Injects an error on the TX GMII ingress ipg_counter_reg signal Bit 9 - Injects an error on the TX GMII ingress preemption_info_reg signal Bit 10 - Injects an error on the TX GMII ingress preamble_info_reg signal Bit 11 - Injects an error on the TX GMII ingress crc_match_d1 signal Bit 12 - Injects an error on the TX GMII ingress hold_crc_match_reg signal Bit 13 - Injects an error on the TX GMII ingress runt_info_reg signal Bit 14 - Injects an error on the TX GMII ingress pch_info_reg signal Bit 15 - Injects an error on the TX GMII ingress frame_info_reg signal Bit 16 - Injects an error on the TX GMII ingress crc_init_reg signal Bit 30-17 - [TBD] **Values:** ■ 0x1 (MACSEC_TX_GMII_INGRESS_ERROR_INJ_ADDRESS): TX GMII ingress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

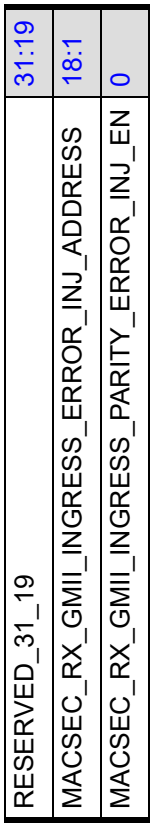
Table 6-1276 Fields for Register: MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX GMII ingress bridges signals. **Values:** ■ 0x1 (MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the TX GMII ingress. ■ 0x0 (MACSEC_TX_GMII_INGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the TX GMII ingress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1270 MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec RX GMII ingress Error Injection Register.
- **Description:** MACsec RX GMII ingress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2044
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1



31:19	18:1	0
RESERVED_31_19	MACSEC_RX_GMII_INGRESS_ERROR_INJ_ADDRESS	MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJ_EN

Table 6-1277 Fields for Register: MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:19	RESERVED_31_19	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1277 Fields for Register: MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
18:1	MACSEC_RX_GMII_INGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RX GMII ingress input_data_parity[0] signal Bit 1 - Injects an error on the RX GMII ingress input_data_parity[1] signal Bit 2 - Injects an error on the RX GMII ingress input_data_parity[2] signal Bit 3 - Injects an error on the RX GMII ingress input_data_parity[3] signal Bit 4 - Injects an error on the RX GMII ingress input_data_parity[4] signal Bit 5 - Injects an error on the RX GMII ingress input_data_parity[5] signal Bit 6 - Injects an error on the RX GMII ingress is_smd_c_reg signal Bit 7 - Injects an error on the RX GMII ingress cfg_min_ipg_reg signal Bit 8 - Injects an error on the RX GMII ingress ipg_counter_reg signal Bit 9 - Injects an error on the RX GMII ingress preemption_info_reg signal Bit 10 - Injects an error on the RX GMII ingress preamble_info_reg signal Bit 11 - Injects an error on the RX GMII ingress crc_match_d1 signal Bit 12 - Injects an error on the RX GMII ingress hold_crc_match_reg signal Bit 13 - Injects an error on the RX GMII ingress runt_fifo_info_reg signal Bit 14 - Injects an error on the RX GMII ingress runt_info_reg signal Bit 15 - Injects an error on the RX GMII ingress pch_info_reg signal Bit 16 - Injects an error on the RX GMII ingress frame_info_reg signal Bit 17 - Injects an error on the RX GMII ingress crc_init_reg signal Bit 30-18 - [TBD] **Values:** ■ 0x1 (MACSEC_RX_GMII_INGRESS_ERROR_INJ_ADDRESS): RX GMII ingress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1277 Fields for Register: MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX GMII ingress bridges signals. **Values:** ■ 0x1 (MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX GMII ingress. ■ 0x0 (MACSEC_RX_GMII_INGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX GMII ingress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1271 MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec TX GMII egress Error Injection Register.
- **Description:** MACsec TX GMII egress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2048
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

31:12	11:1	0
RESERVED_31_12	MACSEC_TX_GMII_EGRESS_ERROR_INJ_ADDRESS	MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJ_EN

Table 6-1278 Fields for Register: MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:12	RESERVED_31_12	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
11:1	MACSEC_TX_GMII_EGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the TX GMII egress insert_preamble_pipe signal Bit 1 - Injects an error on the TX GMII egress output_pipeline_data[1] signal [TBD] Bit 2 - Injects an error on the TX GMII egress output_pipeline_data[2] signal [TBD] Bit 3 - Injects an error on the TX GMII egress output_pipeline_data[3] signal [TBD] Bit 4 - Injects an error on the TX GMII egress output_pipeline_data[4] signal [TBD] Bit 5 - Injects an error on the TX GMII egress output_pipeline_data[5] signal [TBD] Bit 6 - Injects an error on the TX GMII egress output_pipeline_data[6] signal [TBD] Bit 7 - Injects an error on the TX GMII egress output_pipeline_data[7] signal [TBD] Bit 8 - Injects an error on the TX GMII egress frame_info_reg signal Bit 9 - Injects an error on the TX GMII egress crc_reg signal Bit 10 - Injects an error on the TX GMII egress lat_count_reg signal Bit 30-11 - [TBD] **Values:** ■ 0x1 (MACSEC_TX_GMII_EGRESS_ERROR_INJ_ADDRESS): TX GMII egress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1278 Fields for Register: MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX GMII egress bridges signals. **Values:** ■ 0x1 (MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the TX GMII egress. ■ 0x0 (MACSEC_TX_GMII_EGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the TX GMII egress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1272 MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec RX GMII egress Error Injection Register.
- **Description:** MACsec RX GMII egress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x204c
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_GMII_IF==1

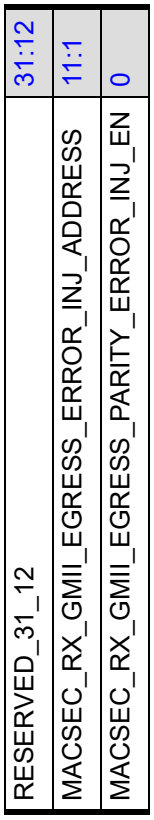


Table 6-1279 Fields for Register: MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:12	RESERVED_31_12	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
11:1	MACSEC_RX_GMII_EGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RX GMII egress insert_preamble_pipe signal Bit 1 - Injects an error on the RX GMII egress output_pipeline_data[1] signal [TBD] Bit 2 - Injects an error on the RX GMII egress output_pipeline_data[2] signal [TBD] Bit 3 - Injects an error on the RX GMII egress output_pipeline_data[3] signal [TBD] Bit 4 - Injects an error on the RX GMII egress output_pipeline_data[4] signal [TBD] Bit 5 - Injects an error on the RX GMII egress output_pipeline_data[5] signal [TBD] Bit 6 - Injects an error on the RX GMII egress output_pipeline_data[6] signal [TBD] Bit 7 - Injects an error on the RX GMII egress output_pipeline_data[7] signal [TBD] Bit 8 - Injects an error on the RX GMII egress frame_info_reg signal Bit 9 - Injects an error on the RX GMII egress crc_reg signal Bit 10 - Injects an error on the RX GMII egress lat_count_reg signal Bit 30-11 - [TBD] **Values:** ■ 0x1 (MACSEC_RX_GMII_EGRESS_ERROR_INJ_ADDRESS): RX GMII egress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1279 Fields for Register: MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the RX GMII egress bridges signals. **Values:** ■ 0x1 (MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX GMII egress. ■ 0x0 (MACSEC_RX_GMII_EGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX GMII egress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1273 MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec TX XGMII ingress Error Injection Register.
- **Description:** MACsec TX XGMII ingress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2050
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_XLGMII_IF==1

31:13	12:1	0
RESERVED_31_13	MACSEC_TX_XGMII_INGRESS_ERROR_INJ_ADDRESS	MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJ_EN

Table 6-1280 Fields for Register: MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:13	RESERVED_31_13	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
12:1	MACSEC_TX_XGMII_INGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the TX XGMII ingress xgmii_data_and_be_reg signal Bit 1 - Injects an error on the TX XGMII ingress frame_decoder_info_reg signal Bit 2 - Injects an error on the TX XGMII ingress frame_data_d1 signal Bit 3 - Injects an error on the TX XGMII ingress preemption_info_reg signal Bit 4 - Injects an error on the TX XGMII ingress runt_info_reg signal Bit 5 - Injects an error on the TX XGMII ingress crc_match_d1 signal Bit 6 - Injects an error on the TX XGMII ingress hold_crc_match_reg signal Bit 7 - Injects an error on the TX XGMII ingress cfg_min_ipg_reg signal Bit 8 - Injects an error on the TX XGMII ingress ipg_counter_reg signal Bit 9 - Injects an error on the TX XGMII ingress pch_info_reg signal Bit 10 - Injects an error on the TX XGMII ingress frame_info_reg signal Bit 11 - Injects an error on the TX XGMII ingress crc_init_reg signal Bit 30-12 - [TBD] **Values:** ■ 0x1 (MACSEC_TX_XGMII_INGRESS_ERROR_INJ_ADDRESS): TX XGMII ingress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1280 Fields for Register: MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX XGMII ingress bridges signals. **Values:** ■ 0x1 (MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the TX XGMII ingress. ■ 0x0 (MACSEC_TX_XGMII_INGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the TX XGMII ingress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1274 MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec RX XGMII ingress Error Injection Register.
- **Description:** MACsec RX XGMII ingress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2054
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_XLGMII_IF==1

31:14	13:1	0
RESERVED_31_14	MACSEC_RX_XGMII_INGRESS_ERROR_INJ_ADDRESS	MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJ_EN

Table 6-1281 Fields for Register: MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:14	RESERVED_31_14	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
13:1	MACSEC_RX_XGMII_INGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RX XGMII ingress xgmii_data_and_be_reg signal Bit 1 - Injects an error on the RX XGMII ingress frame_decoder_info_reg signal Bit 2 - Injects an error on the RX XGMII ingress frame_data_d1 signal Bit 3 - Injects an error on the RX XGMII ingress preemption_info_reg signal Bit 4 - Injects an error on the RX XGMII ingress runt_info_reg signal Bit 5 - Injects an error on the RX XGMII ingress crc_match_d1 signal Bit 6 - Injects an error on the RX XGMII ingress hold_crc_match_reg signal Bit 7 - Injects an error on the RX XGMII ingress cfg_min_ipg_reg signal Bit 8 - Injects an error on the RX XGMII ingress ipg_counter_reg signal Bit 9 - Injects an error on the RX XGMII ingress runt_fifo_info_reg signal Bit 10 - Injects an error on the TX XGMII ingress pch_info_reg signal Bit 11 - Injects an error on the TX XGMII ingress frame_info_reg signal Bit 12 - Injects an error on the TX XGMII ingress crc_init_reg signal Bit 30-13 - [TBD] **Values:** ■ 0x1 (MACSEC_RX_XGMII_INGRESS_ERROR_INJ_ADDRESS): RX XGMII ingress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1281 Fields for Register: MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
0	MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the RX XGMII ingress bridges signals. **Values:** ■ 0x1 (MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX XGMII ingress. ■ 0x0 (MACSEC_RX_XGMII_INGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX XGMII ingress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1275 MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec TX XGMII egress Error Injection Register.
- **Description:** MACsec TX XGMII egress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2058
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_XLGMII_IF==1

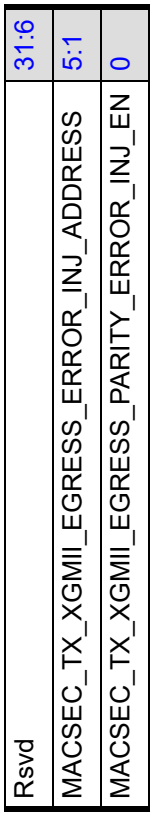


Table 6-1282 Fields for Register: MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:6			**Reserved Field:** Yes

Table 6-1282 Fields for Register: MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
5:1	MACSEC_TX_XGMII_EGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the TX XGMII egress output_data_pipe signal Bit 1 - Injects an error on the TX XGMII egress frame_decoder_info_reg signal Bit 2 - Injects an error on the TX XGMII egress crc_init_reg signal Bit 3 - Injects an error on the TX XGMII egress crc_eop_reg signal Bit 4 - Injects an error on the TX XGMII egress crc_info_reg signal Bit 30-5 - [TBD] **Values:** ■ 0x1 (MACSEC_TX_XGMII_EGRESS_ERROR_INJ_ADDRESS): TX XGMII egress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the TX XGMII egress bridges signals. **Values:** ■ 0x1 (MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the TX XGMII egress. ■ 0x0 (MACSEC_TX_XGMII_EGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the TX XGMII egress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1276 MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJECTION

- **Name:** MACsec RX XGMII egress Error Injection Register.
- **Description:** MACsec RX XGMII egress Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x205c
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && DWC_MACSEC_XLGMII_IF==1

31:6	5:1	0
RESERVED_31_6	MACSEC_RX_XGMII_EGRESS_ERROR_INJ_ADDRESS	MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJ_EN

Table 6-1283 Fields for Register: MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:6	RESERVED_31_6	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
5:1	MACSEC_RX_XGMII_EGRESS_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the error: Bit 0 - Injects an error on the RX XGMII egress output_data_pipe signal Bit 1 - Injects an error on the RX XGMII egress frame_decoder_info_reg signal Bit 2 - Injects an error on the RX XGMII egress crc_init_reg signal Bit 3 - Injects an error on the RX XGMII egress crc_eop_reg signal Bit 4 - Injects an error on the RX XGMII egress crc_info_reg signal Bit 30-5 - [TBD] **Values:** ■ 0x1 (MACSEC_RX_XGMII_EGRESS_ERROR_INJ_ADDRESS): RX XGMII egress Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the RX XGMII egress bridges signals. **Values:** ■ 0x1 (MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJ_ENABLE): Enables error injection on the RX XGMII egress. ■ 0x0 (MACSEC_RX_XGMII_EGRESS_PARITY_ERROR_INJ_DISABLE): Disables error injection on the RX XGMII egress. **Value After Reset:** 0x0 **Exists:** Always

6.1.1277 MACSEC_MAC_ERROR_INJ_EN

- **Name:** MACSEC_MAC_ERROR_INJ_EN
- **Description:** Enable register for the error injection on the MAC on the rx and tx paths of the AES module.
- **Size:** 32 bits
- **Offset:** 0x2060
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

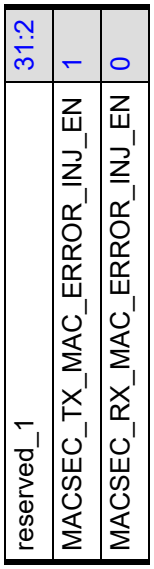


Table 6-1284 Fields for Register: MACSEC_MAC_ERROR_INJ_EN

Bits	Name	Memory Access	Description
31:2	reserved_1	R	Reserved bits **Values:** ■ 0x0 (Reserved): Reserved **Value After Reset:** 0x0 **Exists:** Always

Table 6-1284 Fields for Register: MACSEC_MAC_ERROR_INJ_EN (Continued)

Bits	Name	Memory Access	Description
1	MACSEC_TX_MAC_ERROR_INJ_EN	R/W	Enables the MAC error injection on the TX path. **Values:** ■ 0x1 (MACSEC_TX_MAC_ERROR_INJ_ENABLE): Enables the MAC error injection on the TX path. ■ 0x0 (MACSEC_TX_MAC_ERROR_INJ_DISABLE): Disables the MAC error injection on the TX path. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_RX_MAC_ERROR_INJ_EN	R/W	Enables the MAC error injection on the RX path. **Values:** ■ 0x1 (MACSEC_RX_MAC_ERROR_INJ_ENABLE): Enables the MAC error injection on the RX path. ■ 0x0 (MACSEC_RX_MAC_ERROR_INJ_DISABLE): Disables the MAC error injection on the RX path. **Value After Reset:** 0x0 **Exists:** Always

6.1.1278 MACSEC_CRC_ERROR_INJECTION

- **Name:** MACsec CRC Error Injection Register.
- **Description:** MACsec CRC Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2064
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1)

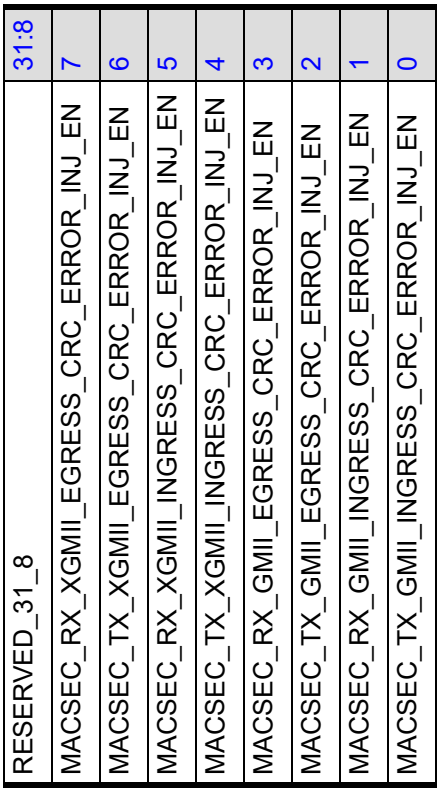


Table 6-1285 Fields for Register: MACSEC_CRC_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:8	RESERVED_31_8	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1285 Fields for Register: MACSEC_CRC_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
7	MACSEC_RX_XGMII_EGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the RX XGMII egress module. **Values:** ■ 0x1 (MACSEC_RX_XGMII_EGRESS_CRC_ERROR_INJ_ENABLE): Enables CRC error injection on the RX XGMII egress module. ■ 0x0 (MACSEC_RX_XGMII_EGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the RX XGMII egress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
6	MACSEC_TX_XGMII_EGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the TX XGMII egress module. **Values:** ■ 0x1 (MACSEC_TX_XGMII_EGRESS_CRC_ERROR_INJ_ENABLE): Enables CRC error injection on the TX XGMII egress module. ■ 0x0 (MACSEC_TX_XGMII_EGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the TX XGMII egress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
5	MACSEC_RX_XGMII_INGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the RX XGMII ingress module. **Values:** ■ 0x1 (MACSEC_RX_XGMII_INGRESS_CRC_ERROR_INJ_ENABLE): Enables error injection on the RX XGMII ingress module. ■ 0x0 (MACSEC_RX_XGMII_INGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the RX XGMII ingress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1

Table 6-1285 Fields for Register: MACSEC_CRC_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
4	MACSEC_TX_XGMII_INGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the TX XGMII ingress module. **Values:** ■ 0x1 (MACSEC_TX_XGMII_INGRESS_CRC_ERROR_INJ_ENABLE): Enables CRC error injection on the TX XGMII ingress module. ■ 0x0 (MACSEC_TX_XGMII_INGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the TX XGMII ingress module. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_DATA_IF==1))==1
3	MACSEC_RX_GMII_EGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the RX GMII egress module. **Values:** ■ 0x1 (MACSEC_RX_GMII_EGRESS_CRC_ERROR_INJ_ENABLE): Enables CRC error injection on the RX GMII egress module. ■ 0x0 (MACSEC_RX_GMII_EGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the RX GMII egress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
2	MACSEC_TX_GMII_EGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the TX GMII egress module. **Values:** ■ 0x1 (MACSEC_TX_GMII_EGRESS_CRC_ERROR_INJ_ENABLE): Enables CRC error injection on the TX GMII egress module. ■ 0x0 (MACSEC_TX_GMII_EGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the TX GMII egress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1

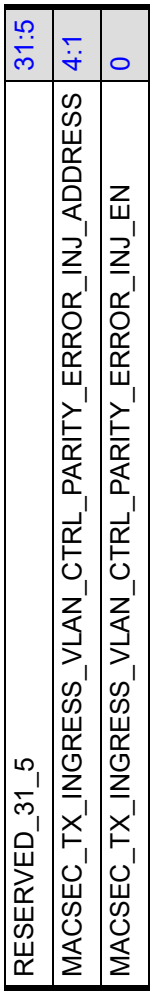
Table 6-1285 Fields for Register: MACSEC_CRC_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
1	MACSEC_RX_GMII_INGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the TX GMII ingress module. **Values:** ■ 0x1 (MACSEC_RX_GMII_INGRESS_CRC_ERROR_INJ_ENABLE): Enables error injection on the RX GMII ingress. ■ 0x0 (MACSEC_RX_GMII_INGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the RX GMII ingress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1
0	MACSEC_TX_GMII_INGRESS_CRC_ERROR_INJ_EN	R/W	Enables CRC error injection on the TX GMII ingress module. **Values:** ■ 0x1 (MACSEC_TX_GMII_INGRESS_CRC_ERROR_INJ_ENABLE): Enables CRC error injection on the TX GMII ingress module. ■ 0x0 (MACSEC_TX_GMII_INGRESS_CRC_ERROR_INJ_DISABLE): Disables CRC error injection on the TX GMII ingress module. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_DATA_IF == 0 || DWC_MACSEC_DUAL_IF == 1)==1

6.1.1279 MACSEC_TX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

- **Name:** MACsec TX ingress VLAN controller Error Injection Register.
- **Description:** MACsec TX ingress VLAN controller Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2068
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1)



31:5	4:1	0
RESERVED_31_5	MACSEC_TX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	MACSEC_TX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN

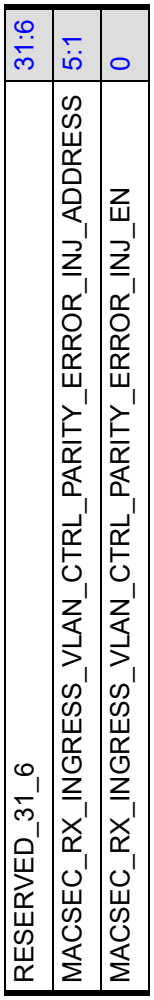
Table 6-1286 Fields for Register: MACSEC_TX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:5	RESERVED_31_5	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
4:1	MACSEC_TX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the parity error: Bit 0 - Injects a parity error on the TX ingress VLAN controller vlan_info signal Bit 1 - Injects a parity error on the TX ingress VLAN controller I_data signal Bit 2 - Injects a parity error on the TX ingress VLAN controller I_sai_idx_ret_code signal Bit 3 - Injects a parity error on the TX ingress VLAN controller I_vlan_tags_found signal Bit 30-4 - [TBD] **Values:** ■ 0x1 (MACSEC_TX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS): TX ingress VLAN controller parity Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_TX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN	R/W	Enables parity error injection on the TX ingress VLAN controller signals. **Values:** ■ 0x1 (MACSEC_TX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_ENABLE): Enables parity error injection on the TX ingress VLAN controller. ■ 0x0 (MACSEC_TX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_DISABLE): Disables parity error injection on the TX ingress VLAN controller. **Value After Reset:** 0x0 **Exists:** Always

6.1.1280 MACSEC_RX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

- **Name:** MACsec RX ingress VLAN controller Error Injection Register.
- **Description:** MACsec RX ingress VLAN controller Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x206c
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1)



31:6	5:1	0
RESERVED_31_6	MACSEC_RX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	MACSEC_RX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN

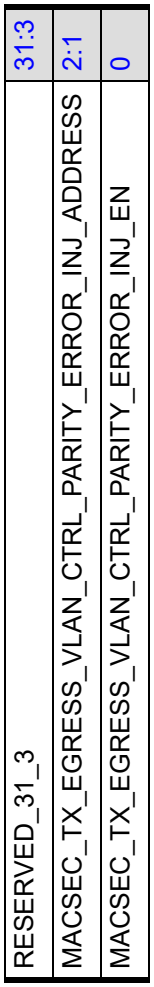
Table 6-1287 Fields for Register: MACSEC_RX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:6	RESERVED_31_6	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
5:1	MACSEC_RX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the parity error: Bit 0 - Injects a parity error on the RX ingress VLAN controller vlan_info signal Bit 1 - Injects a parity error on the RX ingress VLAN controller in_data signal Bit 2 - Injects a parity error on the RX ingress VLAN controller in_sai_idx_ret_code signal Bit 3 - Injects a parity error on the RX ingress VLAN controller in_vlan_tags_found signal Bit 4 - Injects a parity error on the RX ingress VLAN controller last_word_sent signal Bit 30-5 - [TBD] **Values:** ■ 0x1 (MACSEC_RX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS): RX ingress VLAN controller parity Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_RX_INGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN	R/W	Enables parity error injection on the RX ingress VLAN controller signals. **Values:** ■ 0x1 (MACSEC_RX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_ENABLE): Enables parity error injection on the RX ingress VLAN controller. ■ 0x0 (MACSEC_RX_INGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_DISABLE): Disables parity error injection on the RX ingress VLAN controller. **Value After Reset:** 0x0 **Exists:** Always

6.1.1281 MACSEC_TX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

- **Name:** MACsec TX egress VLAN controller Error Injection Register.
- **Description:** MACsec TX egress VLAN controller Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2070
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1)



31:3	2:1	0
RESERVED_31_3	MACSEC_TX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	MACSEC_TX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN

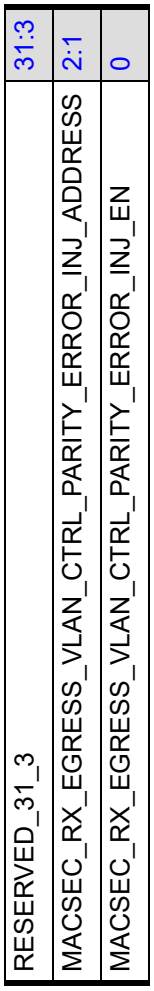
Table 6-1288 Fields for Register: MACSEC_TX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:3	RESERVED_31_3	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
2:1	MACSEC_TX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the parity error: Bit 0 - Injects a parity error on the TX egress VLAN controller vlan_info signal Bit 1 - Injects a parity error on the TX egress VLAN controller vlans signal Bit 30-2 - [TBD] **Values:** ■ 0x1 (MACSEC_TX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS): TX egress VLAN controller parity Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_TX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN	R/W	Enables parity error injection on the TX egress VLAN controller signals. **Values:** ■ 0x1 (MACSEC_TX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_ENABLE): Enables parity error injection on the TX egress VLAN controller. ■ 0x0 (MACSEC_TX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_DISABLE): Disables parity error injection on the TX egress VLAN controller. **Value After Reset:** 0x0 **Exists:** Always

6.1.1282 MACSEC_RX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

- **Name:** MACsec RX egress VLAN controller Error Injection Register.
- **Description:** MACsec RX egress VLAN controller Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2074
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1)



31:3	2:1	0
RESERVED_31_3	MACSEC_RX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	MACSEC_RX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN

Table 6-1289 Fields for Register: MACSEC_RX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:3	RESERVED_31_3	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
2:1	MACSEC_RX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS	R/W	Indicates which corresponding checker to inject the parity error: Bit 0 - Injects a parity error on the RX egress VLAN controller vlan_info signal Bit 1 - Injects a parity error on the RX egress VLAN controller vlans signal Bit 30-2 - [TBD] **Values:** ■ 0x1 (MACSEC_RX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_ADDRESS): RX egress VLAN controller parity Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_RX_EGRESS_VLAN_CTRL_PARITY_ERROR_INJ_EN	R/W	Enables parity error injection on the RX egress VLAN controller signals. **Values:** ■ 0x1 (MACSEC_RX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_ENABLE): Enables parity error injection on the RX egress VLAN controller. ■ 0x0 (MACSEC_RX_EGRESS_VLAN_CONTROLLER_PARITY_ERROR_INJ_DISABLE): Disables parity error injection on the RX egress VLAN controller. **Value After Reset:** 0x0 **Exists:** Always

6.1.1283 MACSEC_RX_PNUP_PARITY_ERROR_INJECTION

- **Name:** MACSEC_RX_PNUP_PARITY_ERROR_INJECTION.
- **Description:** MACsec RX PNUP parity Error Injection Register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x2078
- **Exists:** DWC_MACSEC_AUTOMOTIVE==1

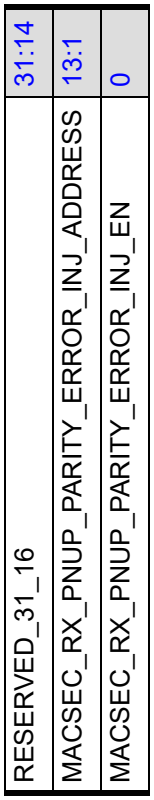


Table 6-1290 Fields for Register: MACSEC_RX_PNUP_PARITY_ERROR_INJECTION

Bits	Name	Memory Access	Description
31:14	RESERVED_31_16	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly

Table 6-1290 Fields for Register: MACSEC_RX_PNUP_PARITY_ERROR_INJECTION (Continued)

Bits	Name	Memory Access	Description
13:1	MACSEC_RX_PNUP_PARITY_ERROR_INJ_ADDRESS	R/W	Indicates which reg parity checker module to inject the error: Bit 0 - Injects an error on in_data_reg_valid_dpp_chk Bit 1 - Injects an error on in_data_reg_cmd_dpp_chk Bit 2 - Injects an error on in_data_reg_fatal_dpp_chk Bit 3 - Injects an error on in_data_reg_bypass_dpp_chk Bit 4 - Injects an error on in_ret_code_n_dpp_chk Bit 5 - Injects an error on sad_rd_data_dpp_chk Bit 6 - Injects an error on sad_done_dpp_chk Bit 7 - Injects an error on sad_rd_done_dpp_chk Bit 8 - Injects an error on arw_done_dpp_chk Bit 9 - Injects an error on arw_rd_data_dpp_chk Bit 10 - Injects an error on sa_idx_dpp_chk Bit 11 - Injects an error on sectag_pn_dpp_chk Bit 12 - Injects an error on new_pn_dpp_chk **Values:** ■ 0x1 (MACSEC_RX_PNUP_PARITY_ERROR_INJ_ADDRESS): RX PNUP Parity Error Injection Address. **Value After Reset:** 0x0 **Exists:** Always
0	MACSEC_RX_PNUP_PARITY_ERROR_INJ_EN	R/W	Enables error injection on the parity checker modules for registers located in the RX PNUP module. **Values:** ■ 0x1 (MACSEC_RX_PNUP_ERROR_INJ_ENABLE): Enables error injection on the parity checker modules for registers located in the RX PNUP module ■ 0x0 (MACSEC_RX_PNUP_ERROR_INJ_DISABLE): Disables error injection on the parity checker modules for registers located in the RX PNUP module **Value After Reset:** 0x0 **Exists:** Always

6.1.1284 MACSEC_SAD_INTERNAL_MEM_PARITY_ERR_INJ

- **Name:** MACSEC_SAD_INTERNAL_MEM_PARITY_ERROR_INJECTION.
- **Description:** MACsec SAD Flop based internal memories parity error injection register.

  Reset Domain: rst_clk_macsec
- **Size:** 32 bits
- **Offset:** 0x207c
- **Exists:** (DWC_MACSEC_AUTOMOTIVE==1) && (DWC_MACSEC_SAD_INTERNAL_FLOPS_IMPL==1)

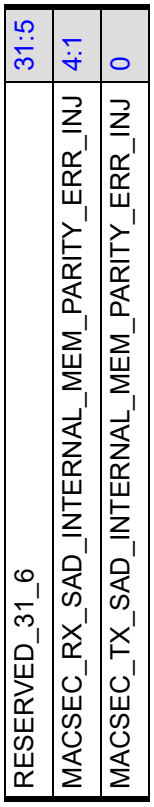


Table 6-1291 Fields for Register: MACSEC_SAD_INTERNAL_MEM_PARITY_ERR_INJ

Bits	Name	Memory Access	Description
31:5	RESERVED_31_6	R	Reserved and read as zero. **Values:** ■ 0x0 (RESERVED): Reserved and read as zero. **Value After Reset:** 0x0 **Exists:** Always **Testable:** readOnly
4:1	MACSEC_RX_SAD_INTERNAL_MEM_PARITY_ERR_INJ	R/W	Enables error injection on the registers located in the RX flop-based internal memory module: Bit 0 - Injects an error on mem_a_parity_chk Bit 1 - Injects an error on mem_b_parity_chk Bit 2 - Injects an error on addr_a_parity_chk Bit 3 - Injects an error on addr_b_parity_chk **Values:** ■ 0x1 (MACSEC_RX_SAD_INTERNAL_MEM_PARITY_ERR_INJ): RX SAD Internal memory Parity Error Injection. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_RX_SAD_MEM_TYPE==0)==1
0	MACSEC_TX_SAD_INTERNAL_MEM_PARITY_ERR_INJ	R/W	Enables error injection on the registers located in the TX flop-based internal memory module: Bit 0 - Injects an error on the memory parity checker **Values:** ■ 0x1 (MACSEC_TX_SAD_INTERNAL_MEM_PARITY_ERR_INJ): TX SAD Internal memory Parity Error Injection. **Value After Reset:** 0x0 **Exists:** (DWC_MACSEC_TX_SAD_MEM_TYPE==0)==1

6.2 DWC_macsec_uaes_gcmp_apb/macsec_id_aes_host_cfg Registers

AES Host . Follow the link for the register to see a detailed description of the register.

Table 6-1292 Registers for Address Block: DWC_macsec_uaes_gcmp_apb/macsec_id_aes_host_cfg

Register	Offset	Description
IRQ_EN on page 3851	0x8	Interrupt Request Enable Register
IRQ_STAT on page 3853	0xc	Interrupt Request Status Register
CONFIG on page 3854	0x10	Config Register
CTRL on page 3856	0x14	Control register
STAT on page 3858	0x18	Status Register
KEY_0 on page 3859	0x20	Key 0 Register.
KEY_1 on page 3860	0x24	Key 1 Register.
KEY_2 on page 3861	0x28	Key 2 Register.
KEY_3 on page 3862	0x2c	Key 3 Register.
KEY_4 on page 3863	0x30	Key 4 Register.
KEY_5 on page 3864	0x34	Key 5 Register.
KEY_6 on page 3865	0x38	Key 6 Register.
KEY_7 on page 3866	0x3c	Key 7 Register.
MISC_CONFIG on page 3867	0x40	Miscellaneous Config Register
FIPS_SELF_TEST_CTL on page 3868	0x50	FIPs Self Test Control Register
FIPS_SELF_TEST_ATT on page 3869	0x54	FIPs Self Test Attribute Register
FIPS_SELF_TEST_STAT on page 3871	0x58	FIPs Self Test Status Register
FIPS_SELF_TEST_MAC_MASK on page 3873	0x5c	FIPs Self Test MAC MASK Register
FIPS_SELF_TEST_VECT_CMD_CTX on page 3874	0x60	FIPs Self Test CTX NUM Register
FIPS_SELF_TEST_AAD_SIZE on page 3875	0x64	FIPs Self Test AAD SIZE Register
FIPS_SELF_TEST_MSG_SIZE on page 3876	0x68	FIPs Self Test MSG SIZE Register
FIPS_SELF_TEST_MAC0 on page 3877	0x70	FIPs Self Test MAC0 Register
FIPS_SELF_TEST_MAC1 on page 3878	0x74	FIPs Self Test MAC1 Register

Table 6-1292 Registers for Address Block: DWC_macsec_uaes_gcmp_apb/macsec_id_aes_host_cfg (Continued)

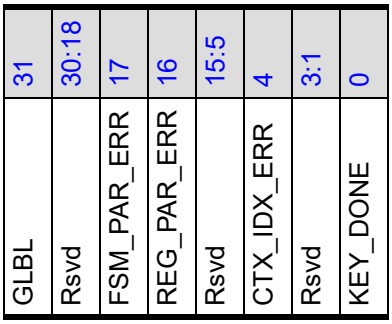
Register	Offset	Description
FIPS_SELF_TEST_MAC2 on page 3879	0x78	FIPs Self Test MAC2 Register
FIPS_SELF_TEST_MAC3 on page 3880	0x7c	FIPs Self Test MAC3 Register
FIPS_SELF_TEST_IV0 on page 3881	0x90	FIPs Self Test IV0 Register
FIPS_SELF_TEST_IV1 on page 3882	0x94	FIPs Self Test IV1 Register
FIPS_SELF_TEST_IV2 on page 3883	0x98	FIPs Self Test IV2 Register
FIPS_SELF_TEST_IV3 on page 3884	0x9c	FIPs Self Test IV3 Register
FIPS_SELF_TEST_IDATA0 on page 3885	0xa0	FIPs Self Test IDATA0 Register
FIPS_SELF_TEST_IDATA1 on page 3886	0xa4	FIPs Self Test IDATA1 Register
FIPS_SELF_TEST_IDATA2 on page 3887	0xa8	FIPs Self Test IDATA2 Register
FIPS_SELF_TEST_IDATA3 on page 3888	0xac	FIPs Self Test IDATA3 Register
FIPS_SELF_TEST_IDATA4 on page 3889	0xb0	FIPs Self Test IDATA4 Register
FIPS_SELF_TEST_IDATA5 on page 3890	0xb4	FIPs Self Test IDATA5 Register
FIPS_SELF_TEST_IDATA6 on page 3891	0xb8	FIPs Self Test IDATA6 Register
FIPS_SELF_TEST_IDATA7 on page 3892	0xbc	FIPs Self Test IDATA7 Register
FIPS_SELF_TEST_IDATA8 on page 3893	0xc0	FIPs Self Test IDATA8 Register
FIPS_SELF_TEST_IDATA9 on page 3894	0xc4	FIPs Self Test IDATA9 Register
FIPS_SELF_TEST_IDATA10 on page 3895	0xc8	FIPs Self Test IDATA10 Register
FIPS_SELF_TEST_IDATA11 on page 3896	0xcc	FIPs Self Test IDATA11 Register
FIPS_SELF_TEST_IDATA12 on page 3897	0xd0	FIPs Self Test IDATA12 Register
FIPS_SELF_TEST_IDATA13 on page 3898	0xd4	FIPs Self Test IDATA13 Register
FIPS_SELF_TEST_IDATA14 on page 3899	0xd8	FIPs Self Test IDATA14 Register
FIPS_SELF_TEST_IDATA15 on page 3900	0xdc	FIPs Self Test IDATA15 Register
FIPS_SELF_TEST_ODATA0 on page 3901	0x1a0	FIPs Self Test ODATA0 Register
FIPS_SELF_TEST_ODATA1 on page 3902	0x1a4	FIPs Self Test ODATA1 Register

Table 6-1292 Registers for Address Block: DWC_macsec_uaes_gcmp_apb/macsec_id_aes_host_cfg (Continued)

Register	Offset	Description
FIPS_SELF_TEST_ODATA2 on page 3903	0x1a8	FIPs Self Test ODATA2 Register
FIPS_SELF_TEST_ODATA3 on page 3904	0x1ac	FIPs Self Test ODATA3 Register
FIPS_SELF_TEST_ODATA4 on page 3905	0x1b0	FIPs Self Test ODATA4 Register
FIPS_SELF_TEST_ODATA5 on page 3906	0x1b4	FIPs Self Test ODATA5 Register
FIPS_SELF_TEST_ODATA6 on page 3907	0x1b8	FIPs Self Test ODATA6 Register
FIPS_SELF_TEST_ODATA7 on page 3908	0x1bc	FIPs Self Test ODATA7 Register
FIPS_SELF_TEST_ODATA8 on page 3909	0x1c0	FIPs Self Test ODATA8 Register
FIPS_SELF_TEST_ODATA9 on page 3910	0x1c4	FIPs Self Test ODATA9 Register
FIPS_SELF_TEST_ODATA10 on page 3911	0x1c8	FIPs Self Test ODATA10 Register
FIPS_SELF_TEST_ODATA11 on page 3912	0x1cc	FIPs Self Test ODATA11 Register
FIPS_SELF_TEST_ODATA12 on page 3913	0x1d0	FIPs Self Test ODATA12 Register
FIPS_SELF_TEST_ODATA13 on page 3914	0x1d4	FIPs Self Test ODATA13 Register
FIPS_SELF_TEST_ODATA14 on page 3915	0x1d8	FIPs Self Test ODATA14 Register
FIPS_SELF_TEST_ODATA15 on page 3916	0x1dc	FIPs Self Test ODATA15 Register
FIPS_SELF_TEST_VECT_CTL on page 3917	0x2a0	FIPs Self Test Vector Control Register
BIST_VECT_MODE on page 3918	0x2a4	BIST Vector Mode Register
BIST_VECT_ERR_INJ on page 3920	0x2a8	BIST Vector Error Injection Register
BIST_VECT_CTL on page 3921	0x2ac	BIST Test Vector Control Register

6.2.1 IRQ_EN

- **Name:** Interrupt Request Enable Register
- **Description:** Interrupt Request Enable Register.
- **Size:** 32 bits
- **Offset:** 0x8
- **Exists:** Always



31	30:18	17	16	15:5	4	3:1	0
GLBL	Rsvd	FSM_PAR_ERR	REG_PAR_ERR	Rsvd	CTX_IDX_ERR	Rsvd	KEY_DONE

Table 6-1293 Fields for Register: IRQ_EN

Bits	Name	Memory Access	Description
31	GLBL	R/W	Global enable control for interrupt sources **Value After Reset:** 0x0 **Exists:** Always
30:18			**Reserved Field:** Yes
17	FSM_PAR_ERR	R/W	Enables interrupt indicating that an FSM parity error has been detected **Value After Reset:** 0x0 **Exists:** Always
16	REG_PAR_ERR	R/W	Enables interrupt indicating that a register parity error has been detected **Value After Reset:** 0x0 **Exists:** Always
15:5			**Reserved Field:** Yes
4	CTX_IDX_ERR	R/W	Status of interrupt indicating that a context index error has been detected **Value After Reset:** 0x0 **Exists:** Always
3:1			**Reserved Field:** Yes

Table 6-1293 Fields for Register: IRQ_EN (Continued)

Bits	Name	Memory Access	Description
0	KEY_DONE	R/W	Enables interrupt indicating that a key load has completed **Value After Reset:** 0x0 **Exists:** ((1)==1)

6.2.2 IRQ_STAT

- **Name:** Interrupt Request Status Register
- **Description:** Interrupt Request Status Register.
- **Size:** 32 bits
- **Offset:** 0xc
- **Exists:** Always

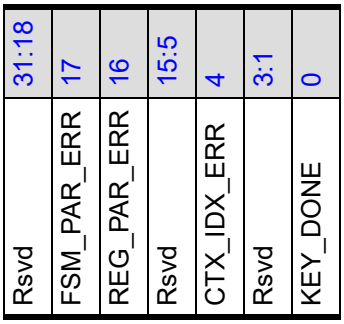


Table 6-1294 Fields for Register: IRQ_STAT

Bits	Name	Memory Access	Description
31:18			**Reserved Field:** Yes
17	FSM_PAR_ERR	R/W1C	Status of interrupt indicating that an FSM parity error has been detected **Value After Reset:** 0x0 **Exists:** Always
16	REG_PAR_ERR	R/W1C	Status of interrupt indicating that a register parity error has been detected **Value After Reset:** 0x0 **Exists:** Always
15:5			**Reserved Field:** Yes
4	CTX_IDX_ERR	R/W1C	Status of interrupt indicating that a context index error has been detected **Value After Reset:** 0x0 **Exists:** Always
3:1			**Reserved Field:** Yes
0	KEY_DONE	R/W1C	Status of interrupt indicating that a key load has completed **Value After Reset:** 0x0 **Exists:** ((1)==1)

6.2.3 CONFIG

- **Name:** Config Register
- **Description:** Configuration Register.
- **Size:** 32 bits
- **Offset:** 0x10
- **Exists:** Always

31:15	14	13	12	11	10	9	8	7	6	5	4	3:0
NUM_CTX	AES_EN	SM4_EN	CTR_STREAM_EN	ILEAVE_EN	DEC_EN	ENC_EN	KEY256_EN	KEY128_EN	CTR_EN	GCM_EN	SK_EN	DP_WIDTH

Table 6-1295 Fields for Register: CONFIG

Bits	Name	Memory Access	Description
31:15	NUM_CTX	R	Number of key contexts **Value After Reset:** DWC_MACSEC_AES_NCTX **Exists:** Always
14	AES_EN	R	AES cypher support **Value After Reset:** 0x1 **Exists:** Always
13	SM4_EN	R	SM4 cypher support **Value After Reset:** 0x0 **Exists:** Always
12	CTR_STREAM_EN	R	Stream support **Value After Reset:** 0x0 **Exists:** Always
11	ILEAVE_EN	R	Packet Interleave support **Value After Reset:** 0x1 **Exists:** Always

Table 6-1295 Fields for Register: CONFIG (Continued)

Bits	Name	Memory Access	Description
10	DEC_EN	R	Decryption support **Value After Reset:** 0x1 **Exists:** Always
9	ENC_EN	R	Encryption support **Value After Reset:** 0x1 **Exists:** Always
8	KEY256_EN	R	256-bit key support **Value After Reset:** DWC_MACSEC_MAX_KEY_SIZE **Exists:** Always
7	KEY128_EN	R	128-bit key support **Value After Reset:** 0x1 **Exists:** Always
6	CTR_EN	R	CTR mode support **Value After Reset:** DWC_MACSEC_AES_FIPS_EN **Exists:** Always
5	GCM_EN	R	GCM mode support **Value After Reset:** 0x1 **Exists:** Always
4	SK_EN	R	Secure Key support **Value After Reset:** 0x0 **Exists:** Always
3:0	DP_WIDTH	R	Indicates the width of the internal AES datapath: - 0: 128-bit - 1: 256-bit - 2: 384-bit **Value After Reset:** {DWC_UAES_CFG_DP_WIDTH == 512 ? 3 : DWC_UAES_CFG_DP_WIDTH == 384 ? 2 : DWC_UAES_CFG_DP_WIDTH == 256 ? 1 : 0} **Exists:** Always

6.2.4 CTRL

- **Name:** Control register
- **Description:** Control Register.
- **Size:** 32 bits
- **Offset:** 0x14
- **Exists:** ((1)==1)

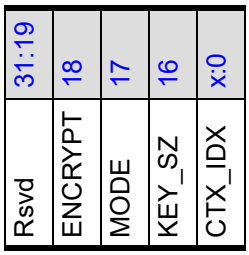


Table 6-1296 Fields for Register: CTRL

Bits	Name	Memory Access	Description
31:19			**Reserved Field:** Yes
18	ENCRYPT	R/W	Encrypt/Decrypt selector for the key **Value After Reset:** 0x0 **Exists:** Always
17	MODE	R/W	Mode for the key: - 0: GCM - 1: CTR **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AES_FIPS_EN
16	KEY_SZ	R/W	Size of the key: - 0:128-bit - 1: 256-bit **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_MAX_KEY_SIZE == 1) ? 1 :0)

Table 6-1296 Fields for Register: CTRL (Continued)

Bits	Name	Memory Access	Description
x:0	CTX_IDX	R/W	Key context index to load the key into **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** (DWC_MACSEC_AES_NCTX_WIDTH) - 1

6.2.5 STAT

- **Name:** Status Register
- **Description:** Status Register.
- **Size:** 32 bits
- **Offset:** 0x18
- **Exists:** Always

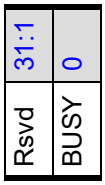


Table 6-1297 Fields for Register: STAT

Bits	Name	Memory Access	Description
31:1			**Reserved Field:** Yes
0	BUSY	R	Indicates that the core is busy installing a key. All other registers must not be written while this field indicates a 1. **Value After Reset:** 0x0 **Exists:** DWC_MACSEC_AES_KEY_IF_APB **Volatile:** true

6.2.6 KEY_0

- **Name:** Key 0 Register.
- **Description:** Key 0 Register.
- **Size:** 32 bits
- **Offset:** 0x20
- **Exists:** Always

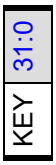


Table 6-1298 Fields for Register: KEY_0

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {00, 01, 02, 03} **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.7 KEY_1

- **Name:** Key 1 Register.
- **Description:** Key 1 Register.
- **Size:** 32 bits
- **Offset:** 0x24
- **Exists:** Always

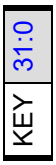


Table 6-1299 Fields for Register: KEY_1

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {04, 05, 06, 07} **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.8 KEY_2

- **Name:** Key 2 Register.
- **Description:** Key 2 Register.
- **Size:** 32 bits
- **Offset:** 0x28
- **Exists:** Always

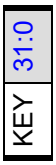


Table 6-1300 Fields for Register: KEY_2

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {08, 09, 10, 11} **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.9 KEY_3

- **Name:** Key 3 Register.
- **Description:** Key 3 Register.
- **Size:** 32 bits
- **Offset:** 0x2c
- **Exists:** Always

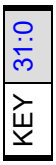


Table 6-1301 Fields for Register: KEY_3

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {12, 13, 14, 15} **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.10 KEY_4

- **Name:** Key 4 Register.
- **Description:** Key 4 Register.
- **Size:** 32 bits
- **Offset:** 0x30
- **Exists:** DWC_MACSEC_MAX_KEY_SIZE==1

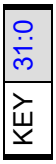


Table 6-1302 Fields for Register: KEY_4

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {16, 17, 18, 19} (256-bit key only) **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.11 KEY_5

- **Name:** Key 5 Register.
- **Description:** Key 5 Register.
- **Size:** 32 bits
- **Offset:** 0x34
- **Exists:** DWC_MACSEC_MAX_KEY_SIZE==1

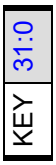


Table 6-1303 Fields for Register: KEY_5

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {20, 21, 22, 23} (256-bit key only) **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.12 KEY_6

- **Name:** Key 6 Register.
- **Description:** Key 6 Register.
- **Size:** 32 bits
- **Offset:** 0x38
- **Exists:** DWC_MACSEC_MAX_KEY_SIZE==1

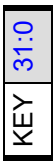


Table 6-1304 Fields for Register: KEY_6

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {24, 25, 26, 27} (256-bit key only) **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.13 KEY_7

- **Name:** Key 7 Register.
- **Description:** Key 7 Register.
- **Size:** 32 bits
- **Offset:** 0x3c
- **Exists:** DWC_MACSEC_MAX_KEY_SIZE==1

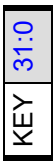


Table 6-1305 Fields for Register: KEY_7

Bits	Name	Memory Access	Description
31:0	KEY	W	Key bytes: {28, 29, 30, 31} (256-bit key only) **Values:** ■ 0x0 (KEY_REG) **Value After Reset:** 0x0 **Exists:** Always

6.2.14 MISC_CONFIG

- **Name:** Miscellaneous Config Register
- **Description:** Miscellaneous Config Register.
- **Size:** 32 bits
- **Offset:** 0x40
- **Exists:** Always

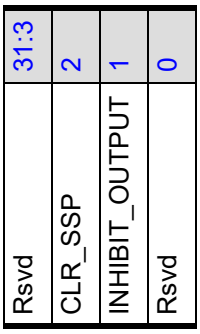


Table 6-1306 Fields for Register: MISC_CONFIG

Bits	Name	Memory Access	Description
31:3			**Reserved Field:** Yes
2	CLR_SSP	R/W	Clear the sensitive security parameters stored in the core. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
1	INHIBIT_OUTPUT	R/W	Inhibit the cmd/data output. **Value After Reset:** 0x0 **Exists:** Always
0			**Reserved Field:** Yes

6.2.15 FIPS_SELF_TEST_CTL

- **Name:** FIPs Self Test Control Register
- **Description:** FIPs Self Test Control Register.
- **Size:** 32 bits
- **Offset:** 0x50
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

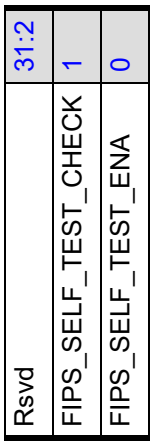
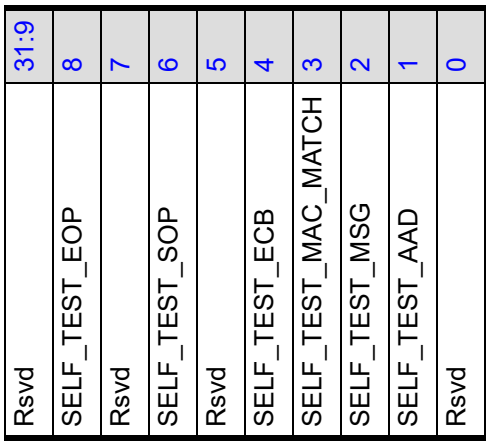


Table 6-1307 Fields for Register: FIPS_SELF_TEST_CTL

Bits	Name	Memory Access	Description
31:2			**Reserved Field:** Yes
1	FIPS_SELF_TEST_CHECK	R/W	Enables the FIPS to check output data. **Value After Reset:** 0x0 **Exists:** Always
0	FIPS_SELF_TEST_ENA	R/W	Enables the self test. The vector injection is not processed if this register is not set. **Value After Reset:** 0x0 **Exists:** Always

6.2.16 FIPS_SELF_TEST_ATT

- **Name:** FIPs Self Test Attribute Register
- **Description:** FIPs Self Test Attribute.
- **Size:** 32 bits
- **Offset:** 0x54
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1



31:9	8	7	6	5	4	3	2	1	0
Rsvd	SELF_TEST_EOP	Rsvd	SELF_TEST_SOP	Rsvd	SELF_TEST_ECB	SELF_TEST_MAC_MATCH	SELF_TEST_MSG	SELF_TEST_AAD	Rsvd

Table 6-1308 Fields for Register: FIPS_SELF_TEST_ATT

Bits	Name	Memory Access	Description
31:9			**Reserved Field:** Yes
8	SELF_TEST_EOP	R/W	When set, this data block is the last block of the data unit (vector).A single block vector will have both SELF_TEST_SOP and SELF_TEST_EOP set. **Value After Reset:** 0x0 **Exists:** Always
7			**Reserved Field:** Yes
6	SELF_TEST_SOP	R/W	When set, this data block is the first block of the data unit (vector). **Value After Reset:** 0x0 **Exists:** Always
5			**Reserved Field:** Yes
4	SELF_TEST_ECB	R/W	When set, it processes FIPS in GCM ECB Mode. **Value After Reset:** 0x0 **Exists:** Always

Table 6-1308 Fields for Register: FIPS_SELF_TEST_ATT (Continued)

Bits	Name	Memory Access	Description
3	SELF_TEST_MAC_MATCH	R/W	1: Contains valid mac match. **Value After Reset:** 0x0 **Exists:** Always
2	SELF_TEST_MSG	R/W	1: Contains valid message. **Value After Reset:** 0x0 **Exists:** Always
1	SELF_TEST_AAD	R/W	1: Contains ADD. **Value After Reset:** 0x0 **Exists:** Always
0			**Reserved Field:** Yes

6.2.17 FIPS_SELF_TEST_STAT

- **Name:** FIPs Self Test Status Register
- **Description:** FIPs Self Test Status.
- **Size:** 32 bits
- **Offset:** 0x58
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

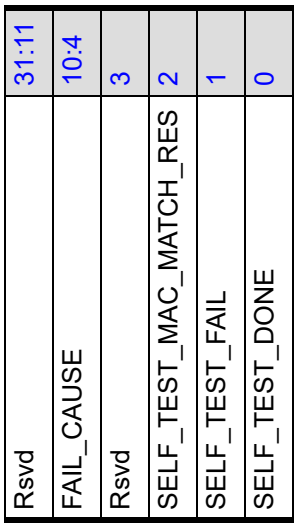


Table 6-1309 Fields for Register: FIPS_SELF_TEST_STAT

Bits	Name	Memory Access	Description
31:11			**Reserved Field:** Yes

Table 6-1309 Fields for Register: FIPS_SELF_TEST_STAT (Continued)

Bits	Name	Memory Access	Description
10:4	FAIL_CAUSE	R/W1C	Each bit of the failure cause has a meaning: ■ 10: ecb, 1=FAIL, 0=PASS ■ 8 : key size, 1: CKEY=256, 0=CKEY=128 ■ 7 : Direction, 1=ENC, 0=DEC ■ 6 : Bist AES GCM test, 1=FAIL, 0=PASS ■ 5 : Bist AES MAC test, 1=FAIL, 0=PASS ■ 4 : Bist AES CTR test, 1=FAIL, 0=PASS **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
3			**Reserved Field:** Yes
2	SELF_TEST_MAC_MATCH_RES	R/W1C	Status of the mac match test indicates that the test failed. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
1	SELF_TEST_FAIL	R/W1C	Status of the self test indicates that the test failed. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
0	SELF_TEST_DONE	R/W1C	Status of the self test indicates that the test is complete. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.18 FIPS_SELF_TEST_MAC_MASK

- **Name:** FIPs Self Test MAC MASK Register
- **Description:** FIPs Self Test MAC MASK Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x5c
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

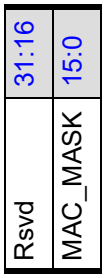


Table 6-1310 Fields for Register: FIPS_SELF_TEST_MAC_MASK

Bits	Name	Memory Access	Description
31:16			**Reserved Field:** Yes
15:0	MAC_MASK	R/W	In GCM decrypt mode, each bit set enables comparison of each of the I_cmd_mac bytes against the MAC value computed by the core. **Value After Reset:** 0x0 **Exists:** Always

6.2.19 FIPS_SELF_TEST_VECT_CMD_CTX

- **Name:** FIPs Self Test CTX NUM Register
- **Description:** FIPs Self Test CTX NUM Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x60
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

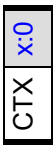


Table 6-1311 Fields for Register: FIPS_SELF_TEST_VECT_CMD_CTX

Bits	Name	Memory Access	Description
x:0	CTX	R	CTX number is needed for the working of the GCM and CTR mode bytes against the MAC value computed by the core. **Value After Reset:** DWC_MACSEC_AES_DEFAULT_FIPS_CTX_IDX **Exists:** Always **Range Variable[x]:** (DWC_MACSEC_AES_NCTX_WIDTH) - 1

6.2.20 FIPS_SELF_TEST_AAD_SIZE

- **Name:** FIPs Self Test AAD SIZE Register
- **Description:** FIPs Self Test AAD SIZE Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x64
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1312 Fields for Register: FIPS_SELF_TEST_AAD_SIZE

Bits	Name	Memory Access	Description
x:0	AAD_SIZE	R/W	Number of AAD bytes valid in the associated data value. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_AES_DP_LOG2_BYTE_WIDTH - 1

6.2.21 FIPS_SELF_TEST_MSG_SIZE

- **Name:** FIPs Self Test MSG SIZE Register
- **Description:** FIPs Self Test MSG SIZE Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x68
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1313 Fields for Register: FIPS_SELF_TEST_MSG_SIZE

Bits	Name	Memory Access	Description
x:0	MSG_SIZE	R/W	Number of MSG bytes valid in the associated data value. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** DWC_MACSEC_AES_DP_LOG2_BYTE_WIDTH - 1

6.2.22 FIPS_SELF_TEST_MAC0

- **Name:** FIPs Self Test MAC0 Register
- **Description:** FIPs Self Test MAC0 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x70
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

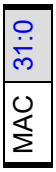


Table 6-1314 Fields for Register: FIPS_SELF_TEST_MAC0

Bits	Name	Memory Access	Description
31:0	MAC	R/W	MAC Value bytes: {00, 01, 02, 03} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.23 FIPS_SELF_TEST_MAC1

- **Name:** FIPs Self Test MAC1 Register
- **Description:** FIPs Self Test MAC1 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x74
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

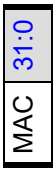


Table 6-1315 Fields for Register: FIPS_SELF_TEST_MAC1

Bits	Name	Memory Access	Description
31:0	MAC	R/W	MAC Value bytes: {04, 05, 06, 07} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.24 FIPS_SELF_TEST_MAC2

- **Name:** FIPs Self Test MAC2 Register
- **Description:** FIPs Self Test MAC2 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x78
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

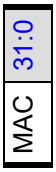


Table 6-1316 Fields for Register: FIPS_SELF_TEST_MAC2

Bits	Name	Memory Access	Description
31:0	MAC	R/W	MAC Value bytes: {08, 09, 10, 11} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.25 FIPS_SELF_TEST_MAC3

- **Name:** FIPs Self Test MAC3 Register
- **Description:** FIPs Self Test MAC3 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x7c
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

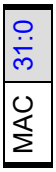


Table 6-1317 Fields for Register: FIPS_SELF_TEST_MAC3

Bits	Name	Memory Access	Description
31:0	MAC	R/W	MAC Value bytes: {12, 13, 14, 15} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.26 FIPS_SELF_TEST_IV0

- **Name:** FIPs Self Test IV0 Register
- **Description:** FIPs Self Test IV0 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0x90
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1318 Fields for Register: FIPS_SELF_TEST_IV0

Bits	Name	Memory Access	Description
31:0	IV	R/W	IV bytes: {00, 01, 02, 03} **Value After Reset:** 0x0 **Exists:** Always

6.2.27 FIPS_SELF_TEST_IV1

- **Name:** FIPs Self Test IV1 Register
- **Description:** FIPs Self Test IV1 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0x94
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1319 Fields for Register: FIPS_SELF_TEST_IV1

Bits	Name	Memory Access	Description
31:0	IV	R/W	IV bytes: {04, 05, 06, 07} **Value After Reset:** 0x0 **Exists:** Always

6.2.28 FIPS_SELF_TEST_IV2

- **Name:** FIPs Self Test IV2 Register
- **Description:** FIPs Self Test IV2 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0x98
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1320 Fields for Register: FIPS_SELF_TEST_IV2

Bits	Name	Memory Access	Description
31:0	IV	R/W	IV bytes: {08, 09, 10, 11} **Value After Reset:** 0x0 **Exists:** Always

6.2.29 FIPS_SELF_TEST_IV3

- **Name:** FIPs Self Test IV3 Register
- **Description:** FIPs Self Test IV3 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0x9c
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1321 Fields for Register: FIPS_SELF_TEST_IV3

Bits	Name	Memory Access	Description
31:0	IV	R/W	IV bytes: {12, 13, 14, 15} **Value After Reset:** 0x0 **Exists:** Always

6.2.30 FIPS_SELF_TEST_IDATA0

- **Name:** FIPs Self Test IDATA0 Register
- **Description:** FIPs Self Test IDATA0 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xa0
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

Table 6-1322 Fields for Register: FIPS_SELF_TEST_IDATA0

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 0, 1, 2, 3} **Value After Reset:** 0x0 **Exists:** Always

6.2.31 FIPS_SELF_TEST_IDATA1

- **Name:** FIPs Self Test IDATA1 Register
- **Description:** FIPs Self Test IDATA1 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xa4
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

Table 6-1323 Fields for Register: FIPS_SELF_TEST_IDATA1

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 4, 5, 6, 7} **Value After Reset:** 0x0 **Exists:** Always

6.2.32 FIPS_SELF_TEST_IDATA2

- **Name:** FIPs Self Test IDATA2 Register
- **Description:** FIPs Self Test IDATA2 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xa8
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

Table 6-1324 Fields for Register: FIPS_SELF_TEST_IDATA2

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 8, 9, 10, 11} **Value After Reset:** 0x0 **Exists:** Always

6.2.33 FIPS_SELF_TEST_IDATA3

- **Name:** FIPs Self Test IDATA3 Register
- **Description:** FIPs Self Test IDATA3 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xac
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

Table 6-1325 Fields for Register: FIPS_SELF_TEST_IDATA3

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 12, 13, 14, 15} **Value After Reset:** 0x0 **Exists:** Always

6.2.34 FIPS_SELF_TEST_IDATA4

- **Name:** FIPs Self Test IDATA4 Register
- **Description:** FIPs Self Test IDATA4 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xb0
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

Table 6-1326 Fields for Register: FIPS_SELF_TEST_IDATA4

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 16, 17, 18, 19} **Value After Reset:** 0x0 **Exists:** Always

6.2.35 FIPS_SELF_TEST_IDATA5

- **Name:** FIPs Self Test IDATA5 Register
- **Description:** FIPs Self Test IDATA5 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xb4
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

Table 6-1327 Fields for Register: FIPS_SELF_TEST_IDATA5

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 20, 21, 22, 23} **Value After Reset:** 0x0 **Exists:** Always

6.2.36 FIPS_SELF_TEST_IDATA6

- **Name:** FIPs Self Test IDATA6 Register
- **Description:** FIPs Self Test IDATA6 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xb8
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

Table 6-1328 Fields for Register: FIPS_SELF_TEST_IDATA6

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 24, 25, 26, 27} **Value After Reset:** 0x0 **Exists:** Always

6.2.37 FIPS_SELF_TEST_IDATA7

- **Name:** FIPs Self Test IDATA7 Register
- **Description:** FIPs Self Test IDATA7 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xbc
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

Table 6-1329 Fields for Register: FIPS_SELF_TEST_IDATA7

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 28, 29, 30, 31} **Value After Reset:** 0x0 **Exists:** Always

6.2.38 FIPS_SELF_TEST_IDATA8

- **Name:** FIPs Self Test IDATA8 Register
- **Description:** FIPs Self Test IDATA8 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xc0
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

Table 6-1330 Fields for Register: FIPS_SELF_TEST_IDATA8

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 32, 33, 34, 35} **Value After Reset:** 0x0 **Exists:** Always

6.2.39 FIPS_SELF_TEST_IDATA9

- **Name:** FIPs Self Test IDATA9 Register
- **Description:** FIPs Self Test IDATA9 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xc4
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

Table 6-1331 Fields for Register: FIPS_SELF_TEST_IDATA9

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 36, 37, 38, 39} **Value After Reset:** 0x0 **Exists:** Always

6.2.40 FIPS_SELF_TEST_IDATA10

- **Name:** FIPs Self Test IDATA10 Register
- **Description:** FIPs Self Test IDATA10 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xc8
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

Table 6-1332 Fields for Register: FIPS_SELF_TEST_IDATA10

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 40, 41, 42, 43} **Value After Reset:** 0x0 **Exists:** Always

6.2.41 FIPS_SELF_TEST_IDATA11

- **Name:** FIPs Self Test IDATA11 Register
- **Description:** FIPs Self Test IDATA11 Register - big endian.
- **Size:** 32 bits
- **Offset:** 0xcc
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

Table 6-1333 Fields for Register: FIPS_SELF_TEST_IDATA11

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 44, 45, 46, 47} **Value After Reset:** 0x0 **Exists:** Always

6.2.42 FIPS_SELF_TEST_IDATA12

- **Name:** FIPs Self Test IDATA12 Register
- **Description:** FIPs Self Test IDATA12 Register - big endian.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0xd0
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)&&(DWC_UAES_CFG_CTR_STREAM_EN==0)

Table 6-1334 Fields for Register: FIPS_SELF_TEST_IDATA12

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 48, 49, 50, 51} **Value After Reset:** 0x0 **Exists:** Always

6.2.43 FIPS_SELF_TEST_IDATA13

- **Name:** FIPs Self Test IDATA13 Register
- **Description:** FIPs Self Test IDATA13 Register - big endian.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0xd4
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)&&(DWC_UAES_CFG_CTR_STREAM_EN==0)

Table 6-1335 Fields for Register: FIPS_SELF_TEST_IDATA13

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 52, 53, 54, 55} **Value After Reset:** 0x0 **Exists:** Always

6.2.44 FIPS_SELF_TEST_IDATA14

- **Name:** FIPs Self Test IDATA14 Register
- **Description:** FIPs Self Test IDATA14 Register - big endian.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0xd8
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)&&(DWC_UAES_CFG_CTR_STREAM_EN==0)

Table 6-1336 Fields for Register: FIPS_SELF_TEST_IDATA14

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 56, 57, 58, 59} **Value After Reset:** 0x0 **Exists:** Always

6.2.45 FIPS_SELF_TEST_IDATA15

- **Name:** FIPs Self Test IDATA15 Register
- **Description:** FIPs Self Test IDATA15 Register - big endian.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0xdc
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)&&(DWC_UAES_CFG_CTR_STREAM_EN==0)

Table 6-1337 Fields for Register: FIPS_SELF_TEST_IDATA15

Bits	Name	Memory Access	Description
31:0	IDATA	R/W	Input Text bytes: { 60, 61, 62, 63} **Value After Reset:** 0x0 **Exists:** Always

6.2.46 FIPS_SELF_TEST_ODATA0

- **Name:** FIPs Self Test ODATA0 Register
- **Description:** FIPs Self Test ODATA0 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1a0
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

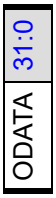


Table 6-1338 Fields for Register: FIPS_SELF_TEST_ODATA0

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 0, 1, 2, 3} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.47 FIPS_SELF_TEST_ODATA1

- **Name:** FIPs Self Test ODATA1 Register
- **Description:** FIPs Self Test ODATA1 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1a4
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

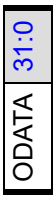


Table 6-1339 Fields for Register: FIPS_SELF_TEST_ODATA1

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 4, 5, 6, 7} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.48 FIPS_SELF_TEST_ODATA2

- **Name:** FIPs Self Test ODATA2 Register
- **Description:** FIPs Self Test ODATA2 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1a8
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

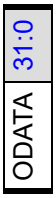


Table 6-1340 Fields for Register: FIPS_SELF_TEST_ODATA2

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 8, 9, 10, 11} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.49 FIPS_SELF_TEST_ODATA3

- **Name:** FIPs Self Test ODATA3 Register
- **Description:** FIPs Self Test ODATA3 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1ac
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=128)

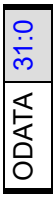


Table 6-1341 Fields for Register: FIPS_SELF_TEST_ODATA3

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 12, 13, 14, 15} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.50 FIPS_SELF_TEST_ODATA4

- **Name:** FIPs Self Test ODATA4 Register
- **Description:** FIPs Self Test ODATA4 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1b0
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

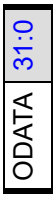


Table 6-1342 Fields for Register: FIPS_SELF_TEST_ODATA4

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 16, 17, 18, 19} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.51 FIPS_SELF_TEST_ODATA5

- **Name:** FIPs Self Test ODATA5 Register
- **Description:** FIPs Self Test ODATA5 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1b4
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

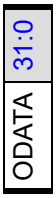


Table 6-1343 Fields for Register: FIPS_SELF_TEST_ODATA5

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 20, 21, 22, 23} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.52 FIPS_SELF_TEST_ODATA6

- **Name:** FIPs Self Test ODATA6 Register
- **Description:** FIPs Self Test ODATA6 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1b8
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

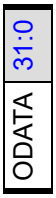


Table 6-1344 Fields for Register: FIPS_SELF_TEST_ODATA6

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 24, 25, 26, 27} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.53 FIPS_SELF_TEST_ODATA7

- **Name:** FIPs Self Test ODATA7 Register
- **Description:** FIPs Self Test ODATA7 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1bc
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=256)

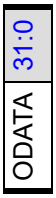


Table 6-1345 Fields for Register: FIPS_SELF_TEST_ODATA7

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 28, 29, 30, 31} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.54 FIPS_SELF_TEST_ODATA8

- **Name:** FIPs Self Test ODATA8 Register
- **Description:** FIPs Self Test ODATA8 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1c0
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

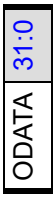


Table 6-1346 Fields for Register: FIPS_SELF_TEST_ODATA8

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 32, 33, 34, 35} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.55 FIPS_SELF_TEST_ODATA9

- **Name:** FIPs Self Test ODATA9 Register
- **Description:** FIPs Self Test ODATA9 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1c4
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

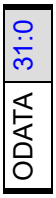


Table 6-1347 Fields for Register: FIPS_SELF_TEST_ODATA9

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 36, 37, 38, 39} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.56 FIPS_SELF_TEST_ODATA10

- **Name:** FIPs Self Test ODATA10 Register
- **Description:** FIPs Self Test ODATA10 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1c8
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

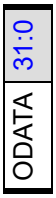


Table 6-1348 Fields for Register: FIPS_SELF_TEST_ODATA10

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 40, 41, 42, 43} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.57 FIPS_SELF_TEST_ODATA11

- **Name:** FIPs Self Test ODATA11 Register
- **Description:** FIPs Self Test ODATA11 Register - big endian ordering.
- **Size:** 32 bits
- **Offset:** 0x1cc
- **Exists:** (DWC_MACSEC_AES_FIPS_EN==1)&&(DWC_MACSEC_AES_DP_WIDTH>=384)

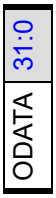


Table 6-1349 Fields for Register: FIPS_SELF_TEST_ODATA11

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 44, 45, 46, 47} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.58 FIPS_SELF_TEST_ODATA12

- **Name:** FIPs Self Test ODATA12 Register
- **Description:** FIPs Self Test ODATA12 Register - big endian ordering.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0x1d0
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)

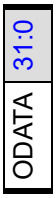


Table 6-1350 Fields for Register: FIPS_SELF_TEST_ODATA12

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 48, 49, 50, 51} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.59 FIPS_SELF_TEST_ODATA13

- **Name:** FIPs Self Test ODATA13 Register
- **Description:** FIPs Self Test ODATA13 Register - big endian ordering.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0x1d4
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)

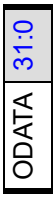


Table 6-1351 Fields for Register: FIPS_SELF_TEST_ODATA13

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 52, 53, 54, 55} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.60 FIPS_SELF_TEST_ODATA14

- **Name:** FIPs Self Test ODATA14 Register
- **Description:** FIPs Self Test ODATA14 Register - big endian ordering.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0x1d8
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)

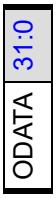


Table 6-1352 Fields for Register: FIPS_SELF_TEST_ODATA14

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 56, 57, 58, 59} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.61 FIPS_SELF_TEST_ODATA15

- **Name:** FIPs Self Test ODATA15 Register
- **Description:** FIPs Self Test ODATA15 Register - big endian ordering.

  Reset Domain: pwr_on_rst
- **Size:** 32 bits
- **Offset:** 0x1dc
- **Exists:** (DWC_UAES_CFG_FIPS_SELF_TEST_INT_EN==1)&&(DWC_UAES_CFG_DP_WIDTH>=512)&&(DWC_UAES_CFG_FIPS_TEST_ALL_BLOCK)

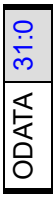


Table 6-1353 Fields for Register: FIPS_SELF_TEST_ODATA15

Bits	Name	Memory Access	Description
31:0	ODATA	R/W	Input Text bytes: { 60, 61, 62, 63} **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.62 FIPS_SELF_TEST_VECT_CTL

- **Name:** FIPs Self Test Vector Control Register
- **Description:** FIPs Self Test Vector Control Register
- **Size:** 32 bits
- **Offset:** 0x2a0
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

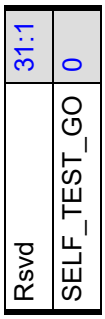


Table 6-1354 Fields for Register: FIPS_SELF_TEST_VECT_CTL

Bits	Name	Memory Access	Description
31:1			**Reserved Field:** Yes
0	SELF_TEST_GO	R/W	Self Clearing field that launches the vector processing. Prior to launching the self test vector, all FIPS self test configuration must be set including the self test keys which are programmed outside of the FIPS register space. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6.2.63 BIST_VECT_MODE

- **Name:** BIST Vector Mode Register
- **Description:** BIST Vector Mode Register
- **Size:** 32 bits
- **Offset:** 0x2a4
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

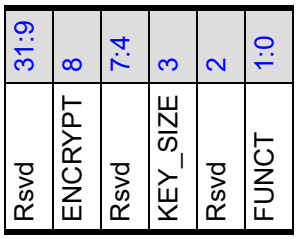


Table 6-1355 Fields for Register: BIST_VECT_MODE

Bits	Name	Memory Access	Description
31:9			**Reserved Field:** Yes
8	ENCRYPT	R/W	Sets the operation mode: ■ 0: Decryption only vectors are run. ■ 1: Encryption only vectors are run. **Value After Reset:** 0x0 **Exists:** Always
7:4			**Reserved Field:** Yes
3	KEY_SIZE	R/W	Size of the key: ■ 0: 128-bit only vectors are run. ■ 1: 256-bit only vectors are run. **Value After Reset:** 0x0 **Exists:** ((DWC_MACSEC_MAX_KEY_SIZE == 1) ? 1 :0)
2			**Reserved Field:** Yes

Table 6-1355 Fields for Register: BIST_VECT_MODE (Continued)

Bits	Name	Memory Access	Description
1:0	FUNCT	R/W	The function has this meaning: ■ 0: Batch mode. All vectors supported by the hardware configuration are run. ■ 1: CTR test is run. ■ 2: GCM test is run. ■ 3: ECB test is run. **Value After Reset:** 0x0 **Exists:** Always

6.2.64 BIST_VECT_ERR_INJ

- **Name:** BIST Vector Error Injection Register
- **Description:** BIST Vector Error Injection Register
- **Size:** 32 bits
- **Offset:** 0x2a8
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

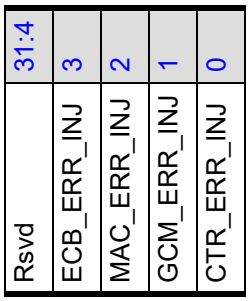


Table 6-1356 Fields for Register: BIST_VECT_ERR_INJ

Bits	Name	Memory Access	Description
31:4			**Reserved Field:** Yes
3	ECB_ERR_INJ	R/W	When set to 1, it forces a bit flip in the injected error to cause a test failure ECB test. **Value After Reset:** 0x0 **Exists:** Always
2	MAC_ERR_INJ	R/W	When set to 1, it forces a bit flip in the injected error to cause a test failure due to MAC Mismatch during the GCM test. **Value After Reset:** 0x0 **Exists:** Always
1	GCM_ERR_INJ	R/W	When set to 1, it forces a bit flip in the injected error to cause a test failure during the GCM test. **Value After Reset:** 0x0 **Exists:** Always
0	CTR_ERR_INJ	R/W	When set to 1, it forces a bit flip in the injected error to cause a test failure during the CTR test. **Value After Reset:** 0x0 **Exists:** Always

6.2.65 BIST_VECT_CTL

- **Name:** BIST Test Vector Control Register
- **Description:** BIST Test Vector Control Register
- **Size:** 32 bits
- **Offset:** 0x2ac
- **Exists:** DWC_MACSEC_AES_FIPS_EN==1

Table 6-1357 Fields for Register: BIST_VECT_CTL

Bits	Name	Memory Access	Description
31:1			**Reserved Field:** Yes
0	BIST_TEST_GO	R/W	This Self Clearing field launches the BIST. Before launching the BIST vector, all BIST test configuration must be set. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

A Acronyms

A.1 List of Acronyms

Table A-1 Acronyms

Acronym	Description
AAD	Additional Authentication Data
AEAD	Authenticated Encryption with Associated Data
AMBA	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus
CTX	ConTeXt
CCTX	Cipher ConTeXt
DP	Dual-Port
CTR	CounTeR Mode
DPA	Differential Power Analysis
DWC	Design Ware Cores
ECB	Electronic Code Book
EOP	End of Packet
FIFO	First-In, First-Out
FIPS	Federal Information Processing Standard (United States)
GCM	Galois-Counter Mode
GF	Galois Field
GUI	Graphical User Interface
HCTX	Hash ConTeXt

Table A-1 Acronyms (Continued)

Acronym	Description
IV	Initialization Vector
ICV	Integrity Check Value
MAC	Message Authentication Code
MSG	Message
NIST	National Institute of Standards and Technology (United States)
SKP	Secure Key Port
SKS	Secure Key Server
SOP	Start of Packet
SP	Single-Port
TP	Two-Port